A simple model for traffic jams

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School of Mathematics

Open Day, University of Bristol, 13 September, 2014

Traffic jams
- Arriving to a traffic jam
- Leaving a traffic jam

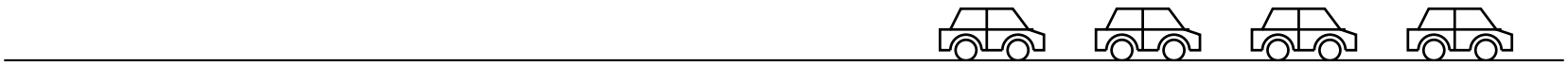
Being ageless

Totally Asymmetric Simple Exclusion Process
- Stationary distribution
- The infinite model

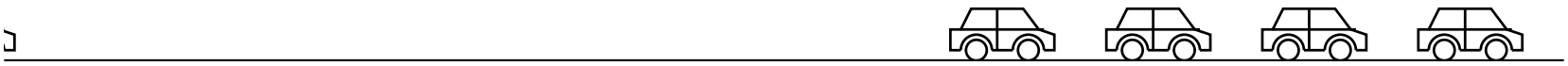
On large scales
- Start of the traffic jam
- End of the traffic jam

Remarks

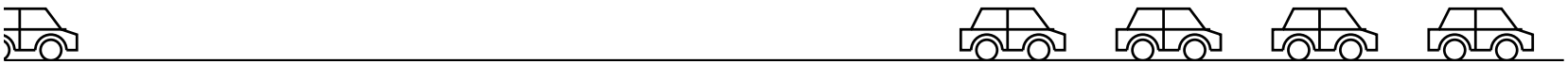
Arriving to a traffic jam



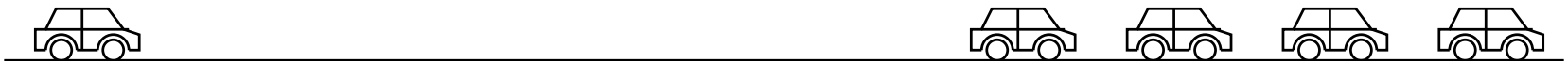
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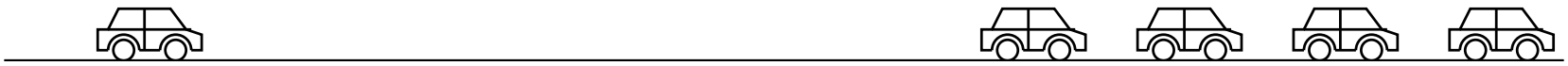
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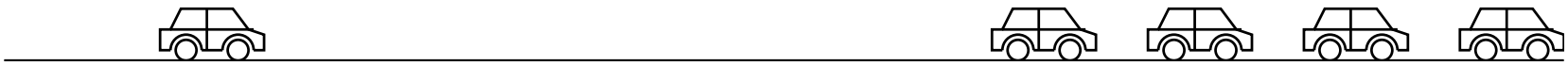
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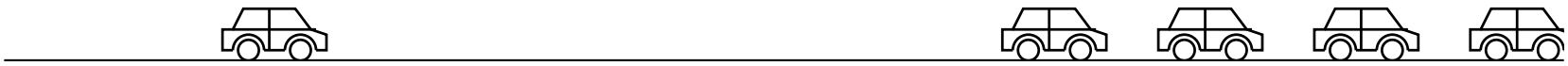
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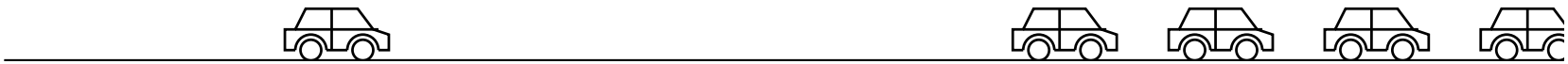
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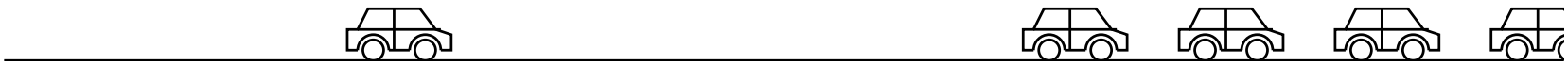
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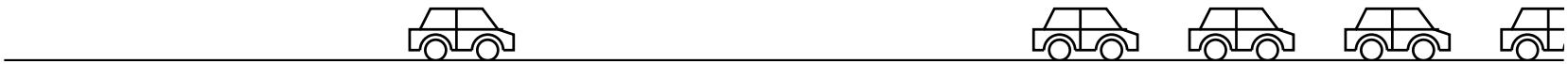
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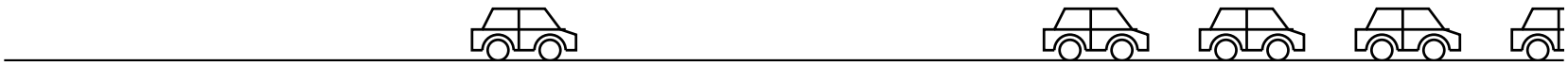
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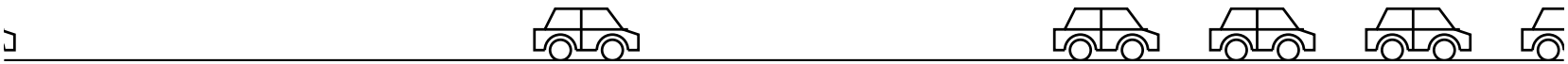
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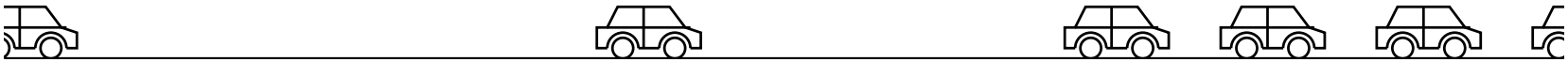
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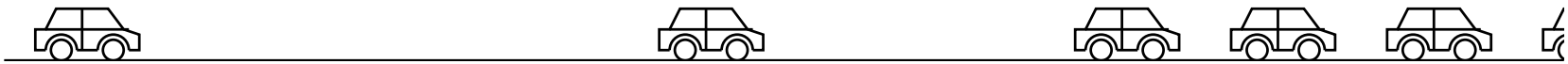
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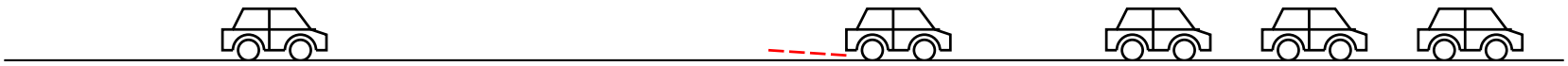
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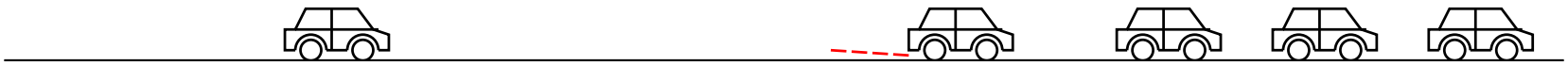
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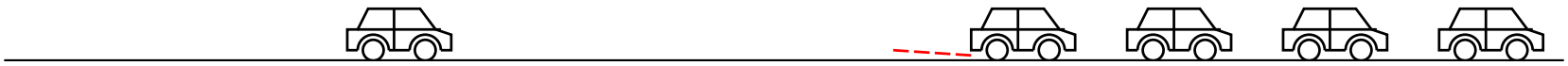
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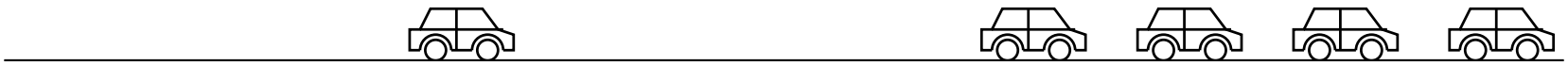
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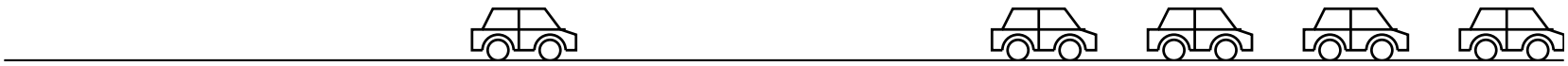
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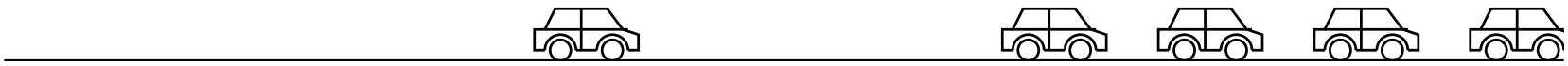
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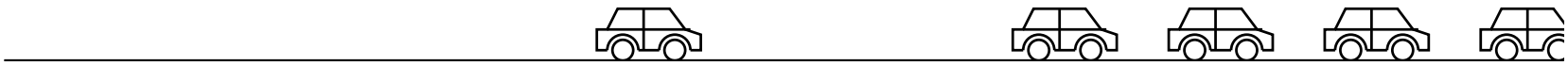
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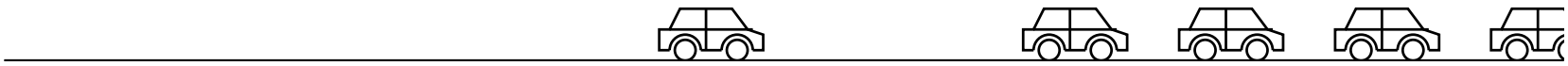
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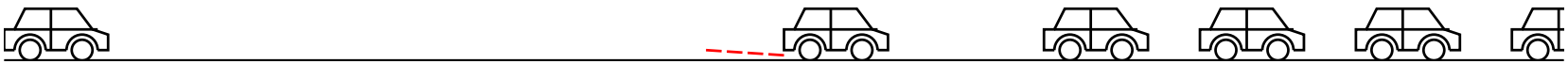
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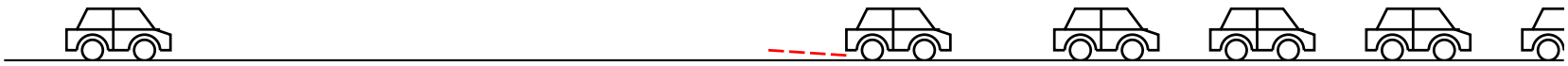
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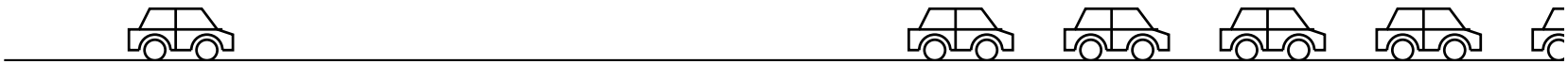
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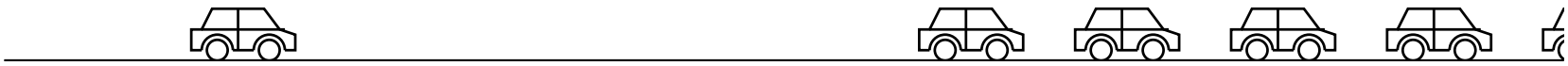
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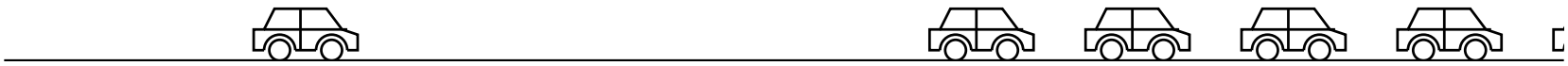
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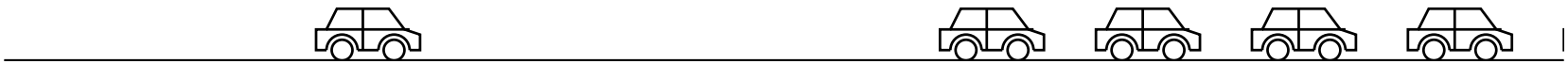
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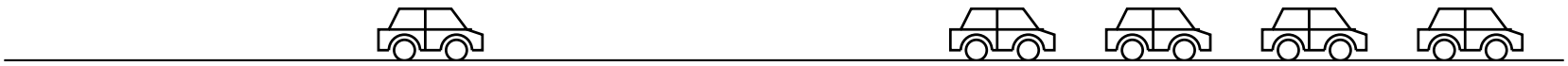
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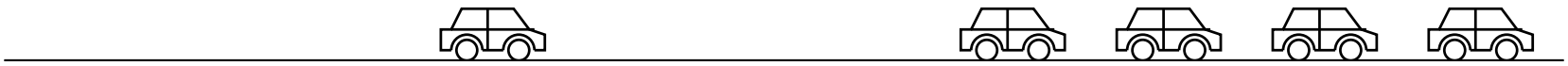
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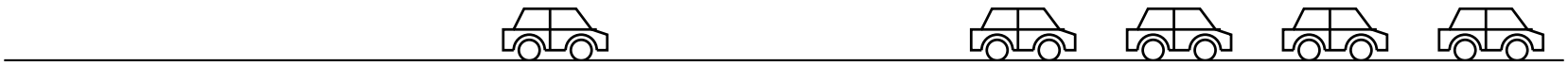
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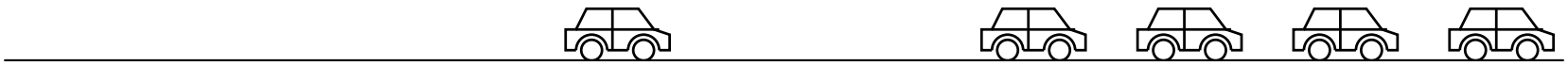
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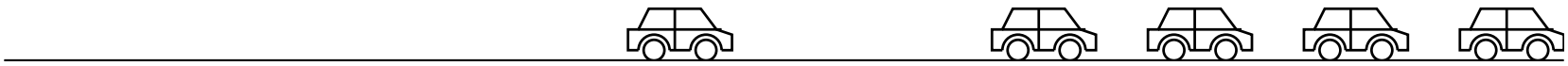
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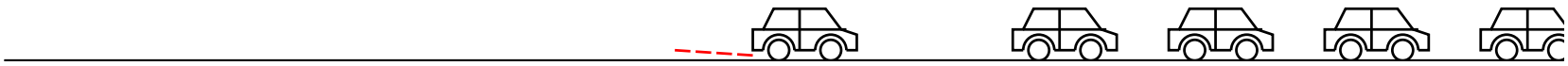
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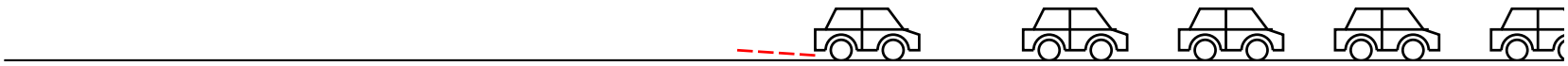
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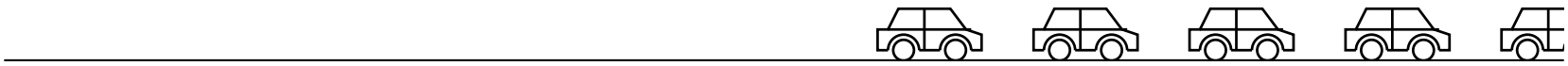
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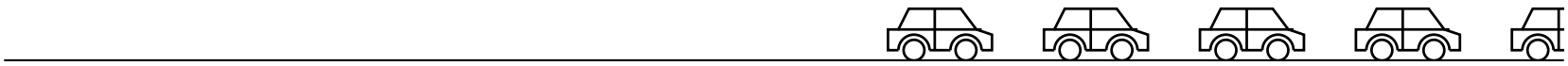
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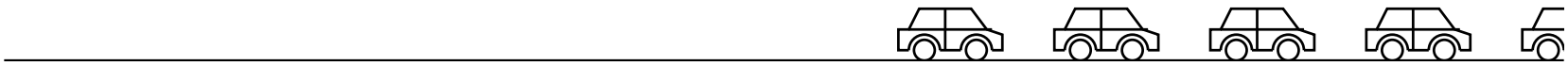
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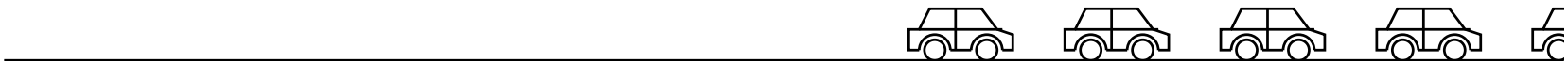
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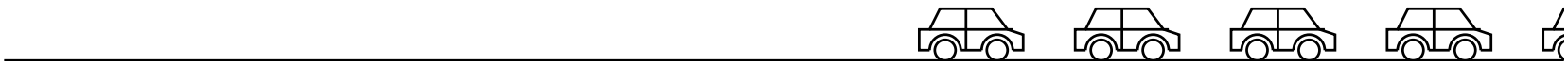
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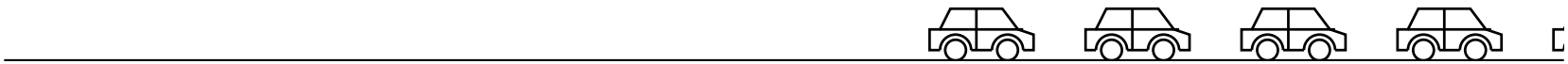
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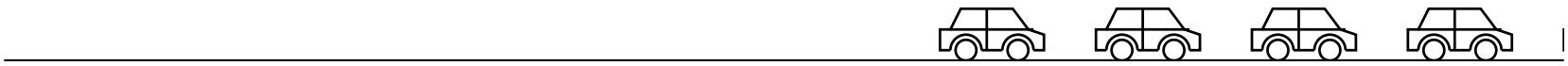
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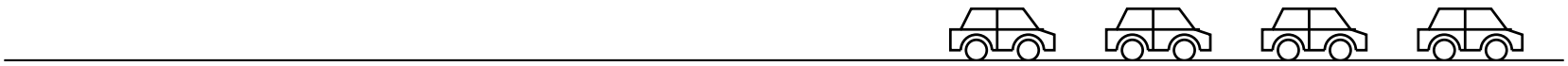
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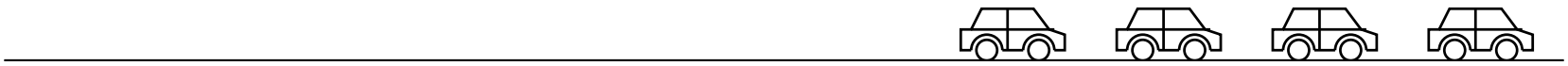
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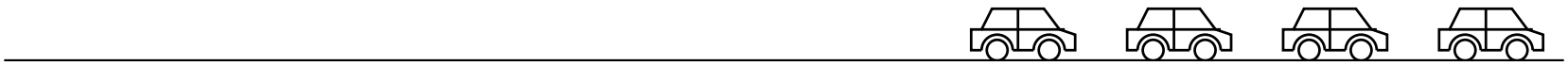
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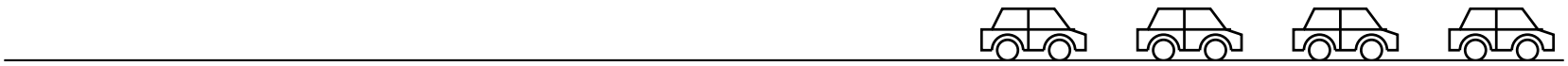
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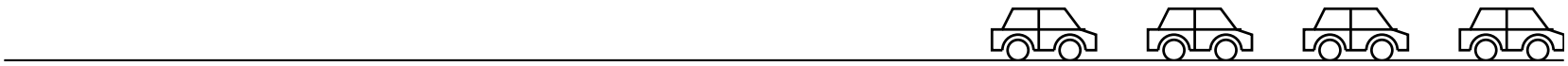
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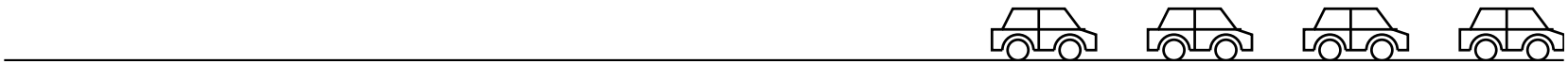
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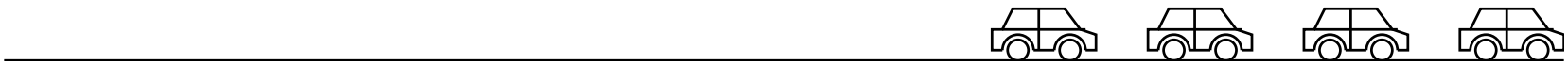
Arriving to a traffic jam



We notice the slow cars $\rightsquigarrow$ strong braking immediately.

Arriving to a traffic jam is always sharp.

Arriving to a traffic jam

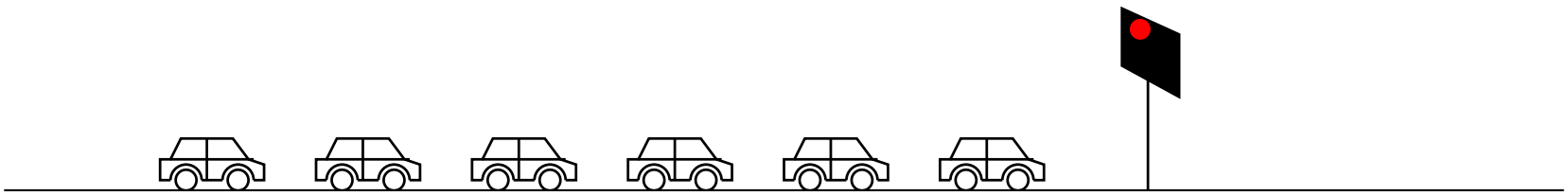


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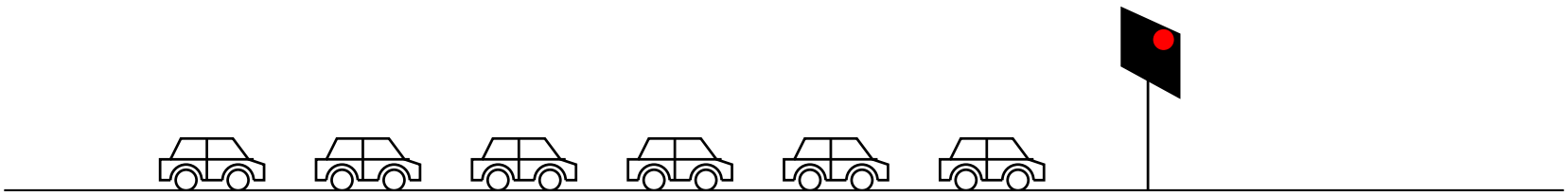
Arriving to a traffic jam is always sharp.

This is one aspect that makes motorways dangerous places.

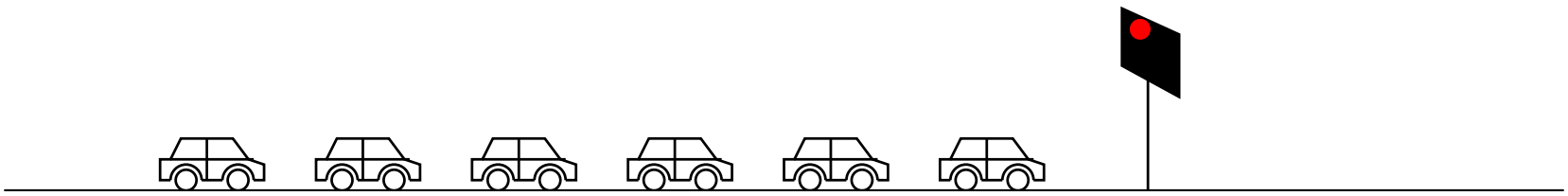
Leaving a traffic jam



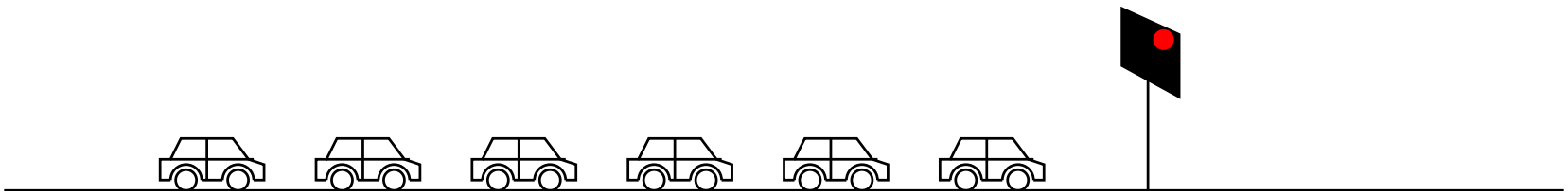
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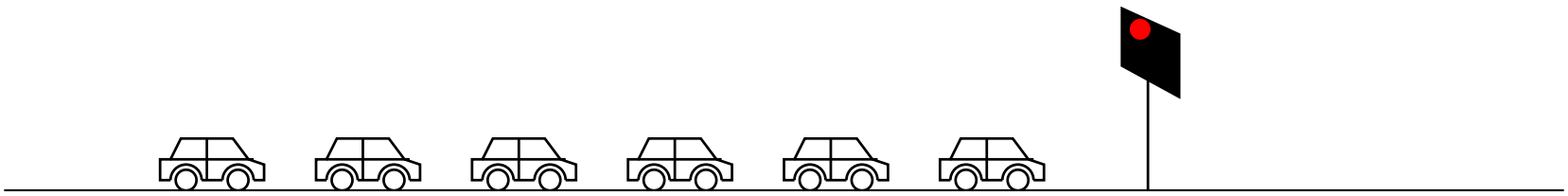
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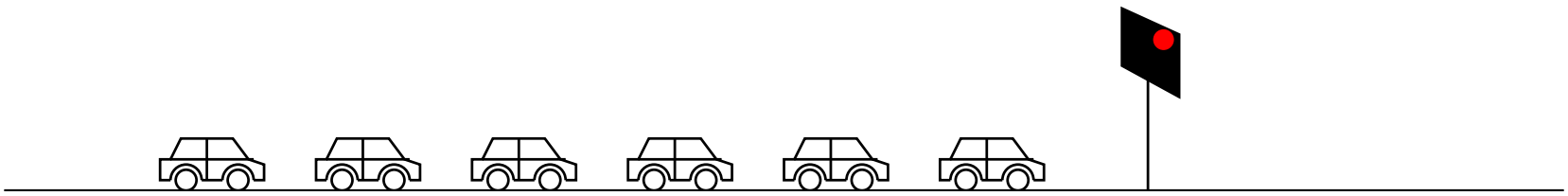
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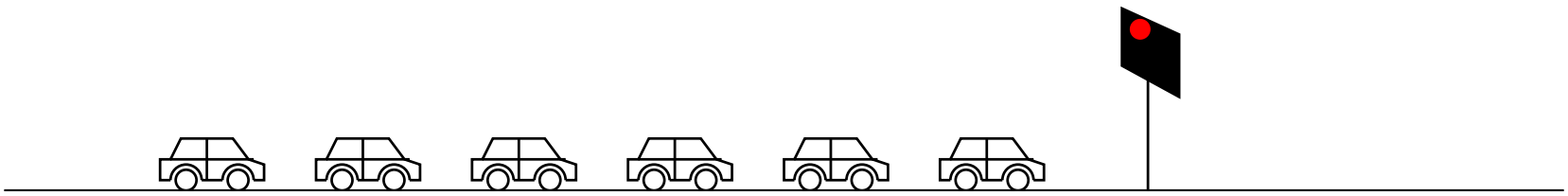
Leaving a traffic jam



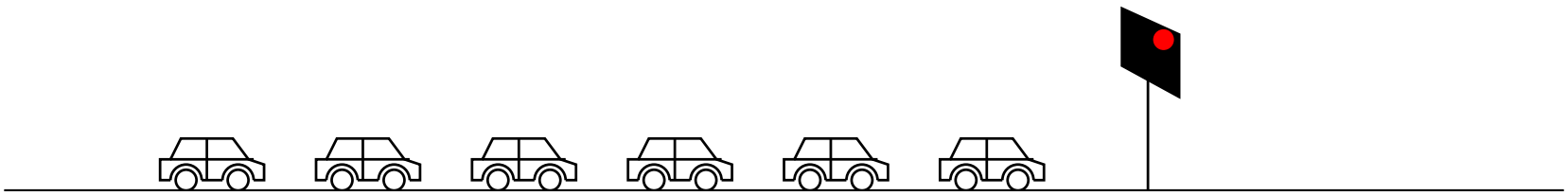
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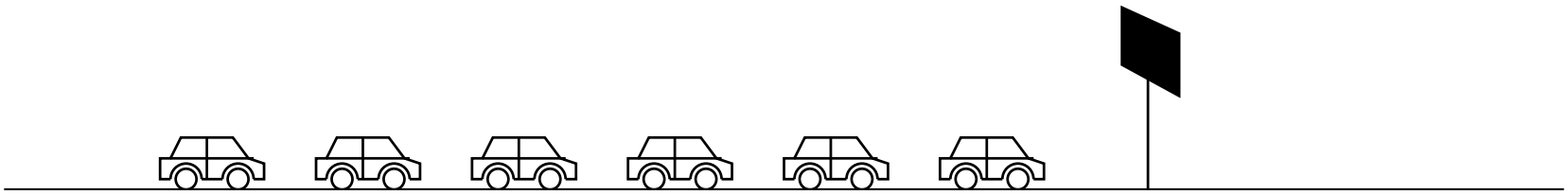
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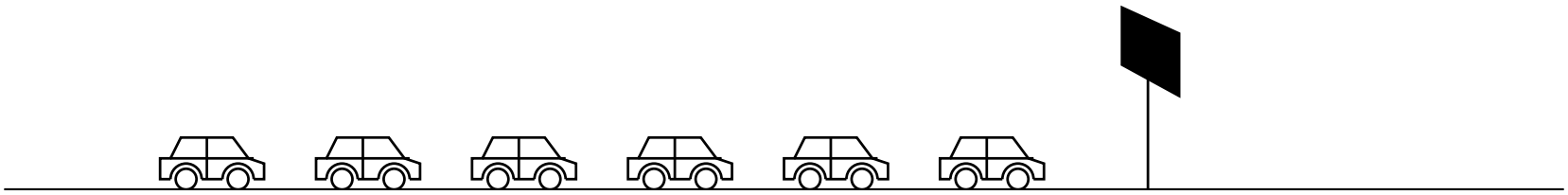
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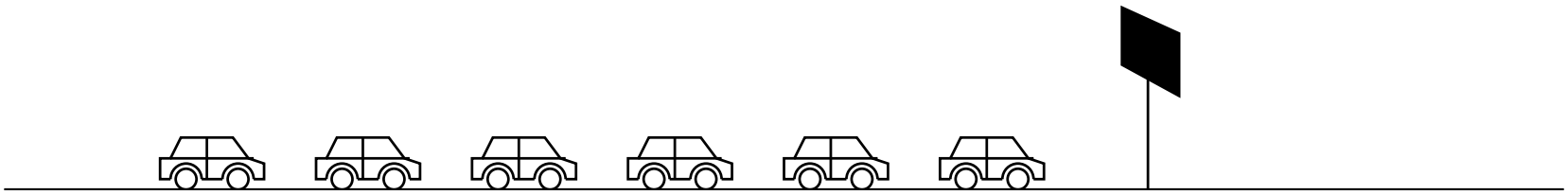
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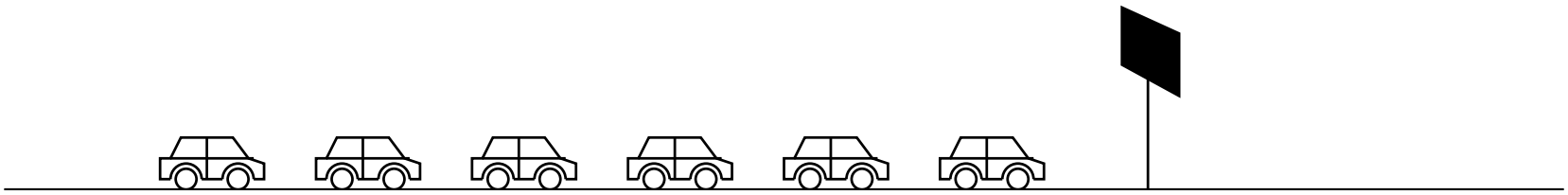
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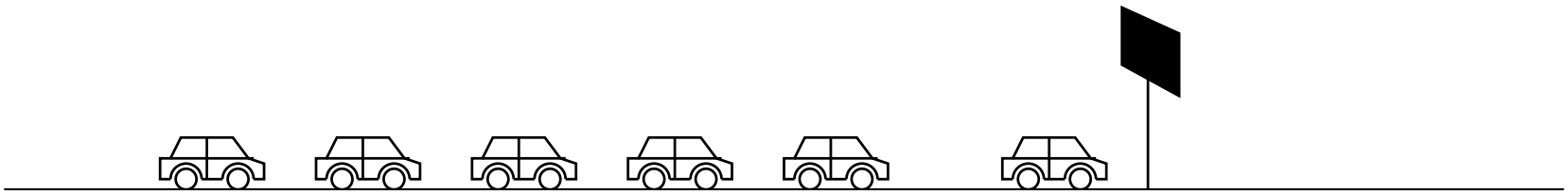
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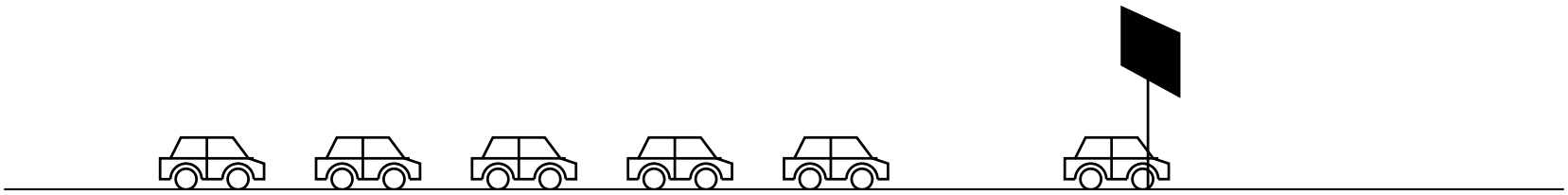
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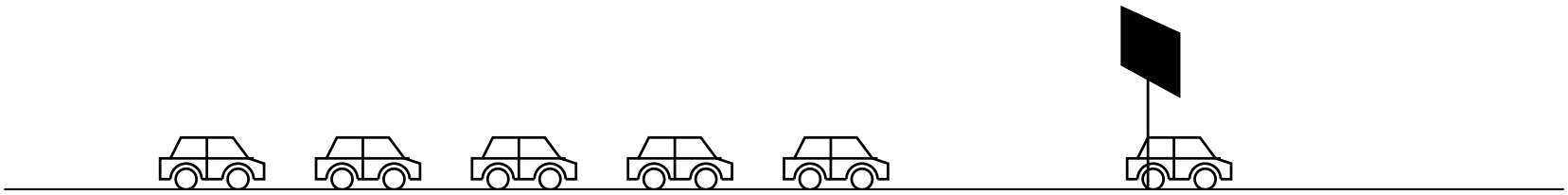
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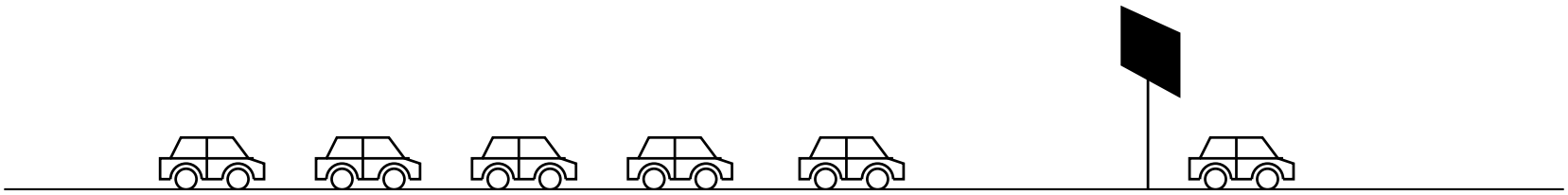
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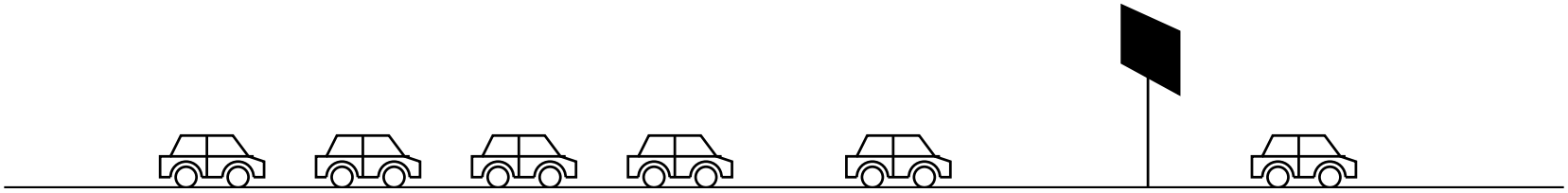
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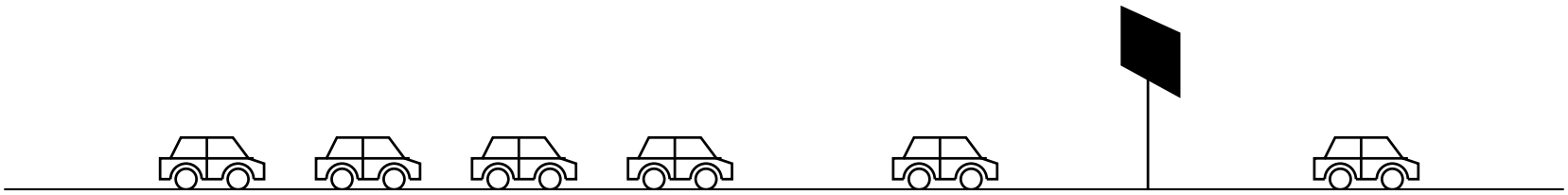
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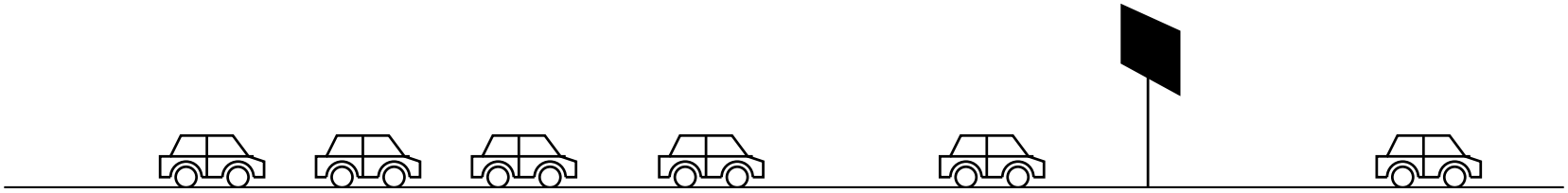
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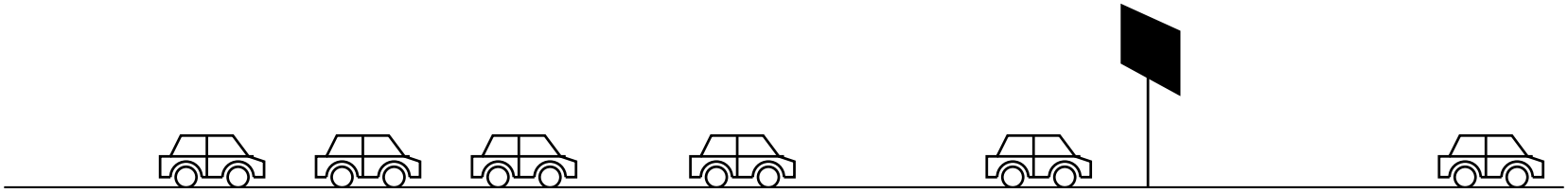
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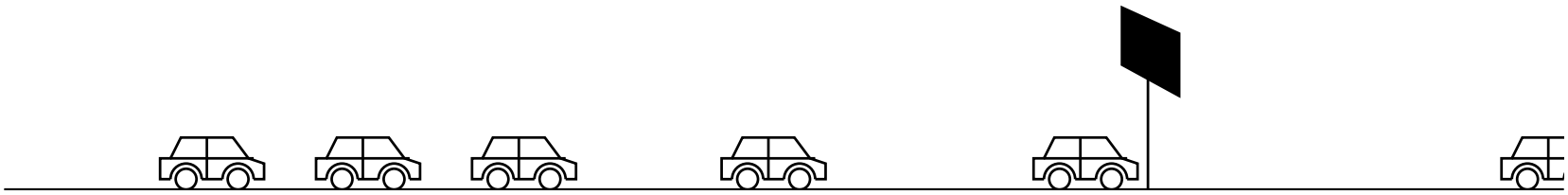
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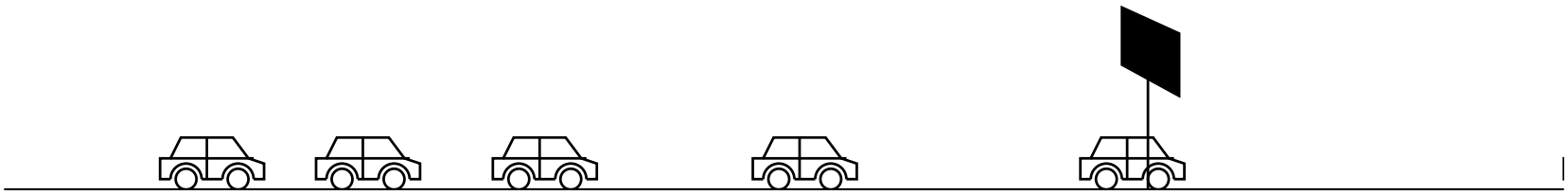
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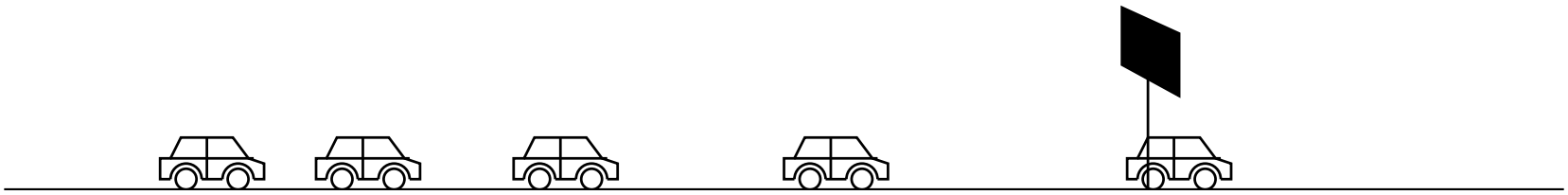
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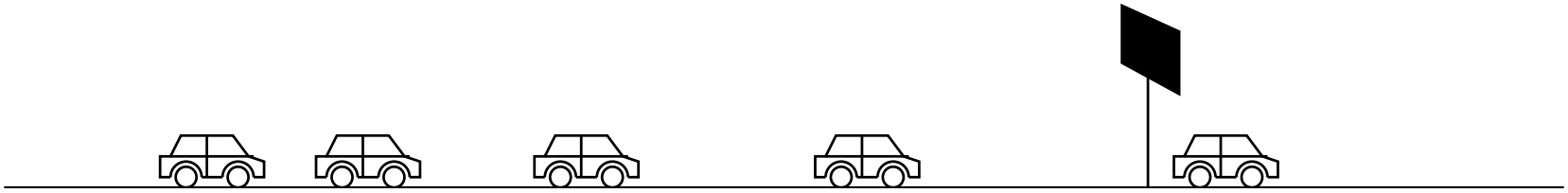
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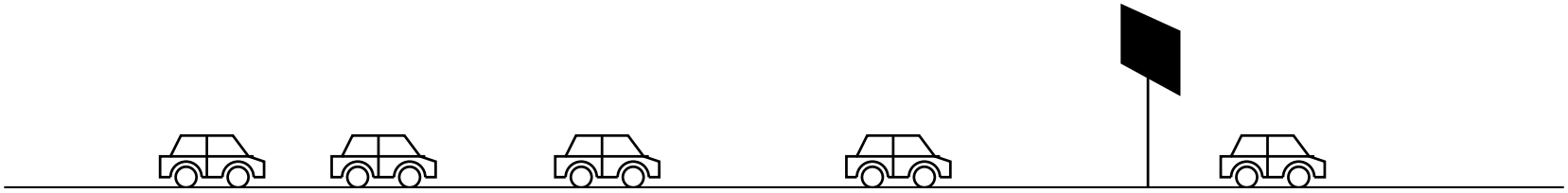
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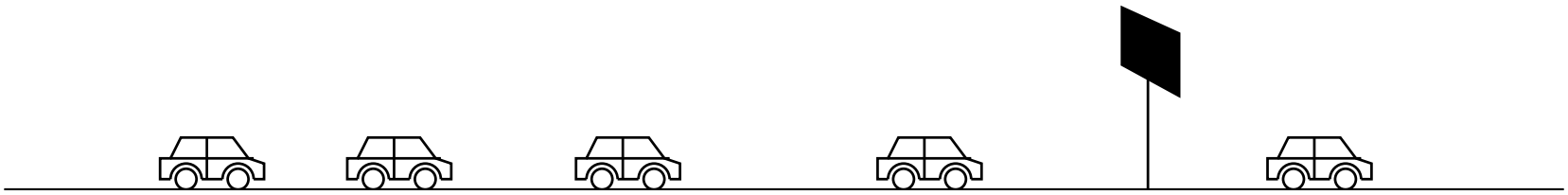
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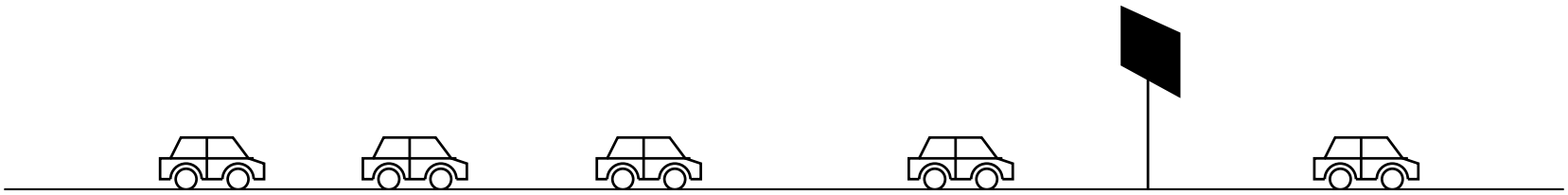
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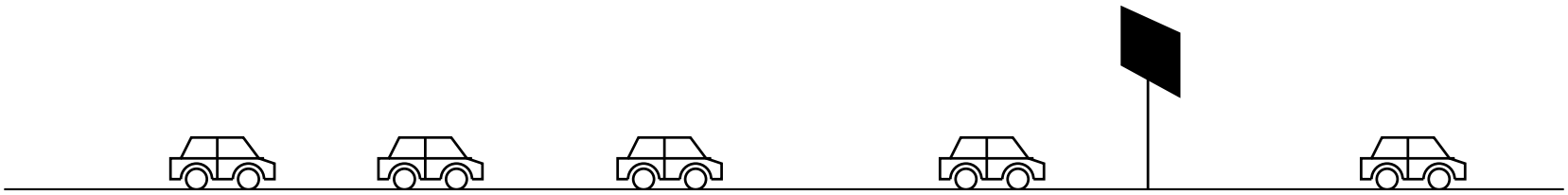
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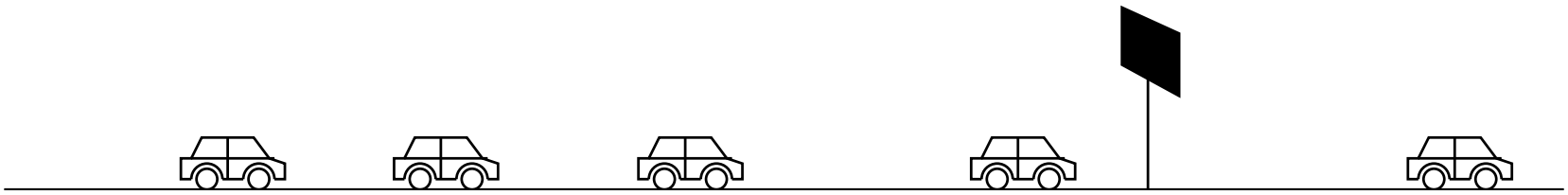
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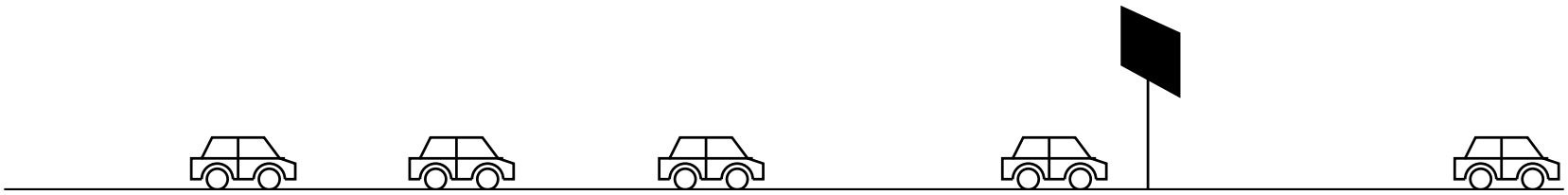
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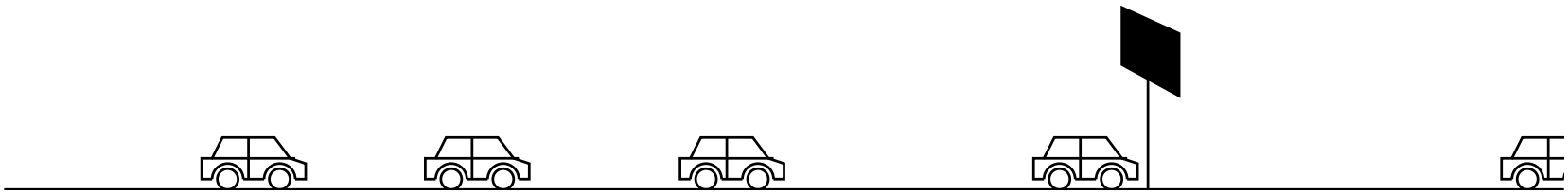
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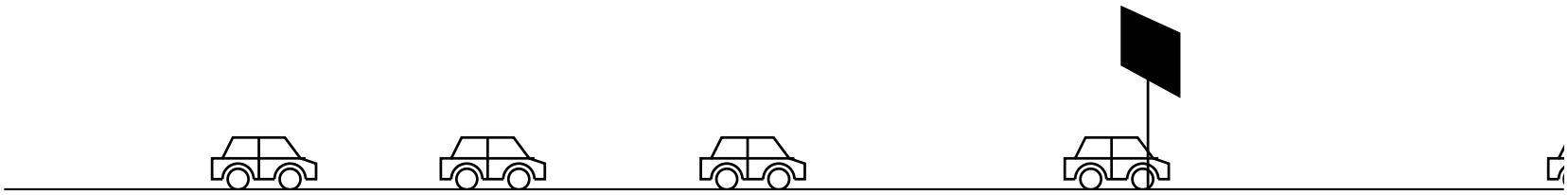
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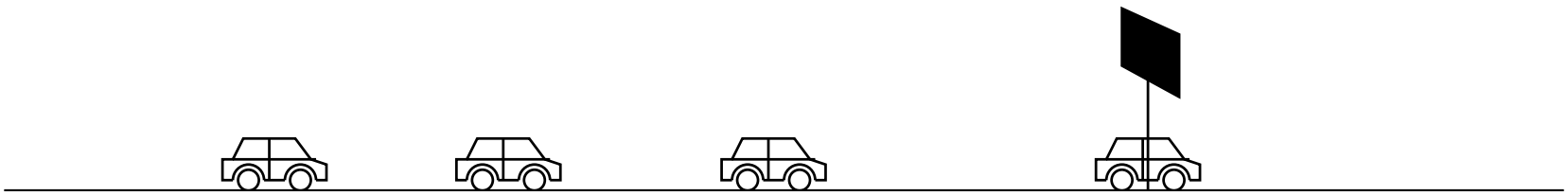
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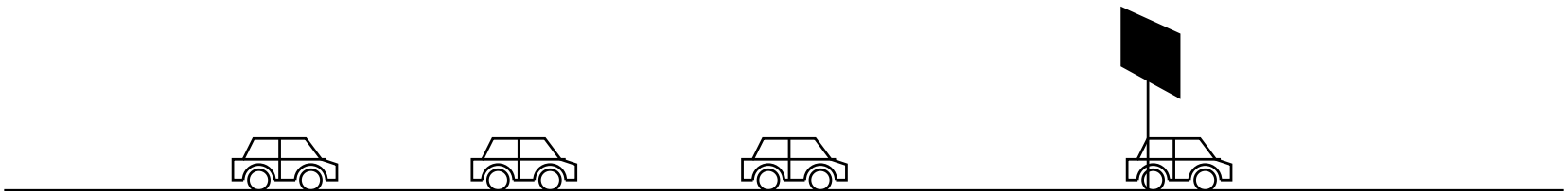
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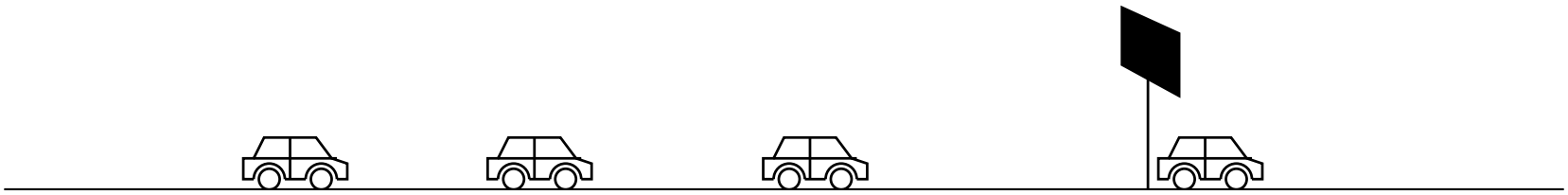
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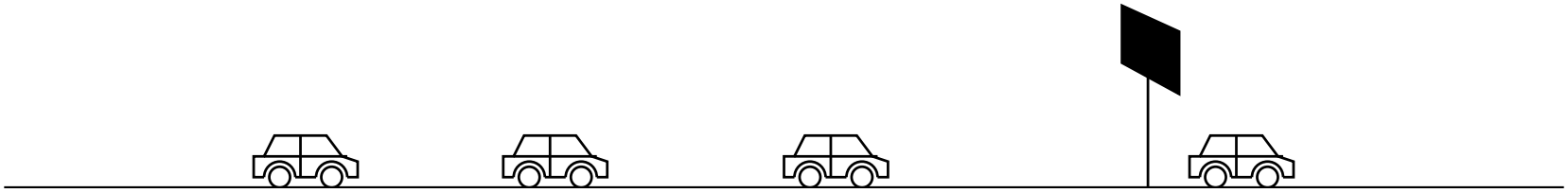
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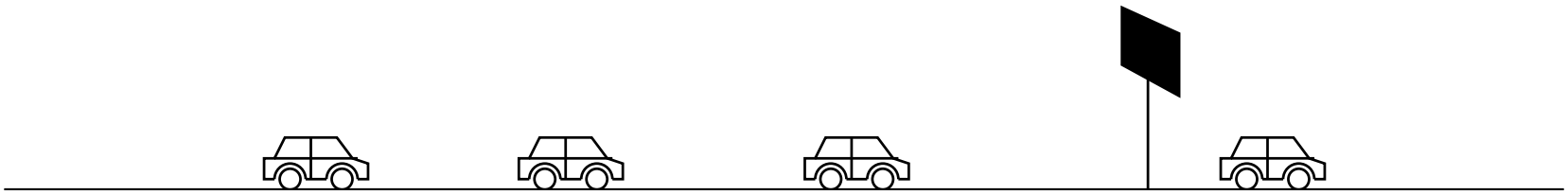
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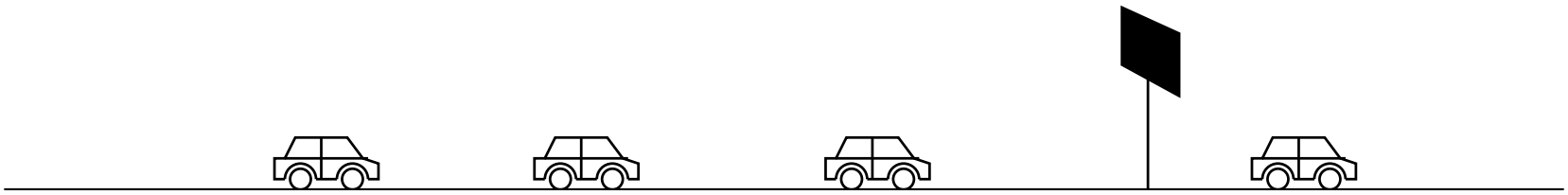
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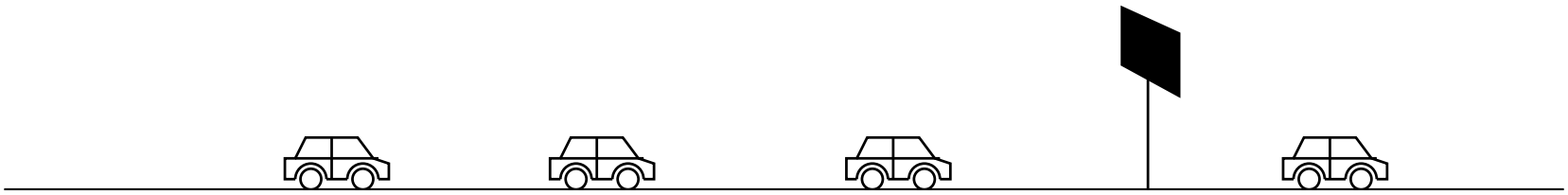
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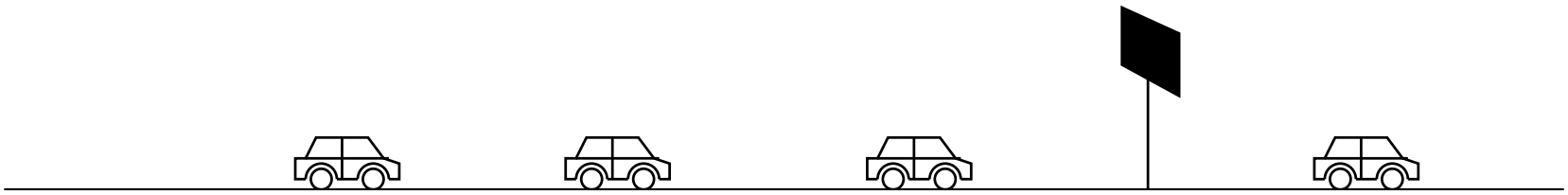
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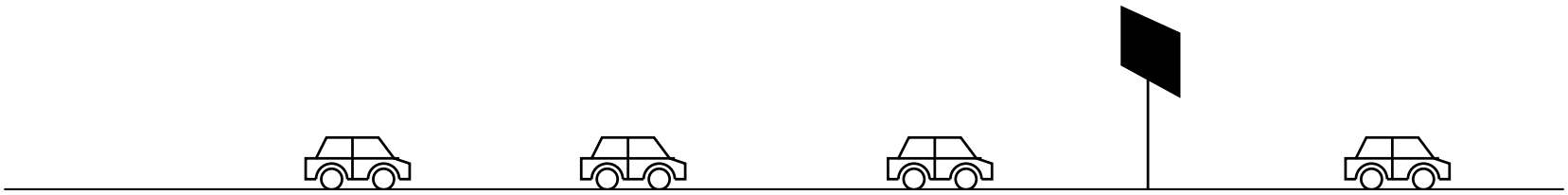
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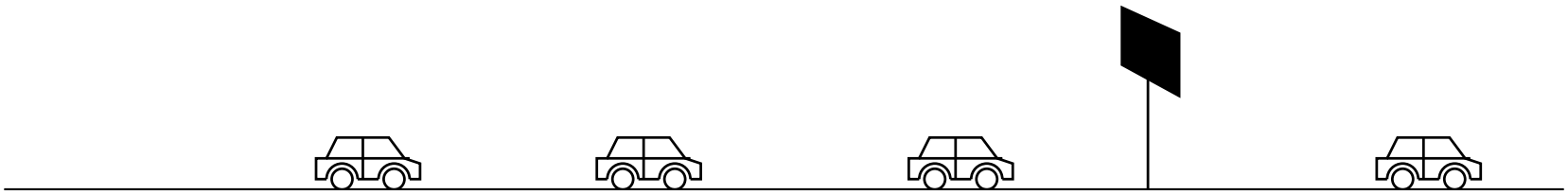
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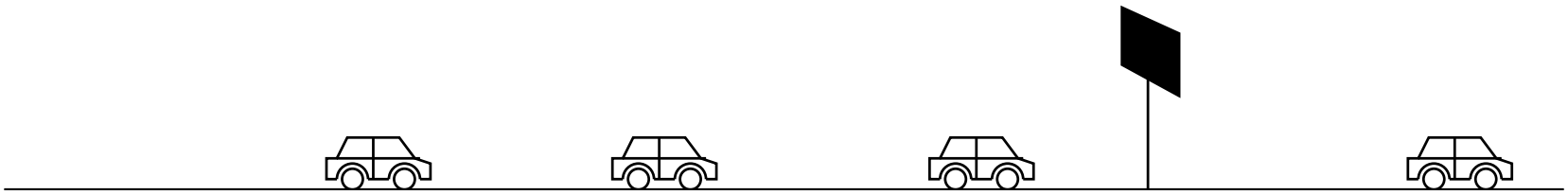
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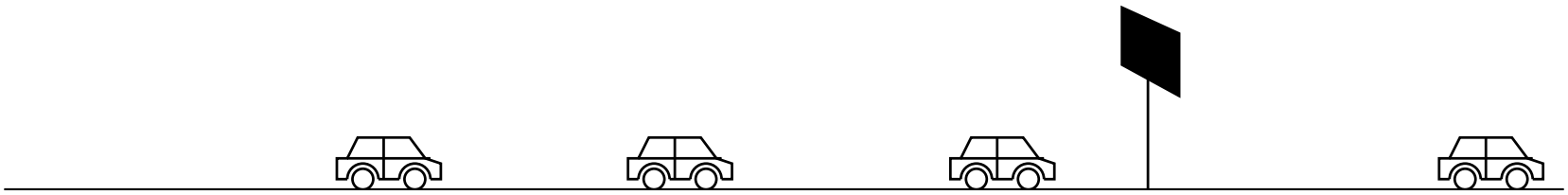
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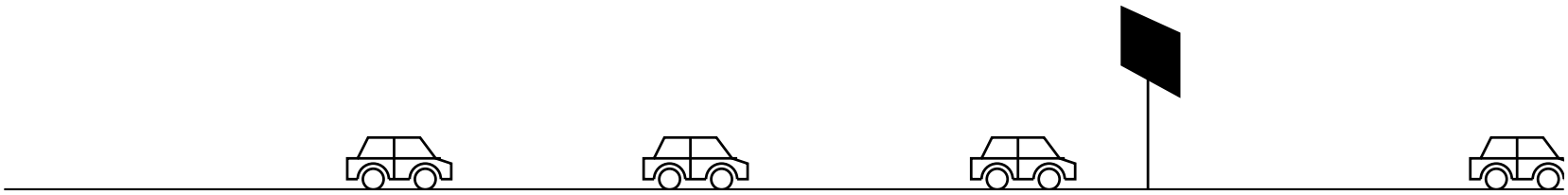
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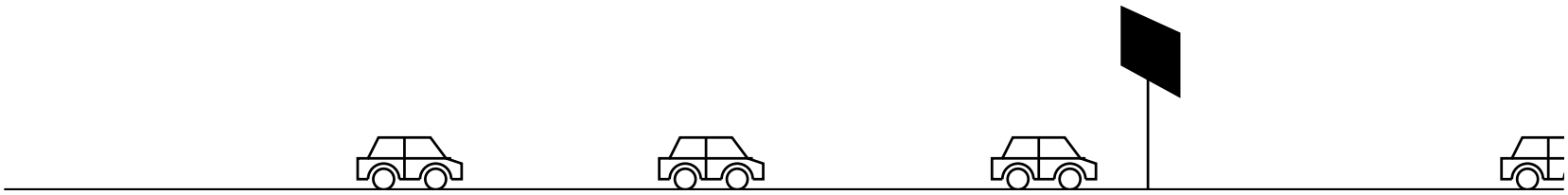
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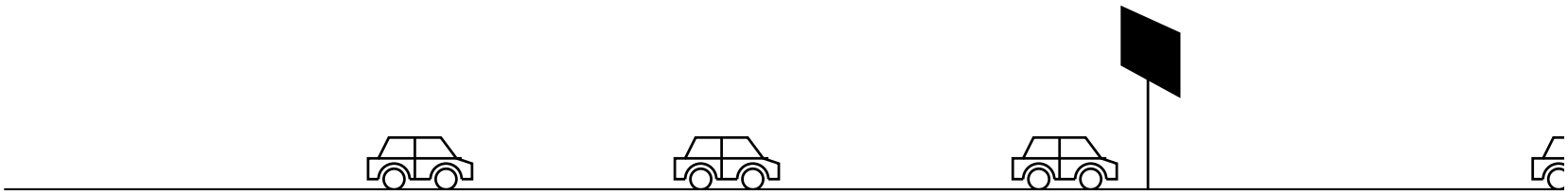
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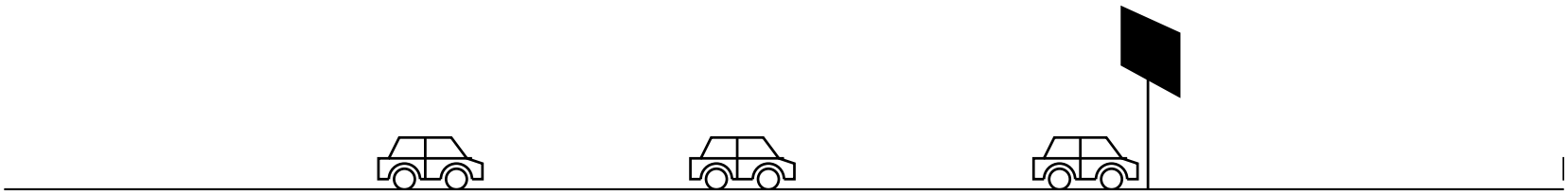
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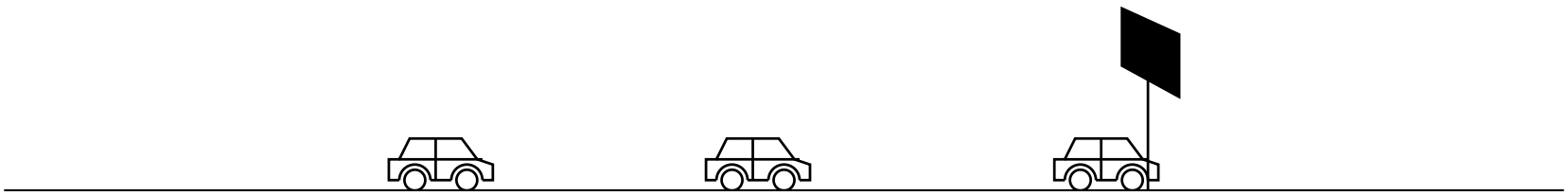
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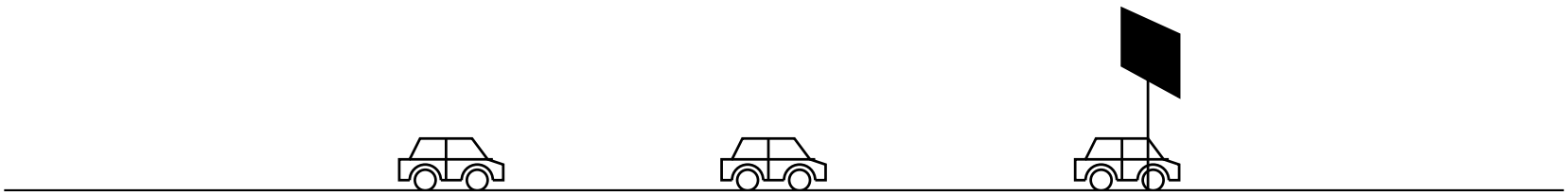
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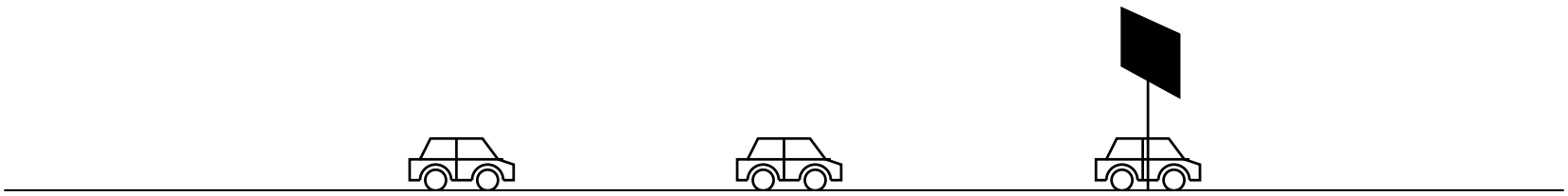
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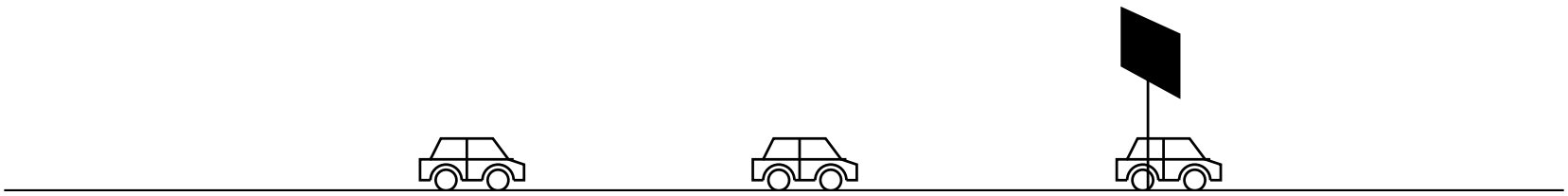
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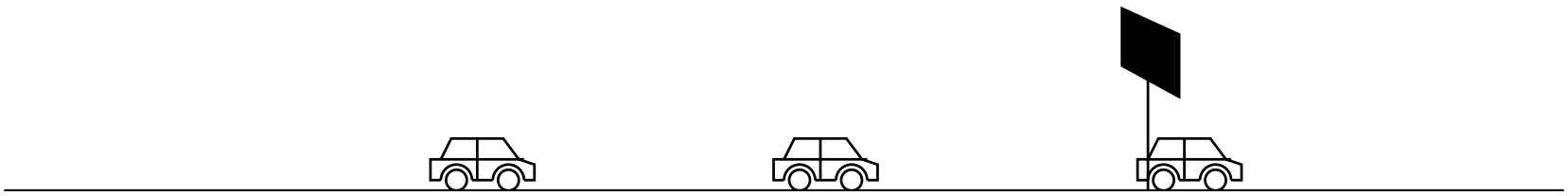
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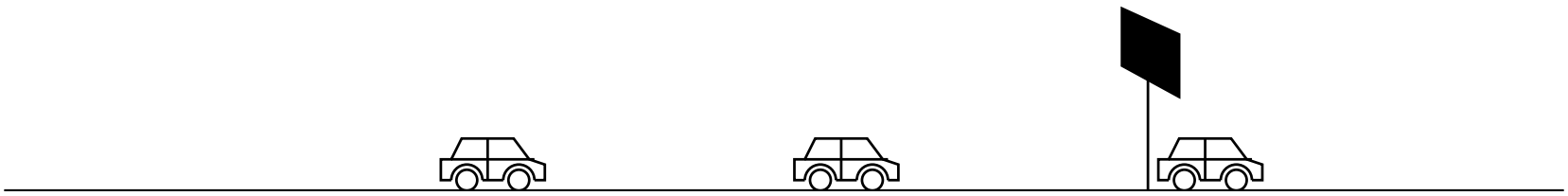
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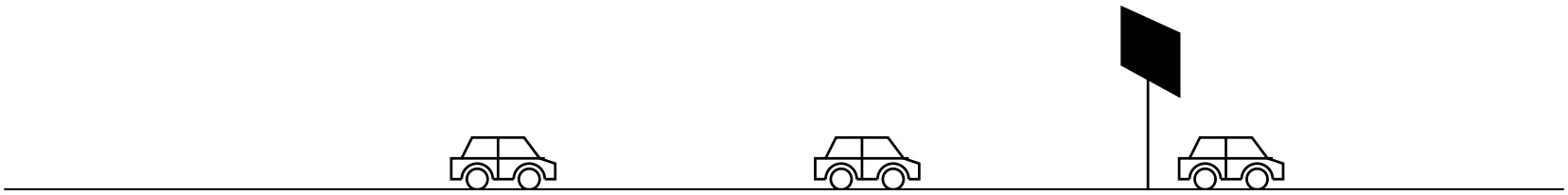
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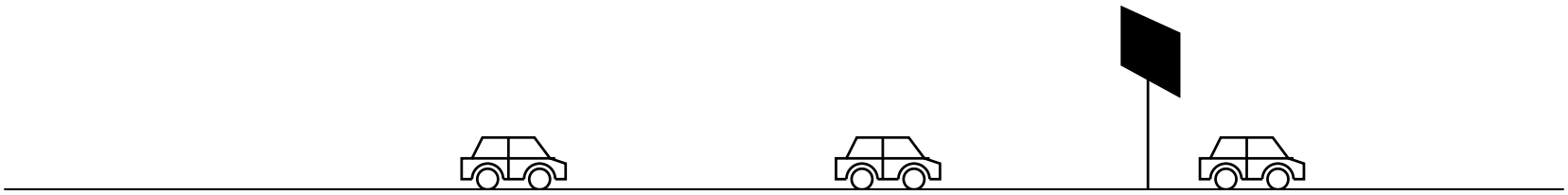
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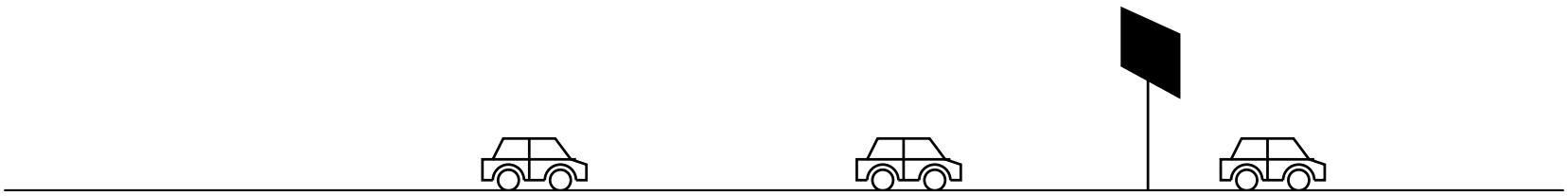
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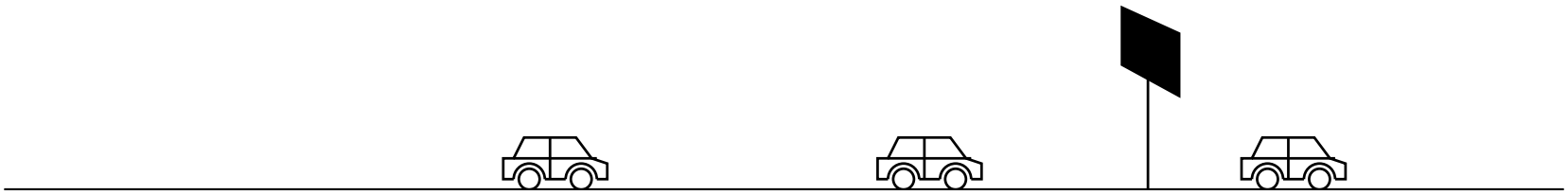
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Leaving a traffic jam



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Leaving a traffic jam



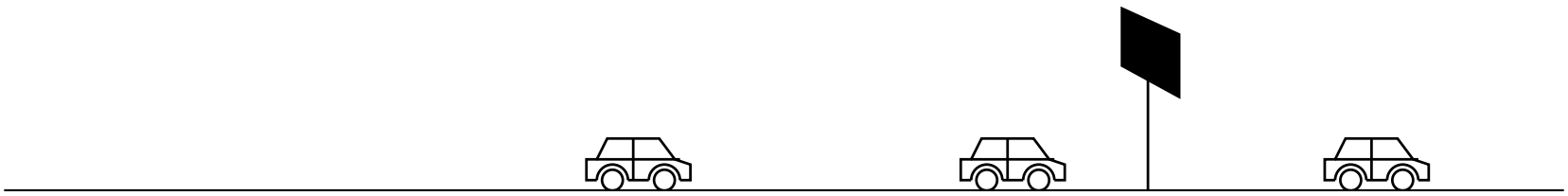
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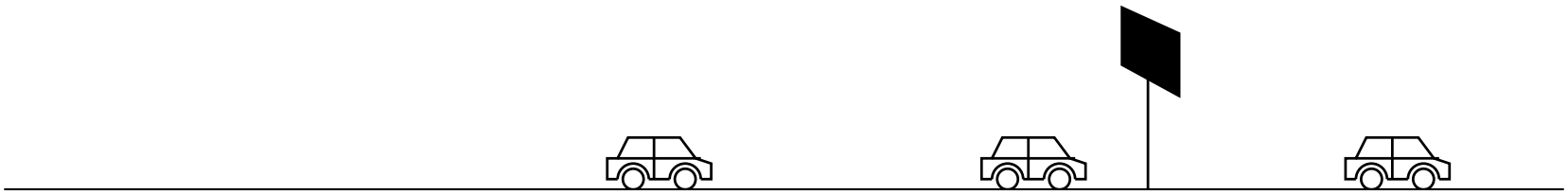
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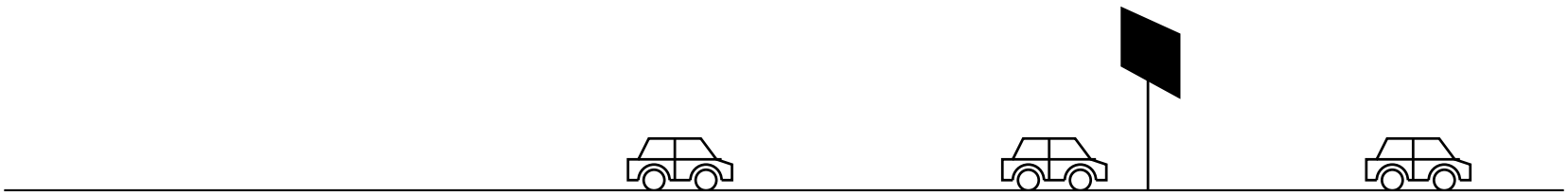
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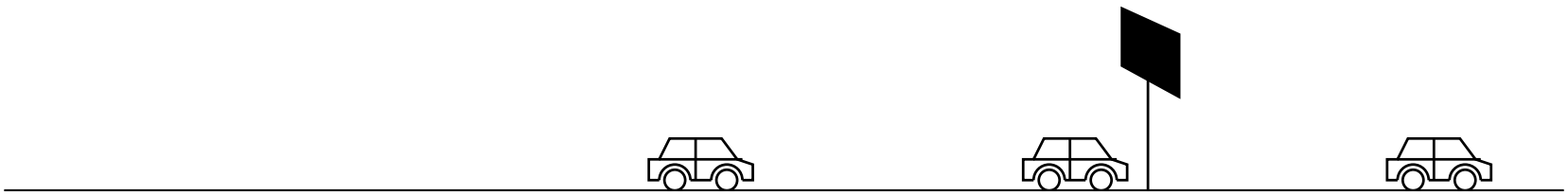
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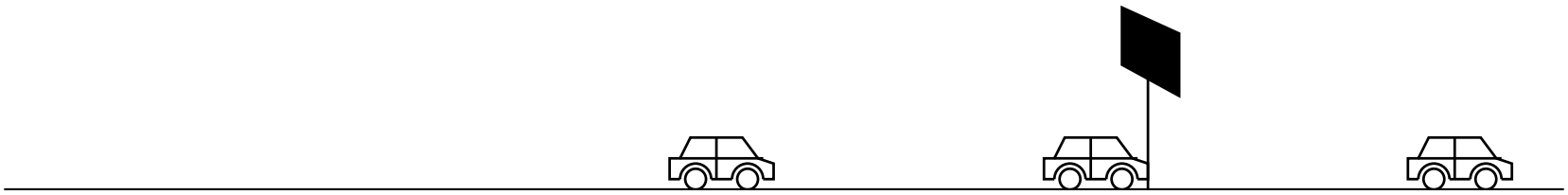
Leaving a traffic jam



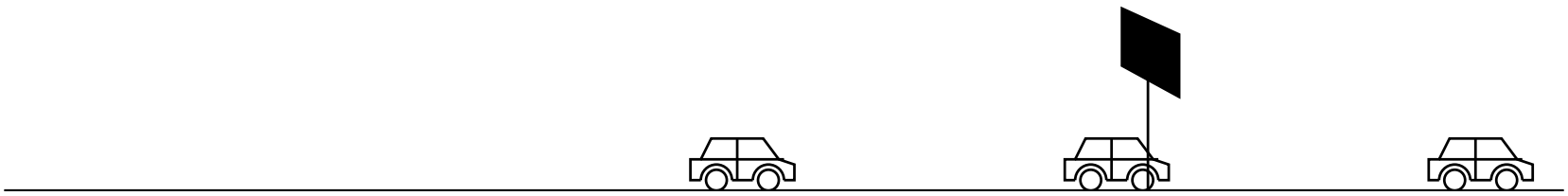
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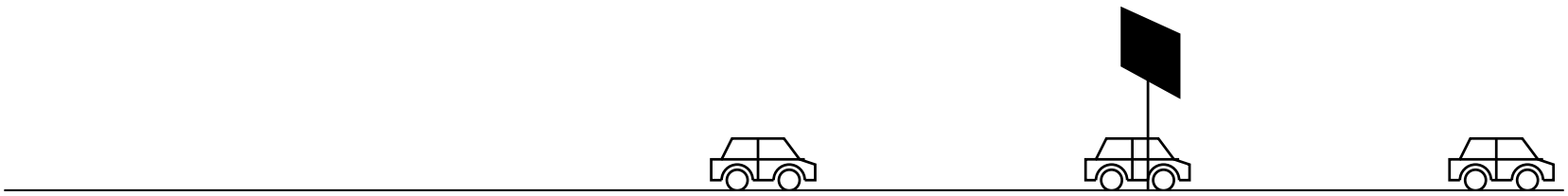
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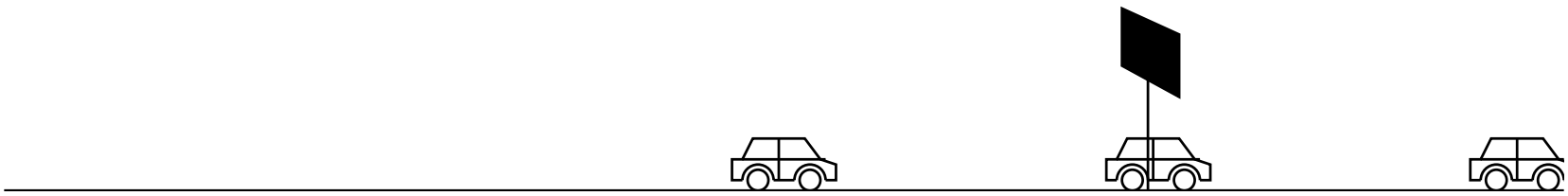
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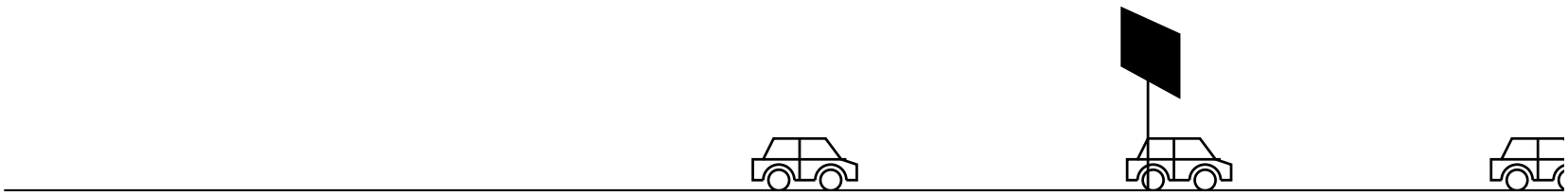
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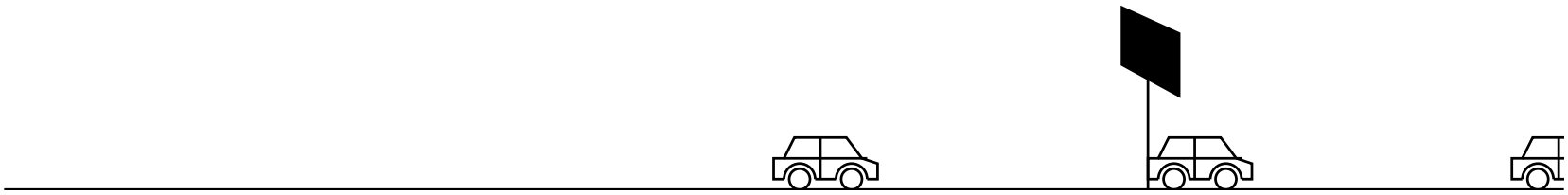
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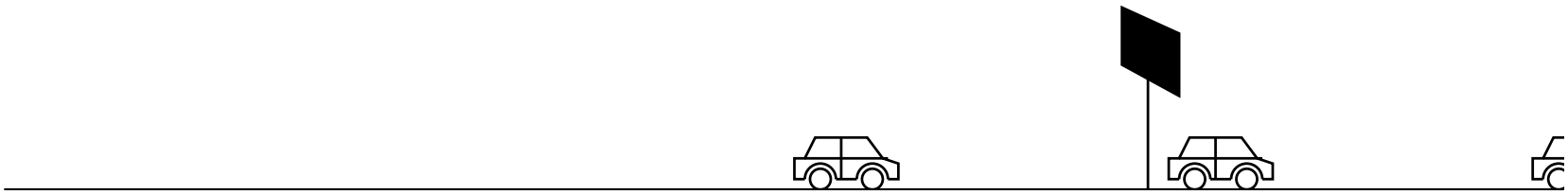
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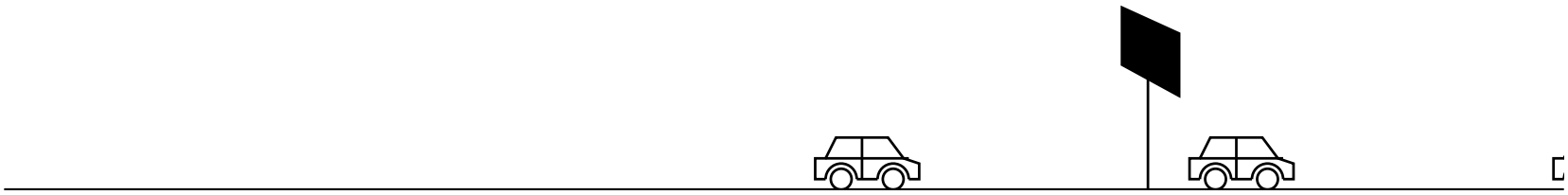
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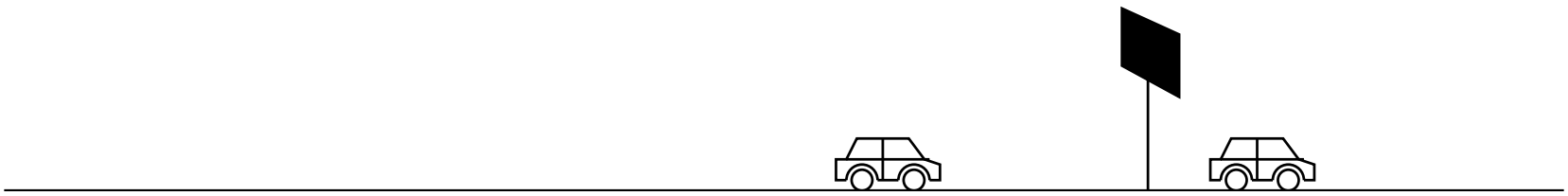
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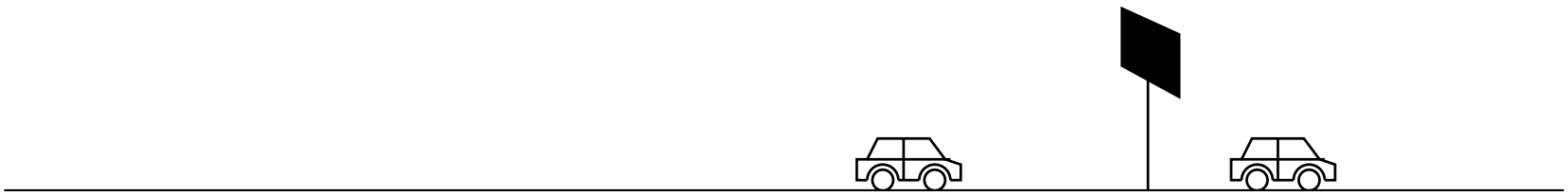
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Leaving a traffic jam



Continuous, long acceleration for those starting from the rear

Leaving a traffic jam



Continuous, long acceleration for those starting from the rear

Leaving a traffic jam is always soft, “blurry”.

Leaving a traffic jam



Continuous, long acceleration for those starting from the rear

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Why is there such a difference between the two ends of a traffic jam?

Leaving a traffic jam



Continuous, long acceleration for those starting from the rear

Leaving a traffic jam is always soft, “blurry”.

Why is there such a difference between the two ends of a traffic jam?

Totally asymmetric simple exclusion process: an explanation

Being ageless

We first seek a random time that does not remember its past.
Let $\tau > 0$ be a random time such that

$$\mathbf{P}\{\tau > t\} = \mathrm{e}^{-t} \qquad \text{for all } t > 0.$$

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Suppose that this time has not passed yet at t. What is the probability that it will not pass before another s seconds? That is, out of those cases when $\tau > t$ occurs, in what percentage will $\tau > t + s$ also occur?

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$$\frac{\mathbf{P}\{\tau > t + s\}}{\mathbf{P}\{\tau > t\}} = \frac{\mathrm{e}^{-(t+s)}}{\mathrm{e}^{-t}} = \mathrm{e}^{-s}.$$

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The same as $\mathbf{P}\{\tau > s\}$, regardless of t!

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We have found the secret of being ageless.

Being ageless

 ← This will be the ageless alarm clock that rings at time τ

Being ageless

⏰ ← This will be the ageless alarm clock that rings at time τ

⇝ What is the probability that an ⏰ rings within a small time t?

Being ageless

 $\leftarrow$ This will be the ageless alarm clock that rings at time τ

$\rightsquigarrow$ What is the probability that an  rings within a small time t?

$$\mathbf{P}\{\tau \leq t\} = 1 - \mathbf{P}\{\tau > t\} = 1 - \mathrm{e}^{-t} \simeq 1 - (1-t) + \text{error} = t + \text{error}.$$

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⇝ What is the probability that *two* independent ⏰⏰ both ring within a small time t?

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→ More ⏰'s, even smaller probability.

Being ageless

$\rightsquigarrow$ What is the probability that *none* of k independent ⏰'s ring within a small time t?

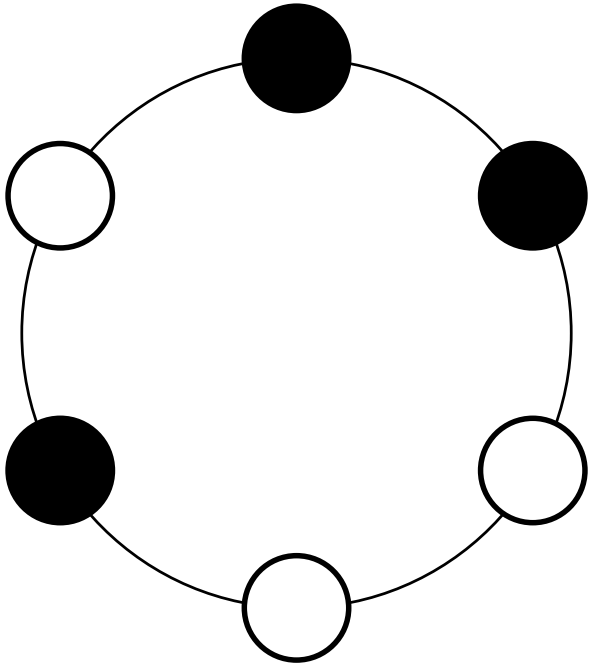
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$$\begin{aligned}\mathbf{P}\{\text{none of them ring}\} &= \mathbf{P}\{\tau > t\}^k \\ &= \mathrm{e}^{-kt} \\ &\simeq (1 - kt) + \text{error.}\end{aligned}$$

The Totally Asymmetric Simple Exclusion Process

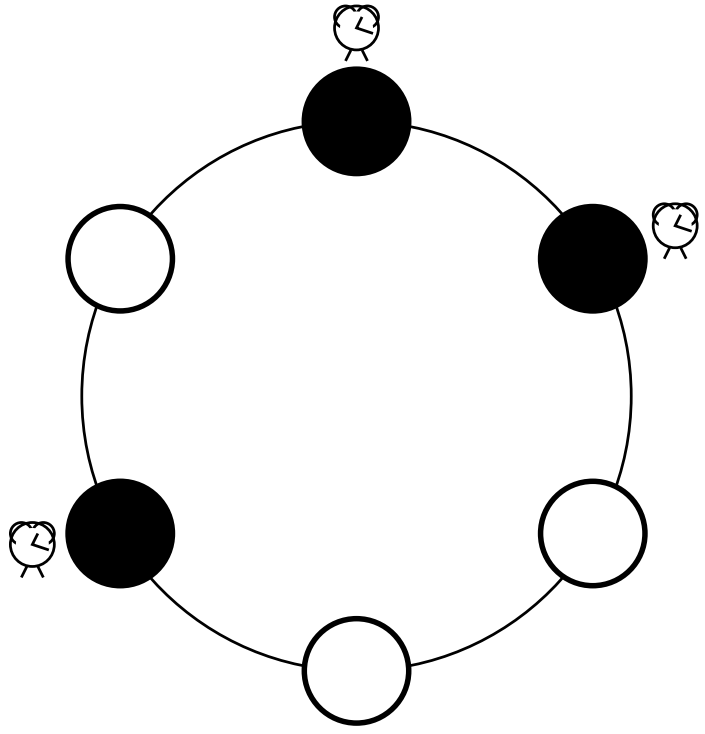
TASEP



m balls in *N* possible slots.

The Totally Asymmetric Simple Exclusion Process

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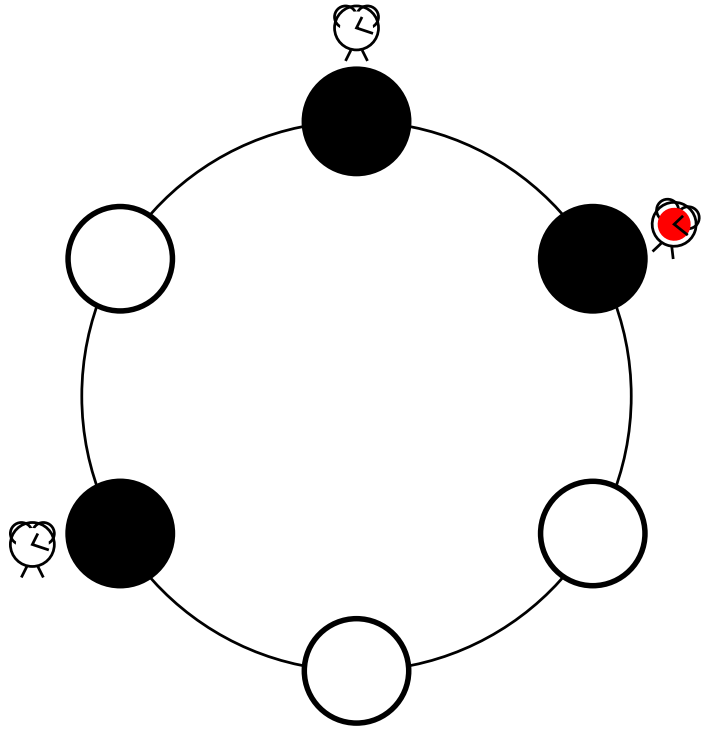


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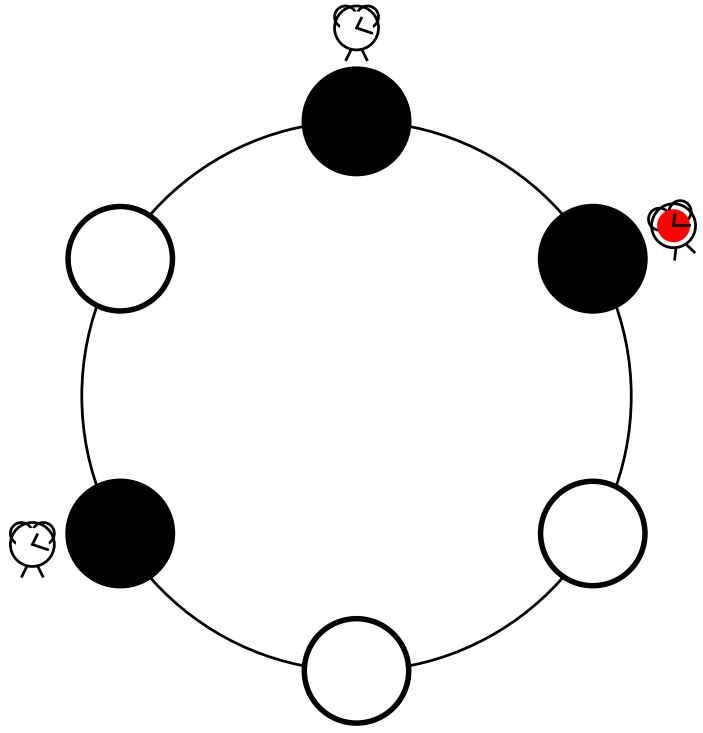


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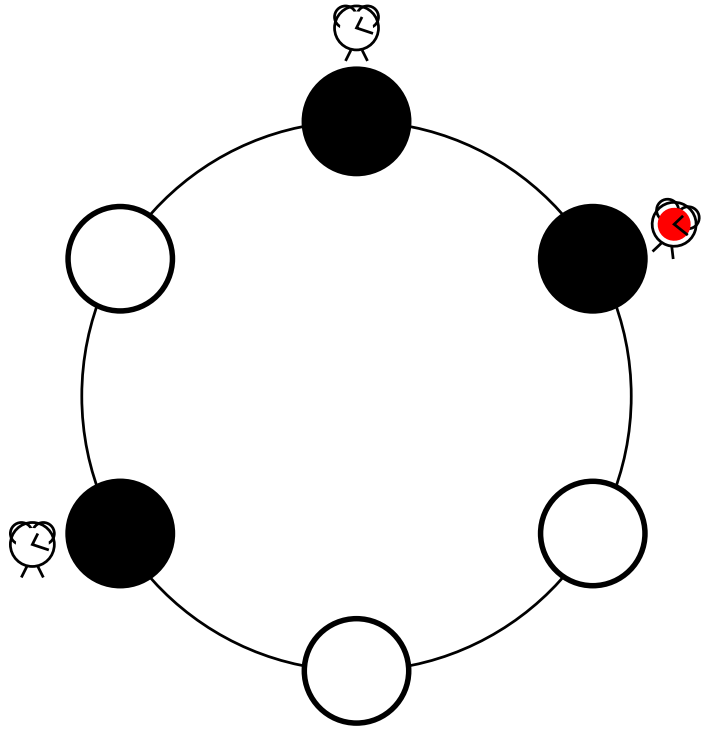


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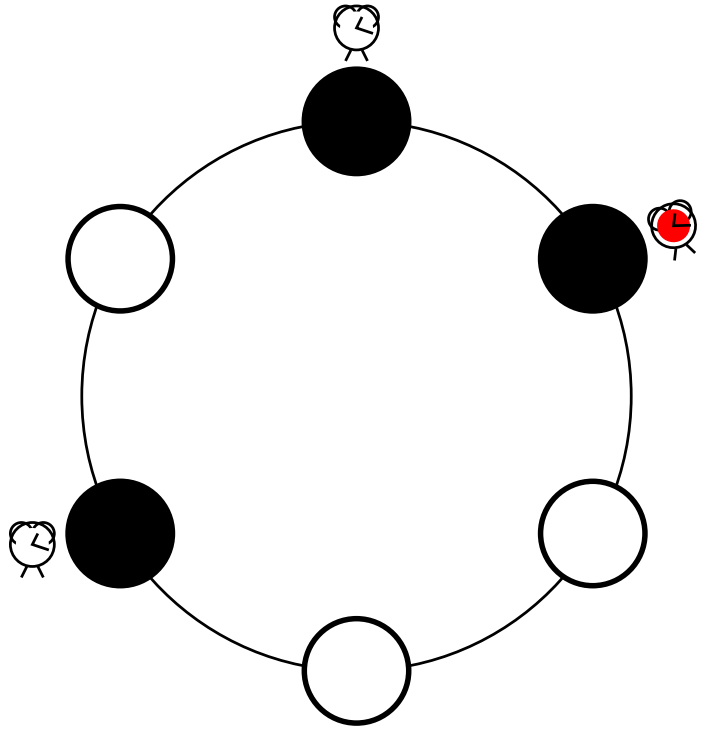


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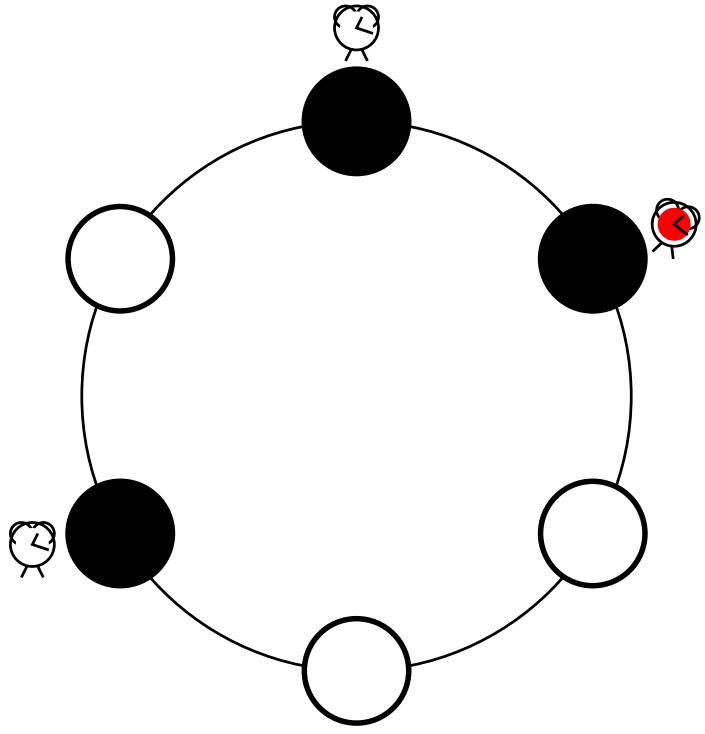


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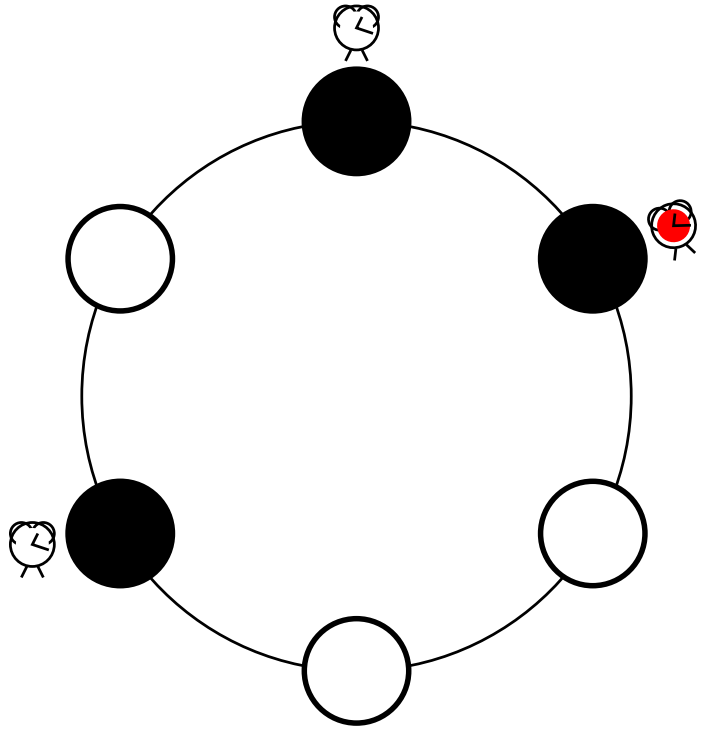


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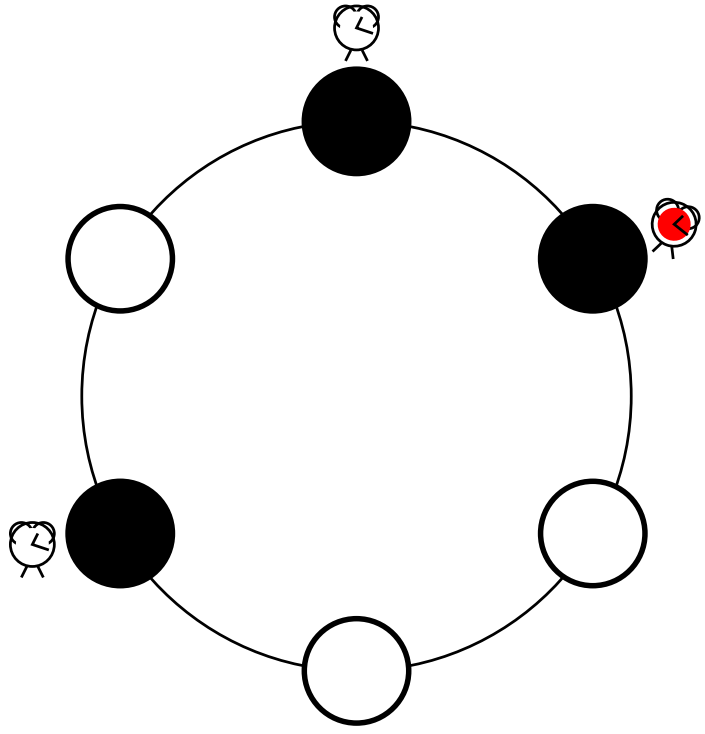


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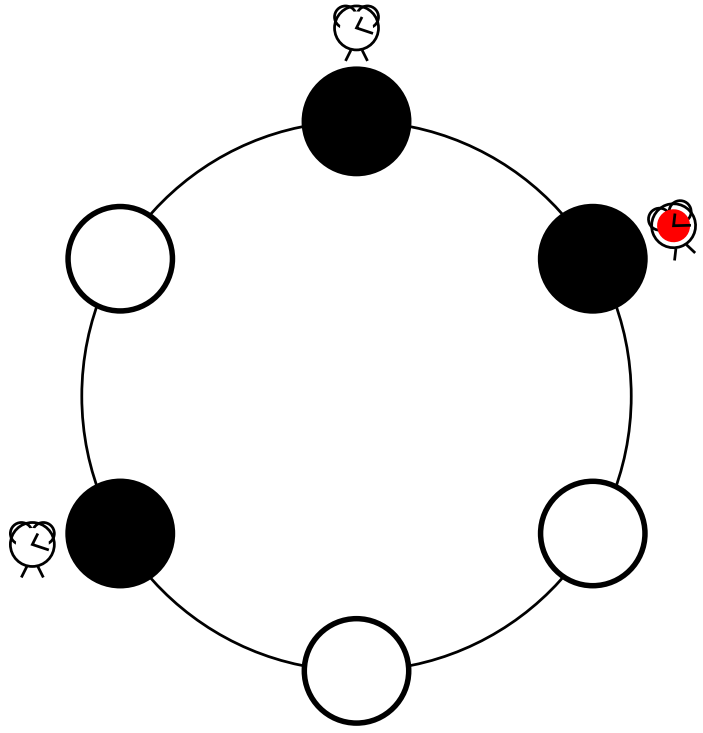


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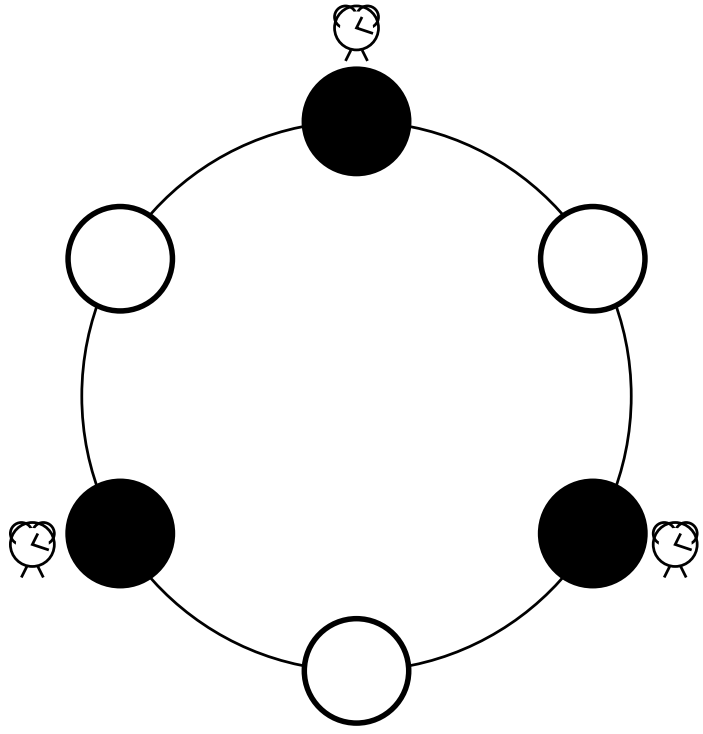


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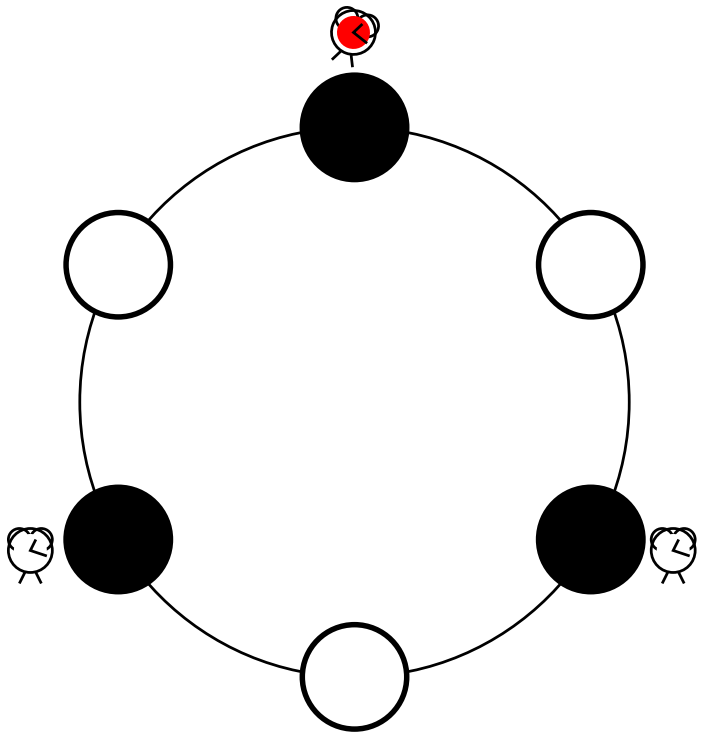


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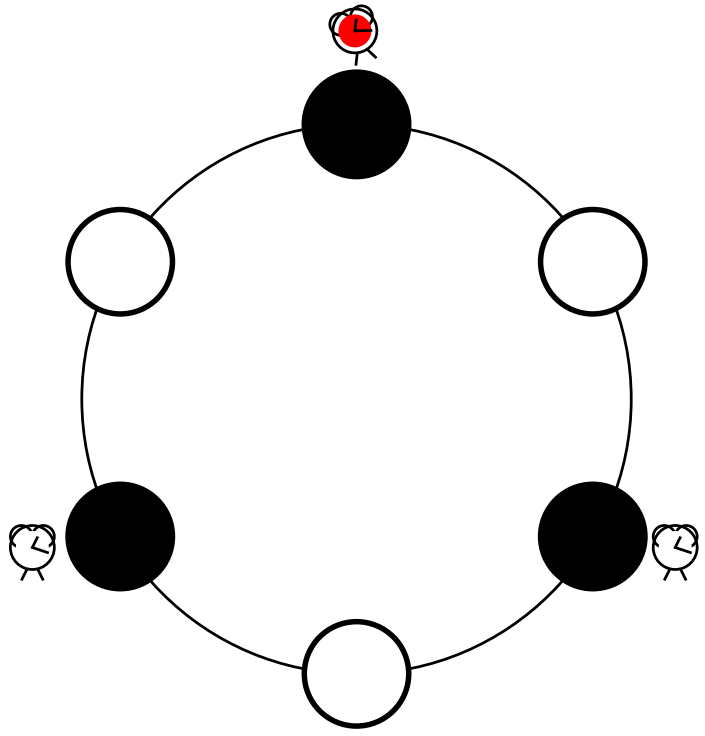


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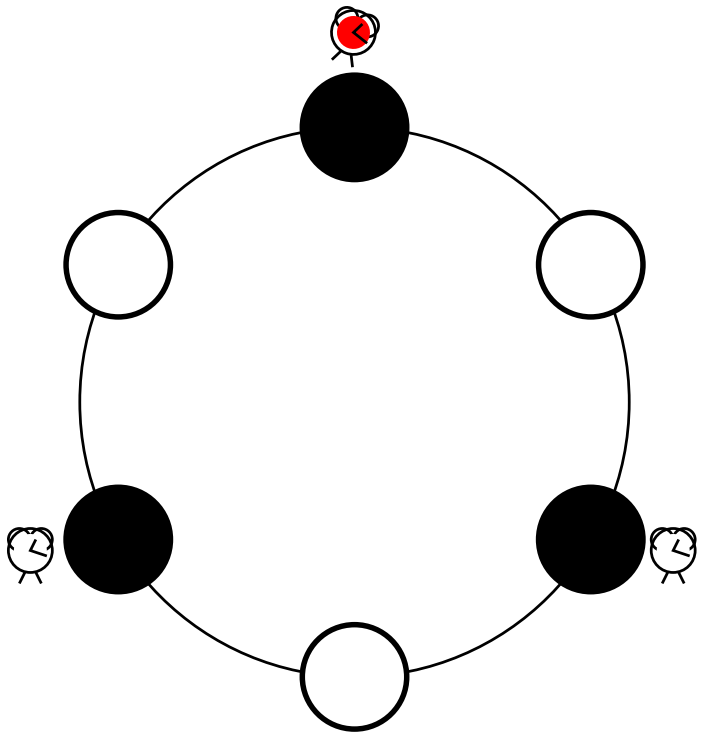


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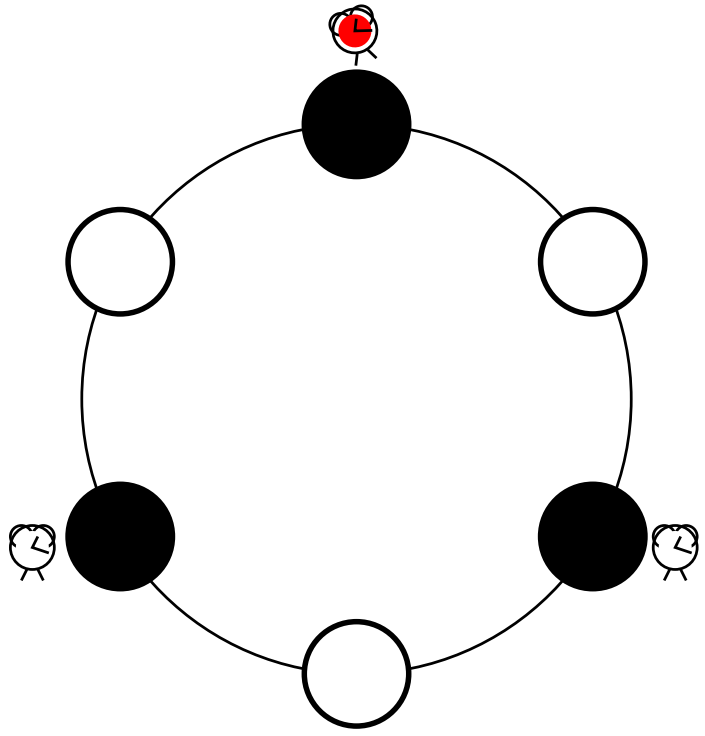


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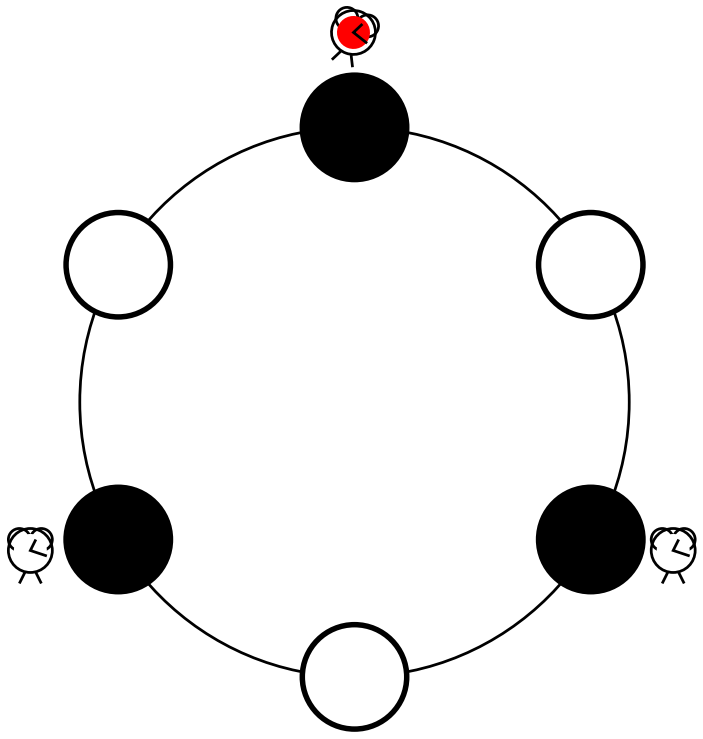


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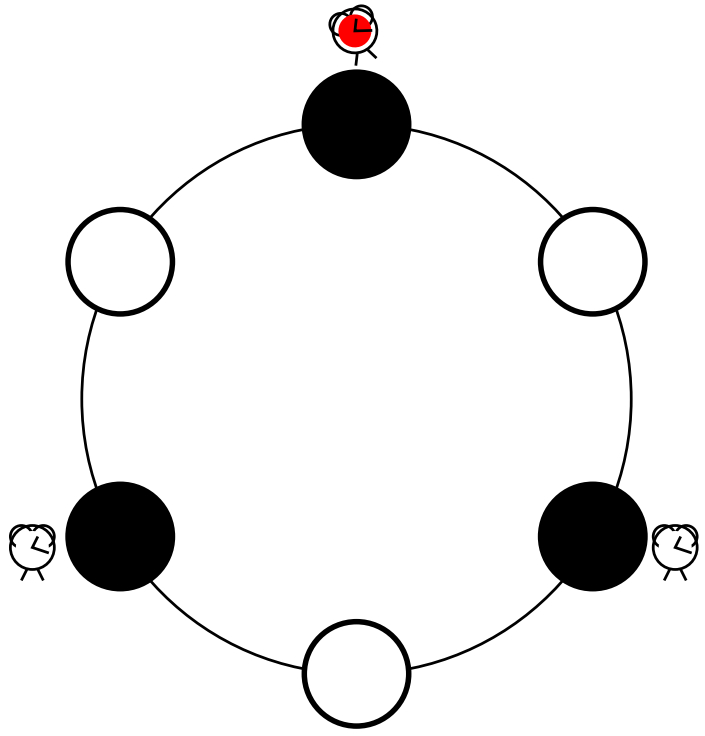


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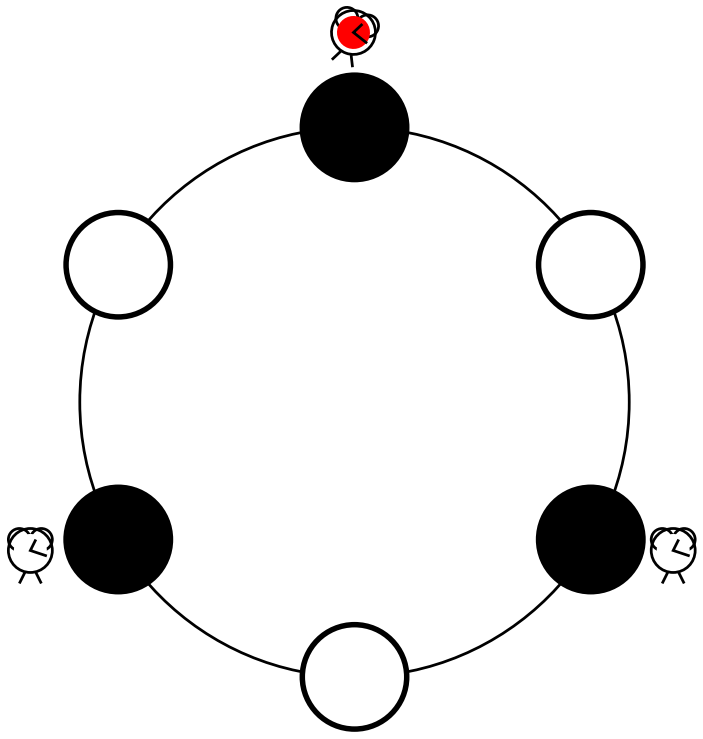


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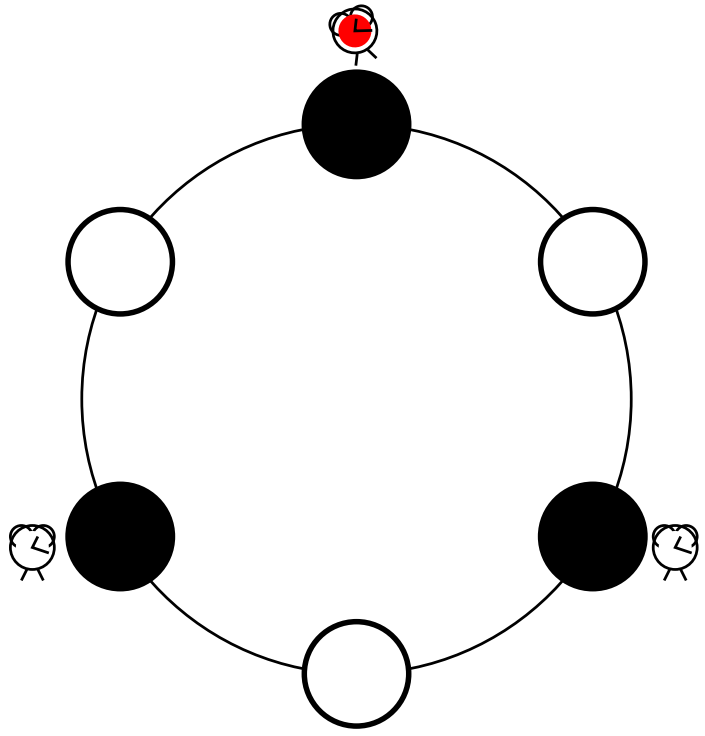


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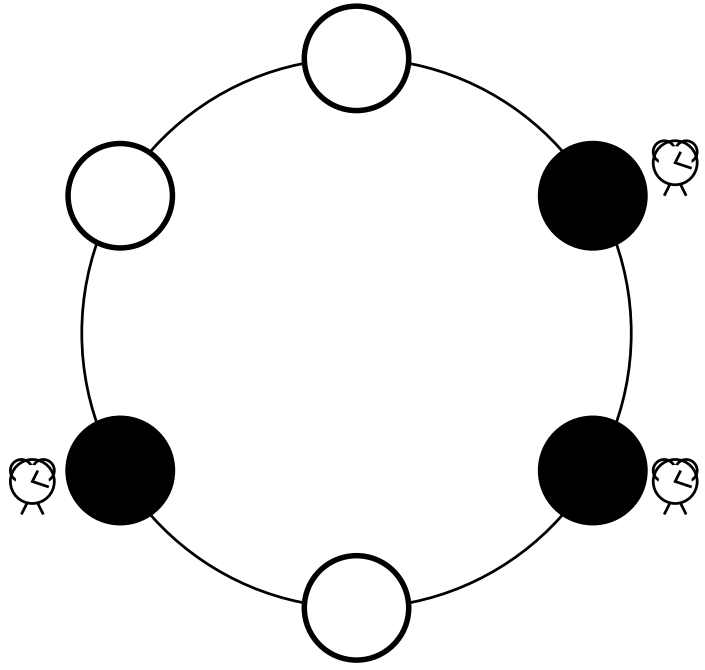


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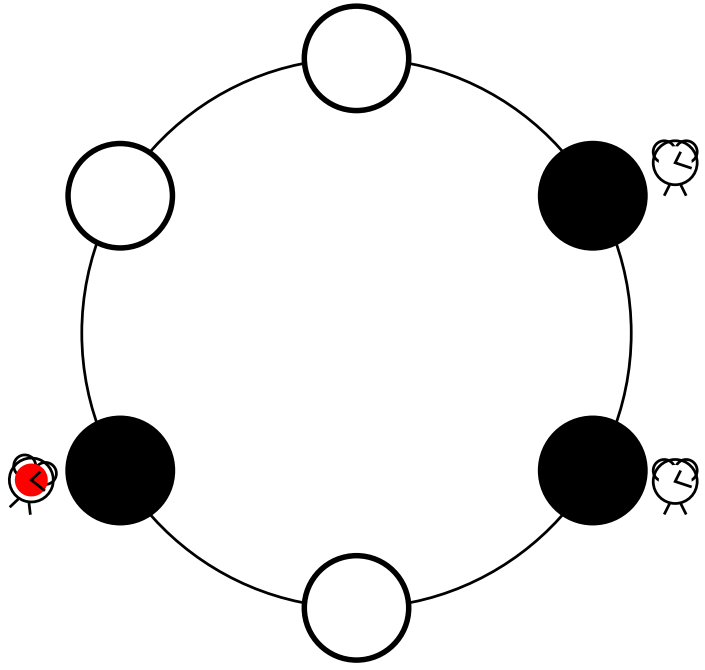


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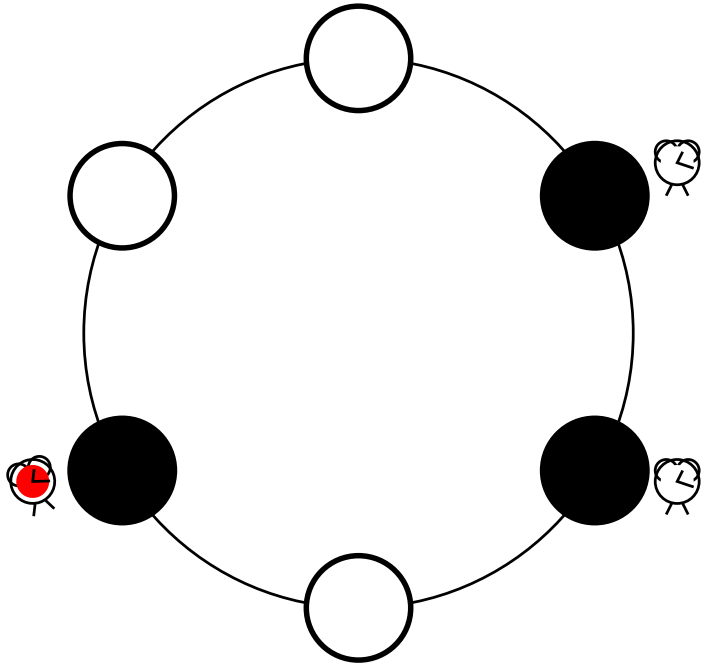


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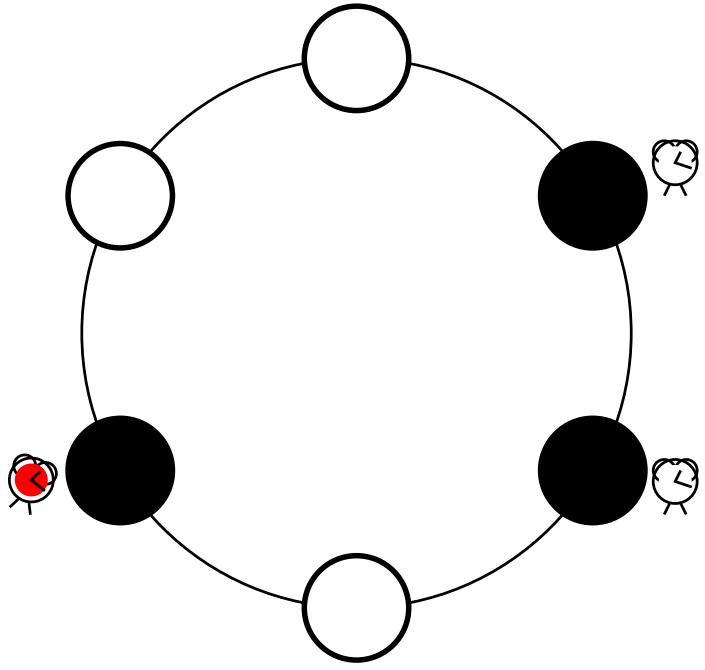


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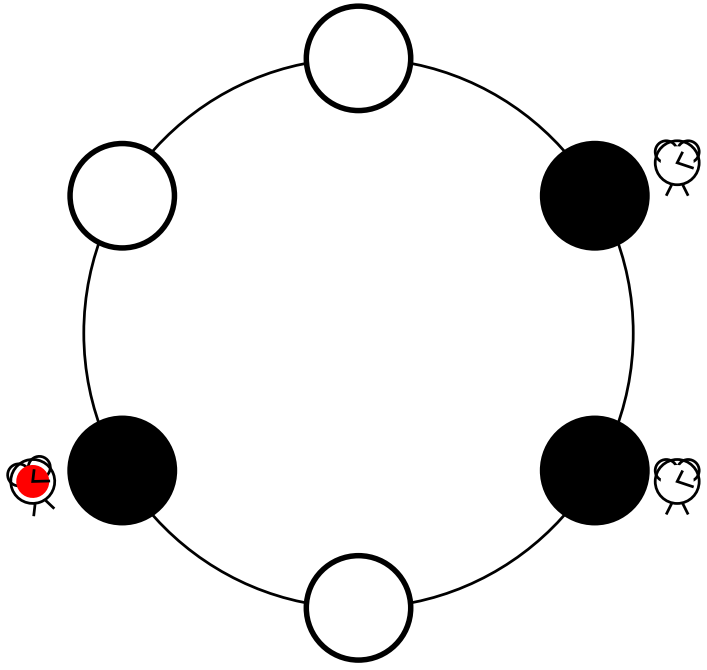


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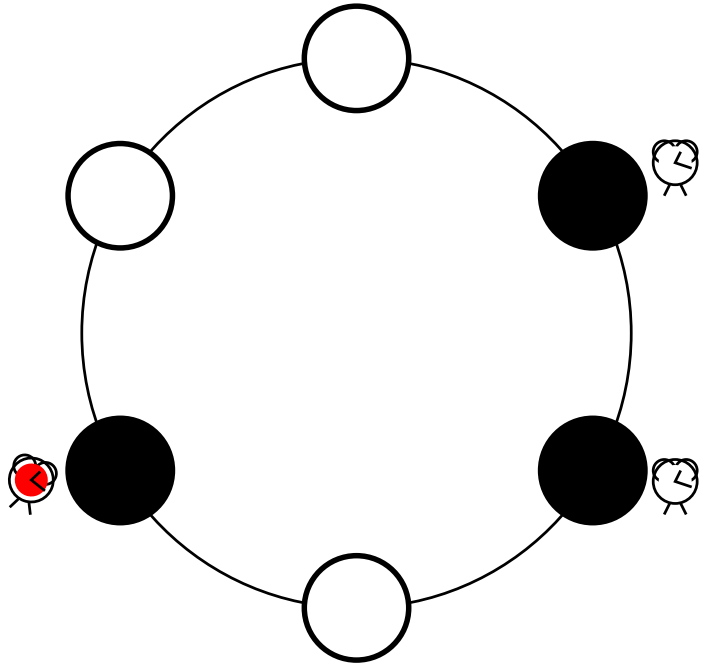


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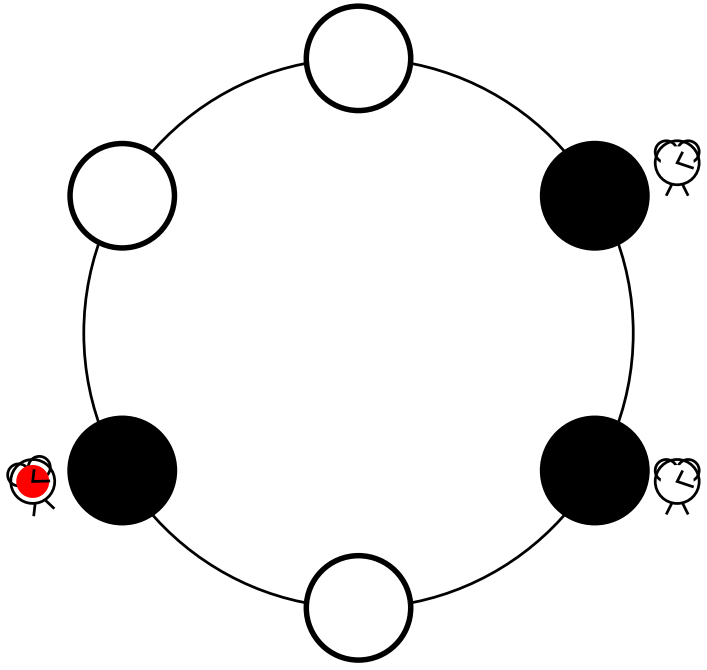


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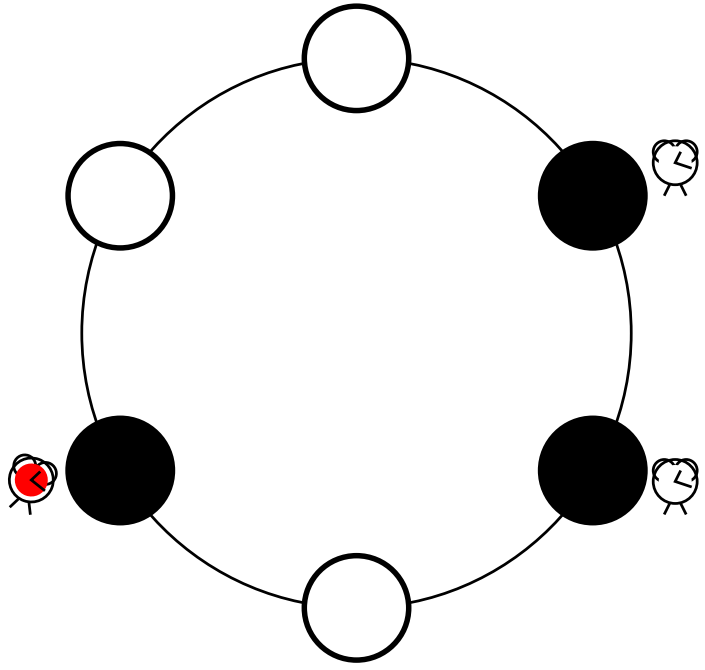


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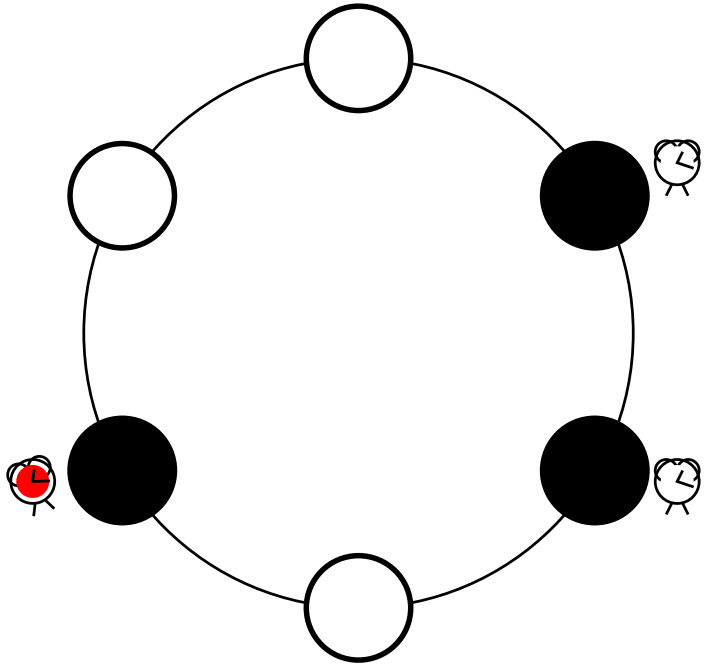


m balls in *N* possible slots.

Each listening to its own  . When that rings, the ball tries to jump to the right.

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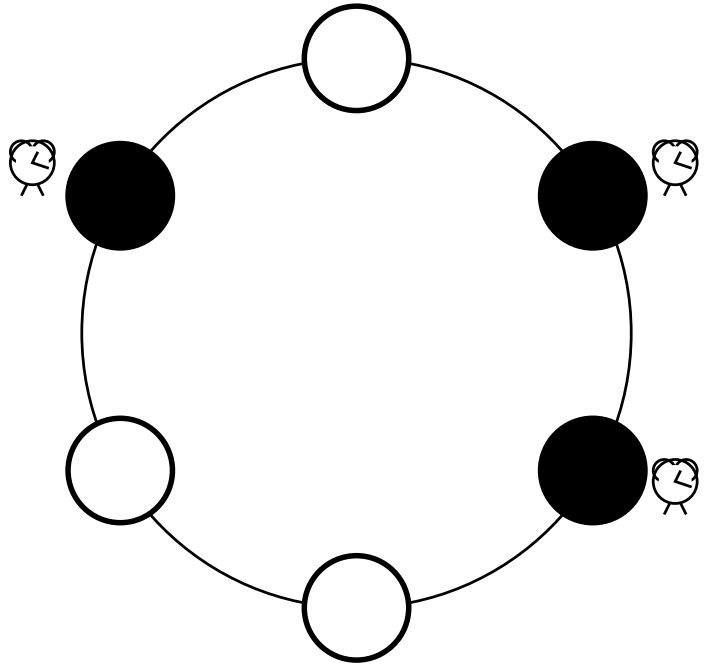


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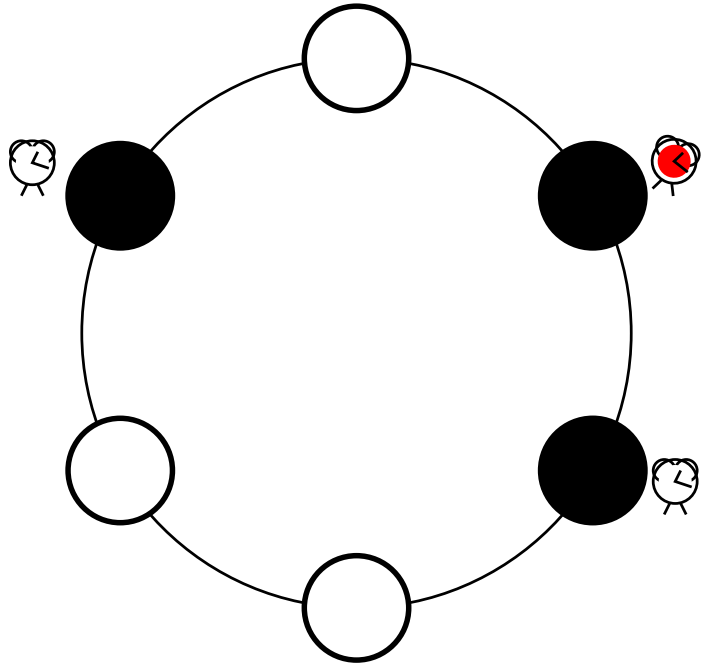


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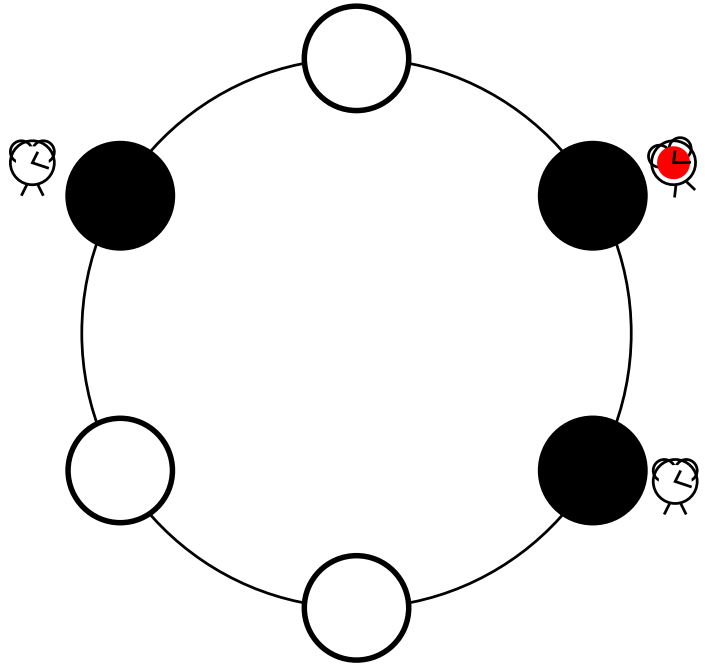


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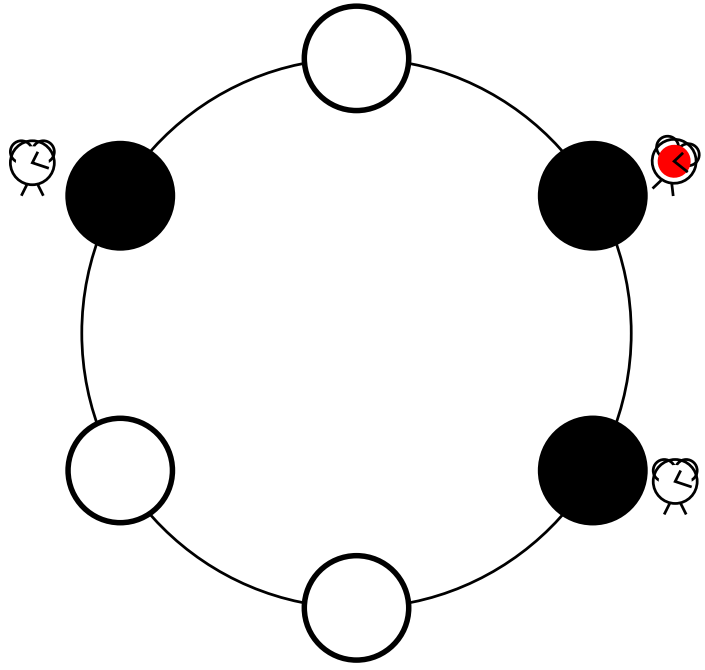


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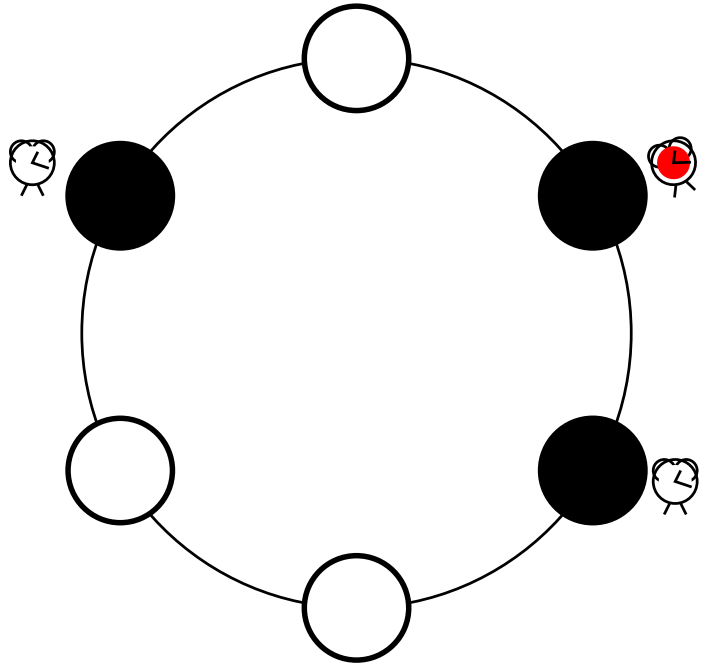


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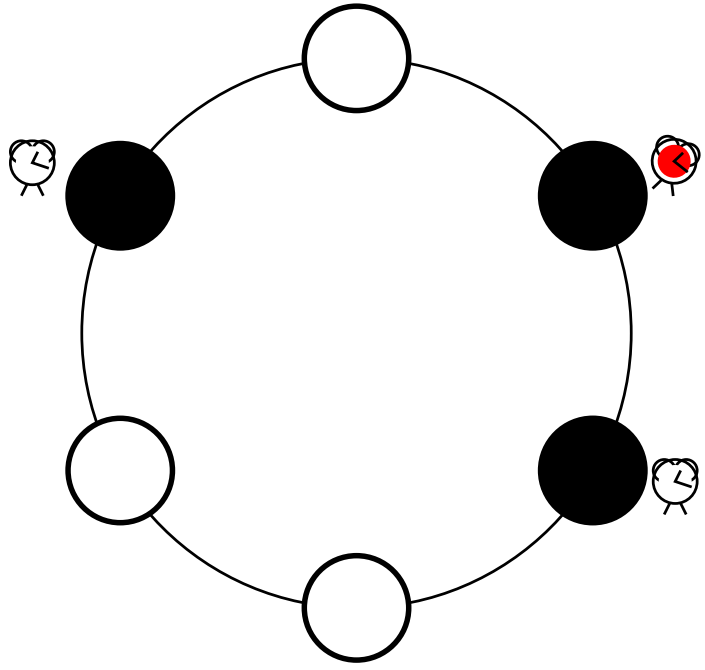


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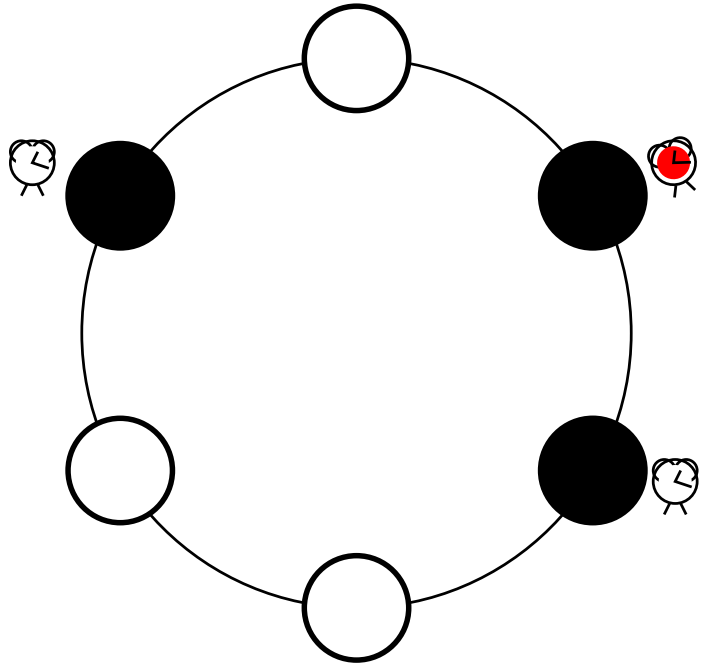


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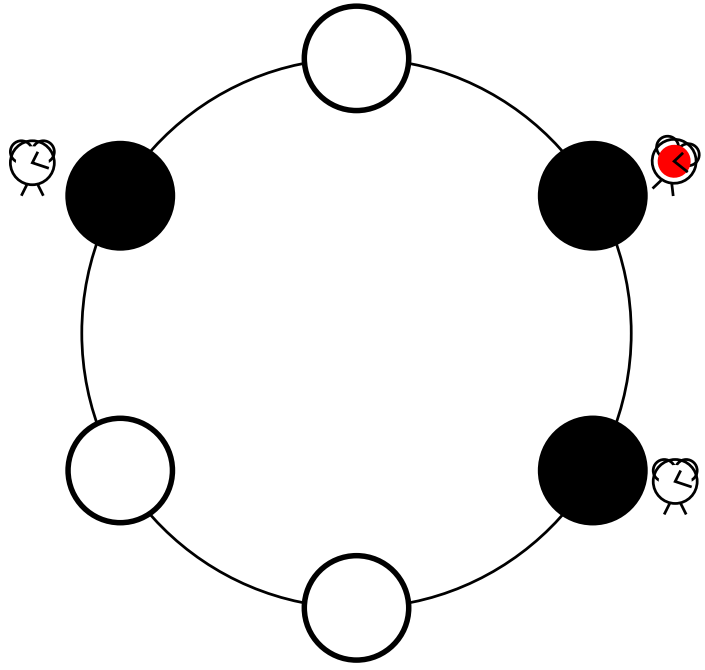


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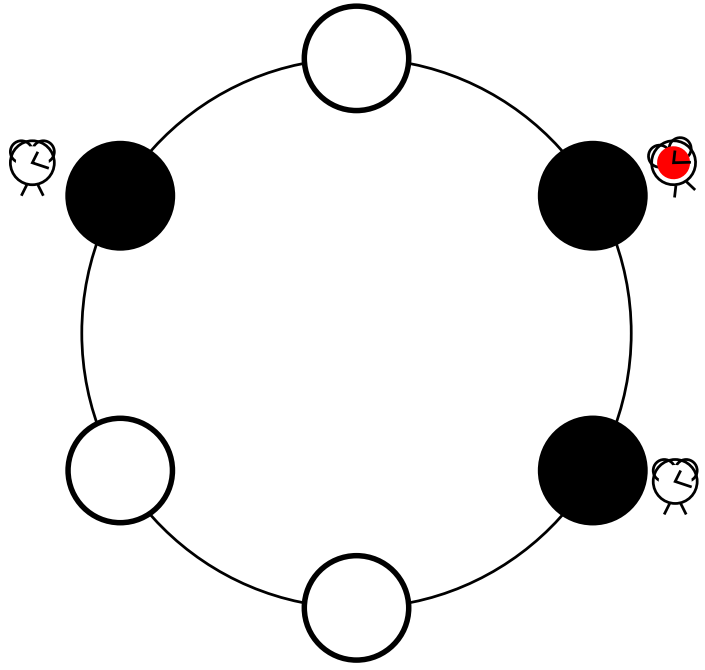


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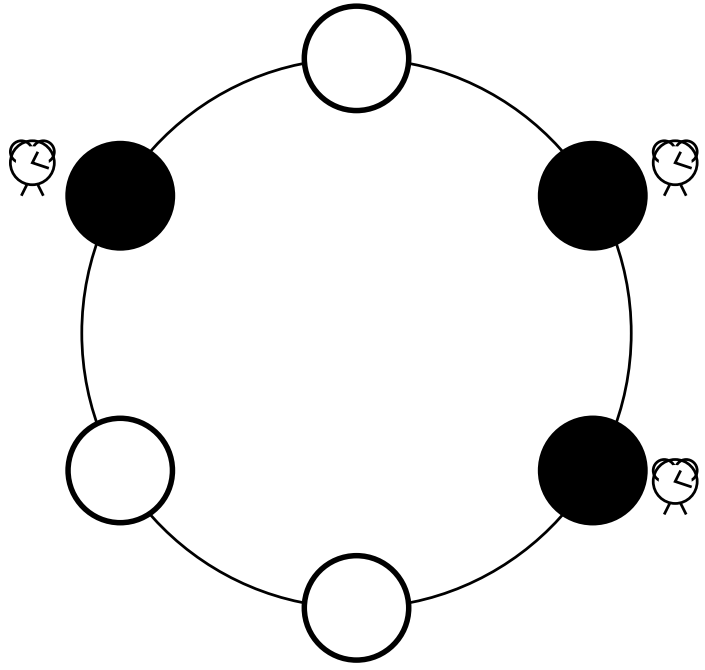


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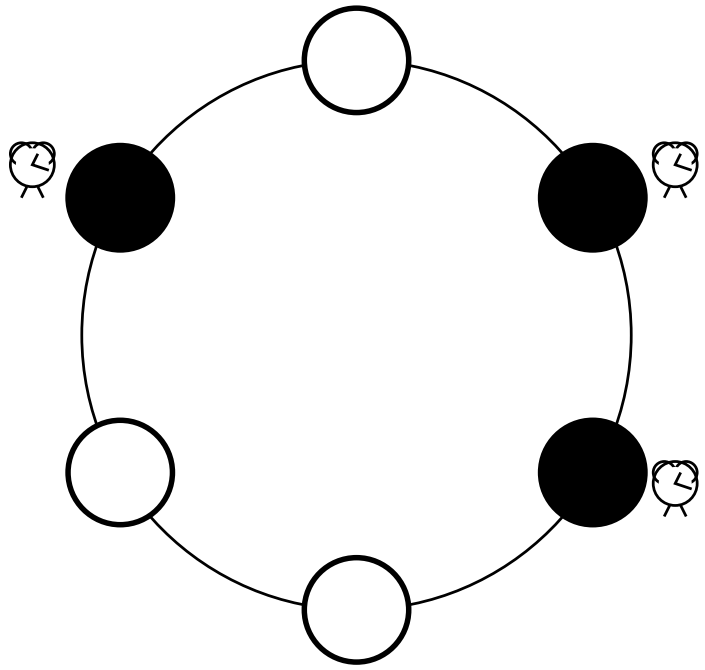


m balls in *N* possible slots.

Each listening to its own . When that rings, the ball tries to jump to the right. But sometimes it's blocked.

The Totally Asymmetric Simple Exclusion Process

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m balls in *N* possible slots.

Each listening to its own ⏰. When that rings, the ball tries to jump to the right. But sometimes it's blocked.
Ageless, independent ⏰'s $\Rightarrow$ if we know the present, no need to know the past. *Markov property*, makes things handy.

Stationary distribution

Random process $\rightsquigarrow$ need to talk about *distributions*.

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With N and m fixed, the distribution that gives equal chance to each (m-ball) configuration, is stationary.

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In this case every configuration occurs with probability $1/\binom{N}{m}$.

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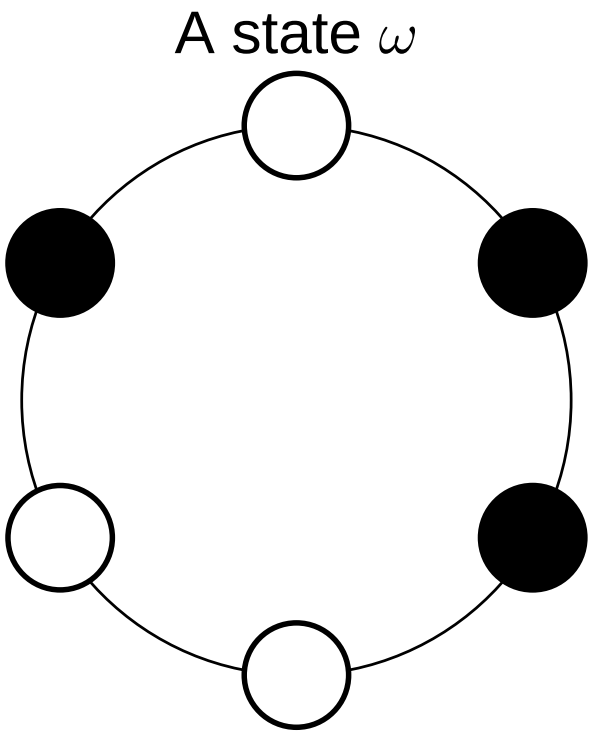
In this case every configuration occurs with probability $1/\binom{N}{m}$.

2nd remark.

With fixed N, m, there is no other stationary distribution.

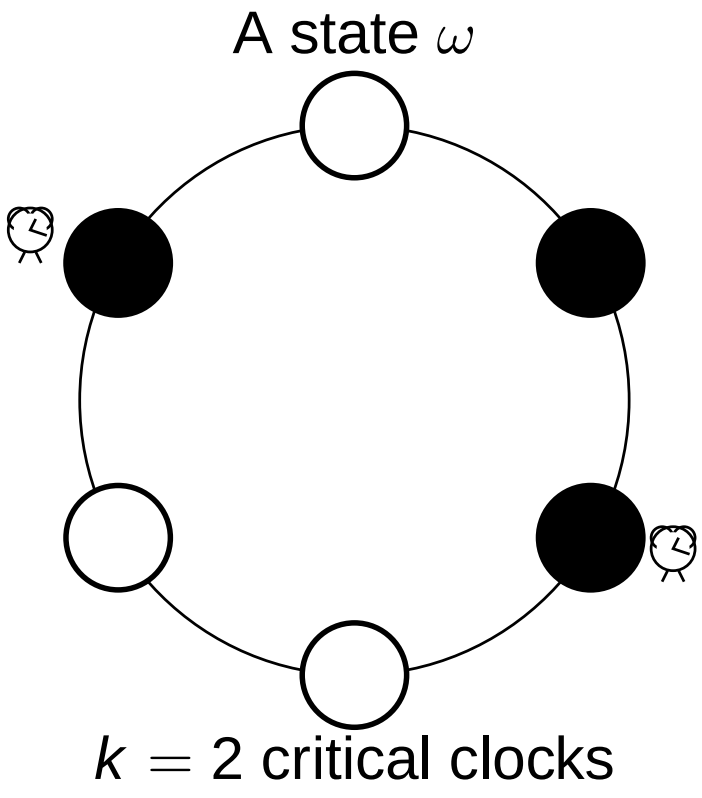
Stationary distribution

Almost proof



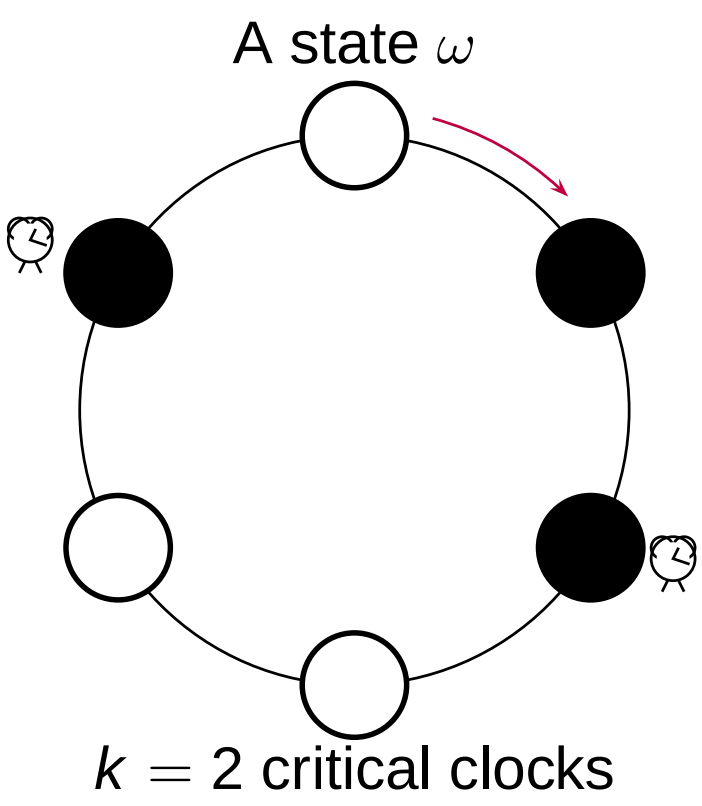
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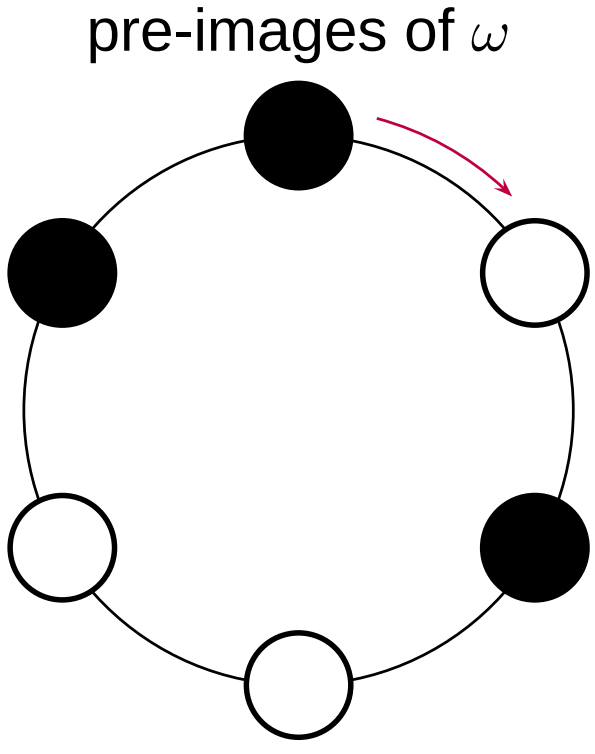
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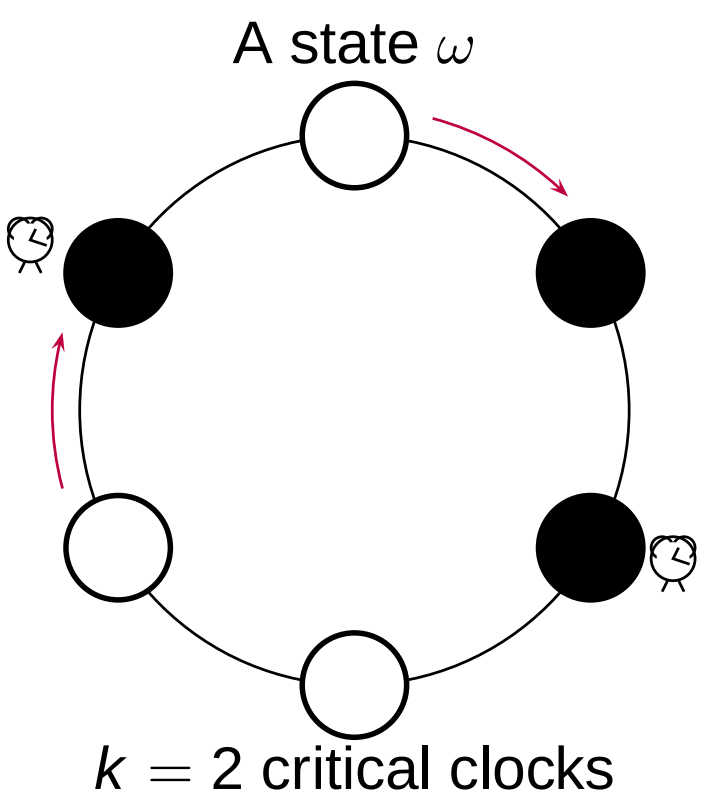
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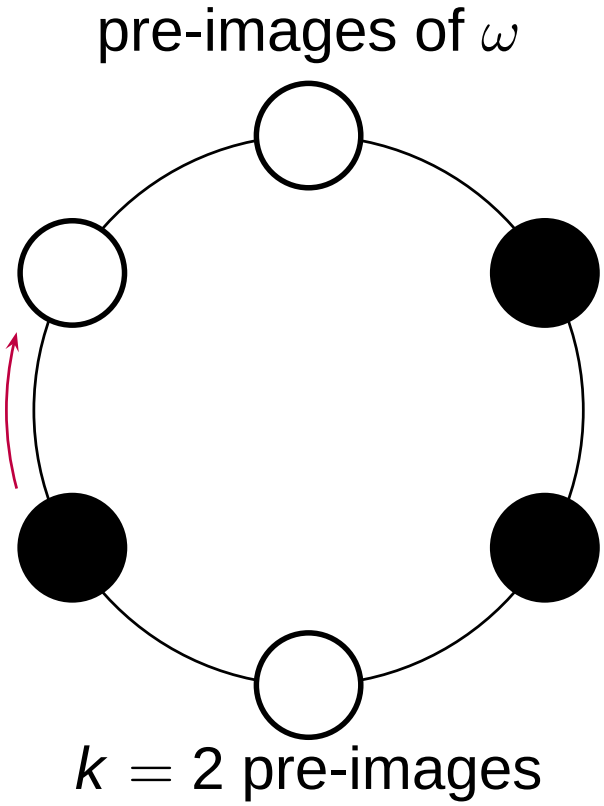




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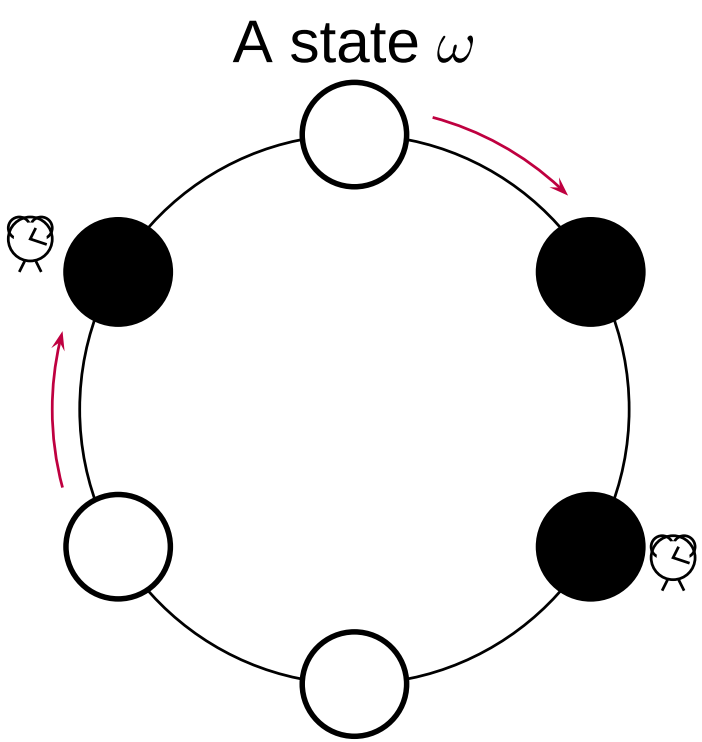
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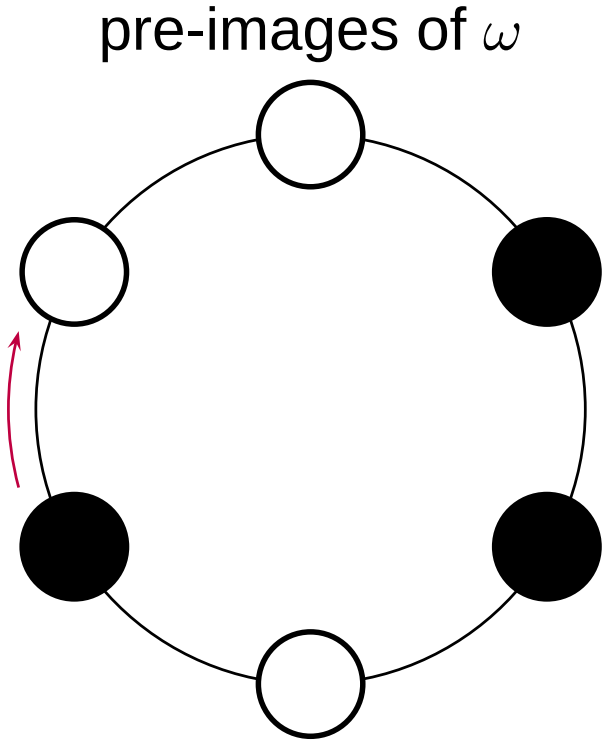




Stationary distribution

Almost proof





The number of critical clocks for ω = the number of pre-images of $\omega = k$

Stationary distribution

Almost proof

Suppose that each configuration has the same probability p at time s. What is the probability of the state ω after a small time t?

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$$\mathbf{P}\{\omega \text{ at time } s+t\}$$

Stationary distribution

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Suppose that each configuration has the same probability p at time s. What is the probability of the state ω after a small time t?

$$
\begin{aligned}
&\mathbf{P}\{\omega \text{ at time } s+t\}\\
&= \mathbf{P}\{\omega \text{ at time } s \text{ and no jumps within time } t\}\\
&\quad + \mathbf{P}\{\text{was a pre-image of } \omega \text{ at time } s, \text{ and jumps to } \omega\}\\
&\quad + \text{error (at least two jumps occur within the small time } t)
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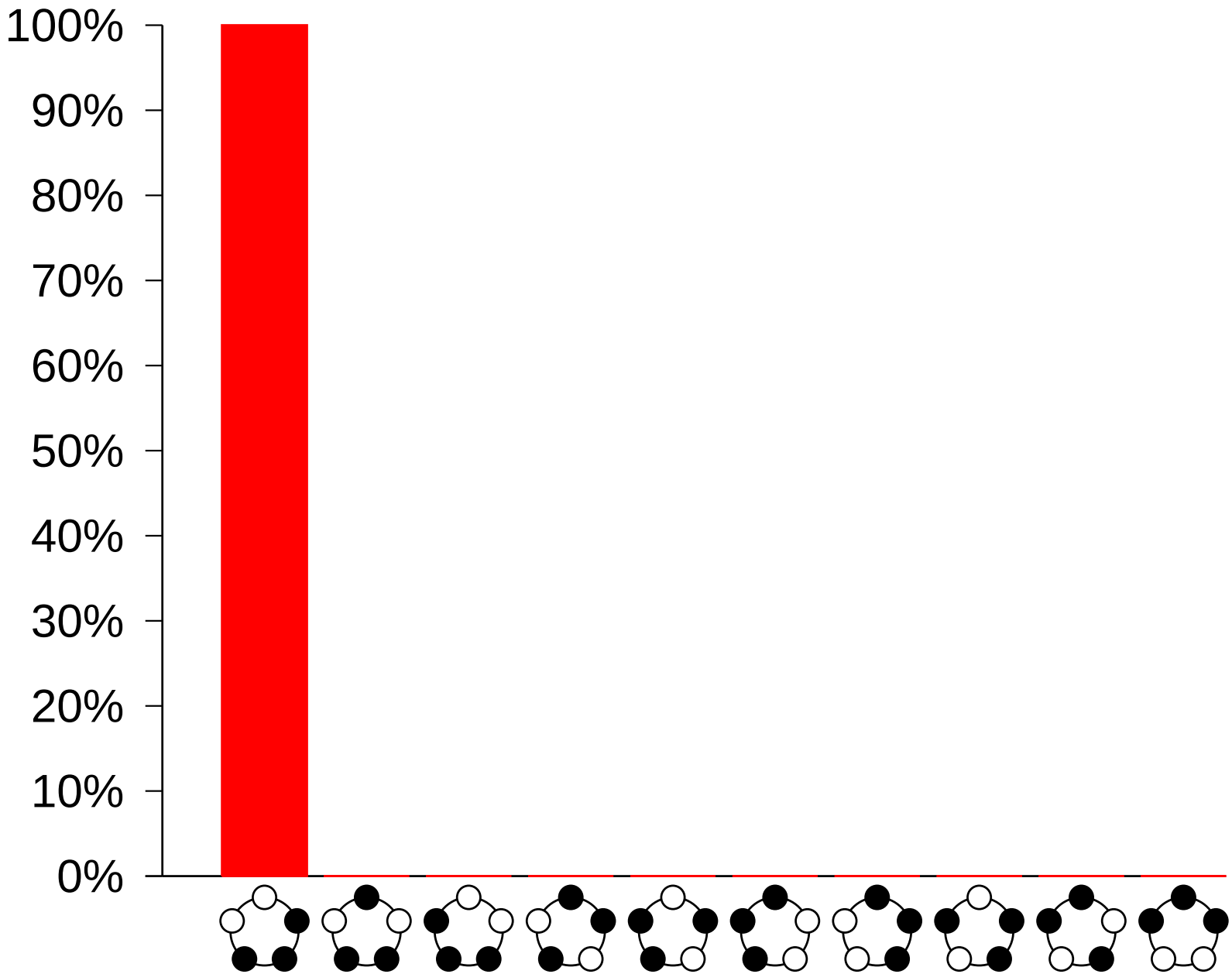
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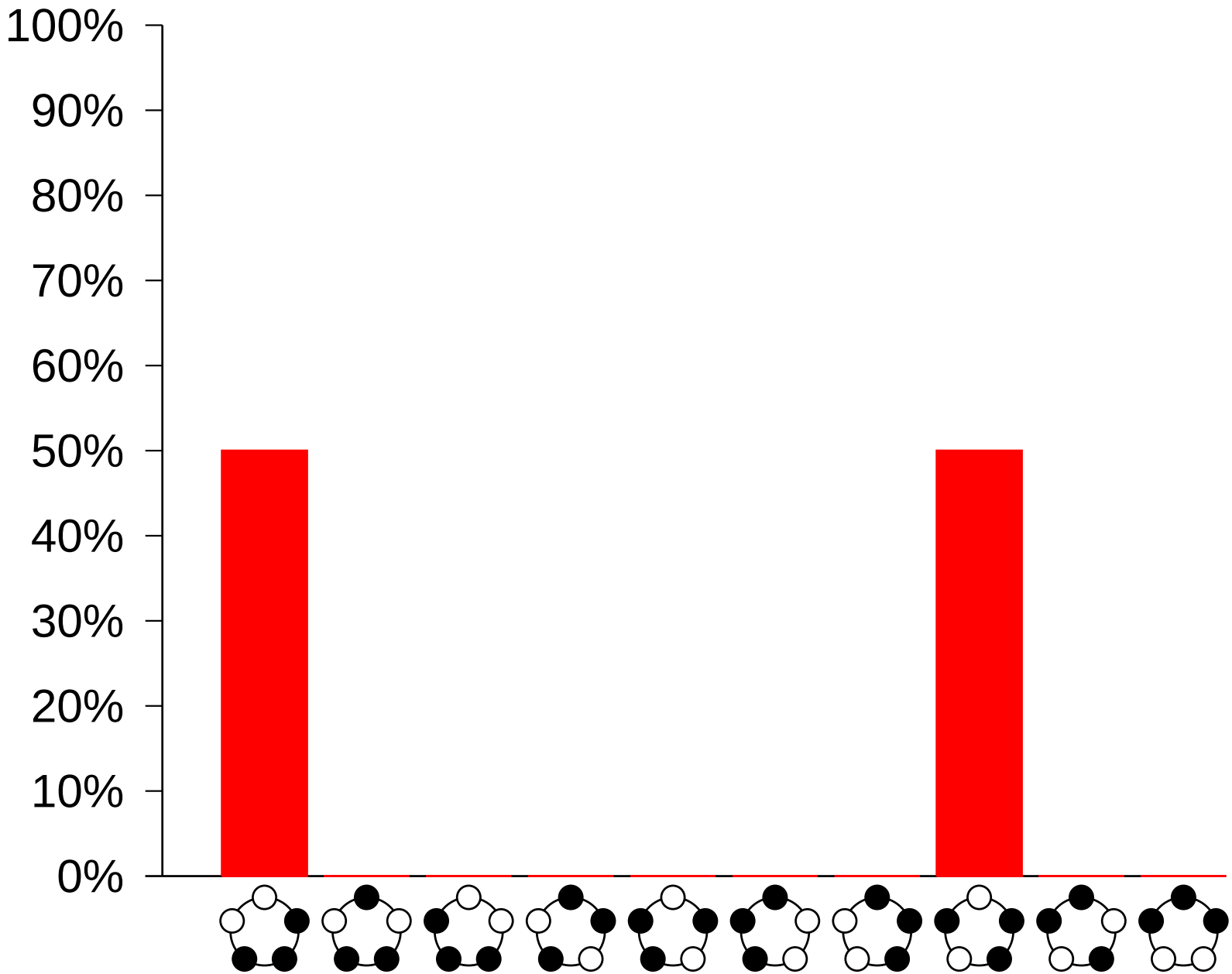
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In fact error $\simeq t^2$, stays small if summed up for more and more smaller and smaller intervals of length t. □

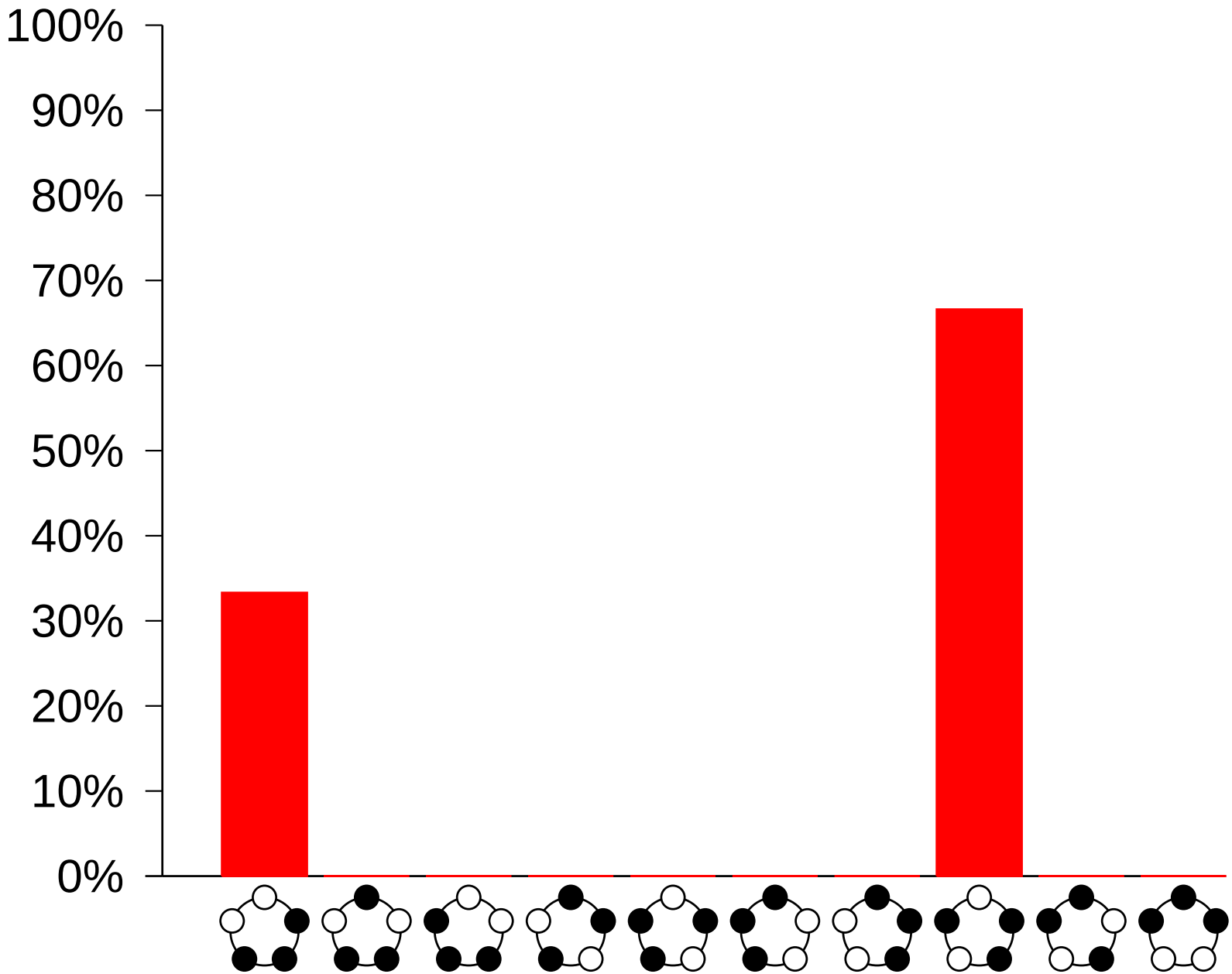
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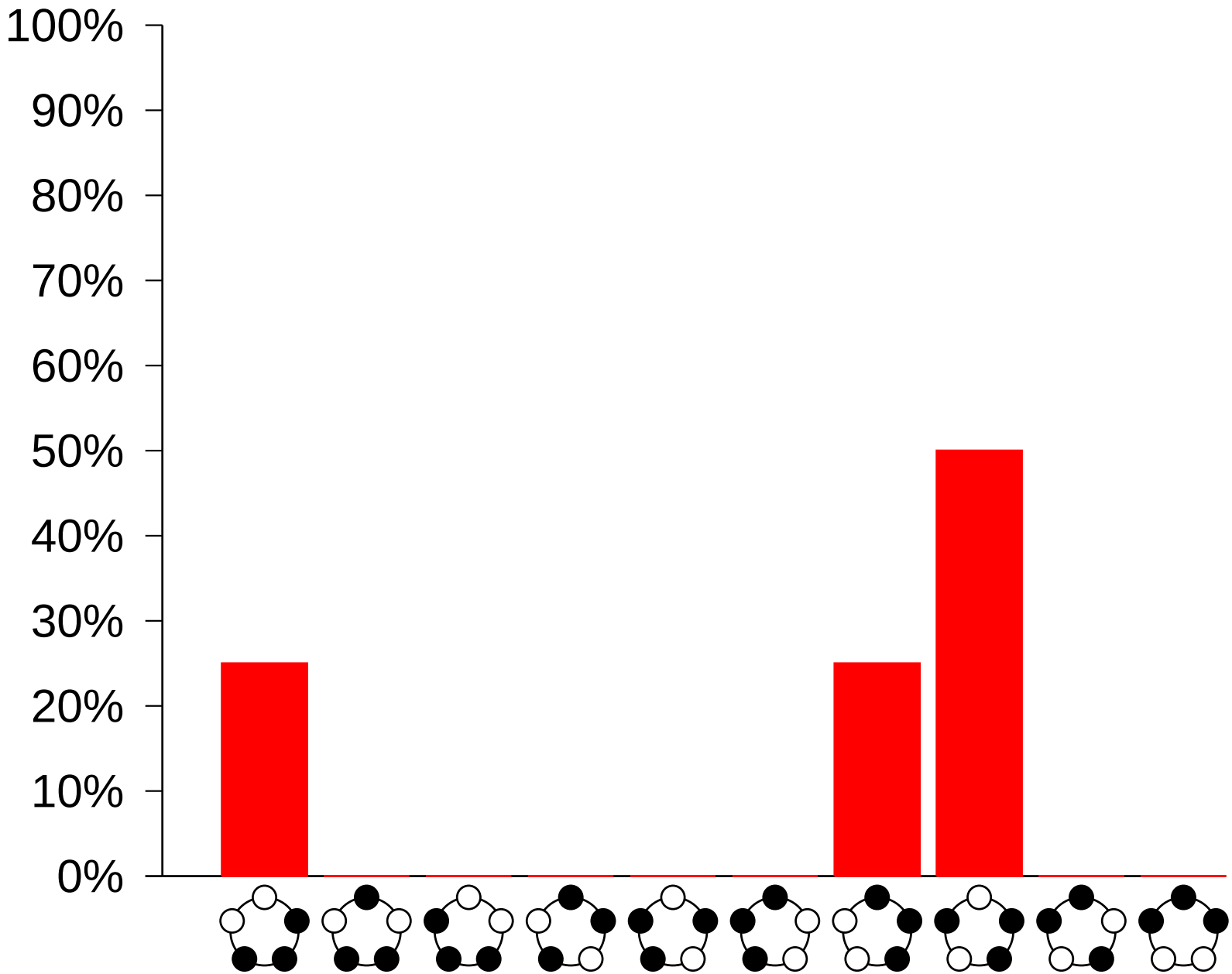
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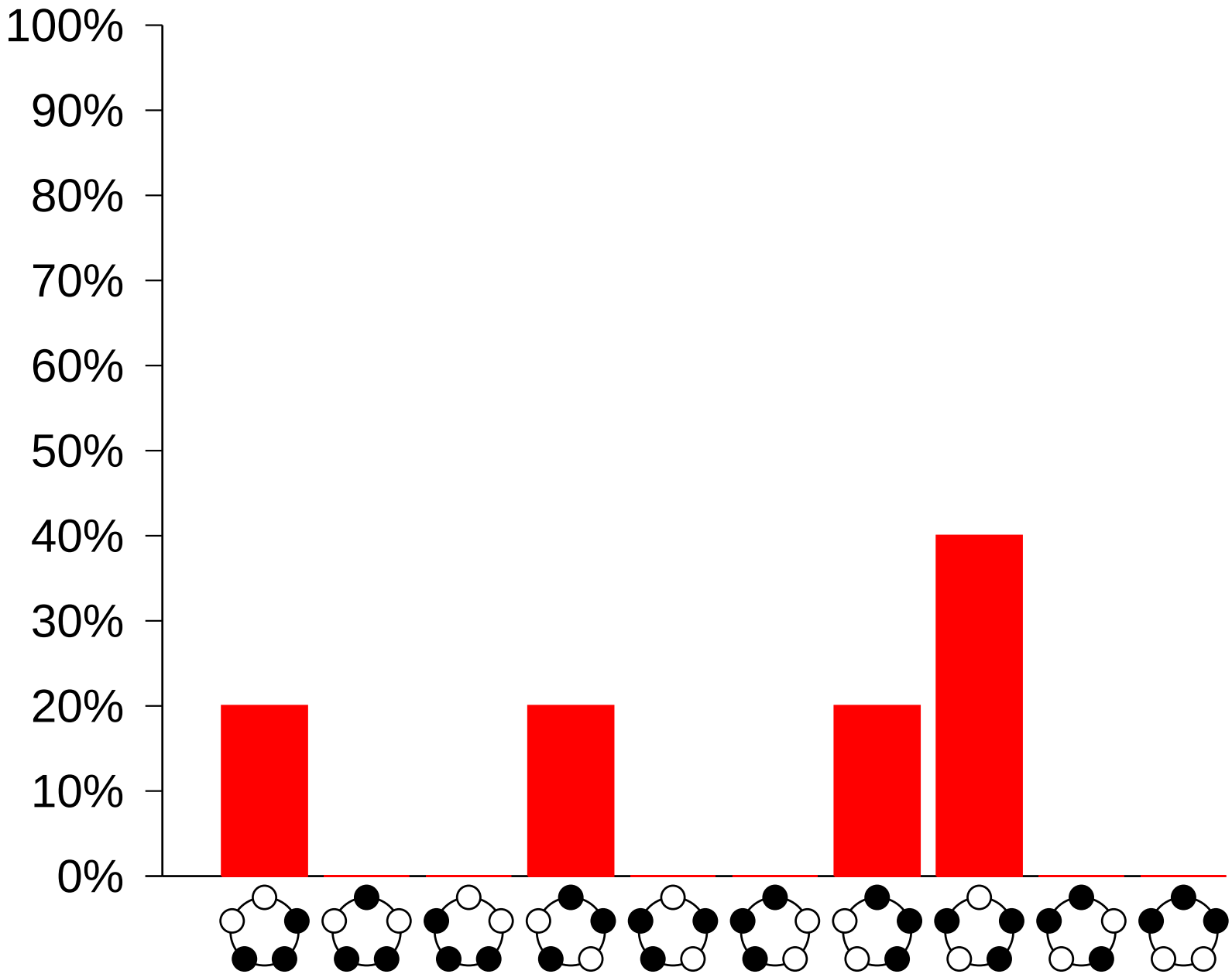
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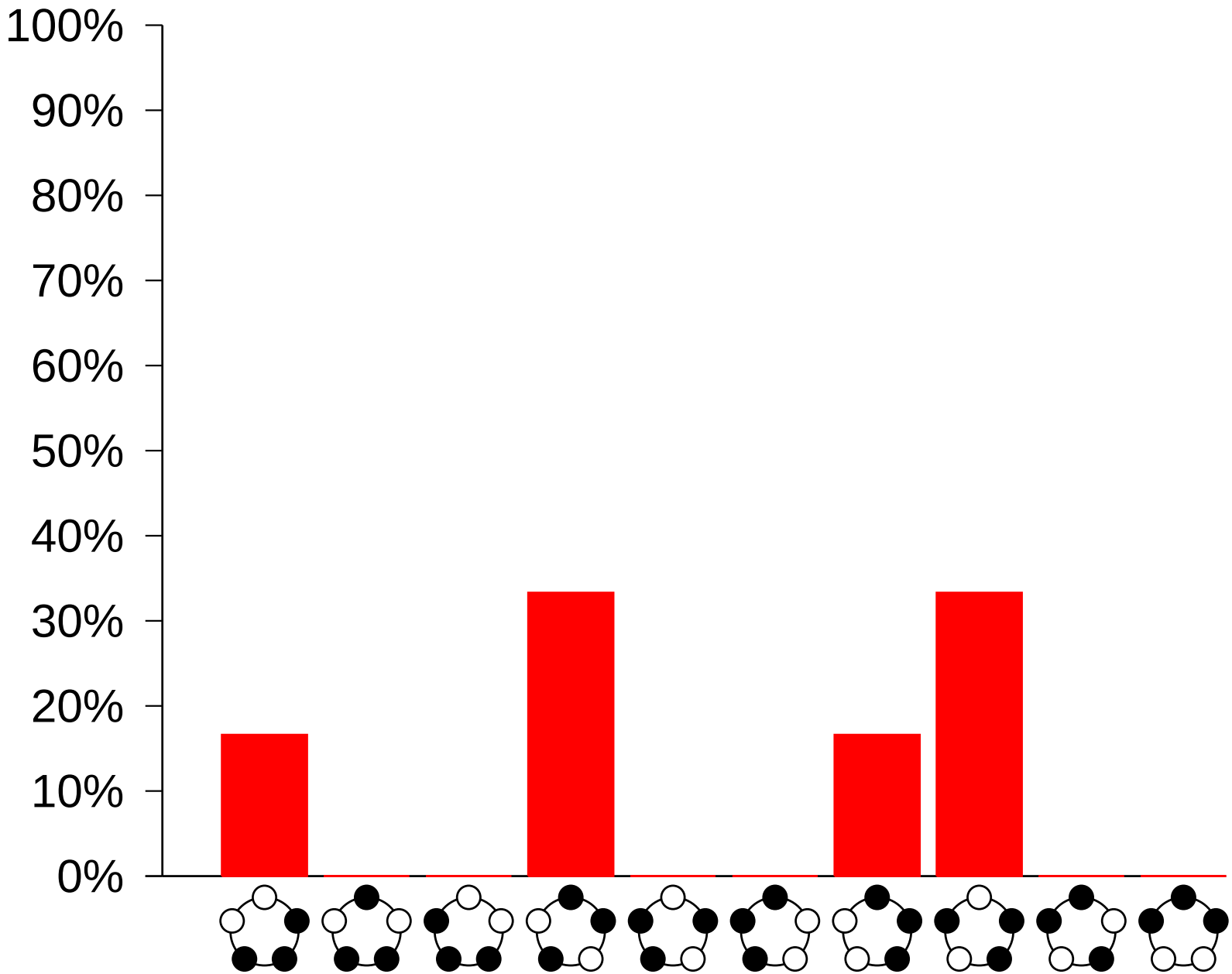
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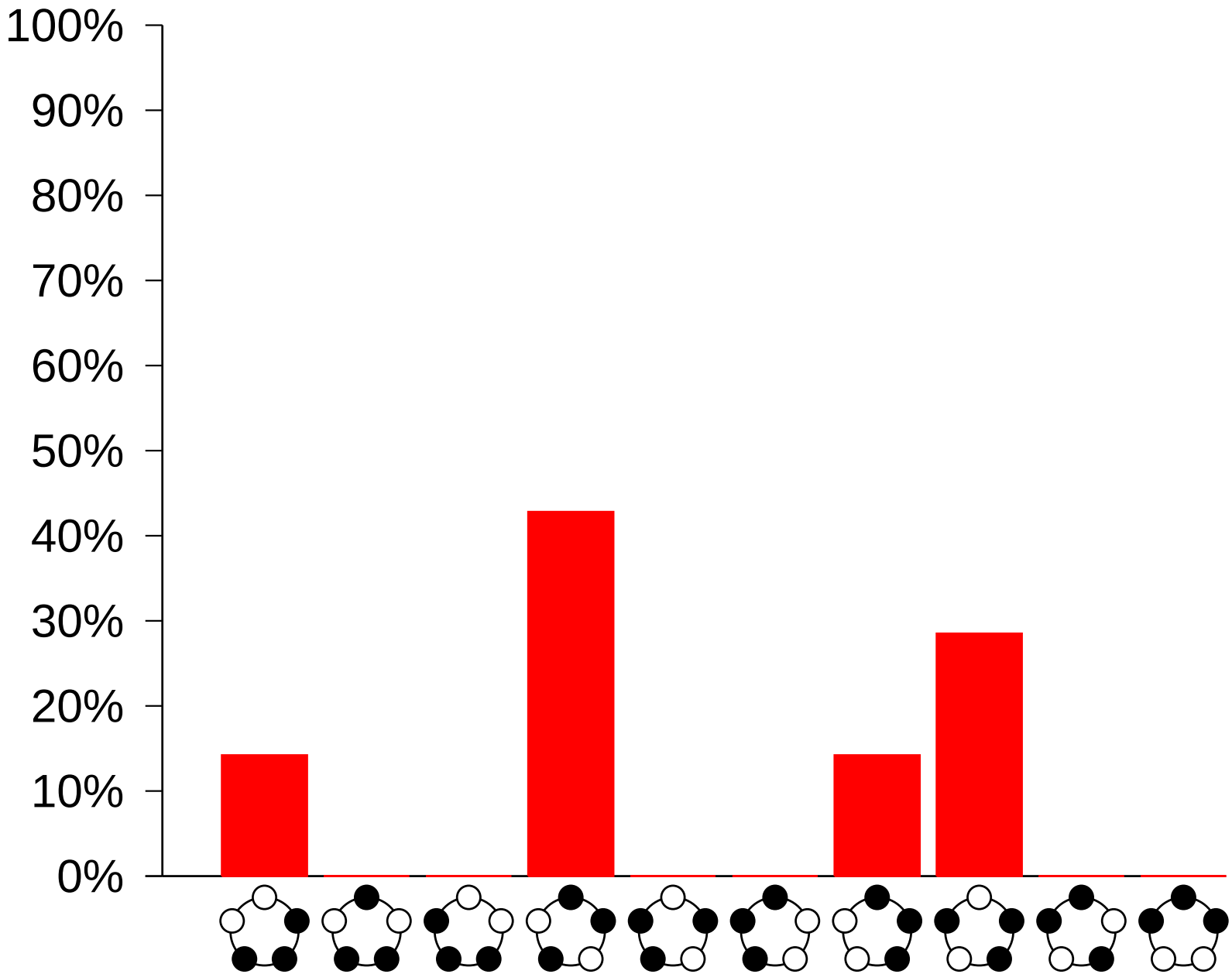
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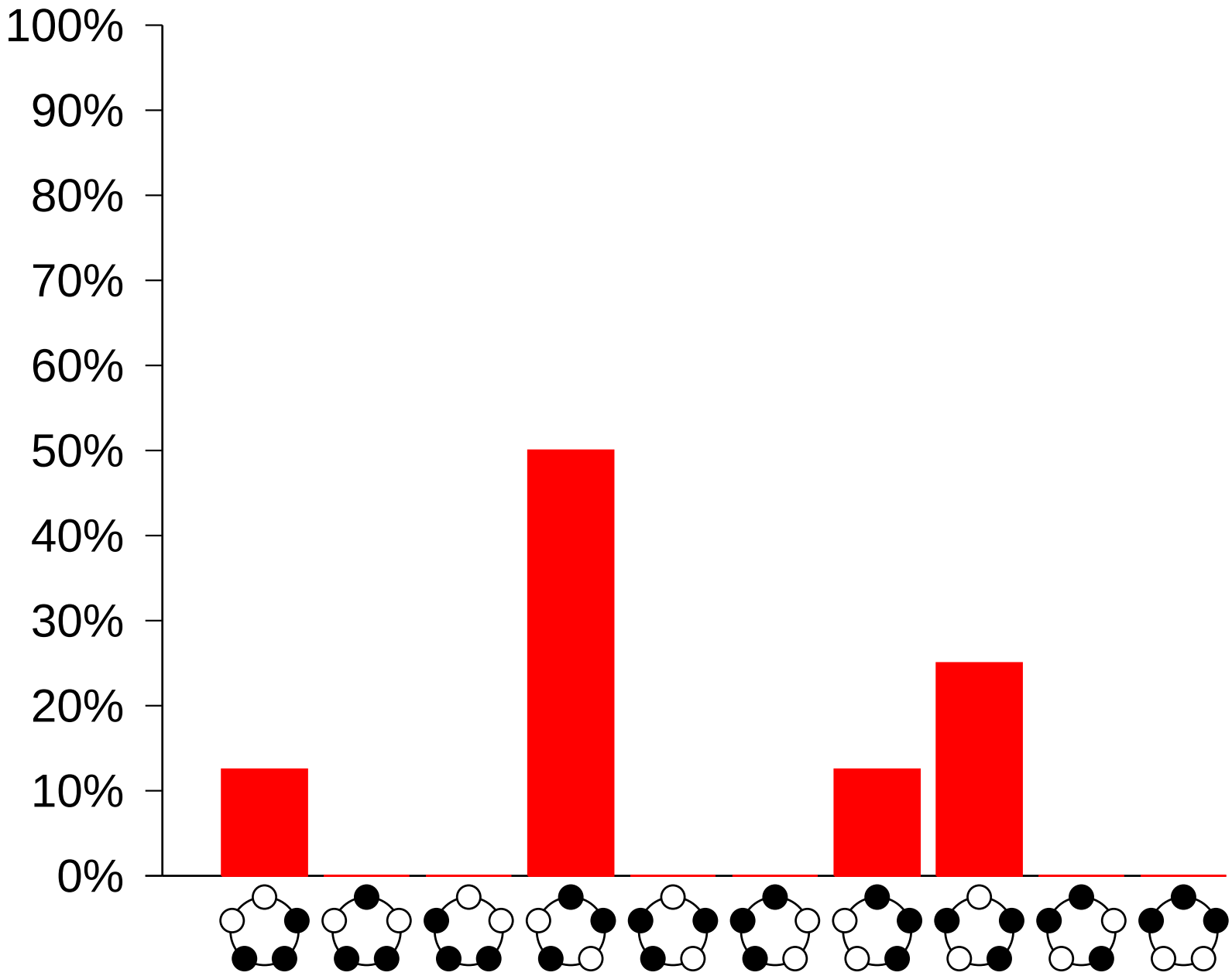
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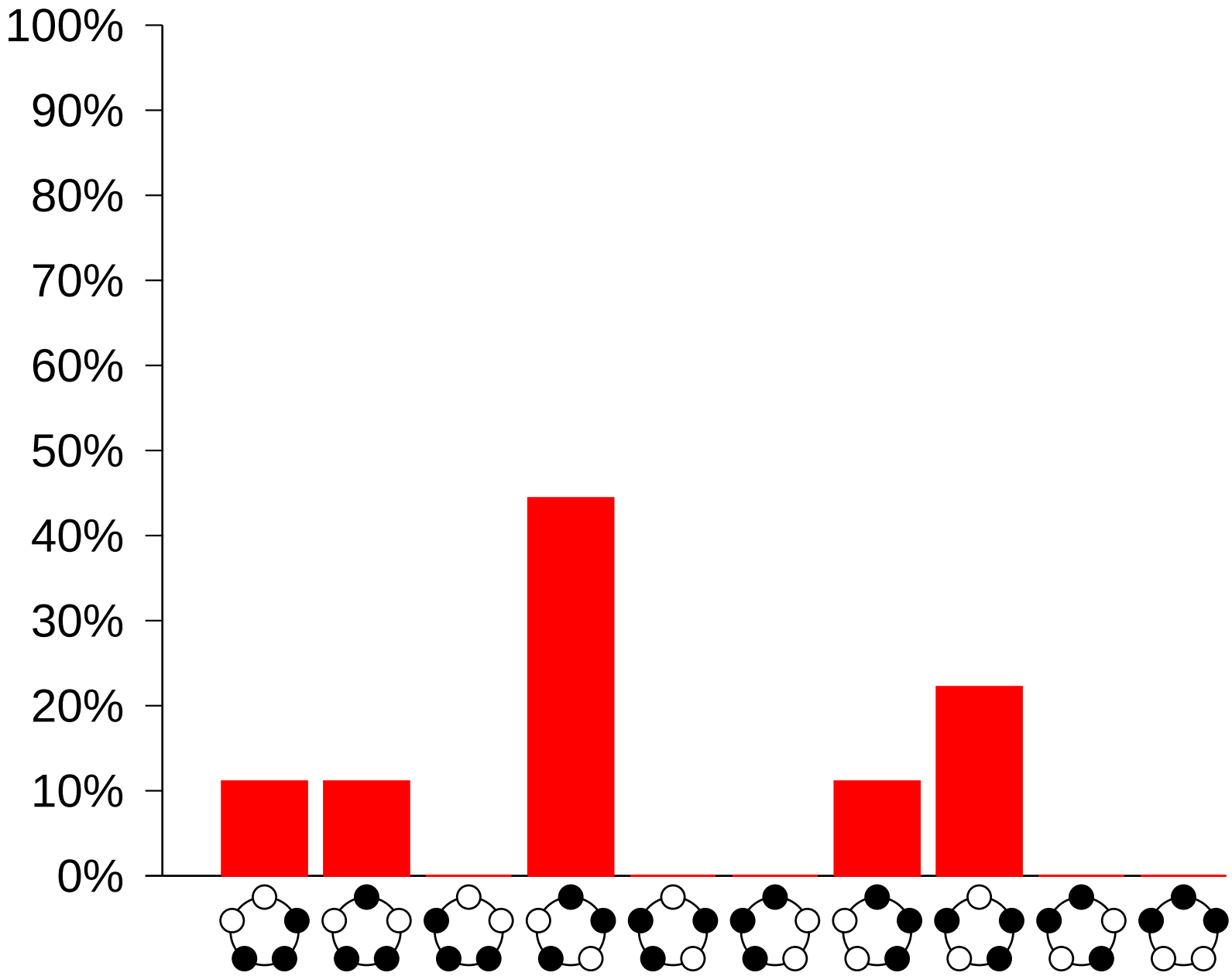
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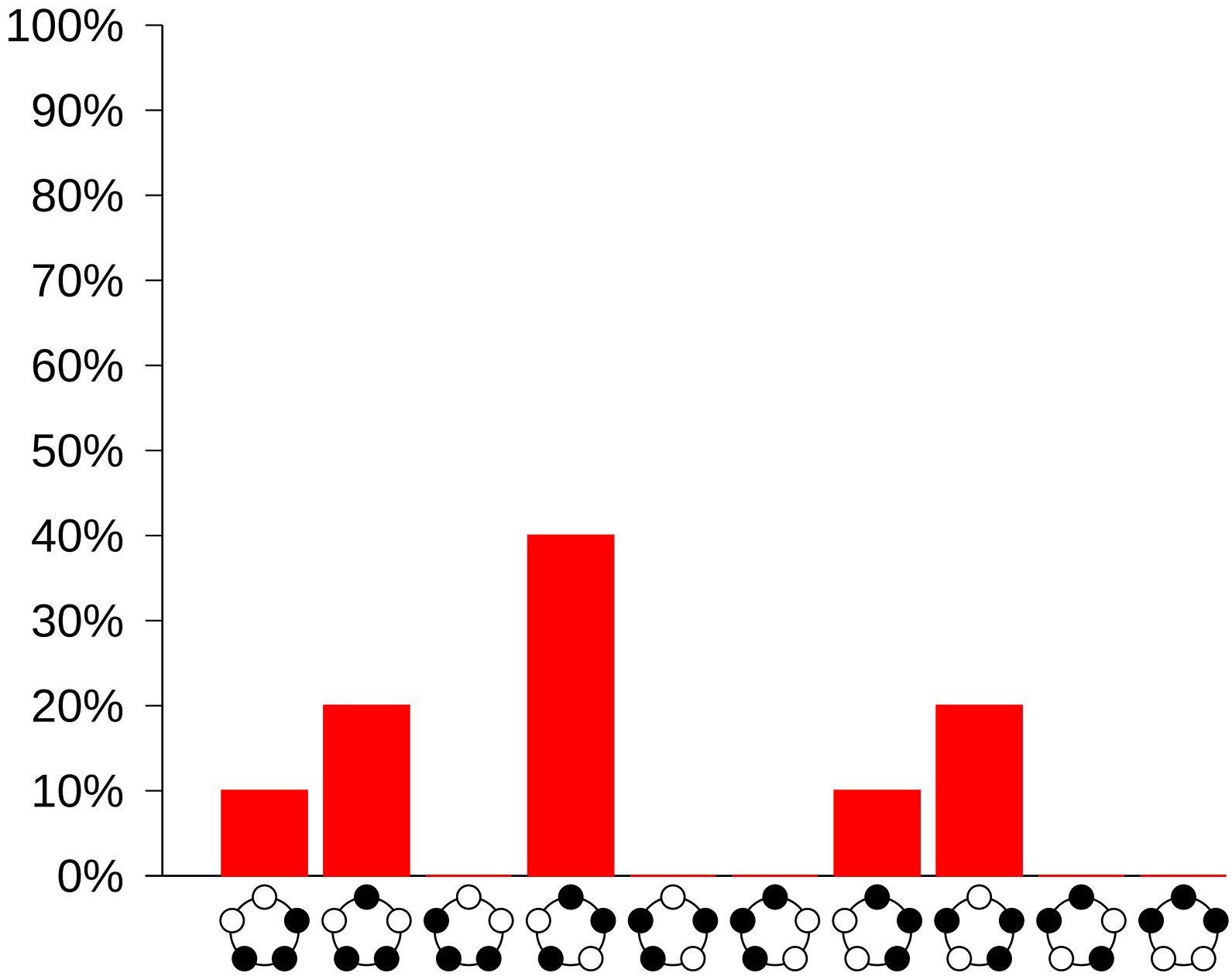
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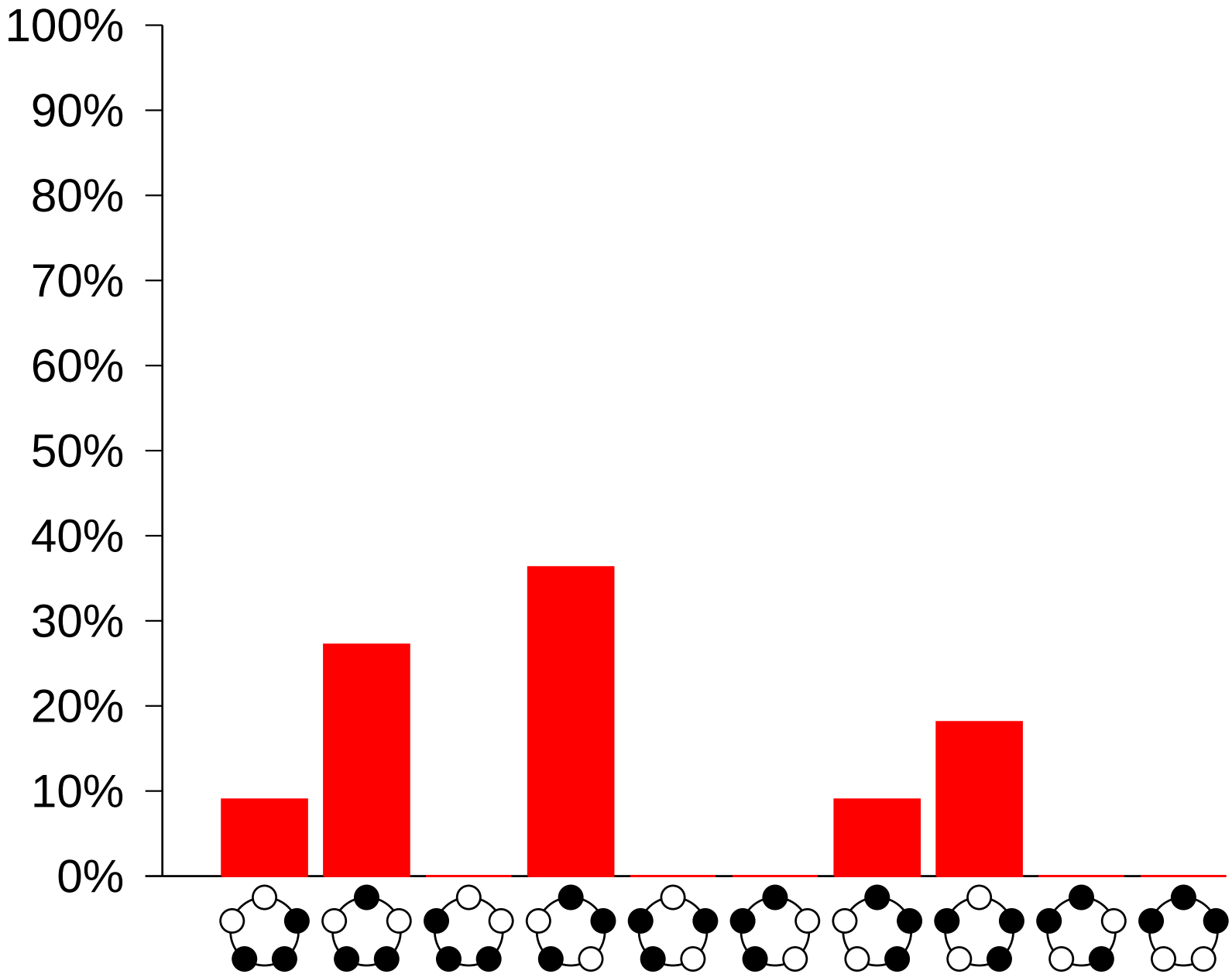
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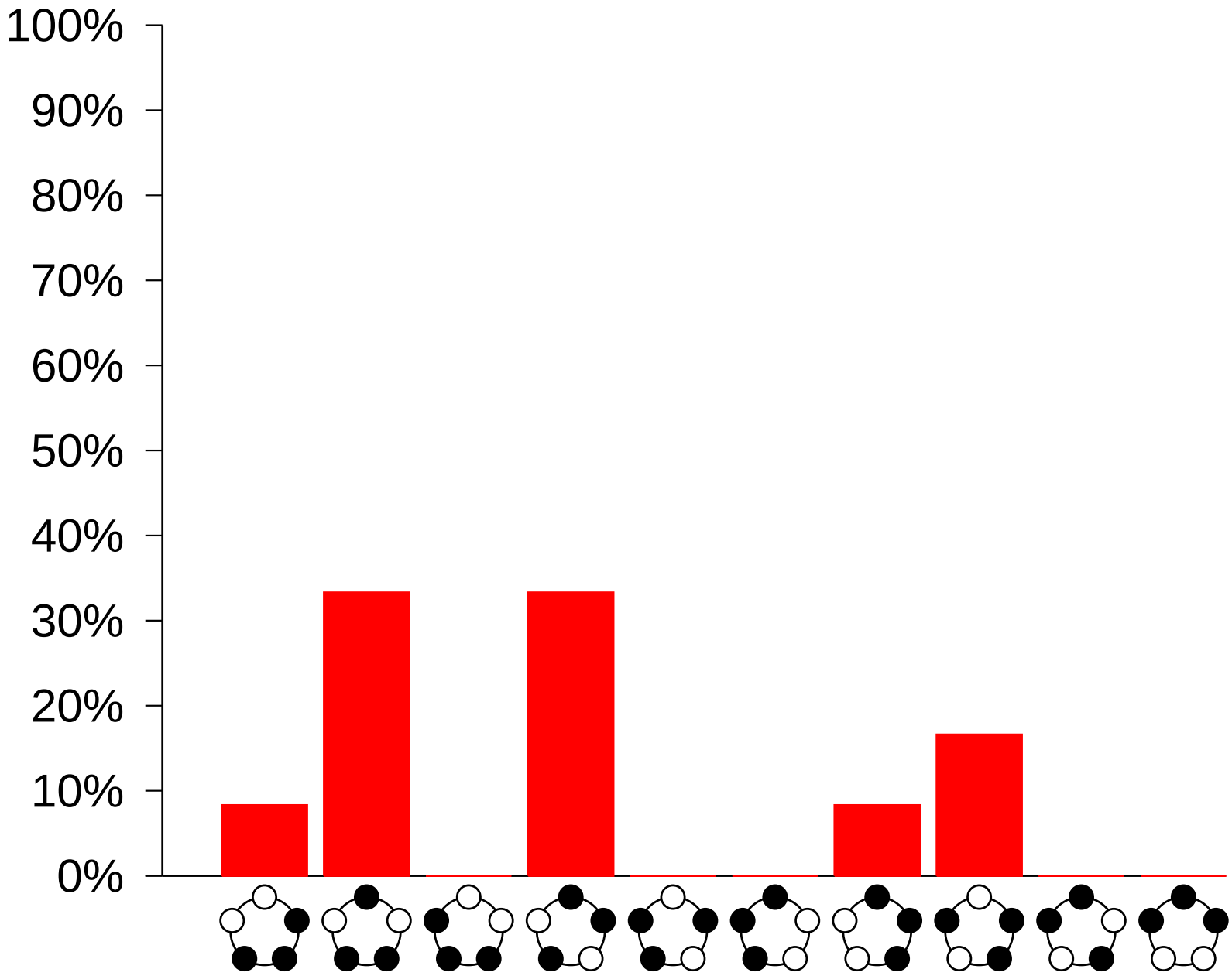
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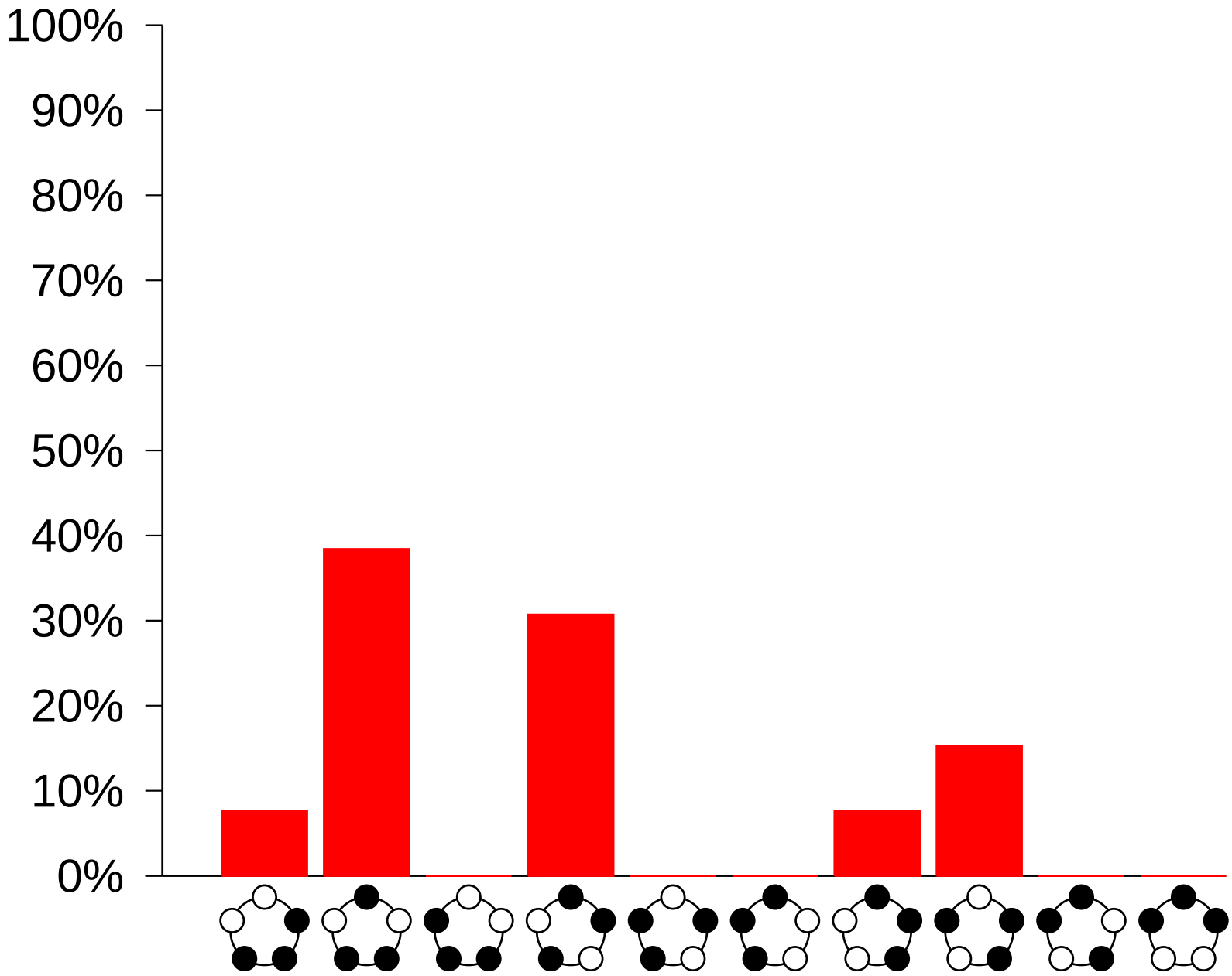
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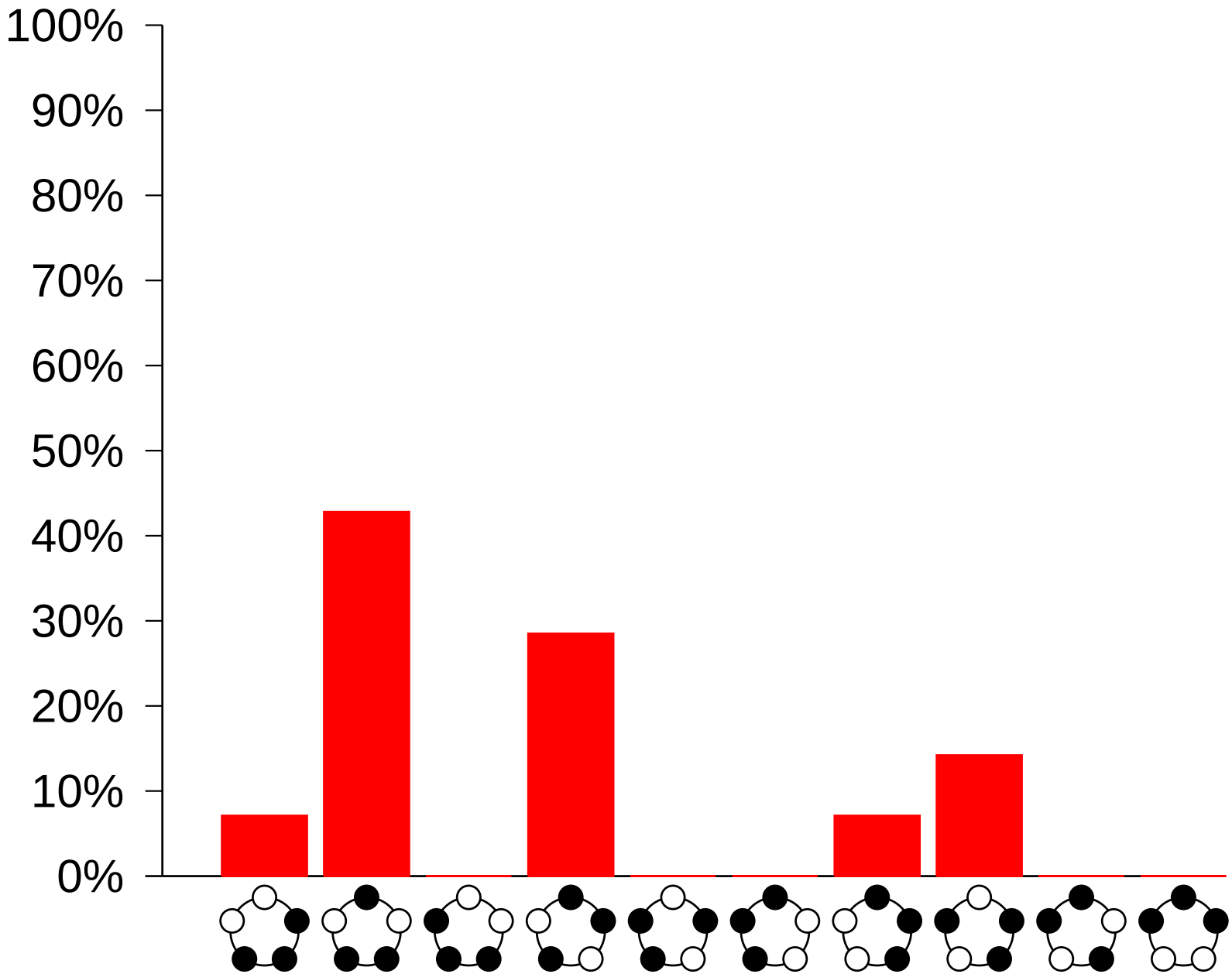
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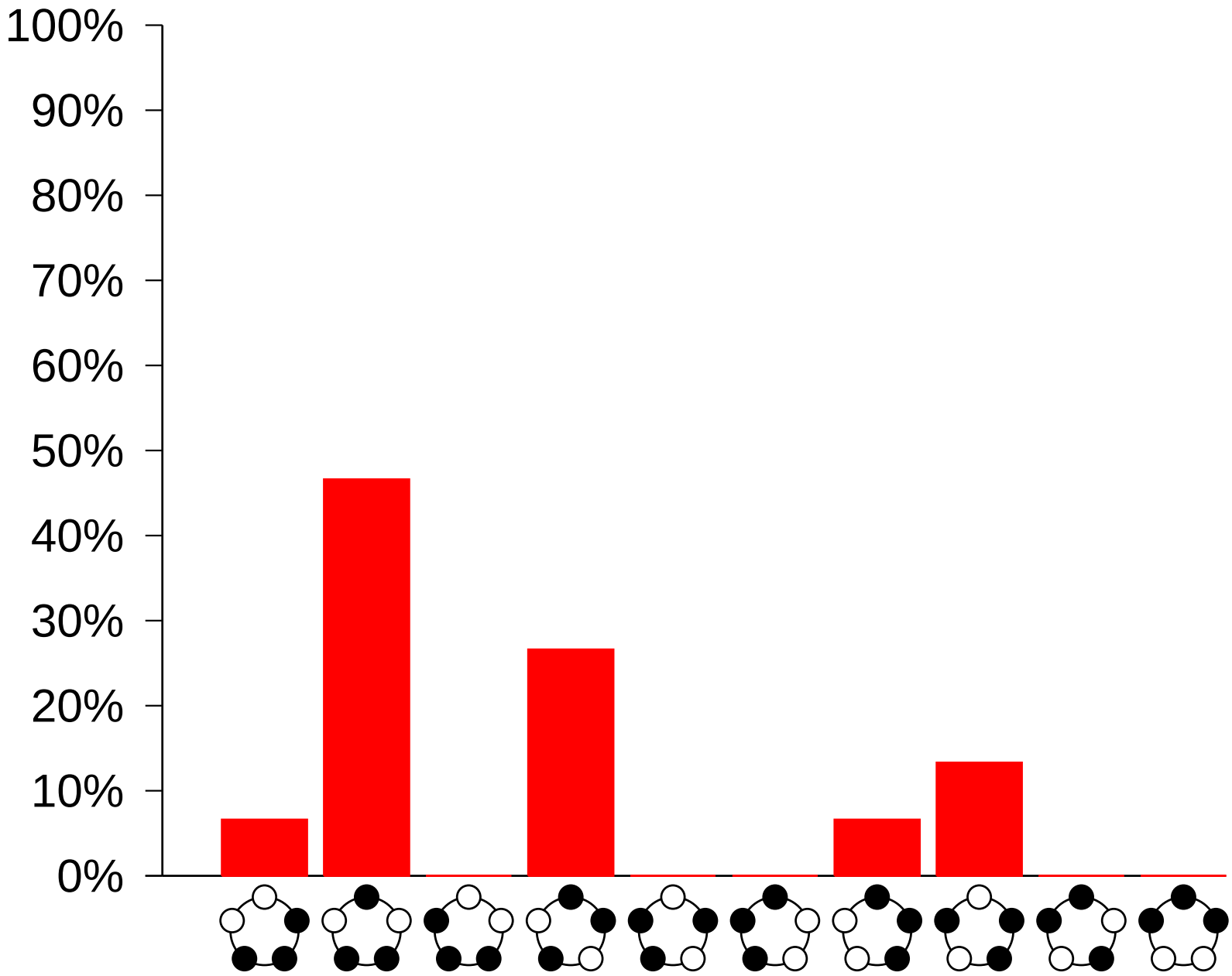
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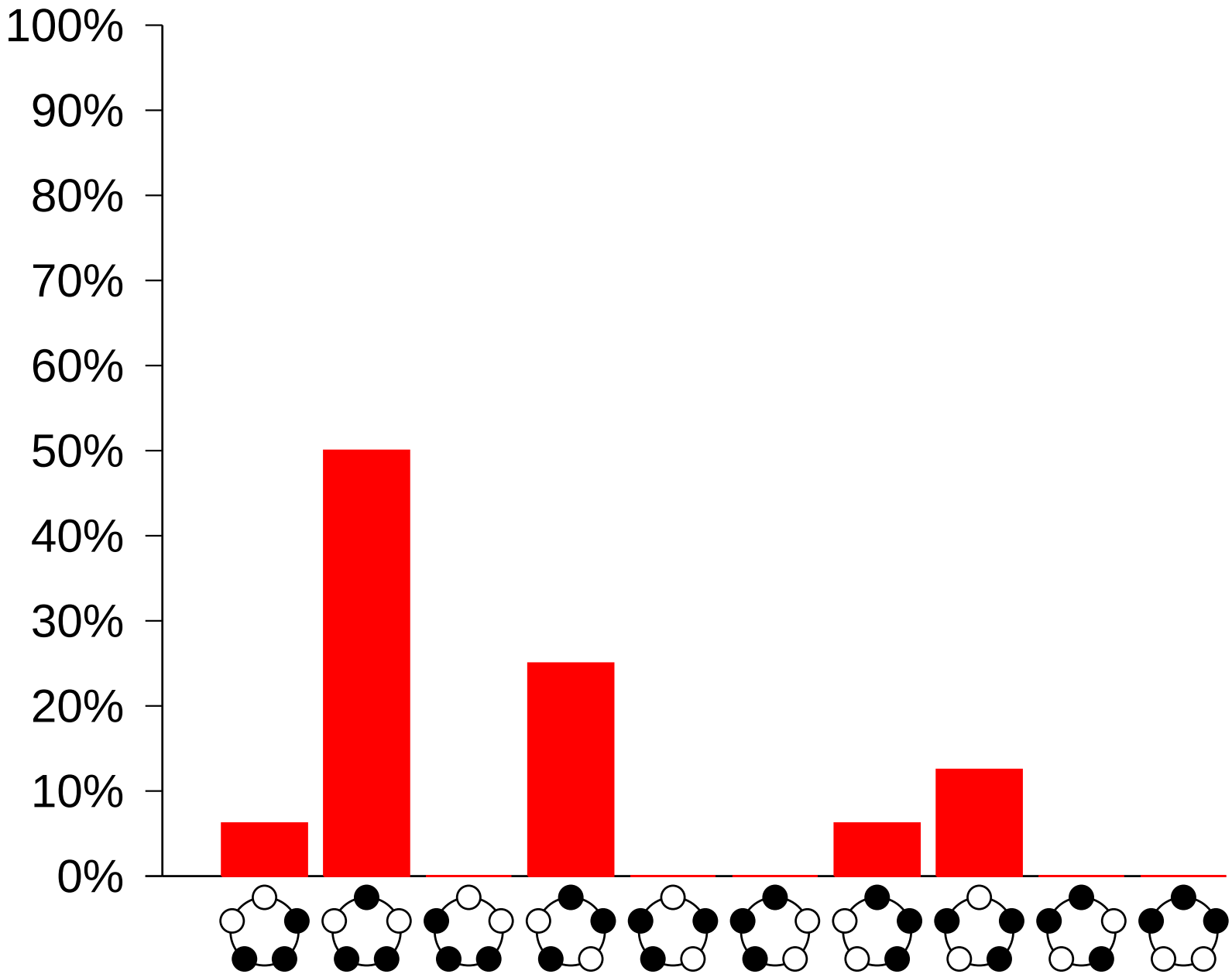
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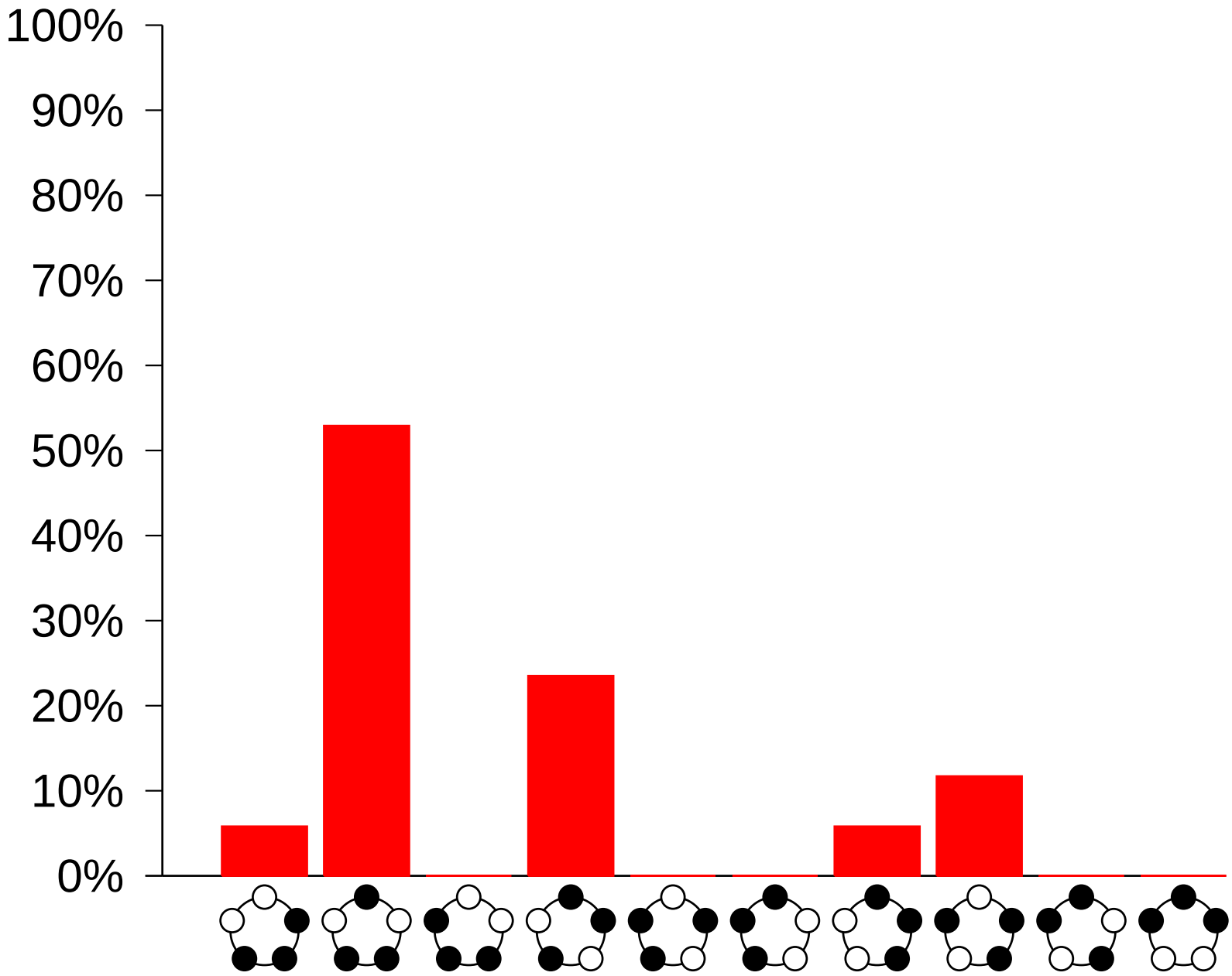
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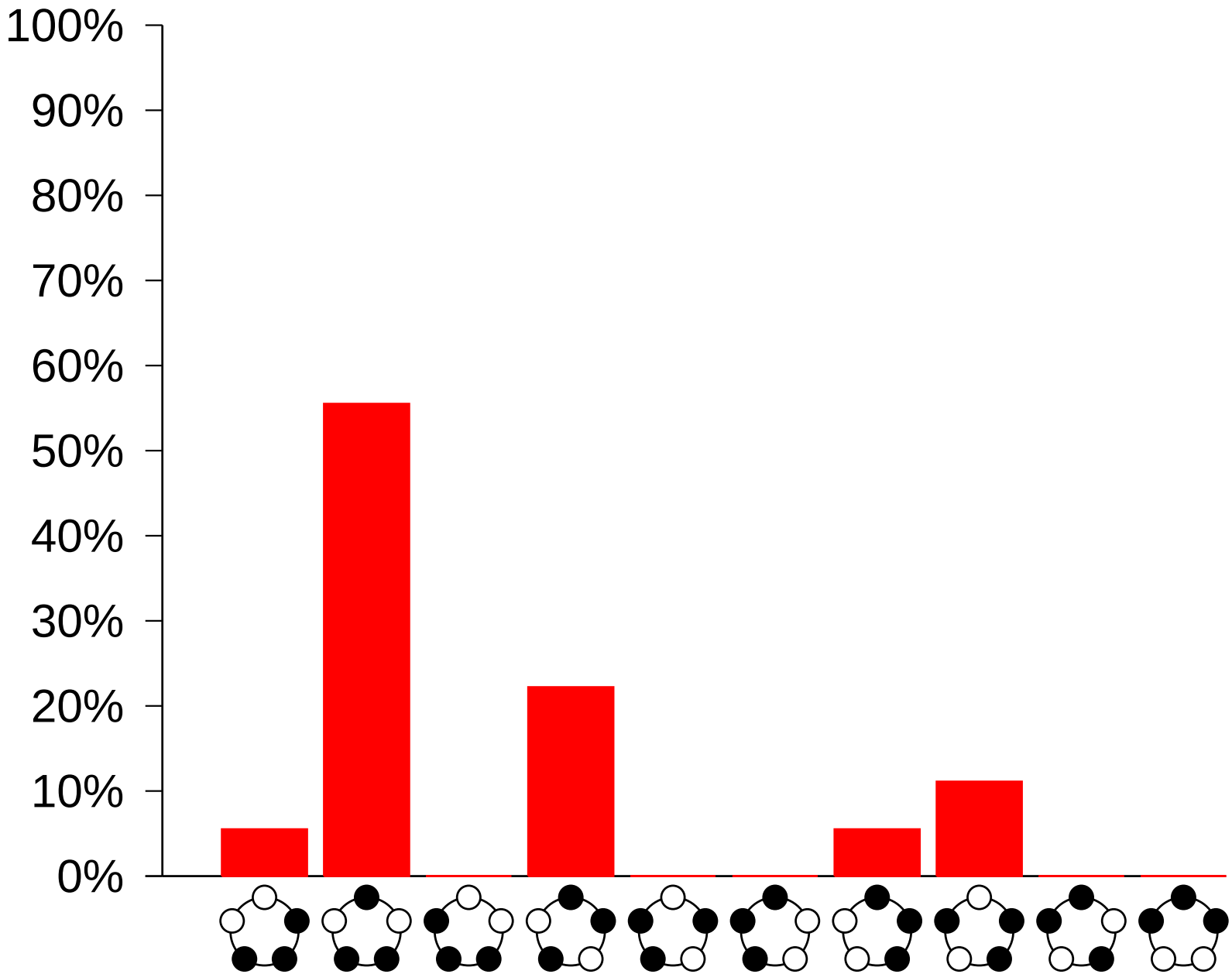
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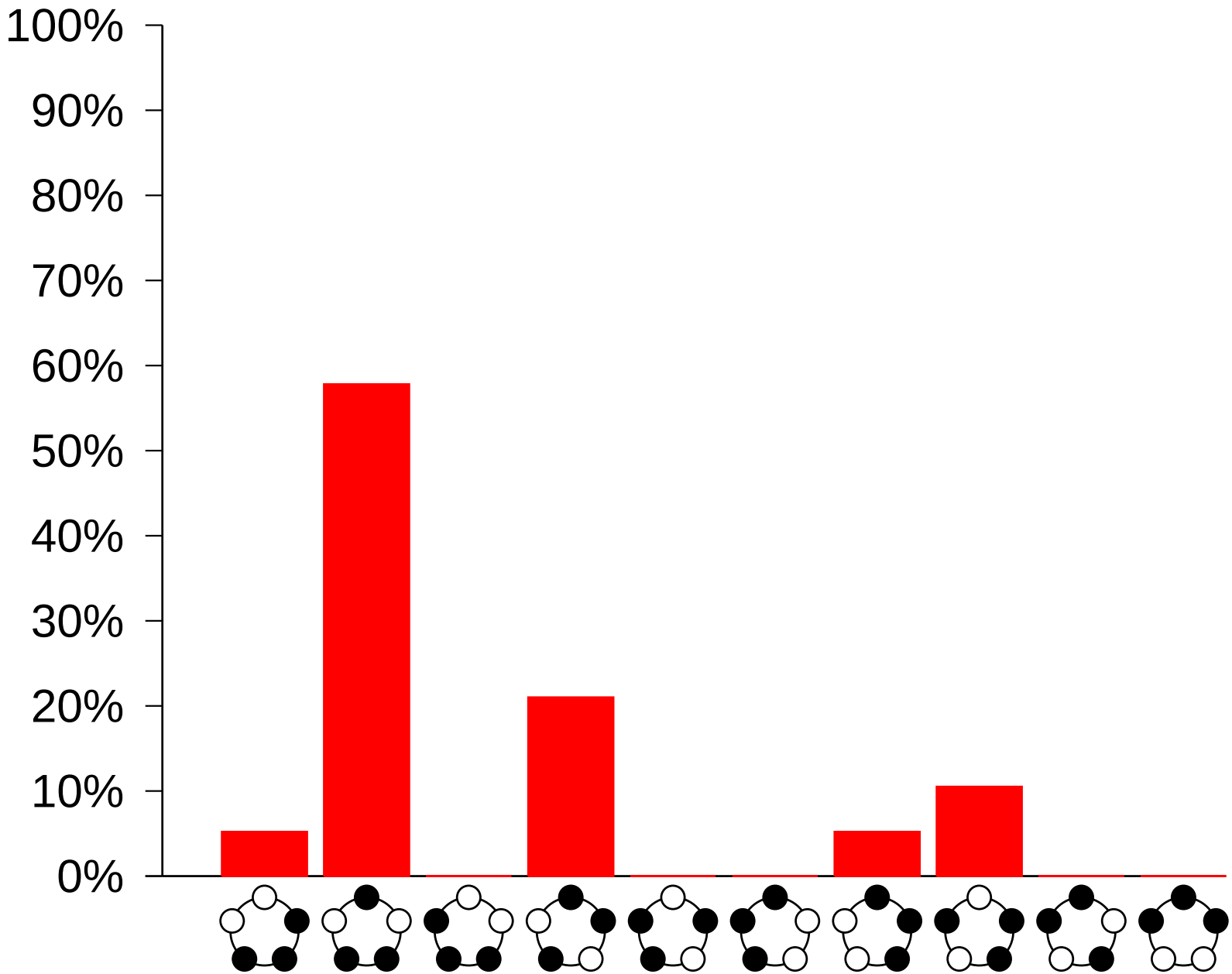
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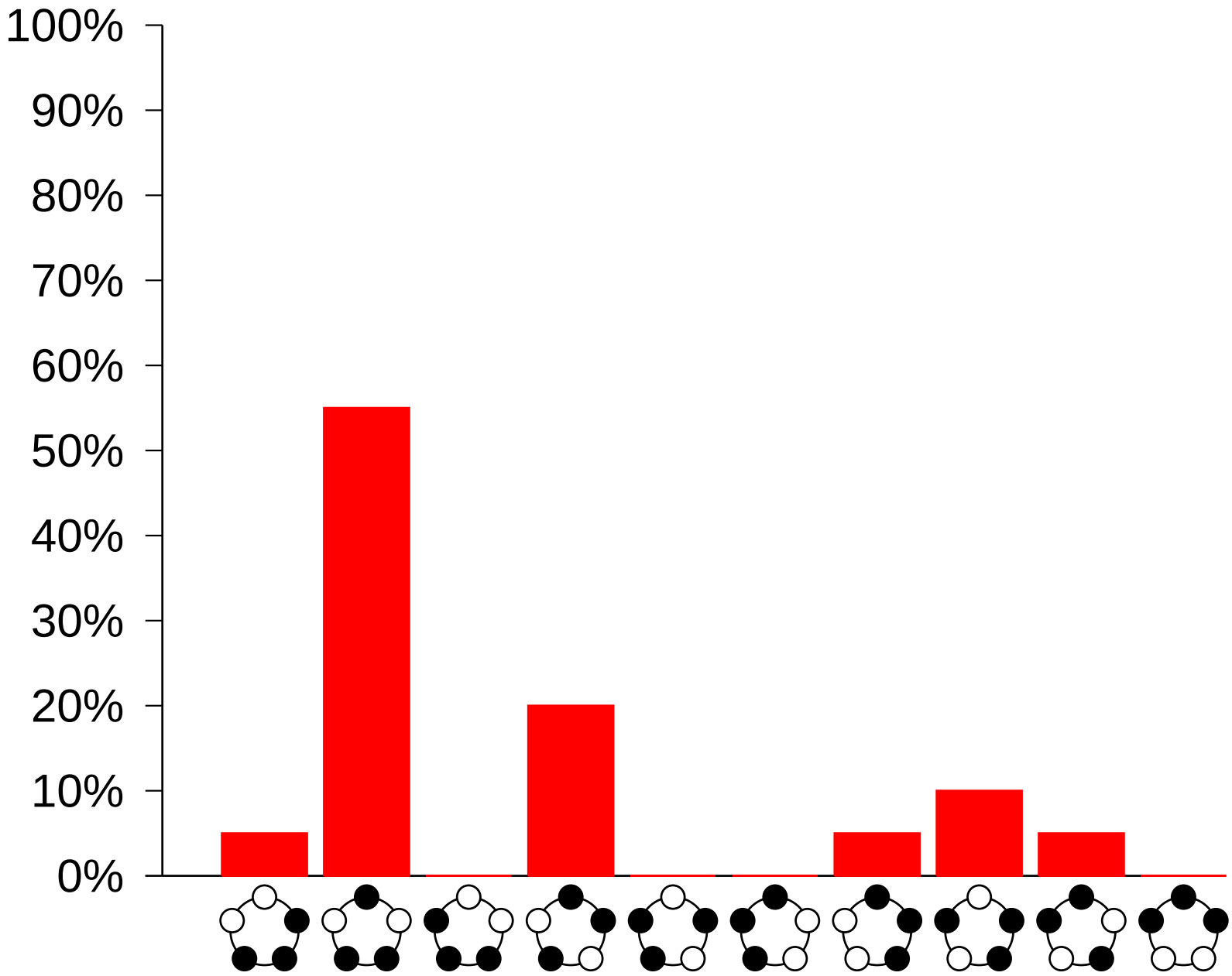
Stationary distribution



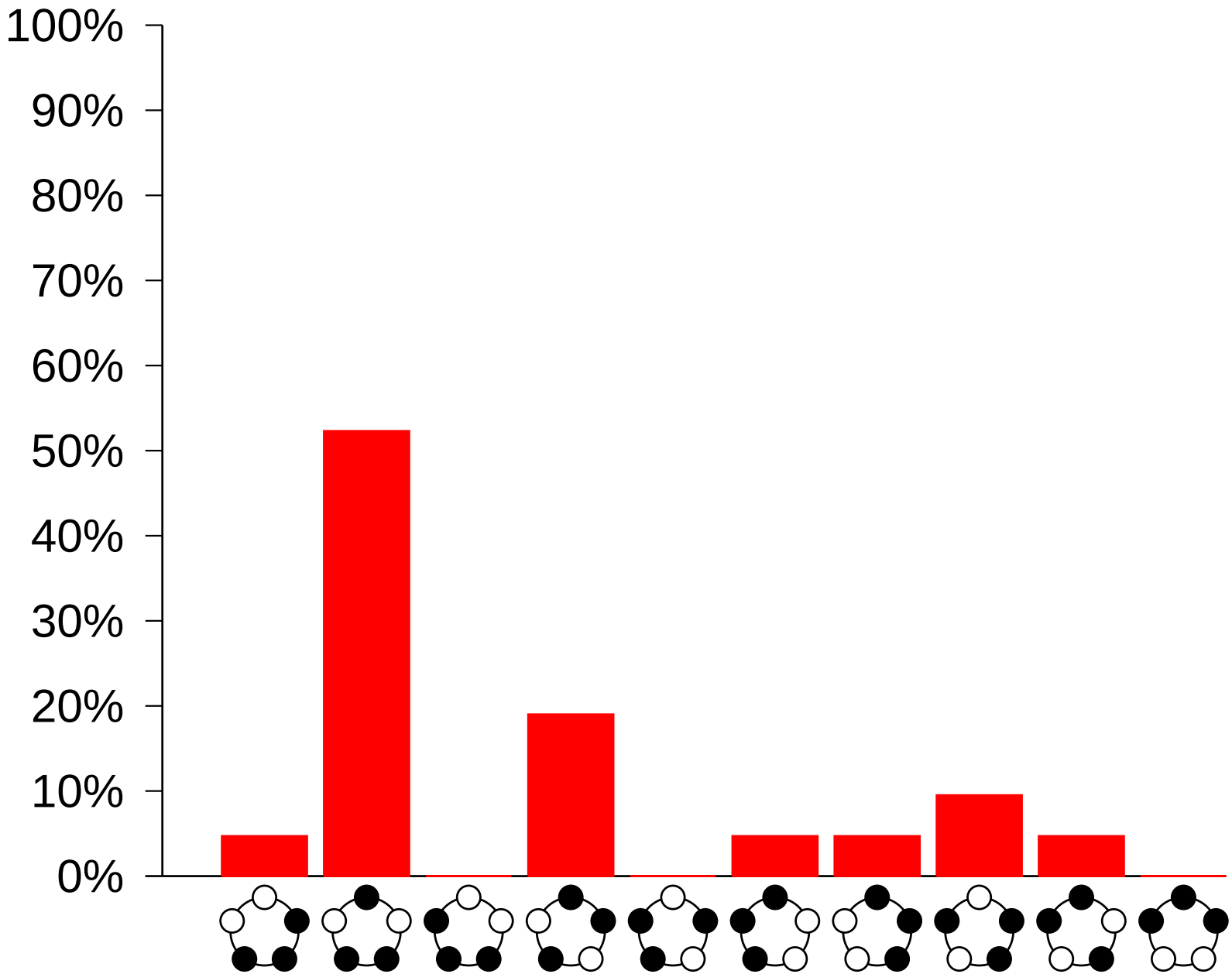
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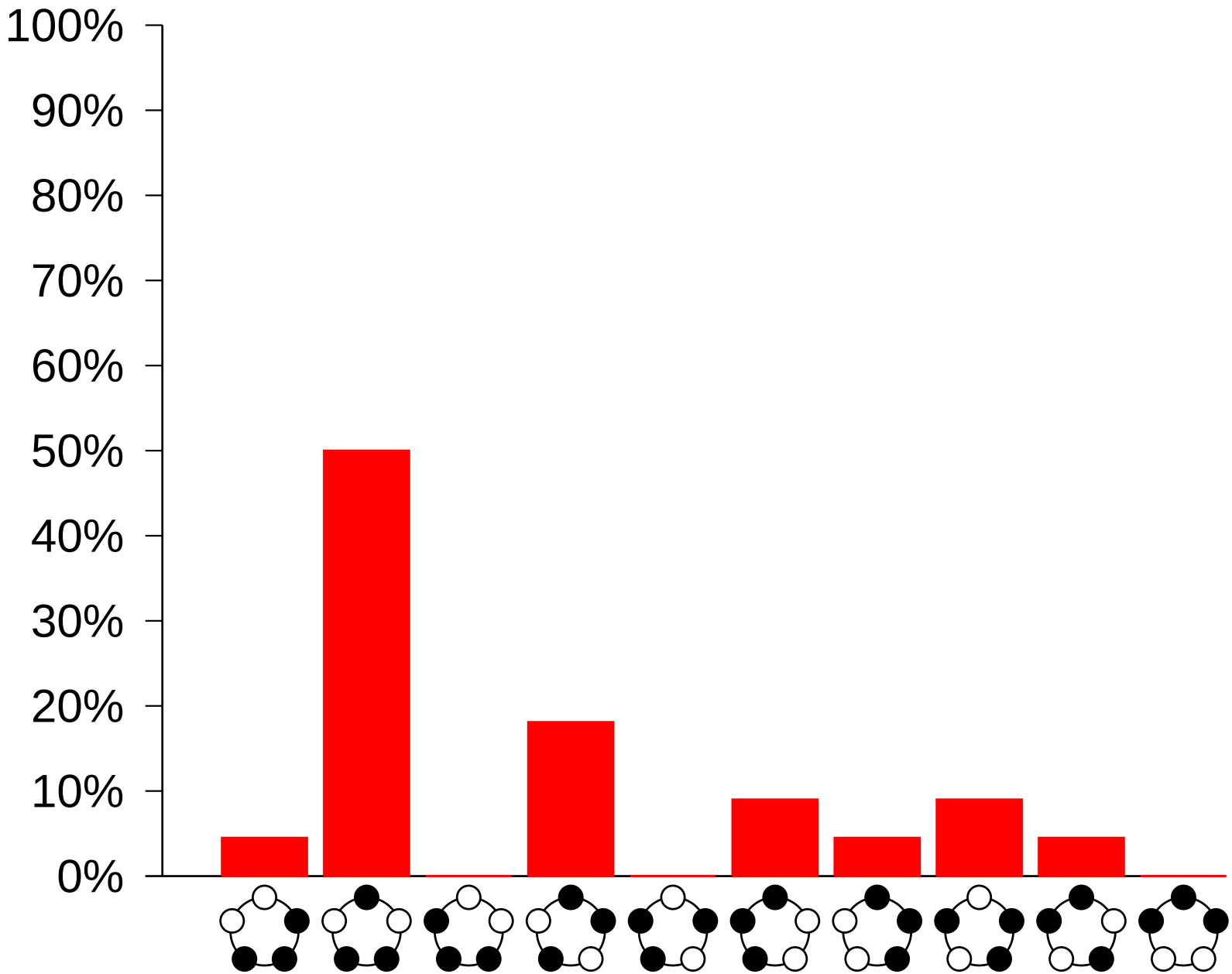
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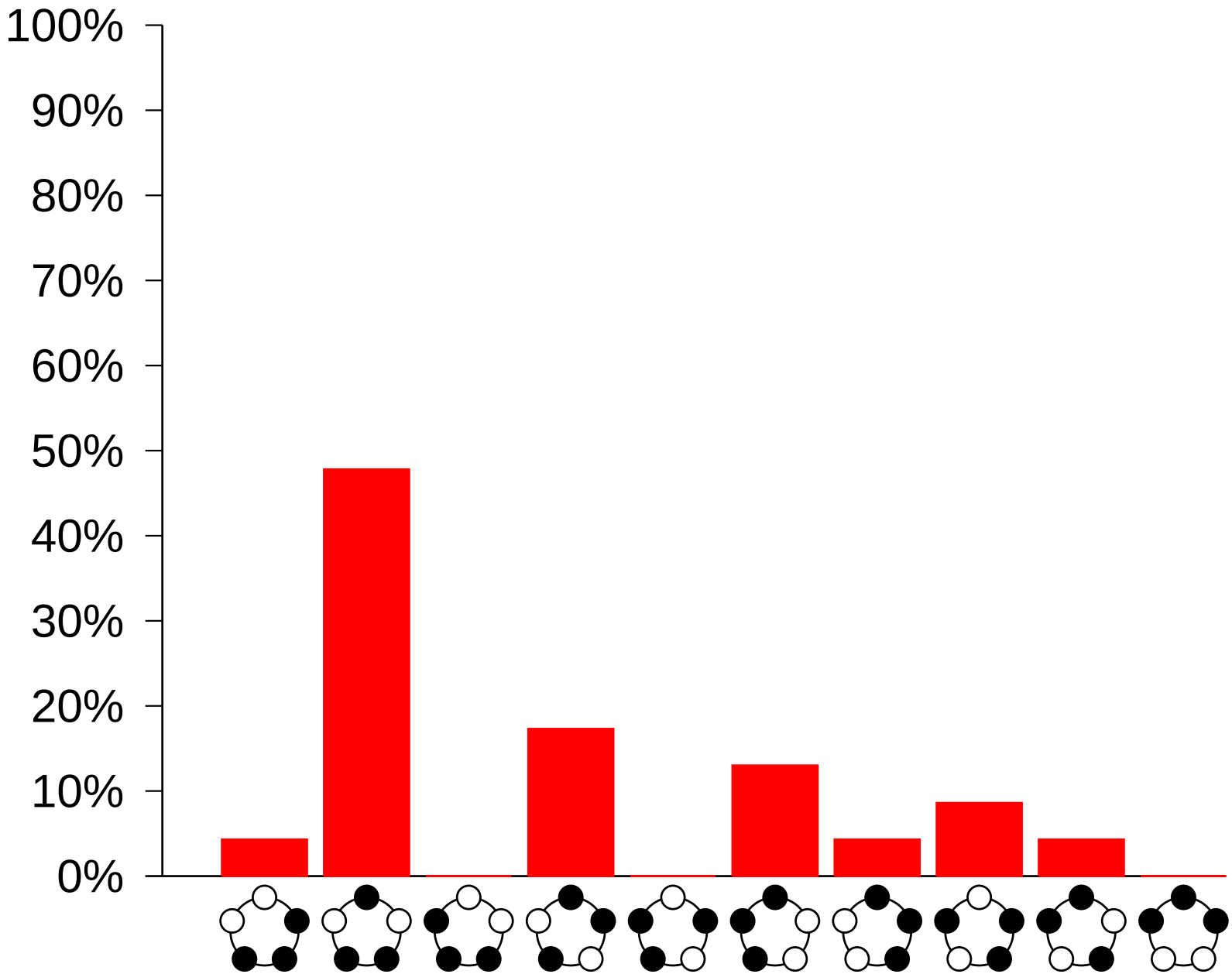
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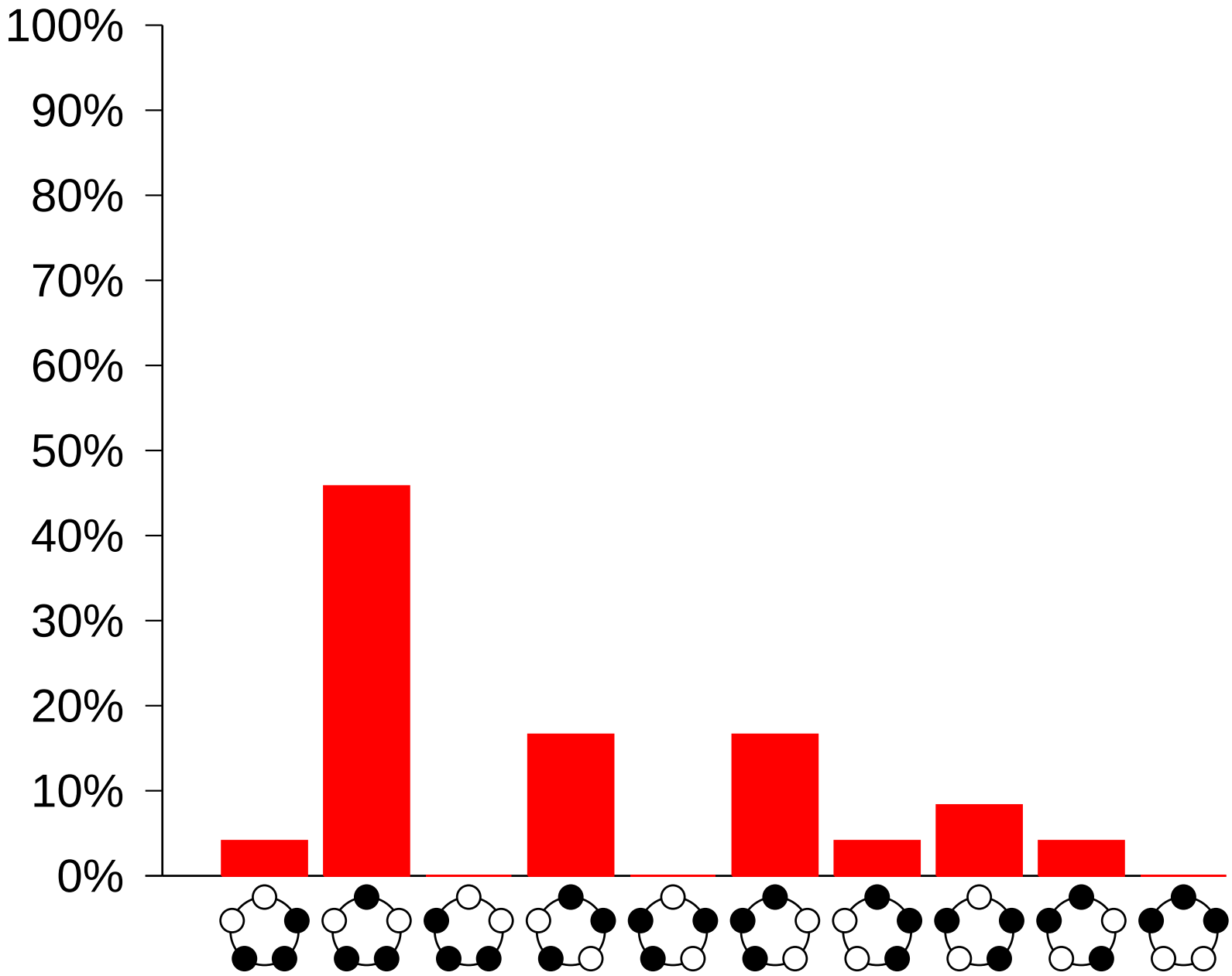
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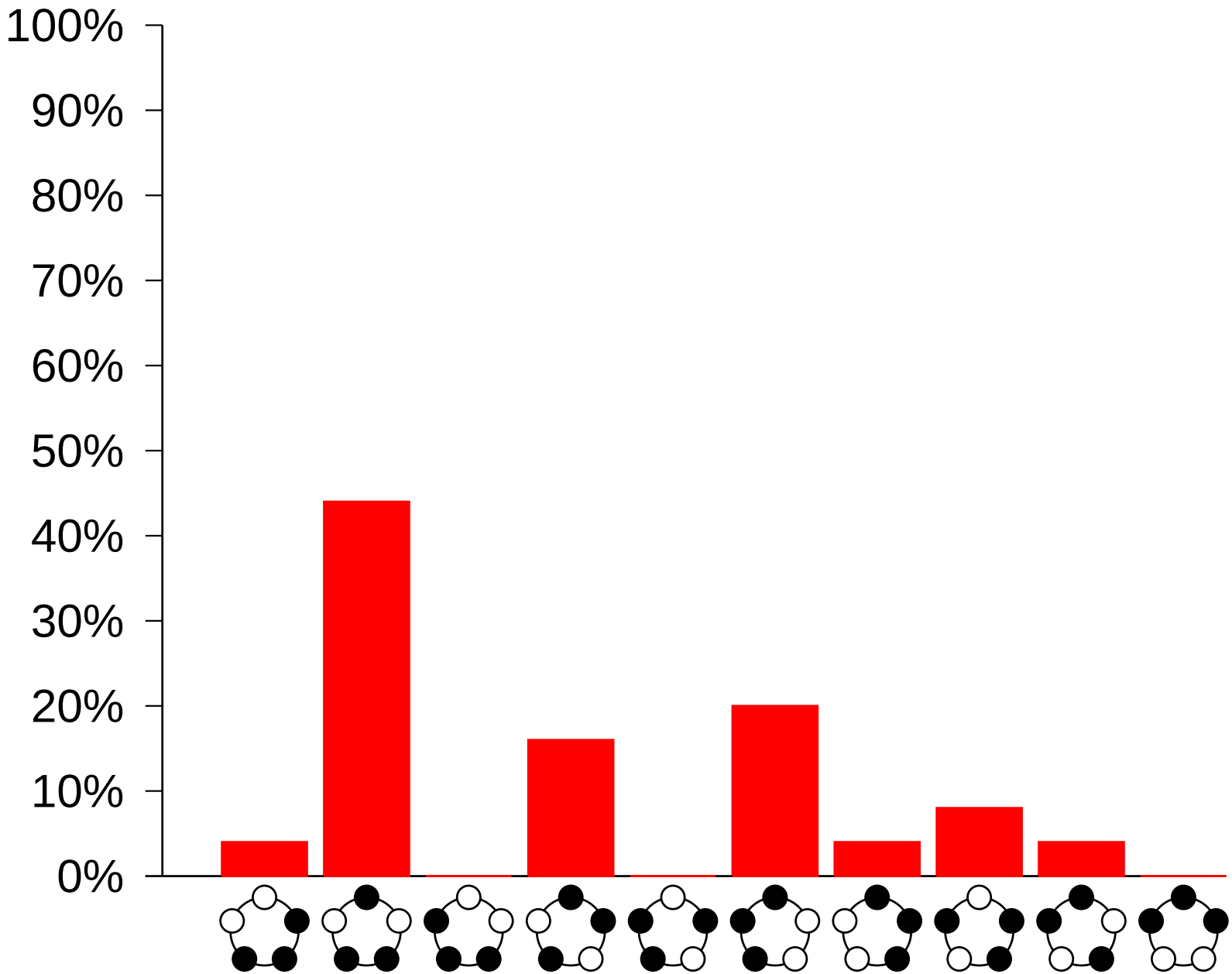
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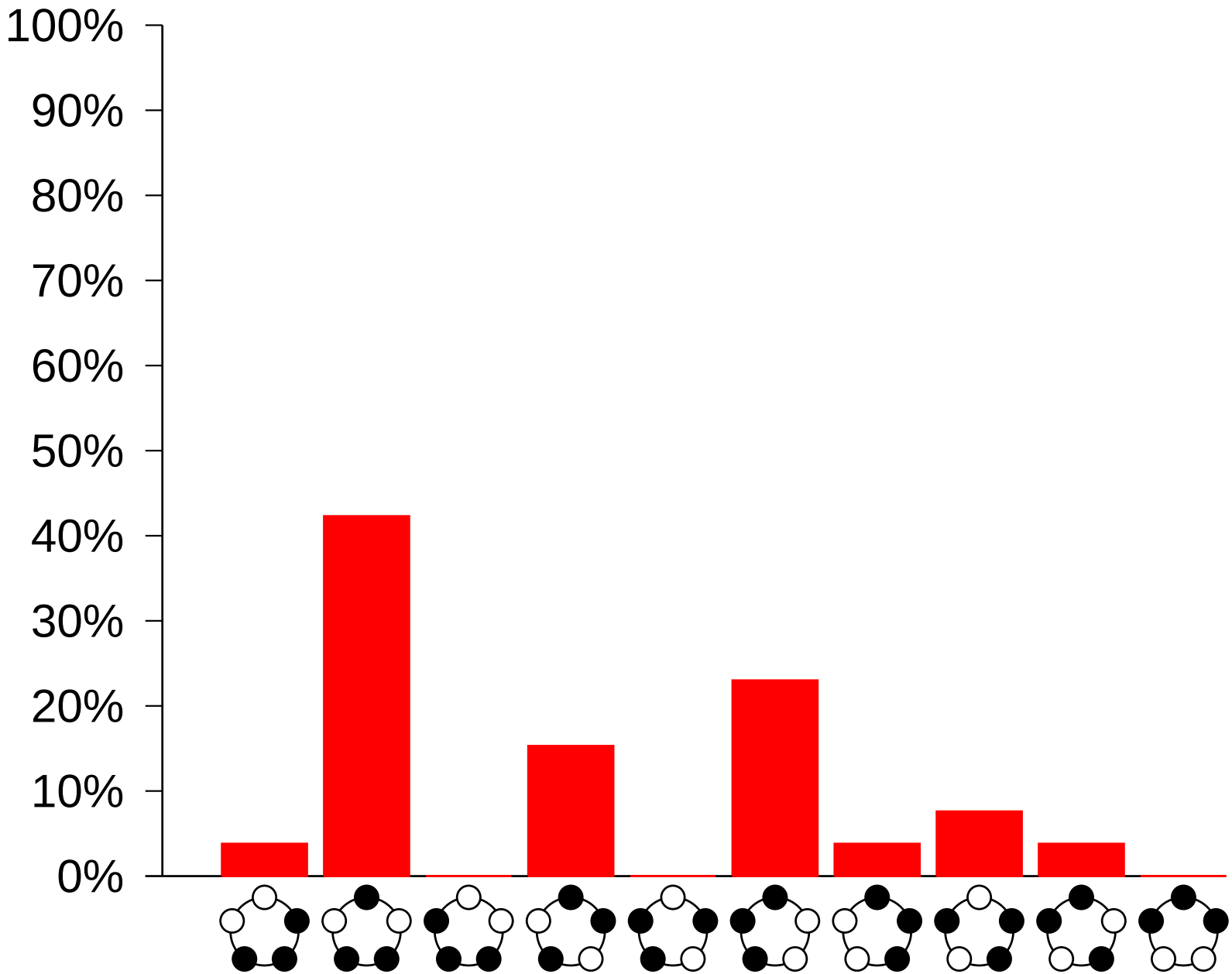
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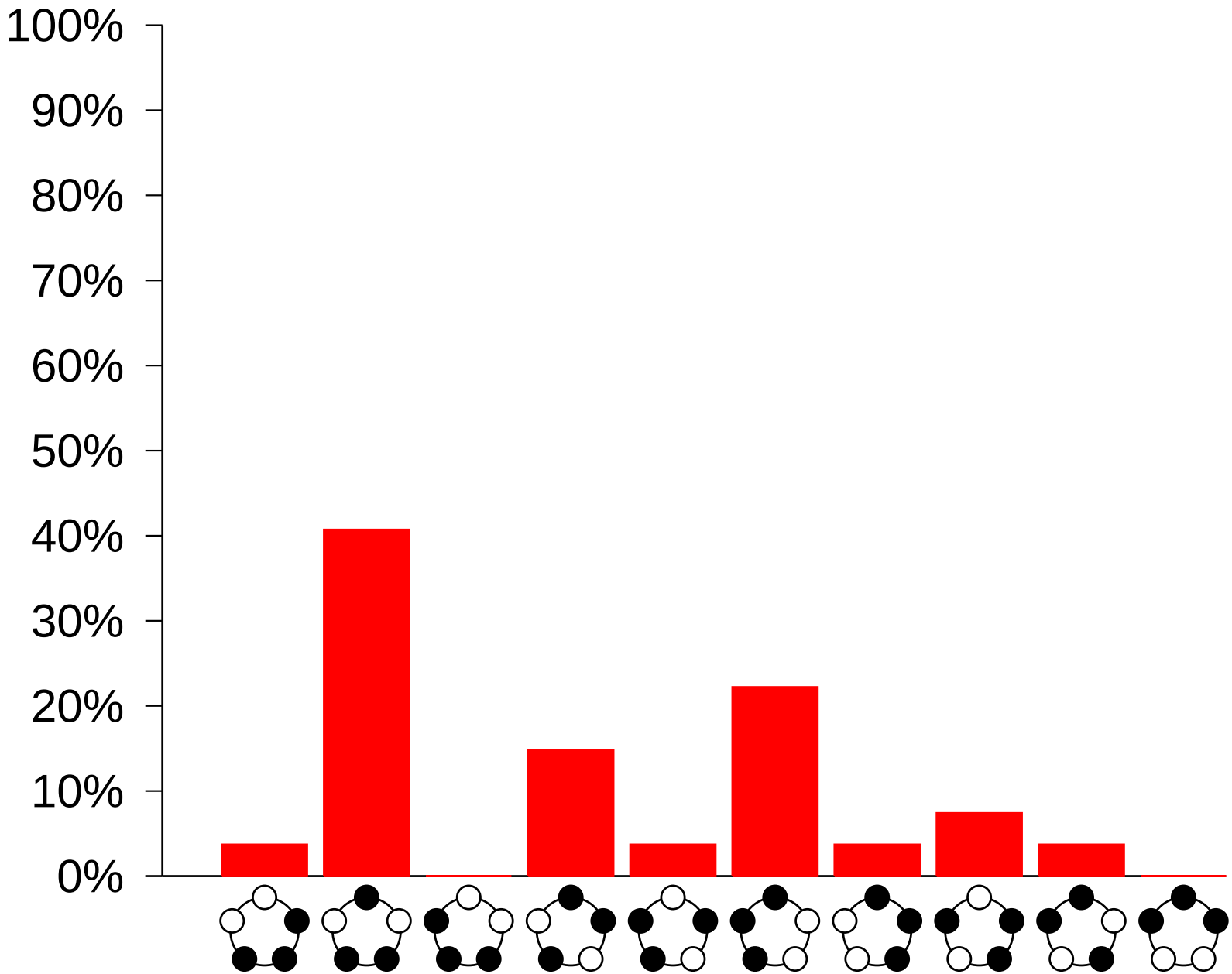
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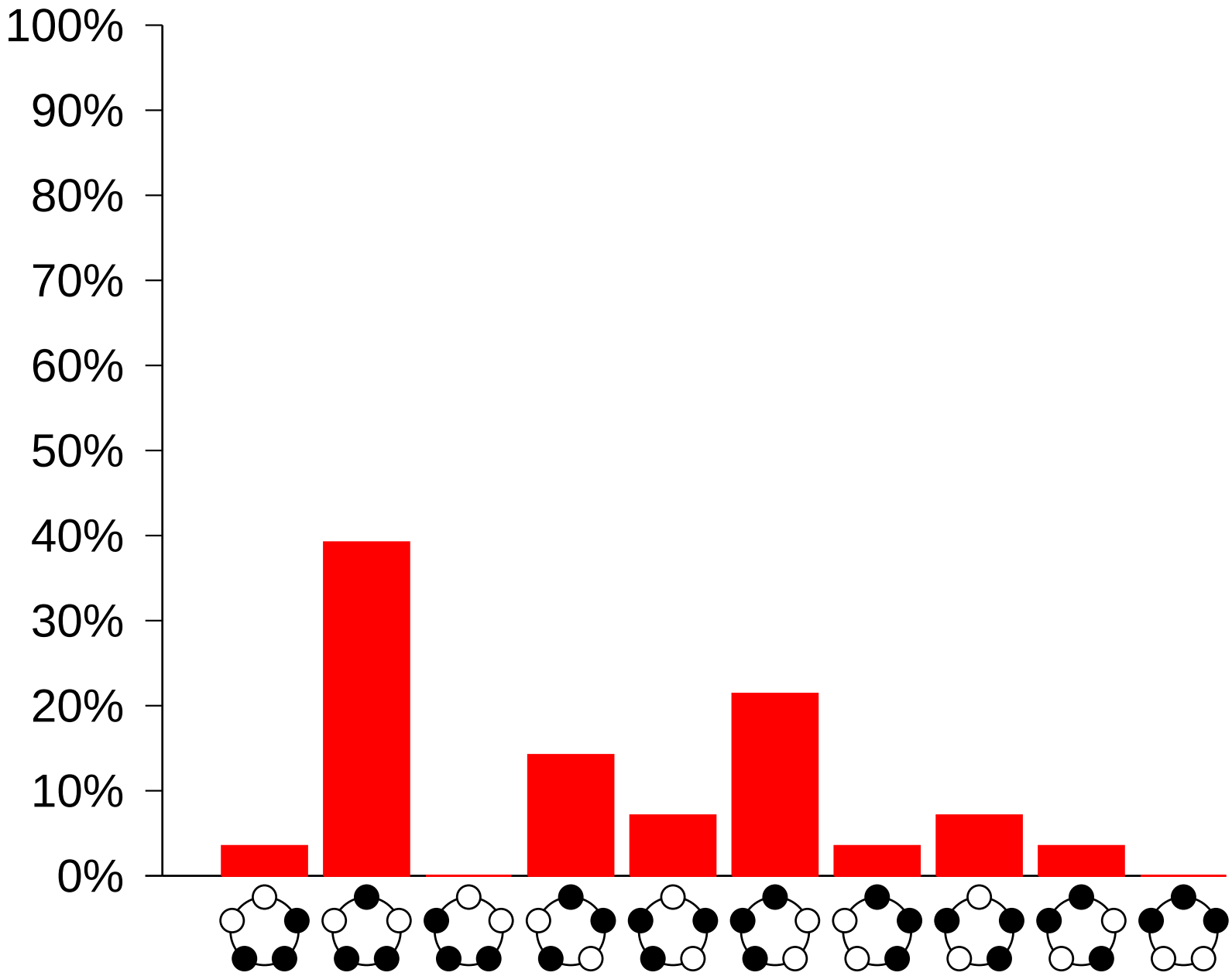
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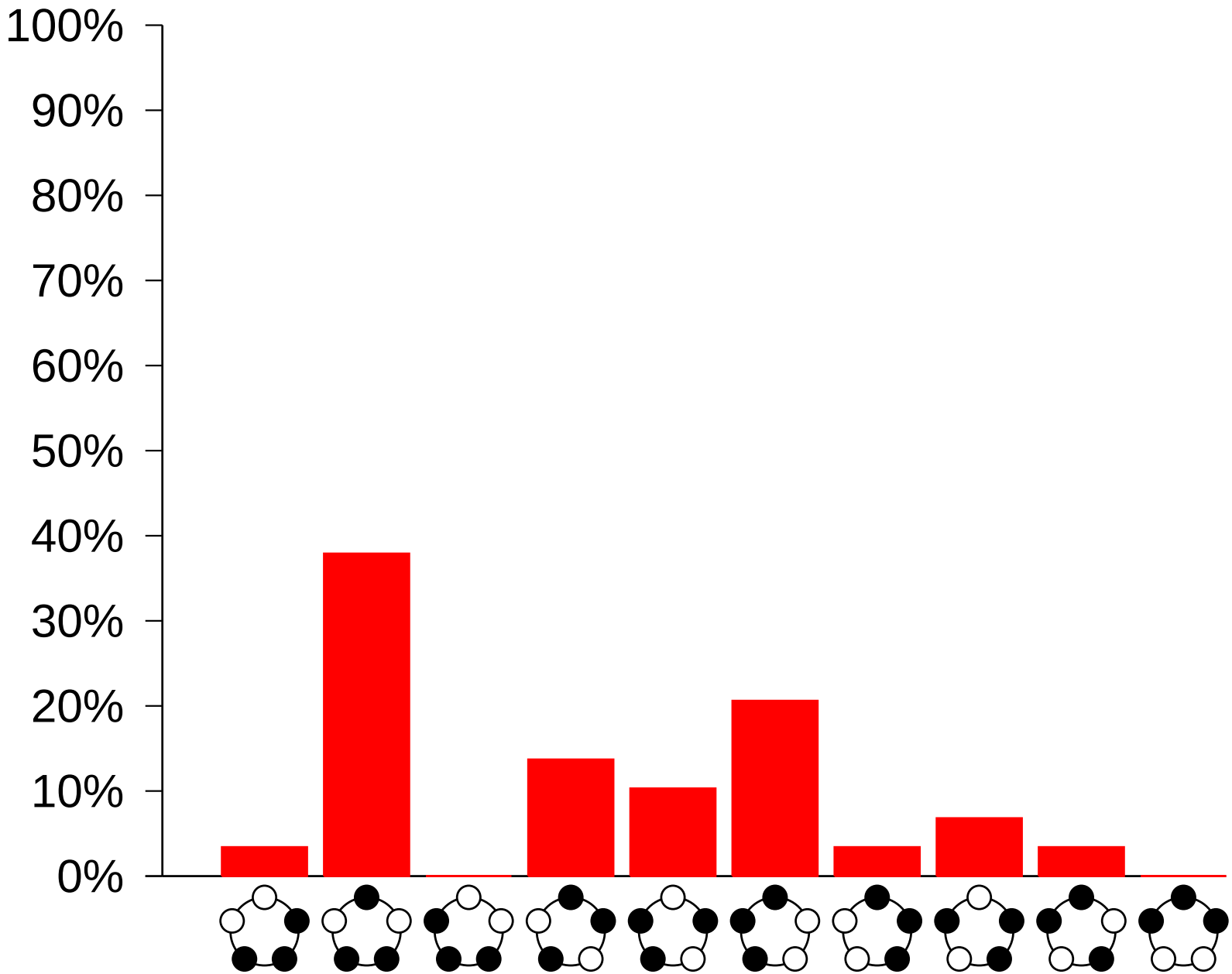
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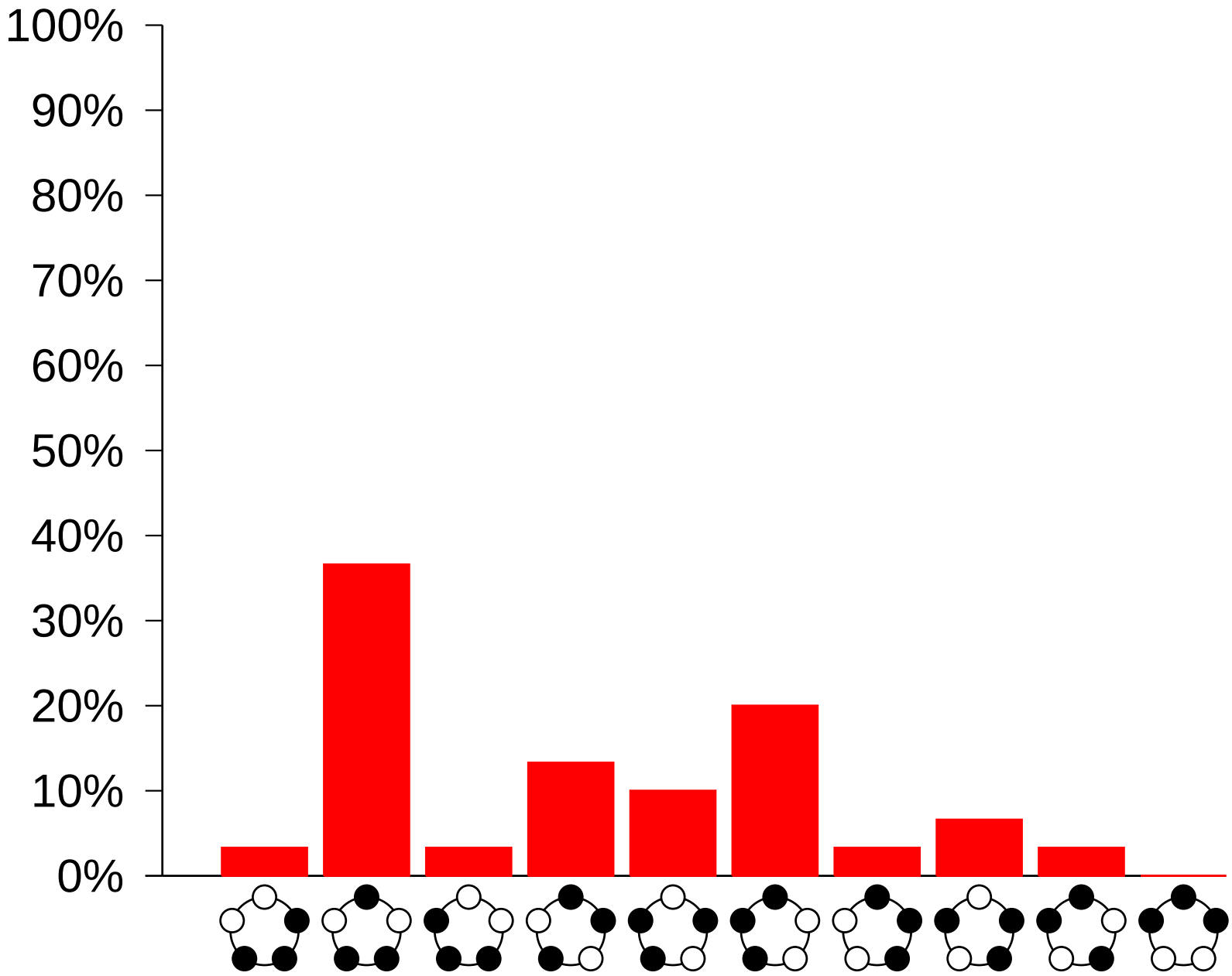
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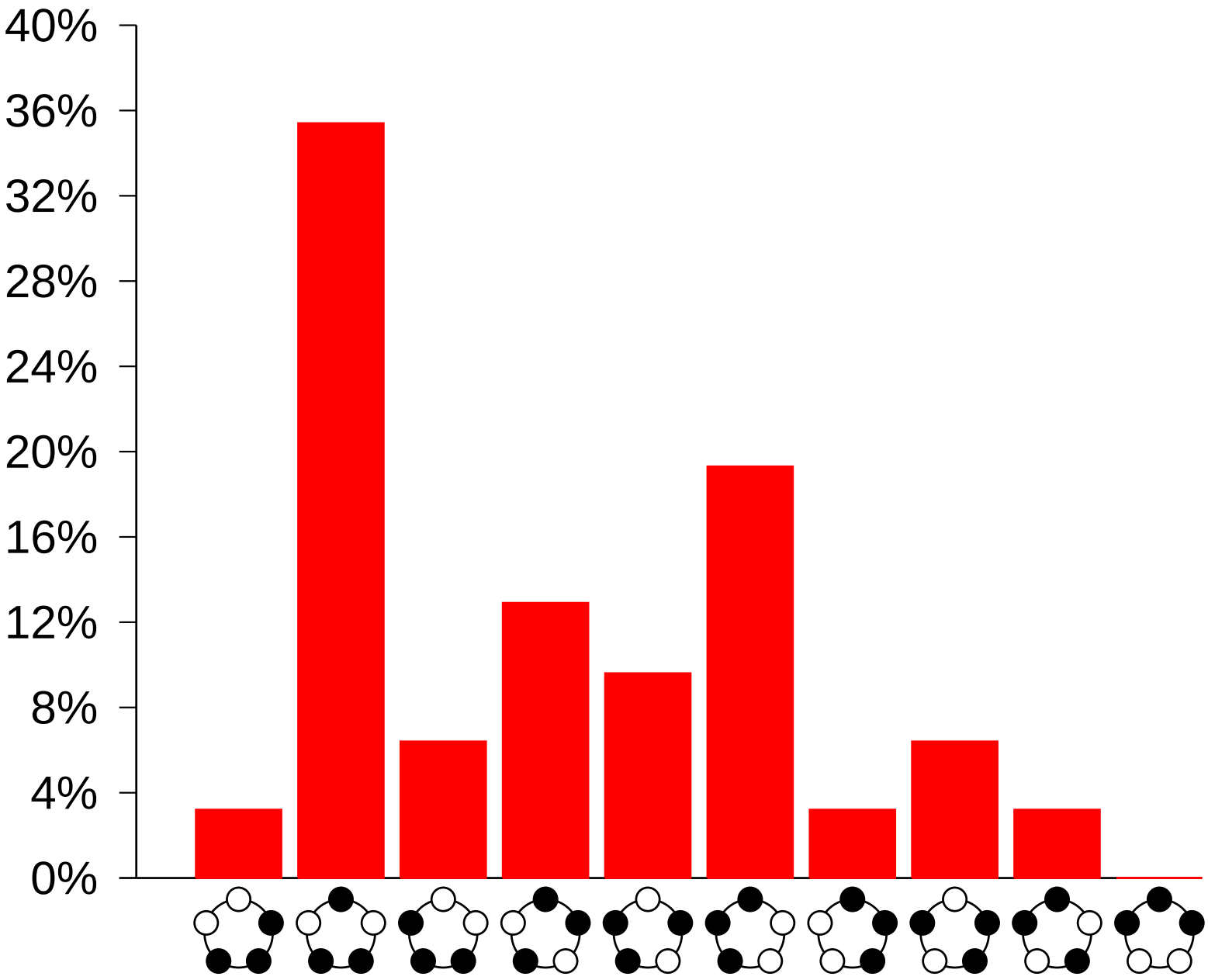
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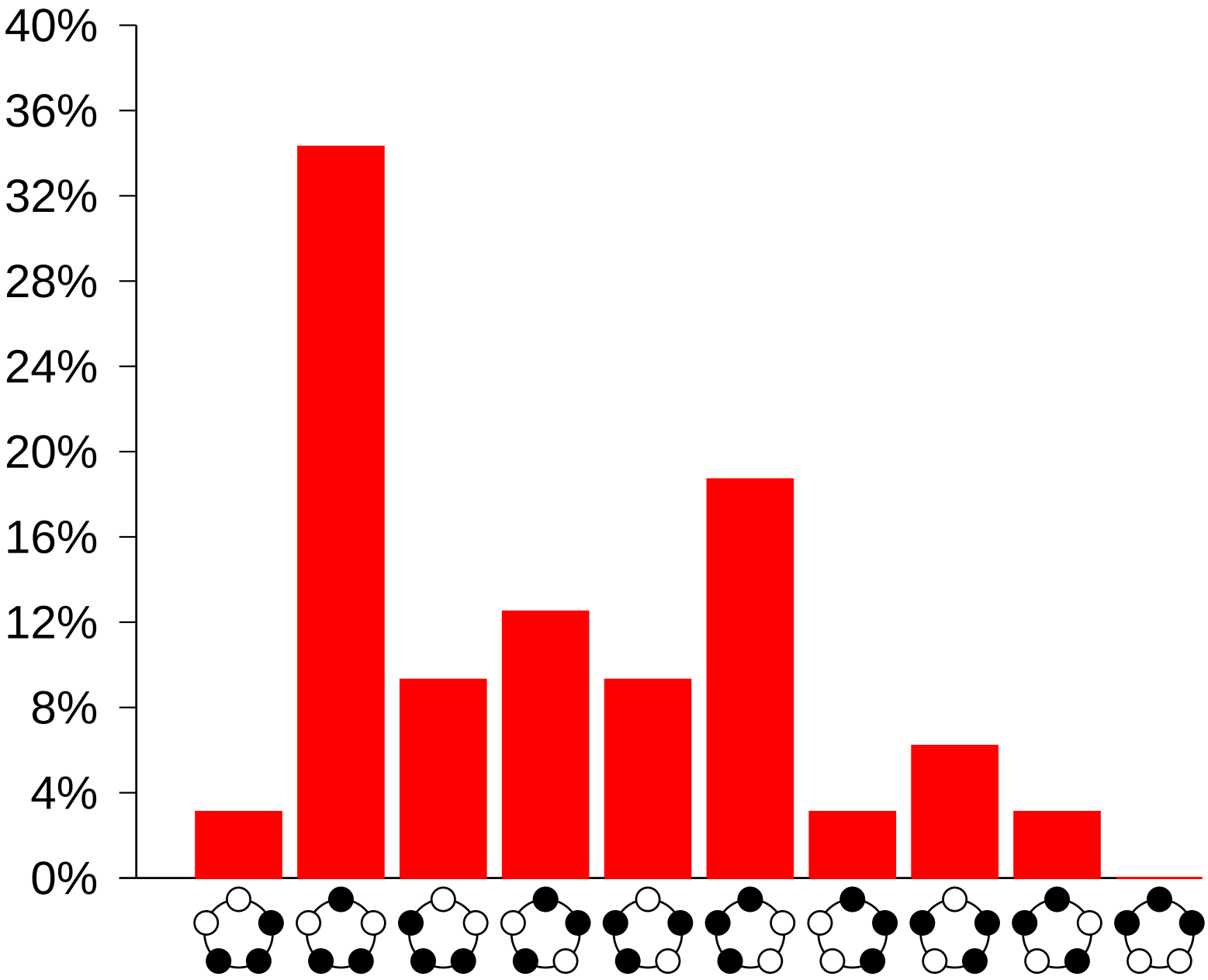
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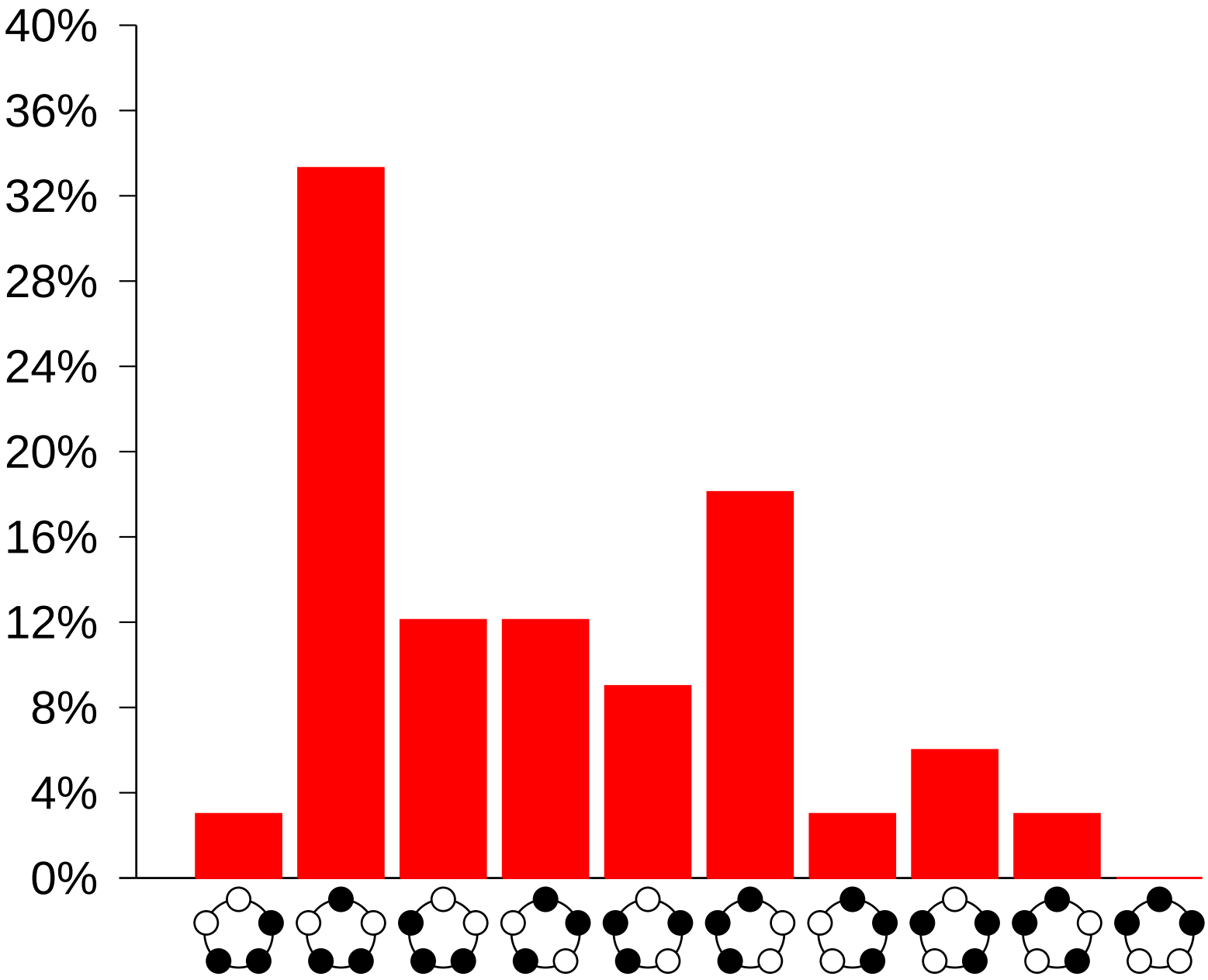
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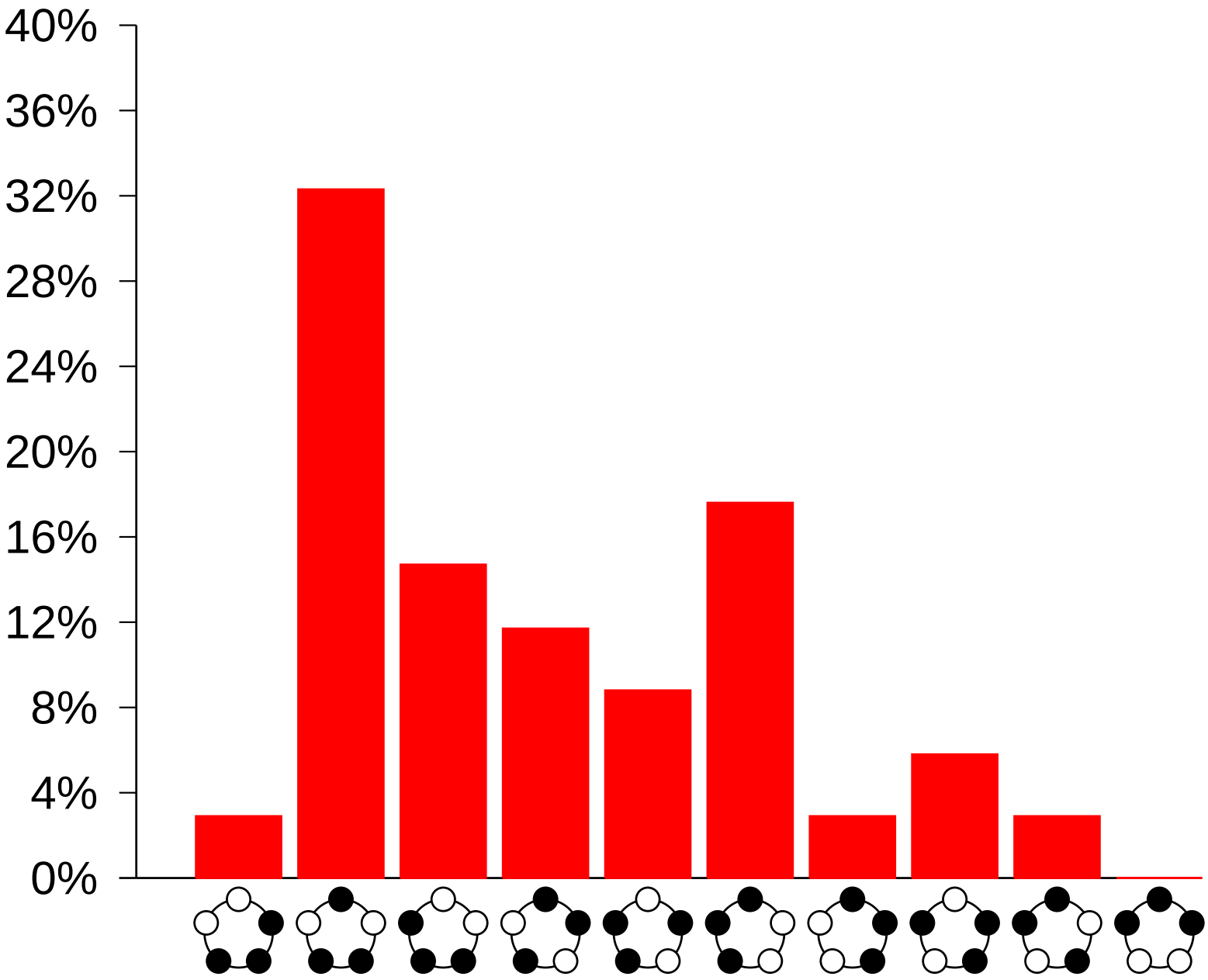
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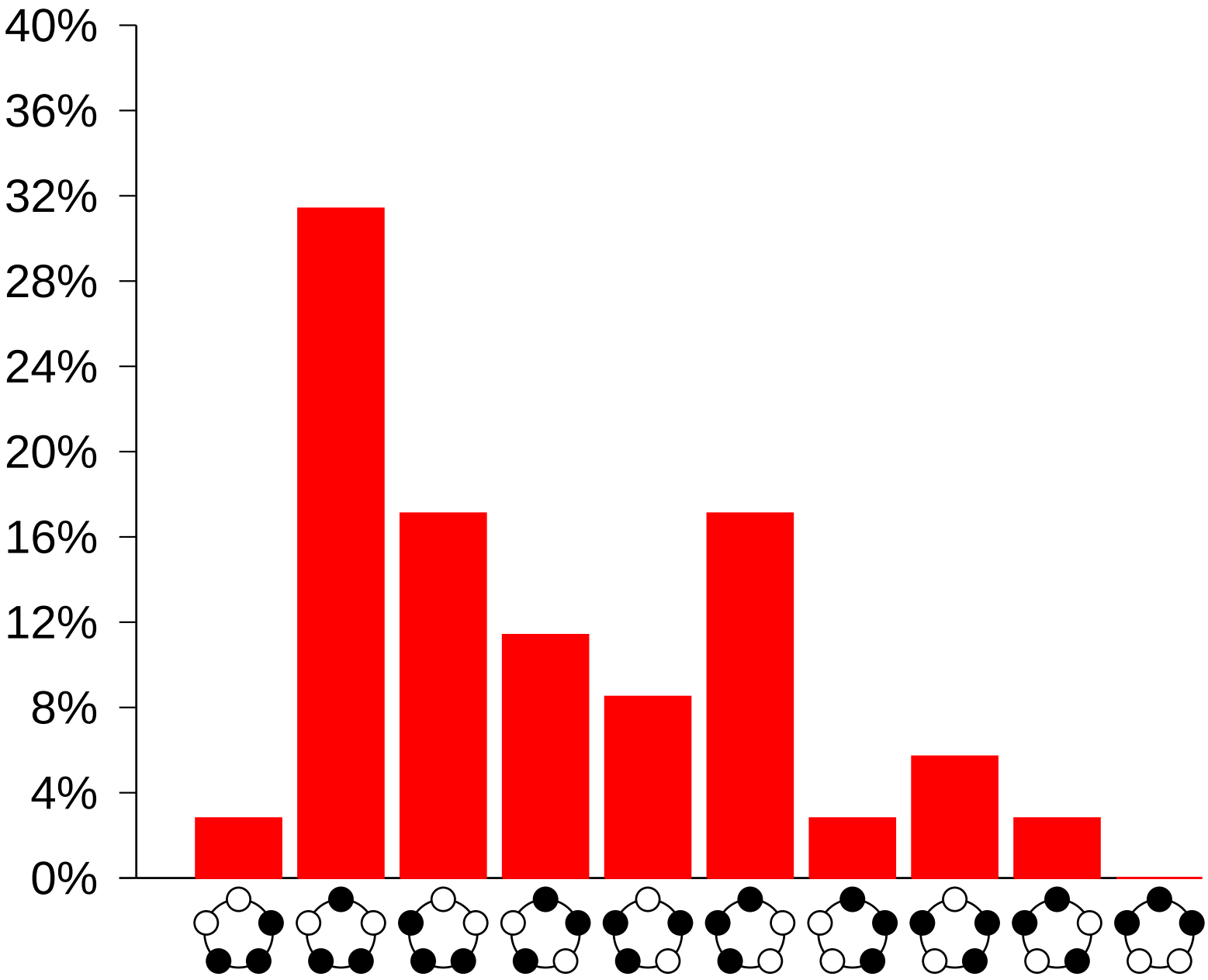
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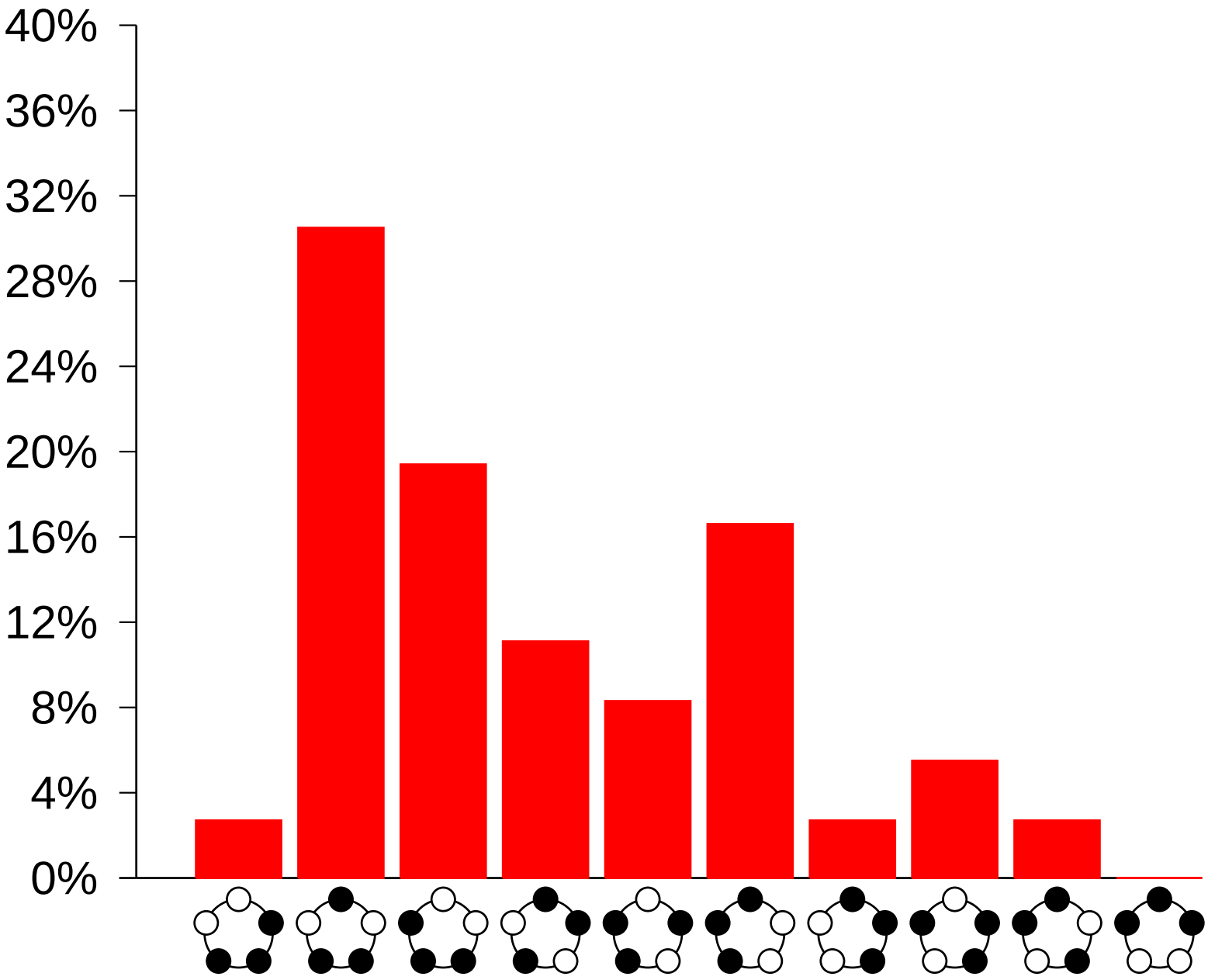
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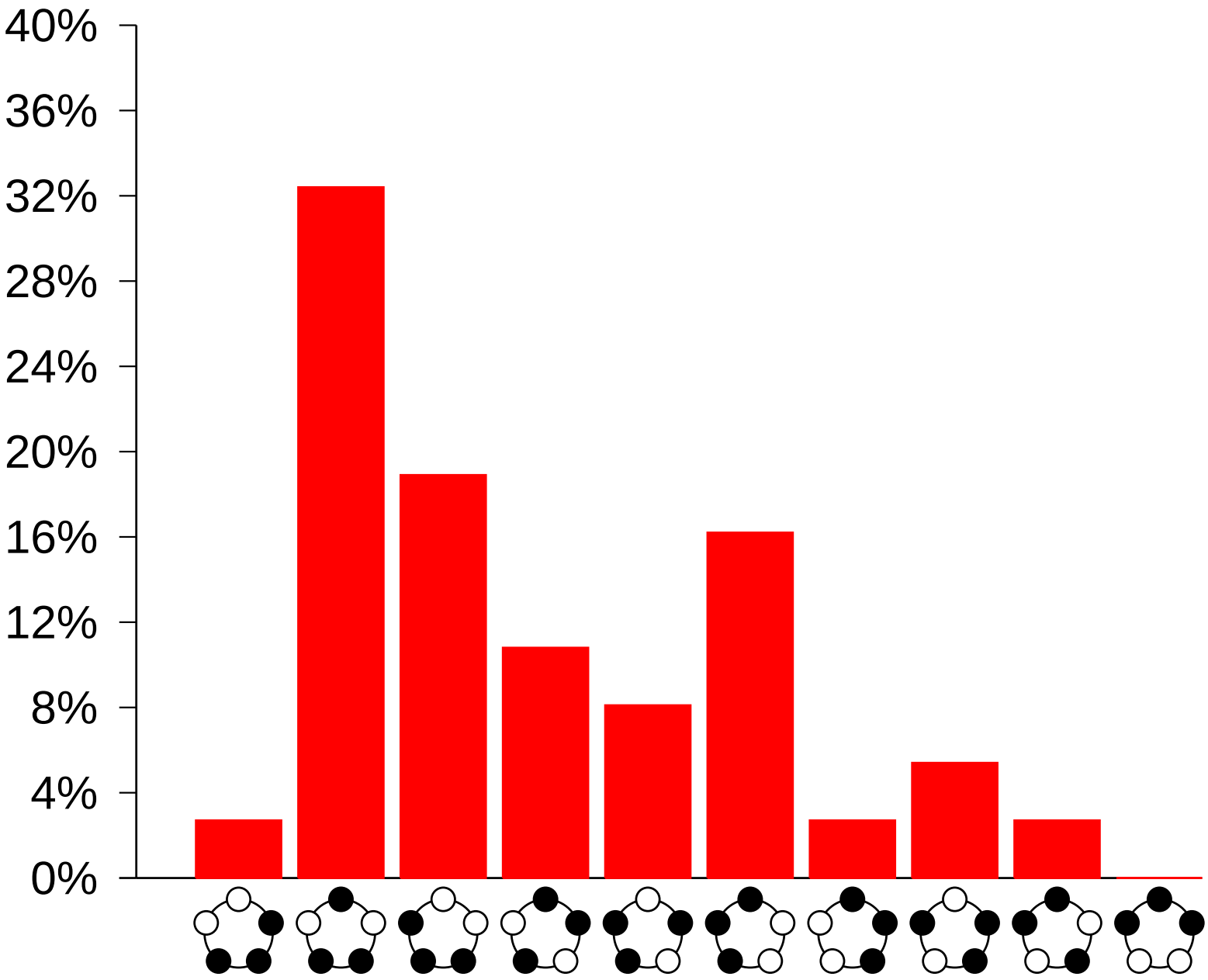
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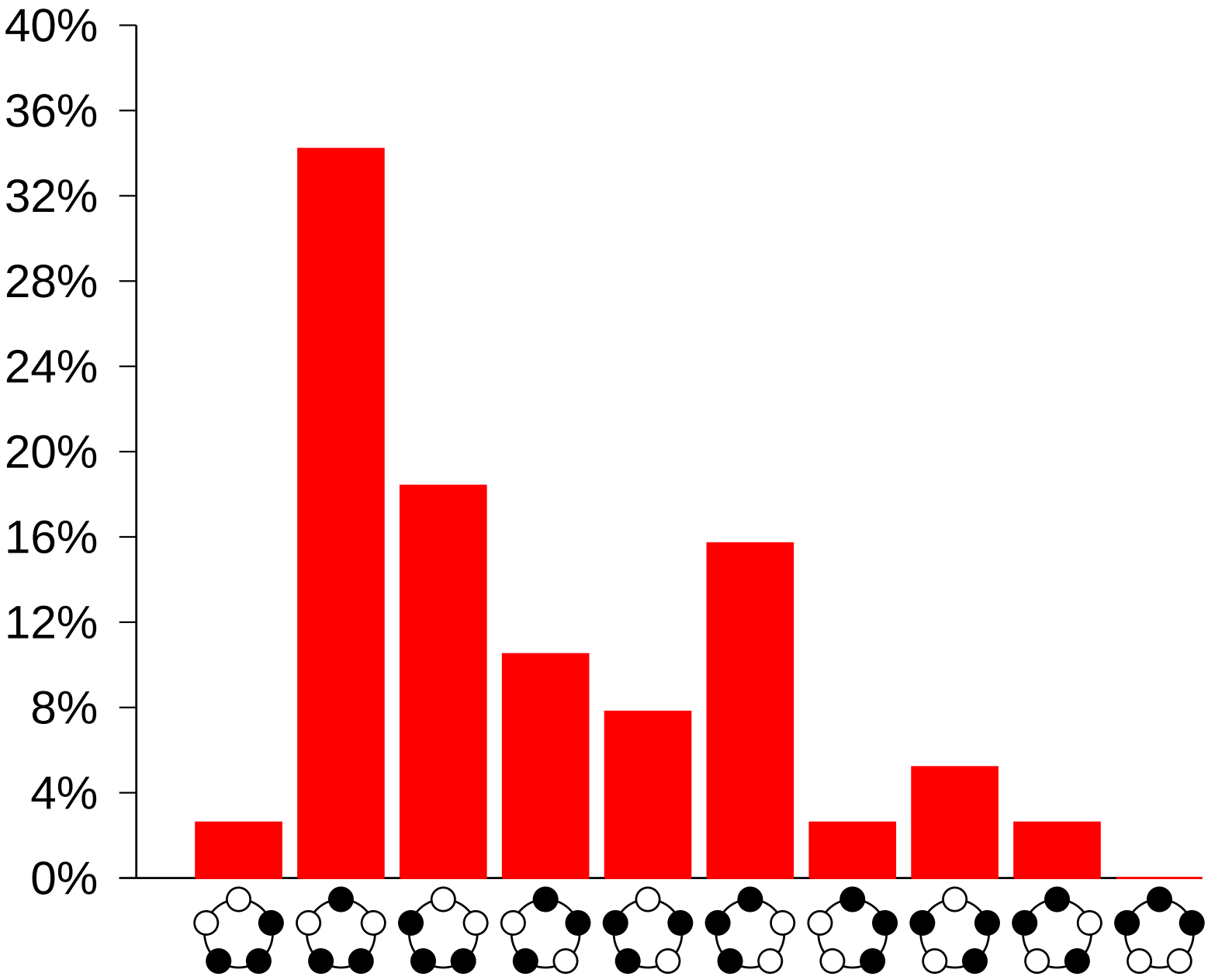
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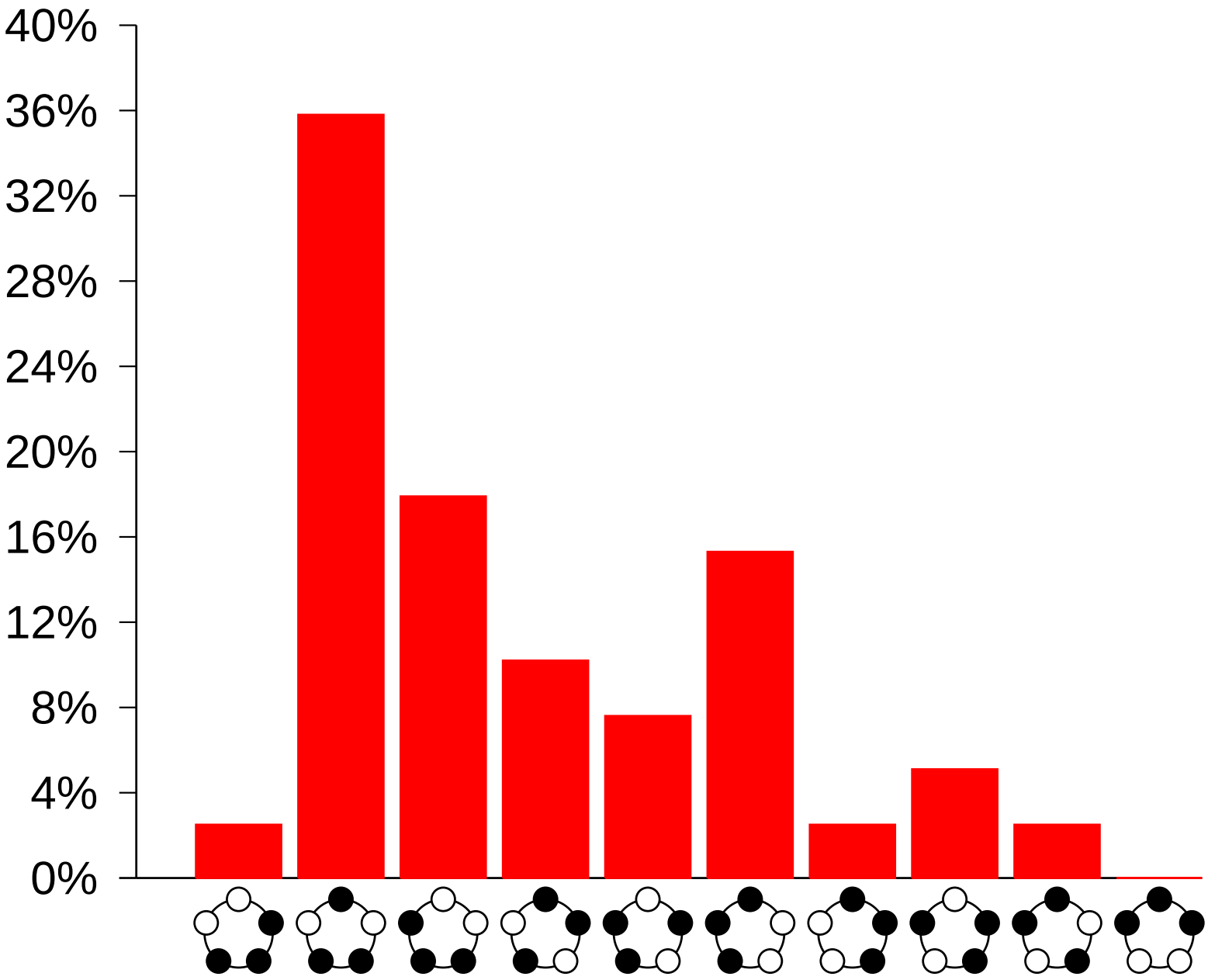
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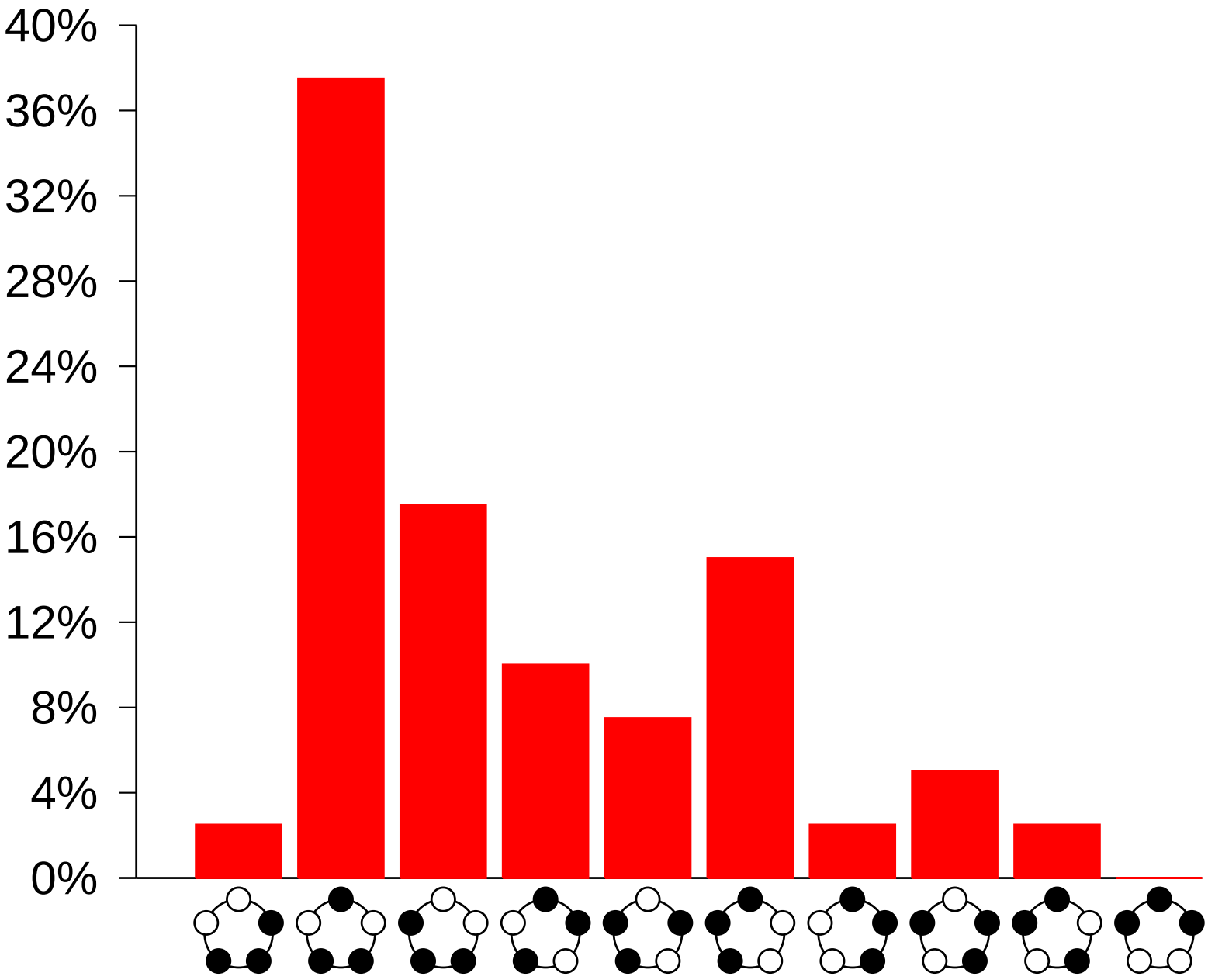
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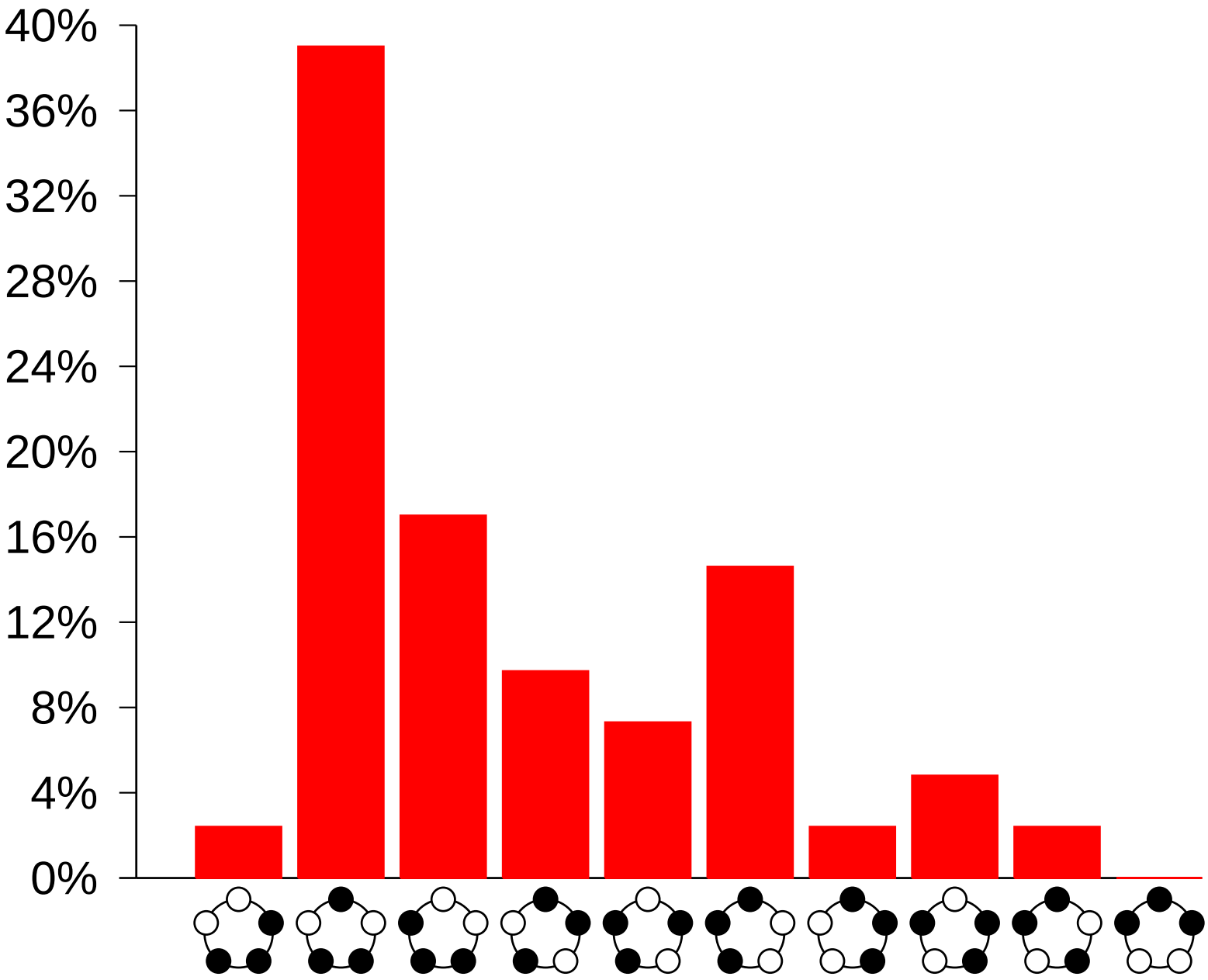
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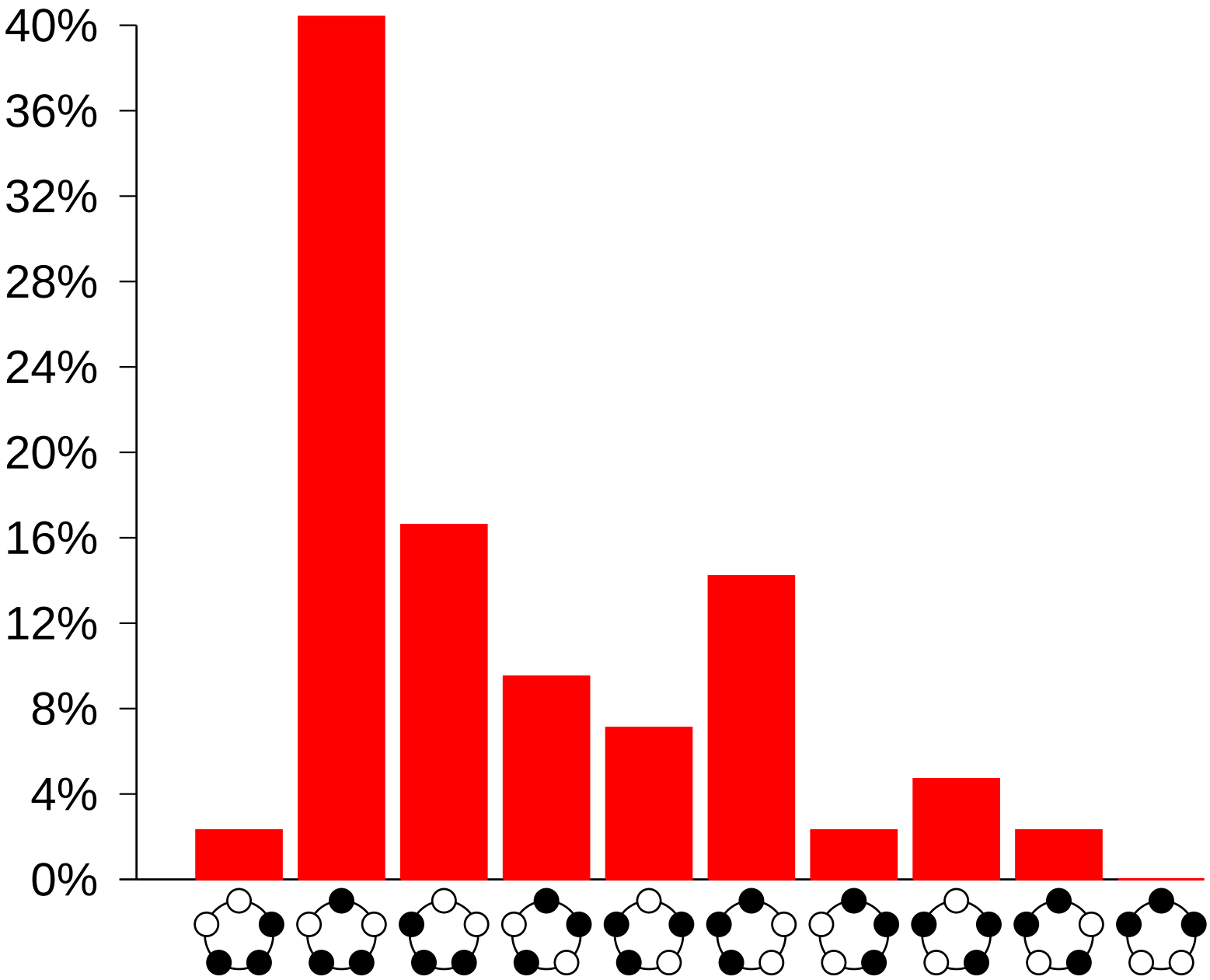
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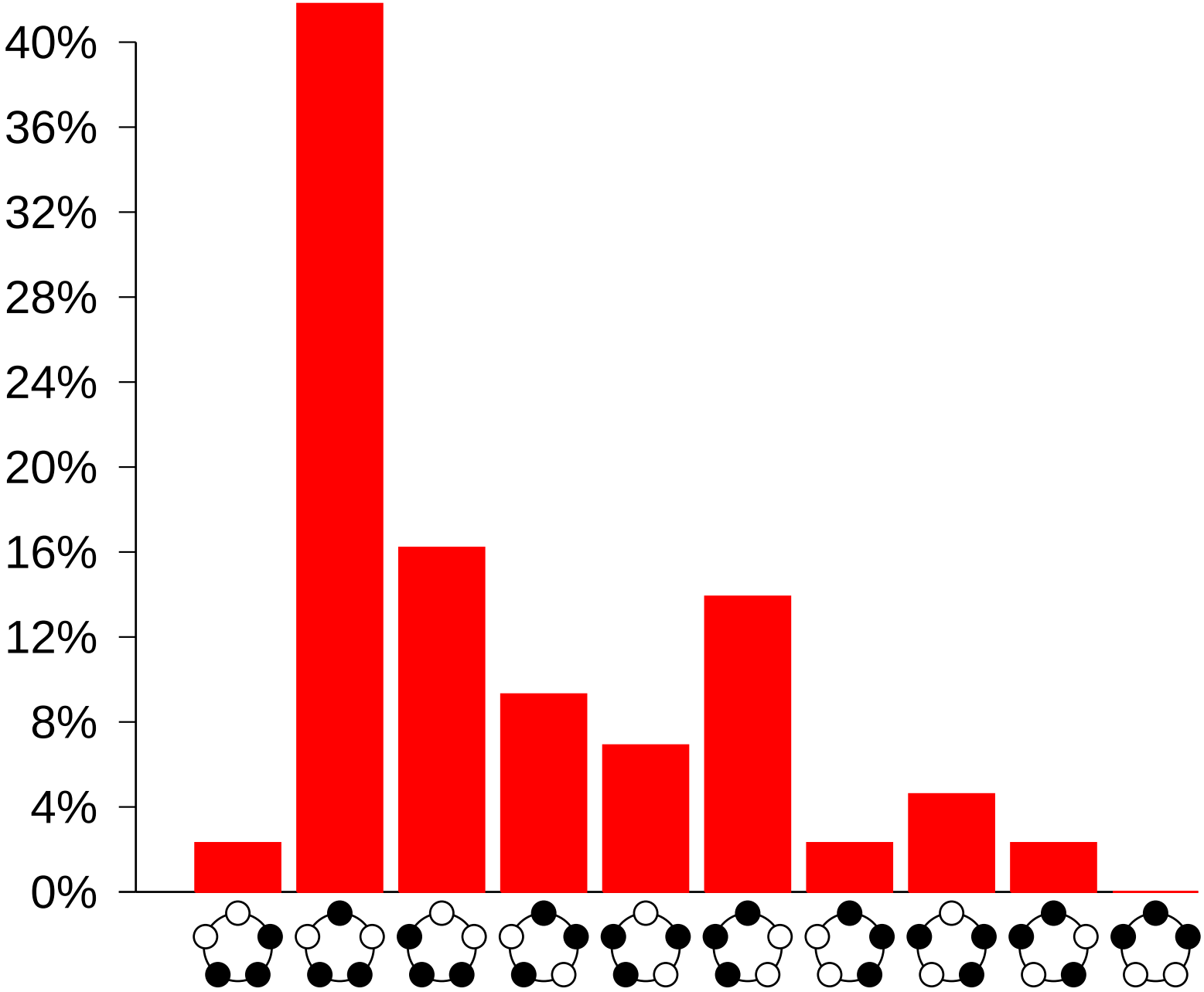
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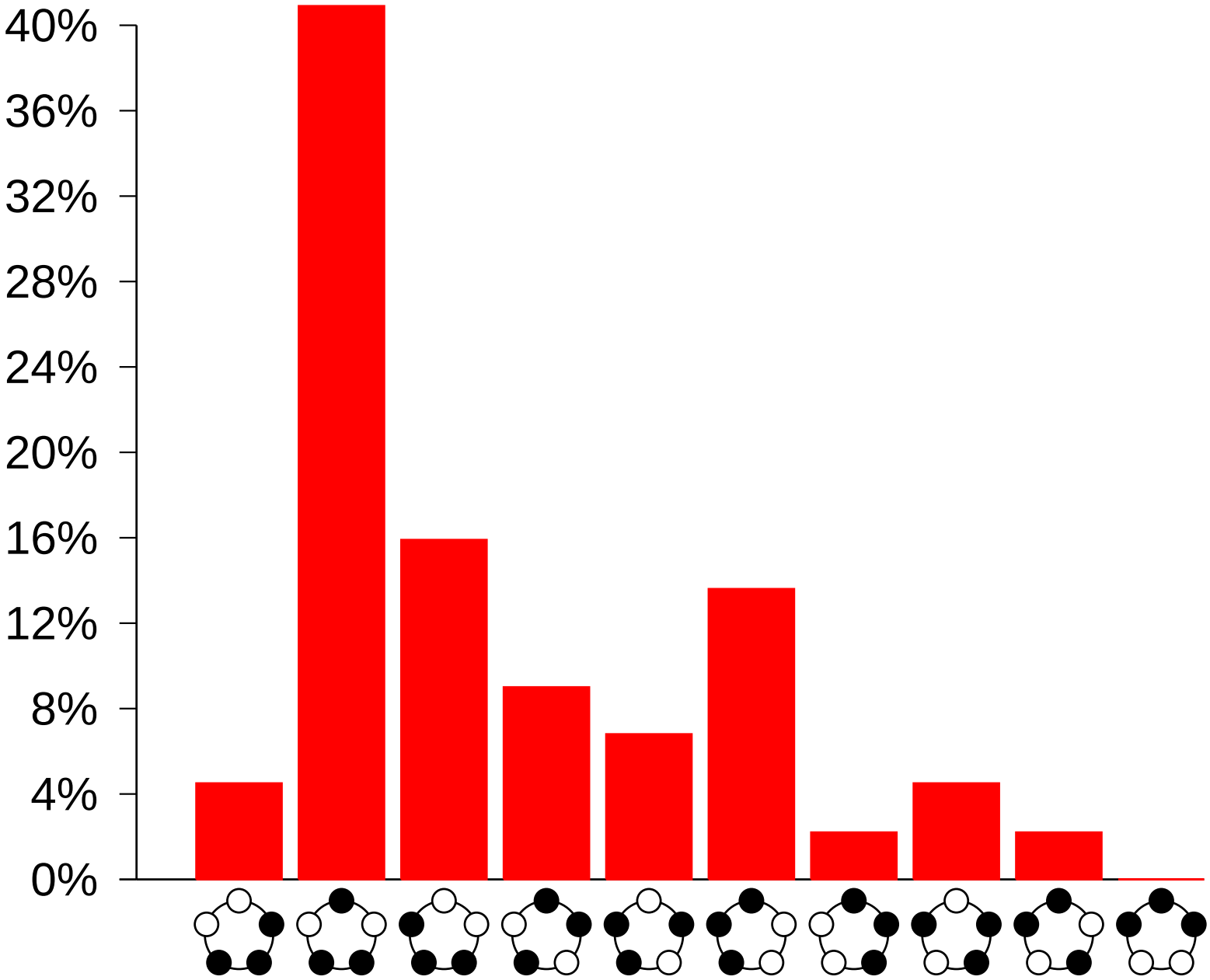
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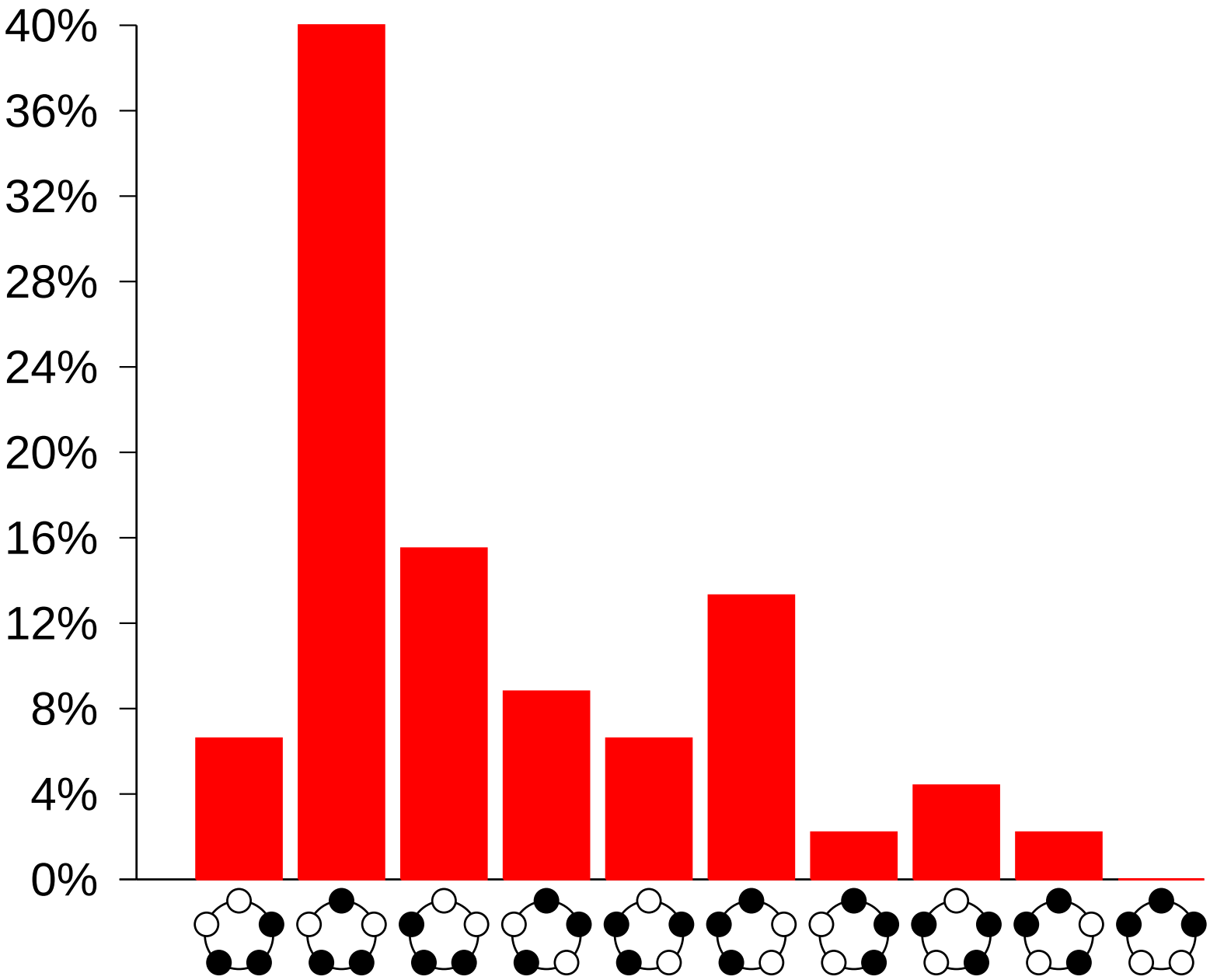
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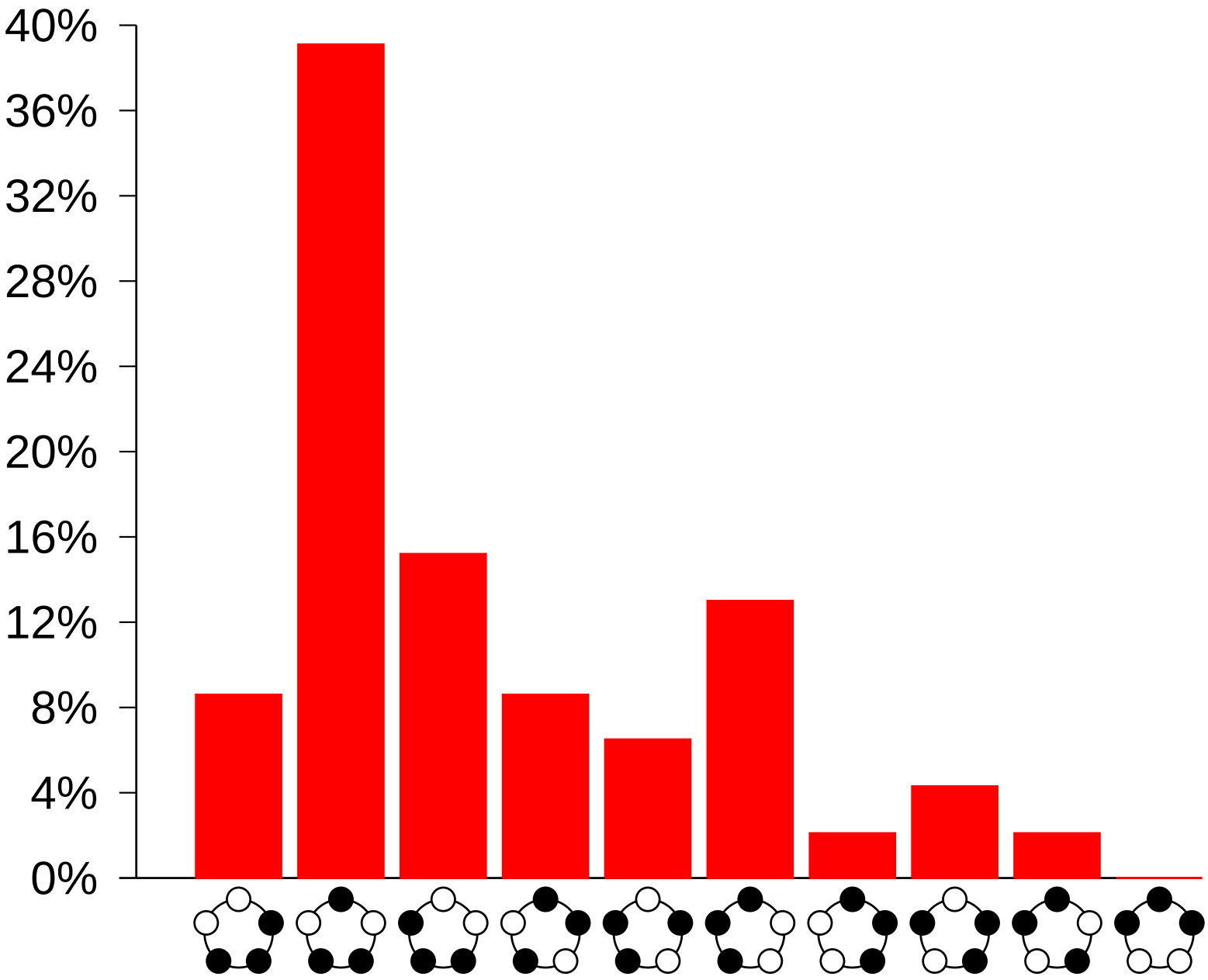
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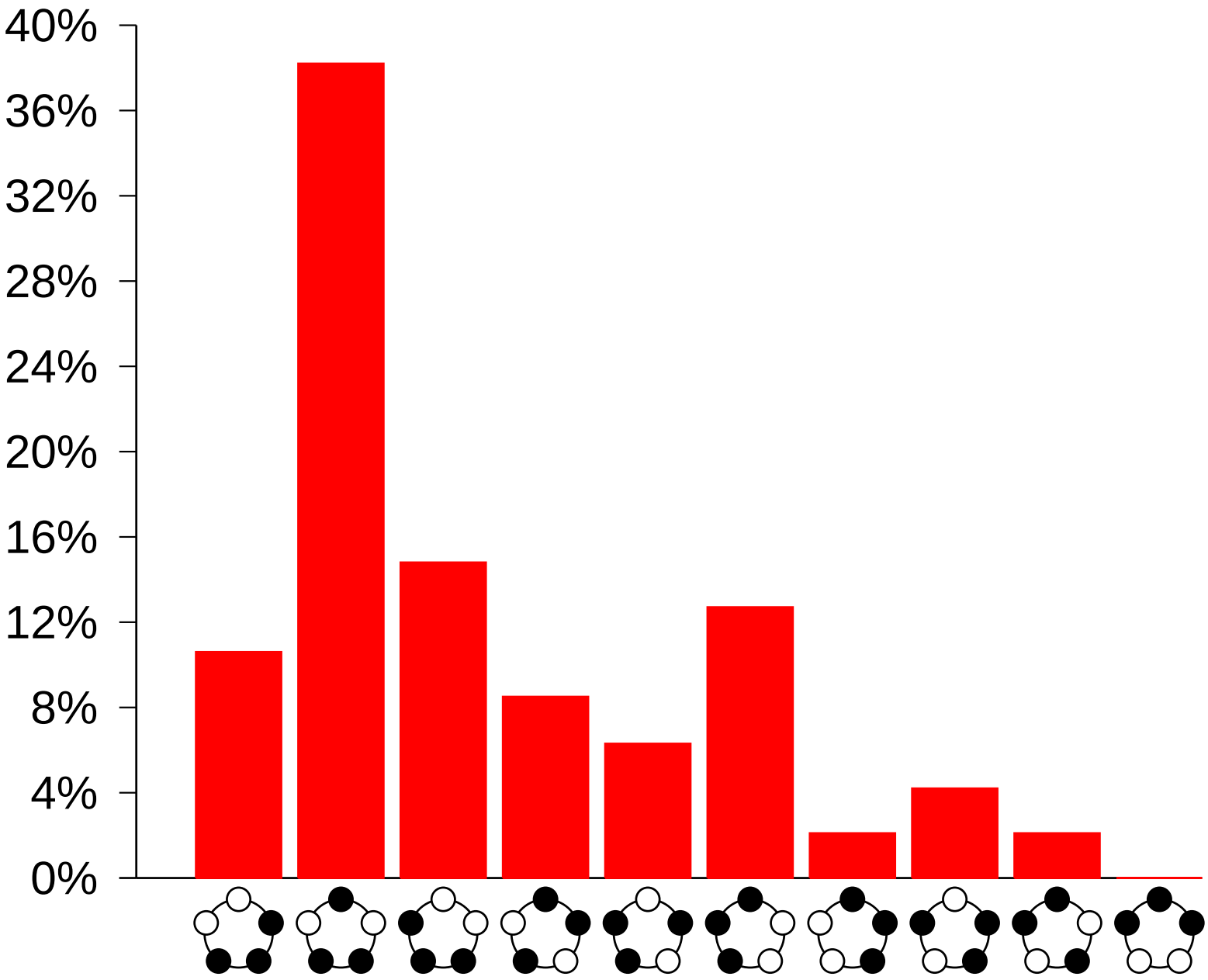
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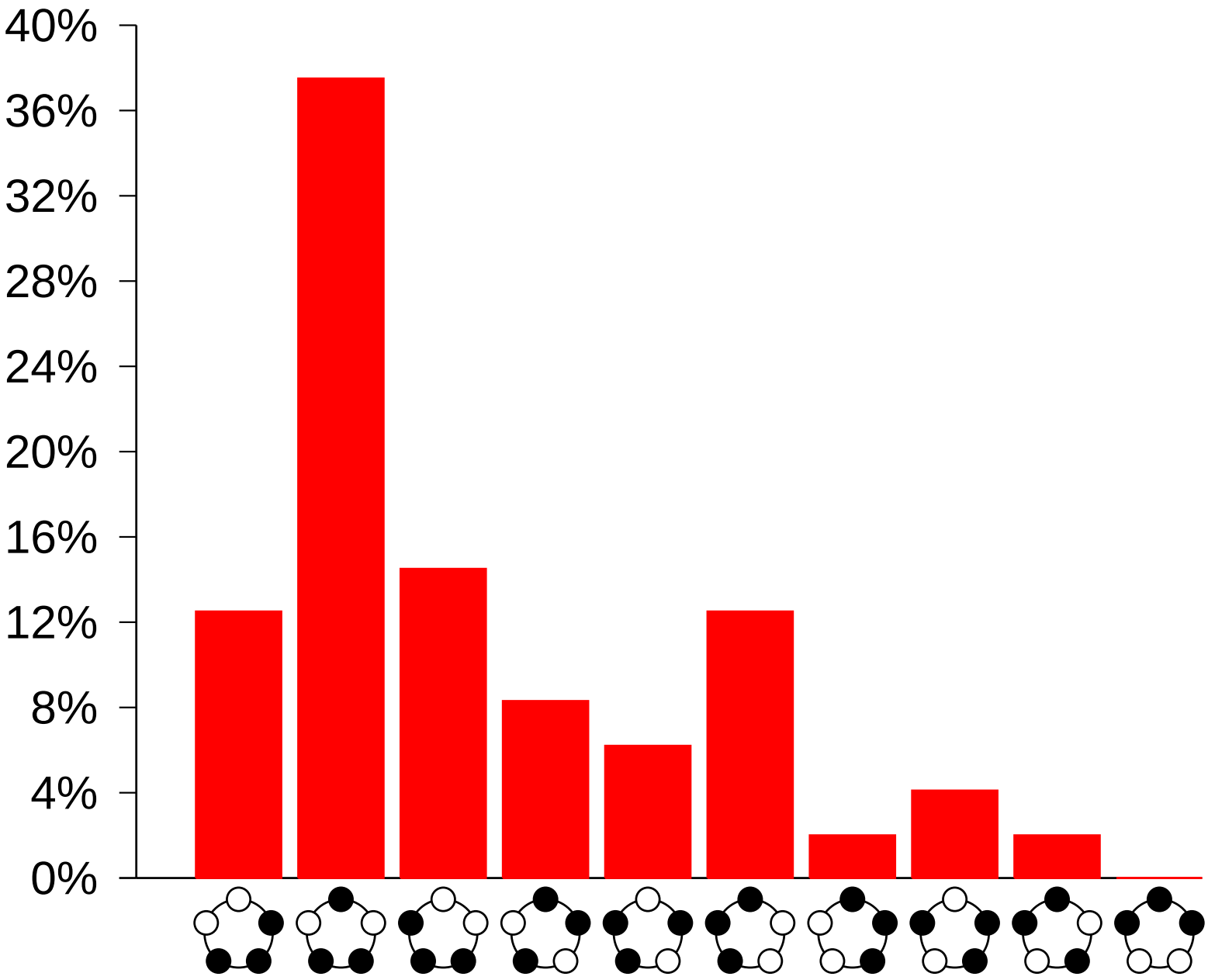
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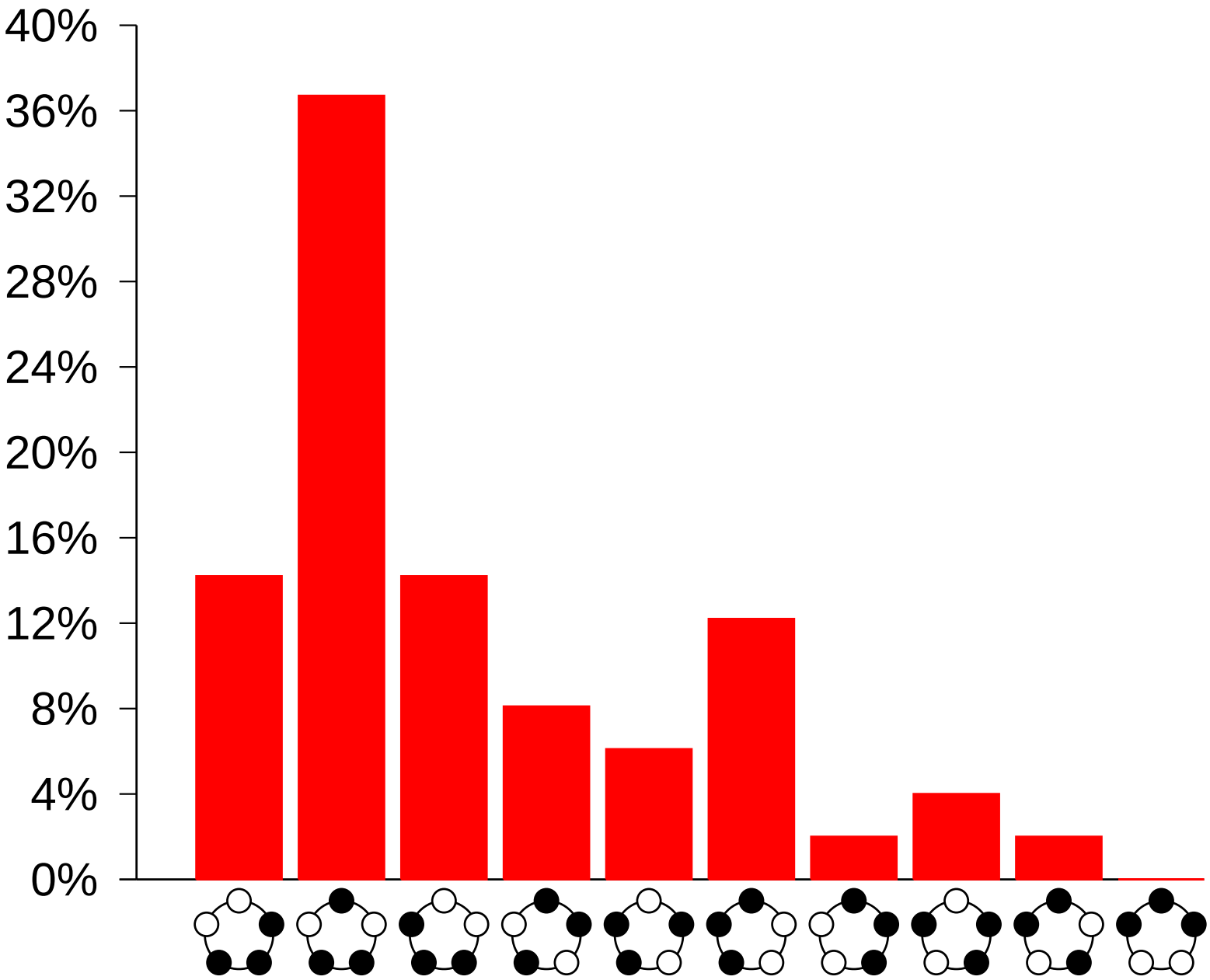
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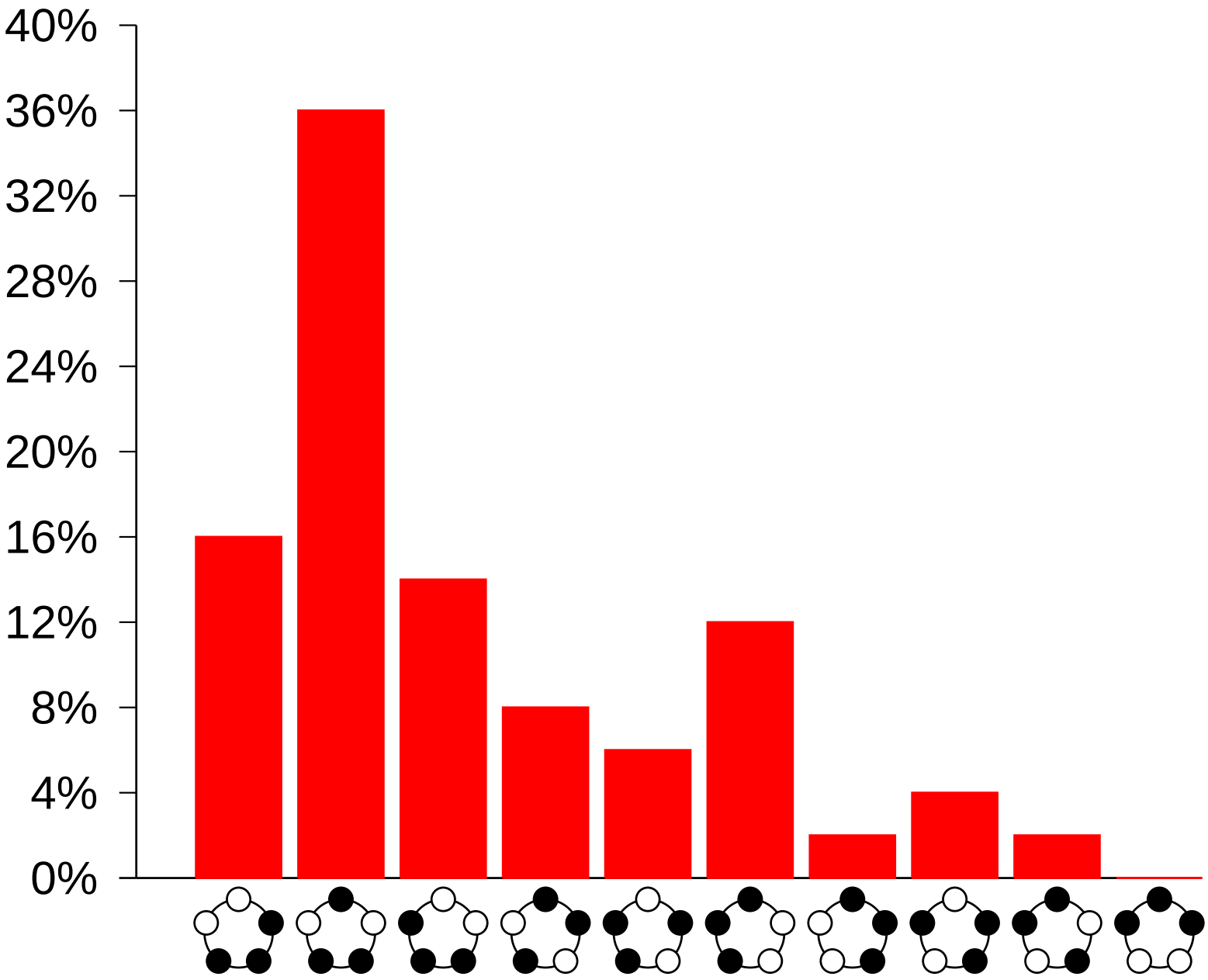
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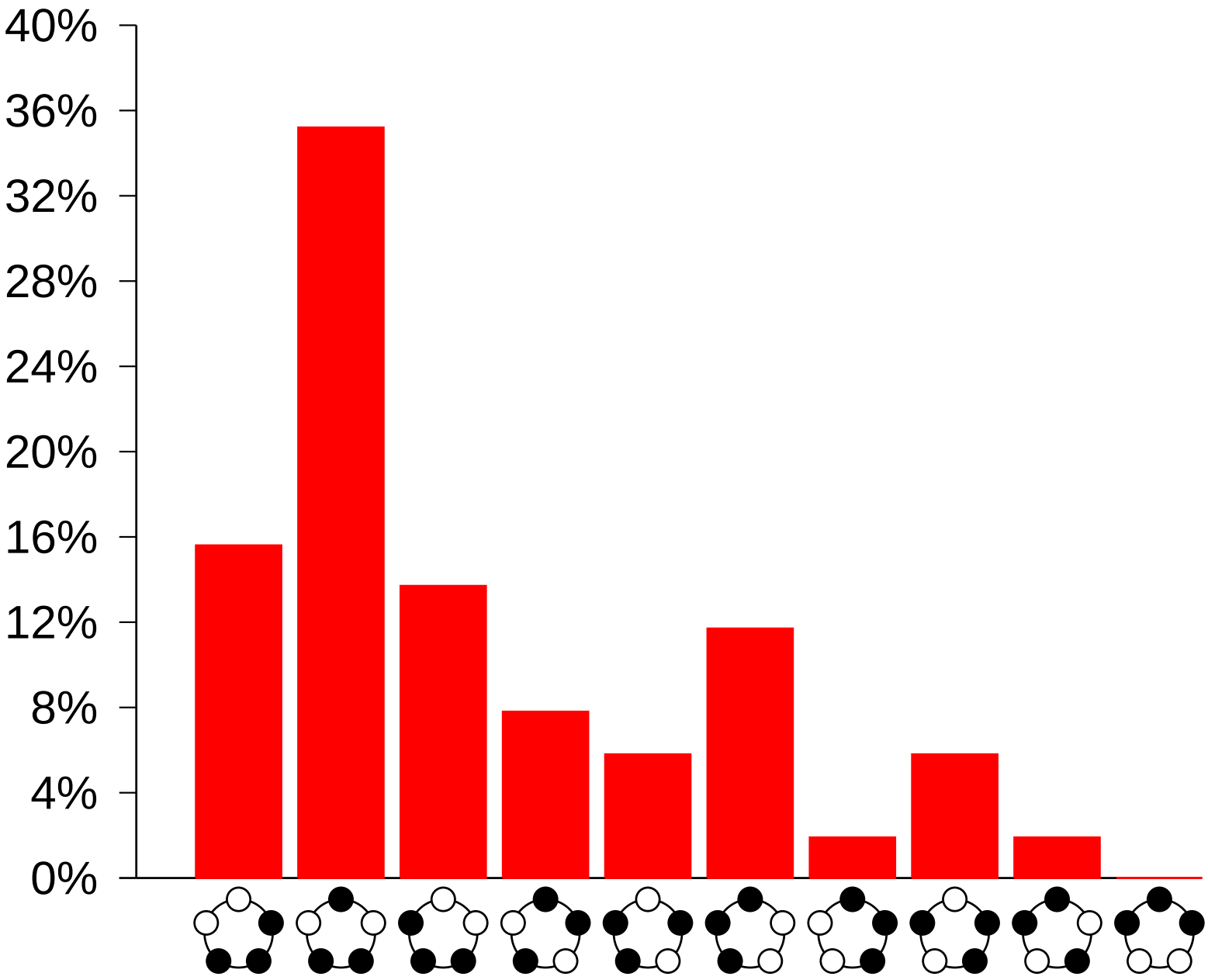
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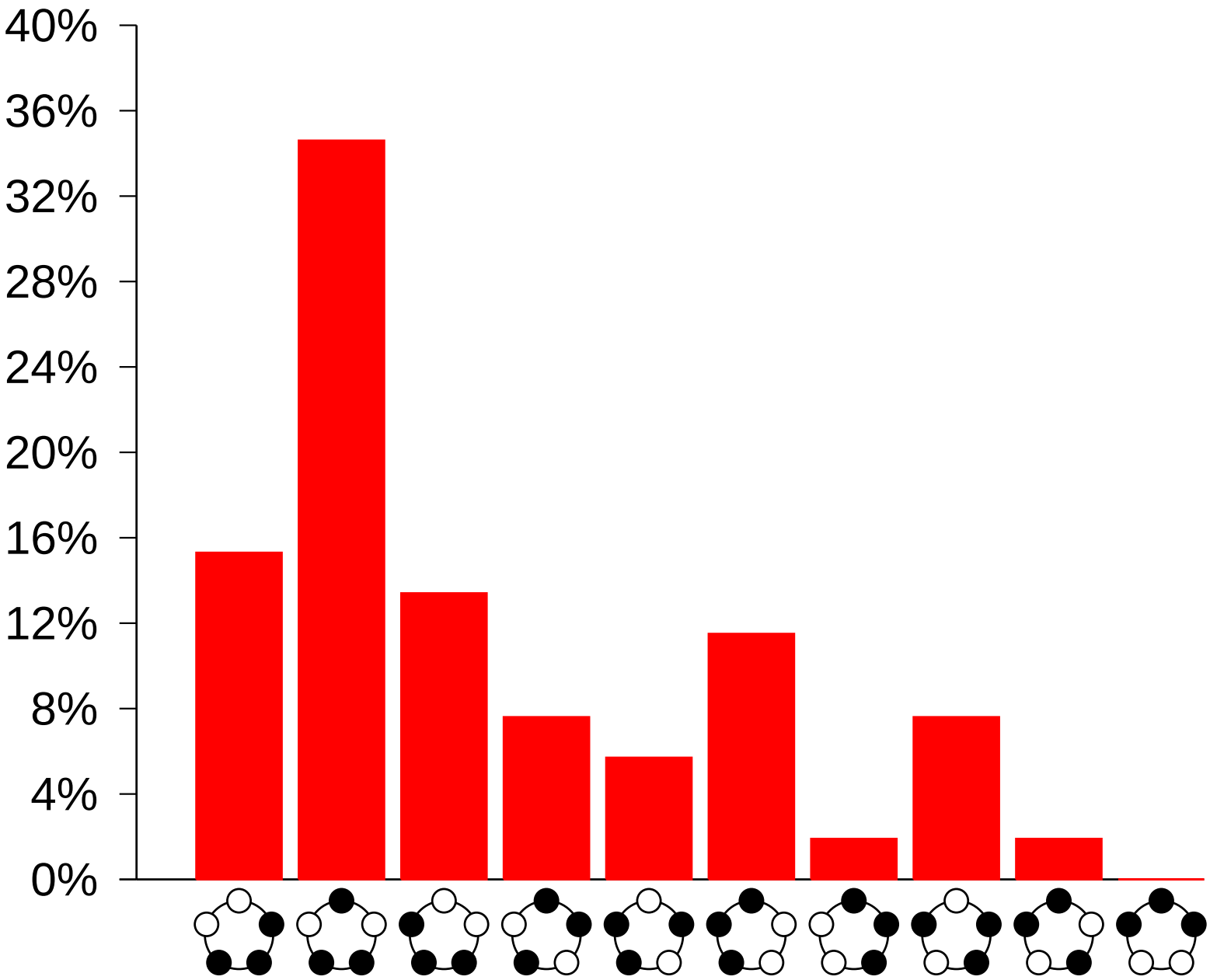
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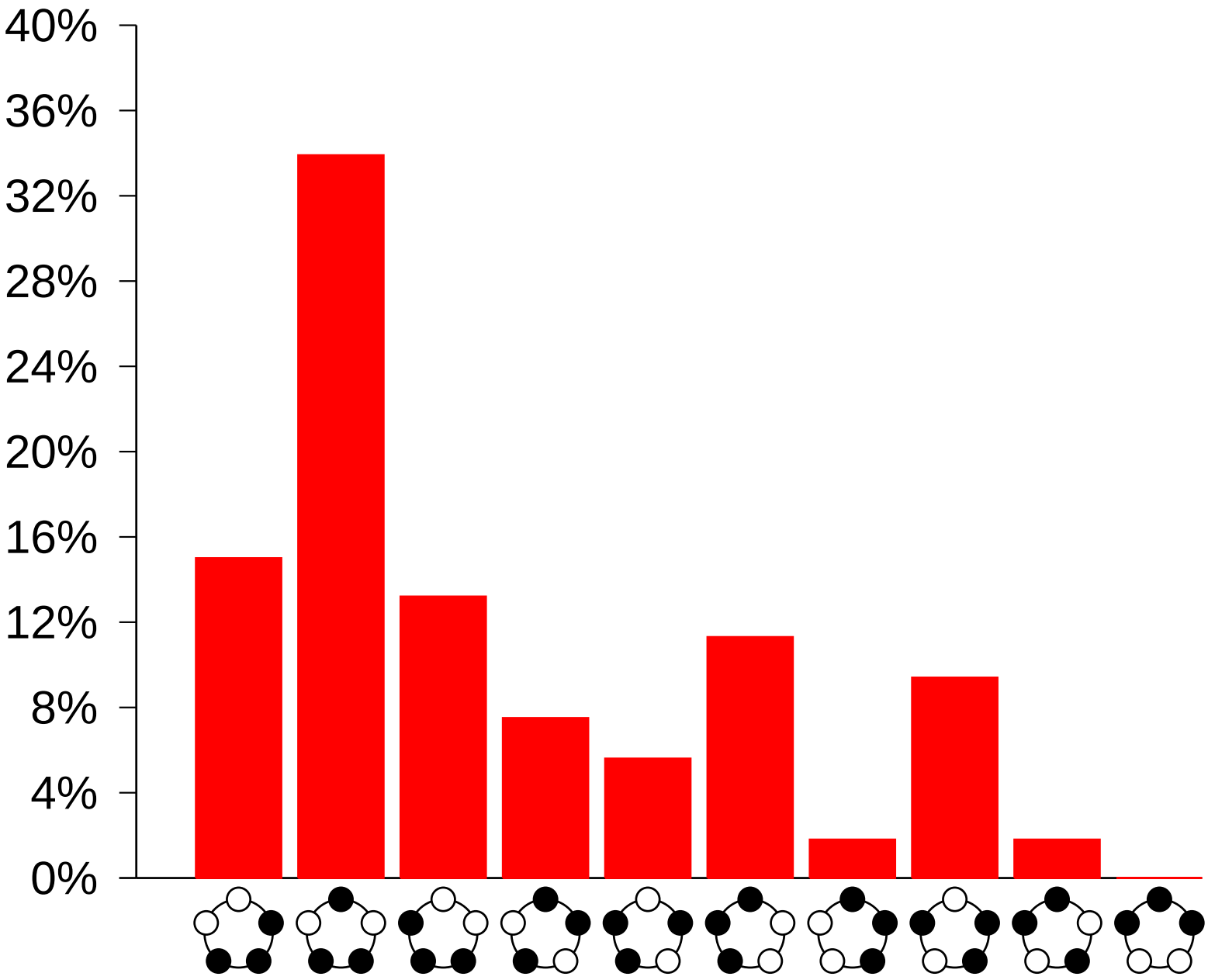
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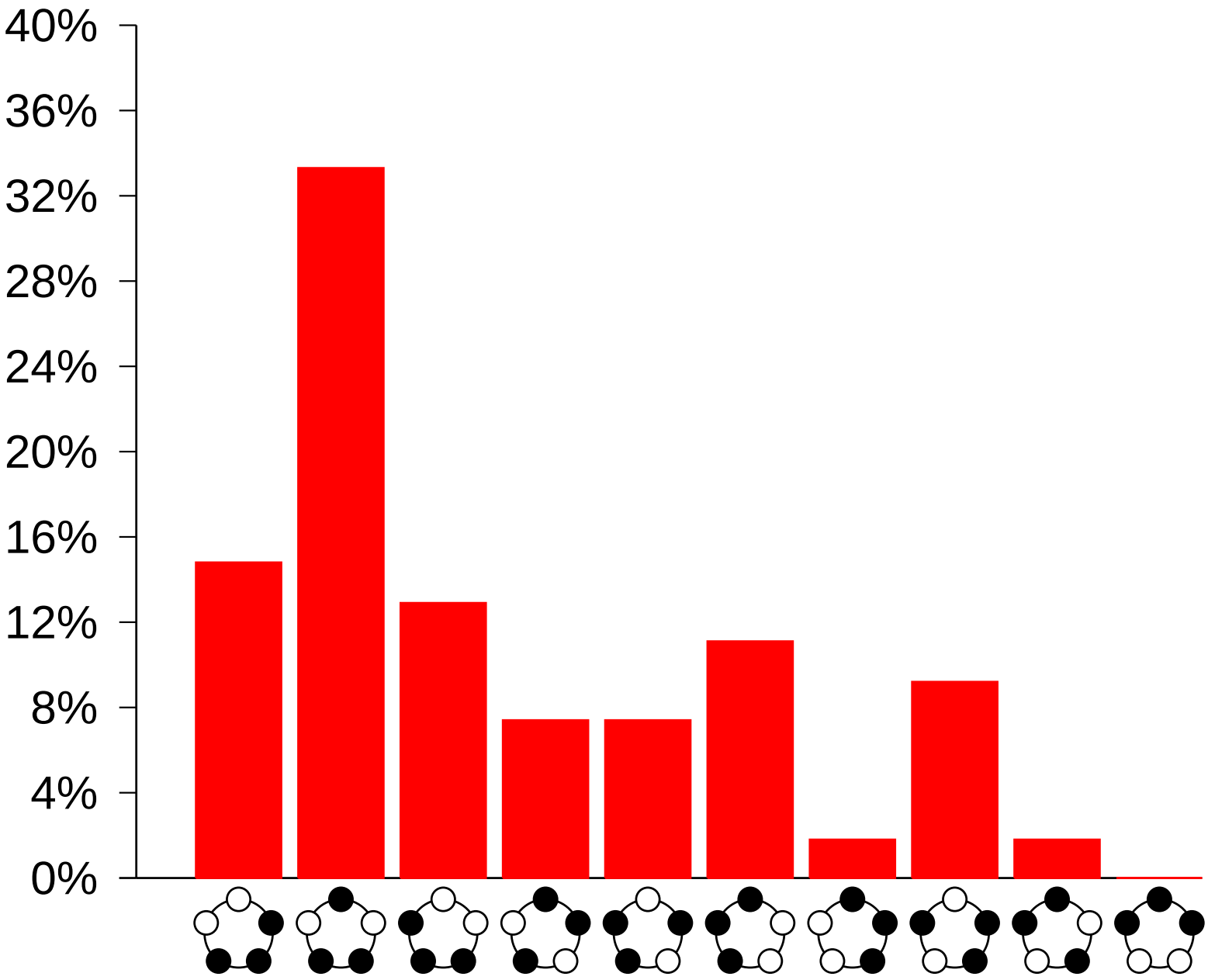
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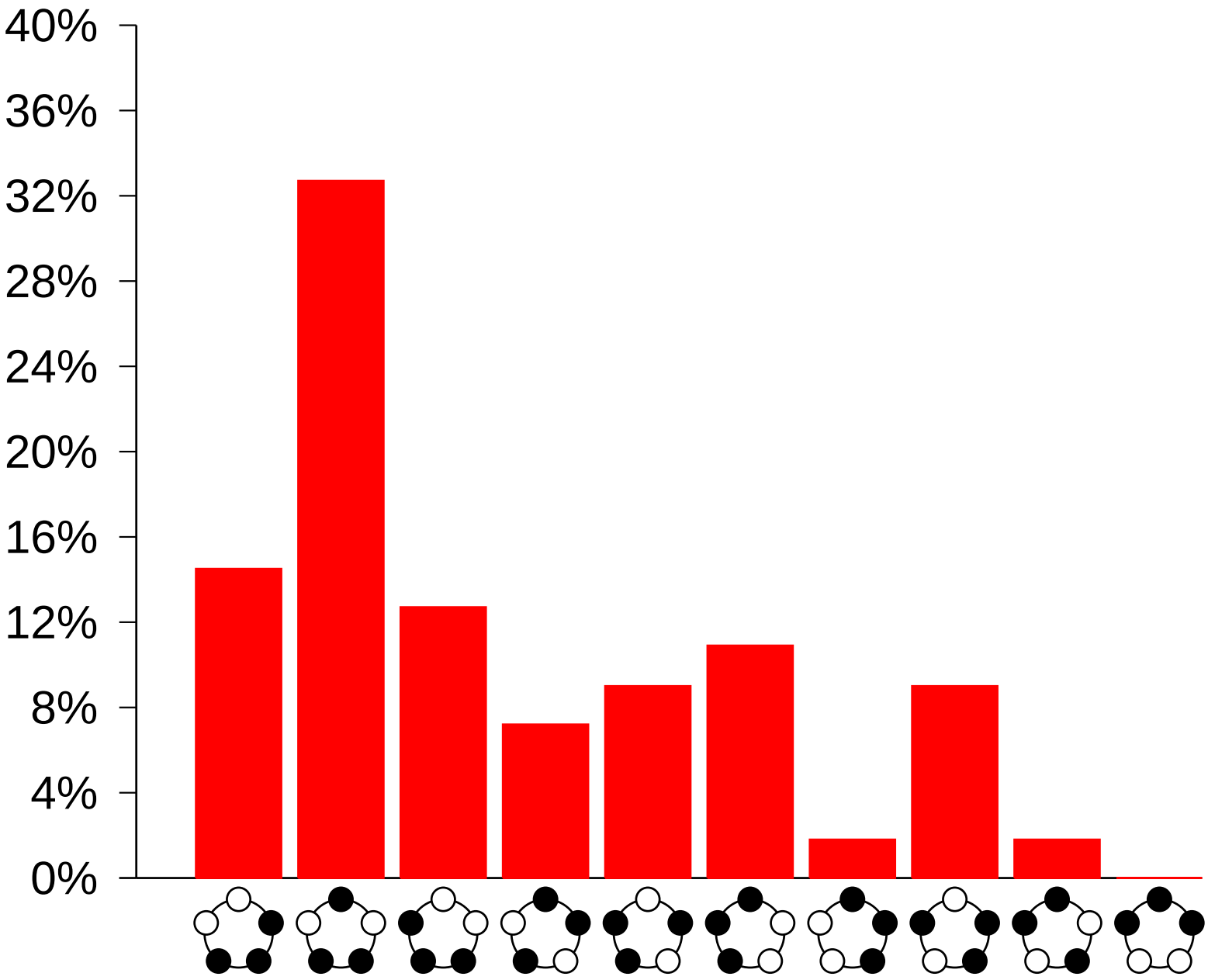
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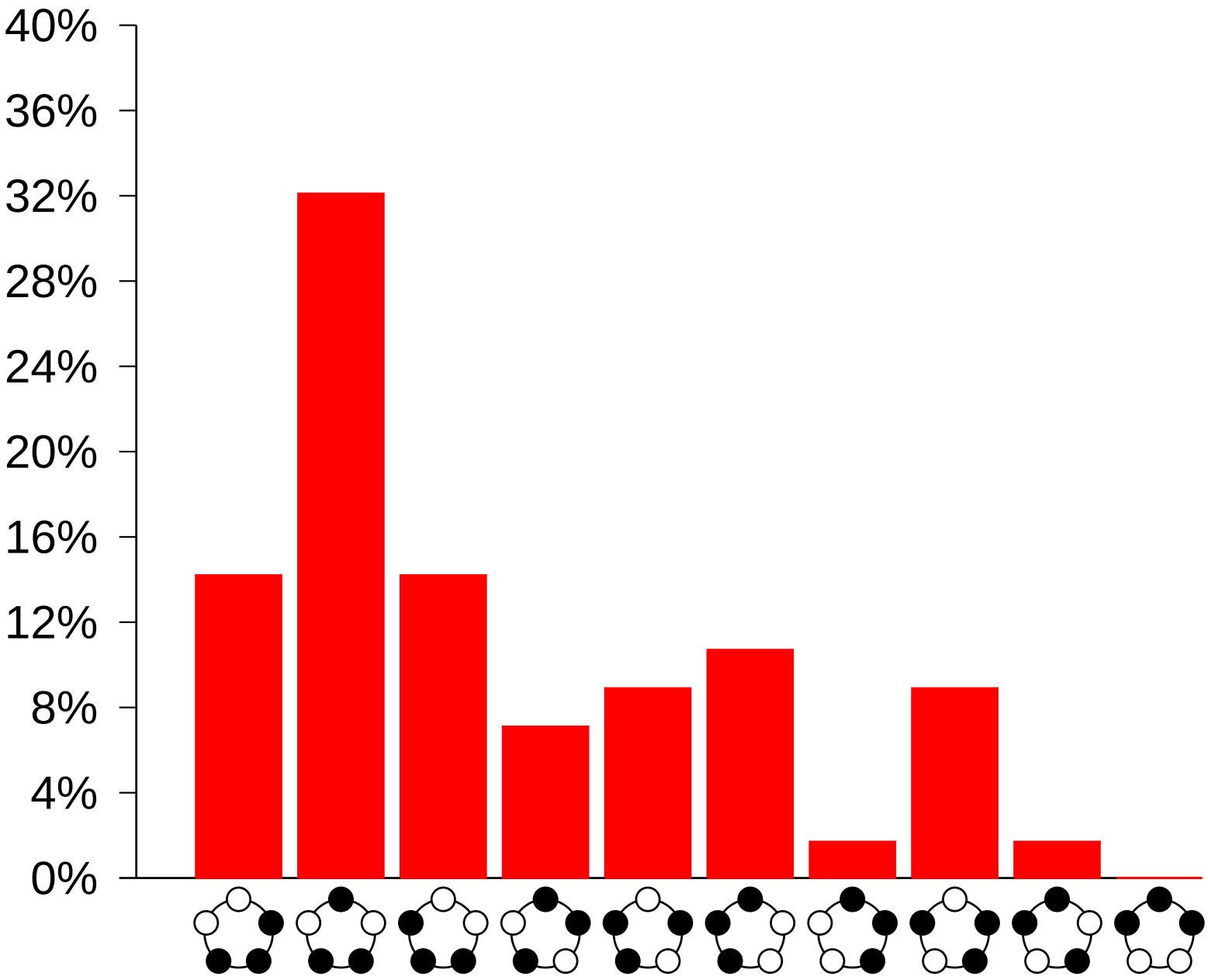
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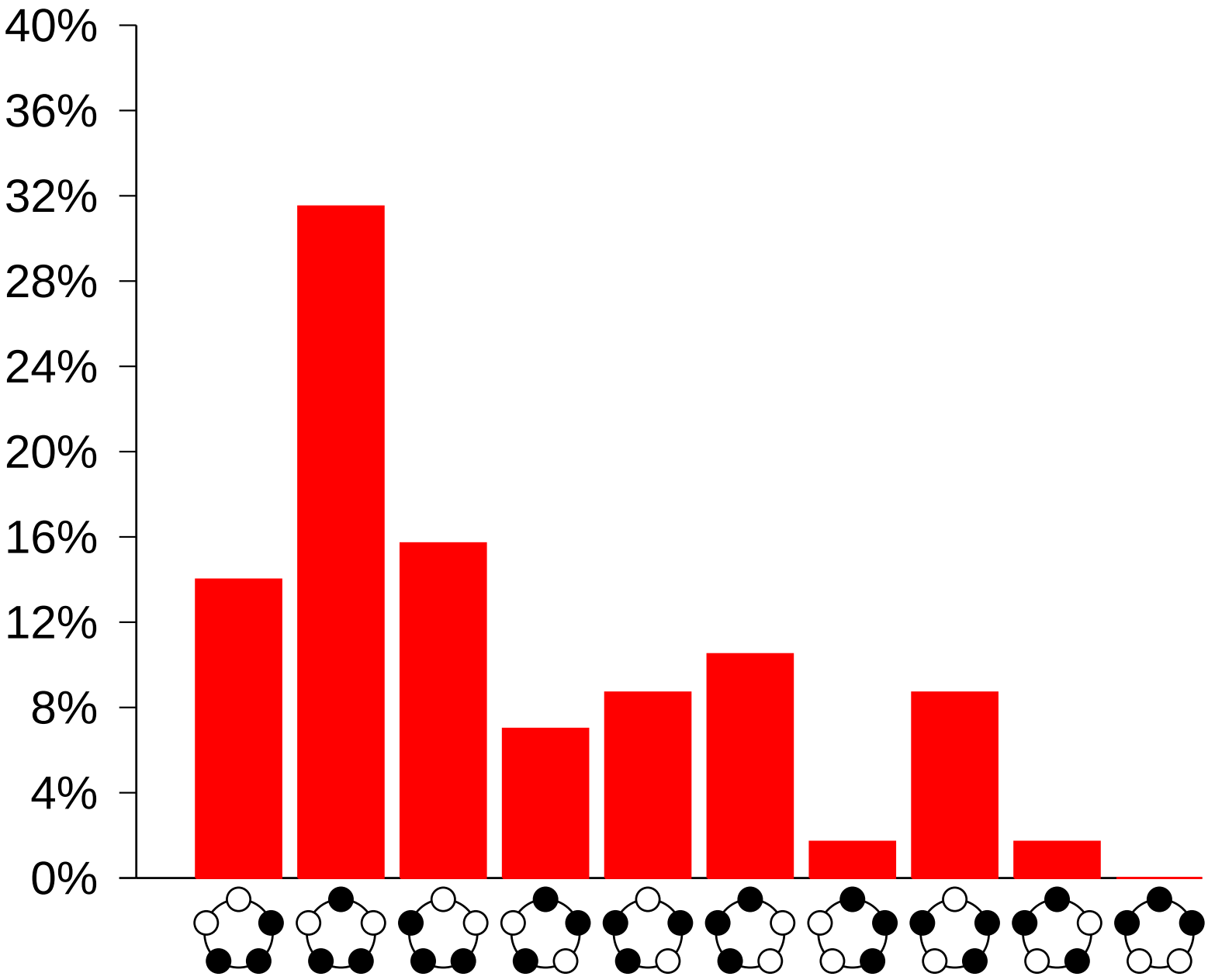
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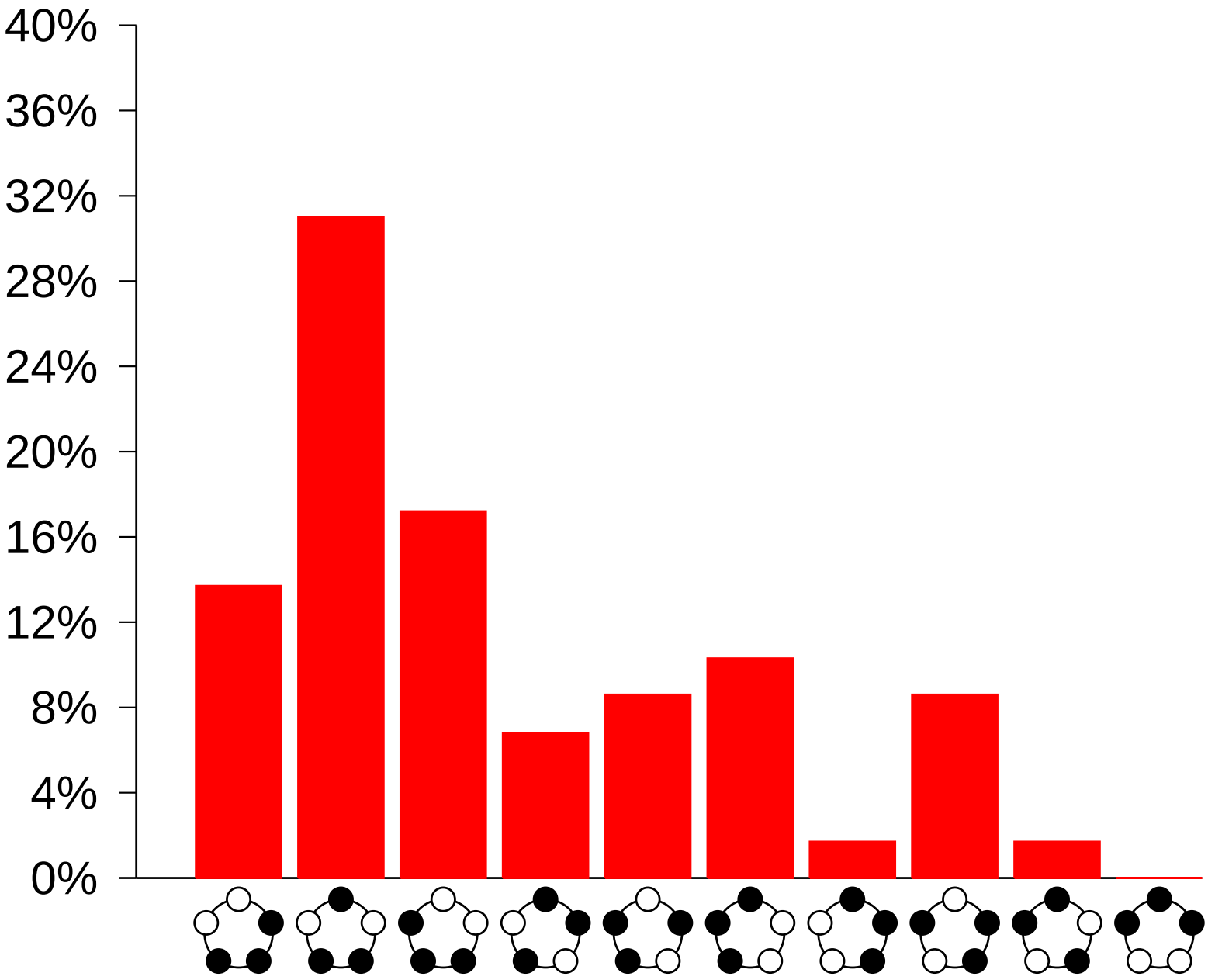
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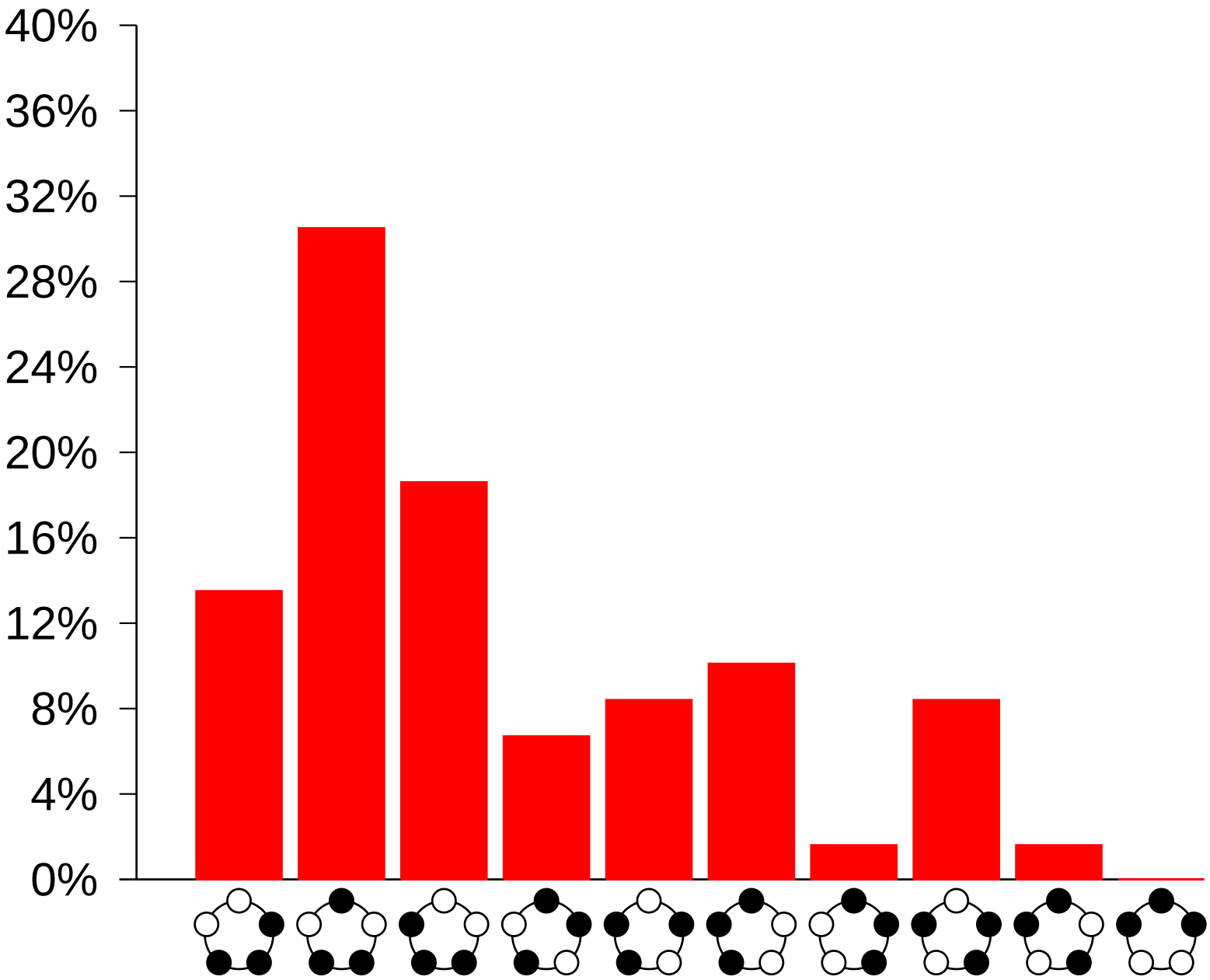
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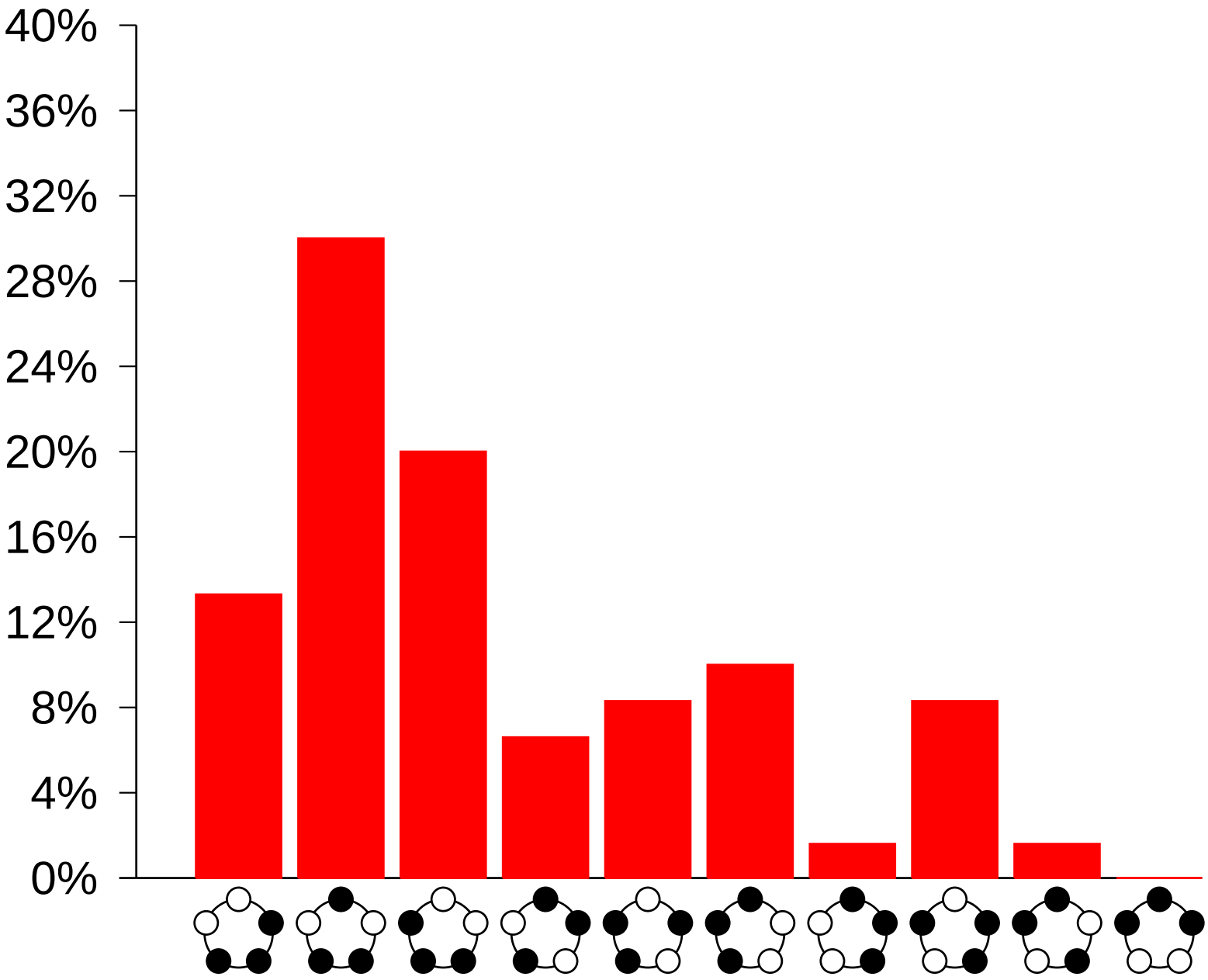
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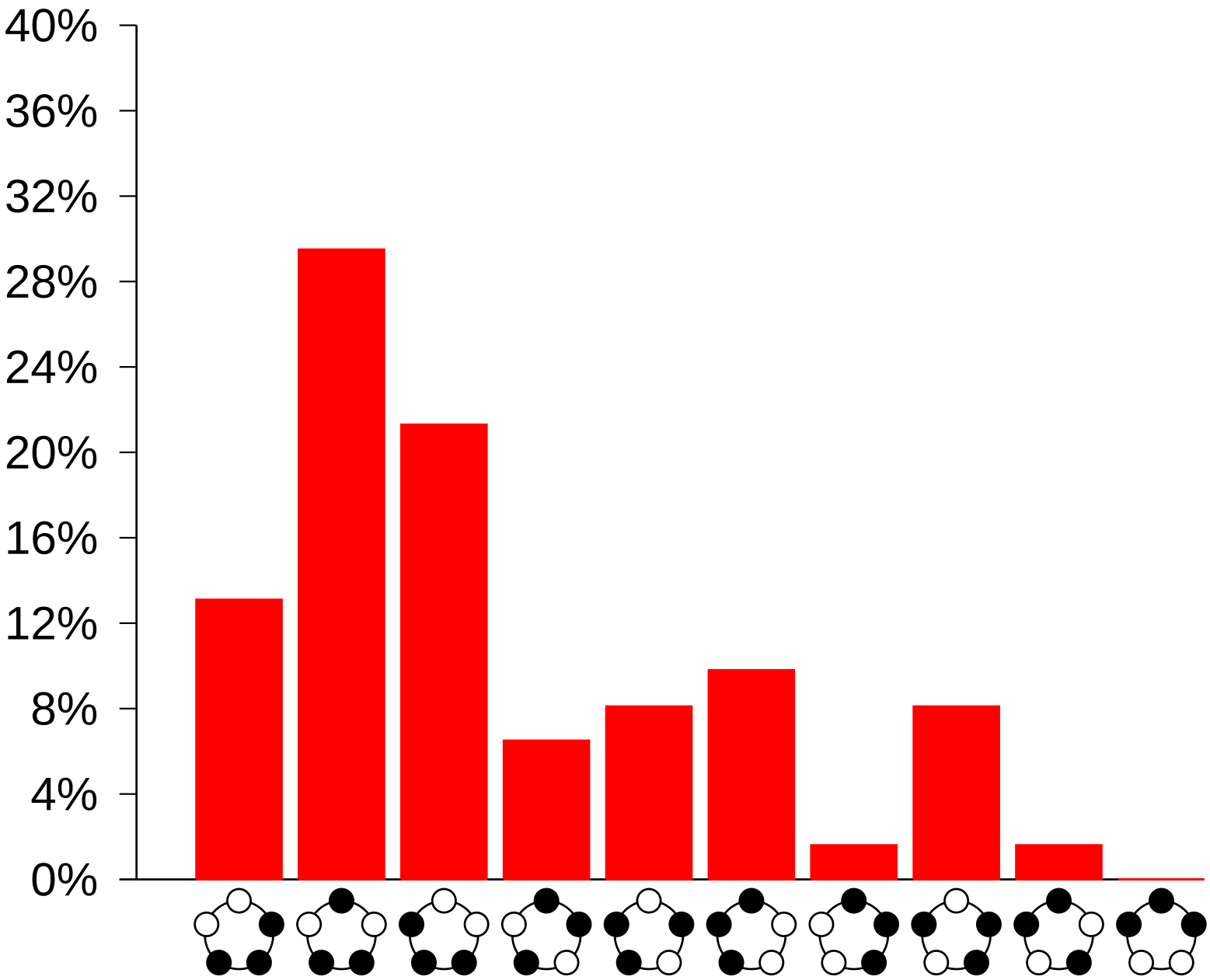
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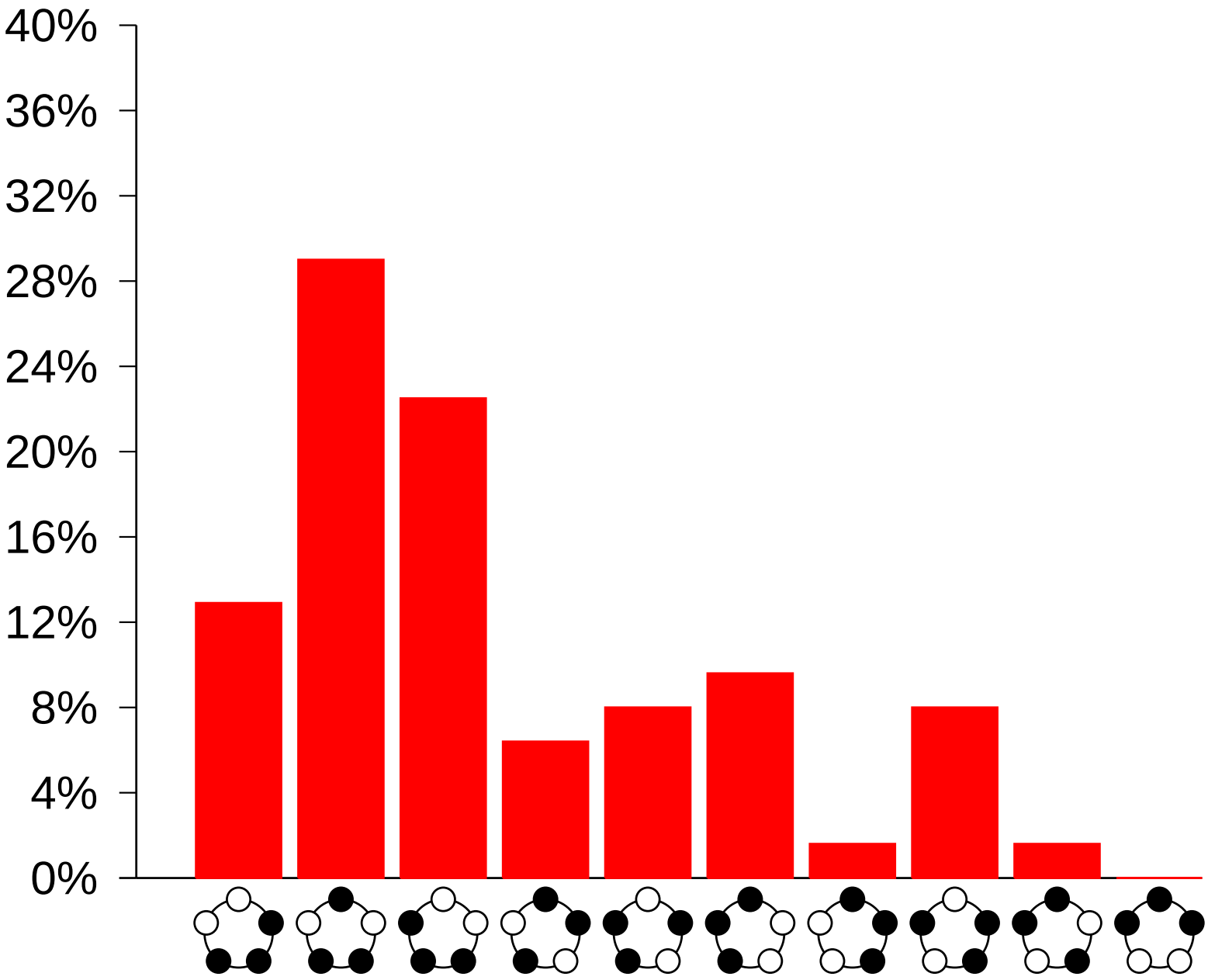
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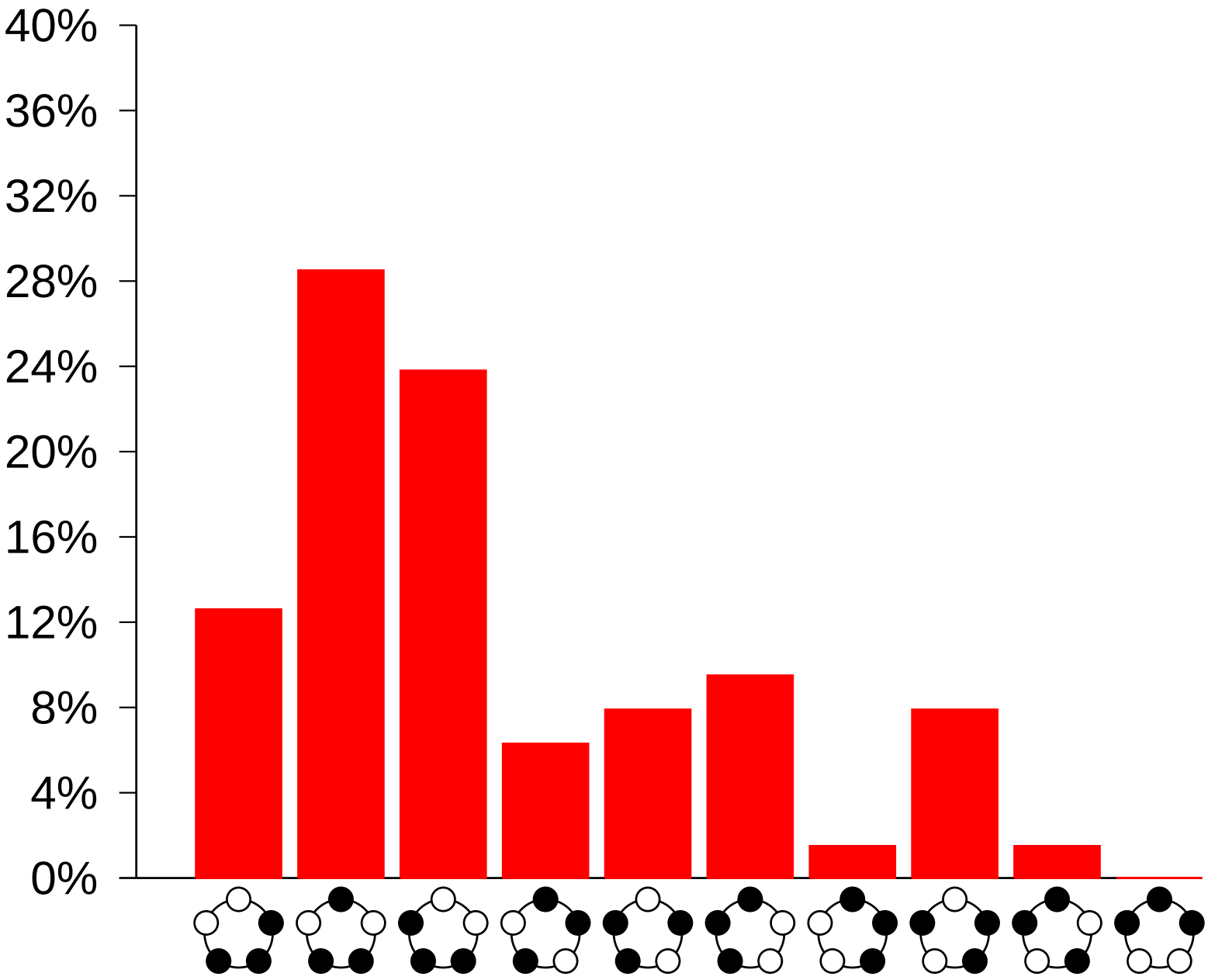
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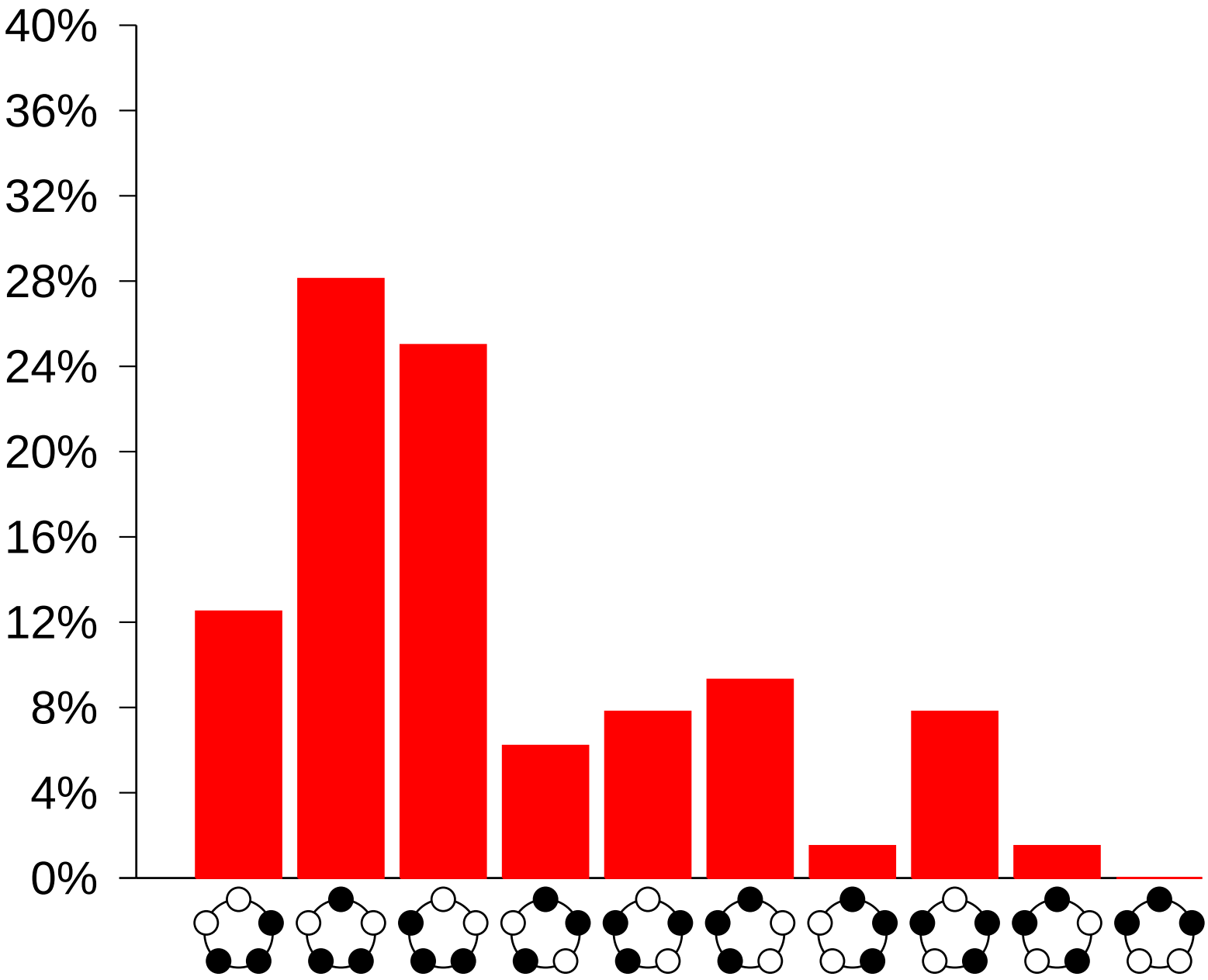
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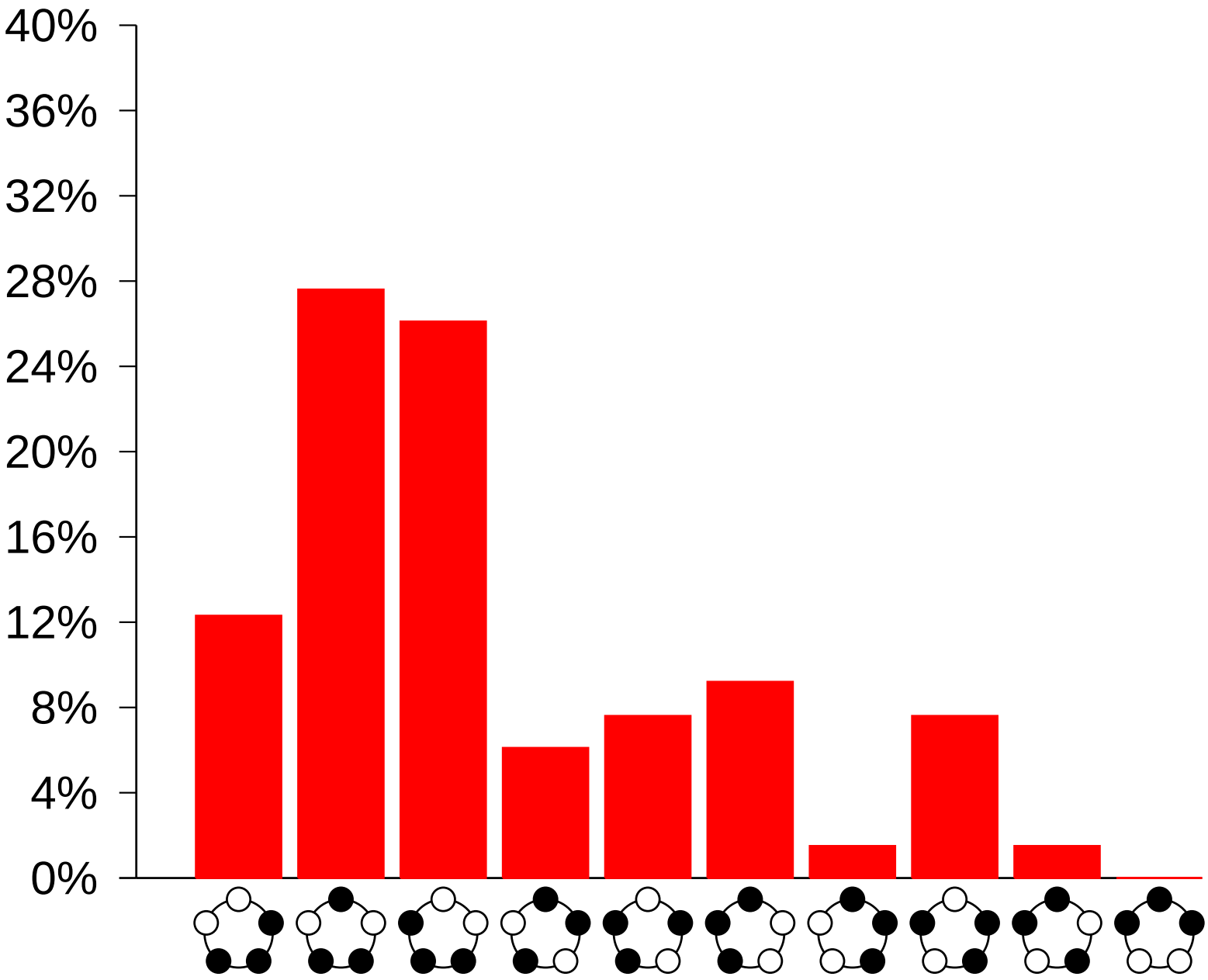
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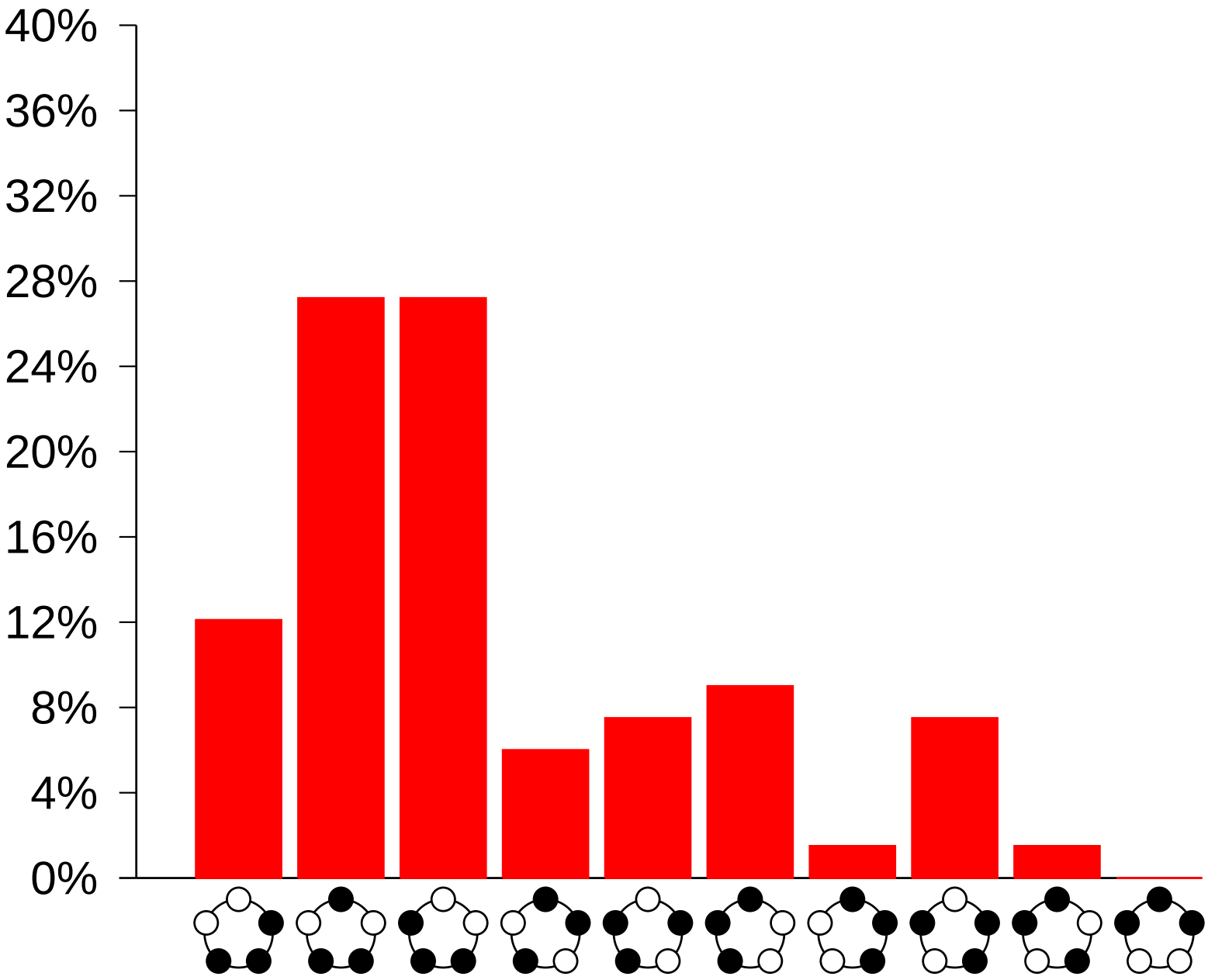
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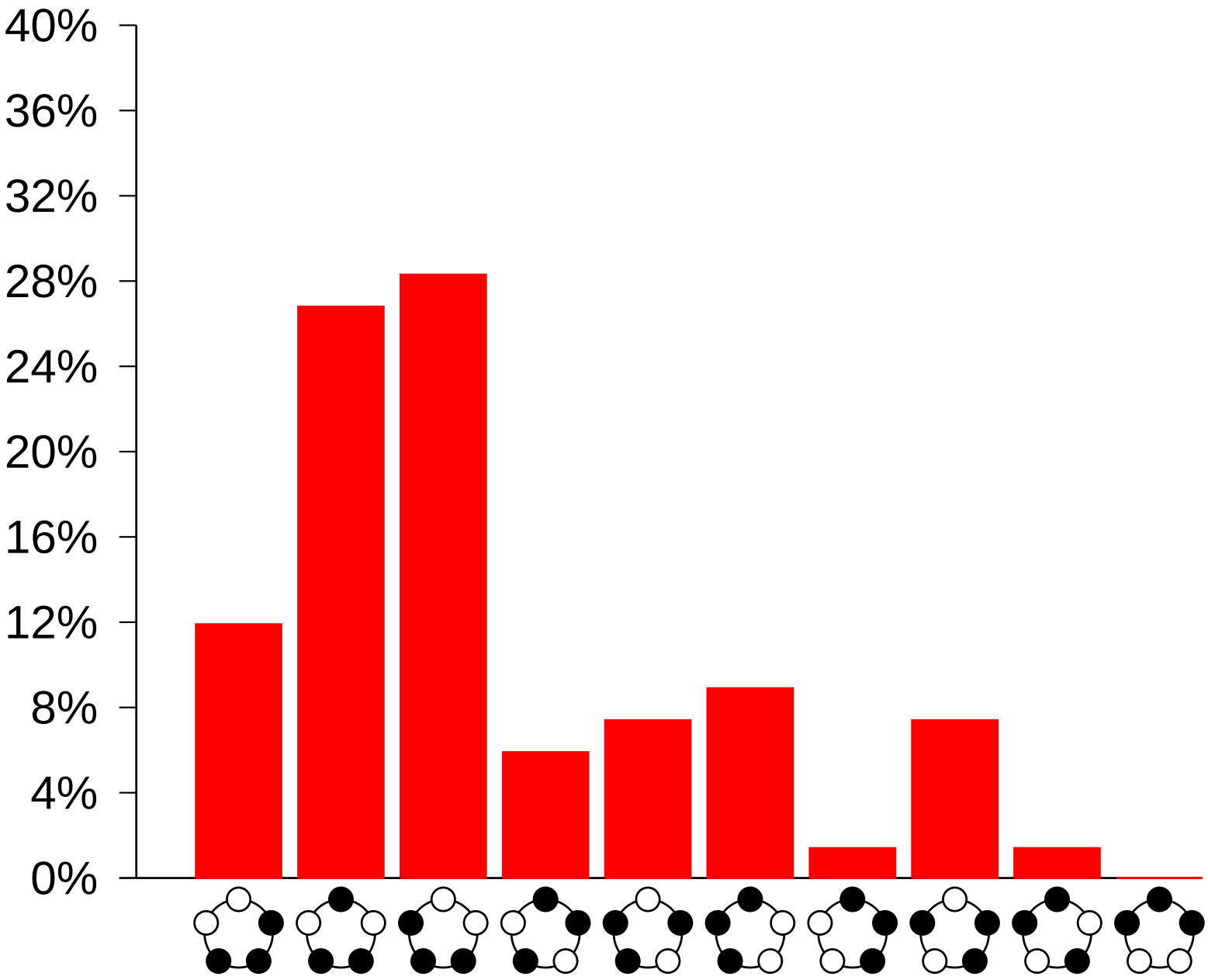
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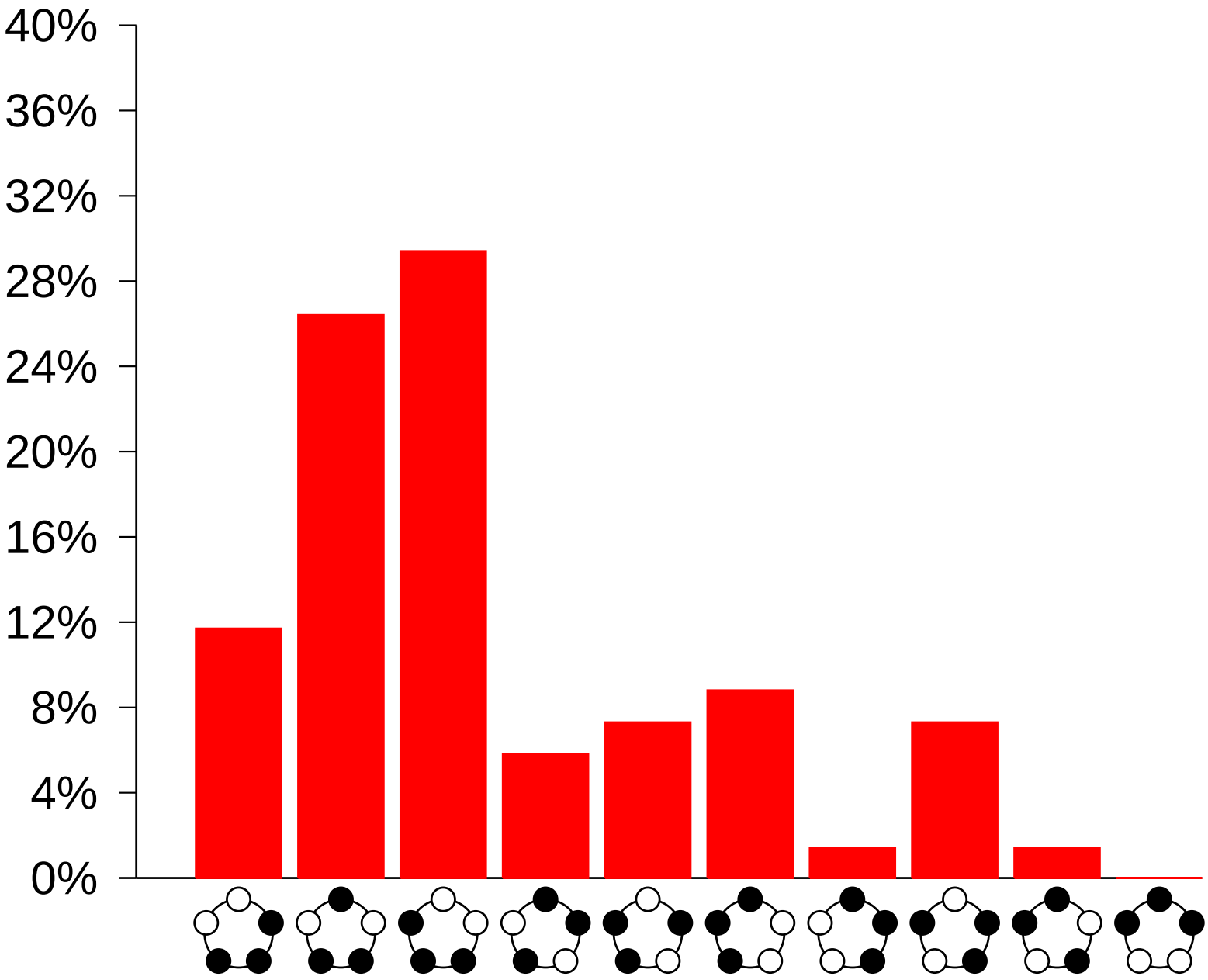
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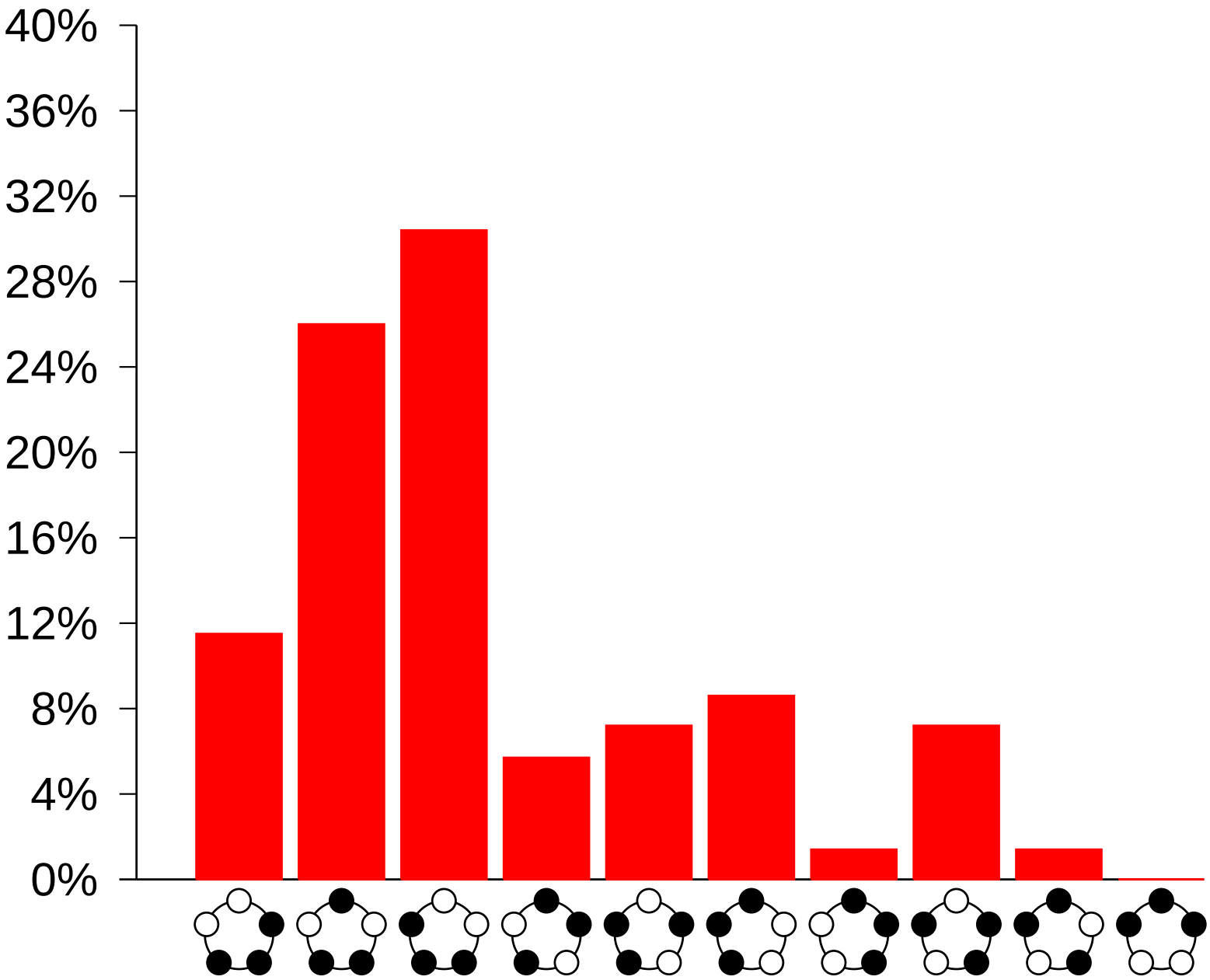
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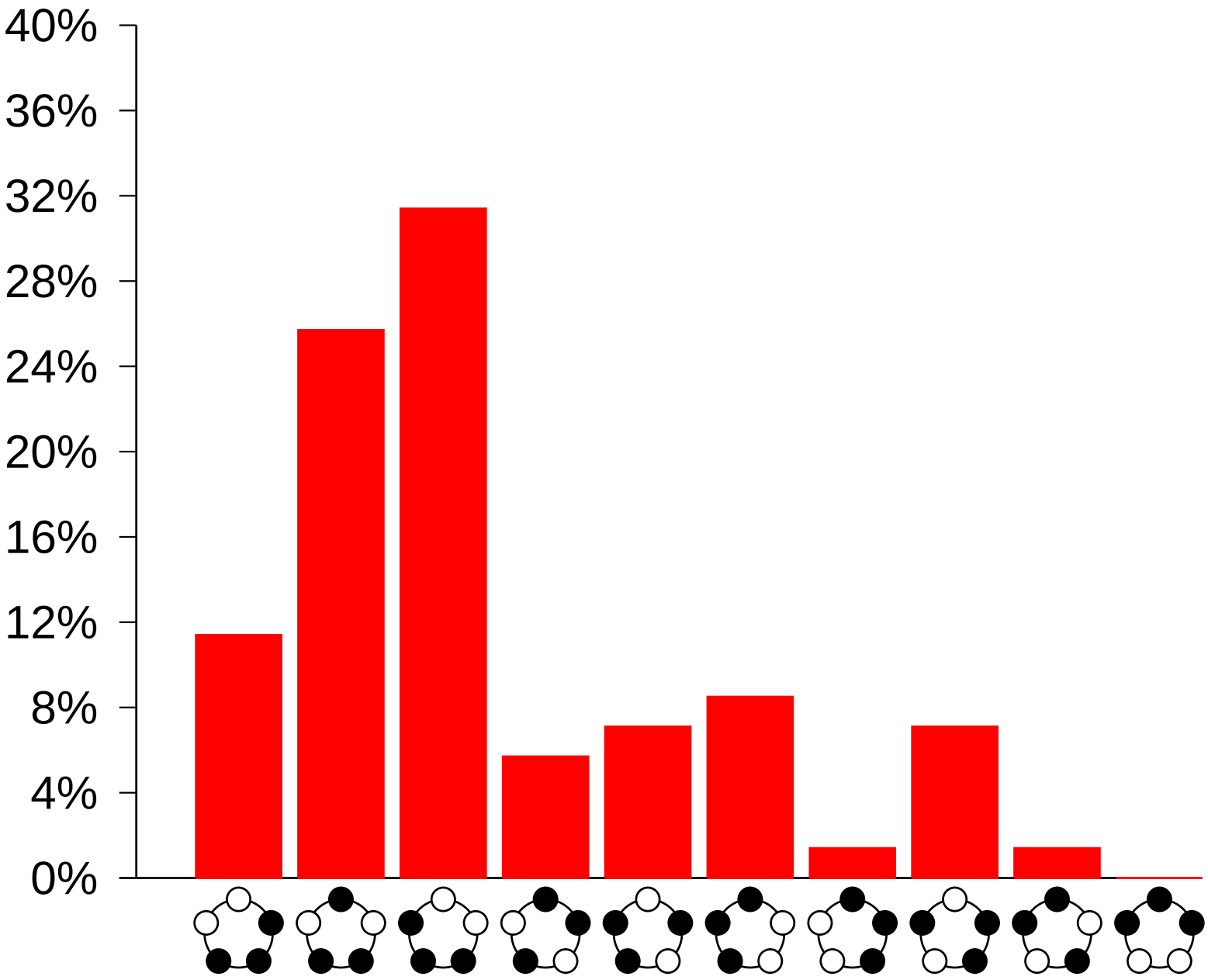
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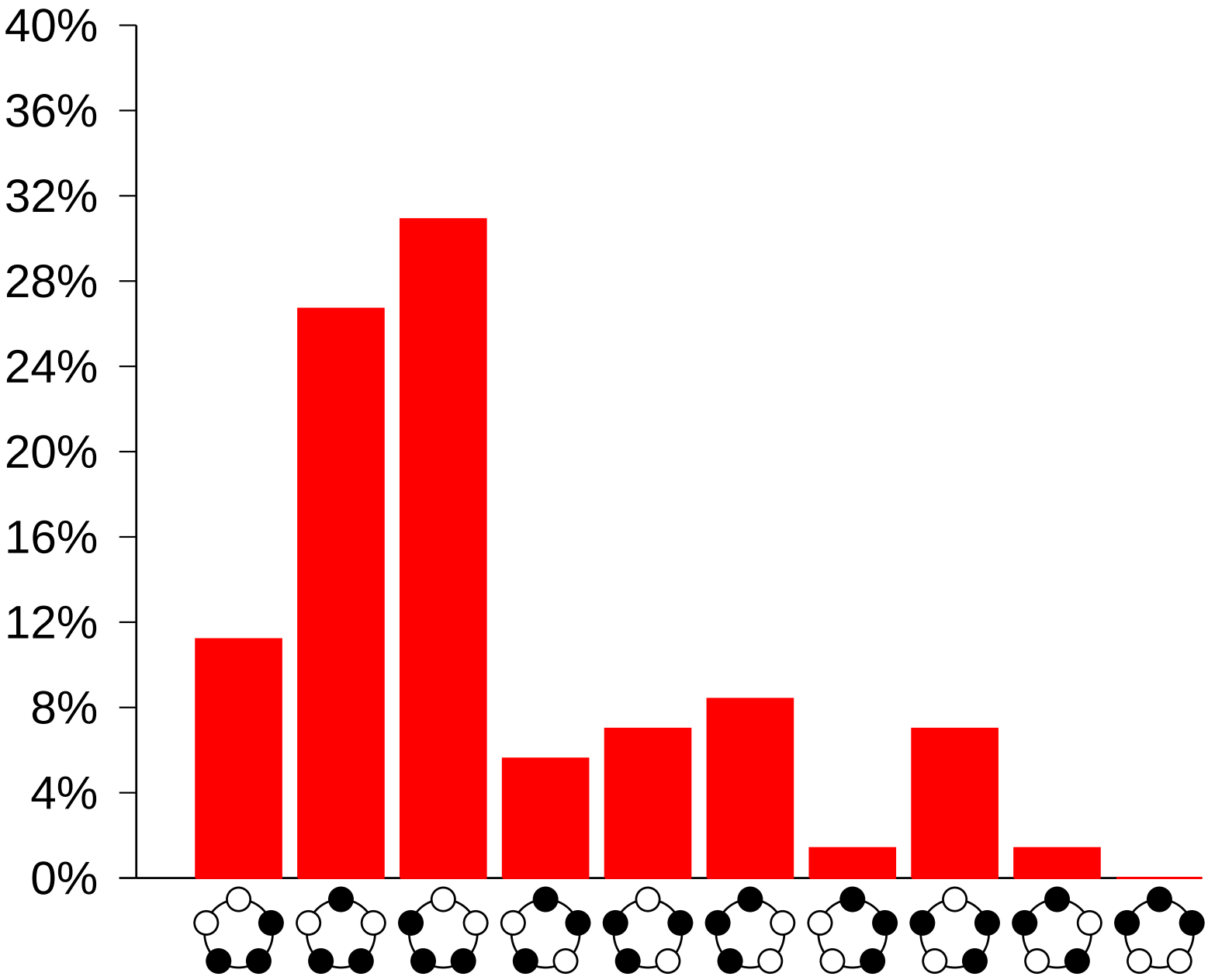
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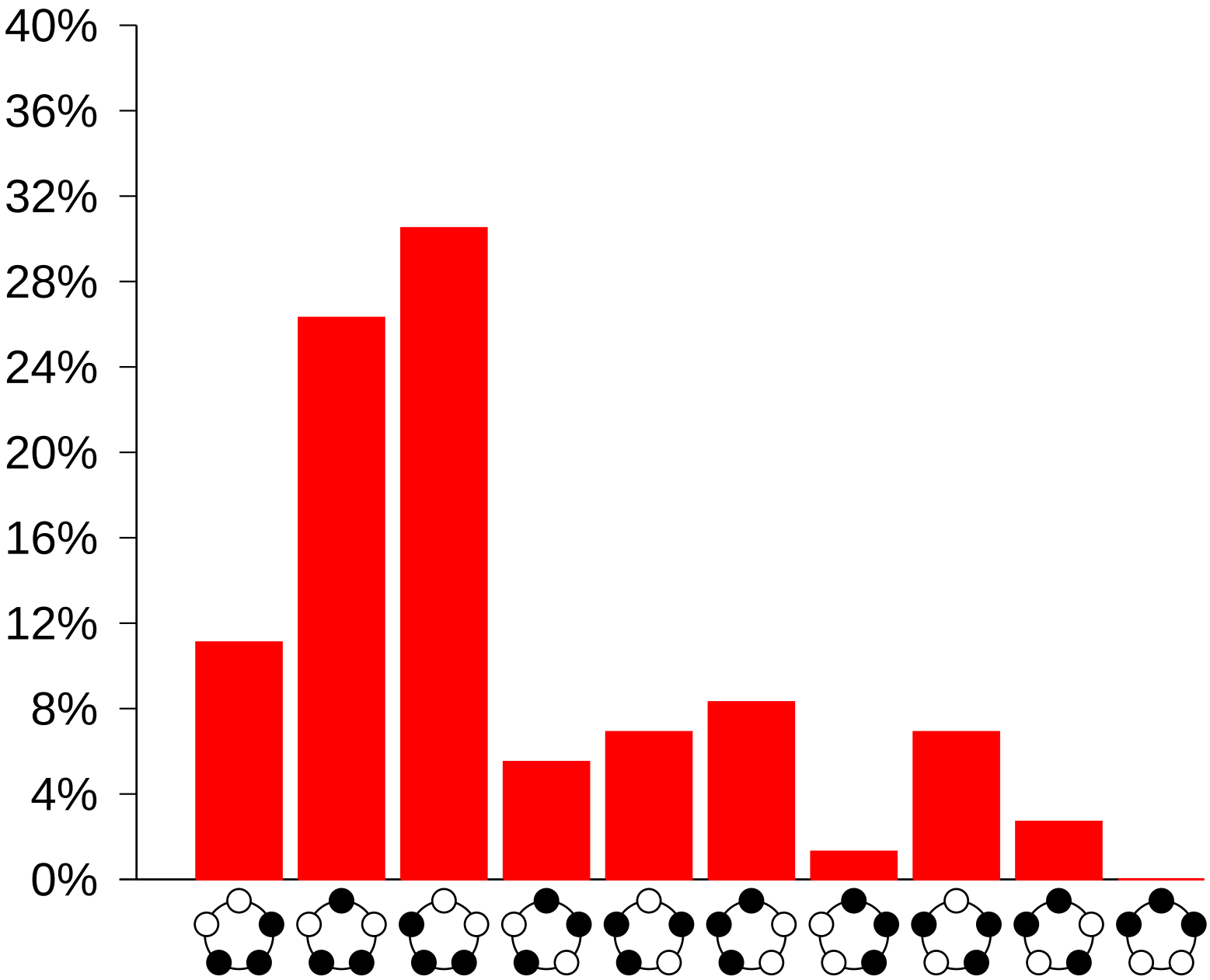
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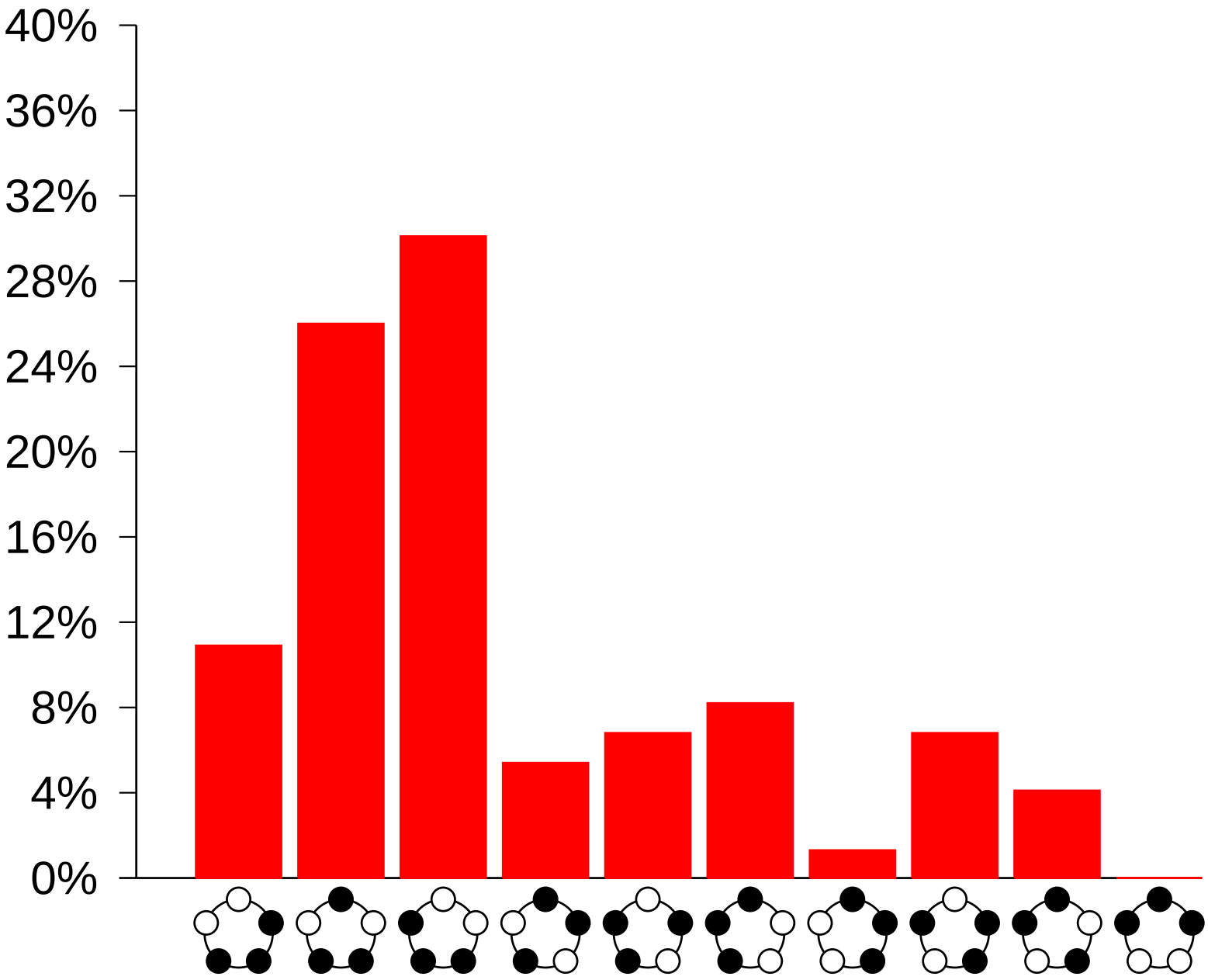
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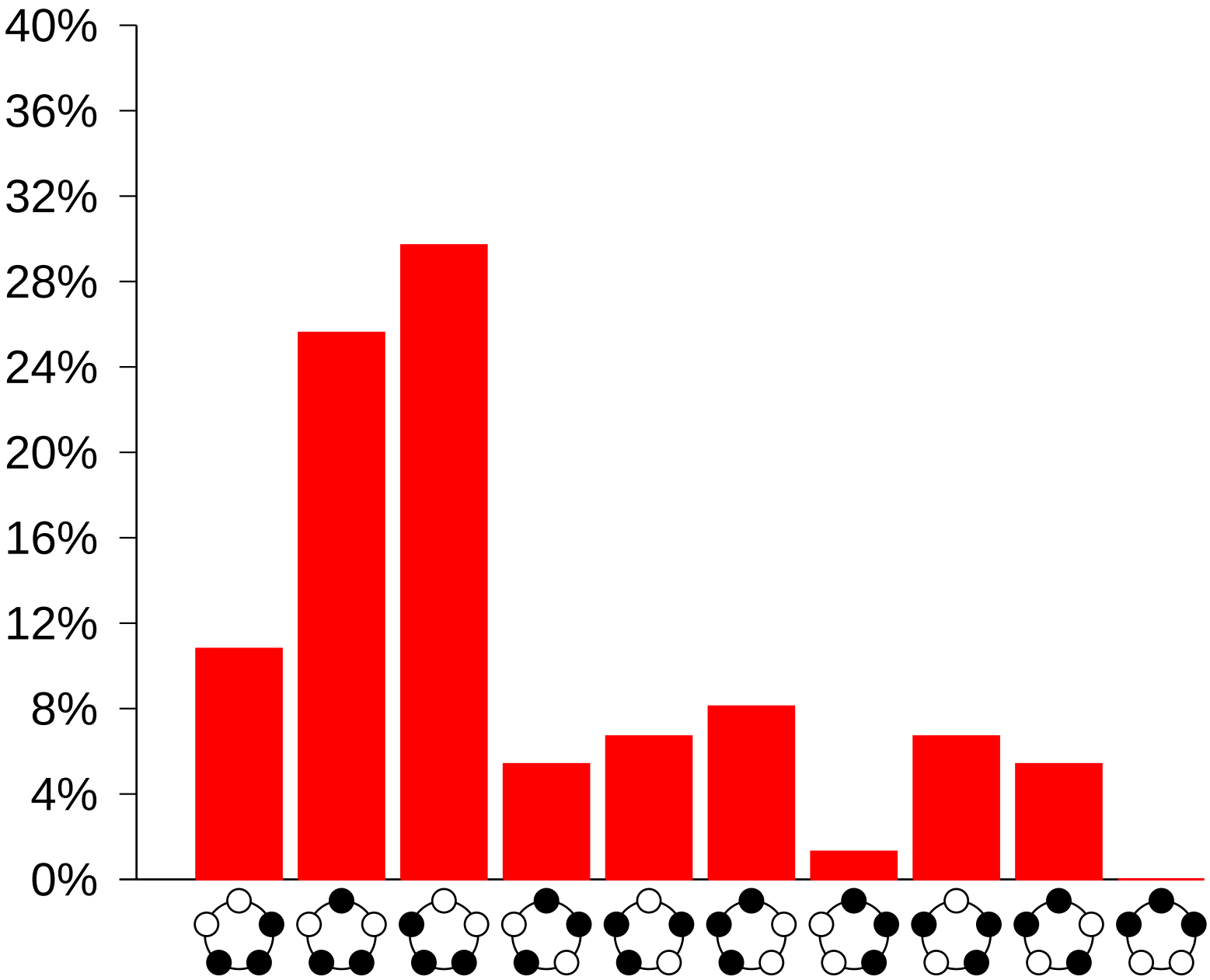
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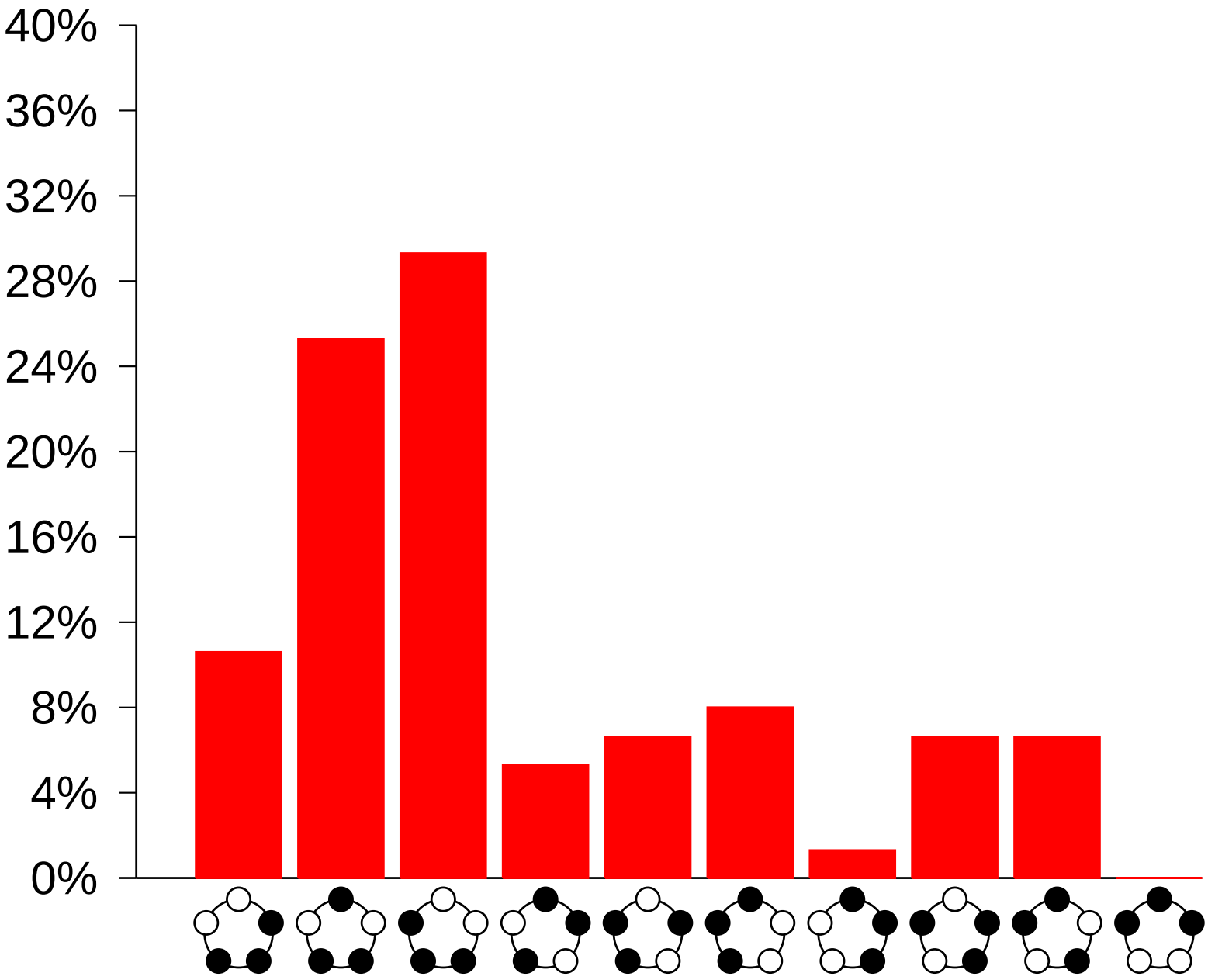
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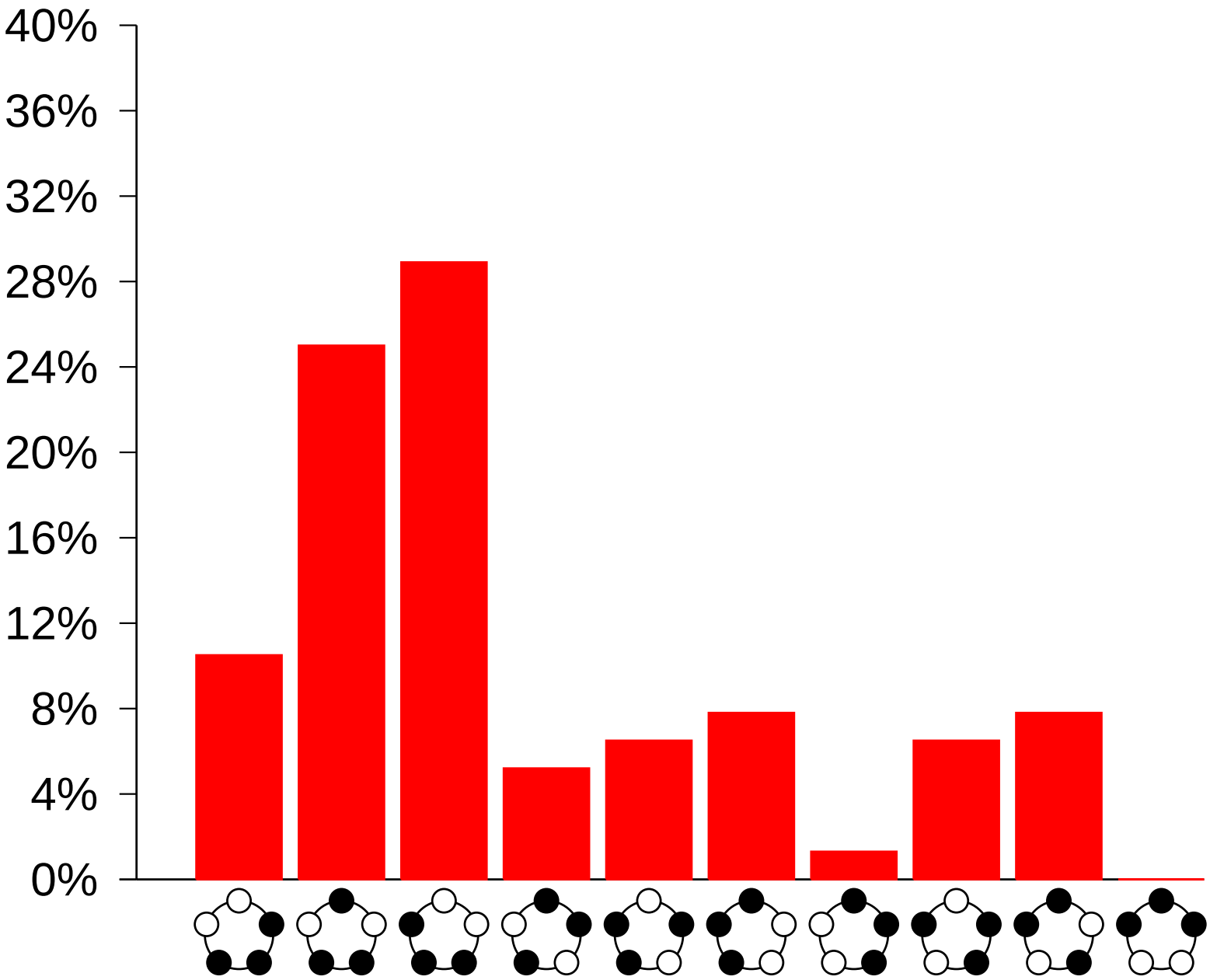
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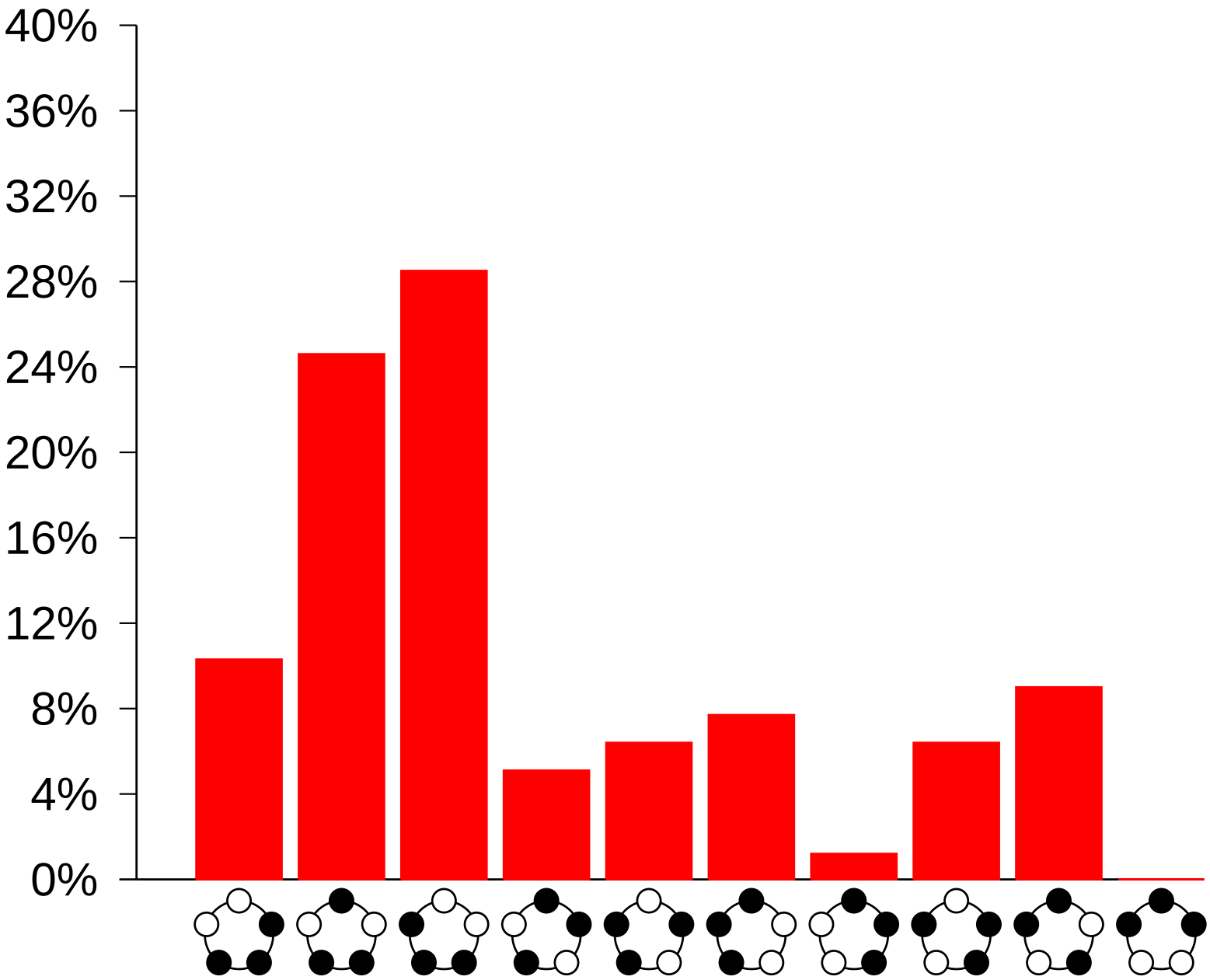
Stationary distribution



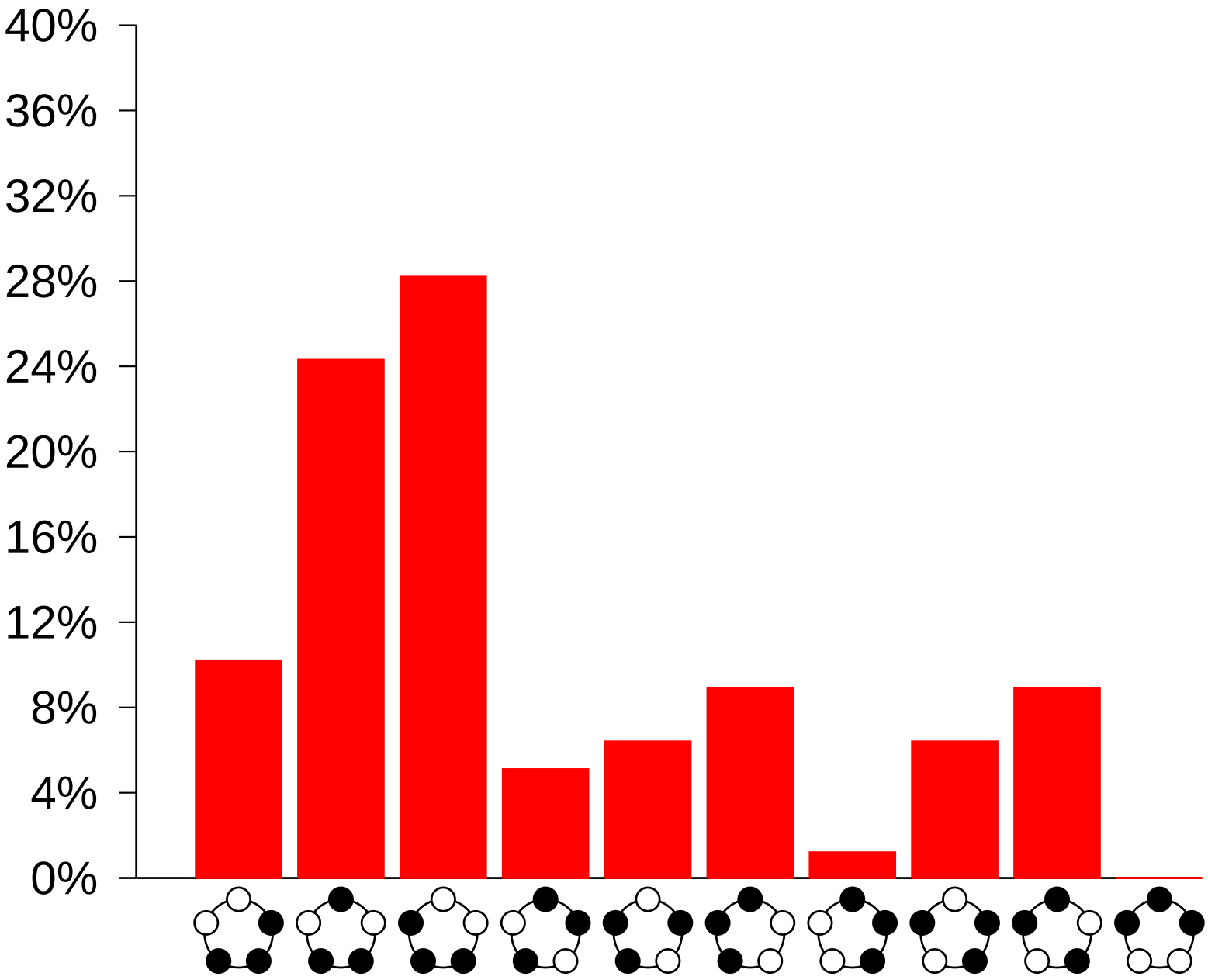
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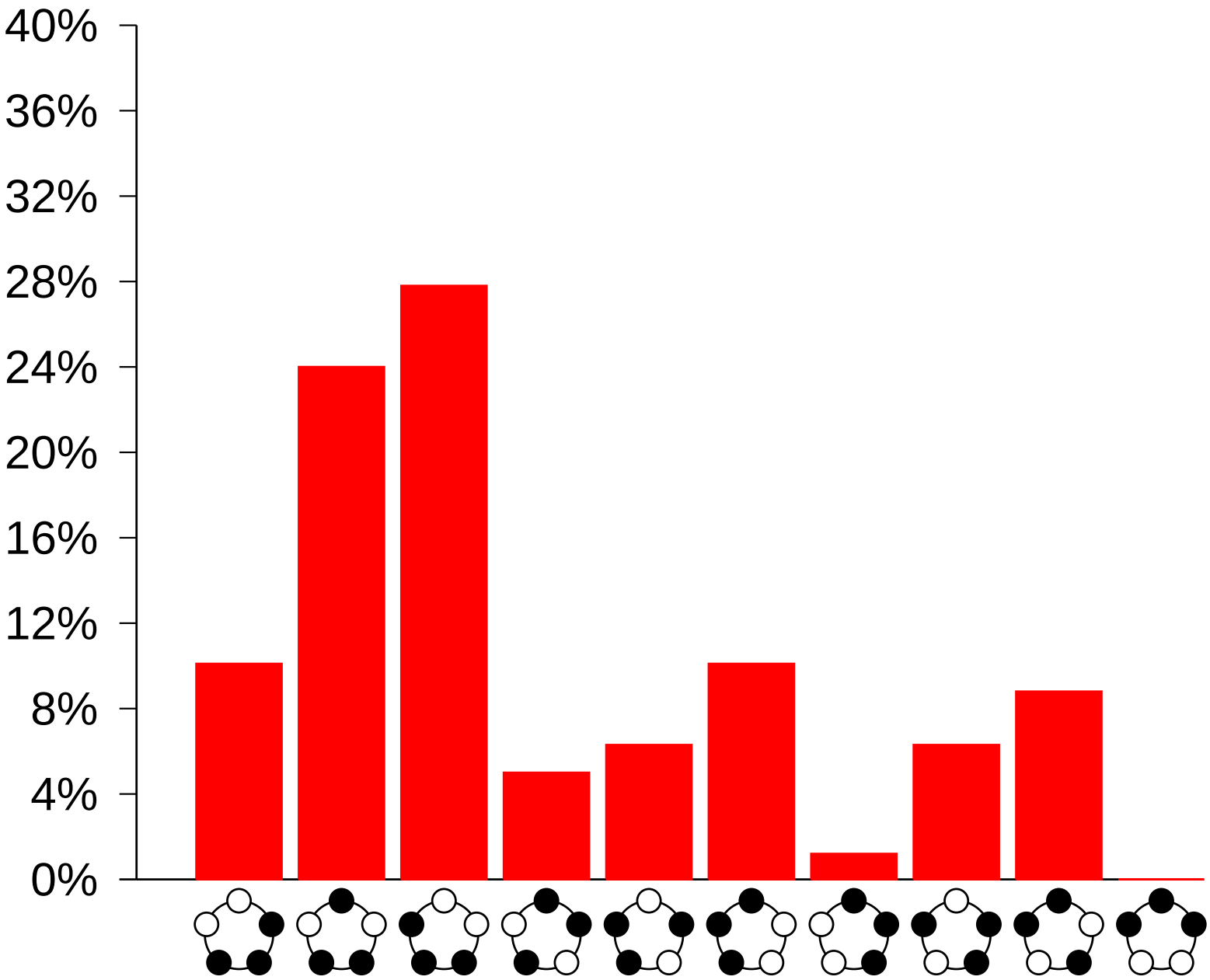
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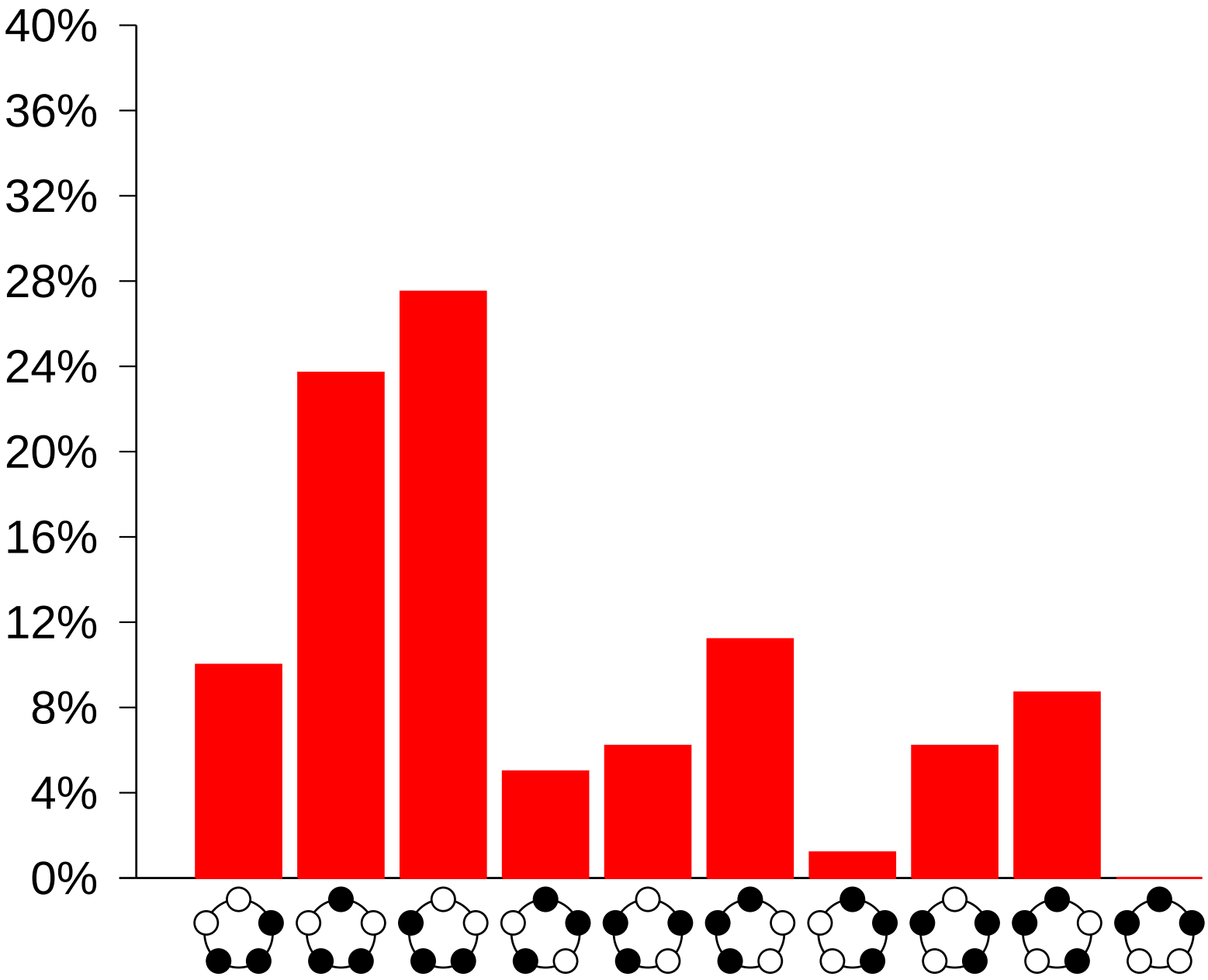
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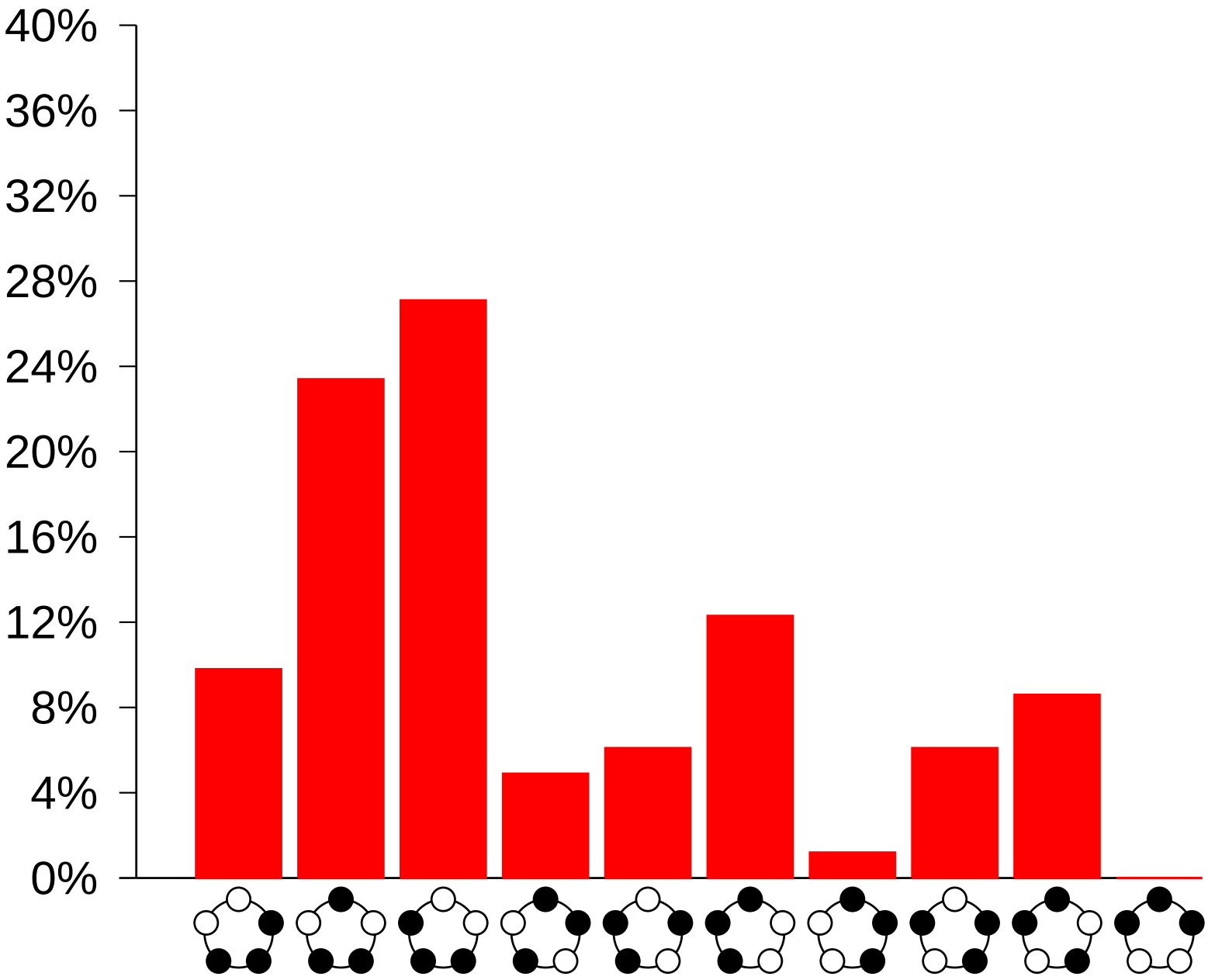
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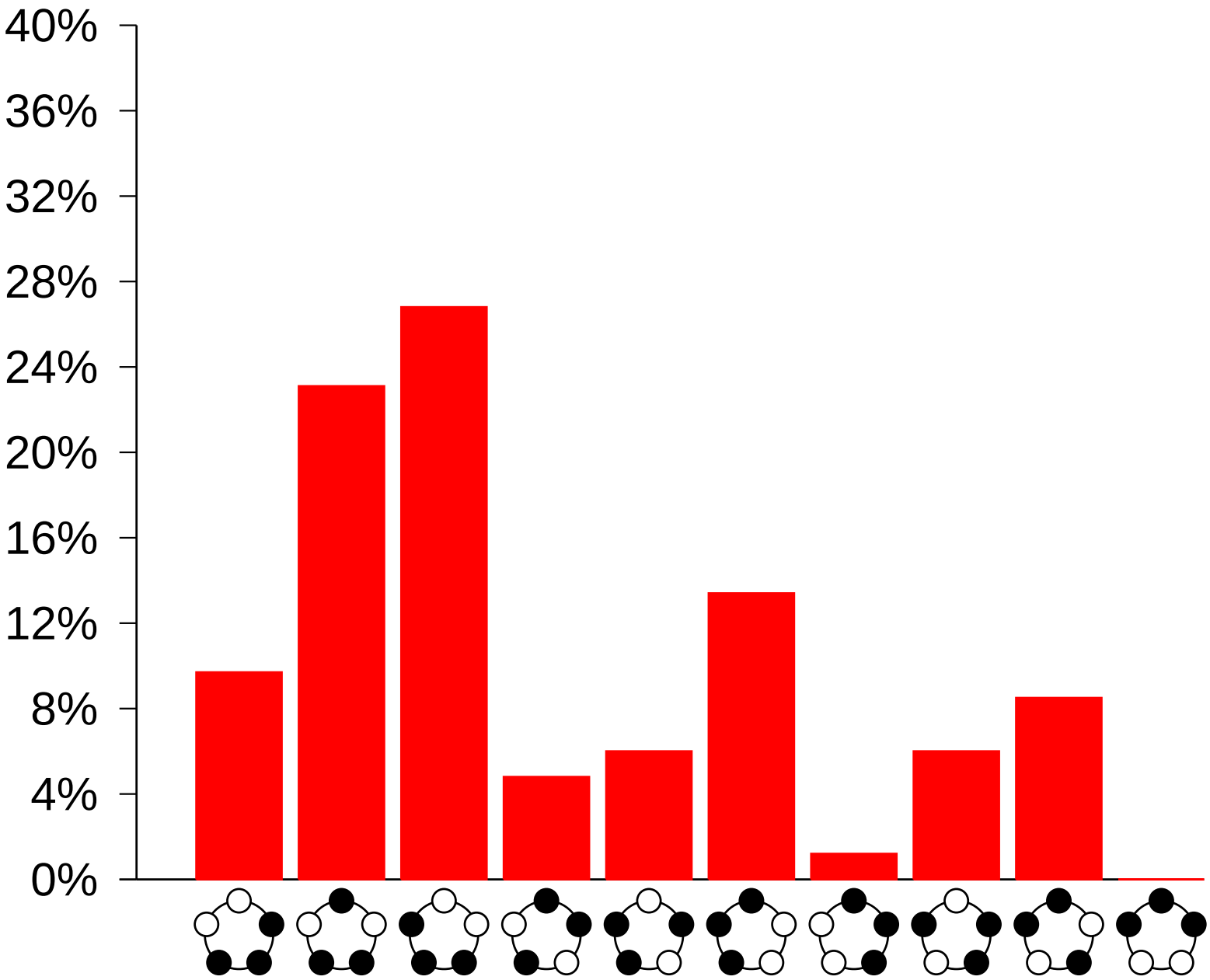
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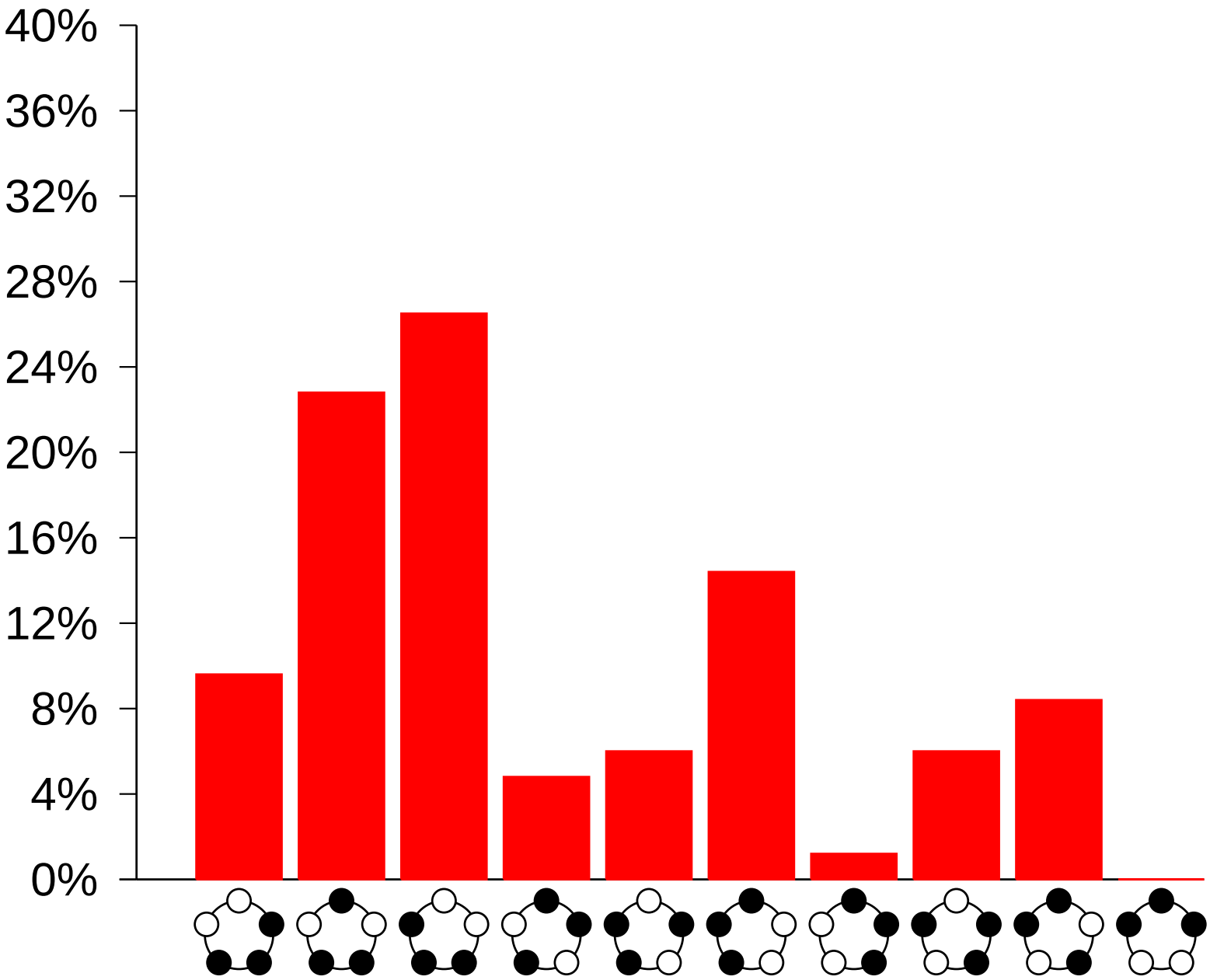
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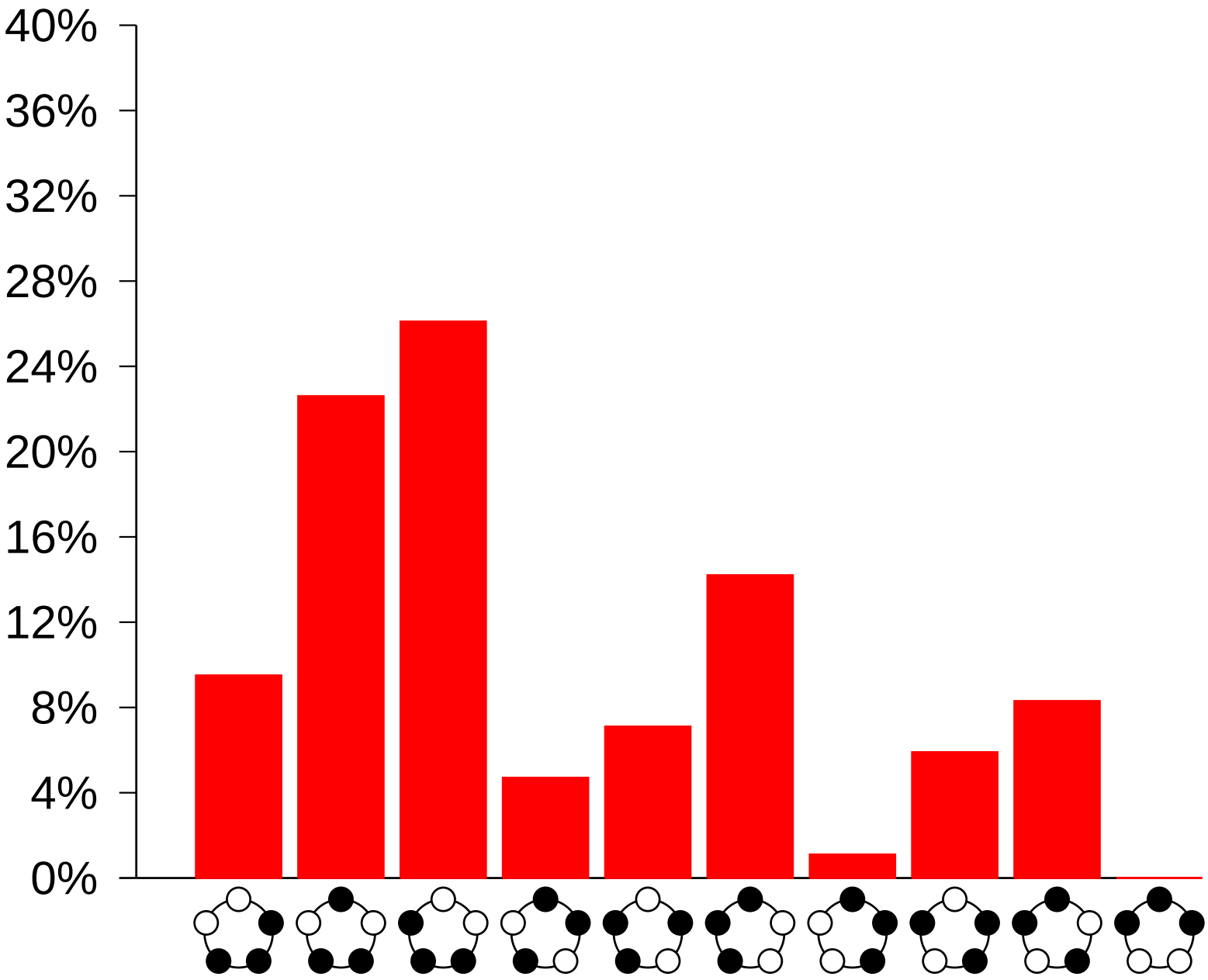
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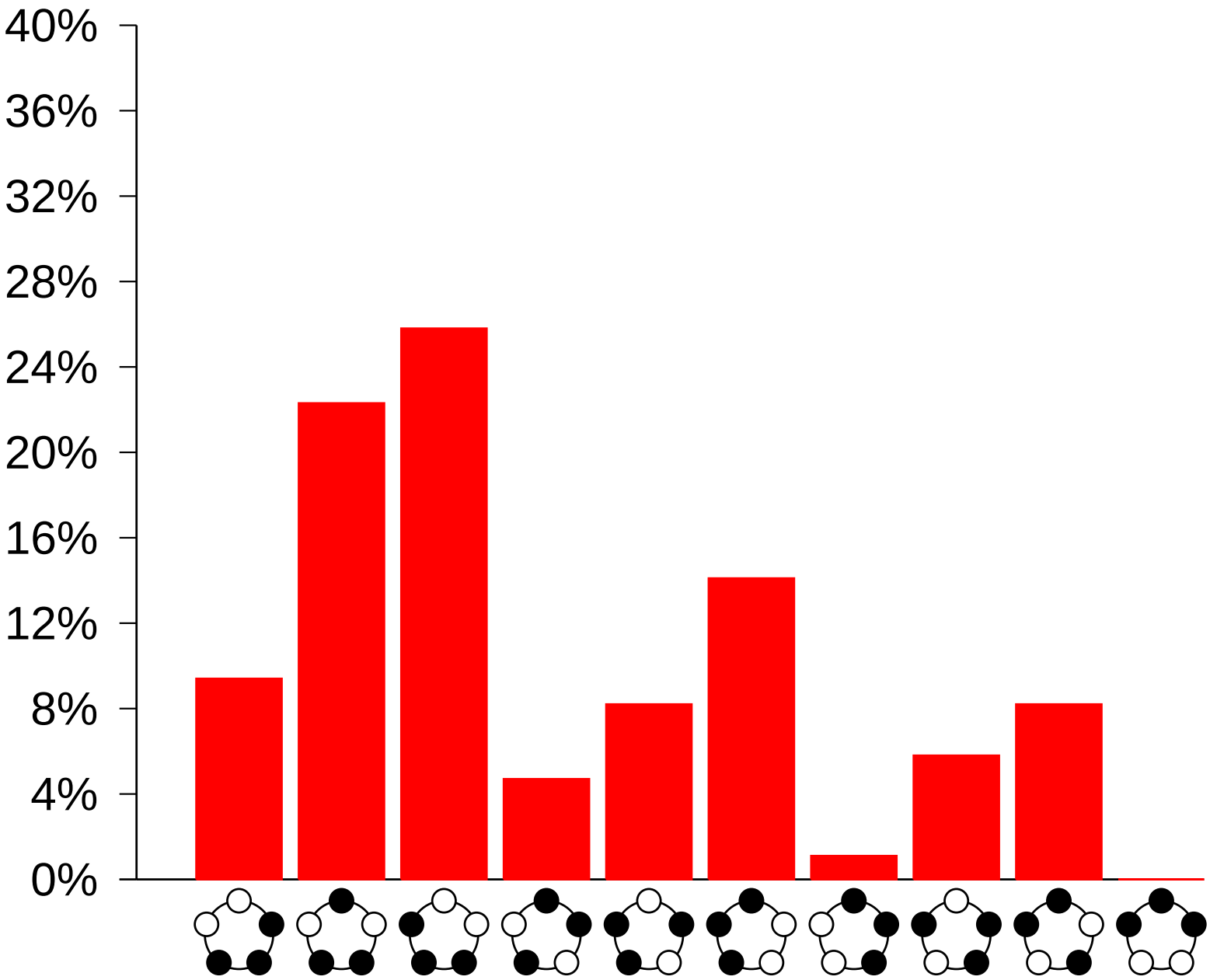
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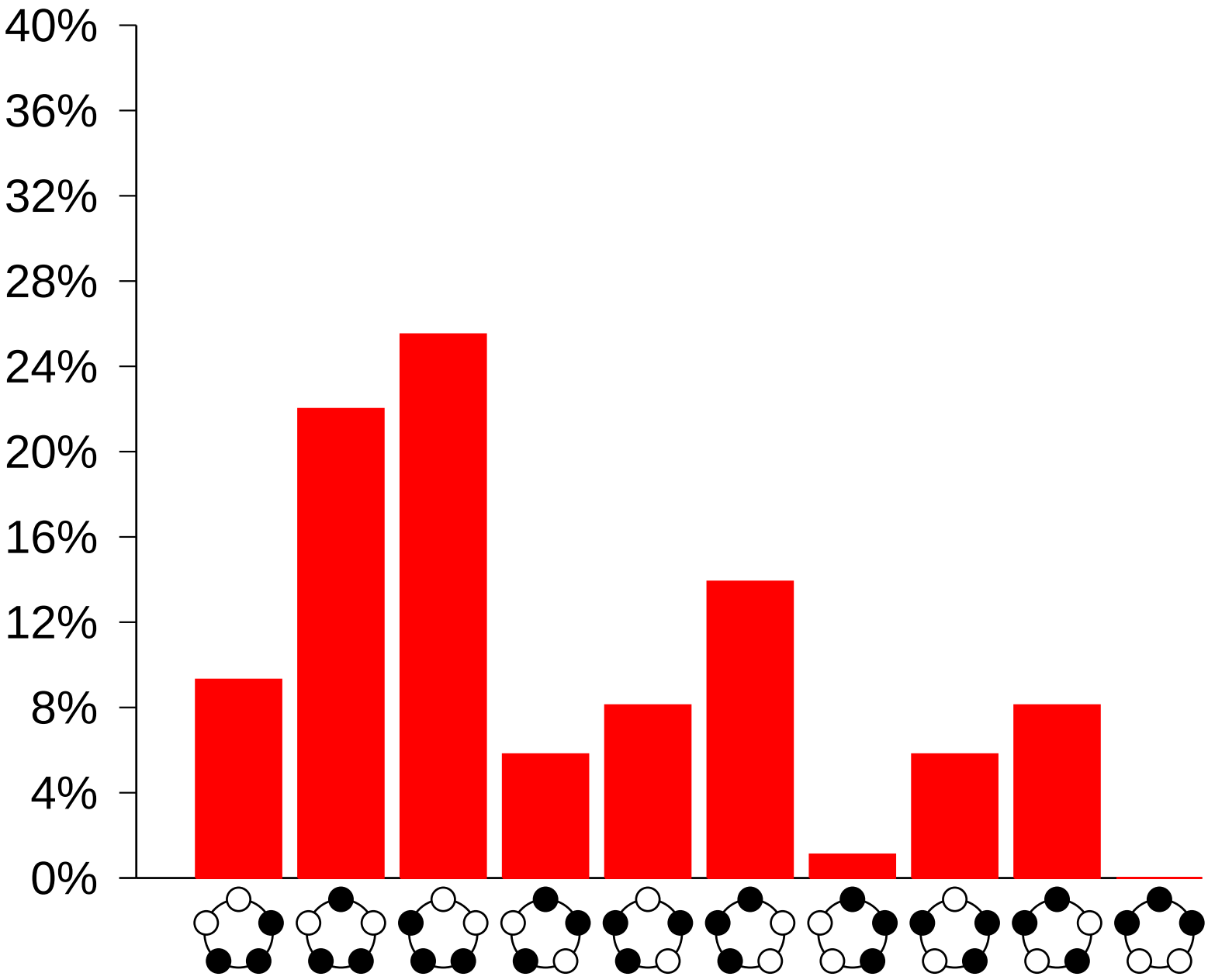
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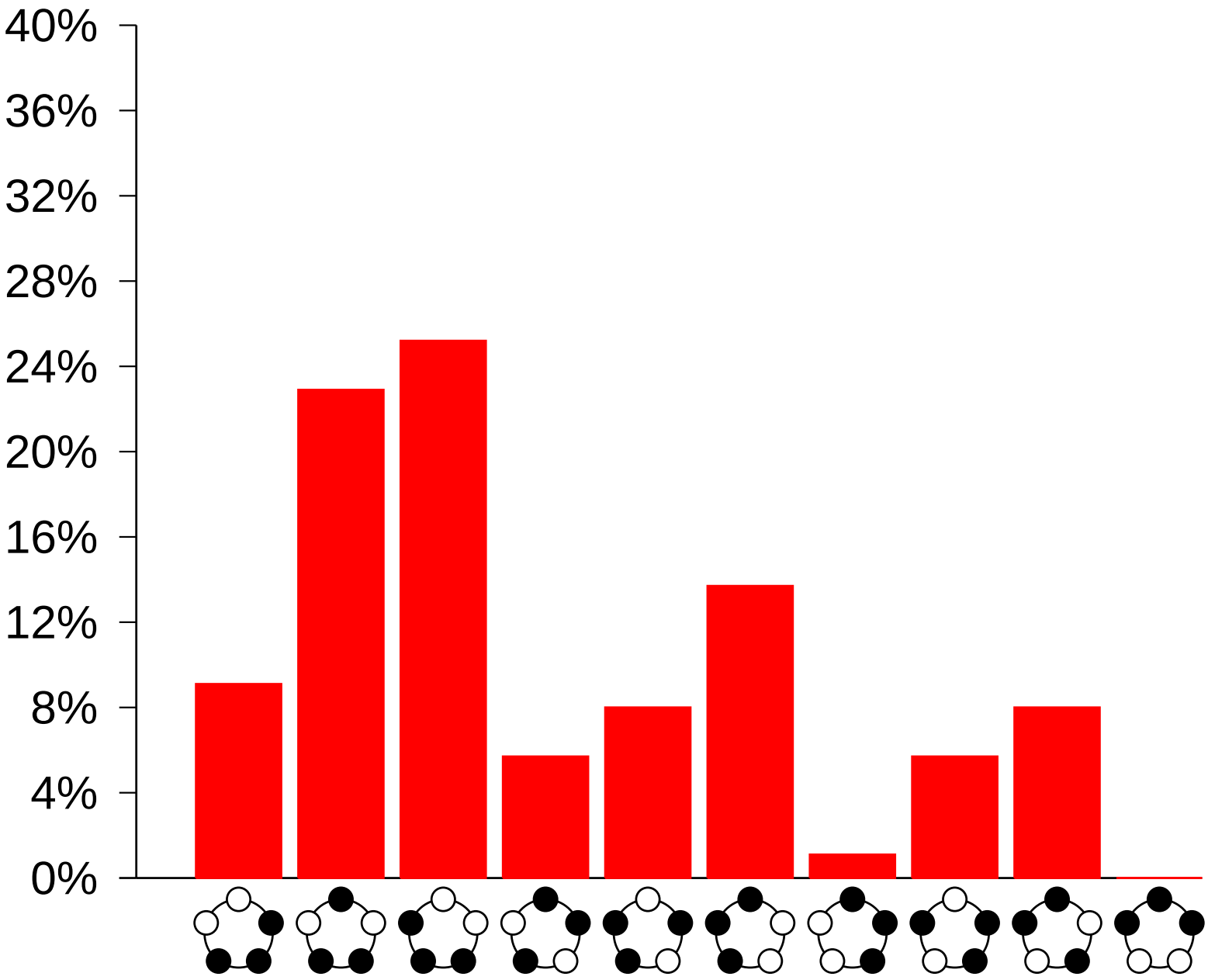
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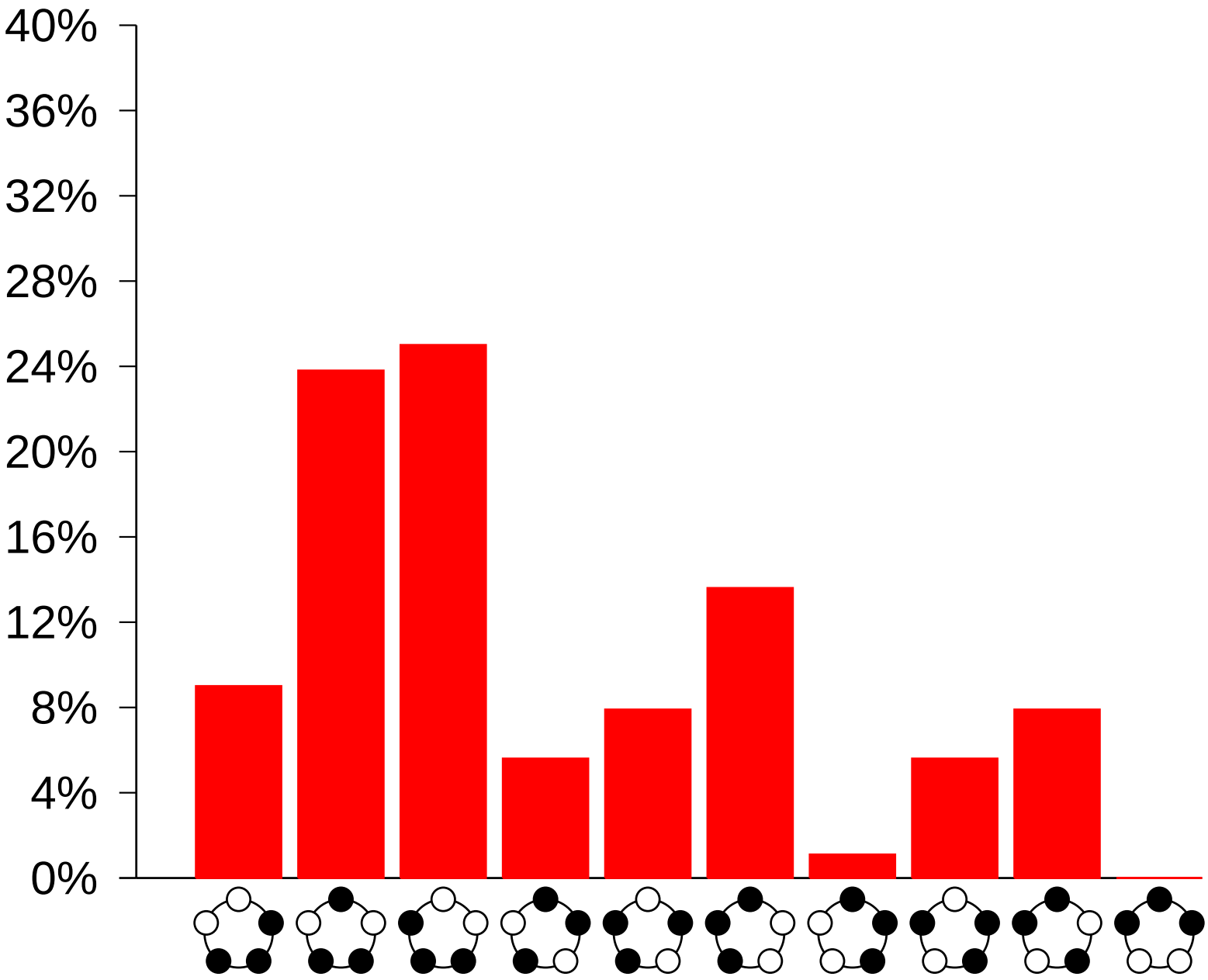
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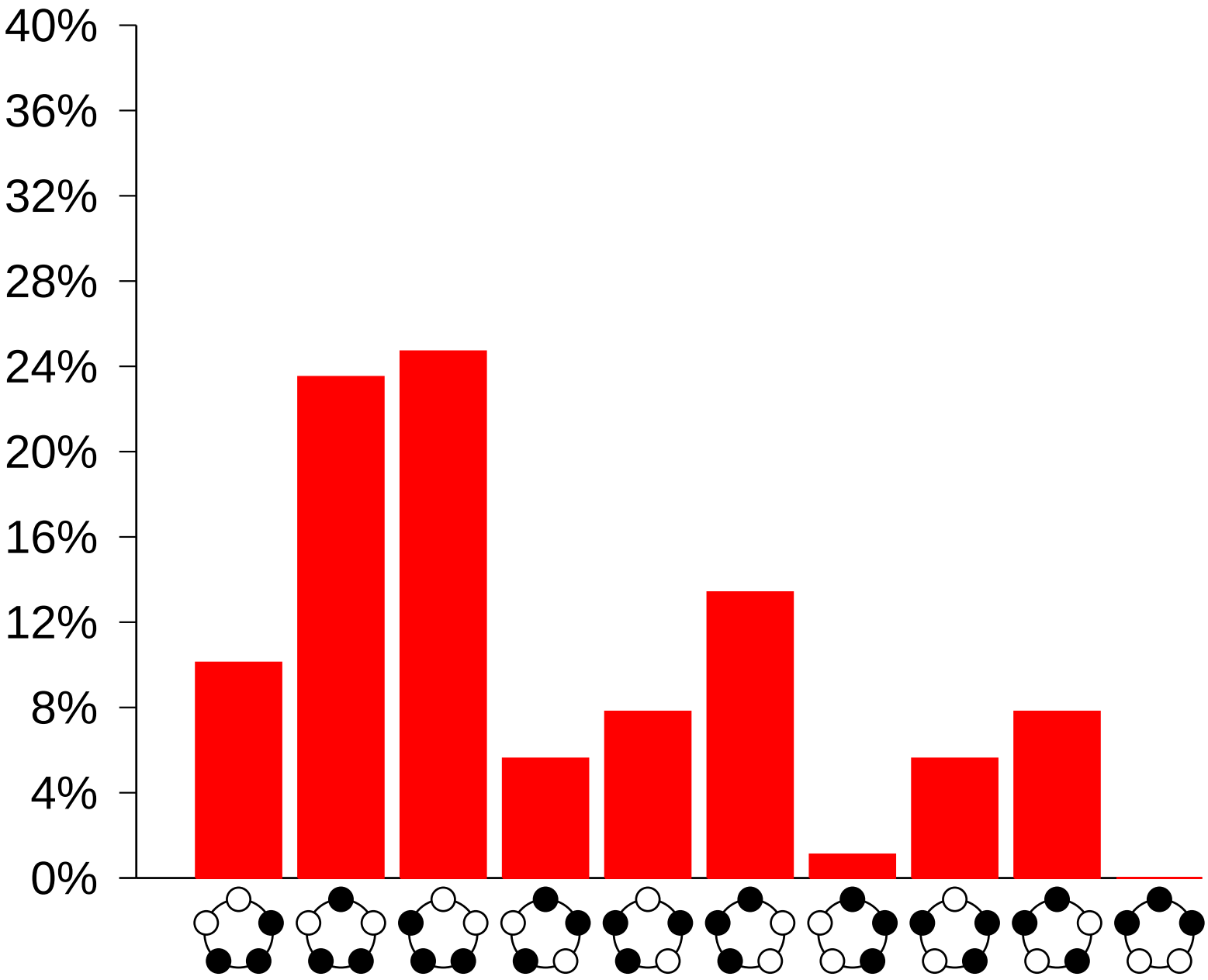
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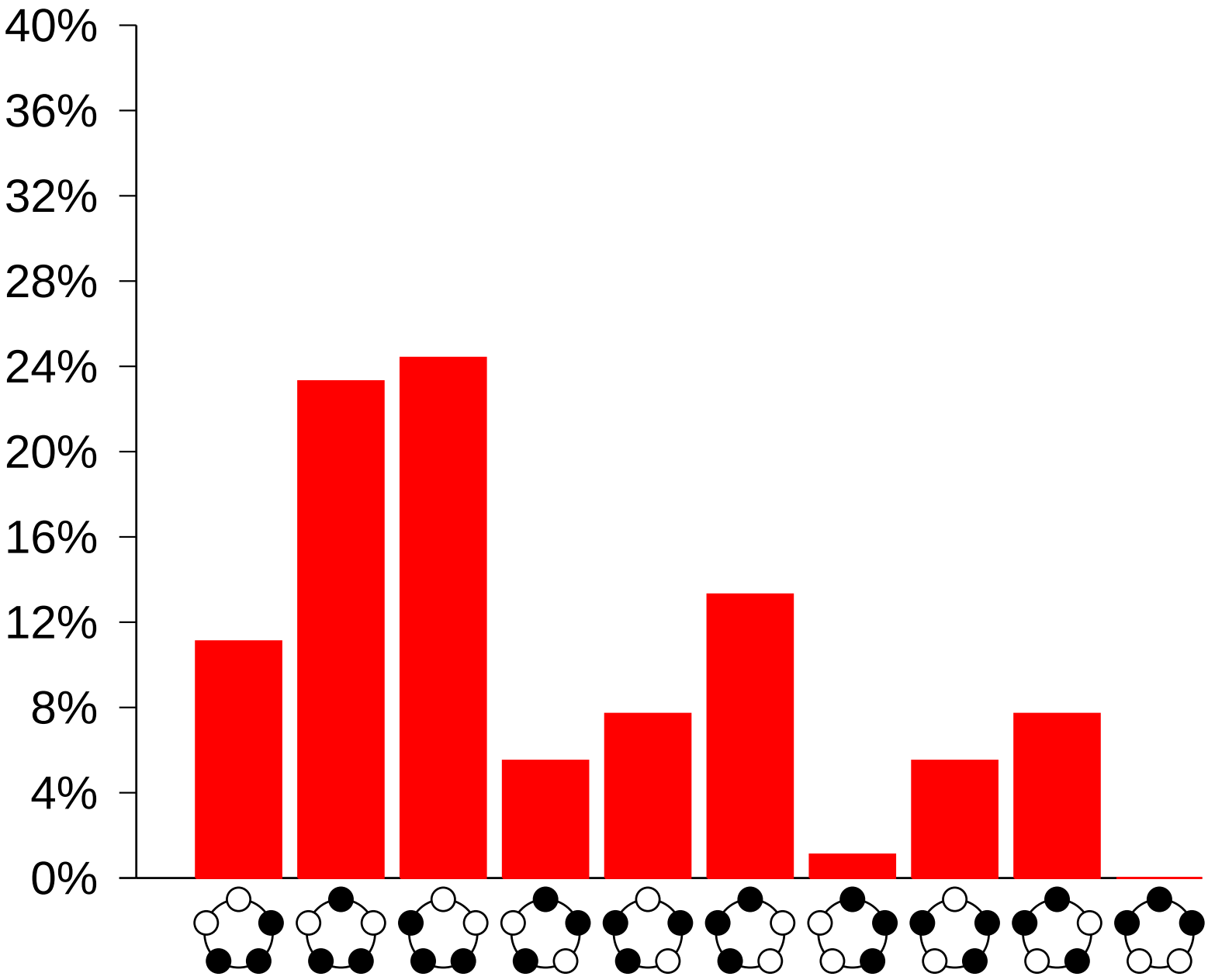
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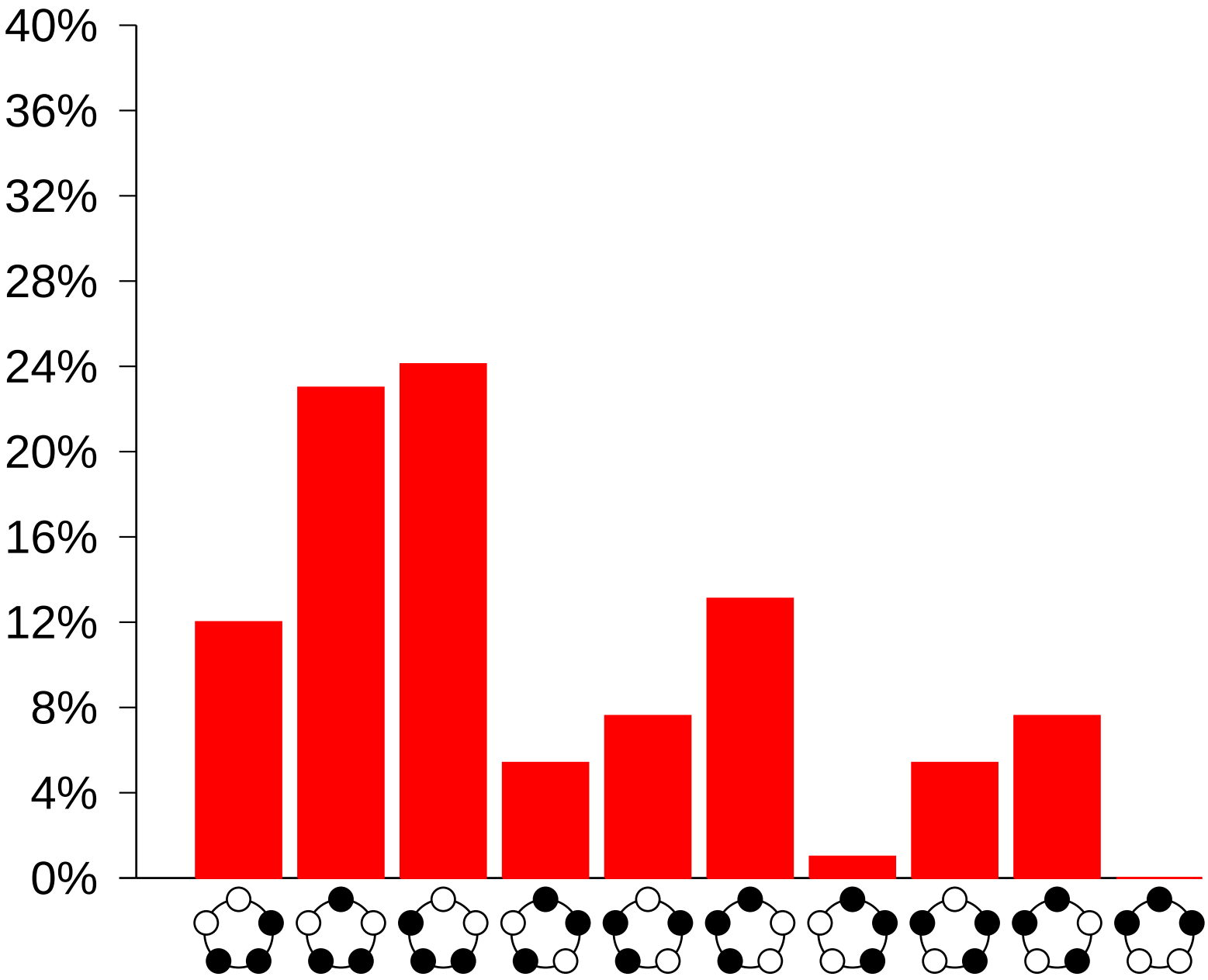
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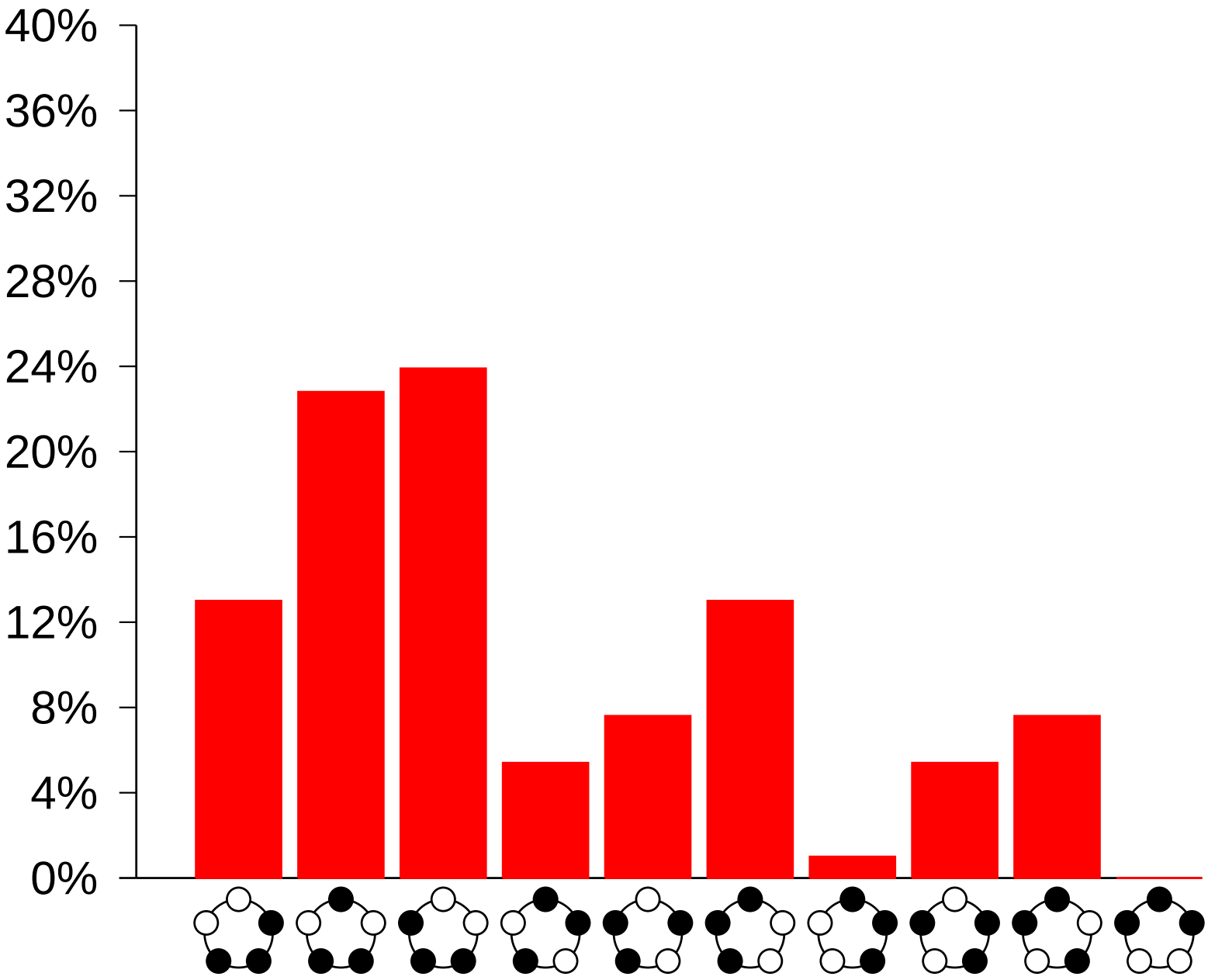
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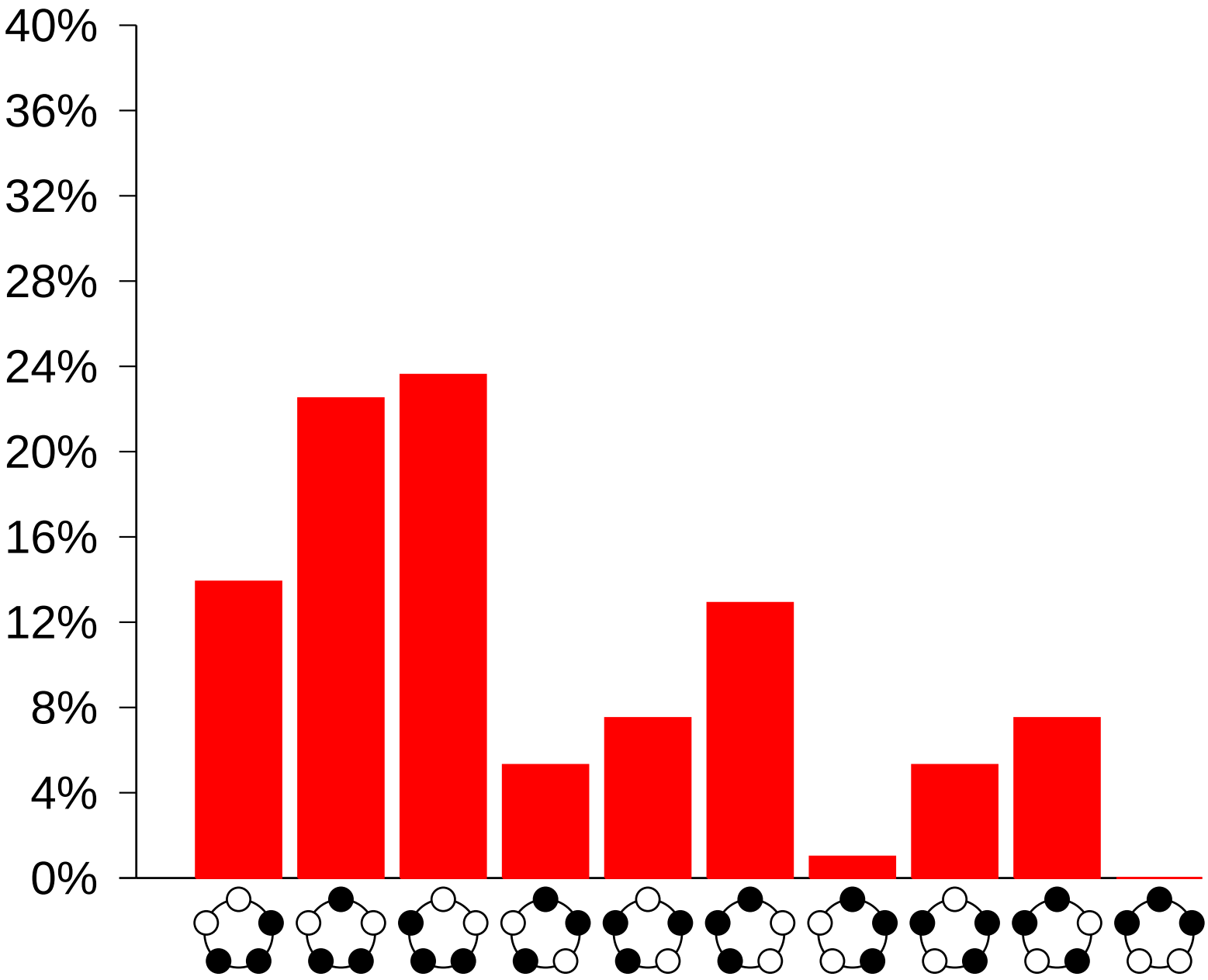
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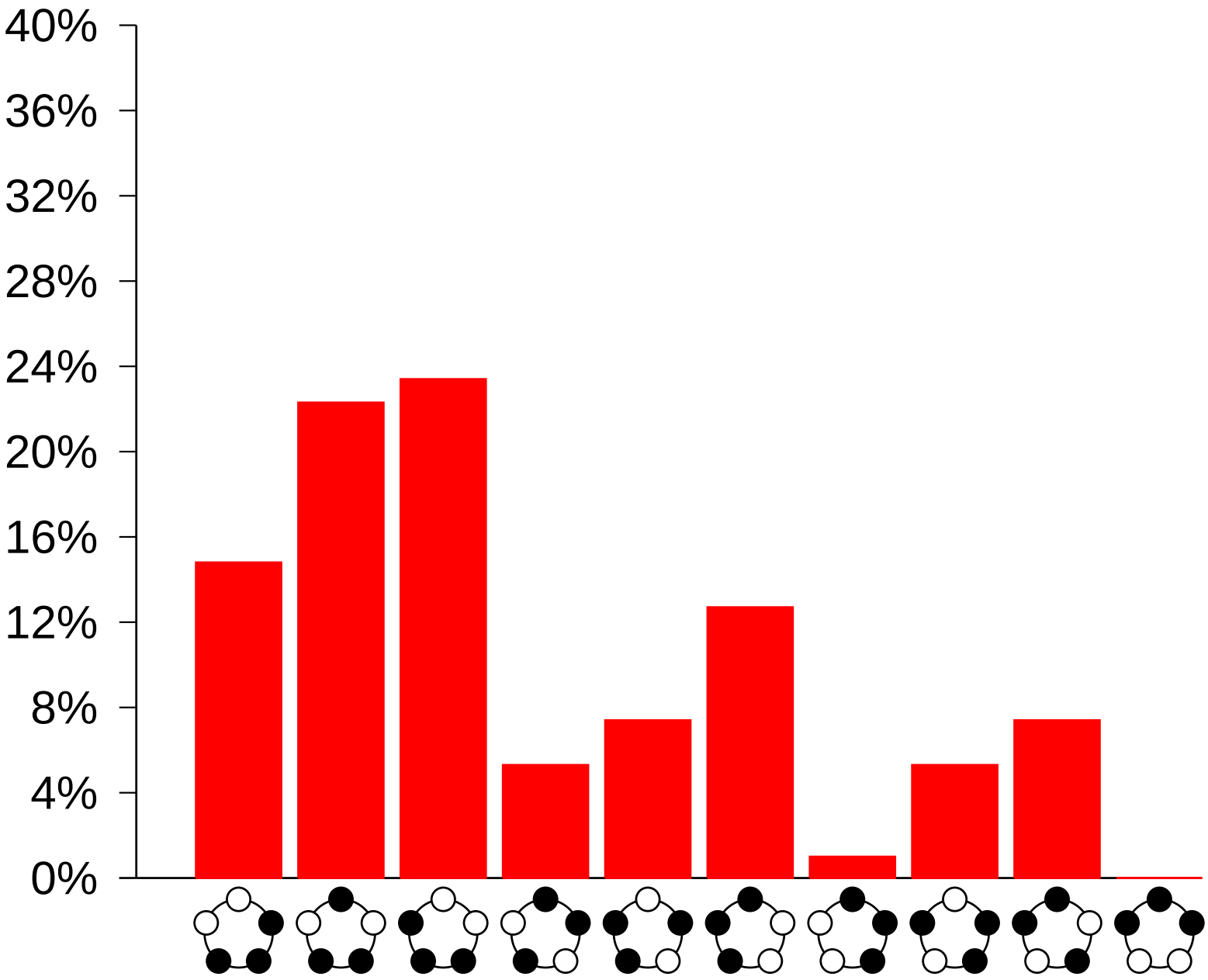
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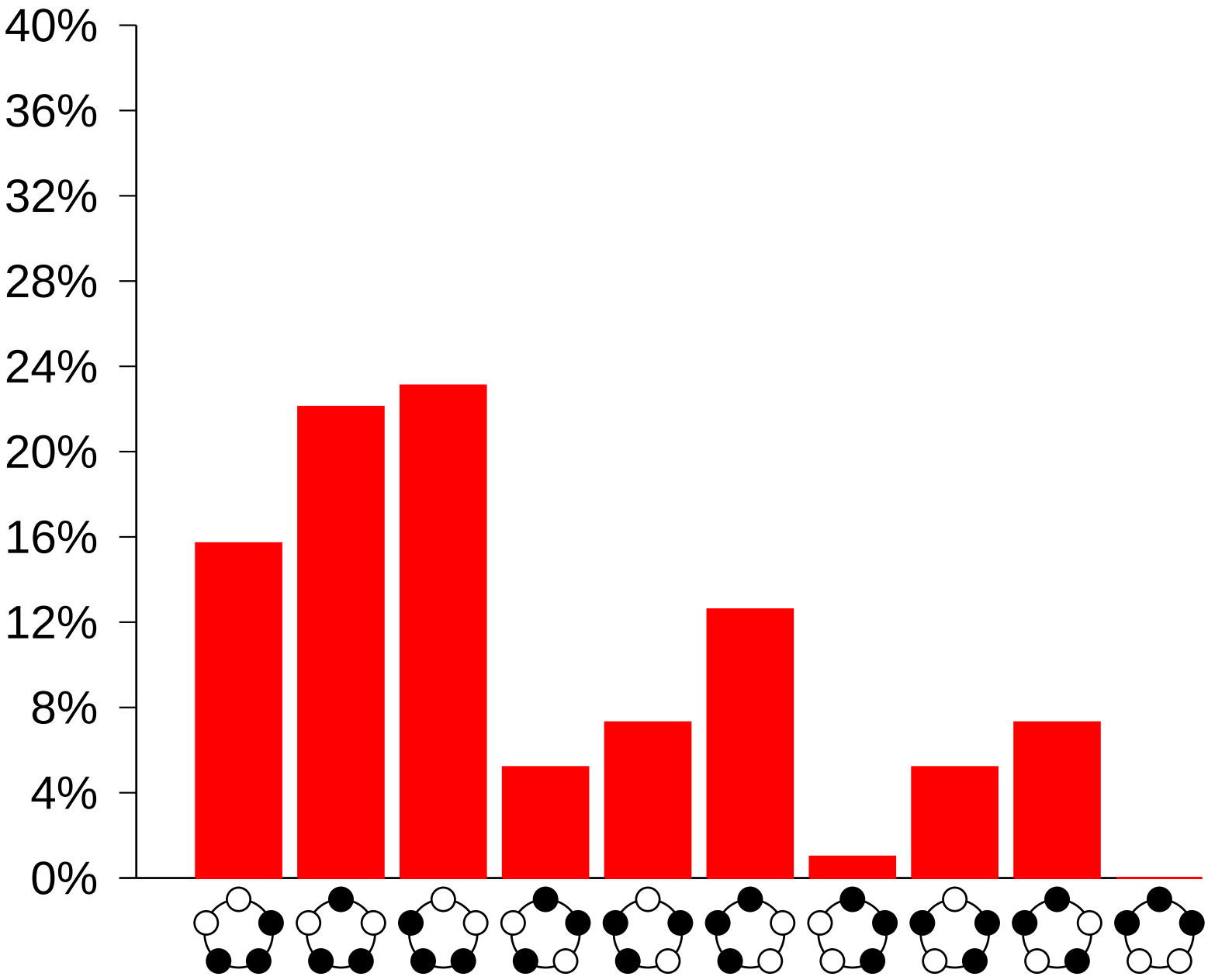
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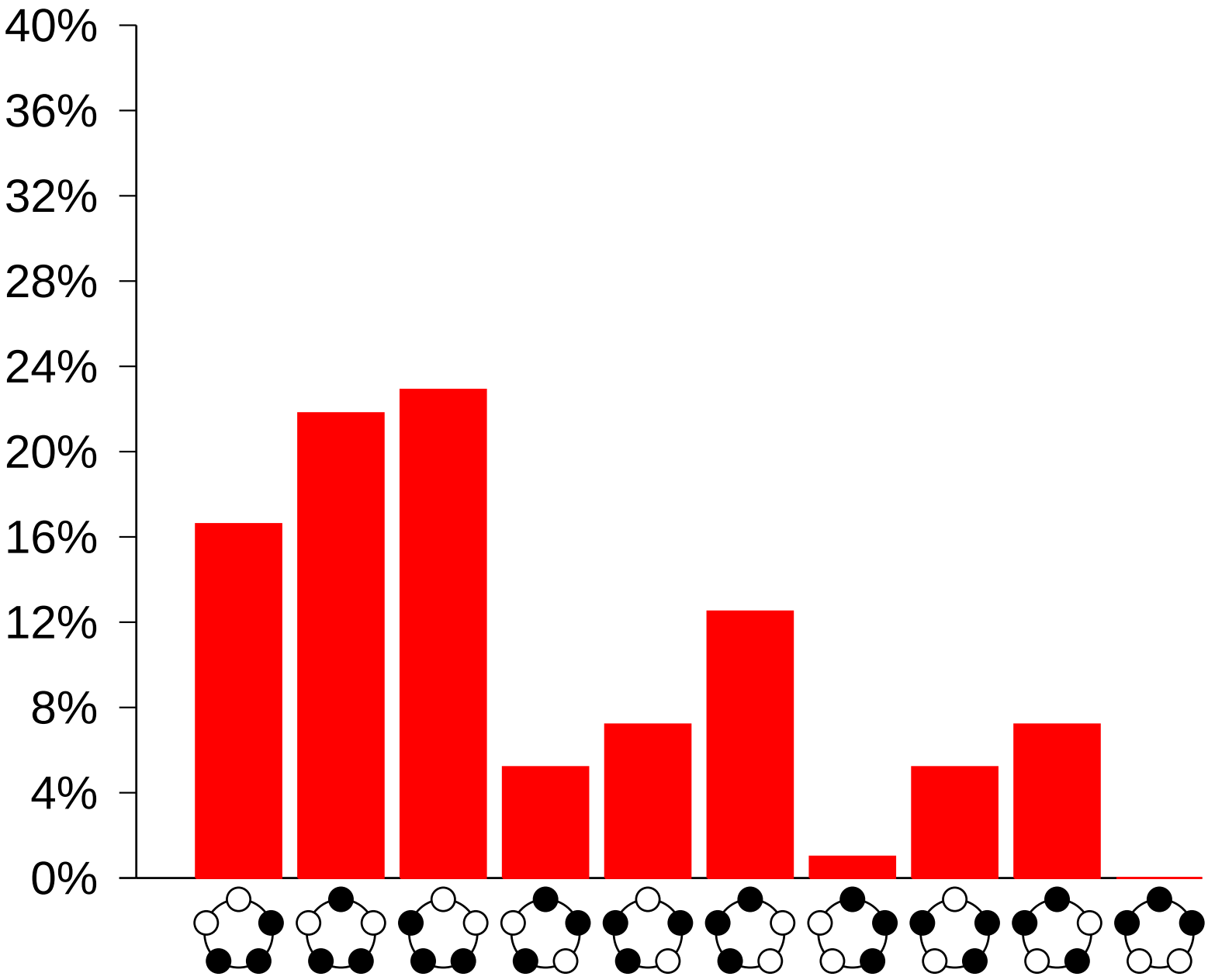
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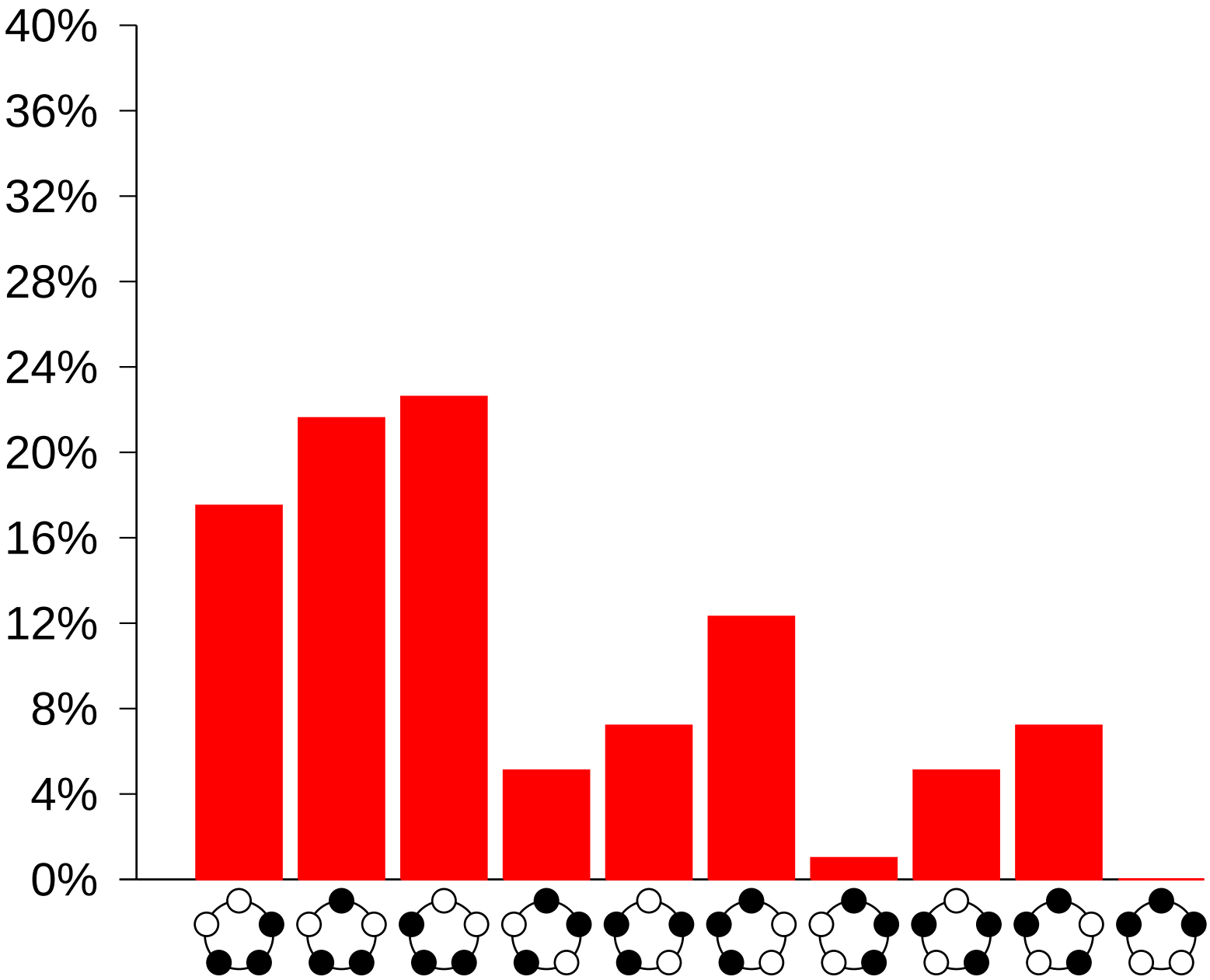
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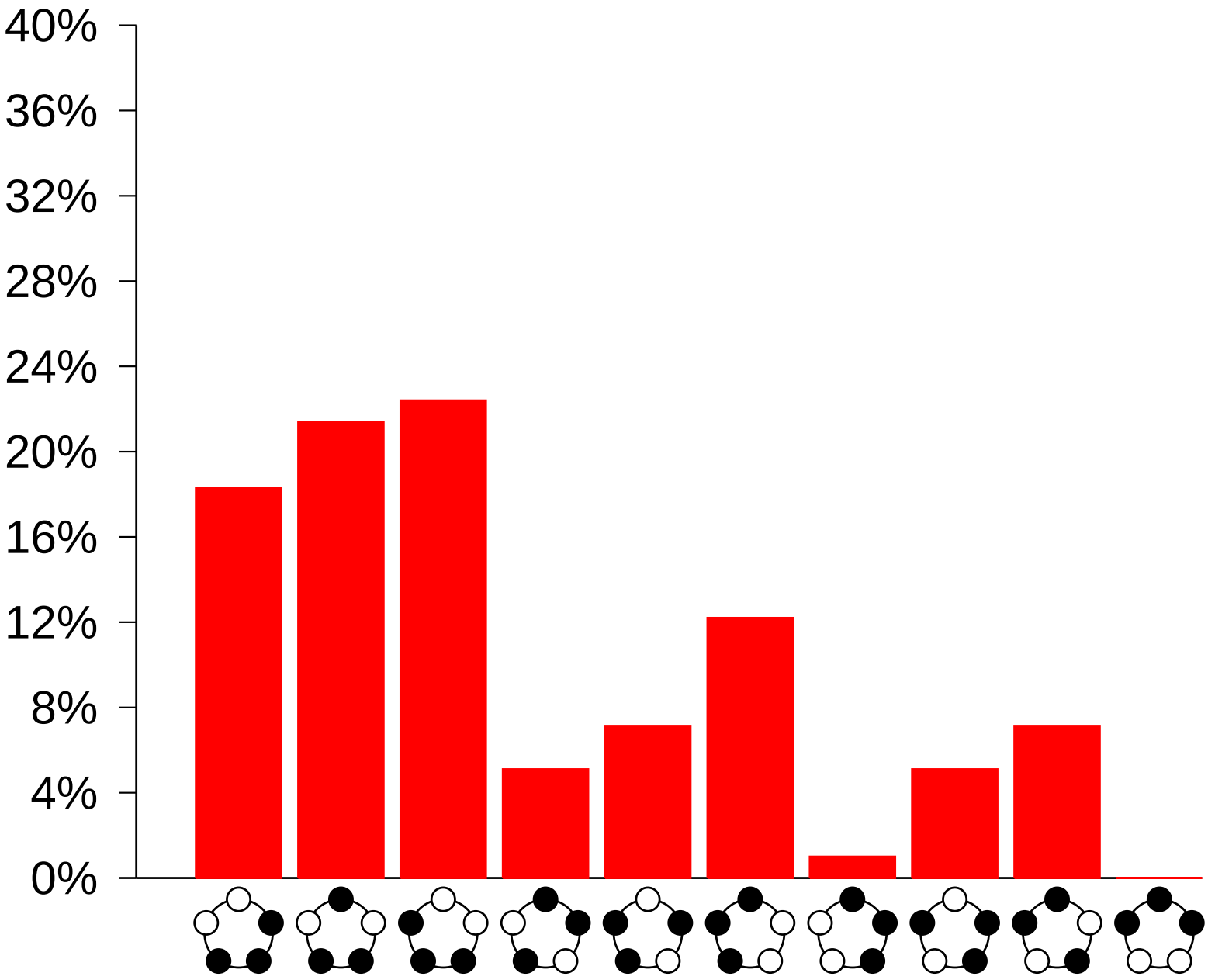
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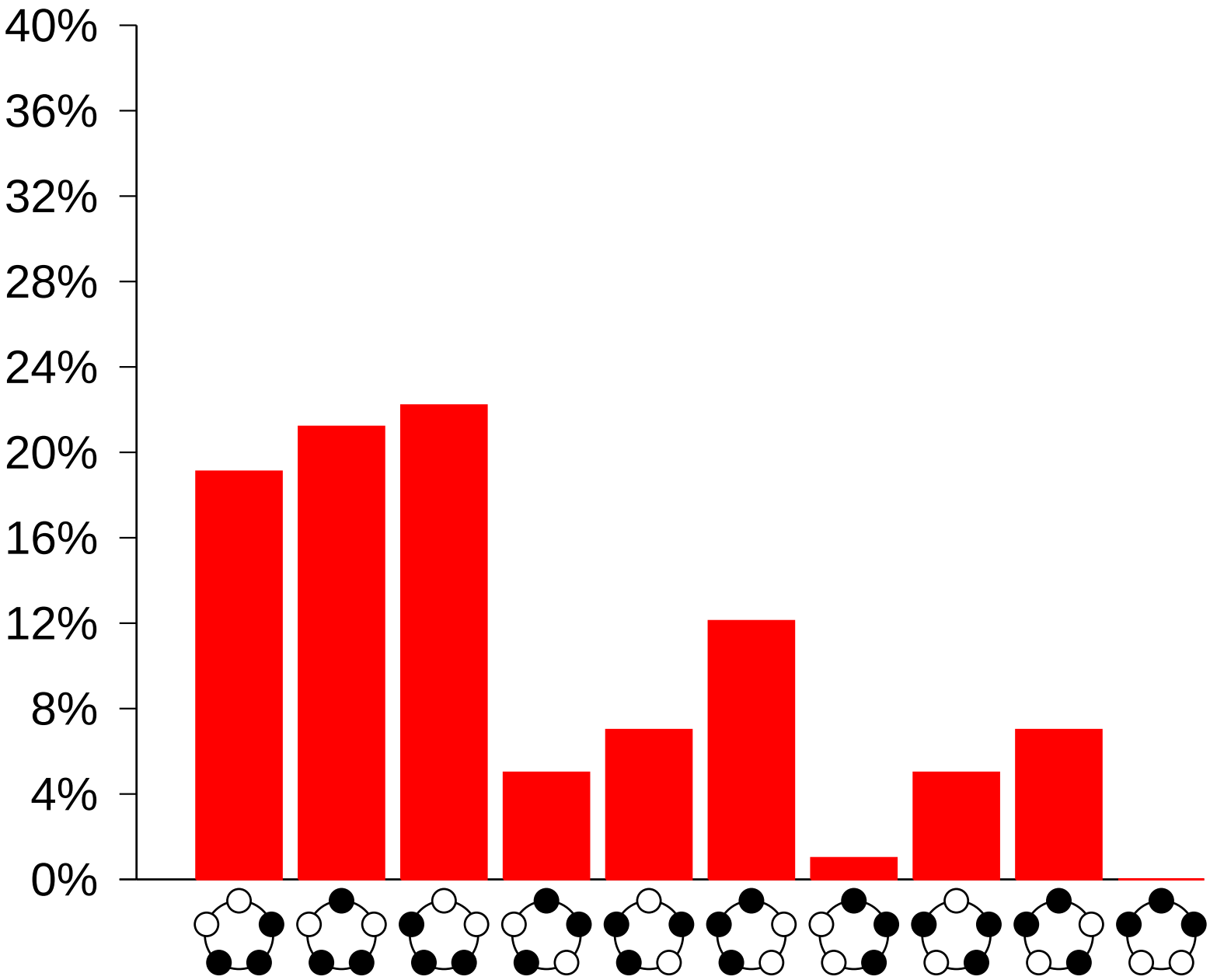
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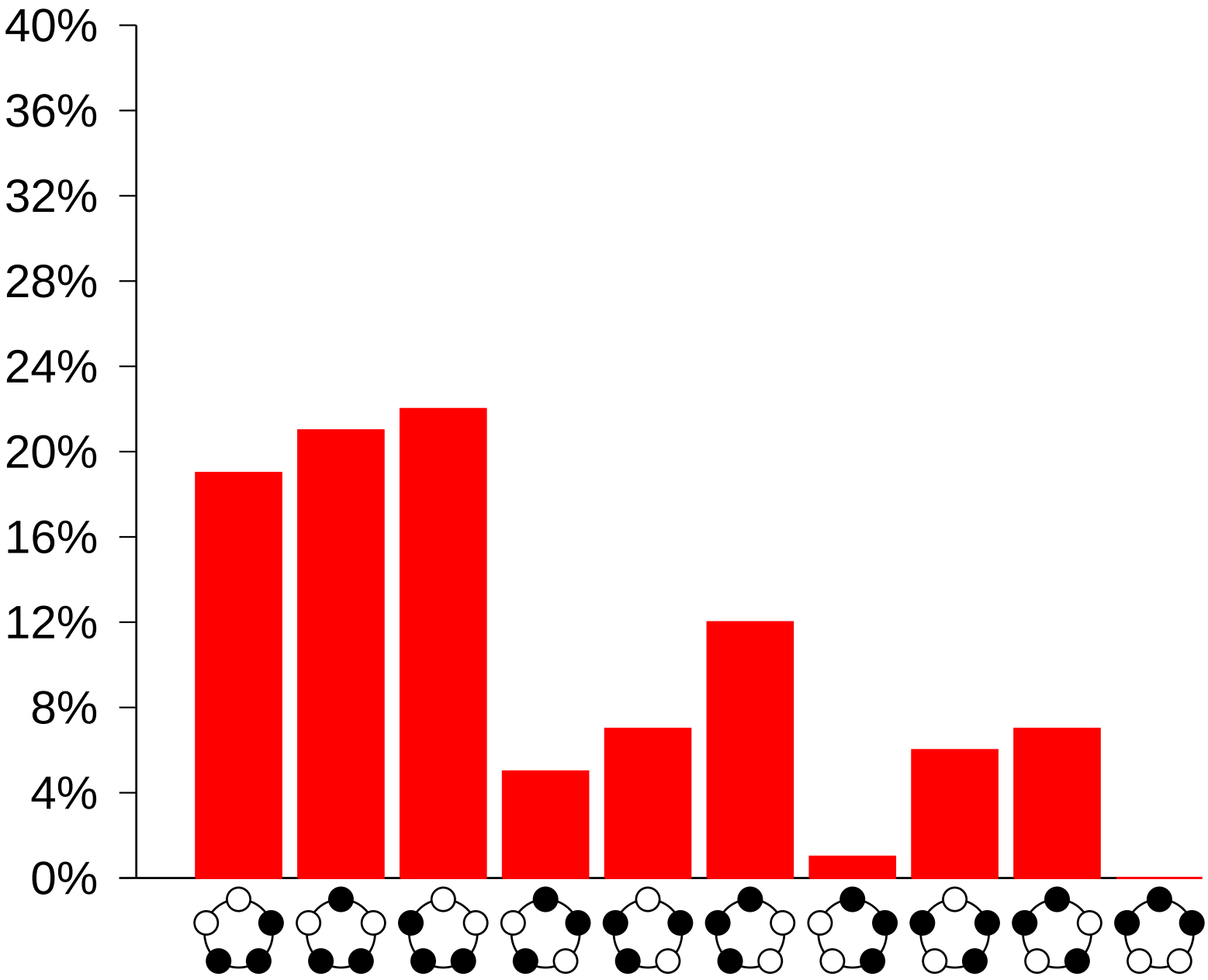
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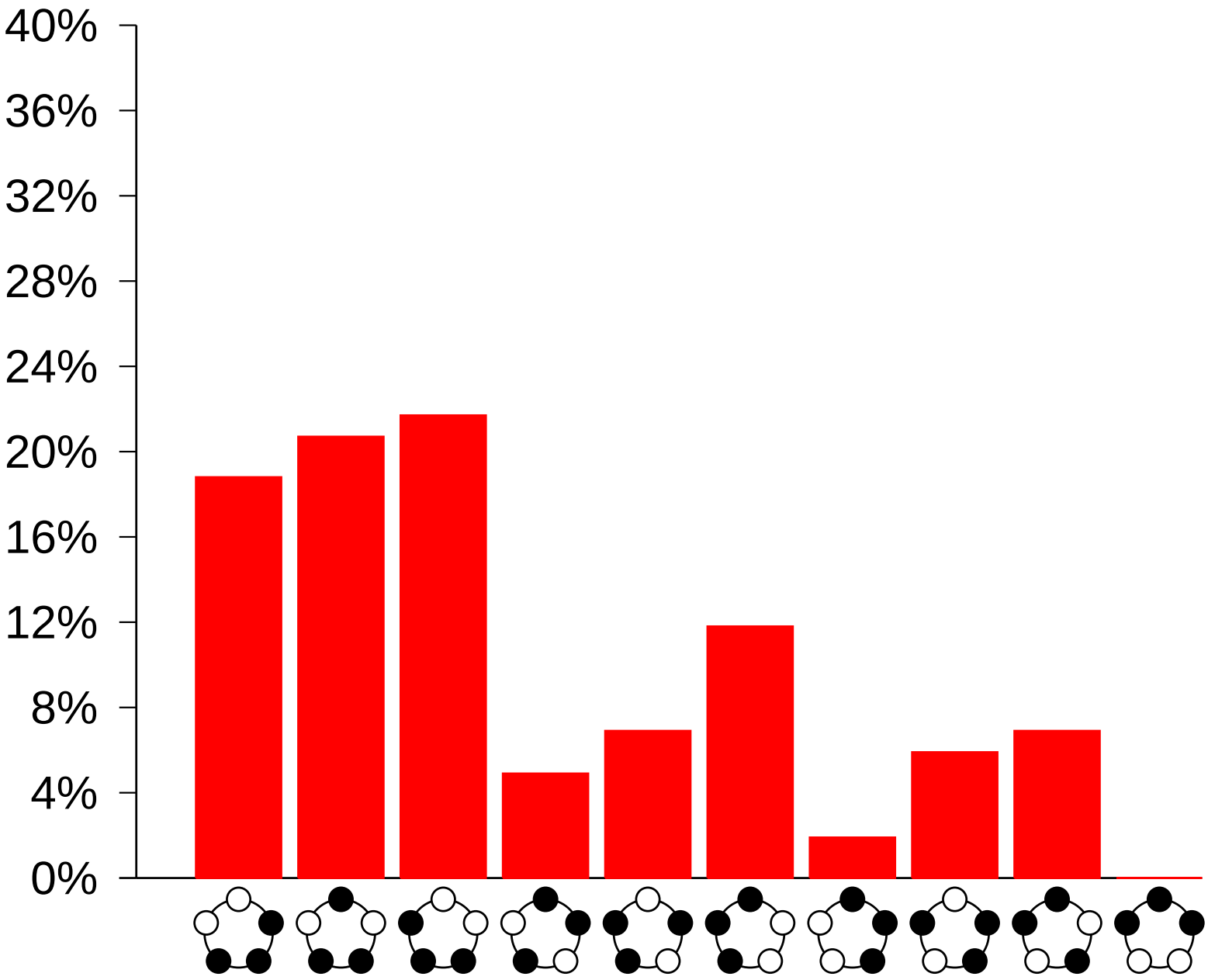
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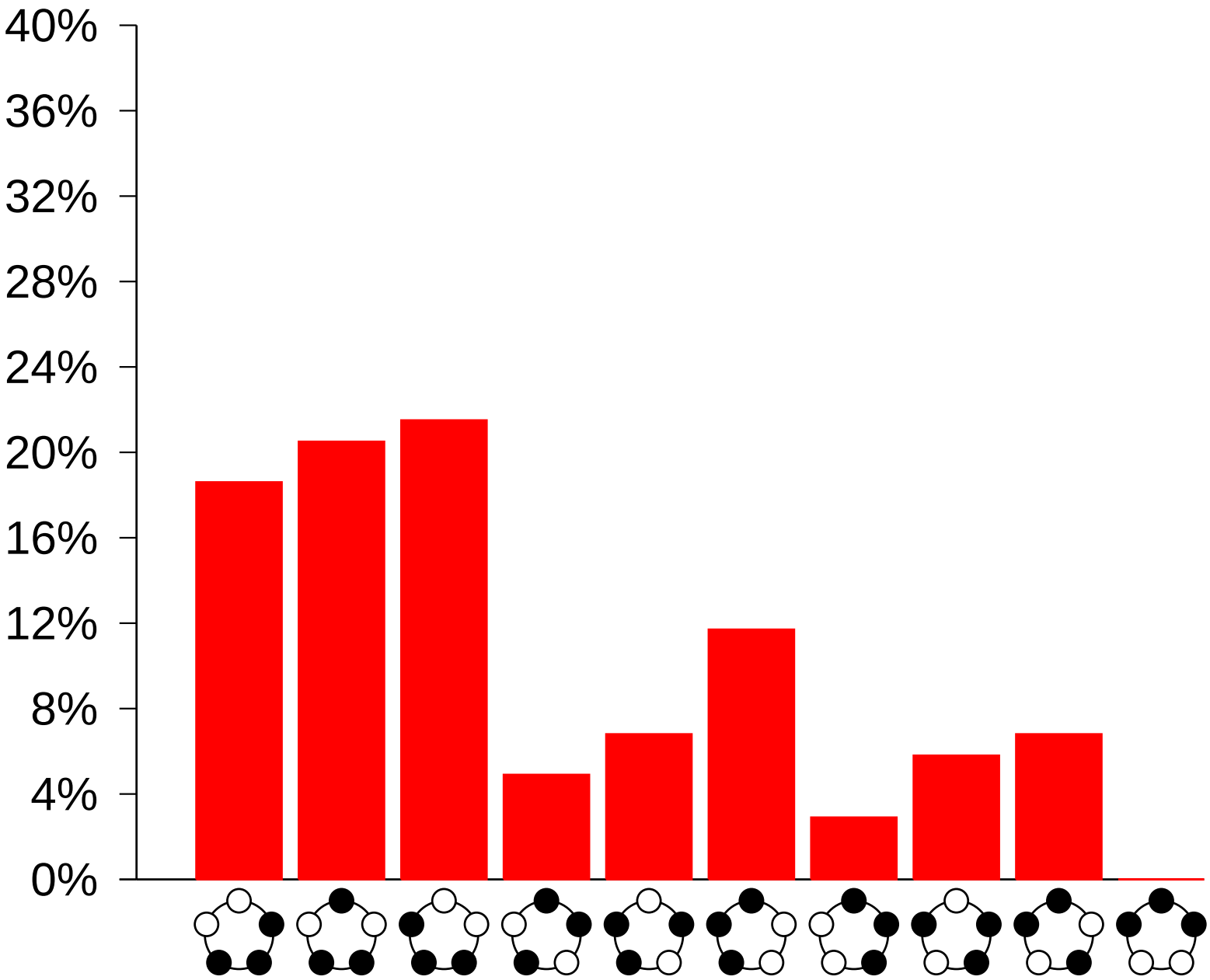
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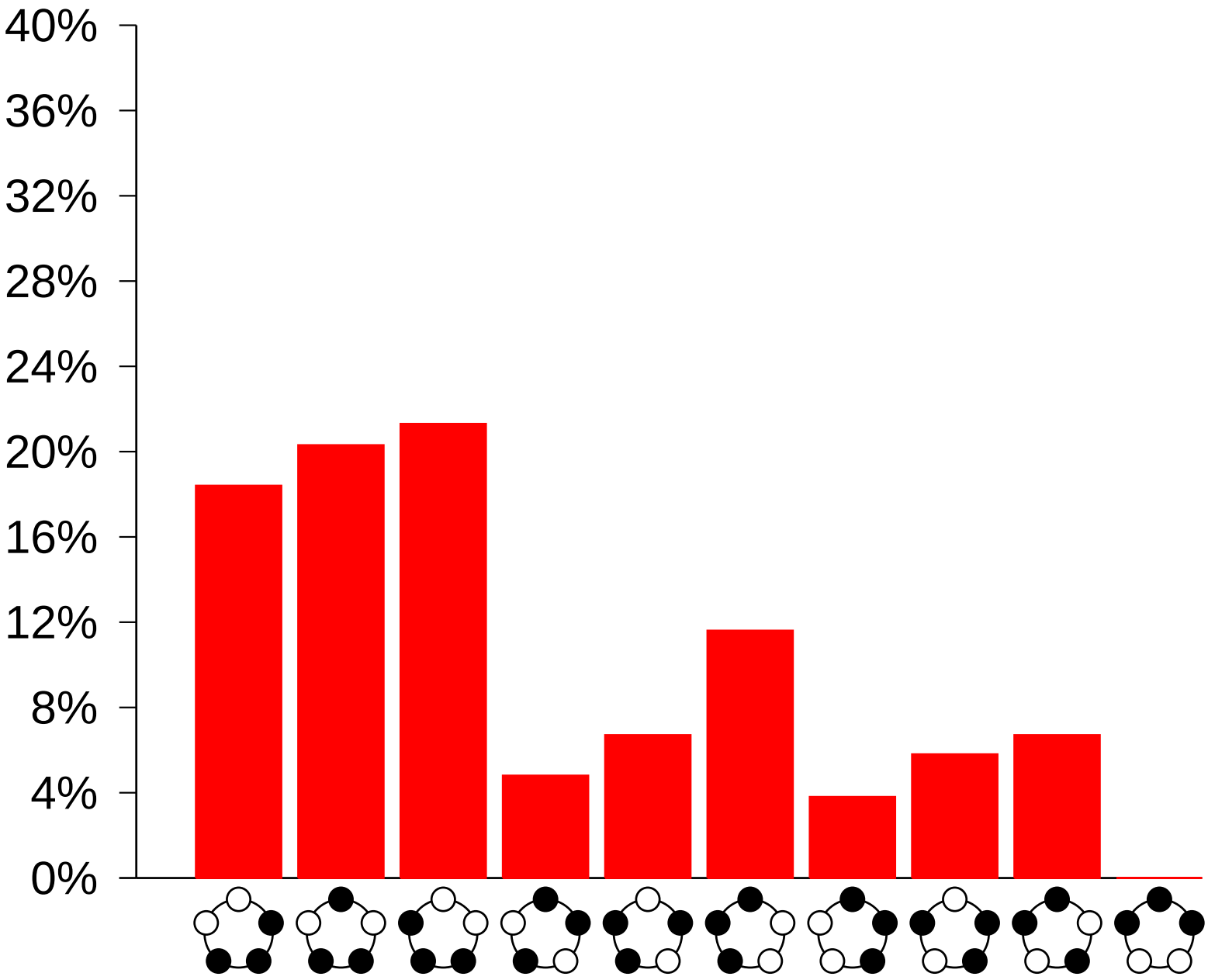
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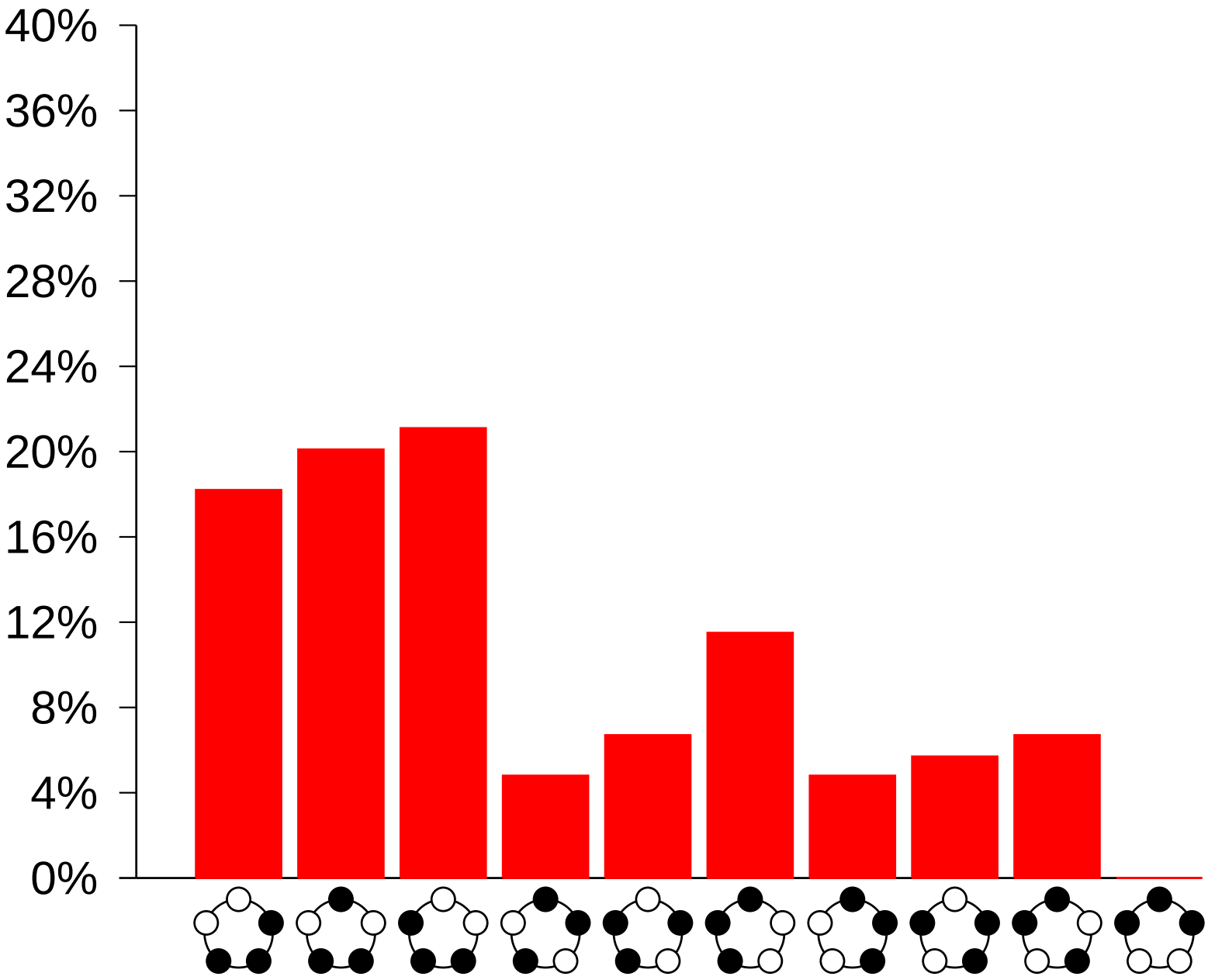
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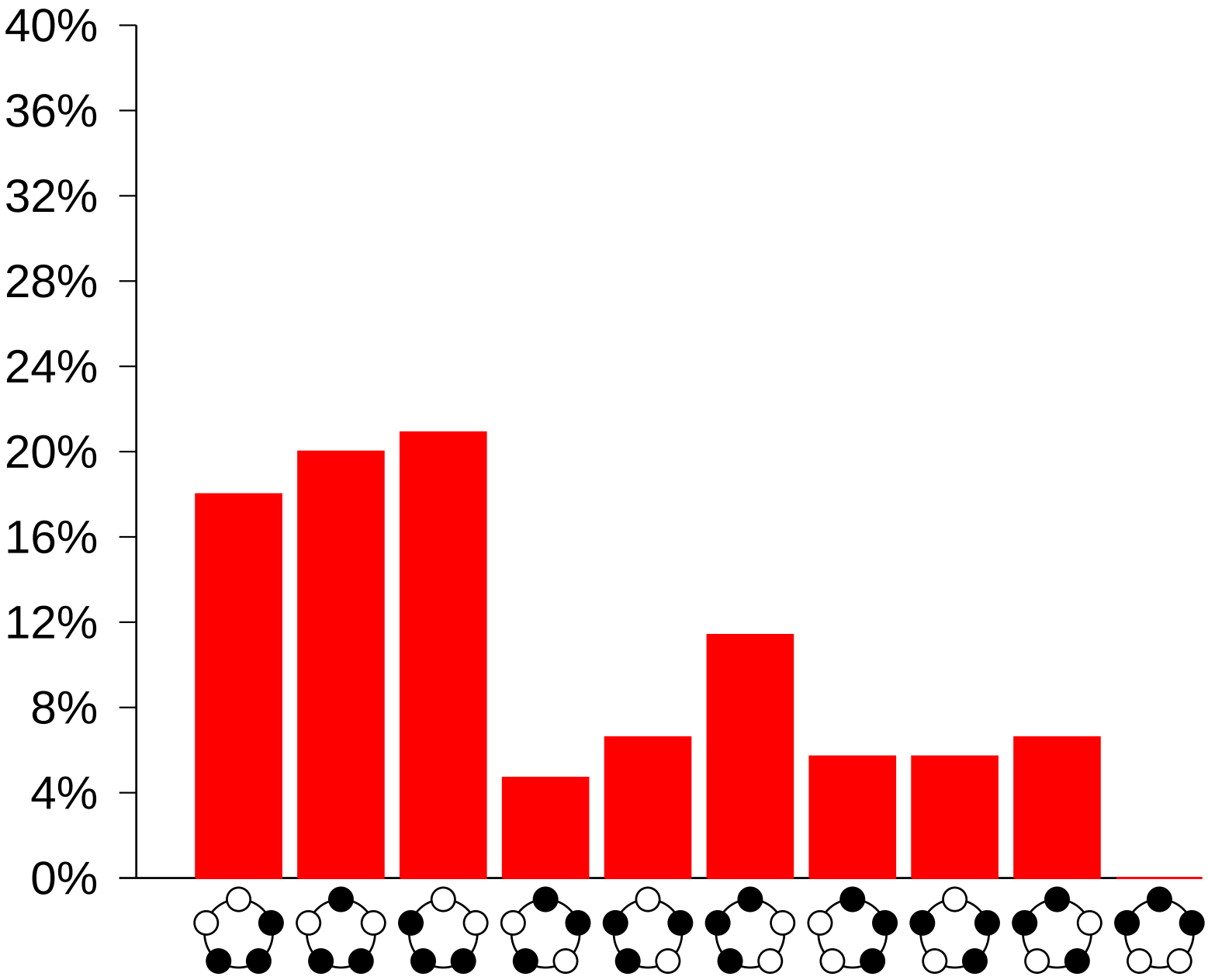
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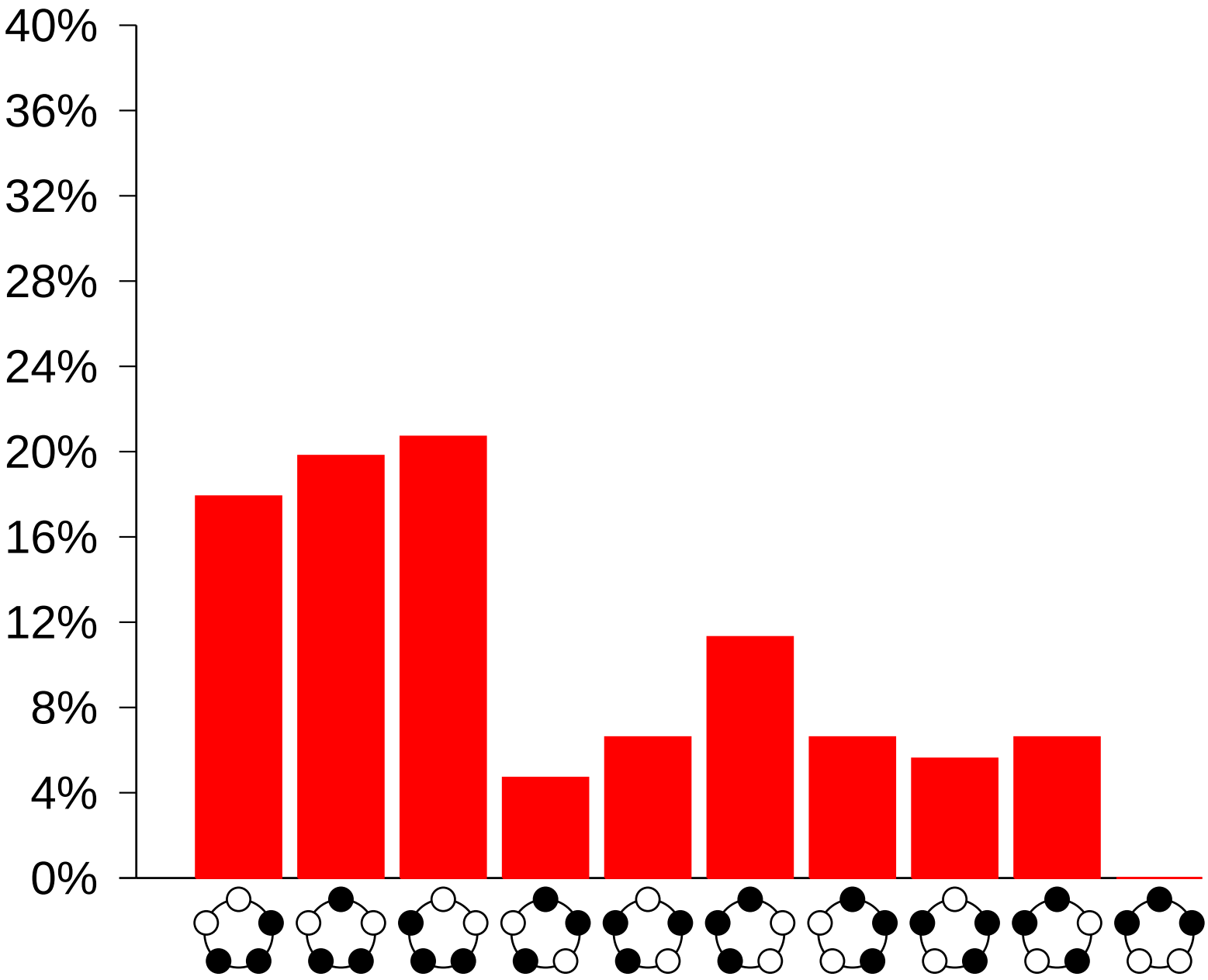
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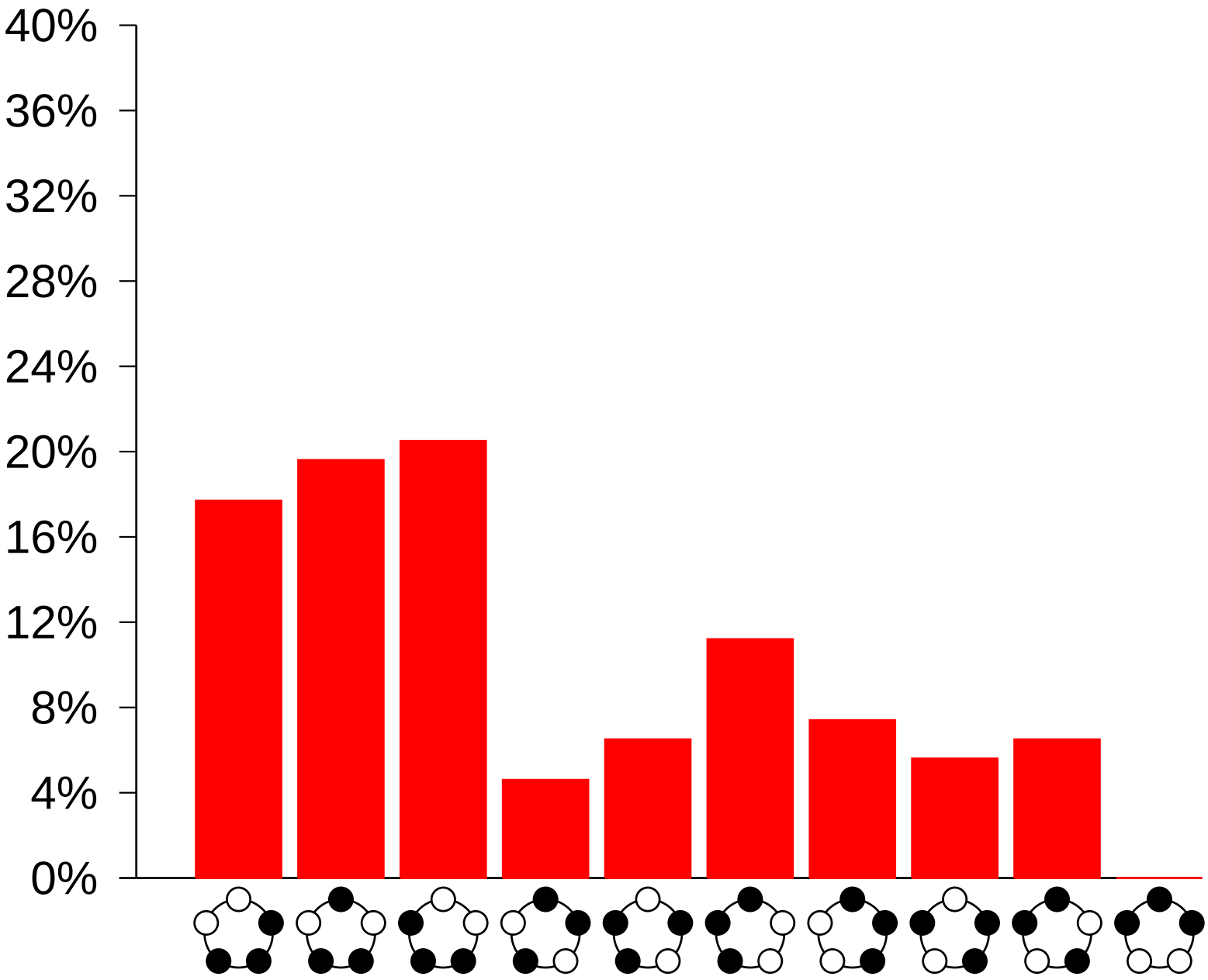
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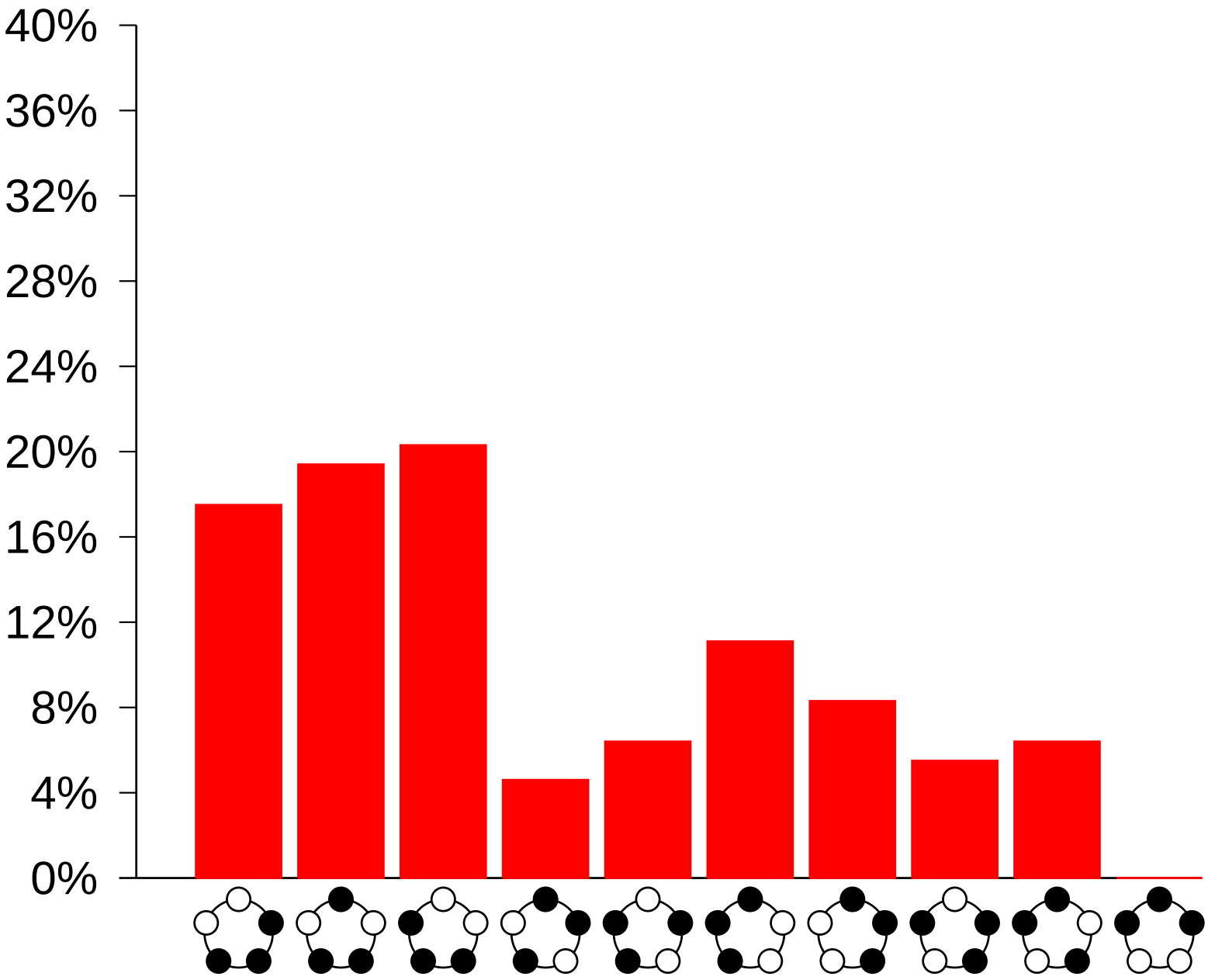
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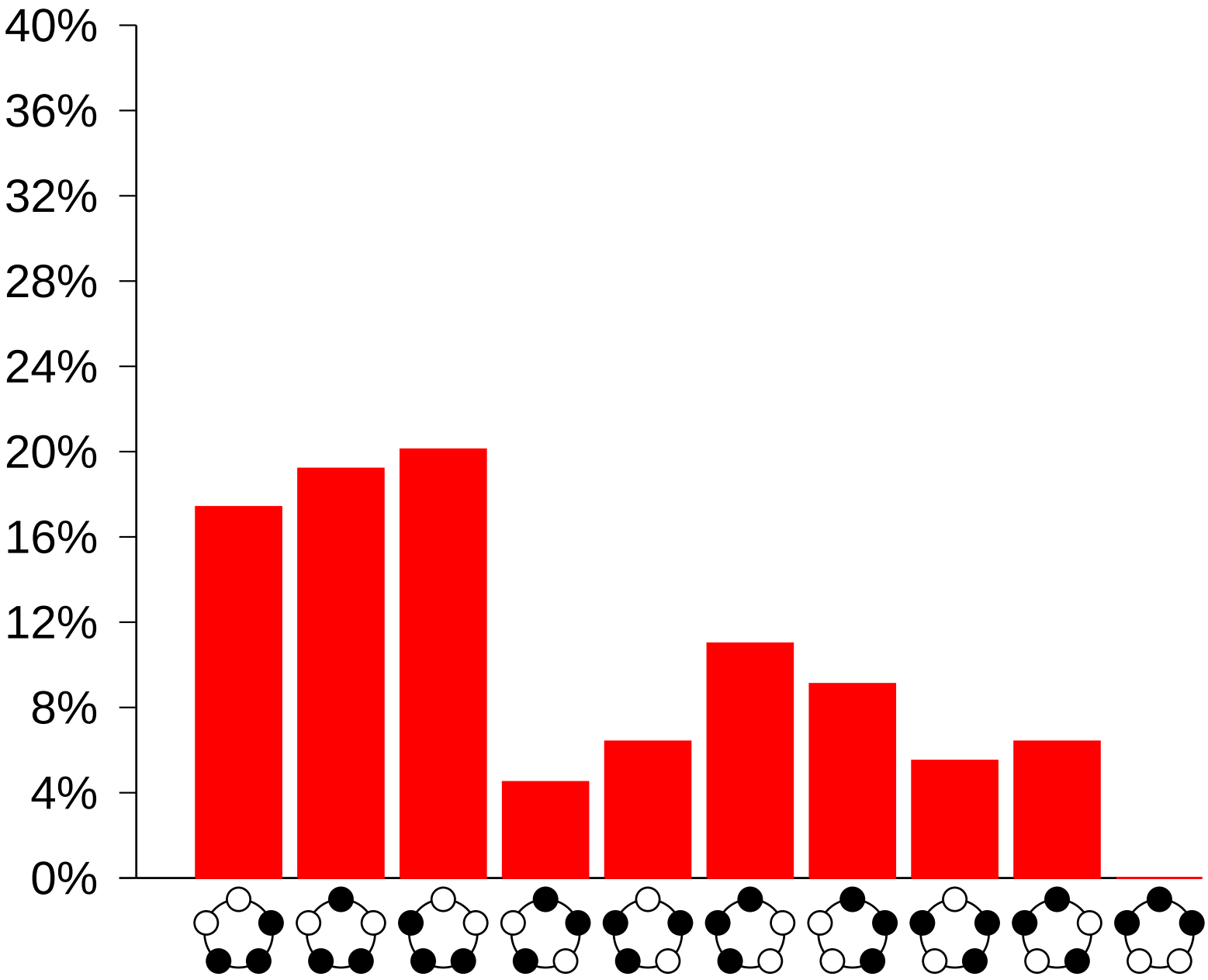
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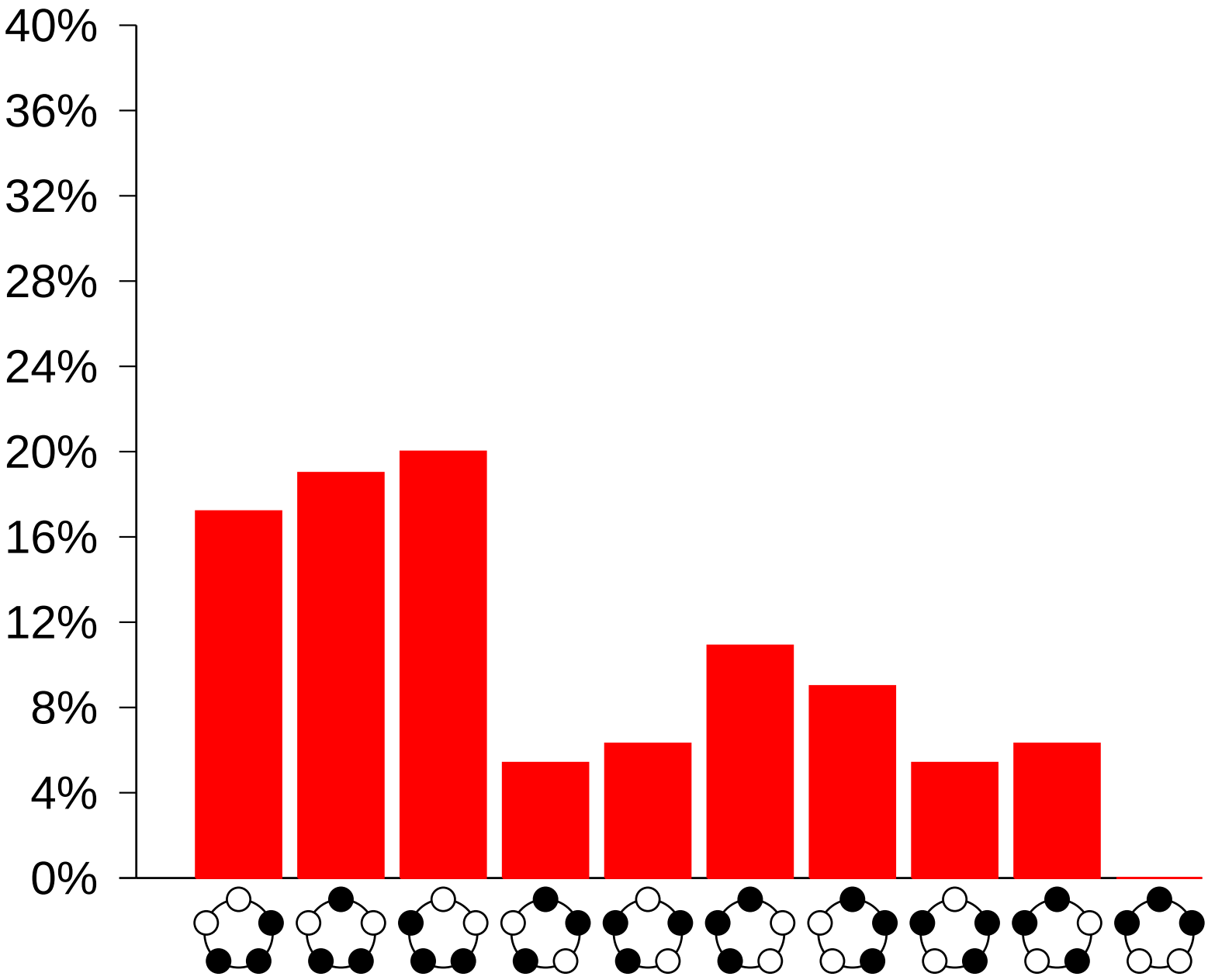
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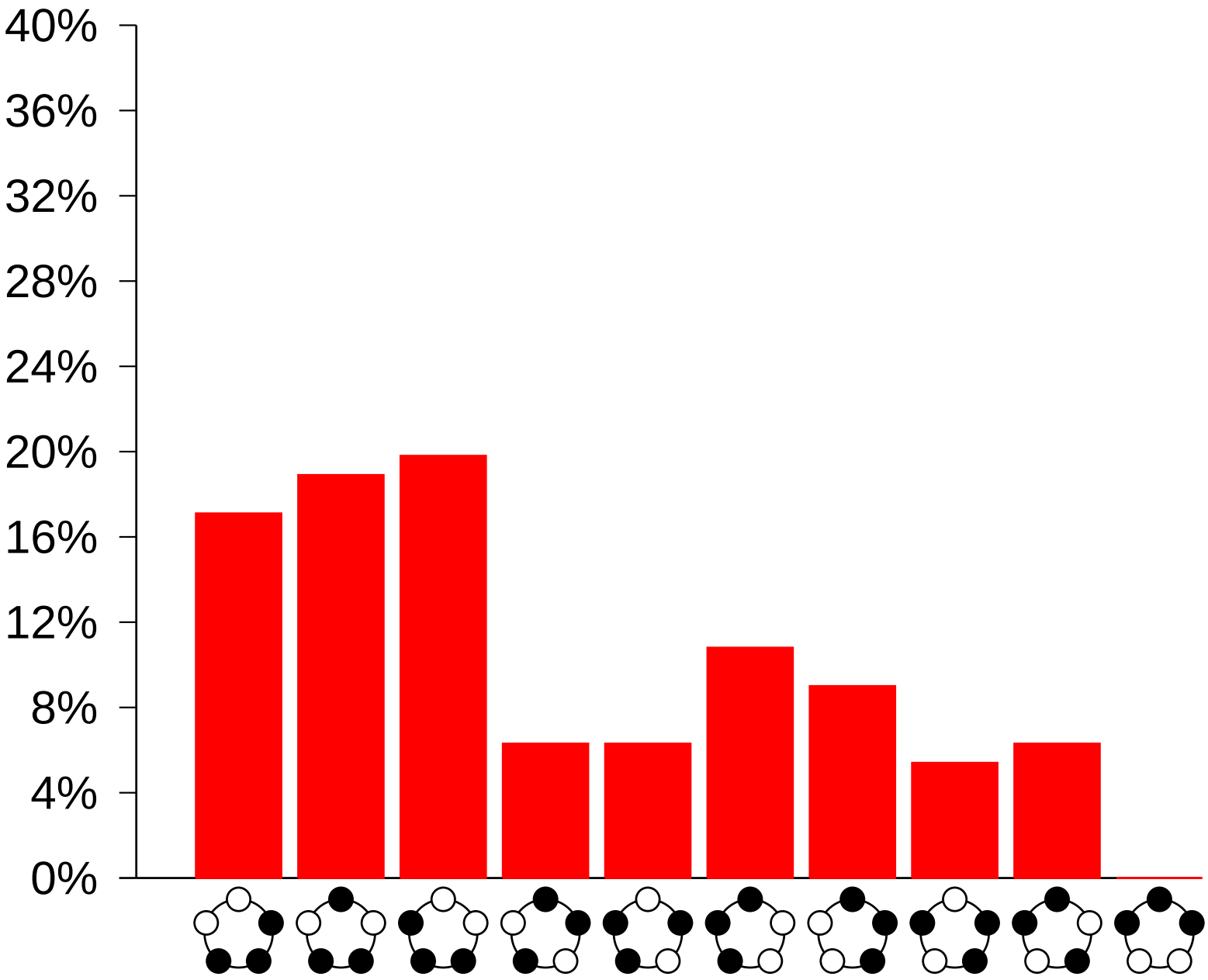
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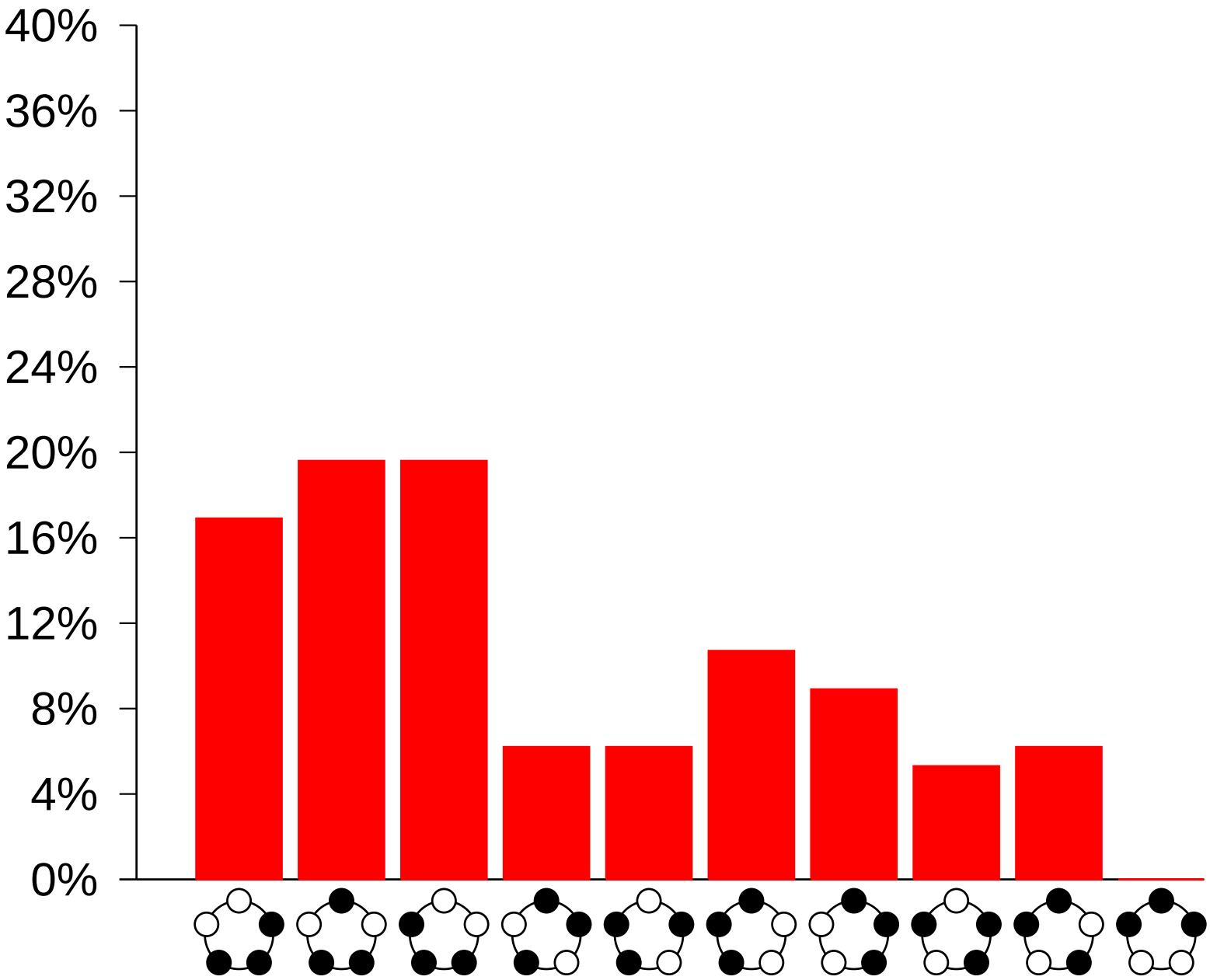
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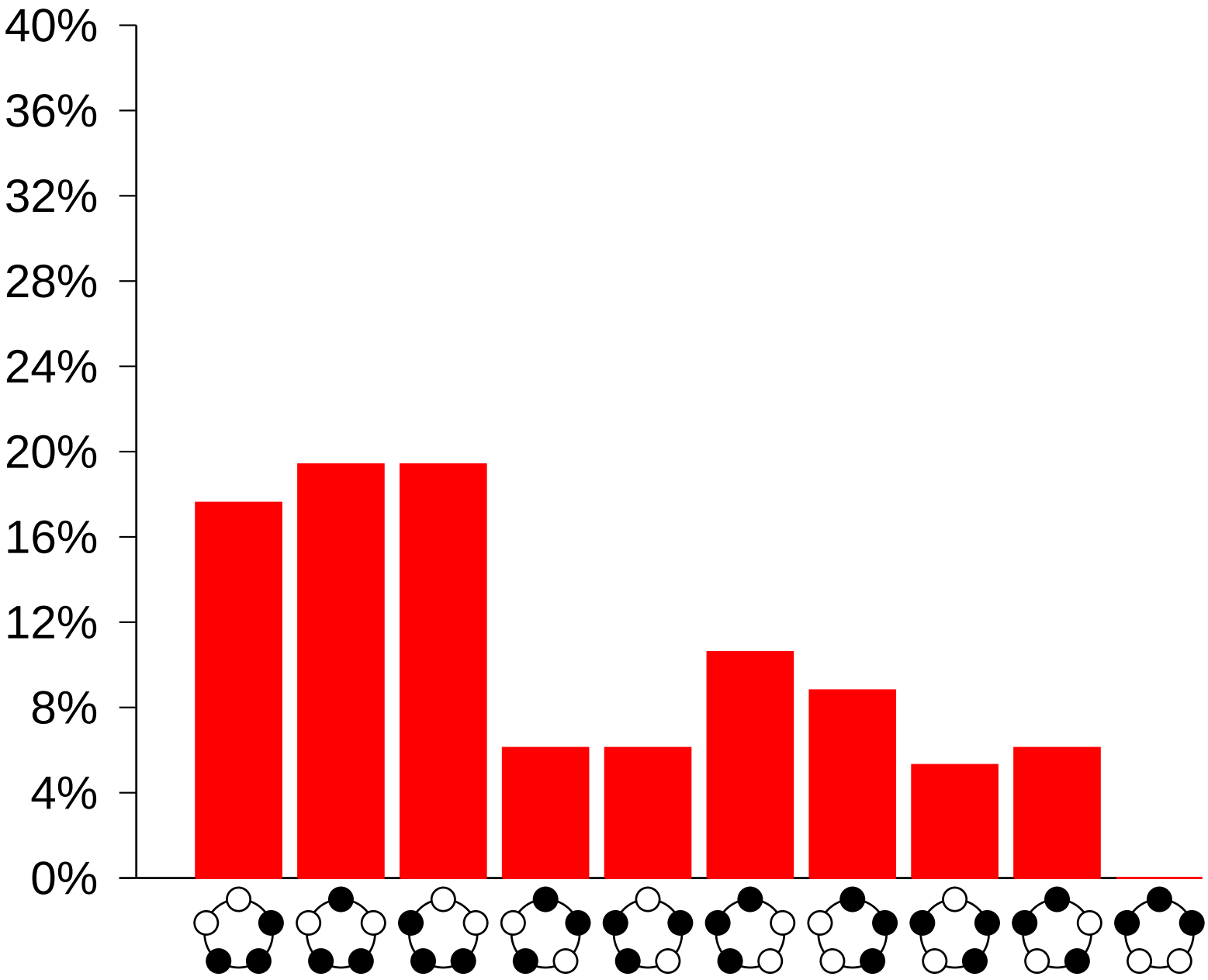
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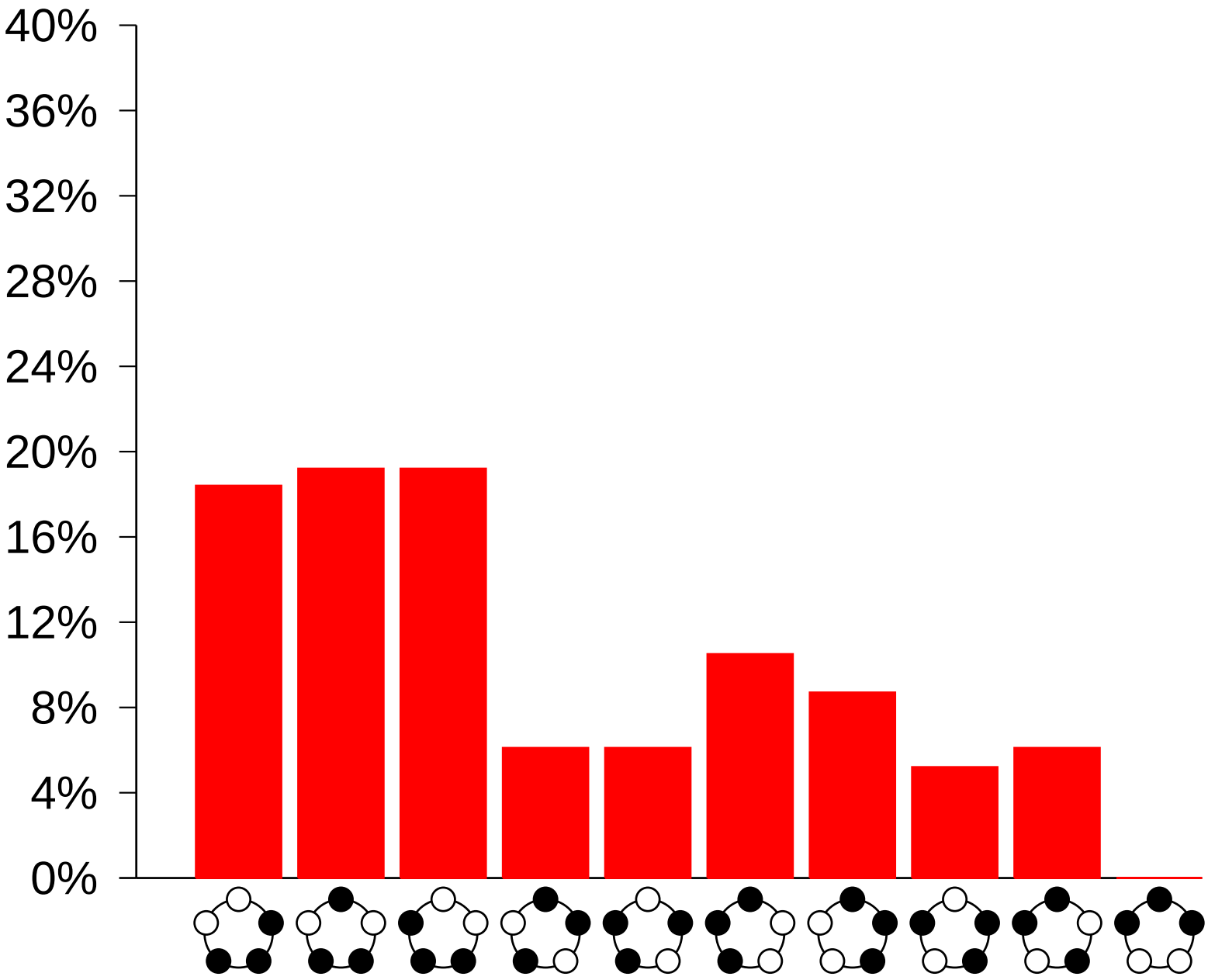
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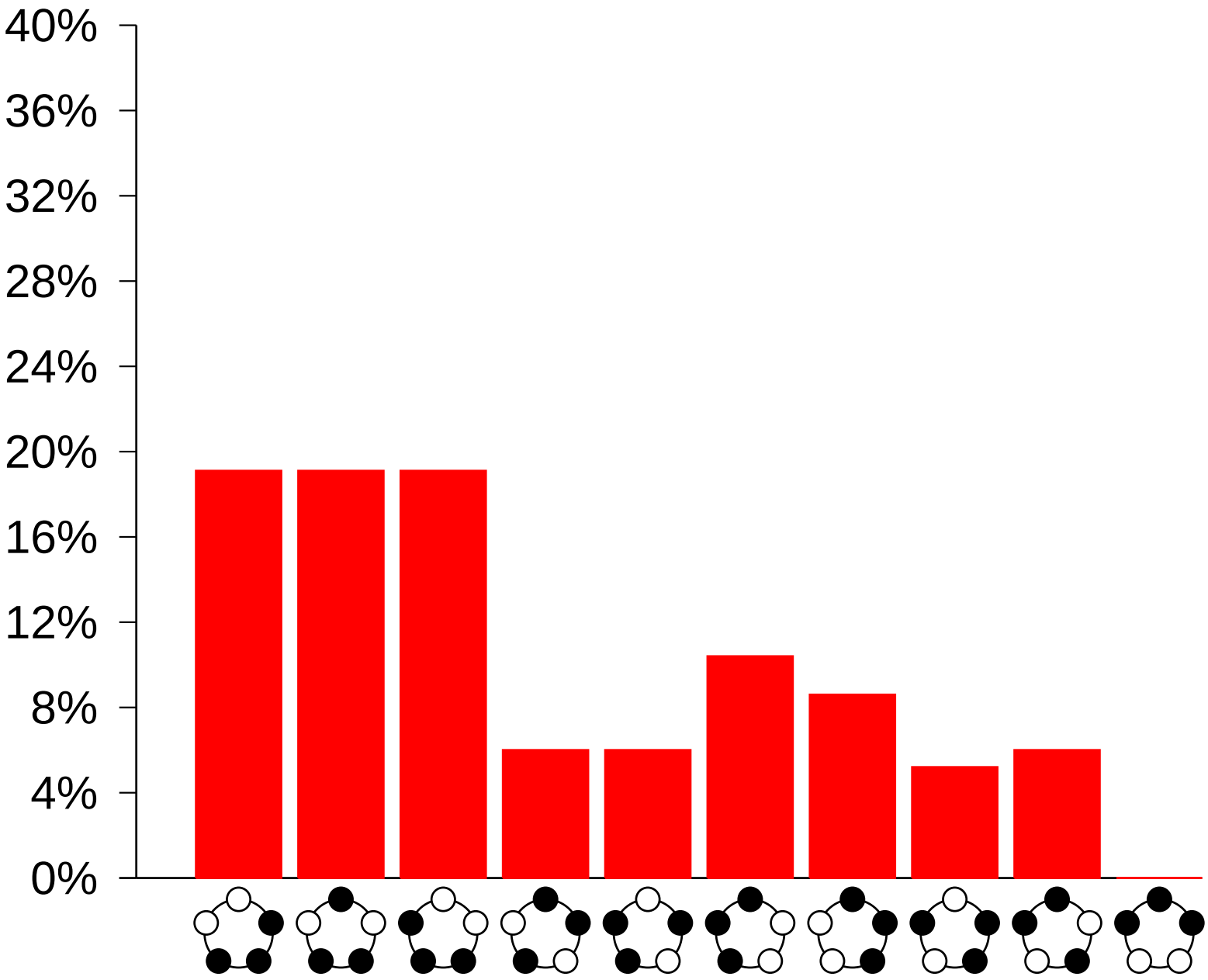
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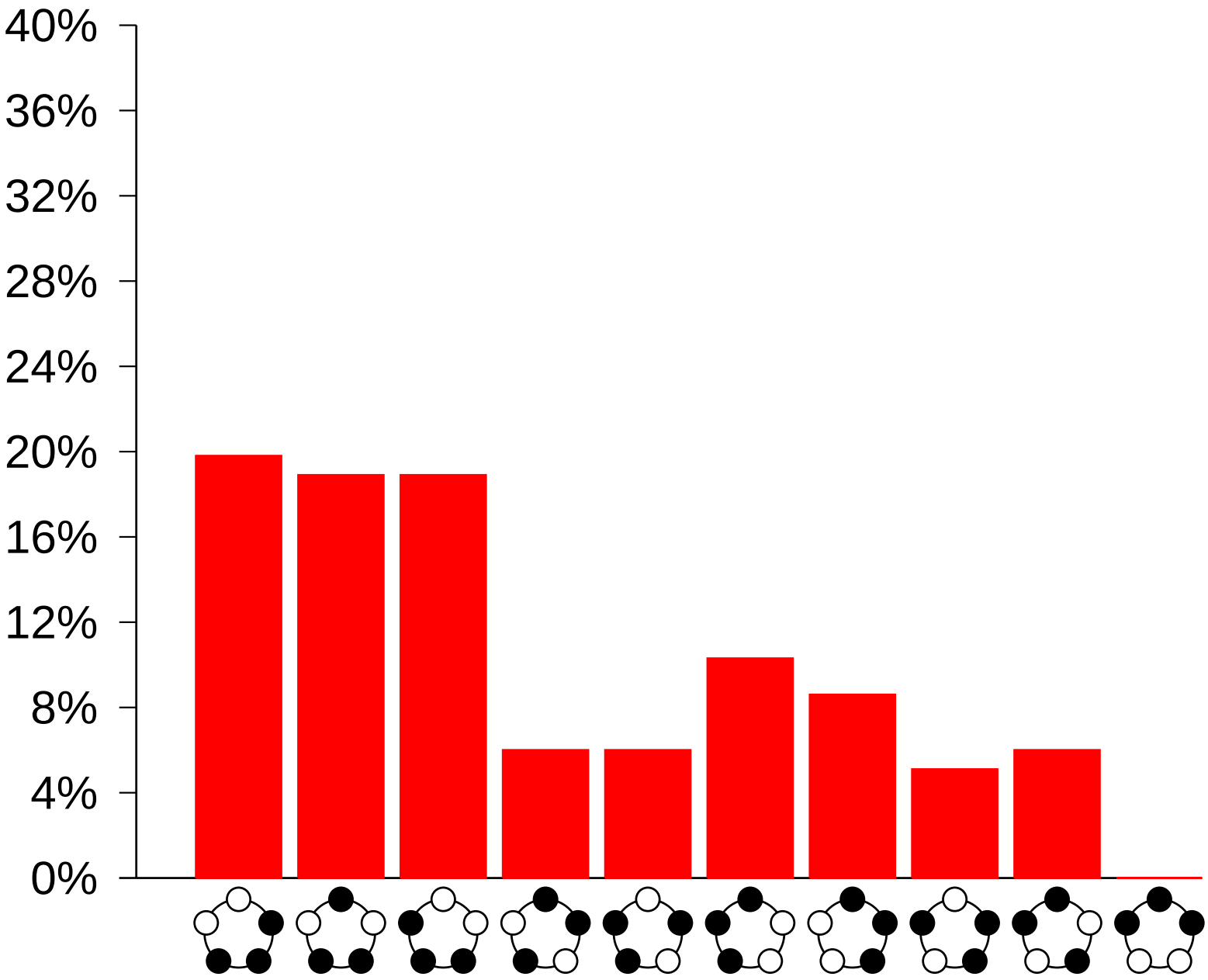
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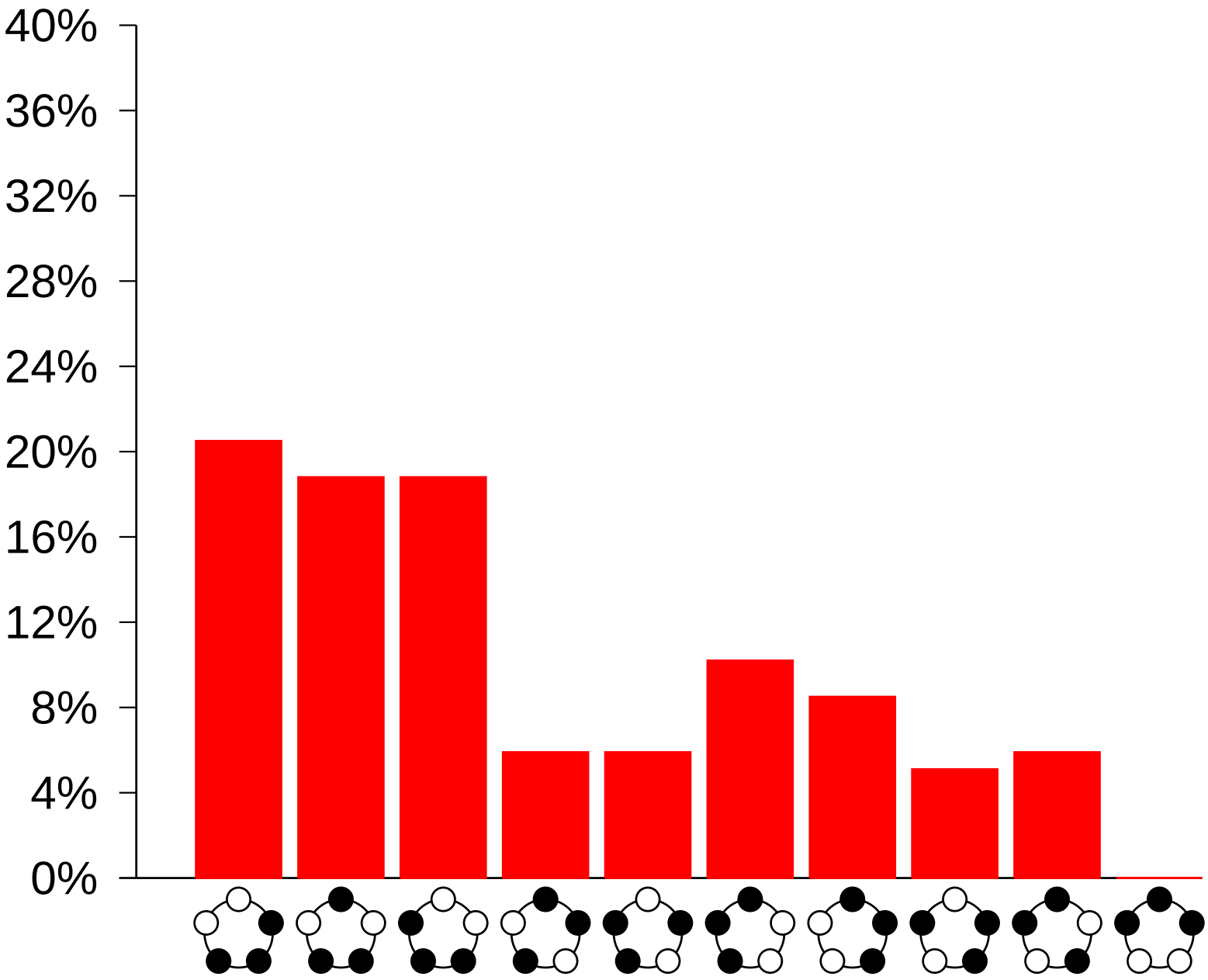
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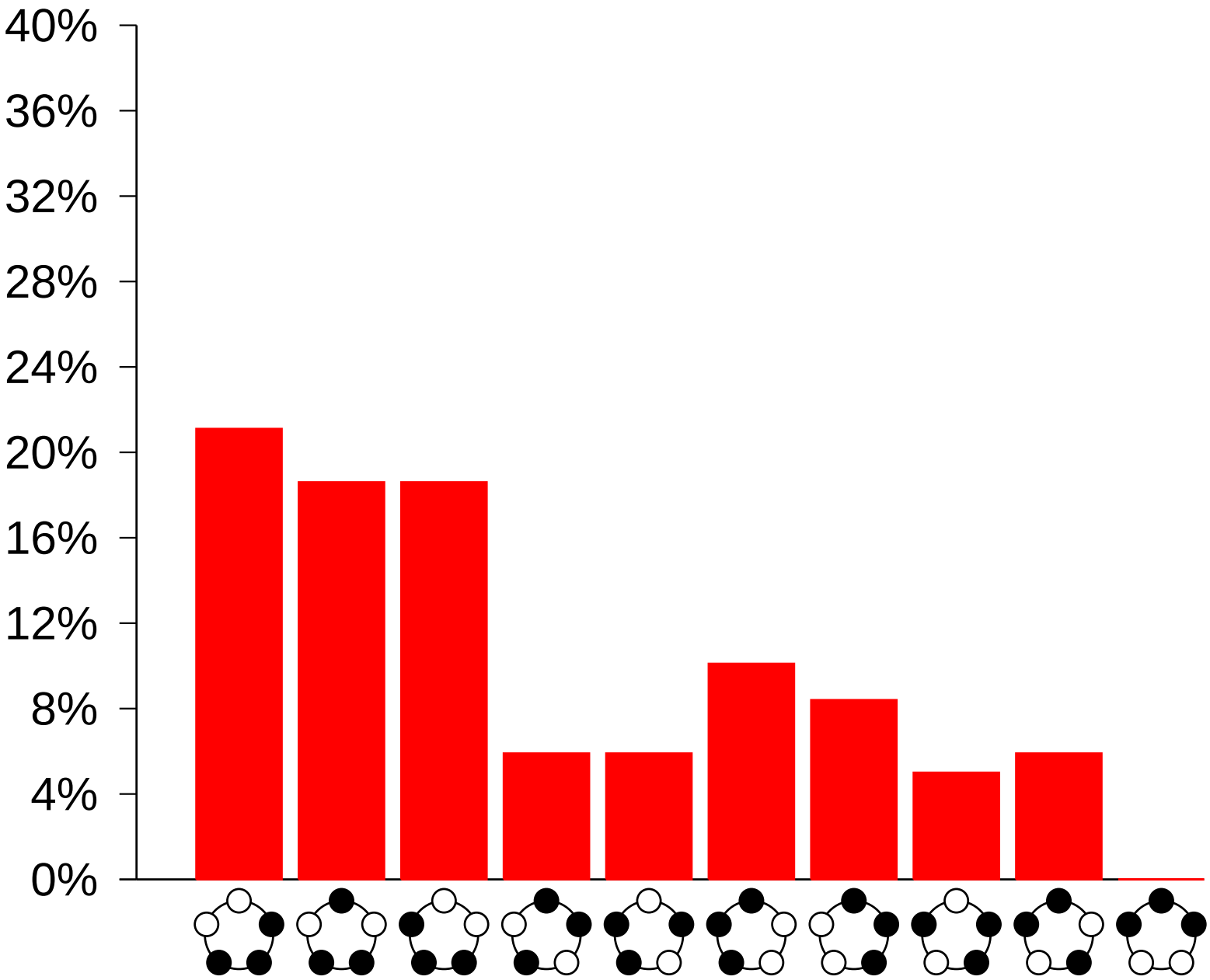
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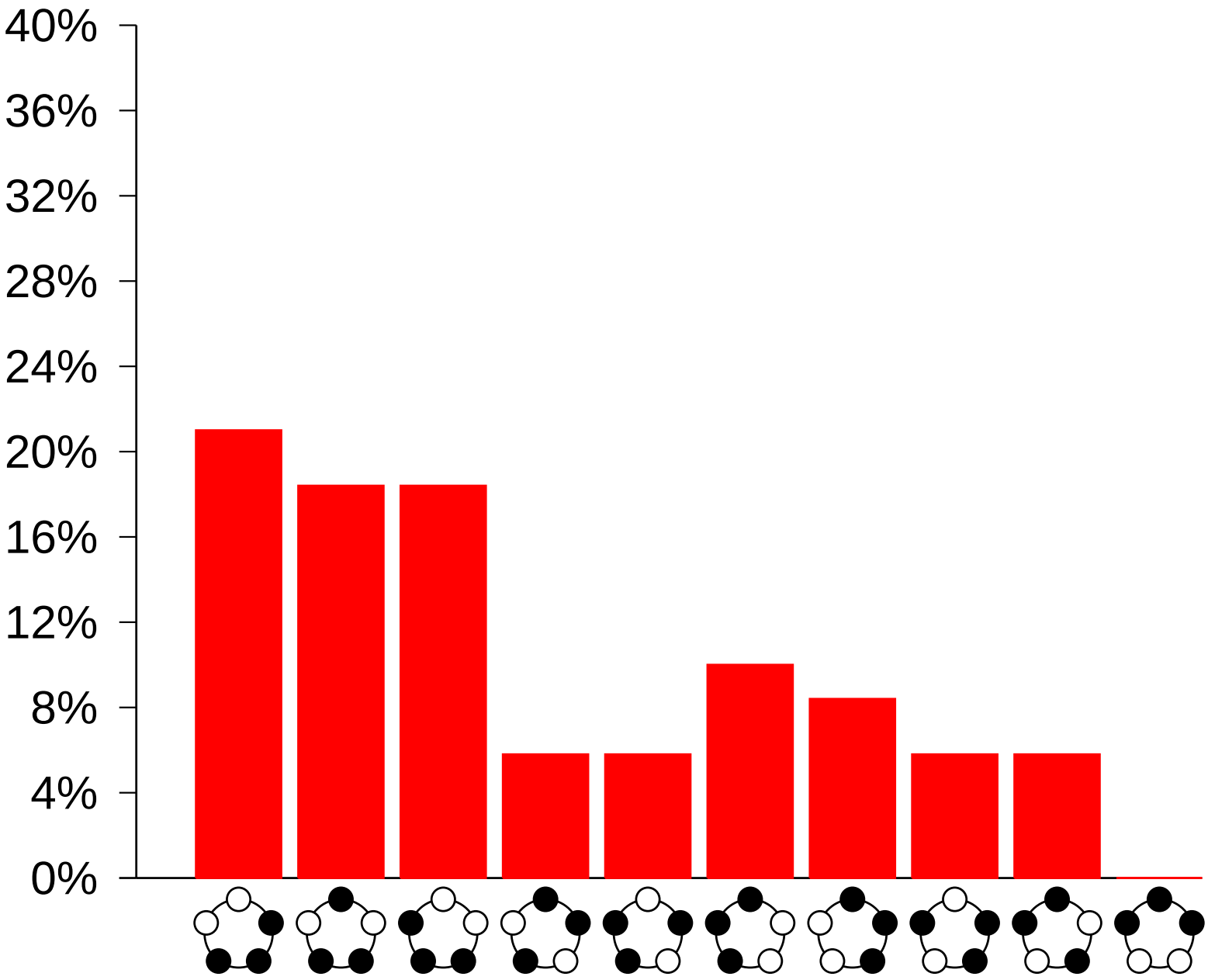
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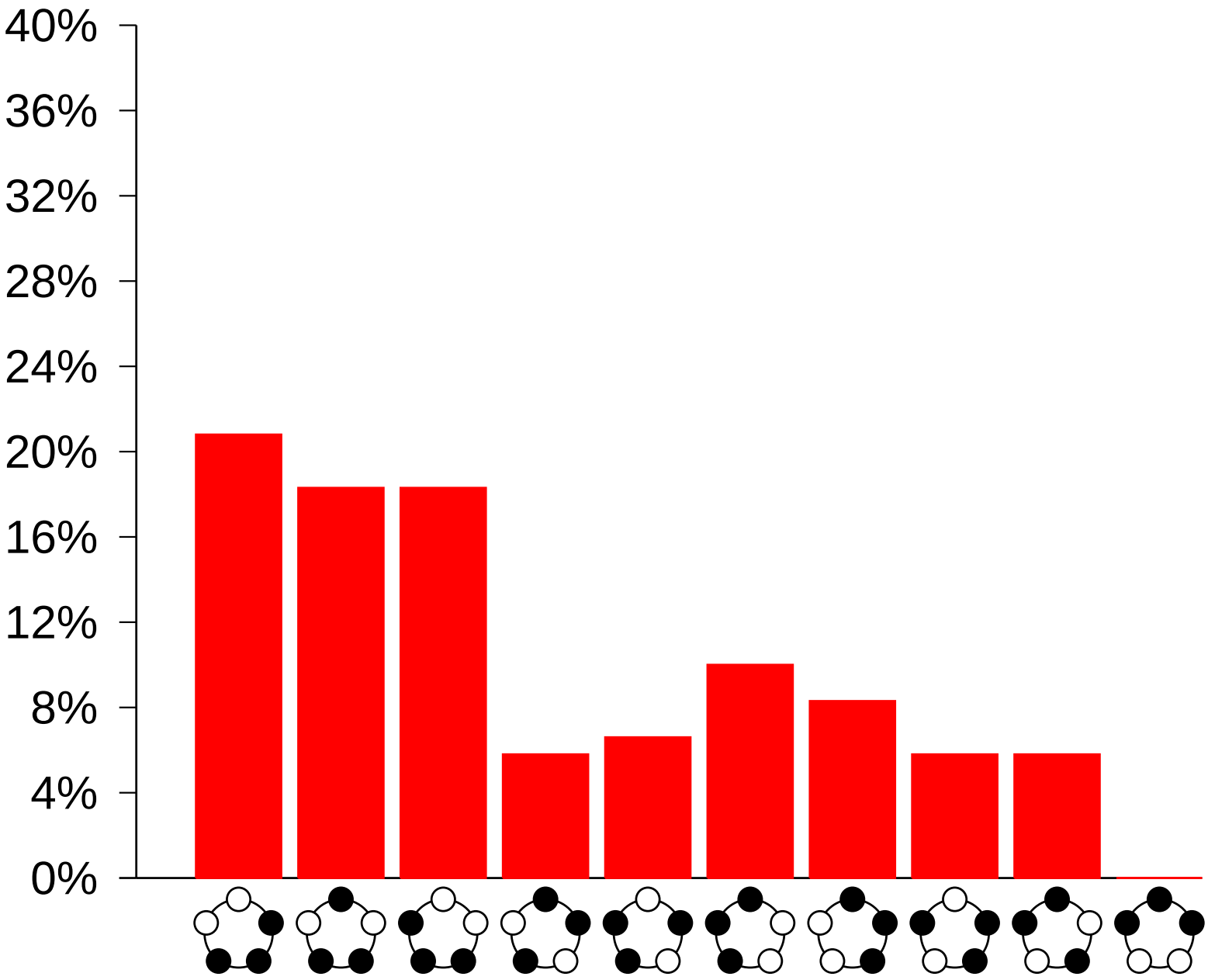
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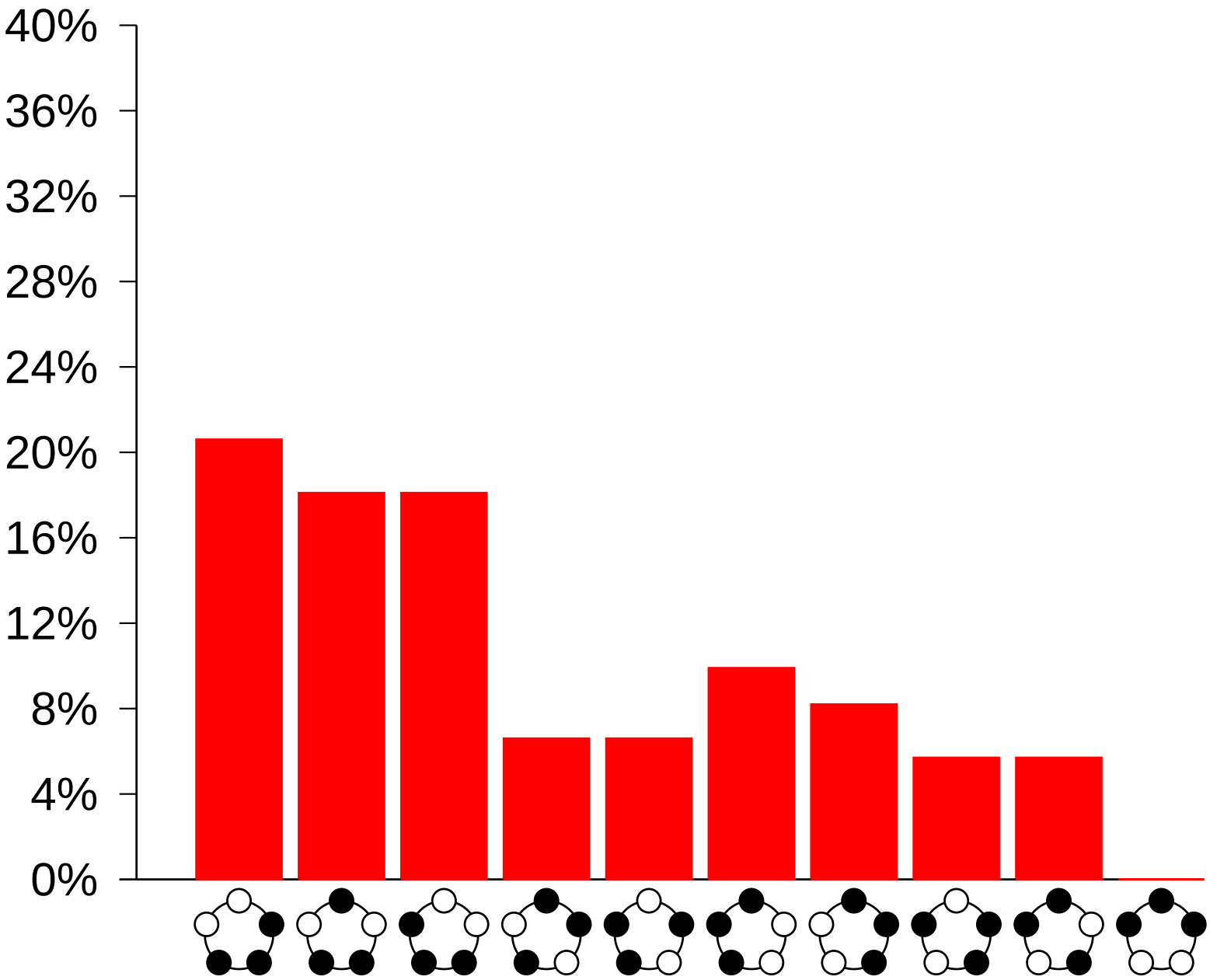
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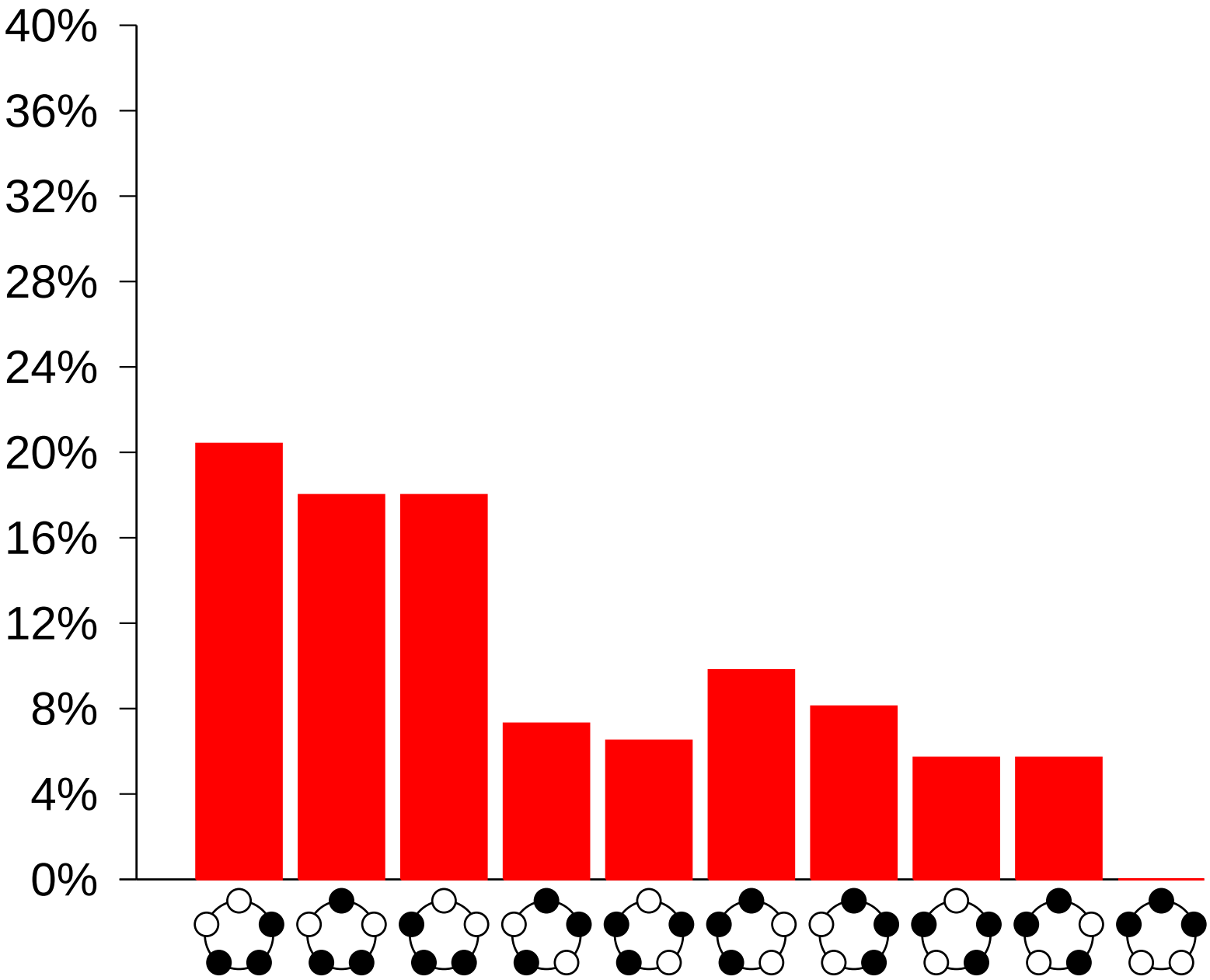
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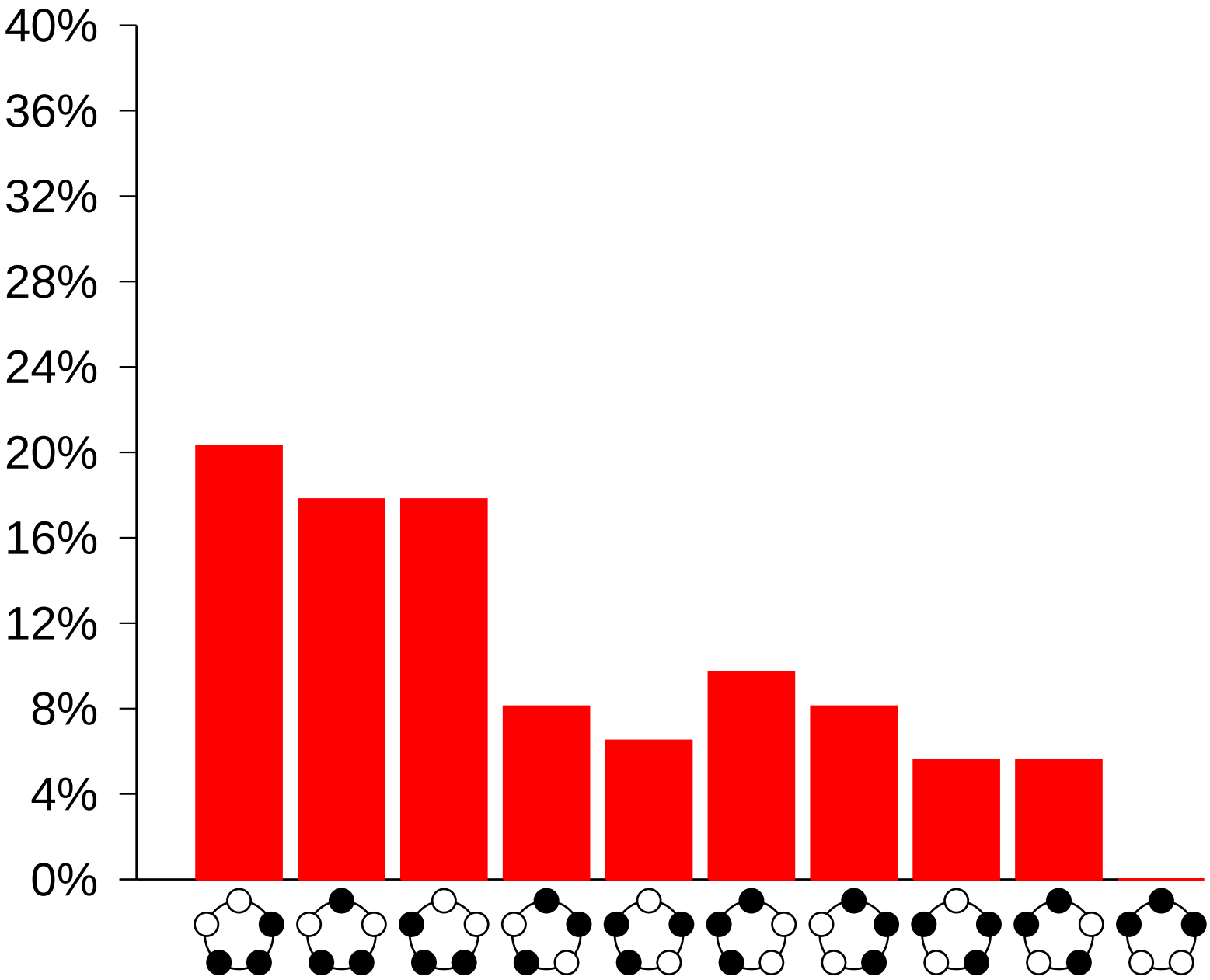
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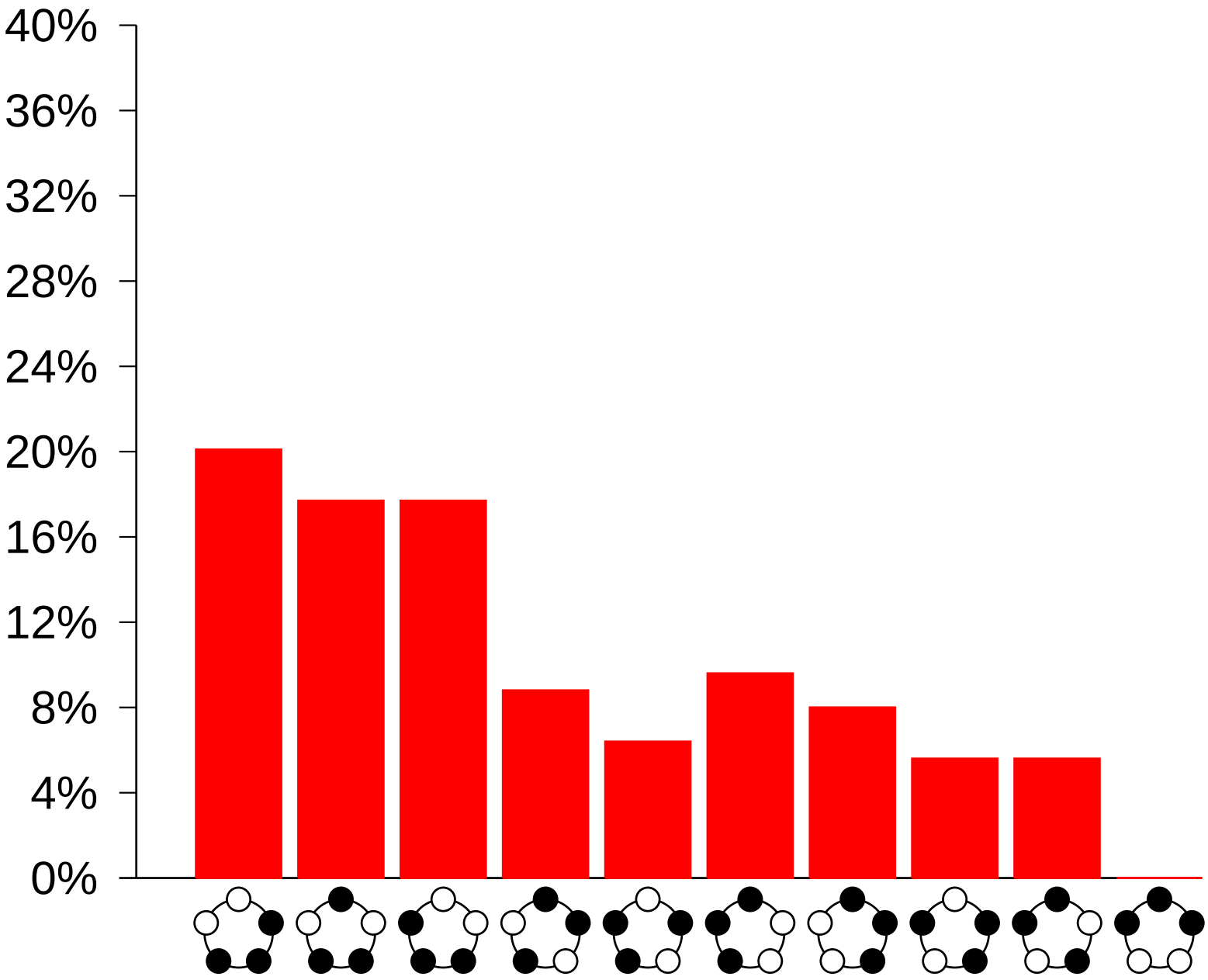
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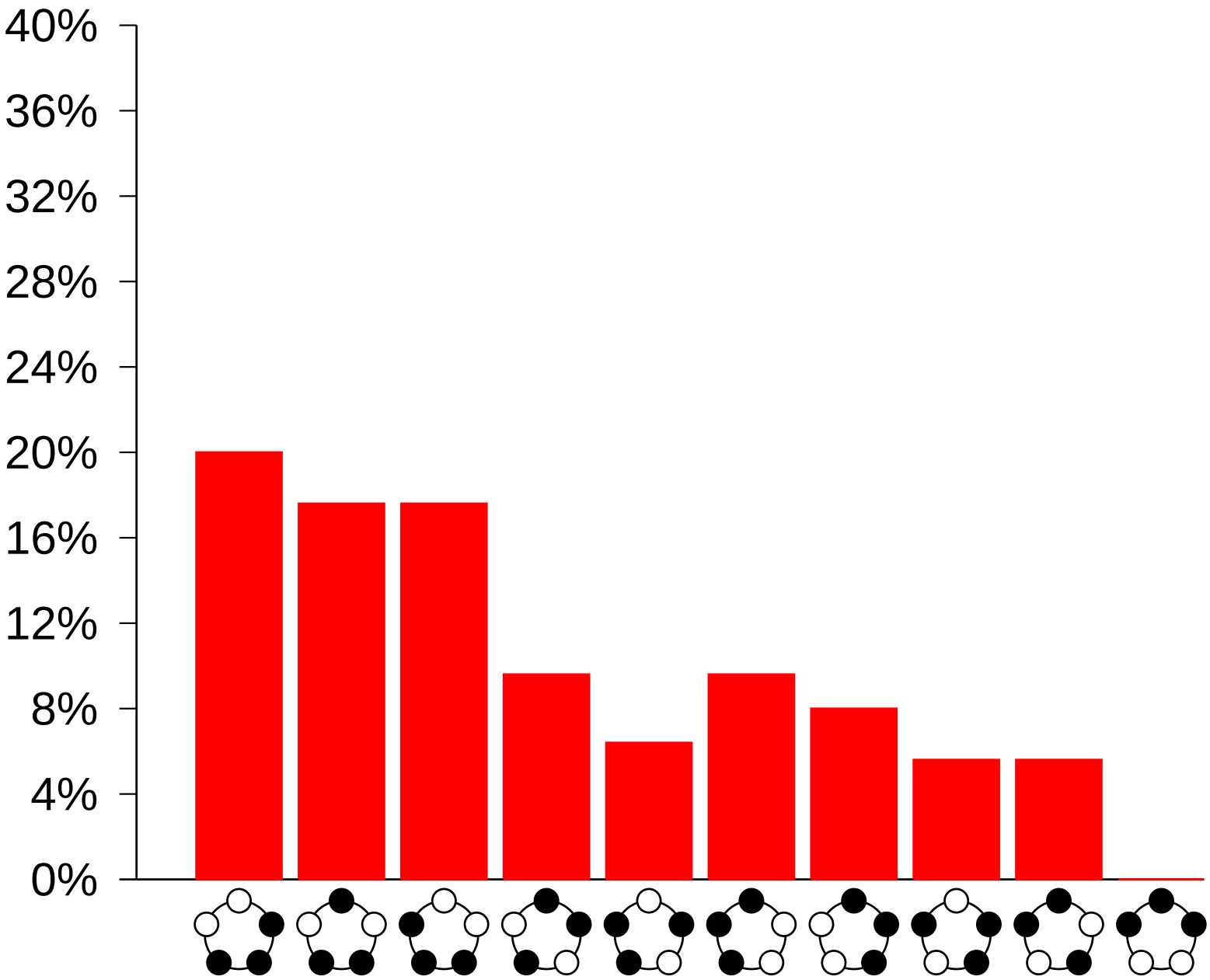
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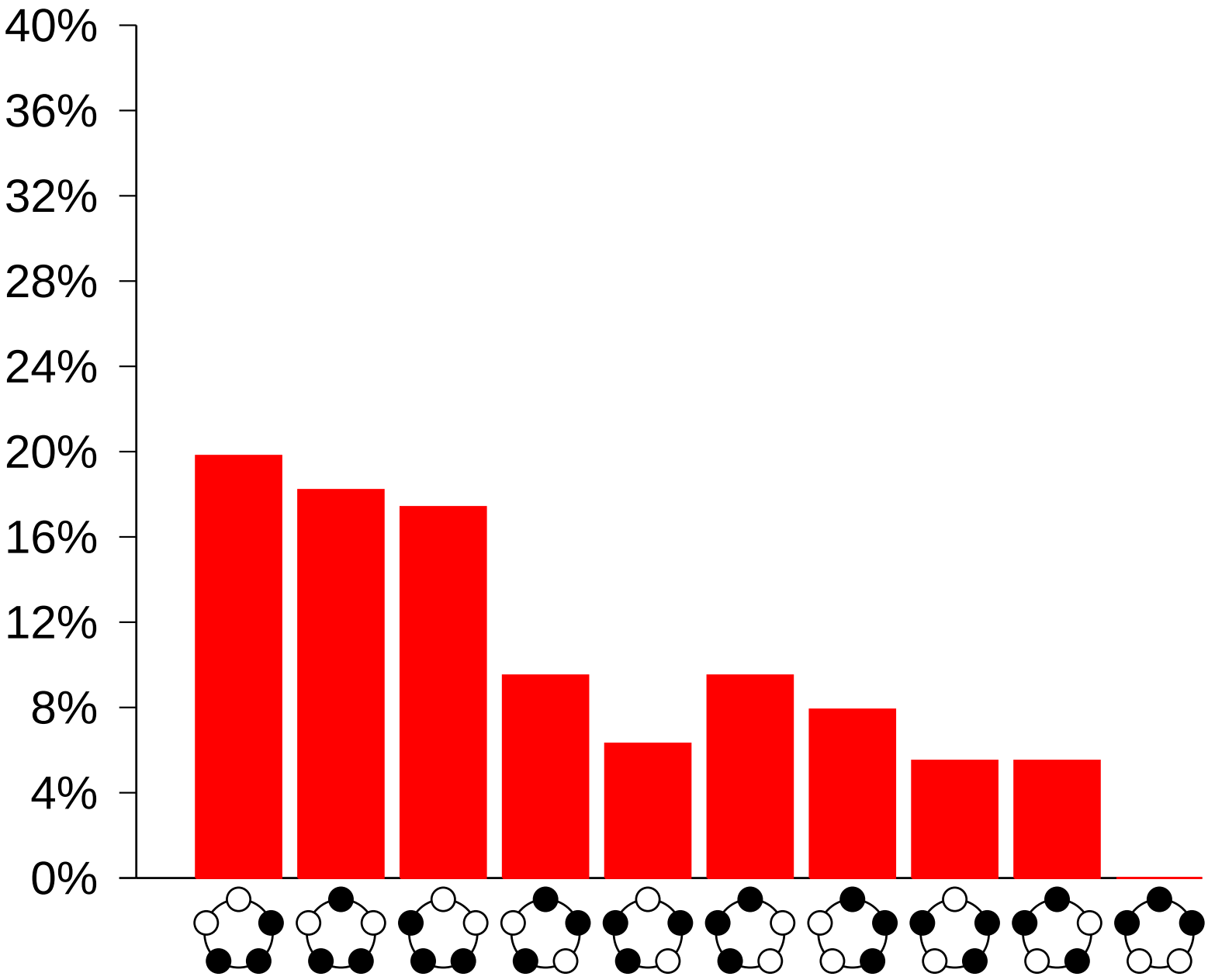
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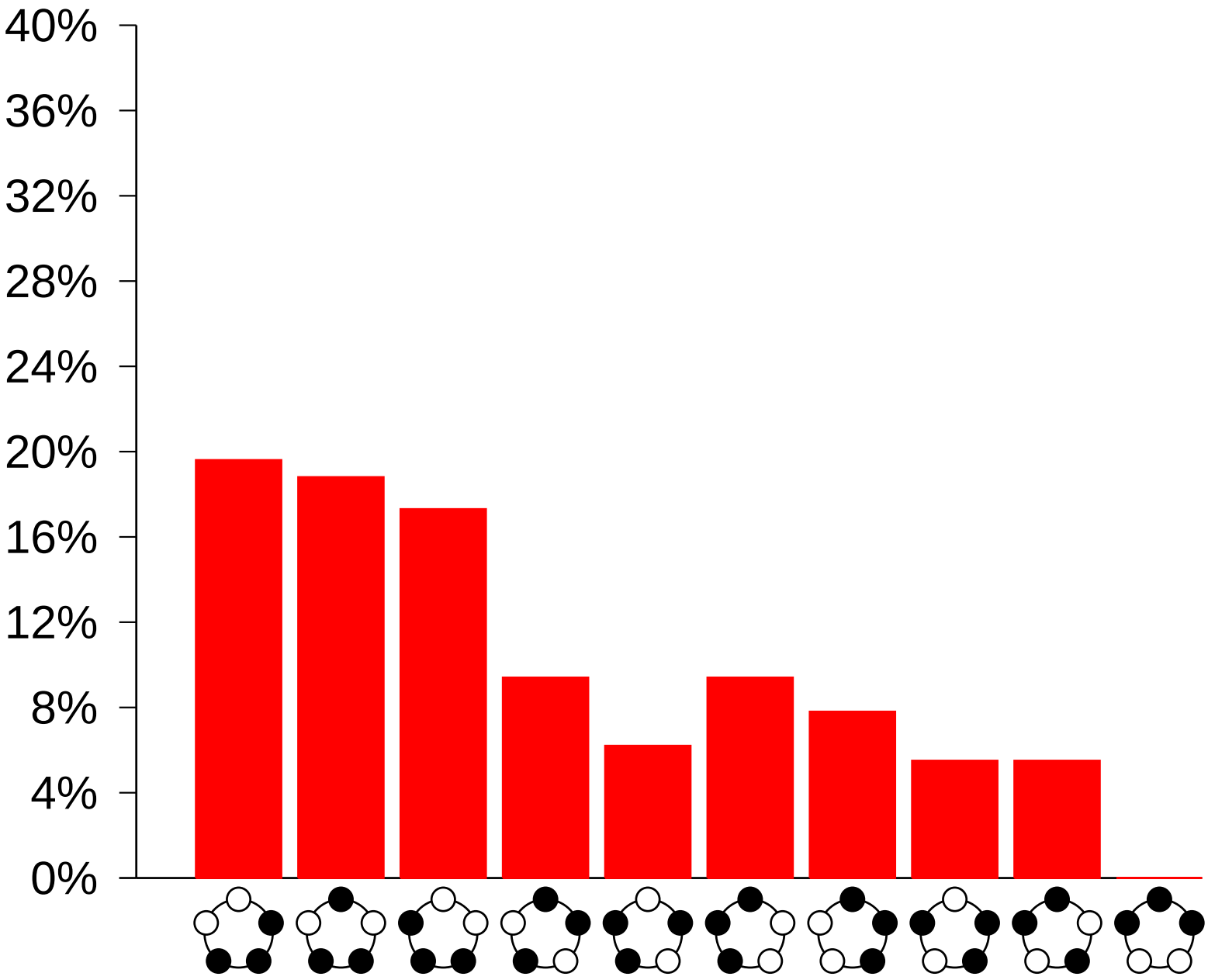
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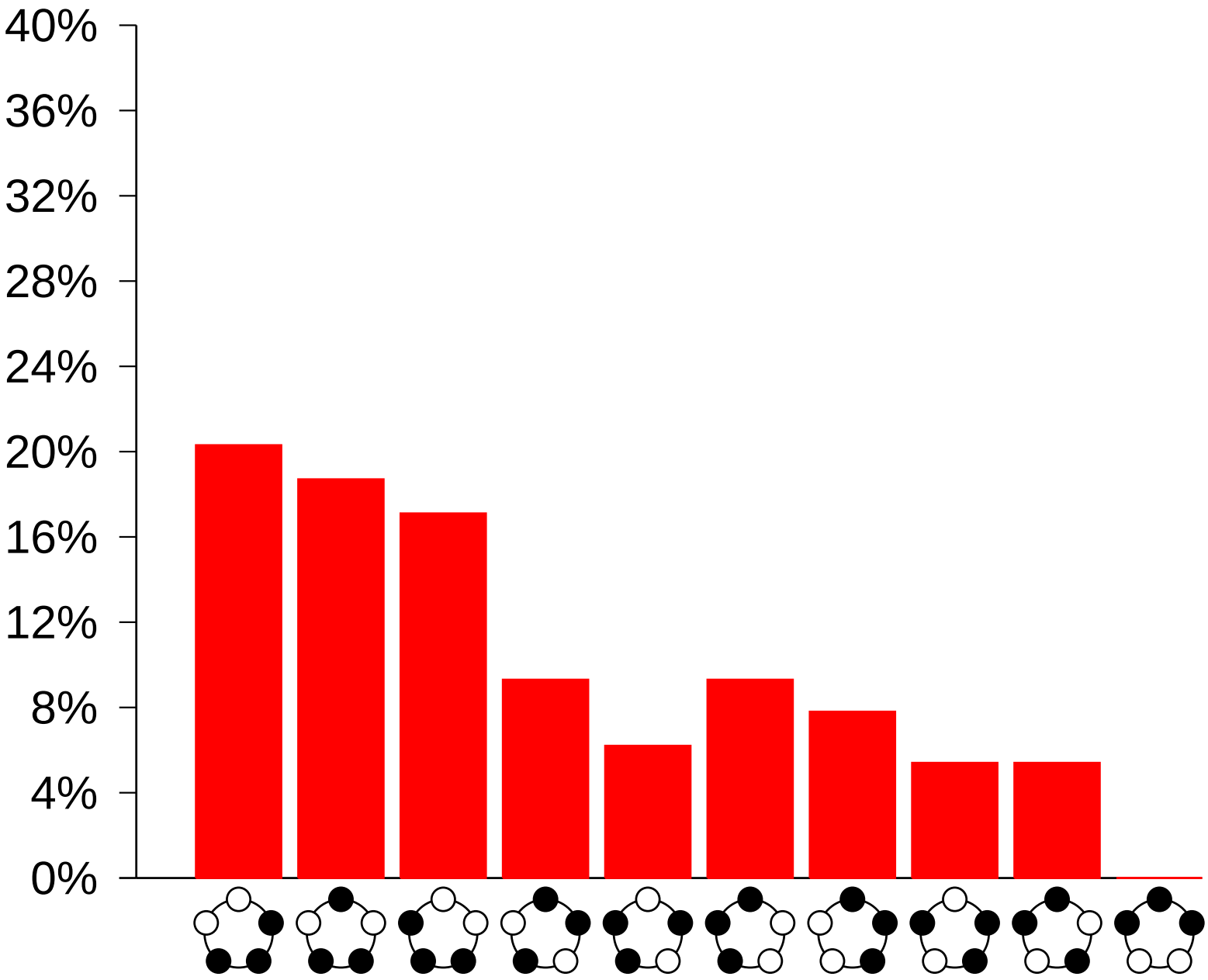
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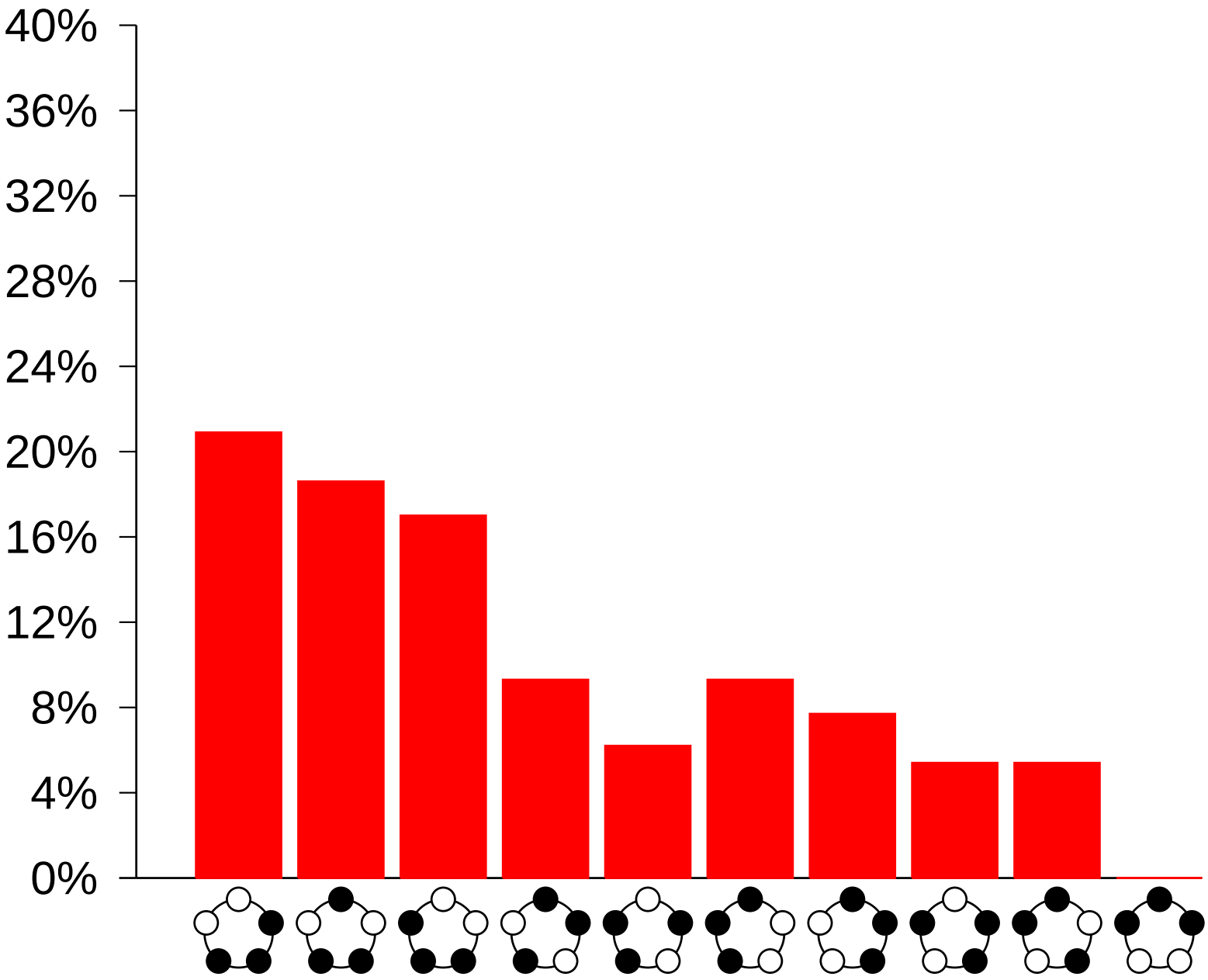
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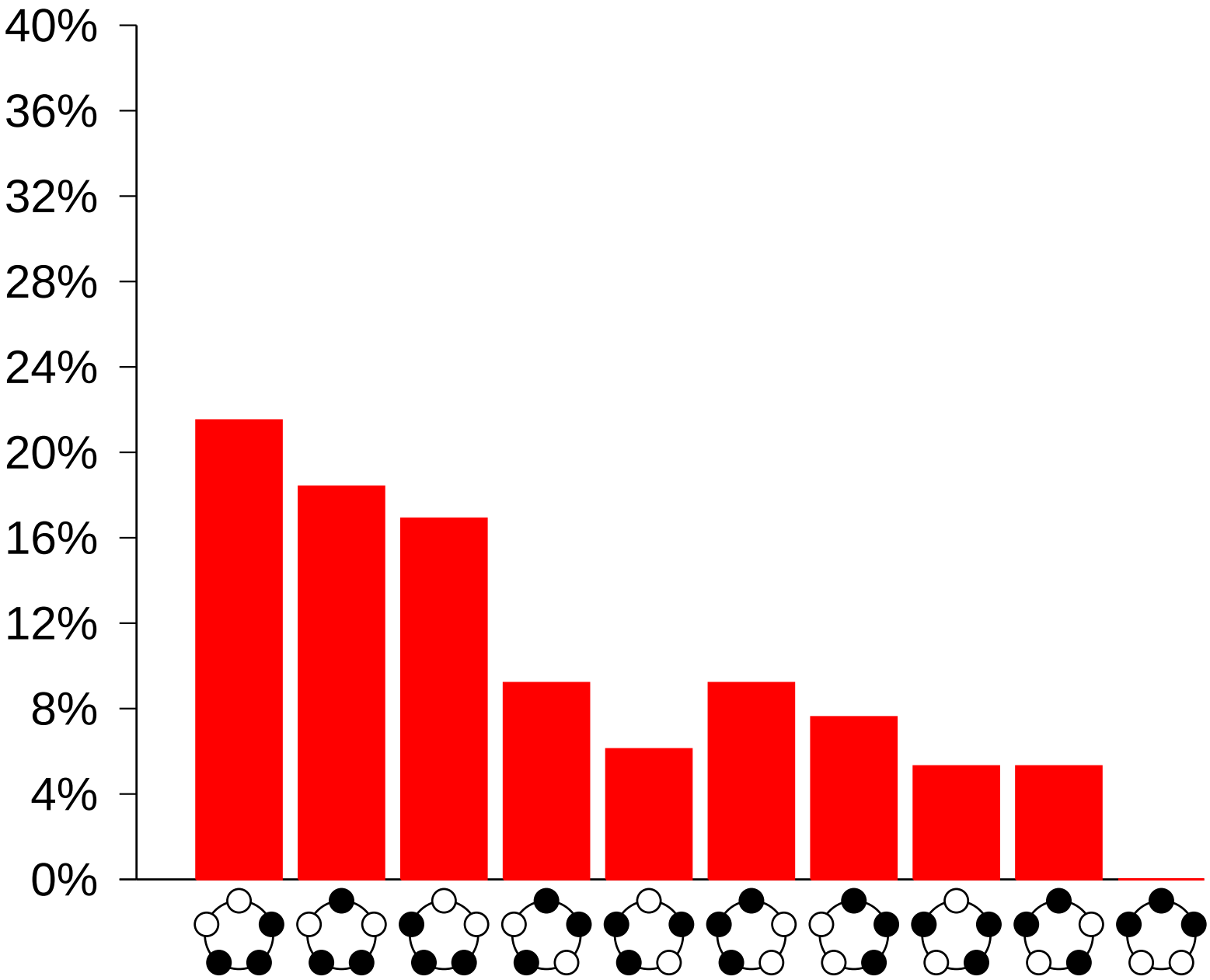
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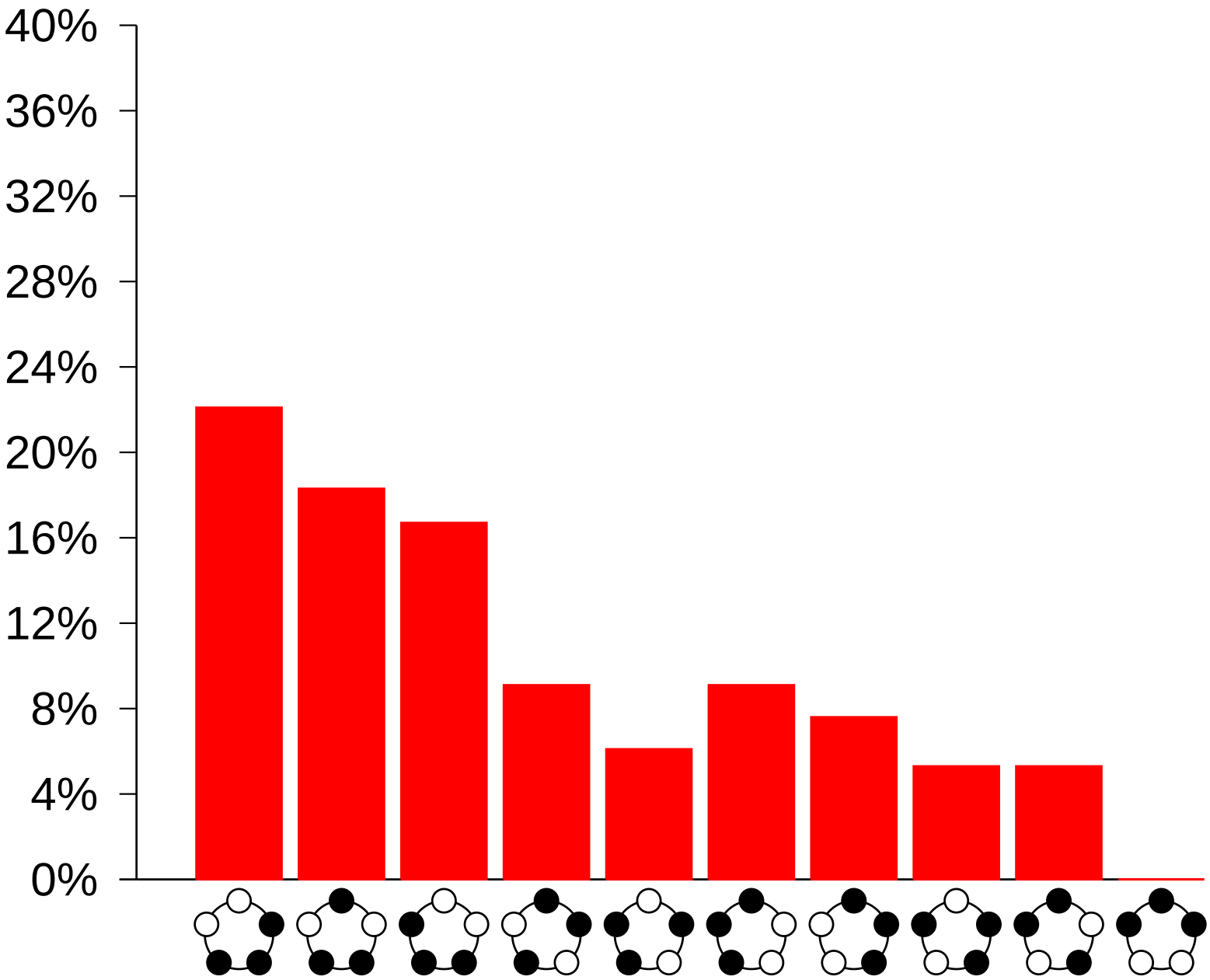
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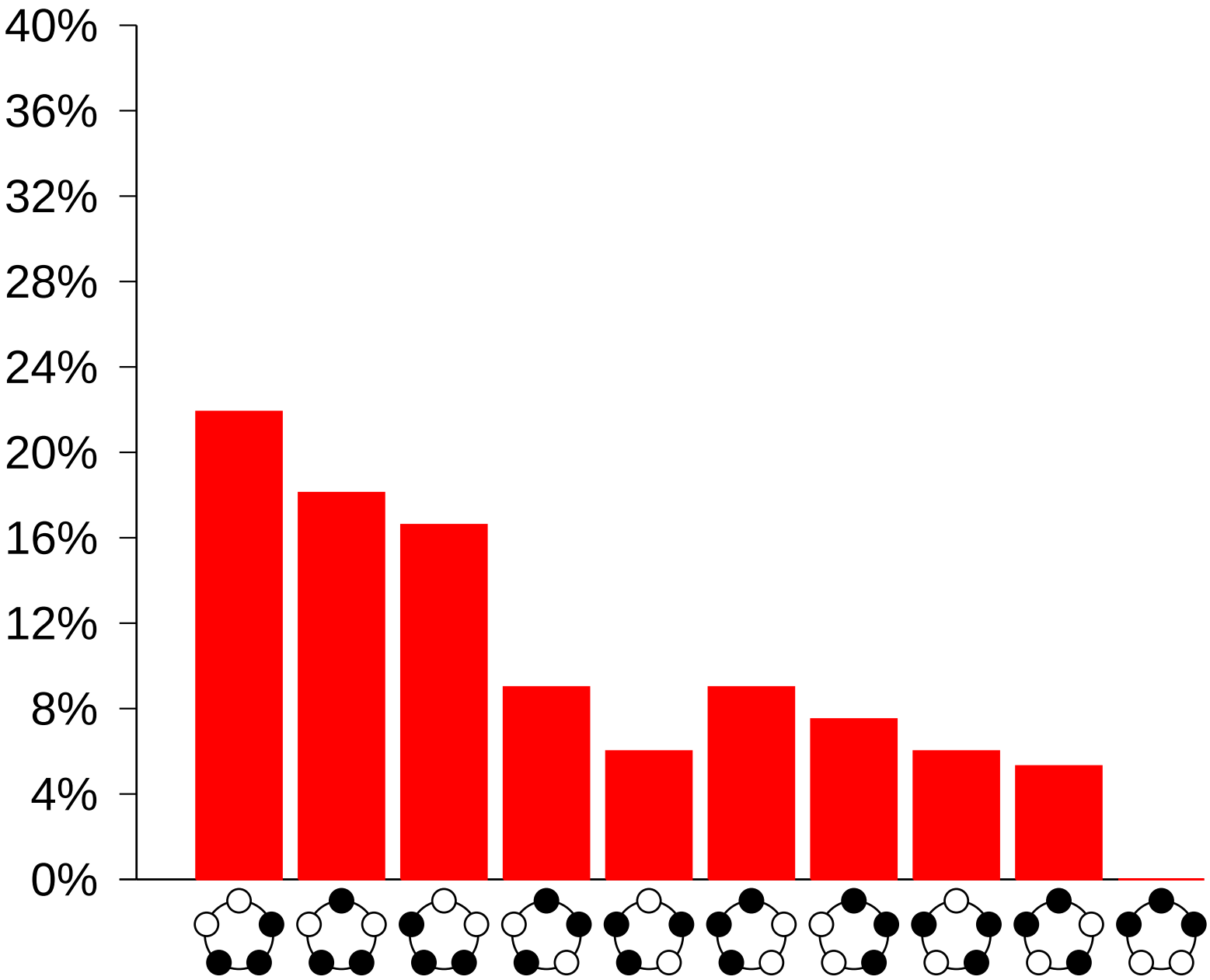
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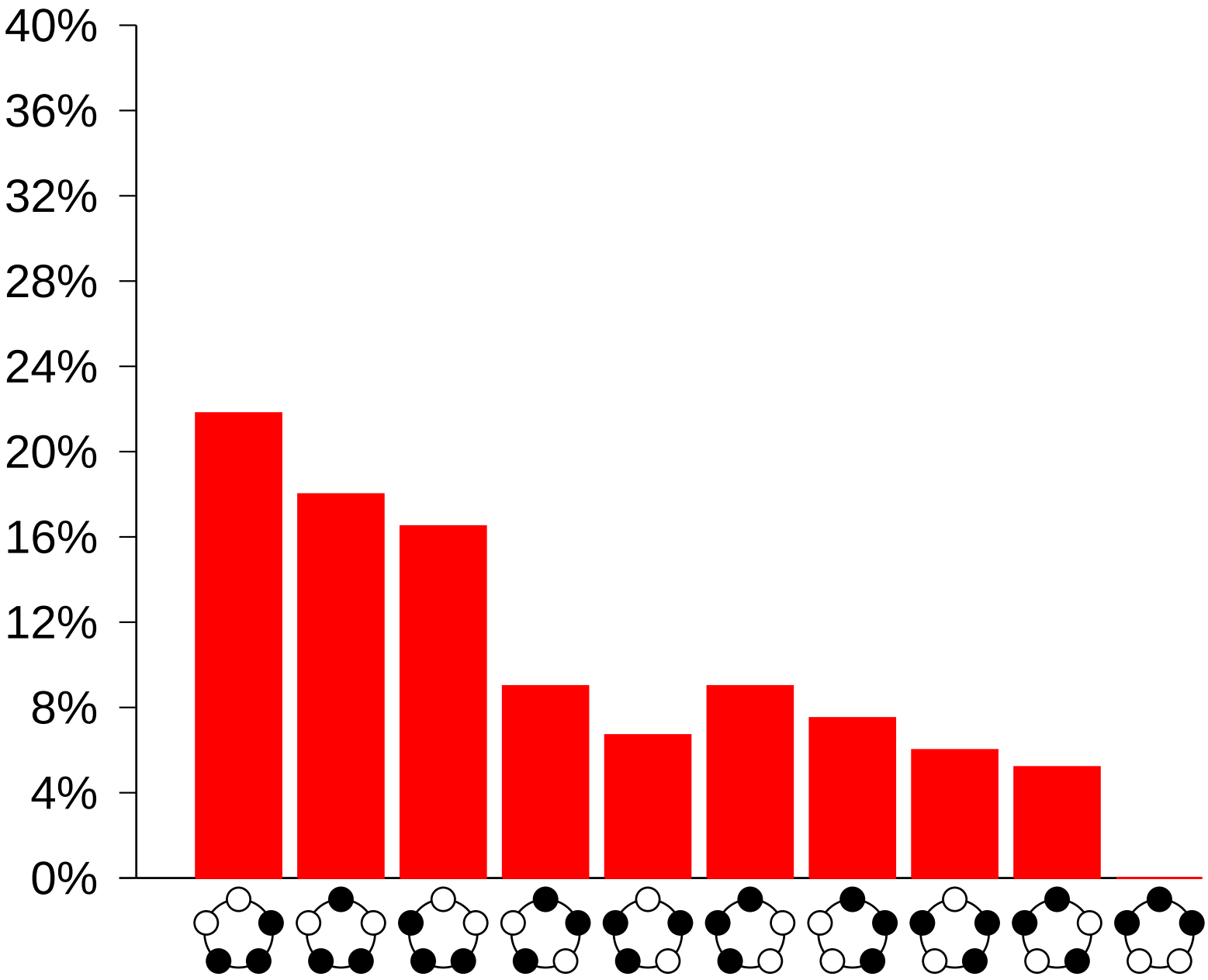
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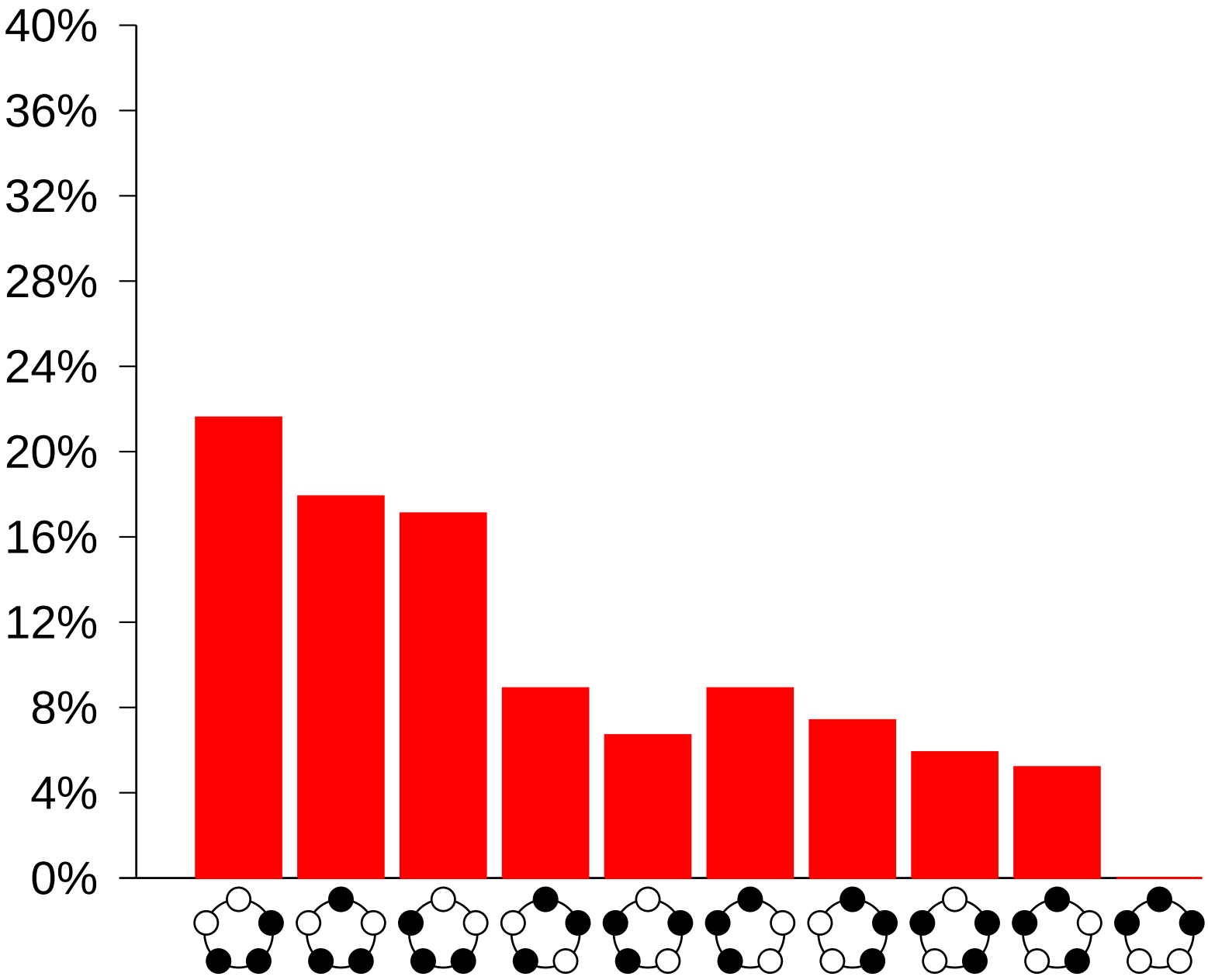
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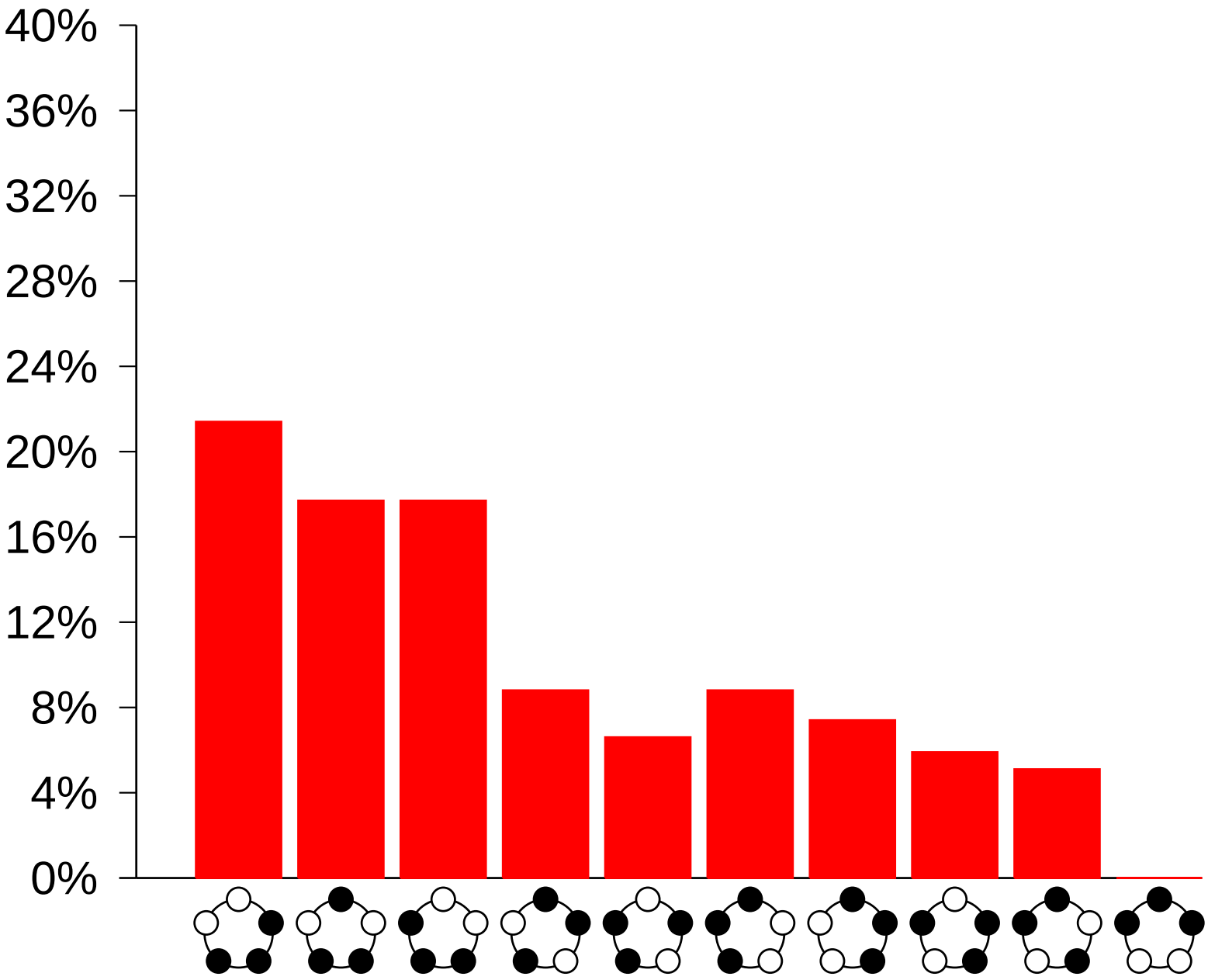
Stationary distribution



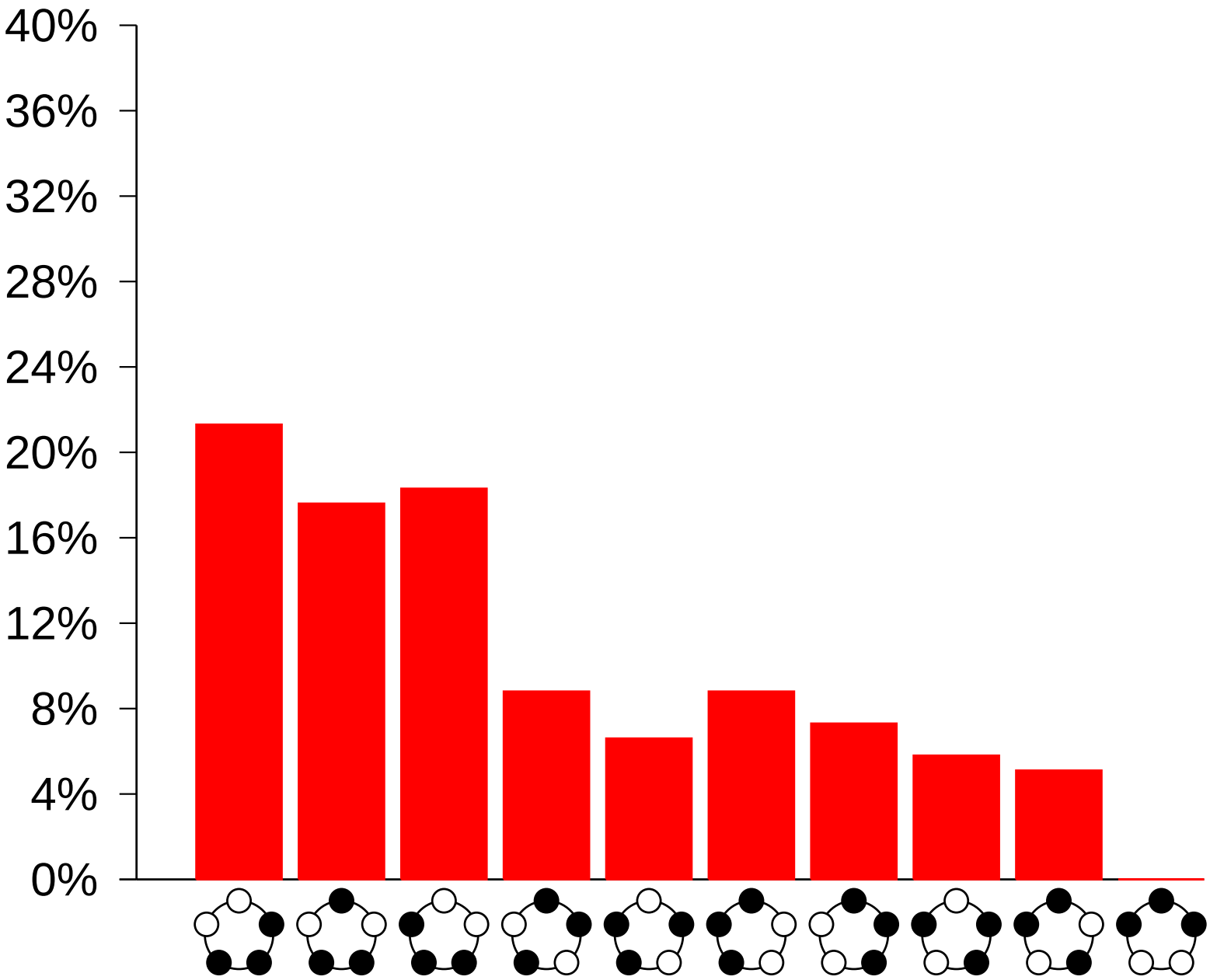
Stationary distribution



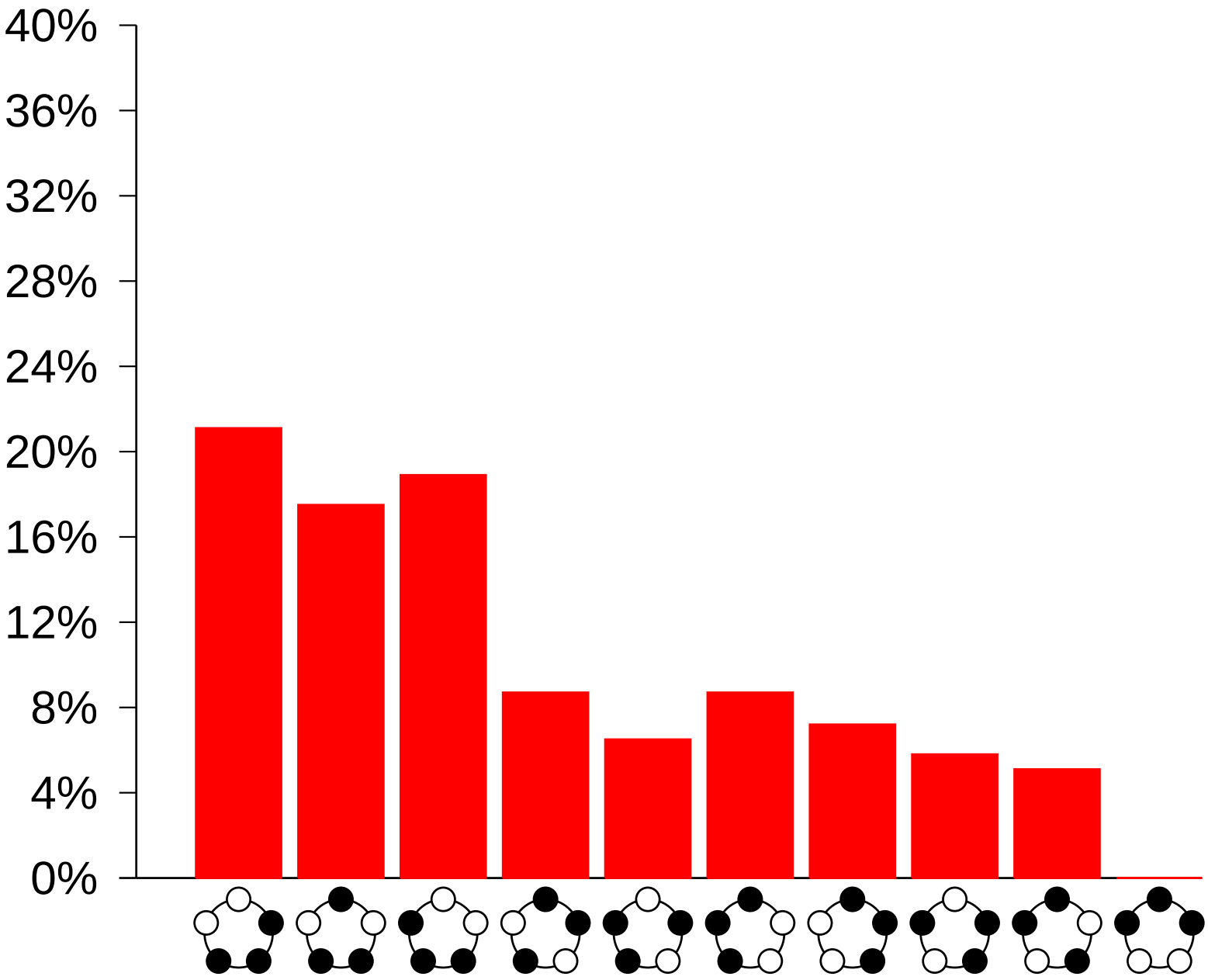
Stationary distribution



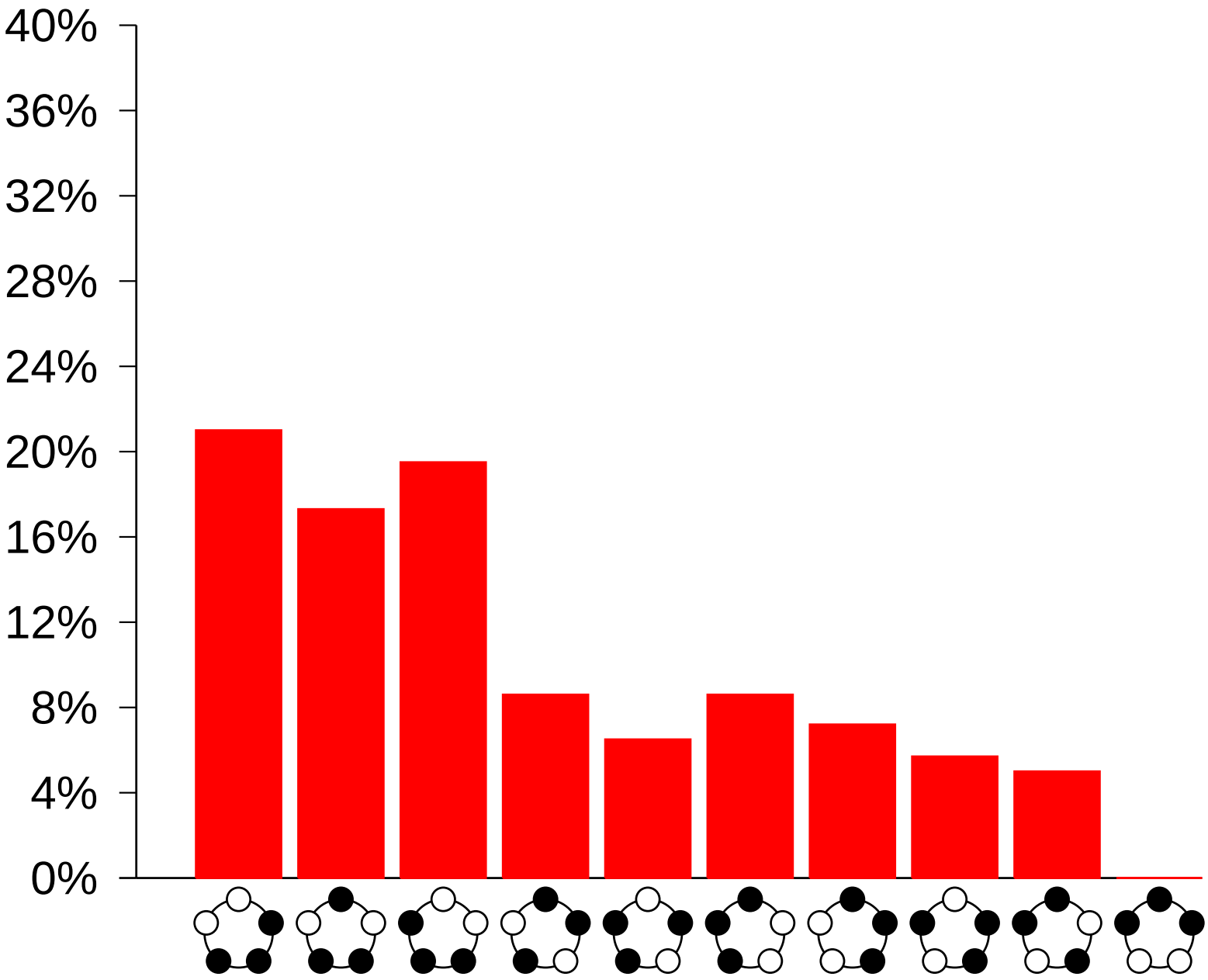
Stationary distribution



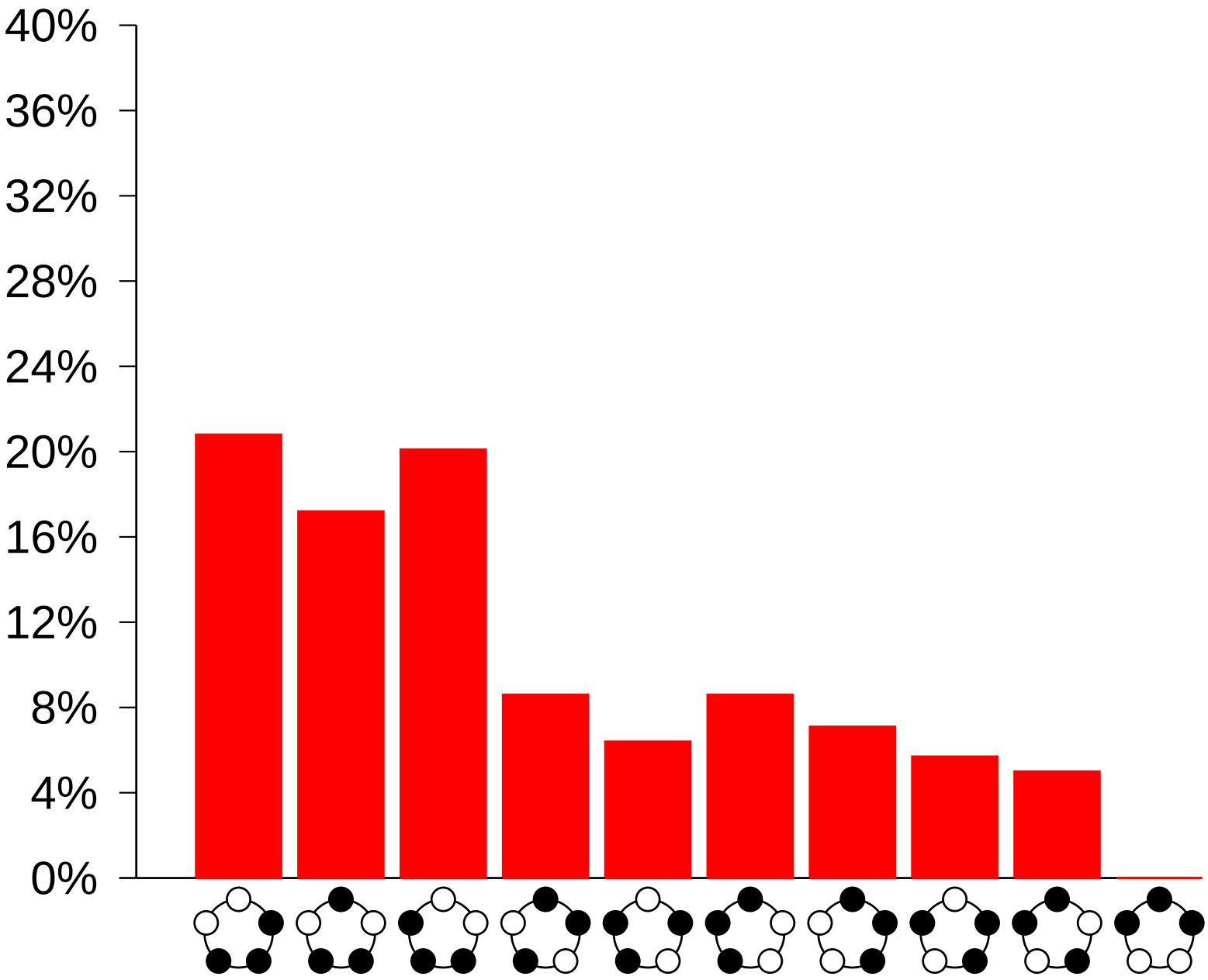
Stationary distribution



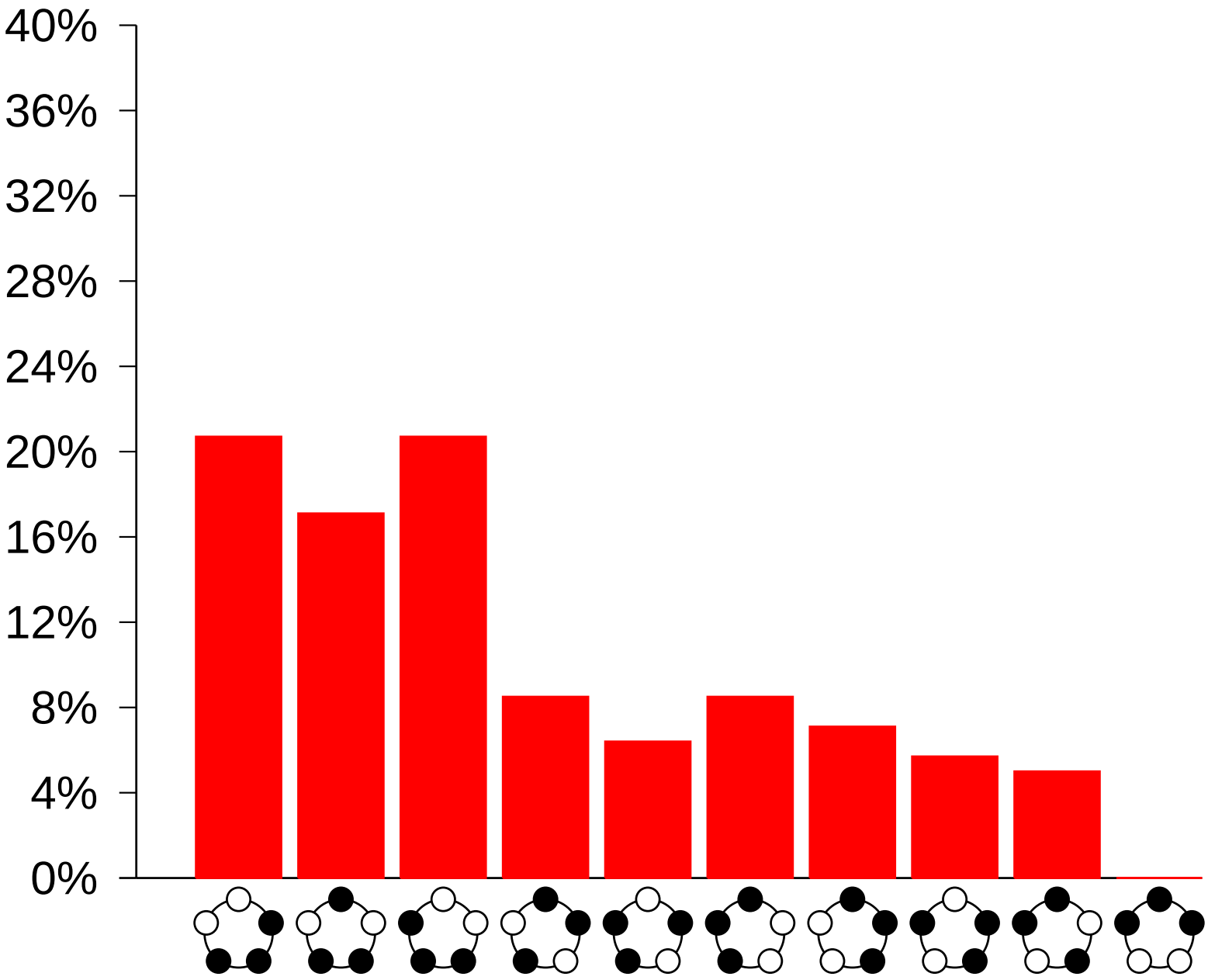
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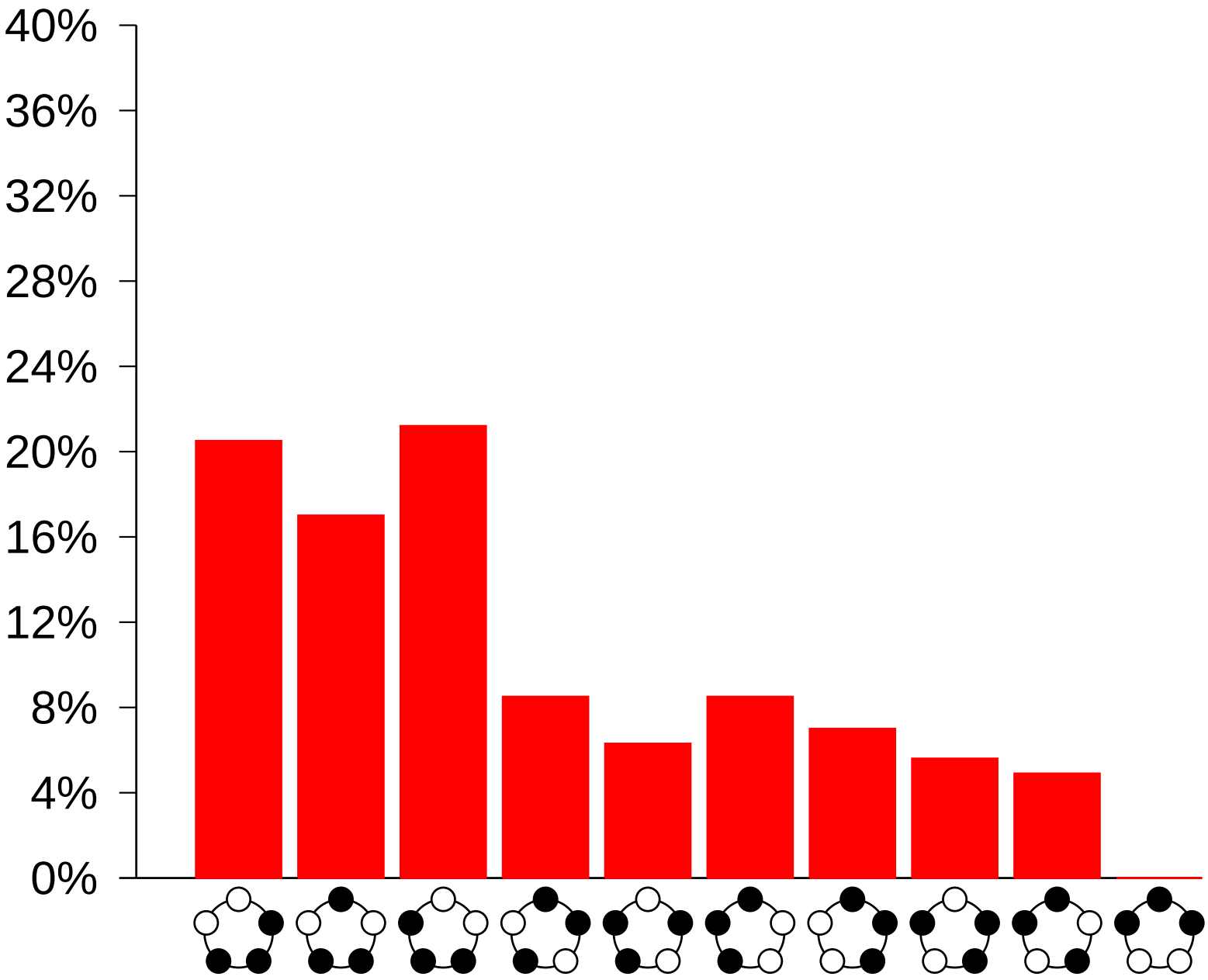
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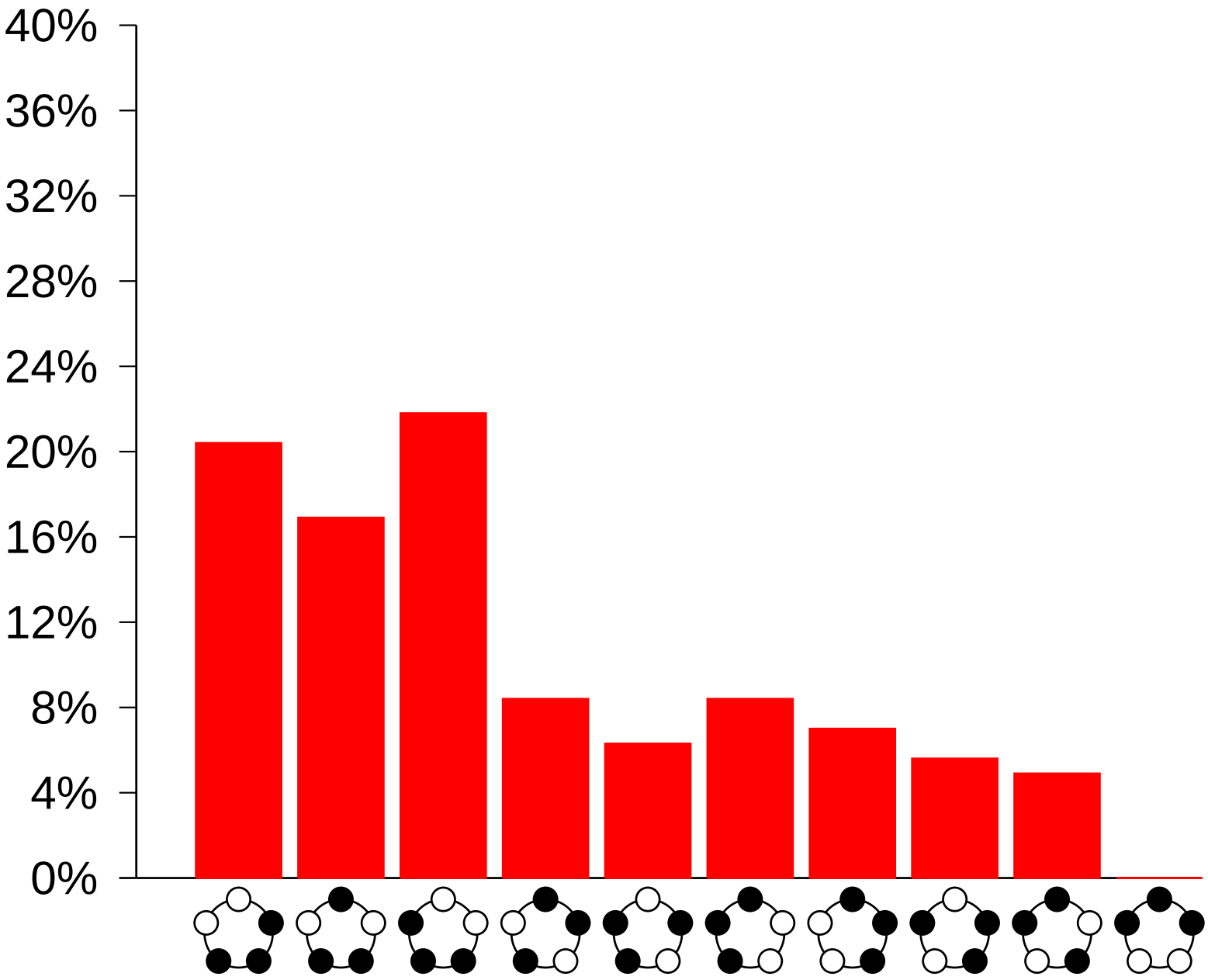
Stationary distribution



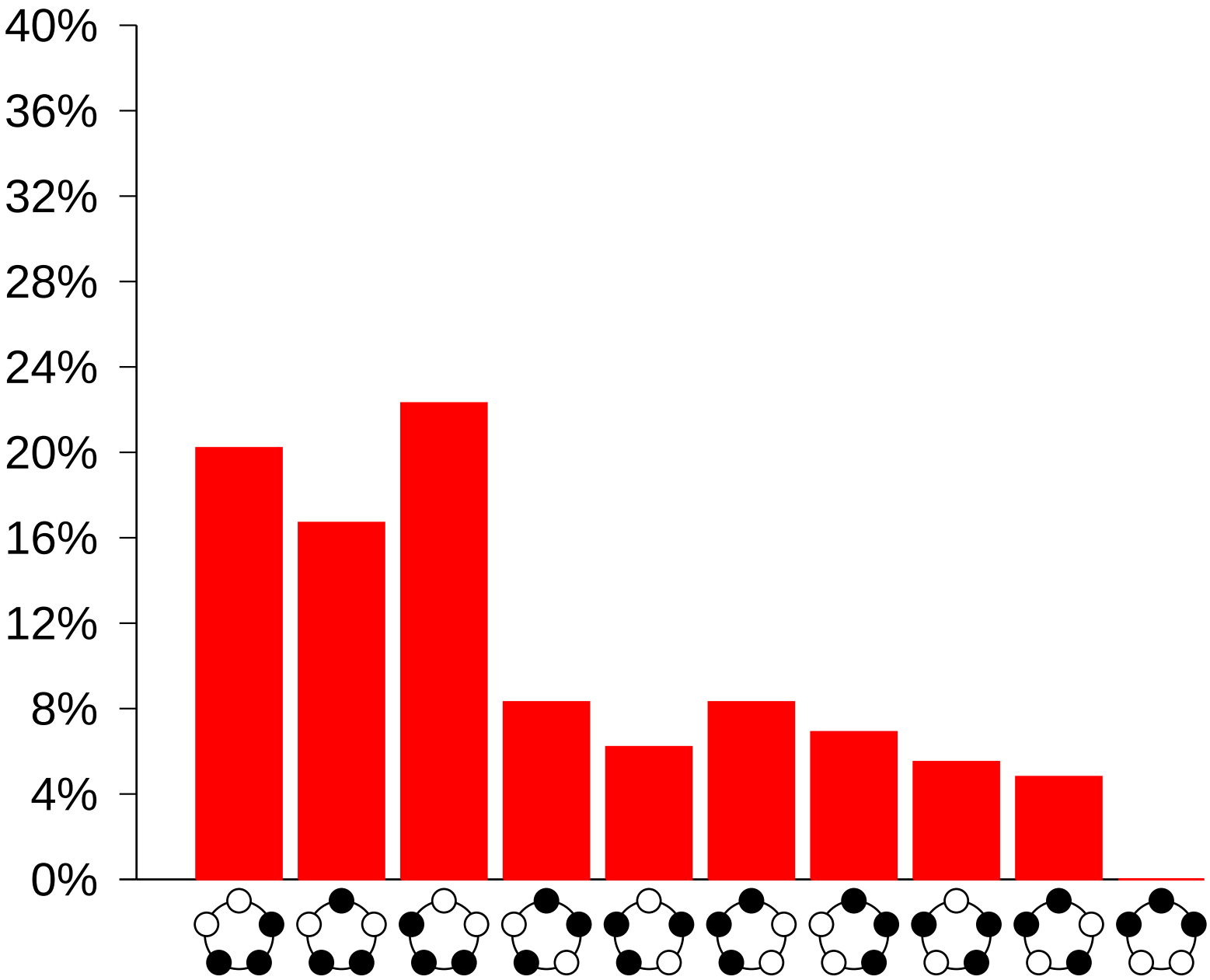
Stationary distribution



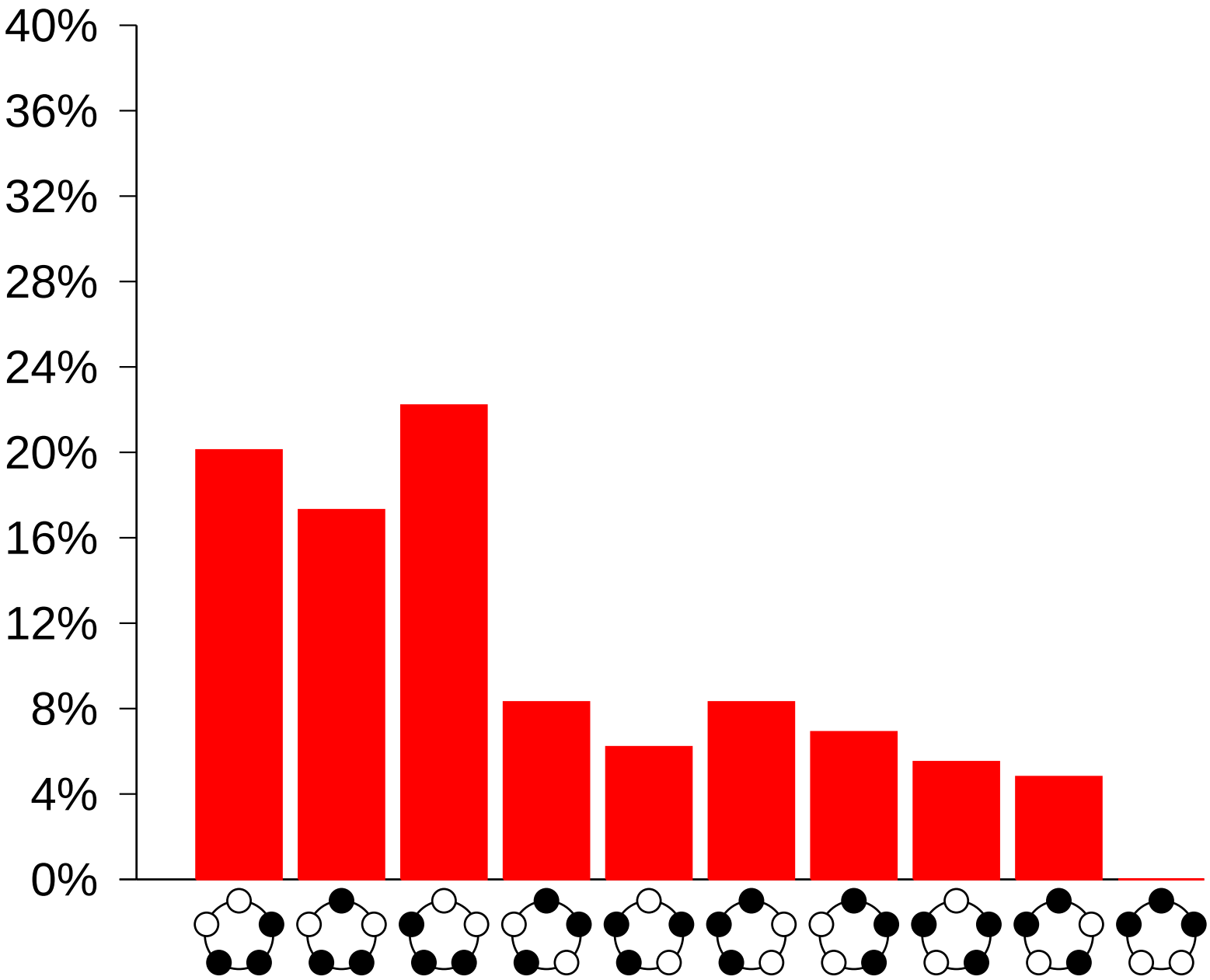
Stationary distribution



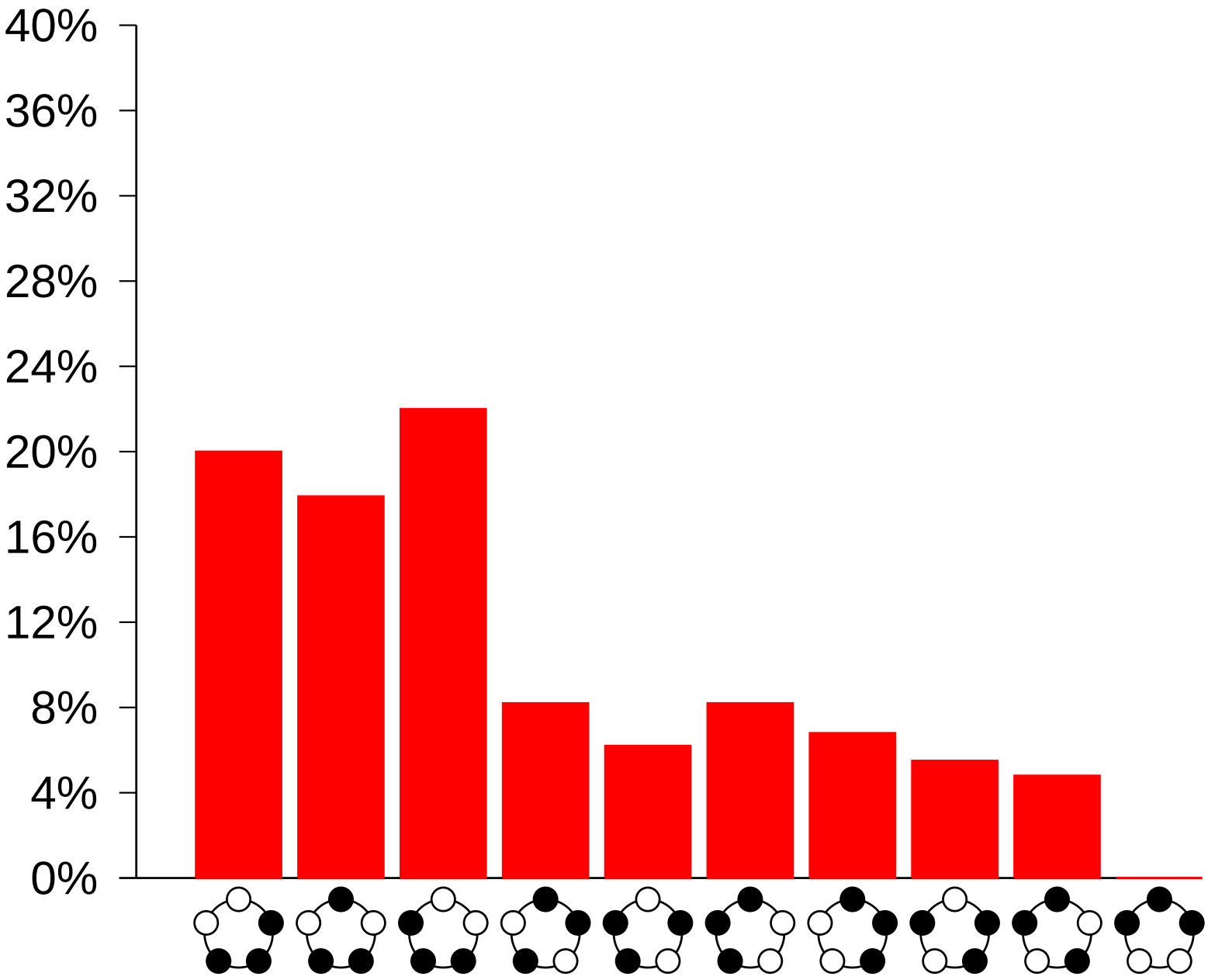
Stationary distribution



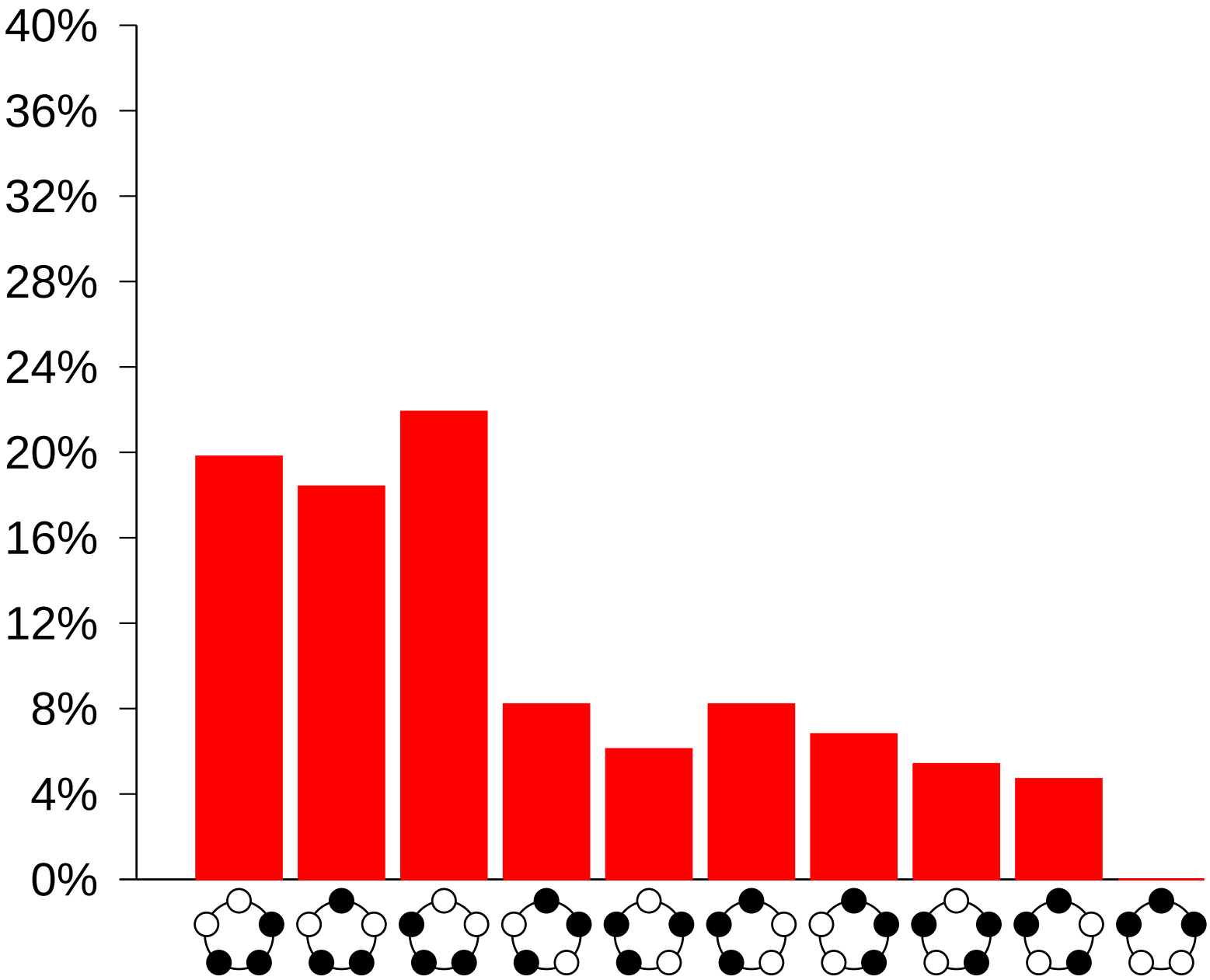
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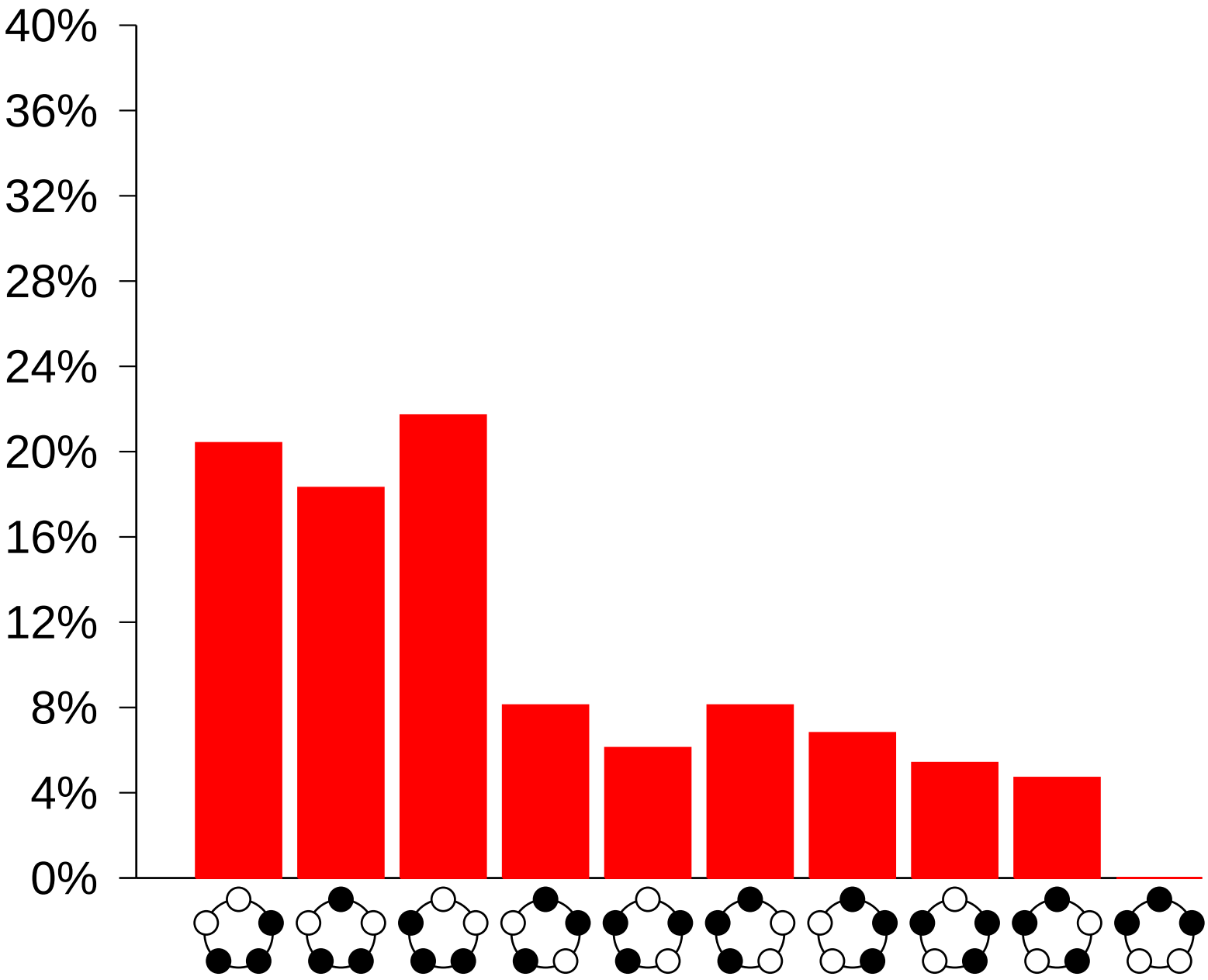
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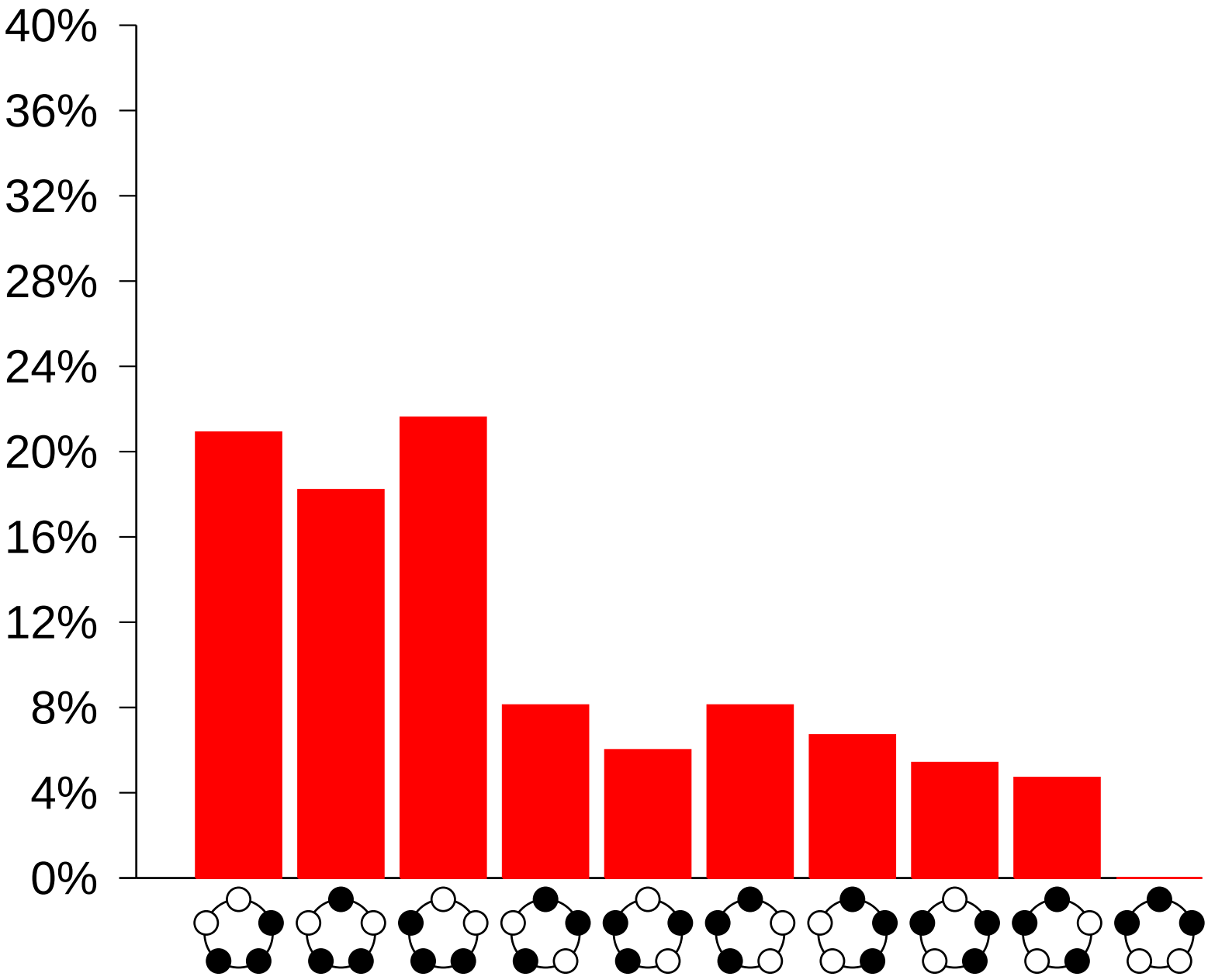
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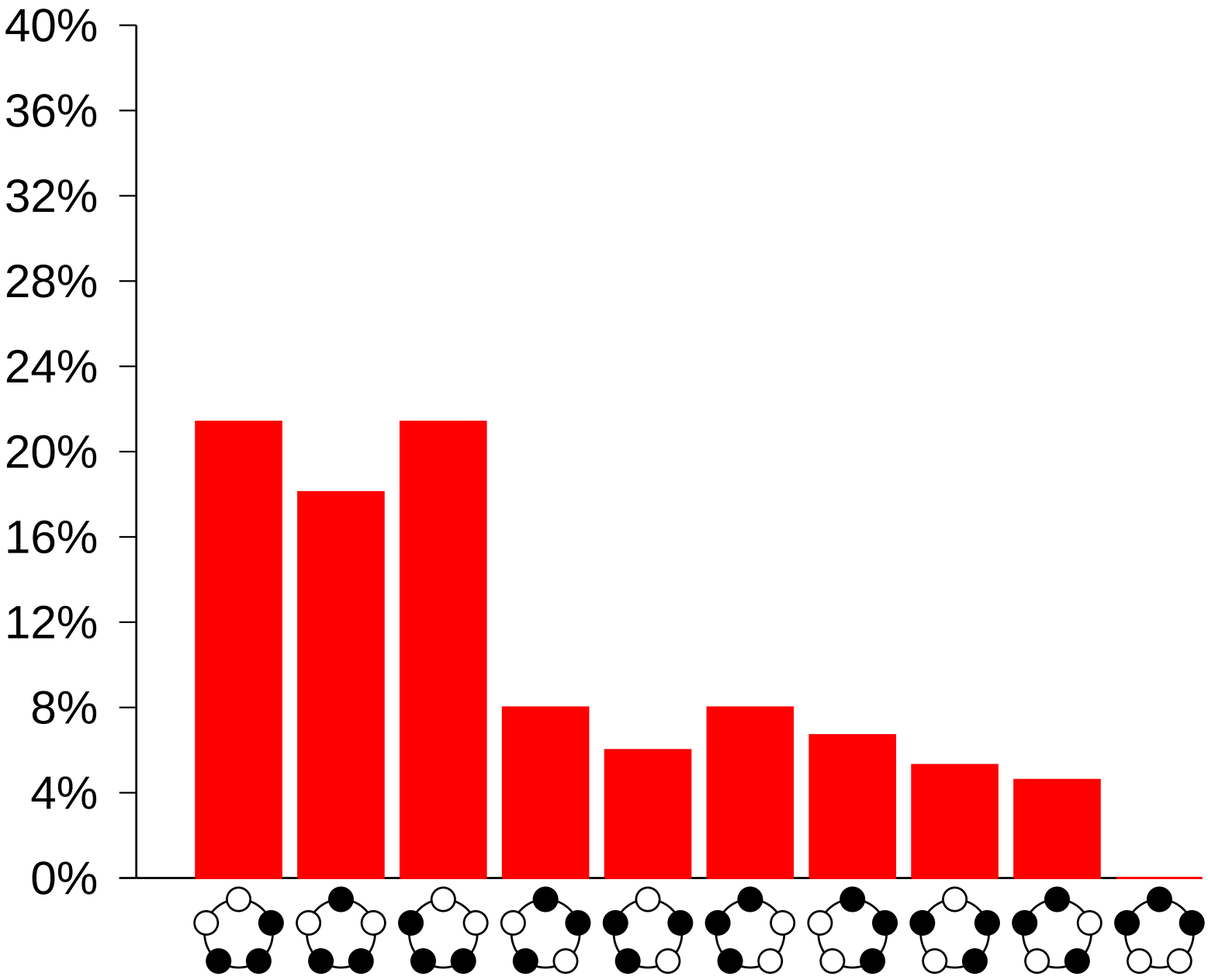
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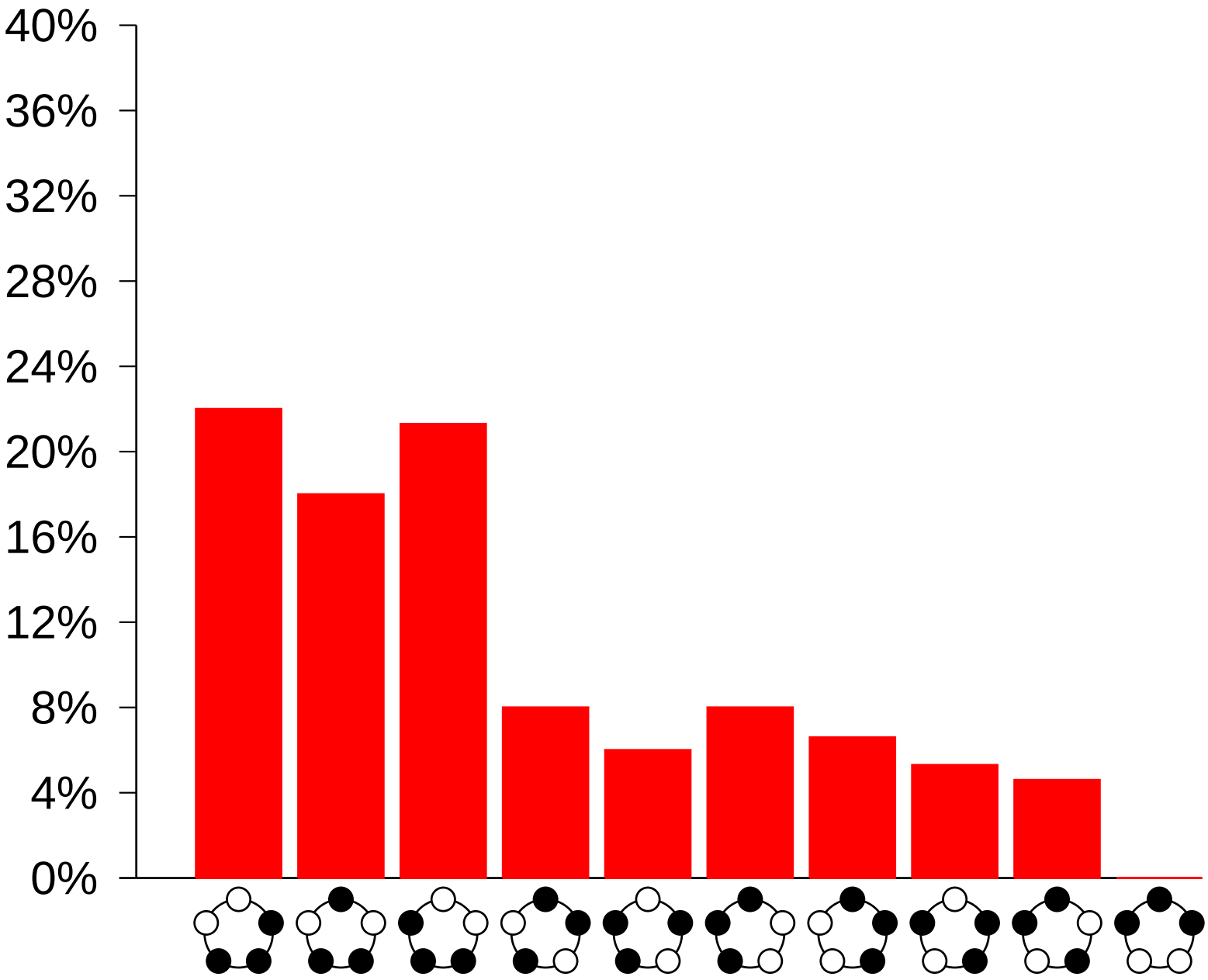
Stationary distribution



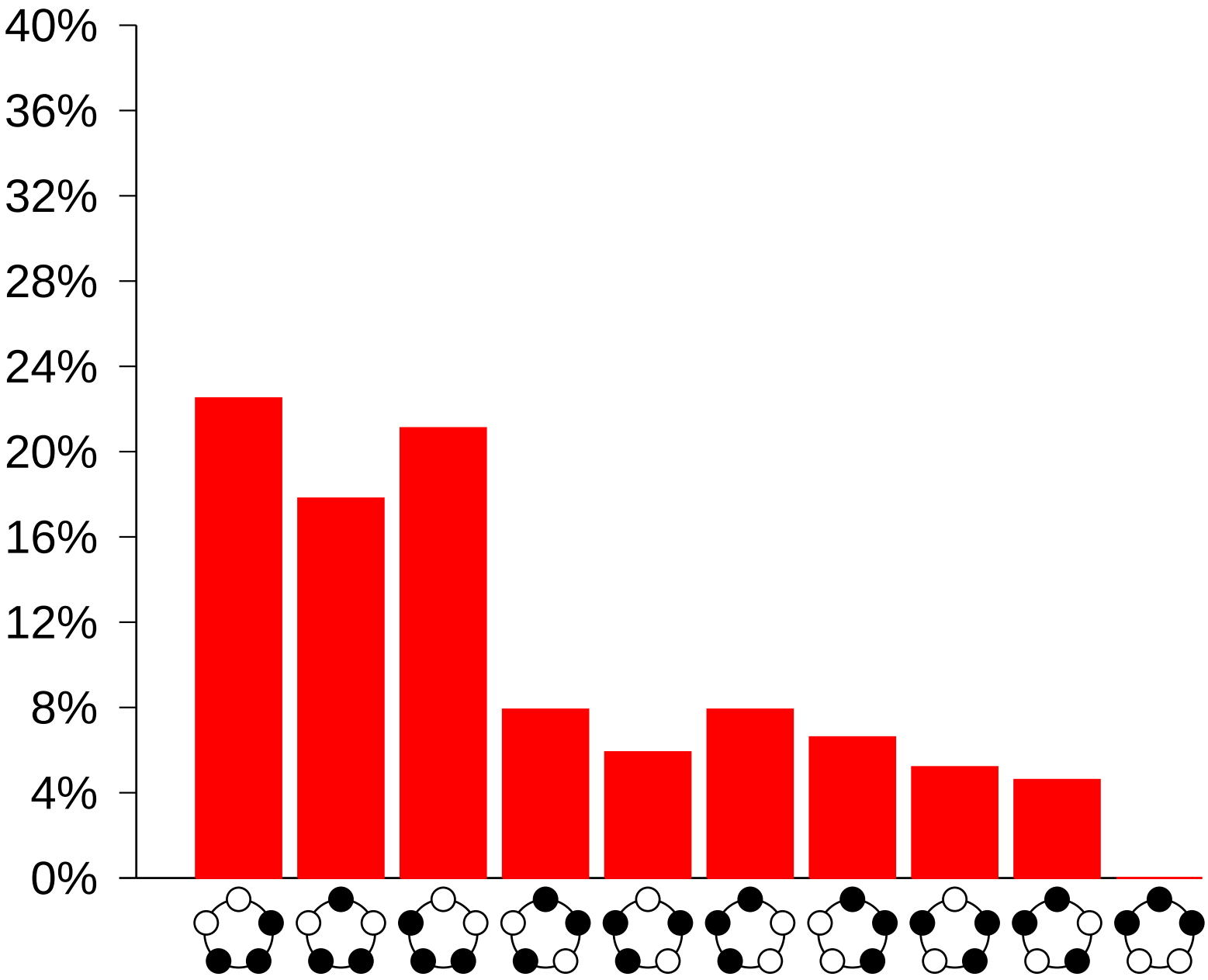
Stationary distribution



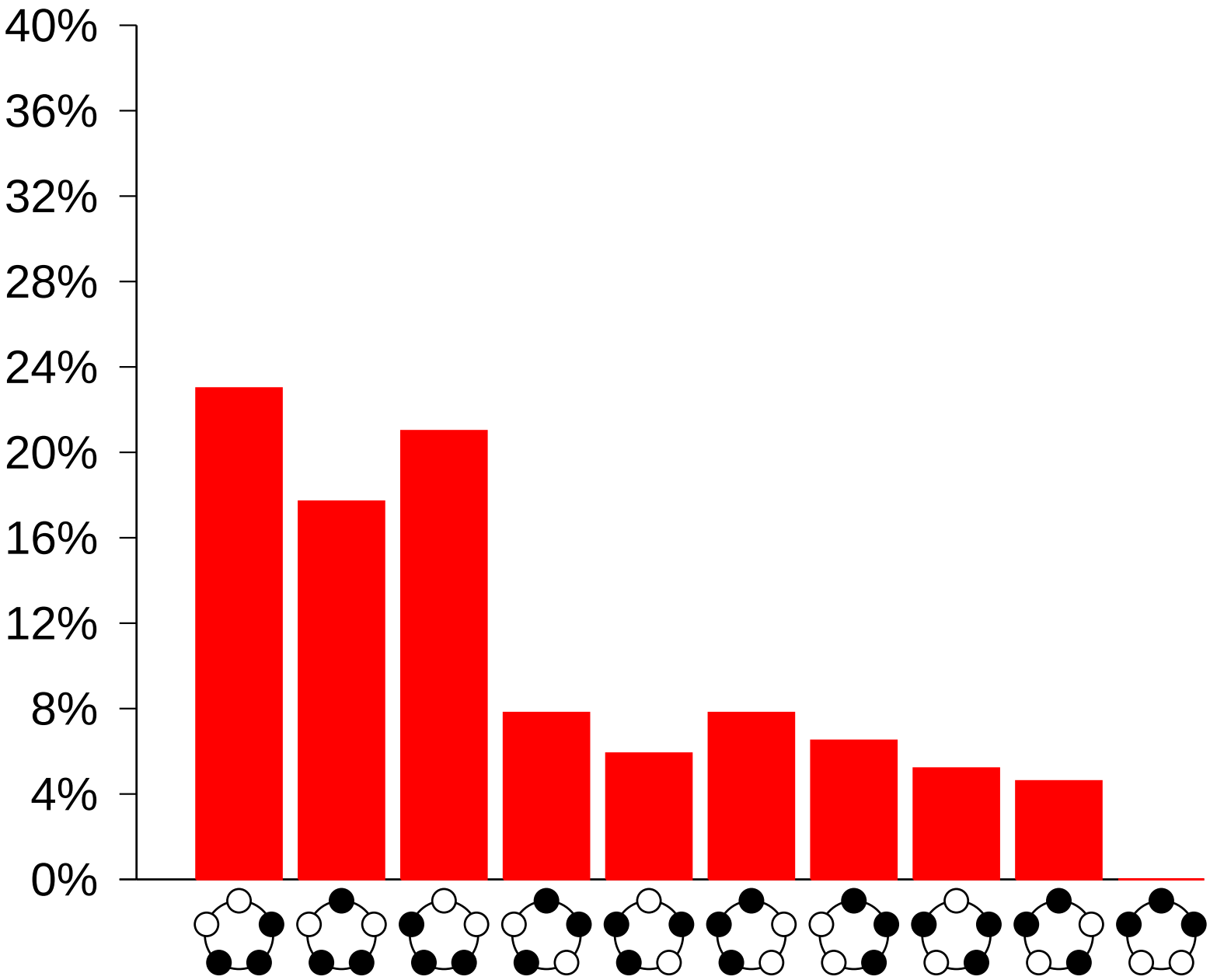
Stationary distribution



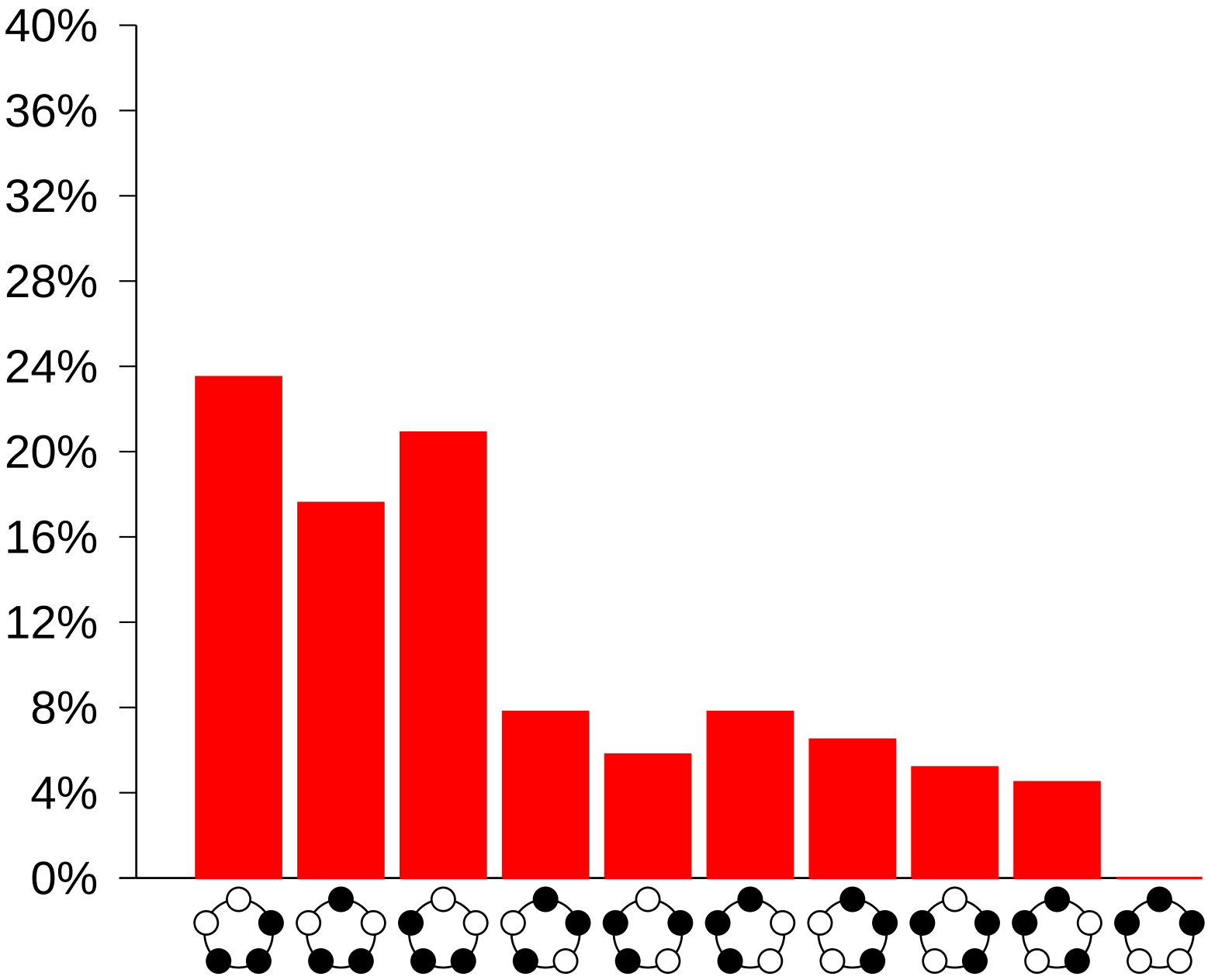
Stationary distribution



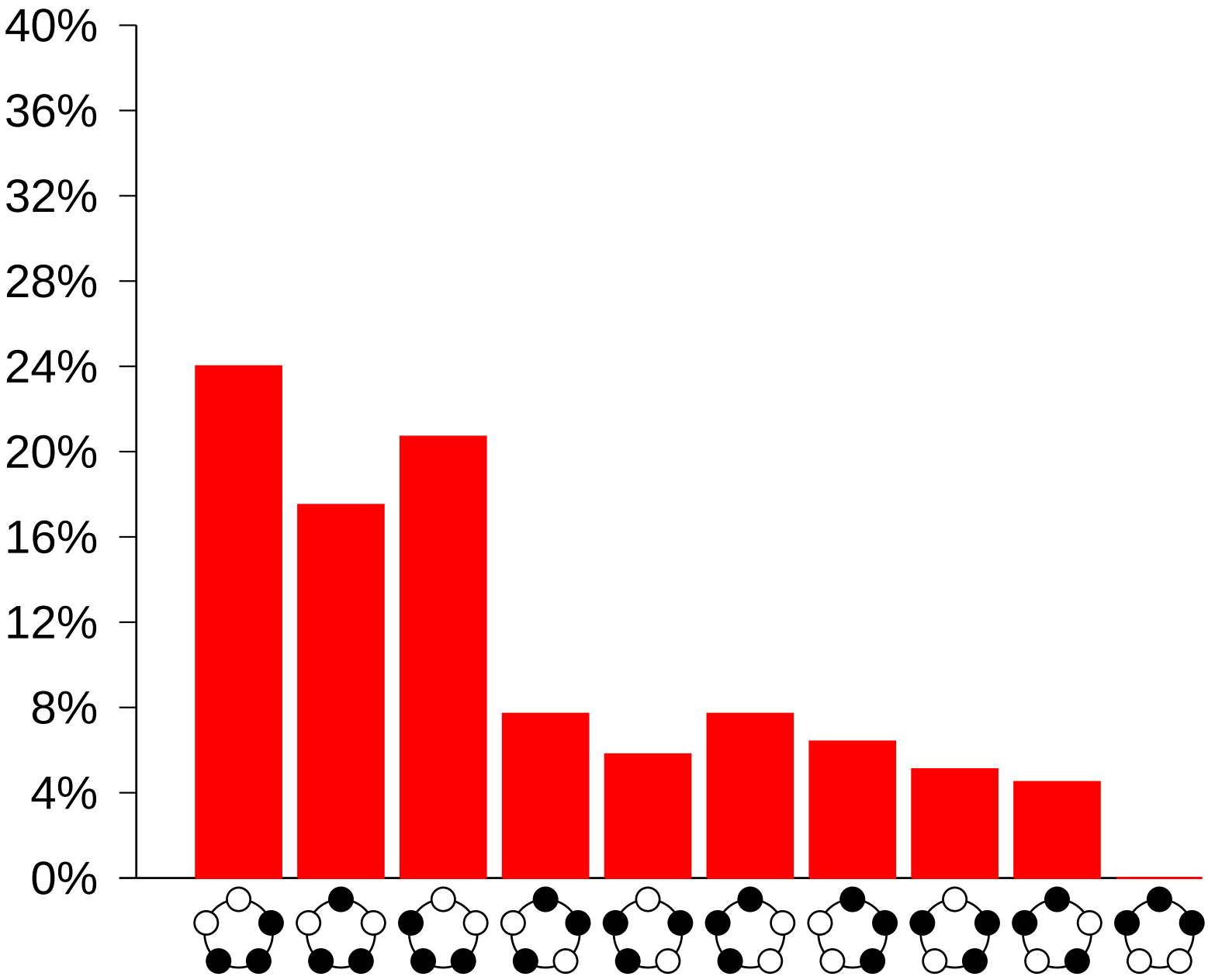
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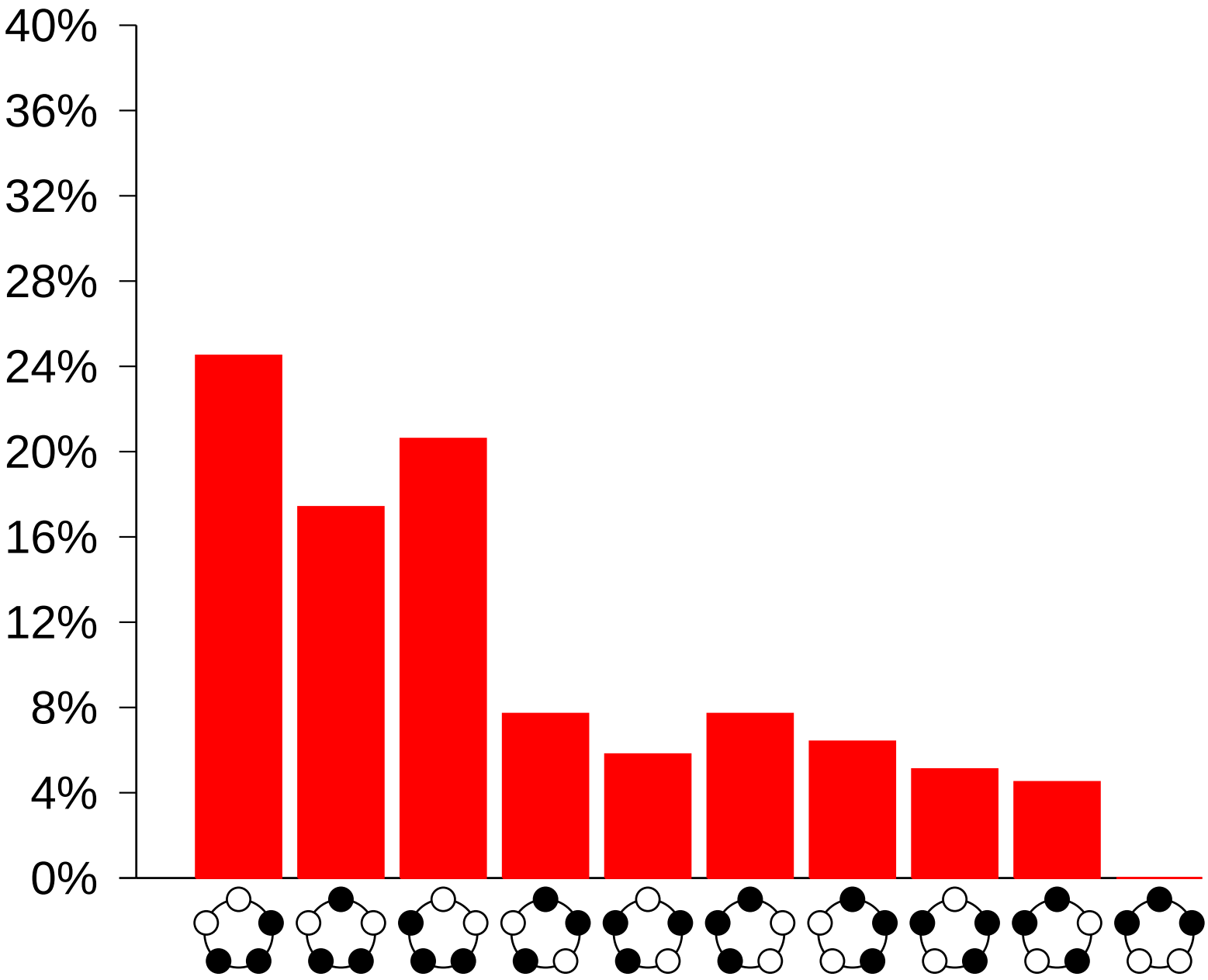
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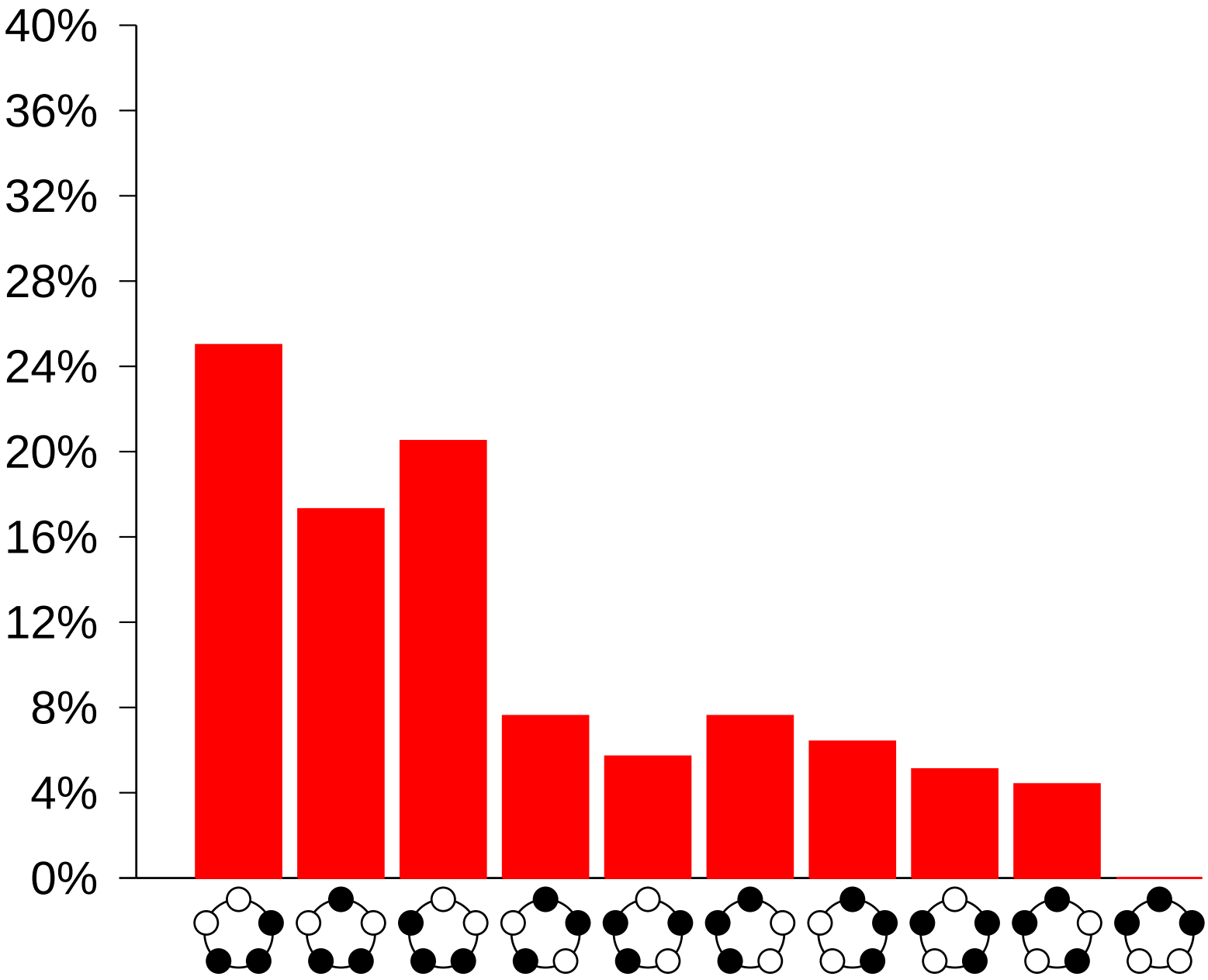
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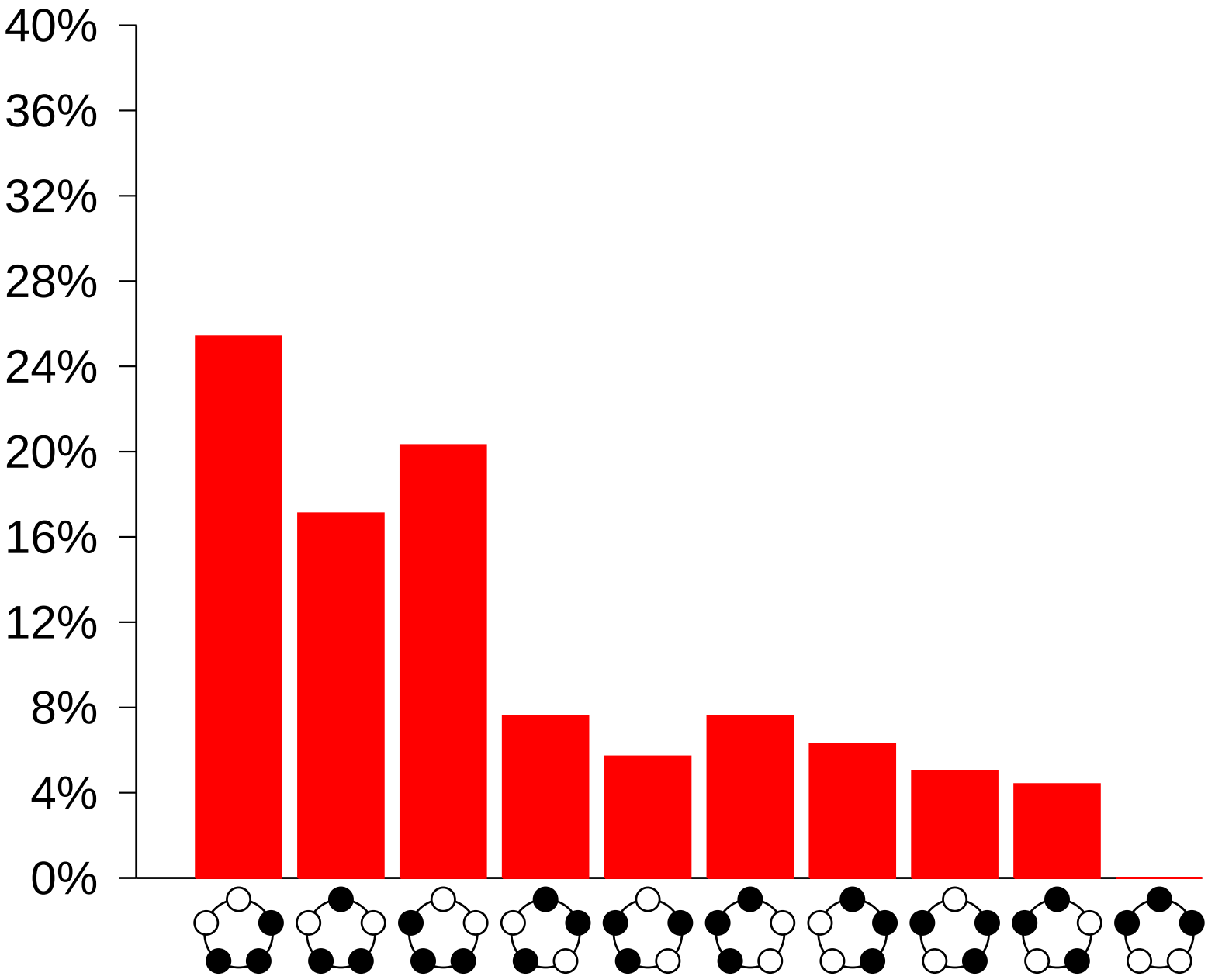
Stationary distribution



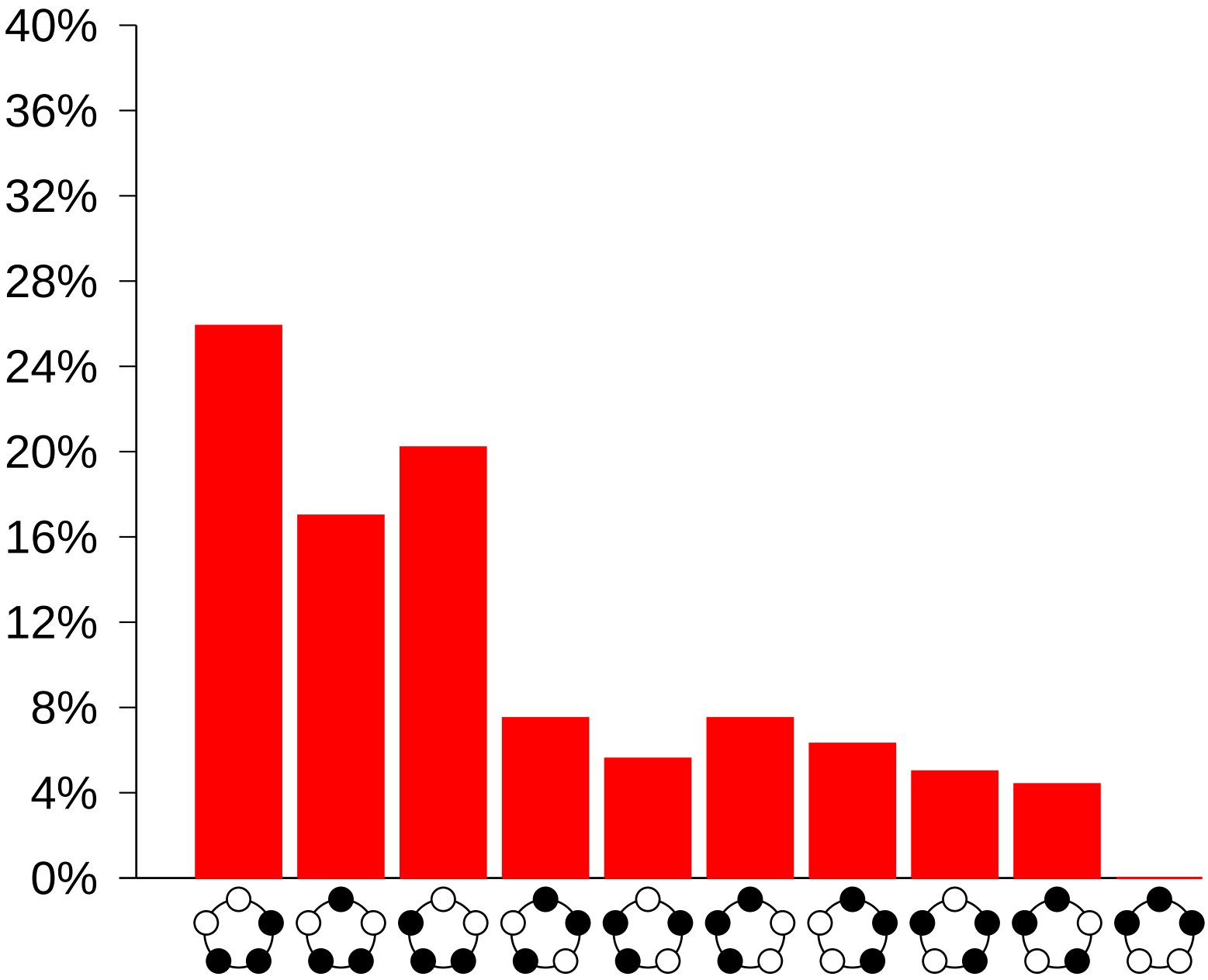
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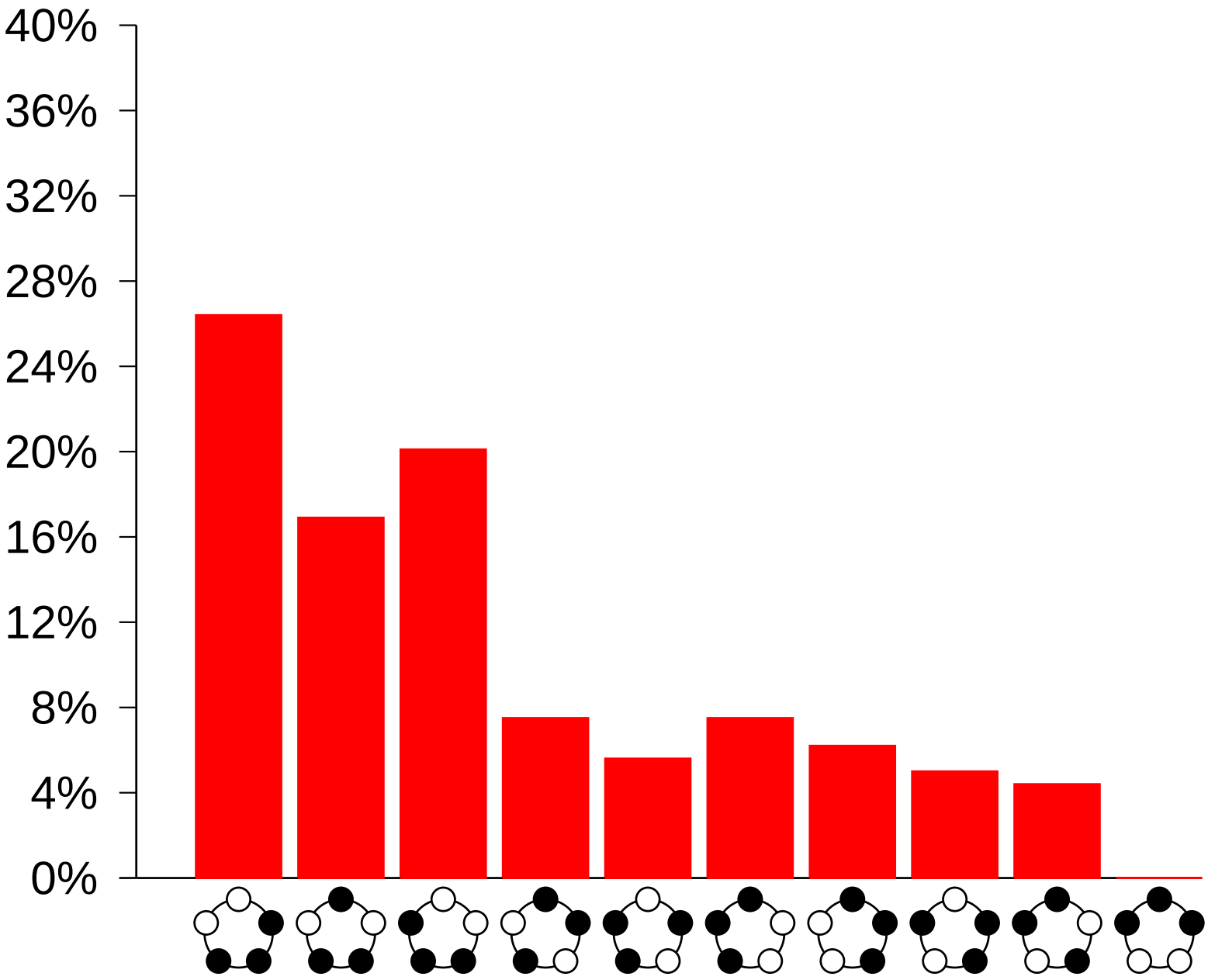
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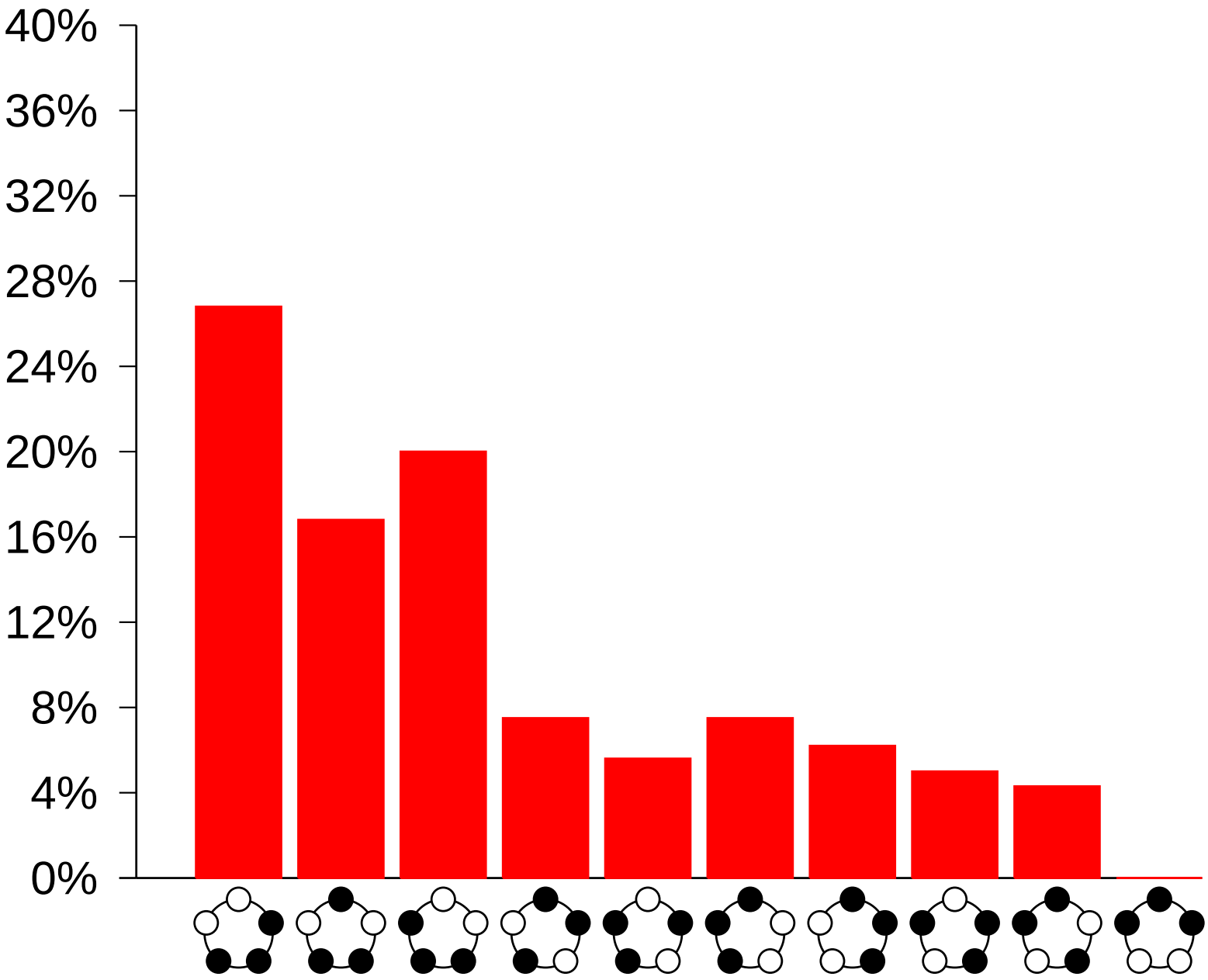
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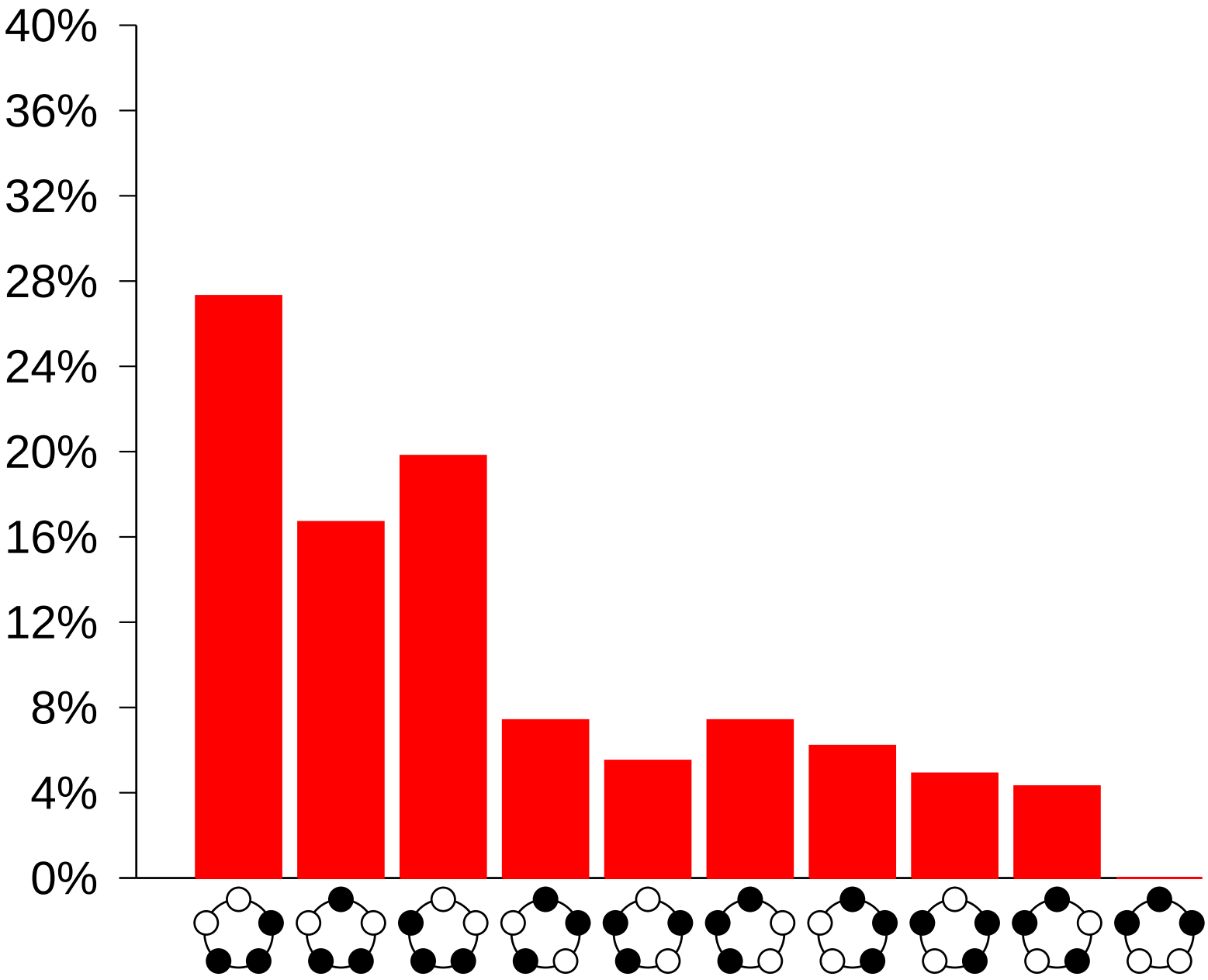
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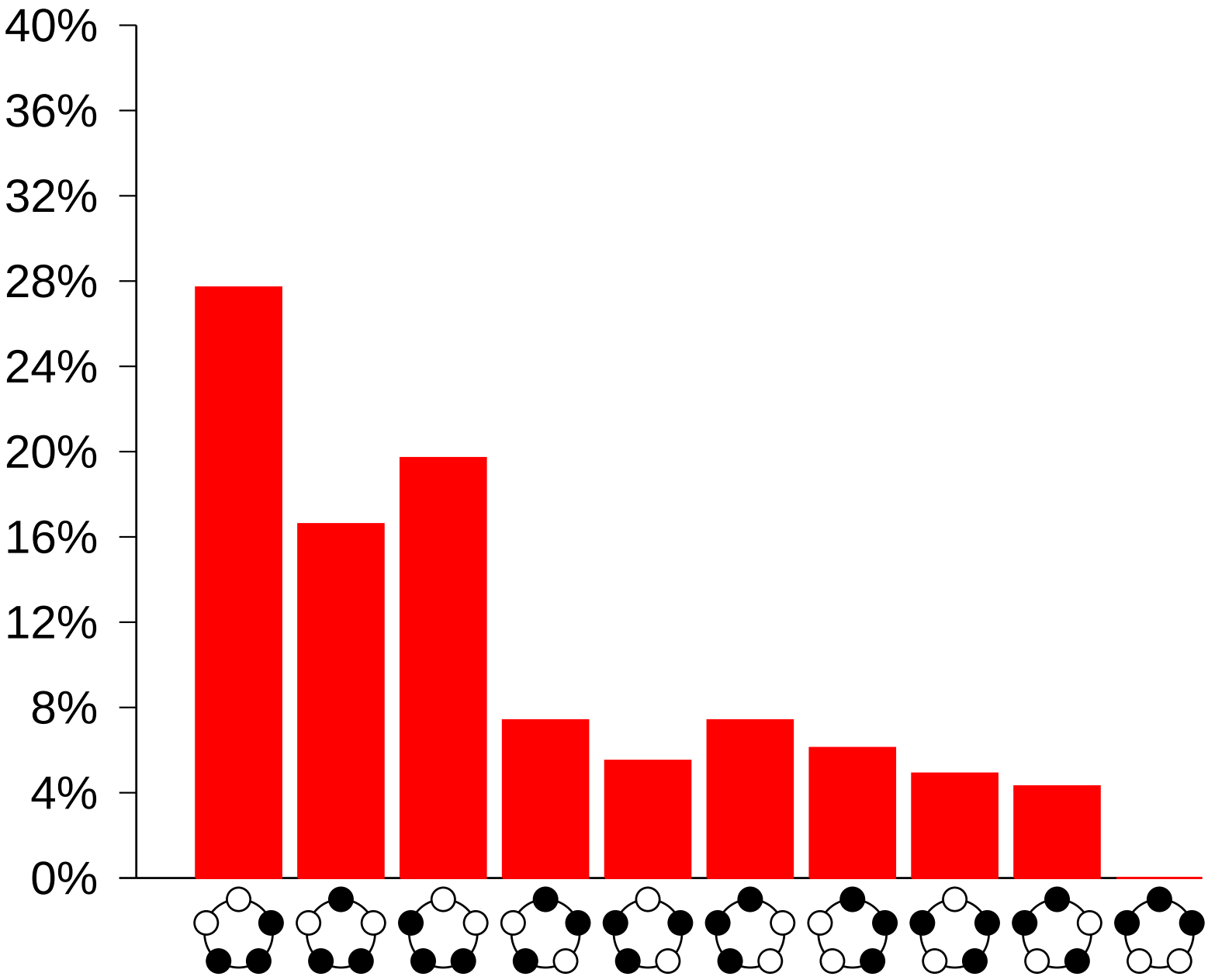
Stationary distribution



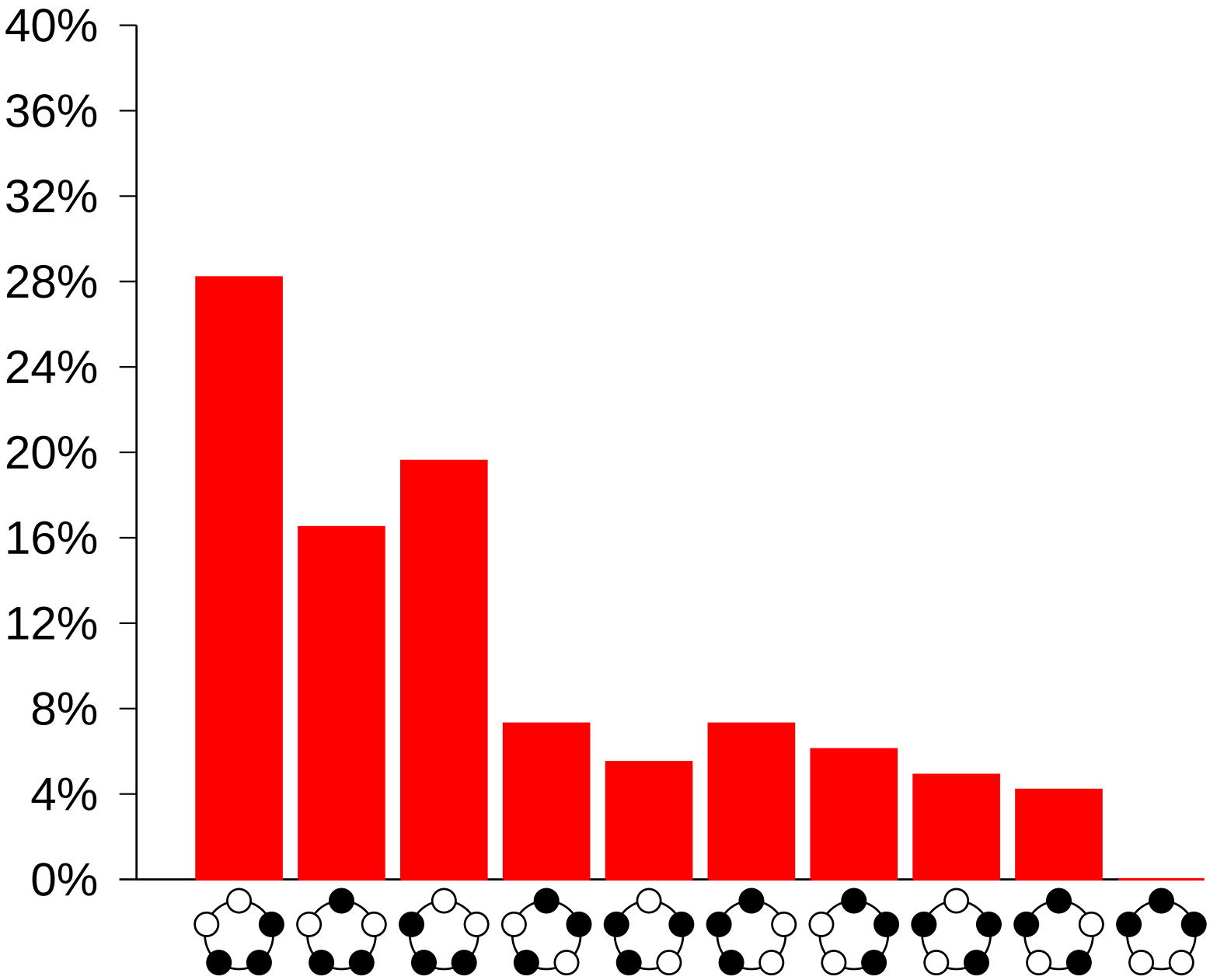
Stationary distribution



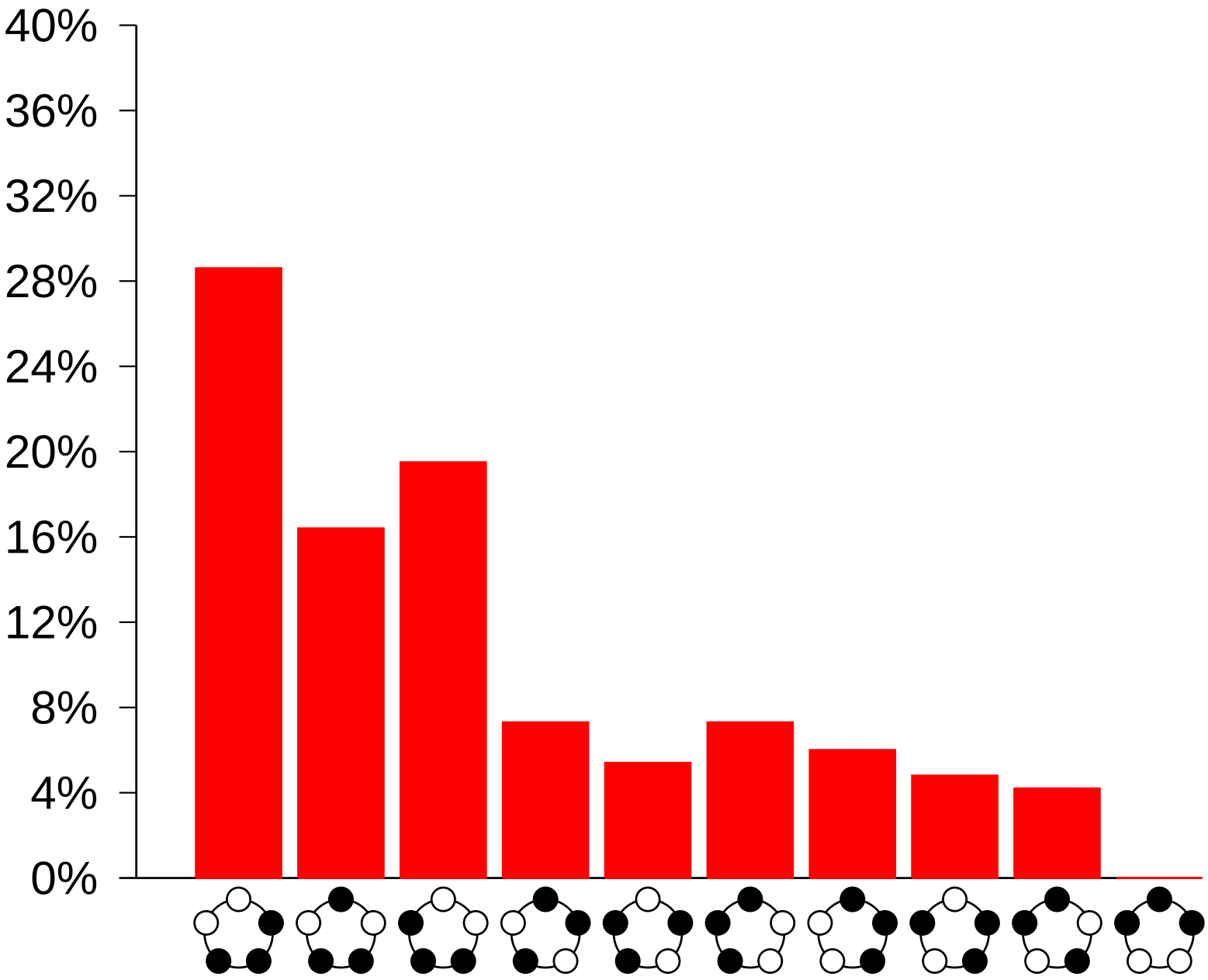
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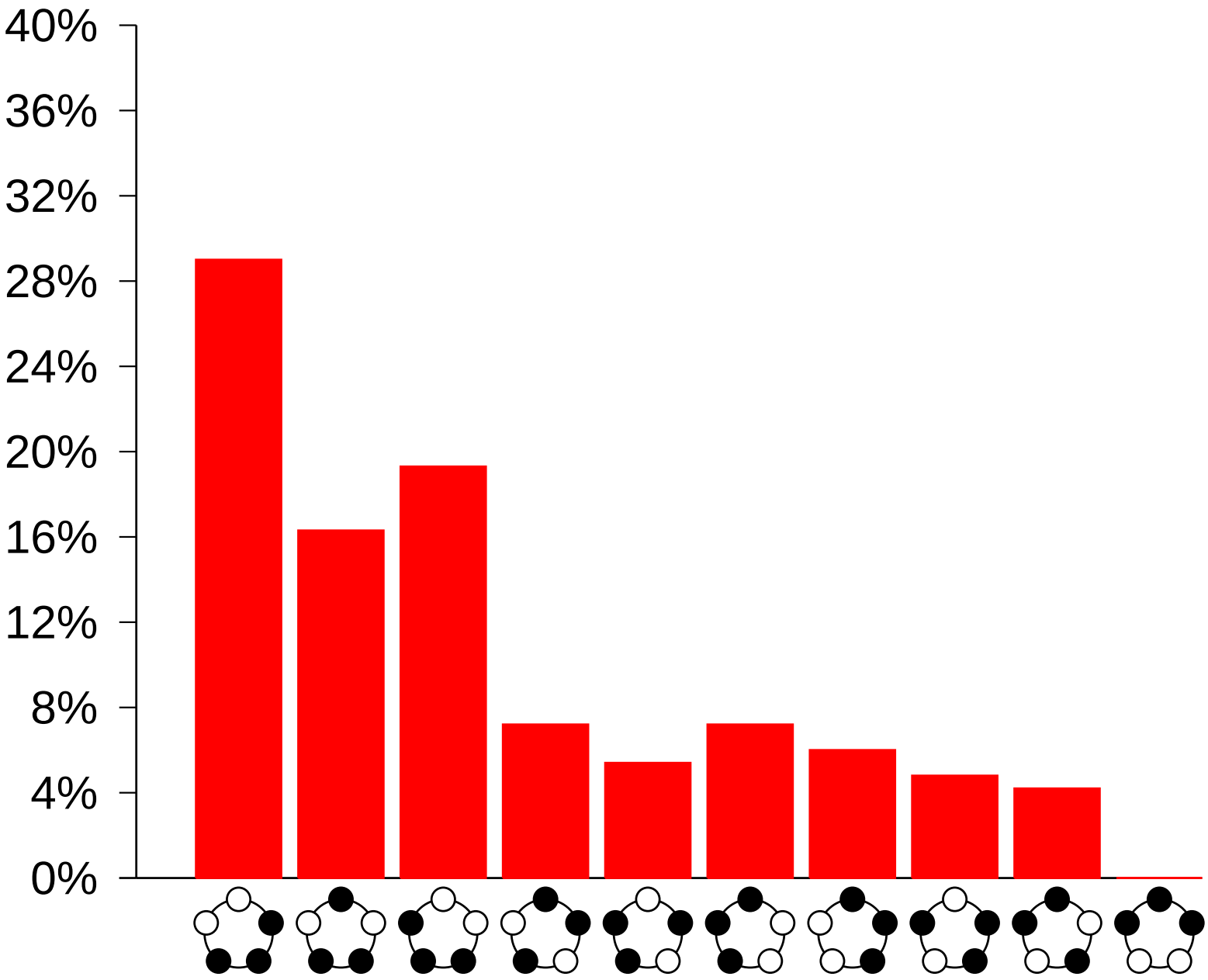
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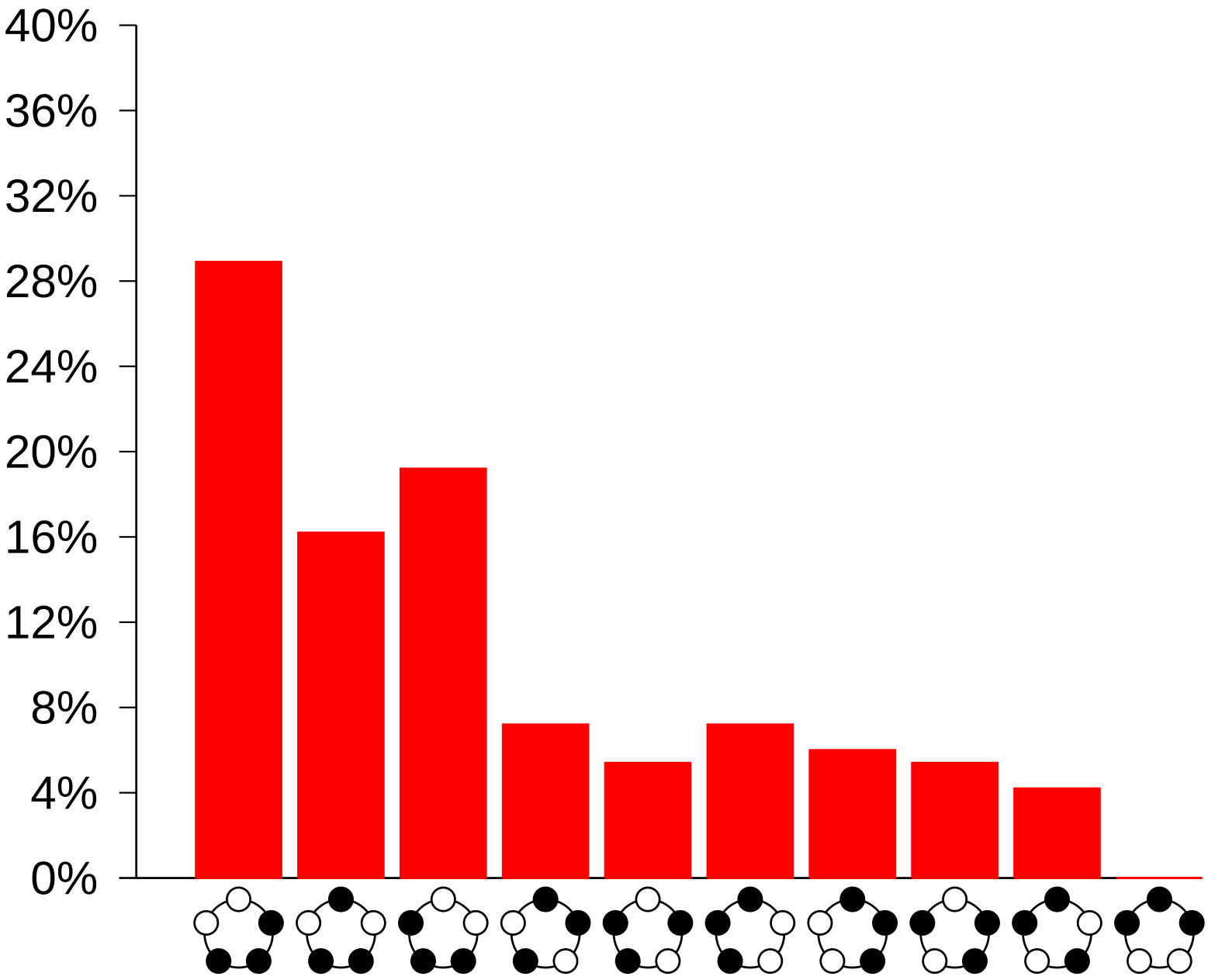
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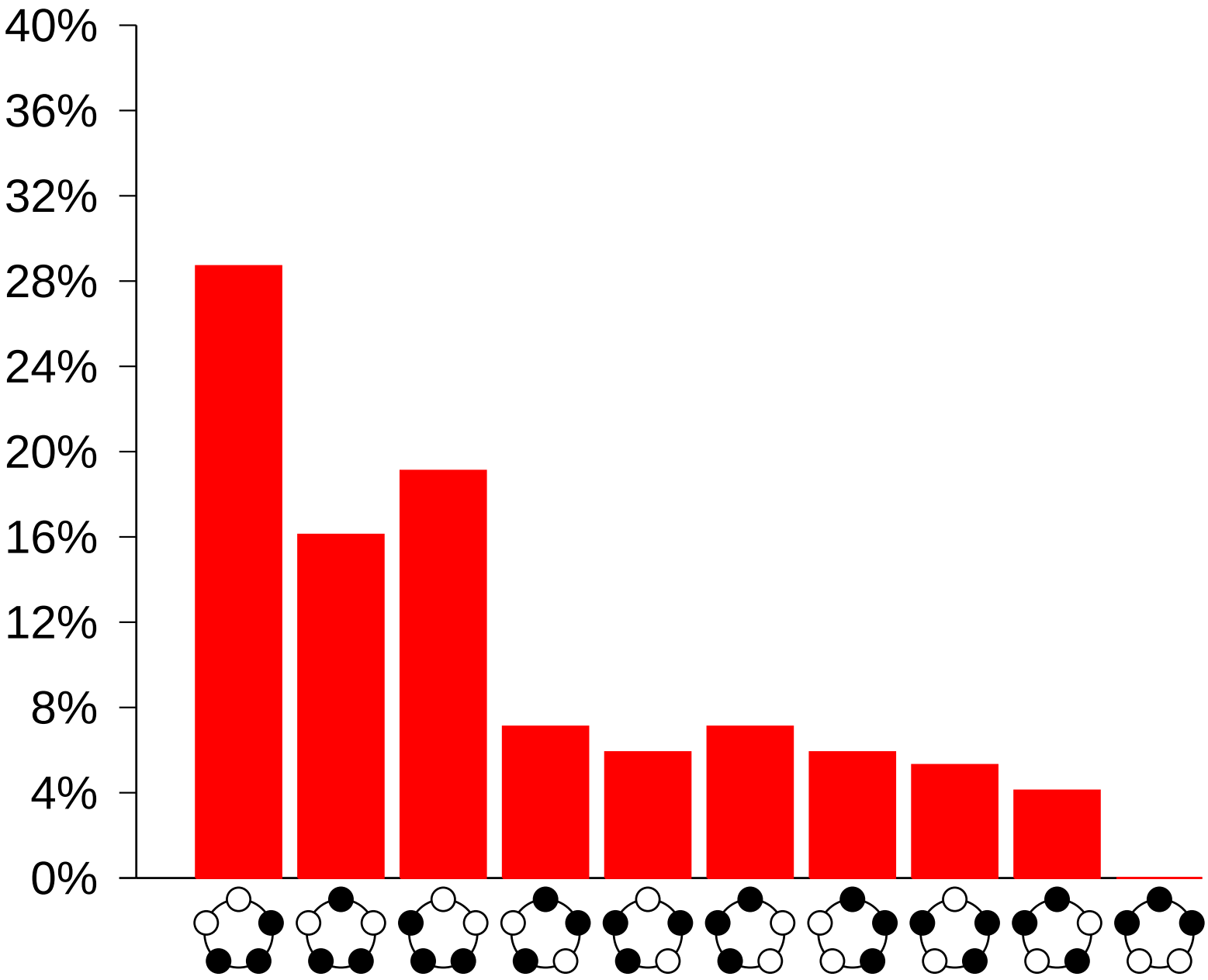
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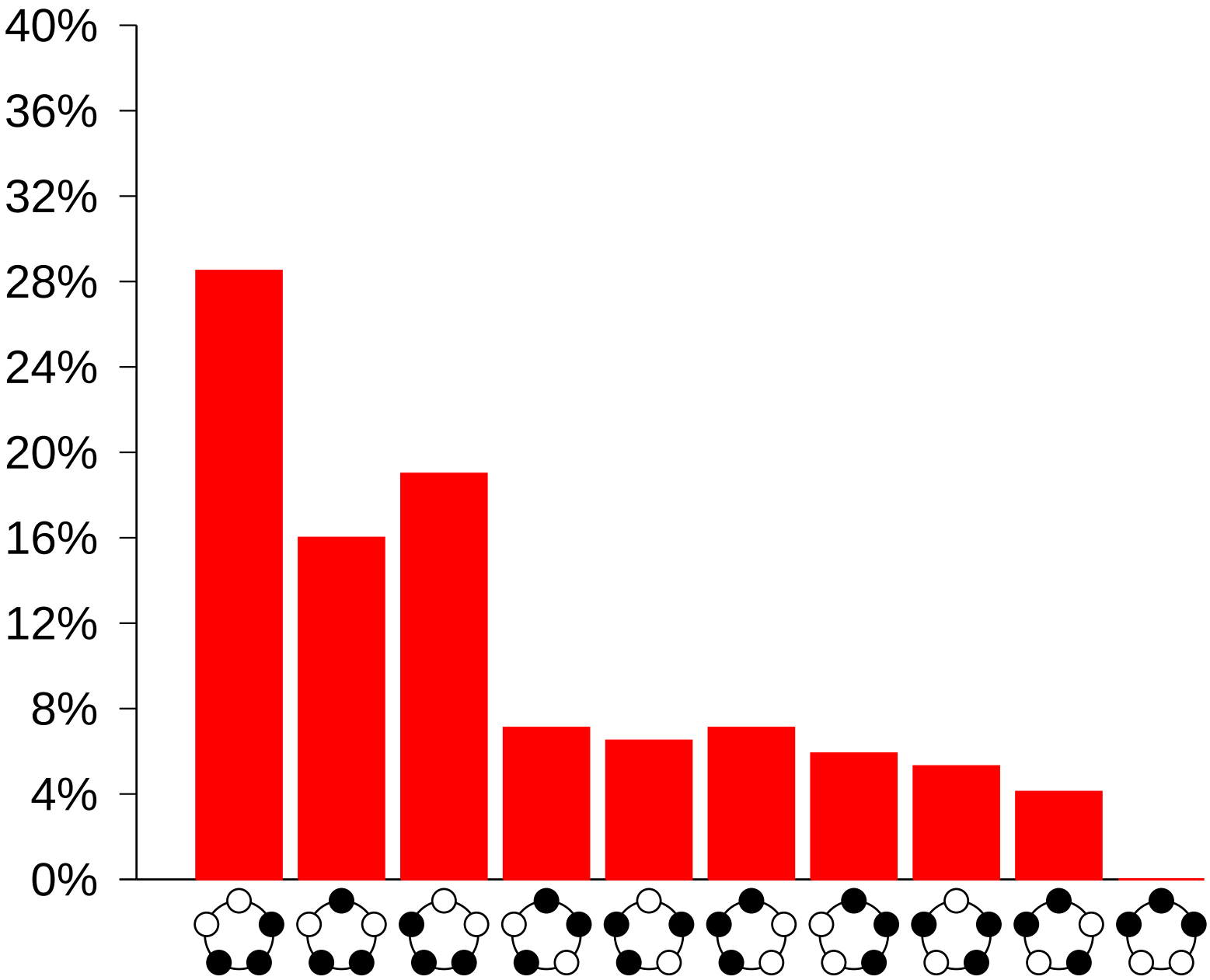
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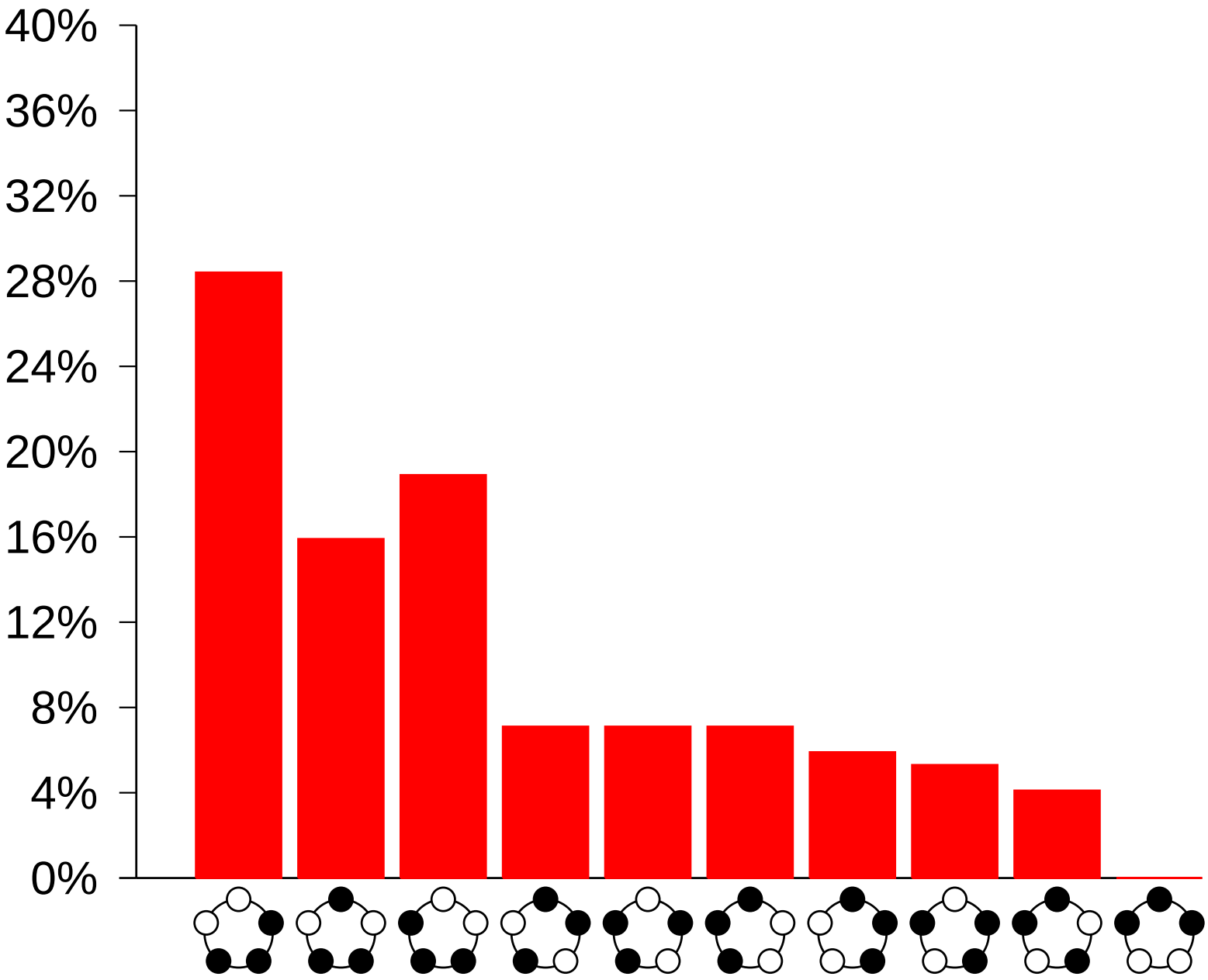
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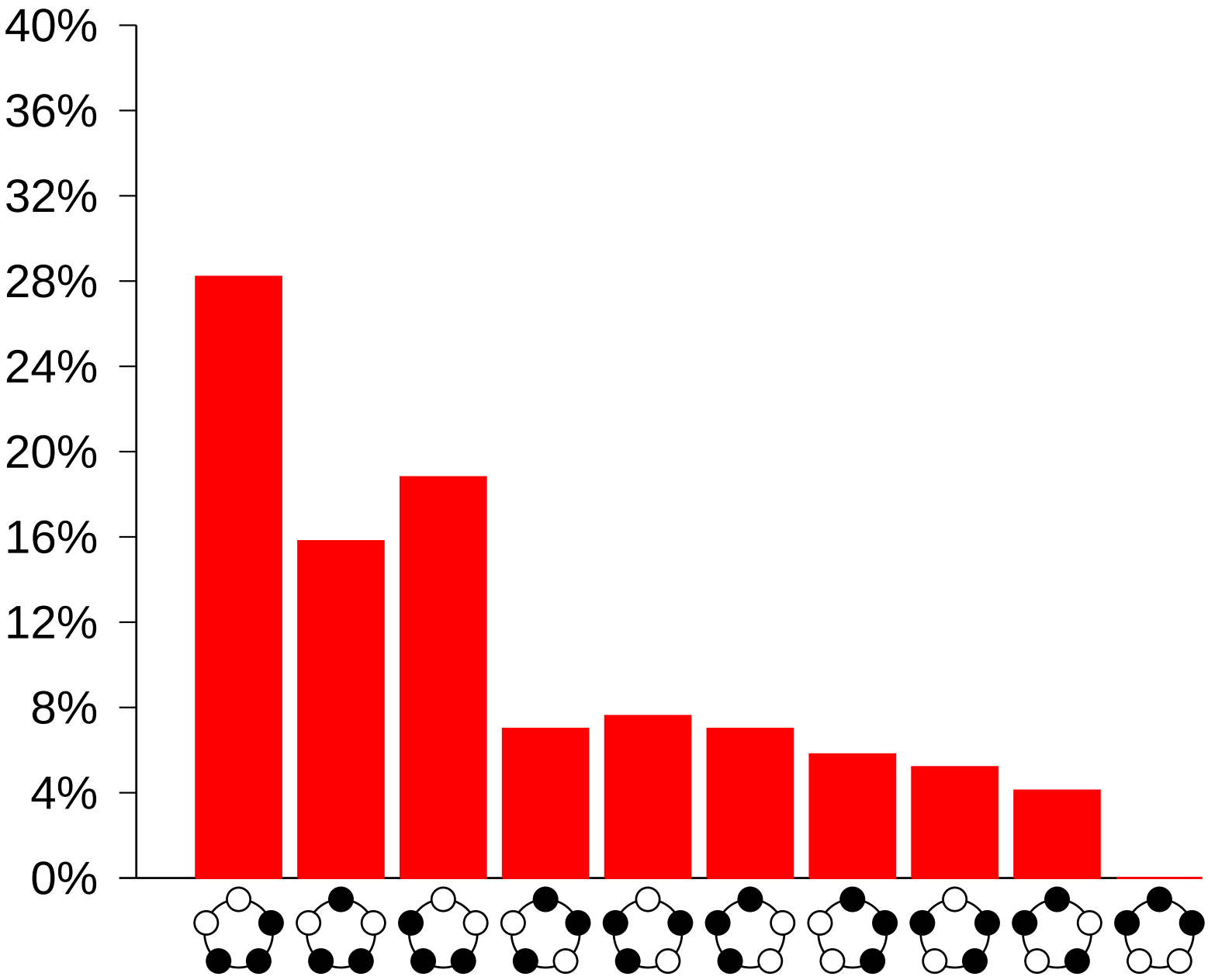
Stationary distribution



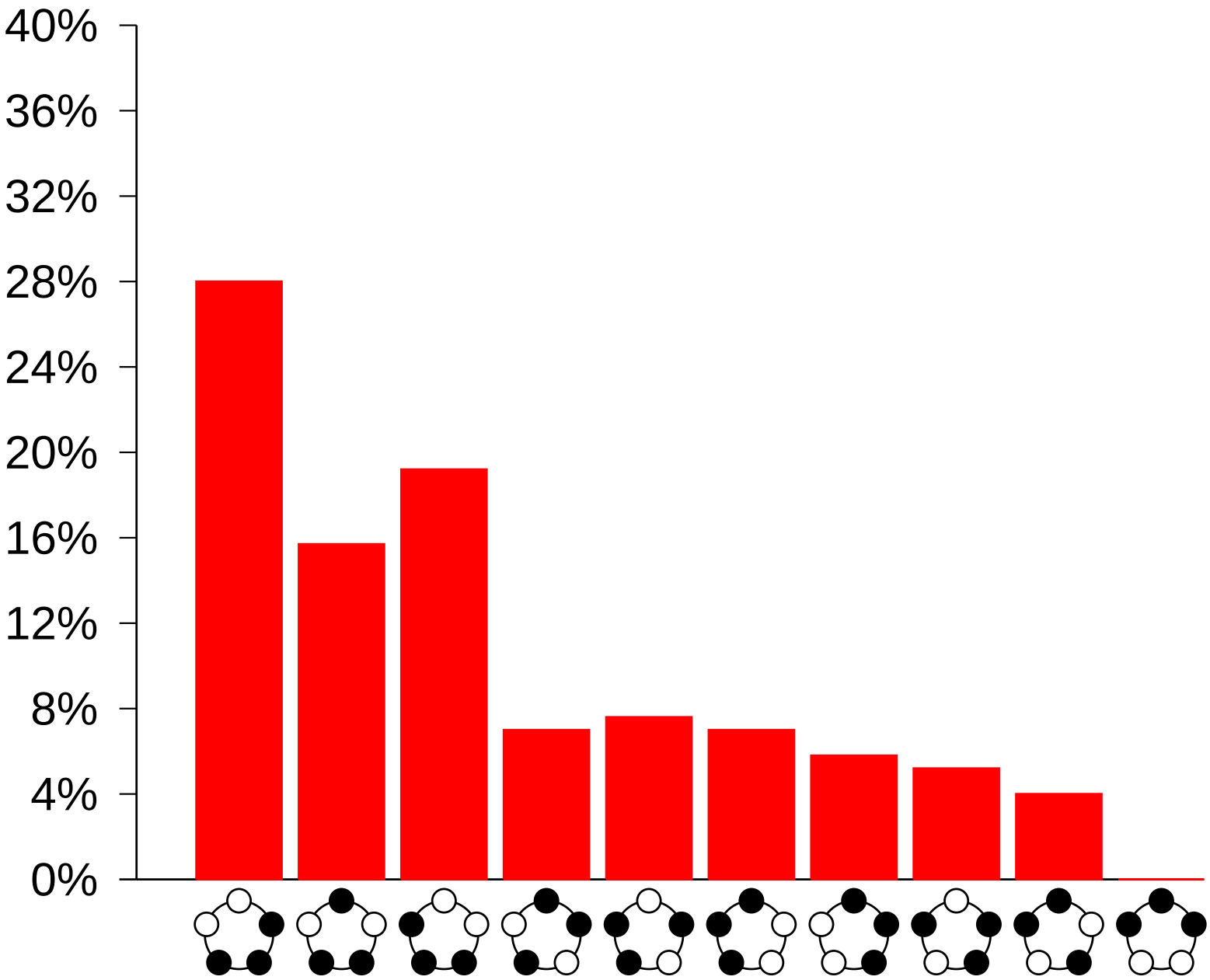
Stationary distribution



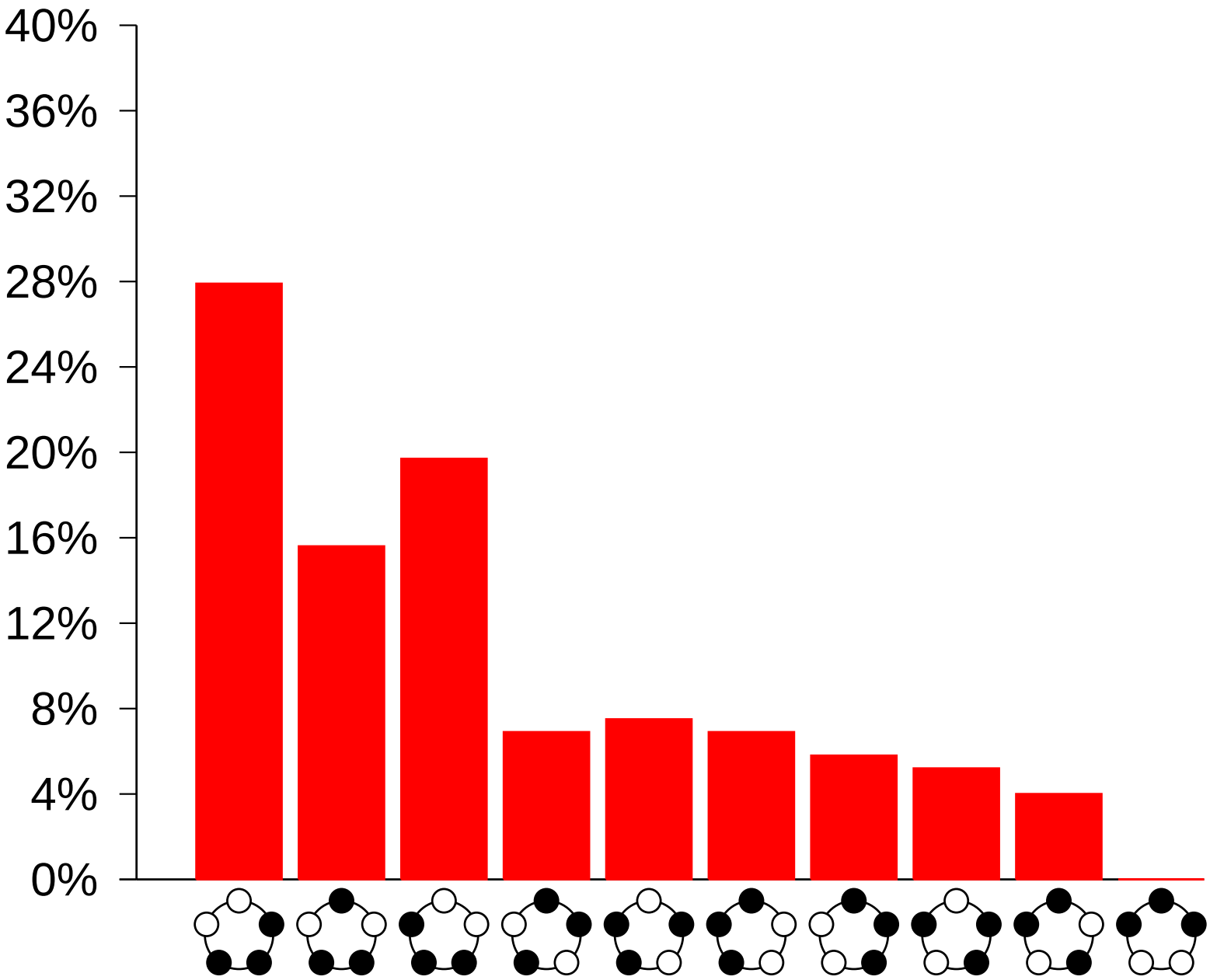
Stationary distribution



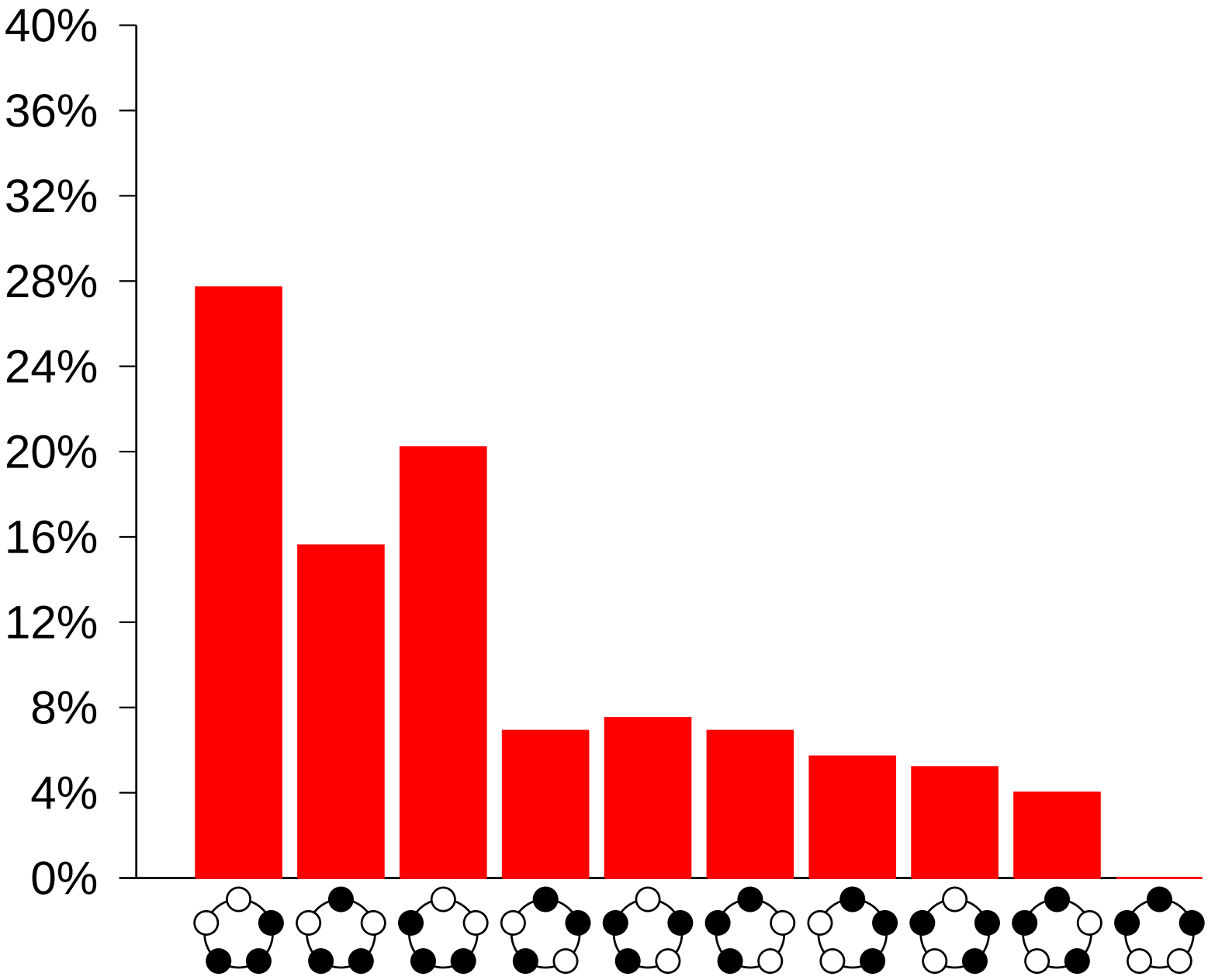
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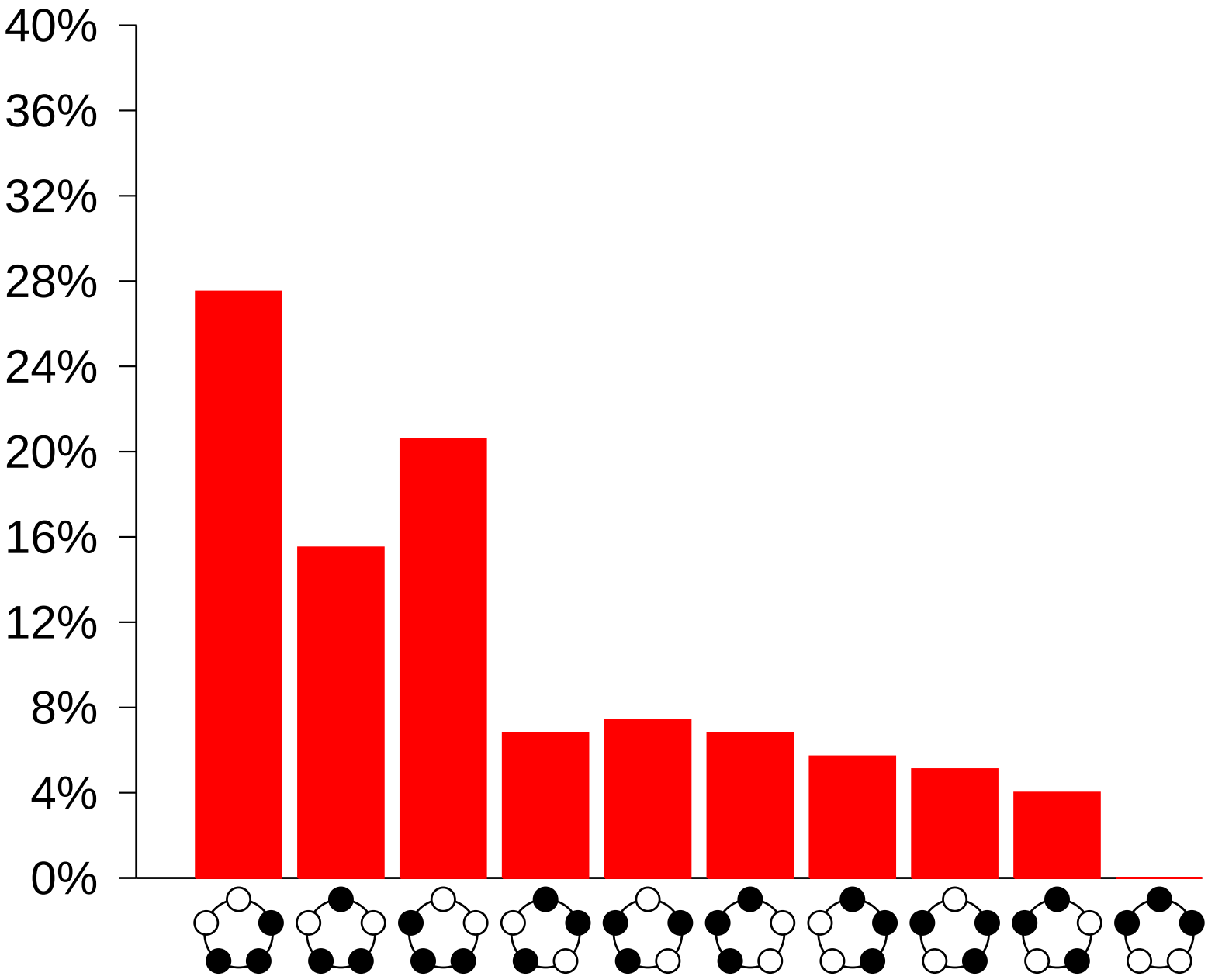
Stationary distribution



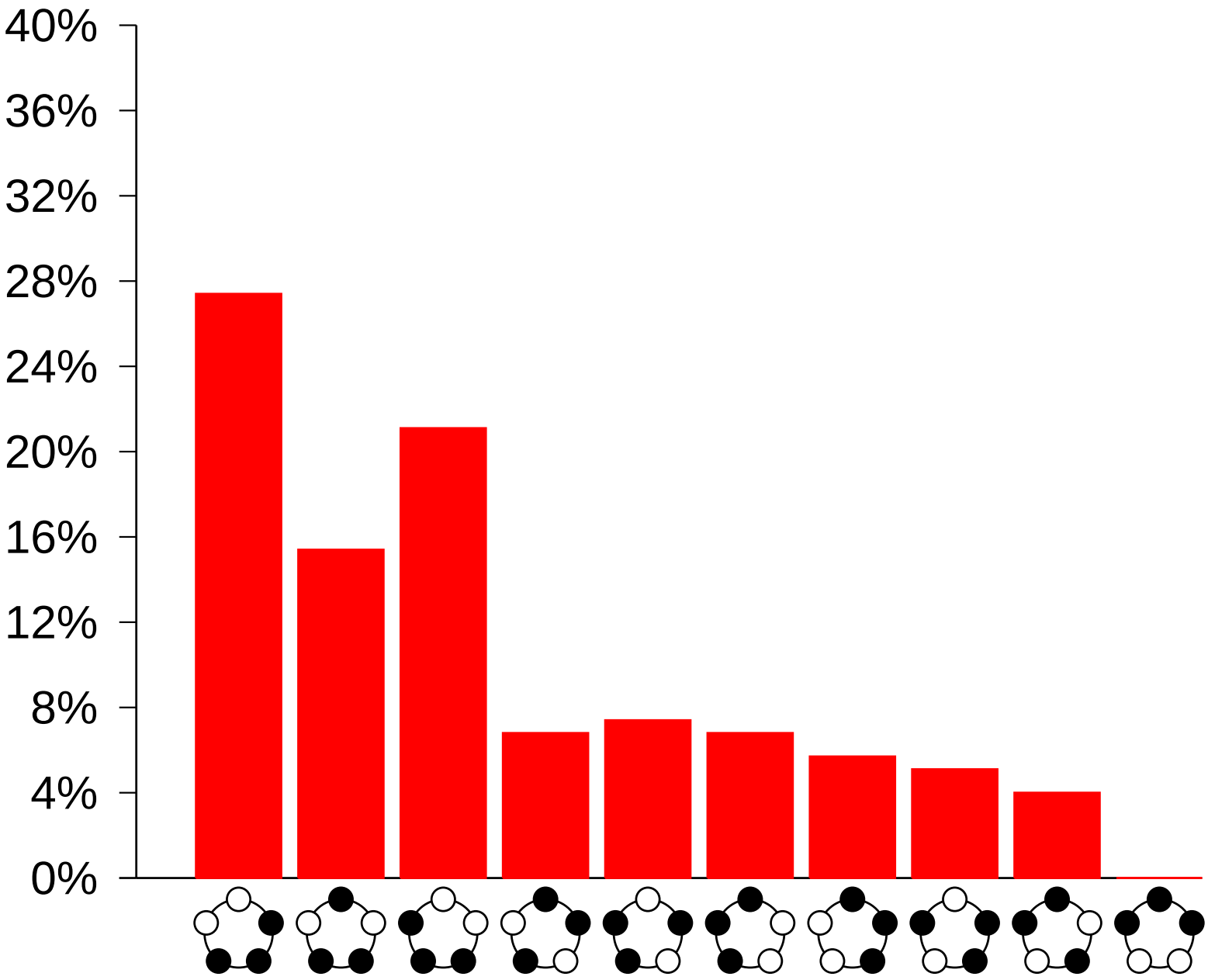
Stationary distribution



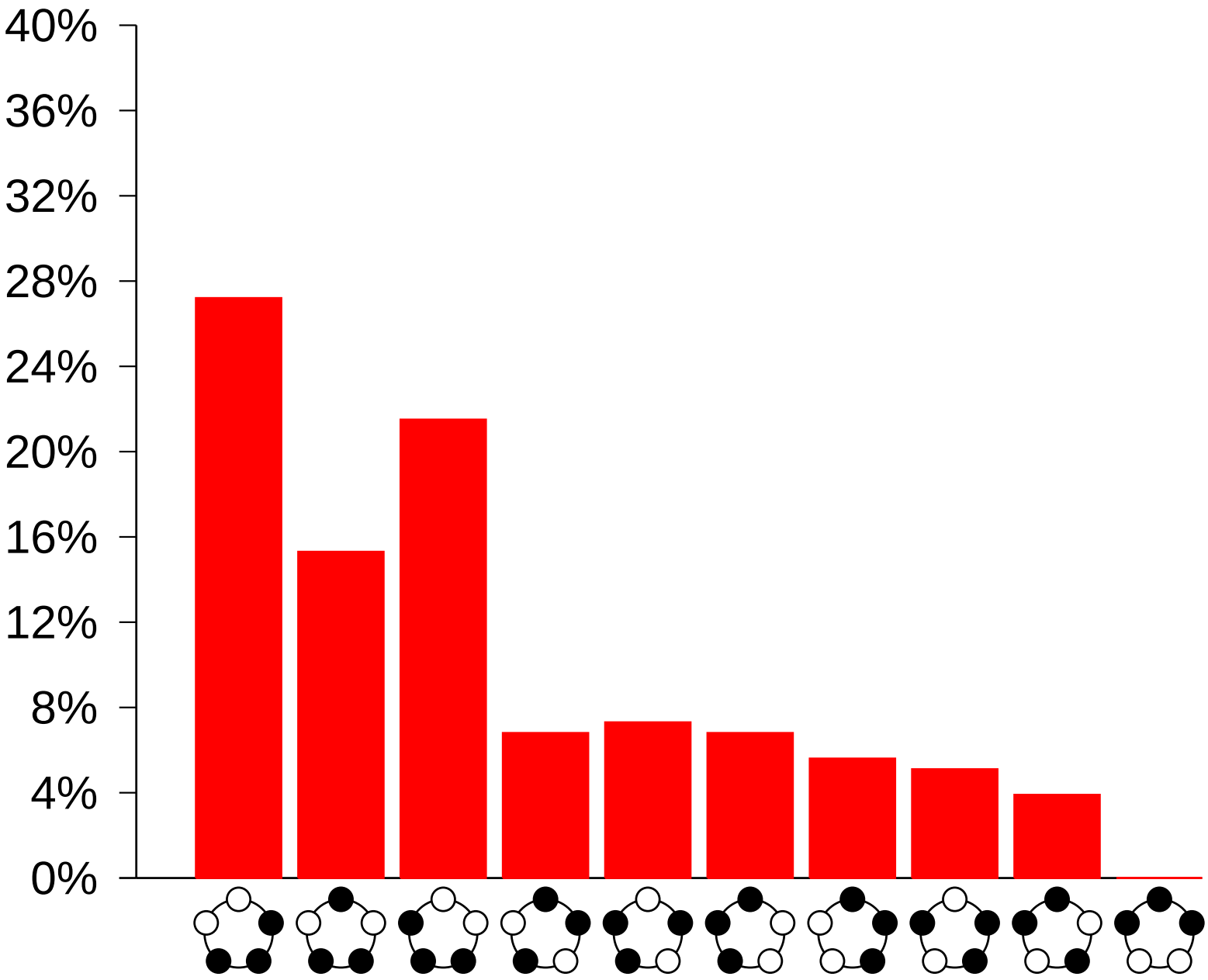
Stationary distribution



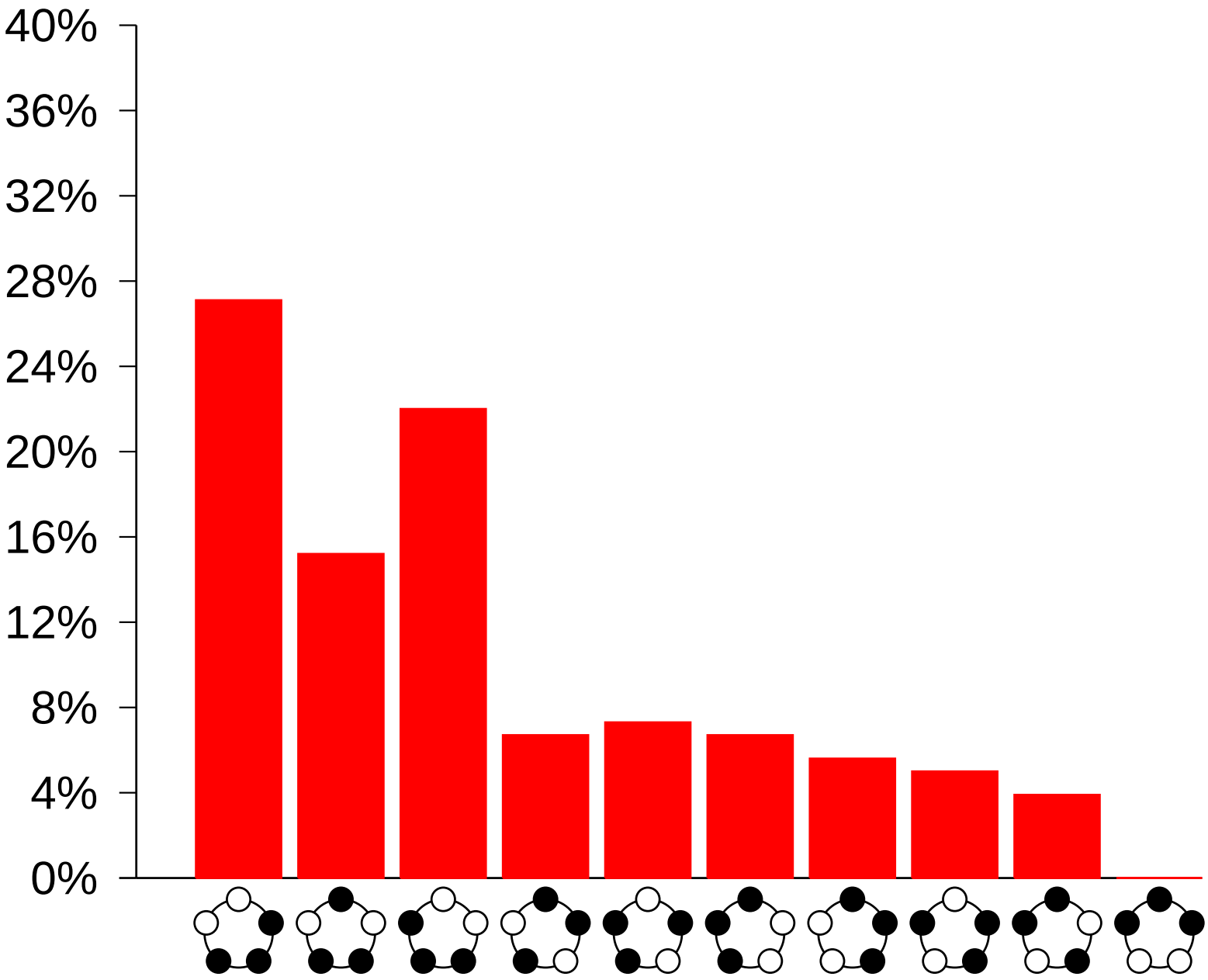
Stationary distribution



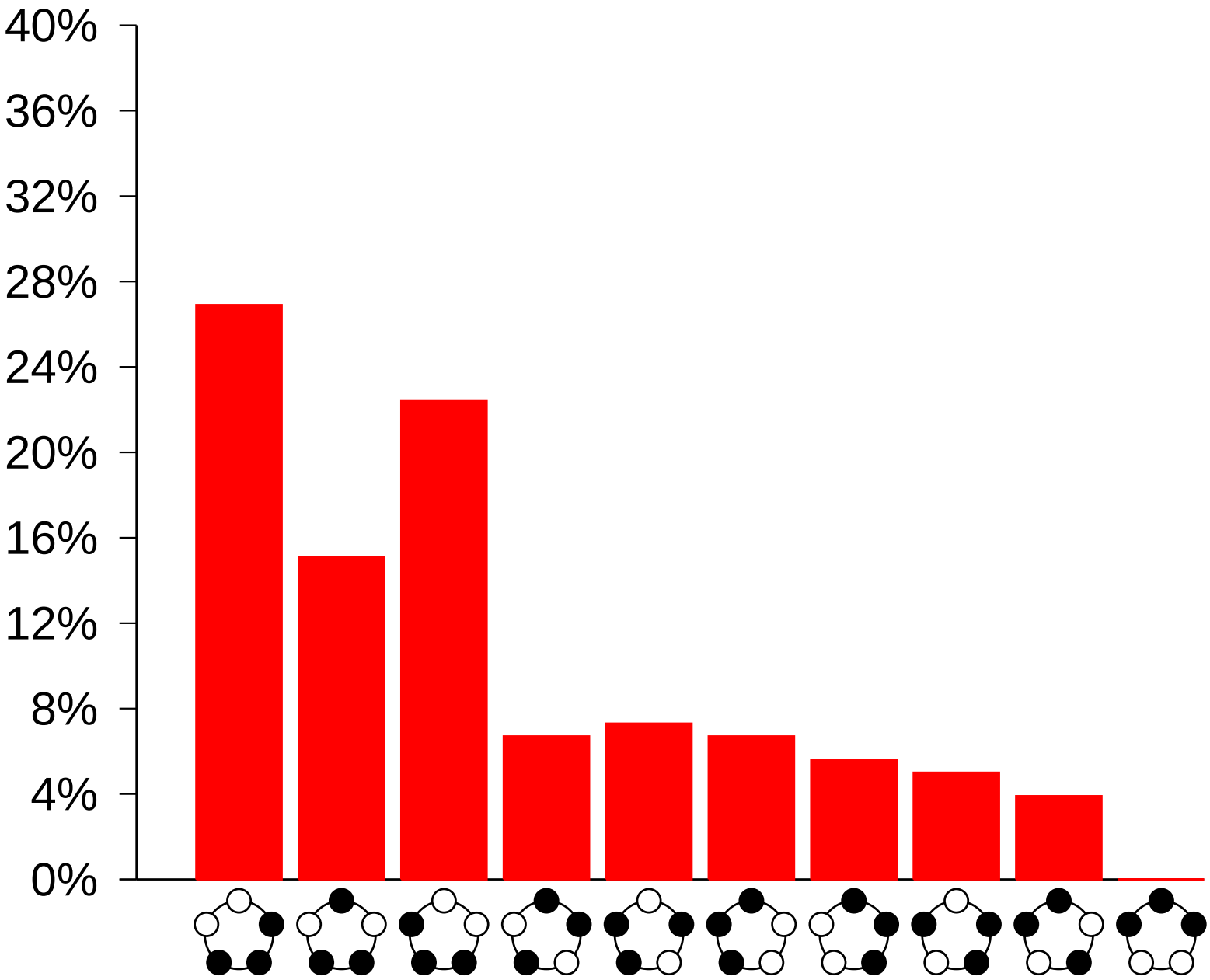
Stationary distribution



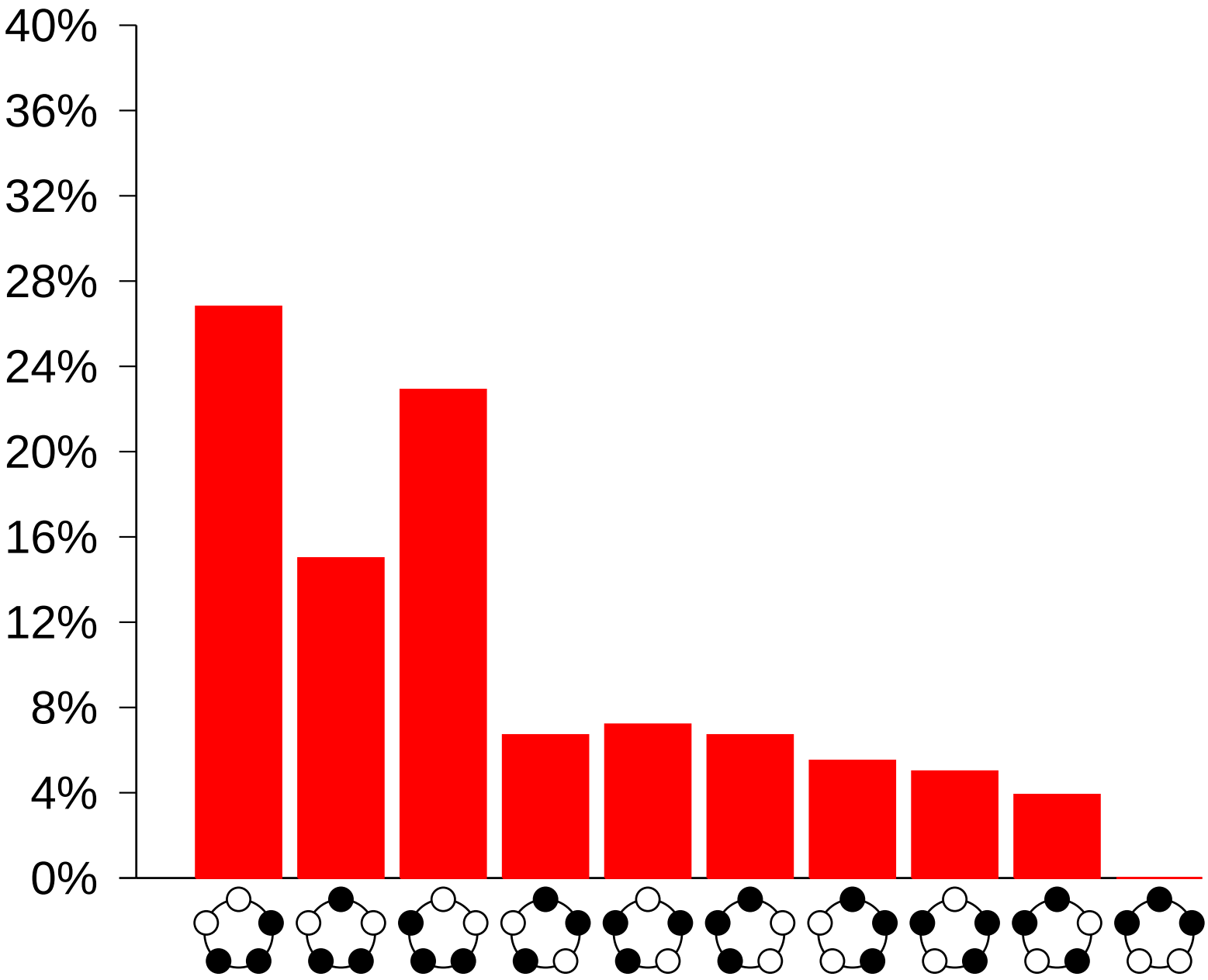
Stationary distribution



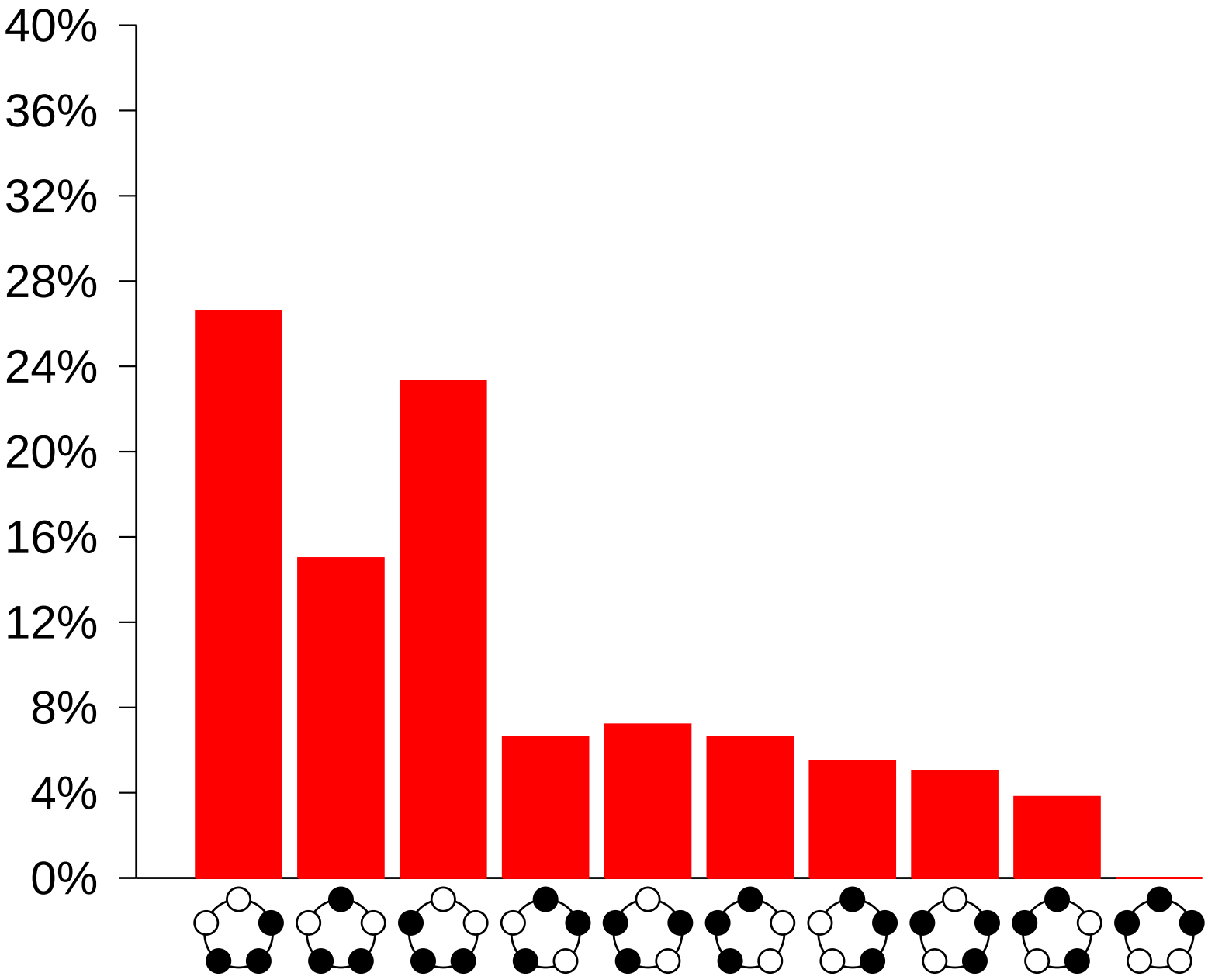
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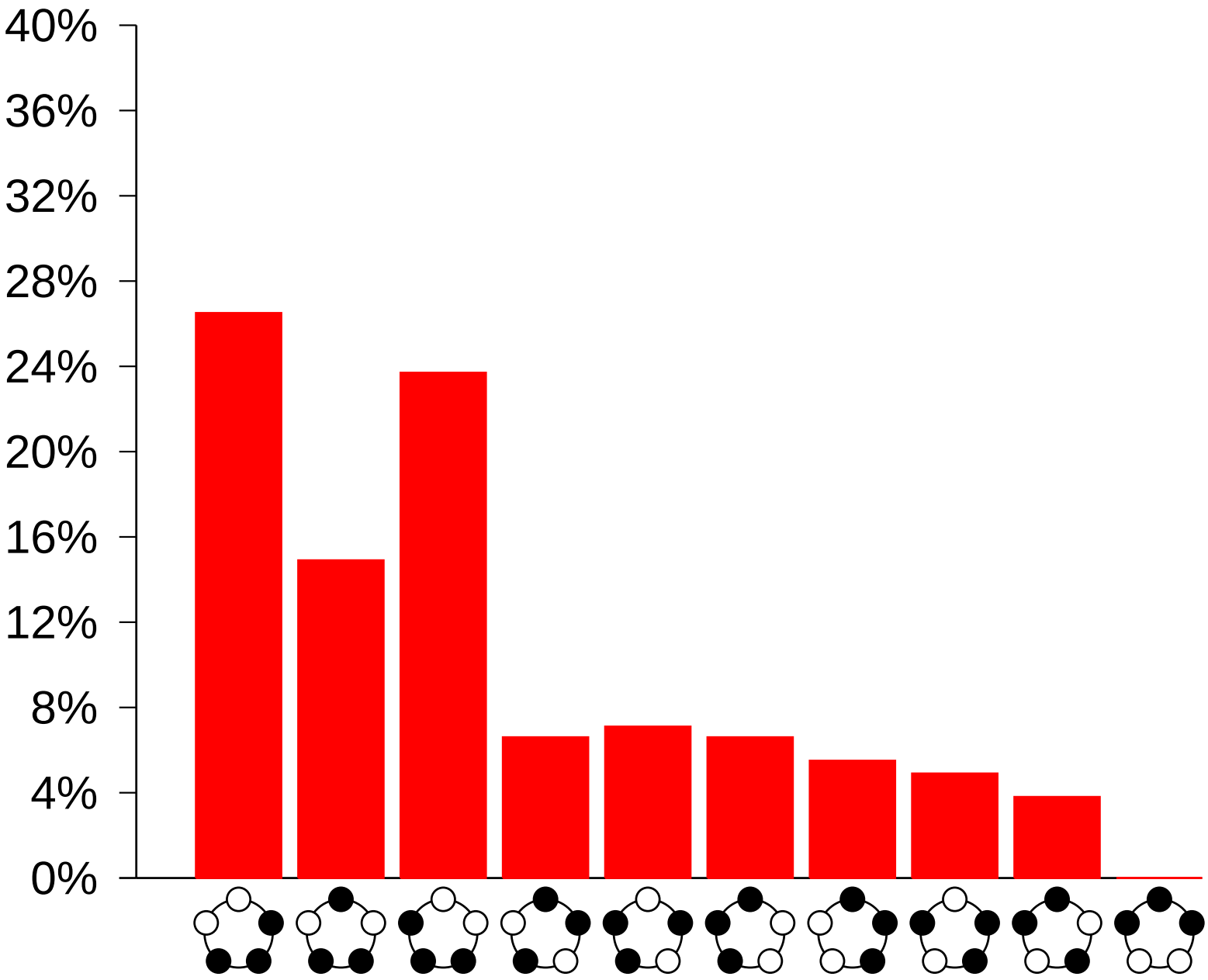
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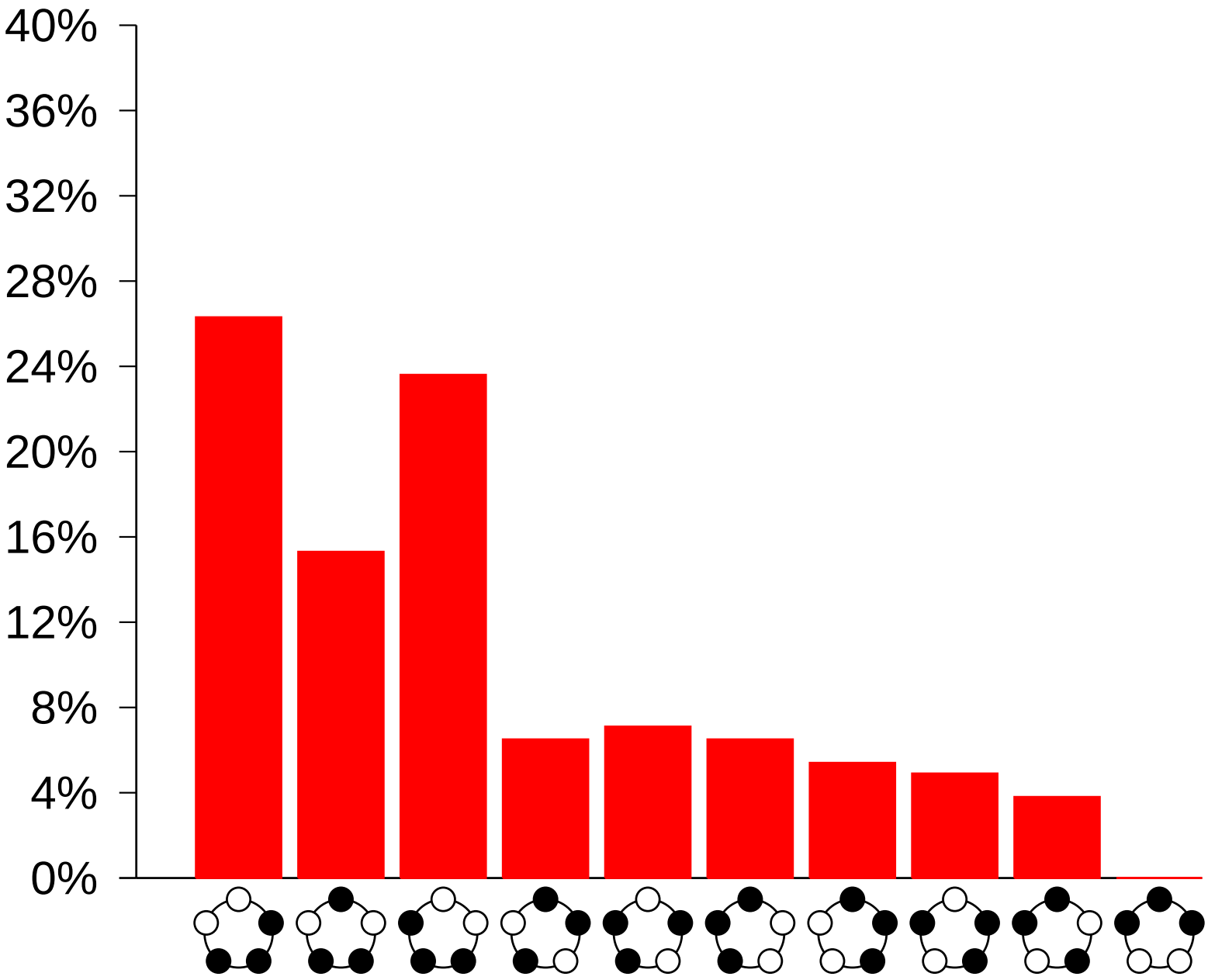
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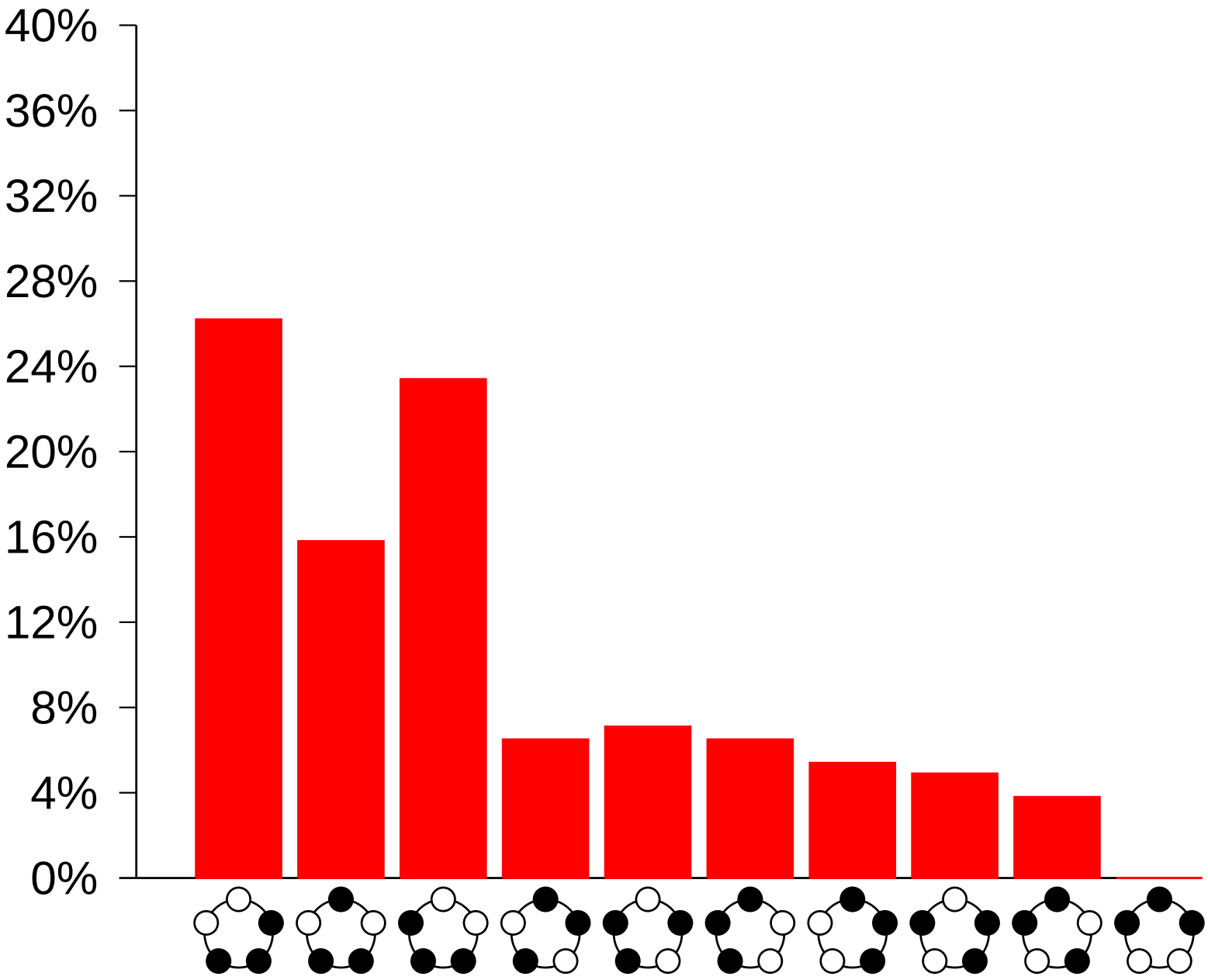
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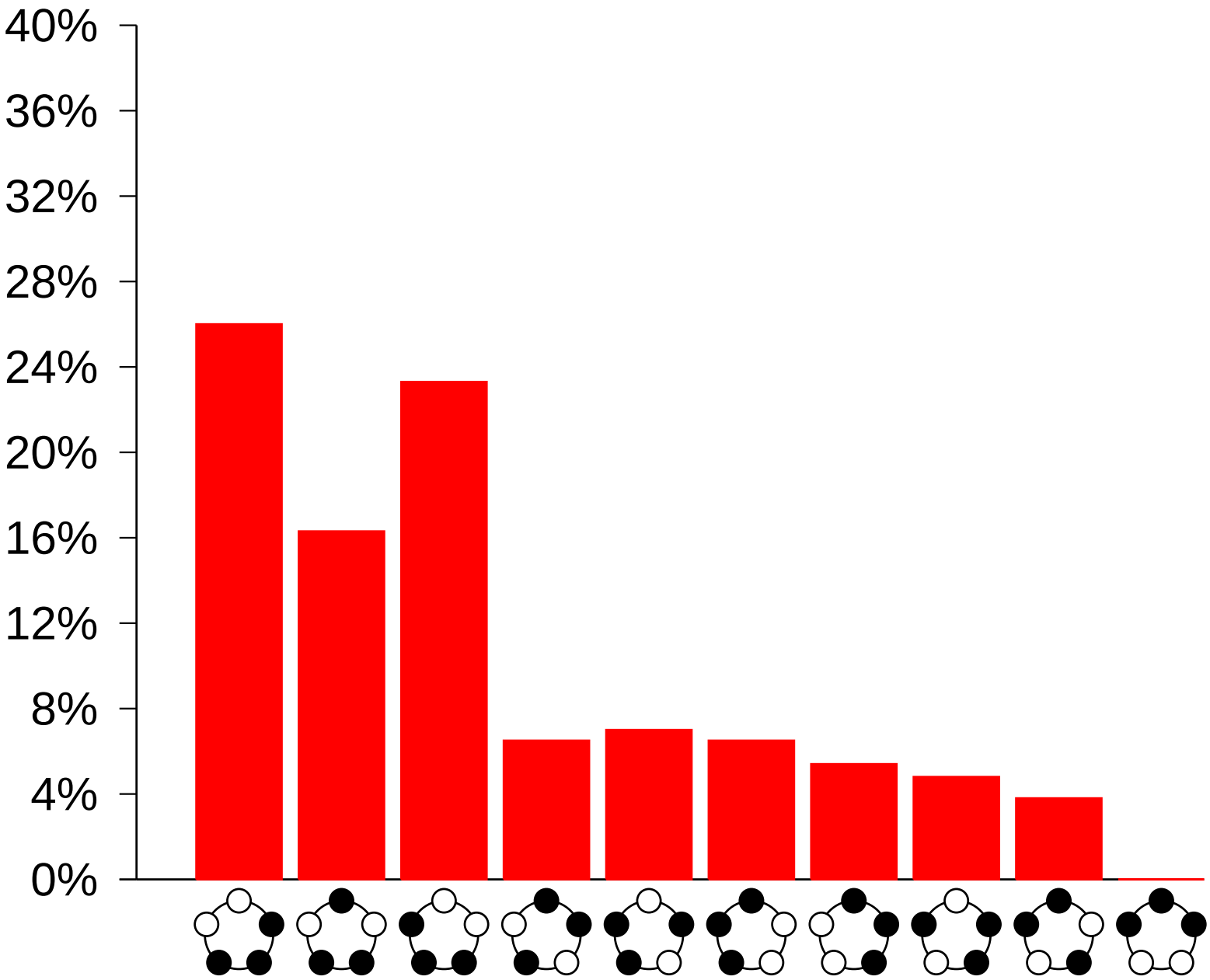
Stationary distribution



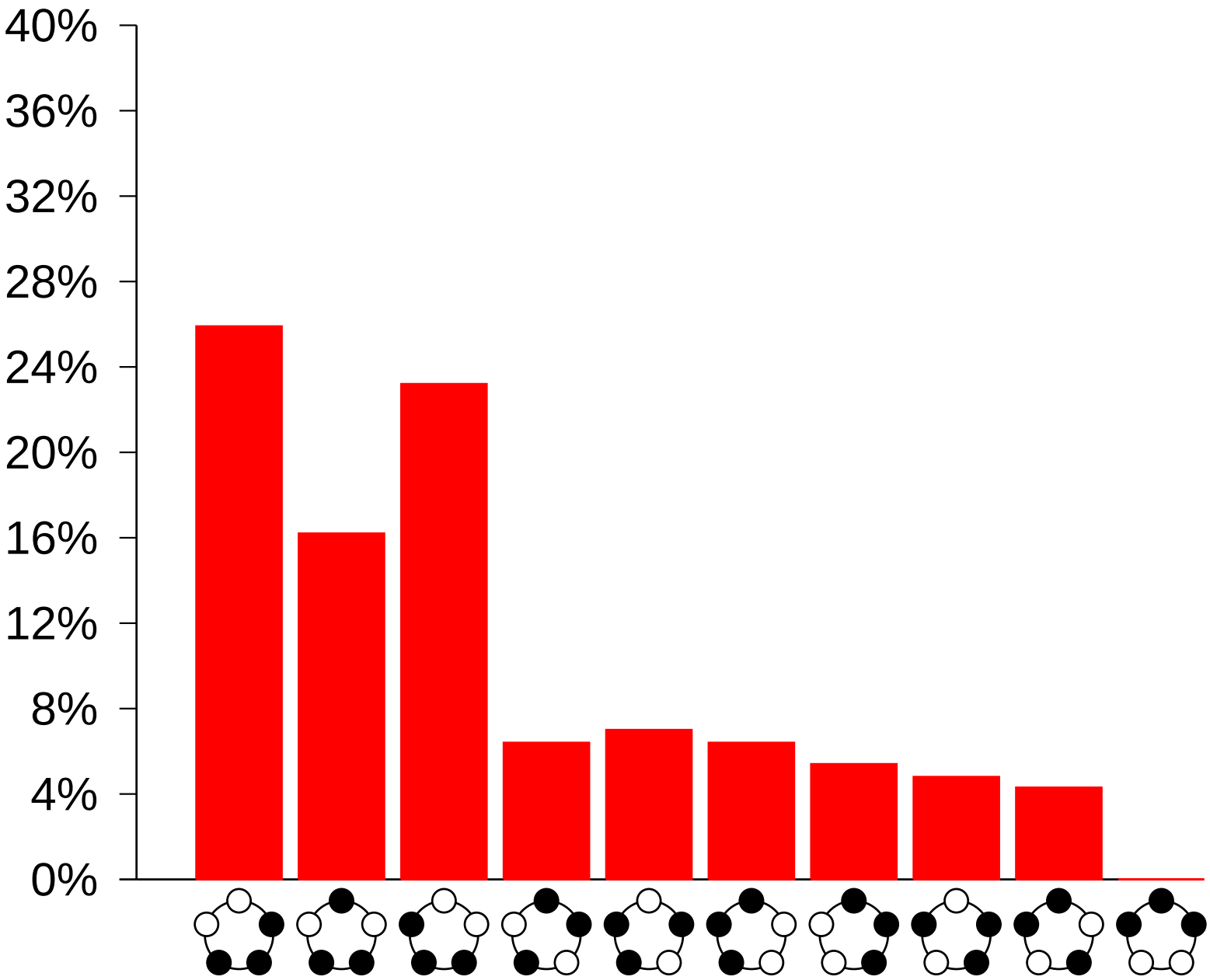
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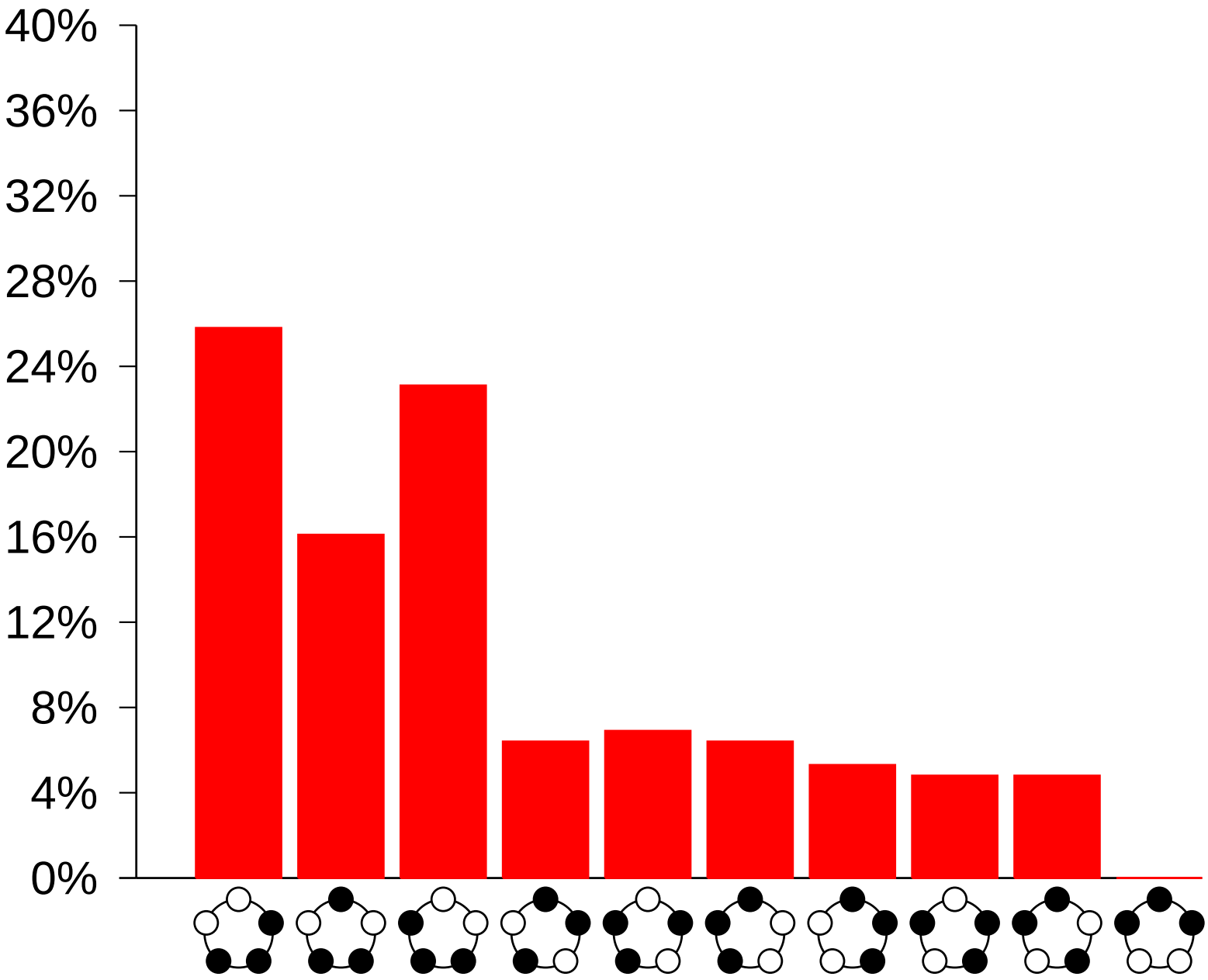
Stationary distribution



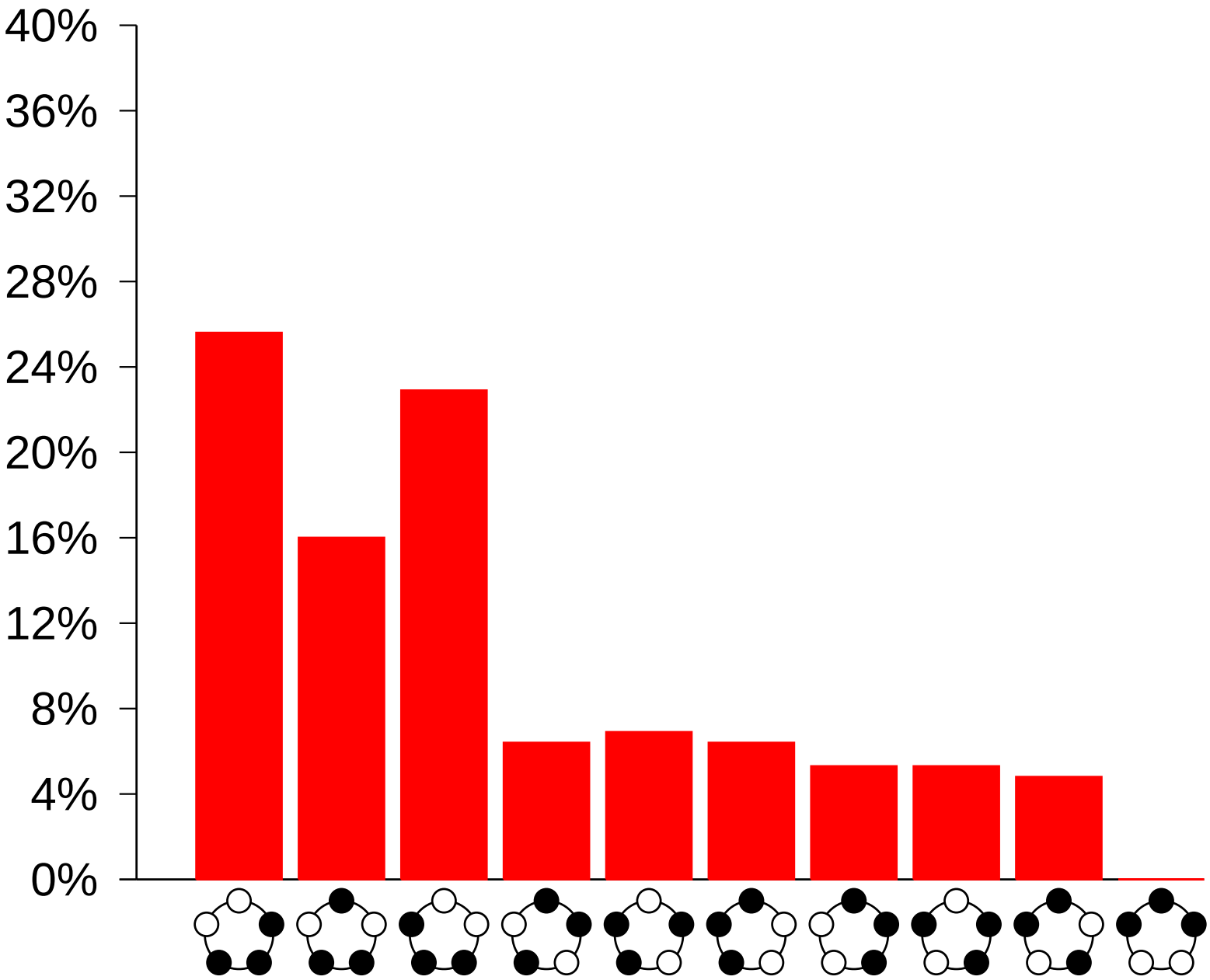
Stationary distribution



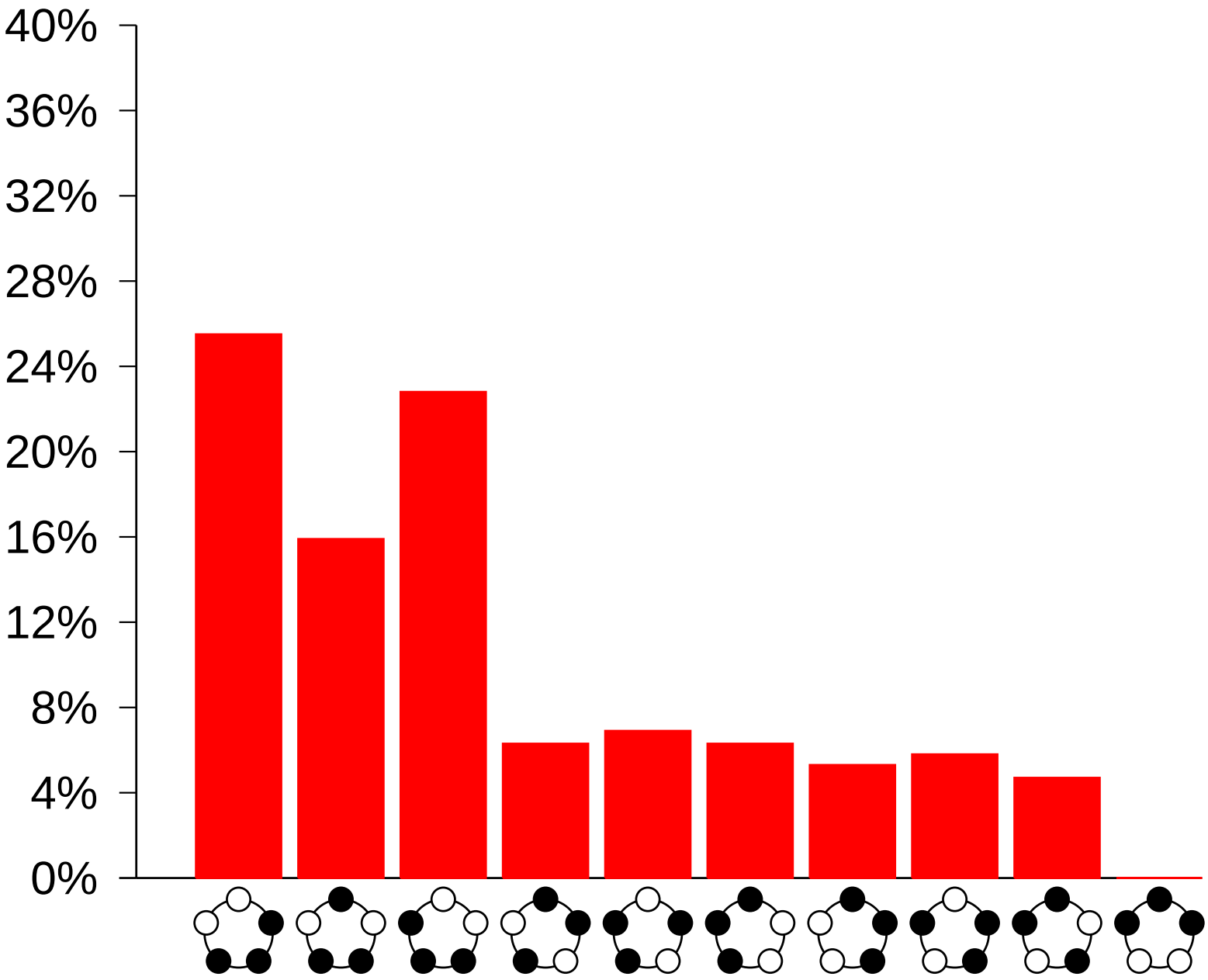
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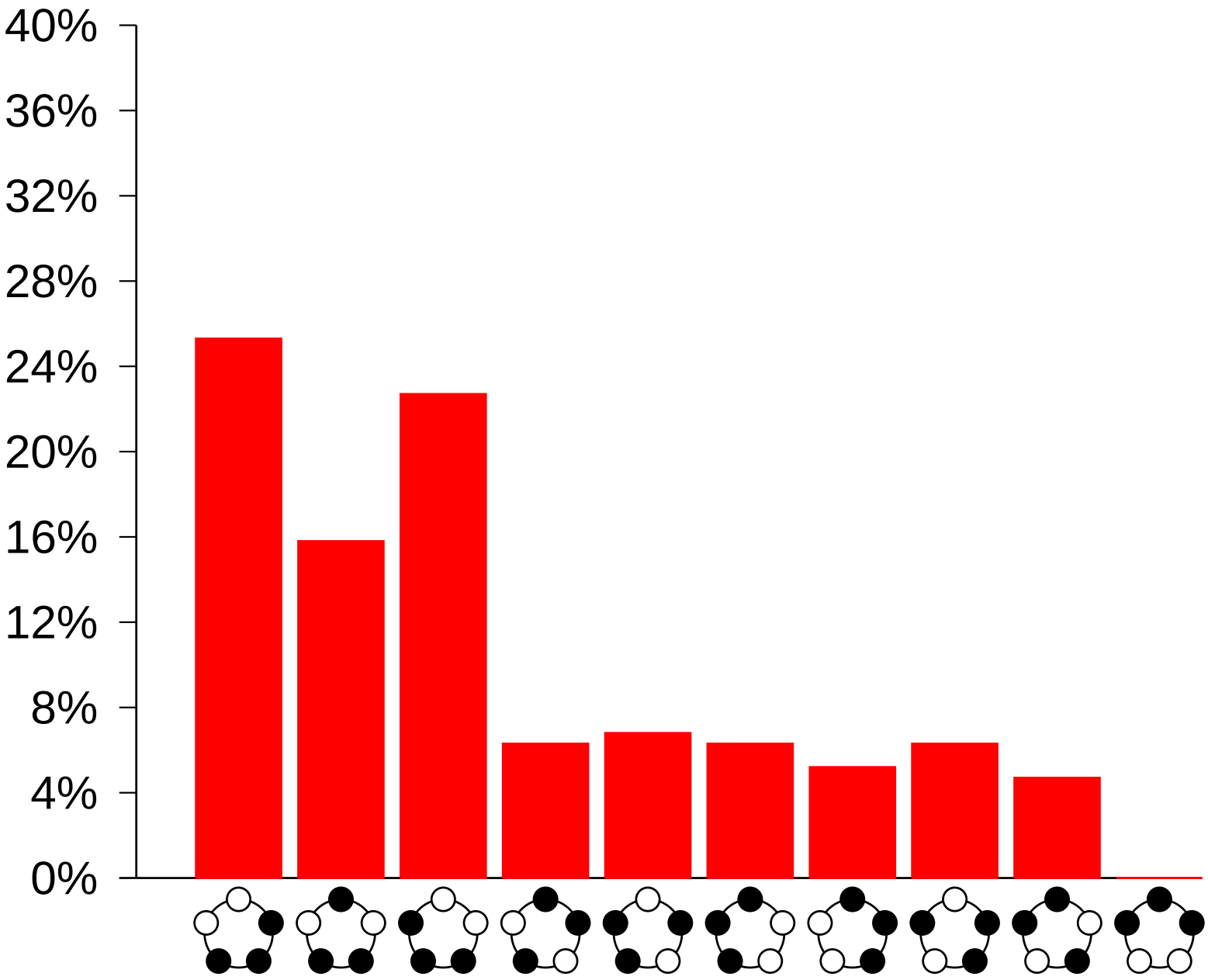
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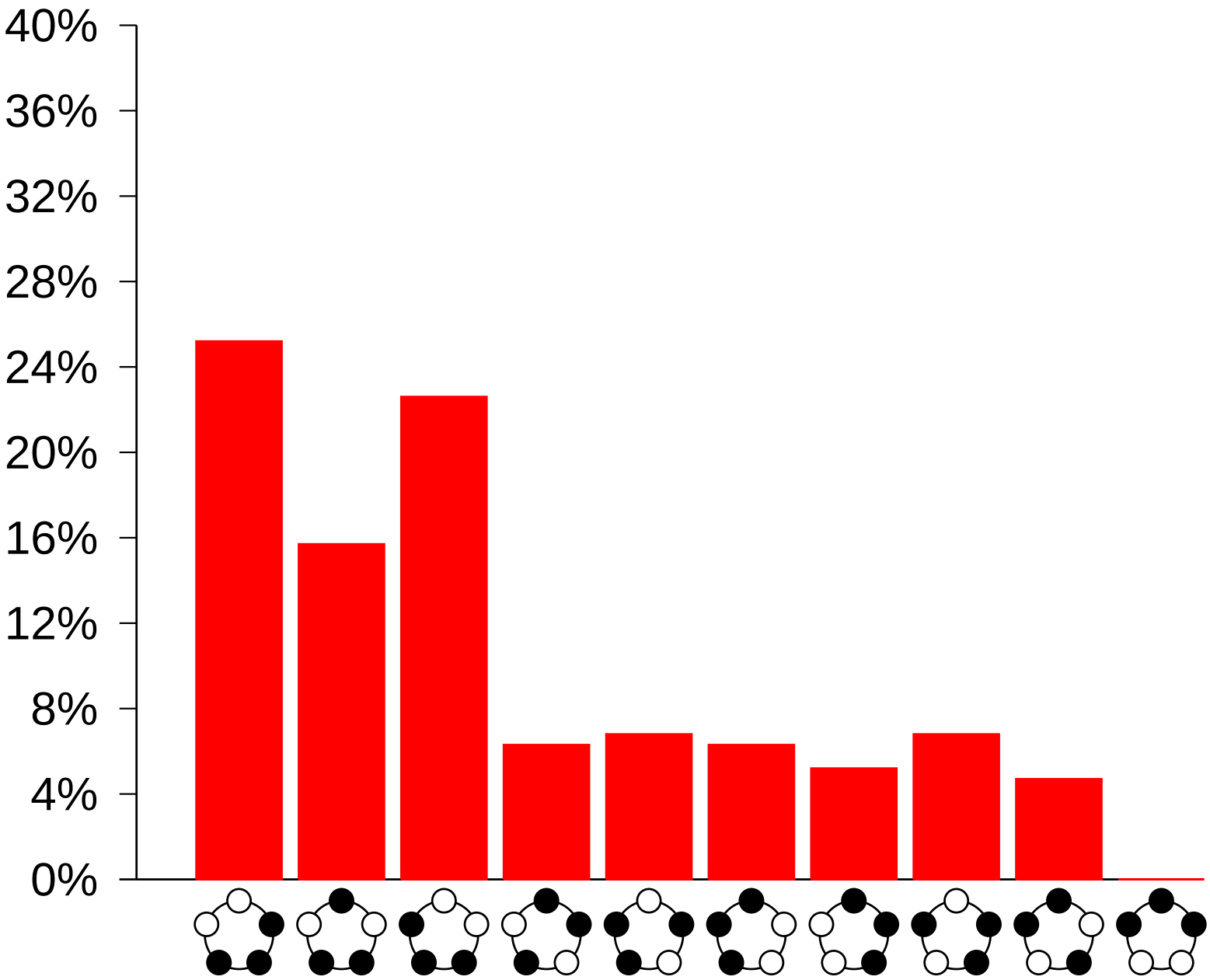
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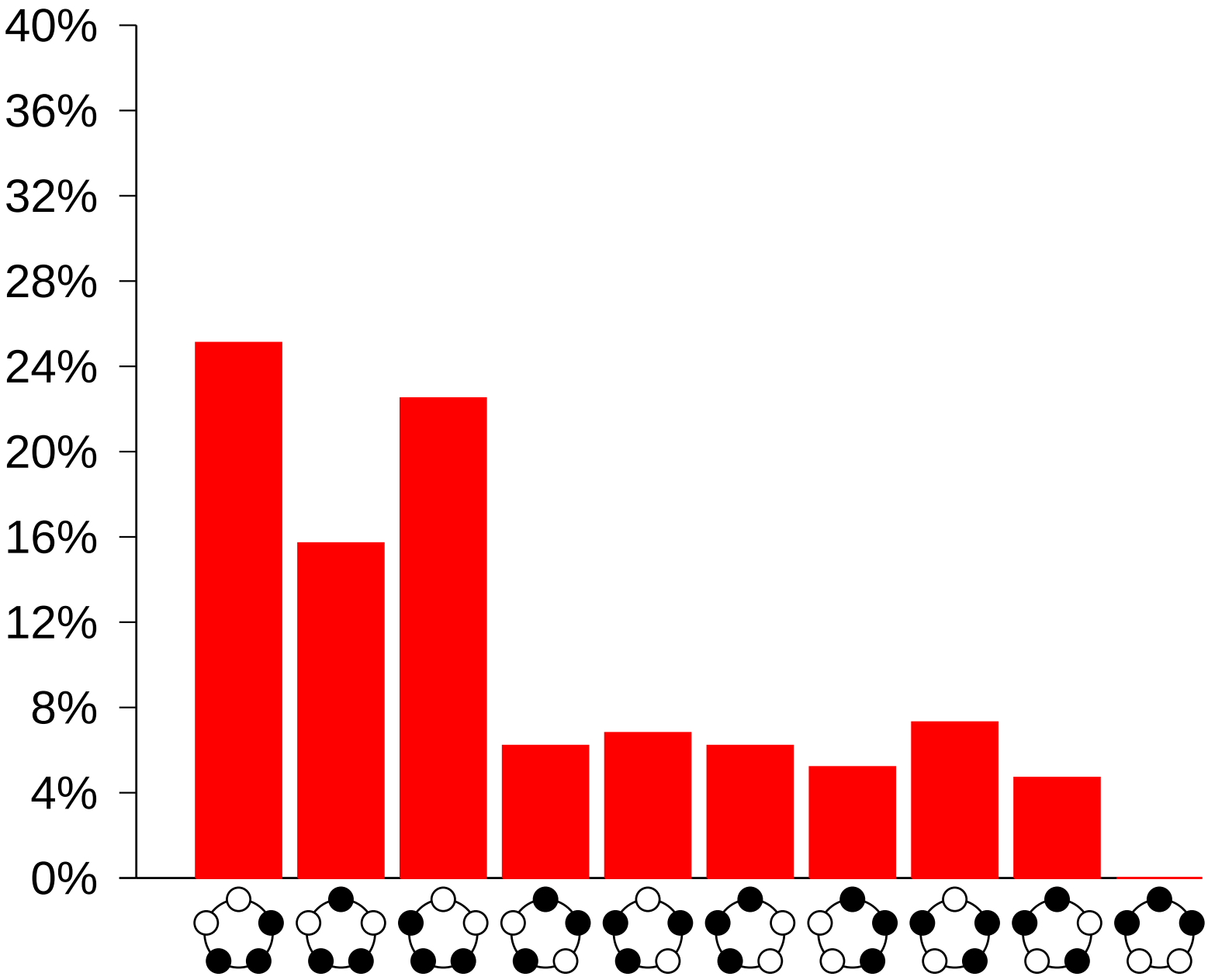
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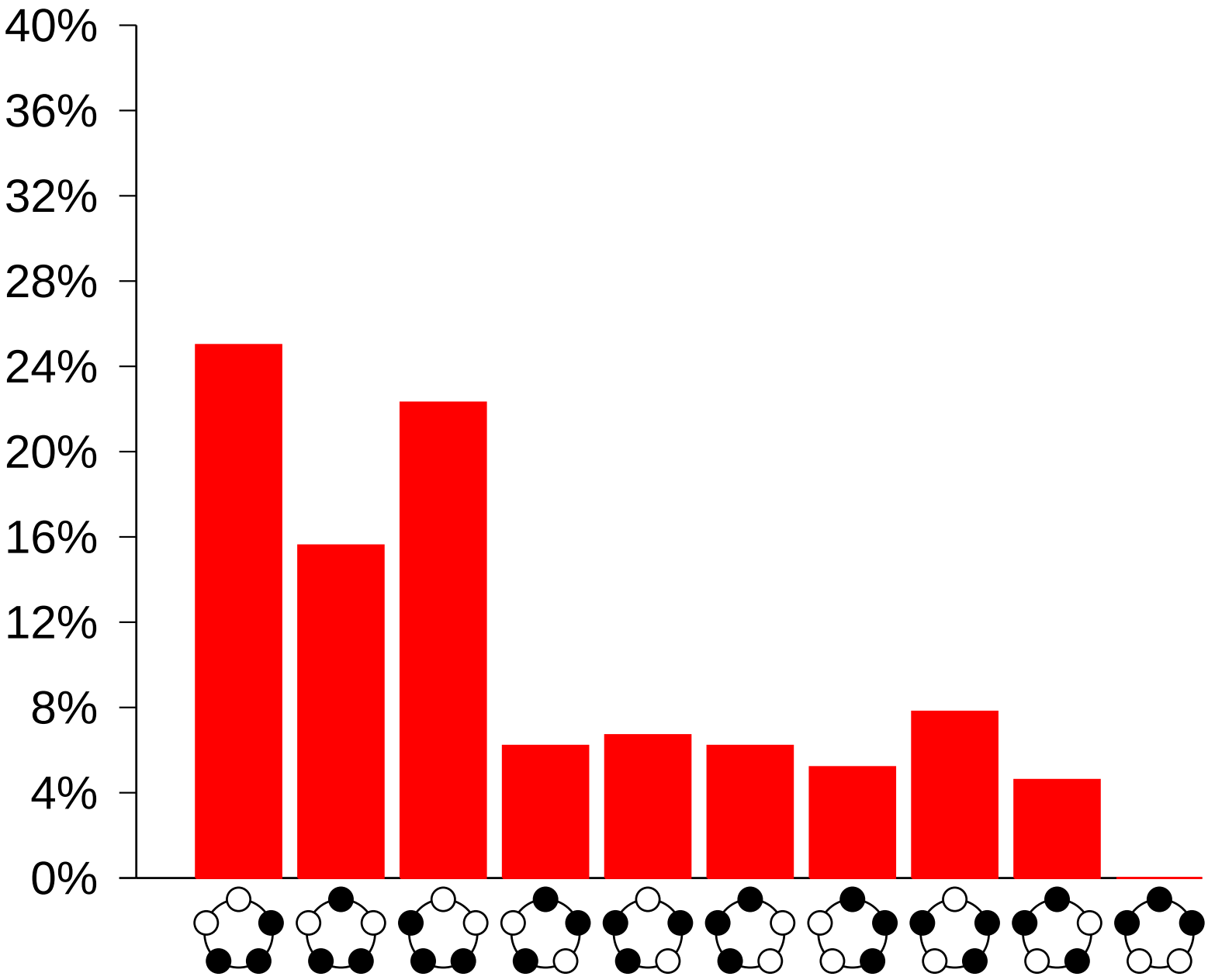
Stationary distribution



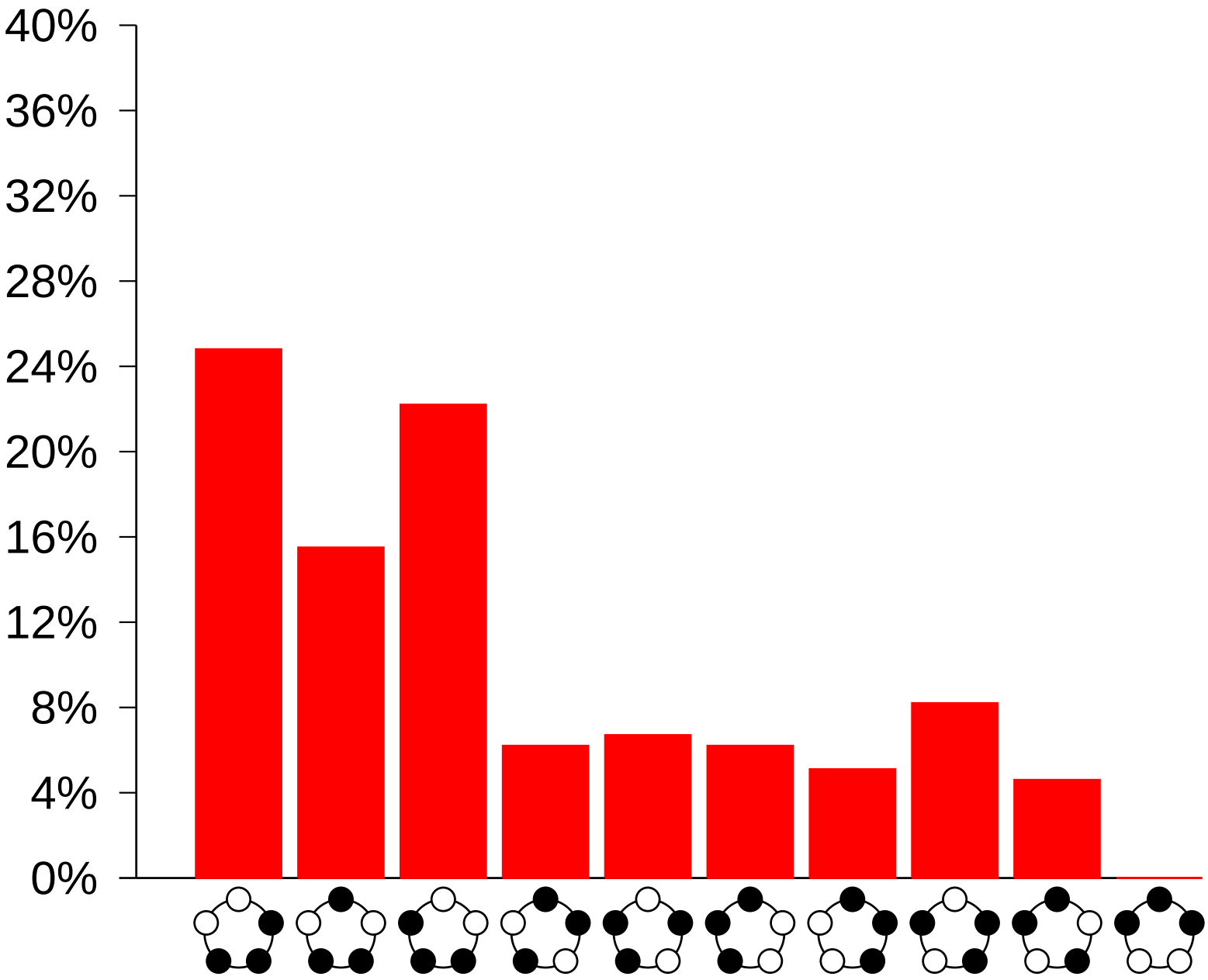
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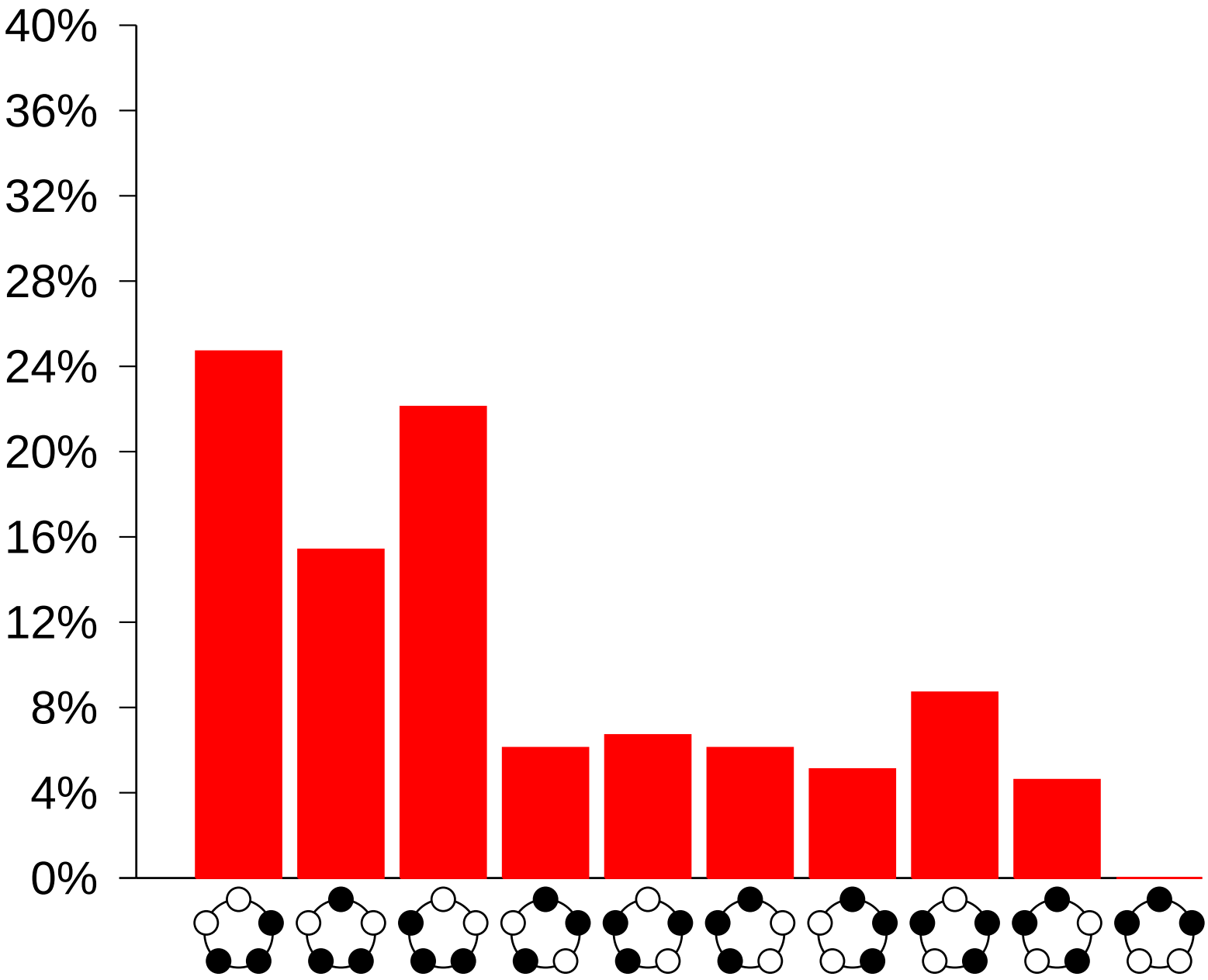
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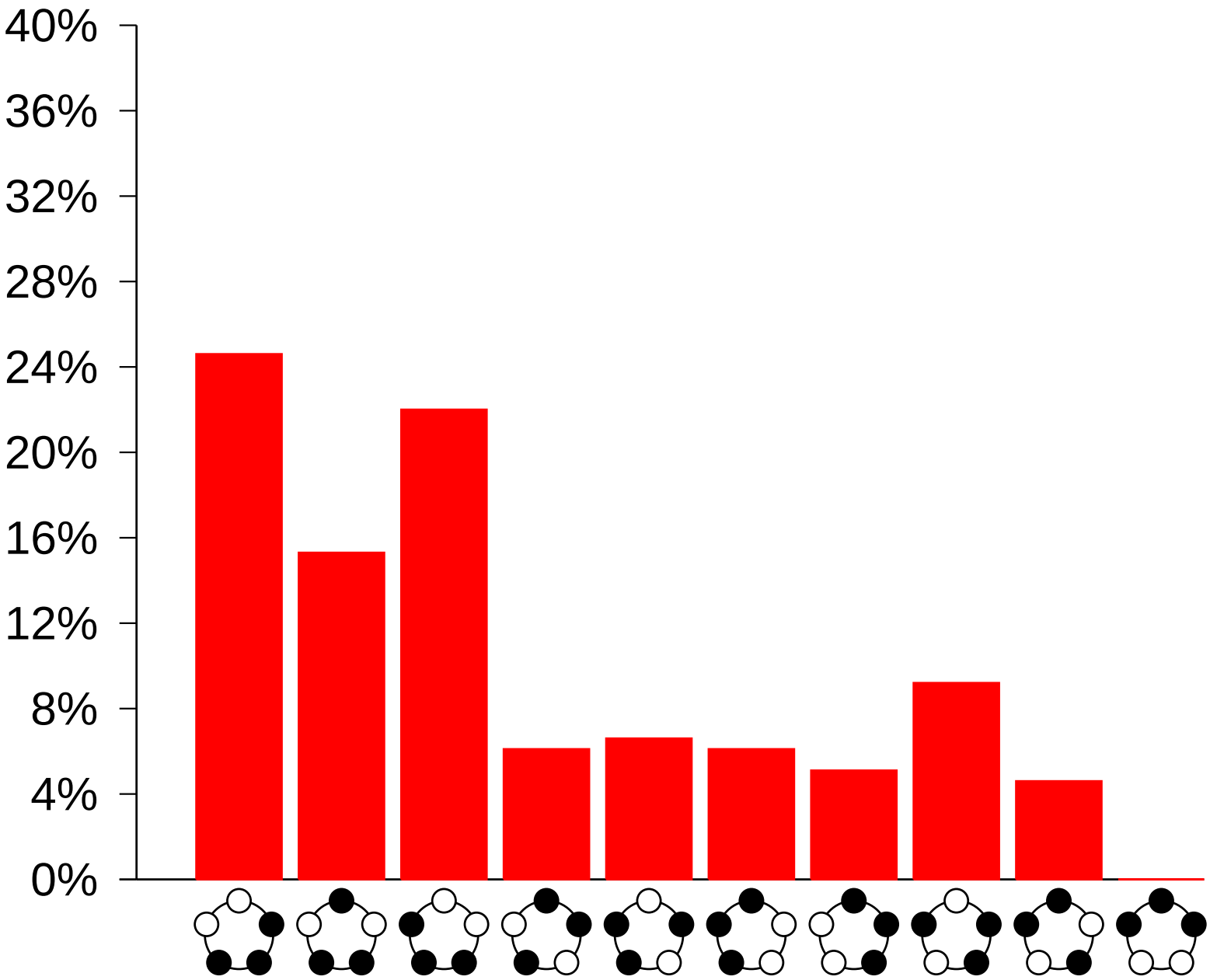
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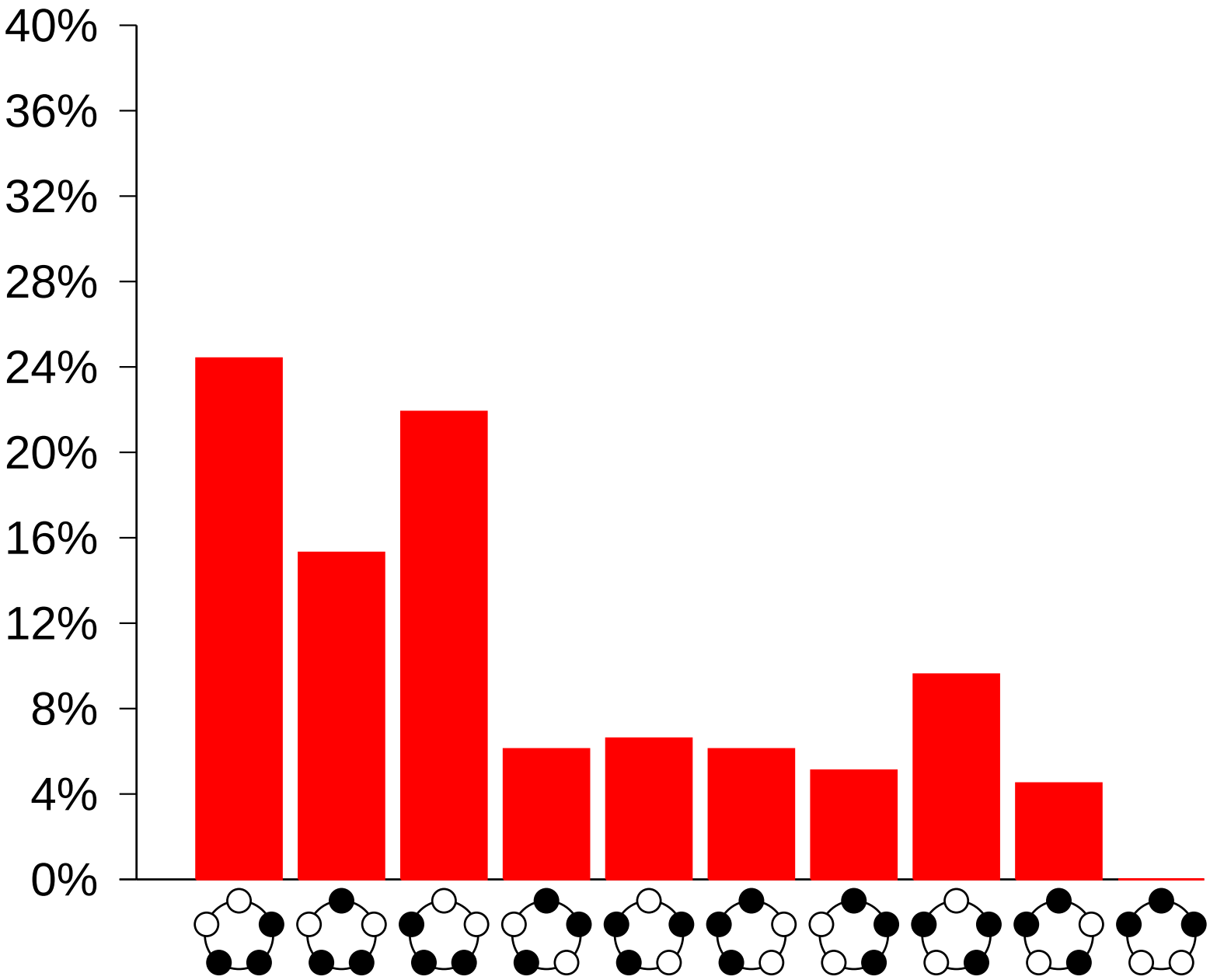
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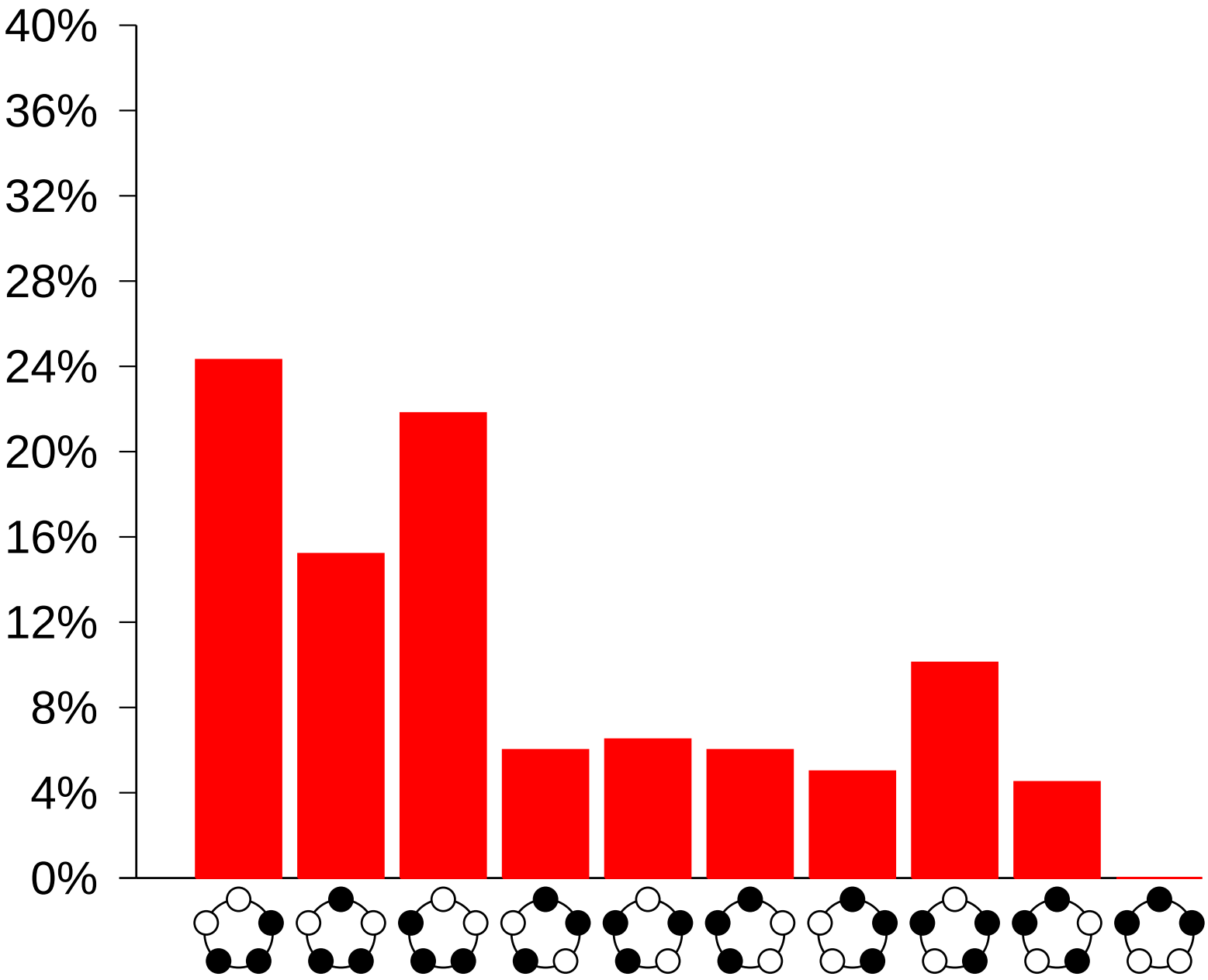
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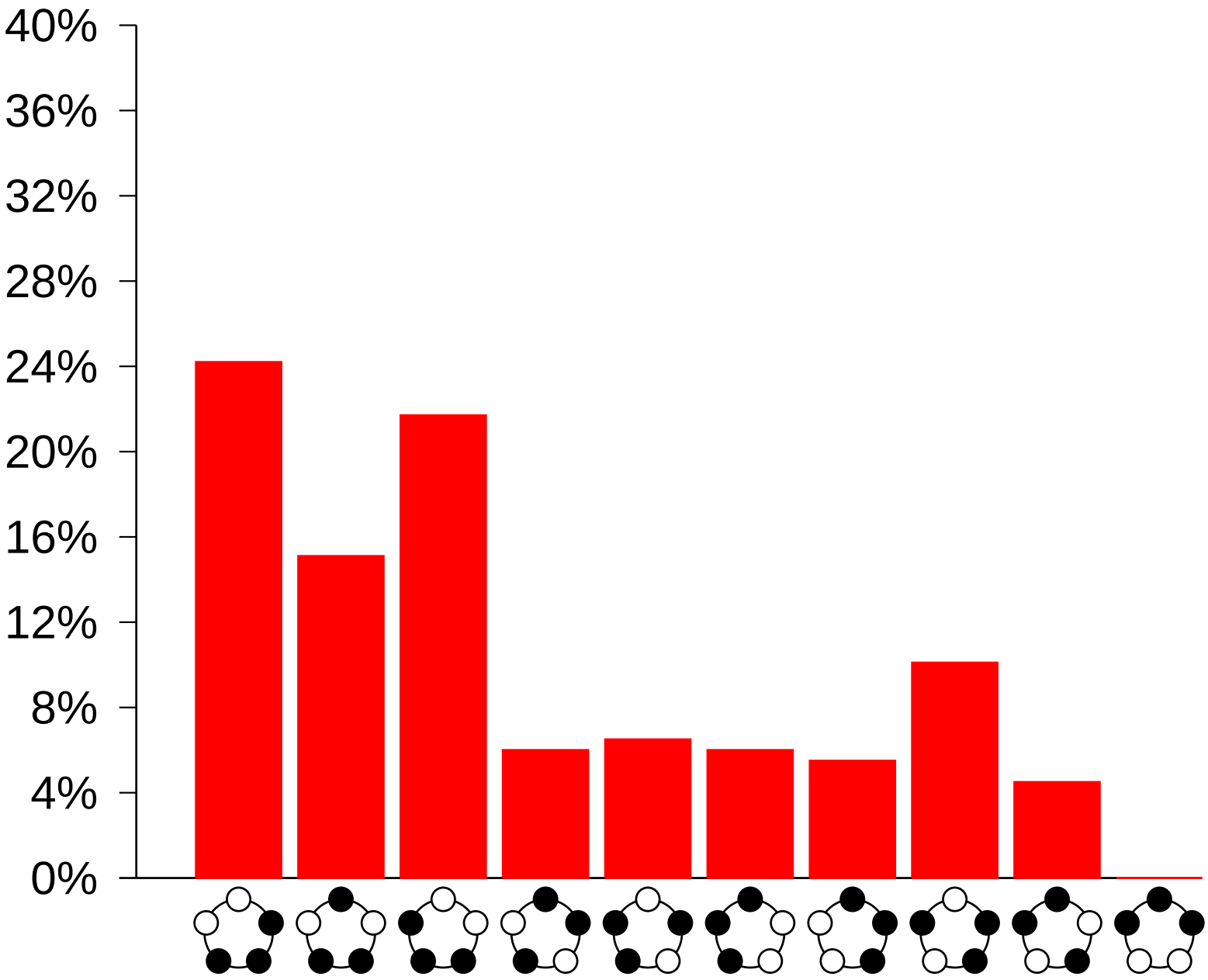
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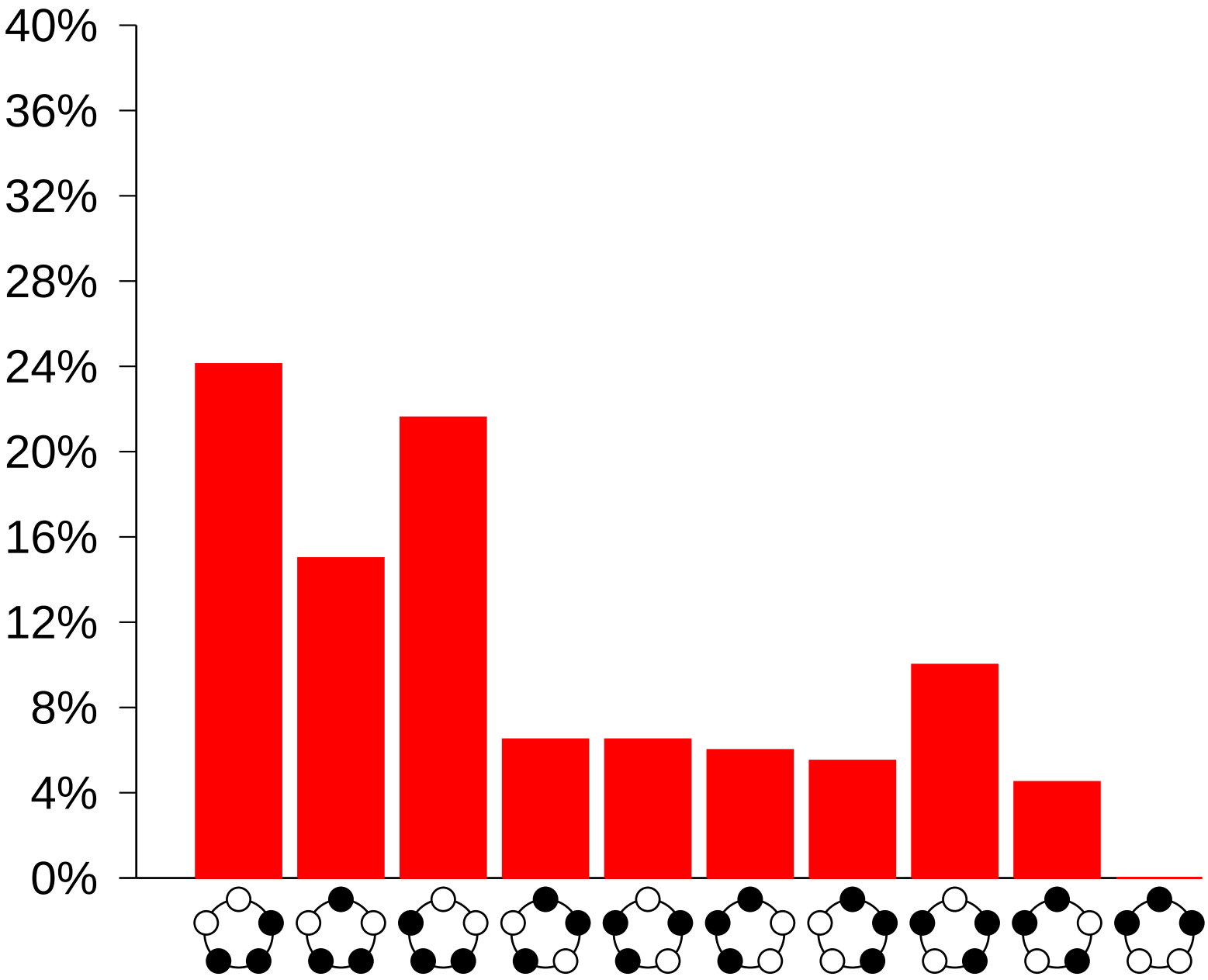
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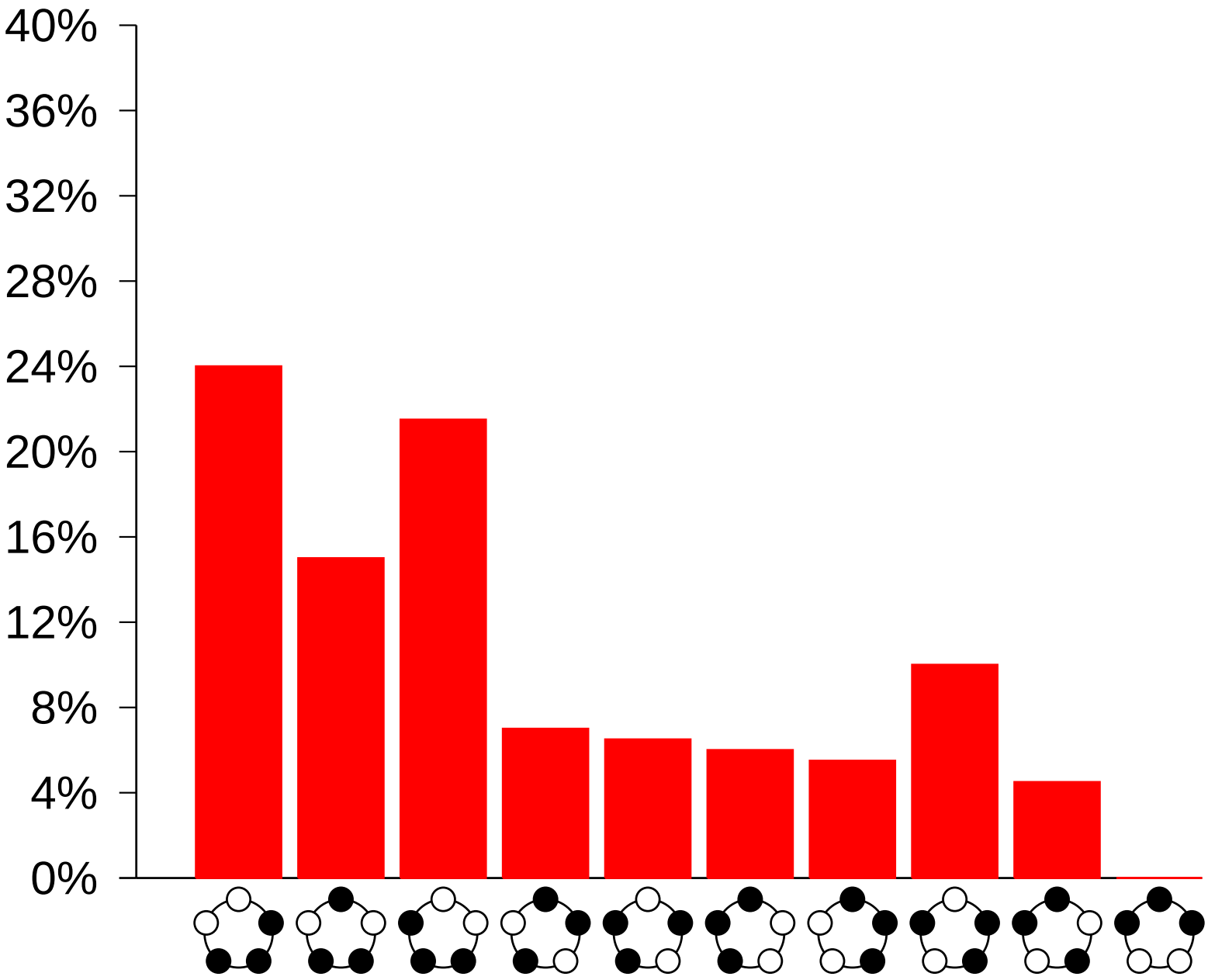
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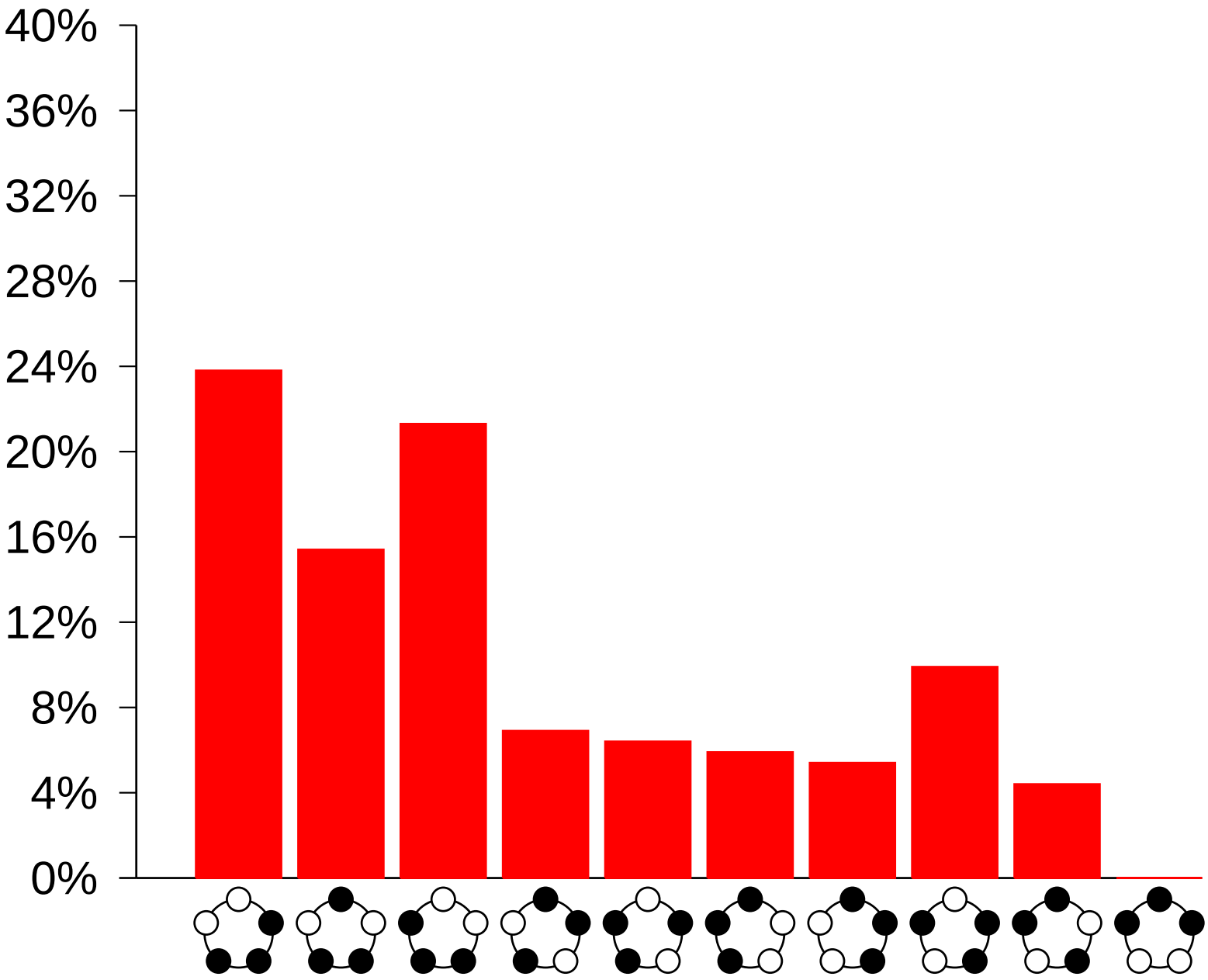
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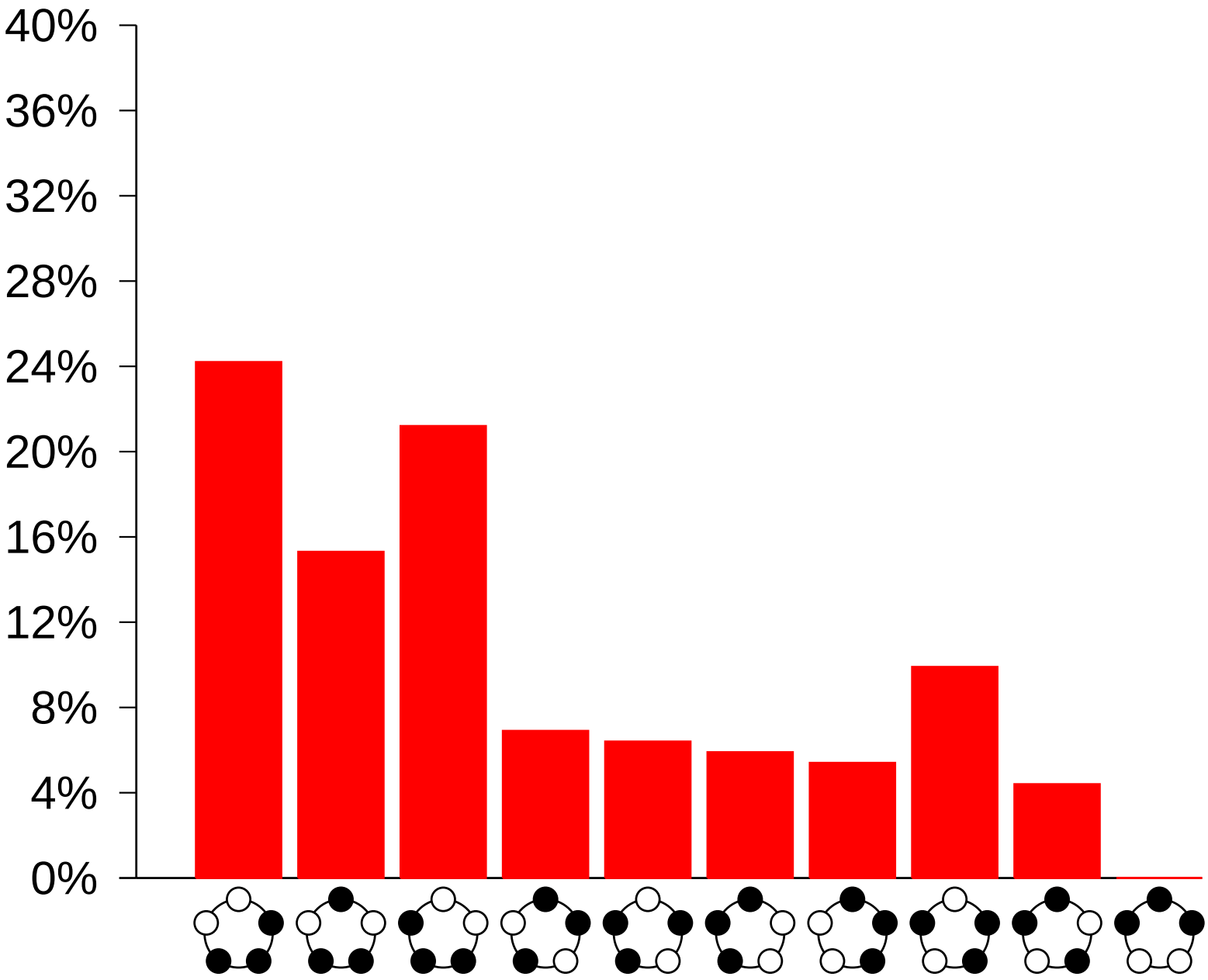
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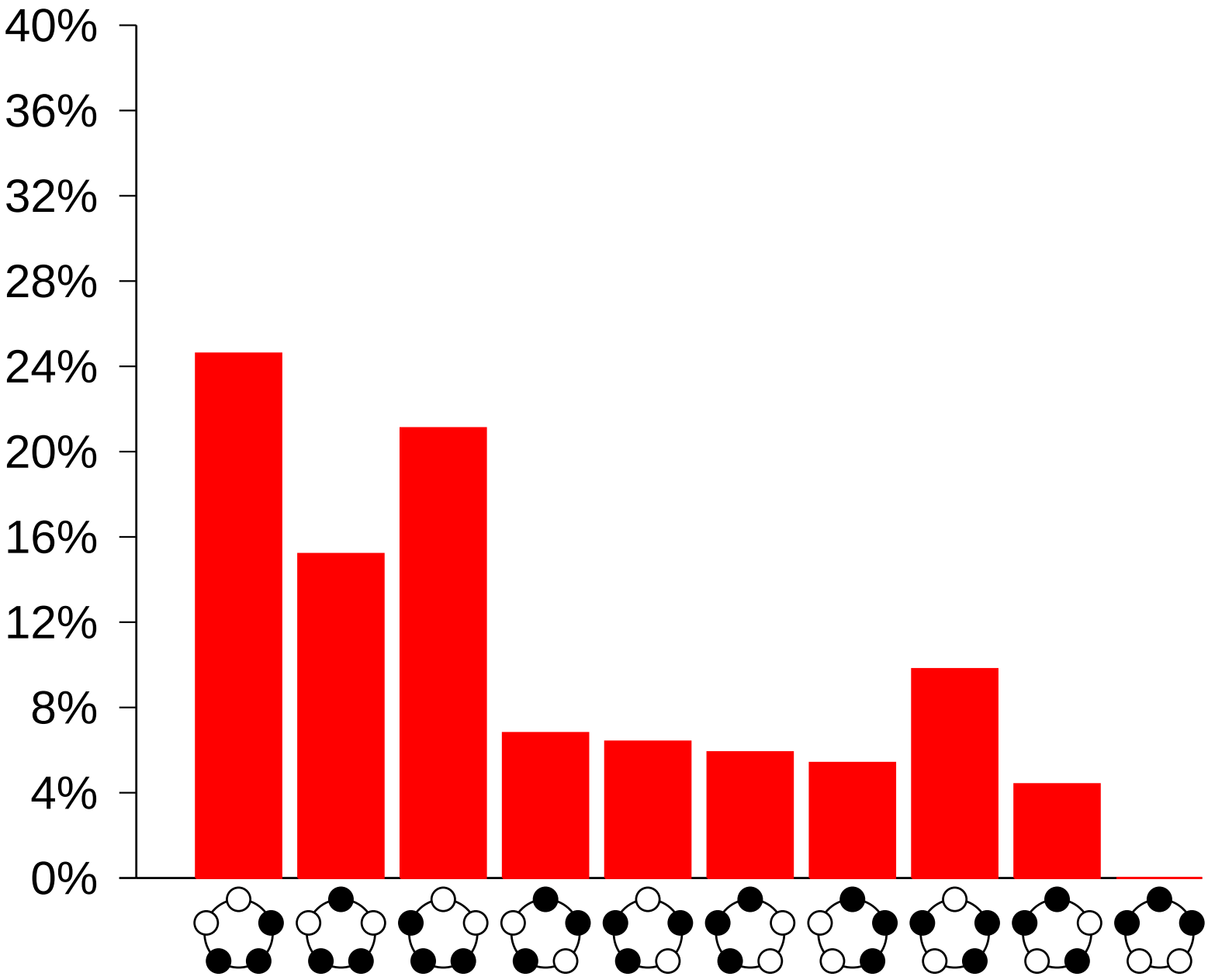
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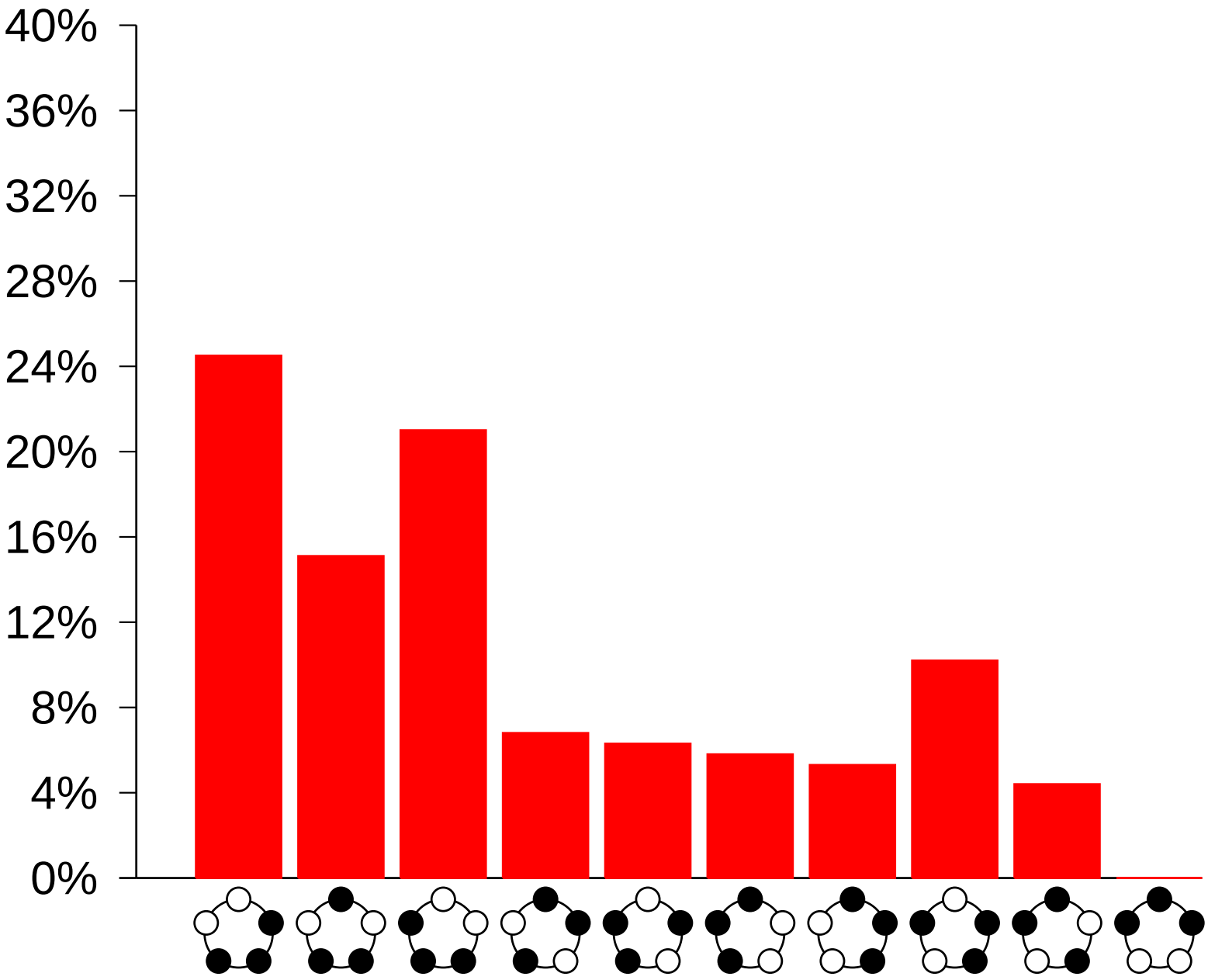
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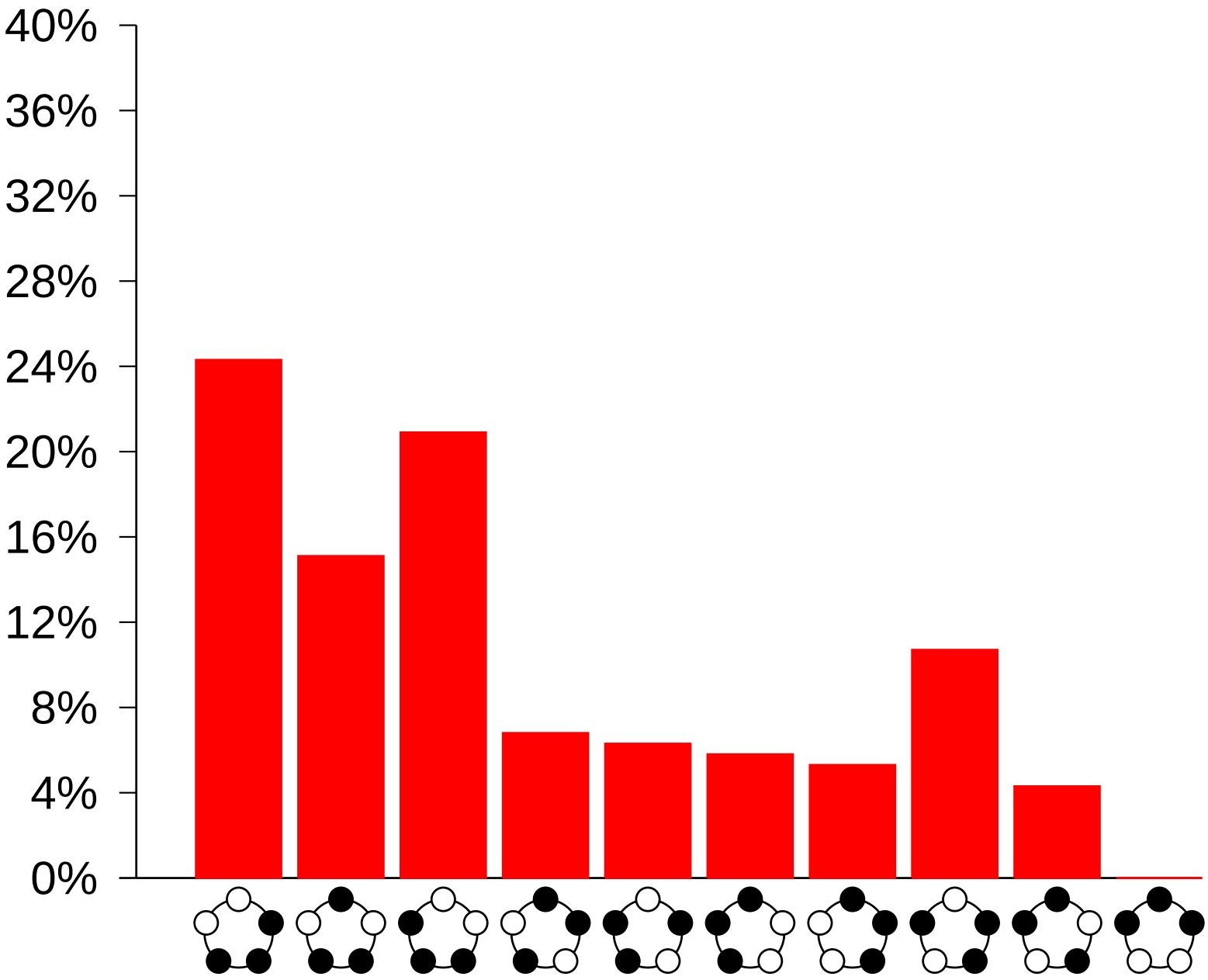
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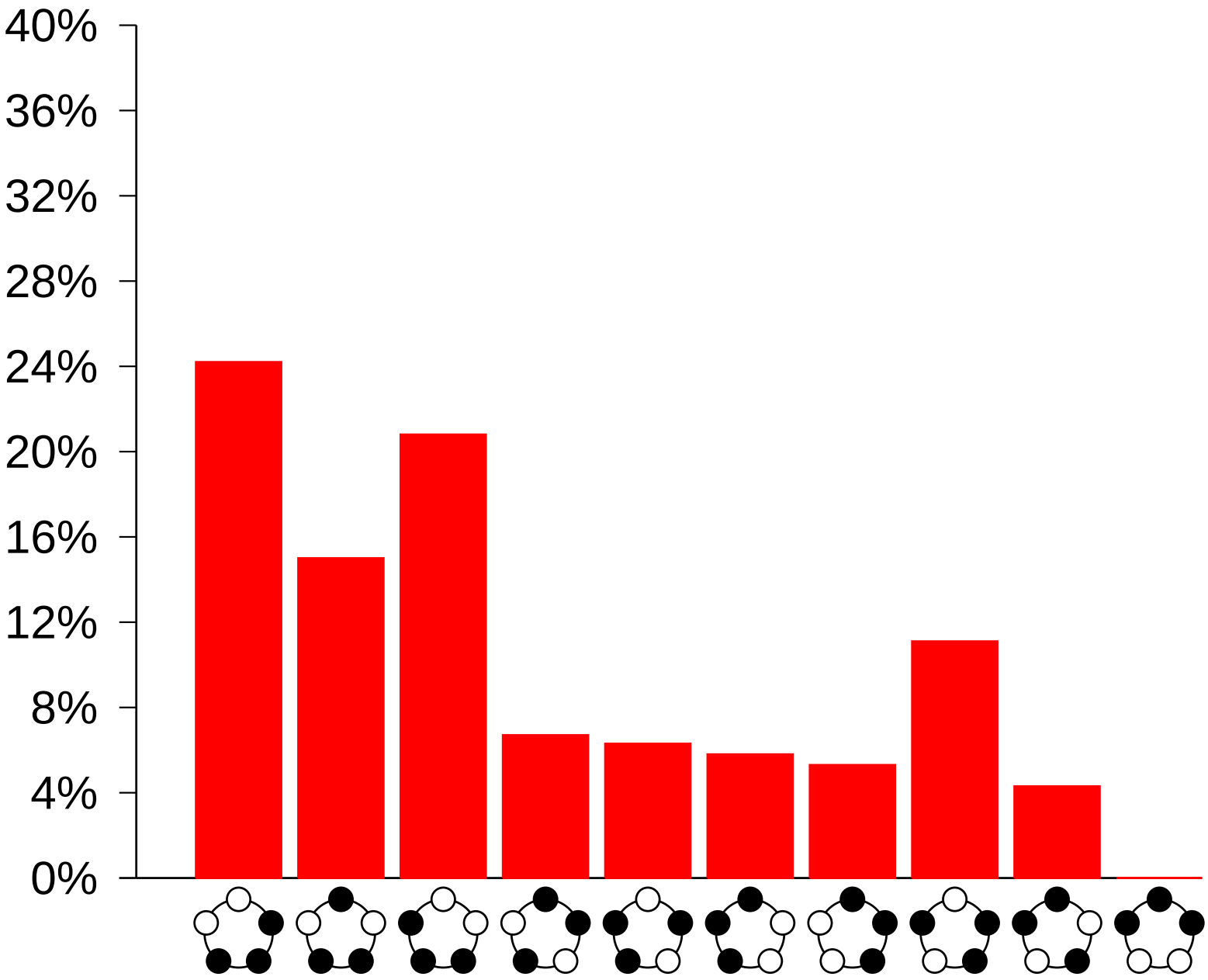
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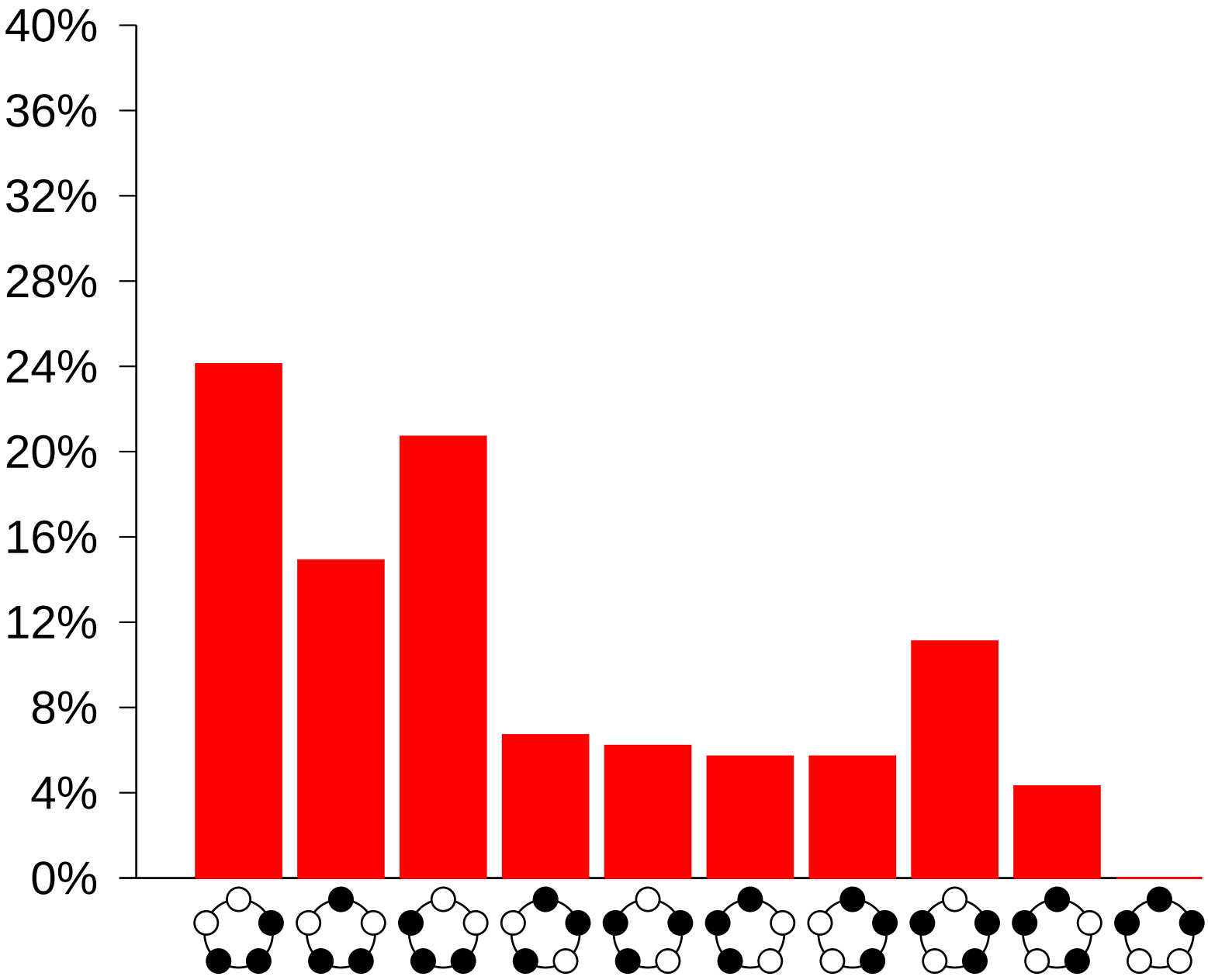
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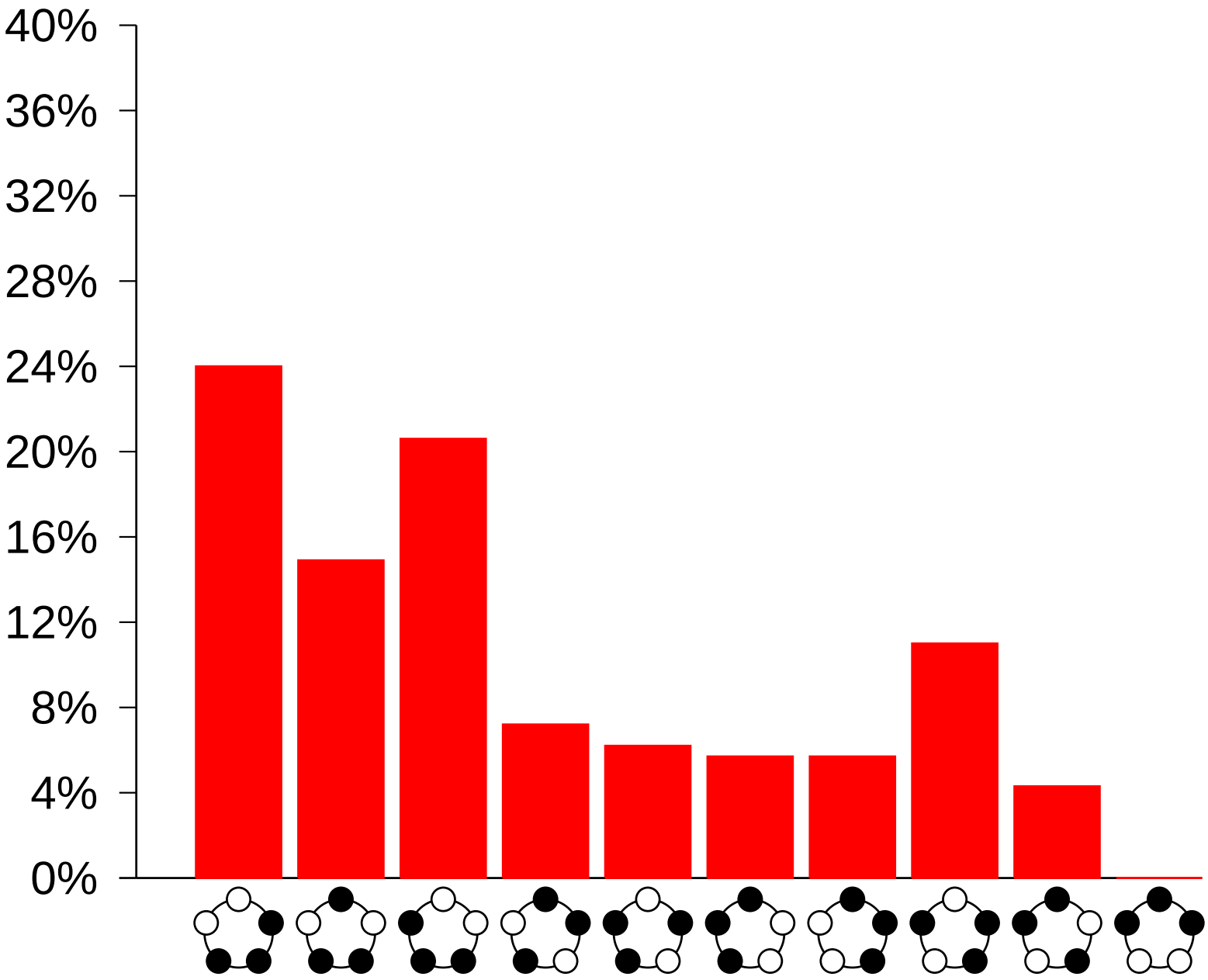
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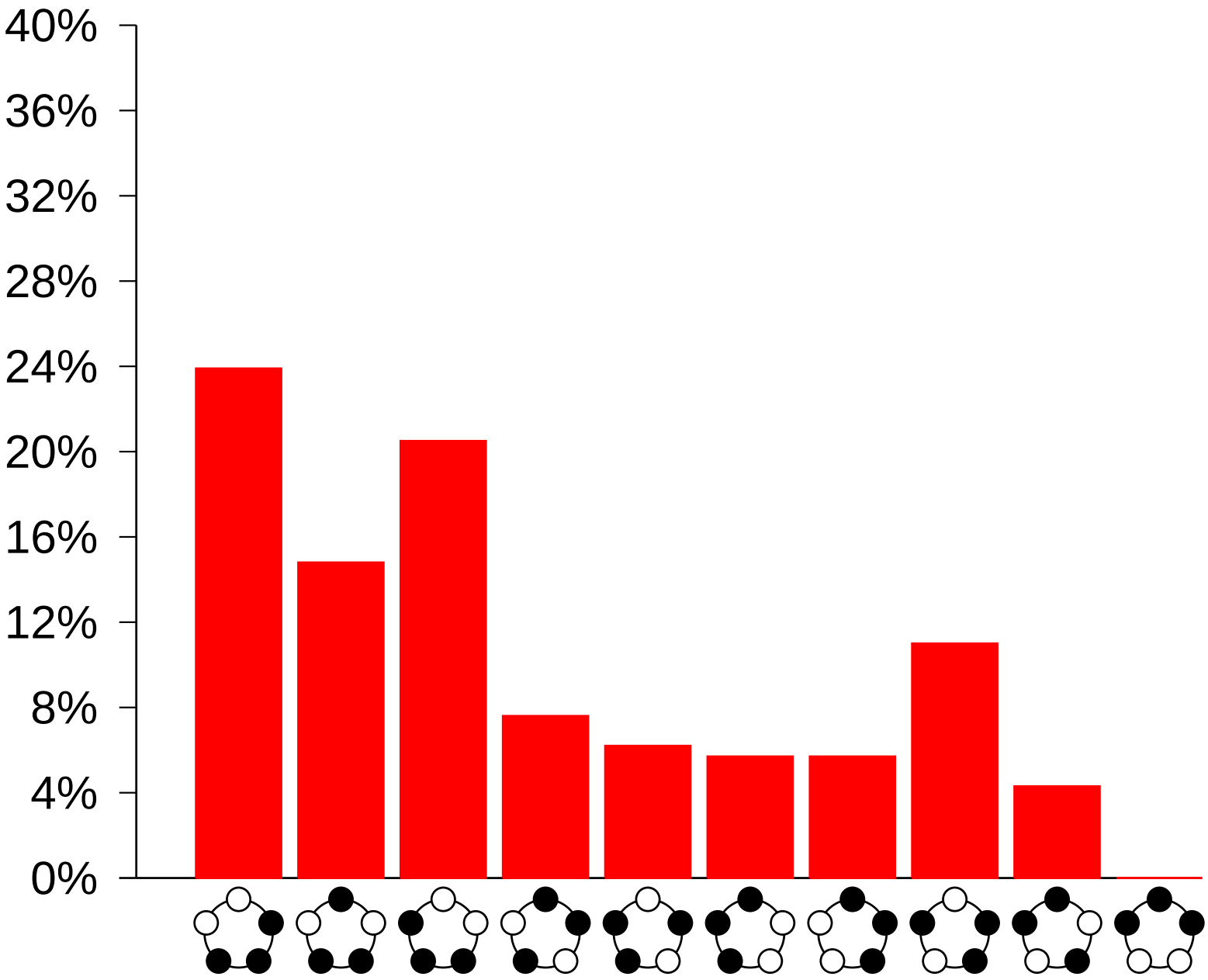
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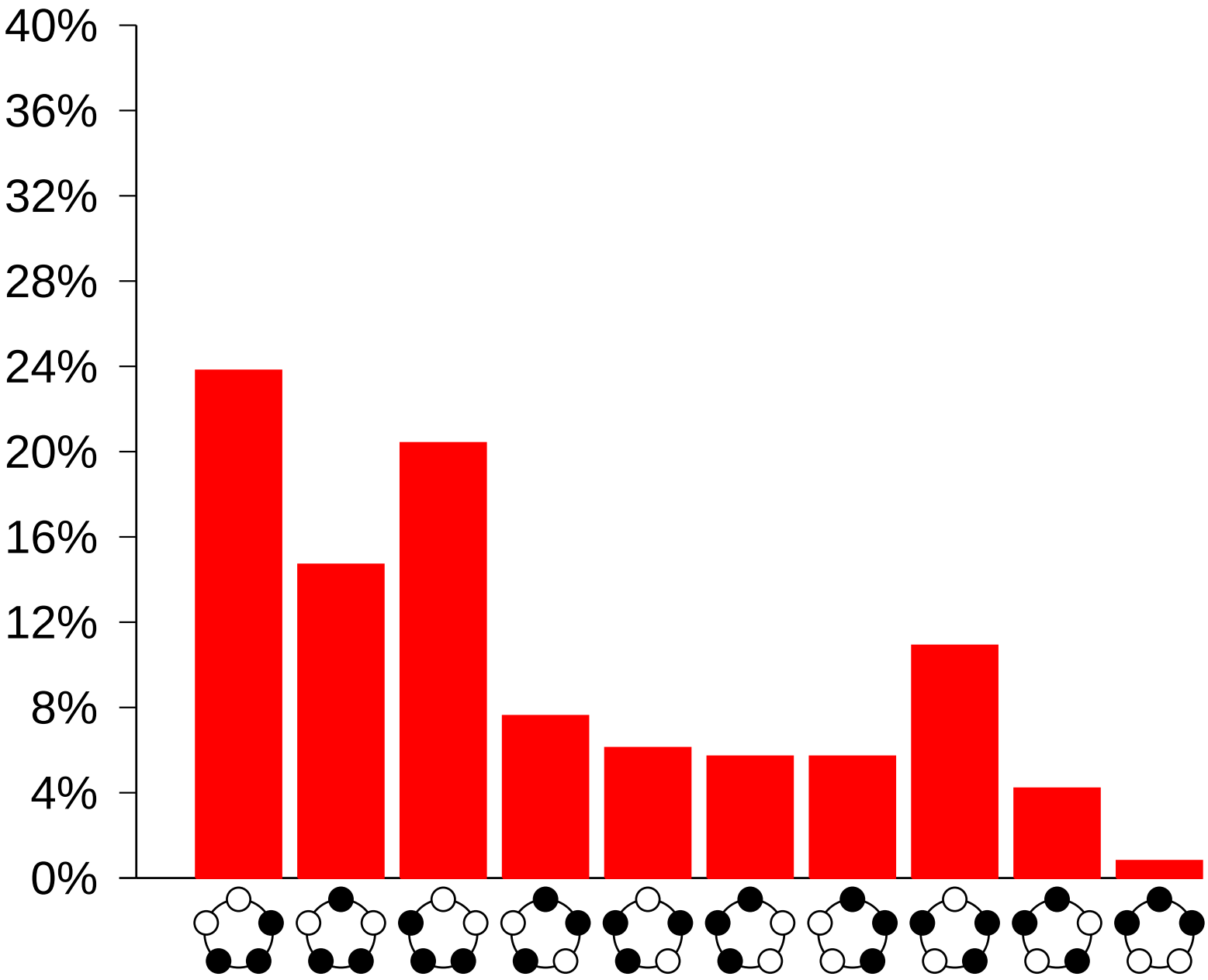
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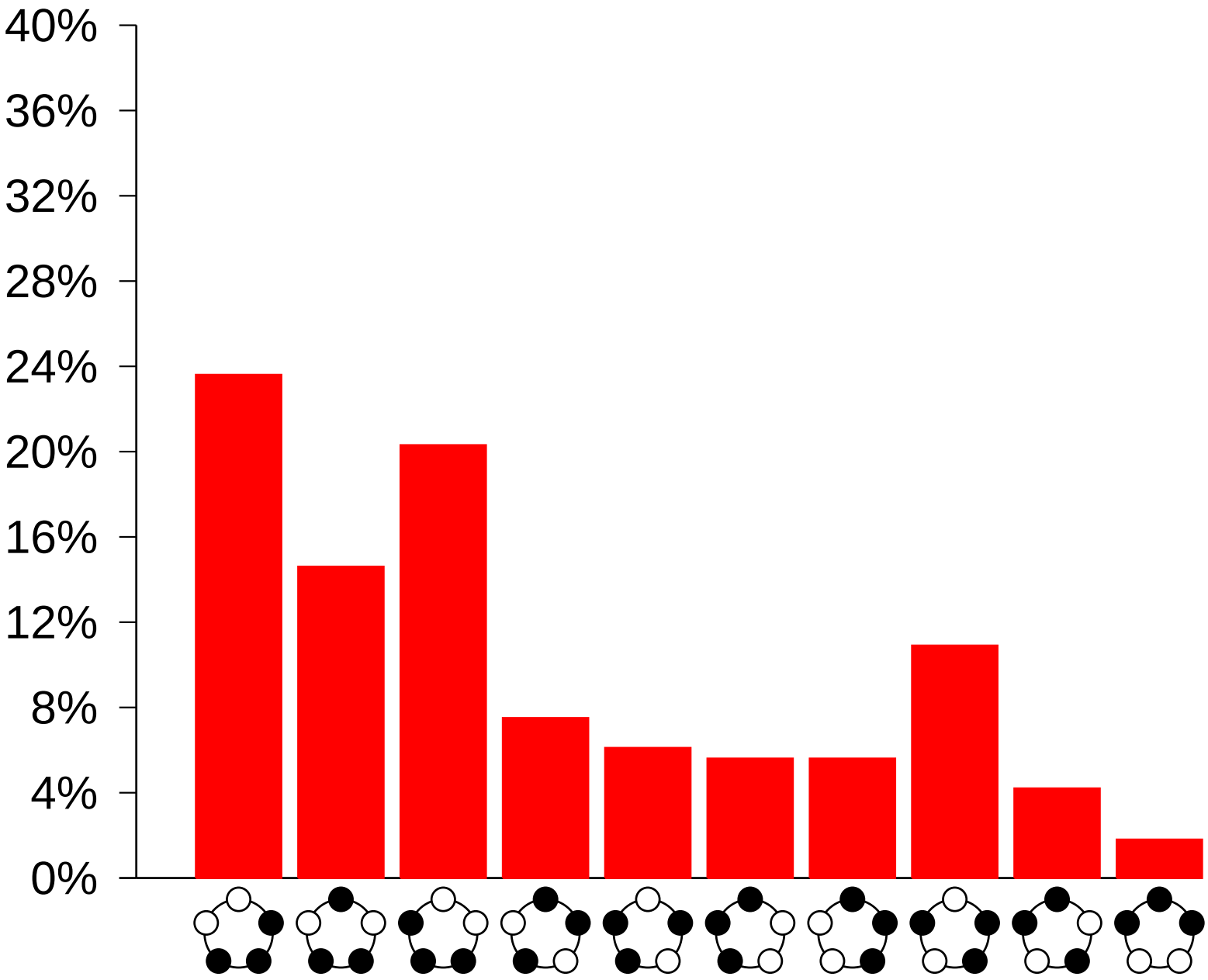
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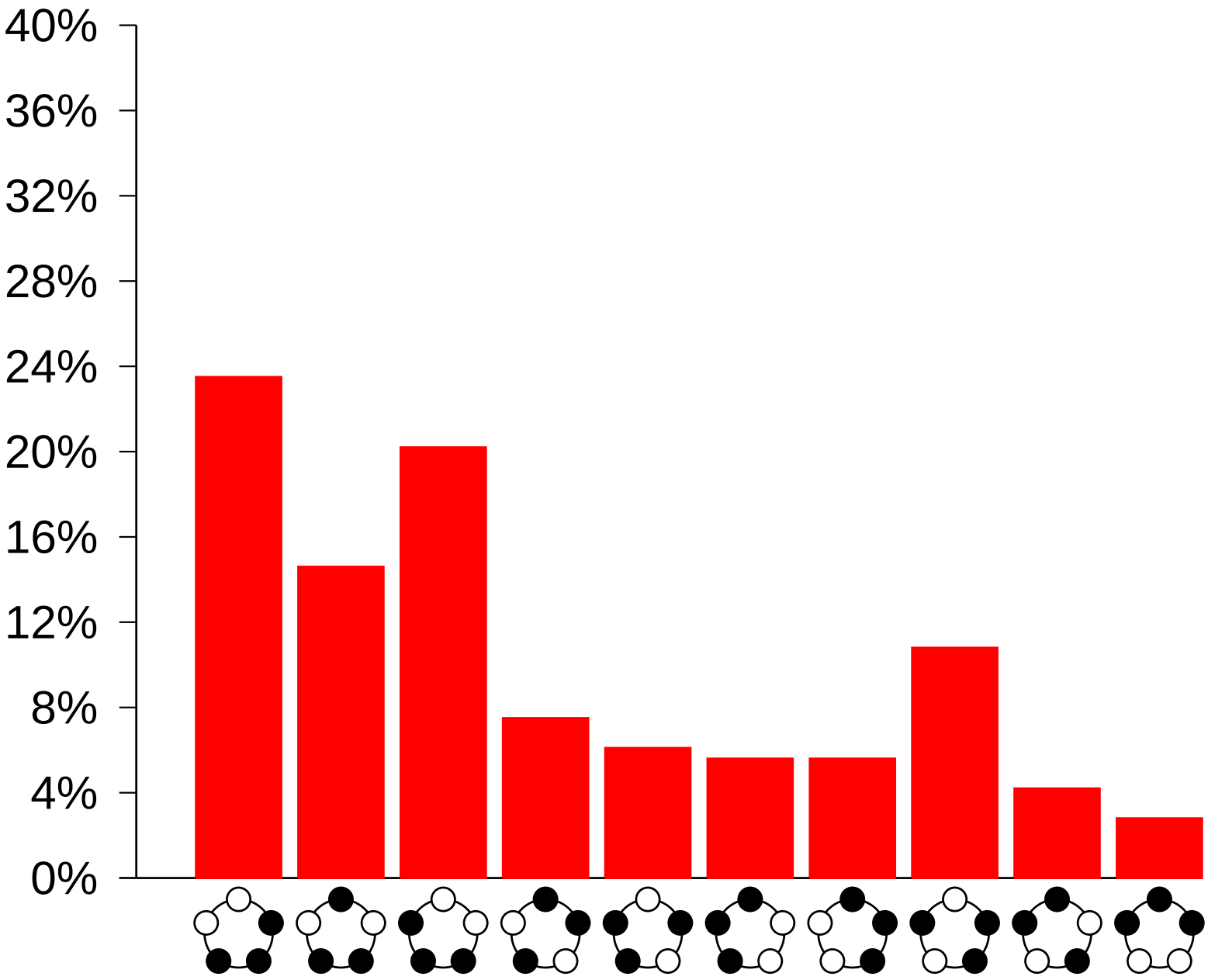
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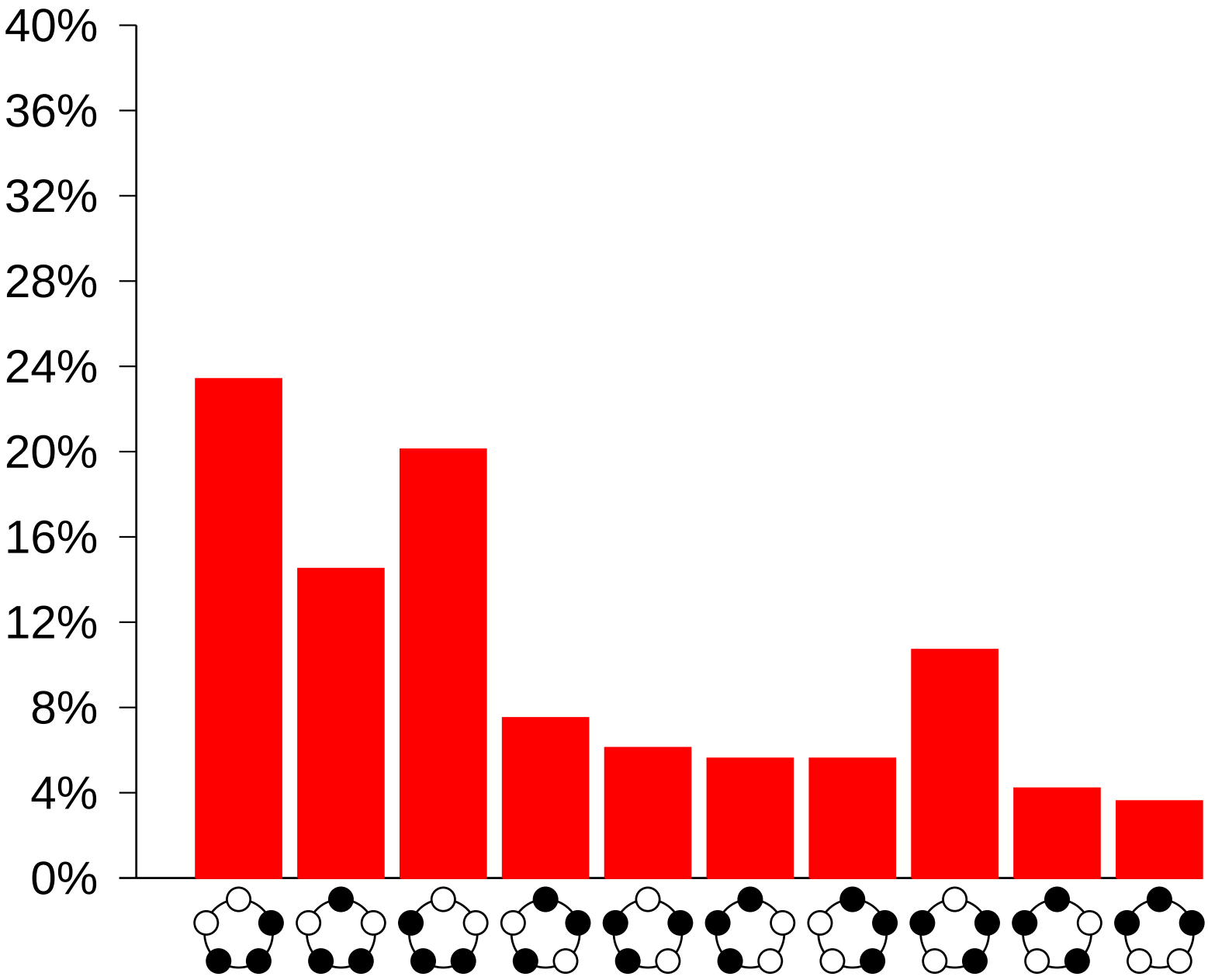
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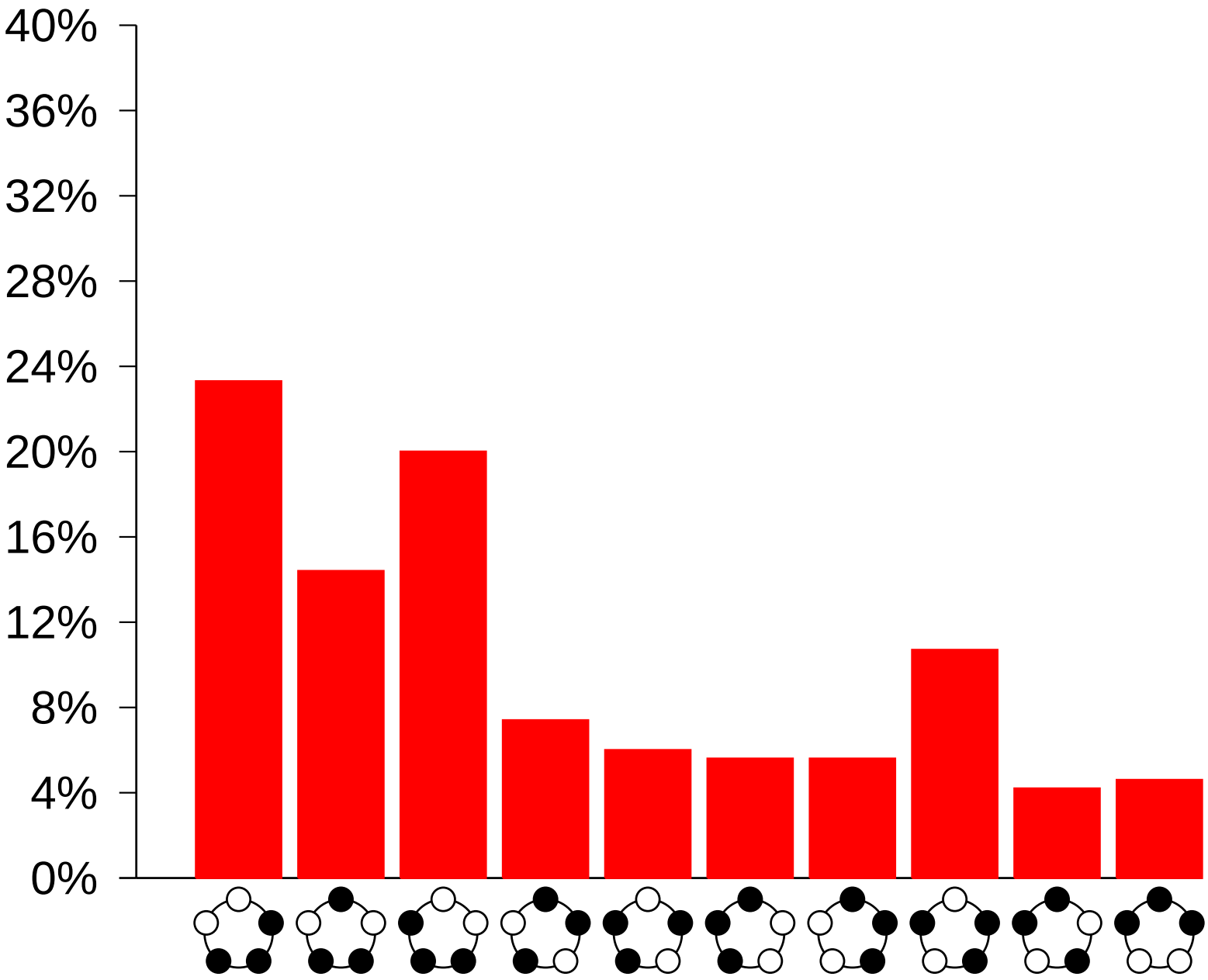
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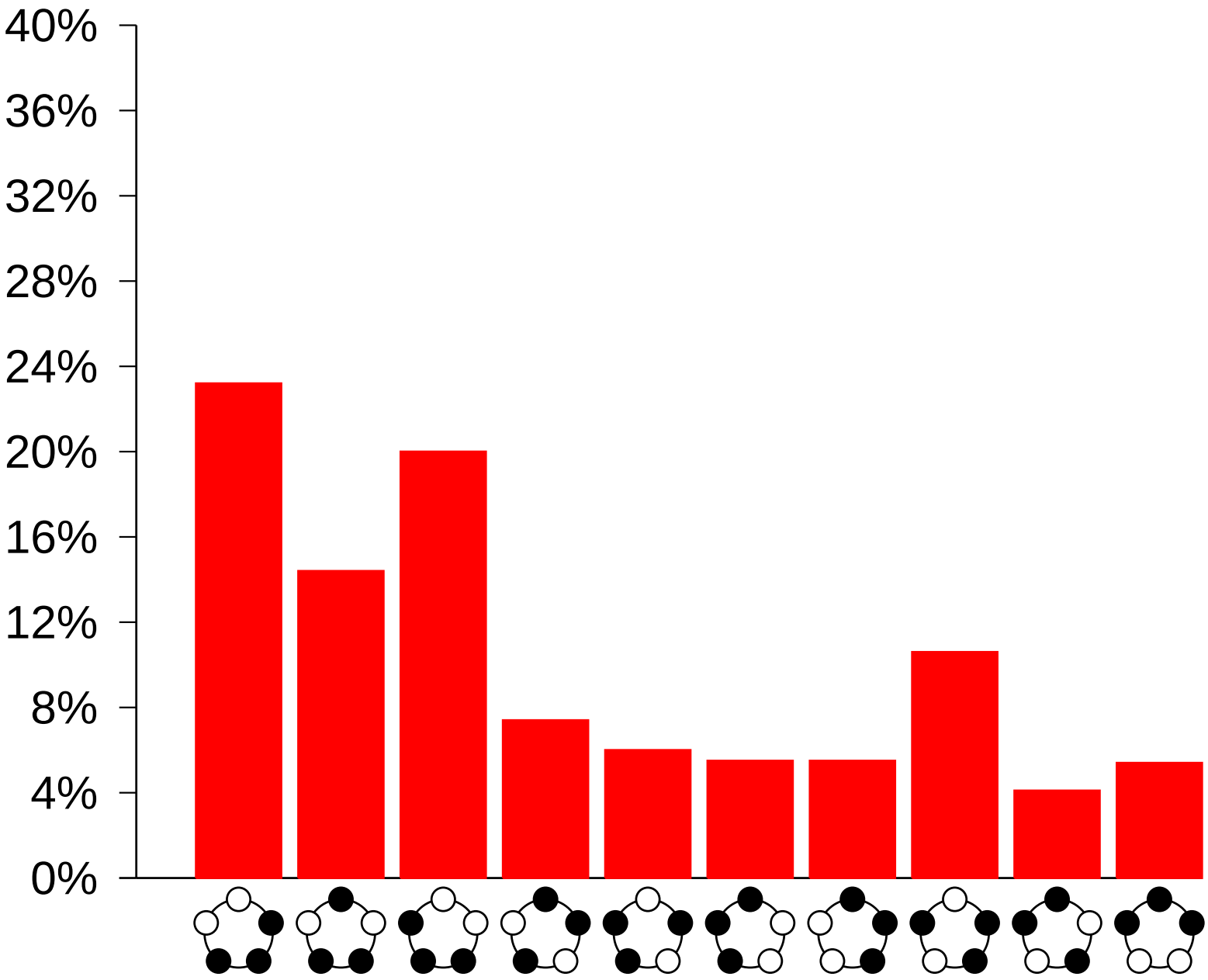
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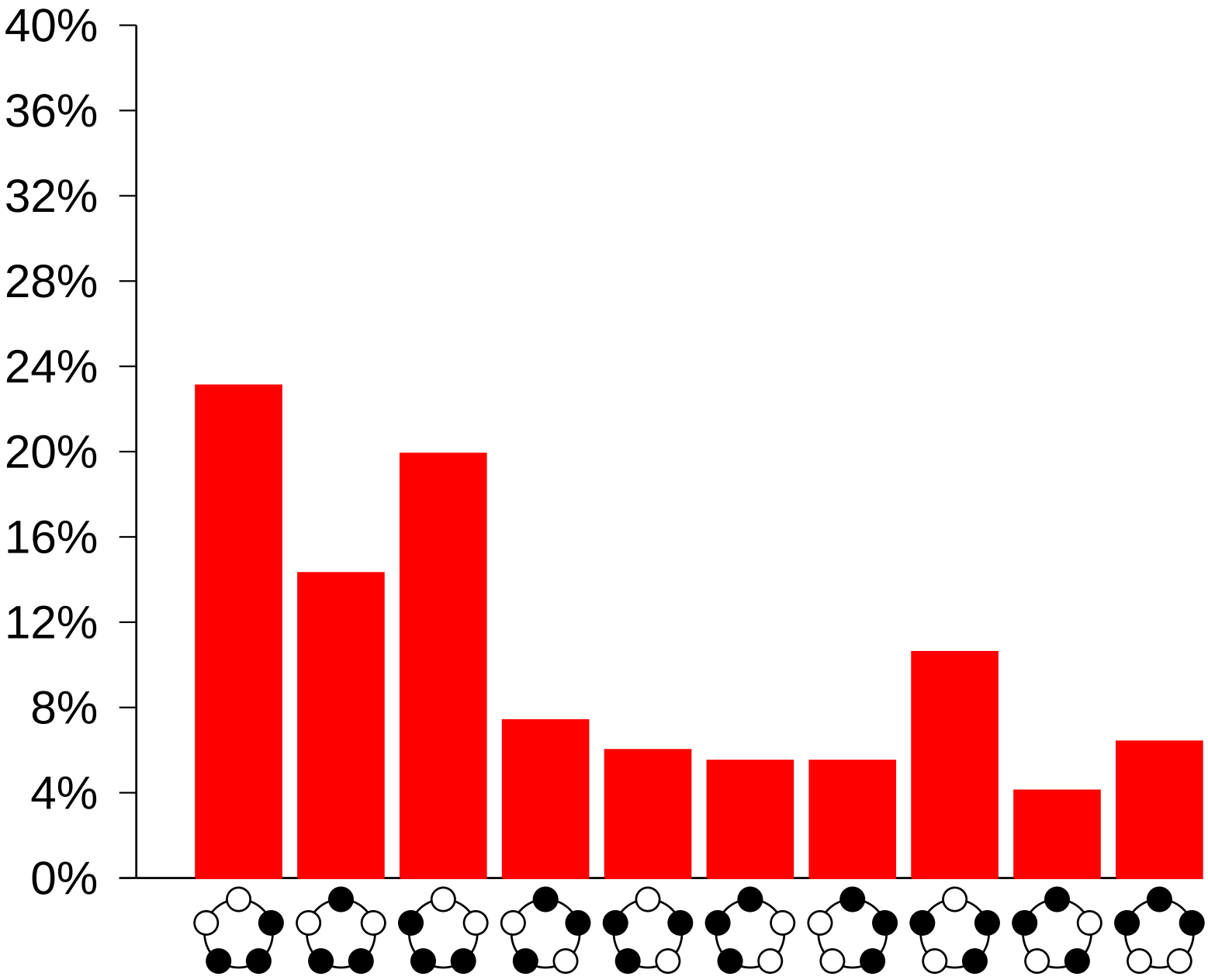
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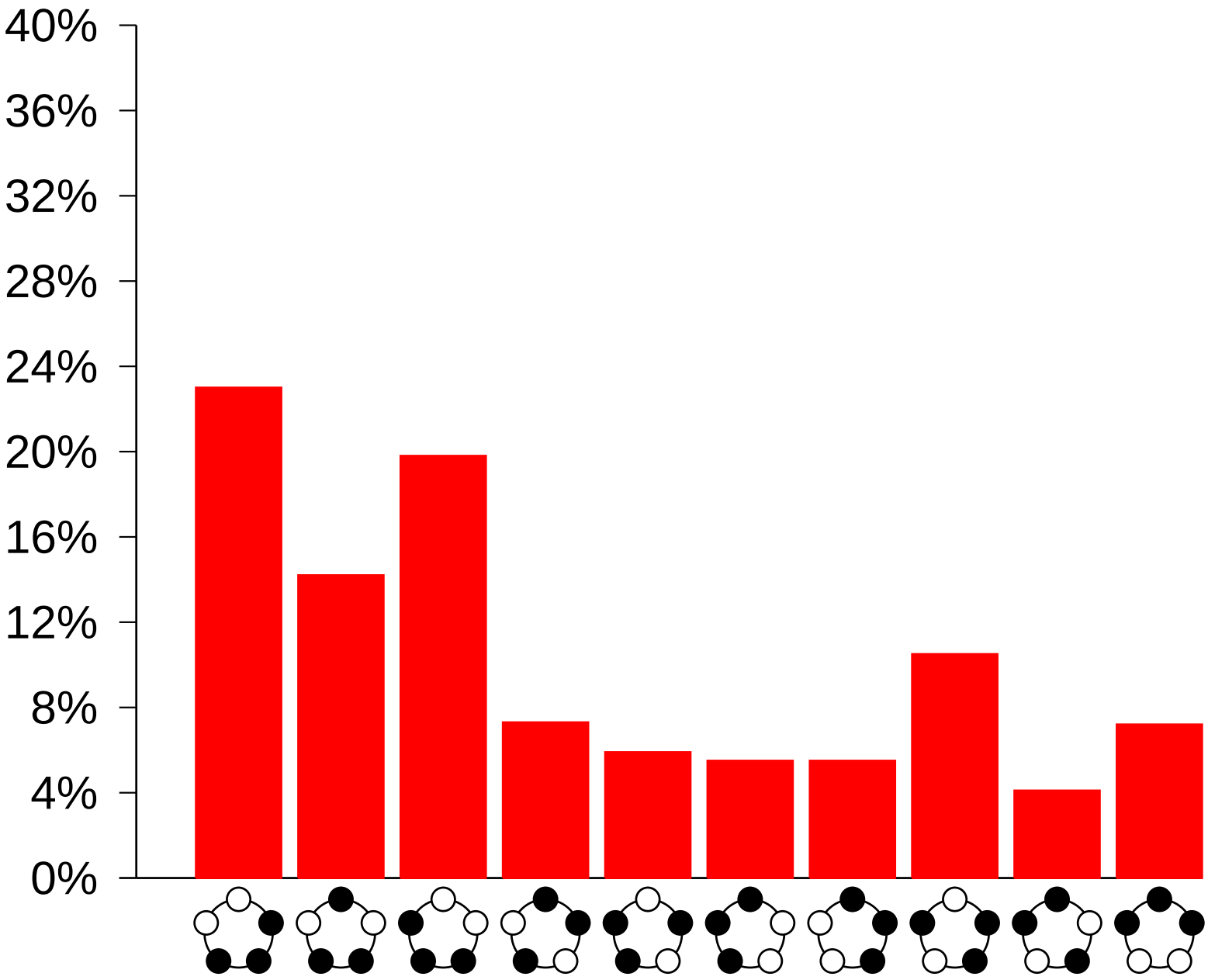
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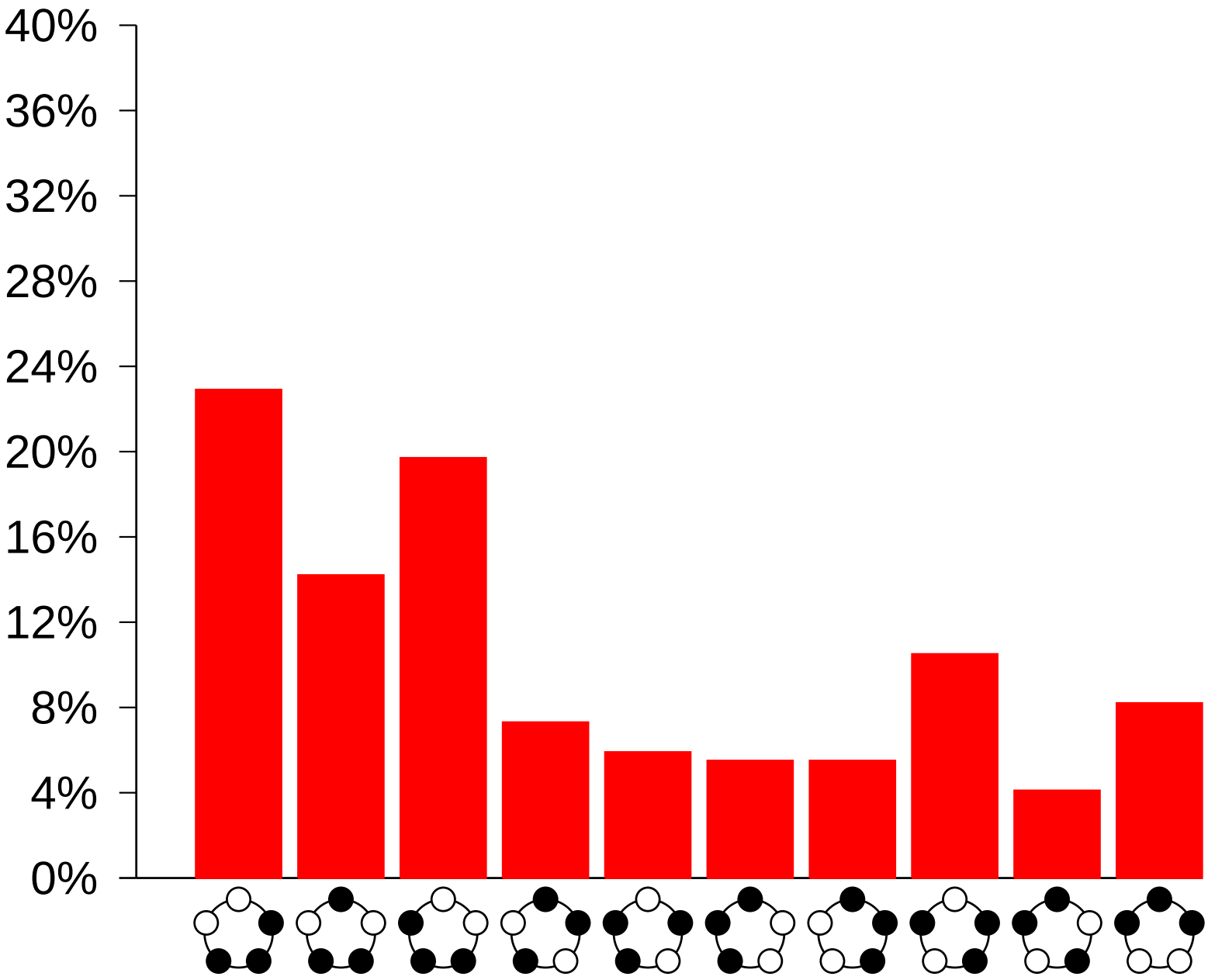
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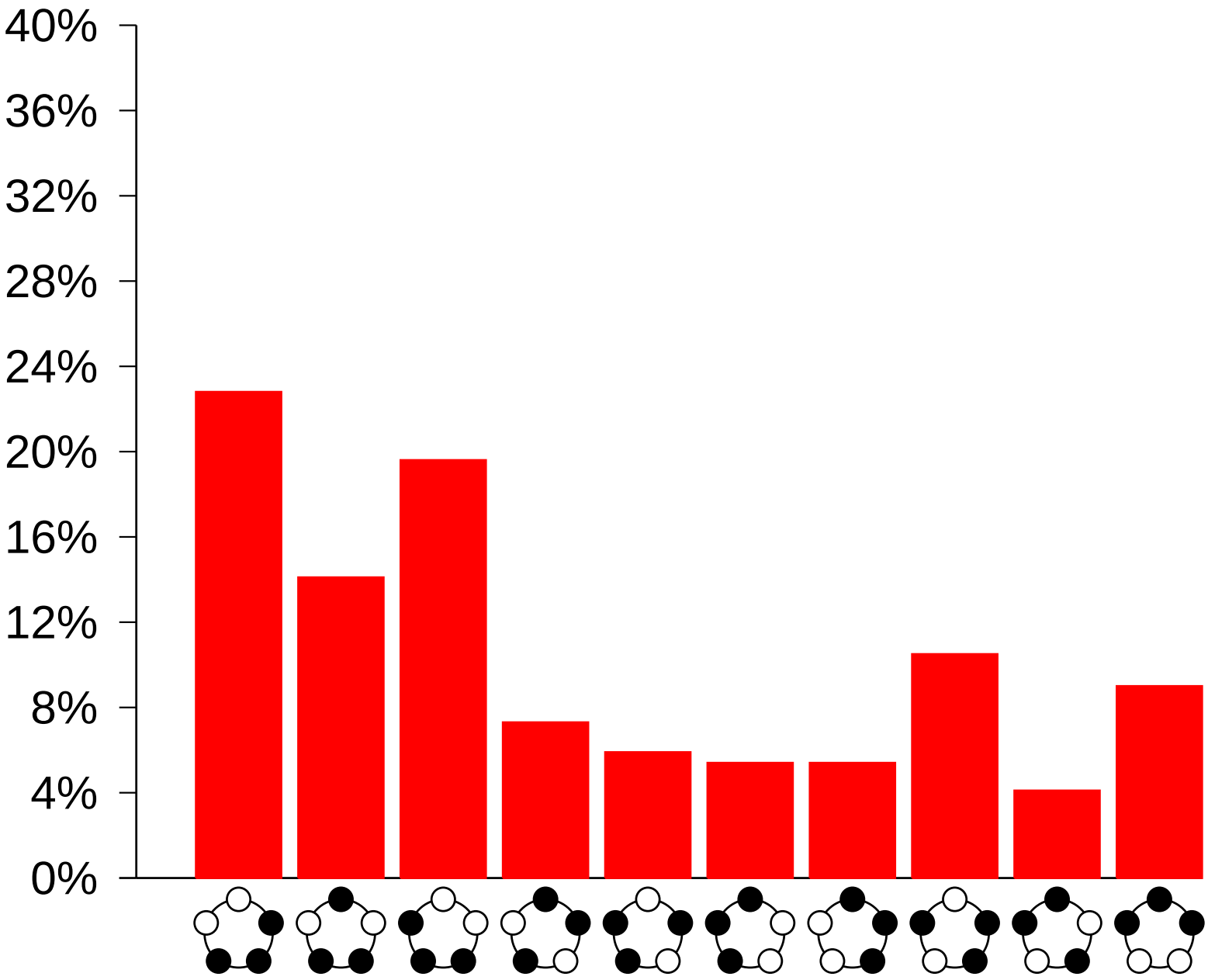
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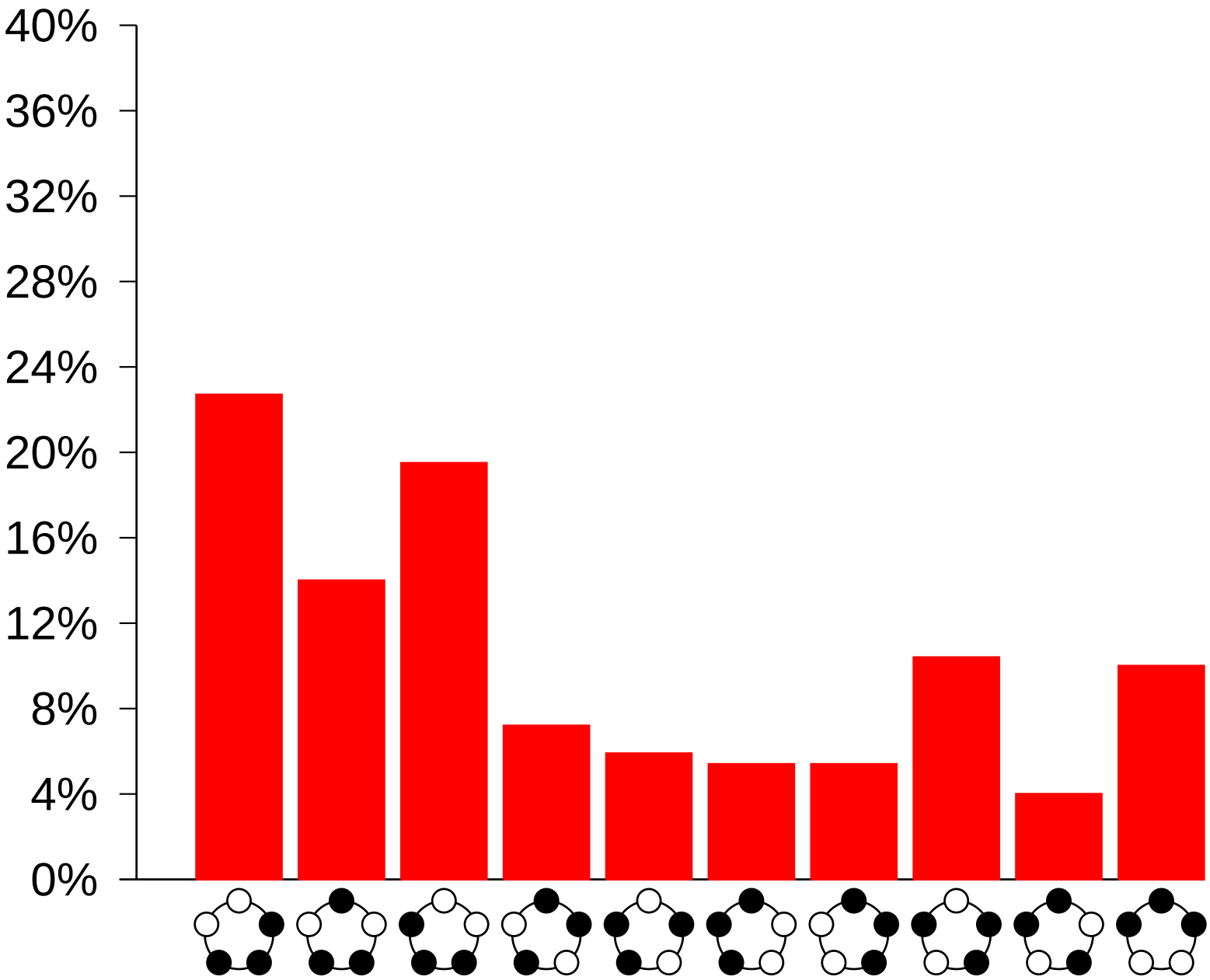
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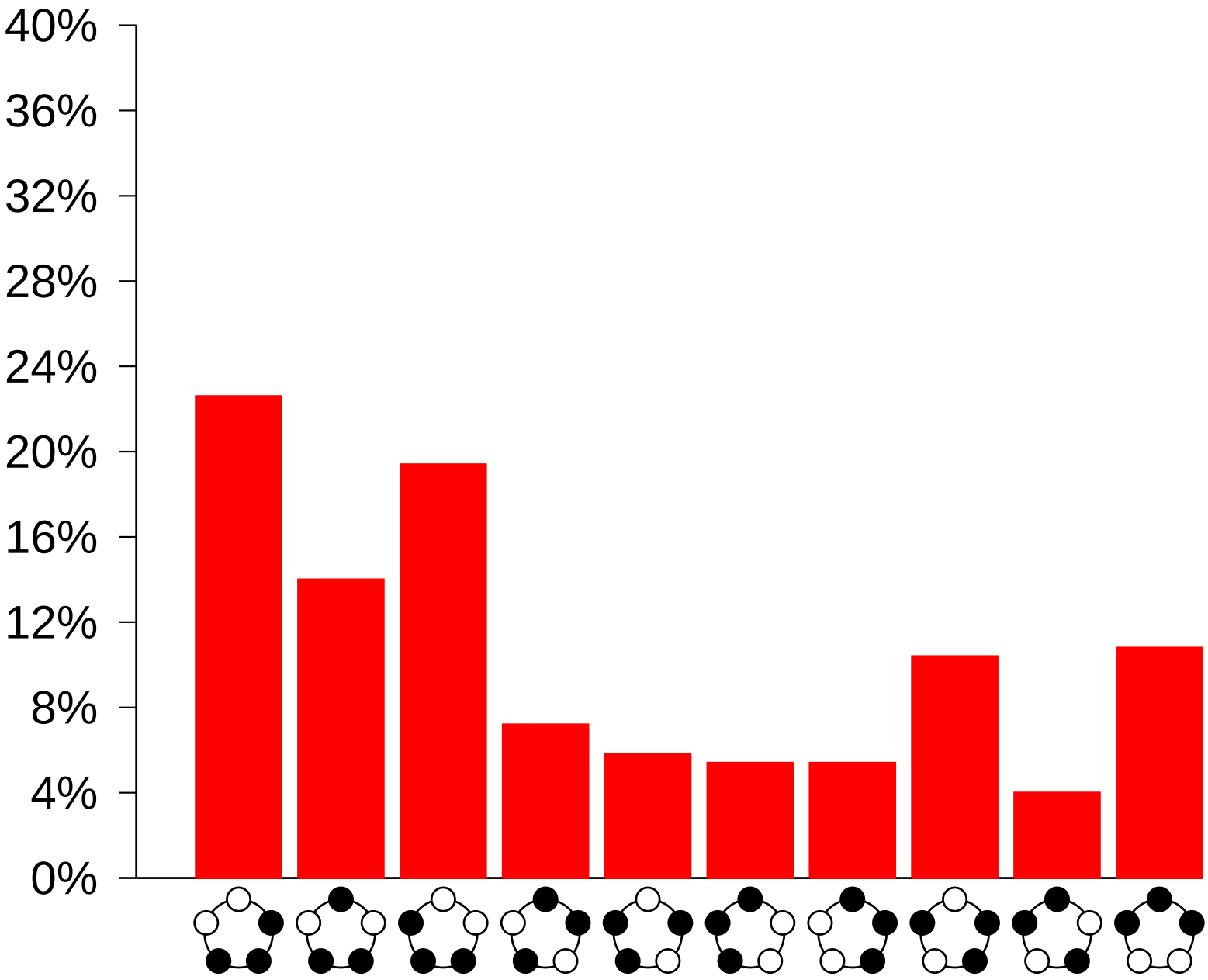
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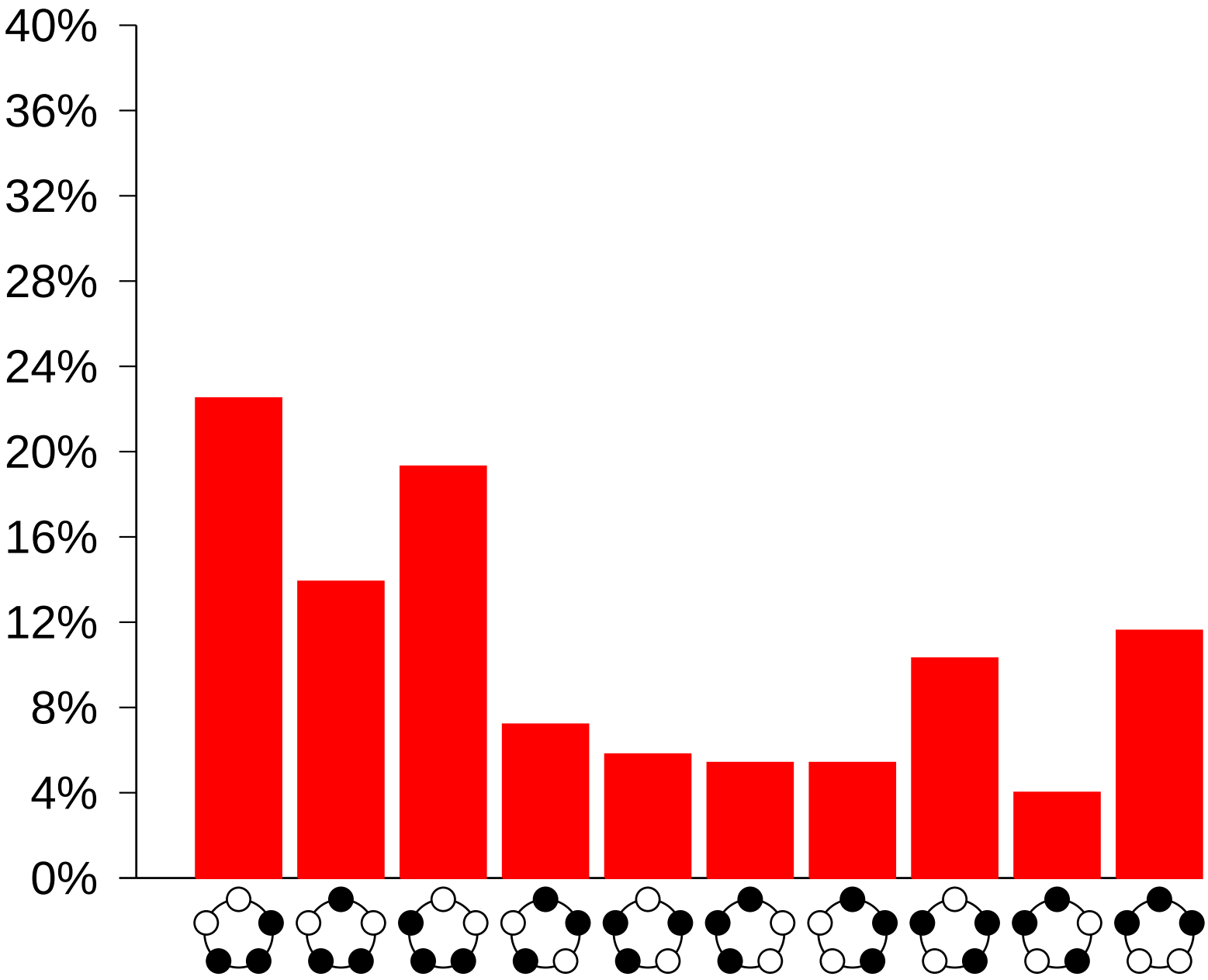
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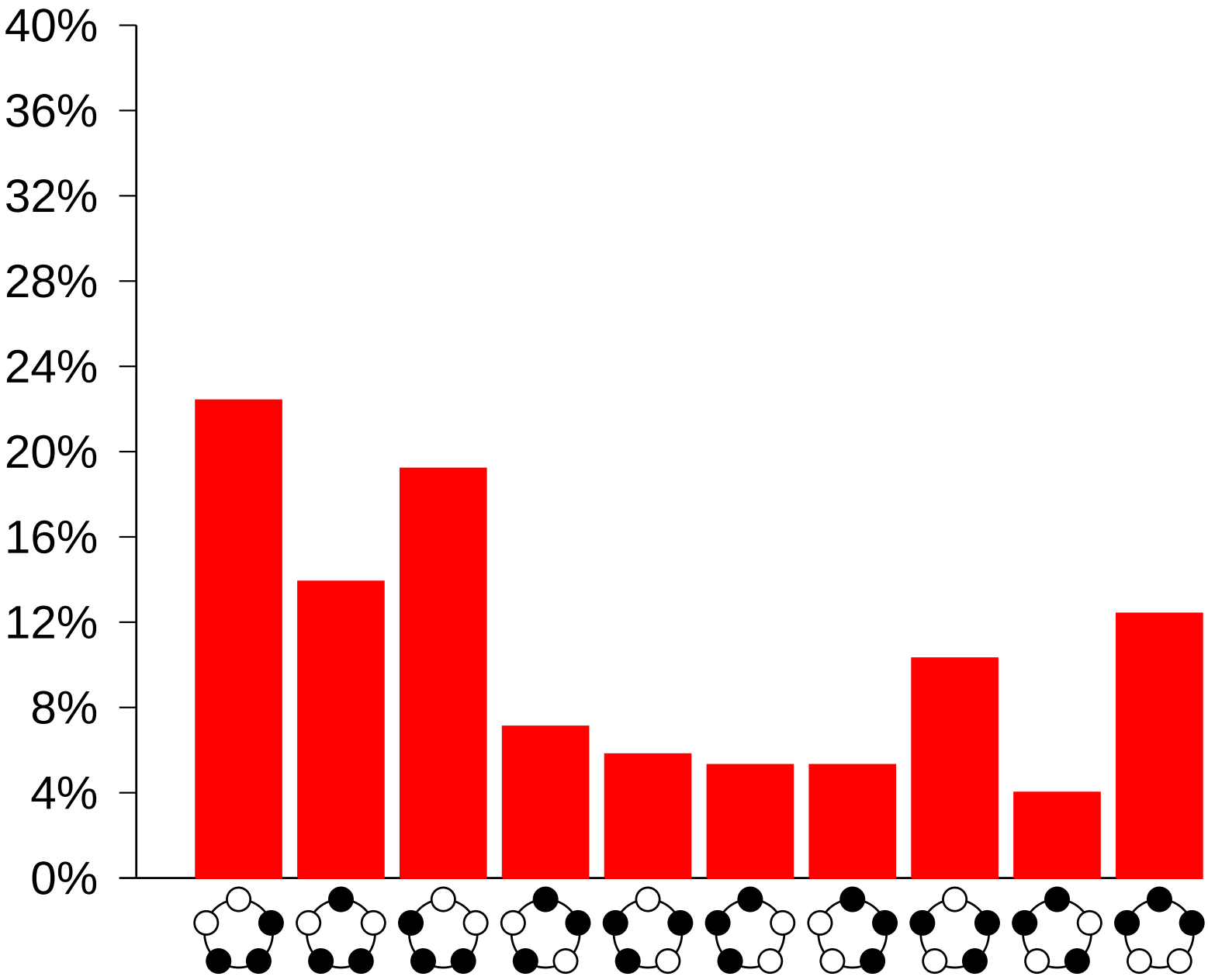
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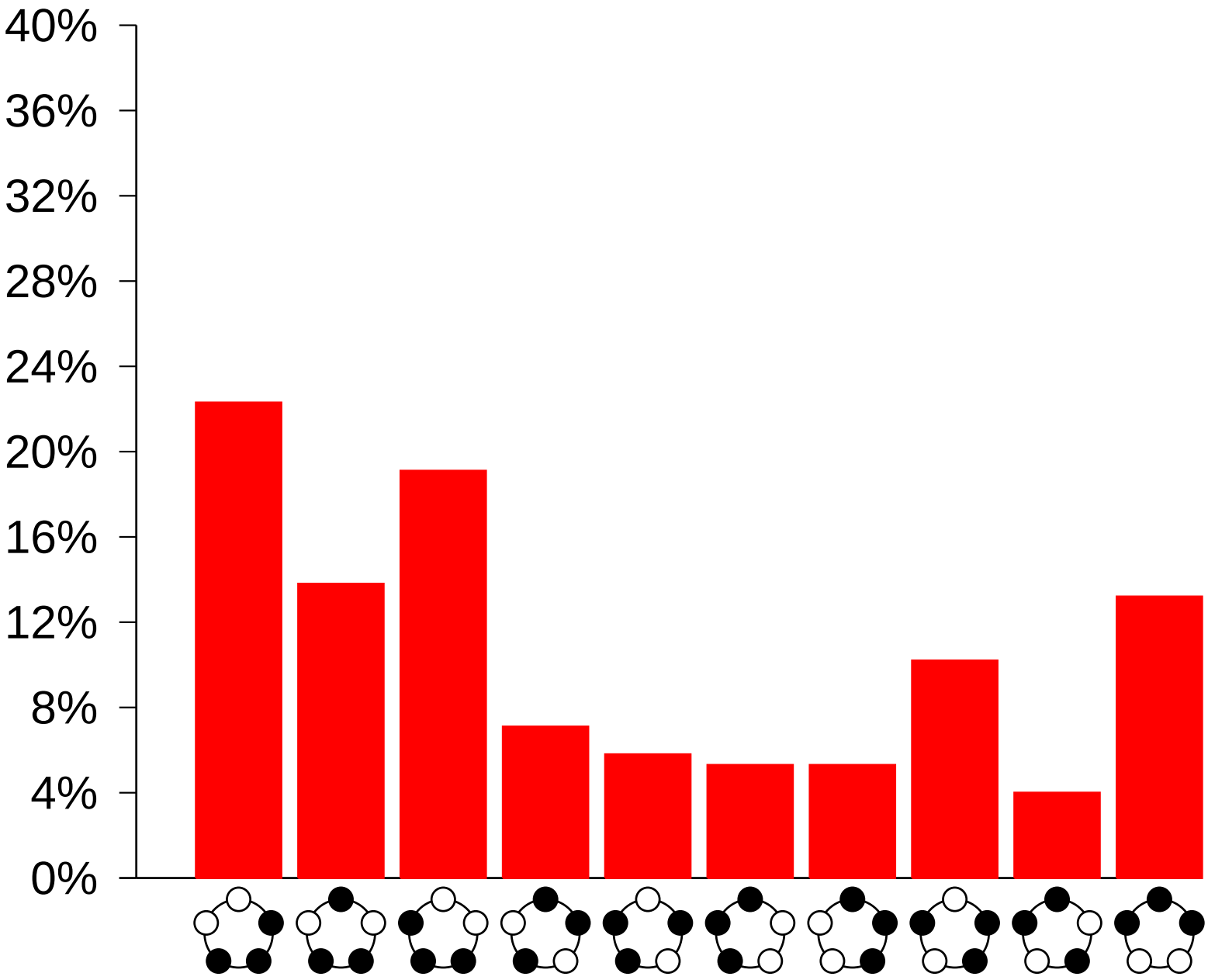
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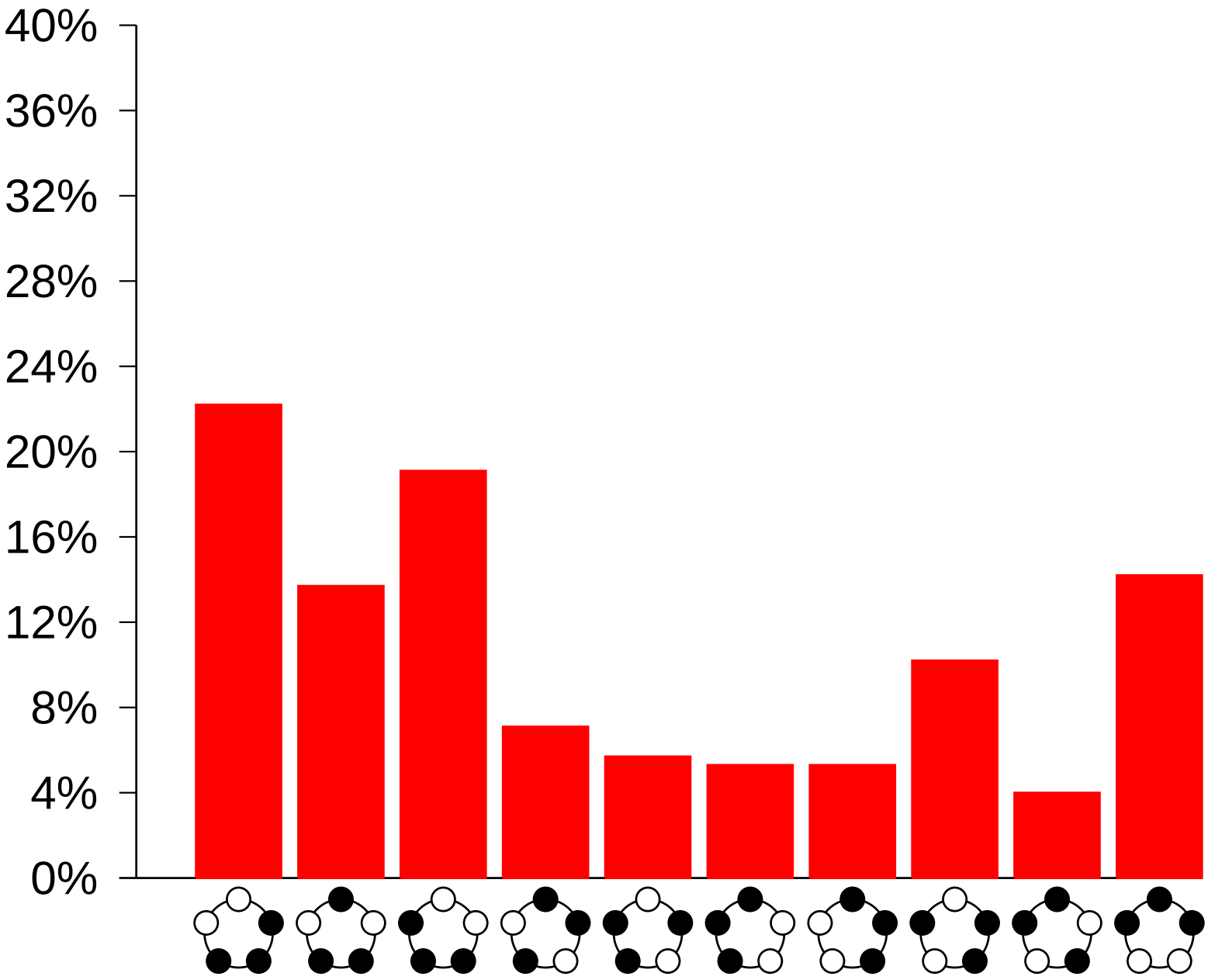
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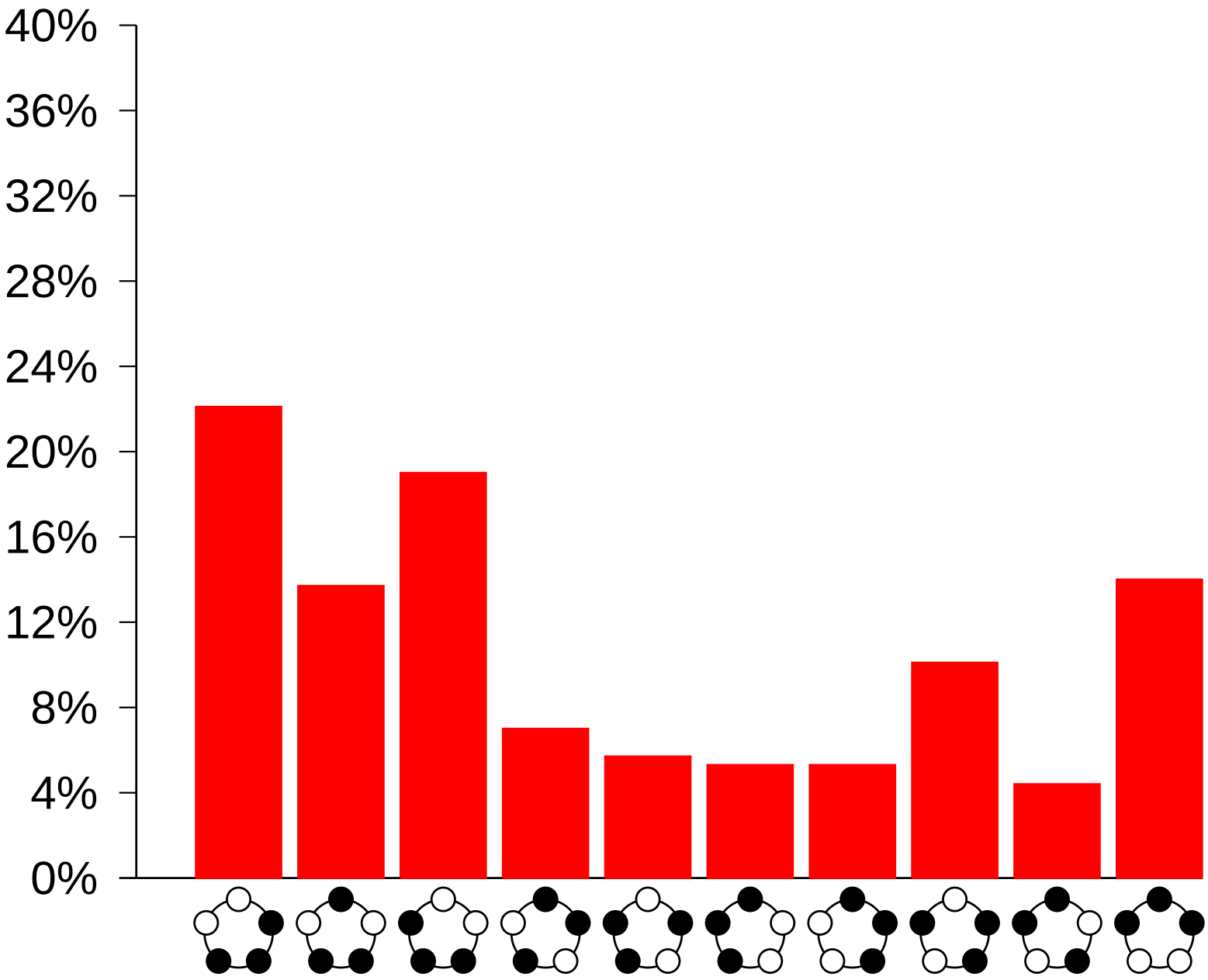
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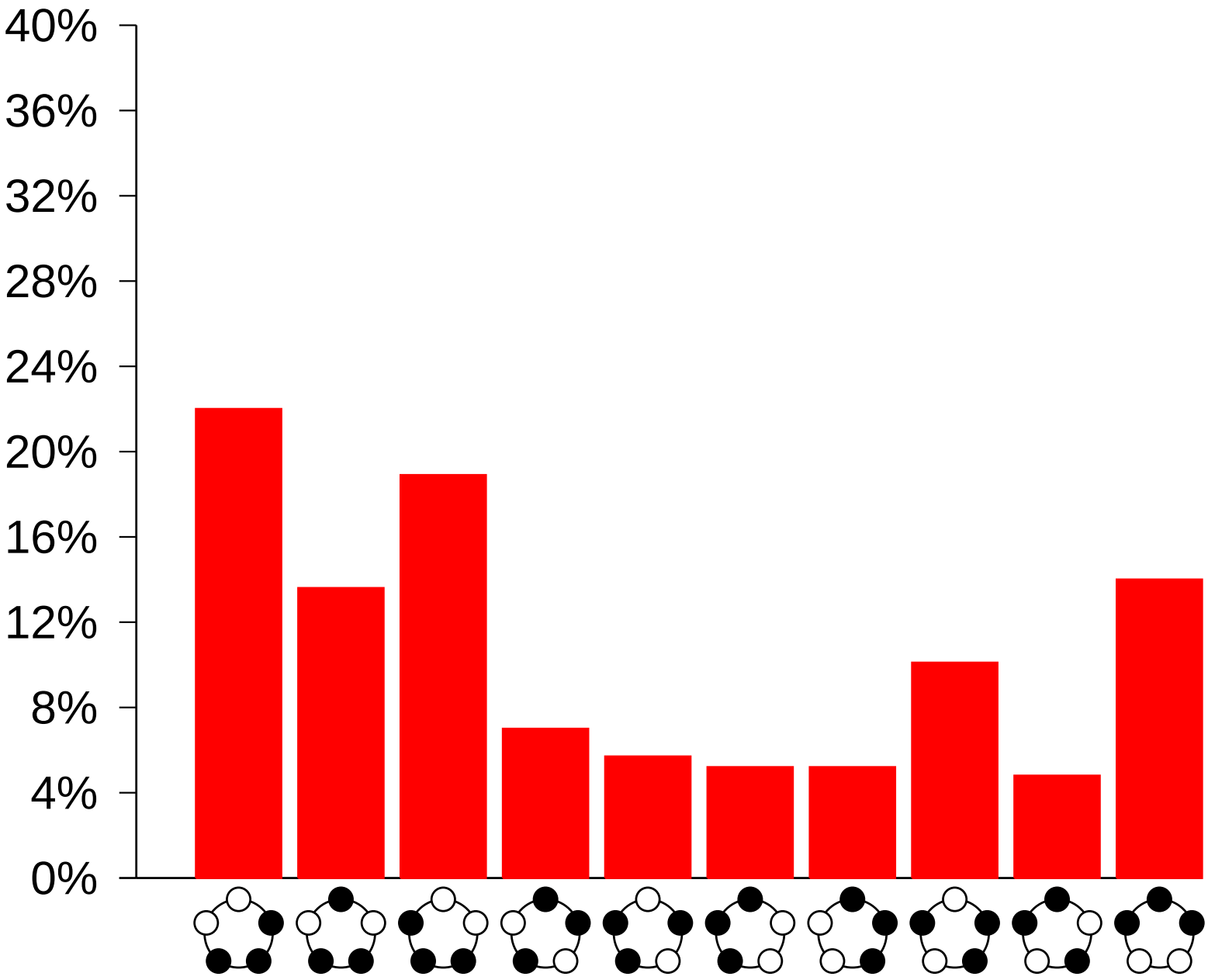
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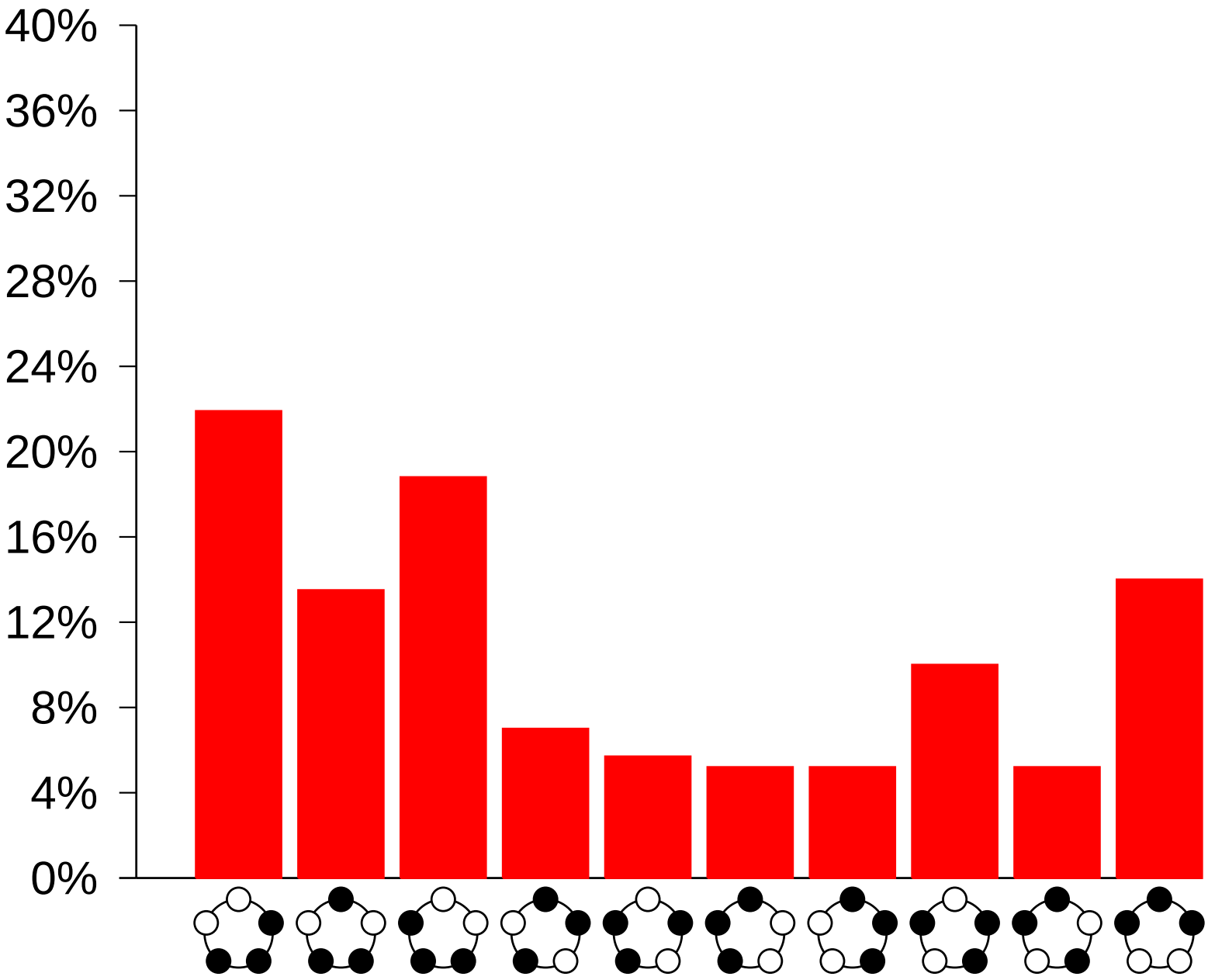
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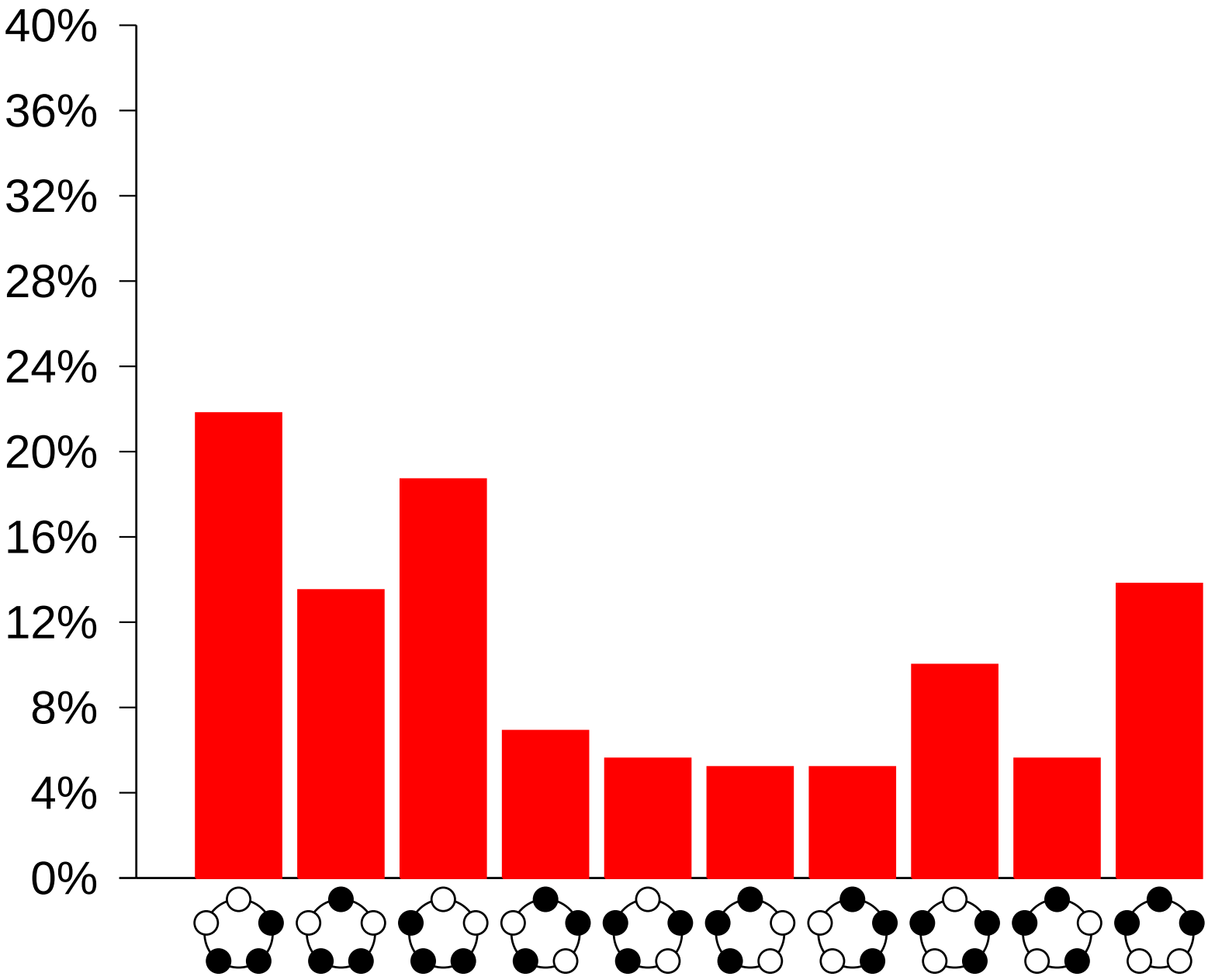
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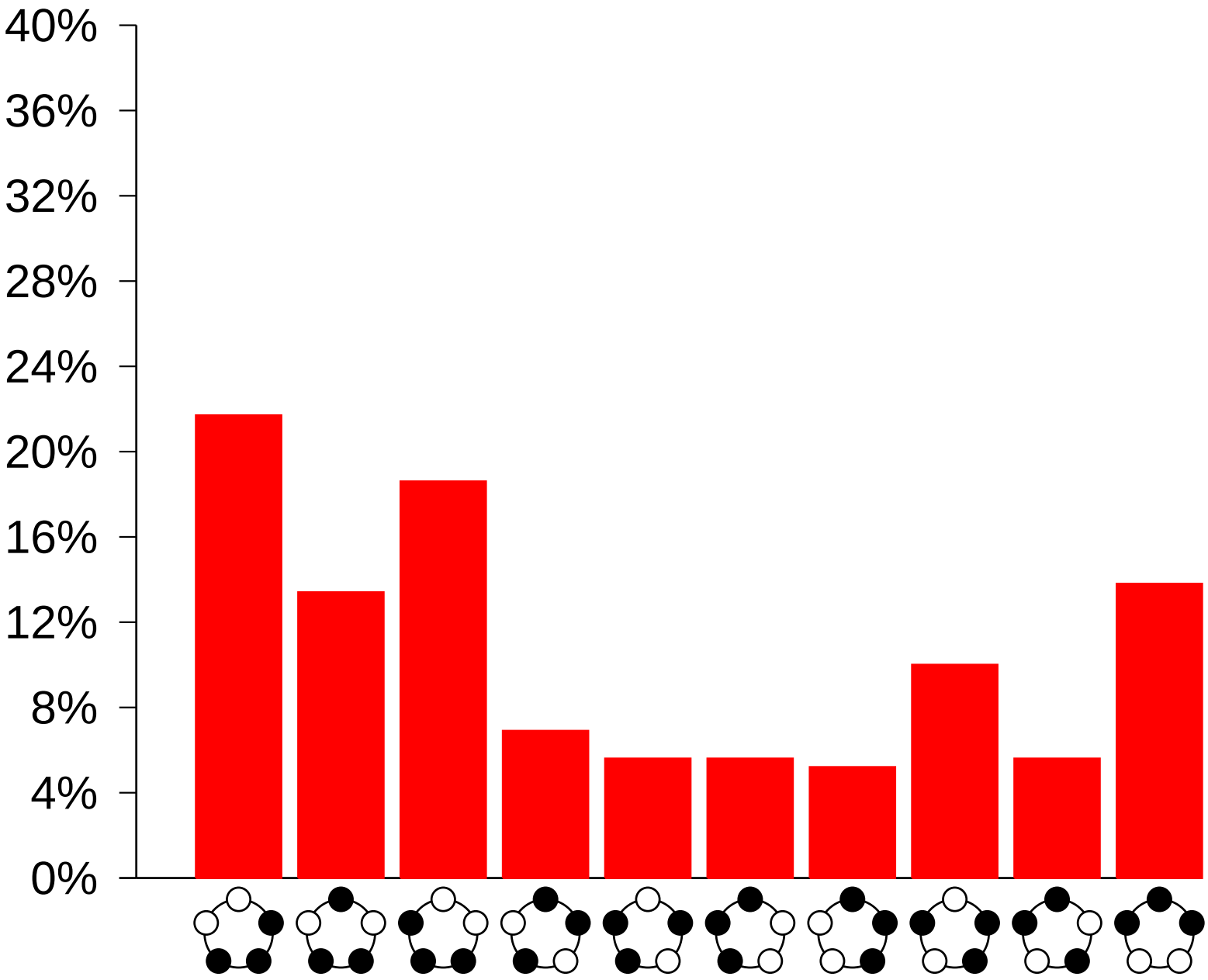
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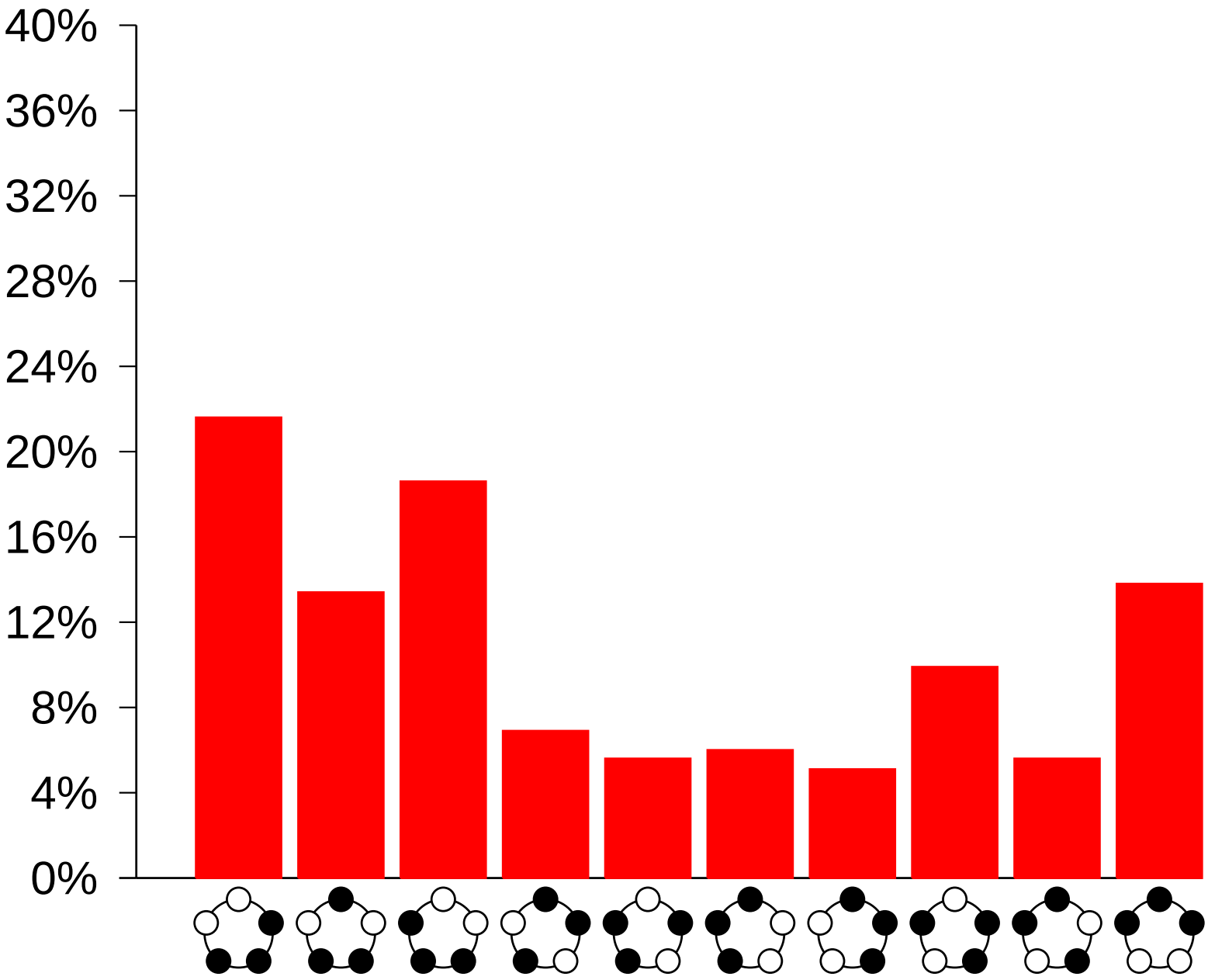
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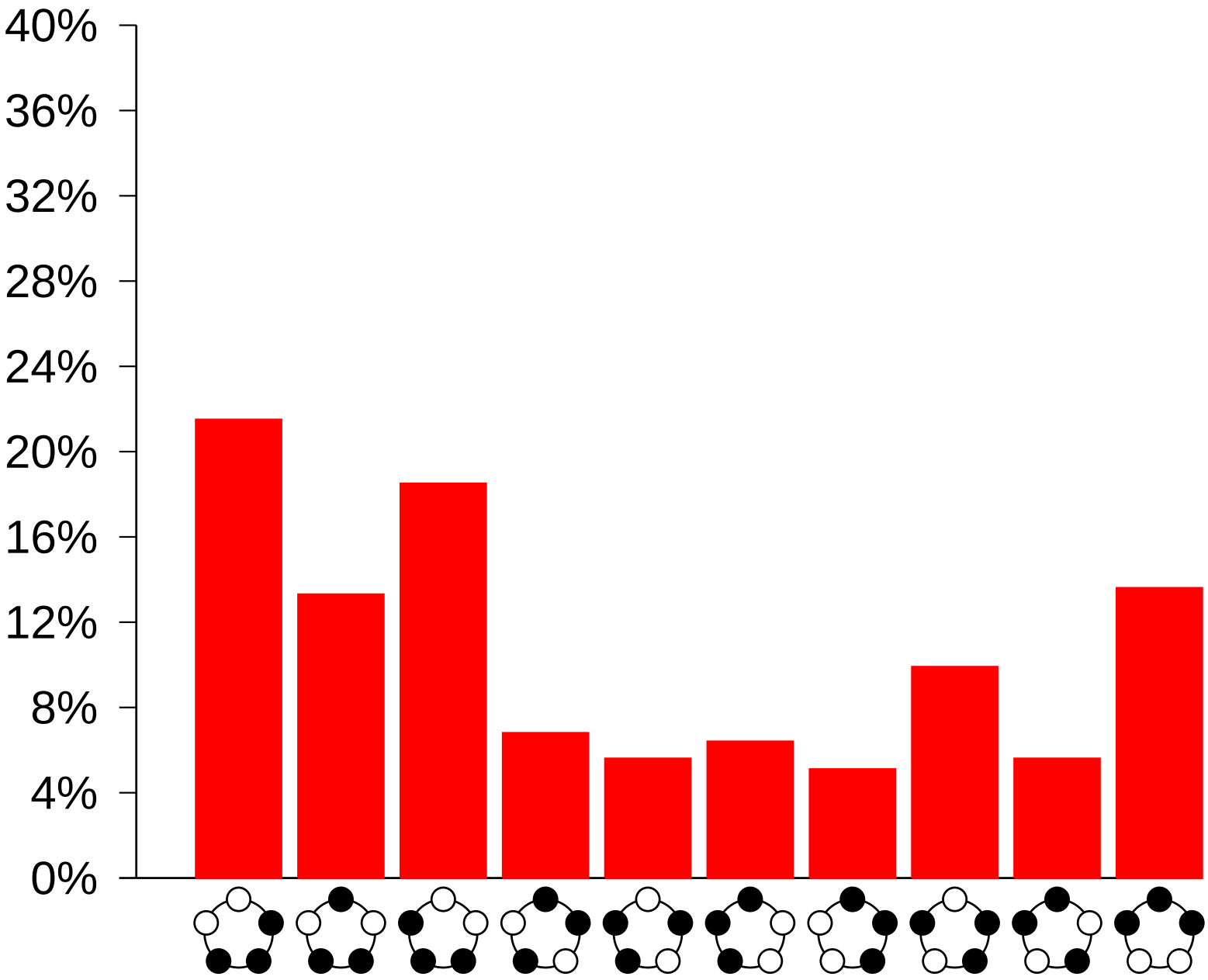
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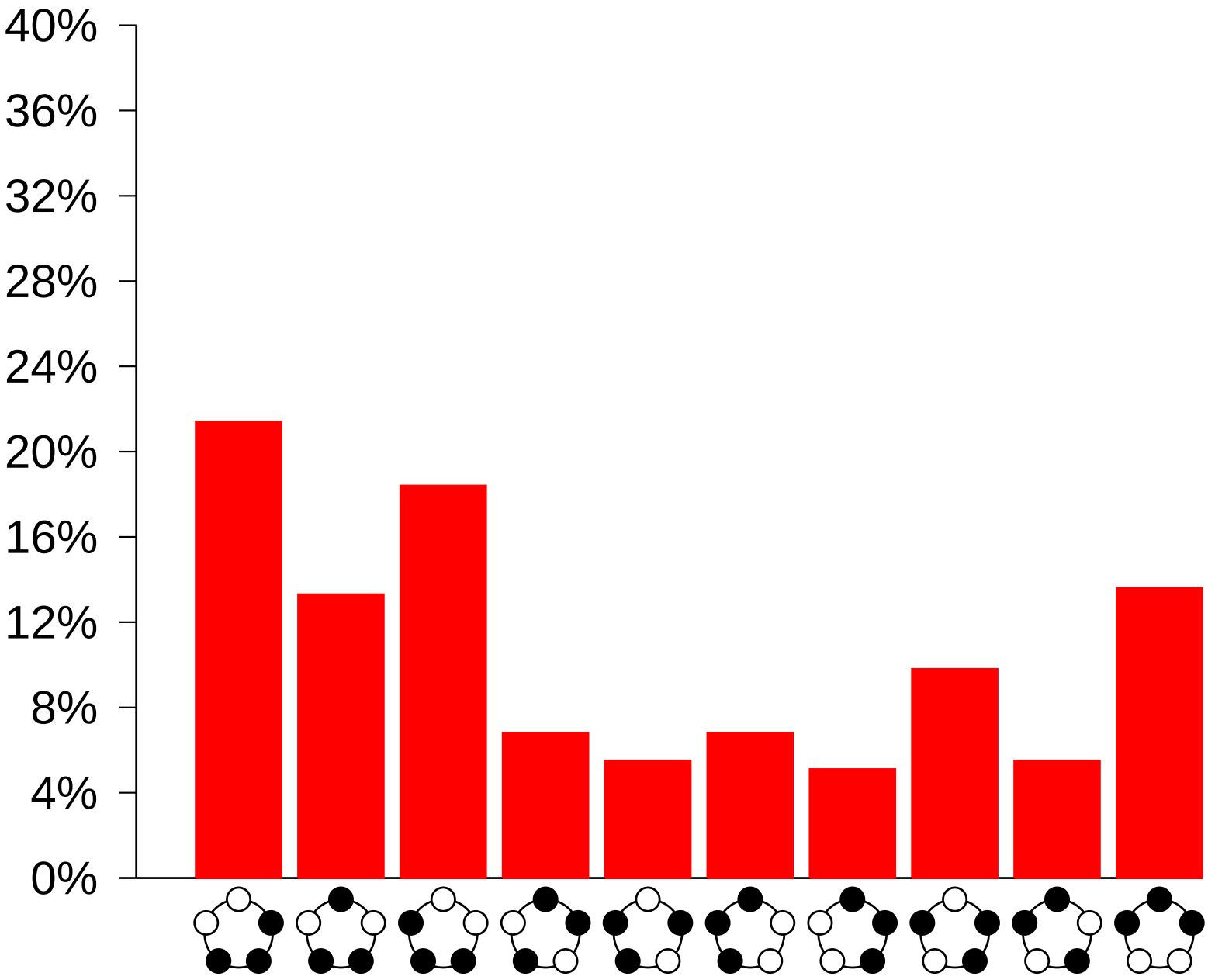
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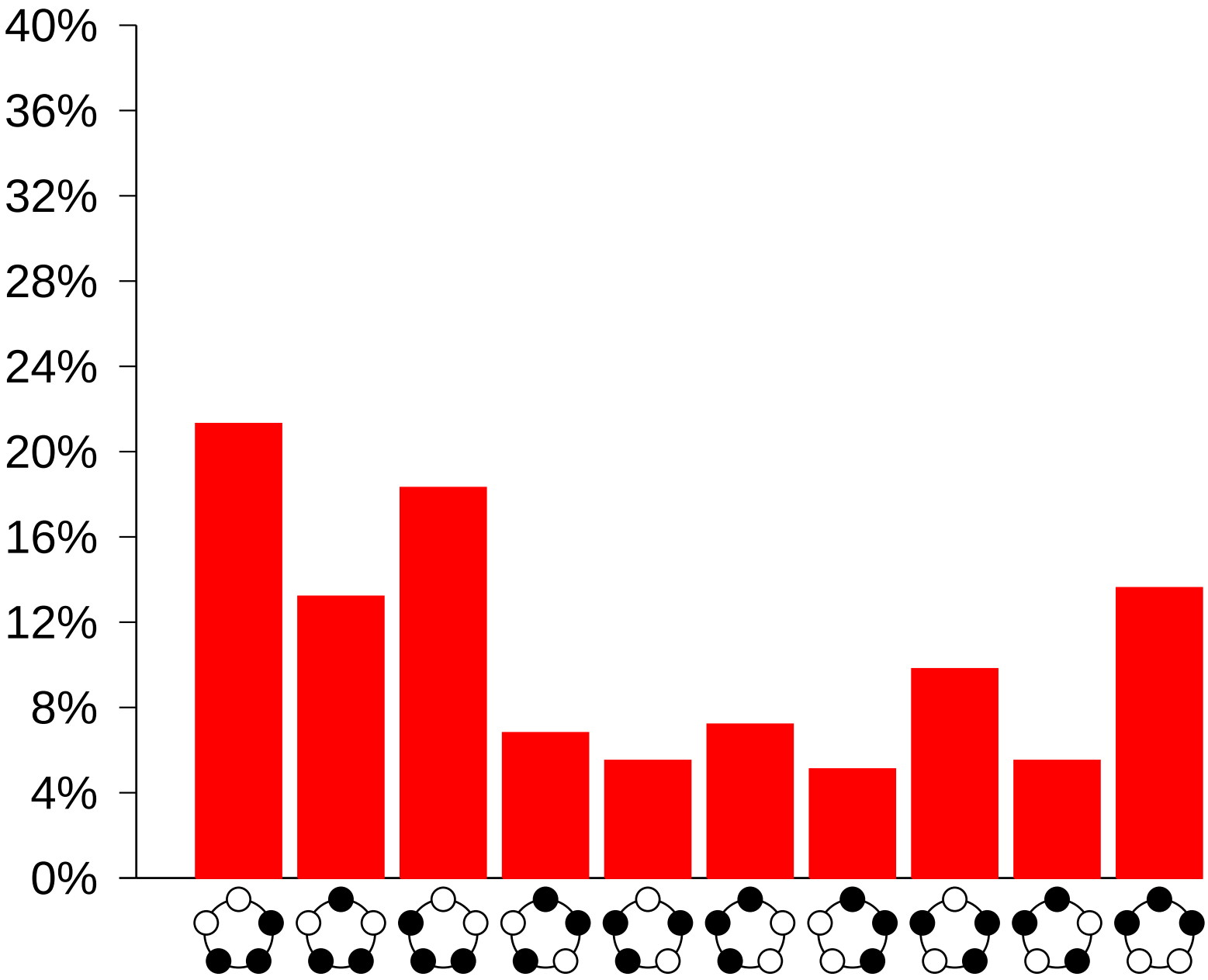
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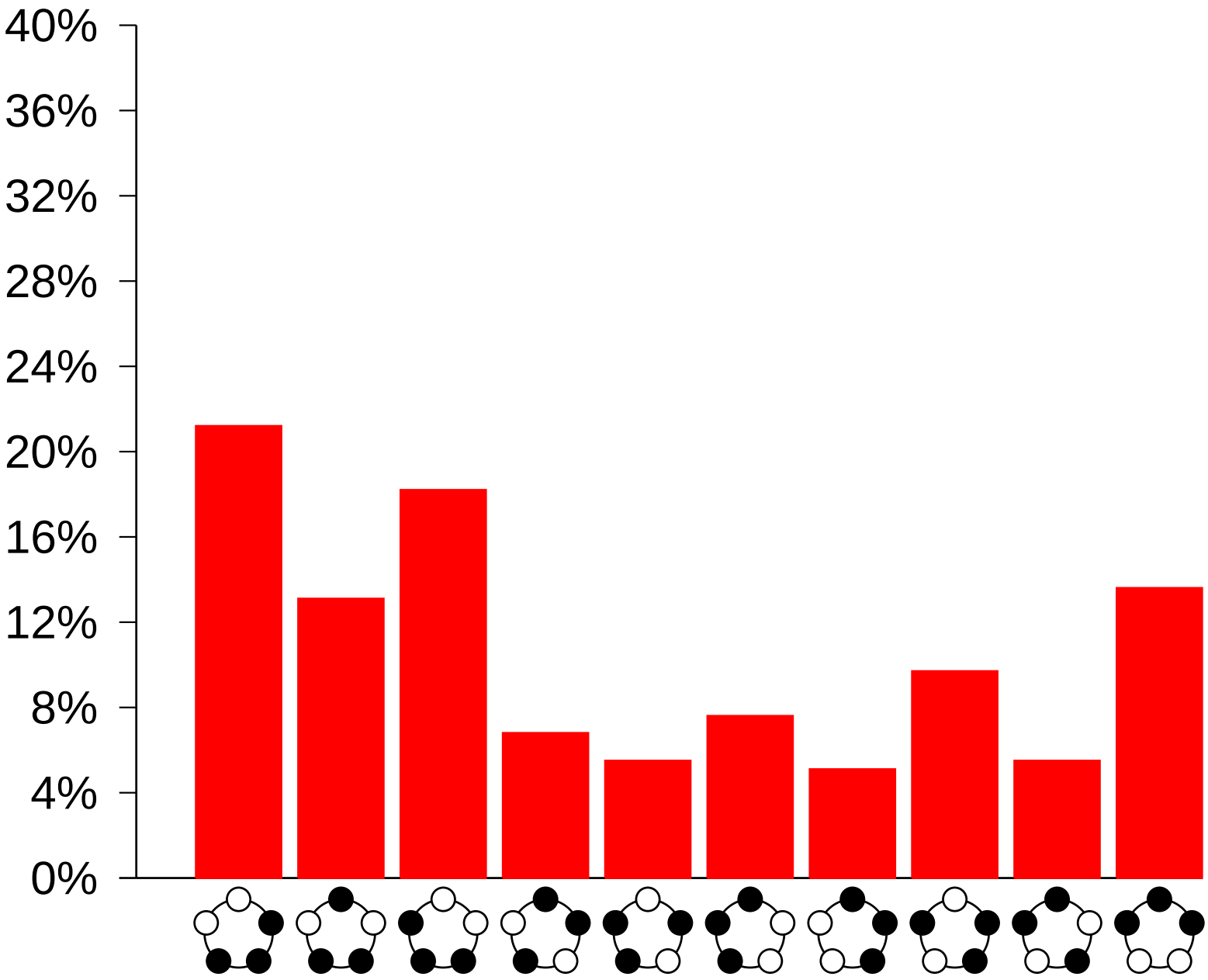
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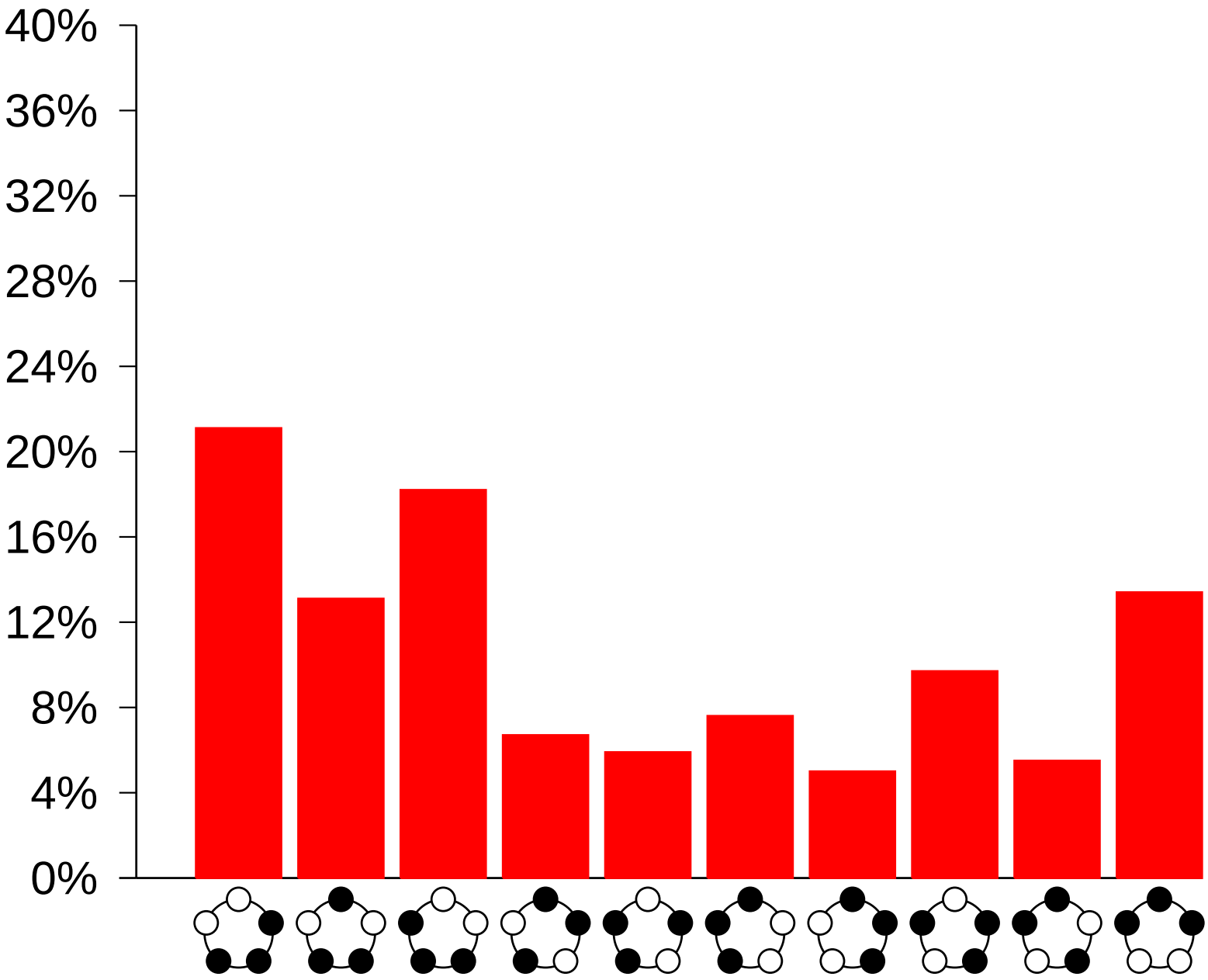
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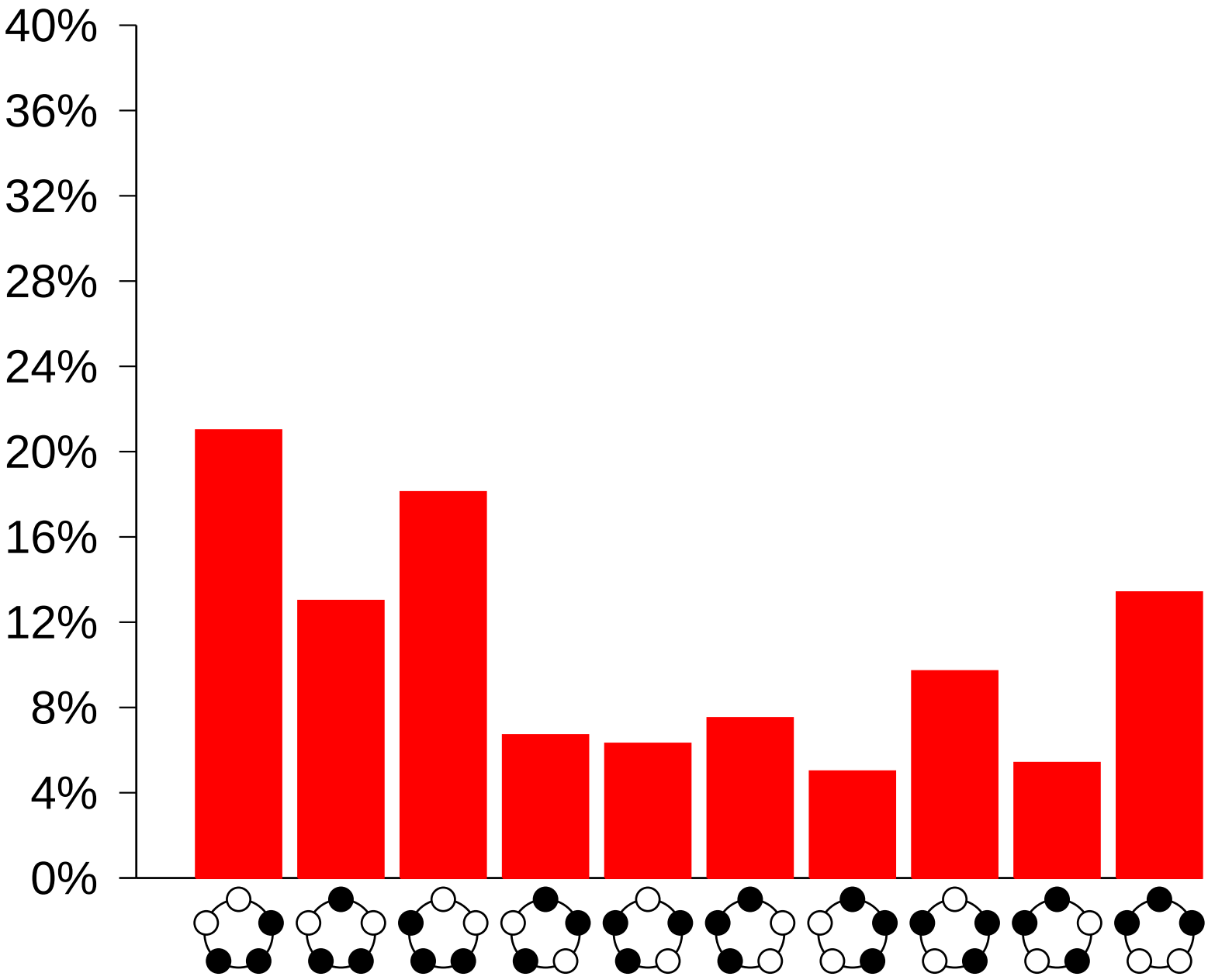
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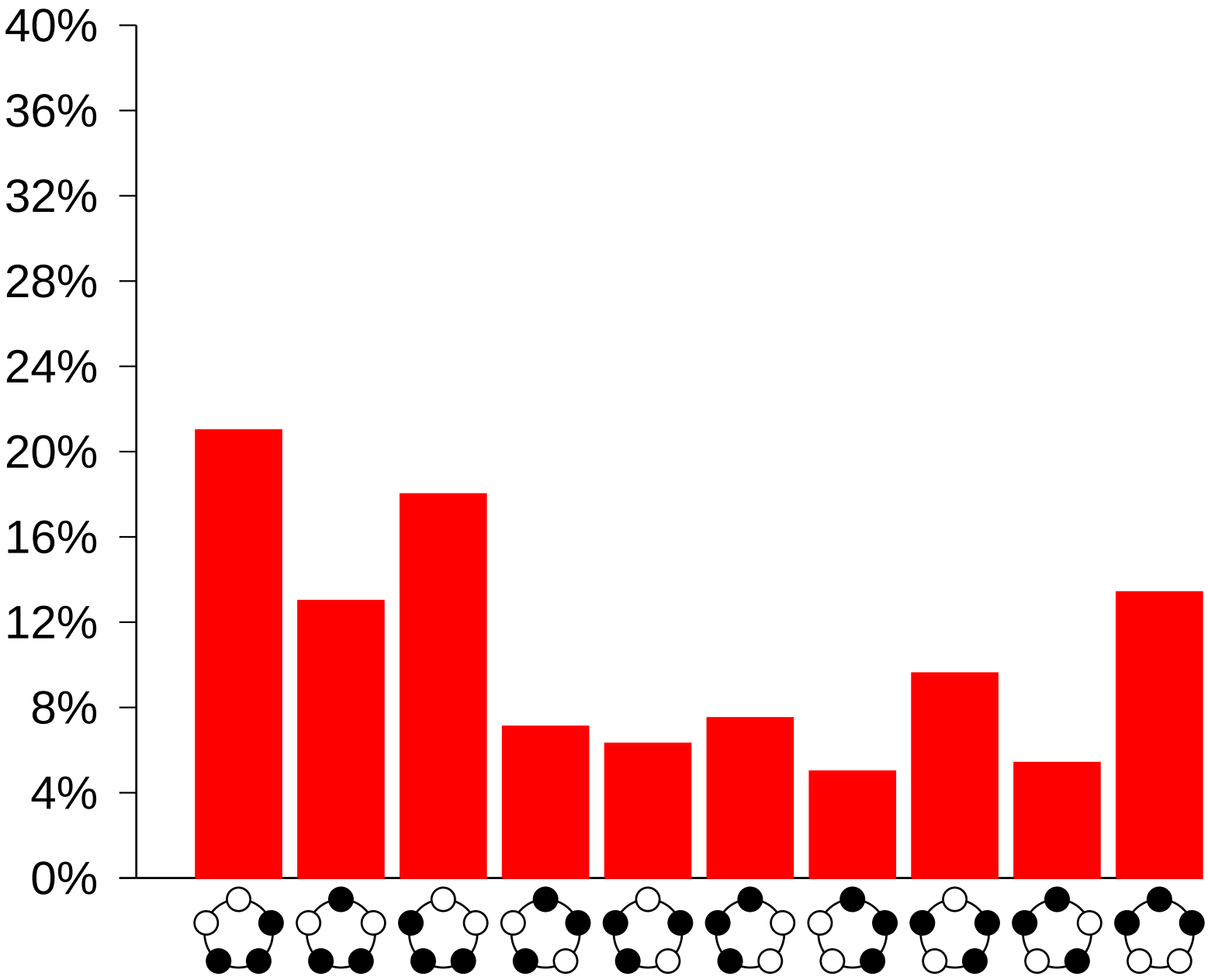
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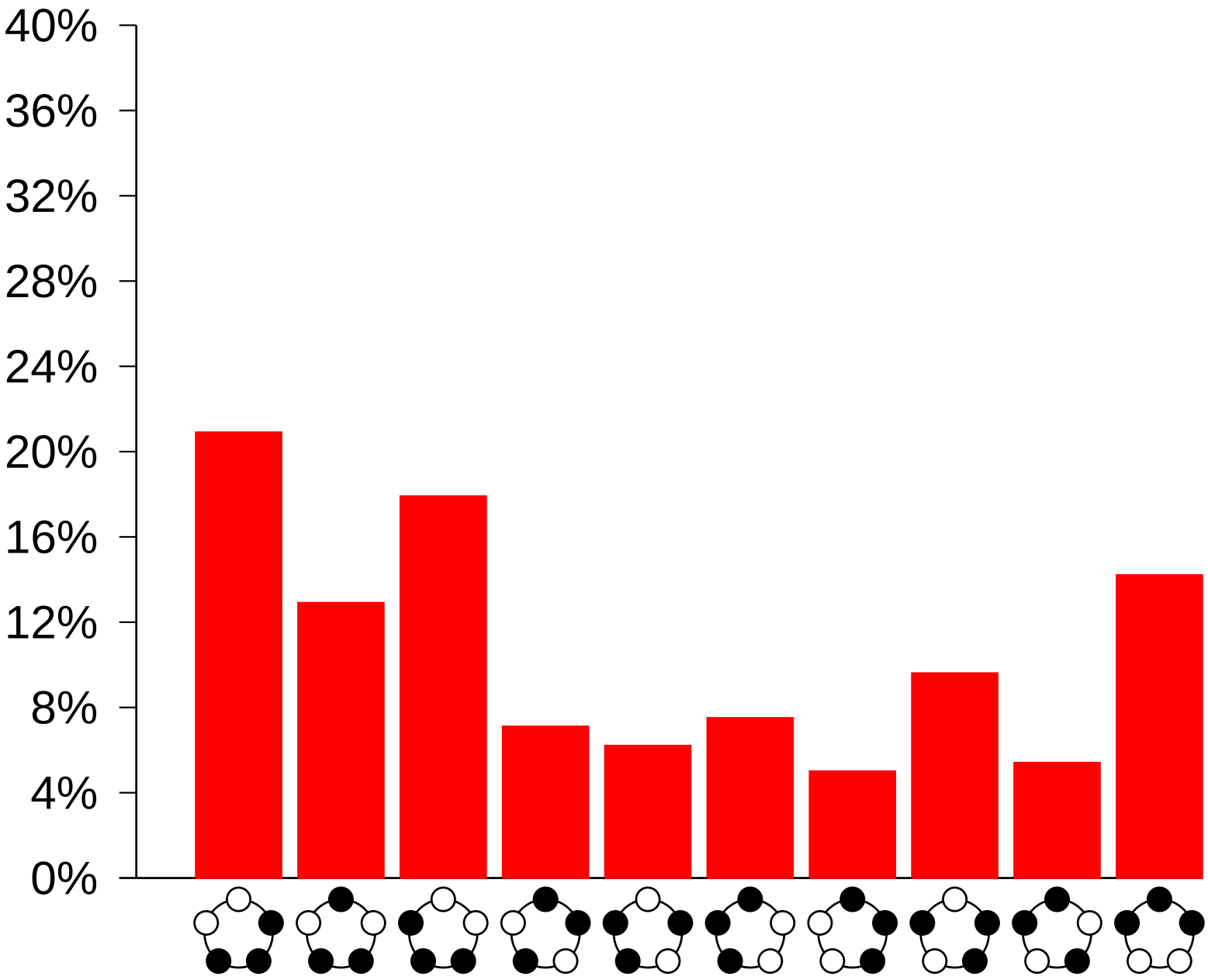
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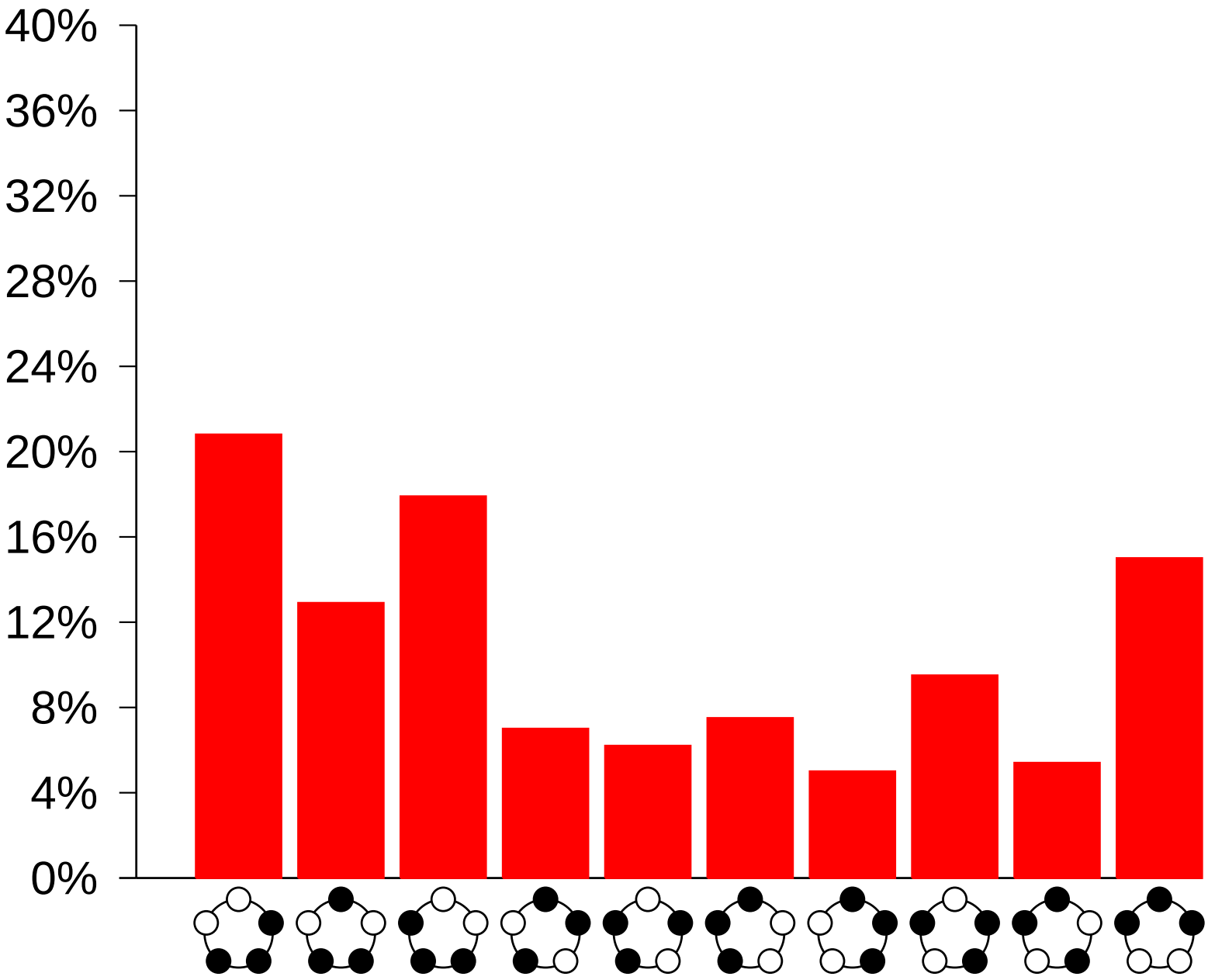
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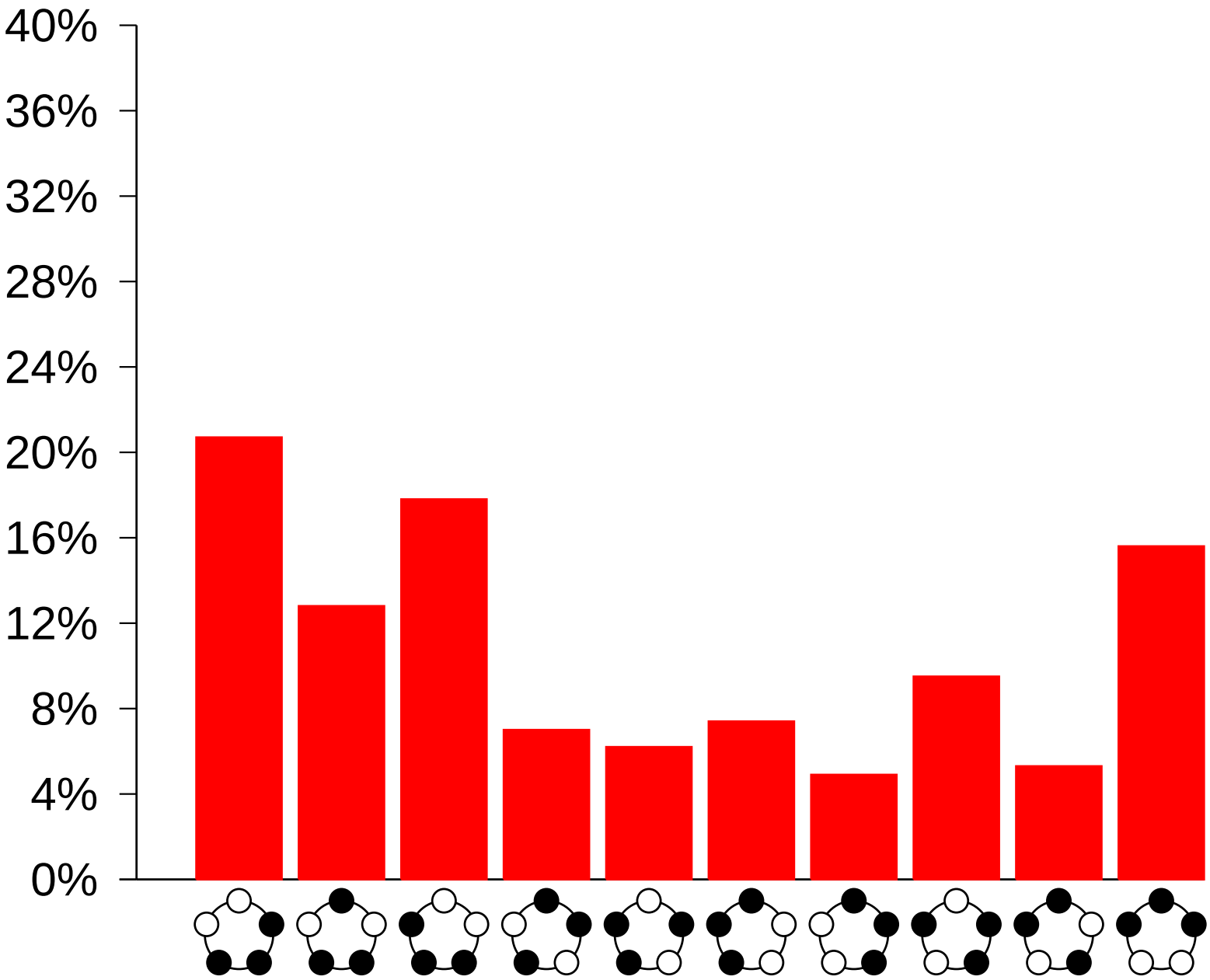
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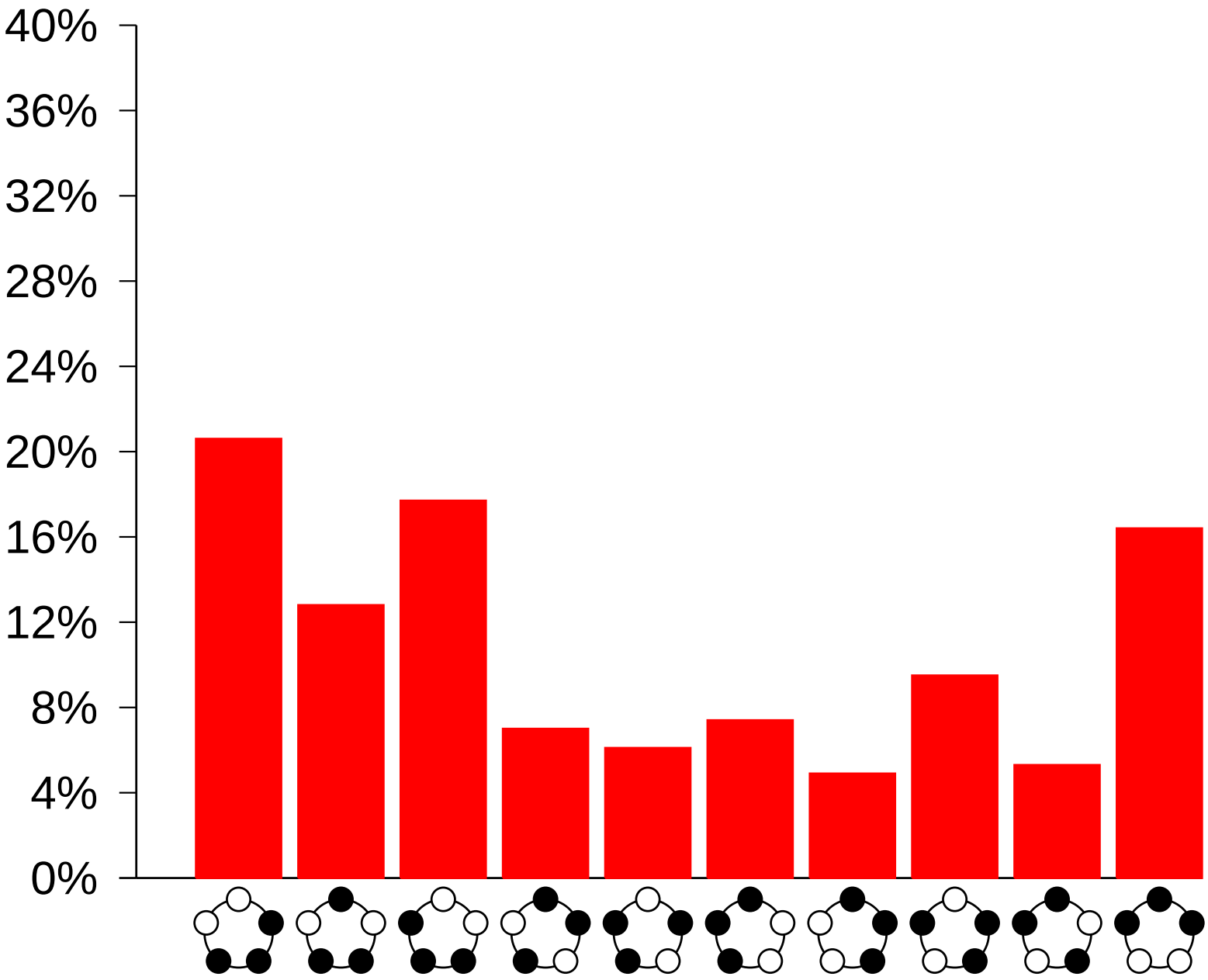
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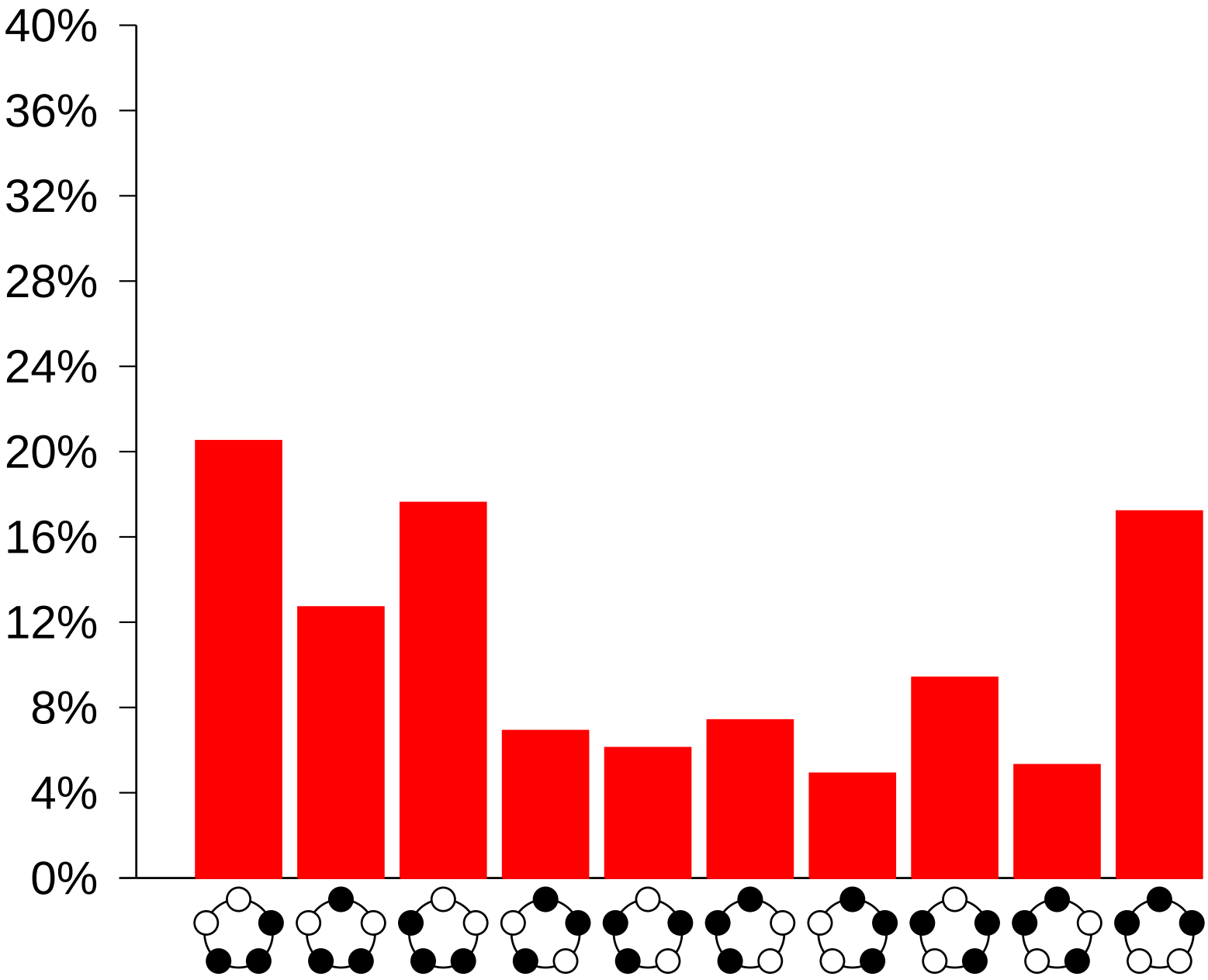
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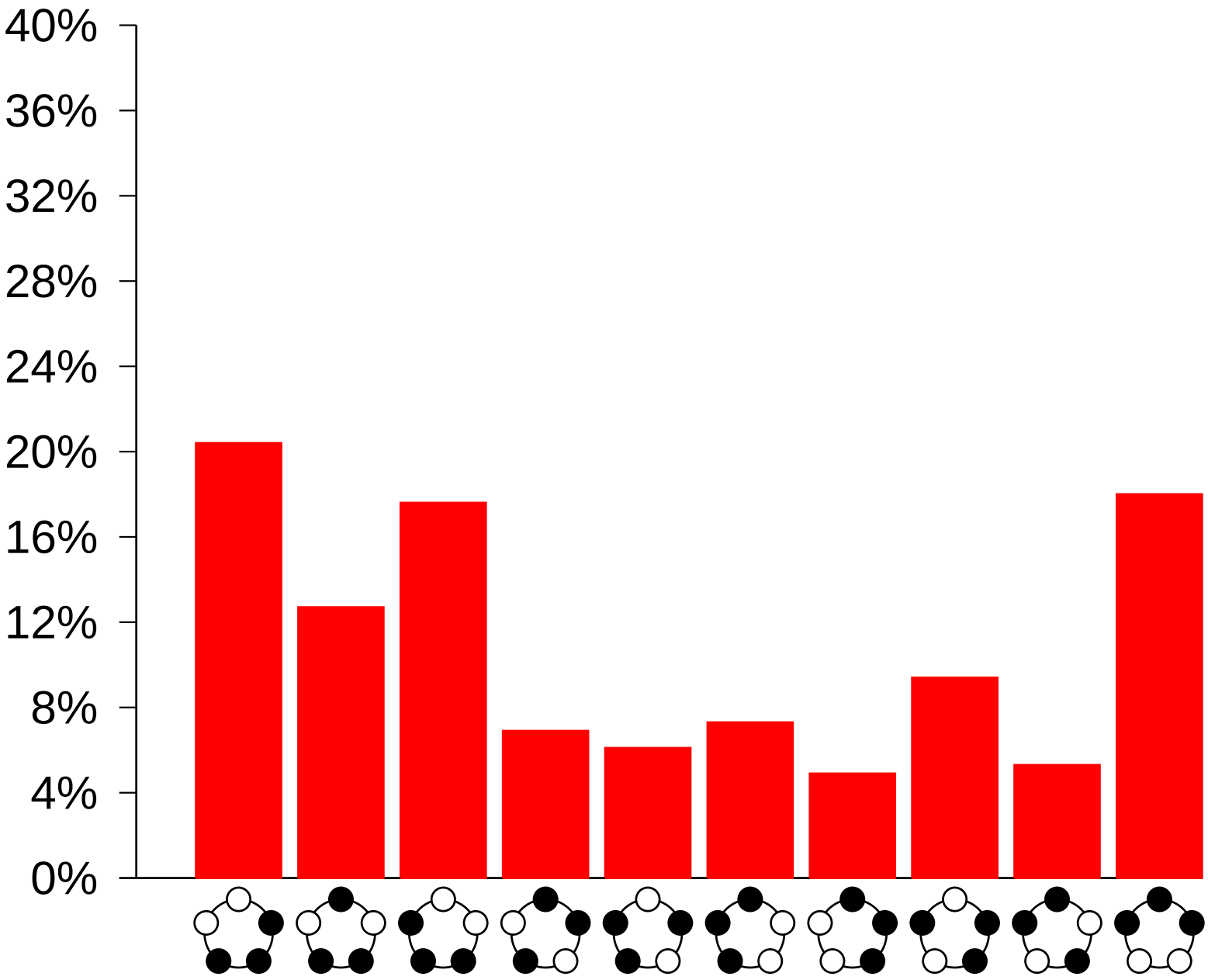
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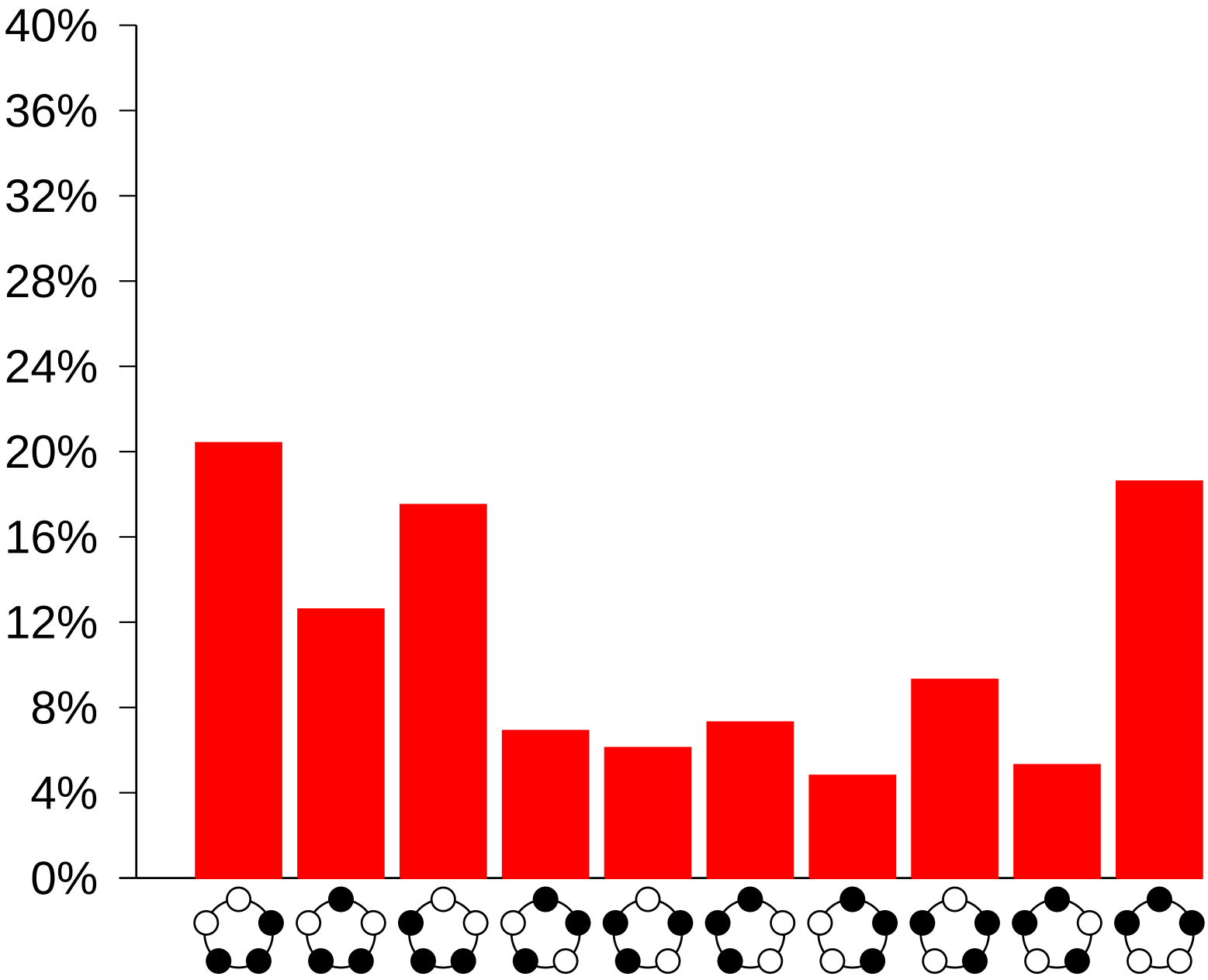
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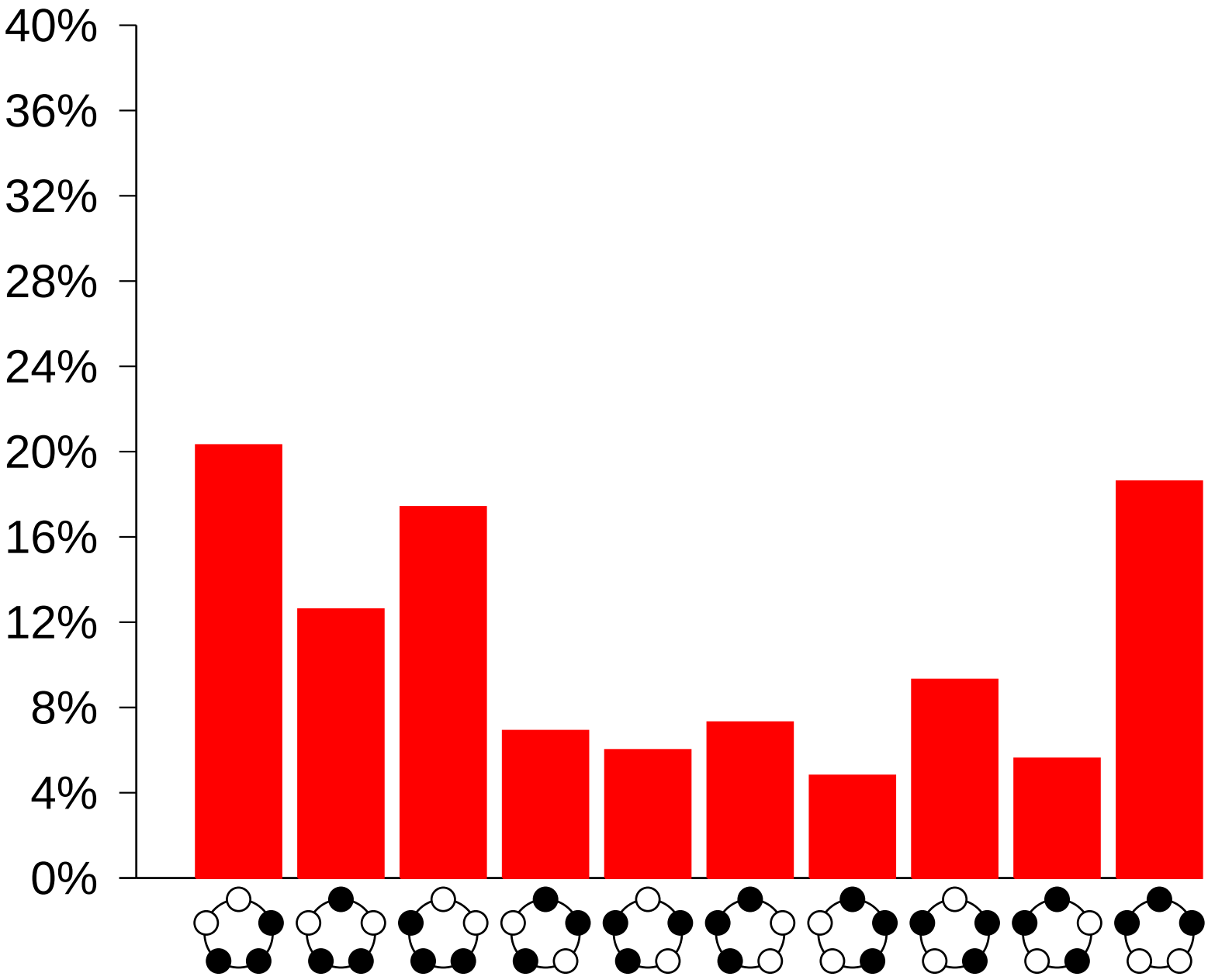
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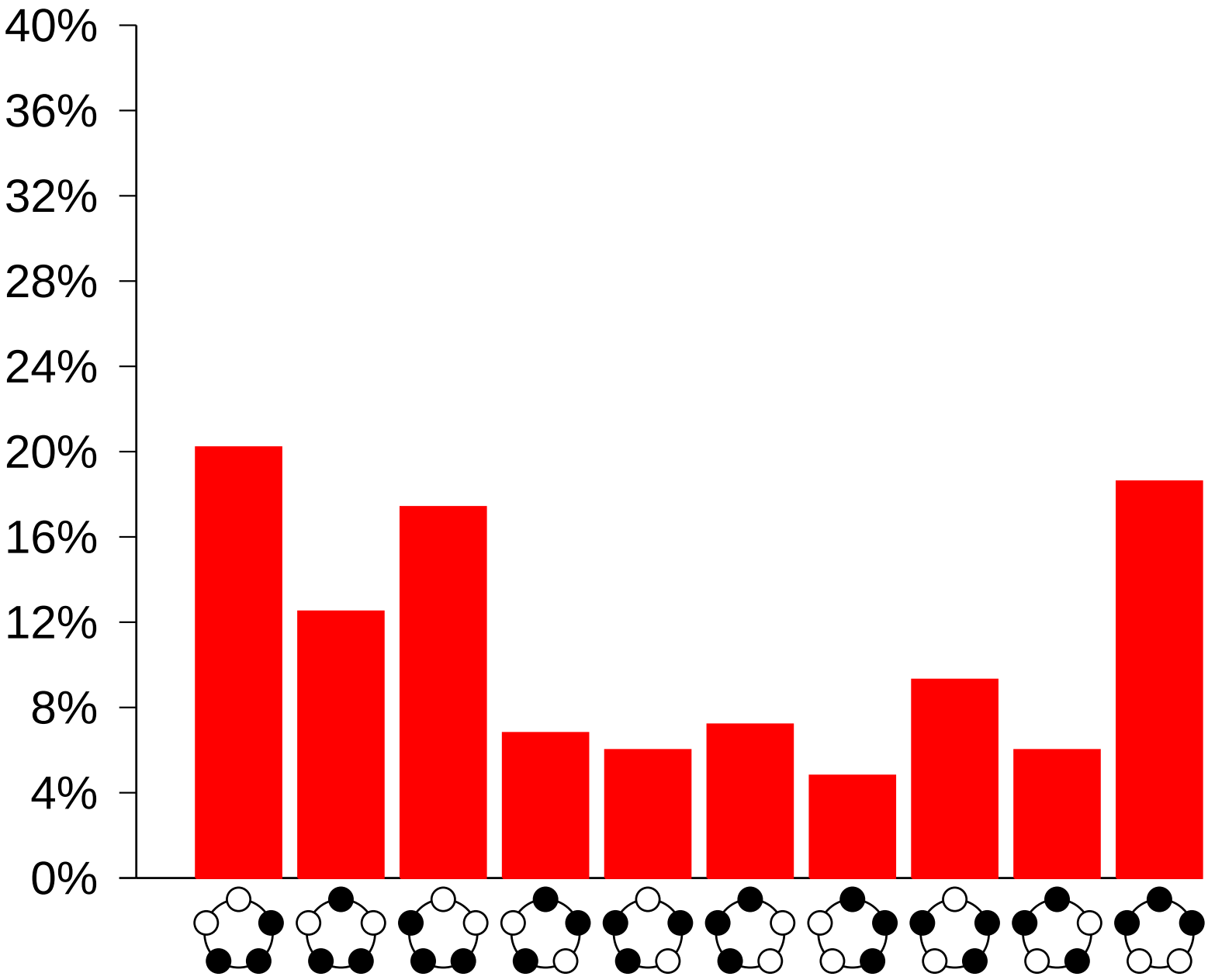
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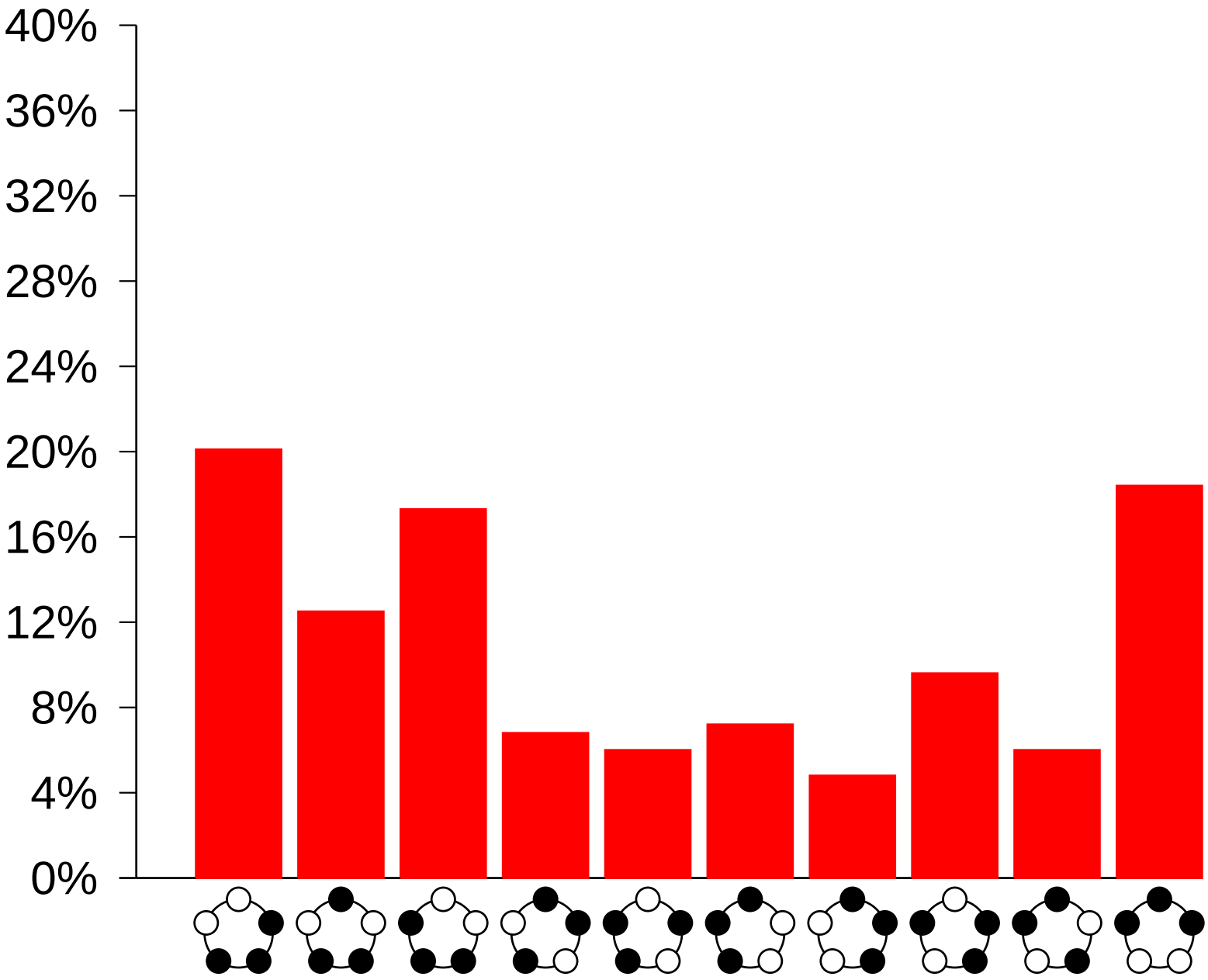
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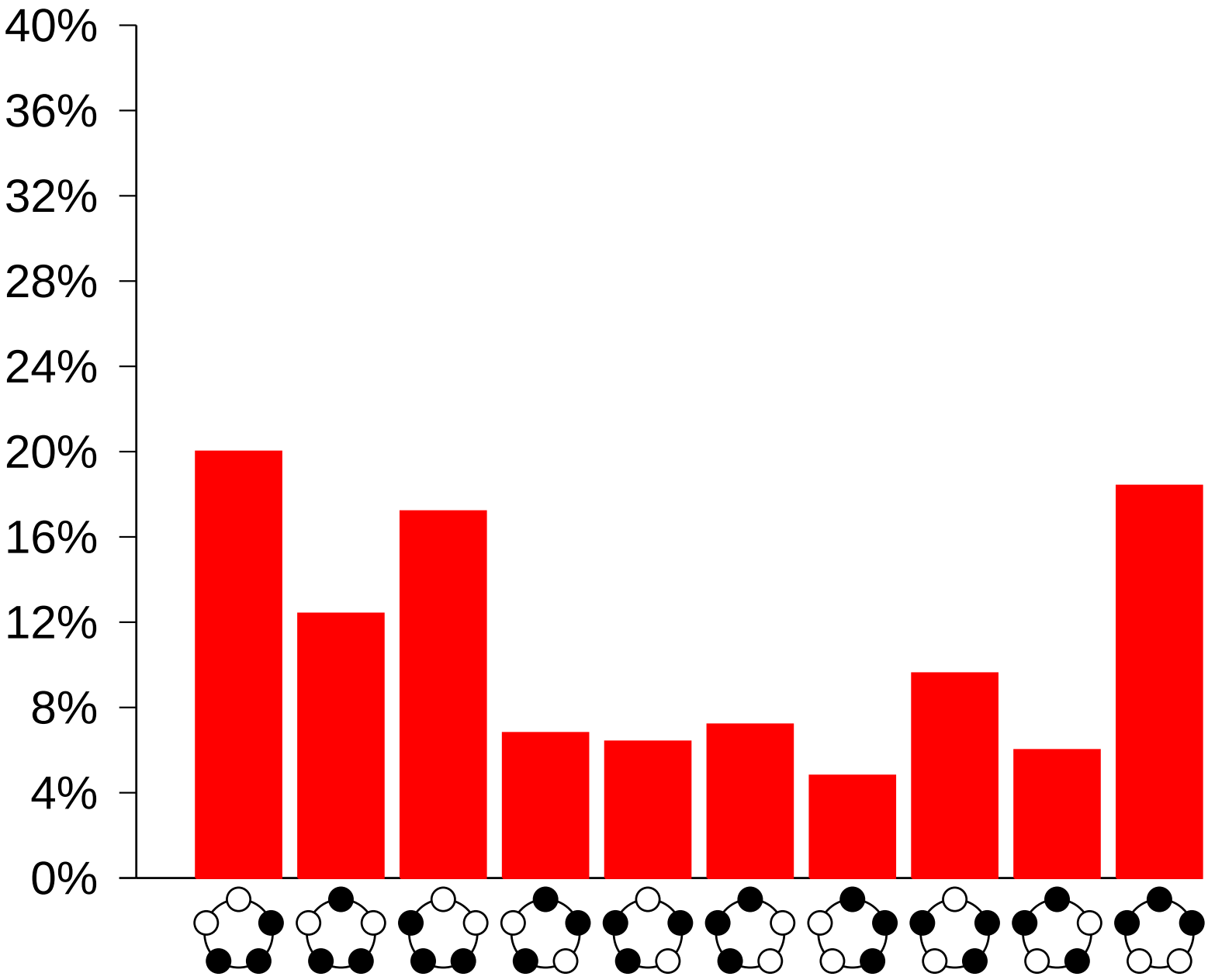
Stationary distribution



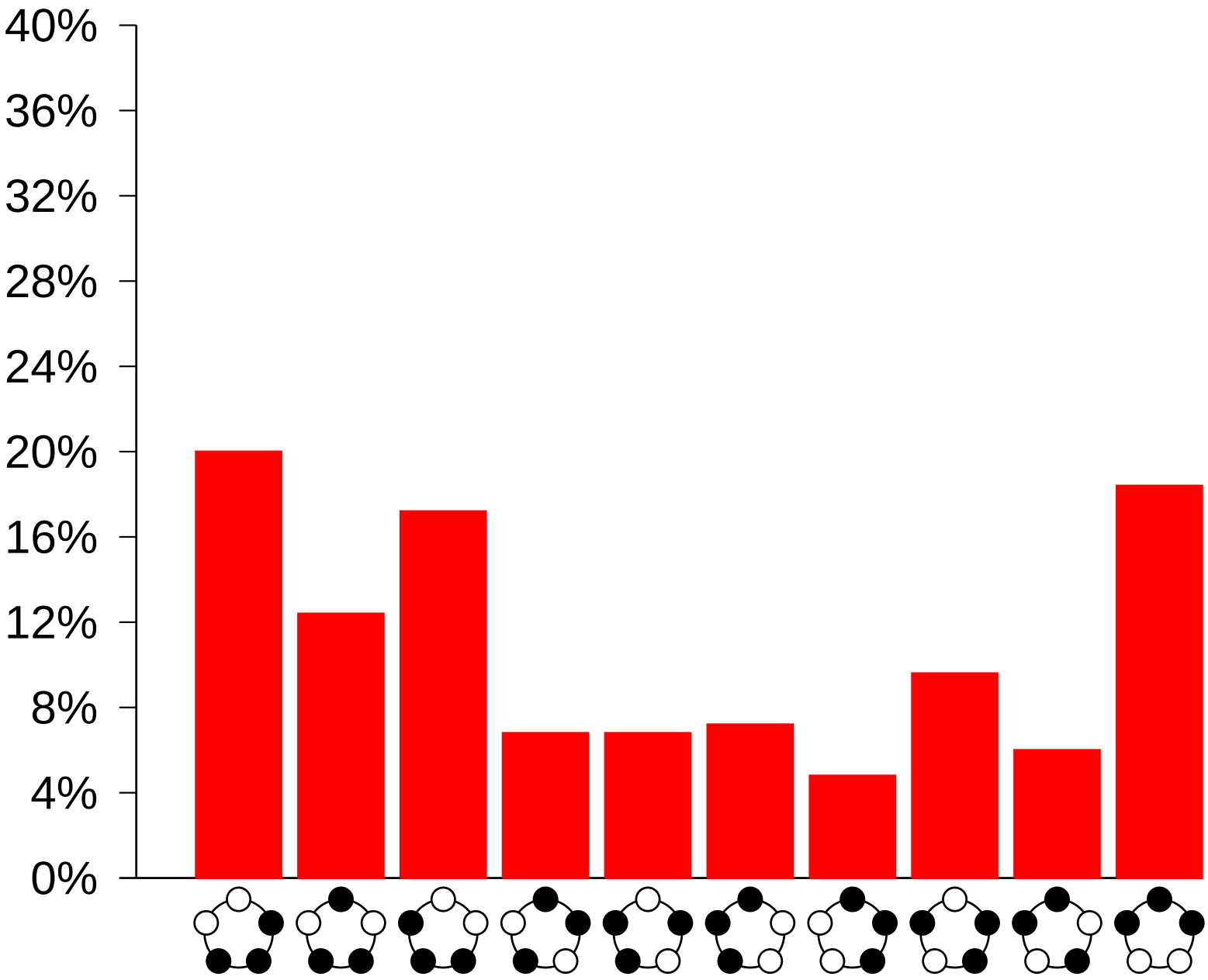
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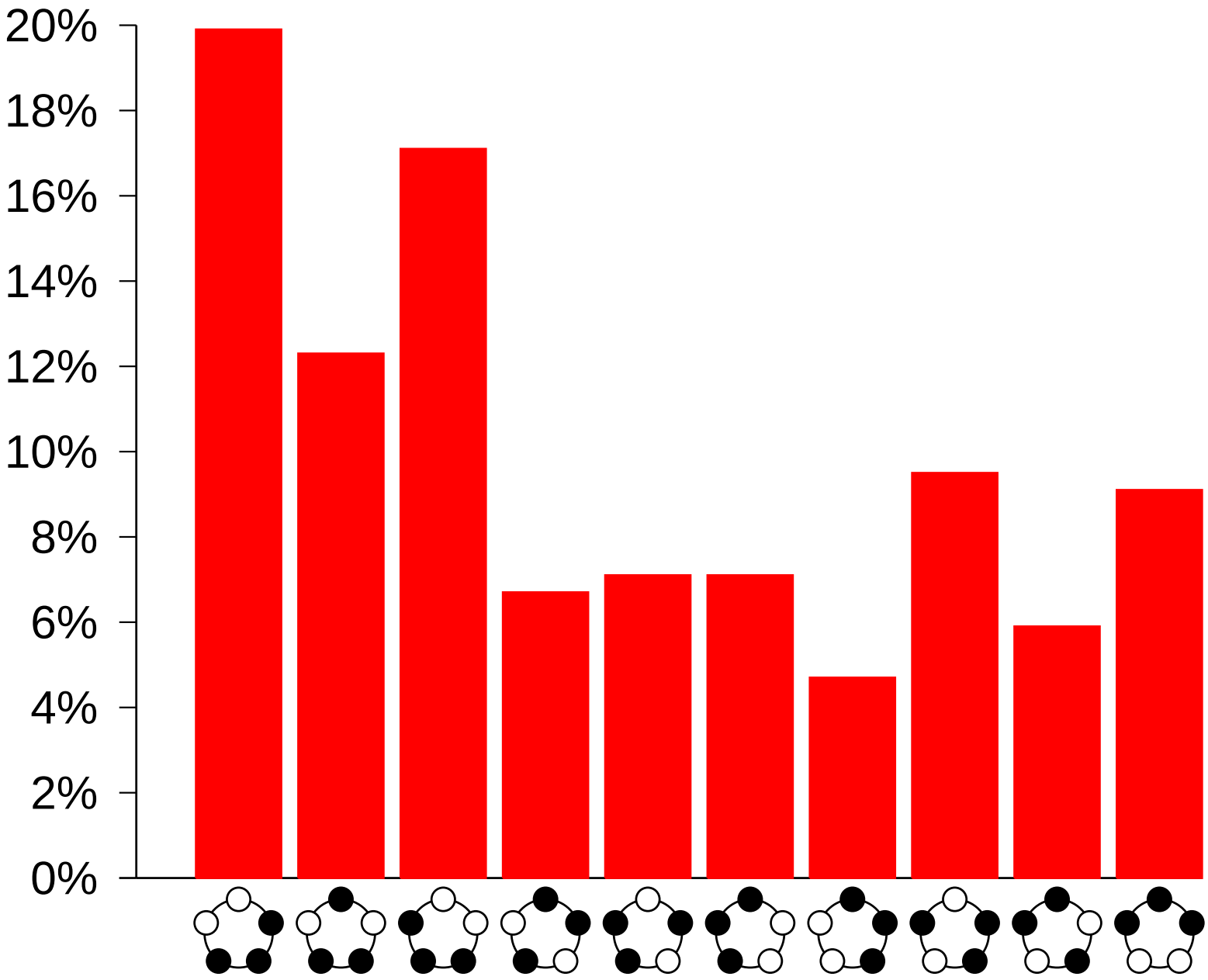
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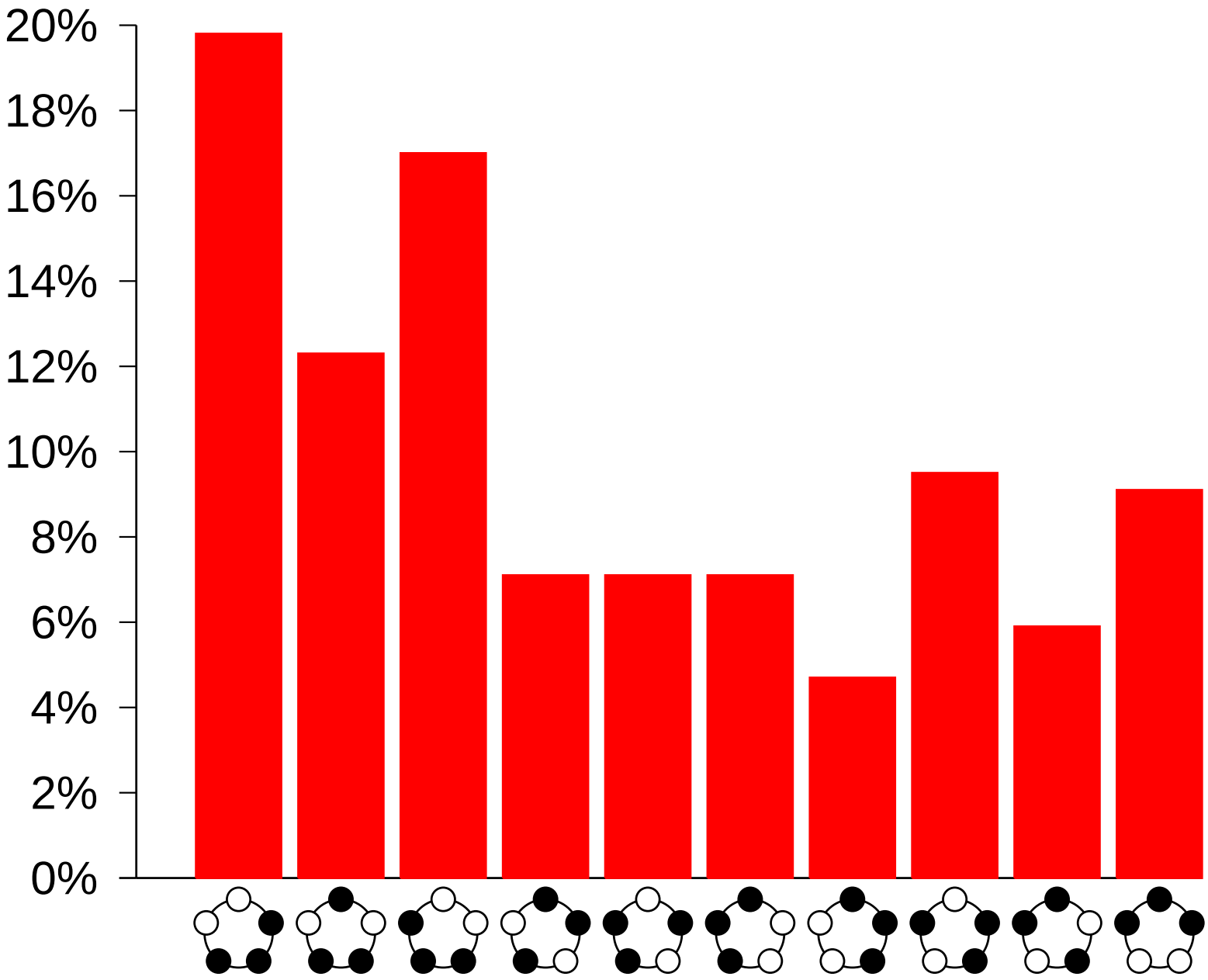
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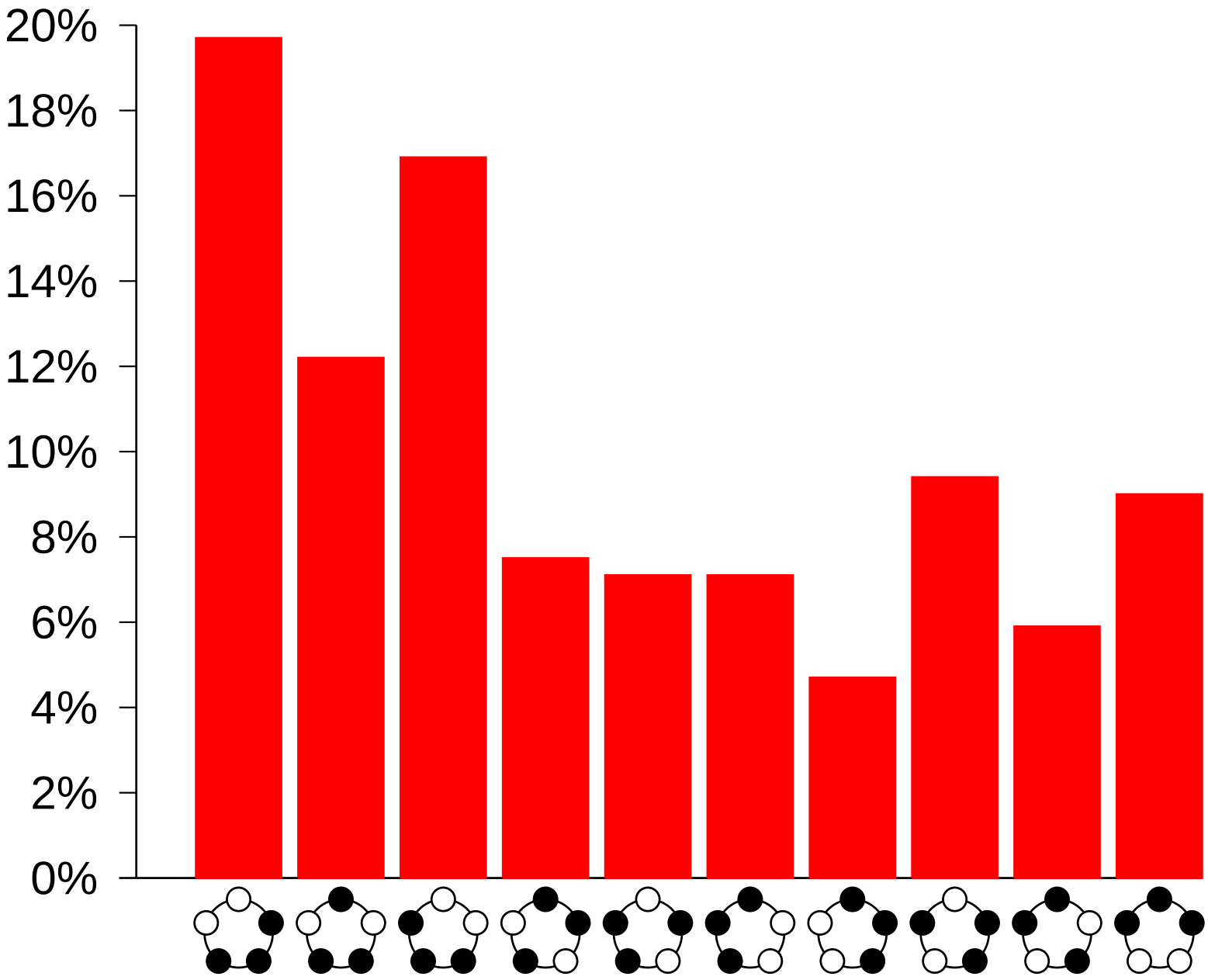
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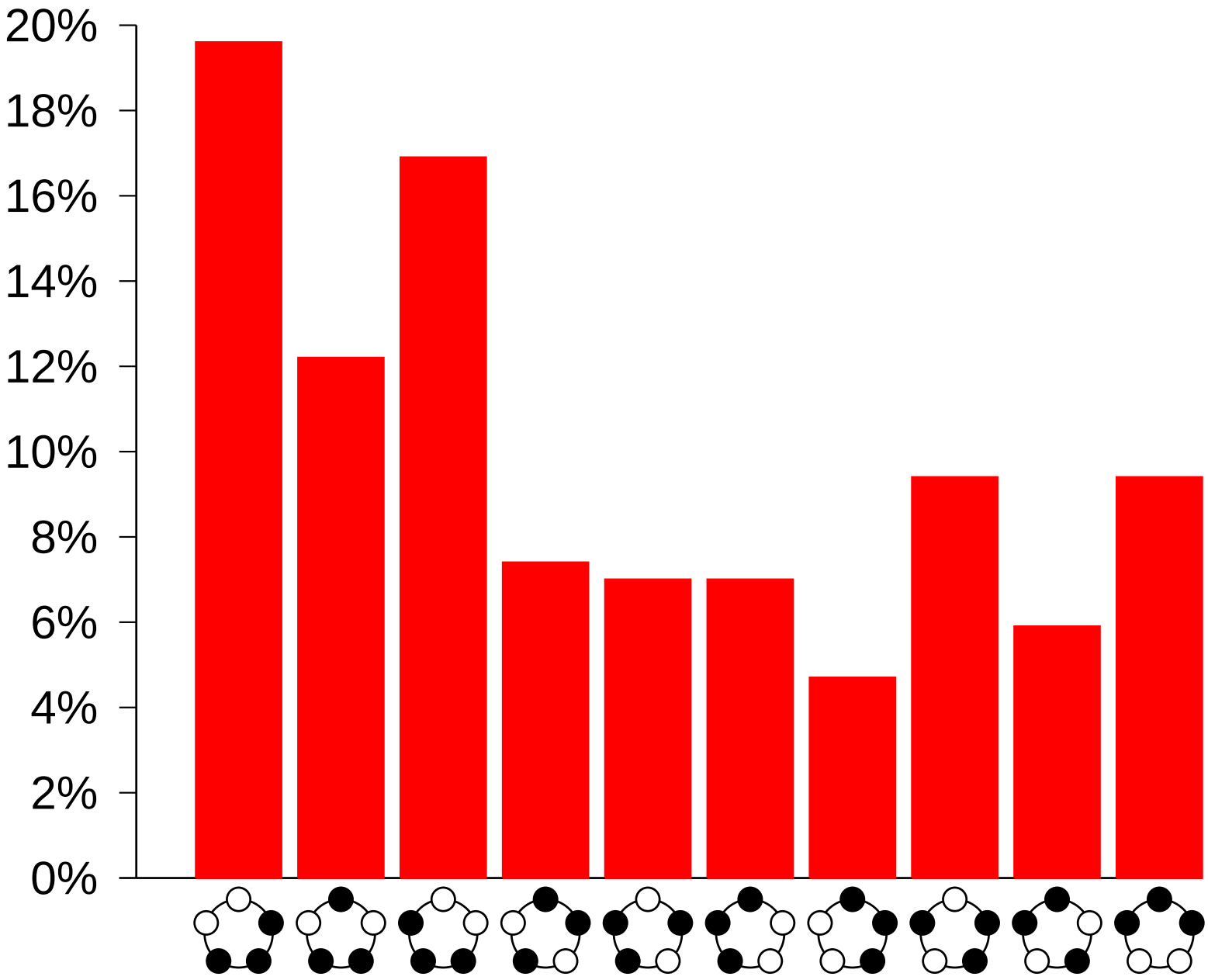
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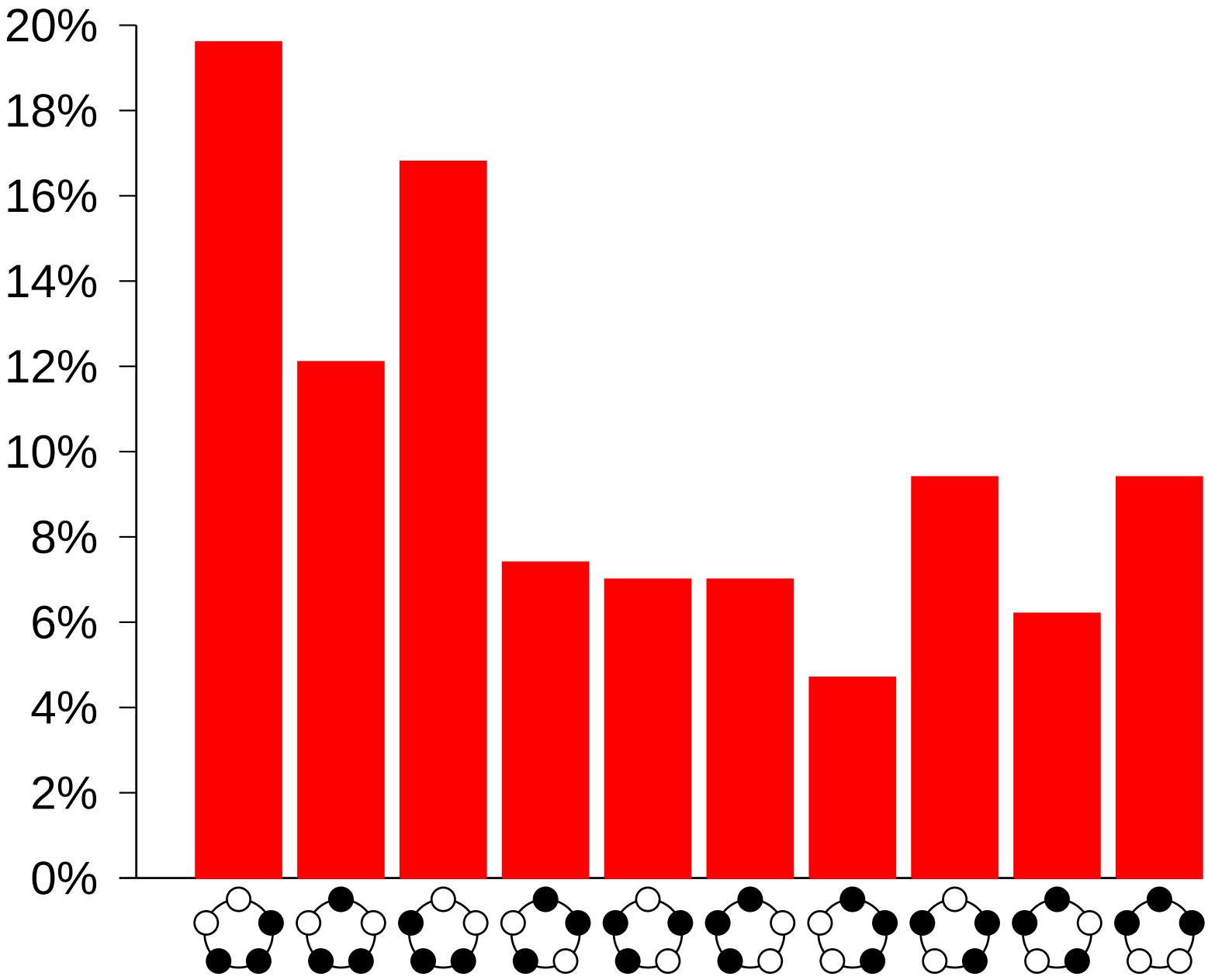
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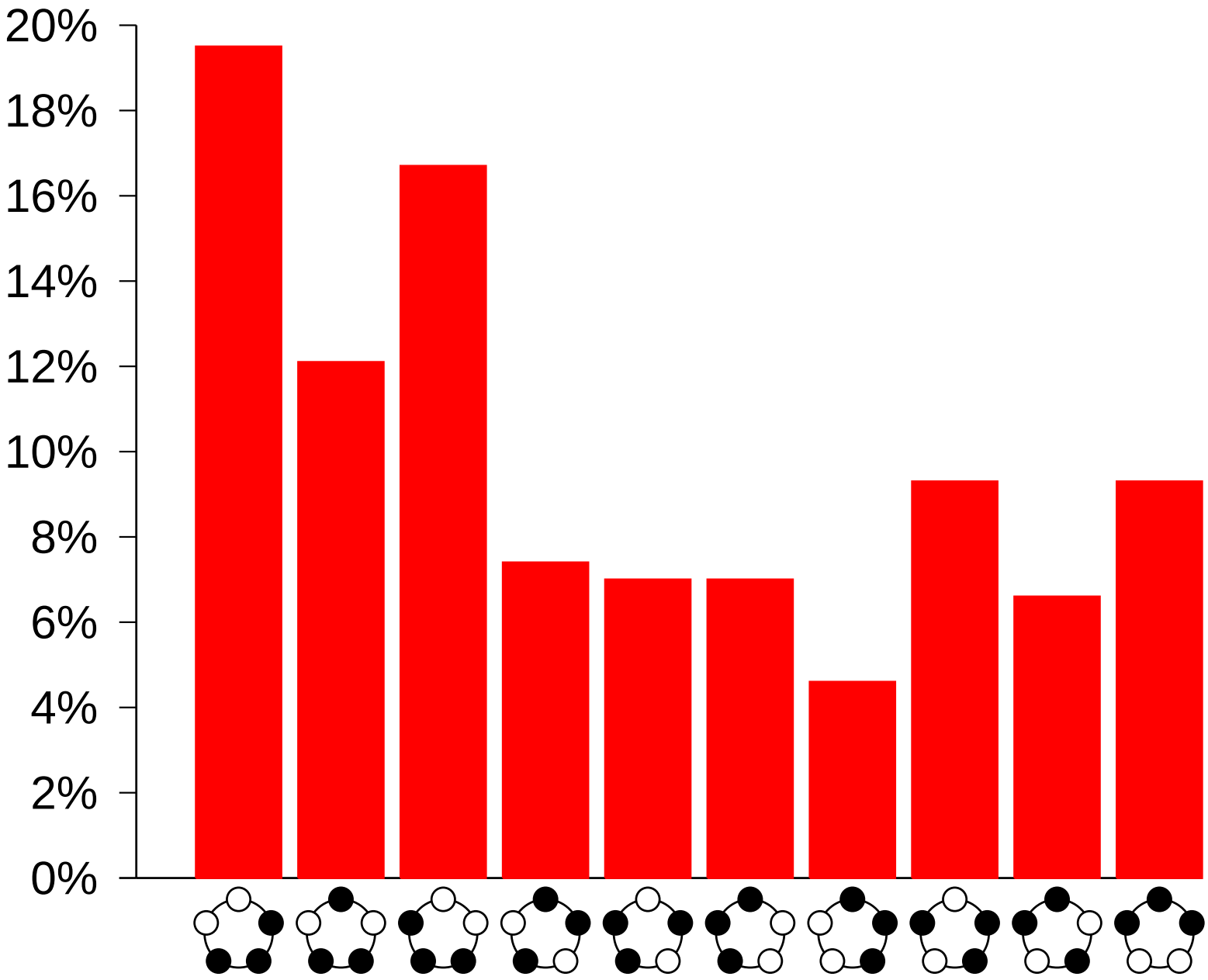
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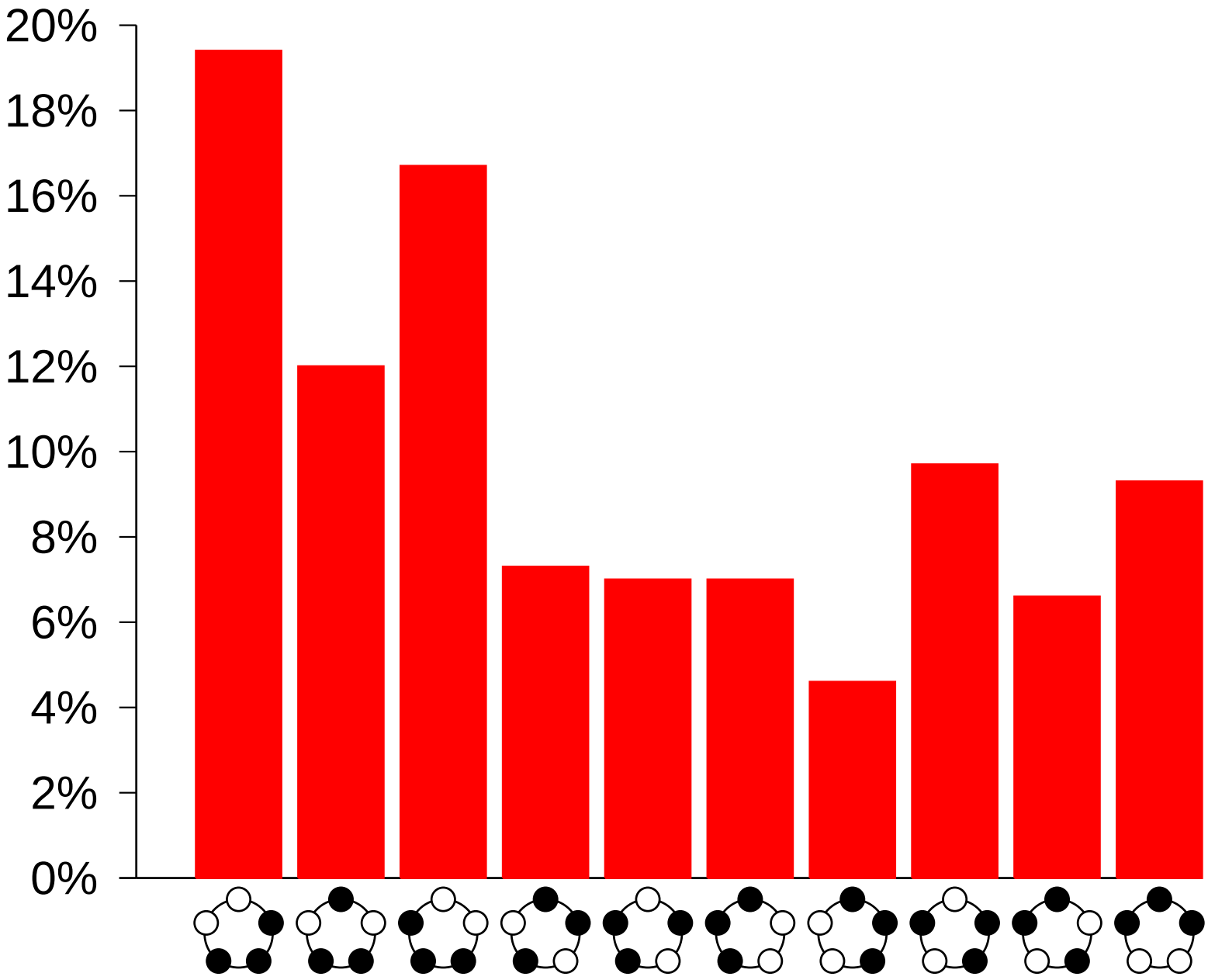
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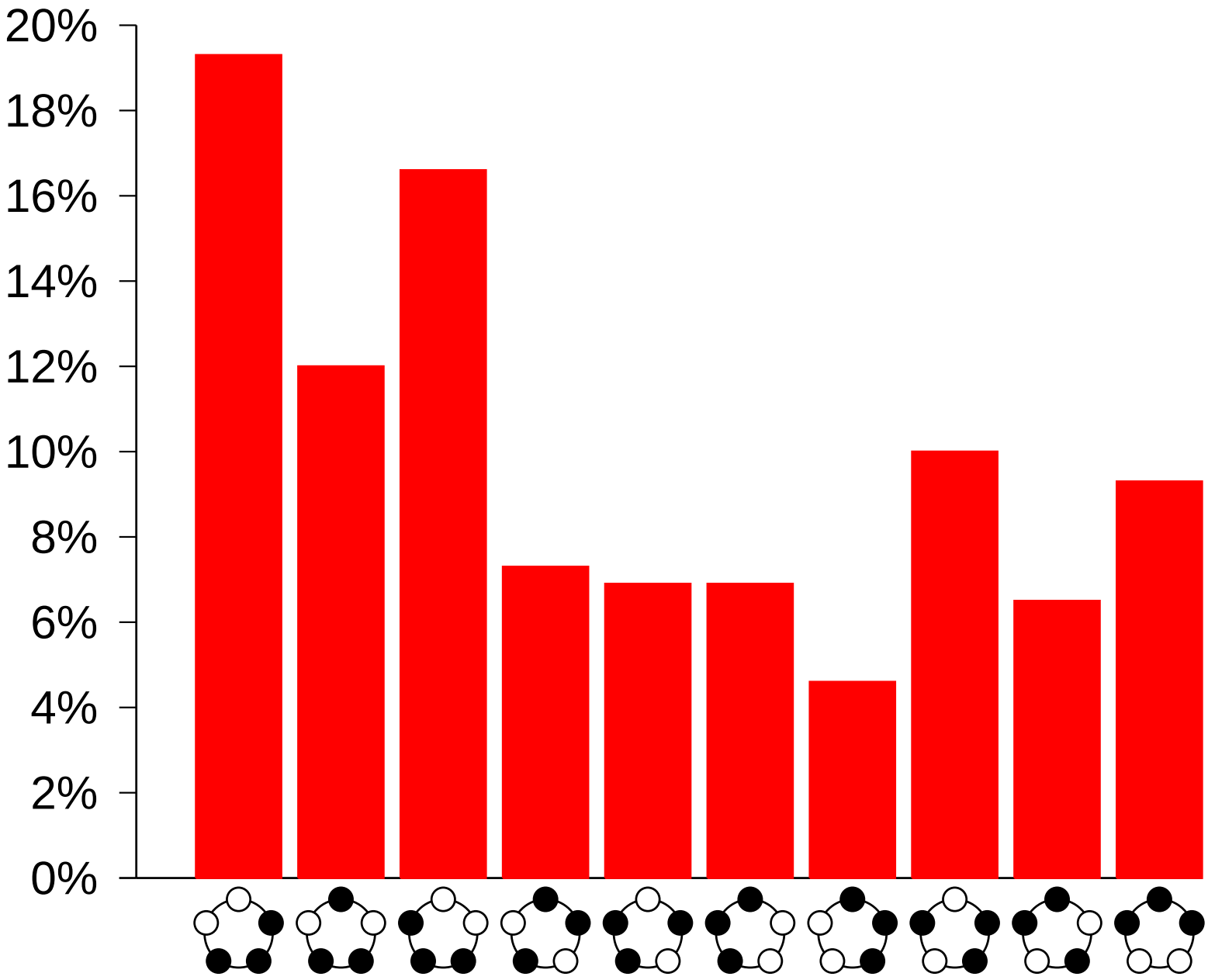
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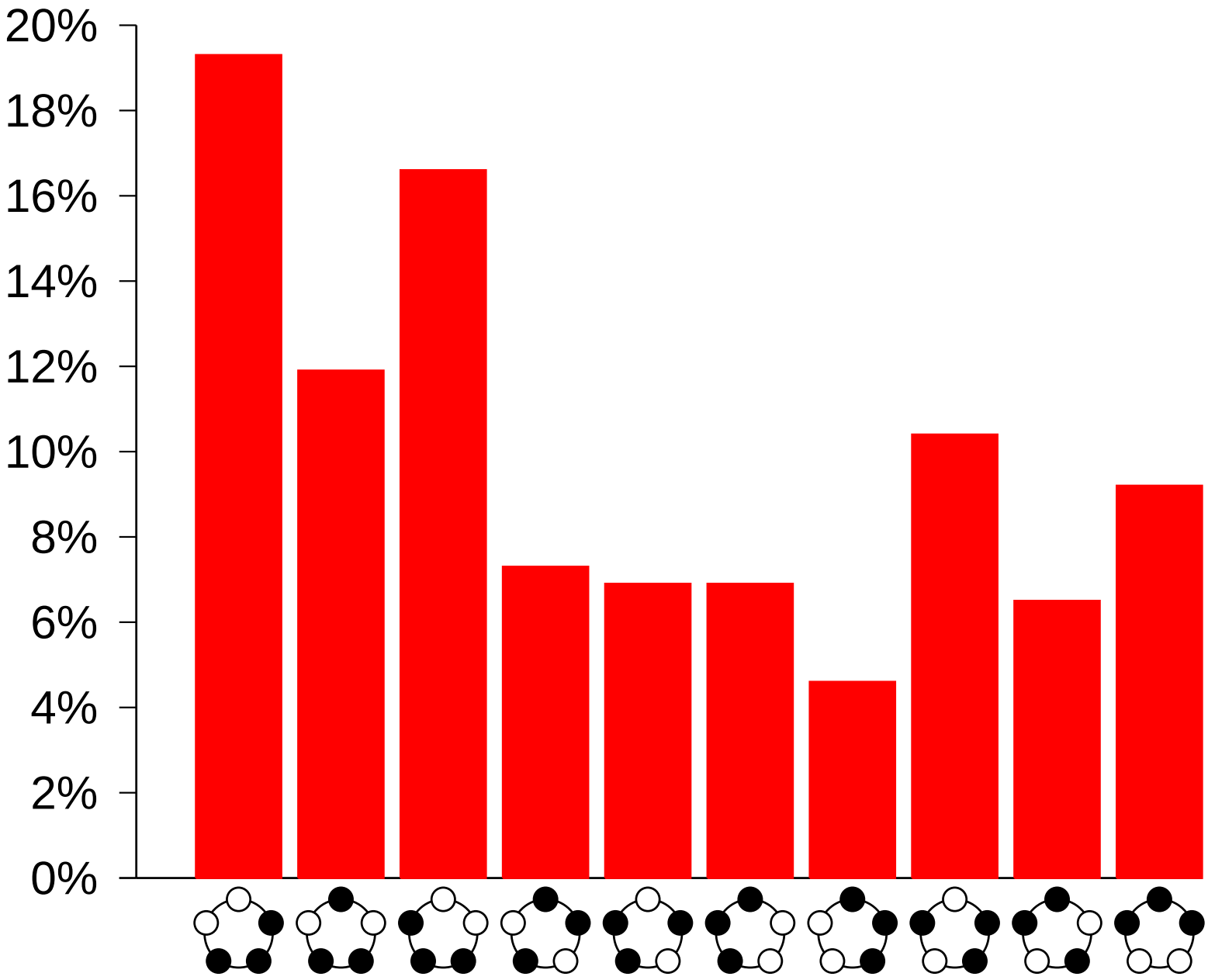
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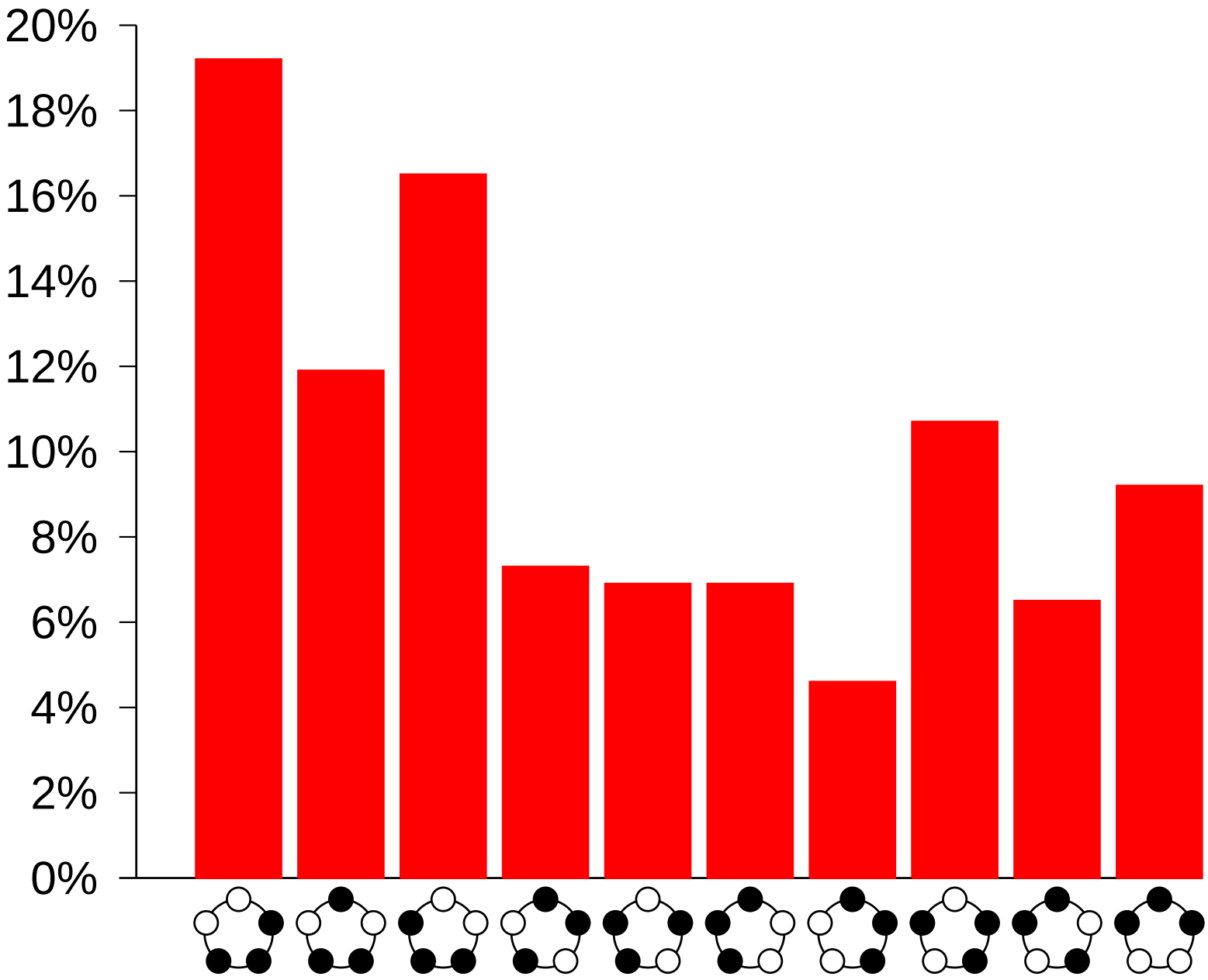
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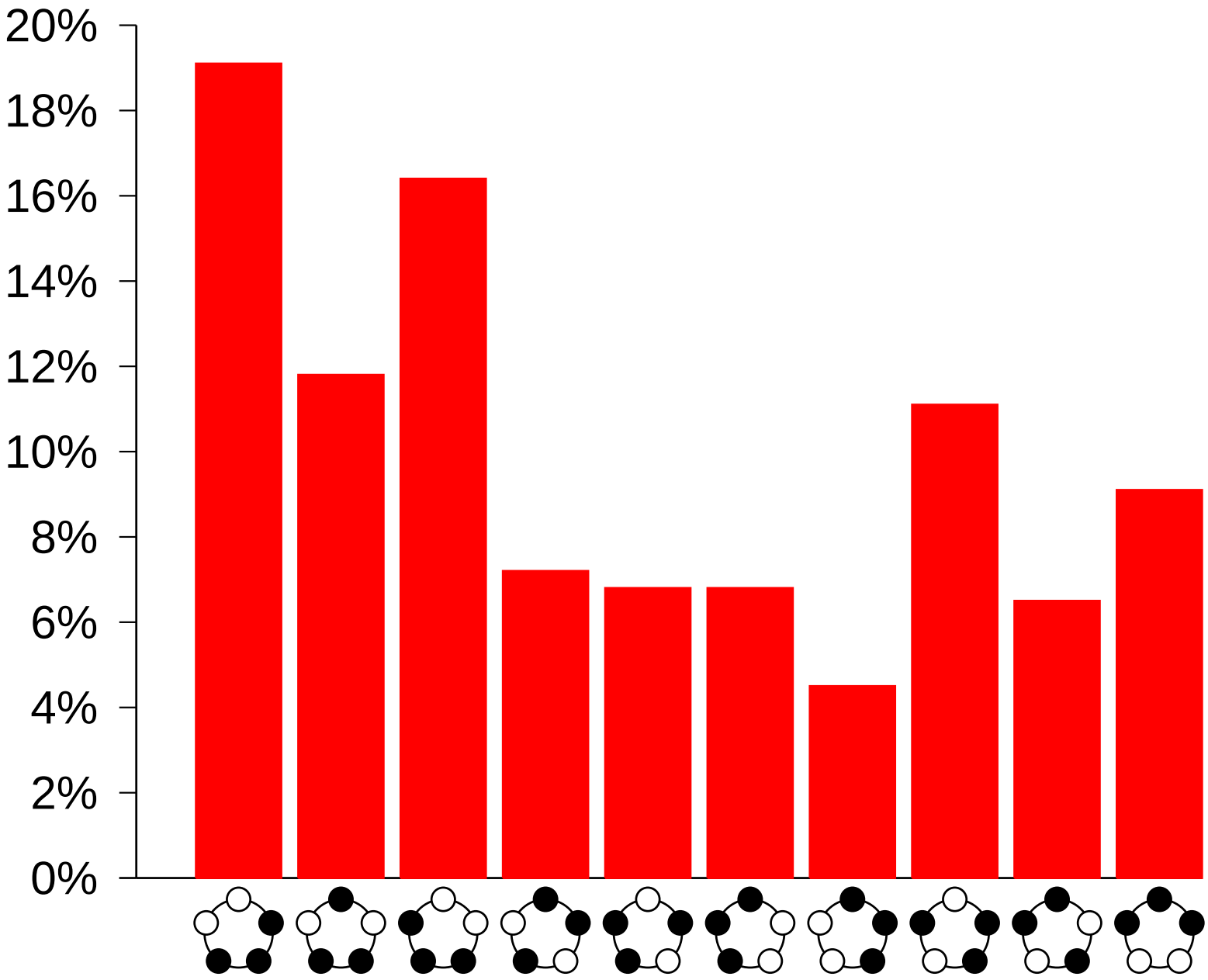
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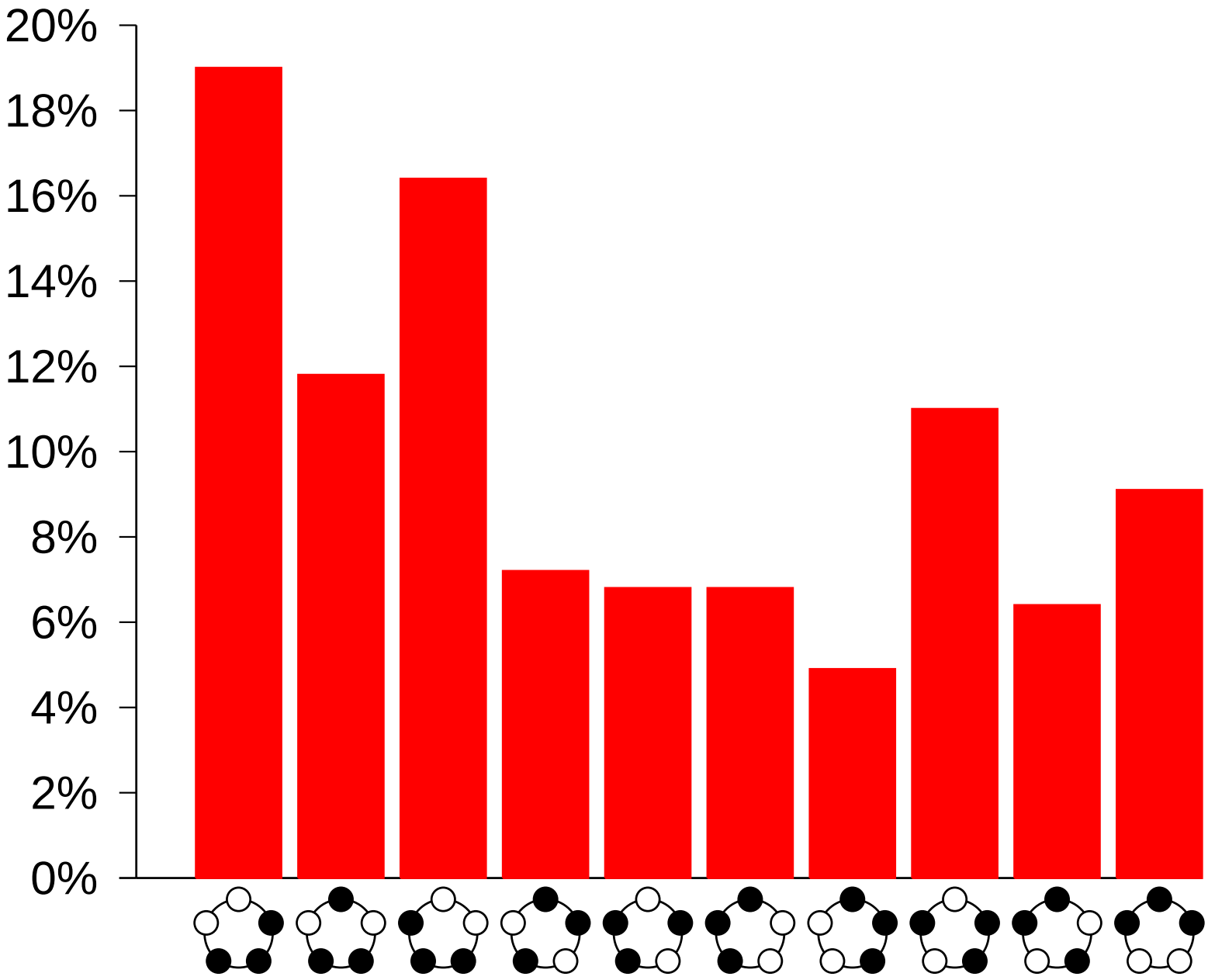
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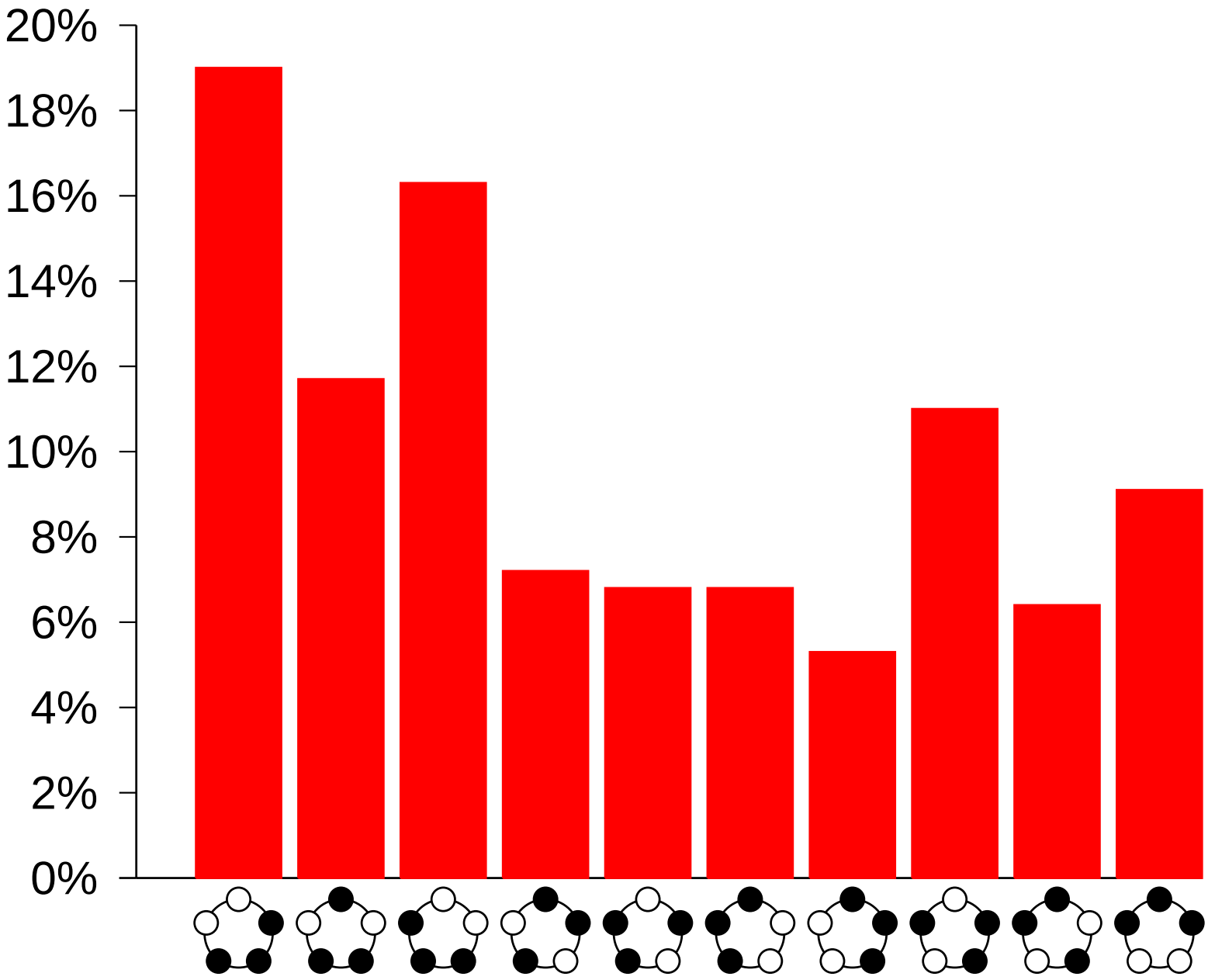
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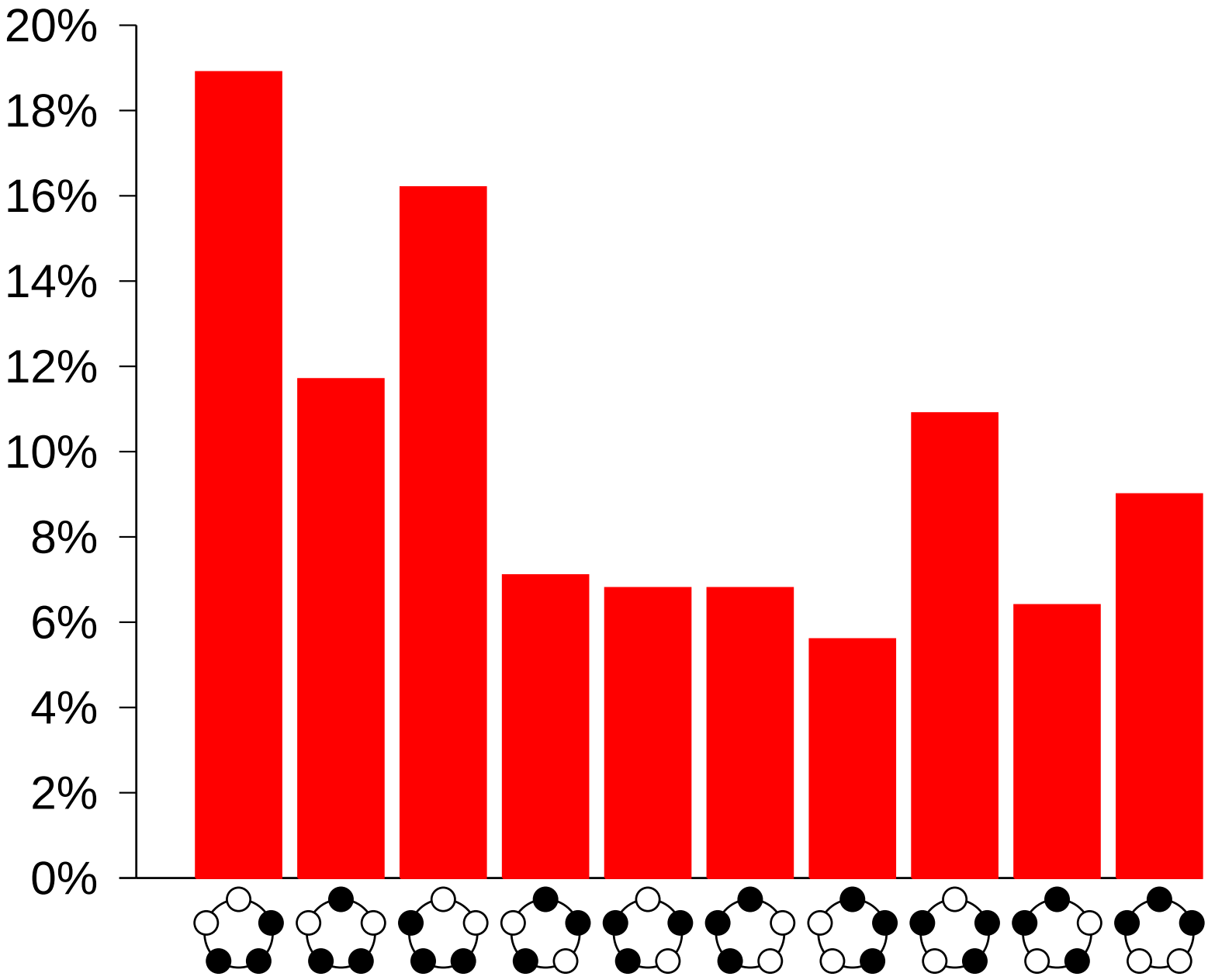
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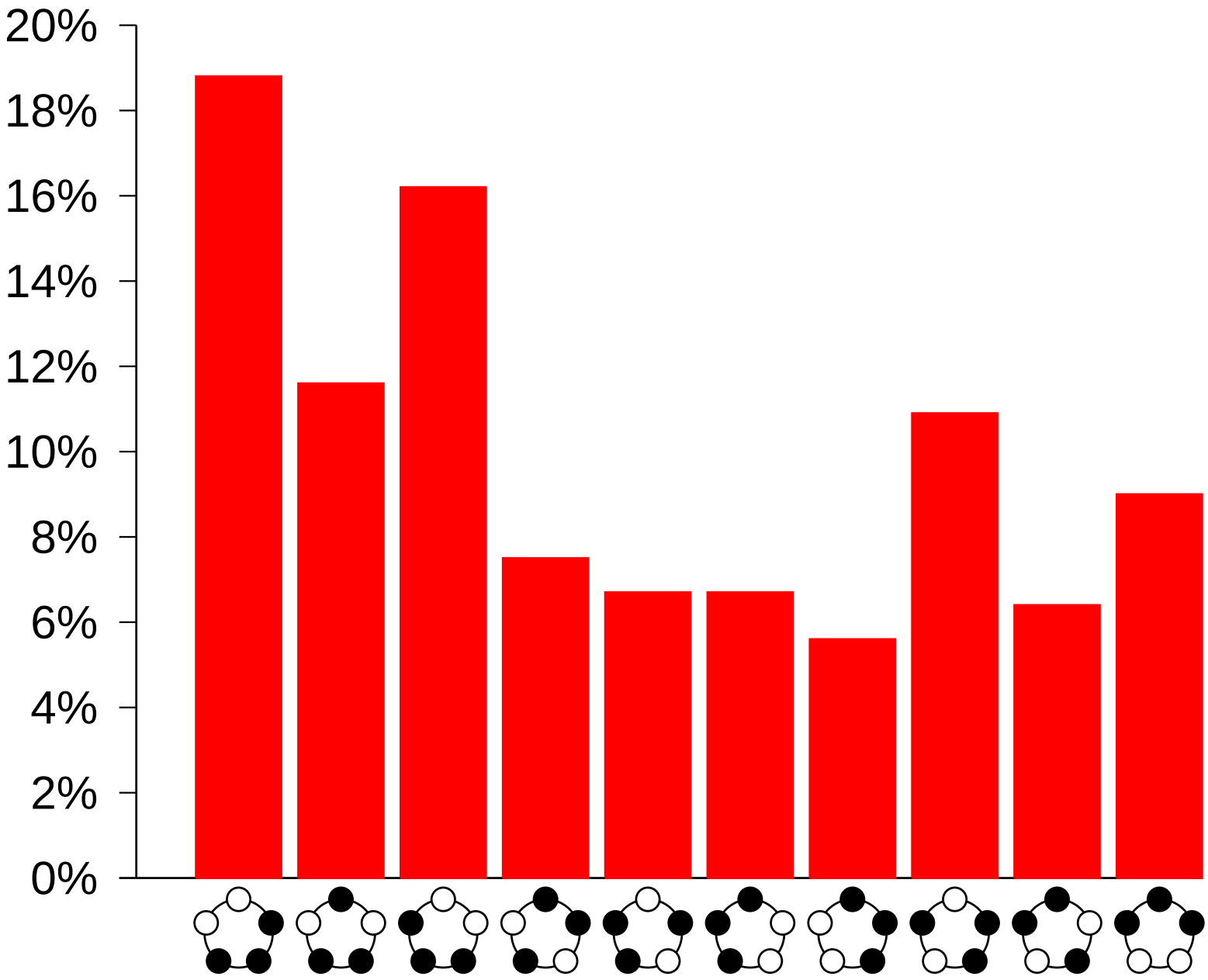
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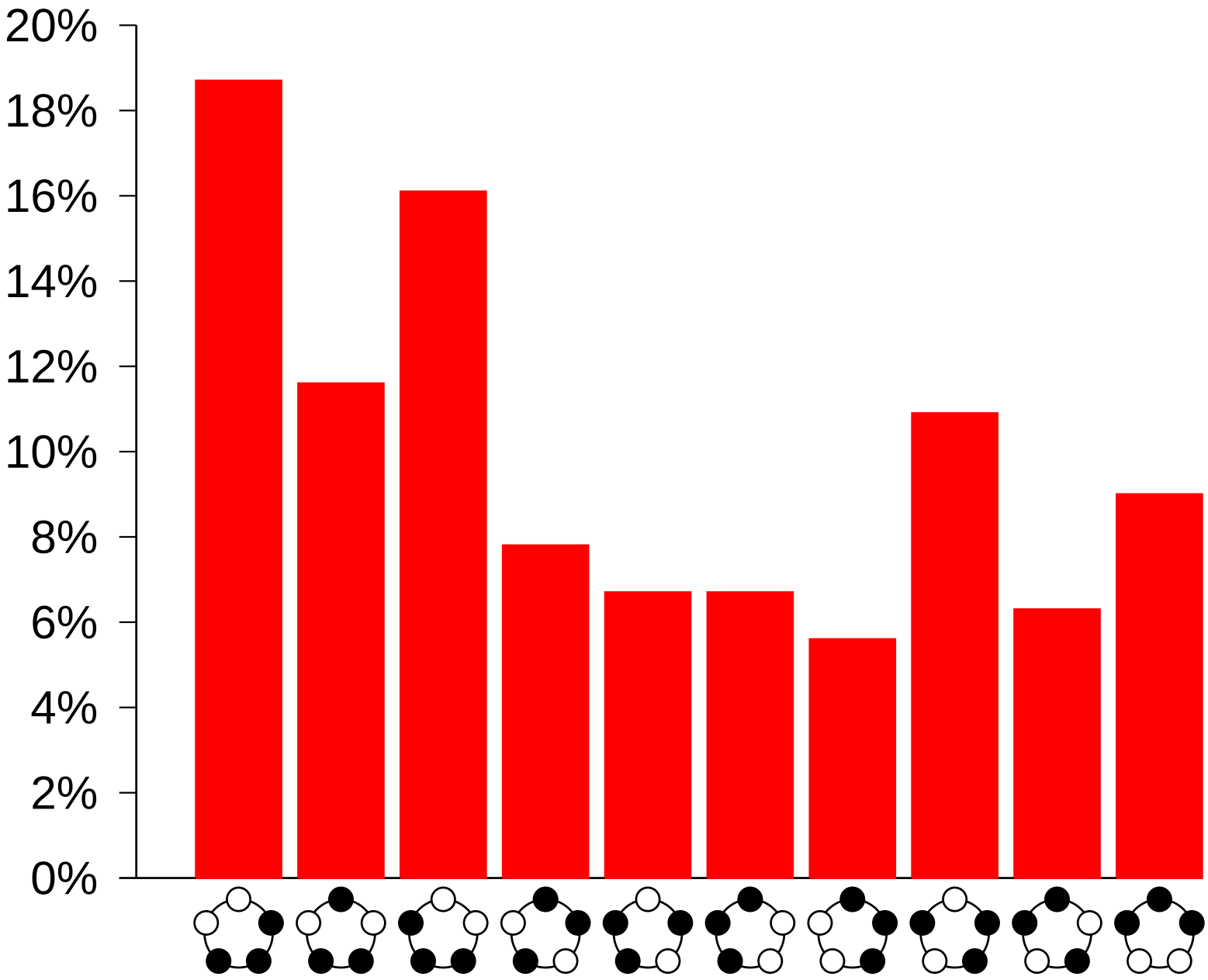
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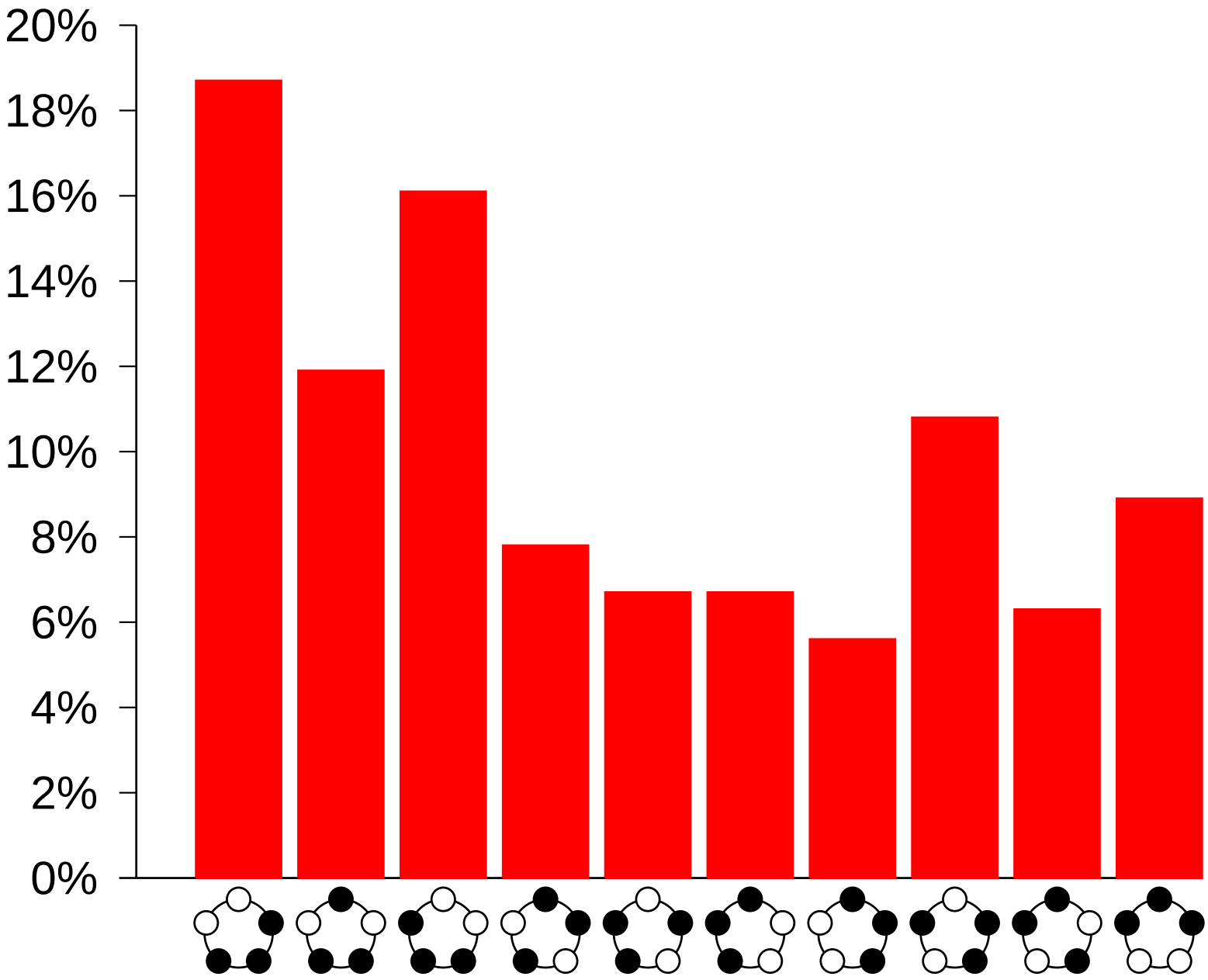
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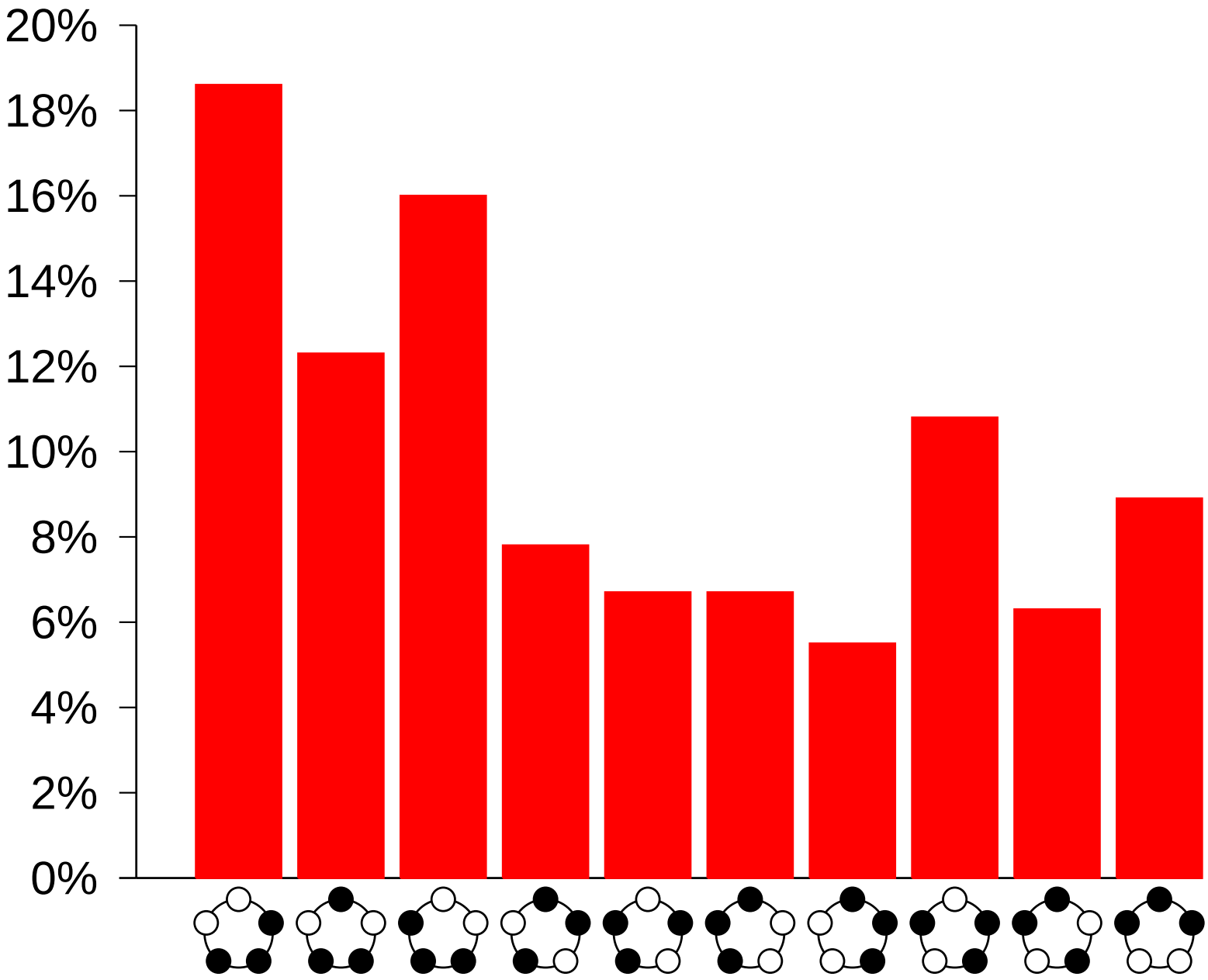
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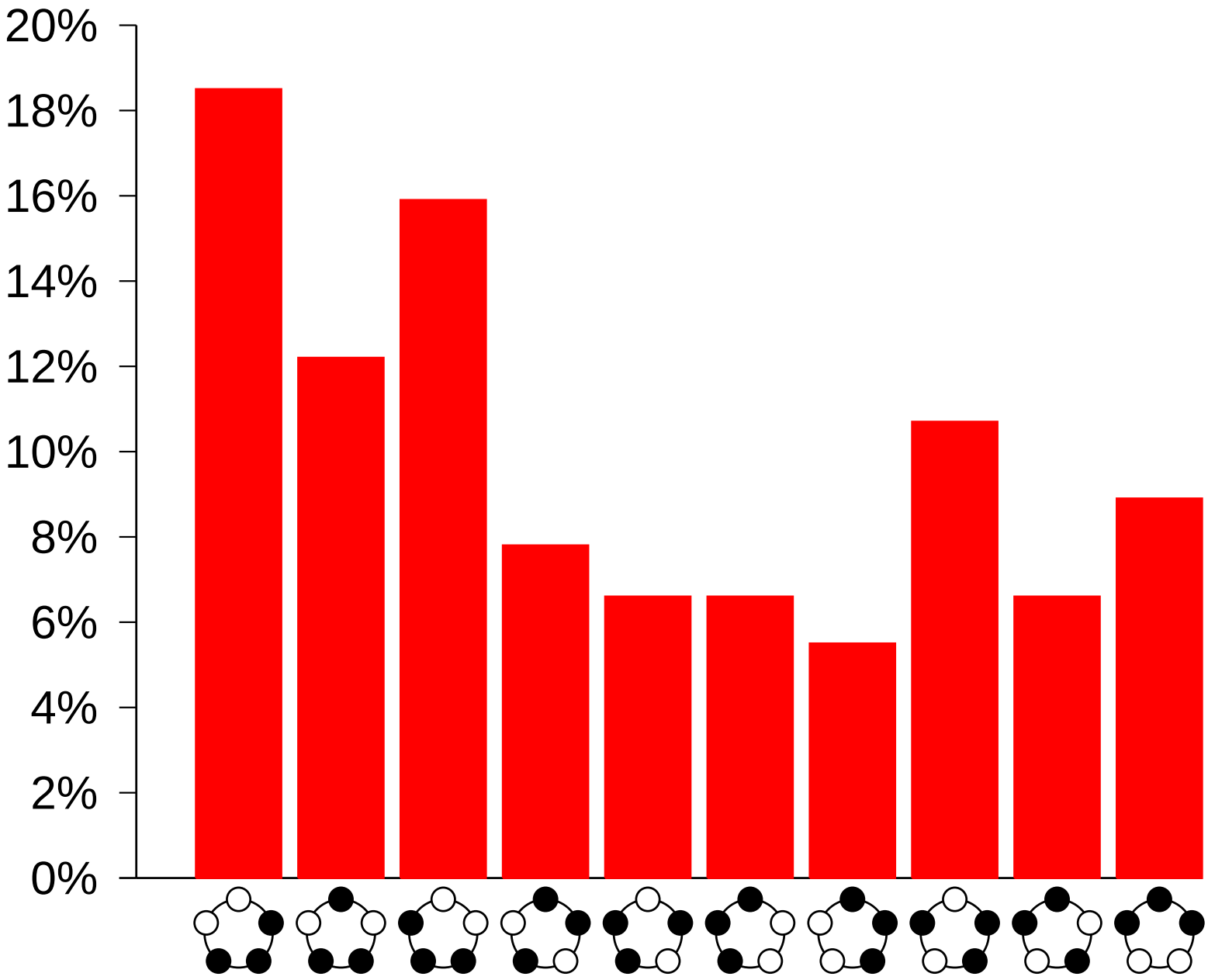
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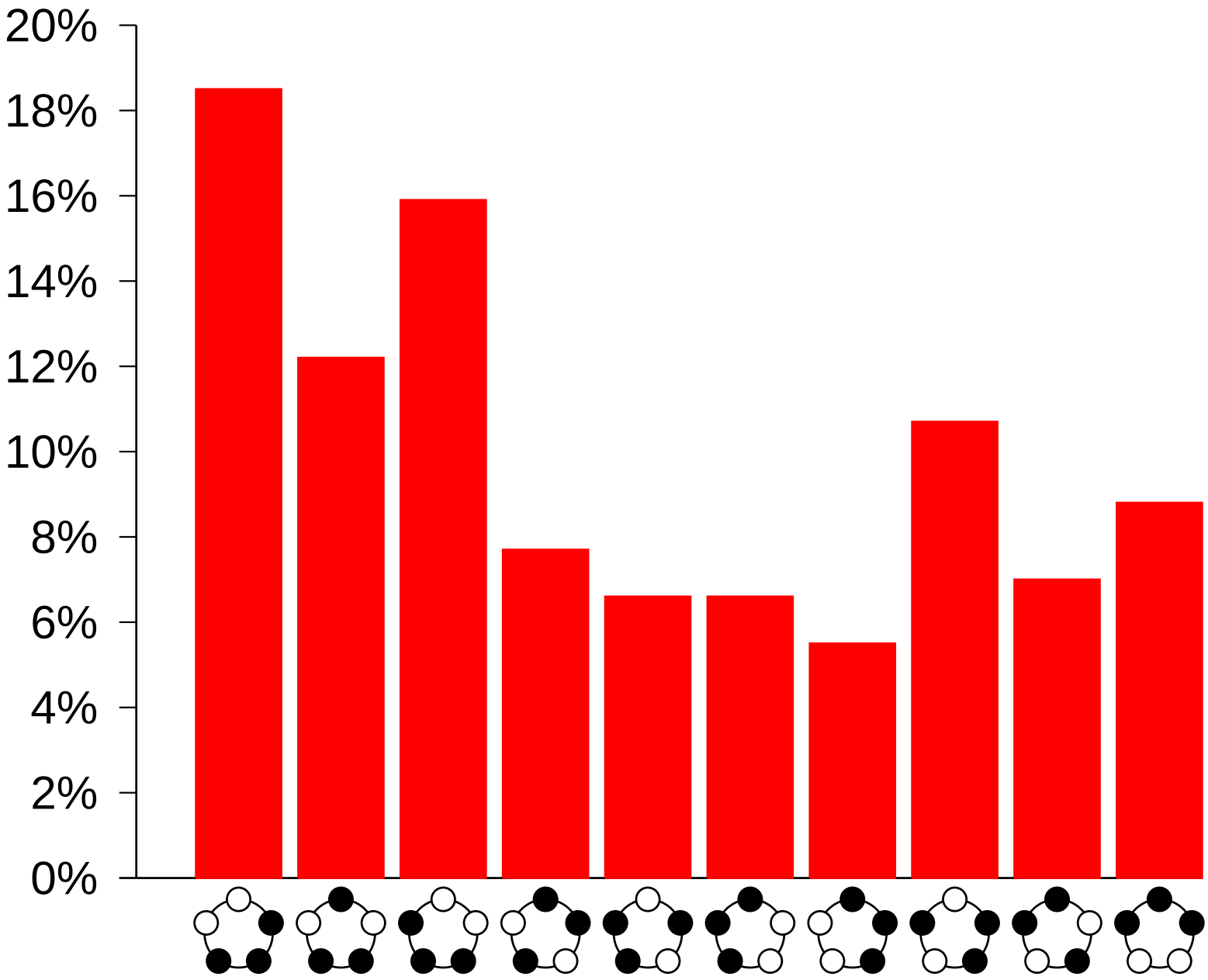
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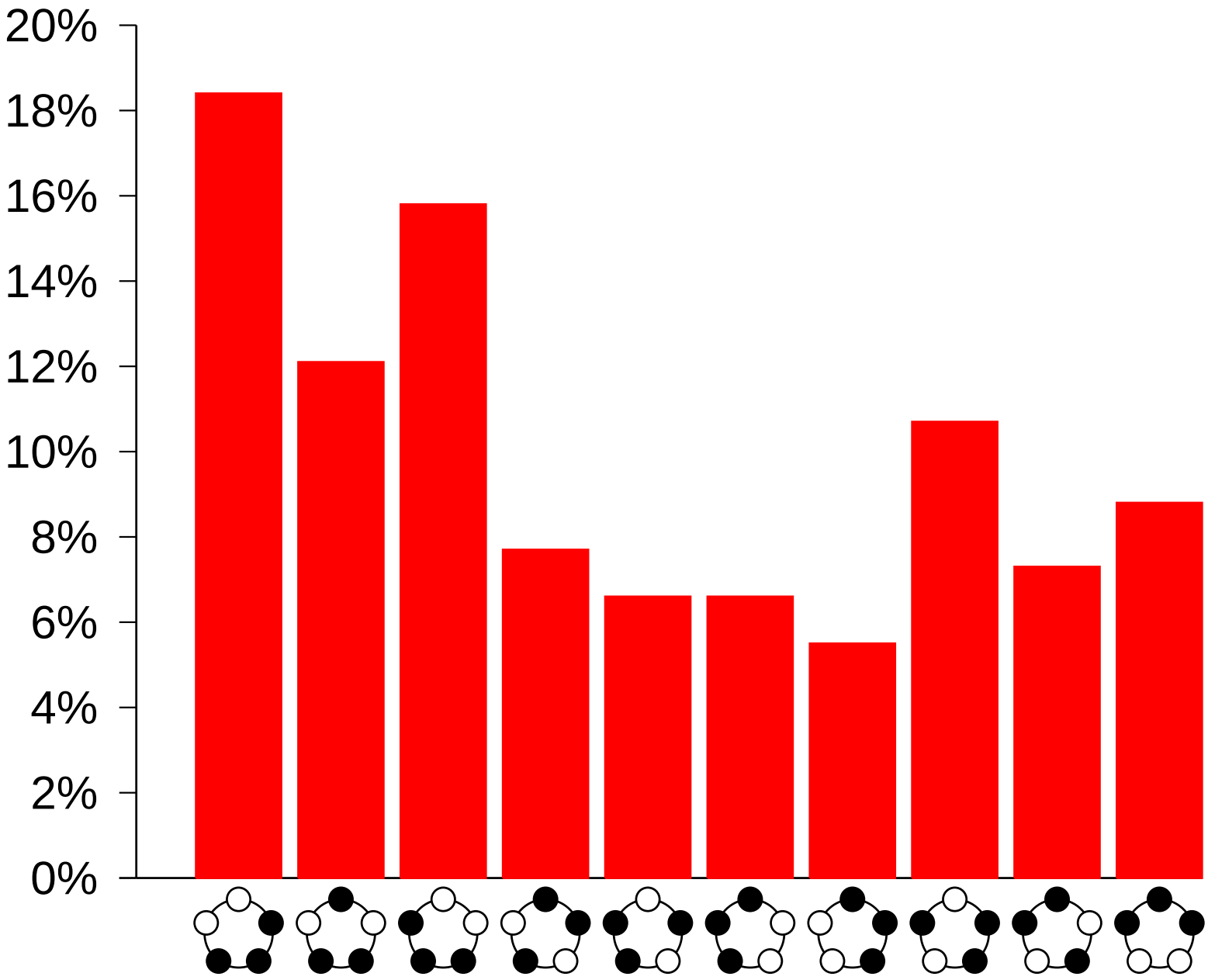
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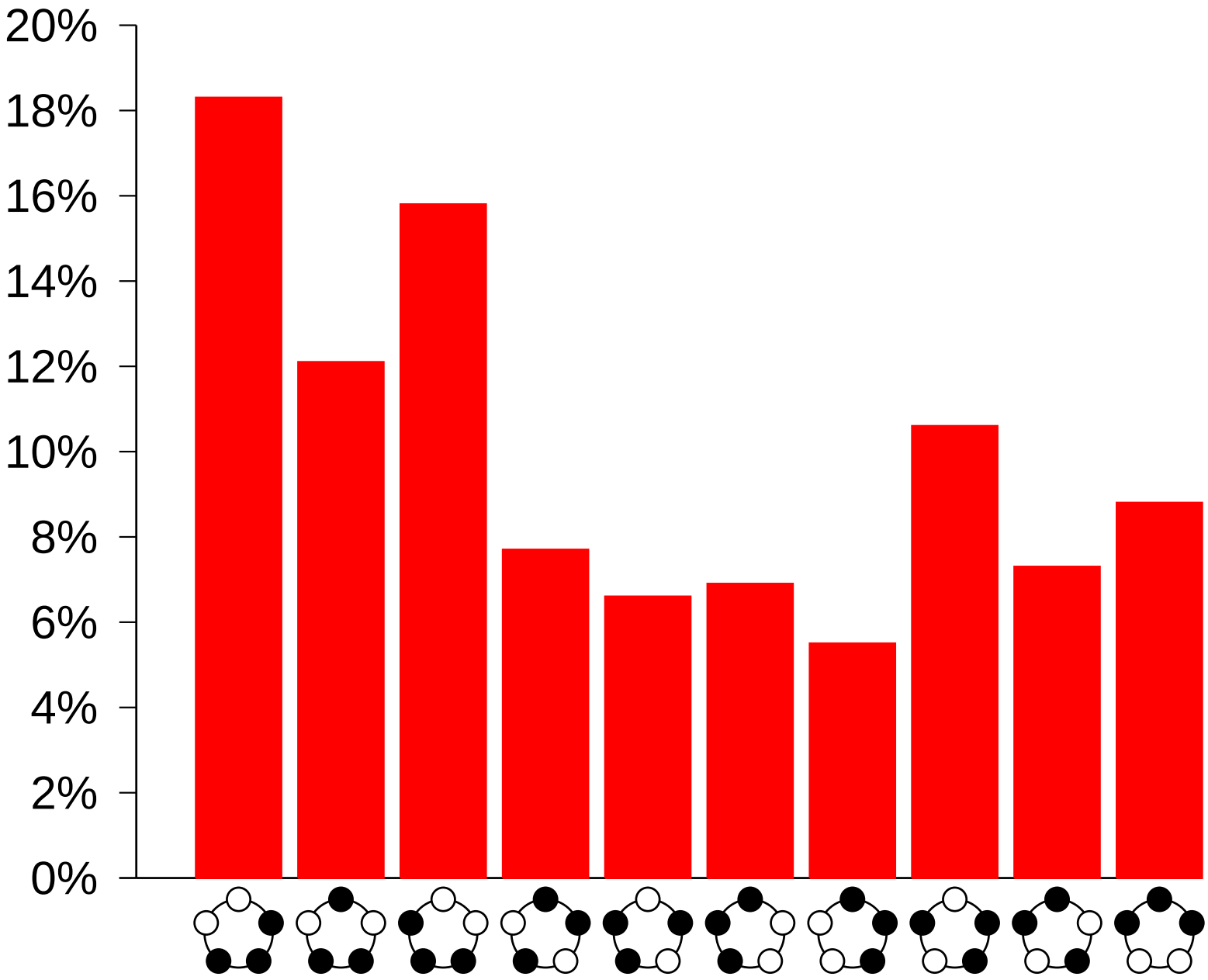
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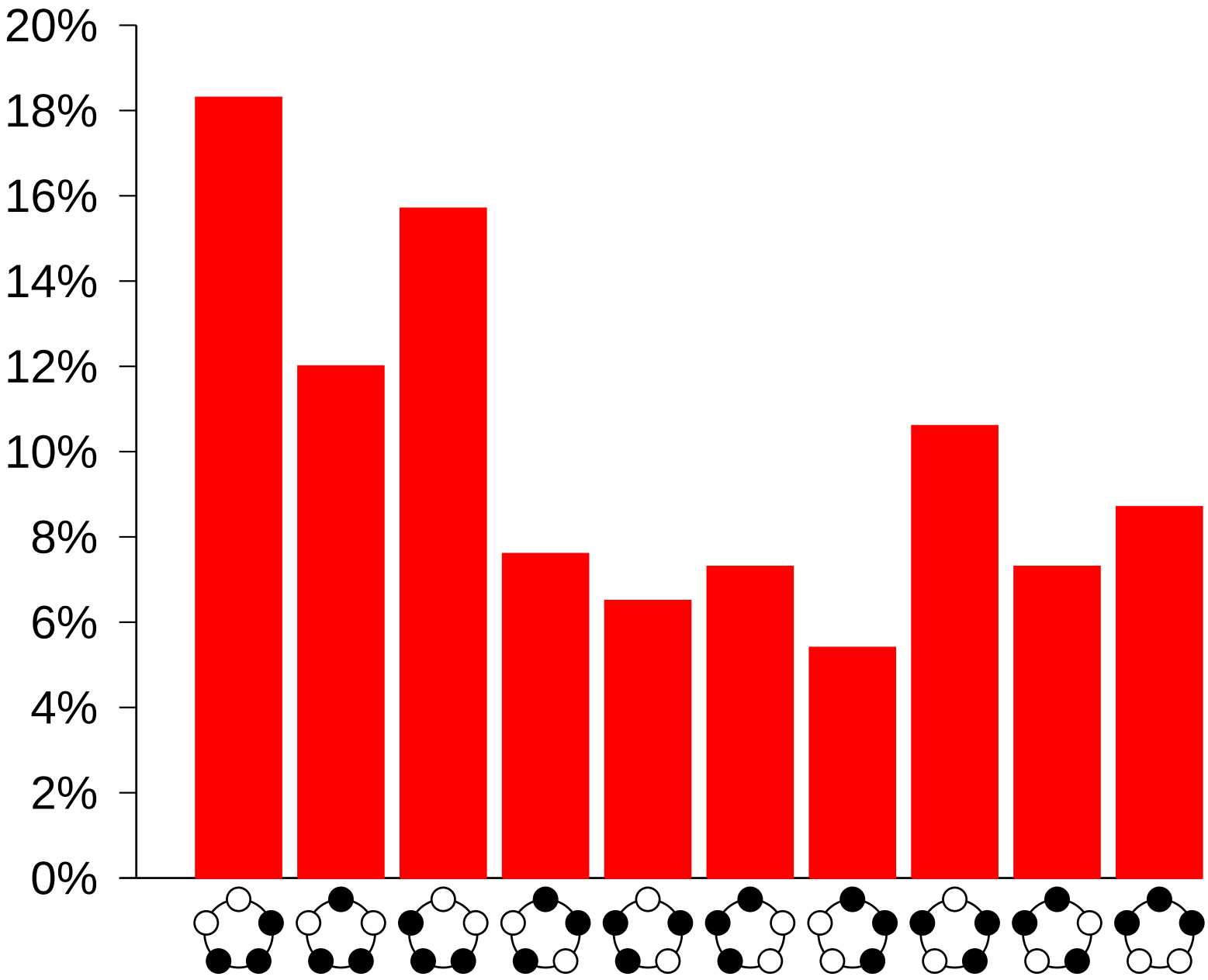
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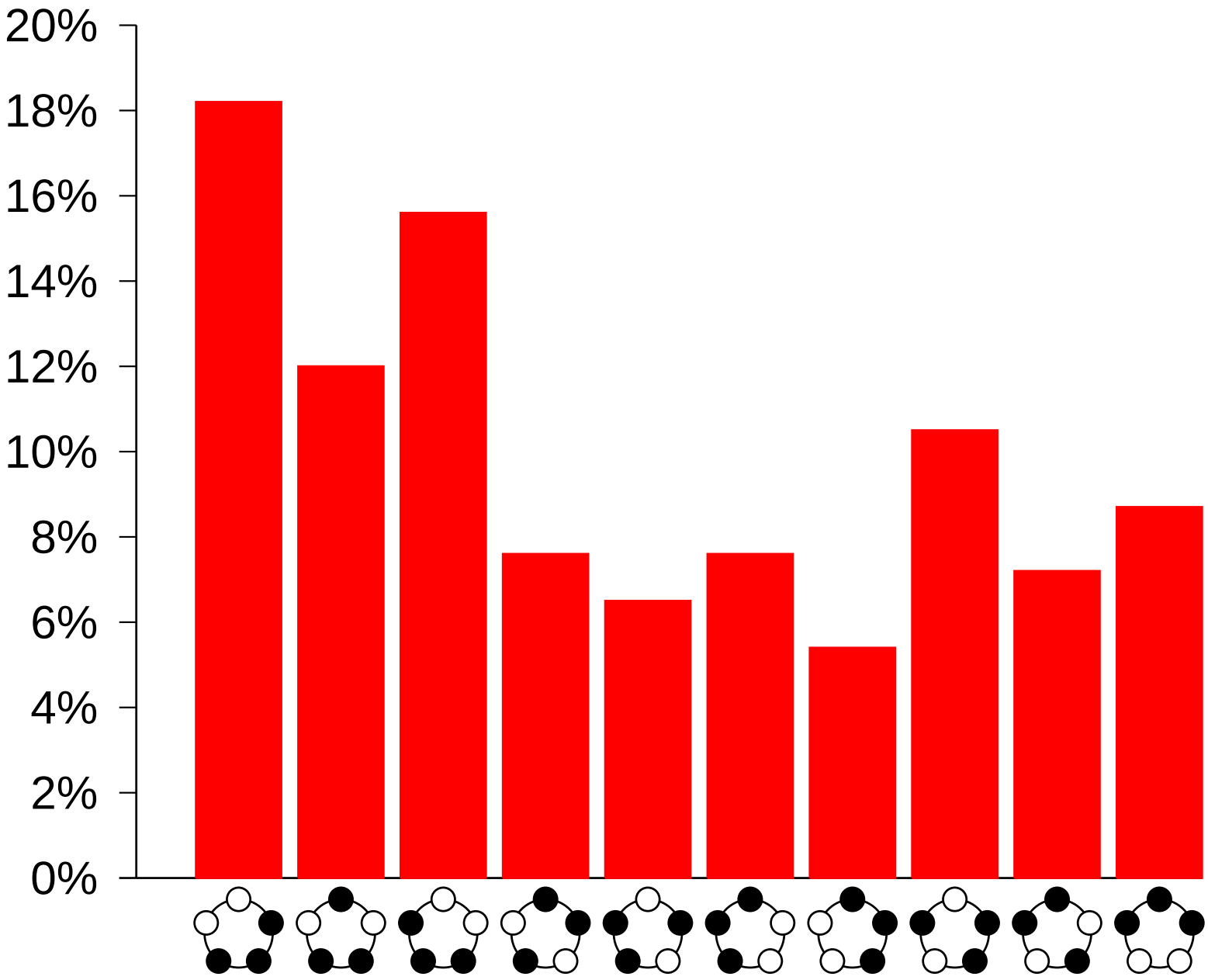
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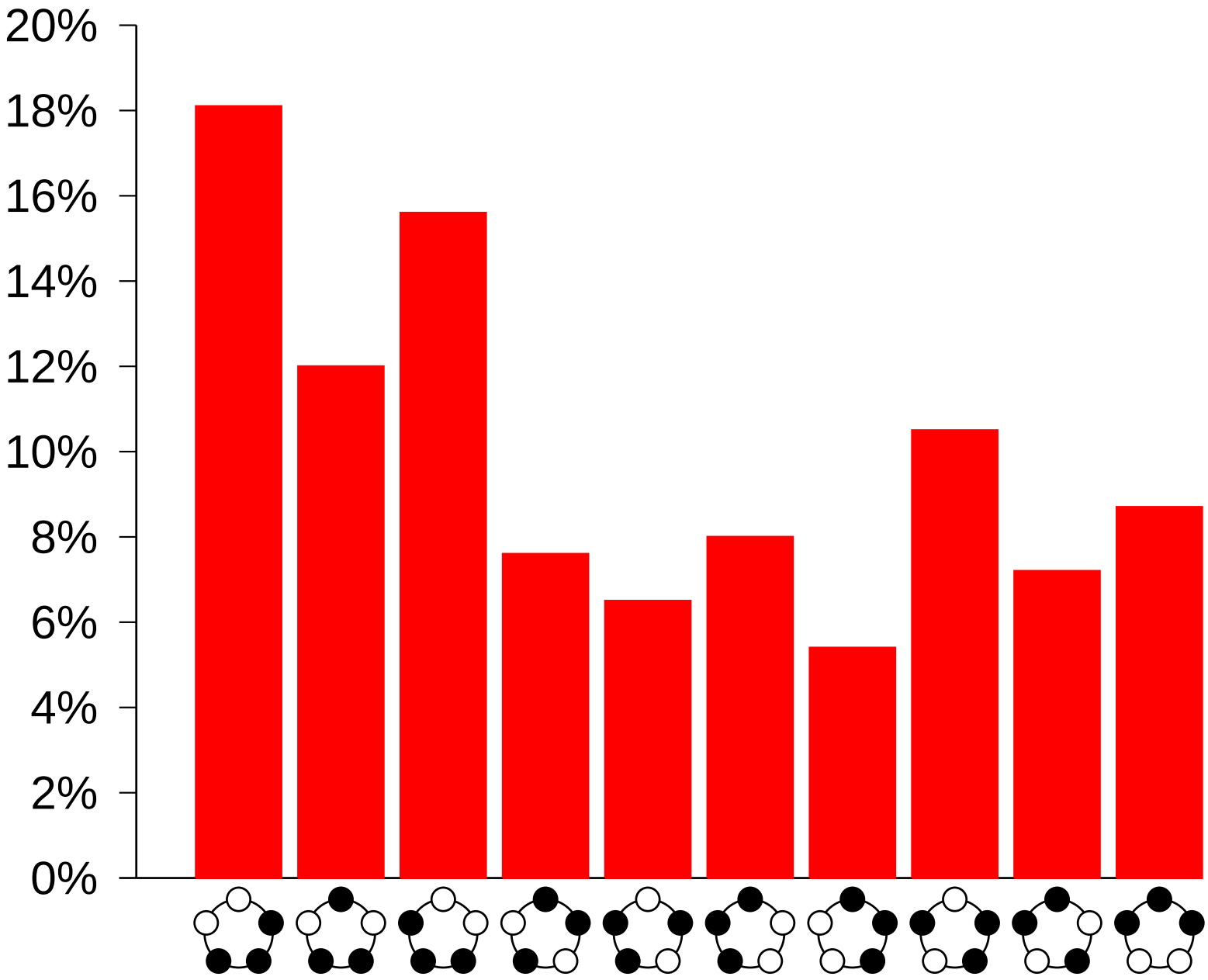
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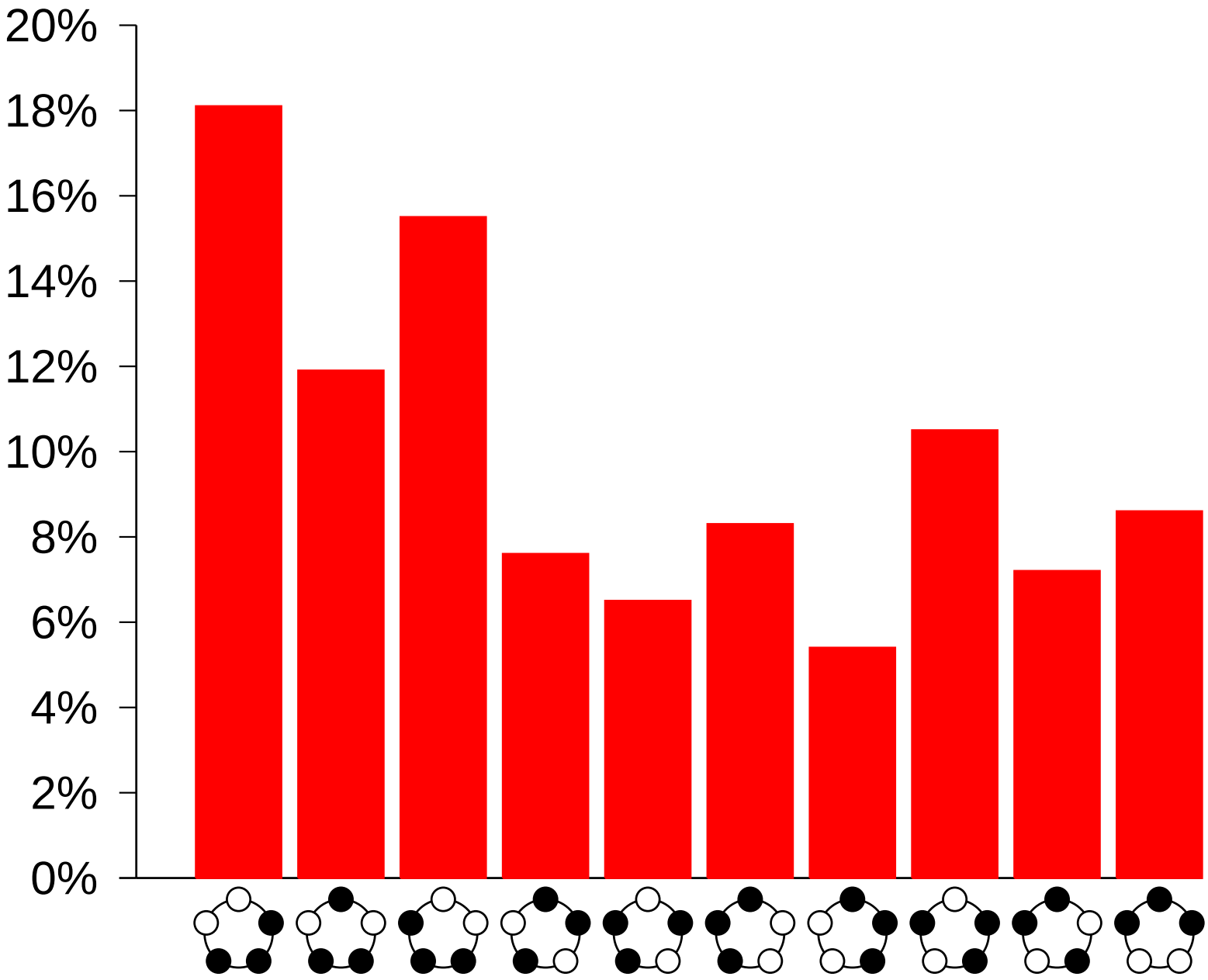
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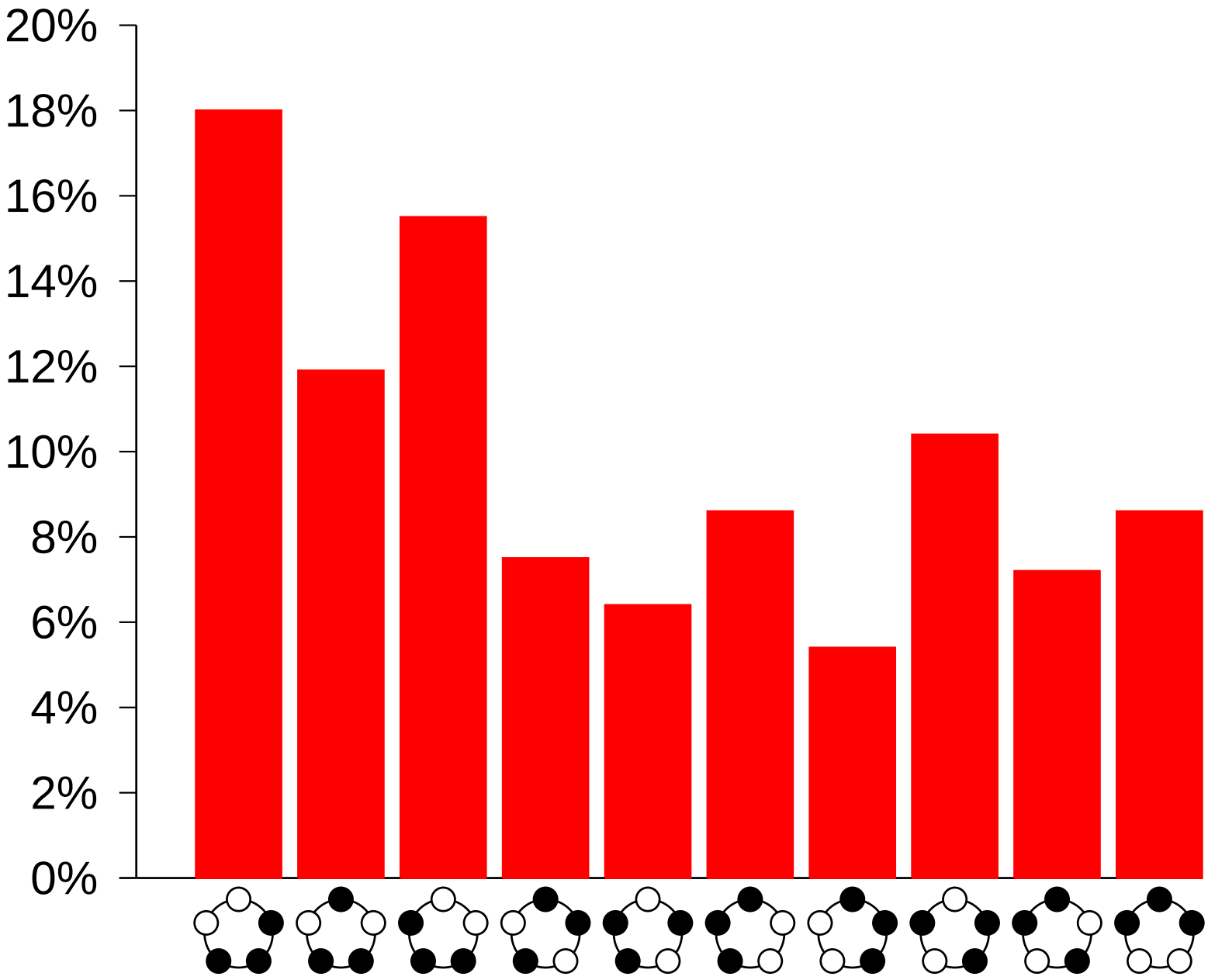
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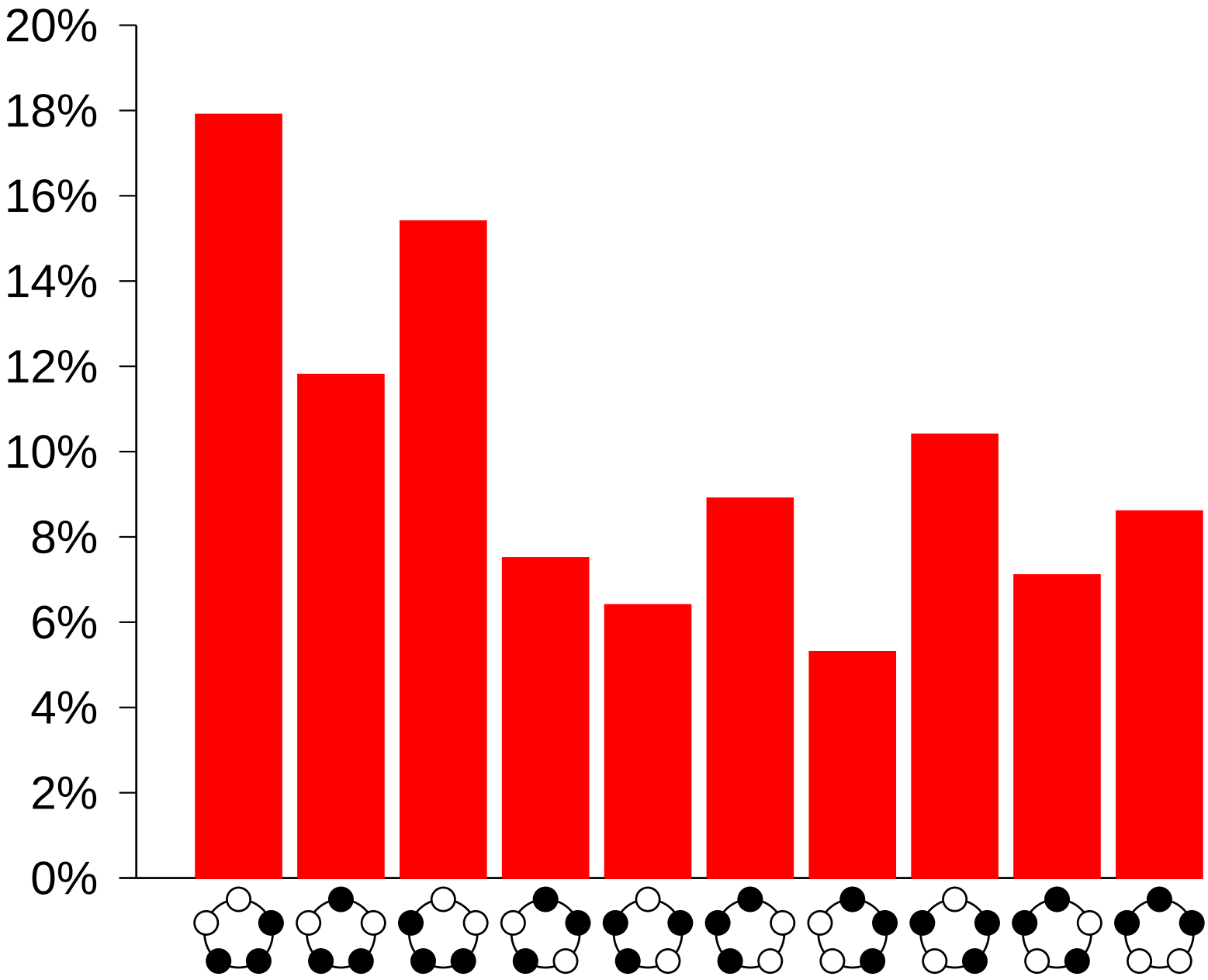
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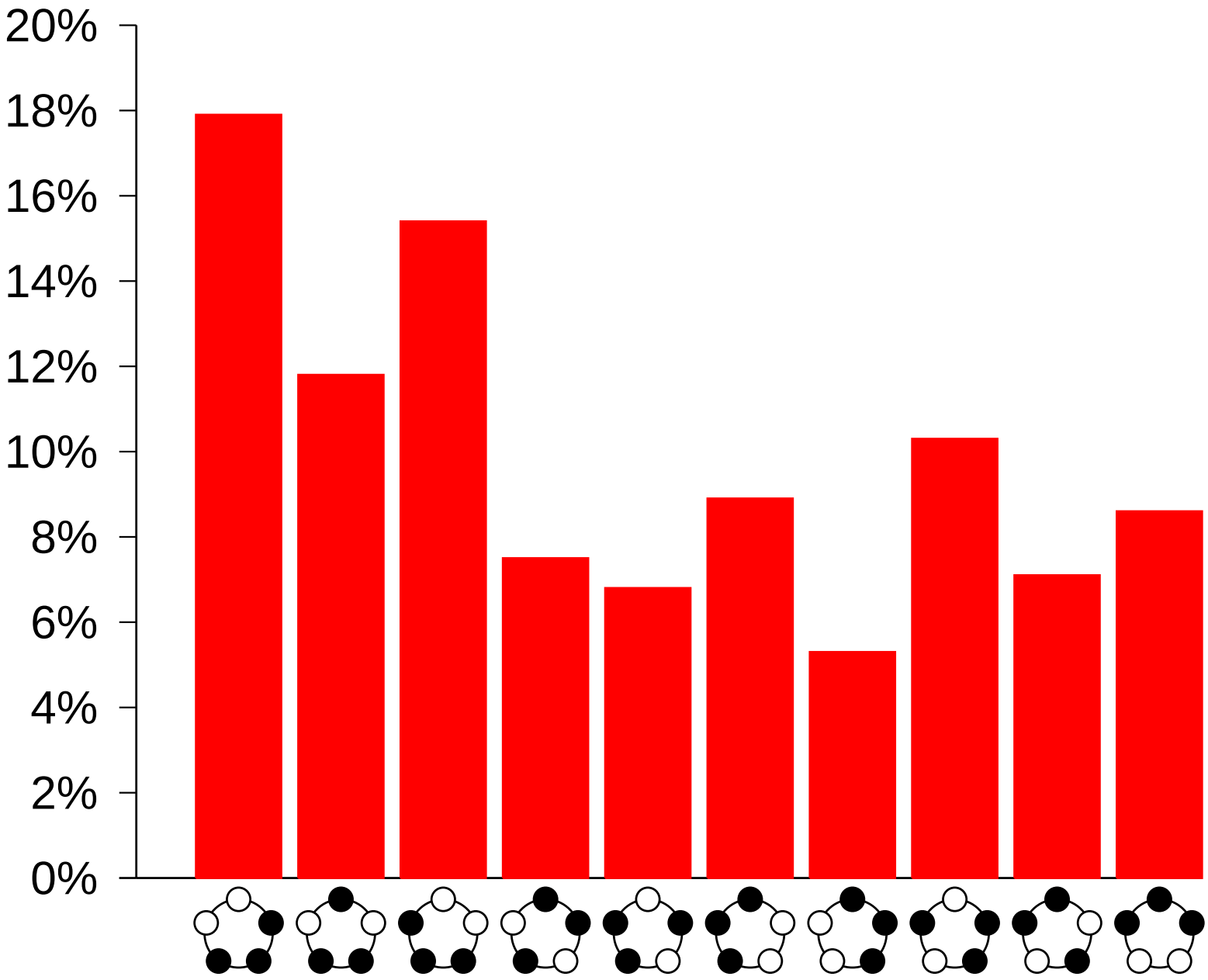
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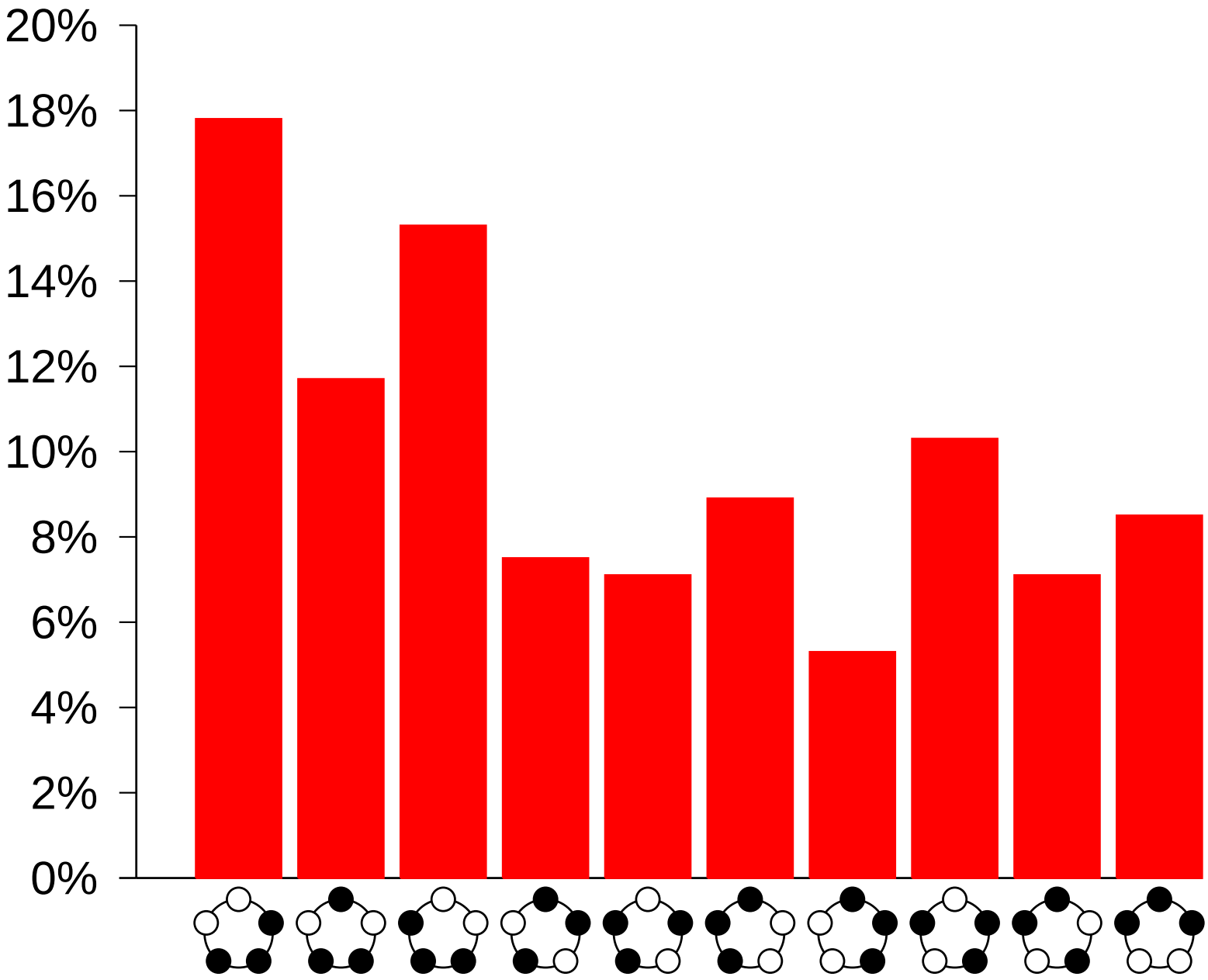
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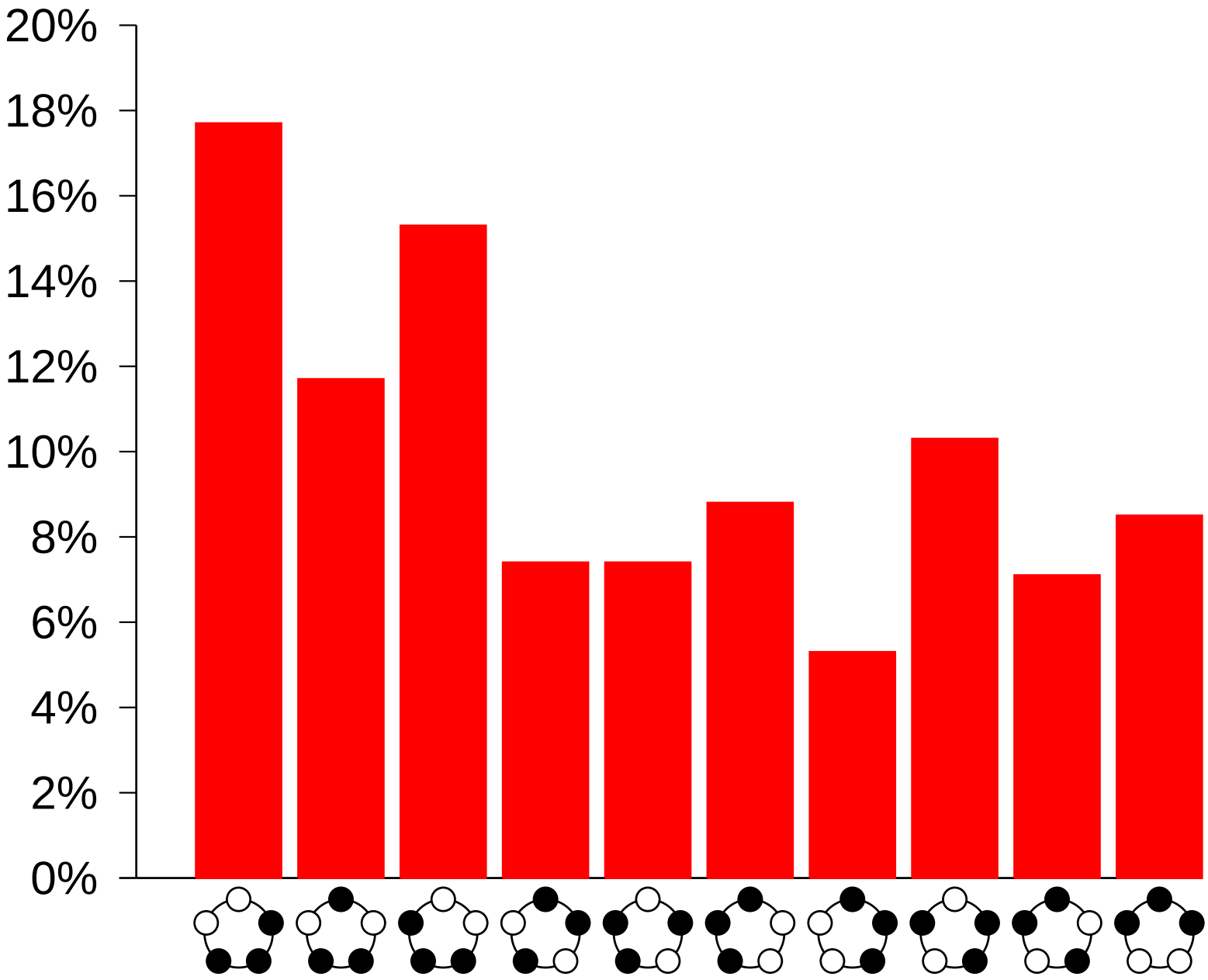
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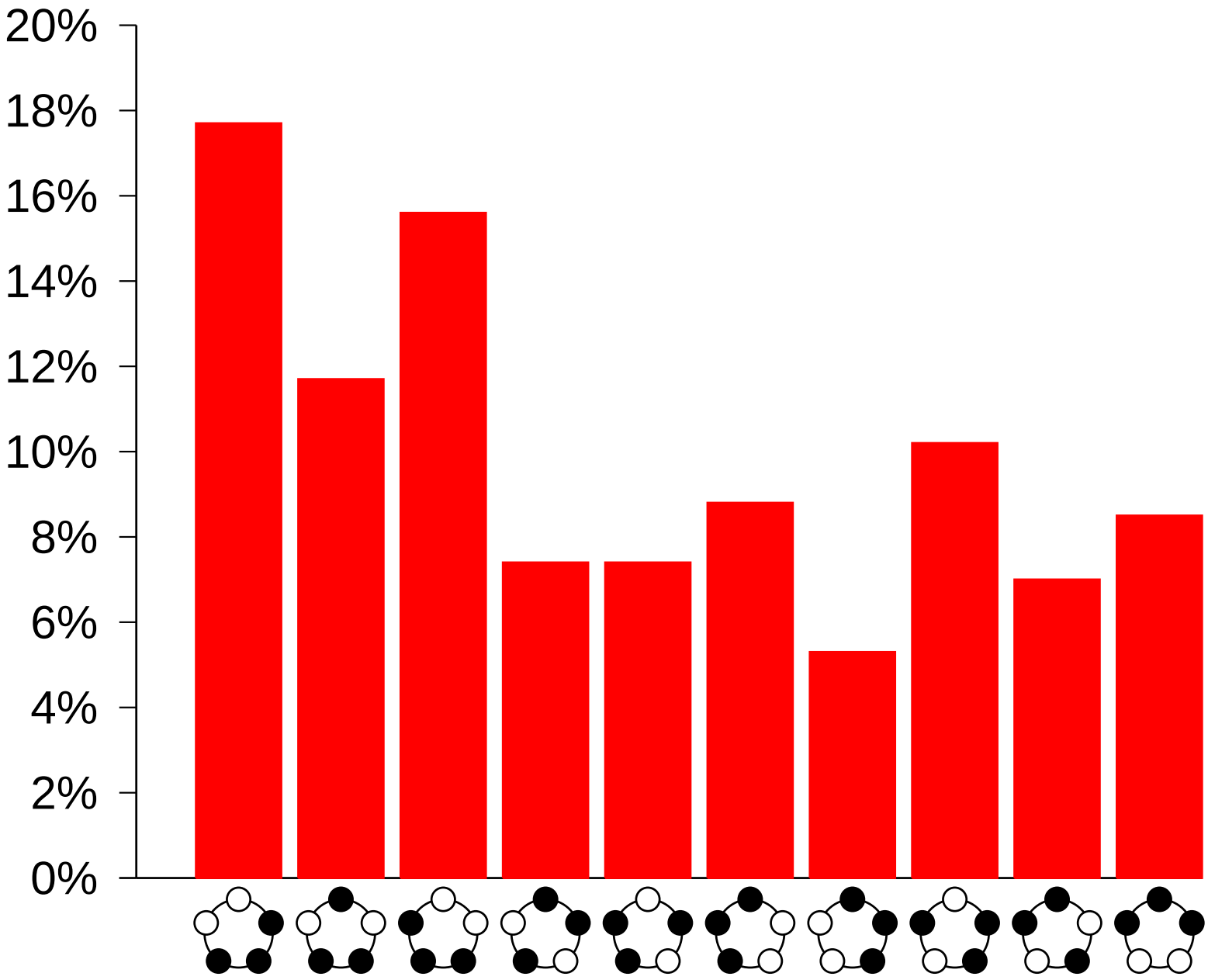
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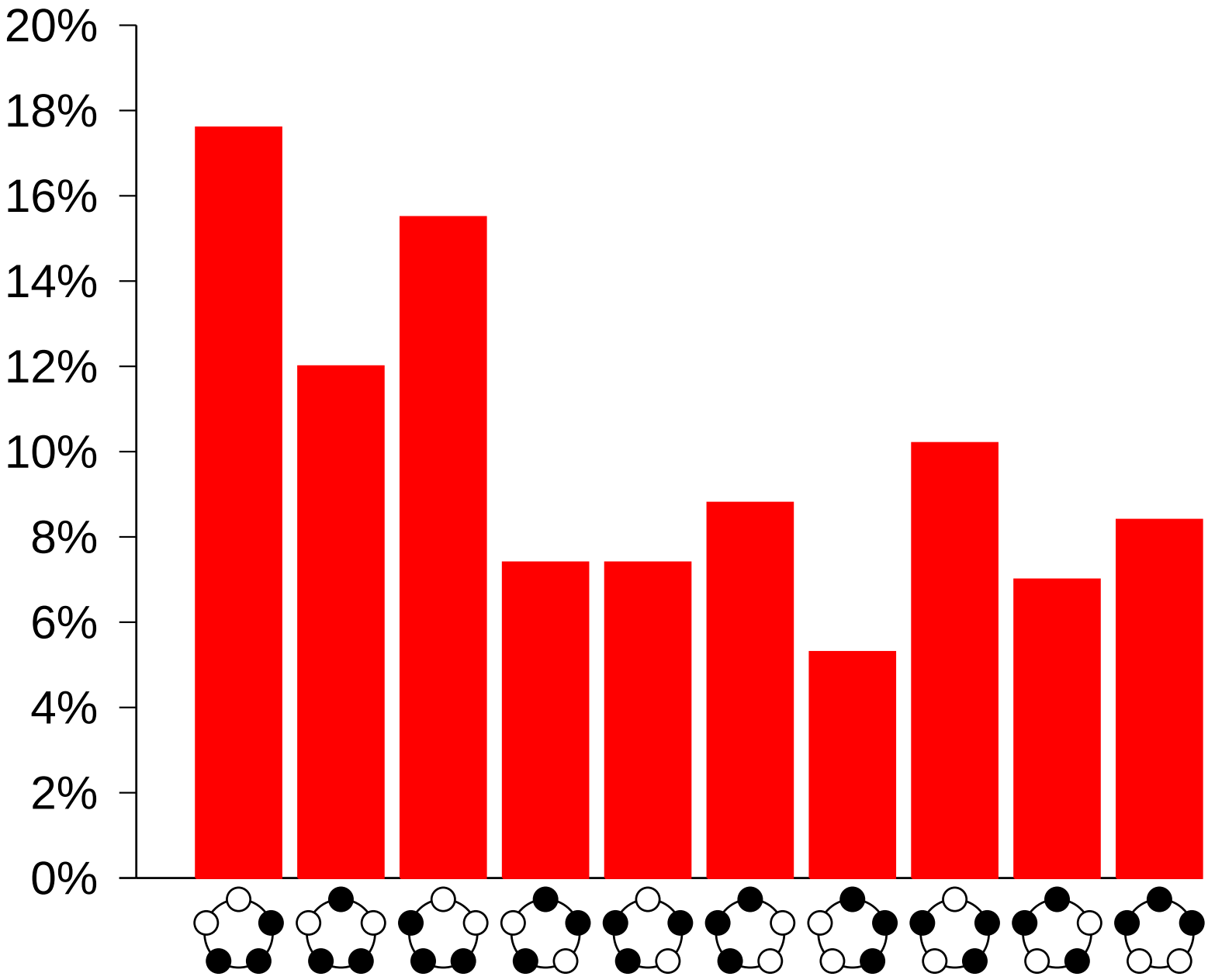
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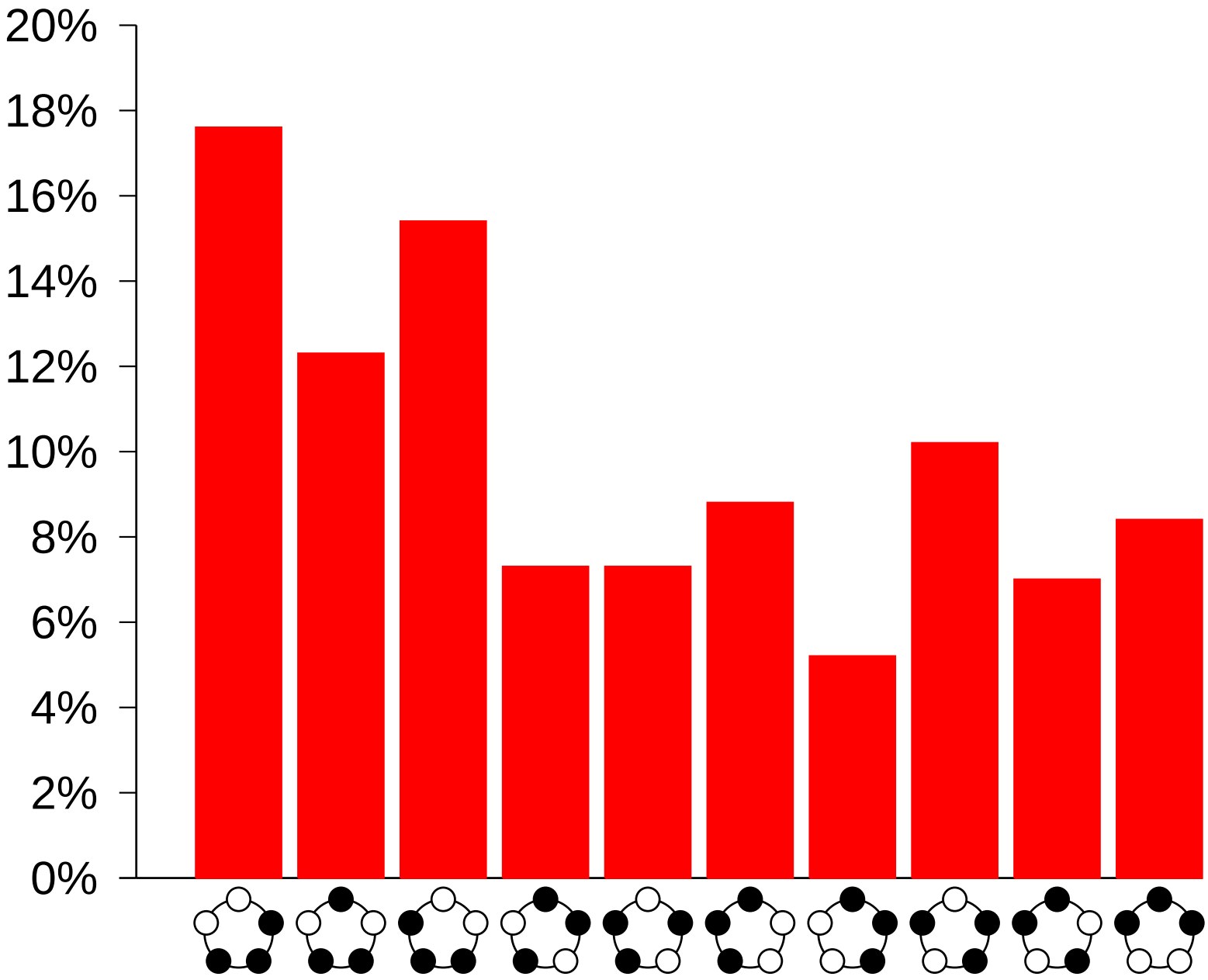
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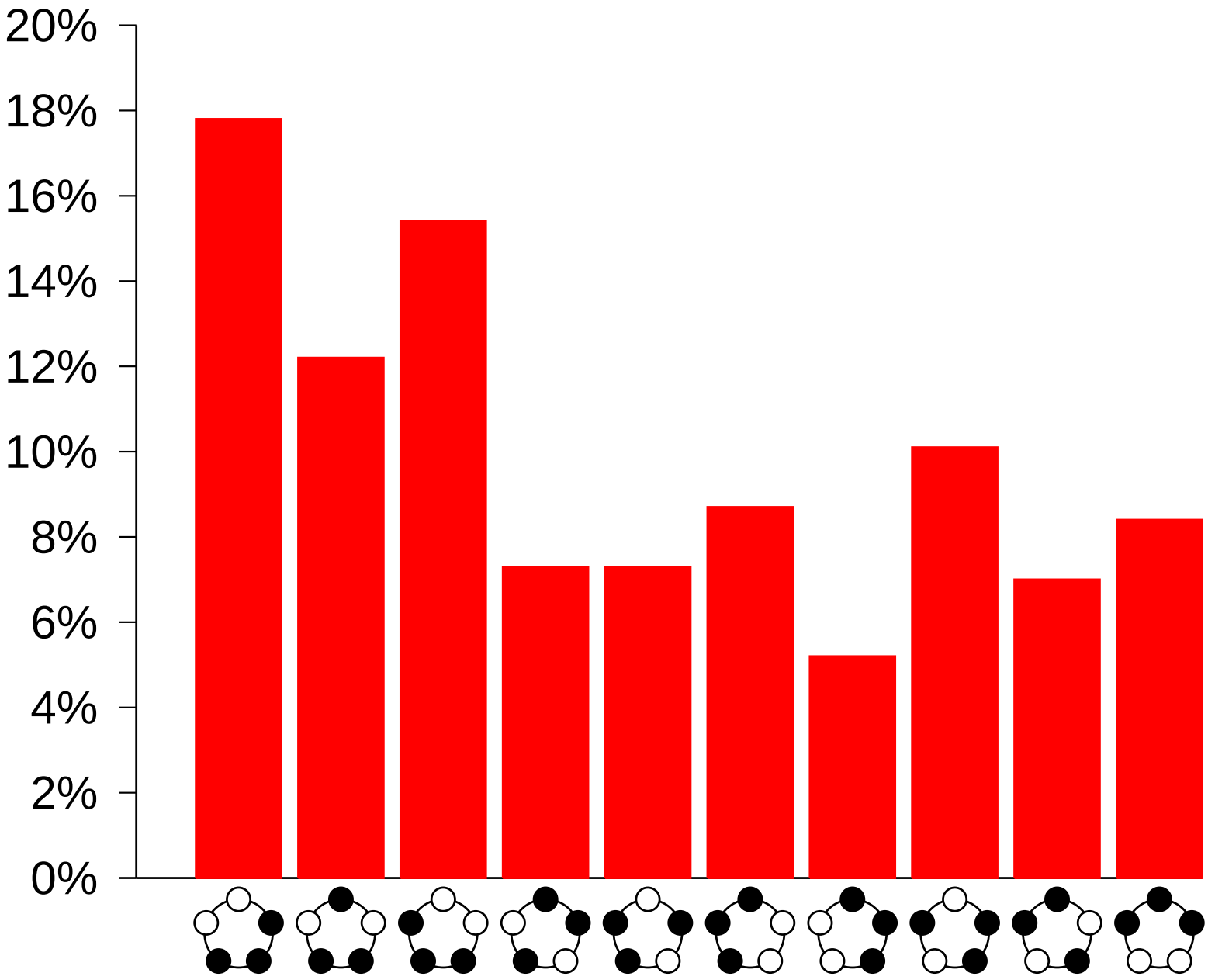
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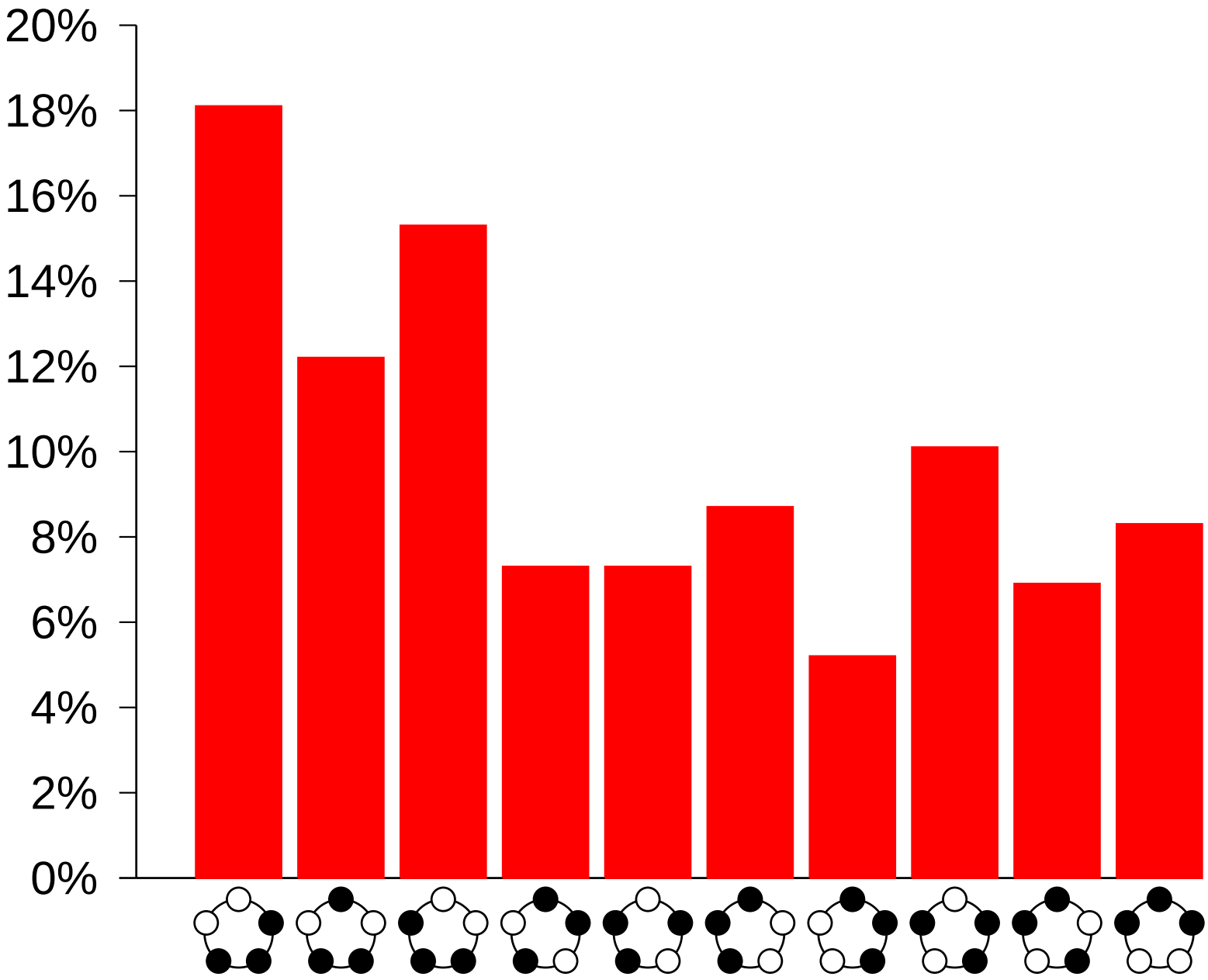
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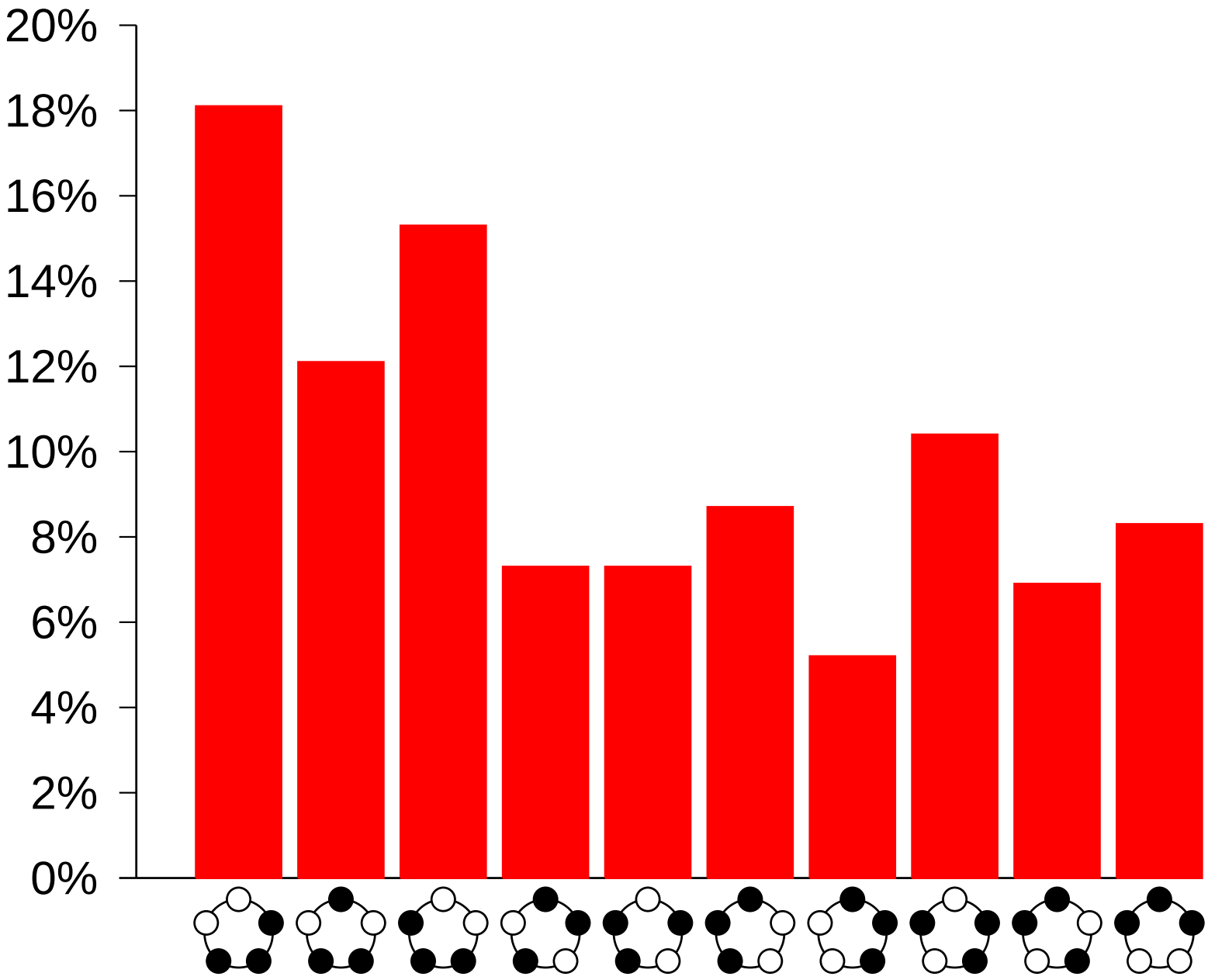
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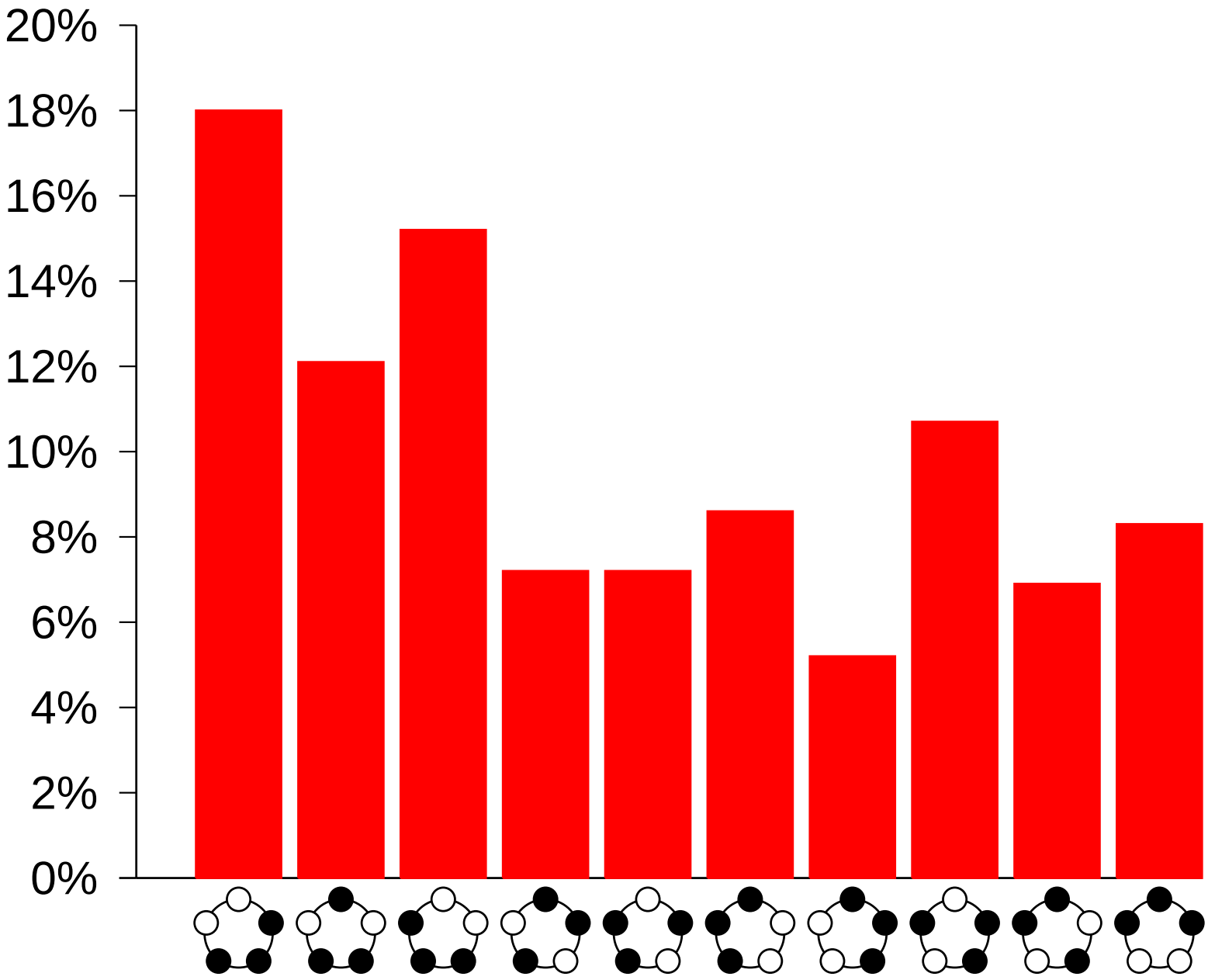
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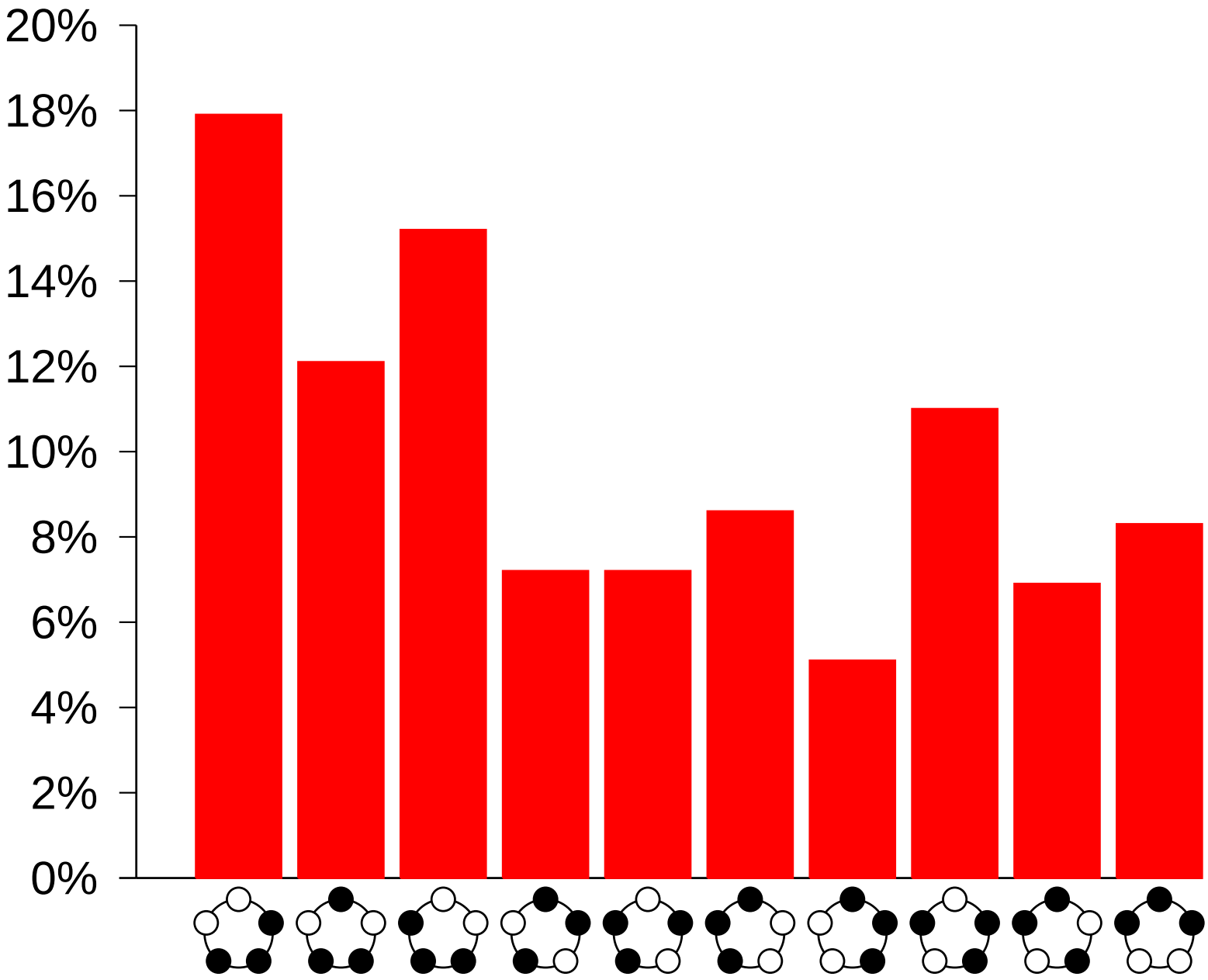
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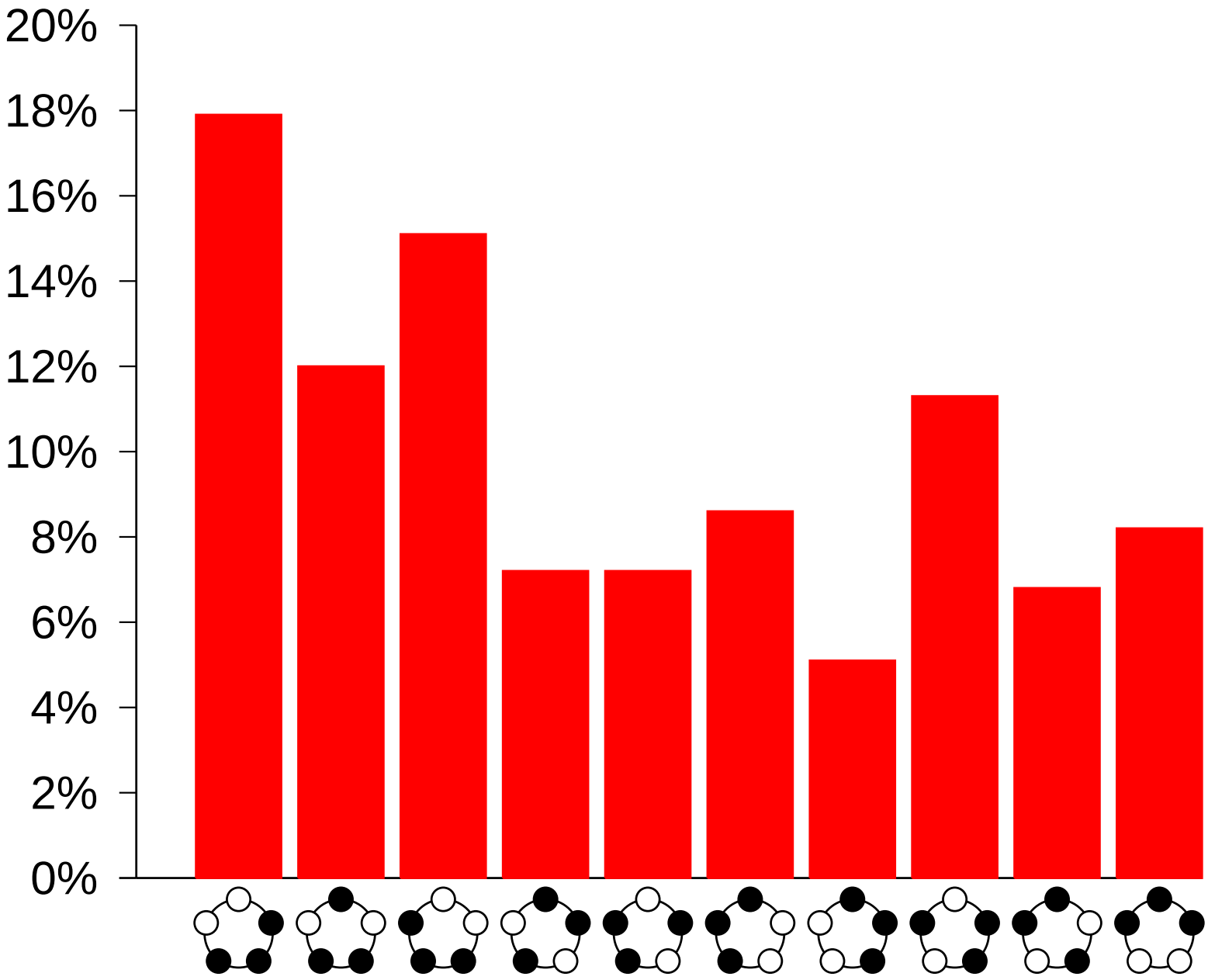
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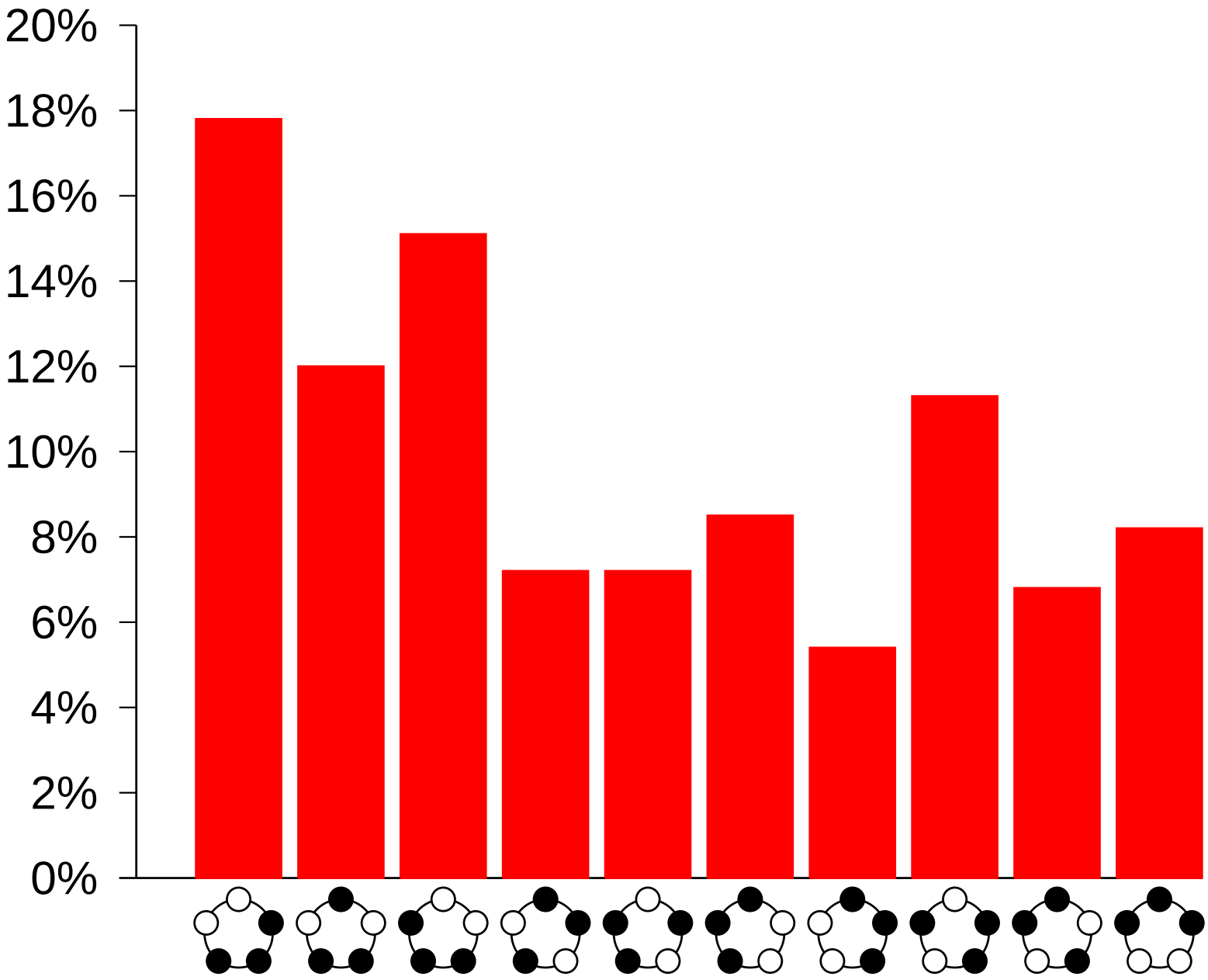
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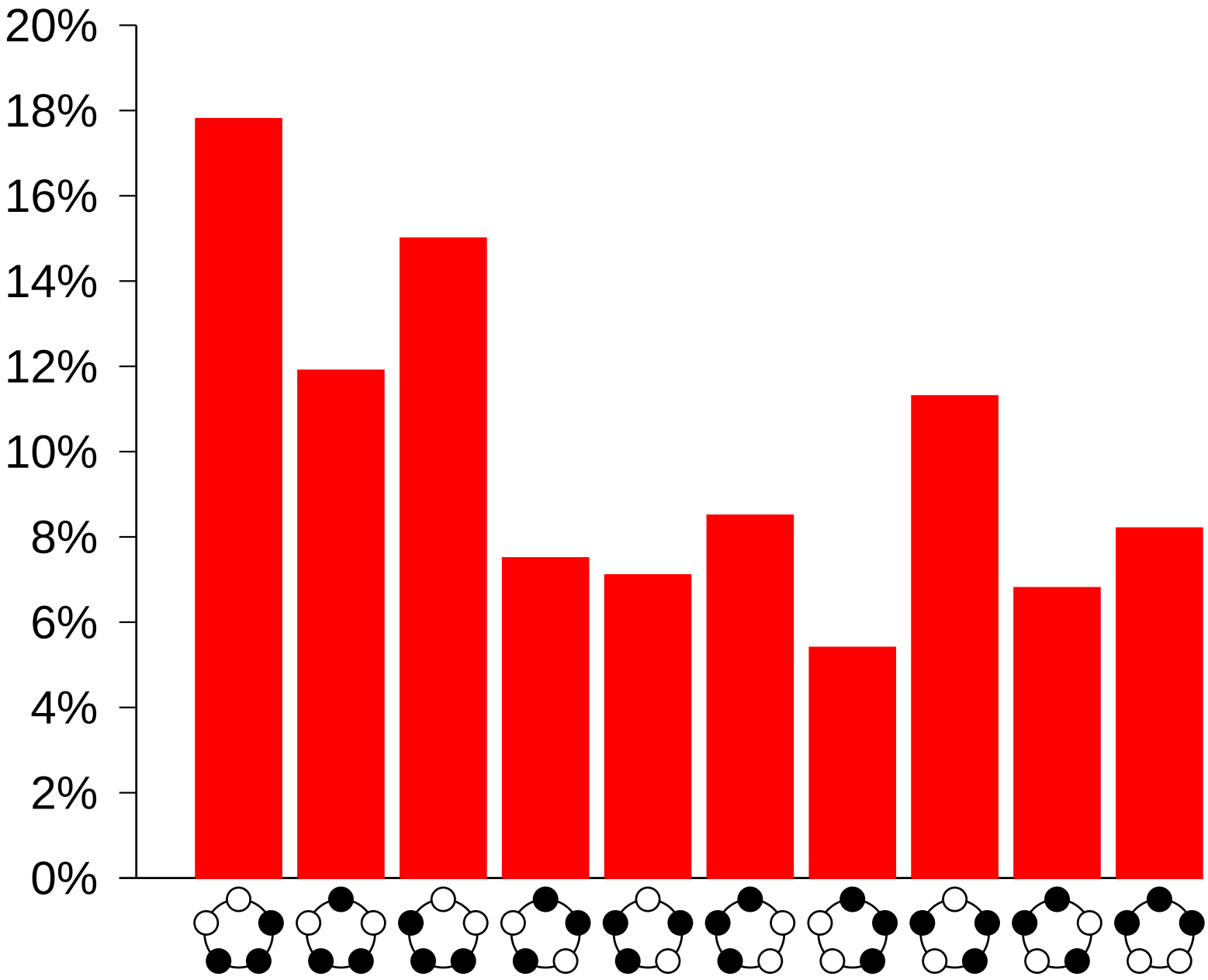
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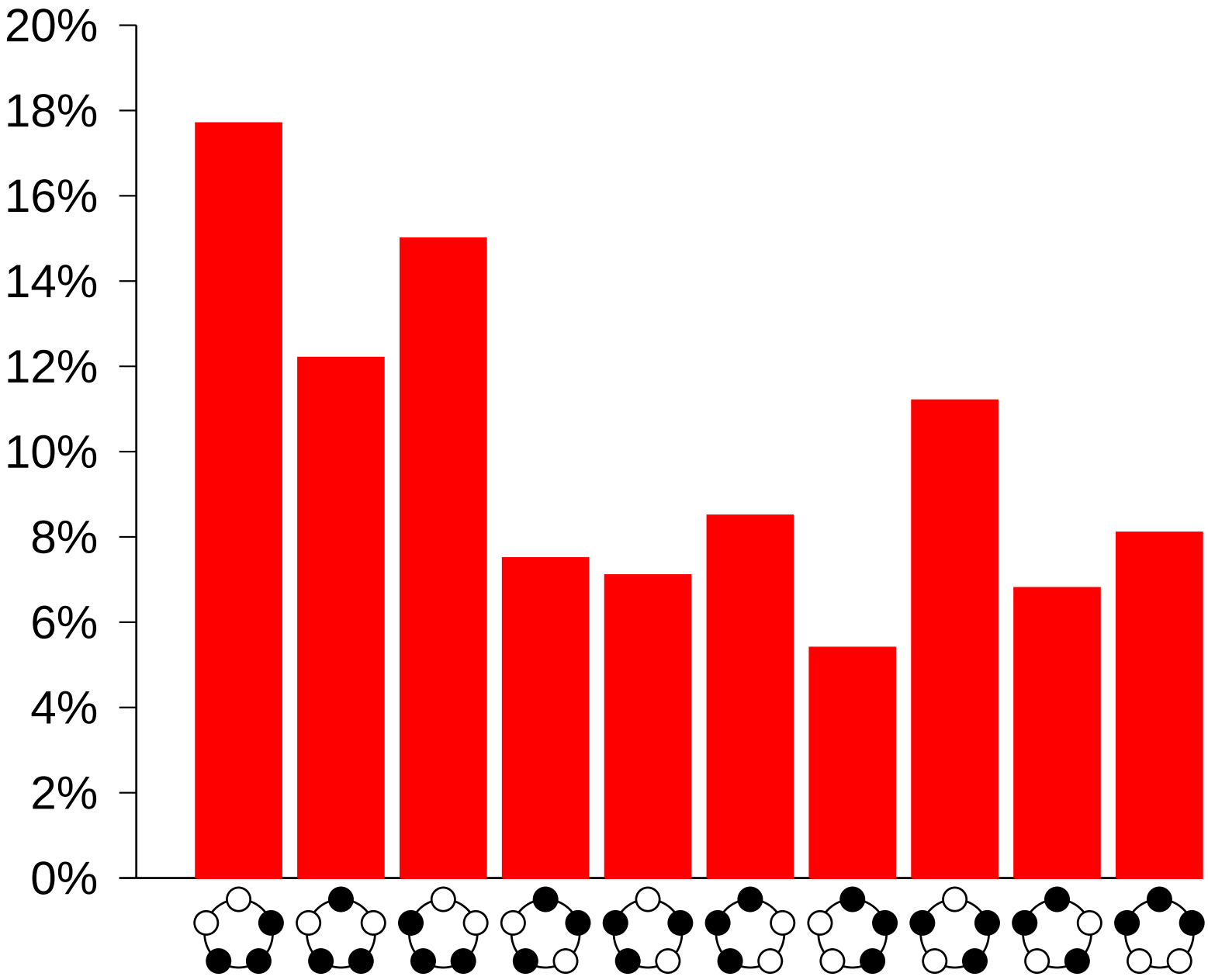
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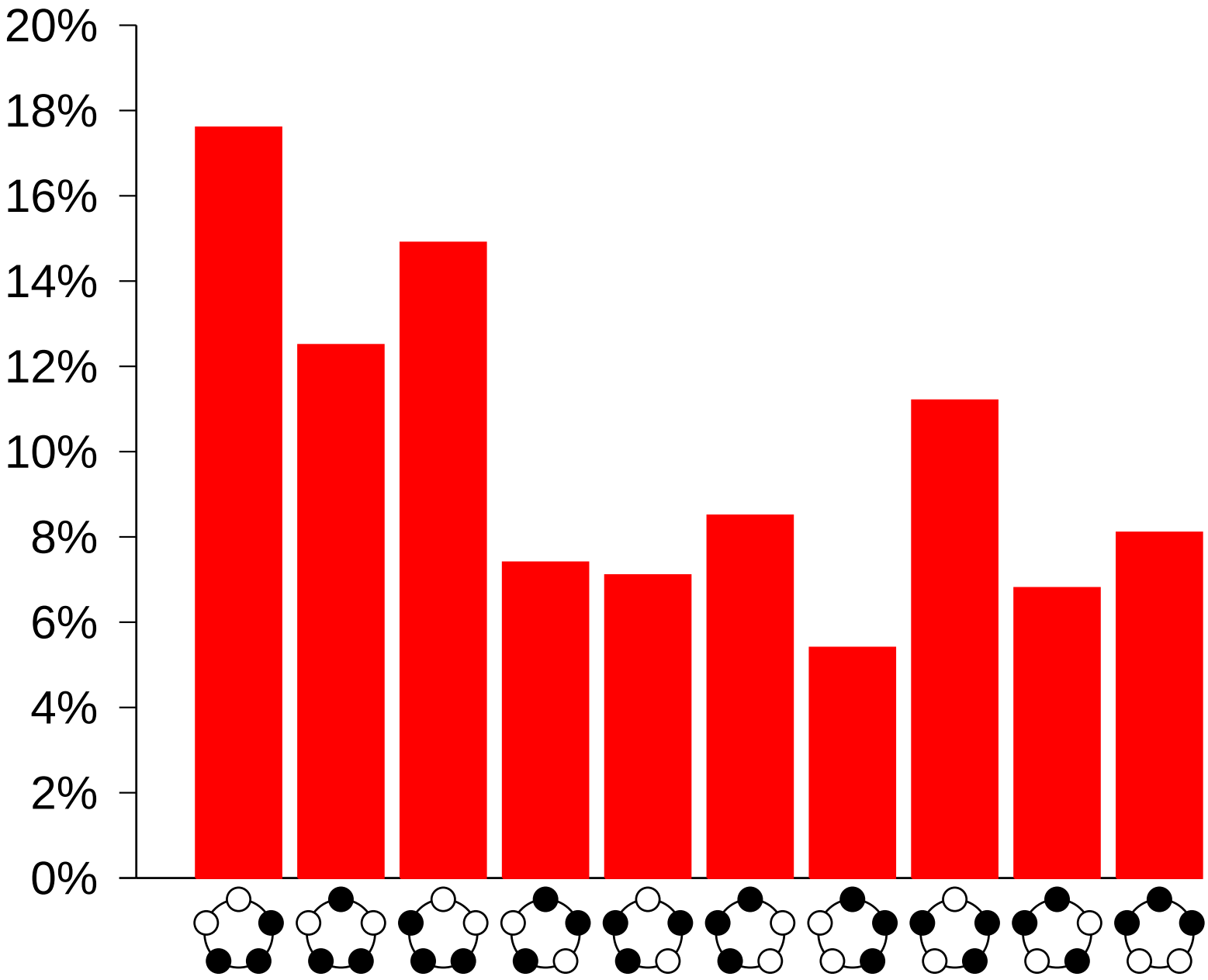
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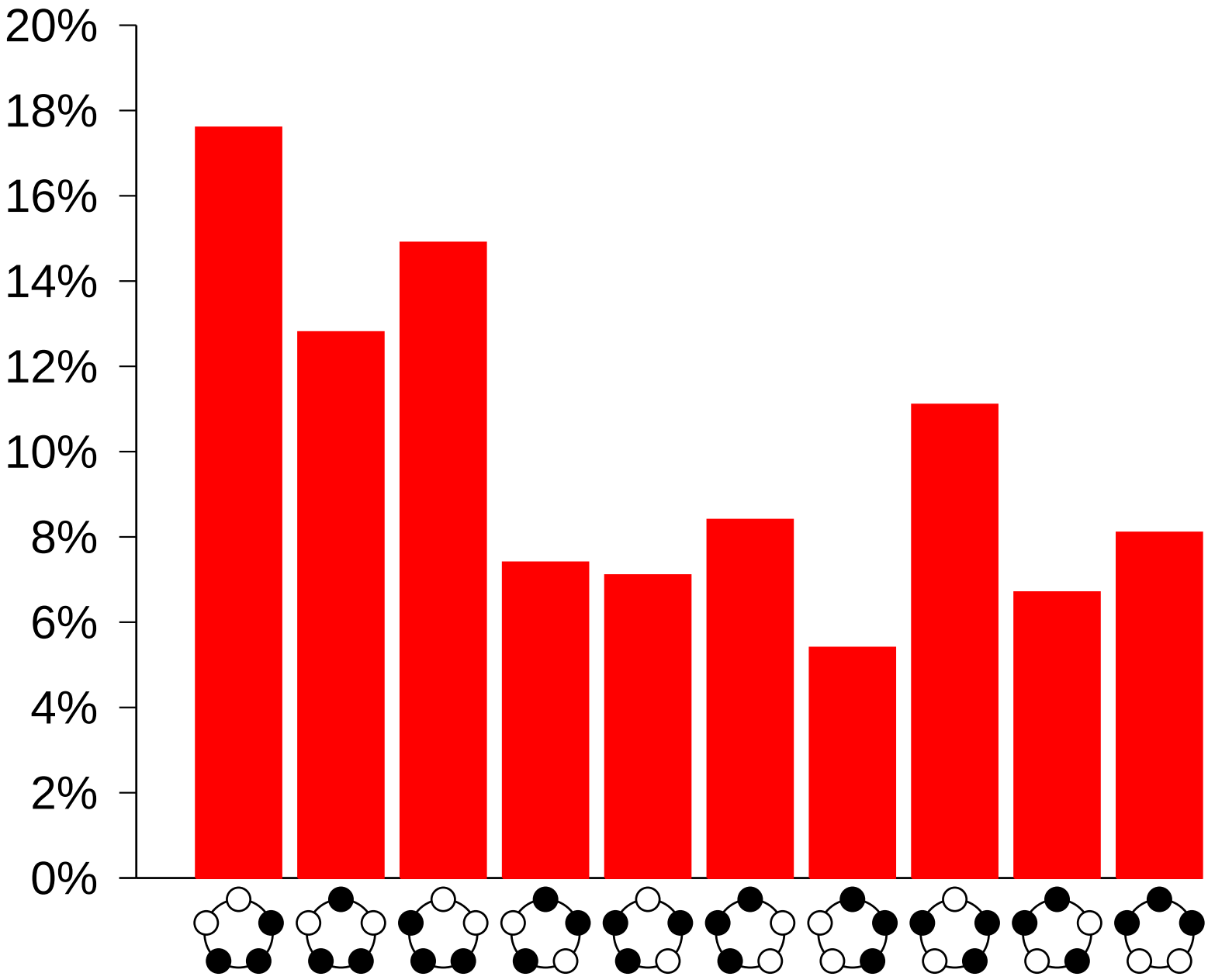
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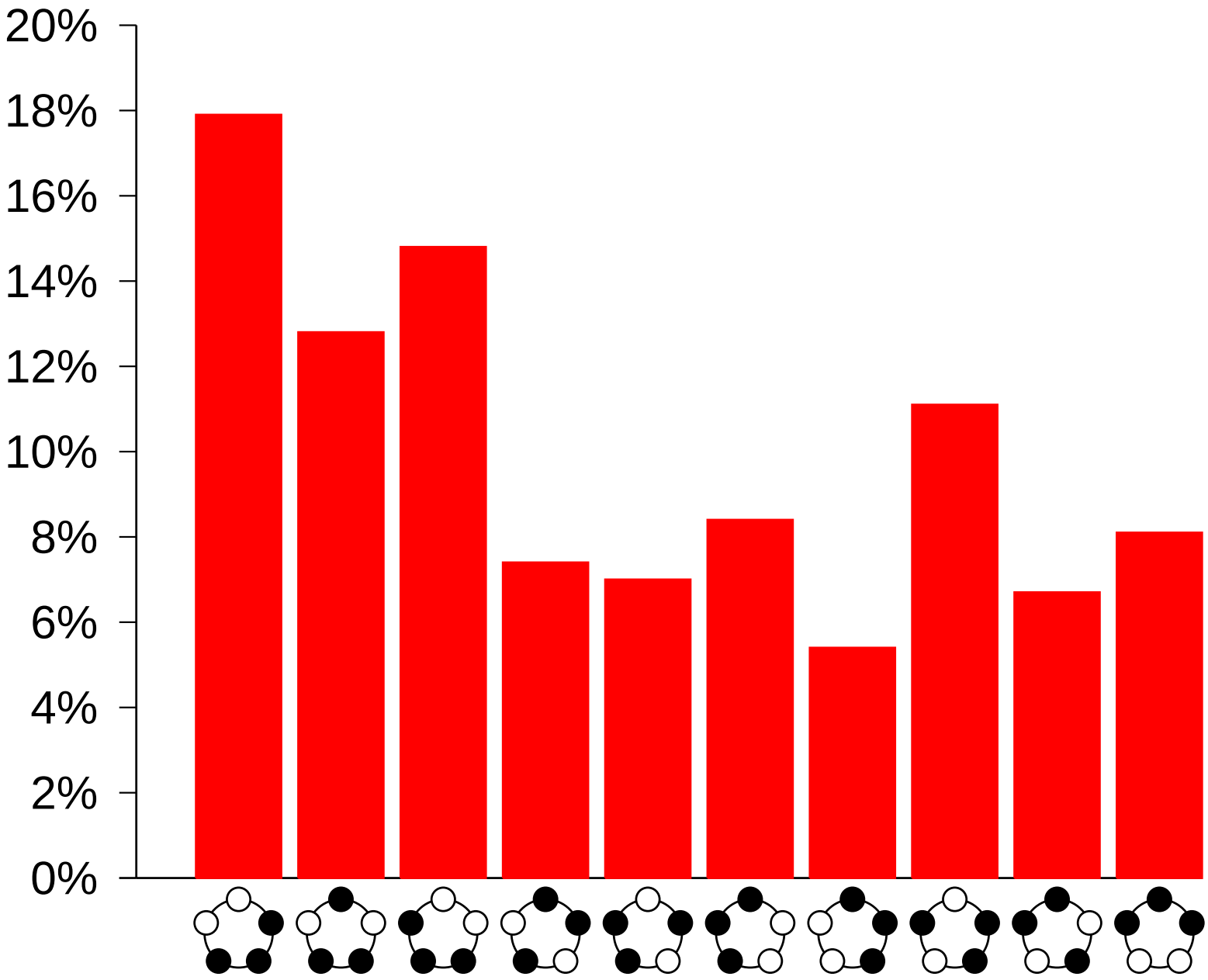
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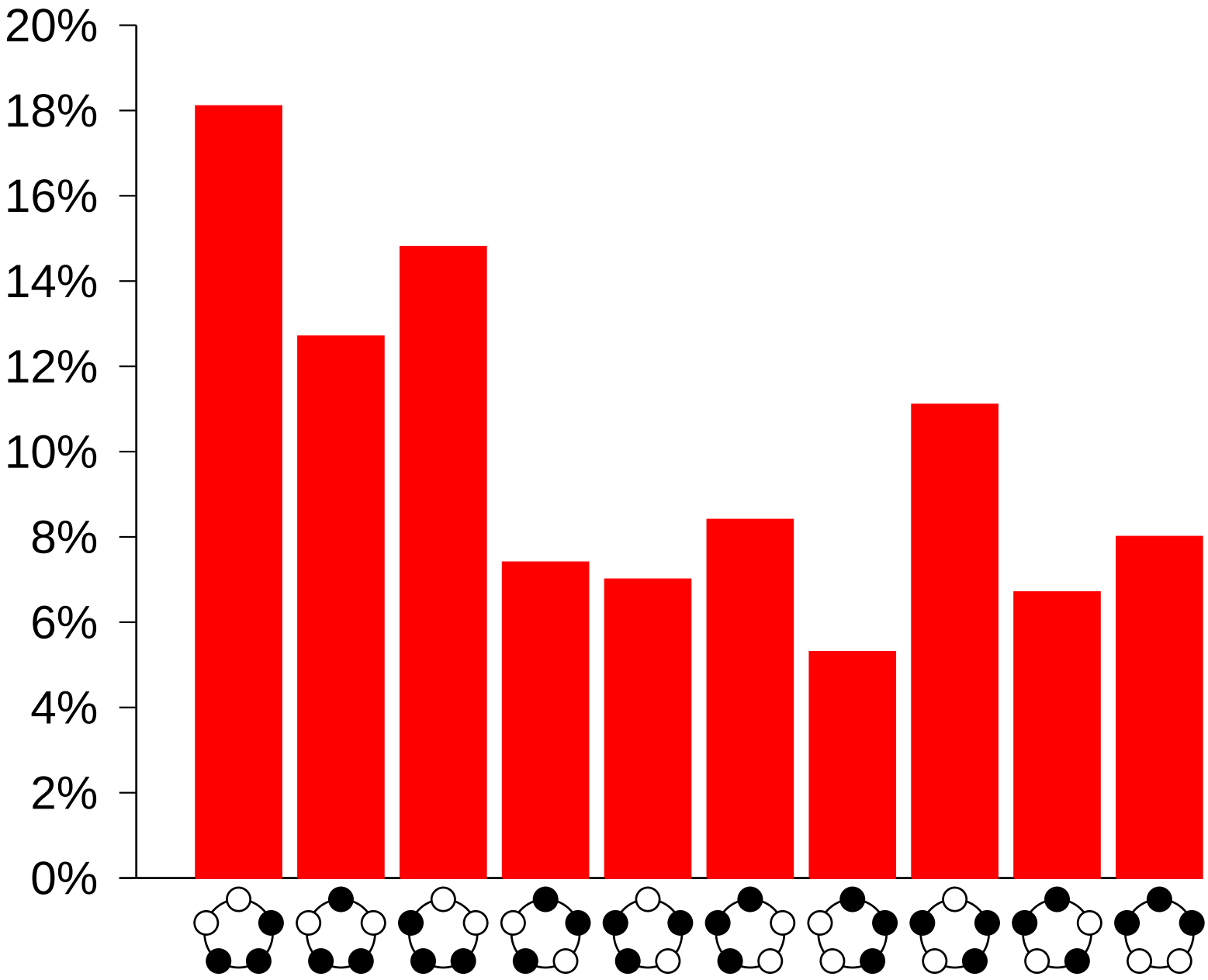
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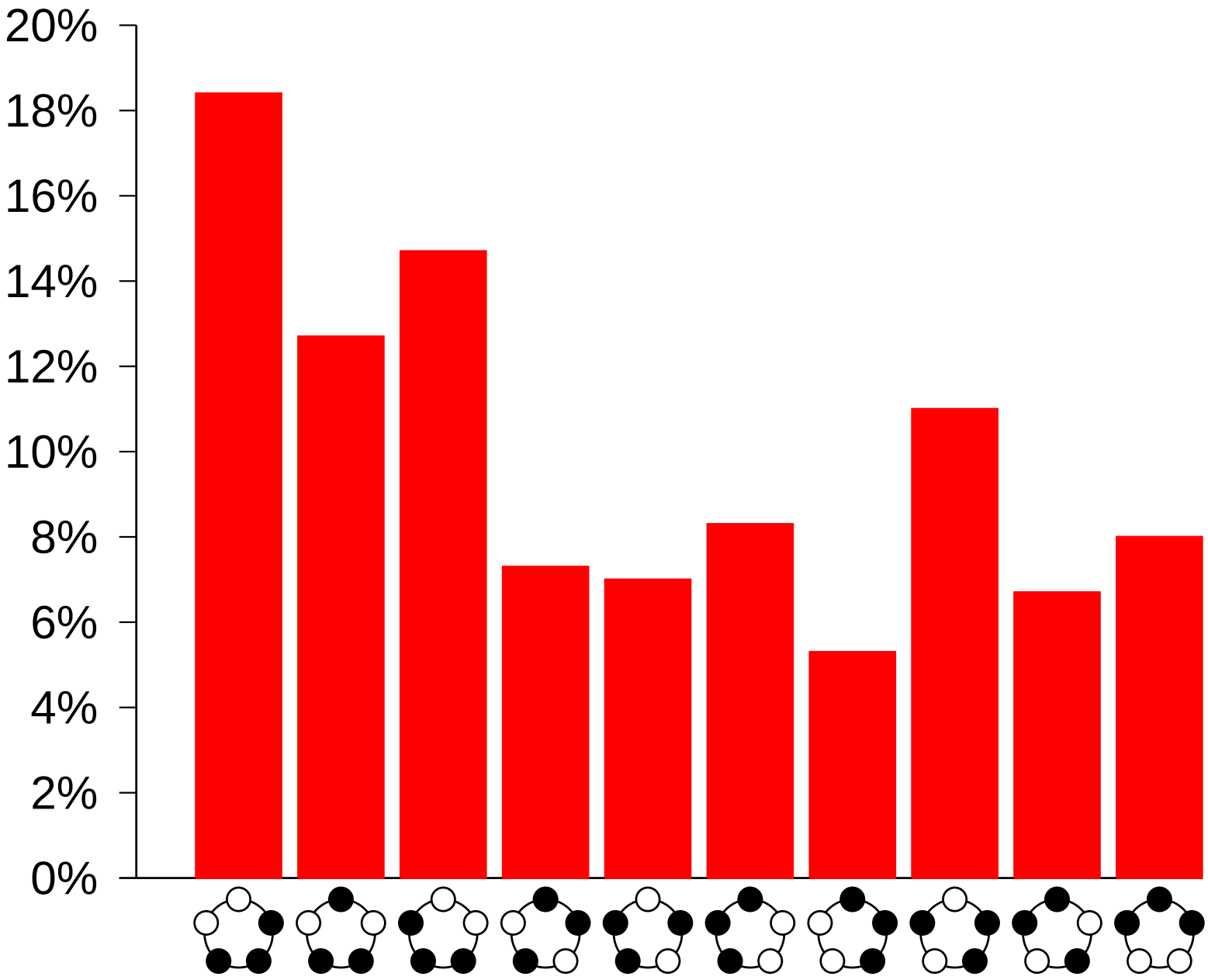
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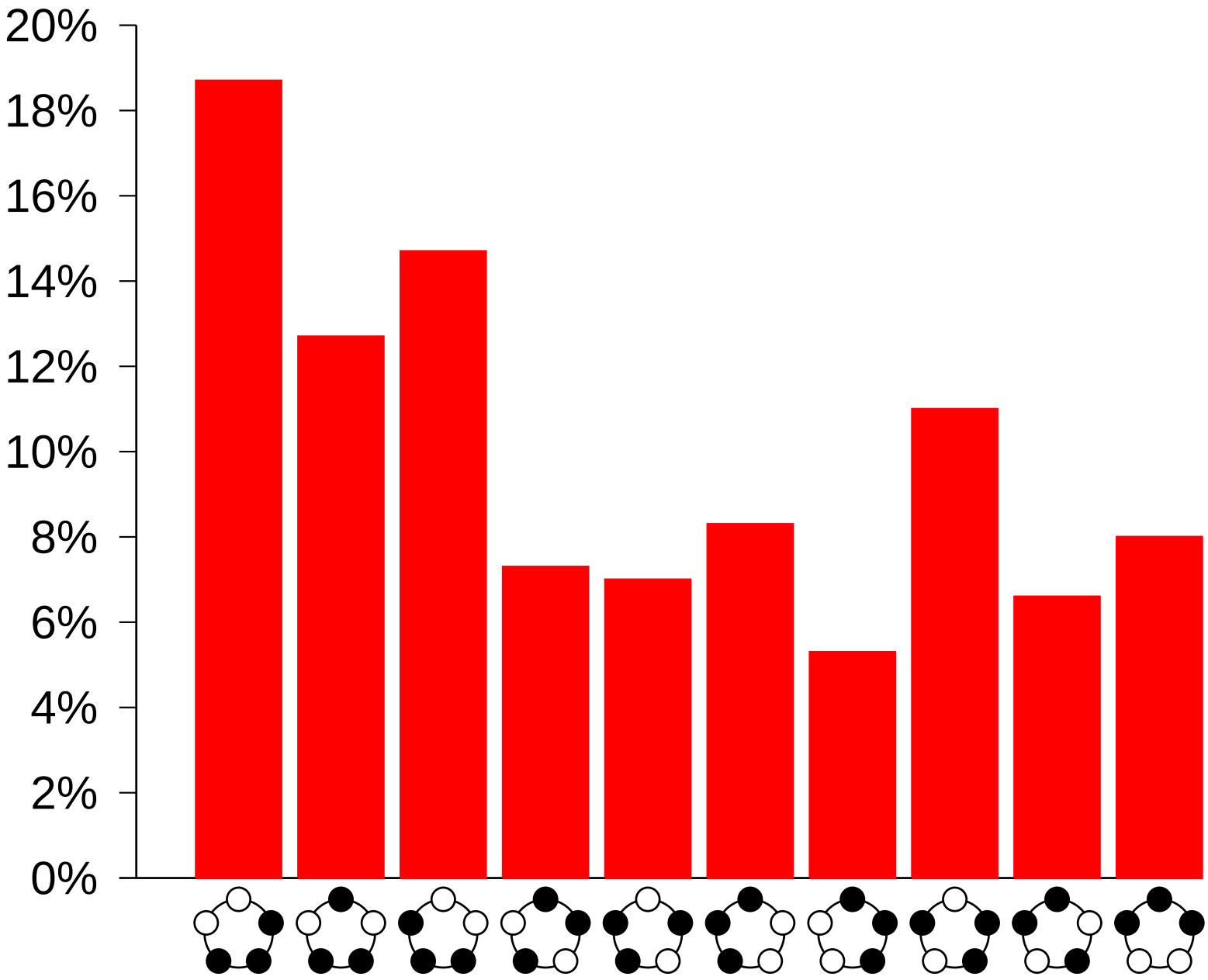
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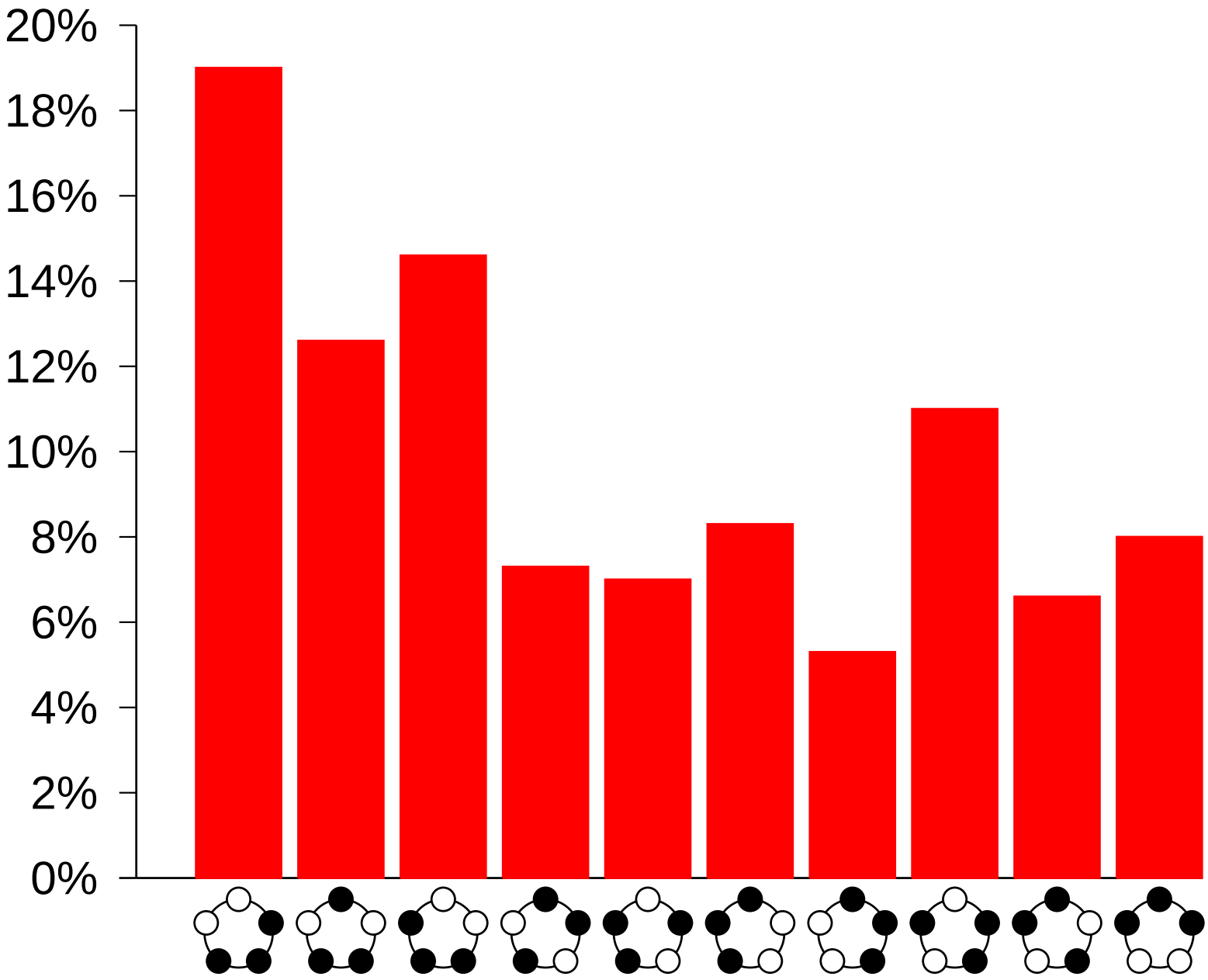
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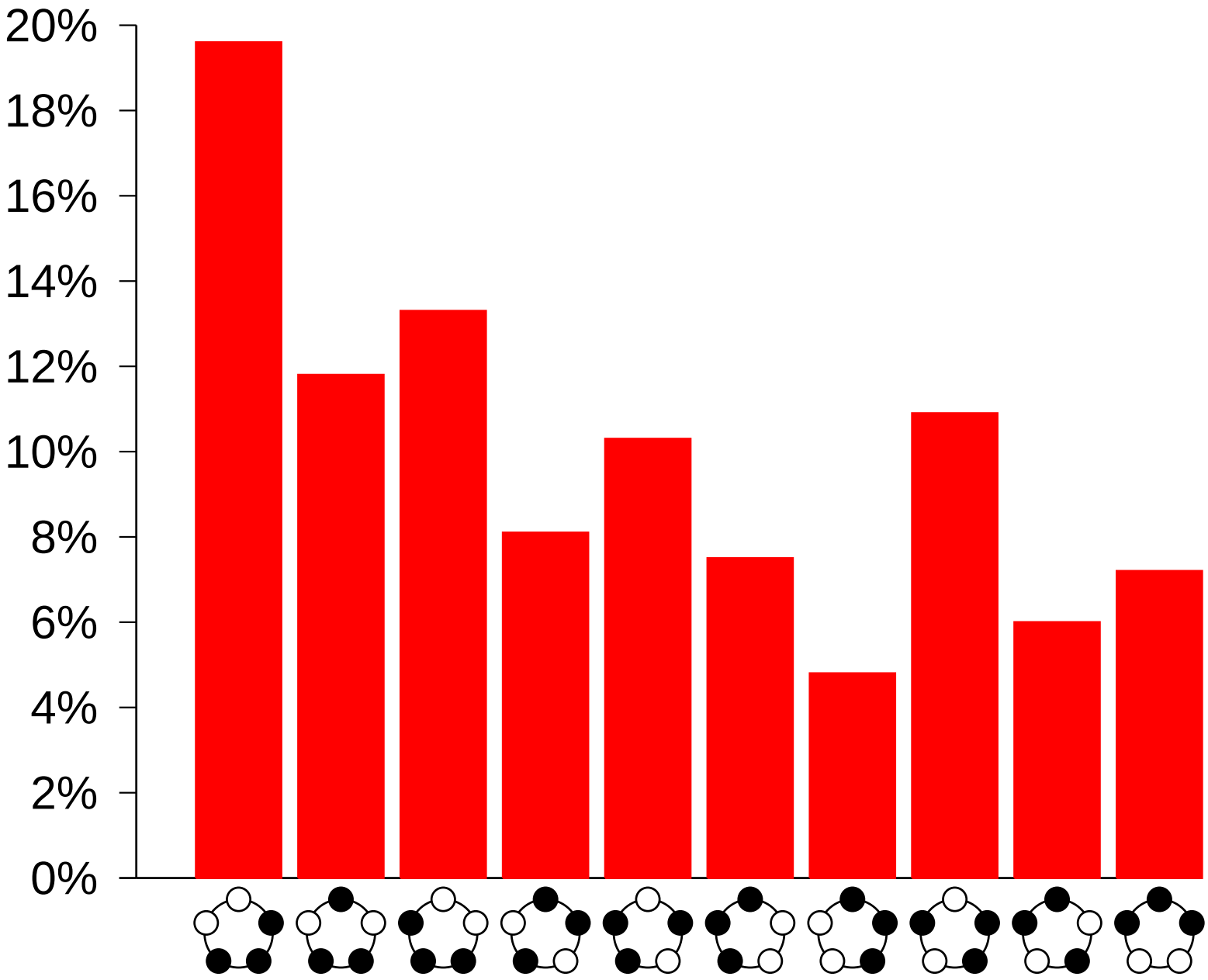
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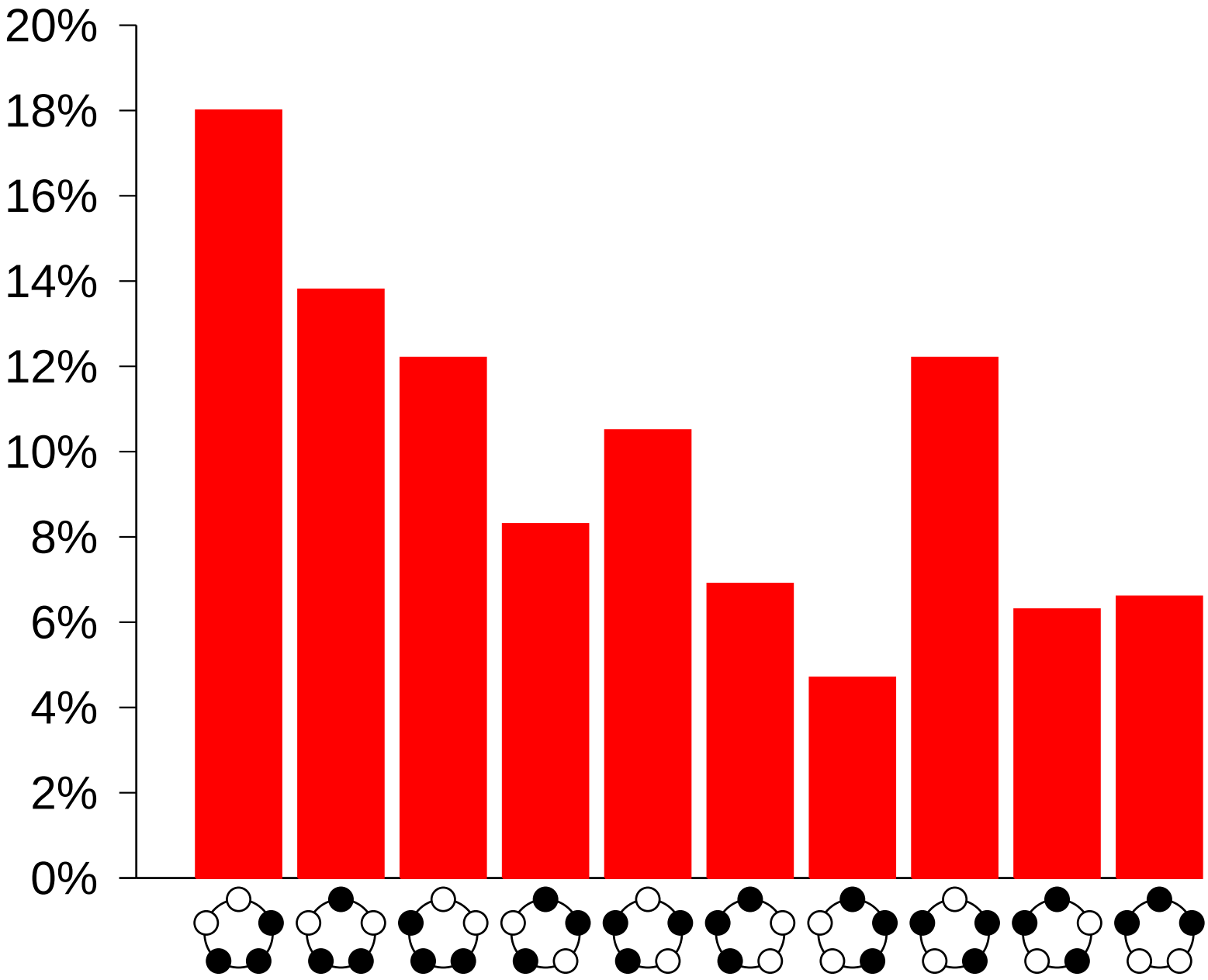
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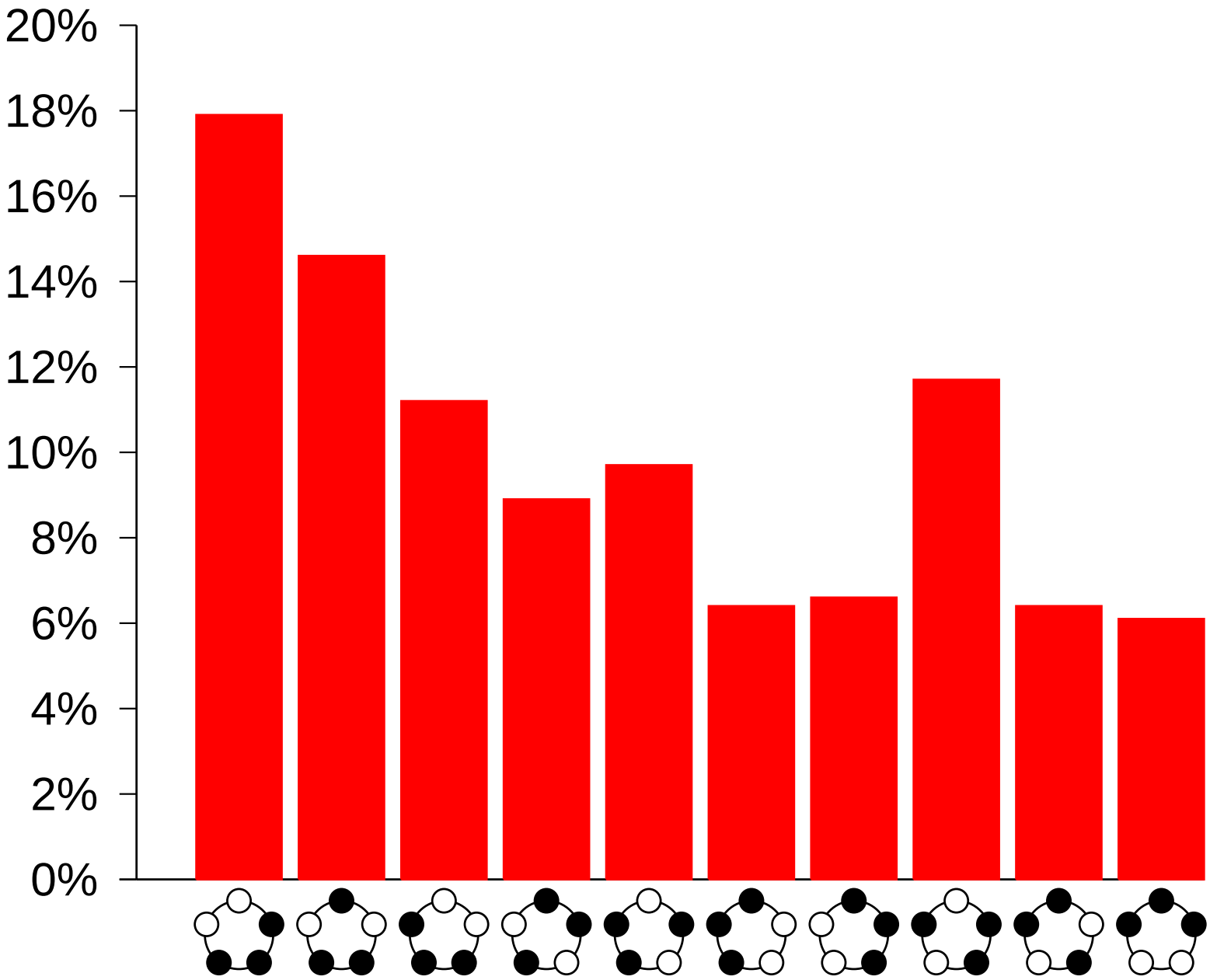
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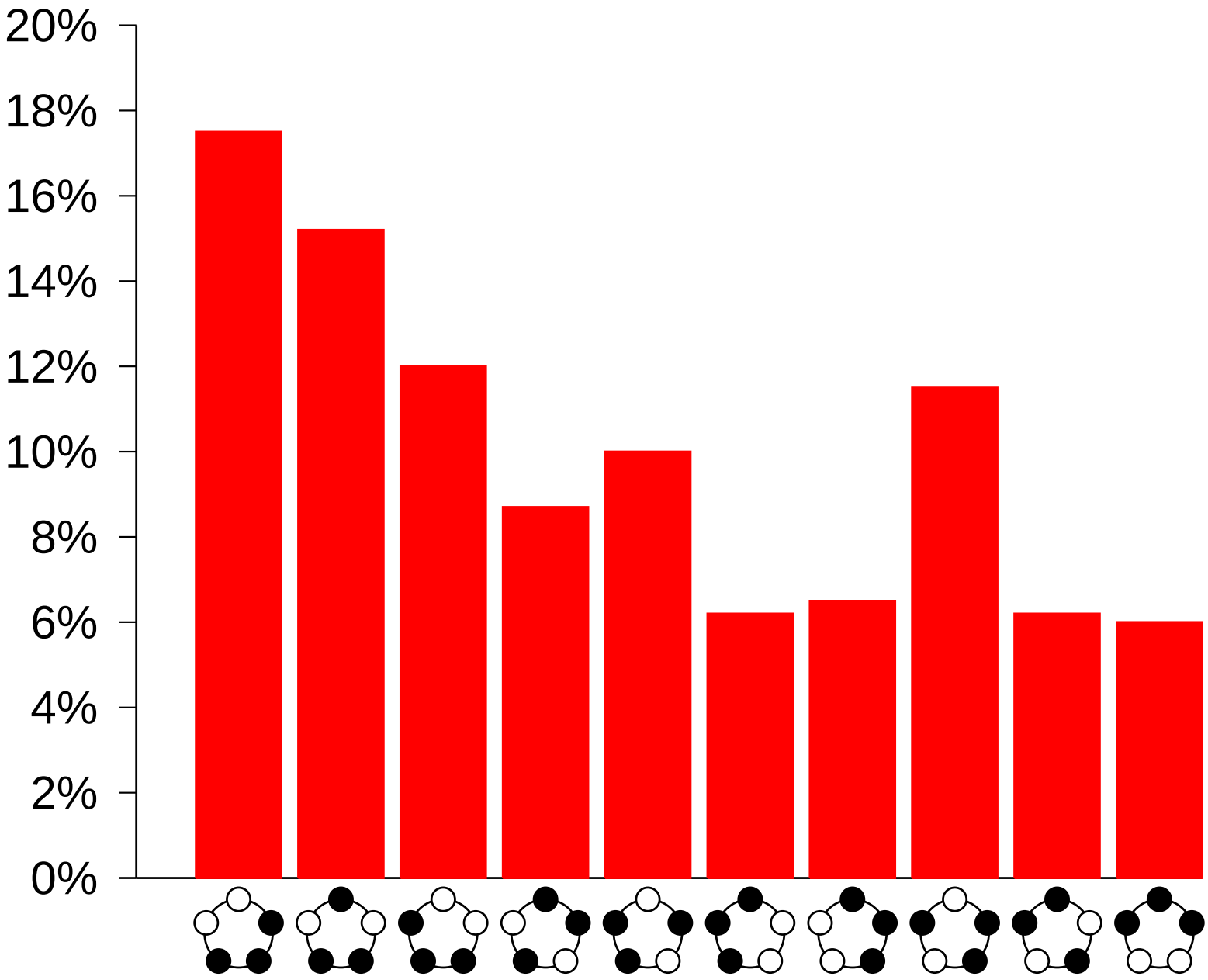
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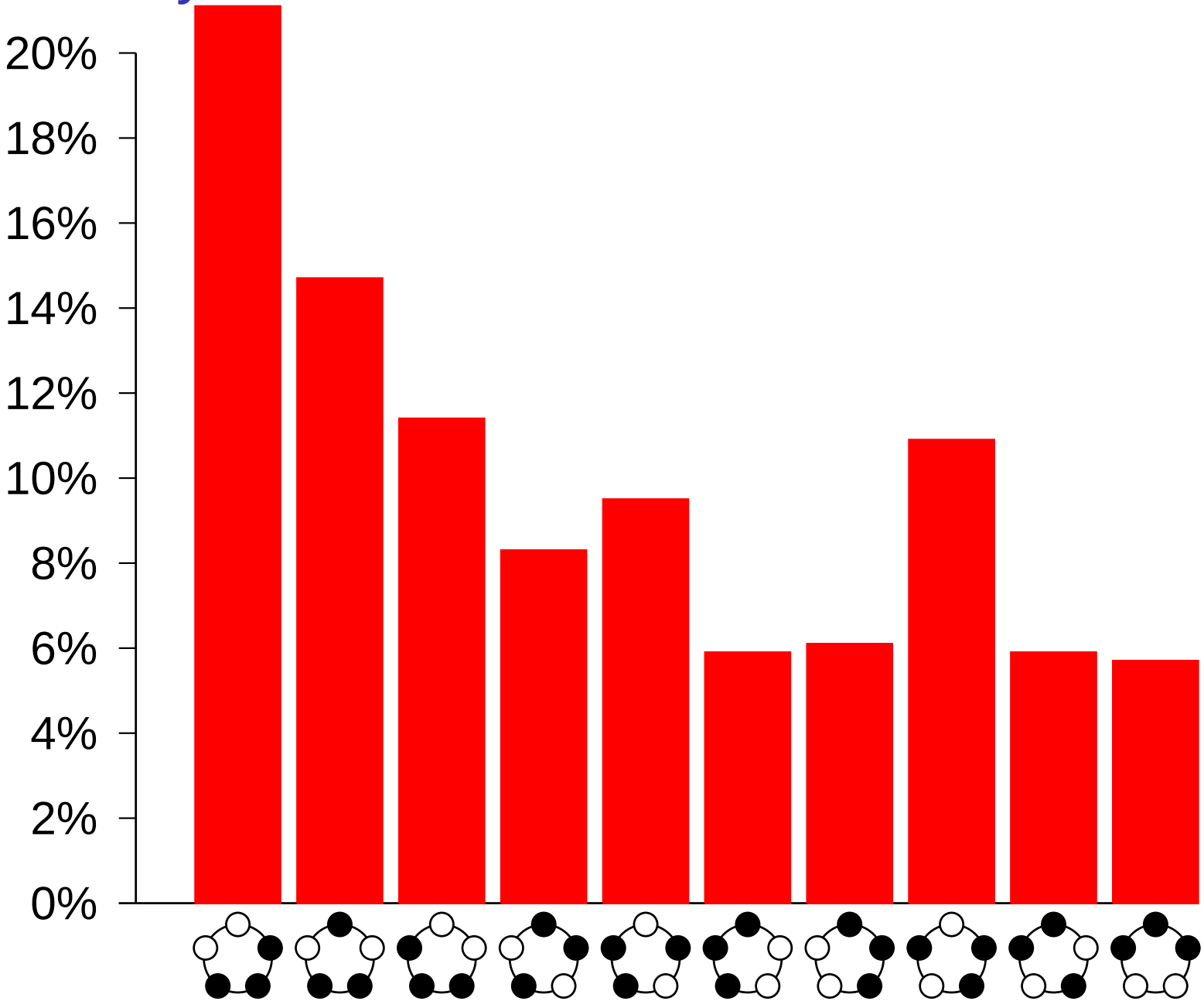
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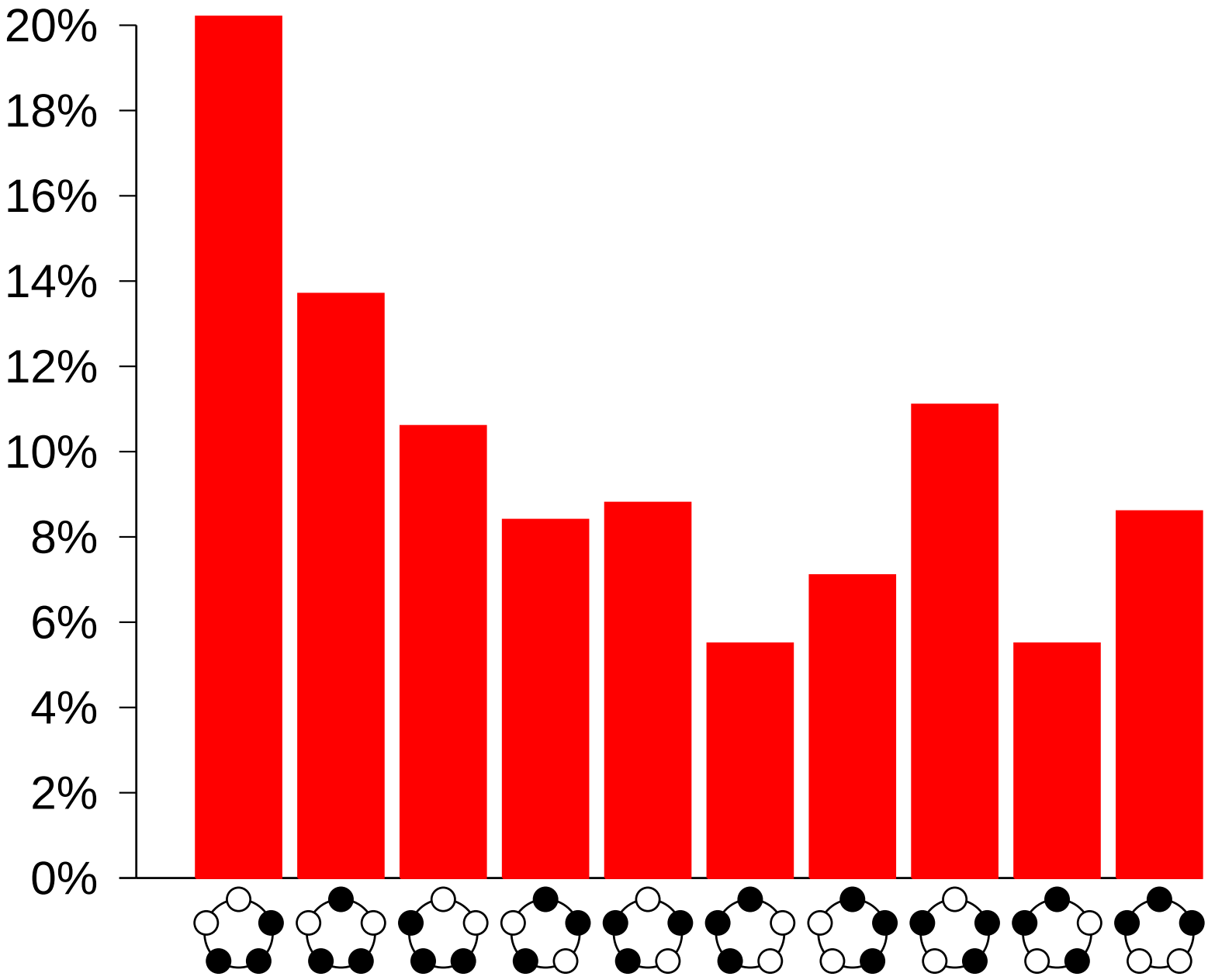
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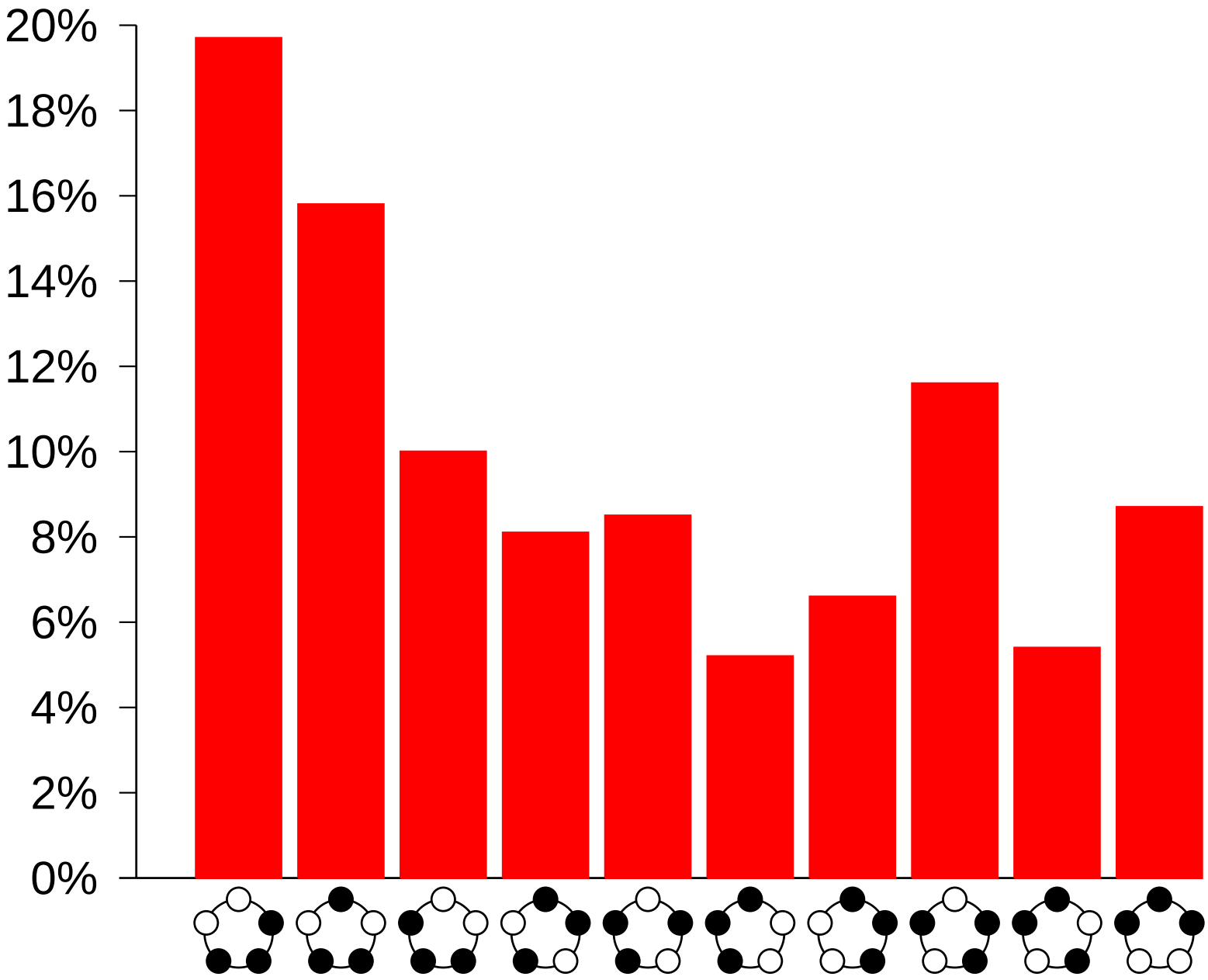
Stationary distribution



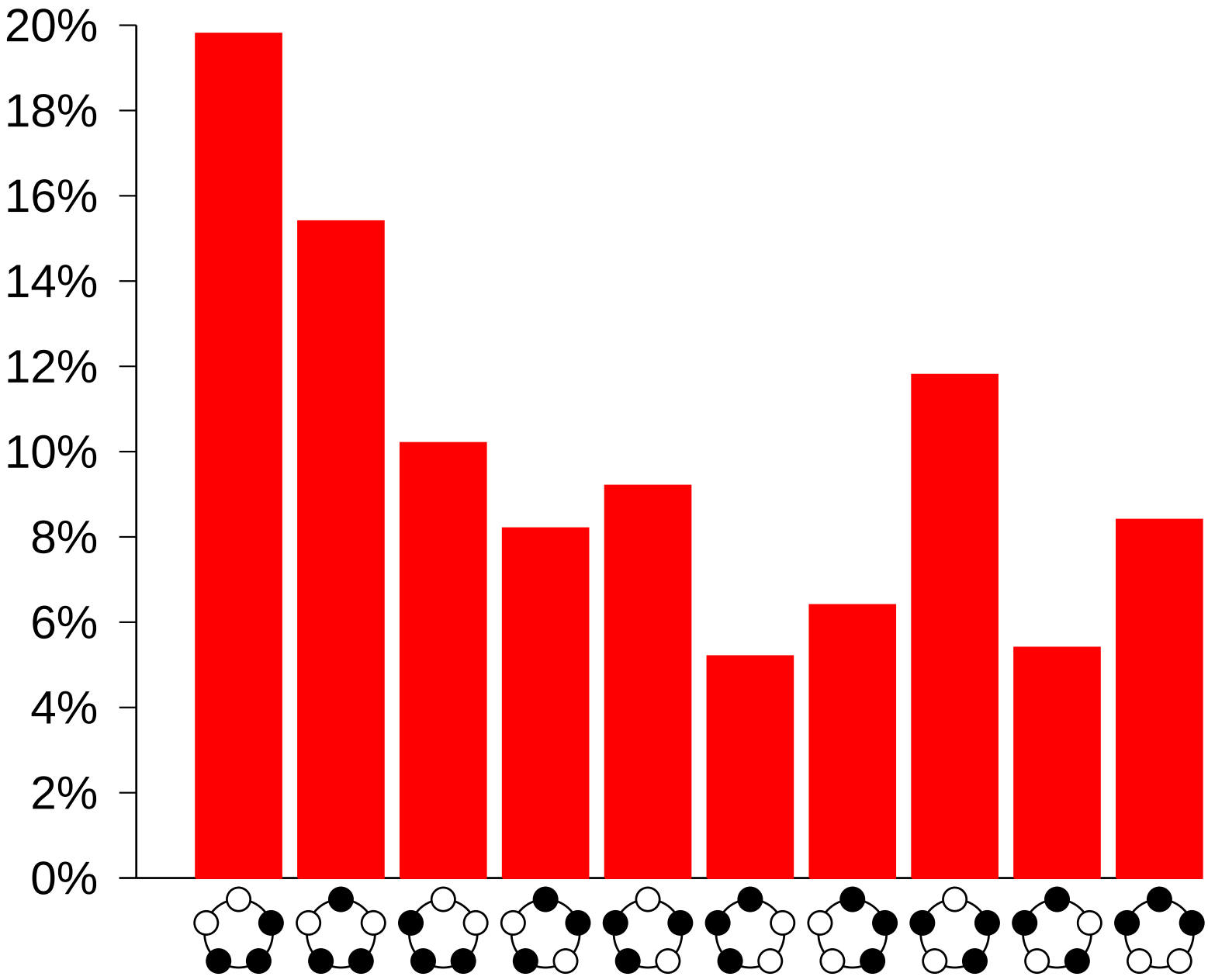
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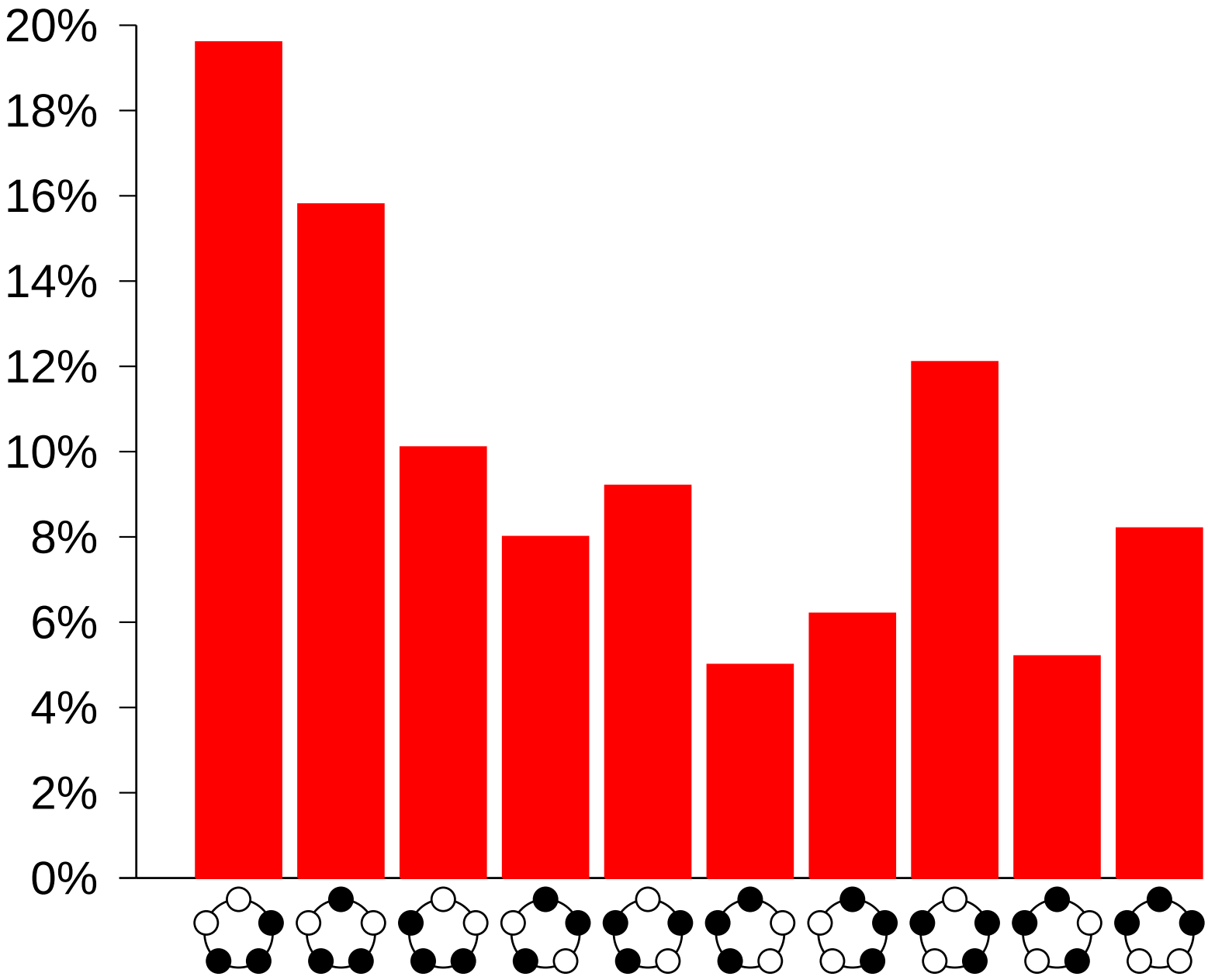
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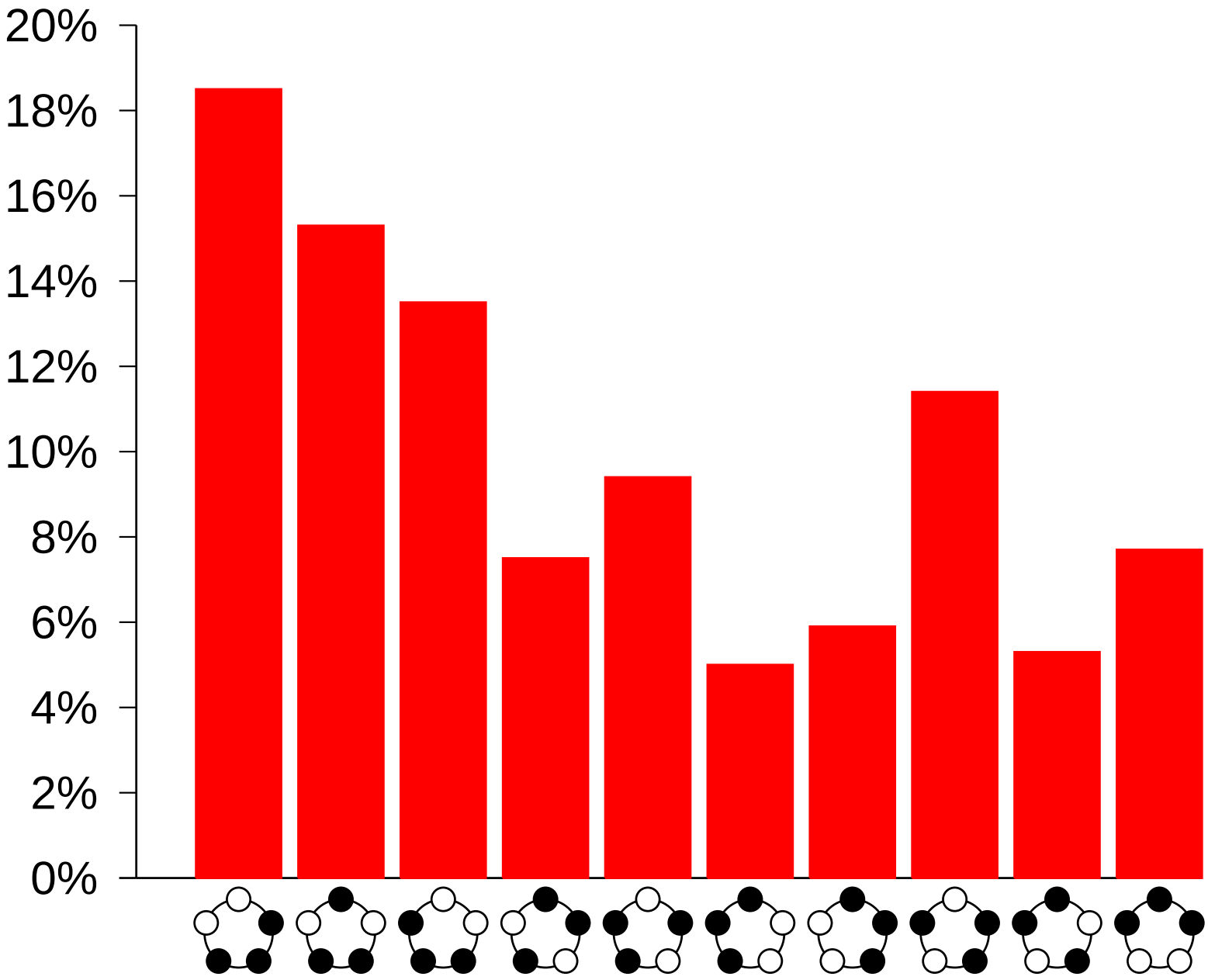
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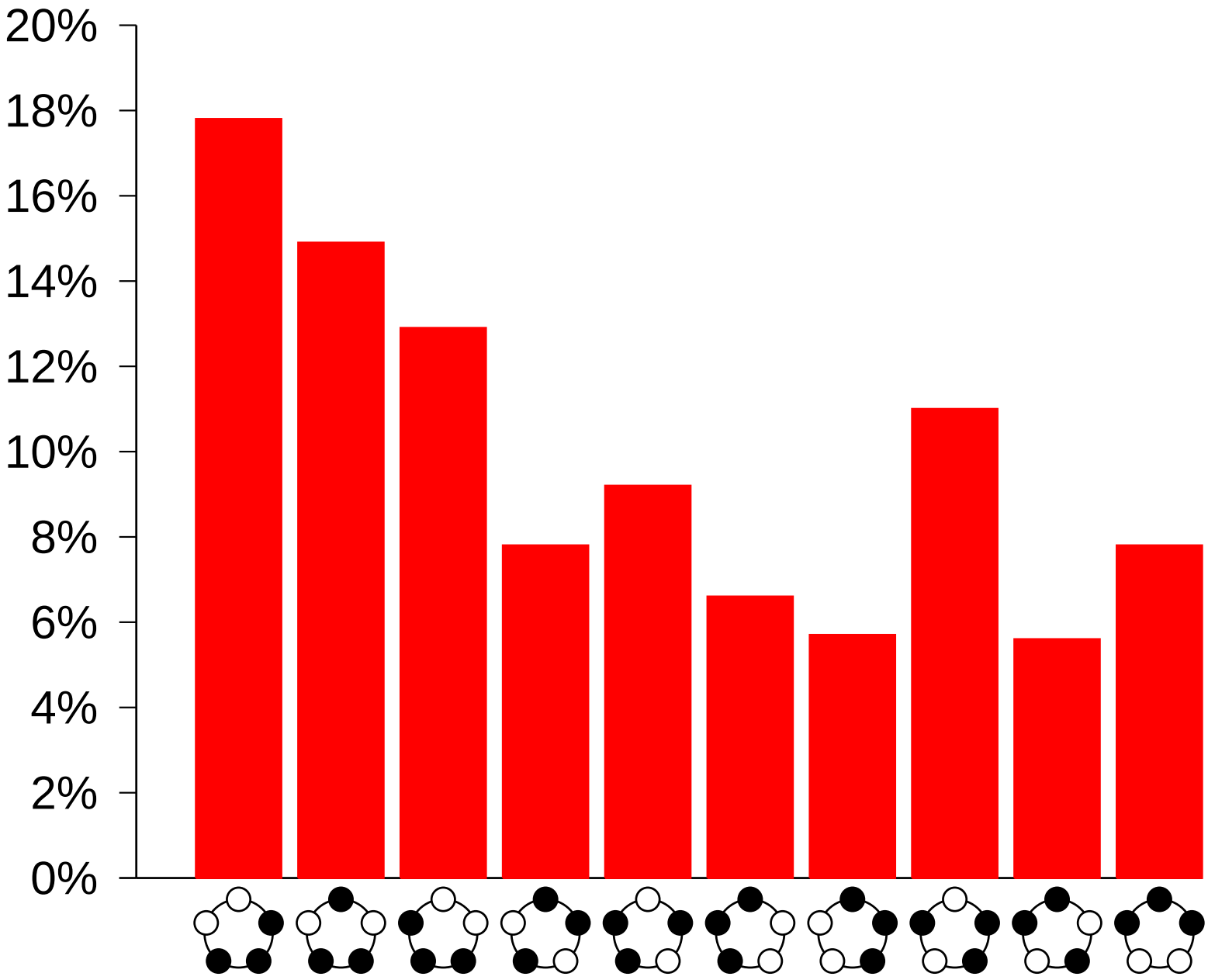
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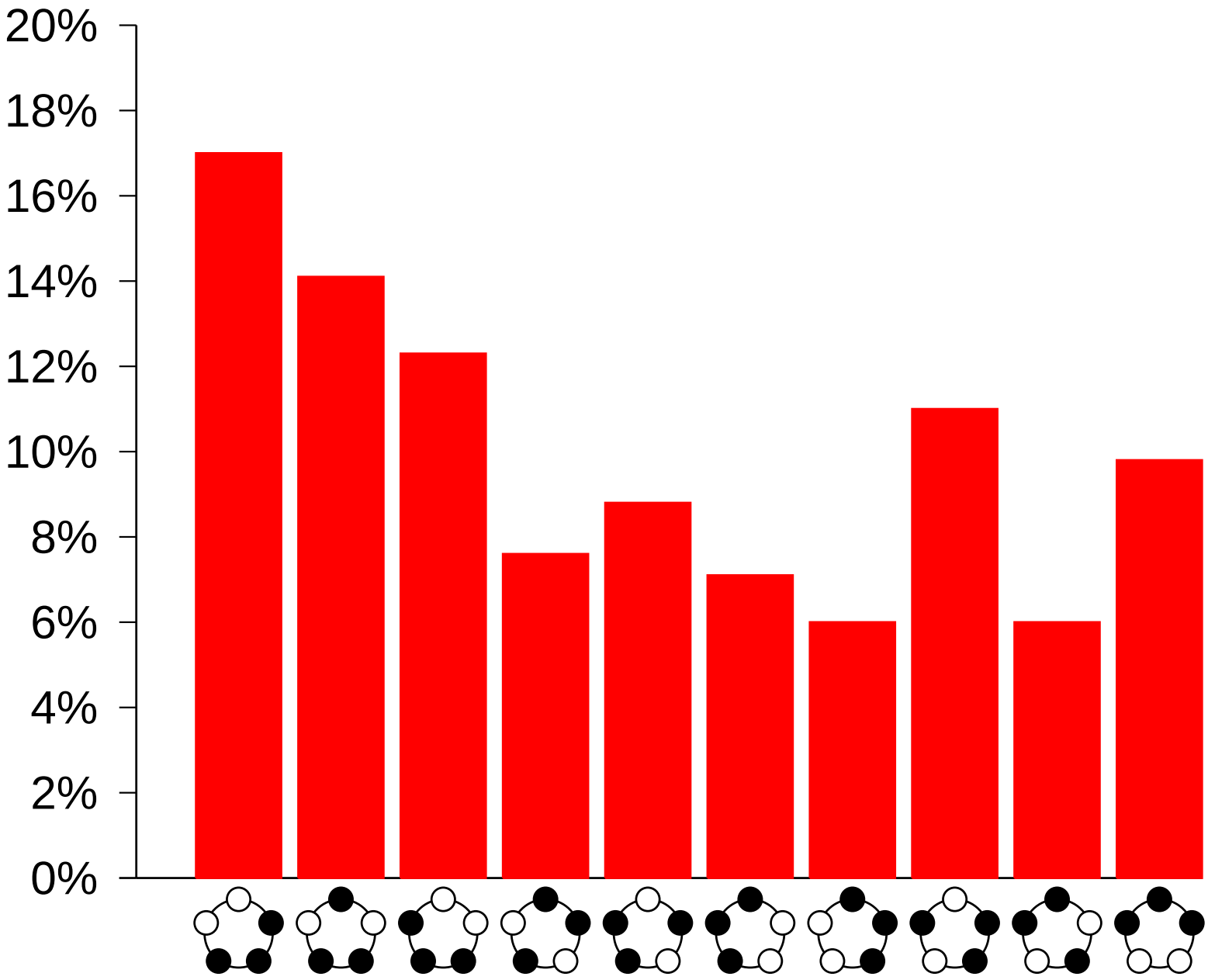
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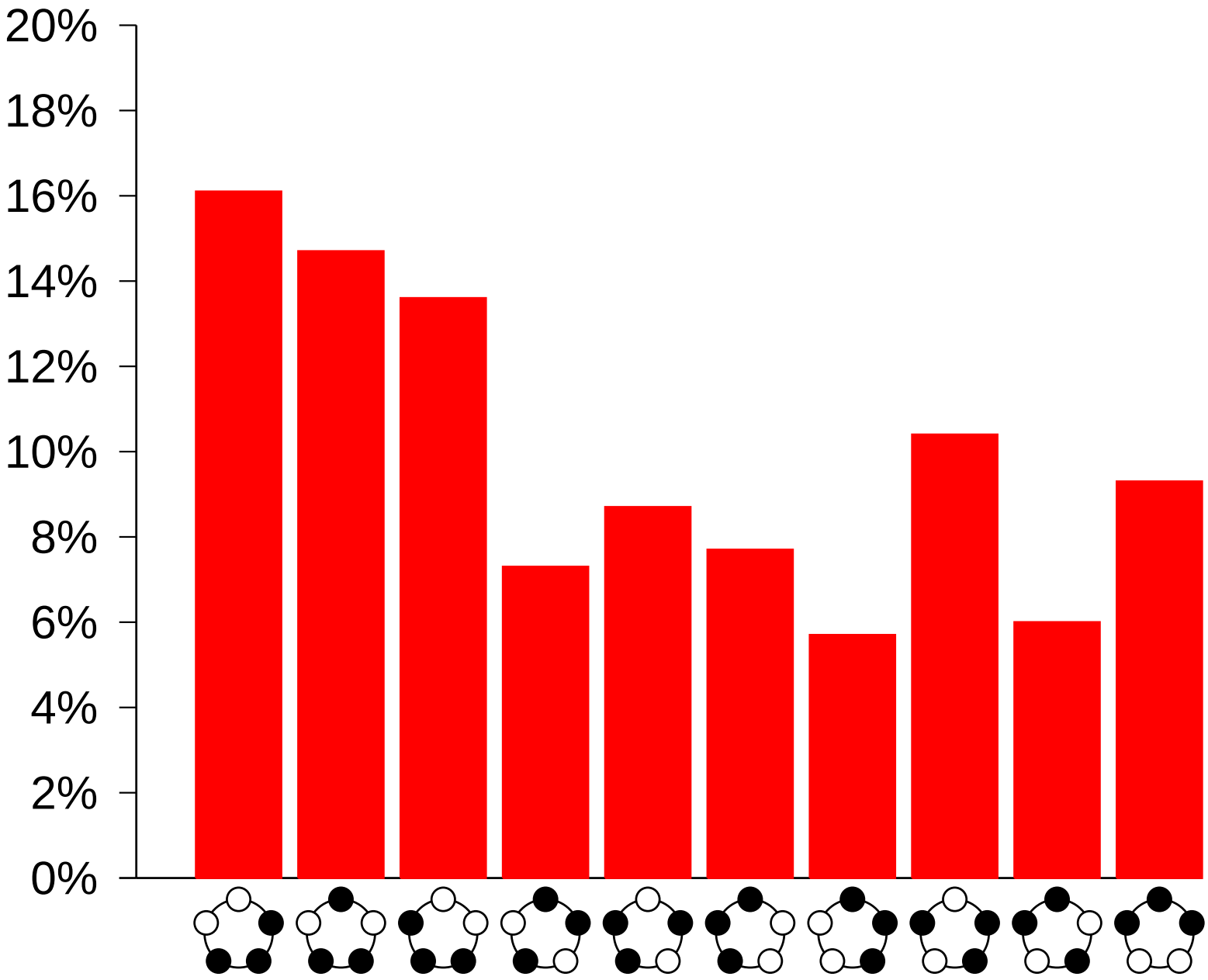
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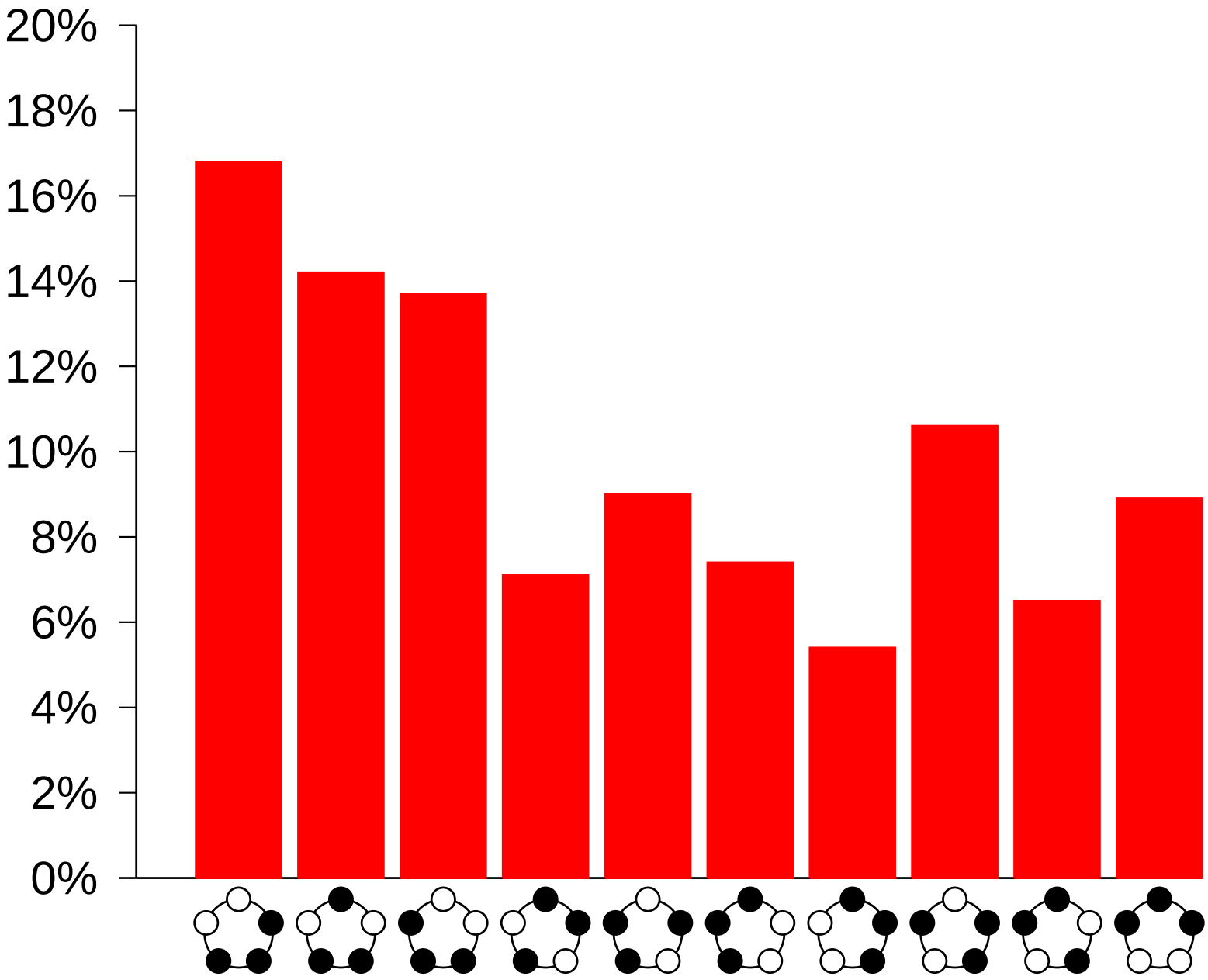
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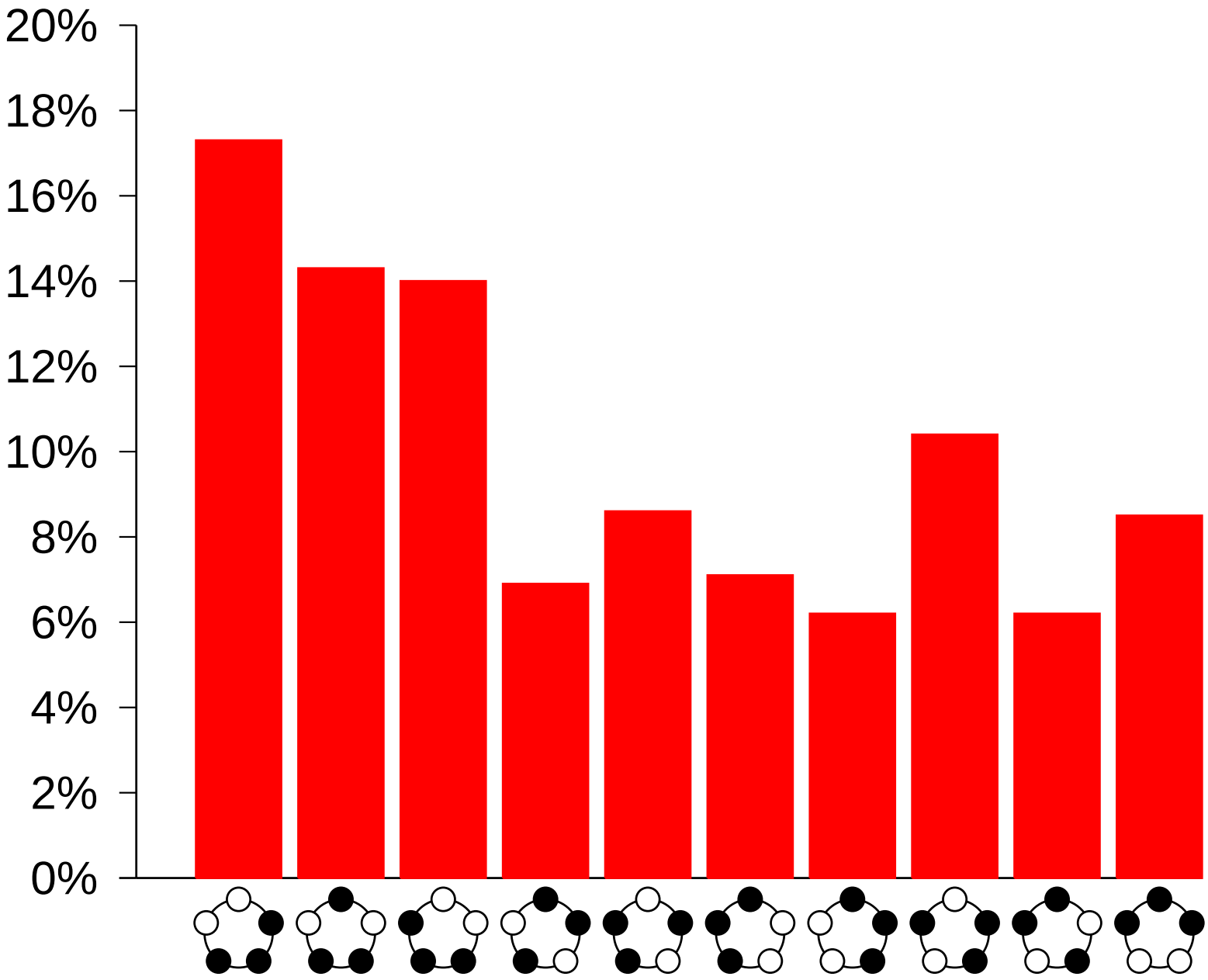
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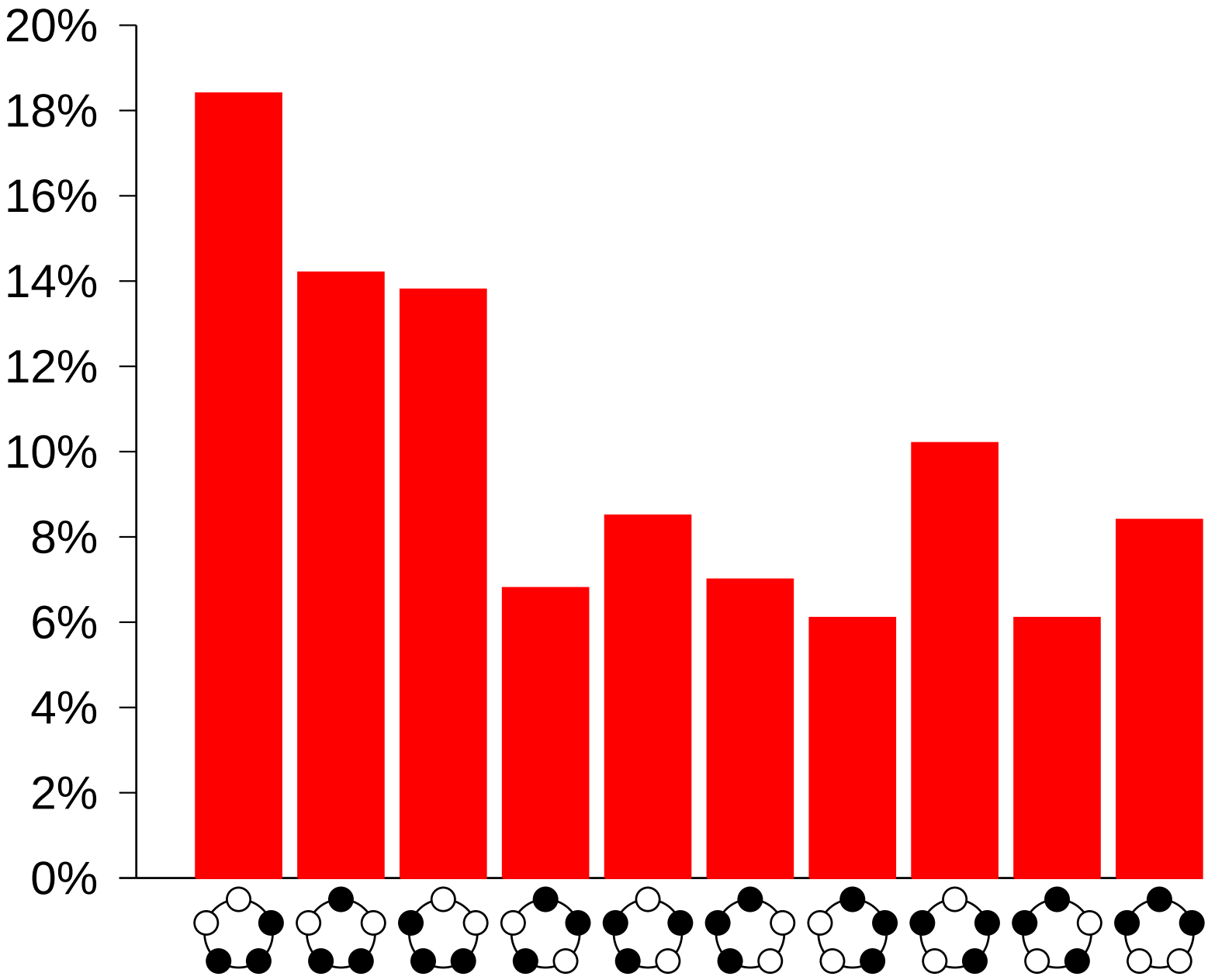
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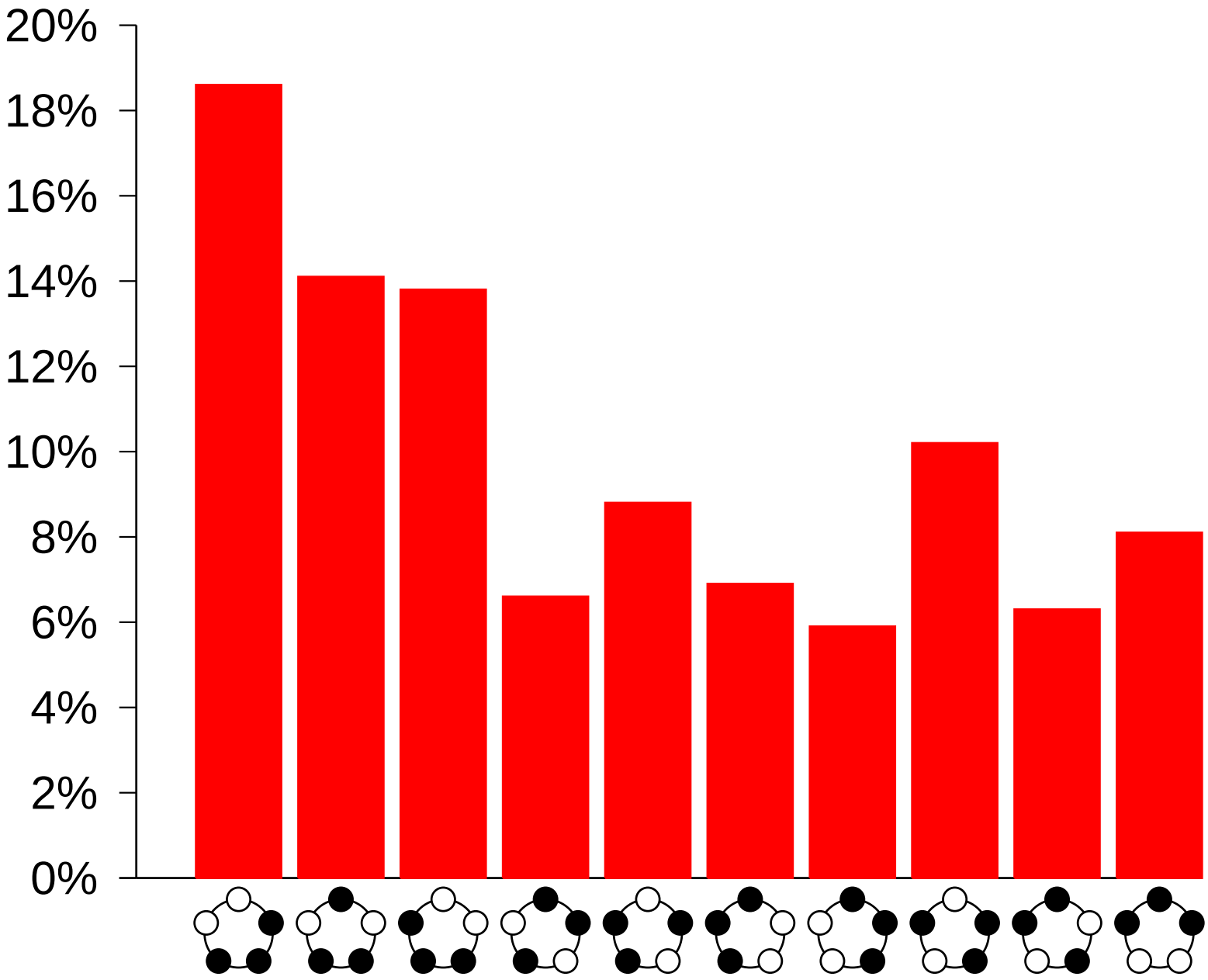
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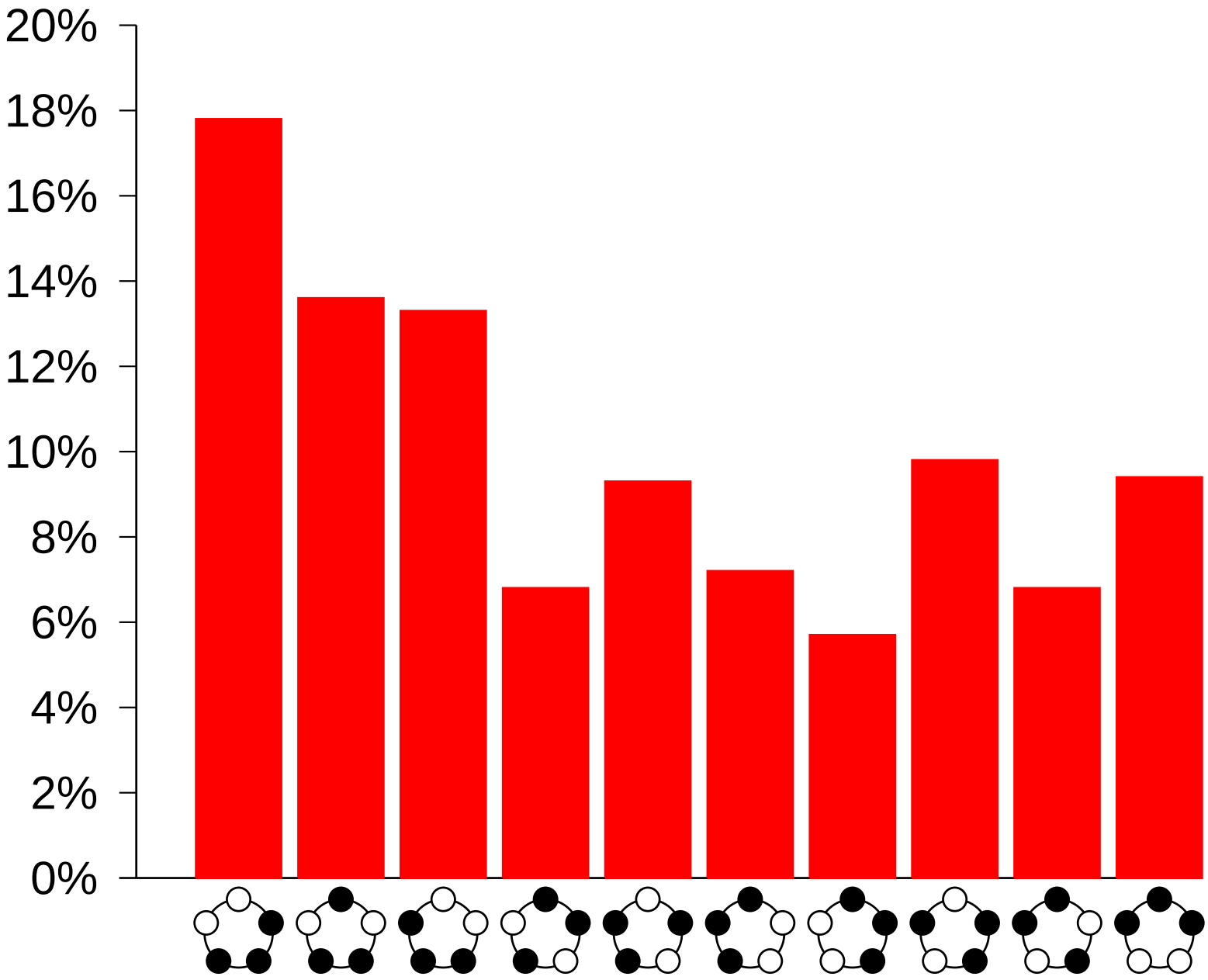
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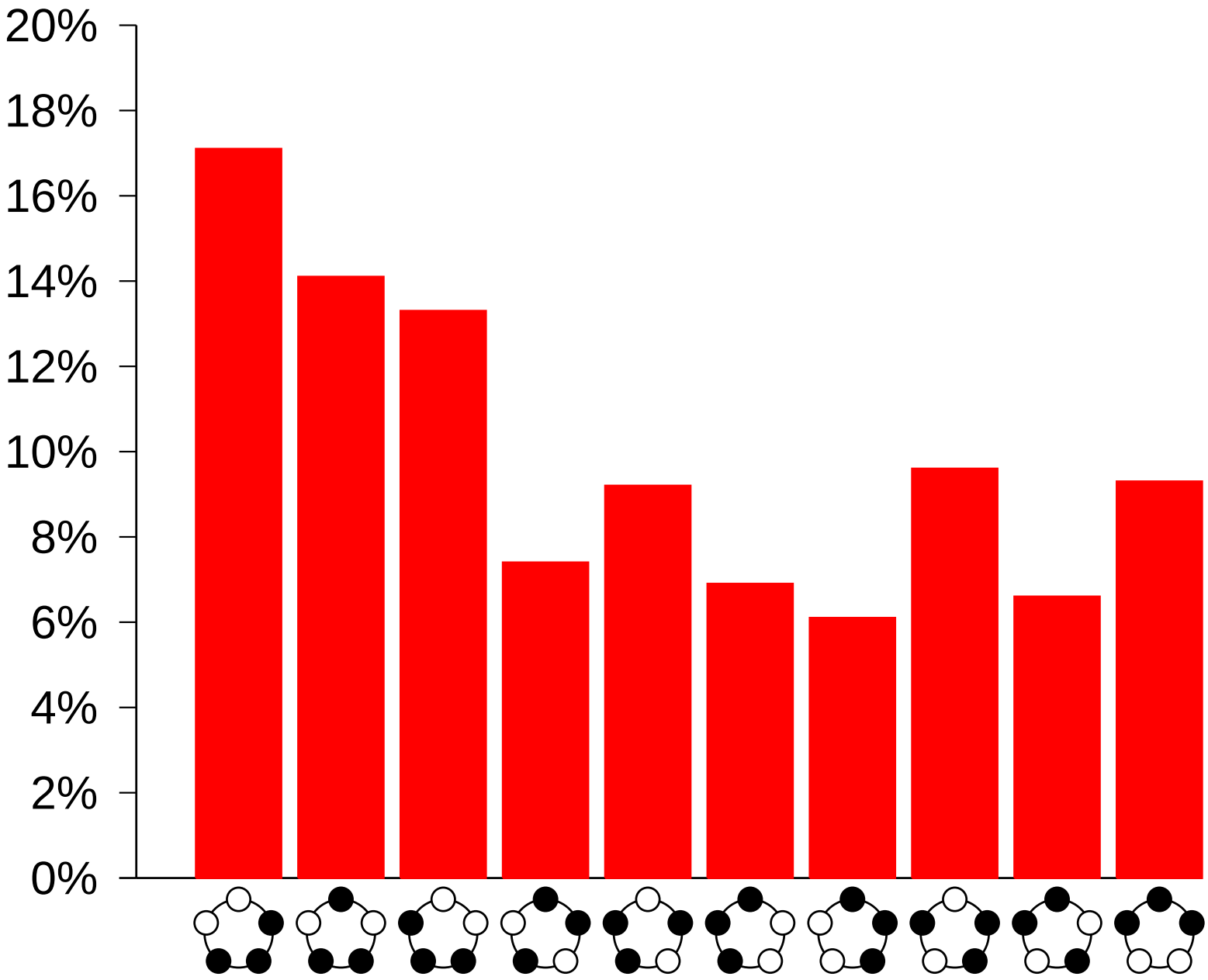
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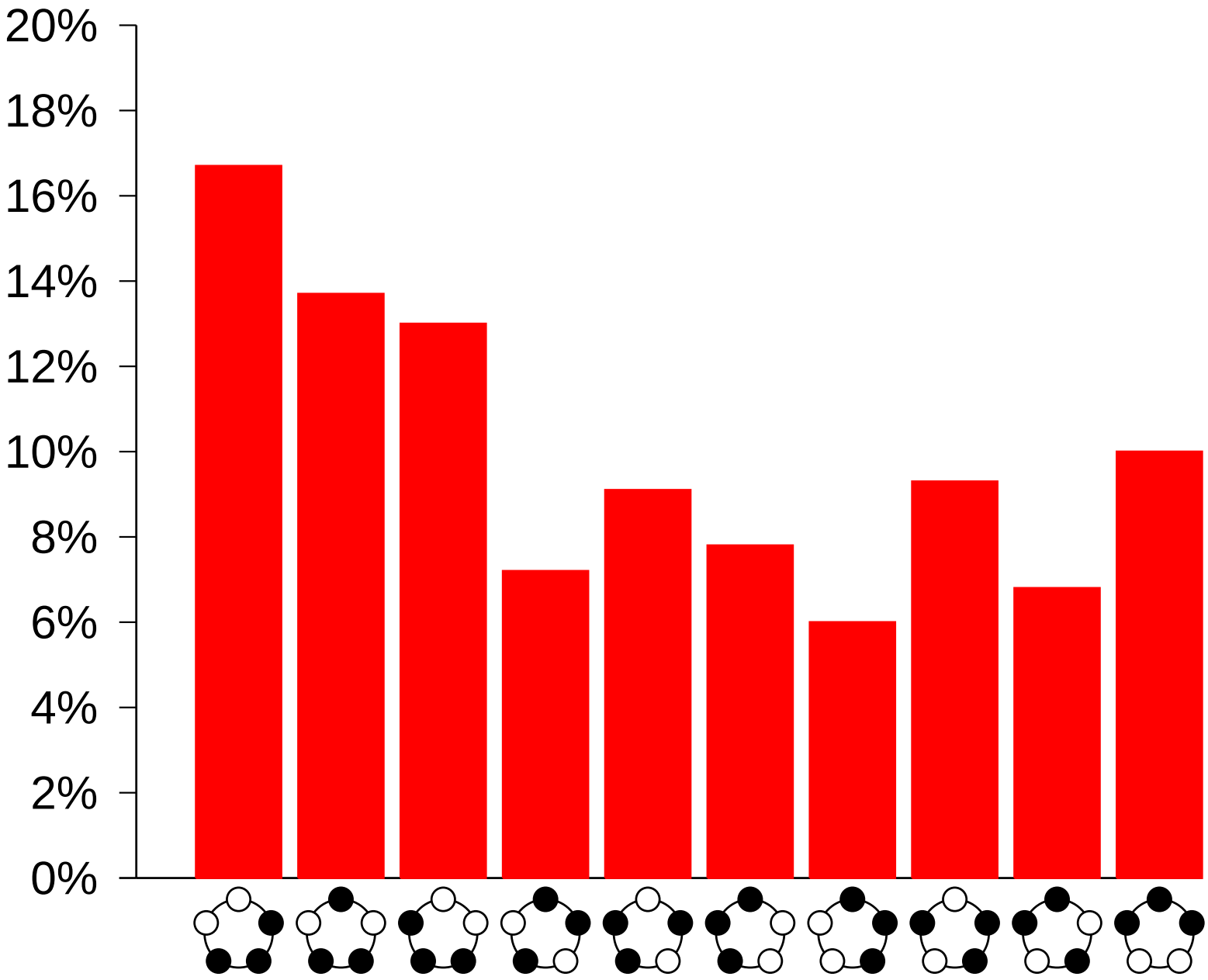
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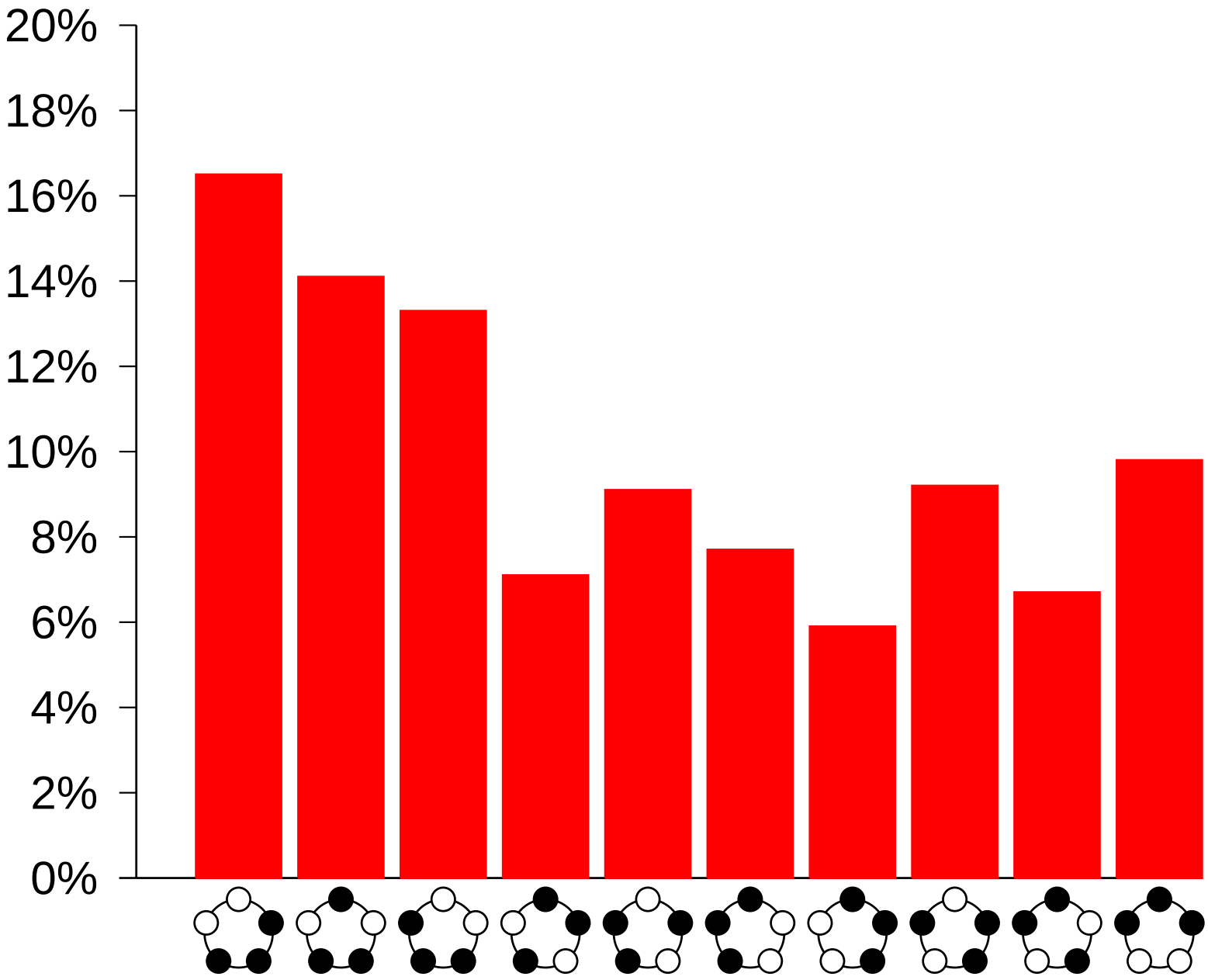
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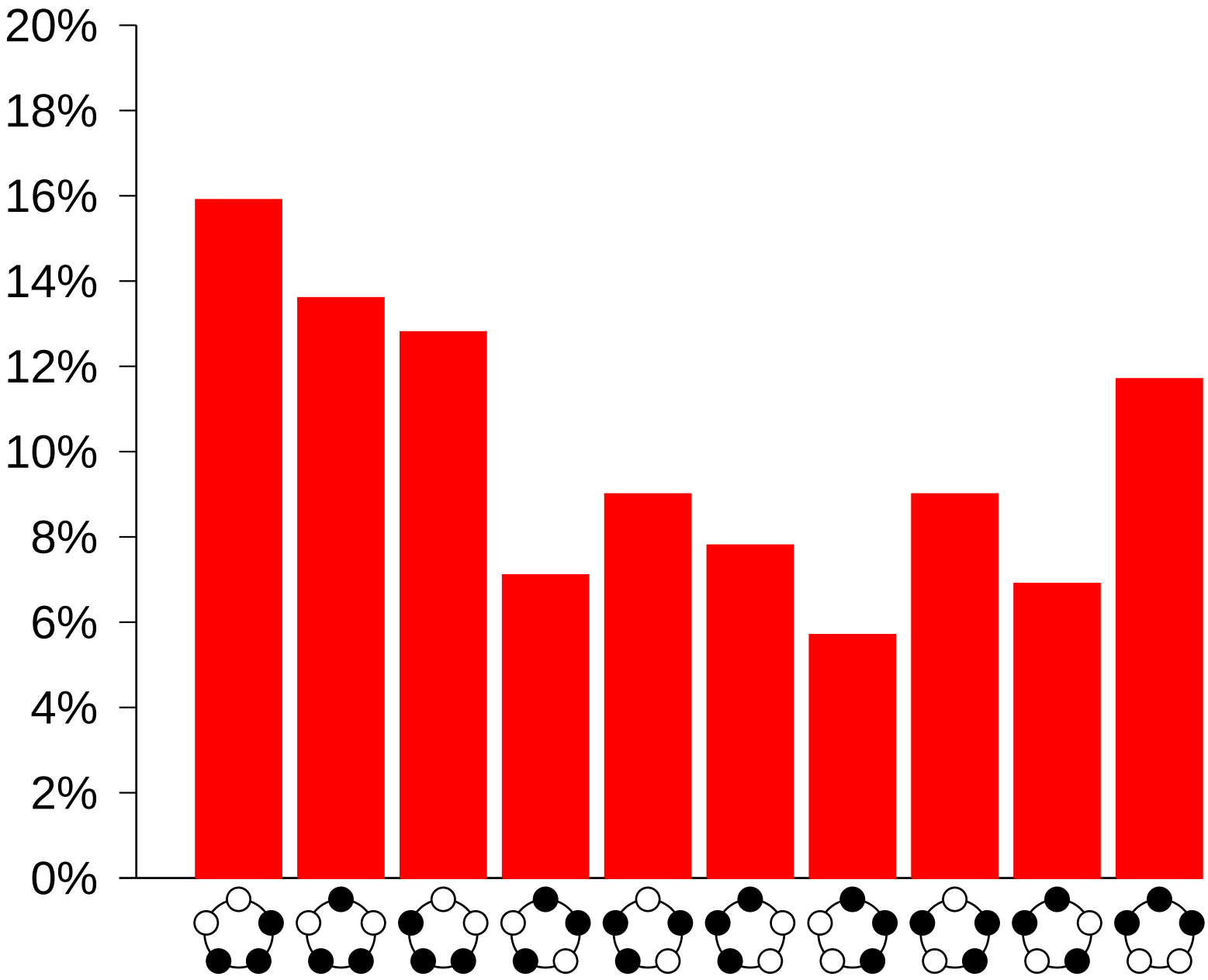
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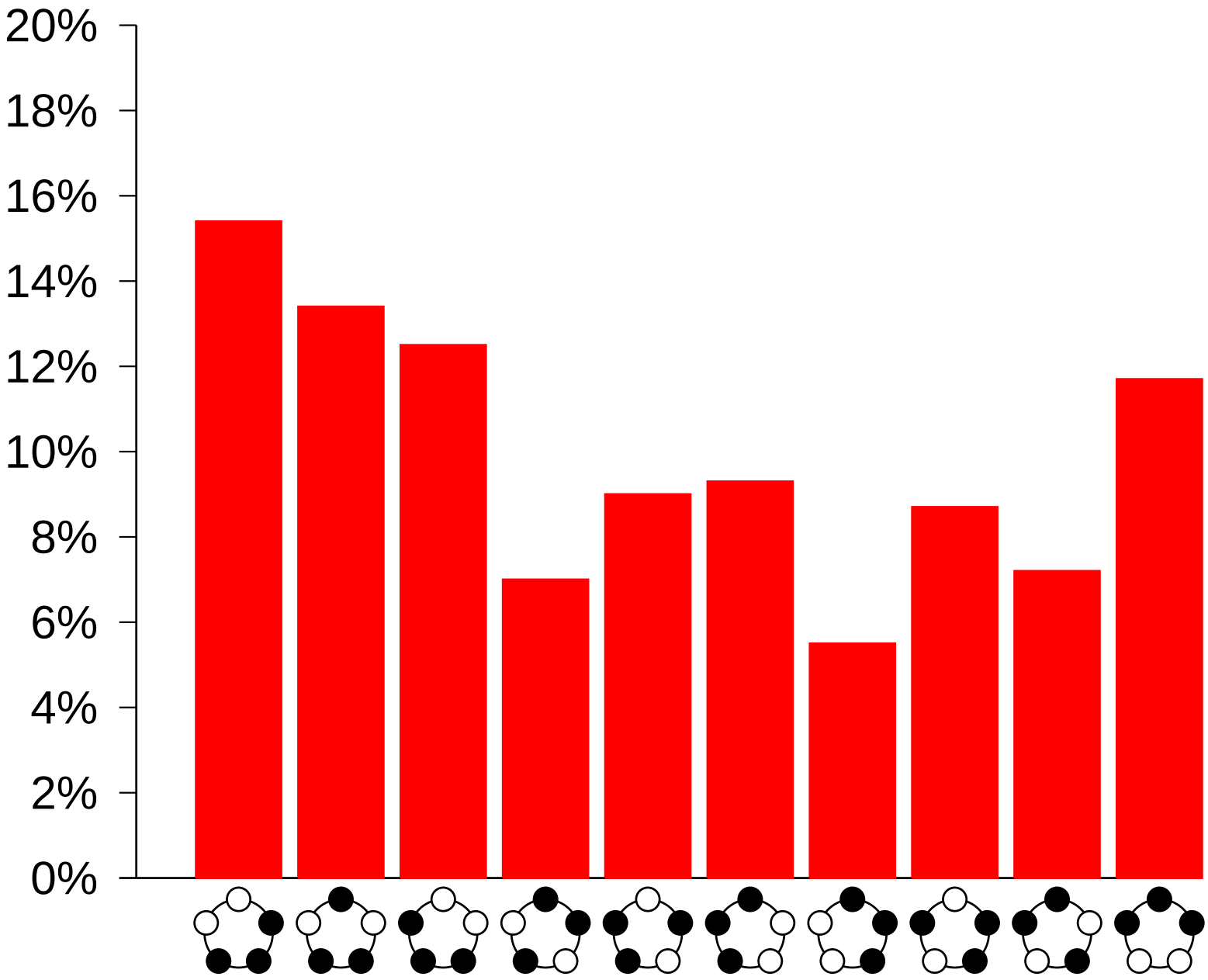
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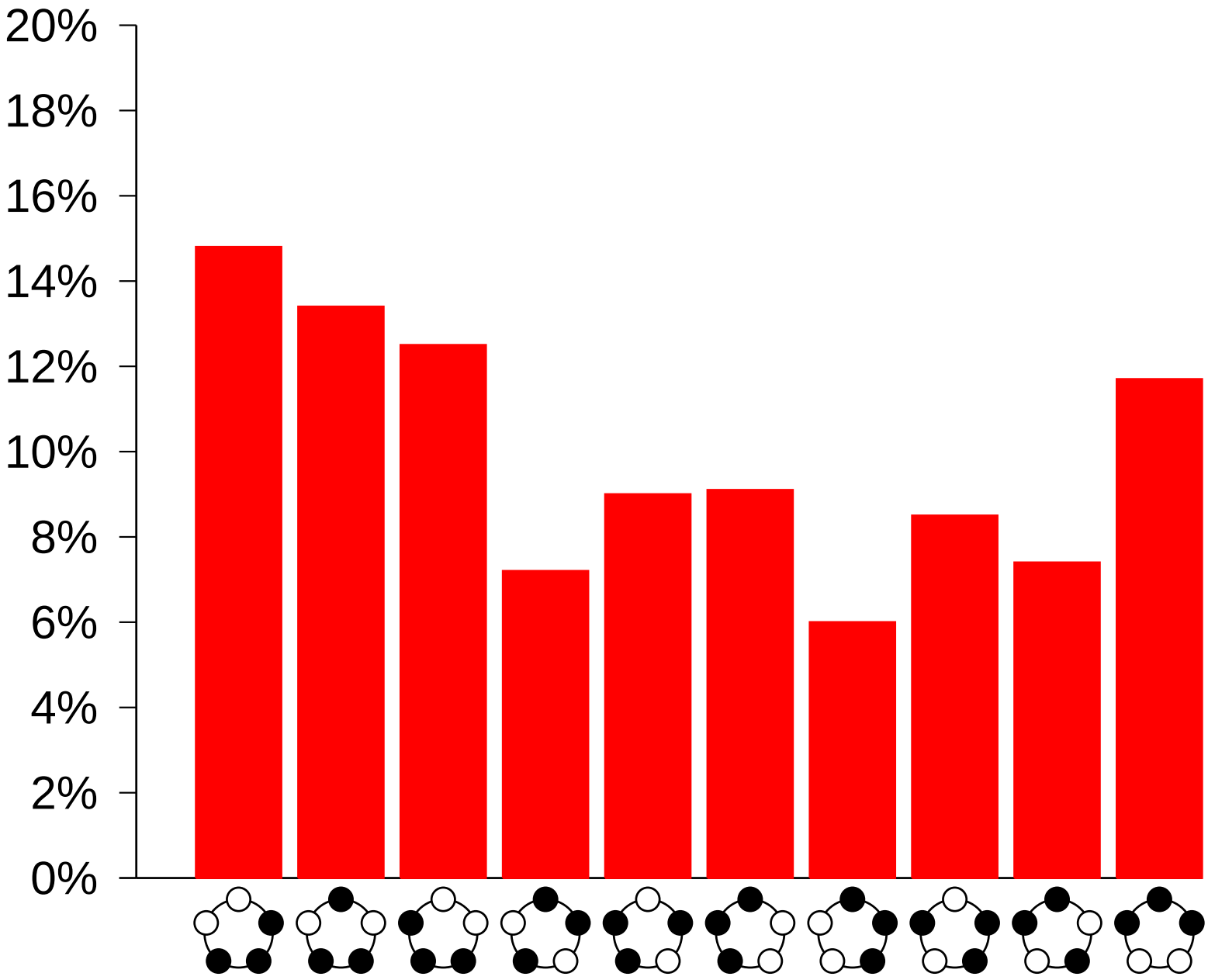
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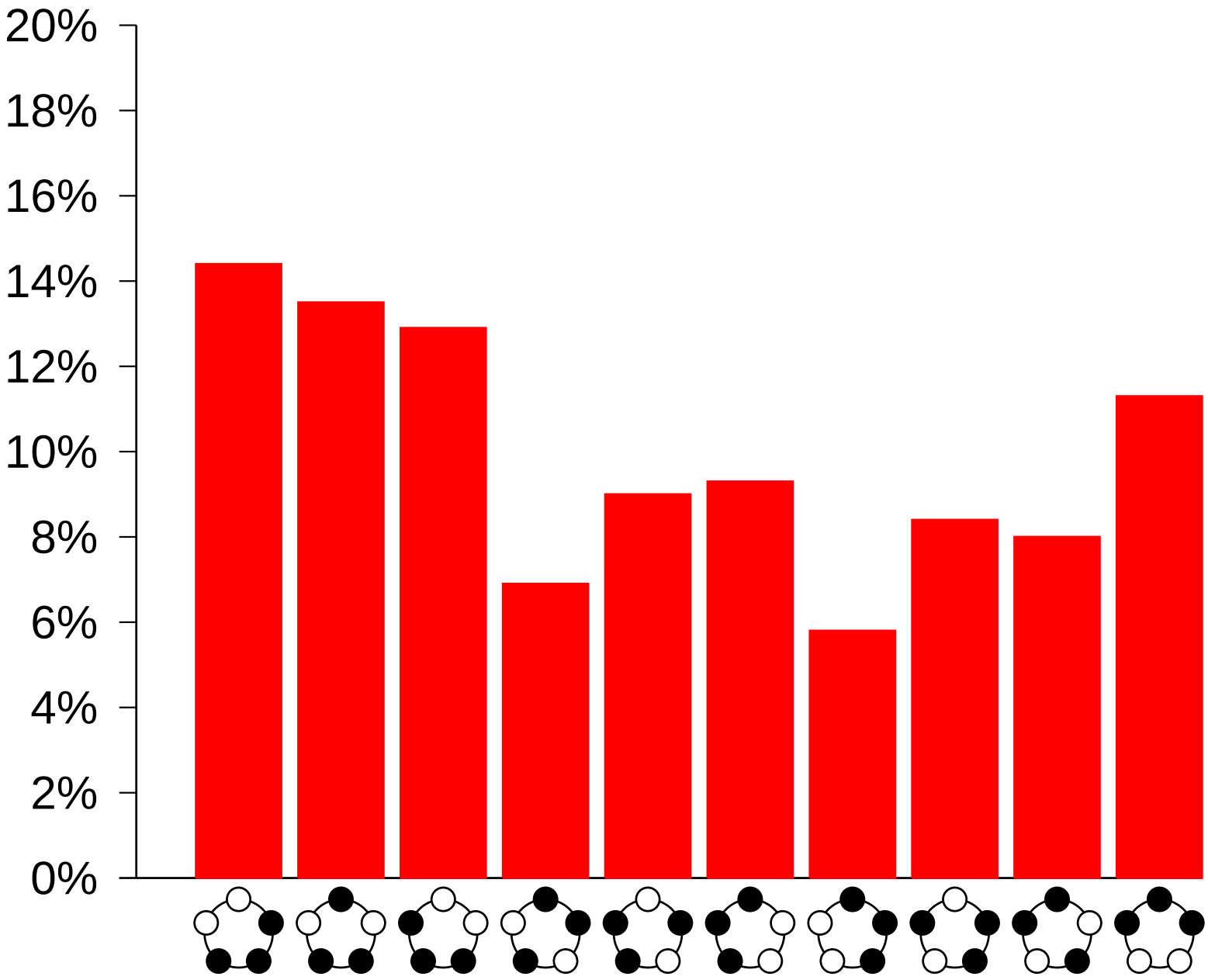
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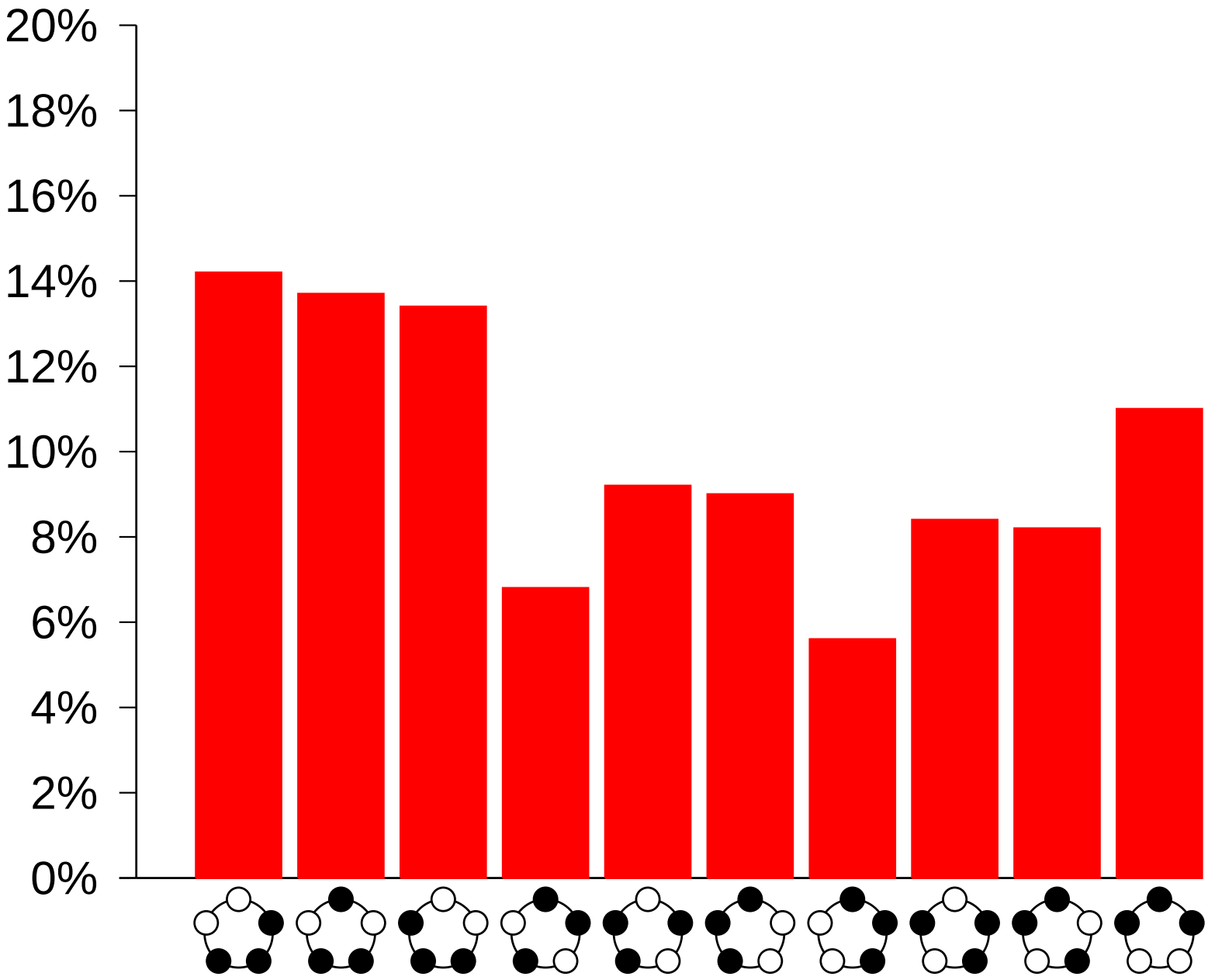
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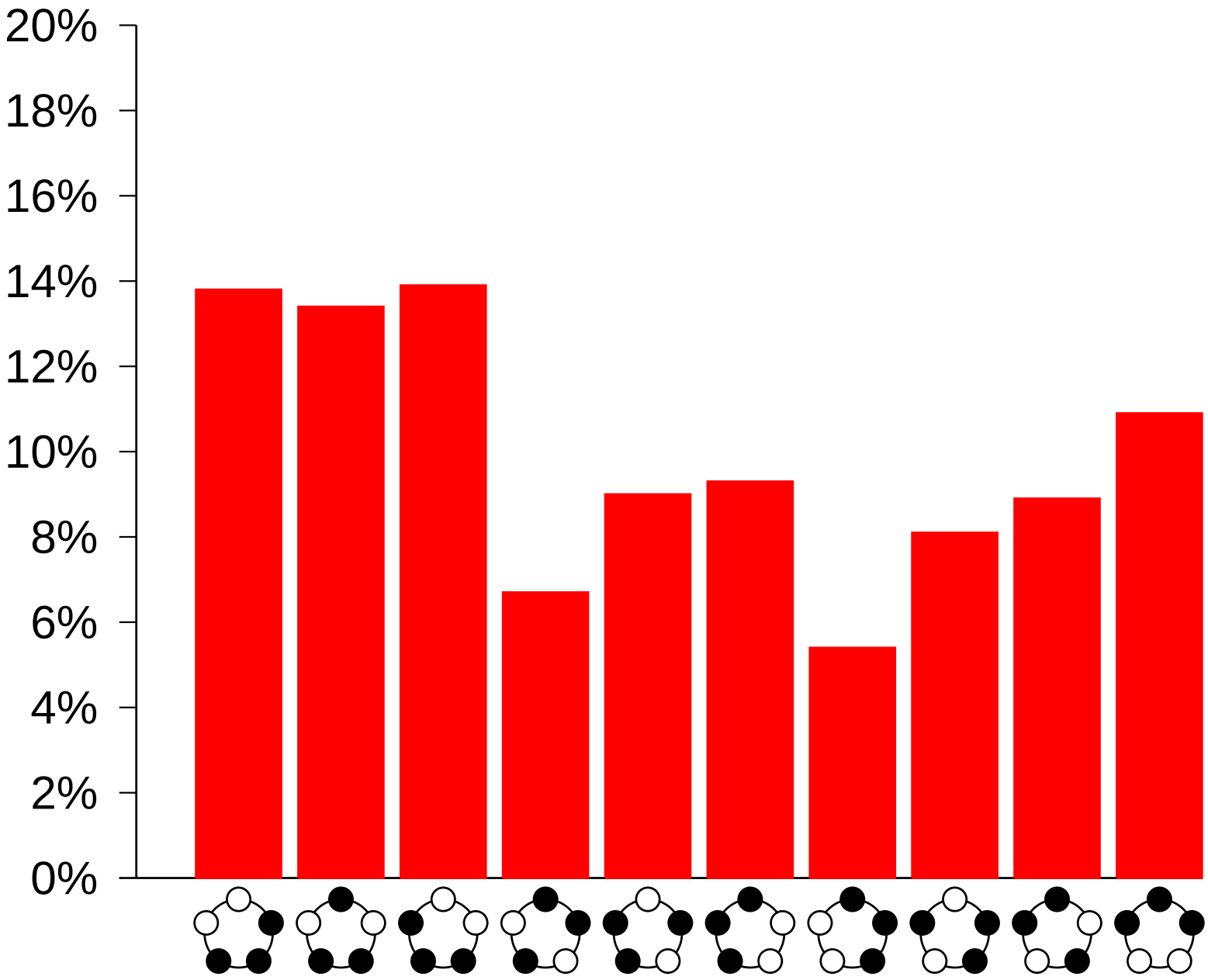
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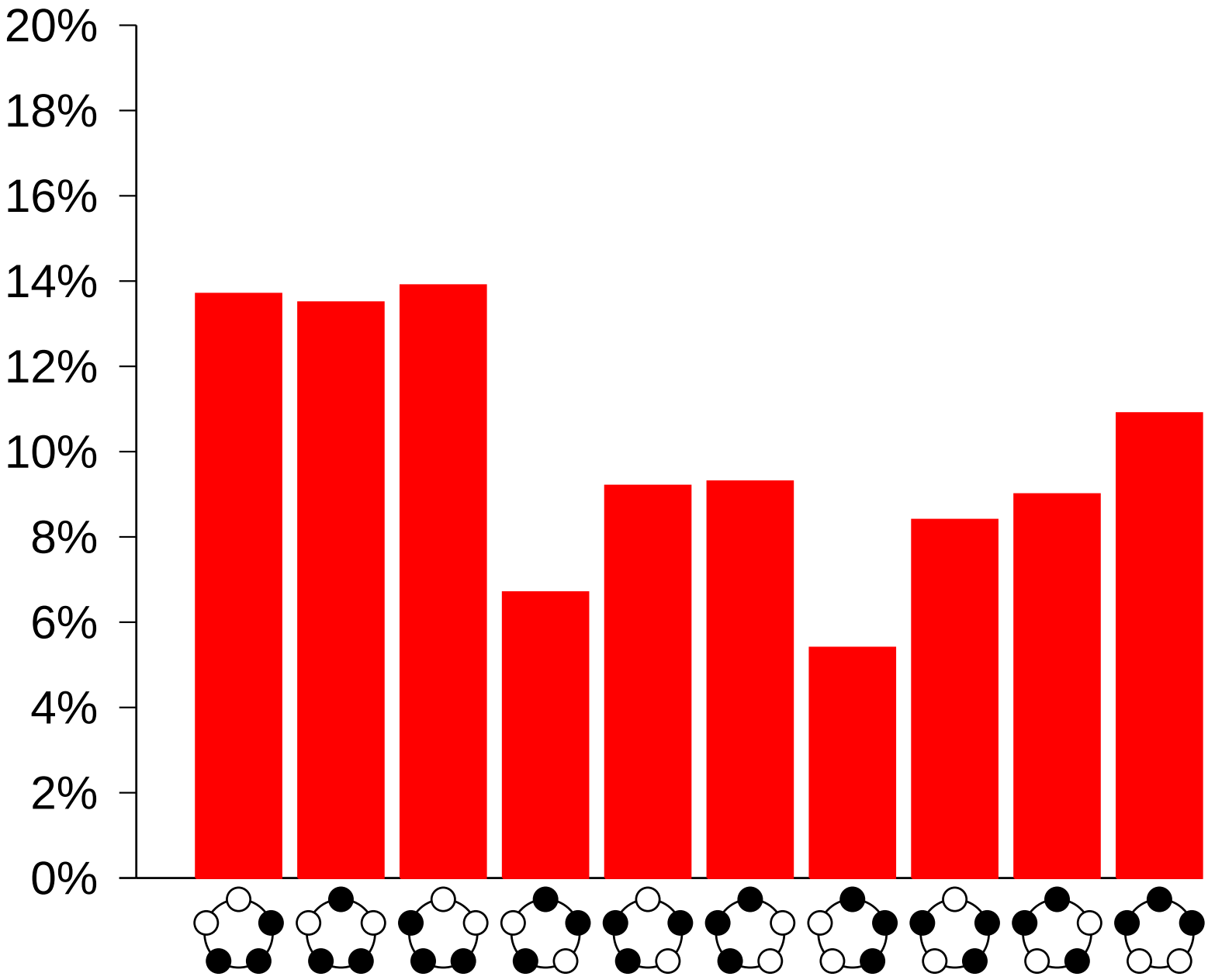
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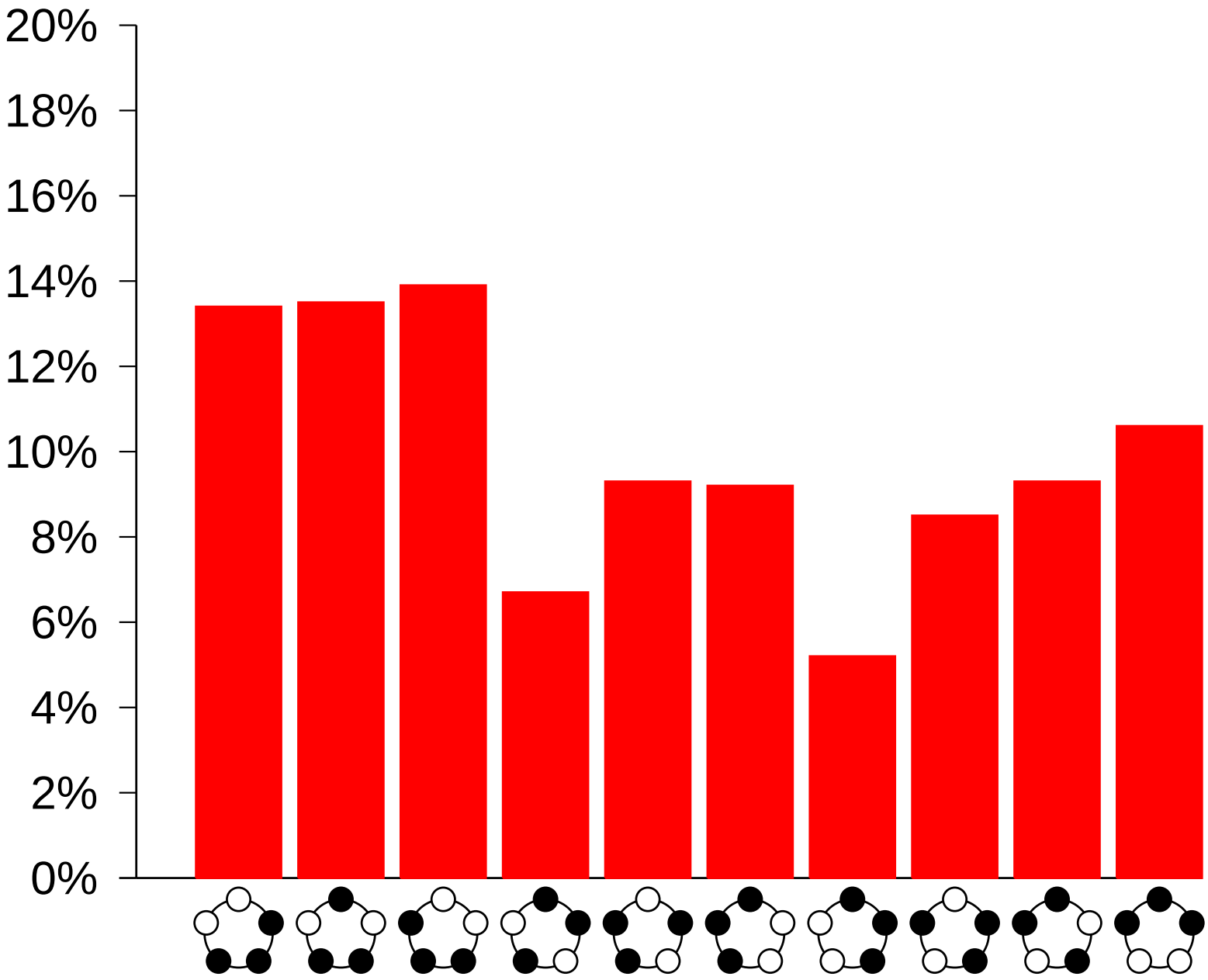
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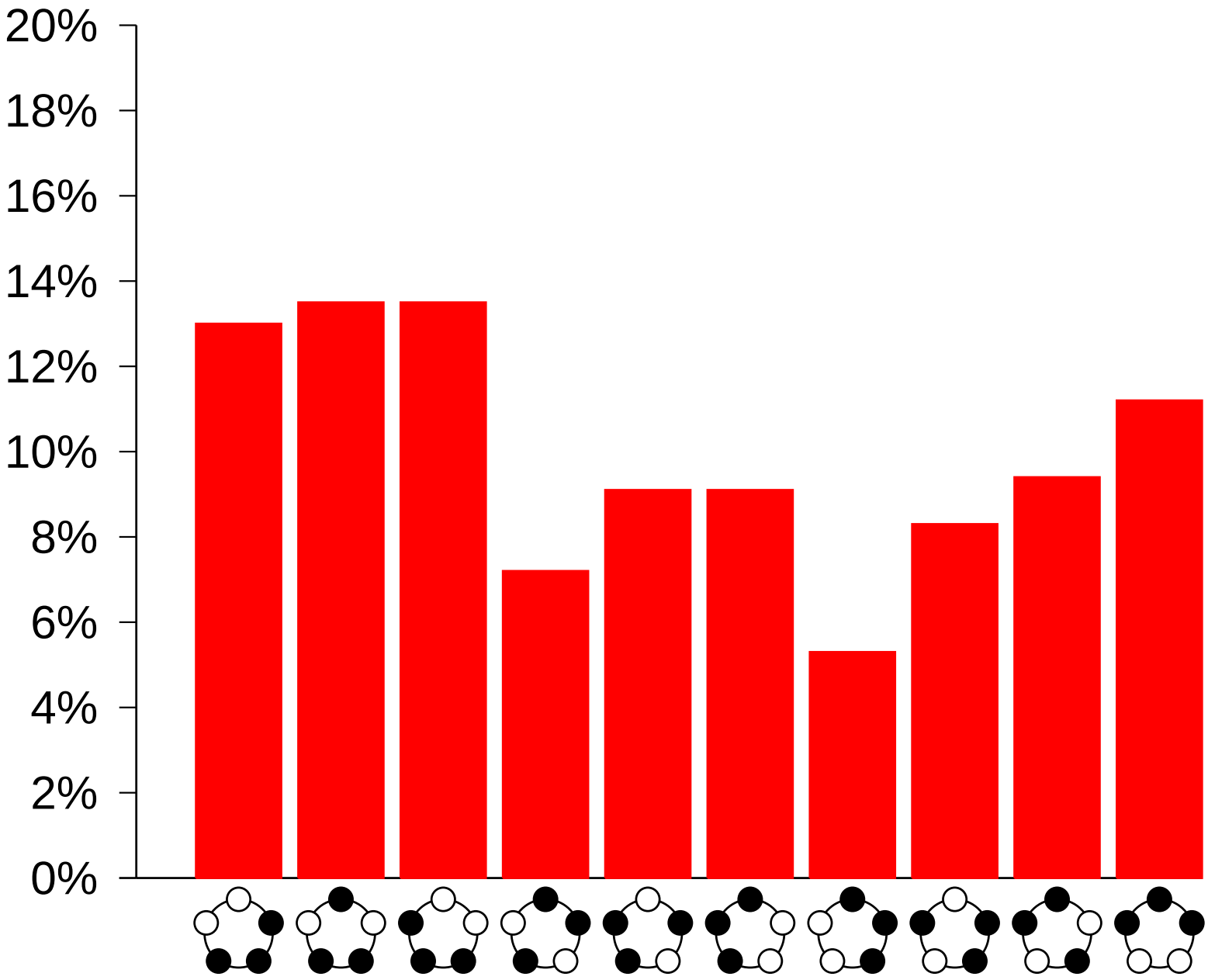
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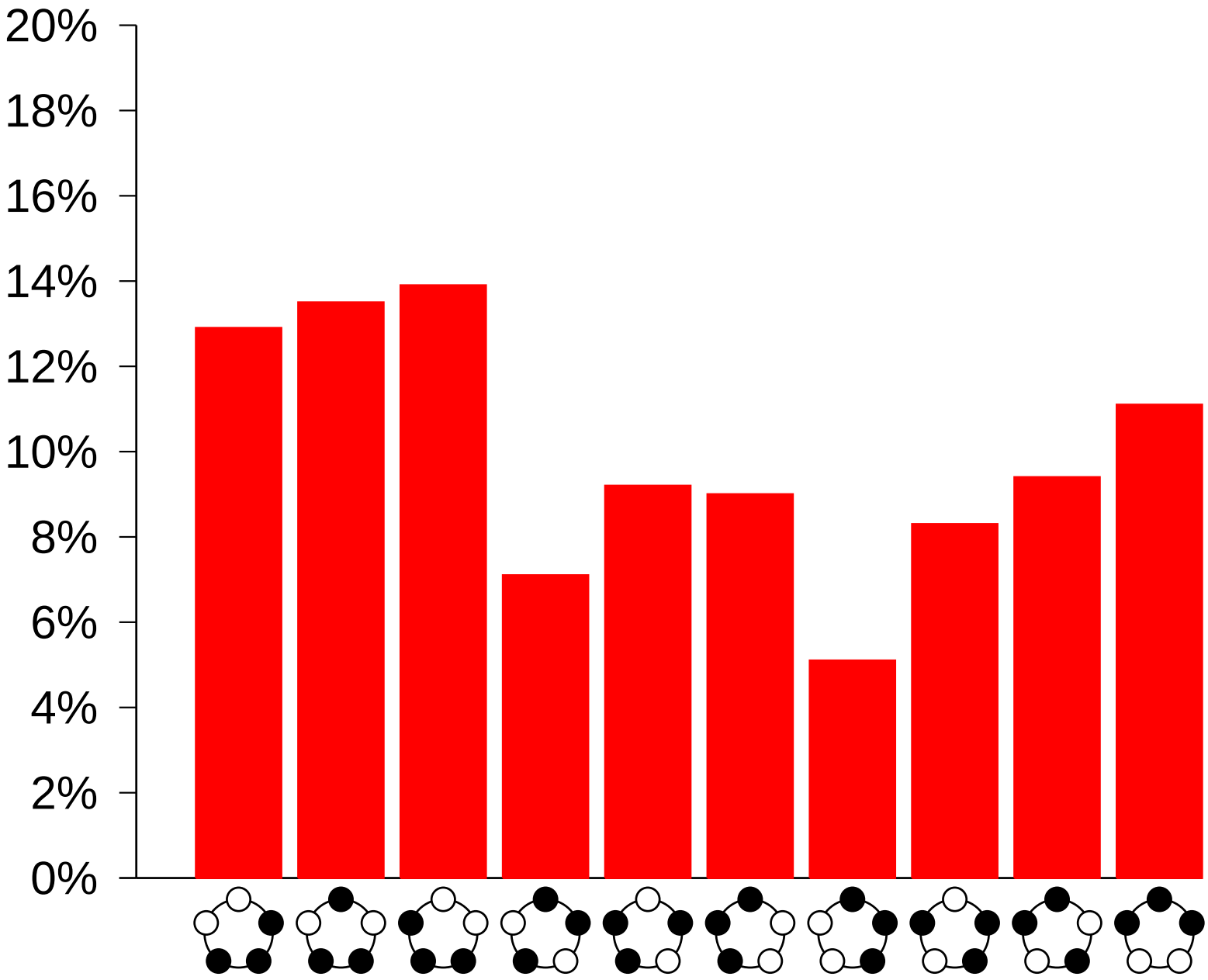
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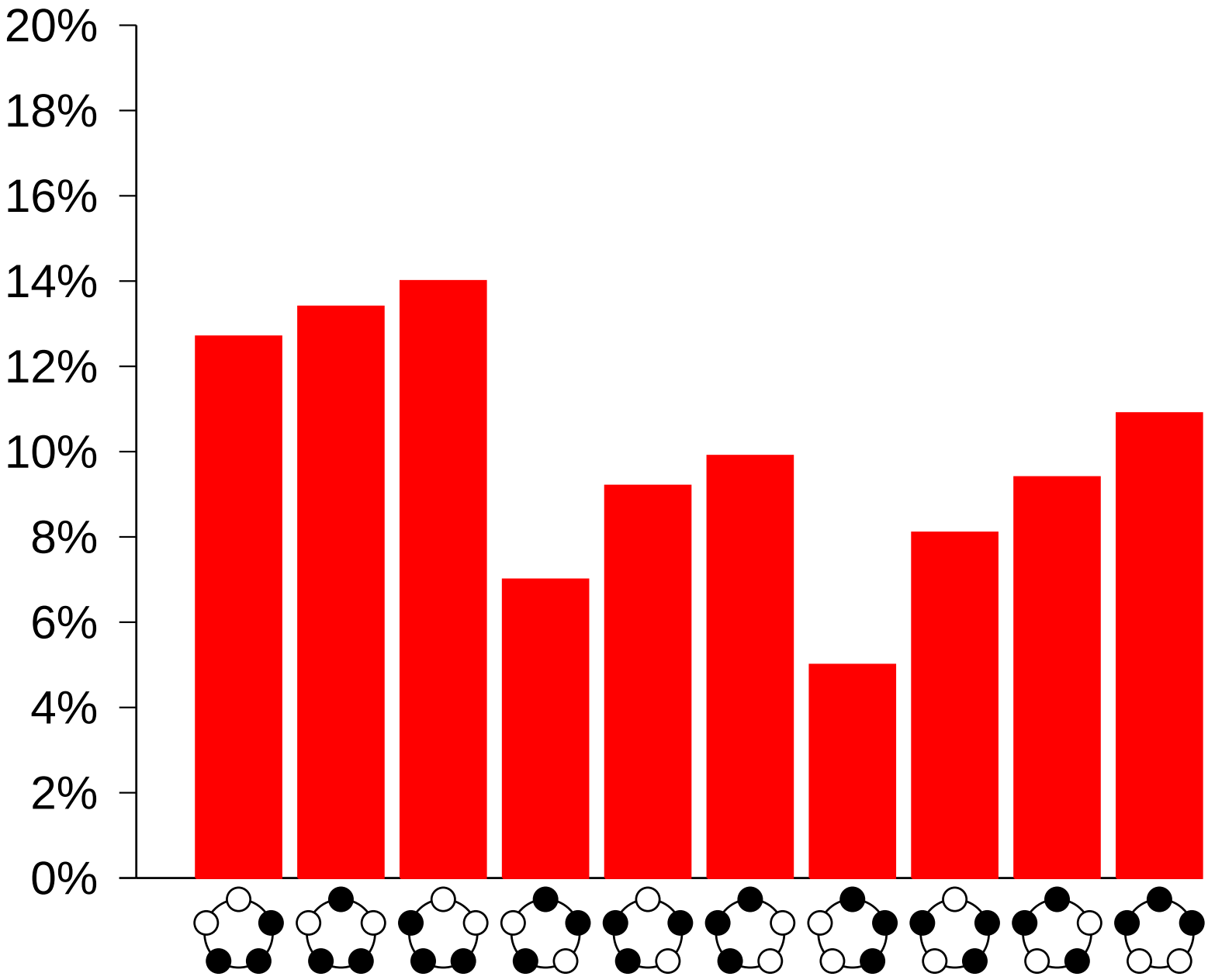
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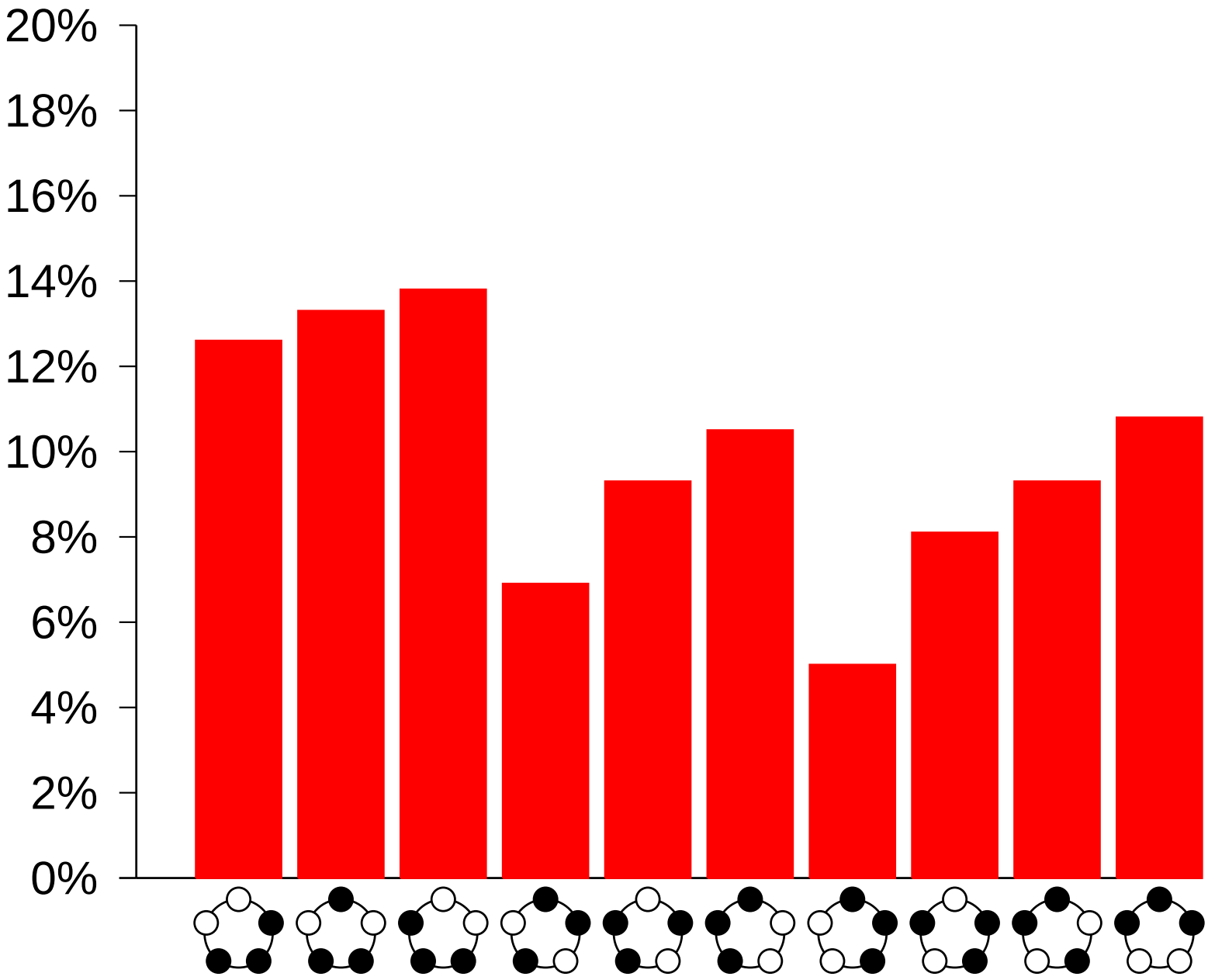
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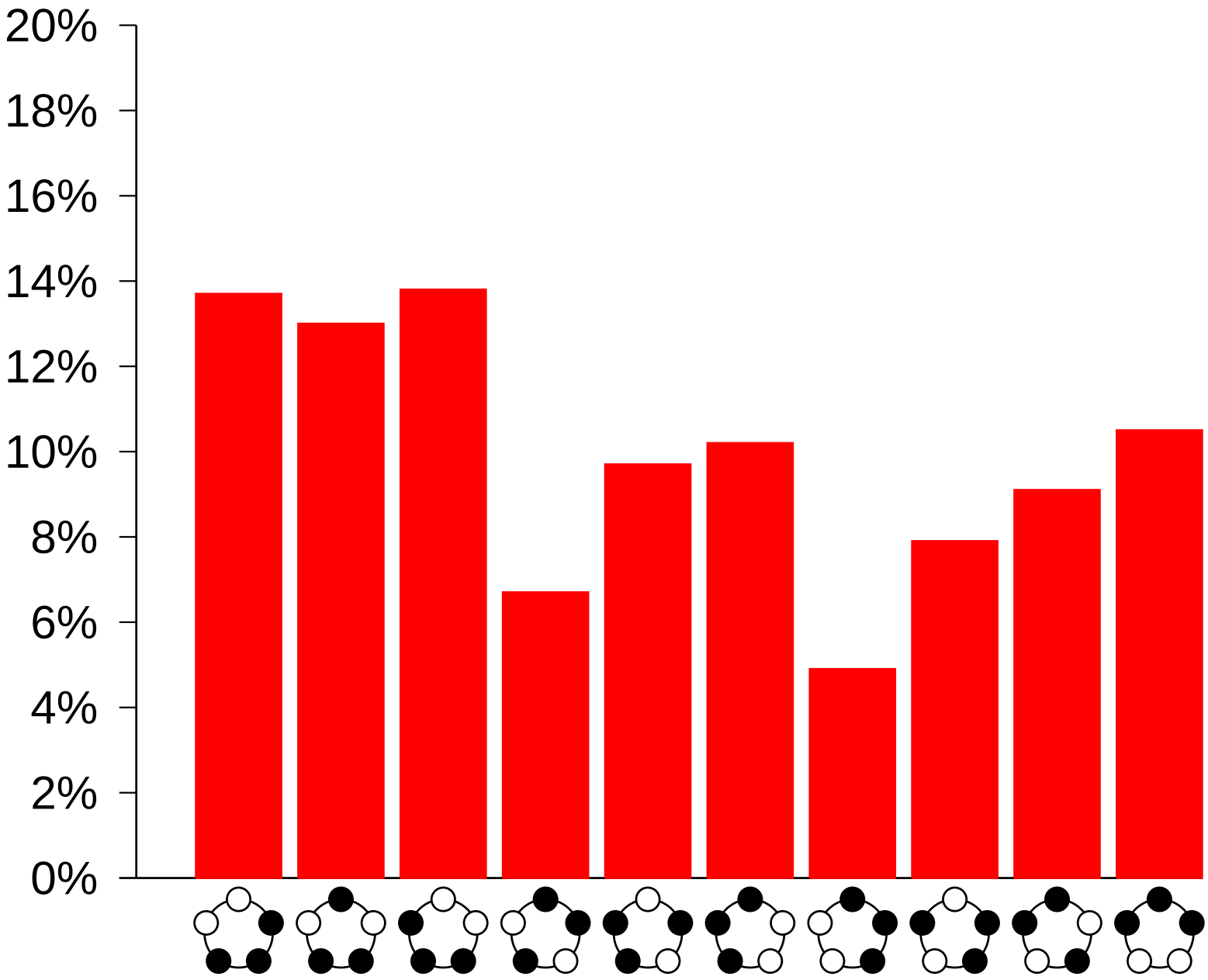
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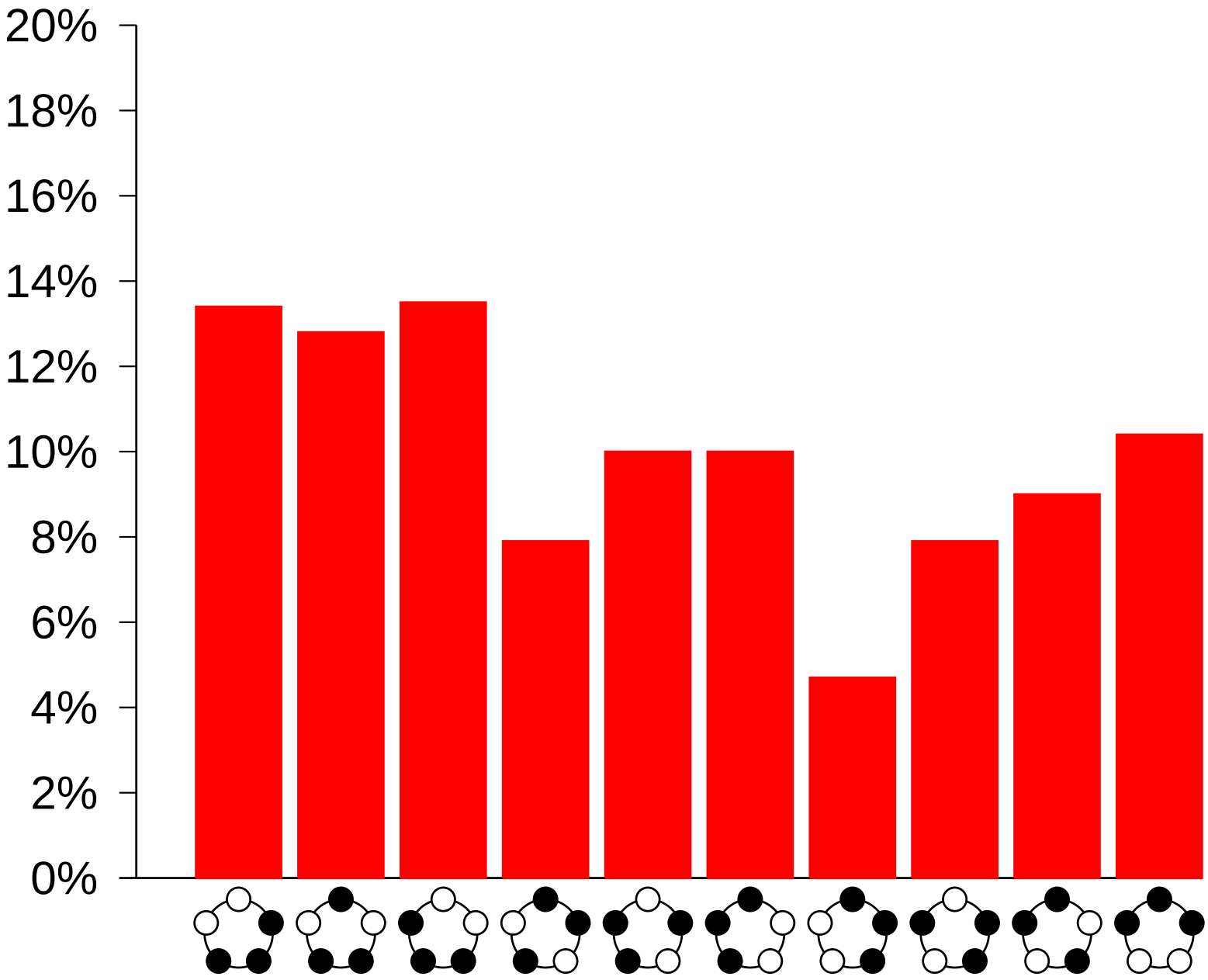
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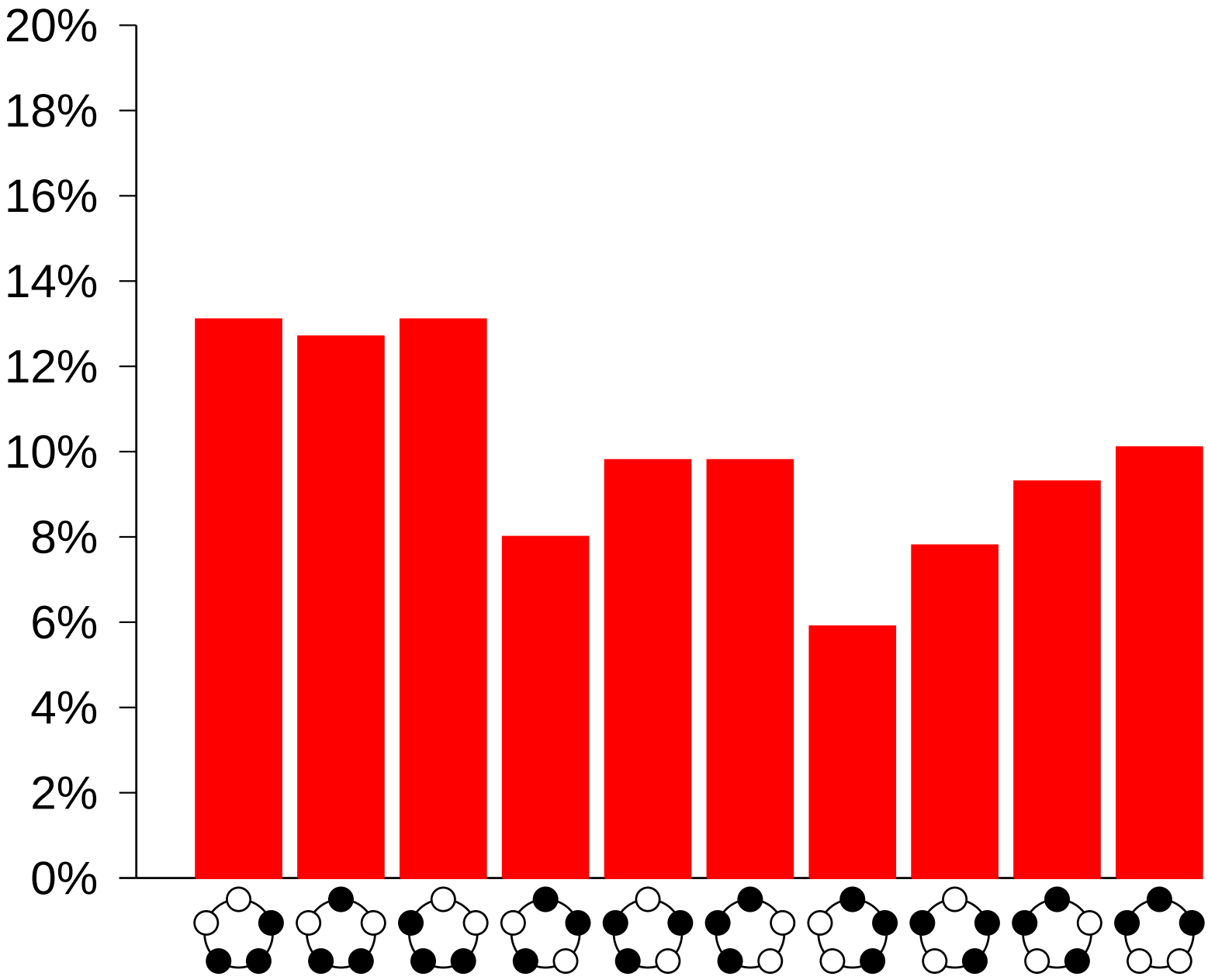
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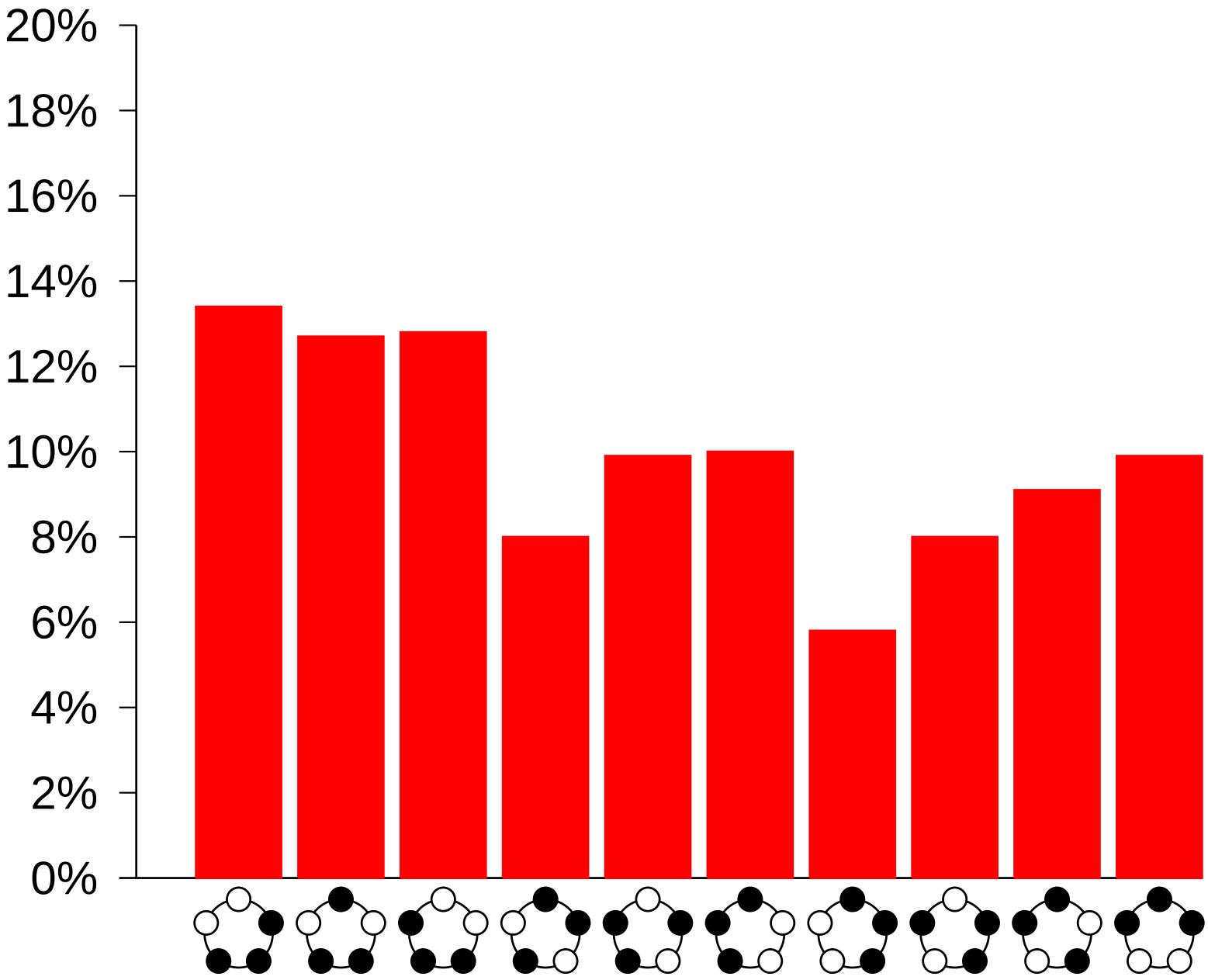
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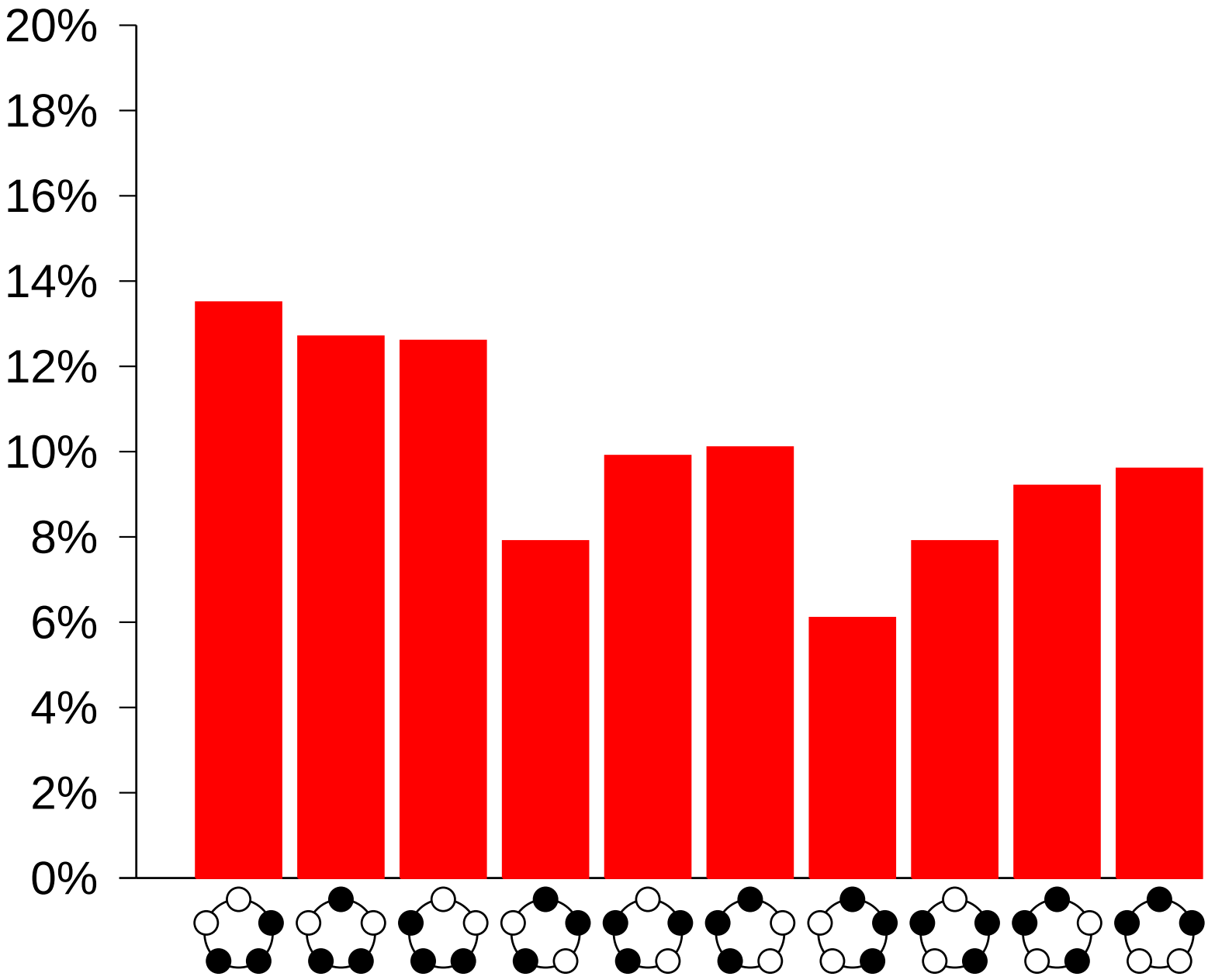
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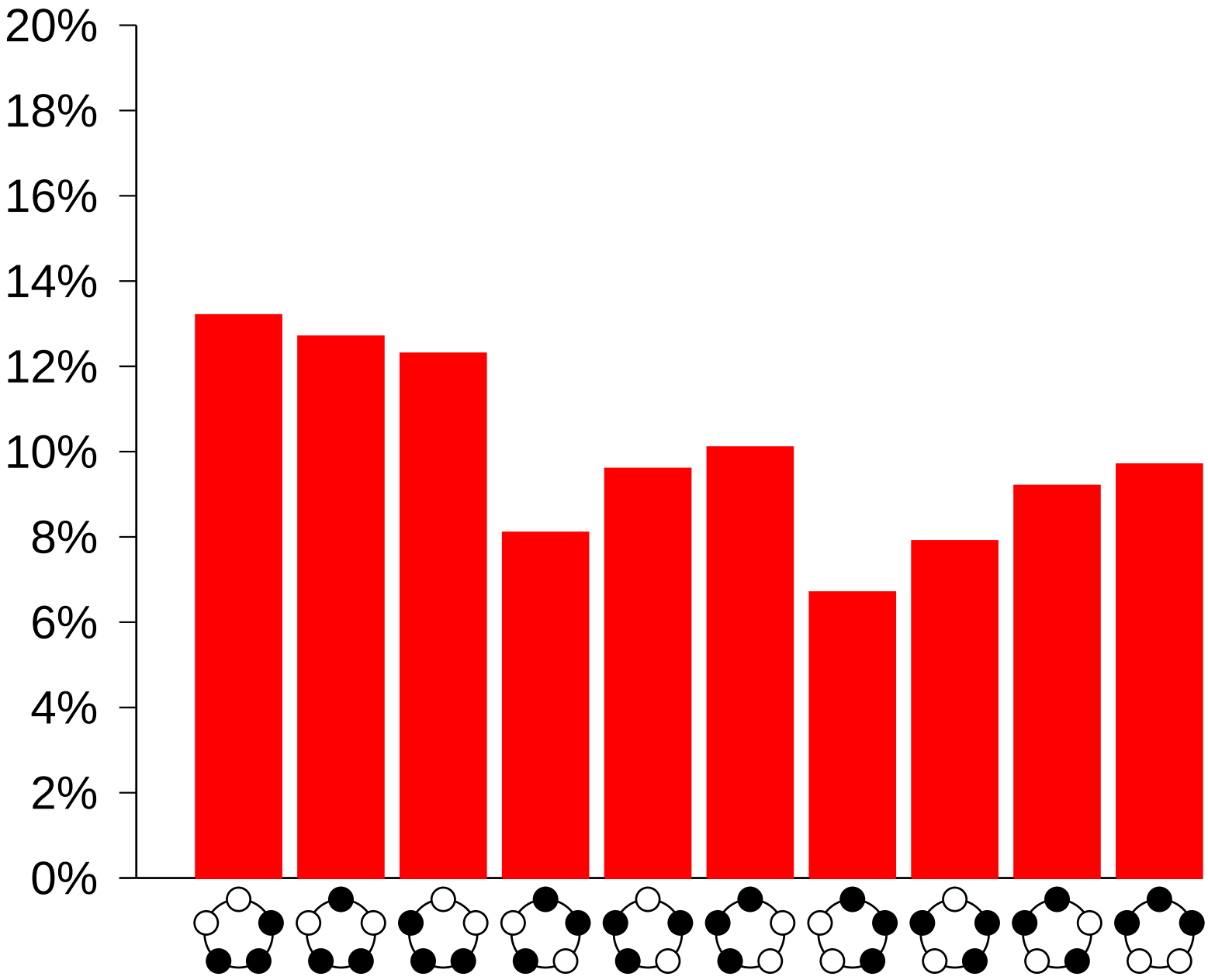
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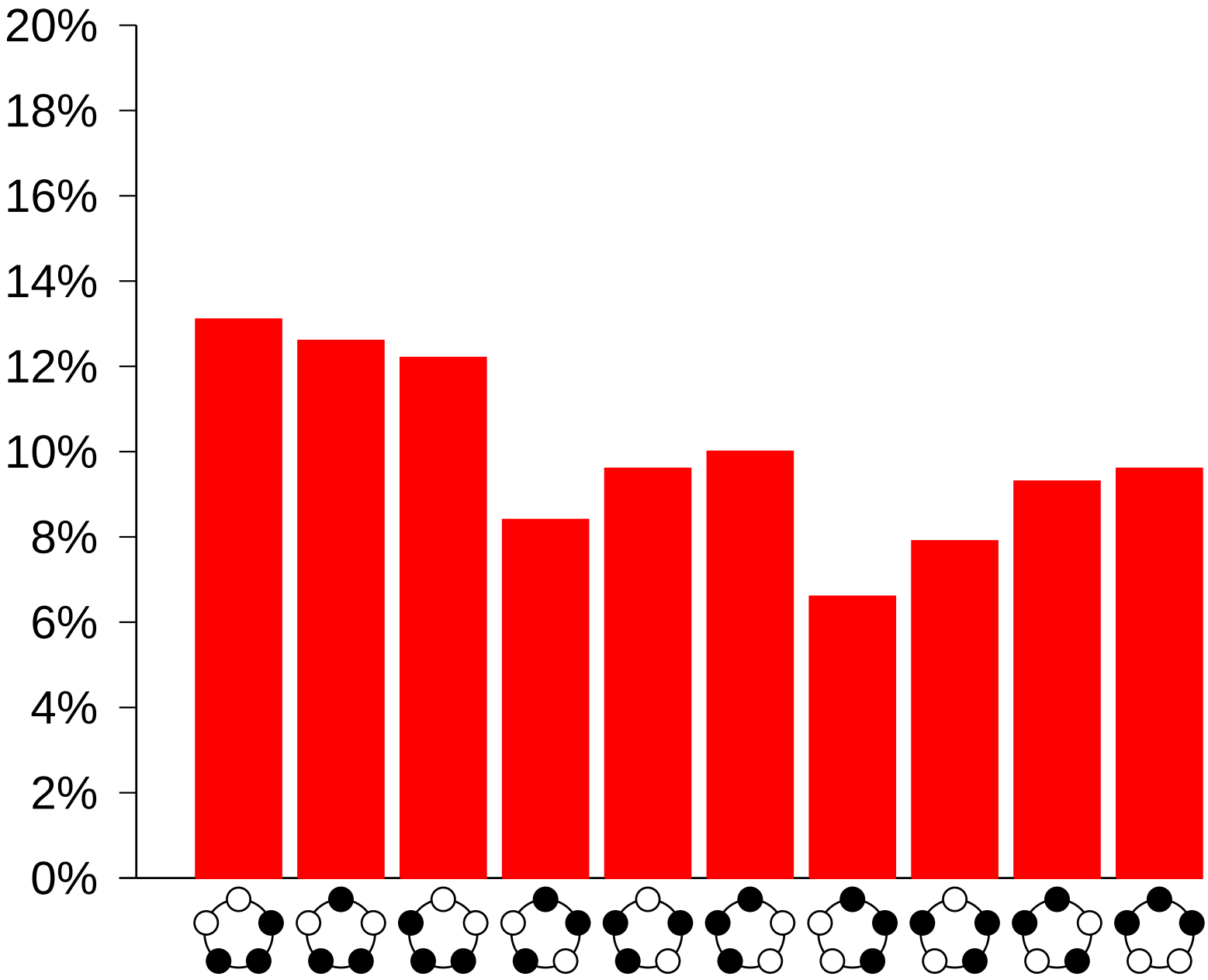
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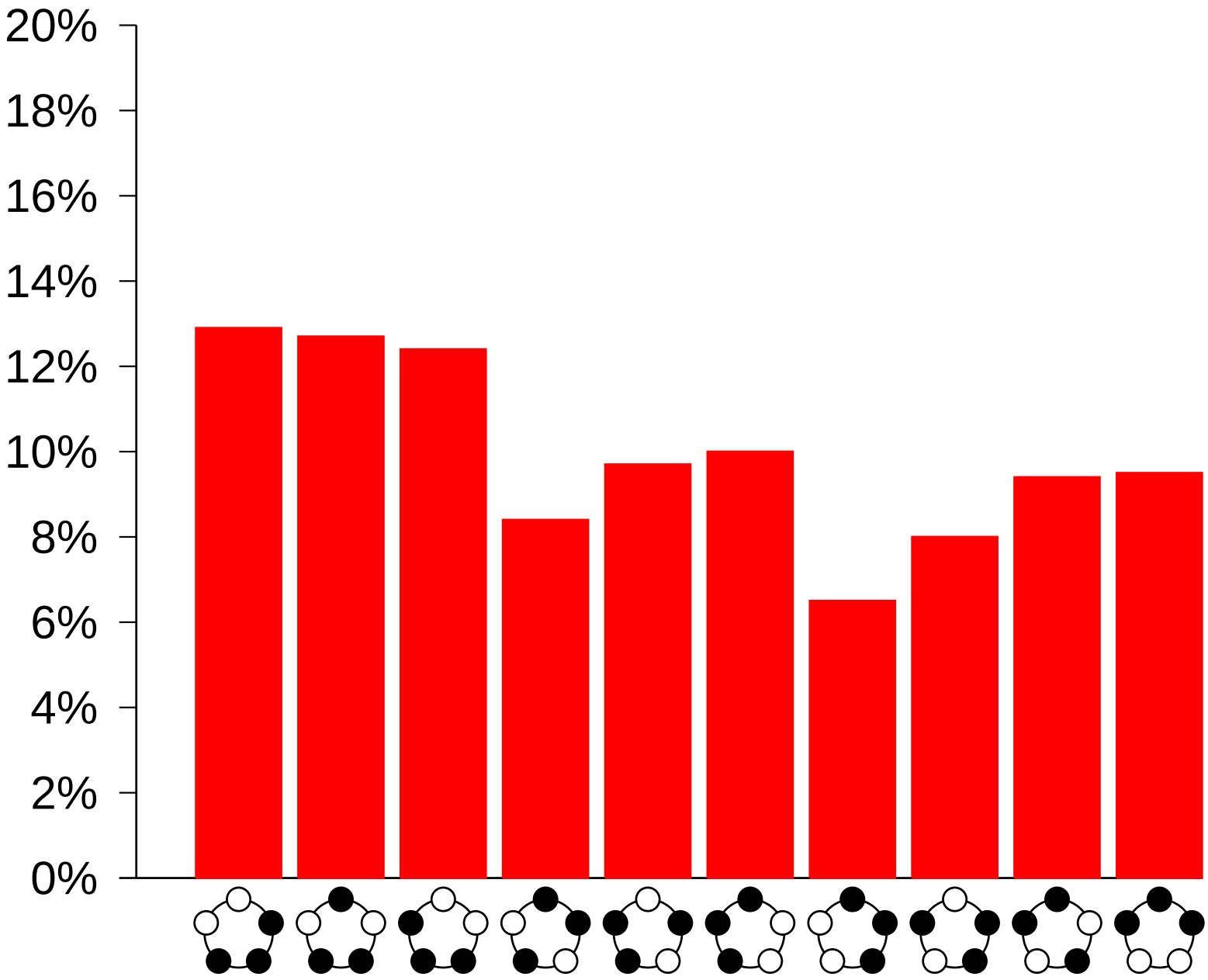
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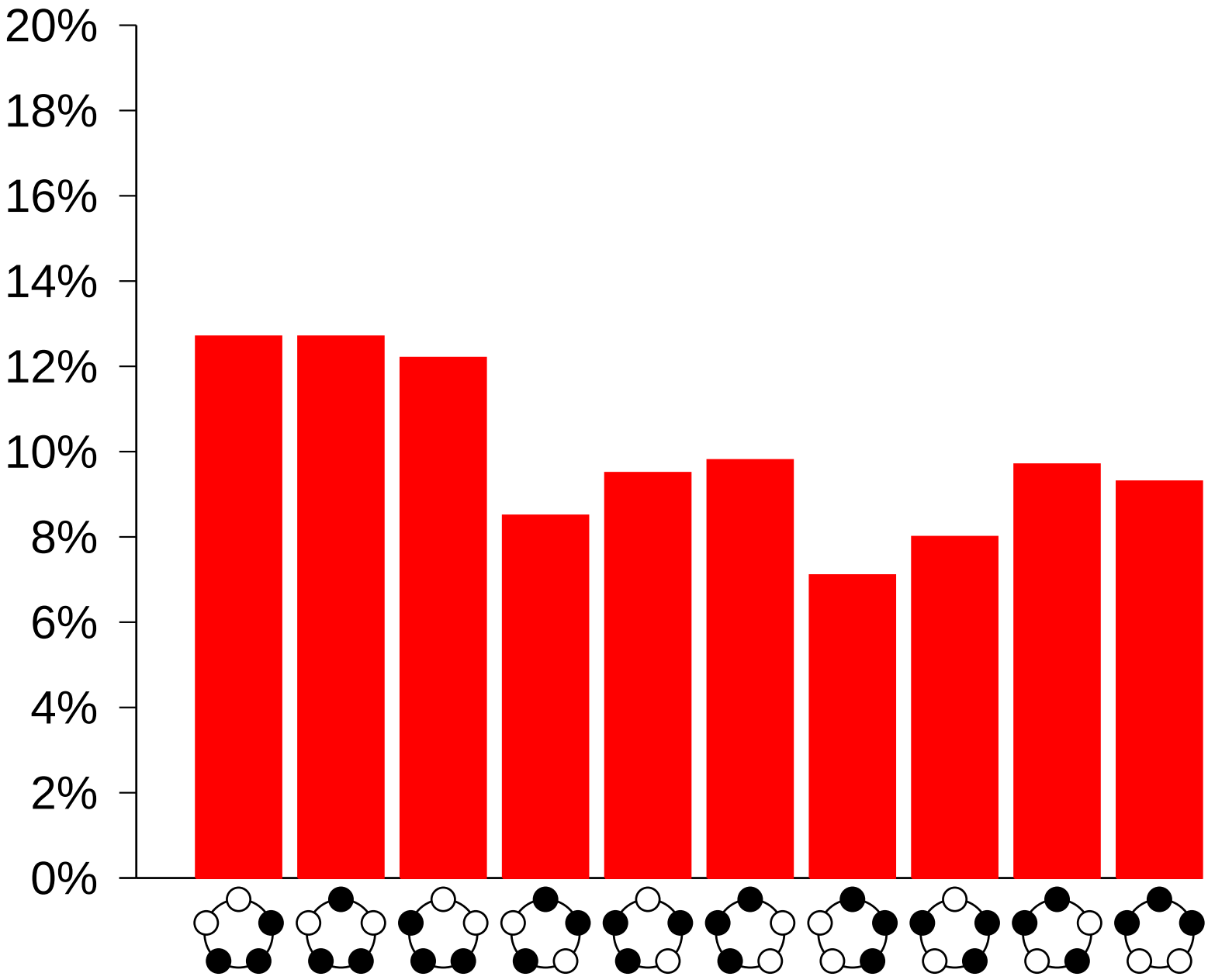
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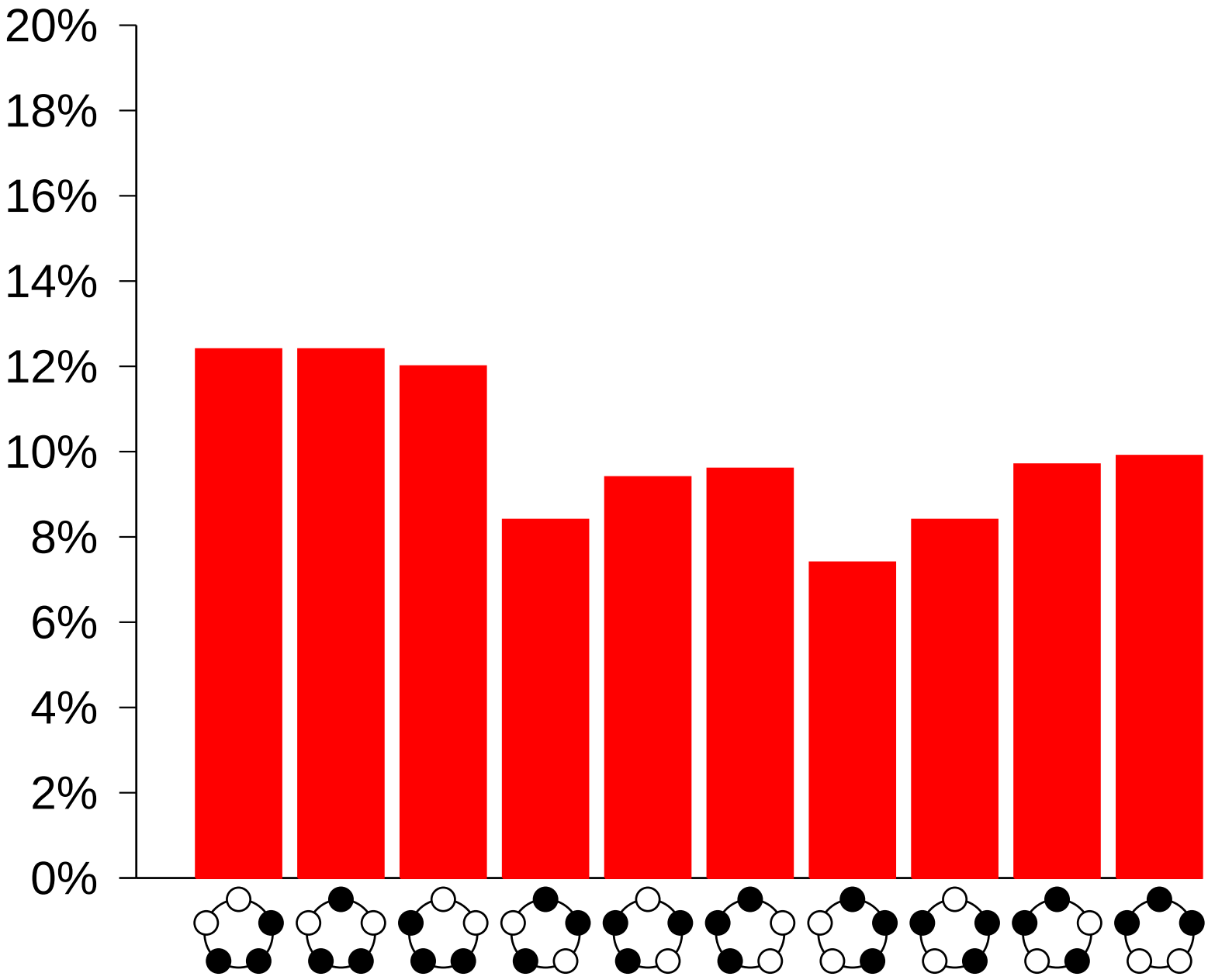
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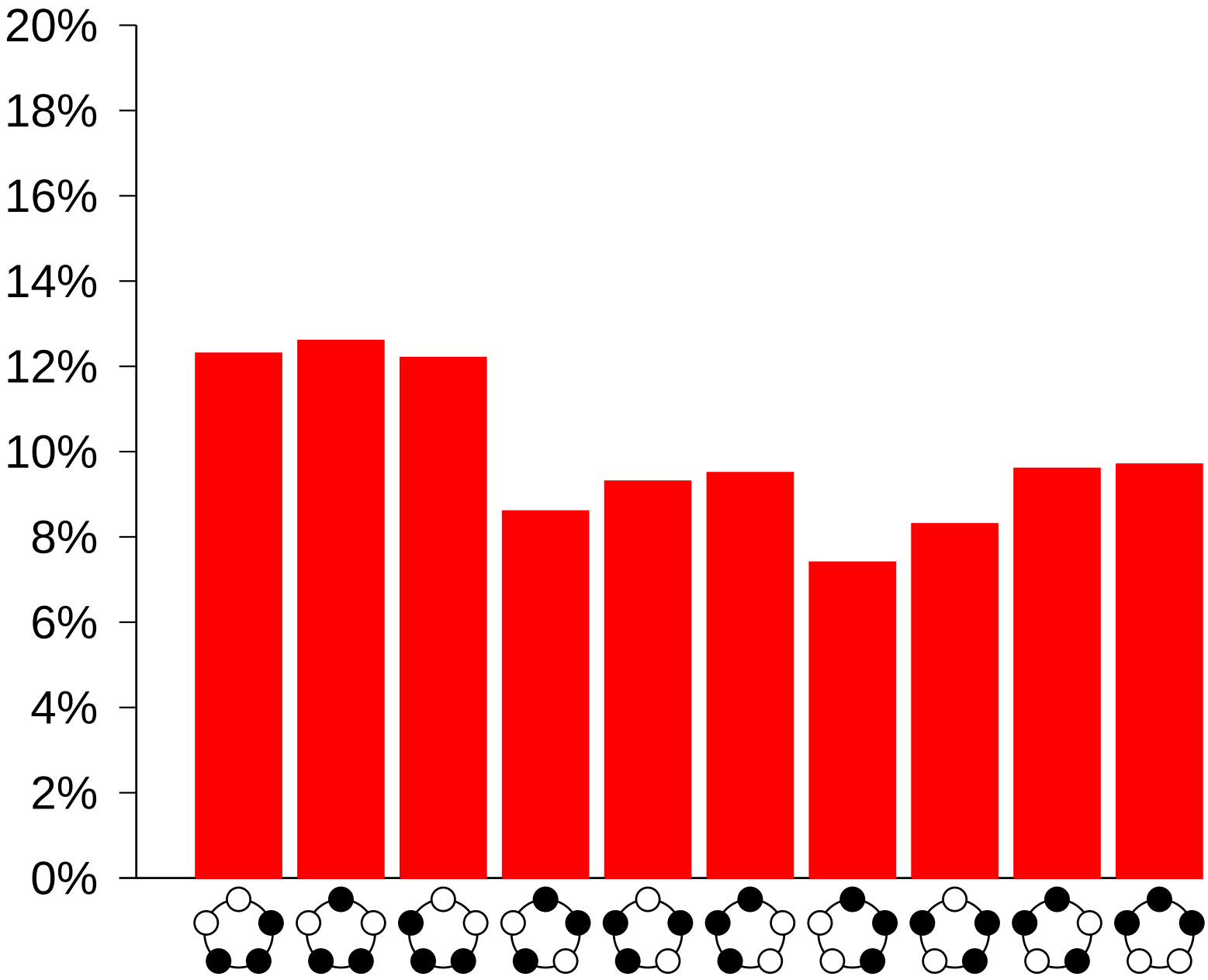
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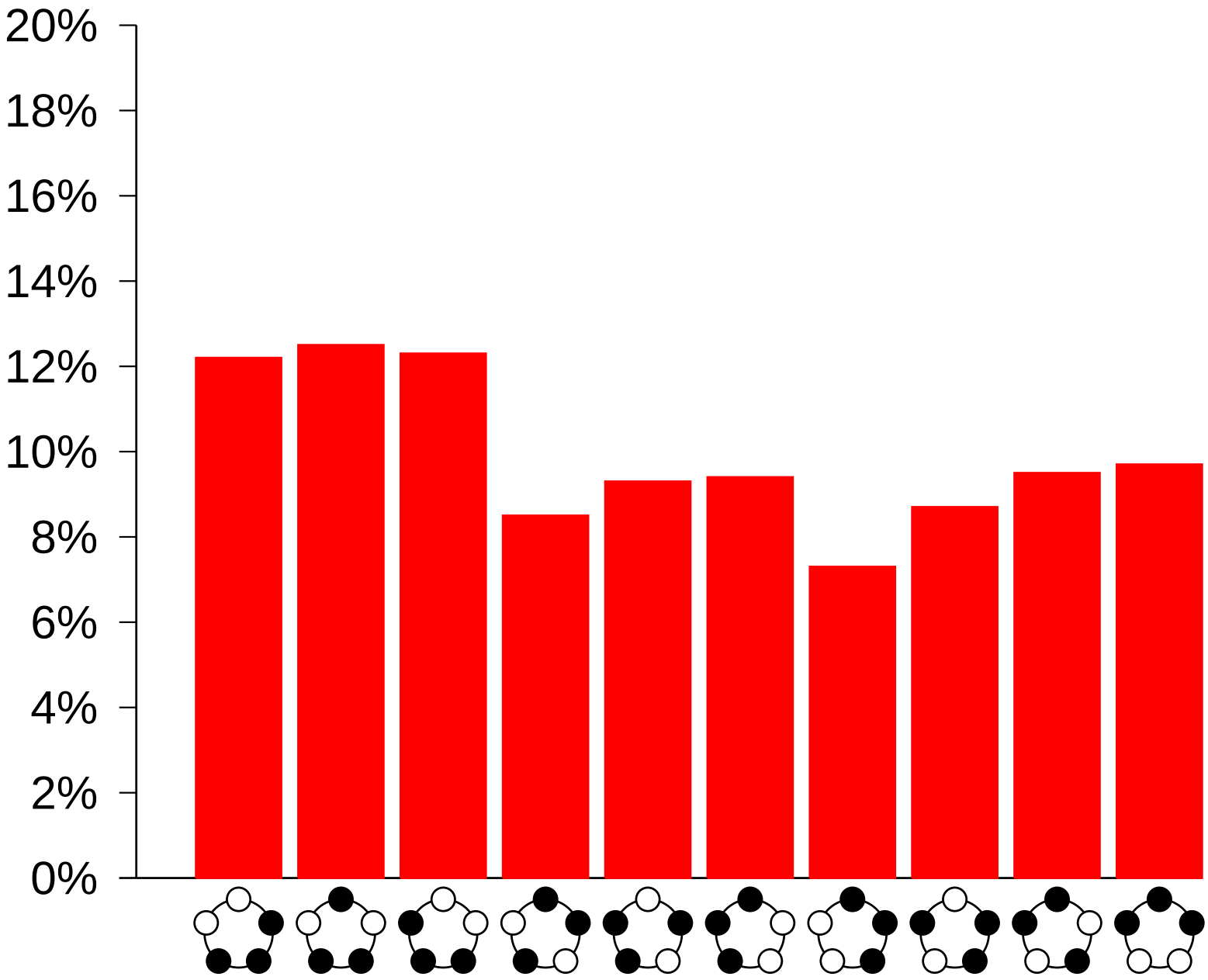
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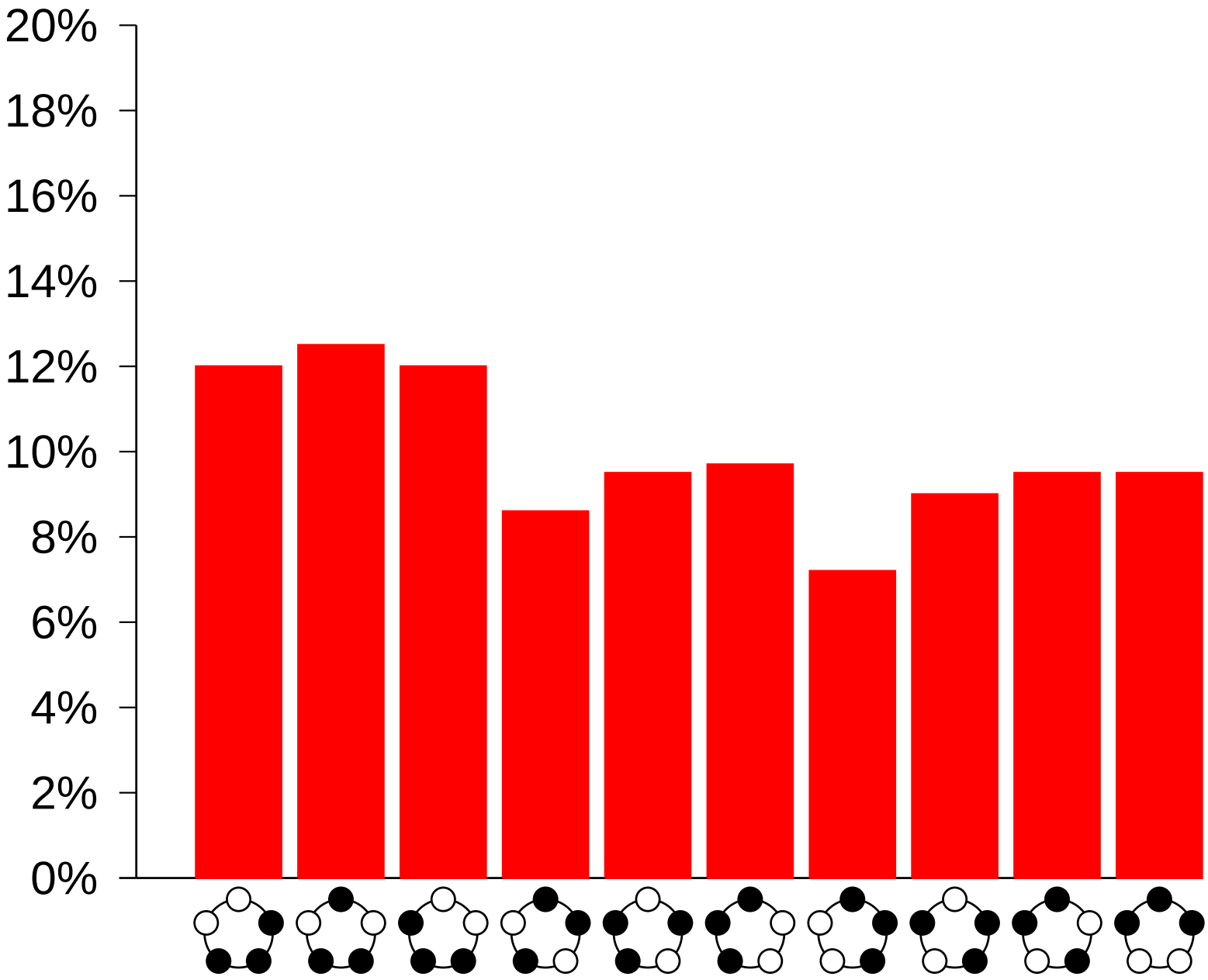
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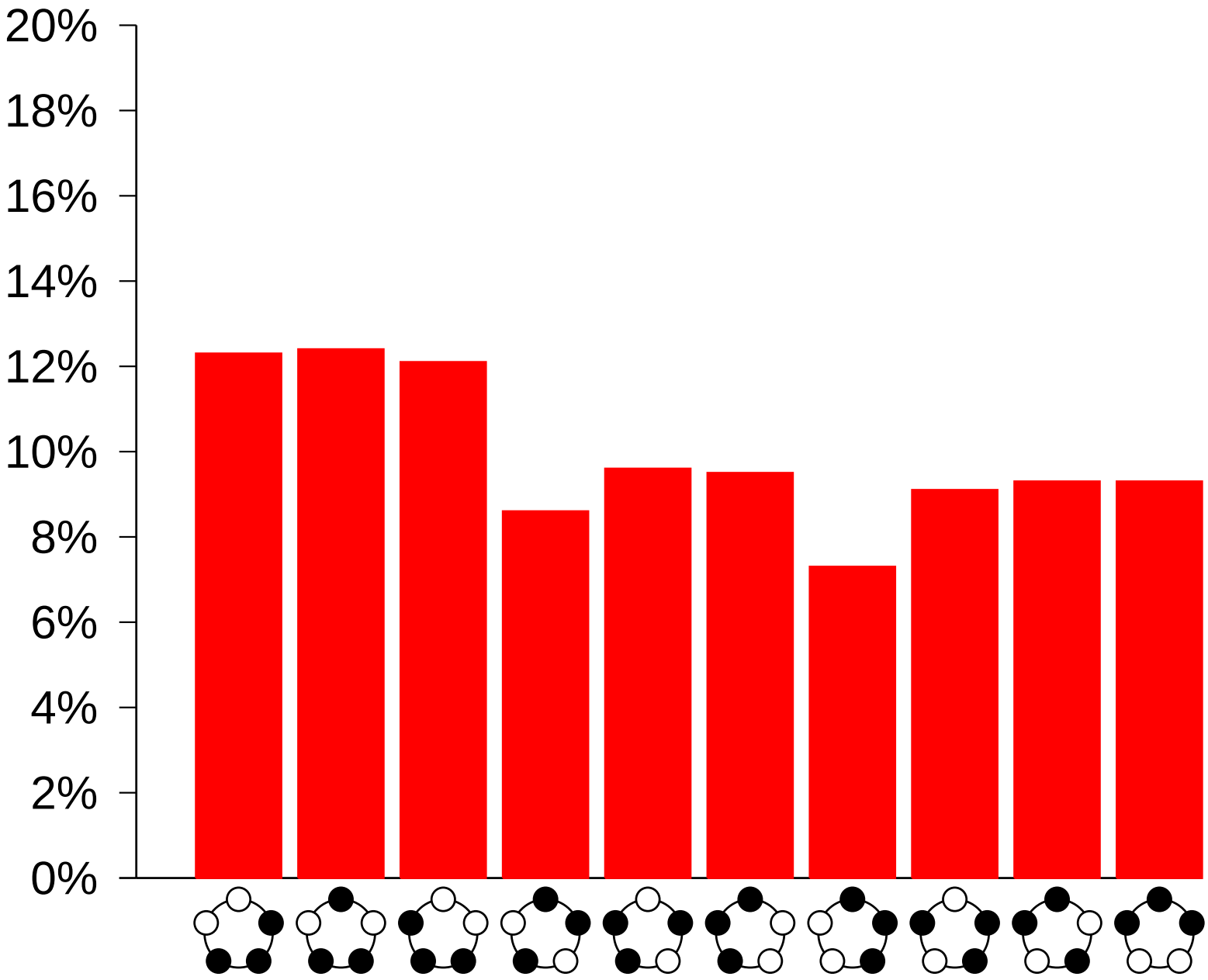
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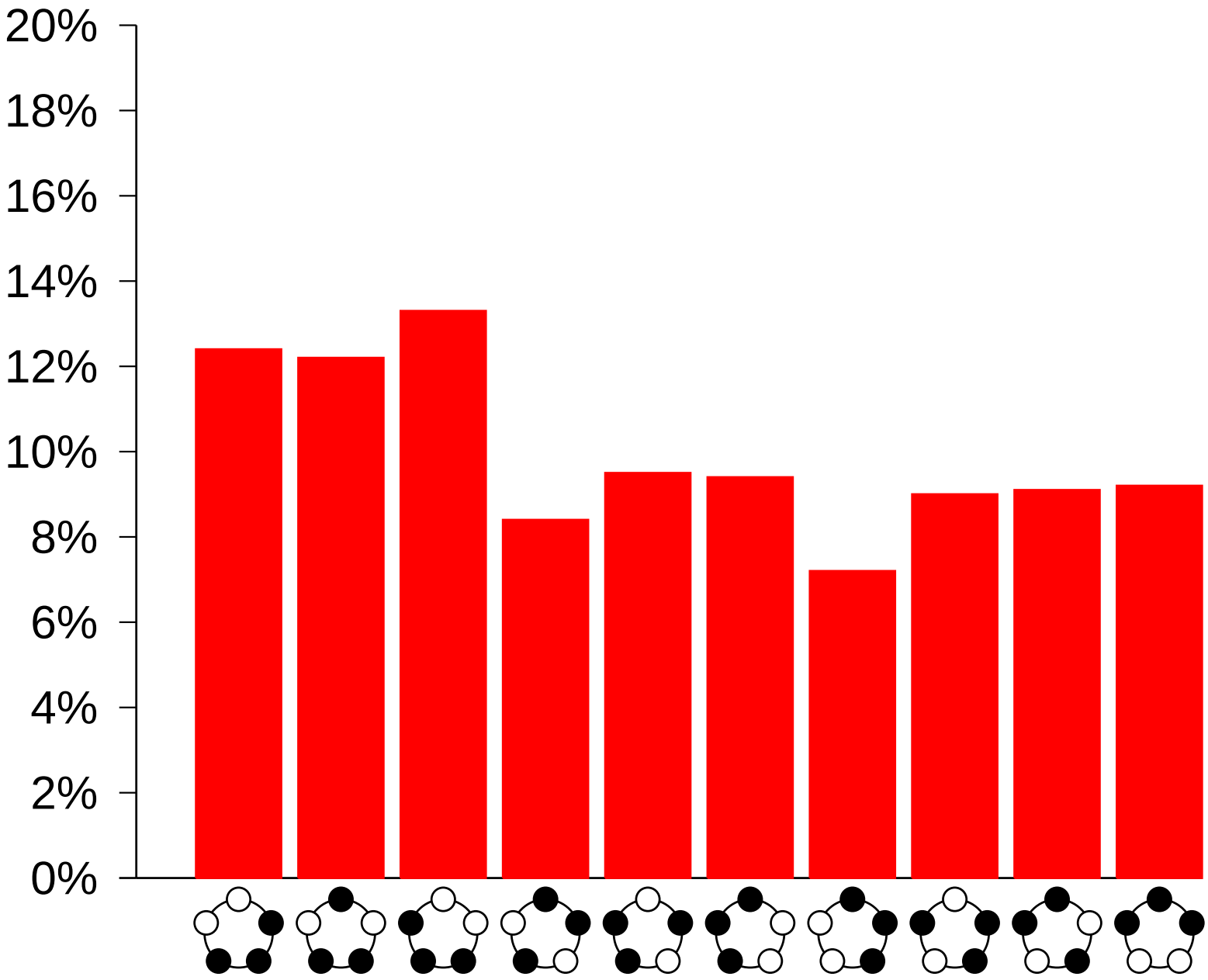
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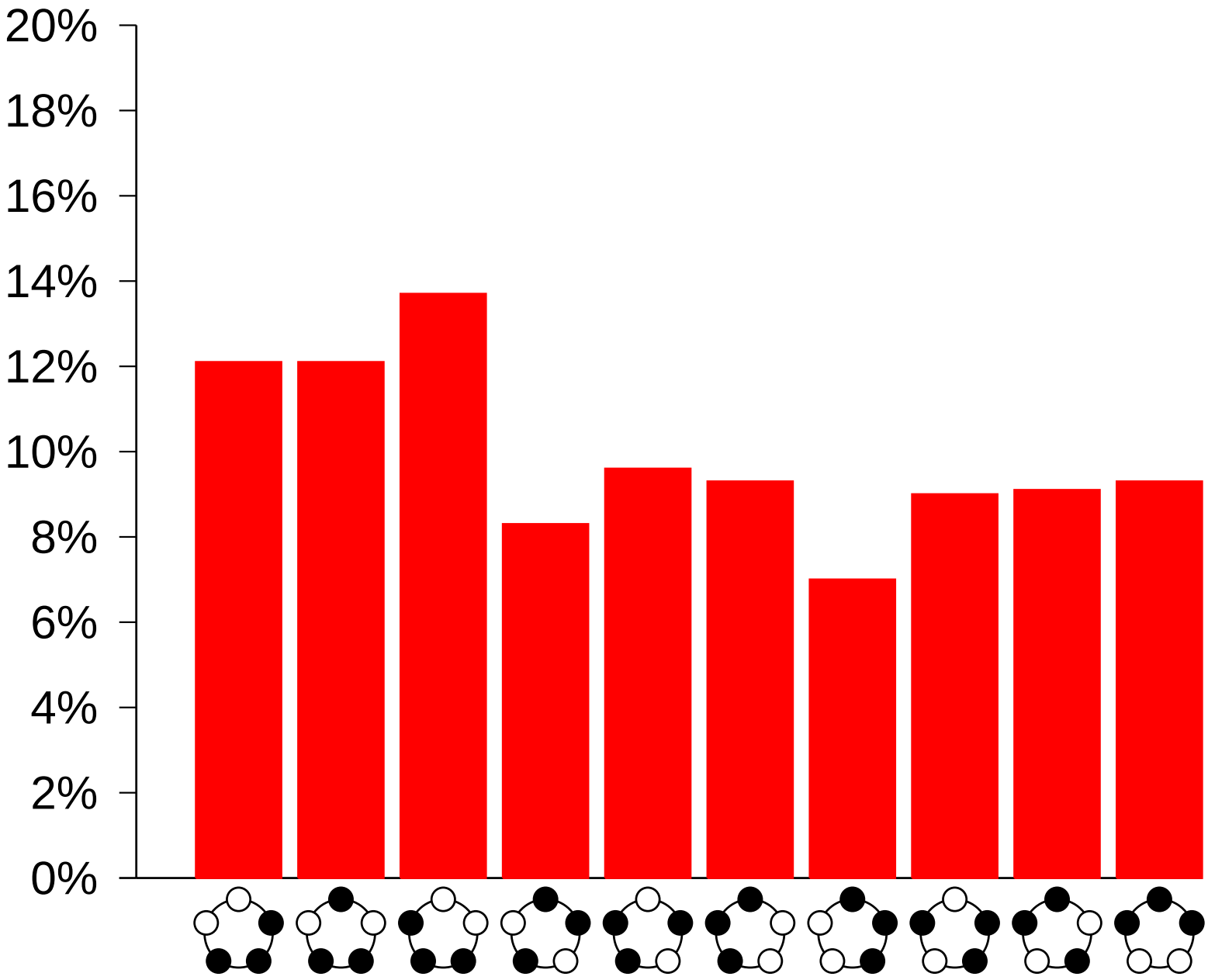
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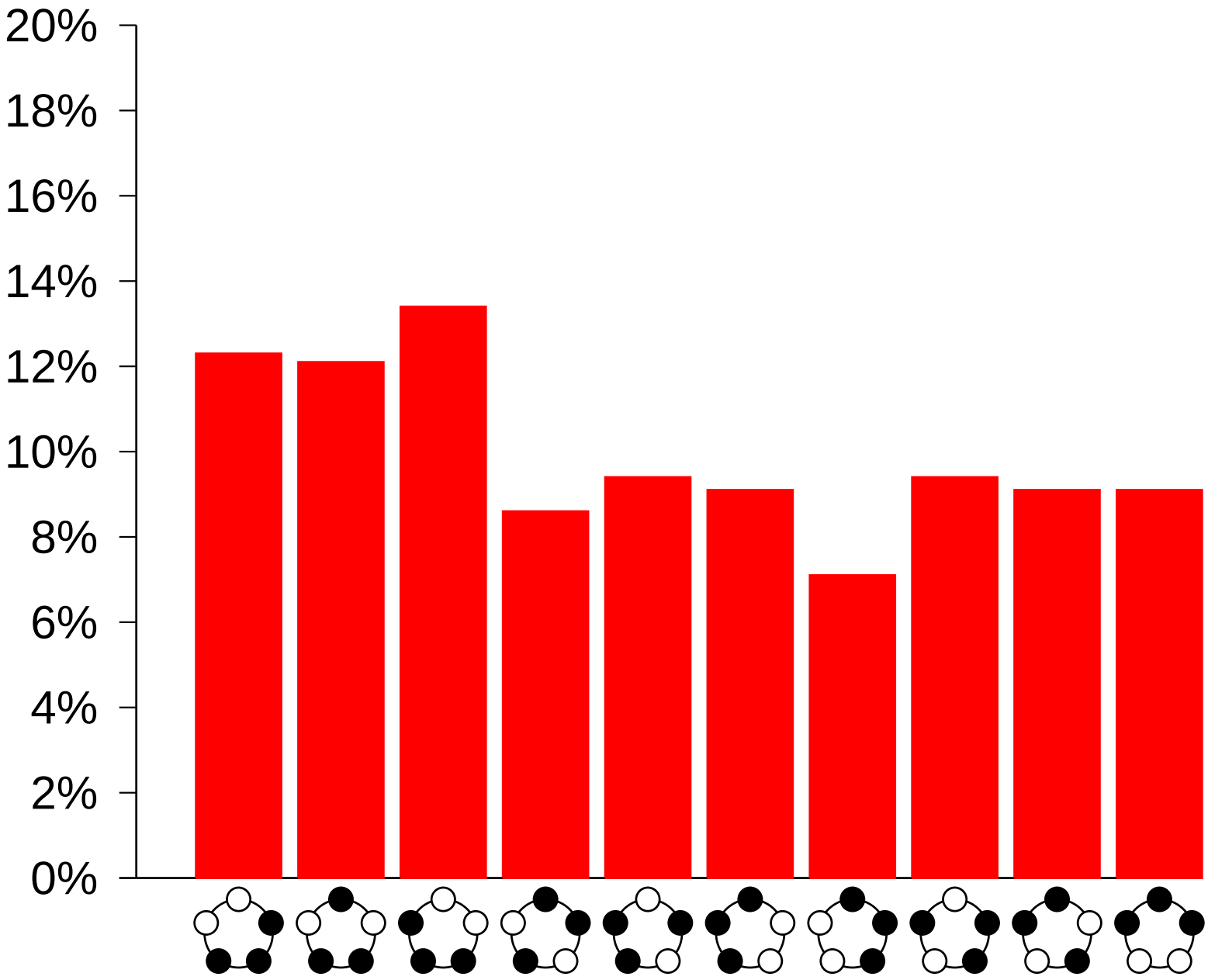
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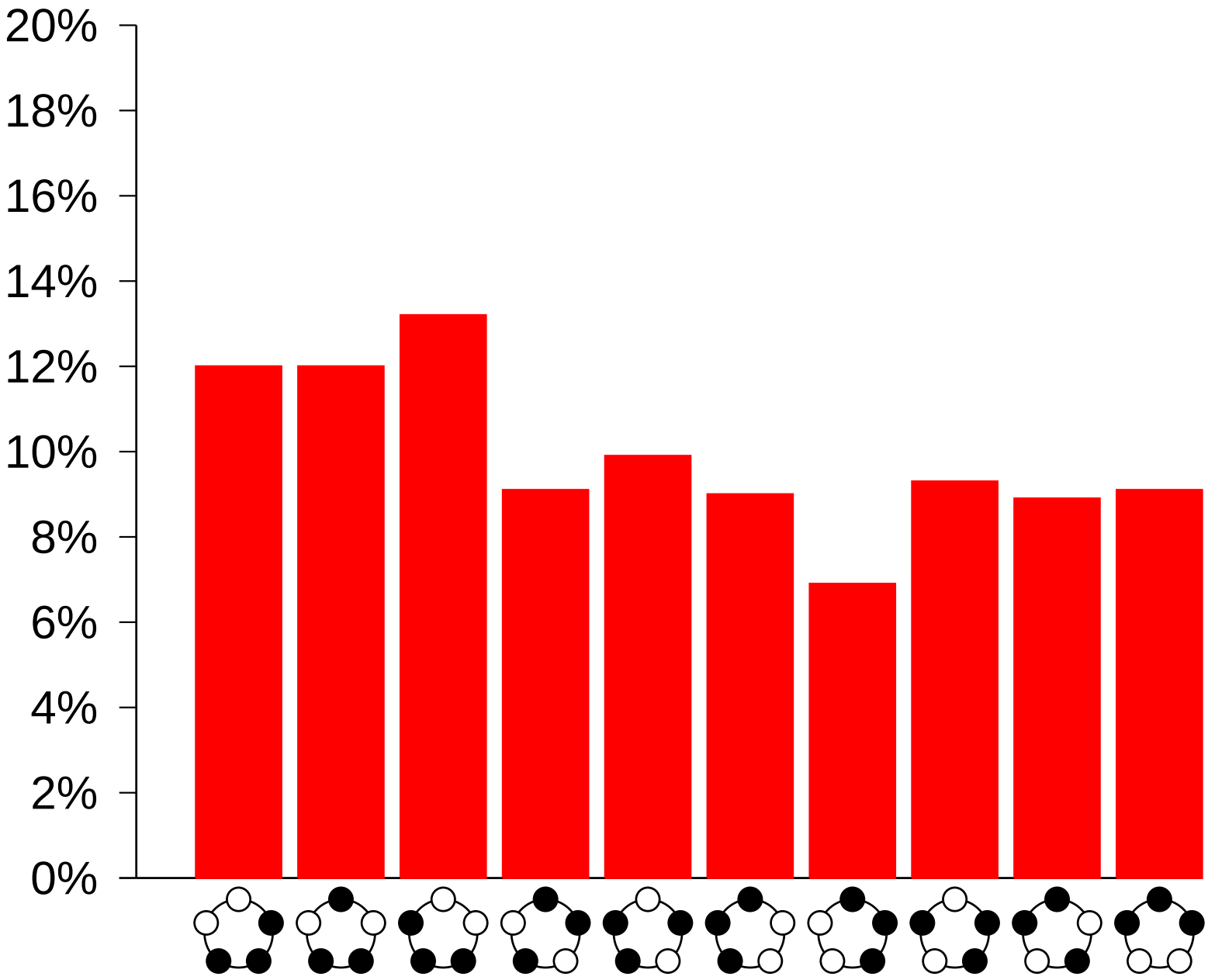
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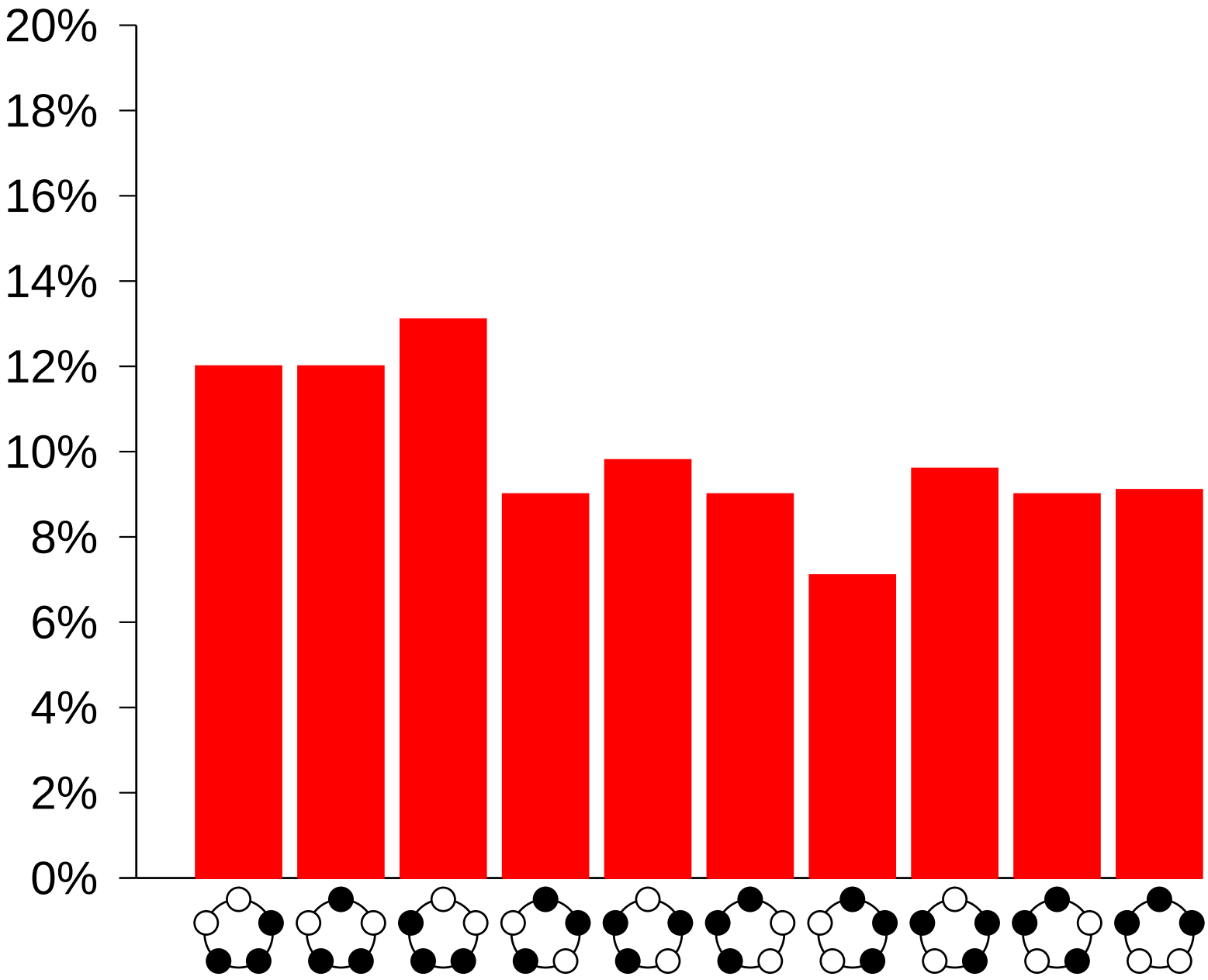
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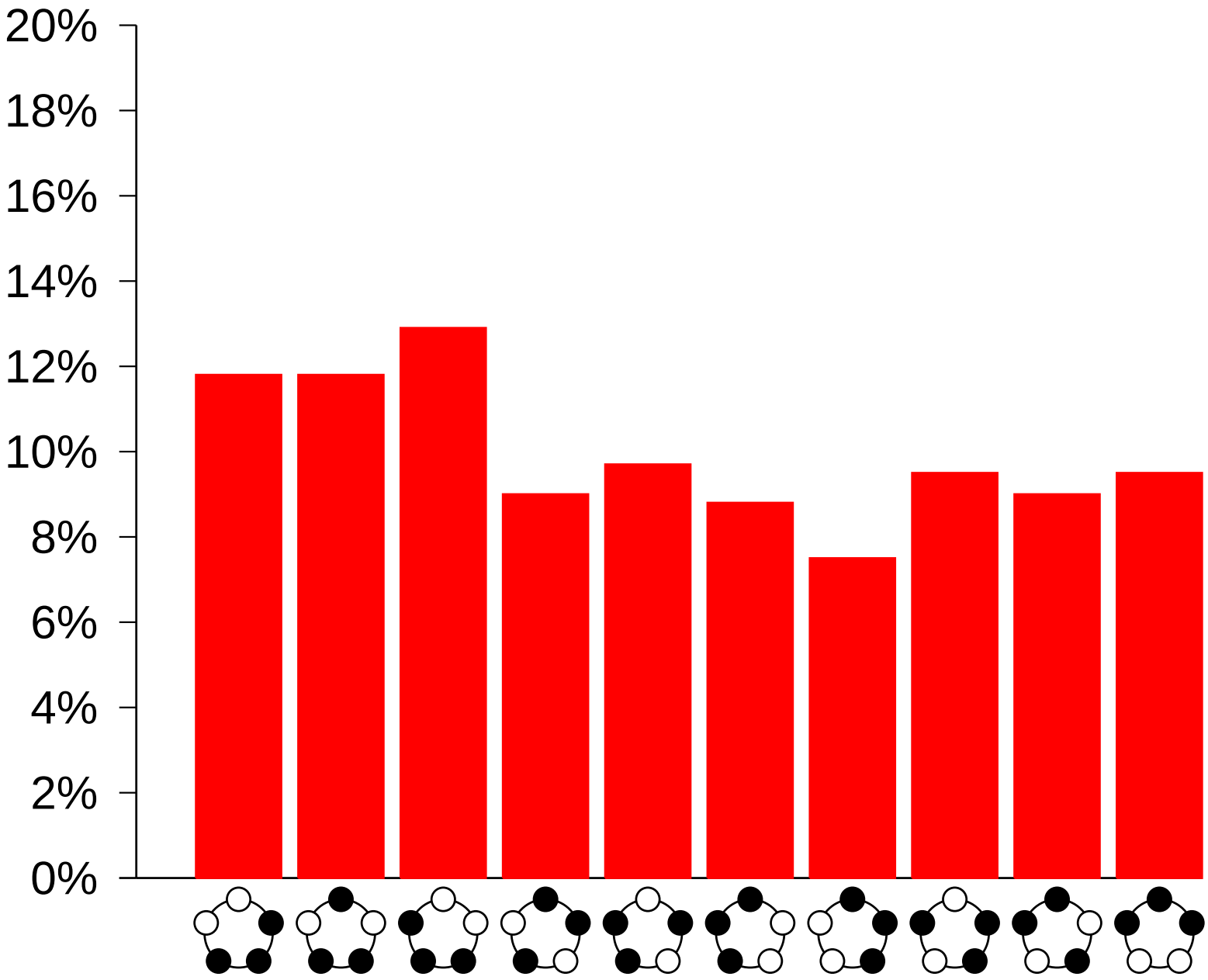
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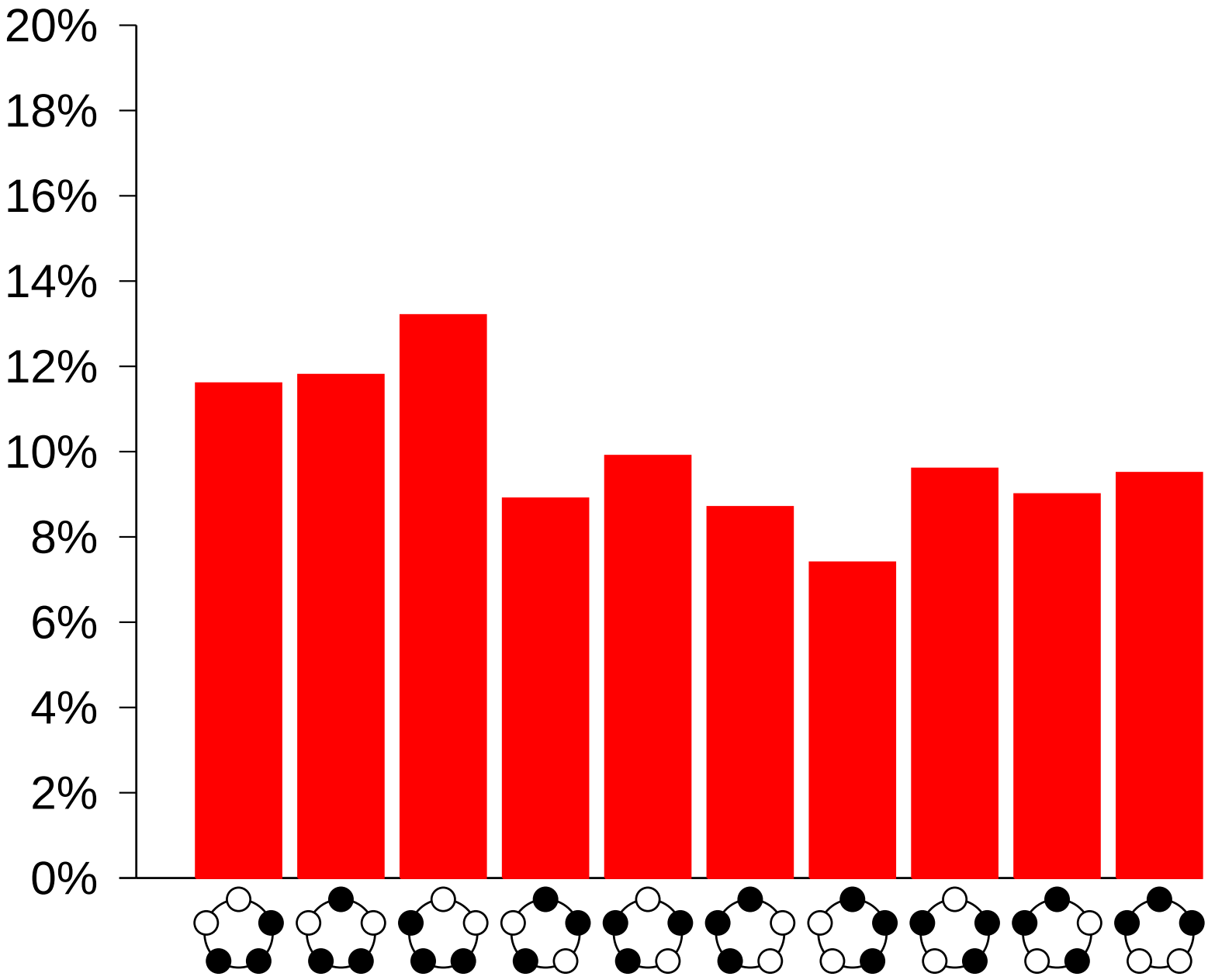
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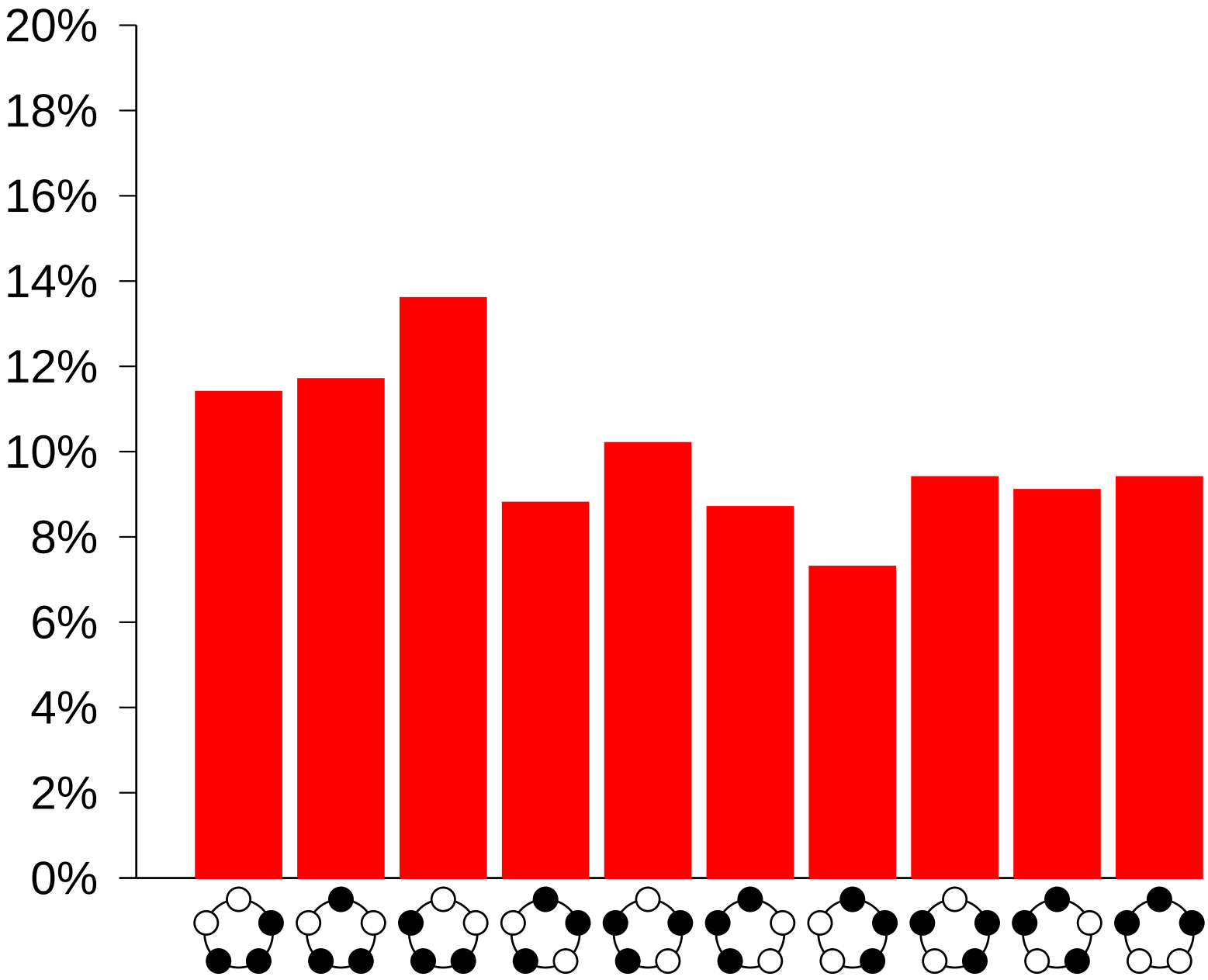
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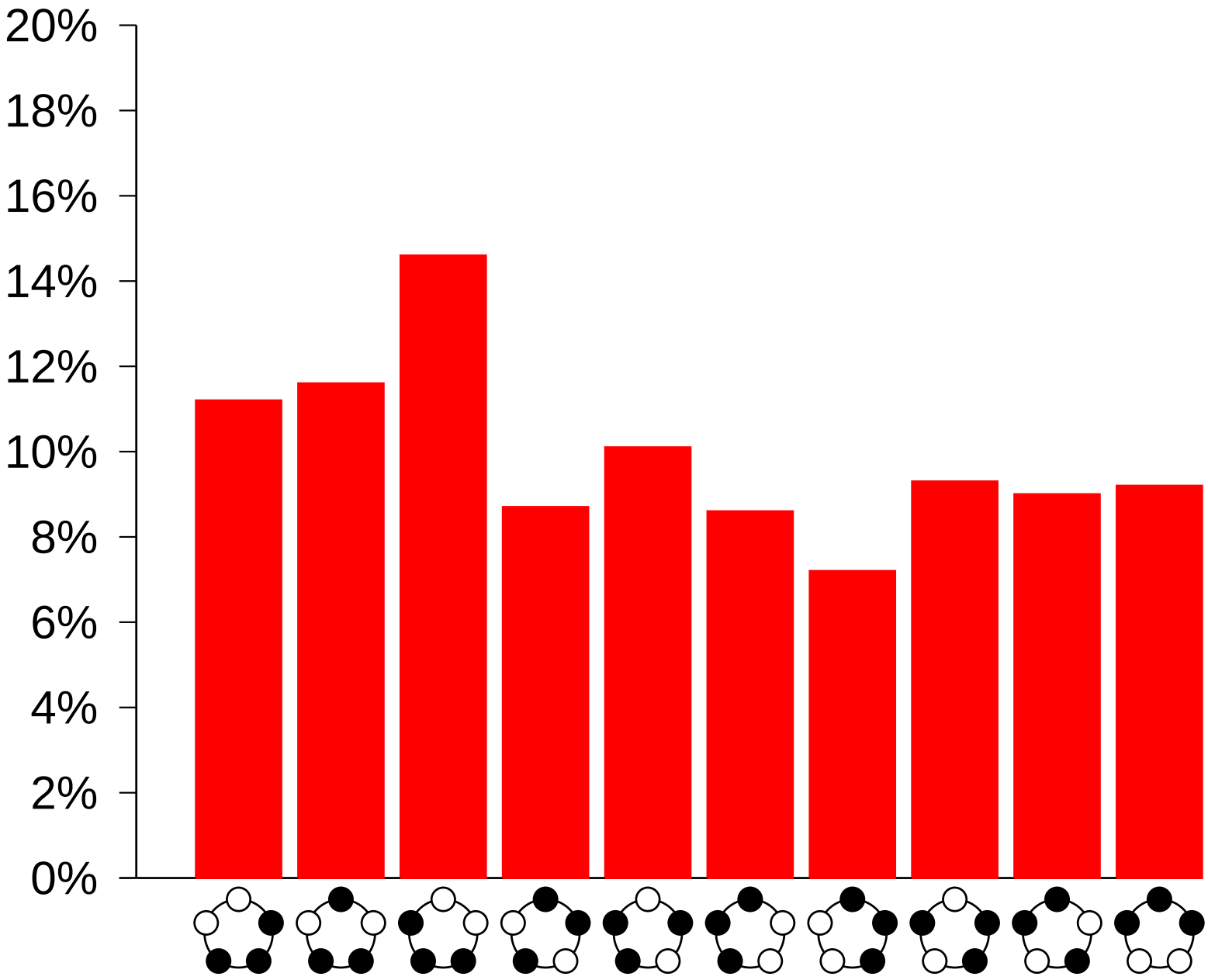
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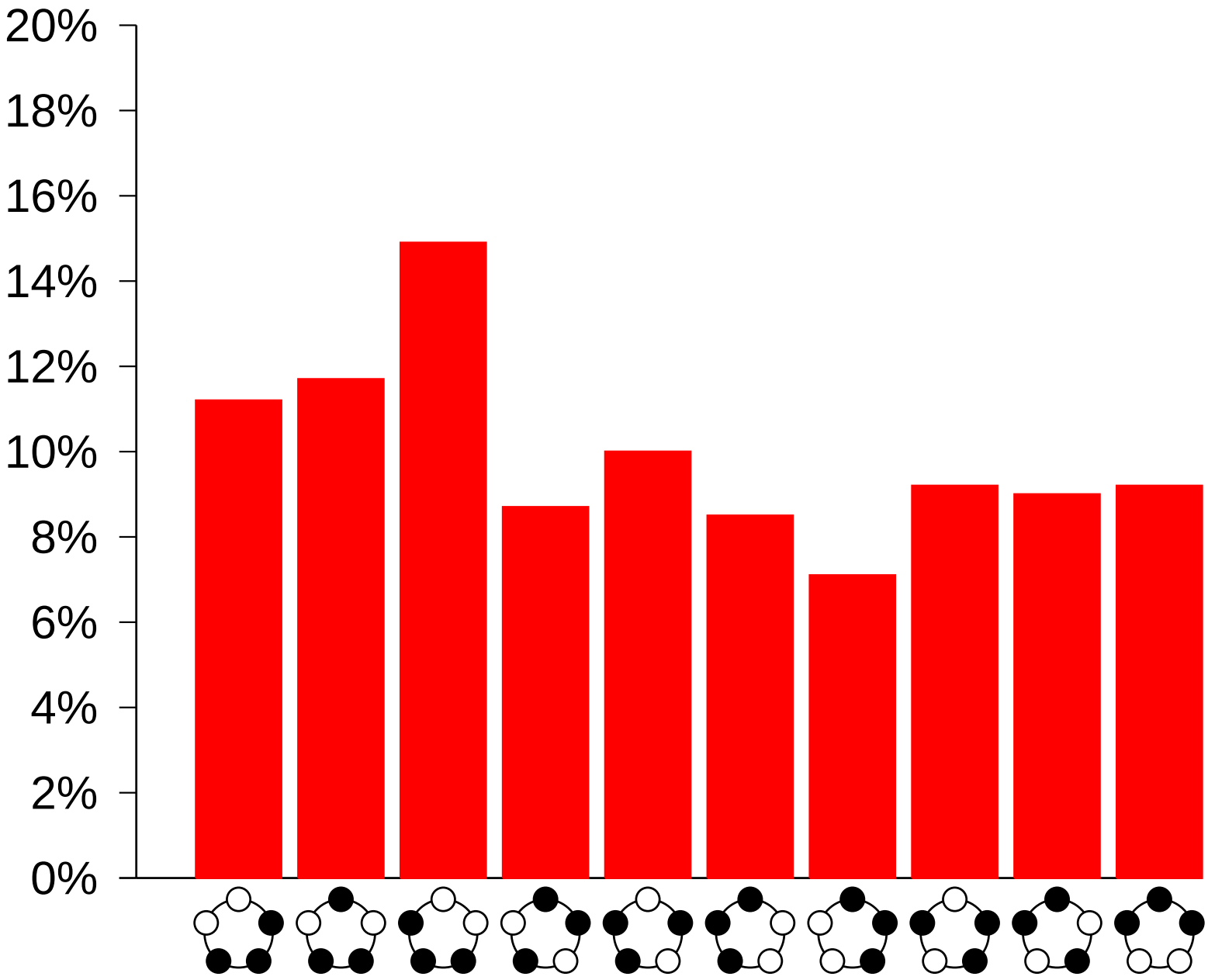
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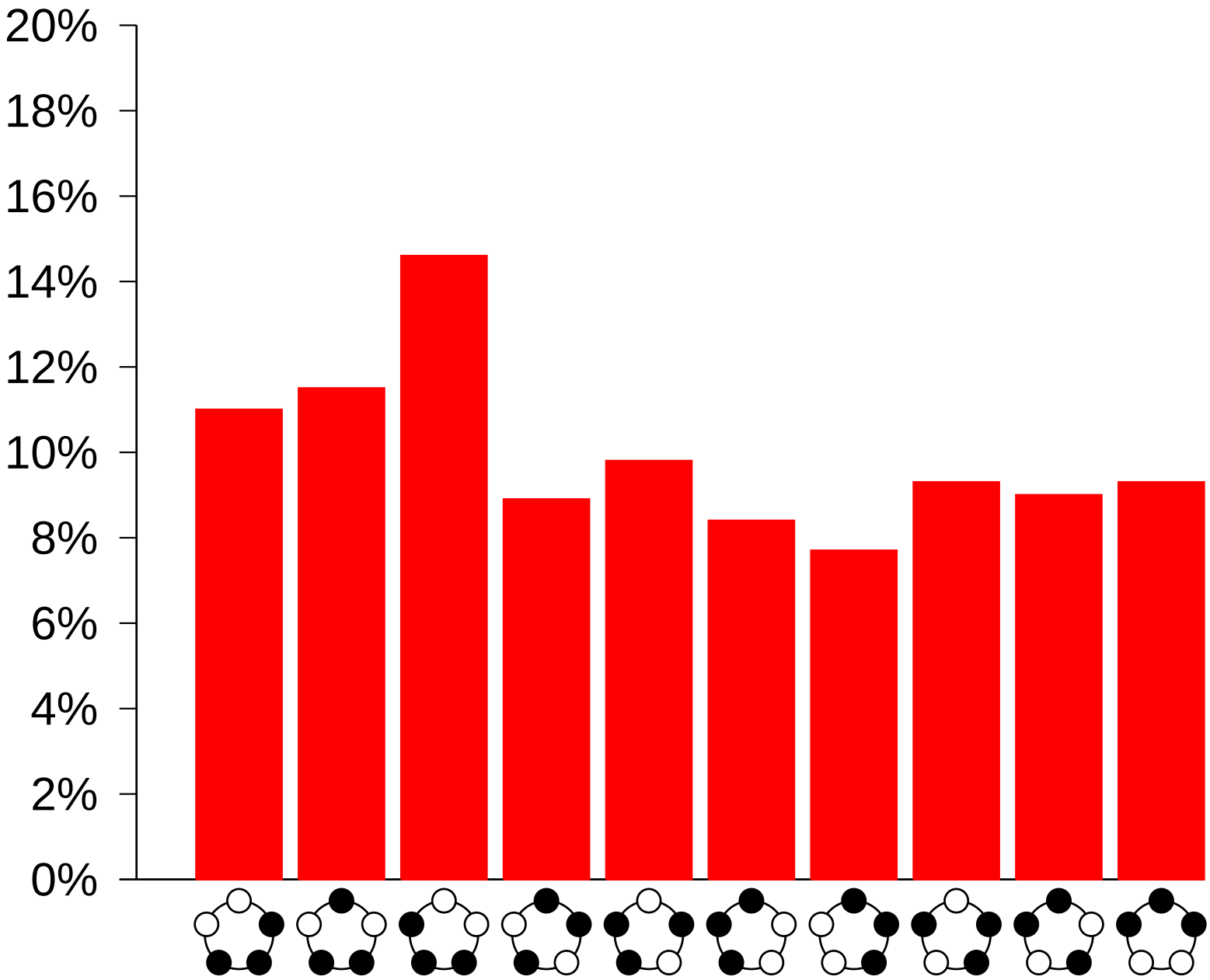
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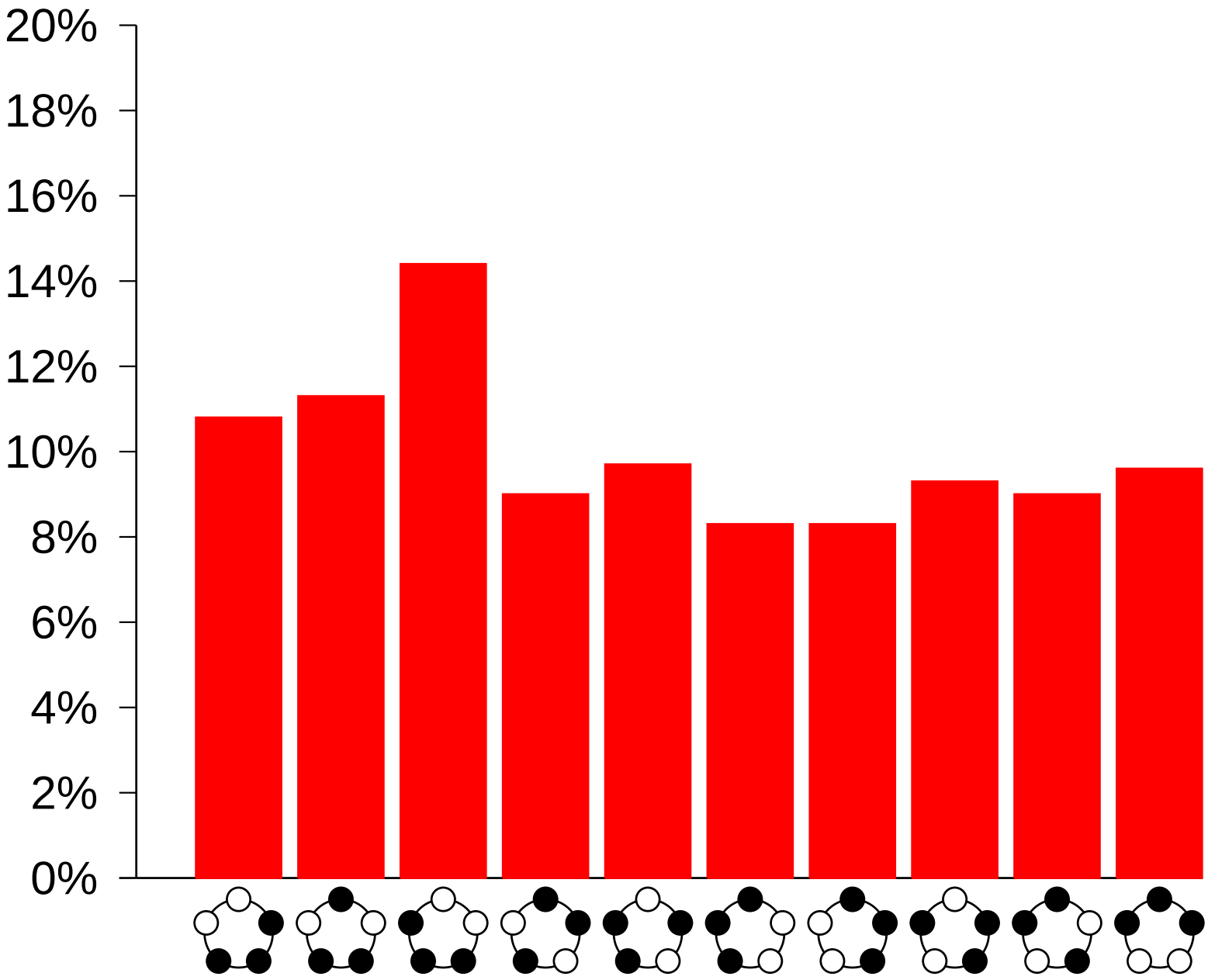
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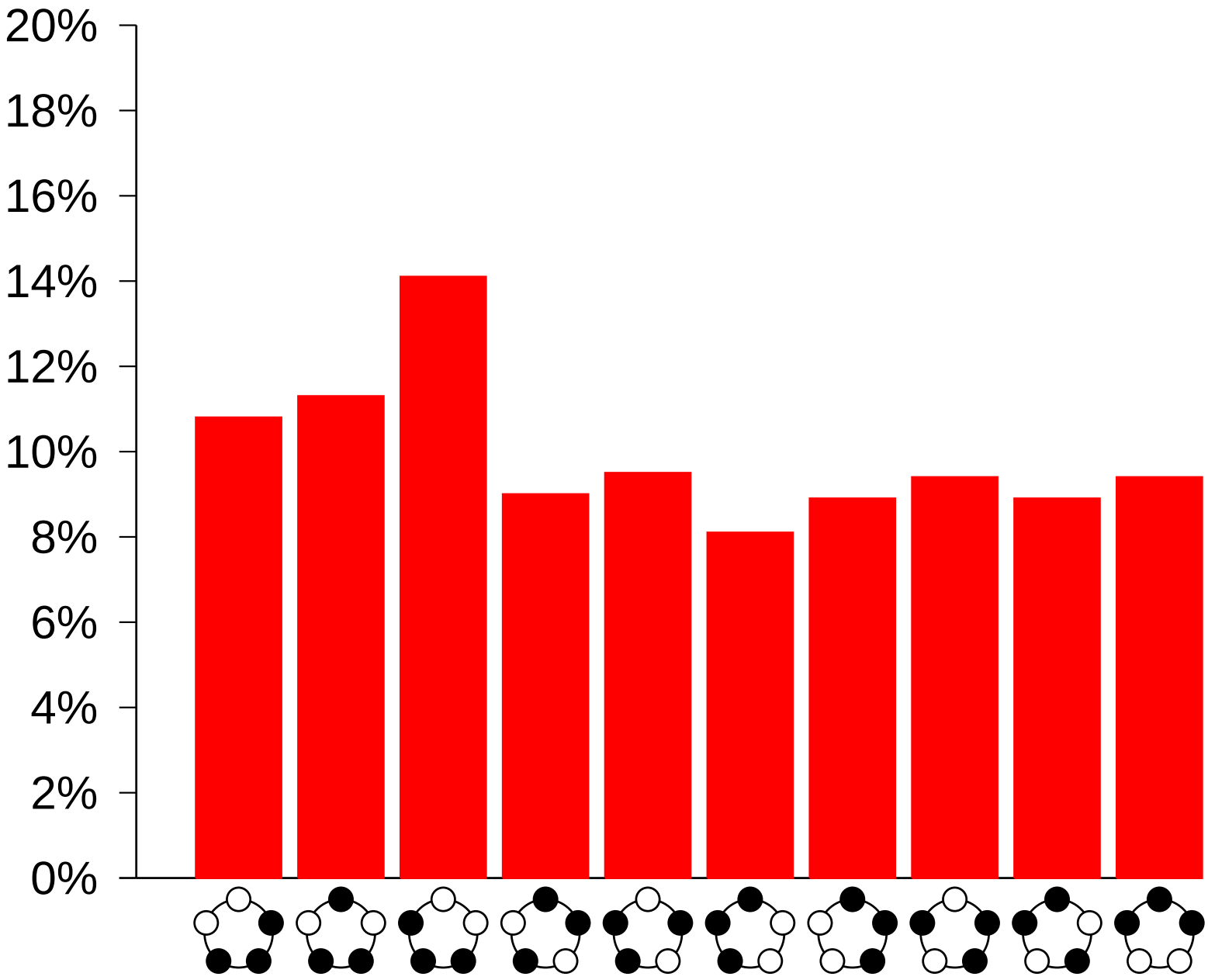
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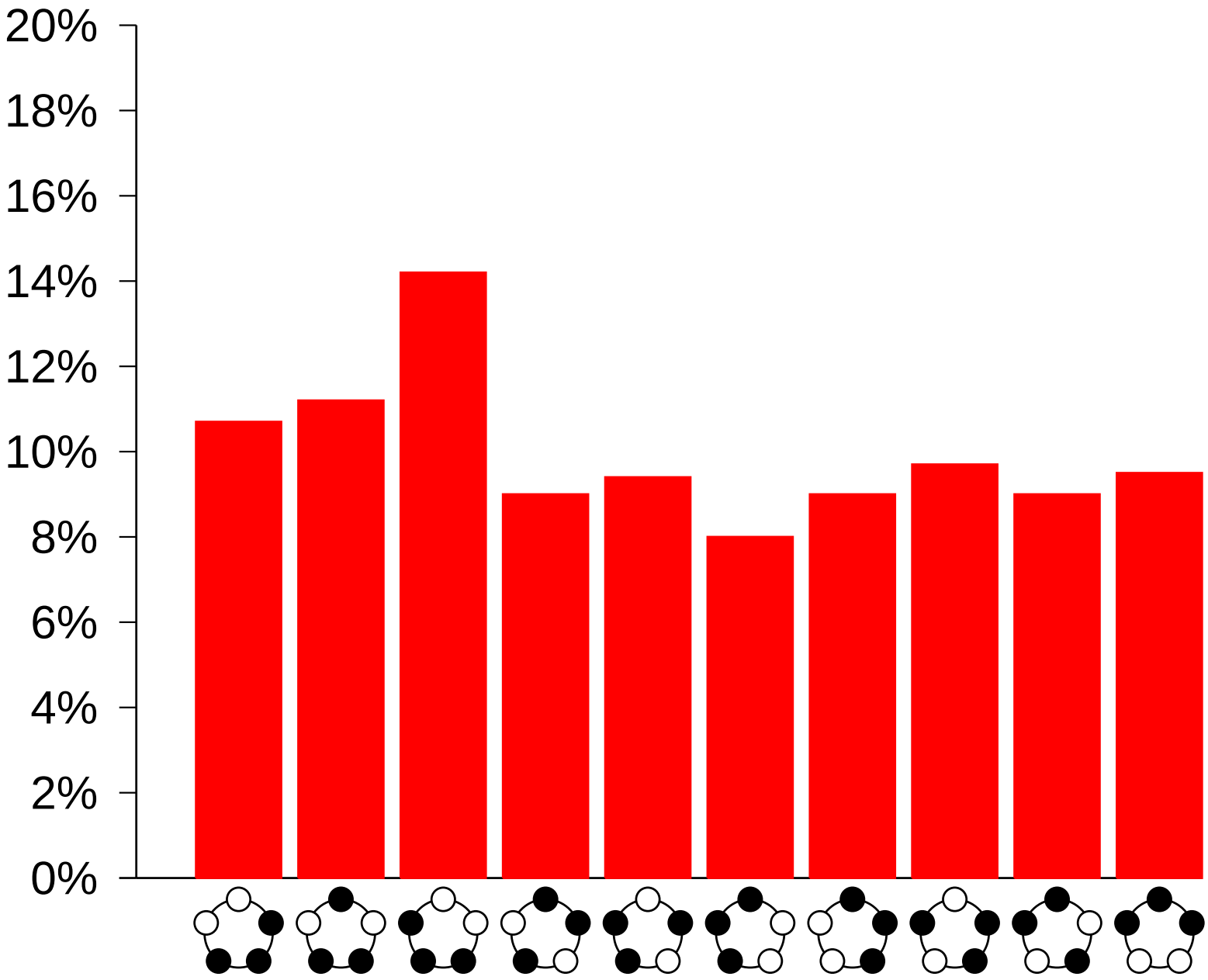
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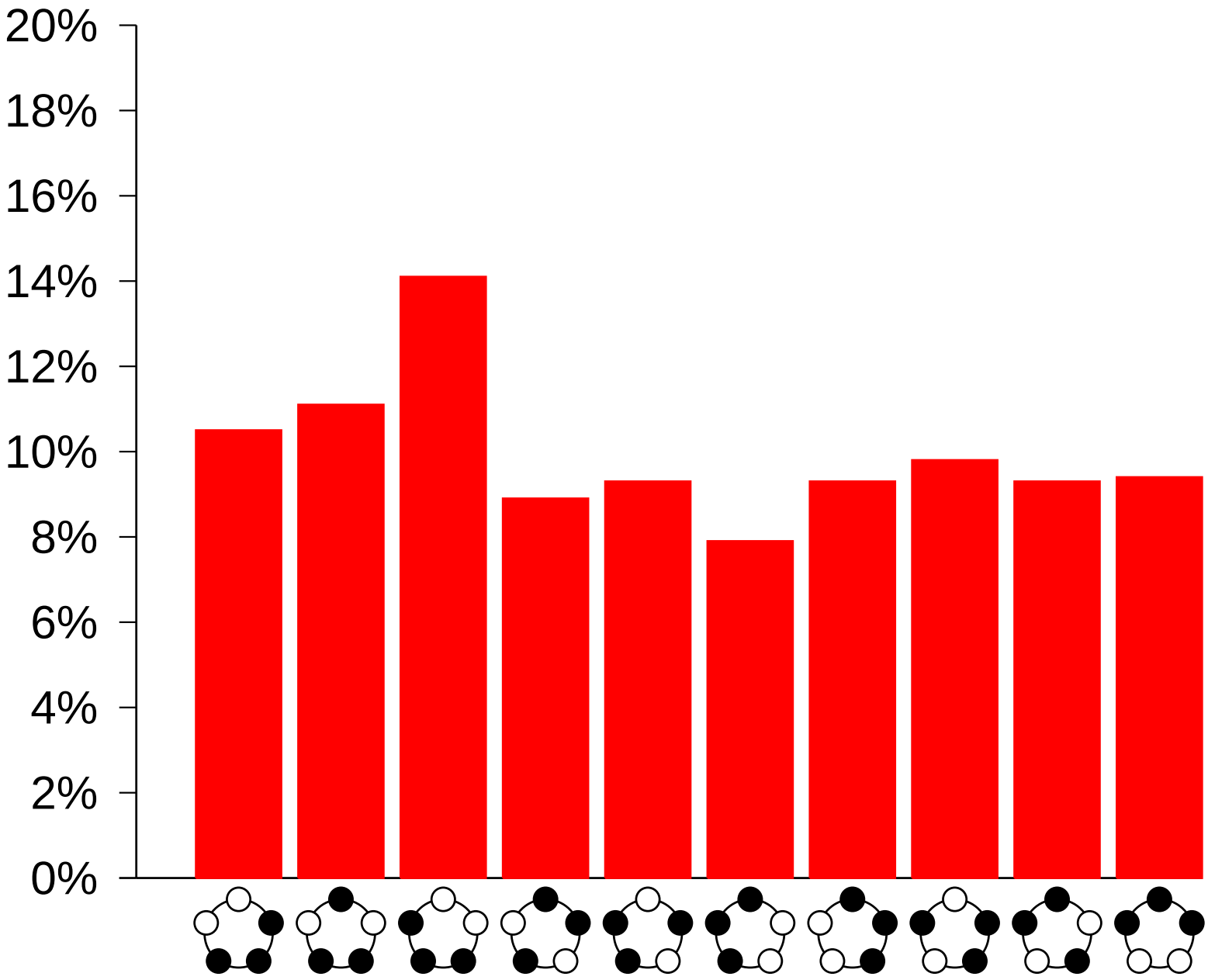
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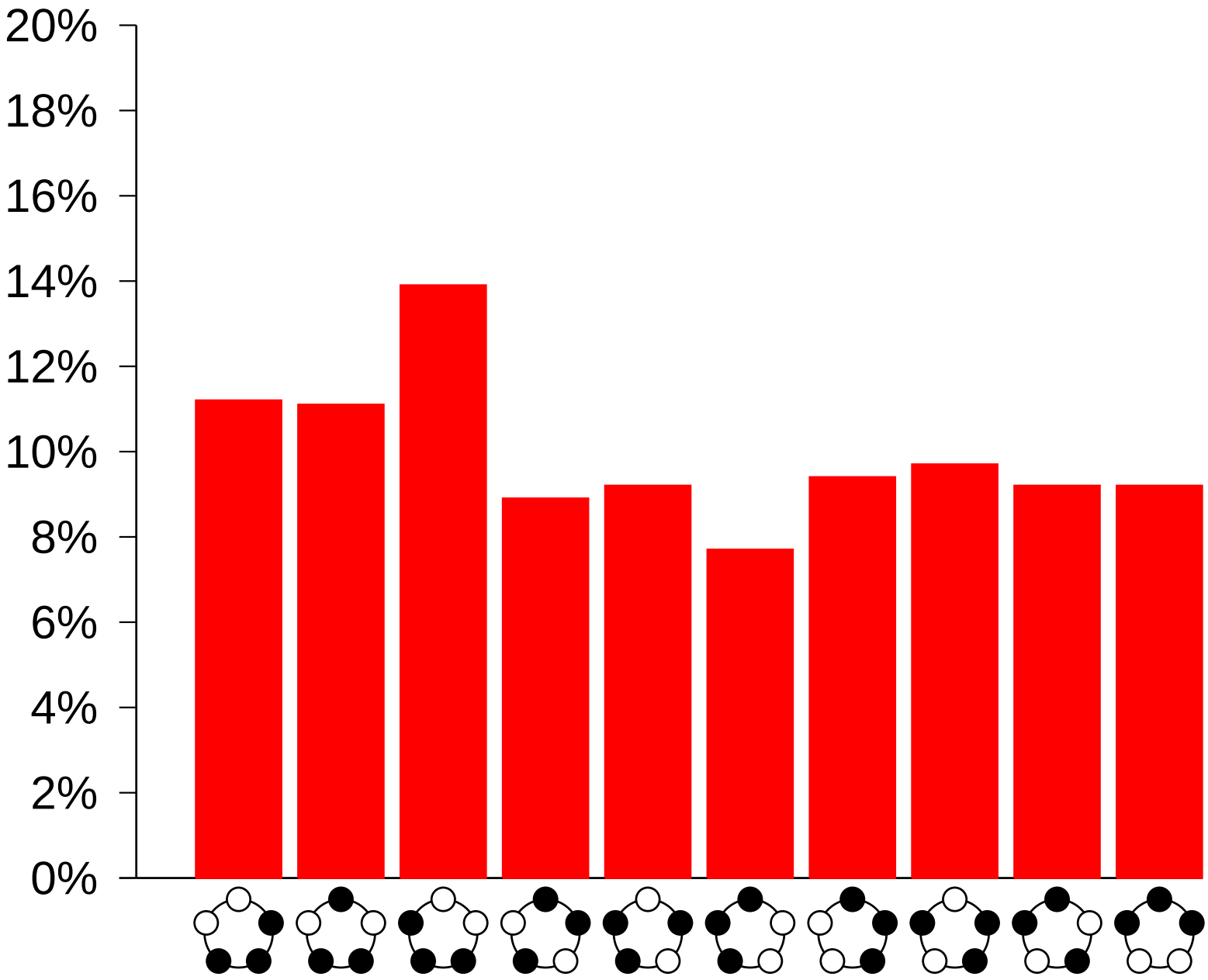
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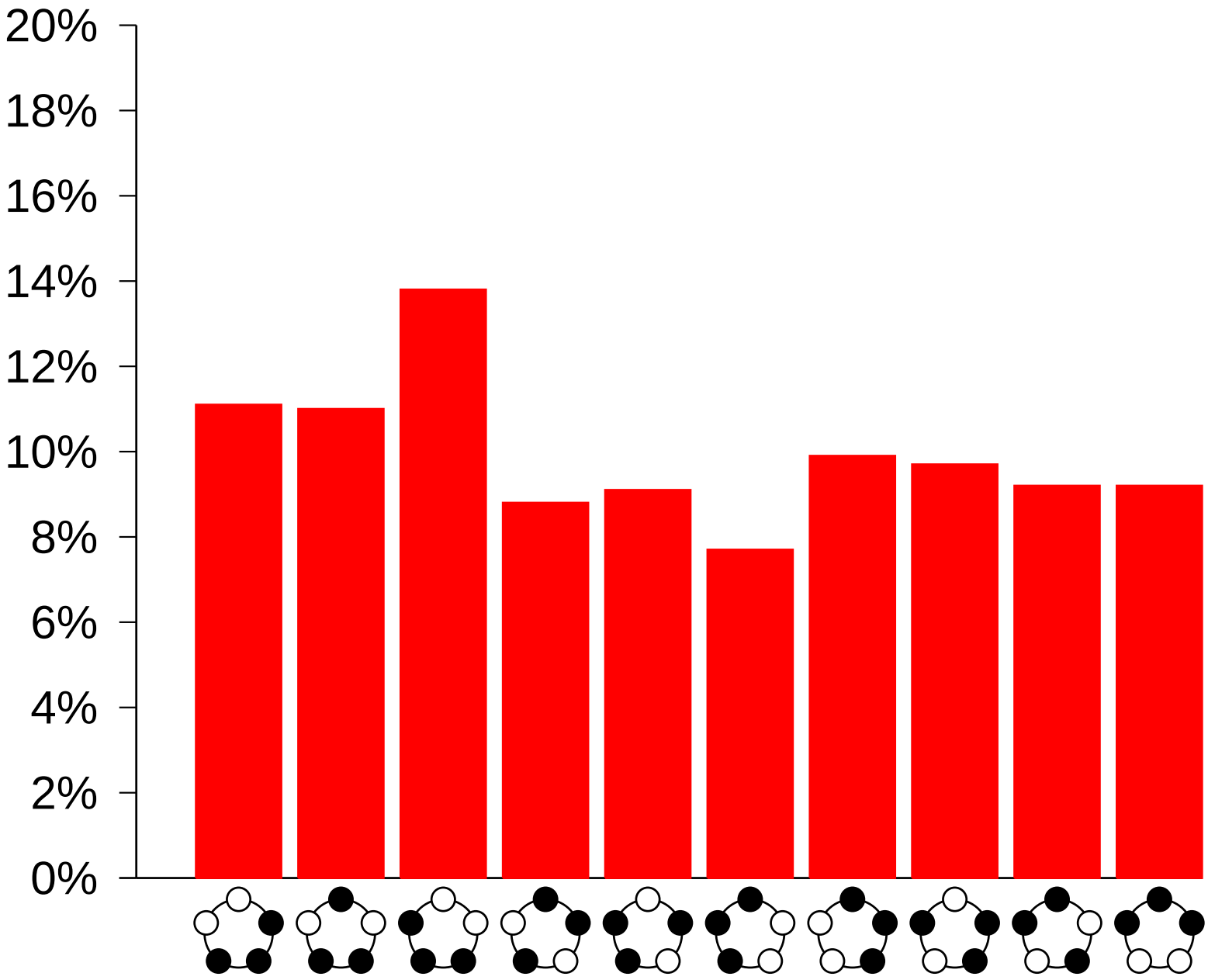
Stationary distribution



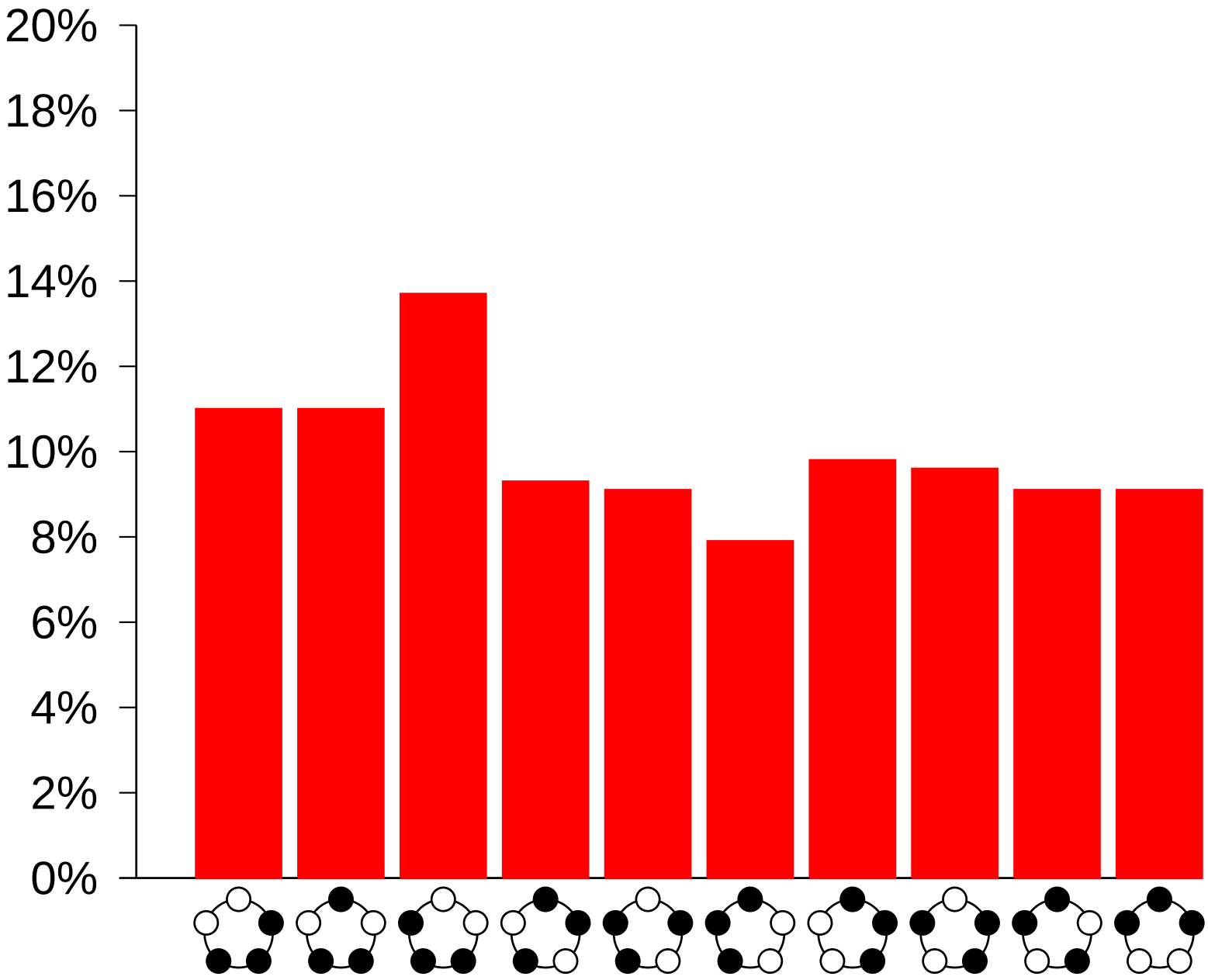
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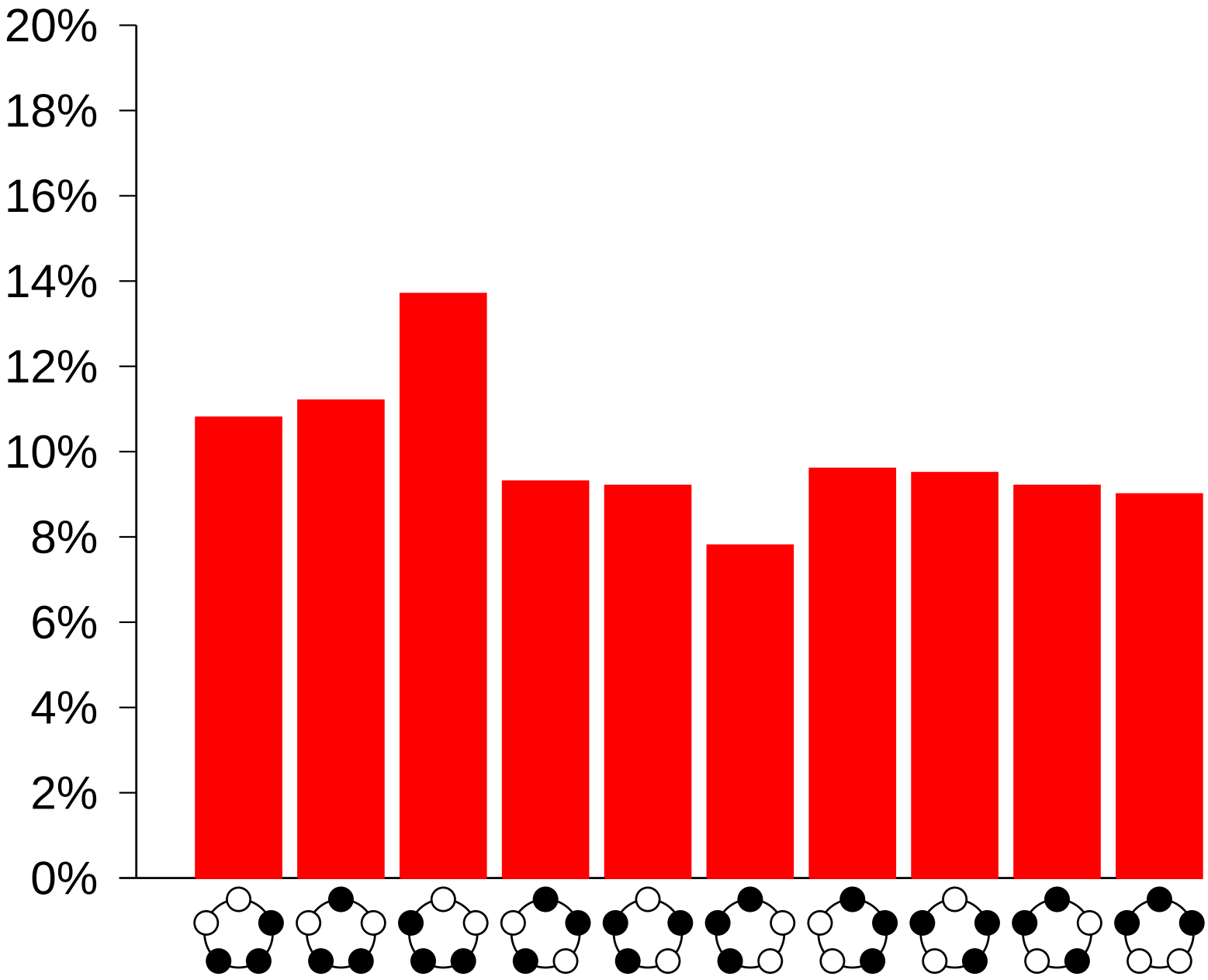
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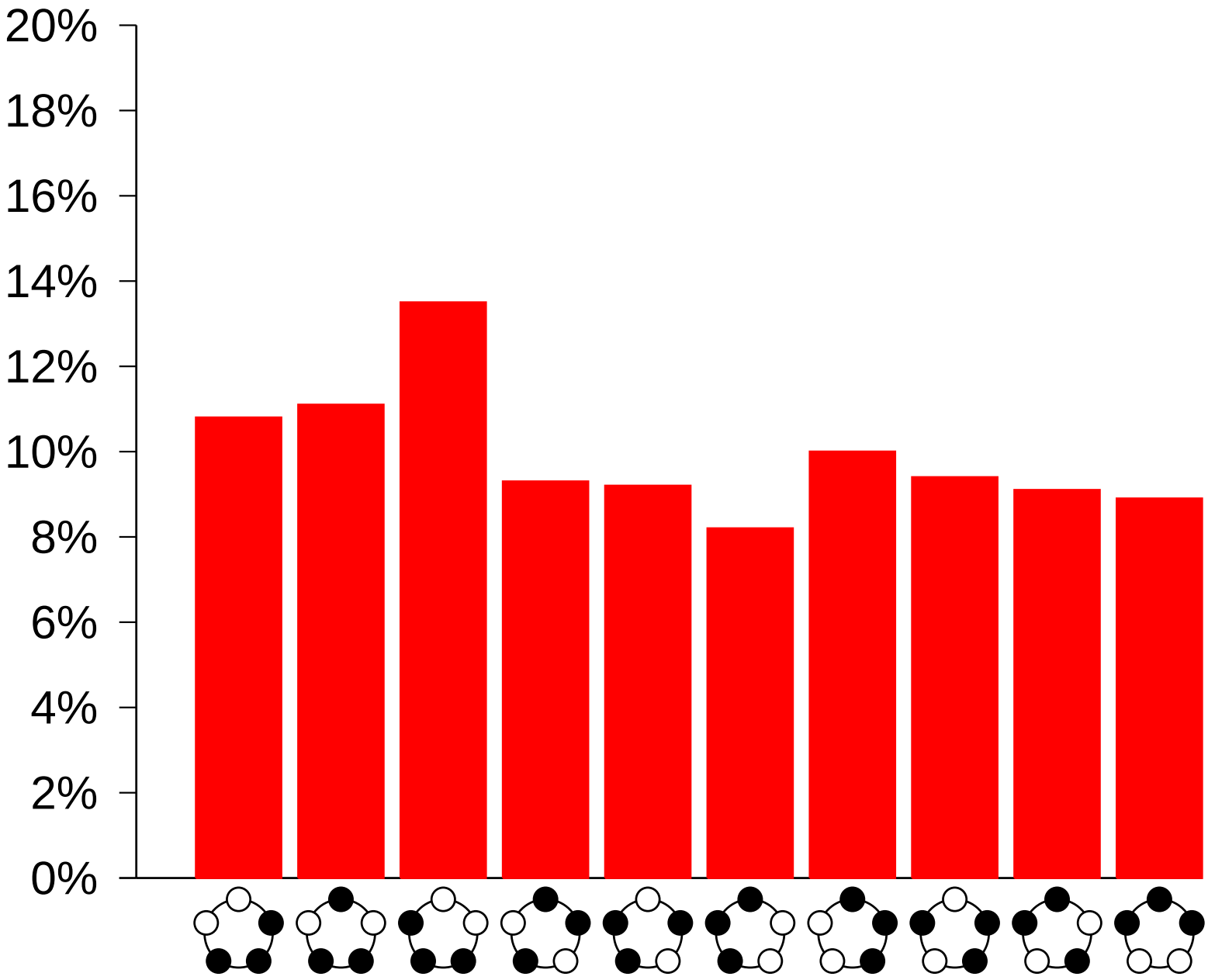
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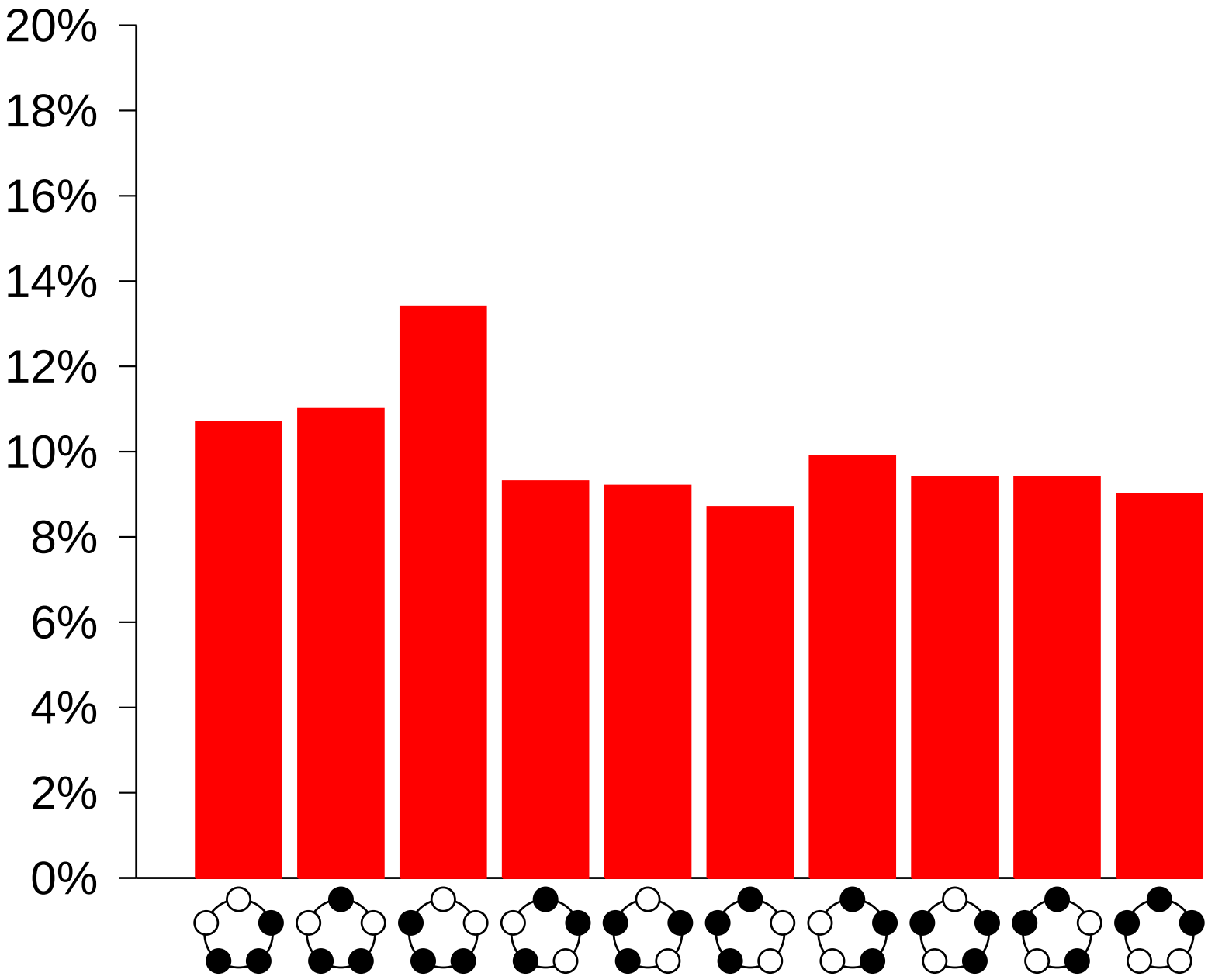
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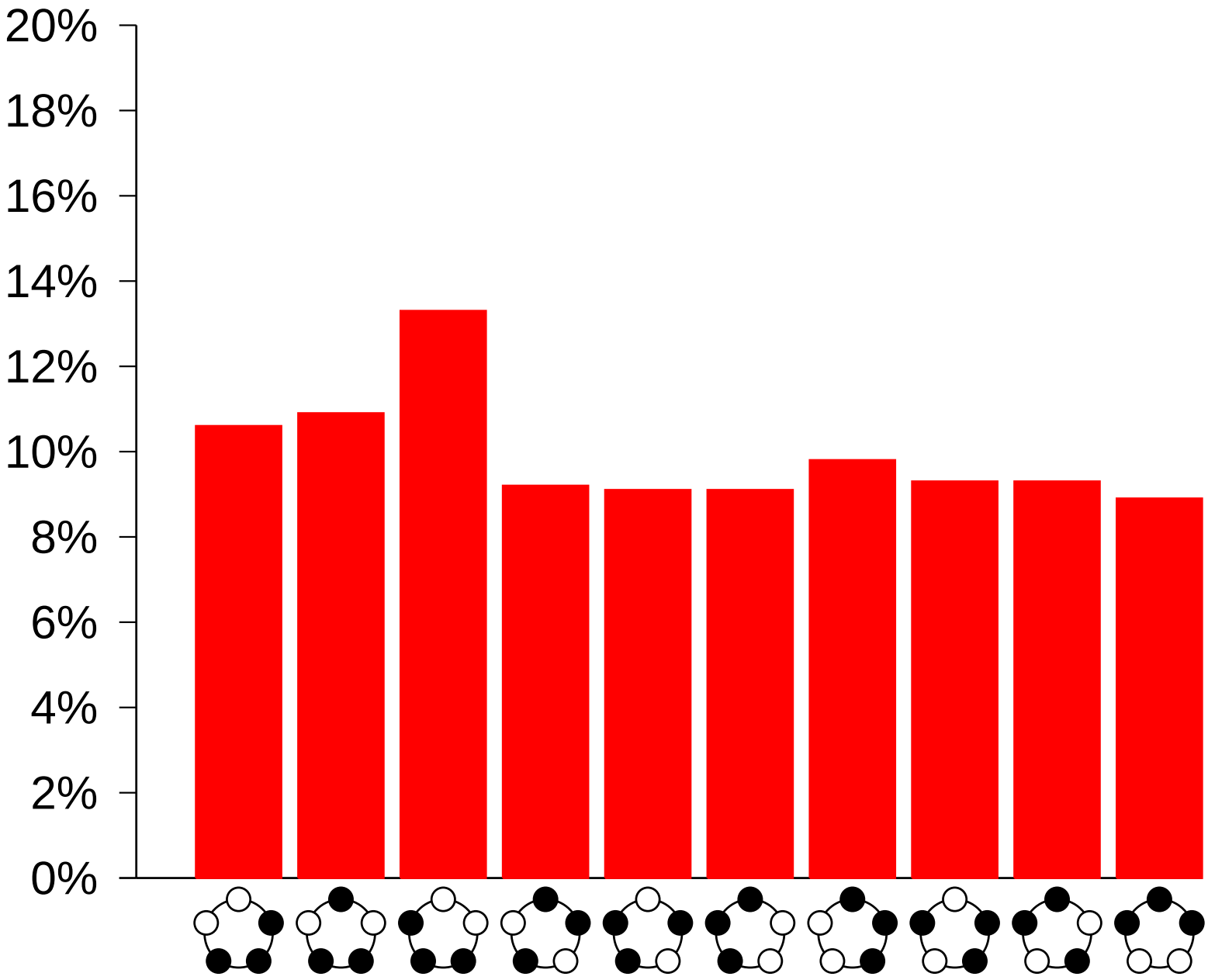
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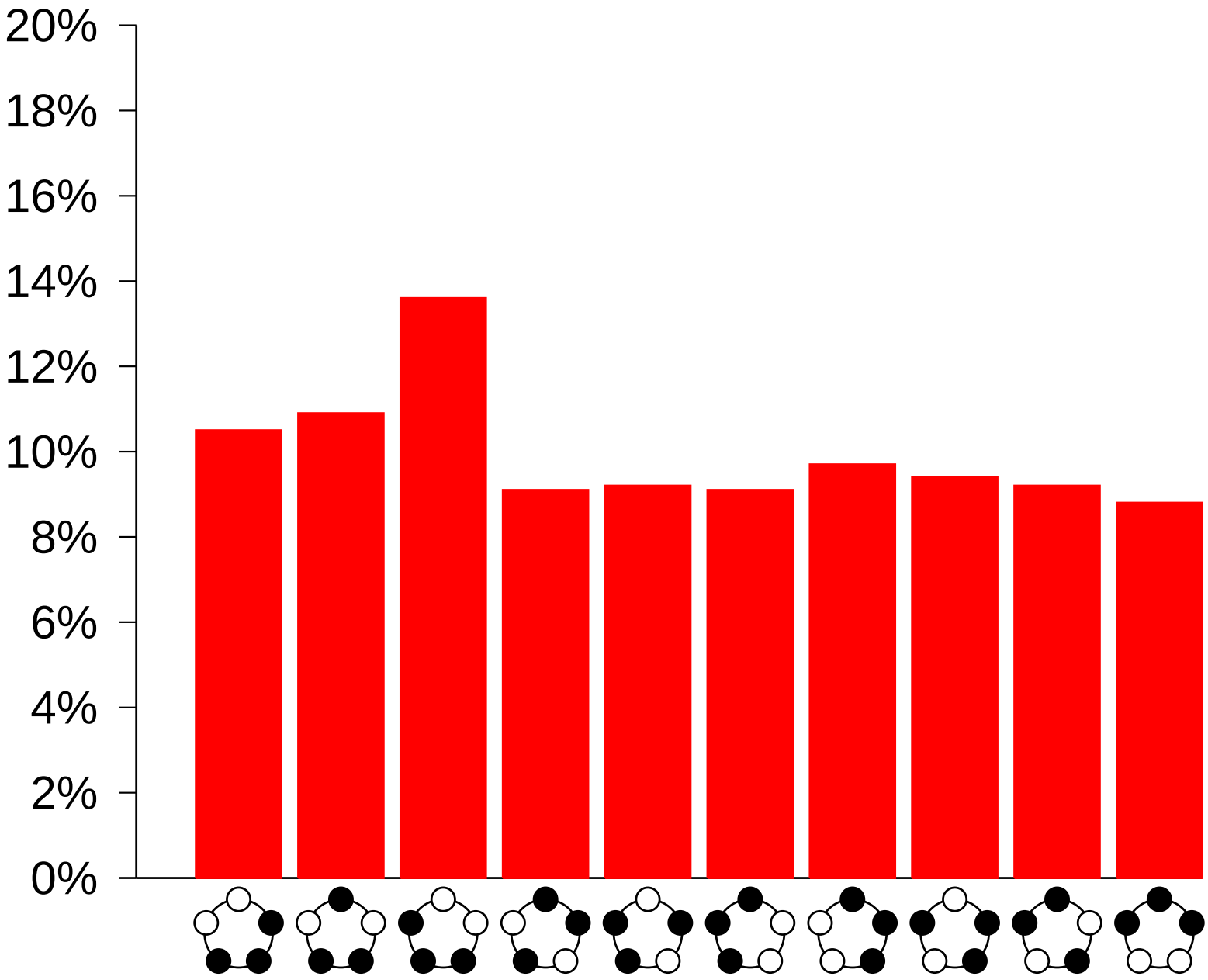
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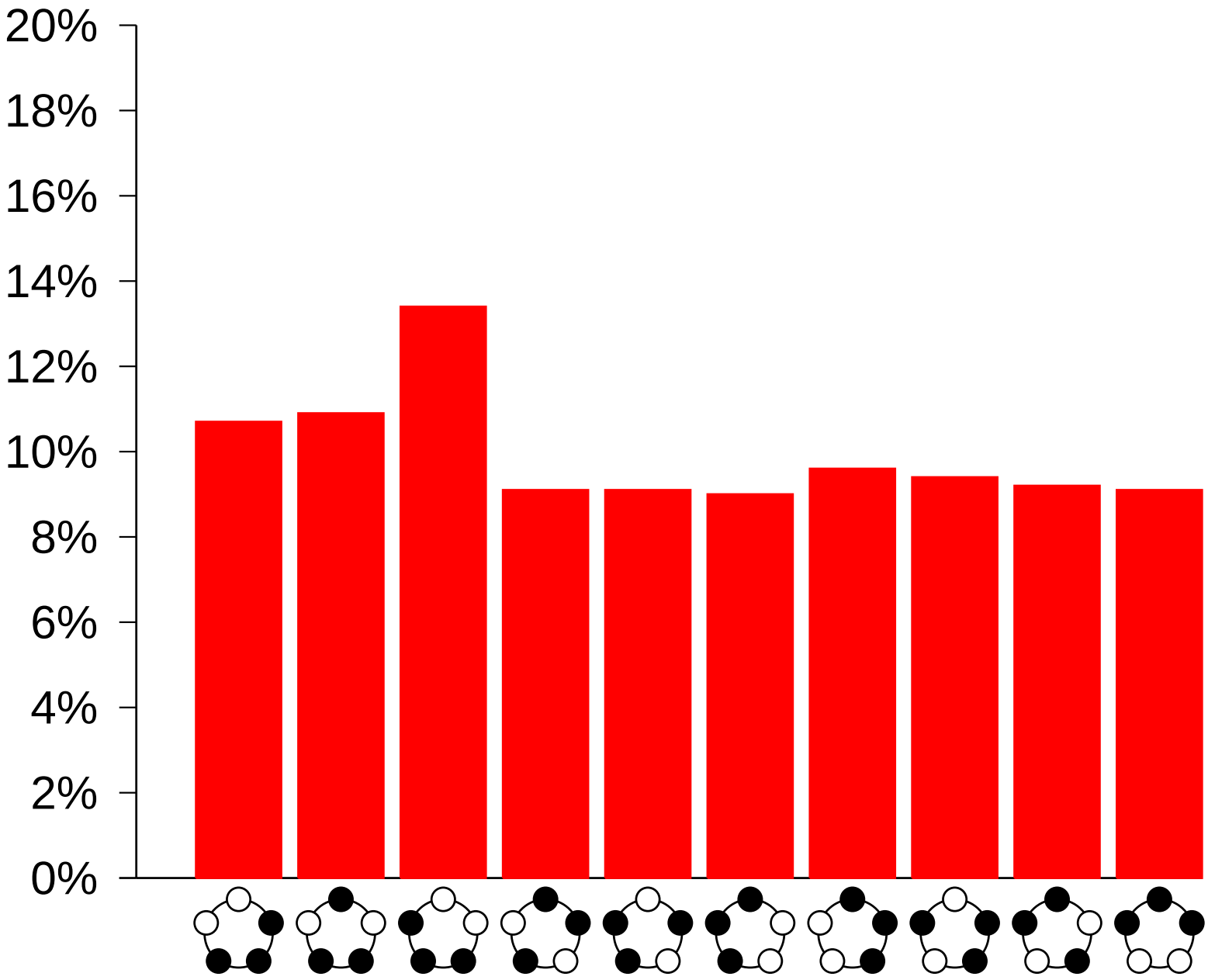
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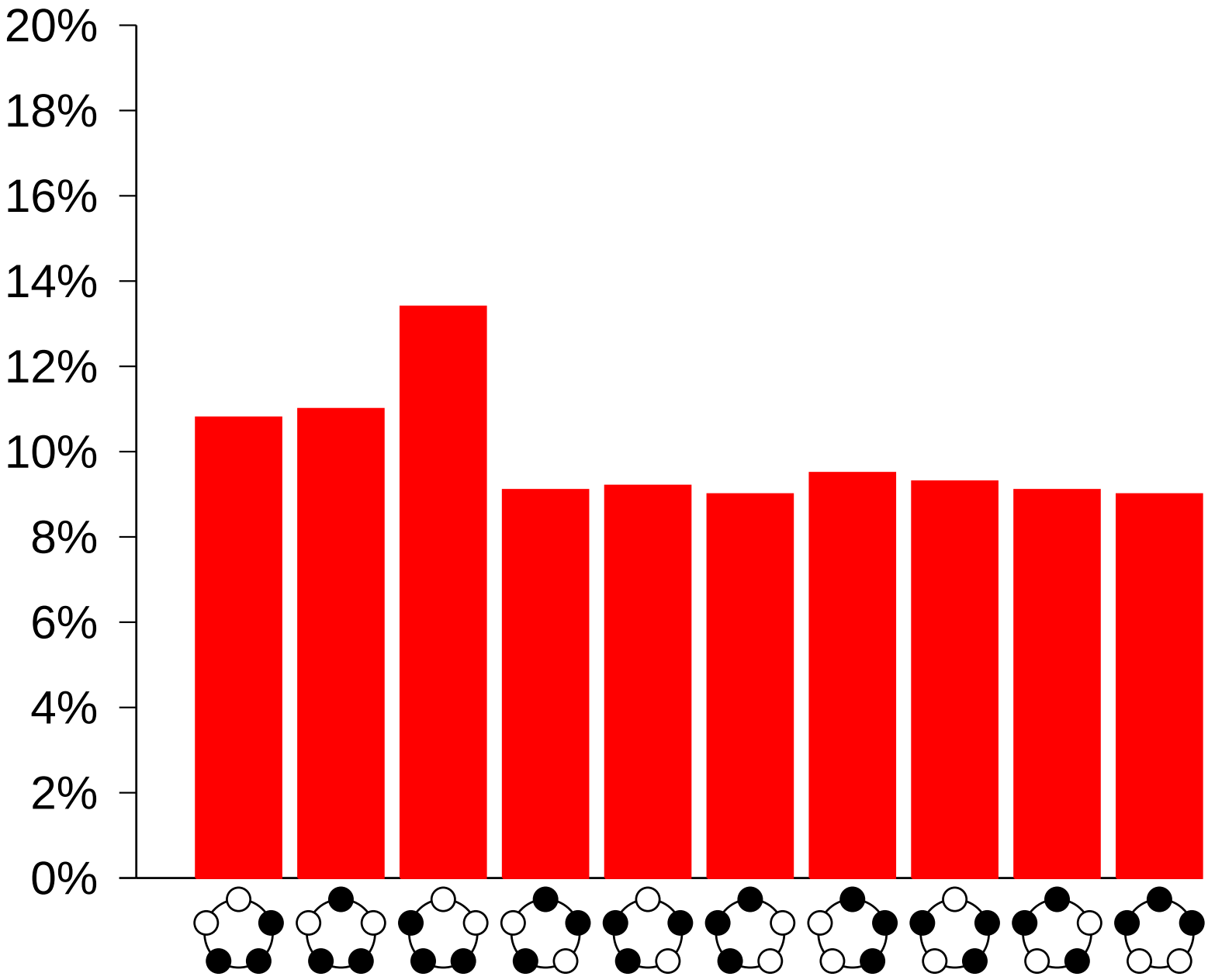
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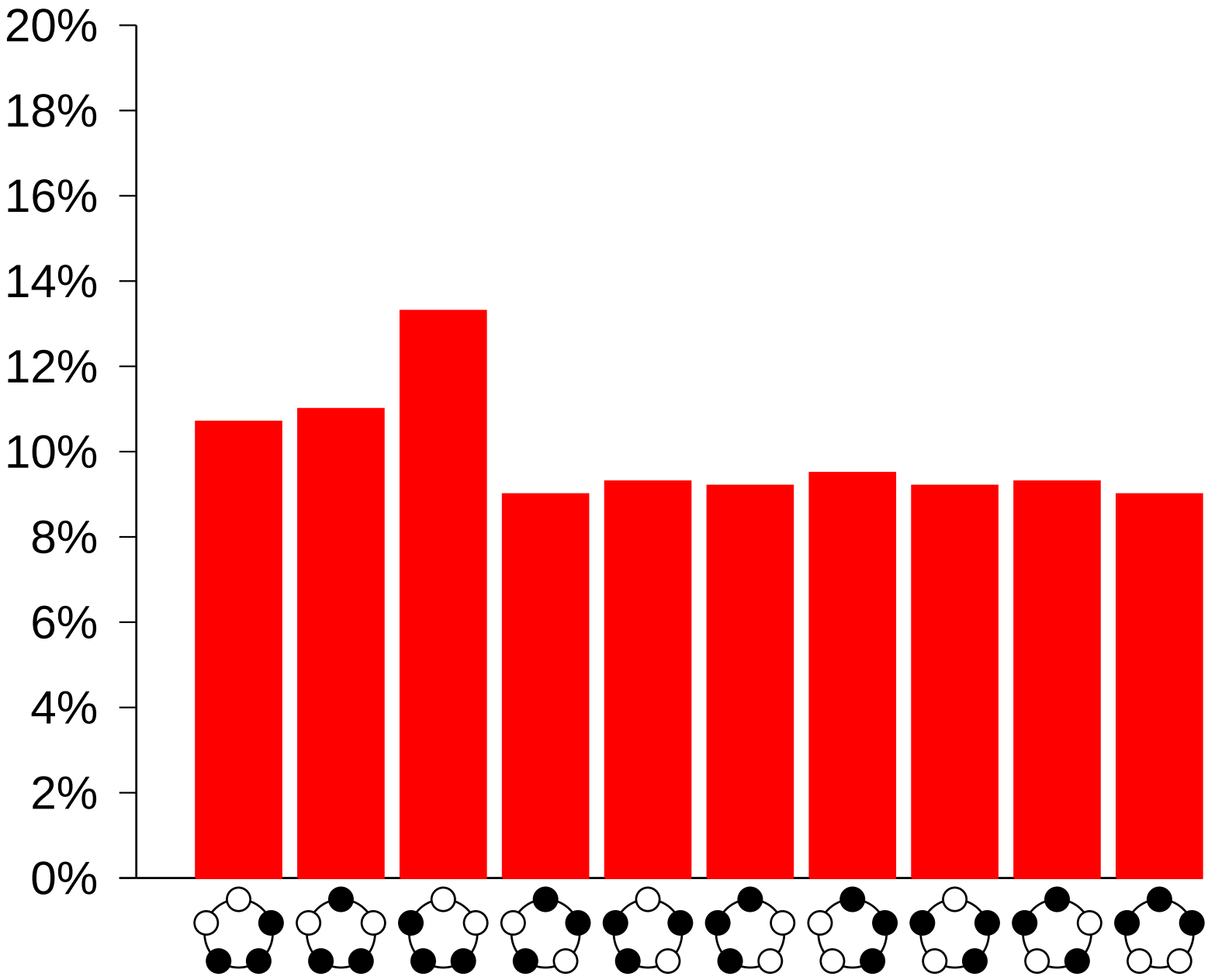
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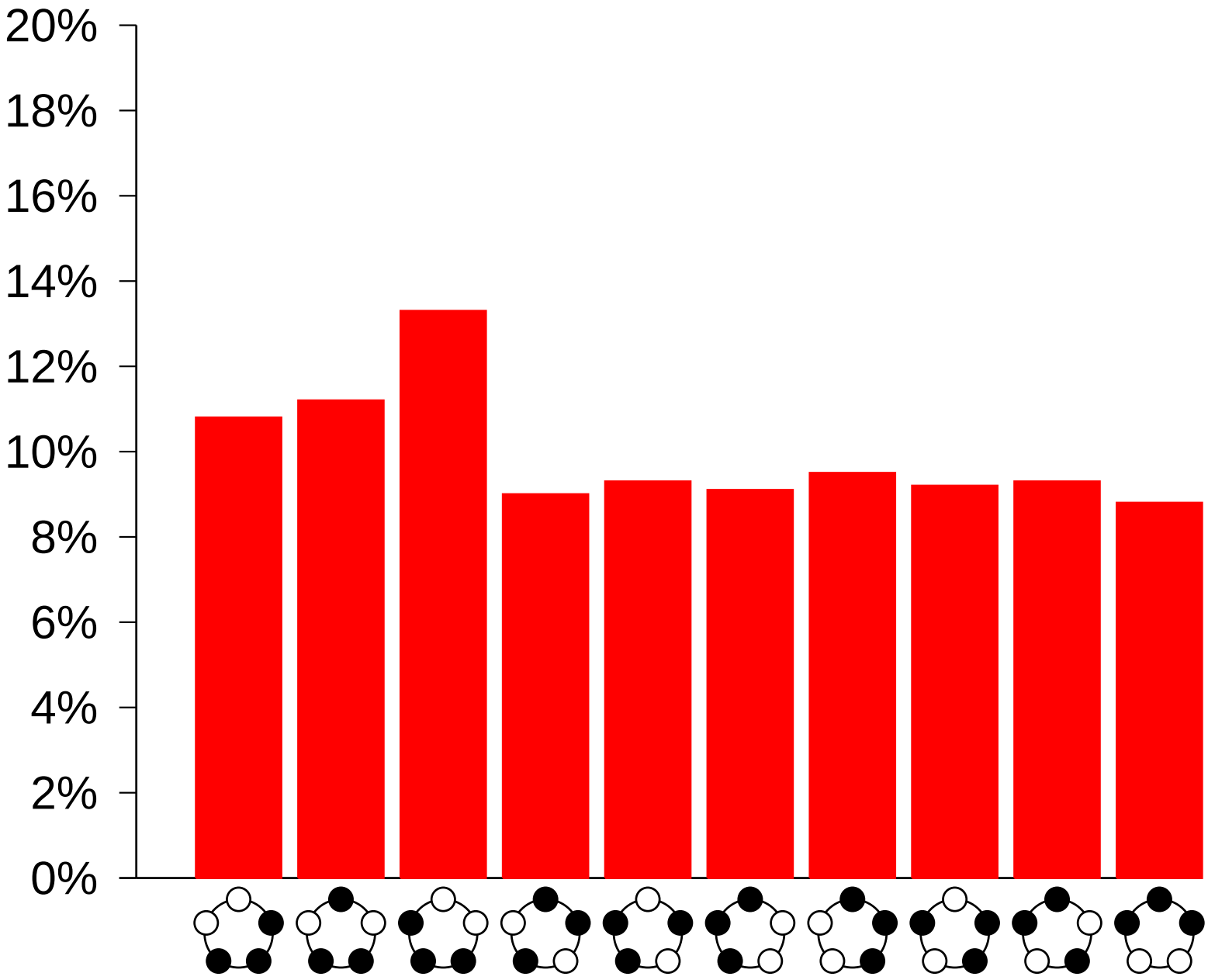
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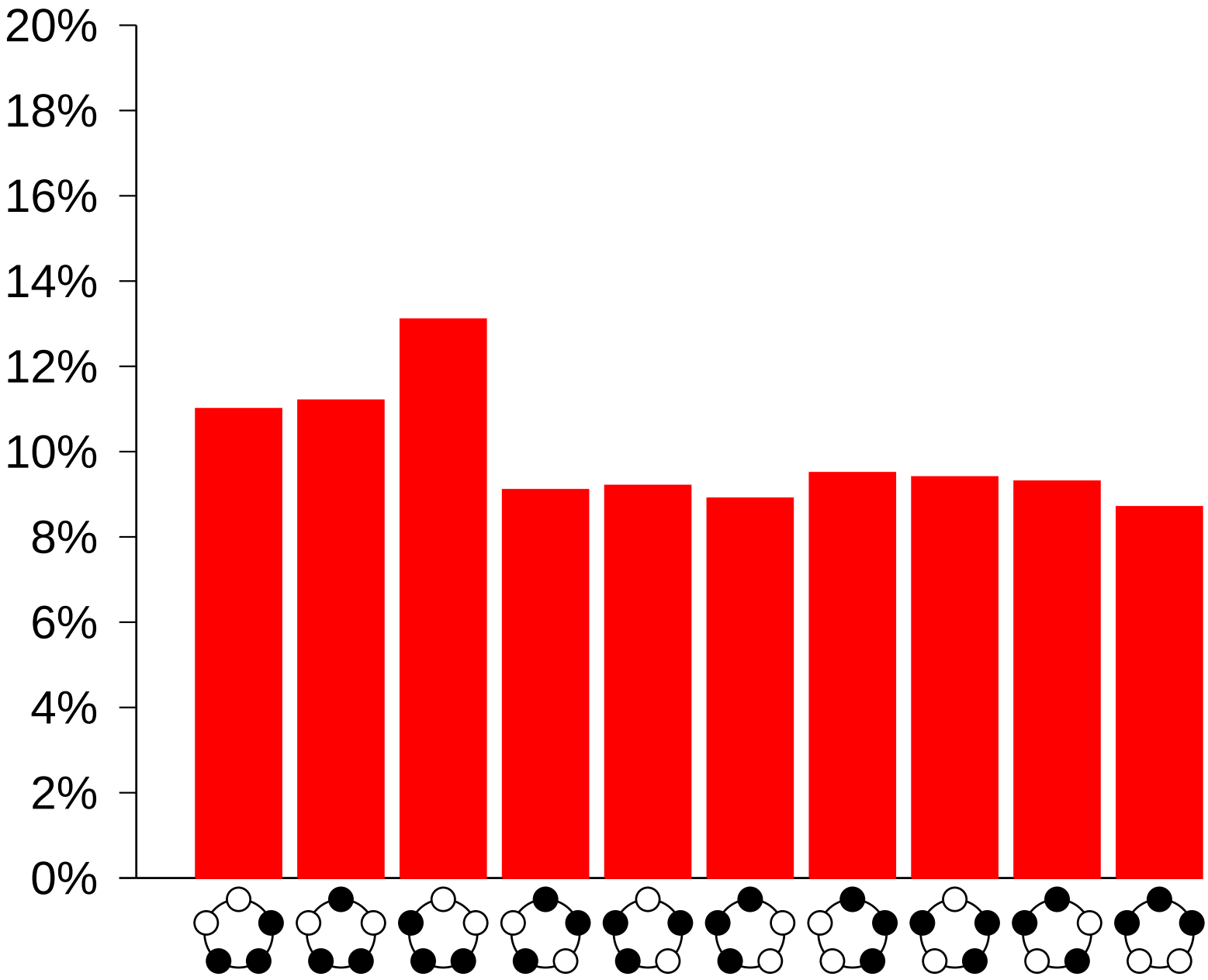
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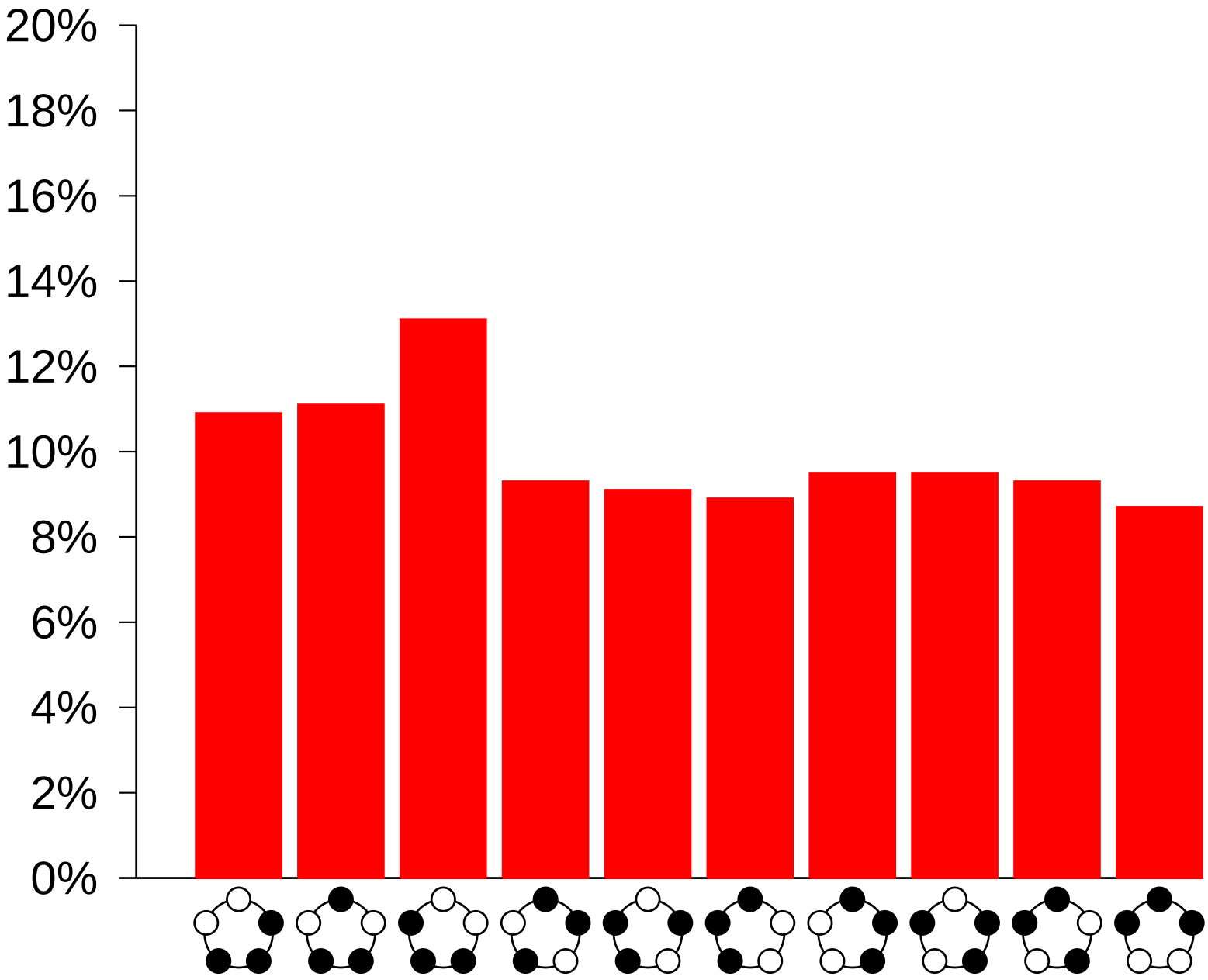
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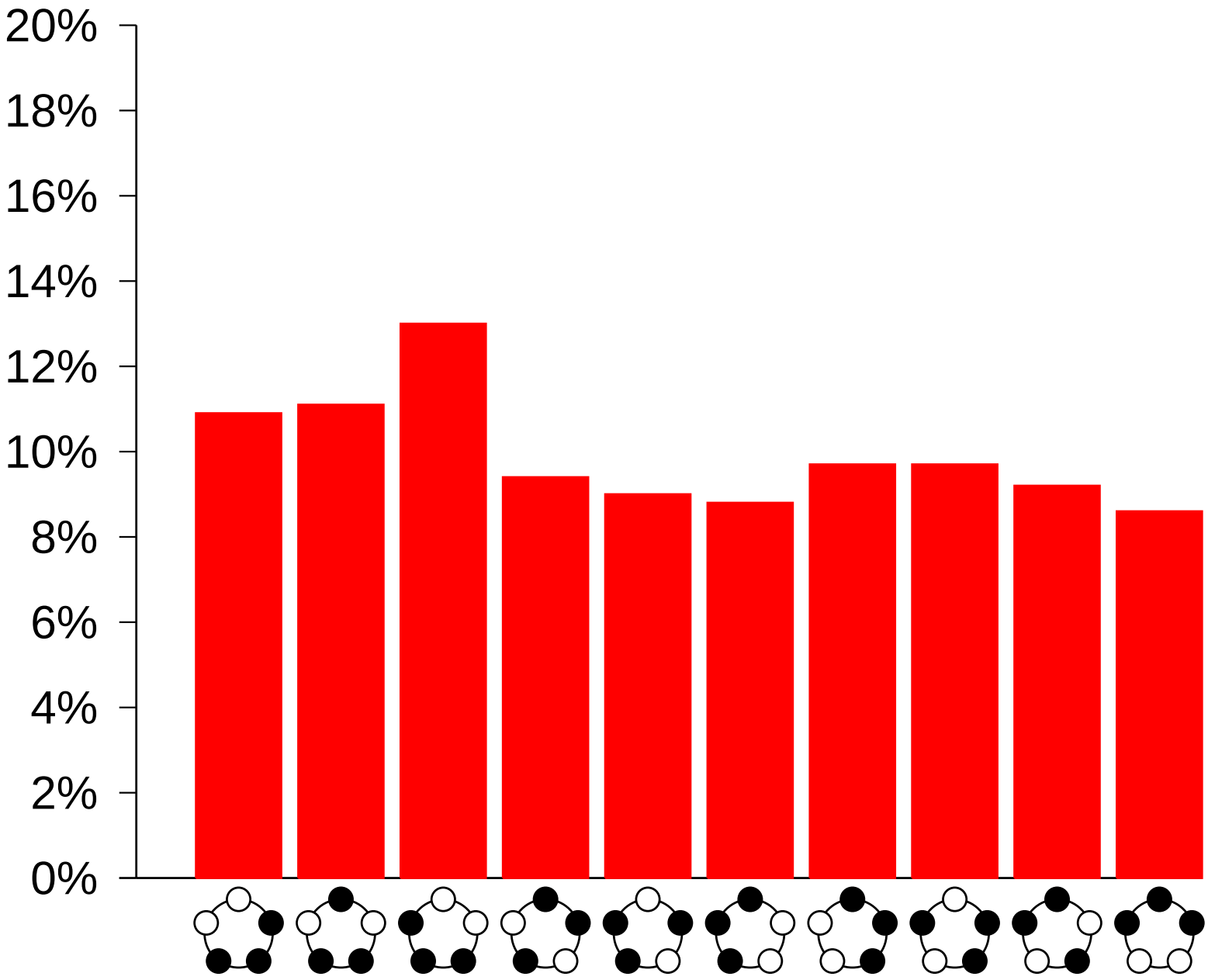
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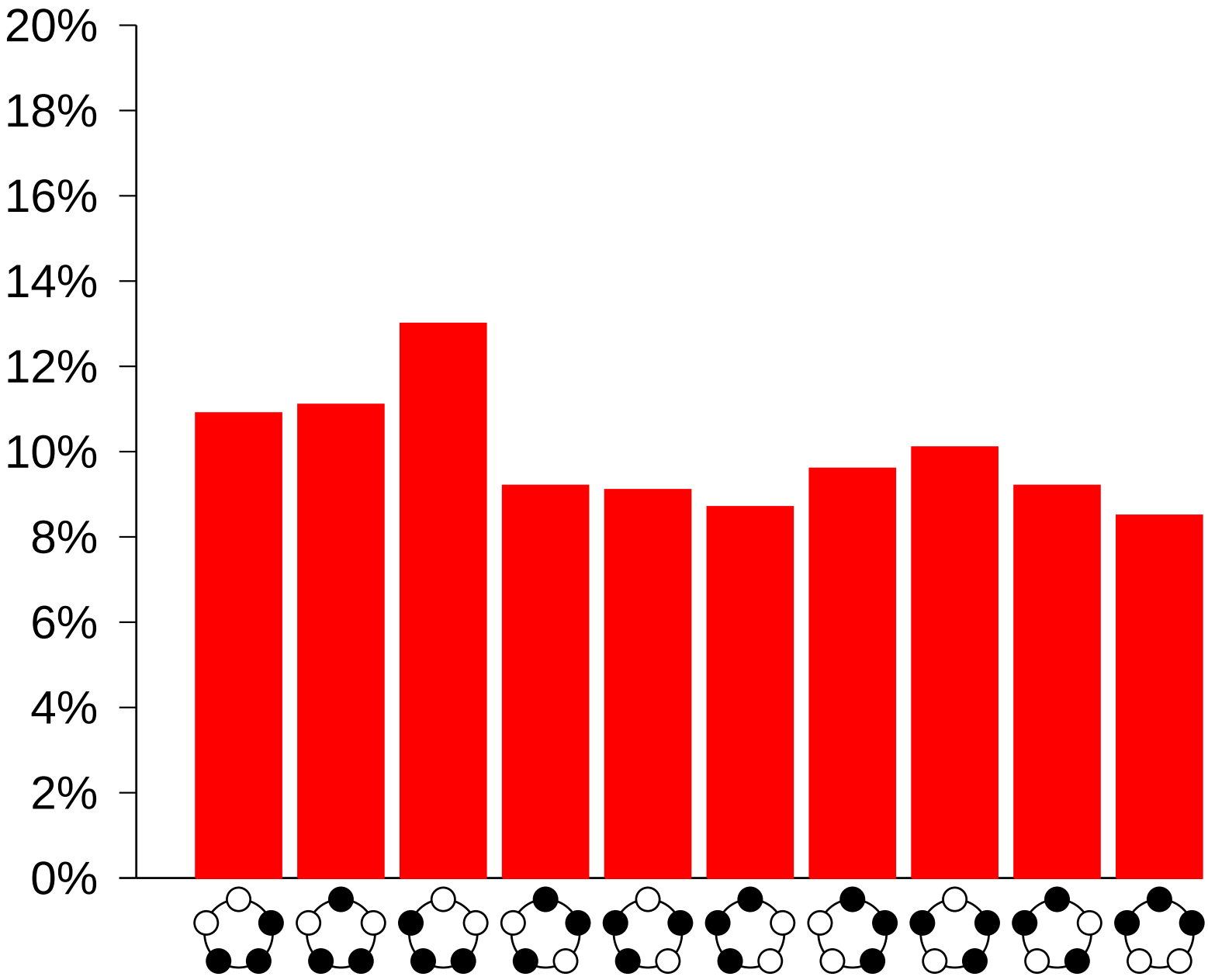
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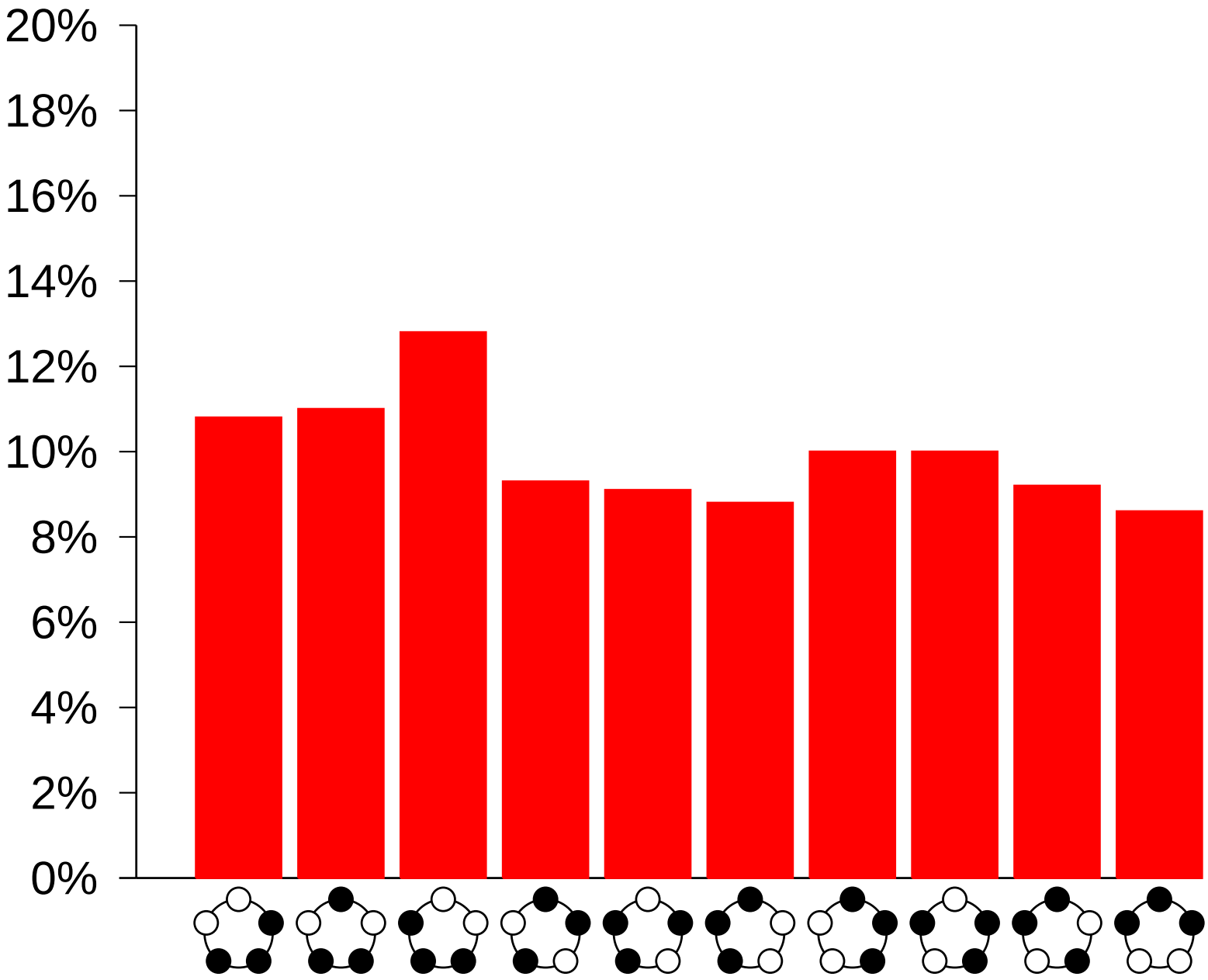
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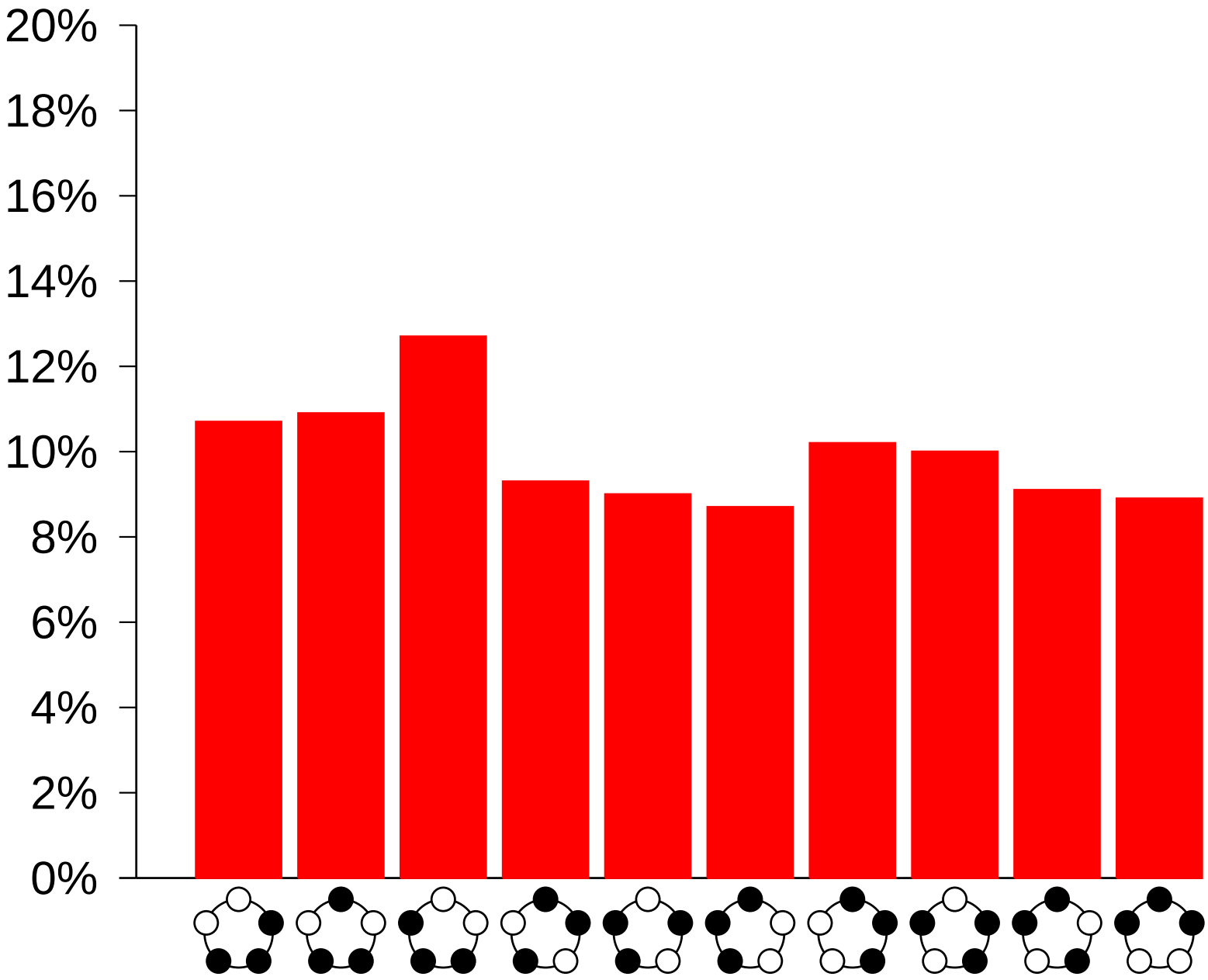
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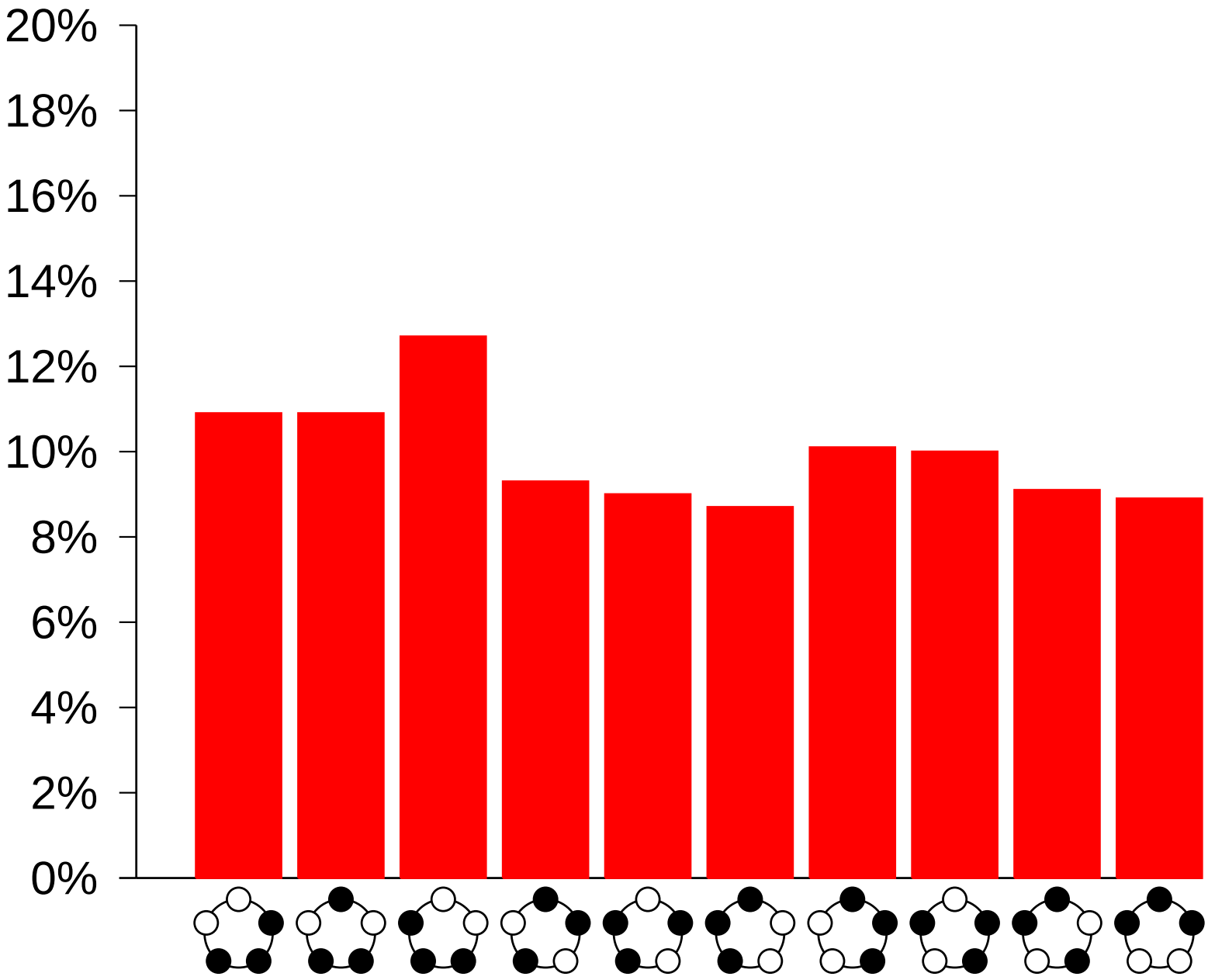
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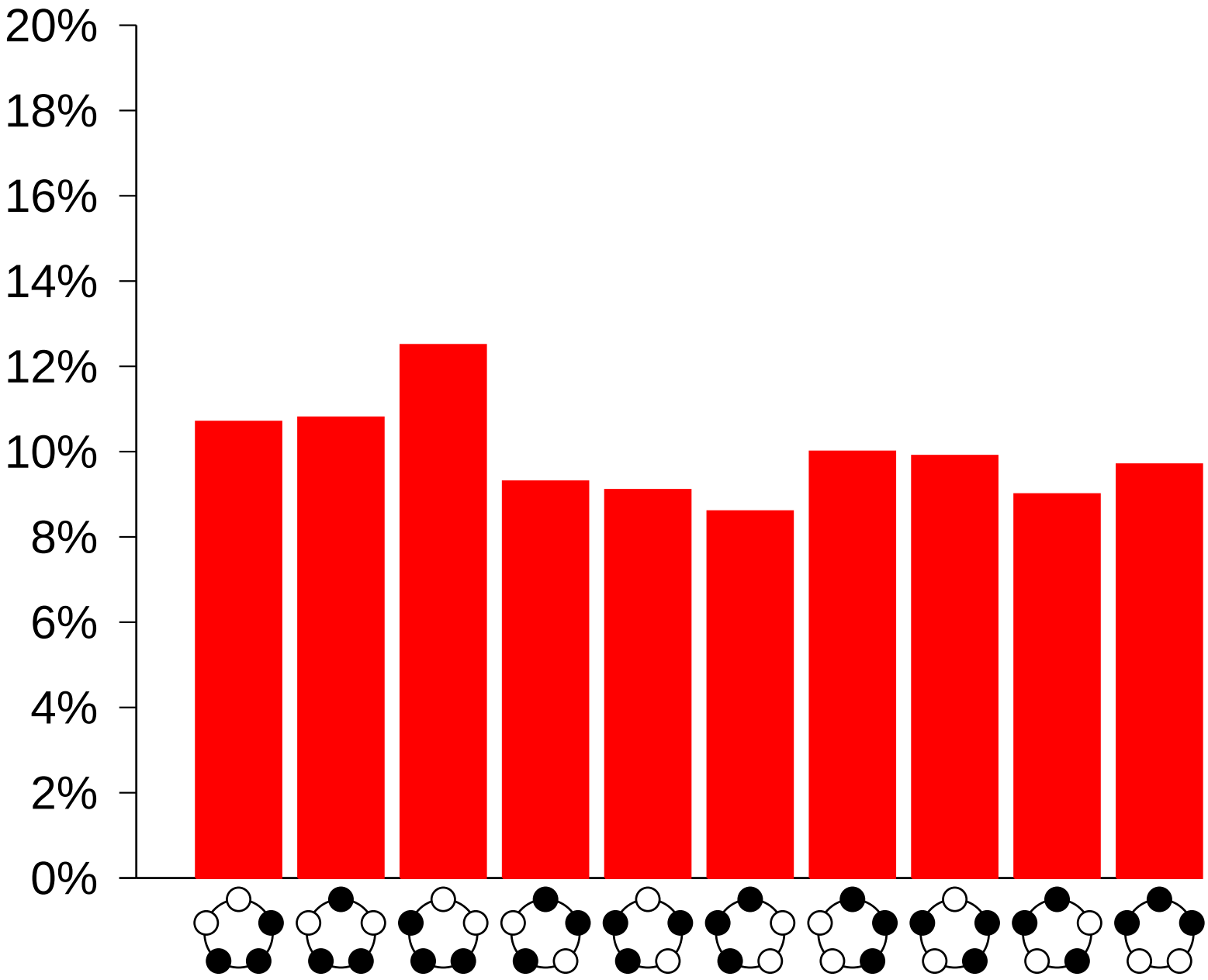
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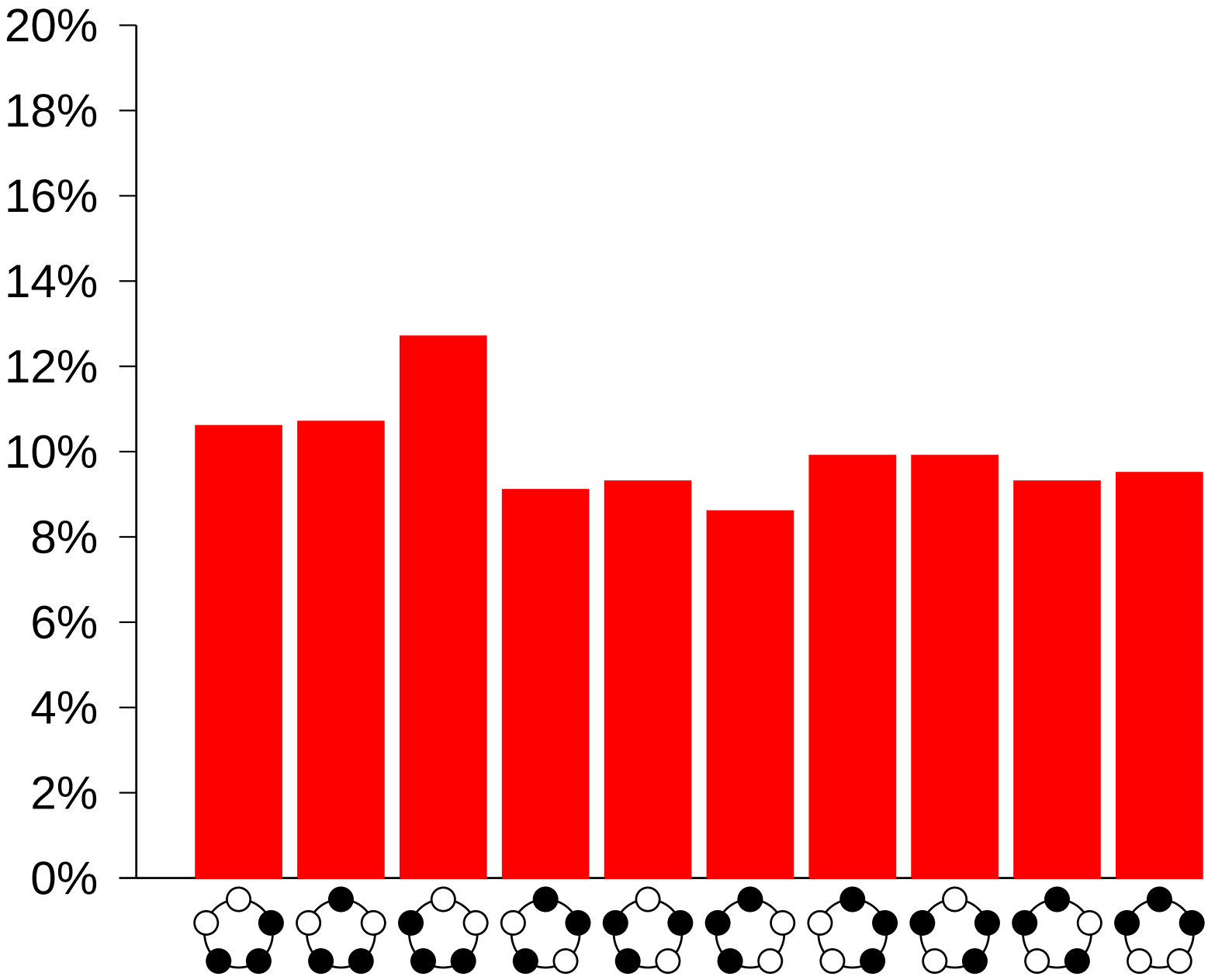
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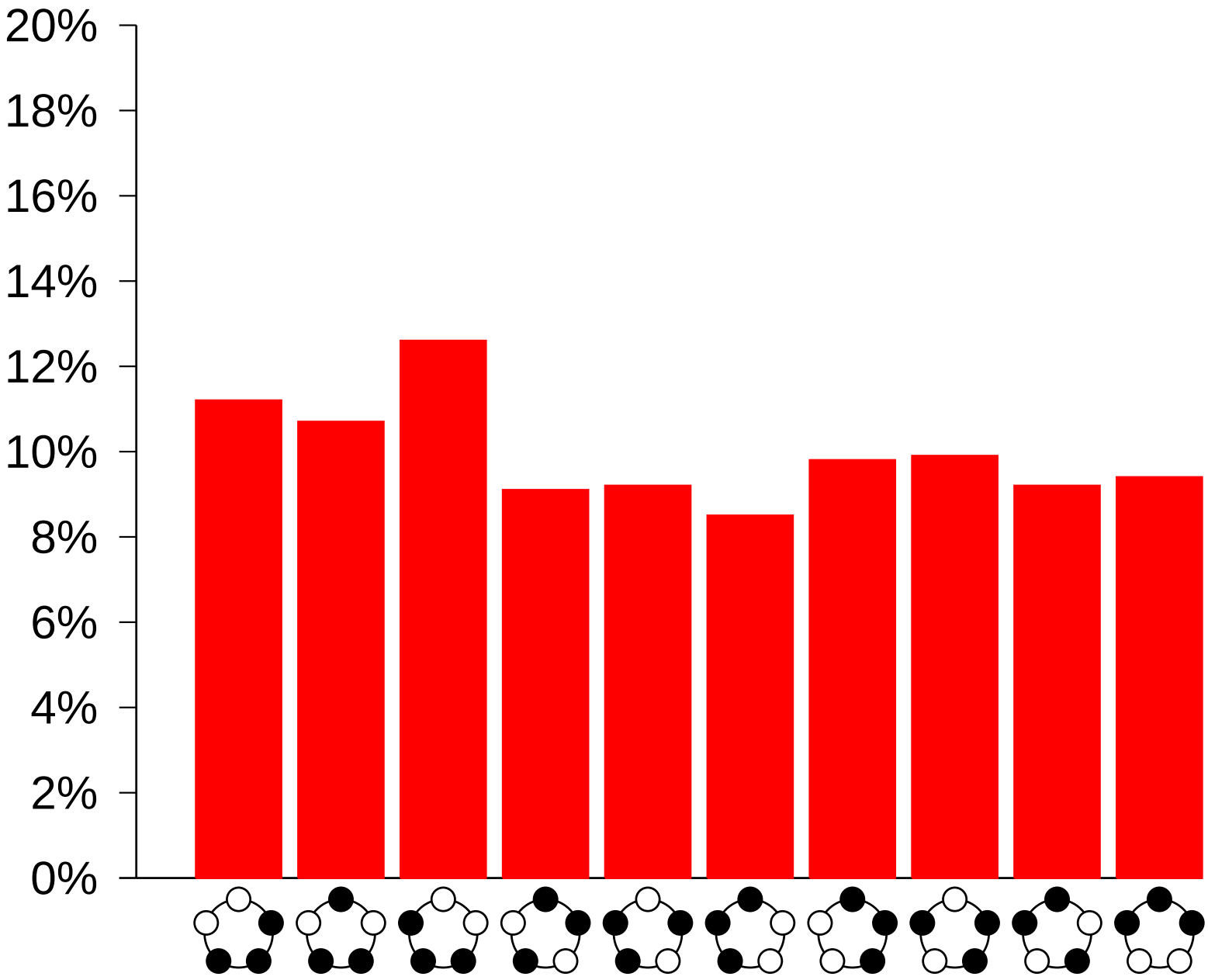
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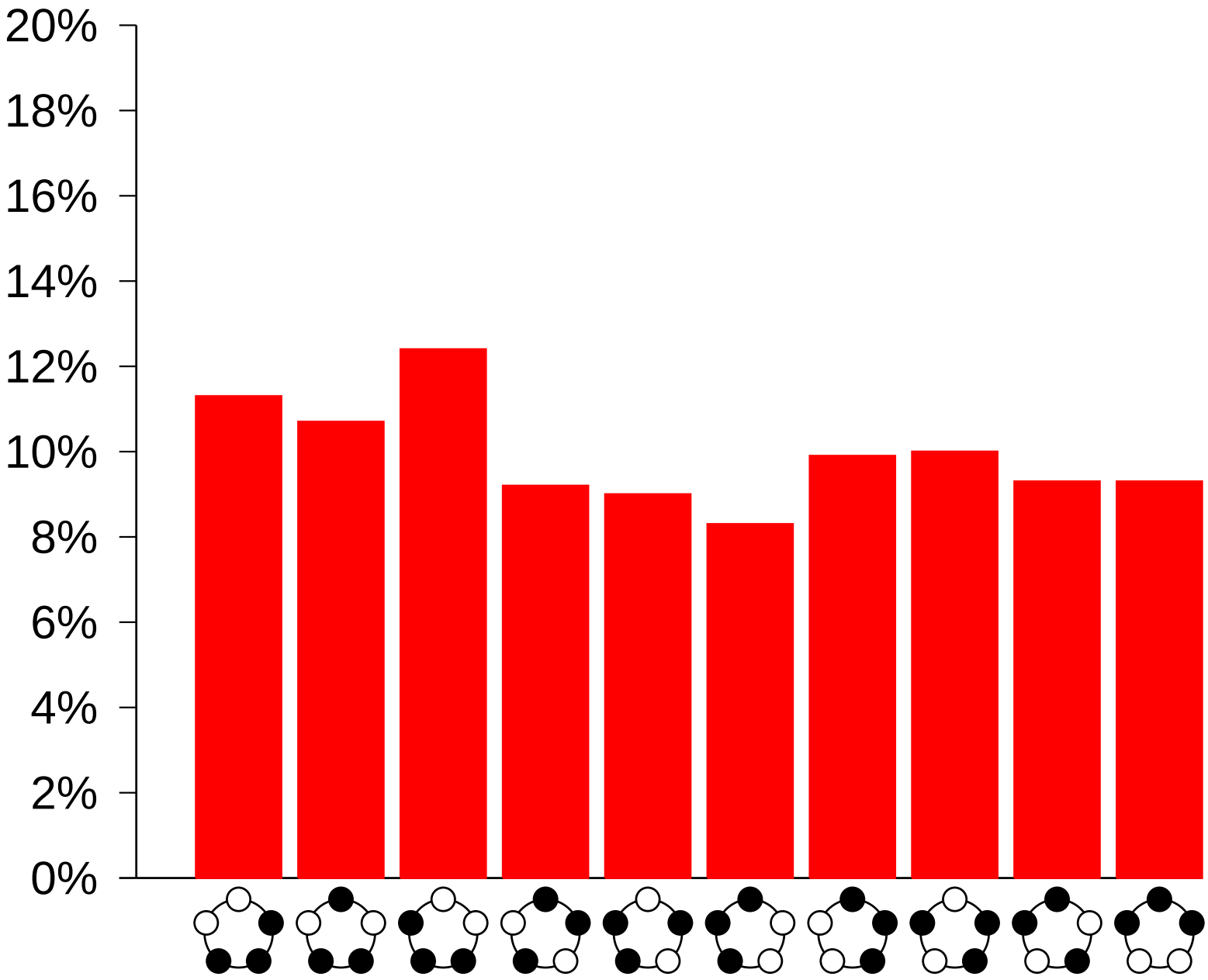
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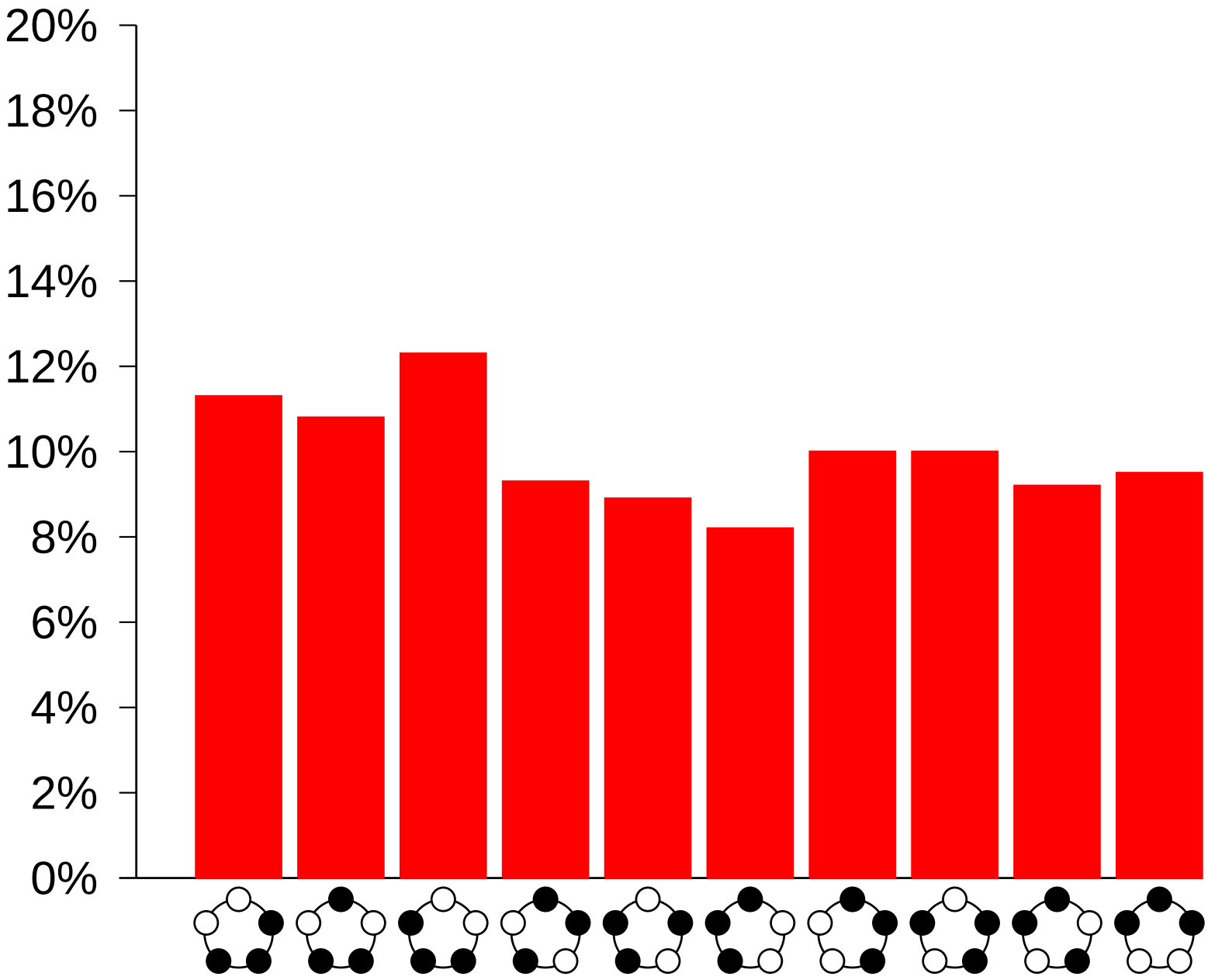
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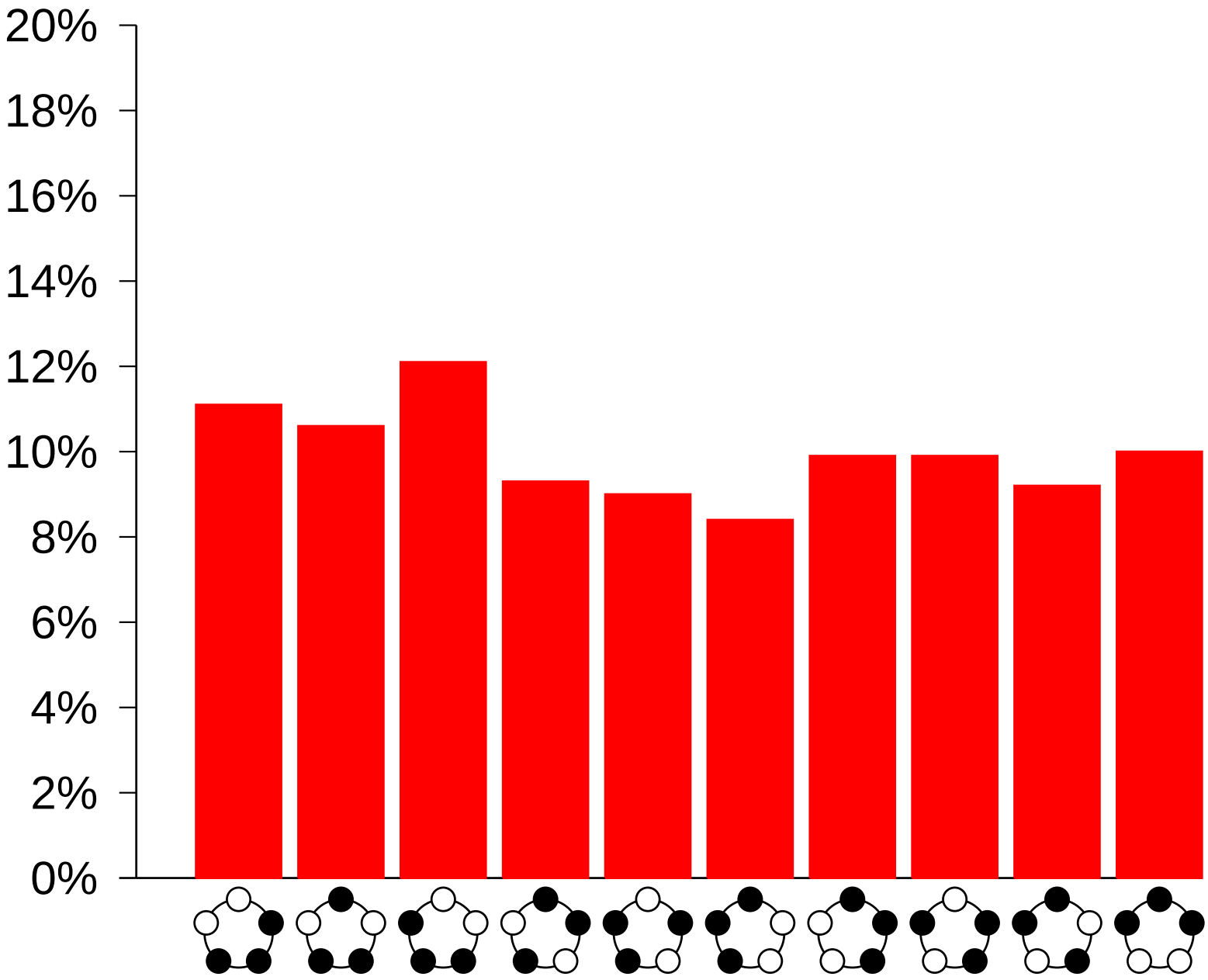
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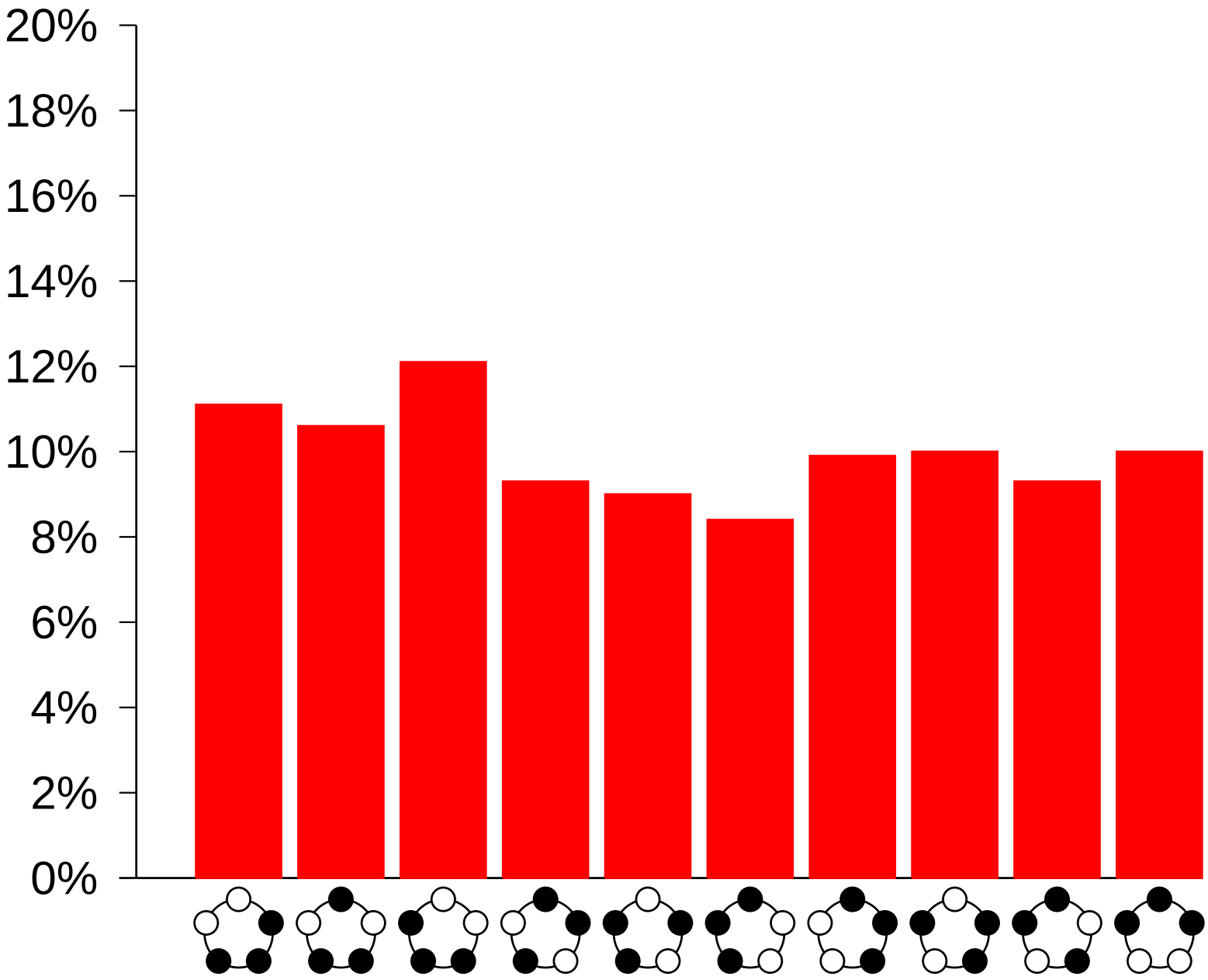
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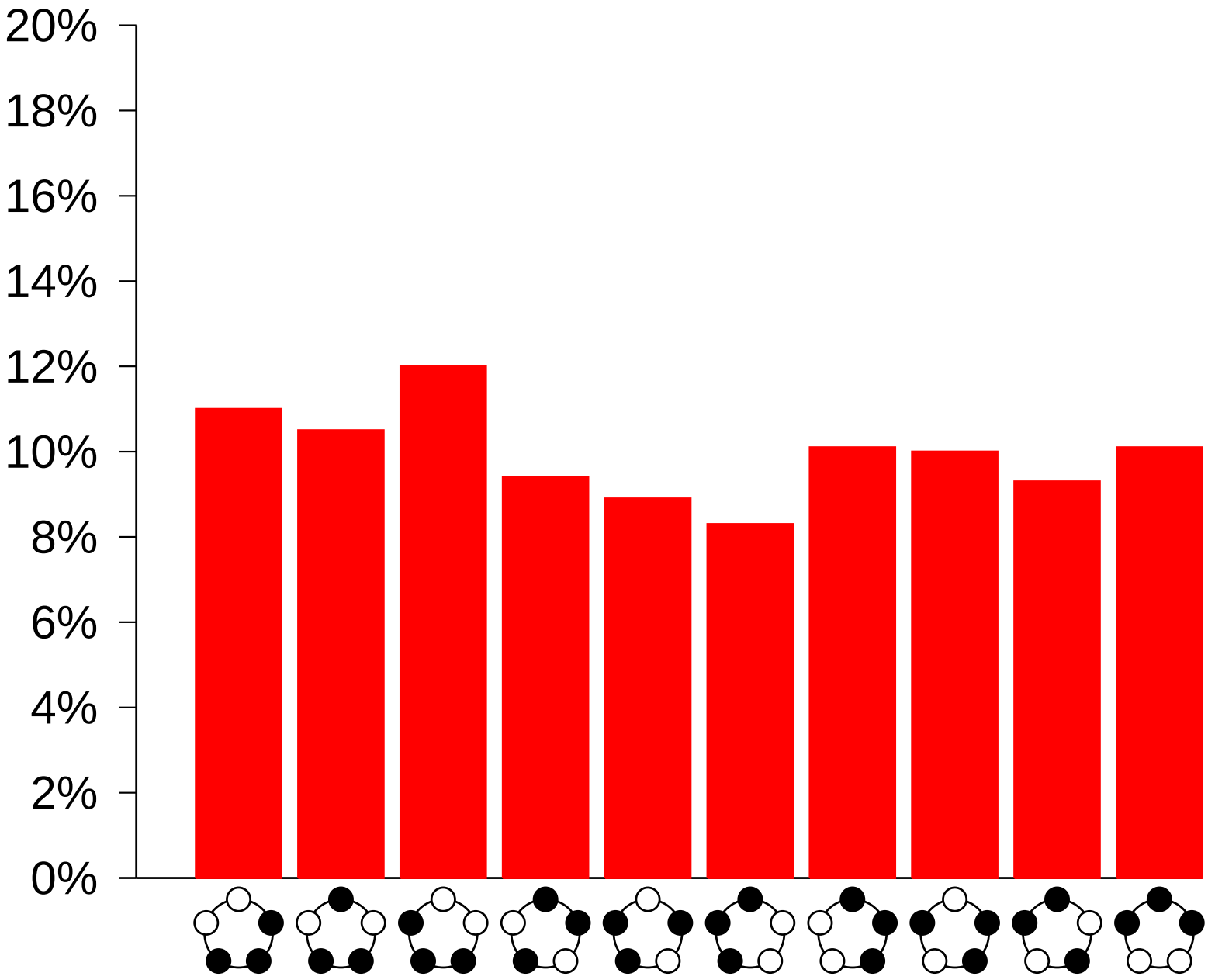
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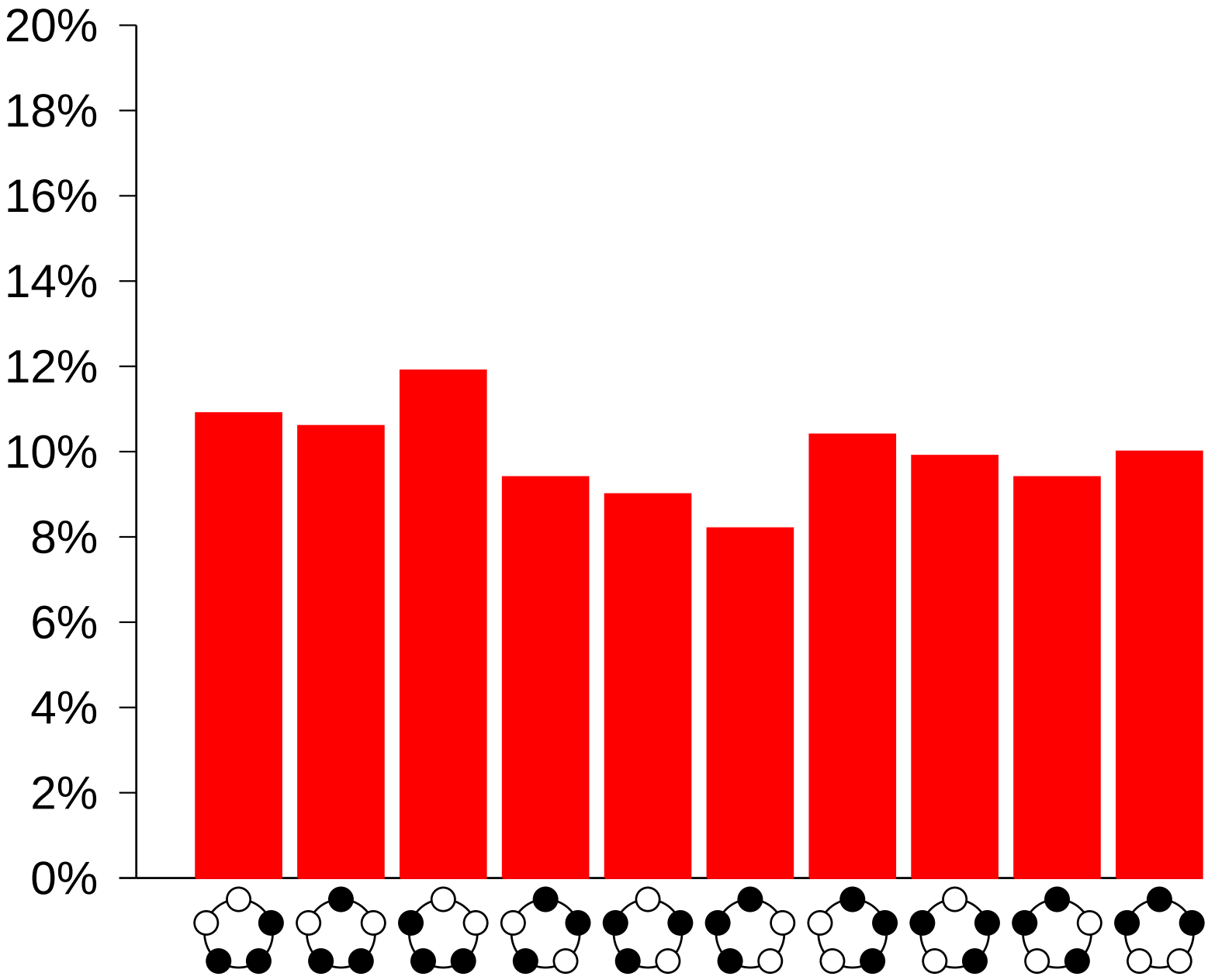
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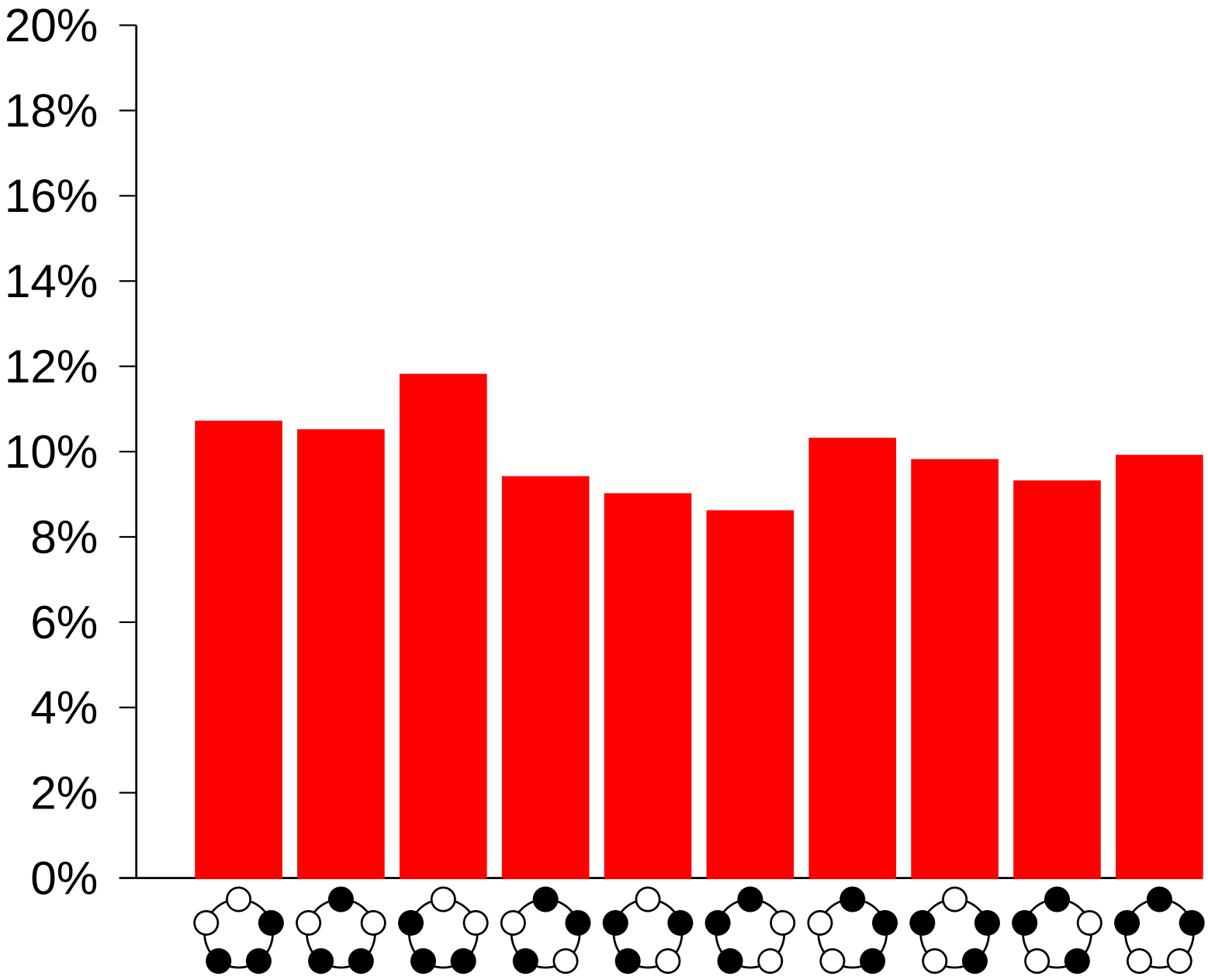
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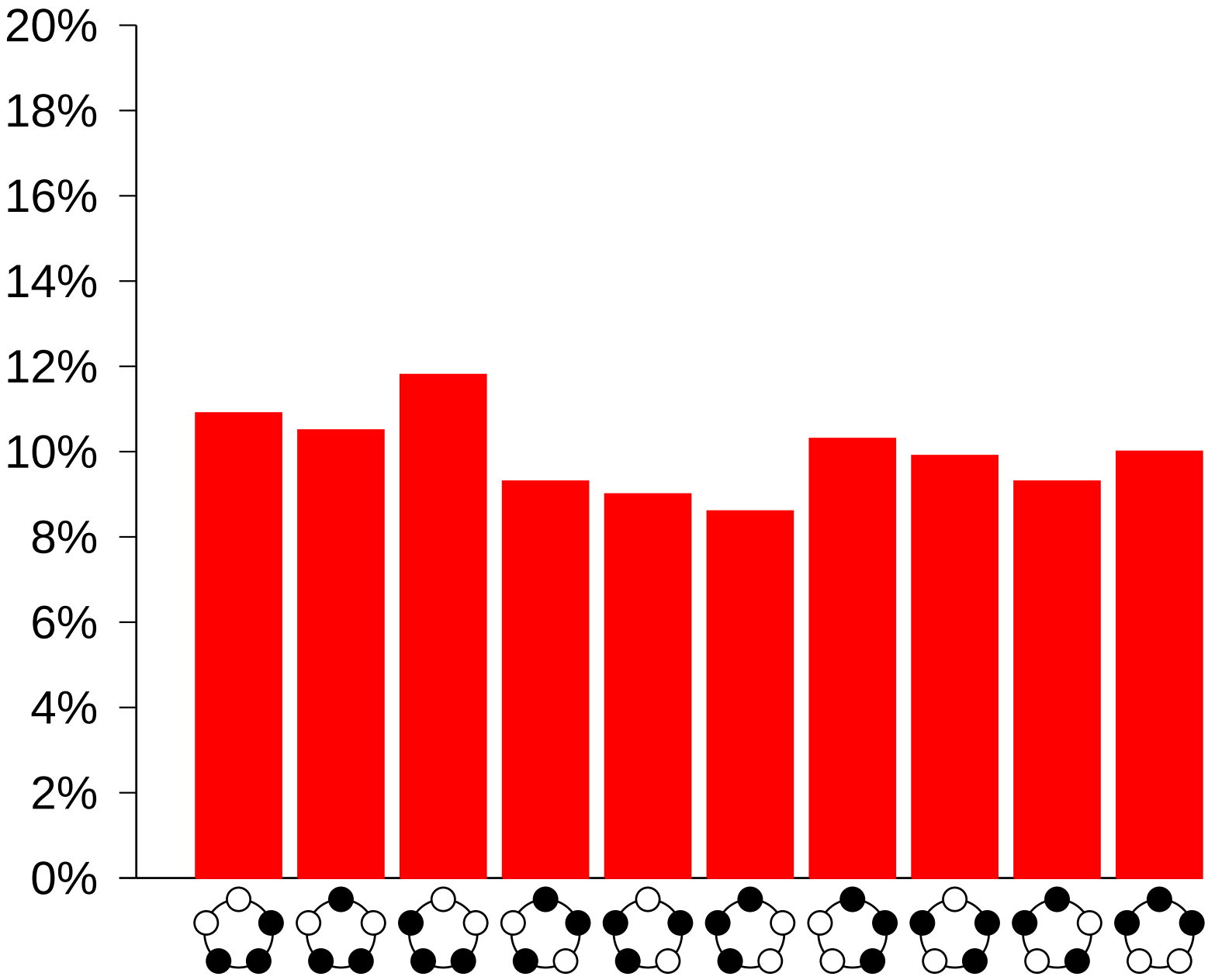
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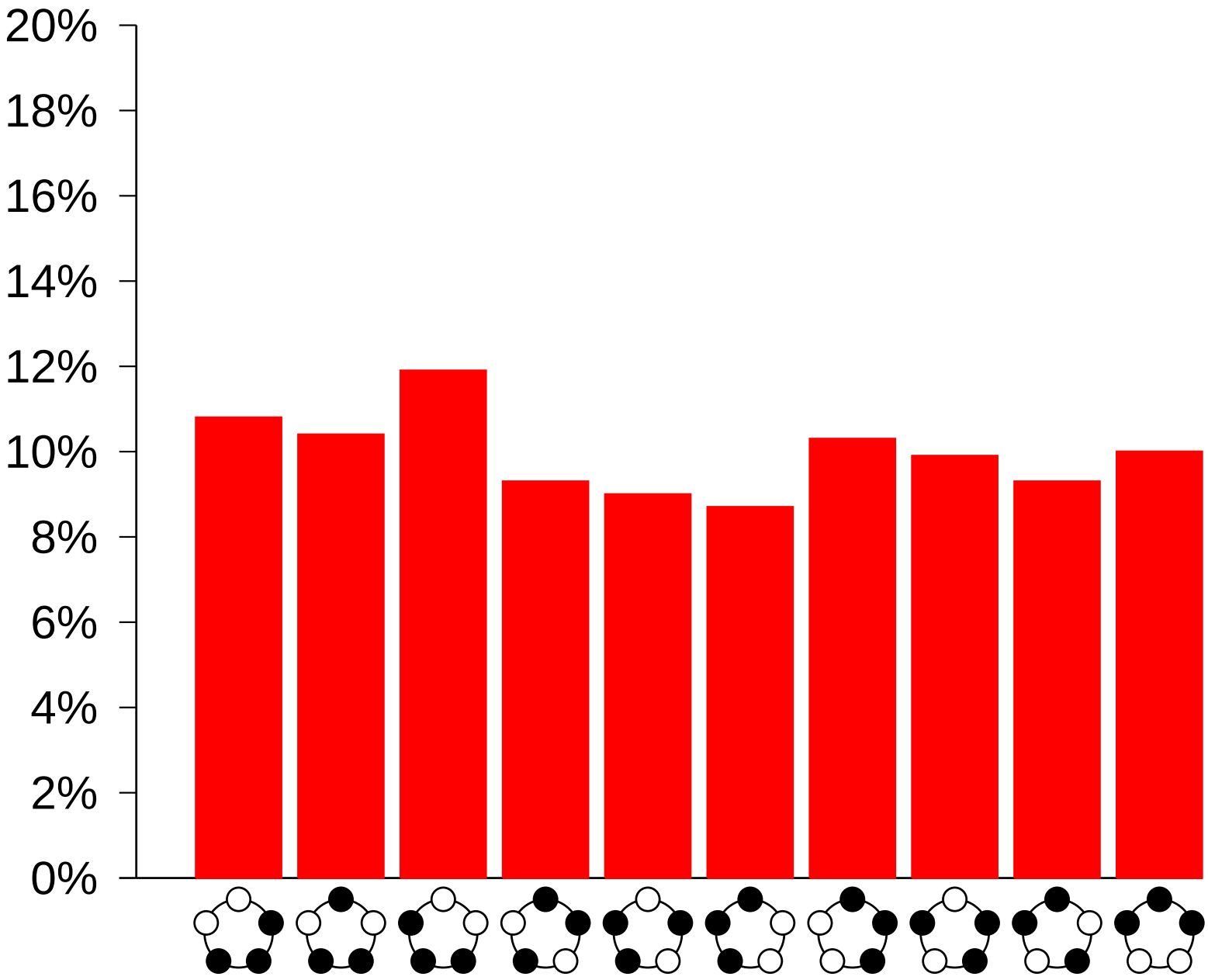
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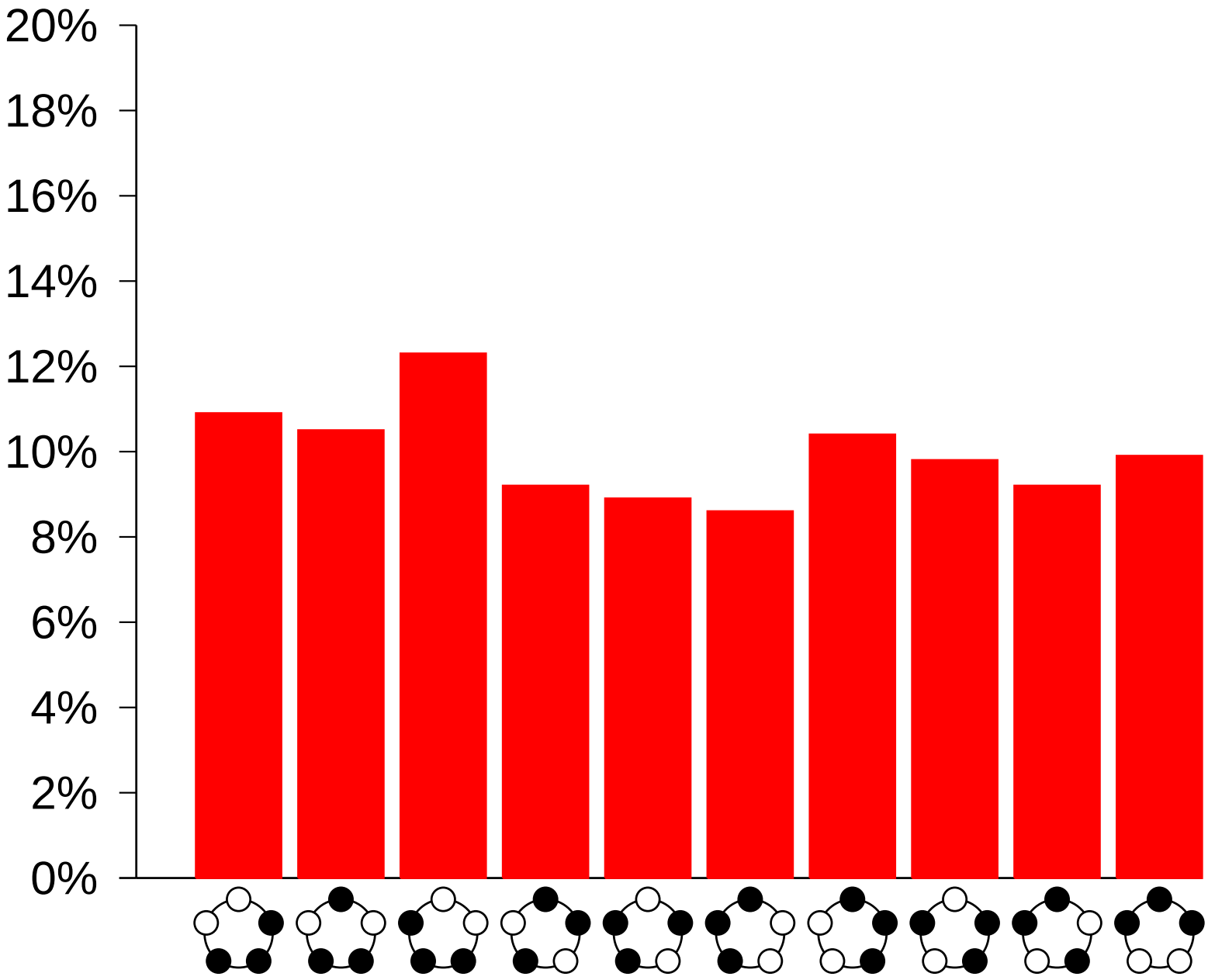
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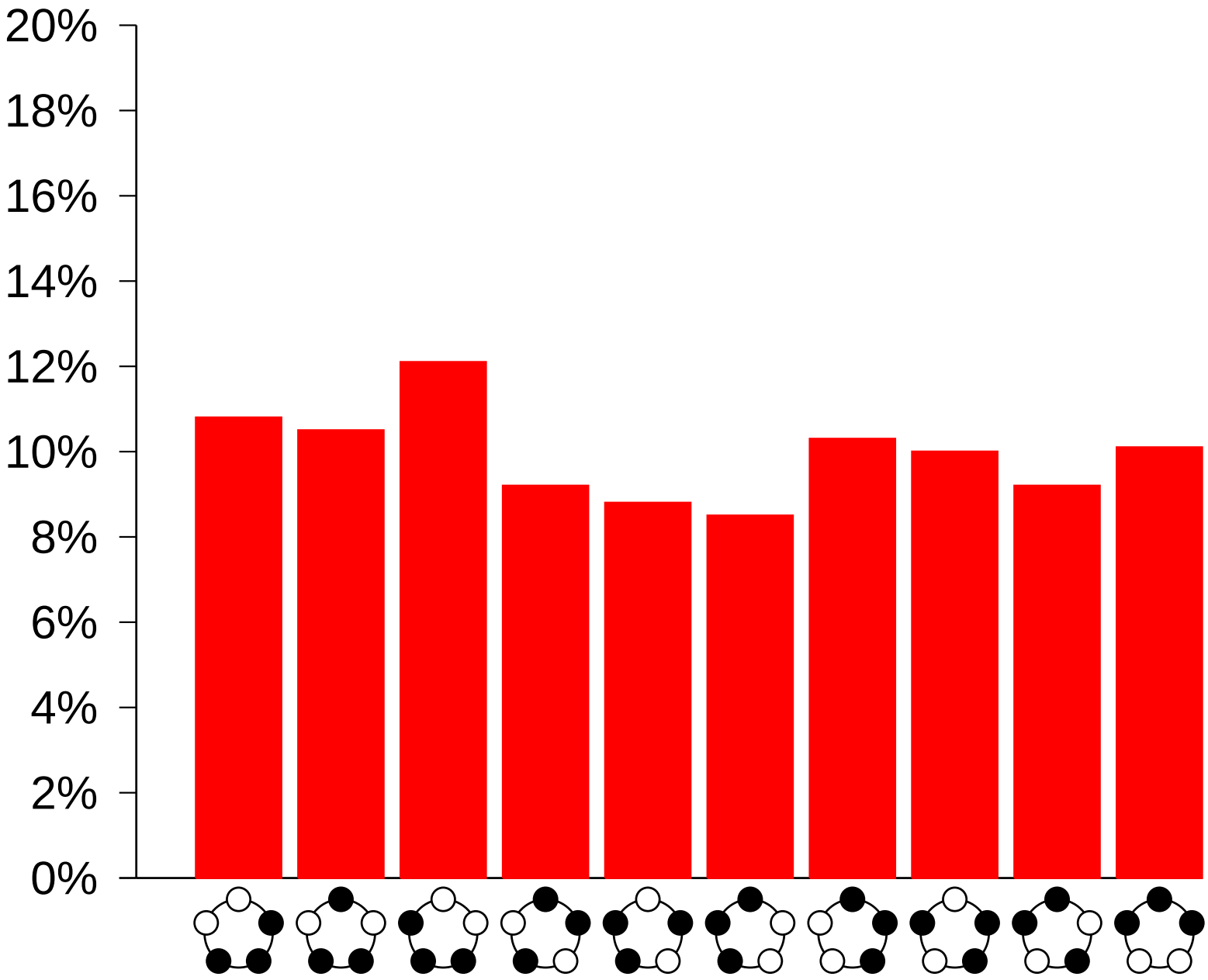
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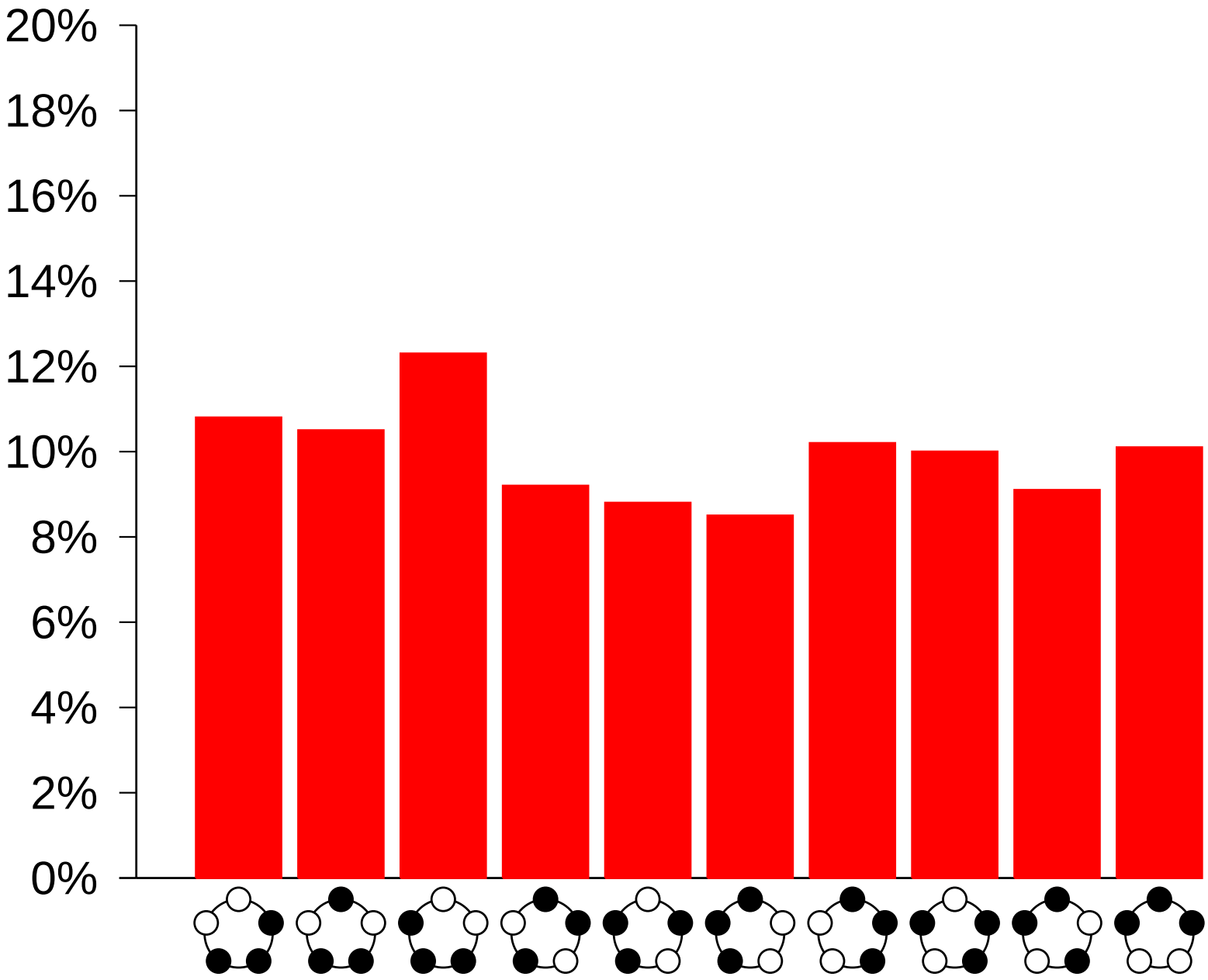
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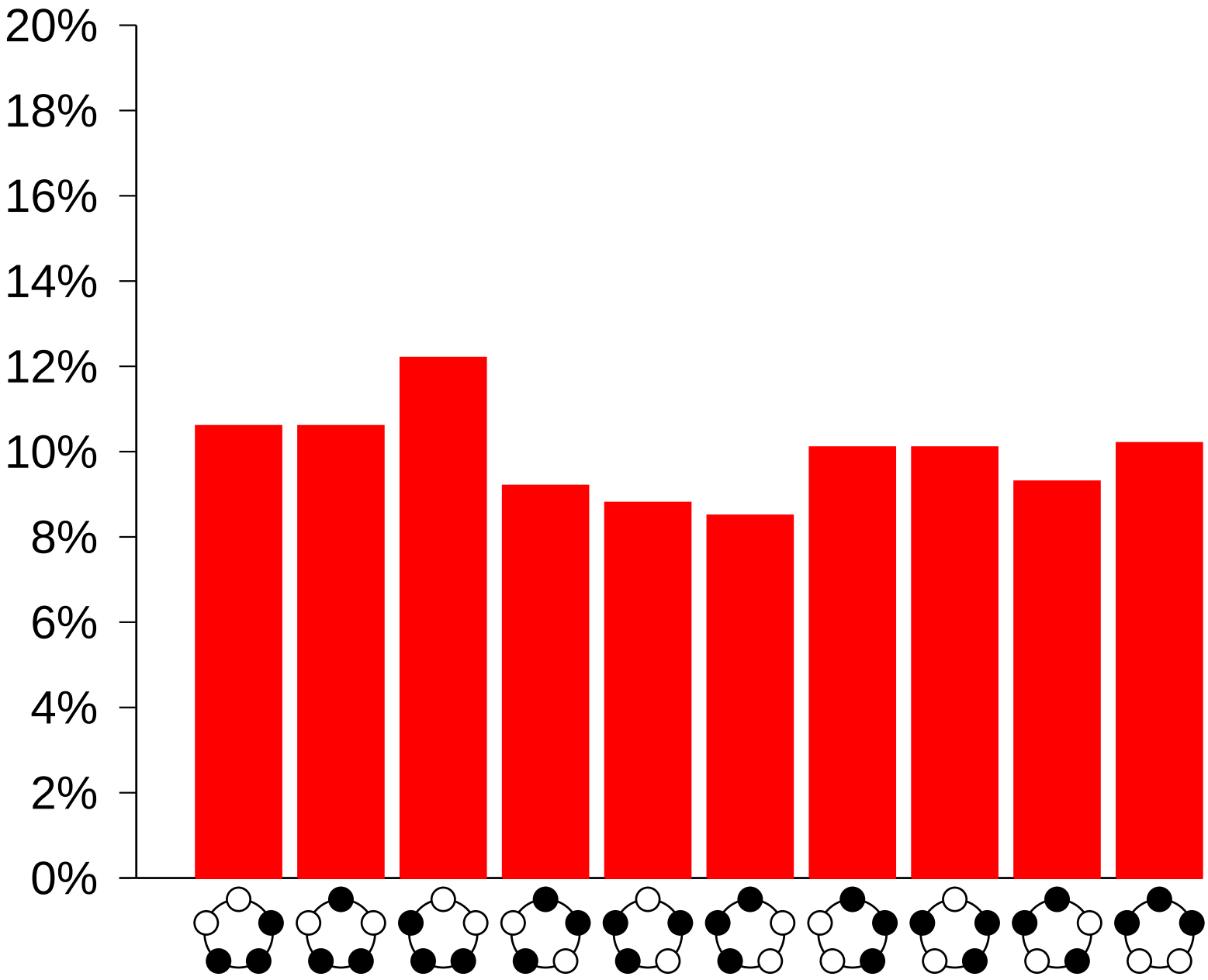
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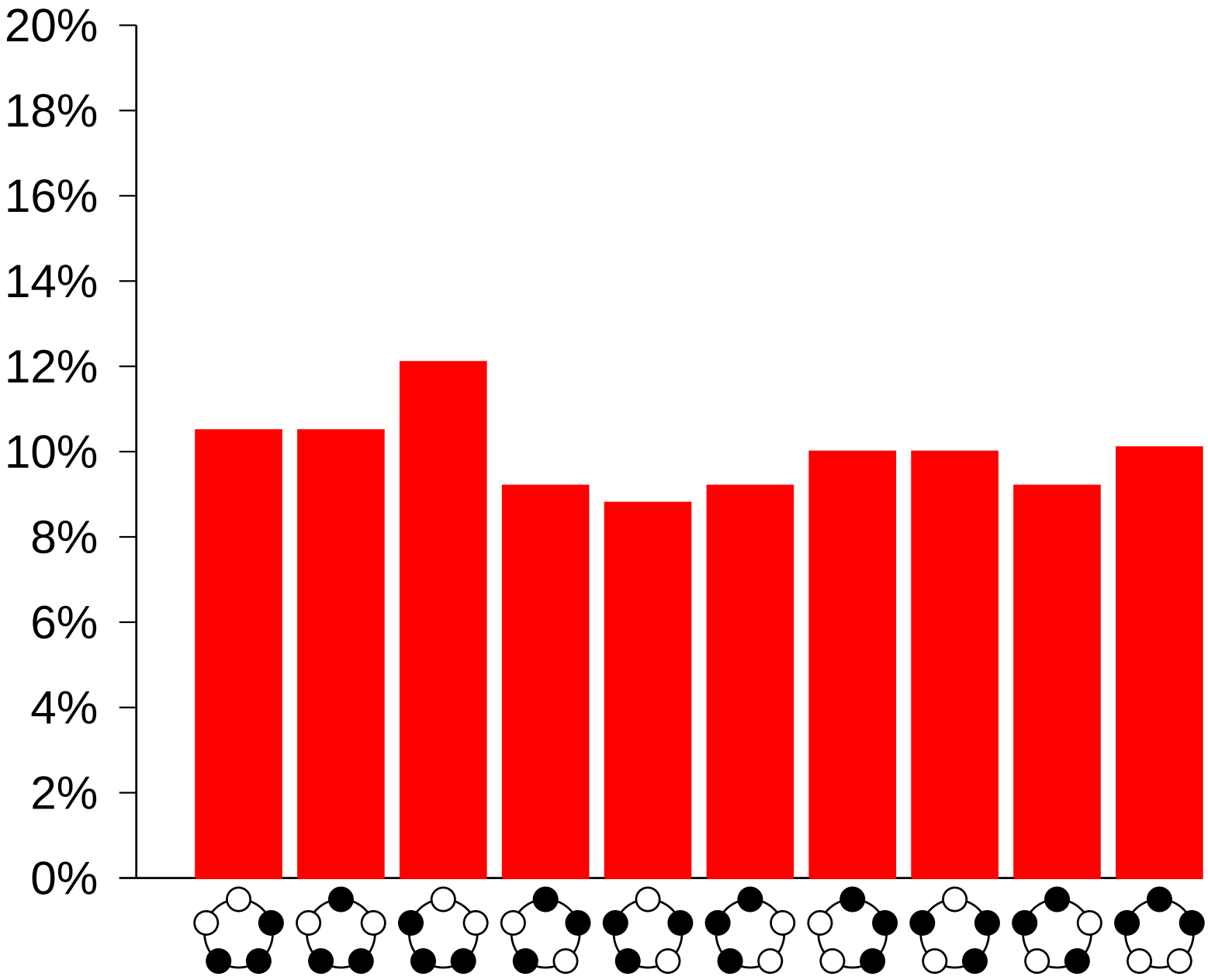
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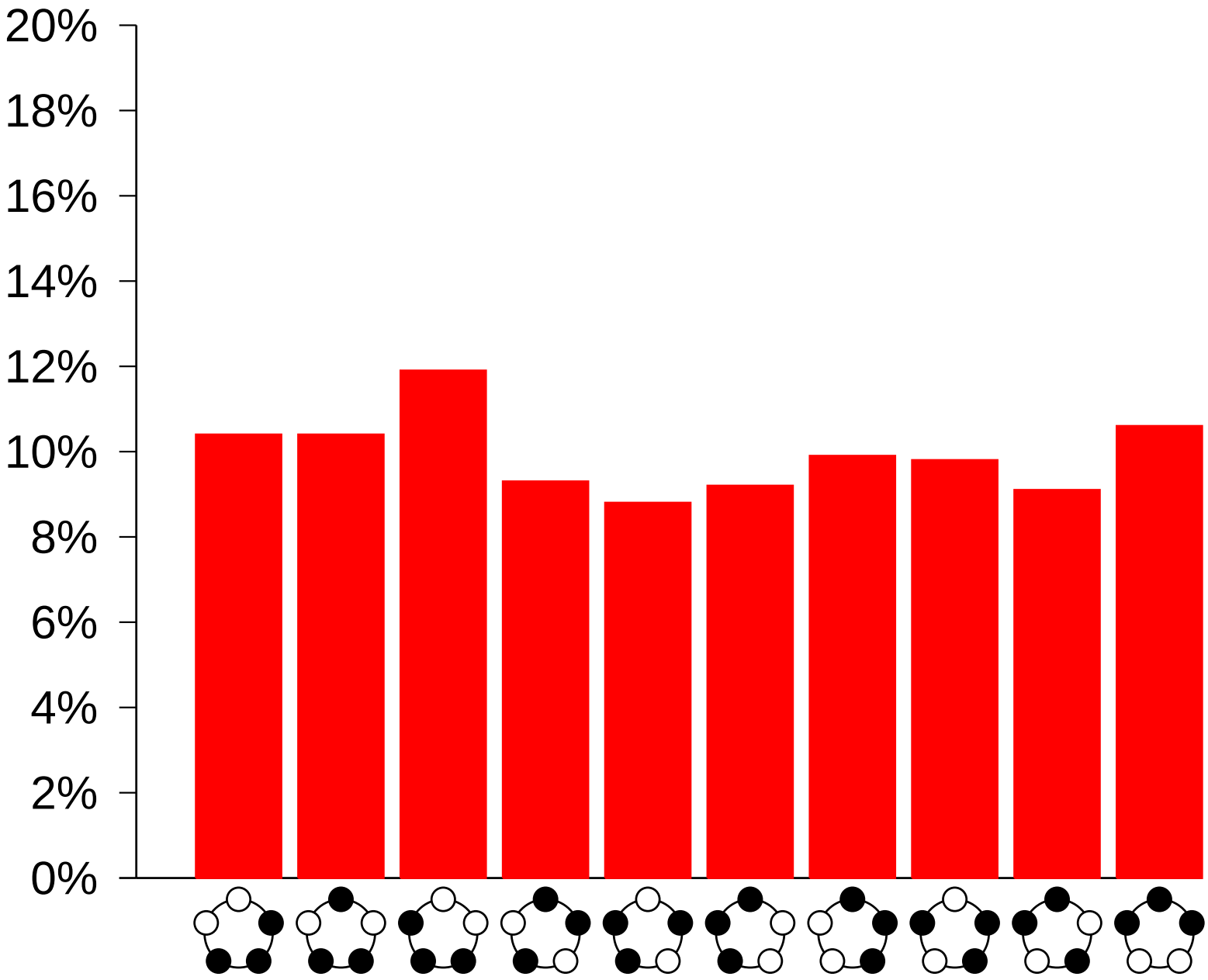
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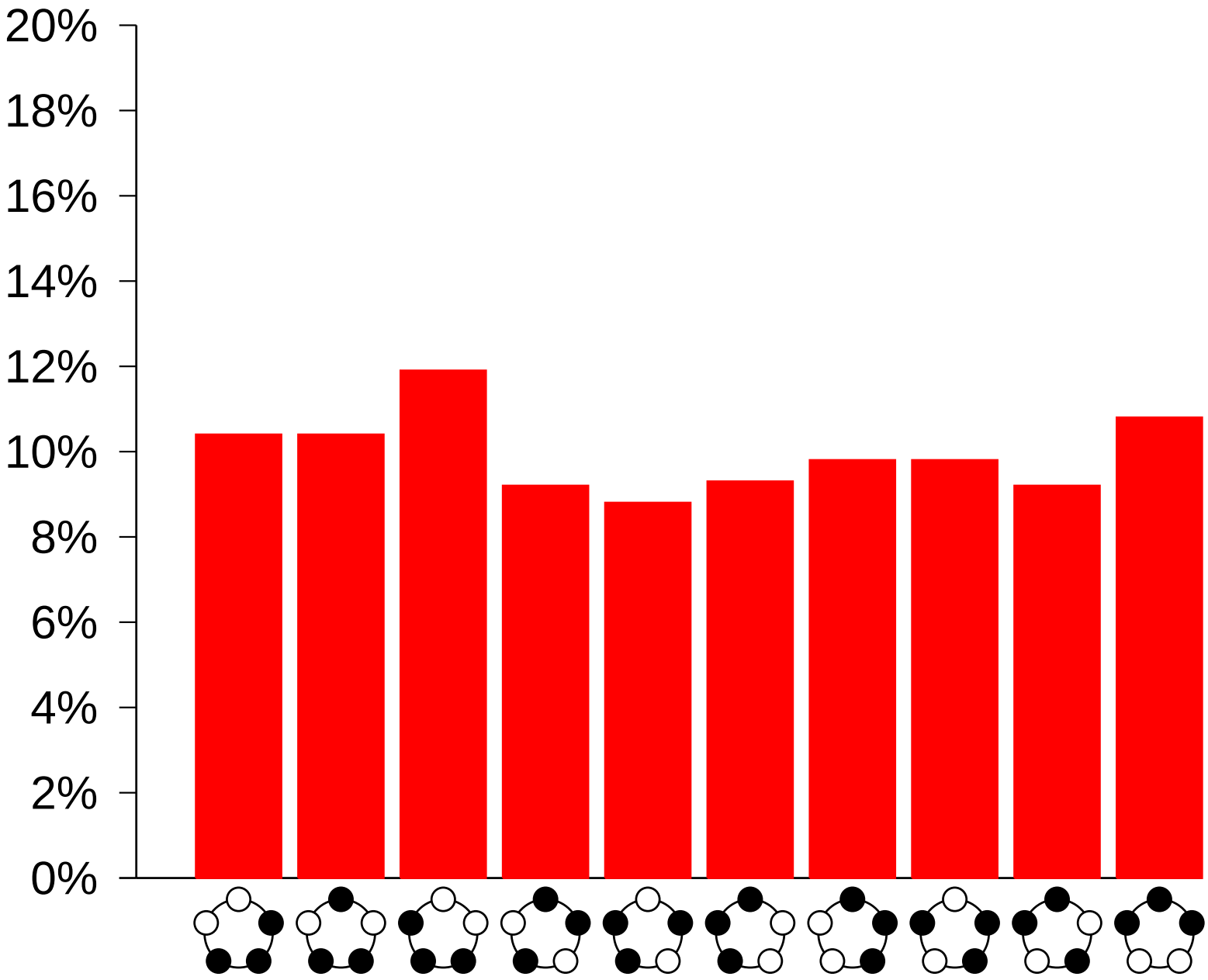
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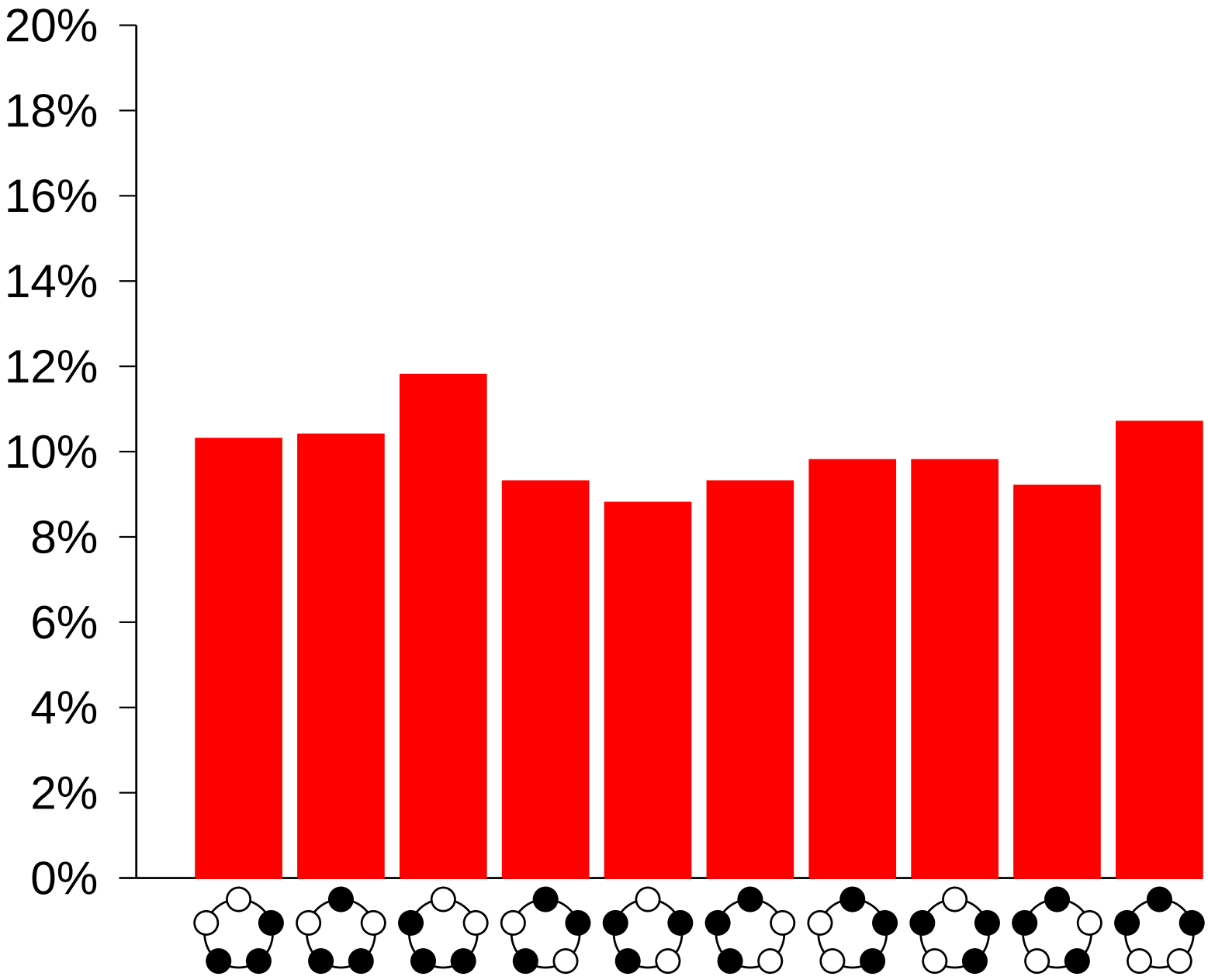
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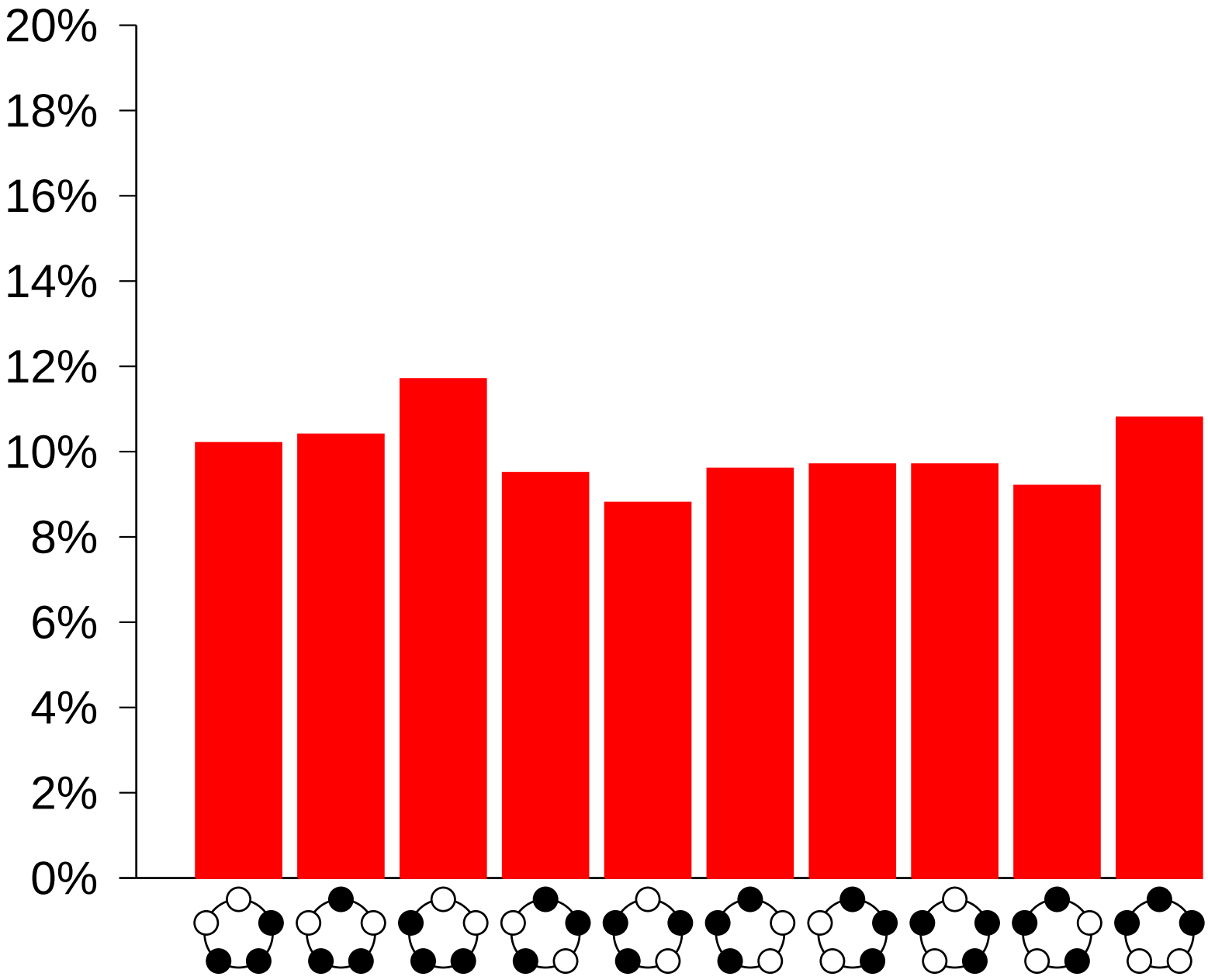
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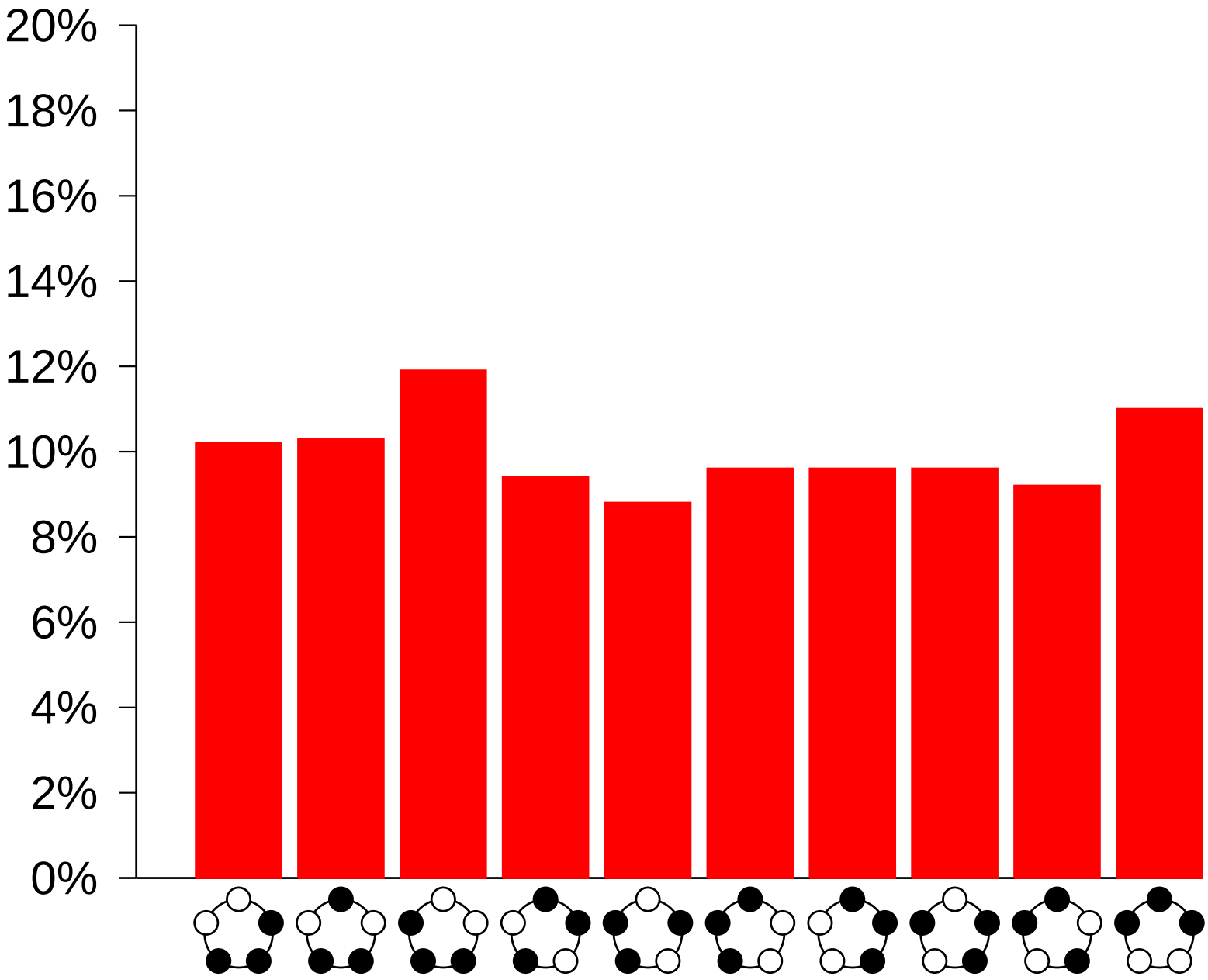
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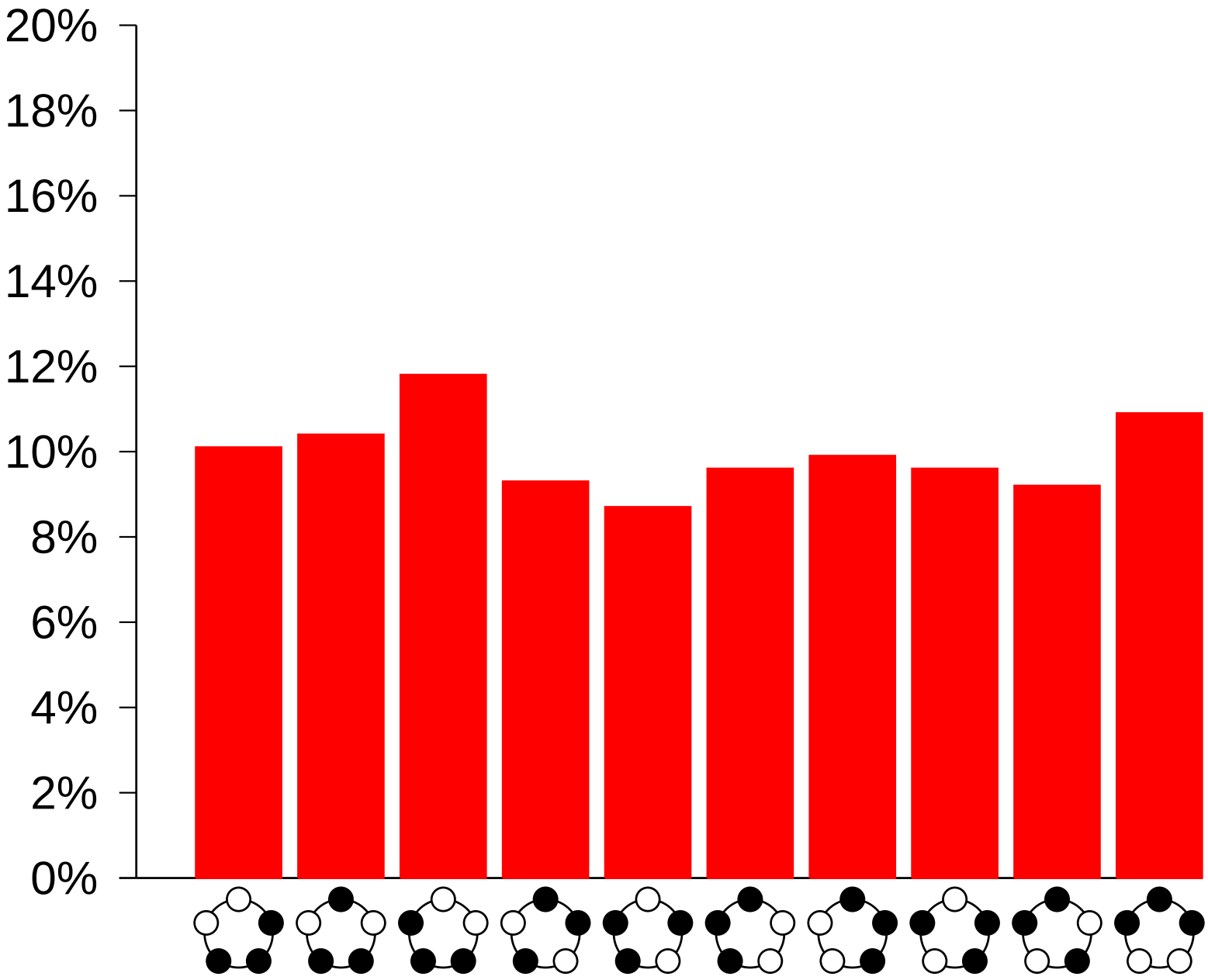
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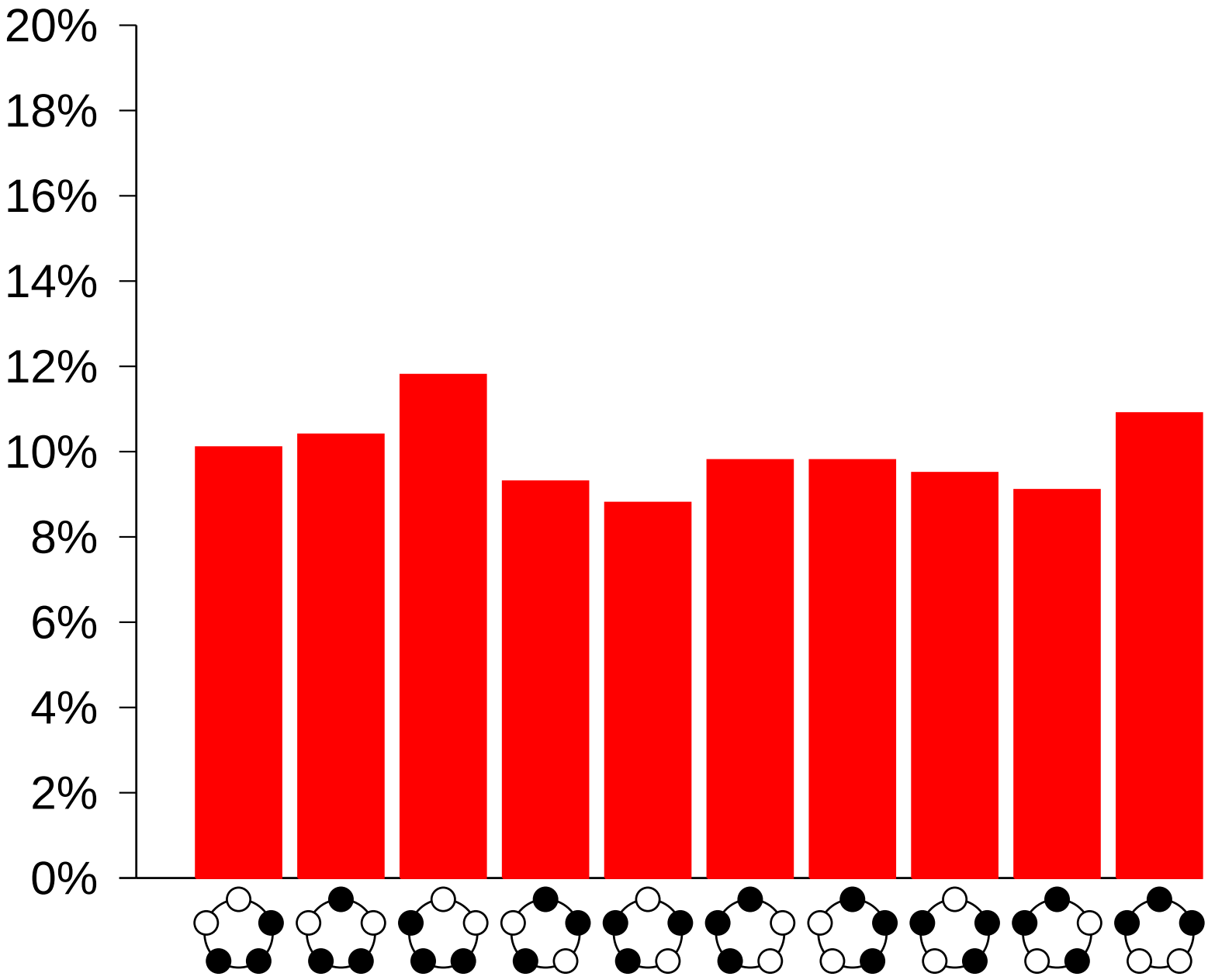
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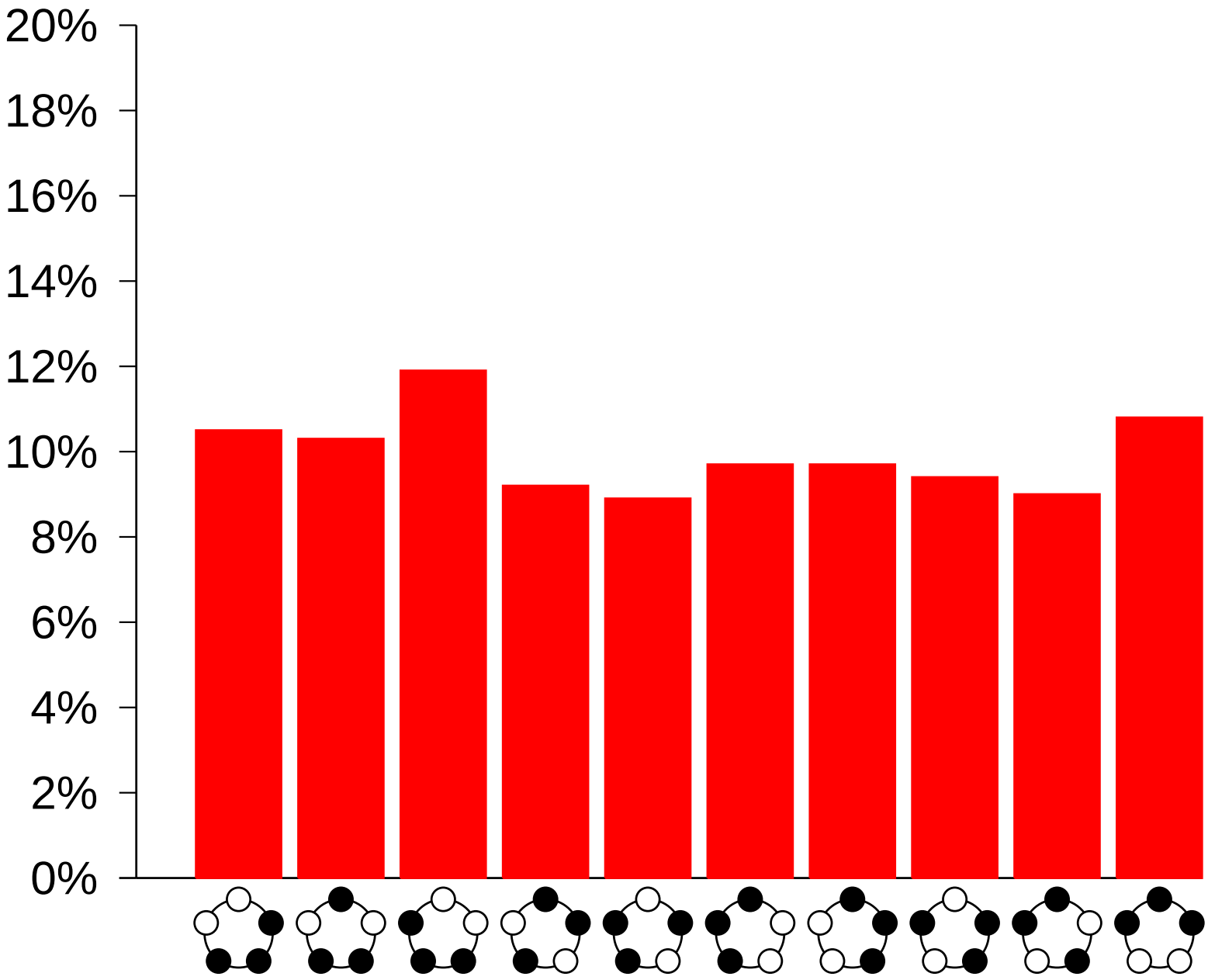
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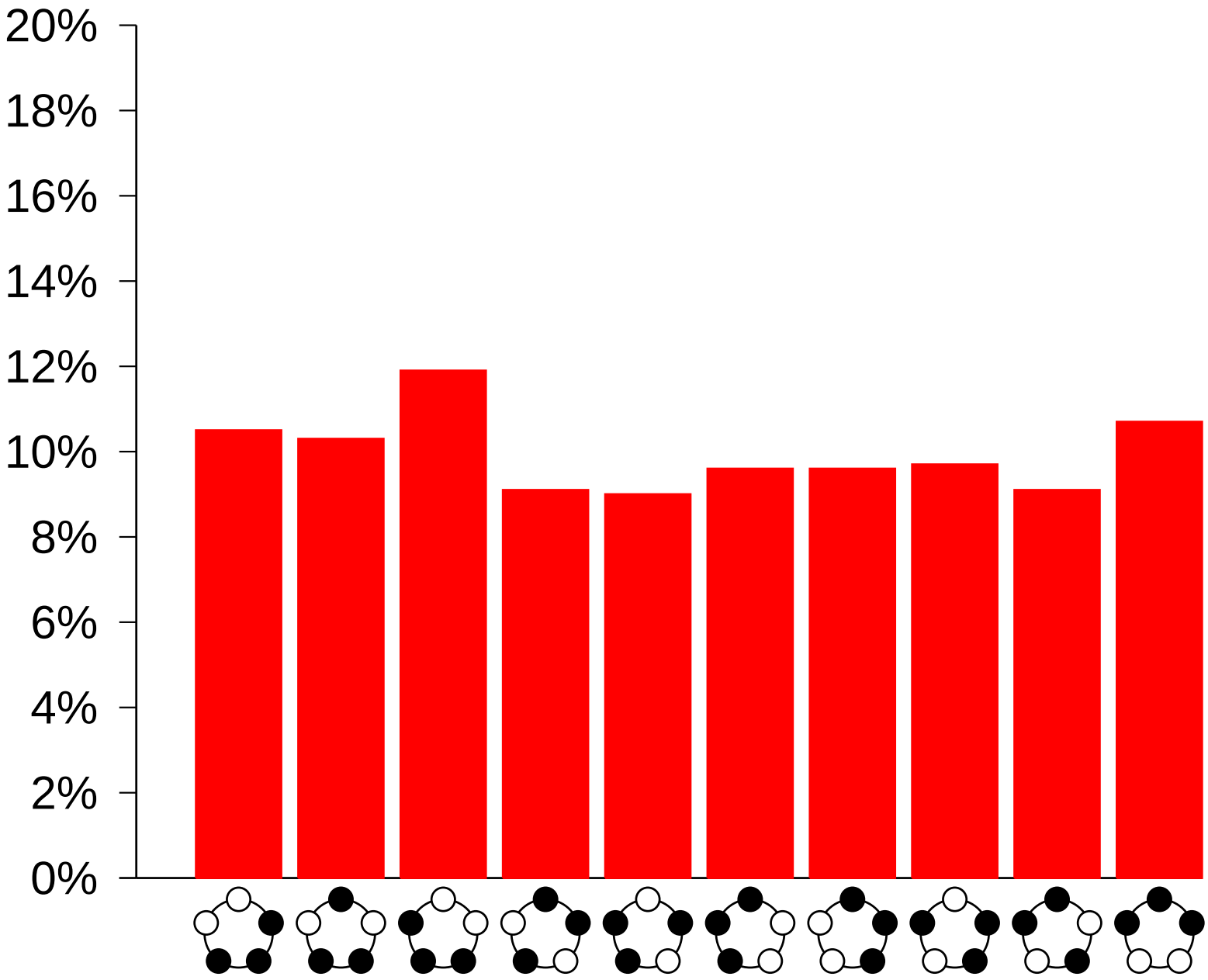
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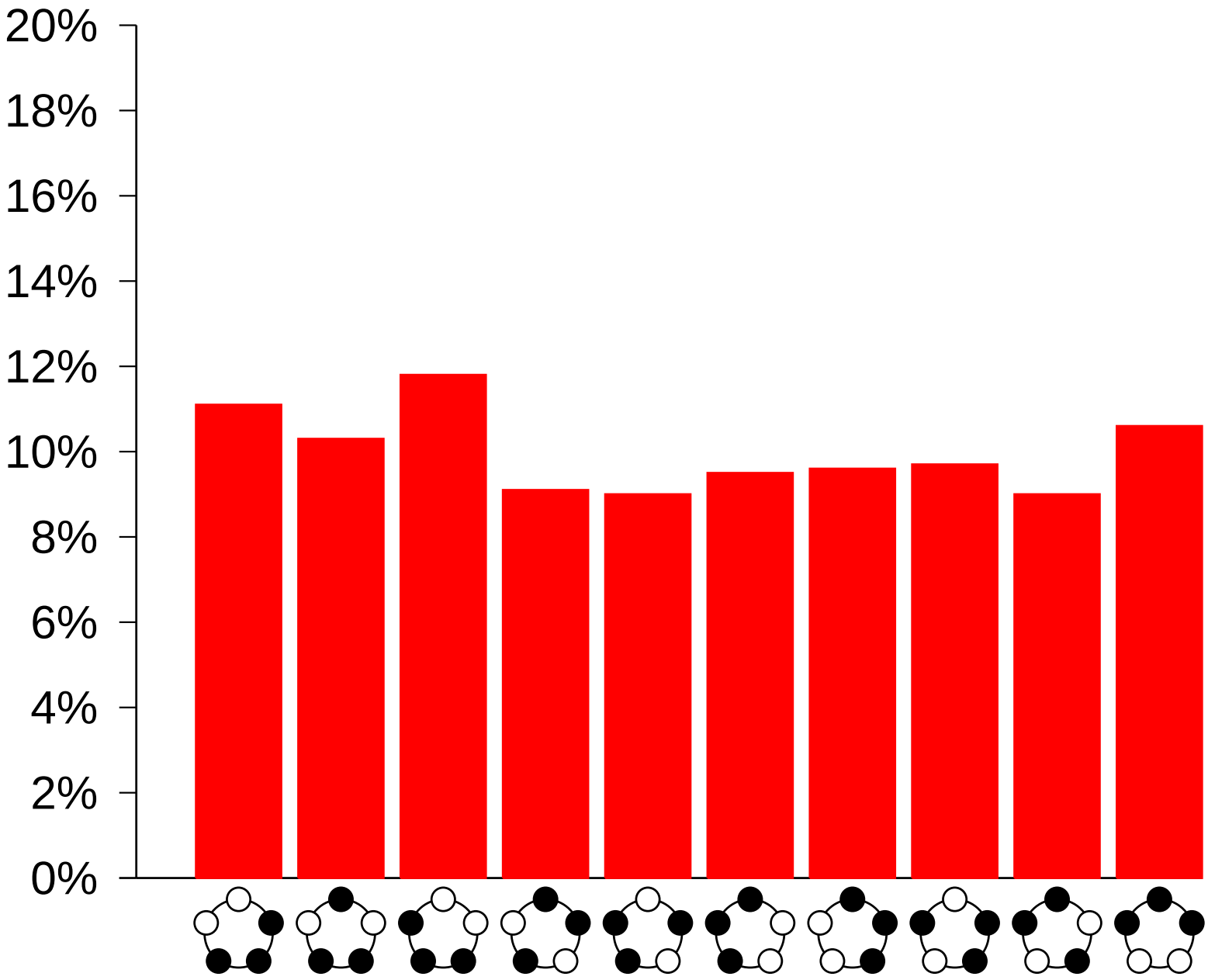
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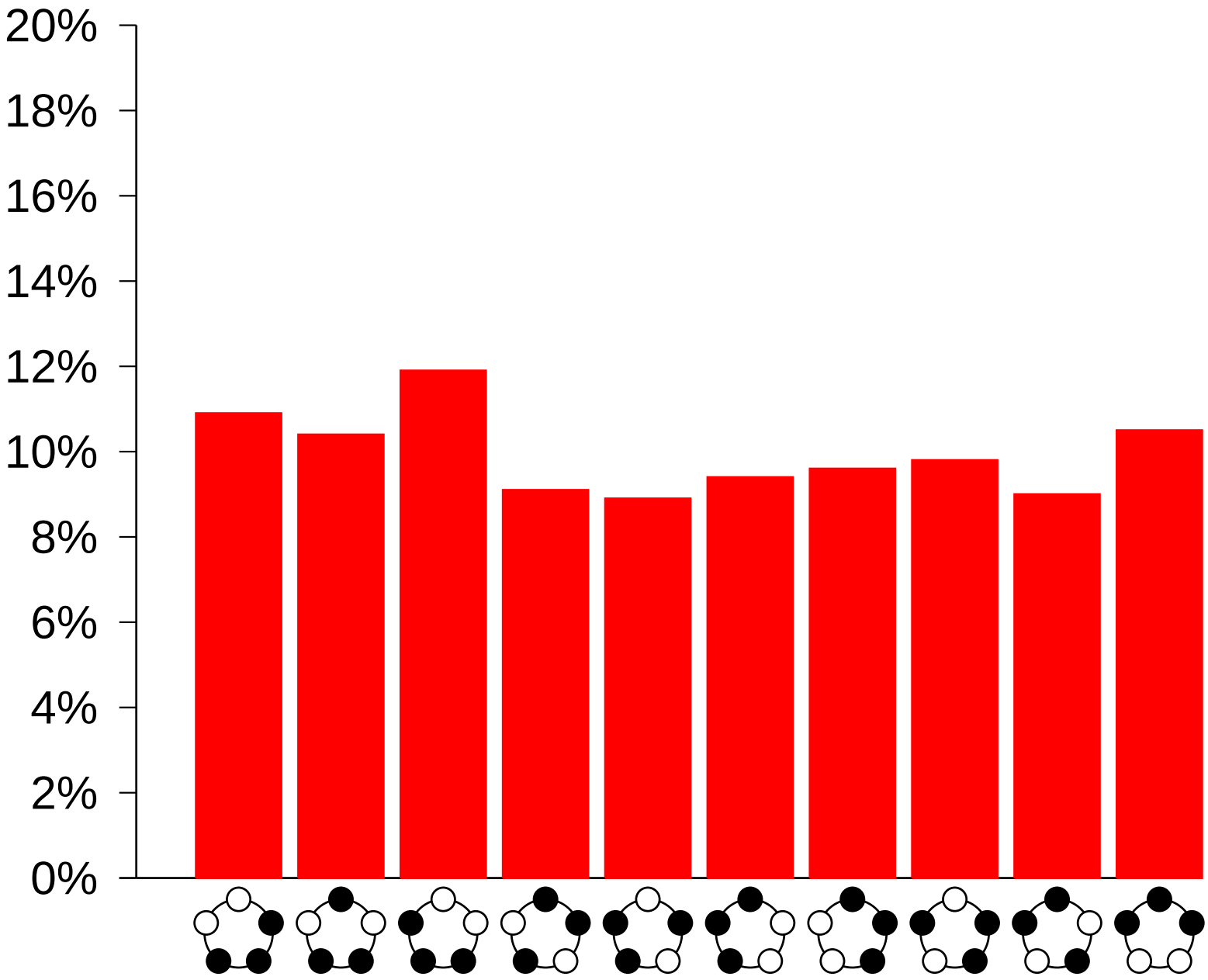
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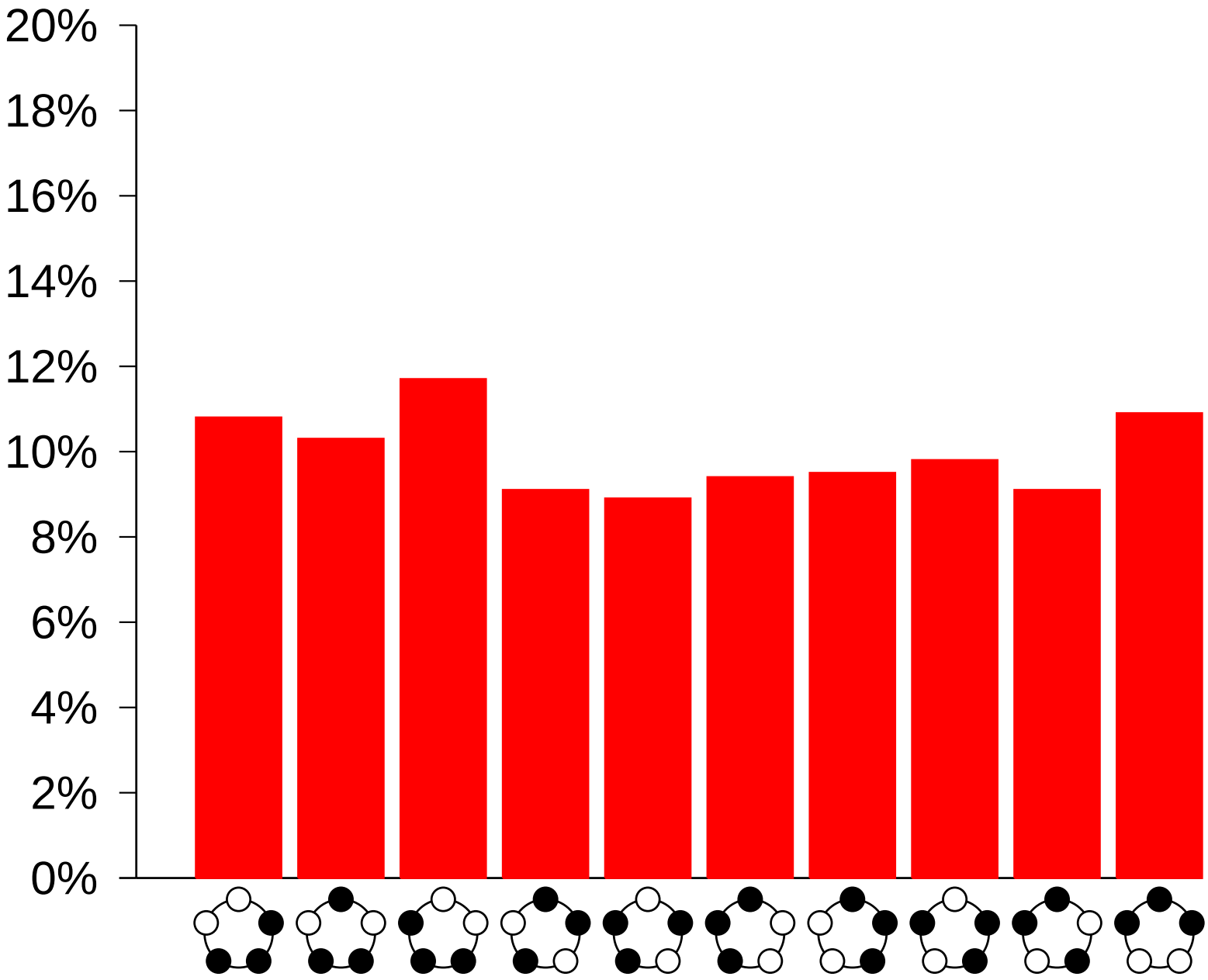
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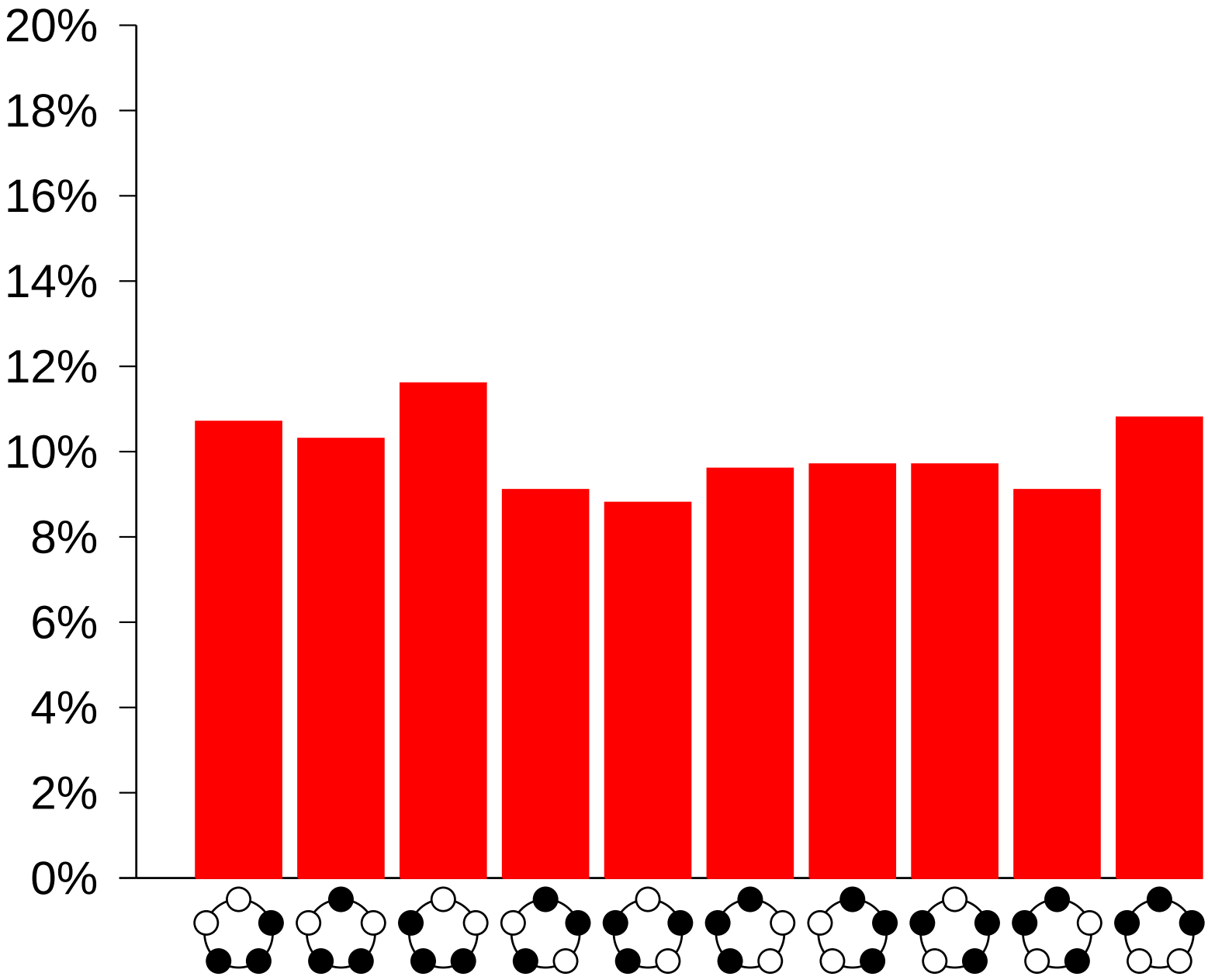
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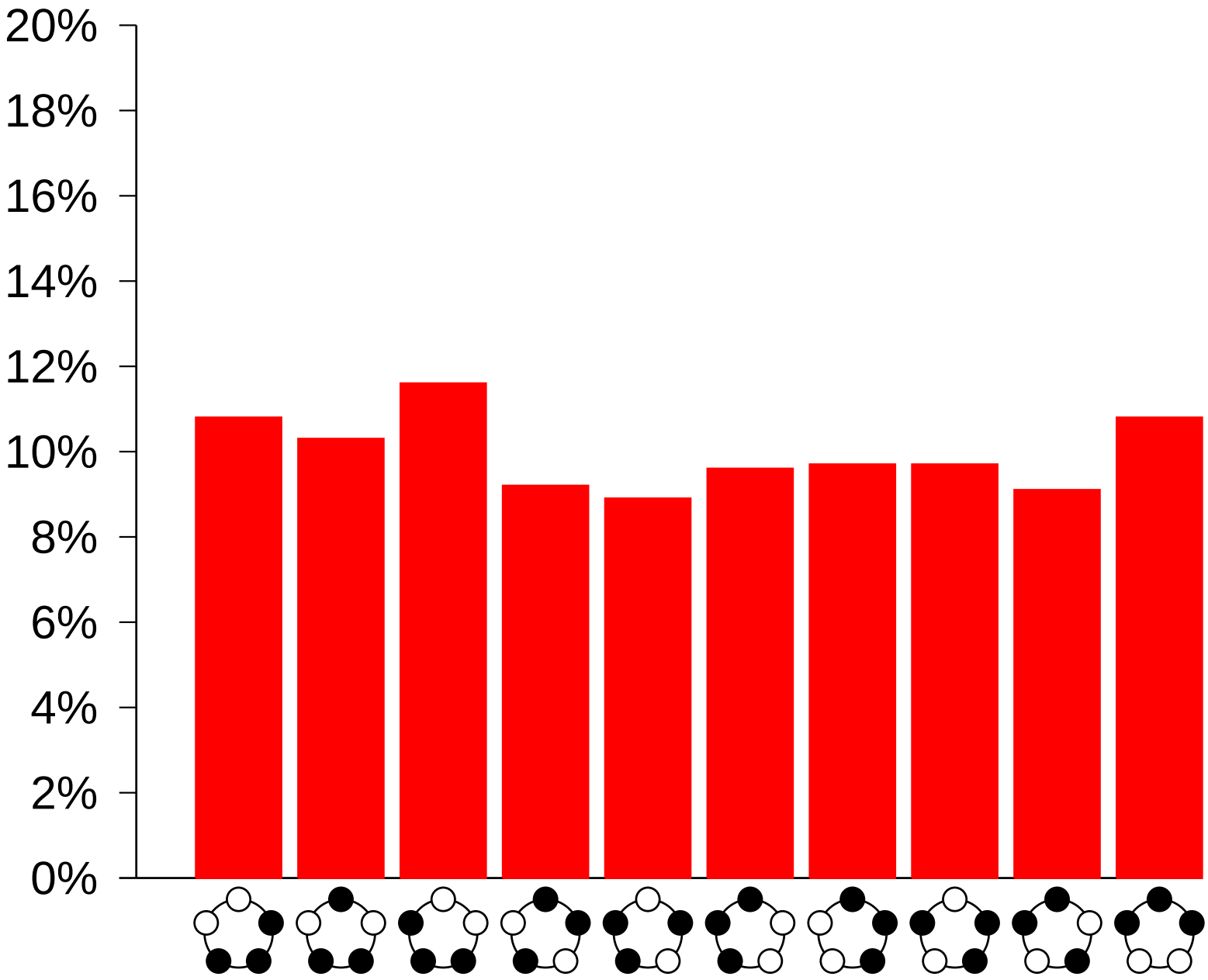
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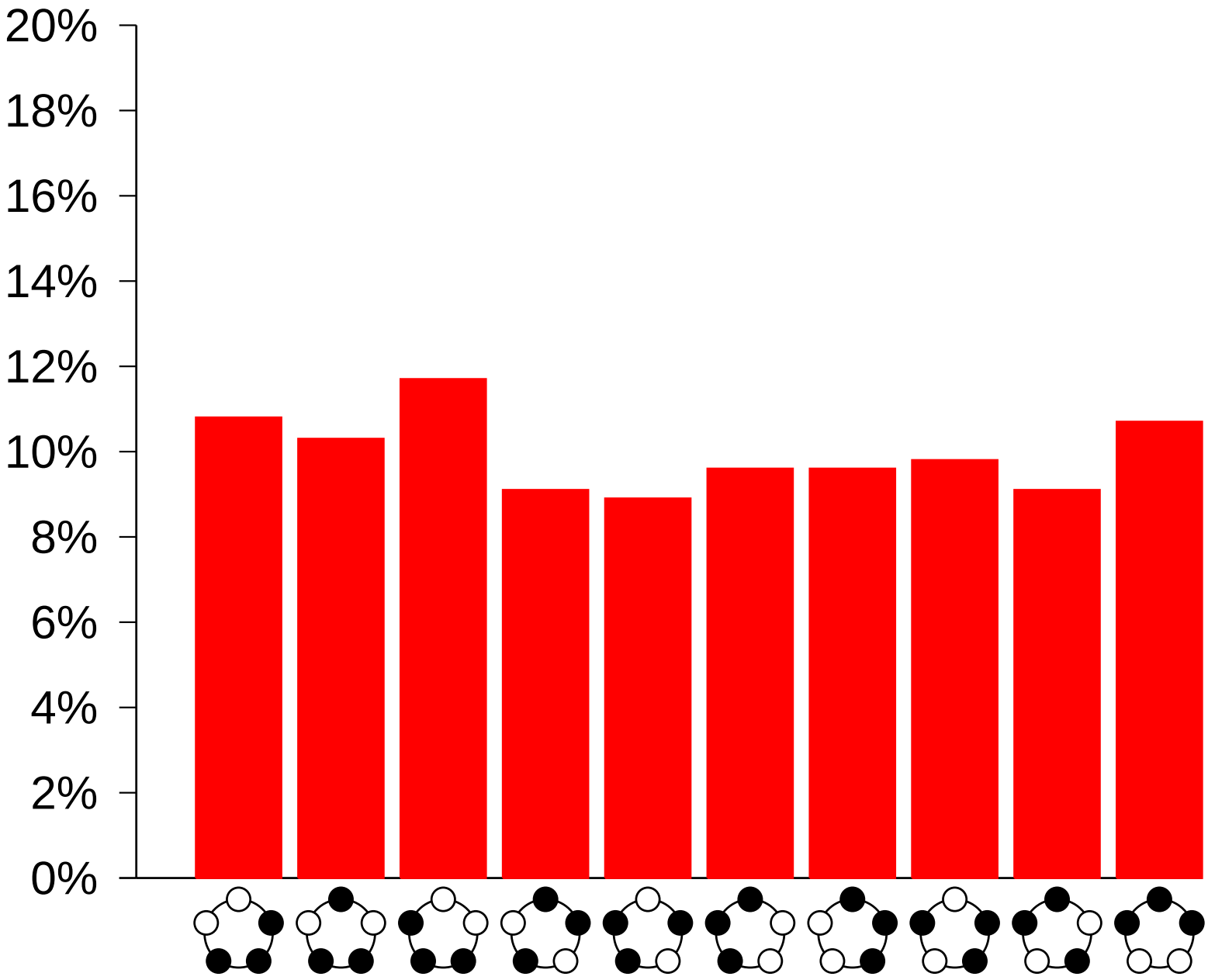
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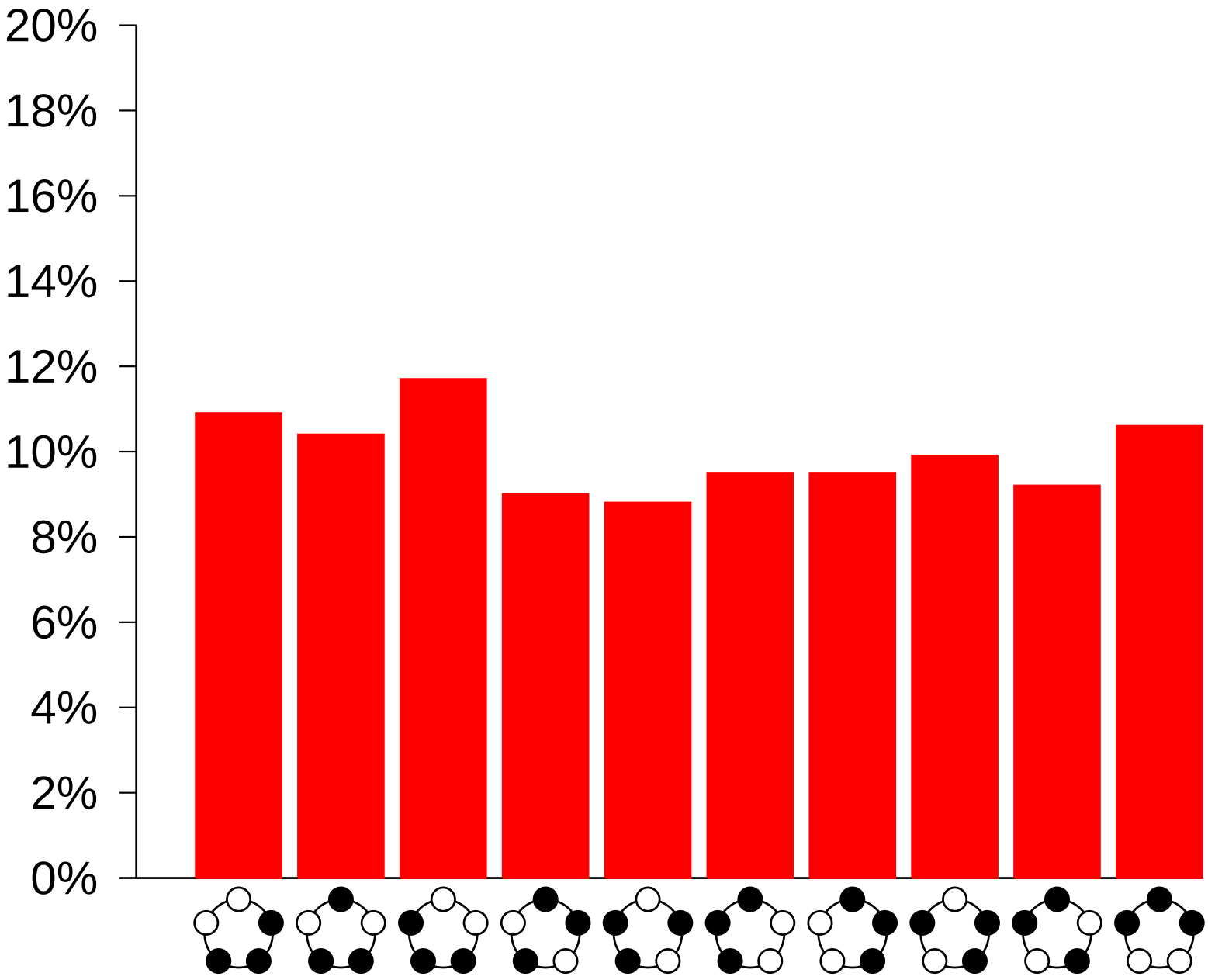
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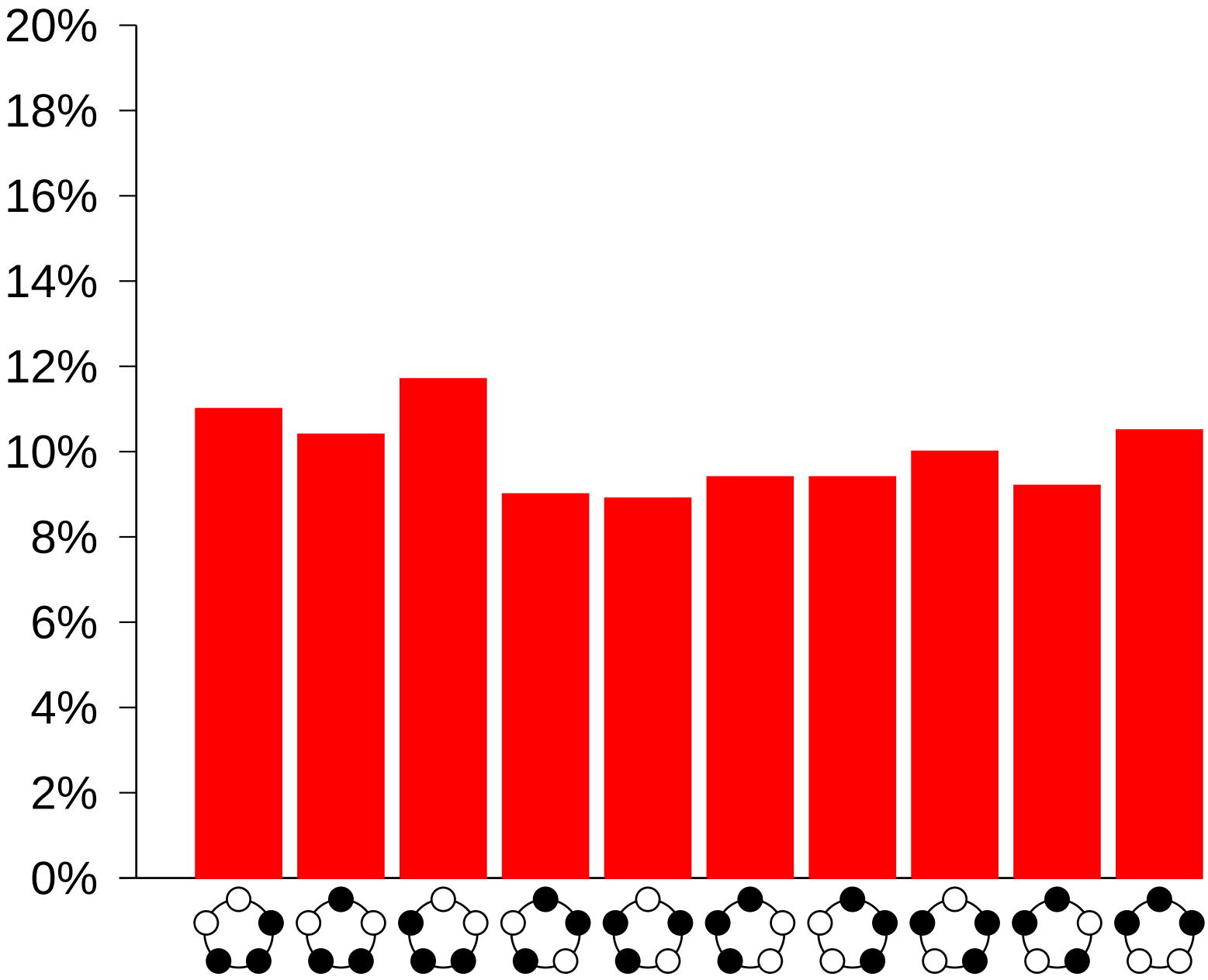
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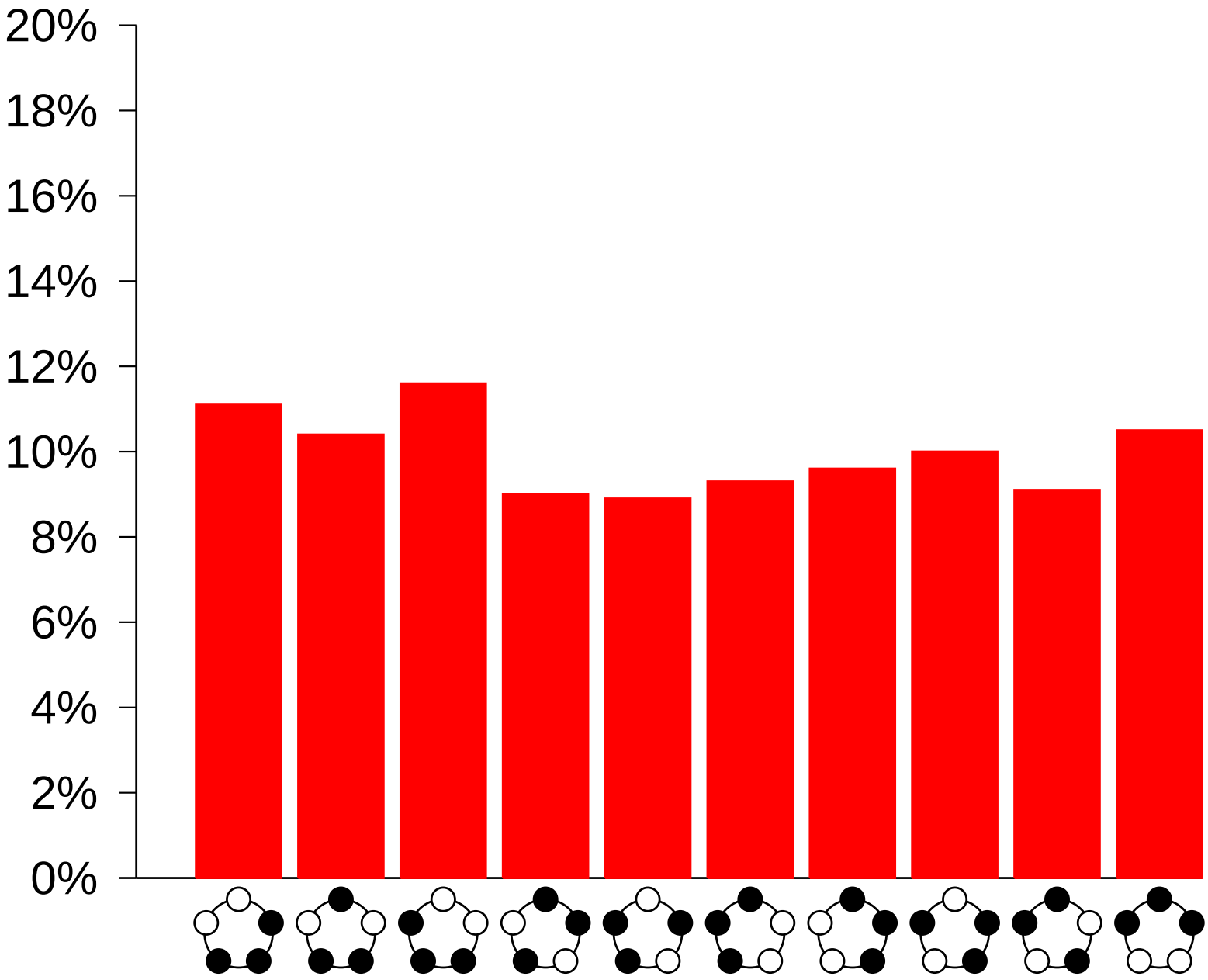
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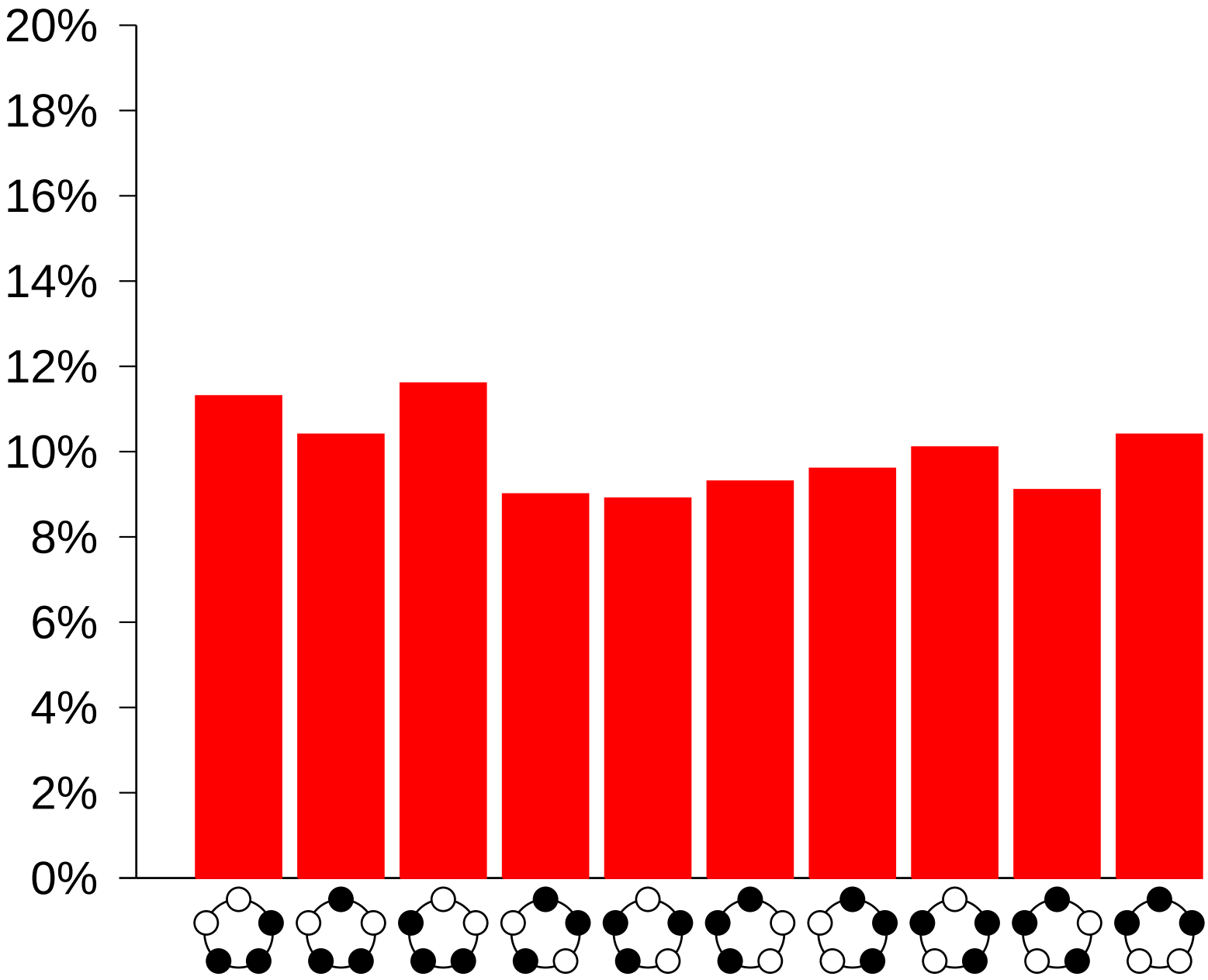
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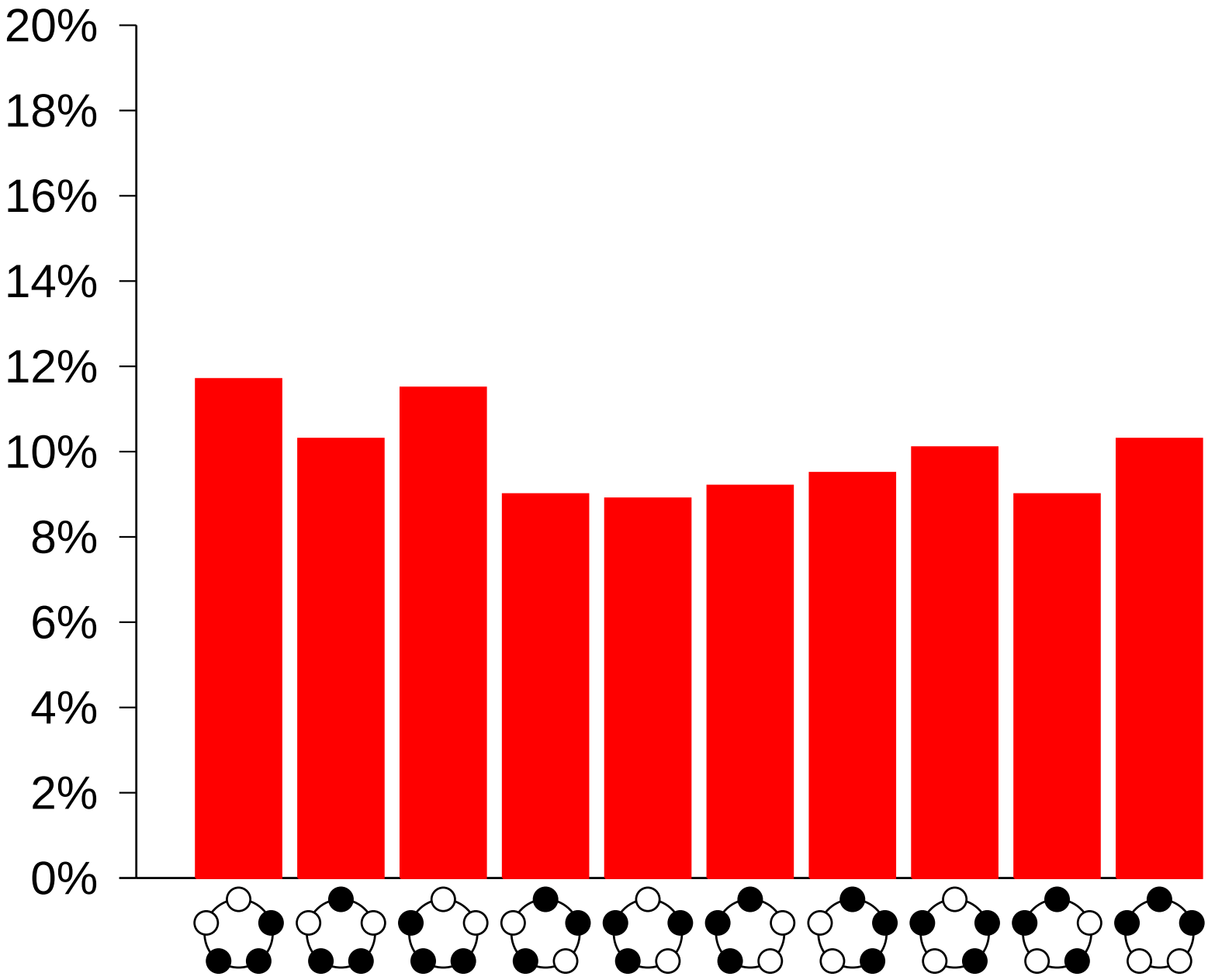
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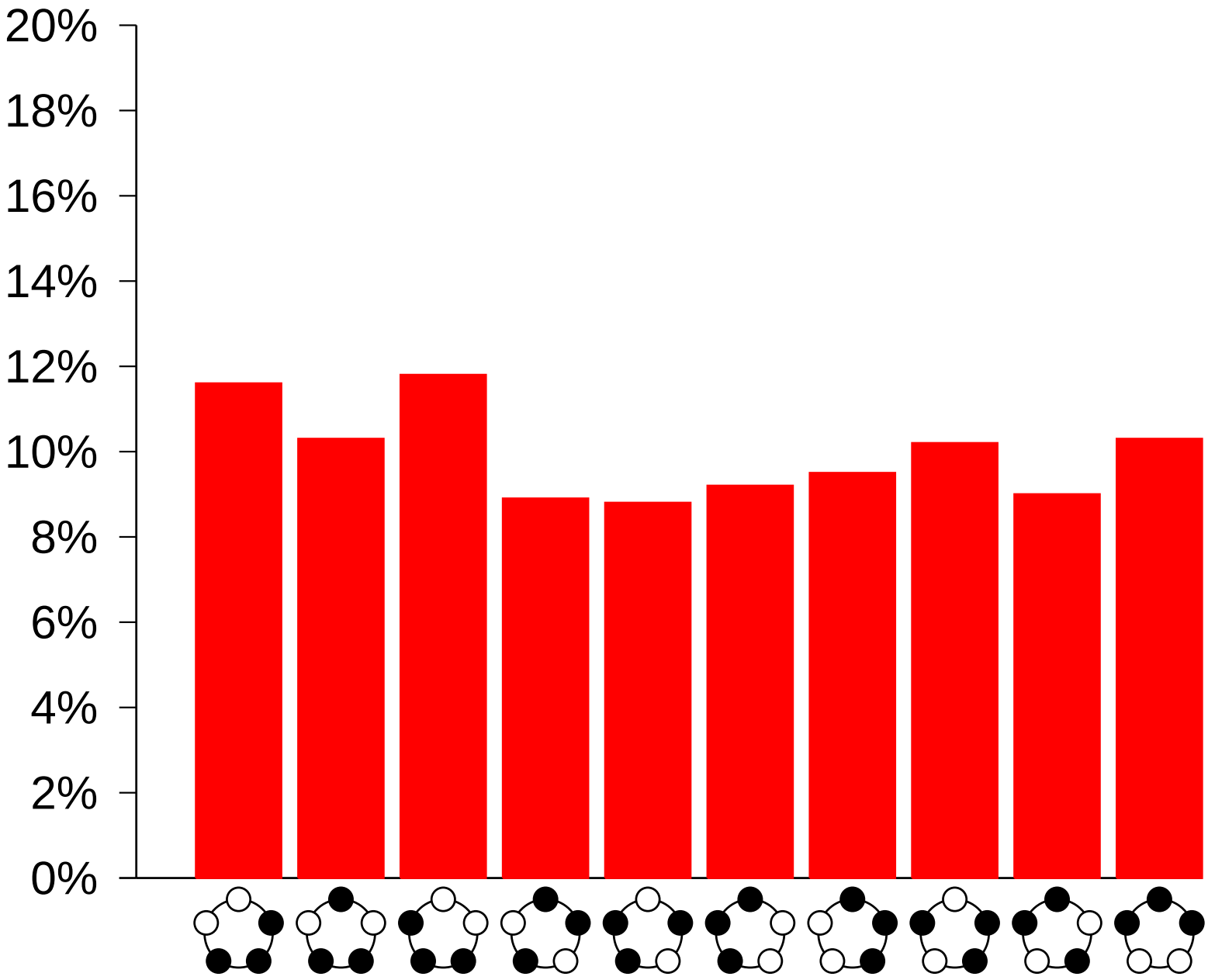
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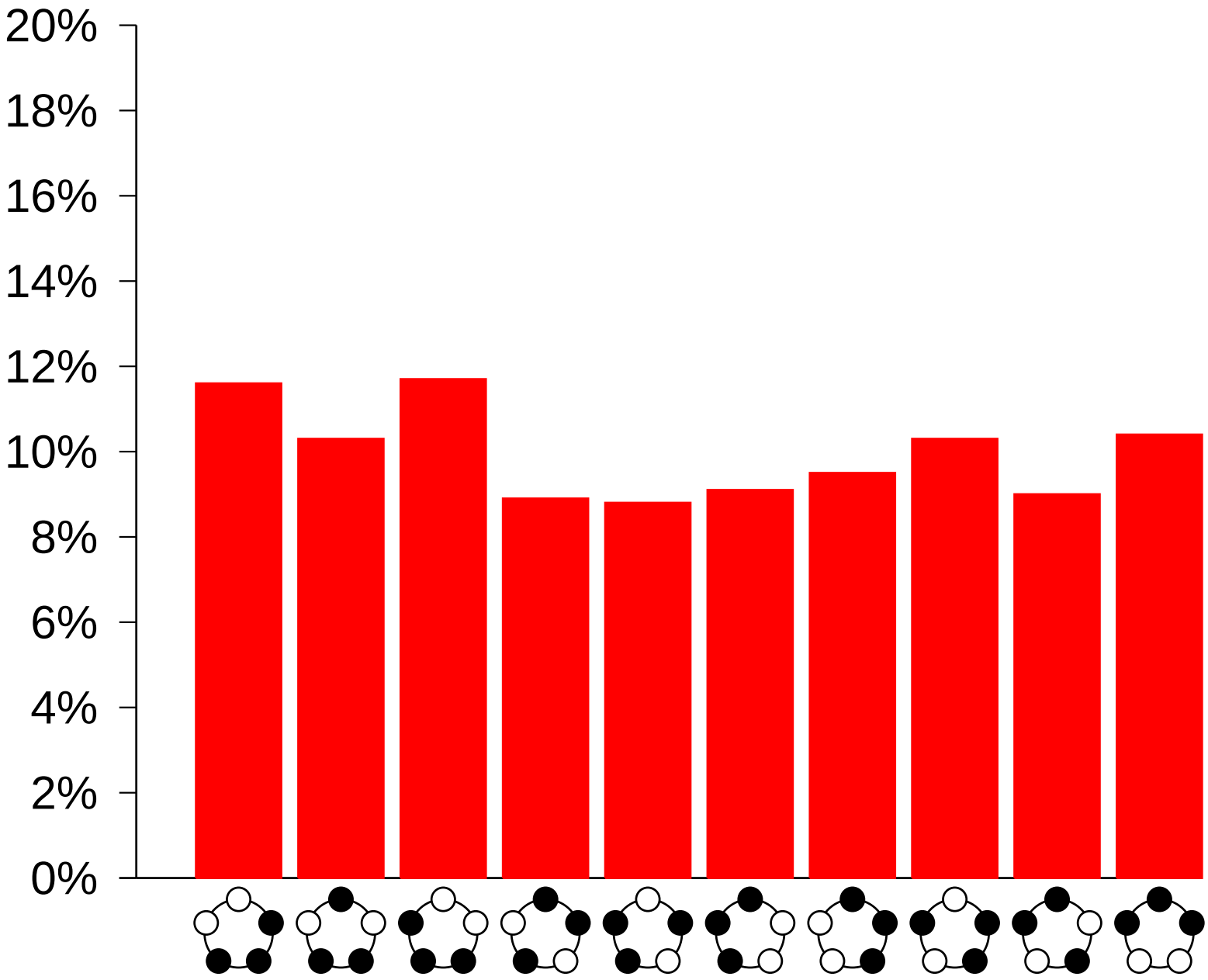
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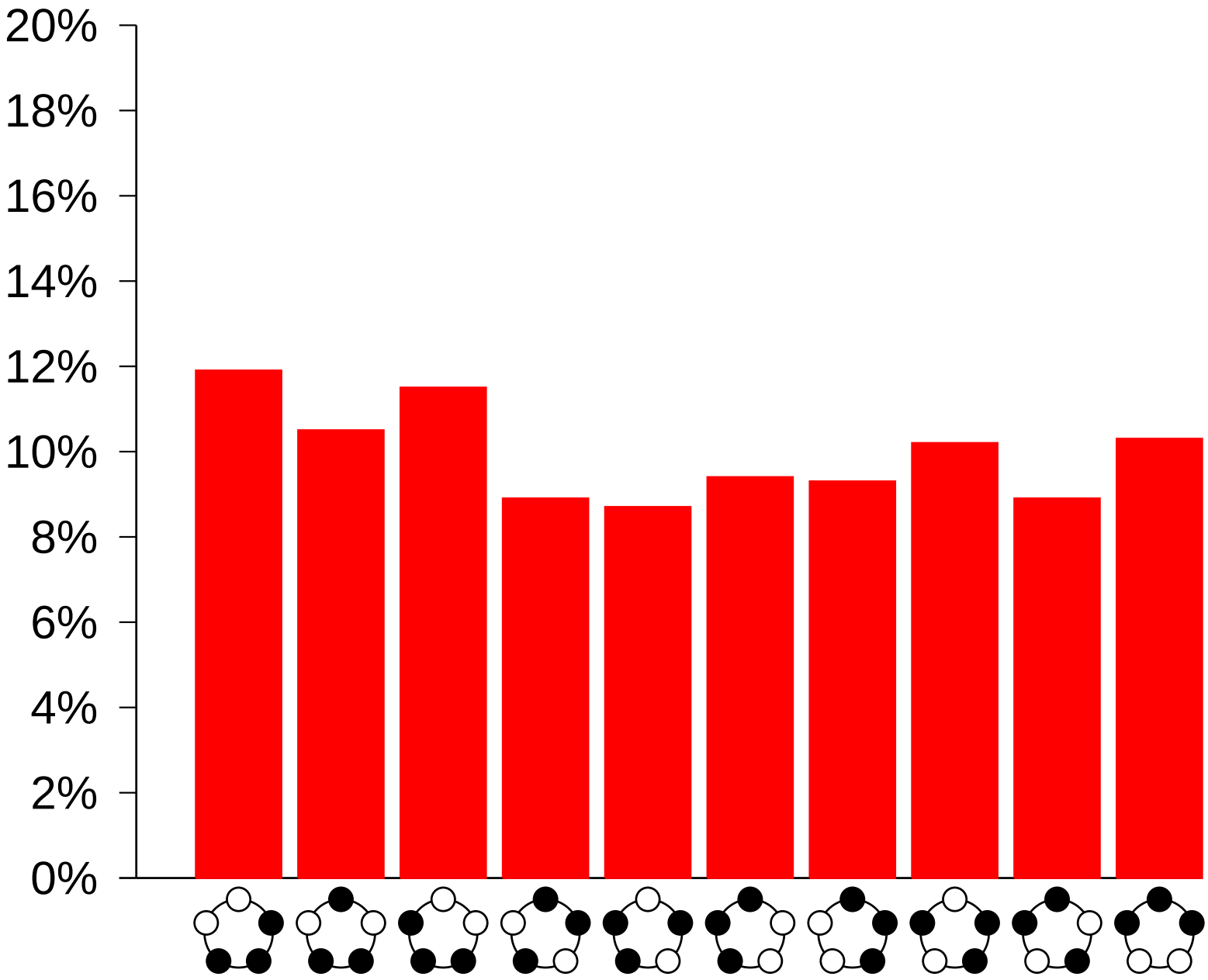
Stationary distribution



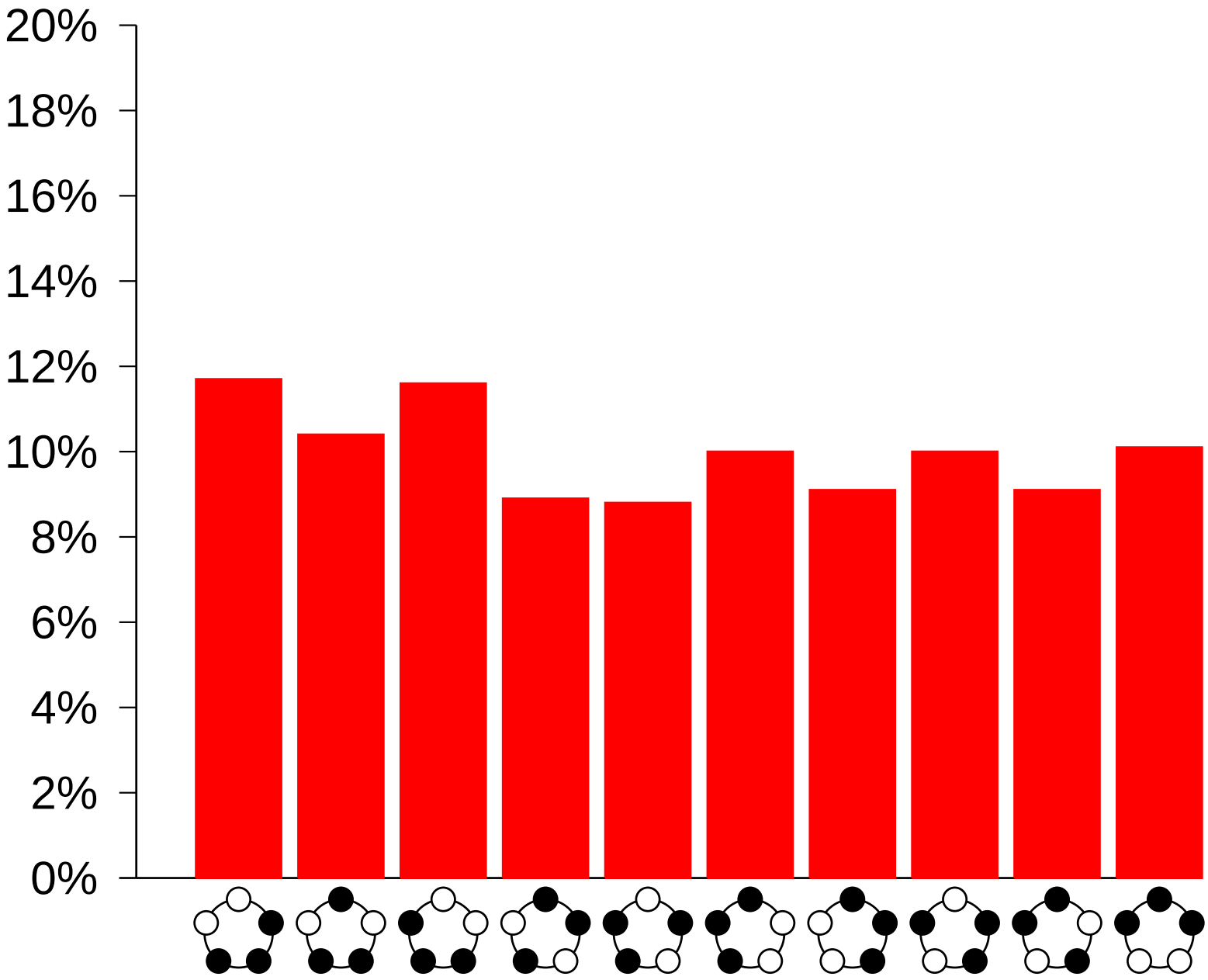
Stationary distribution



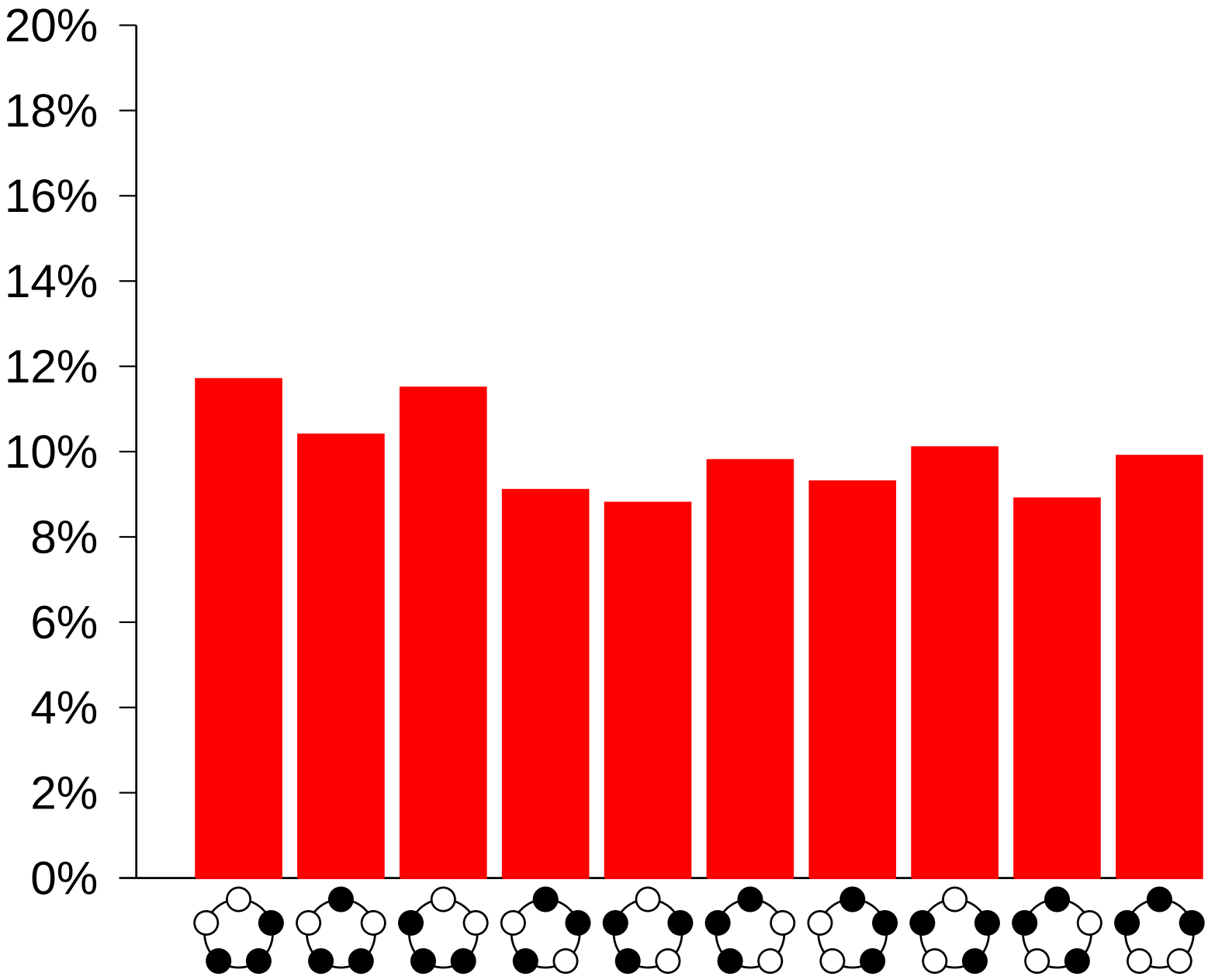
Stationary distribution



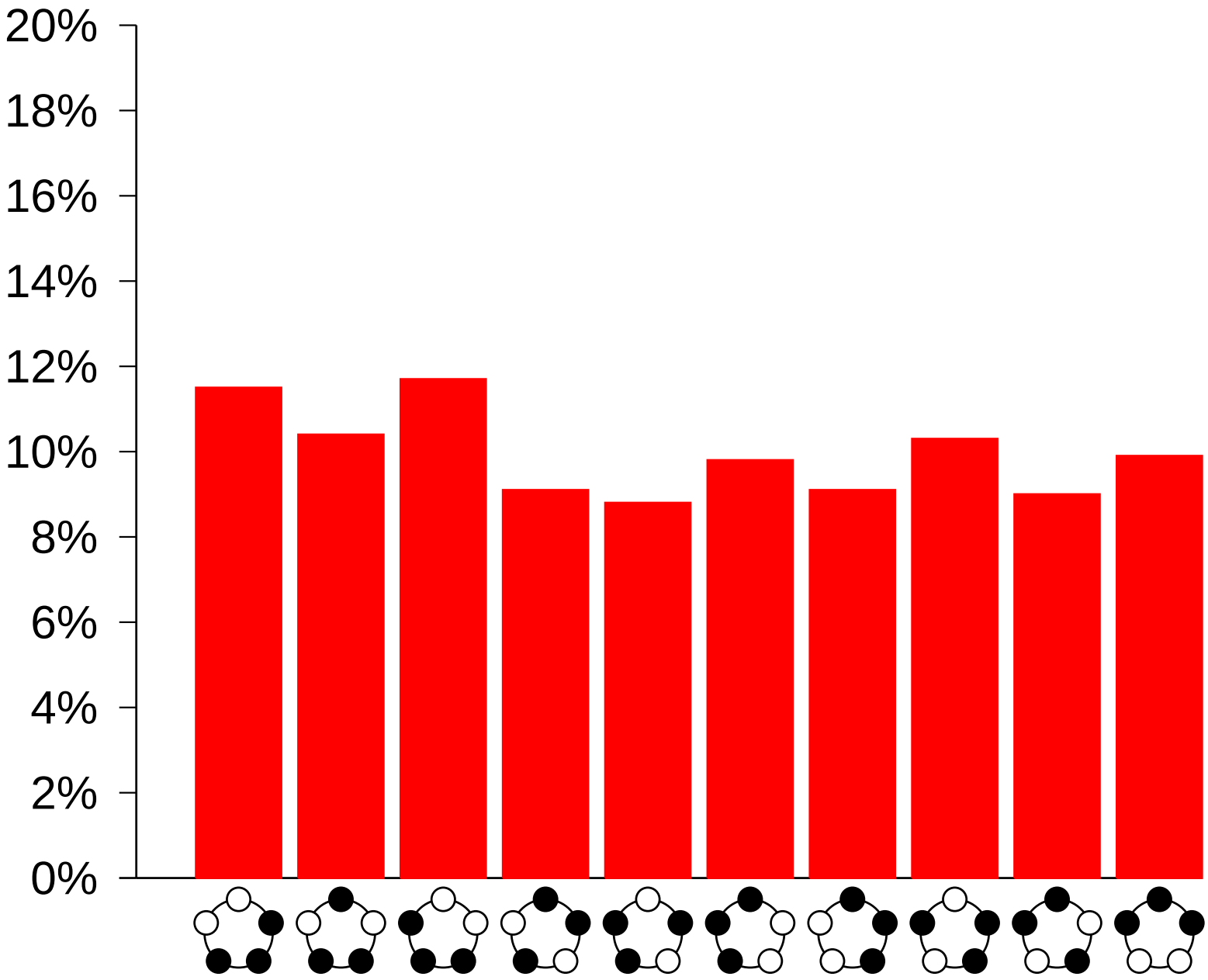
Stationary distribution



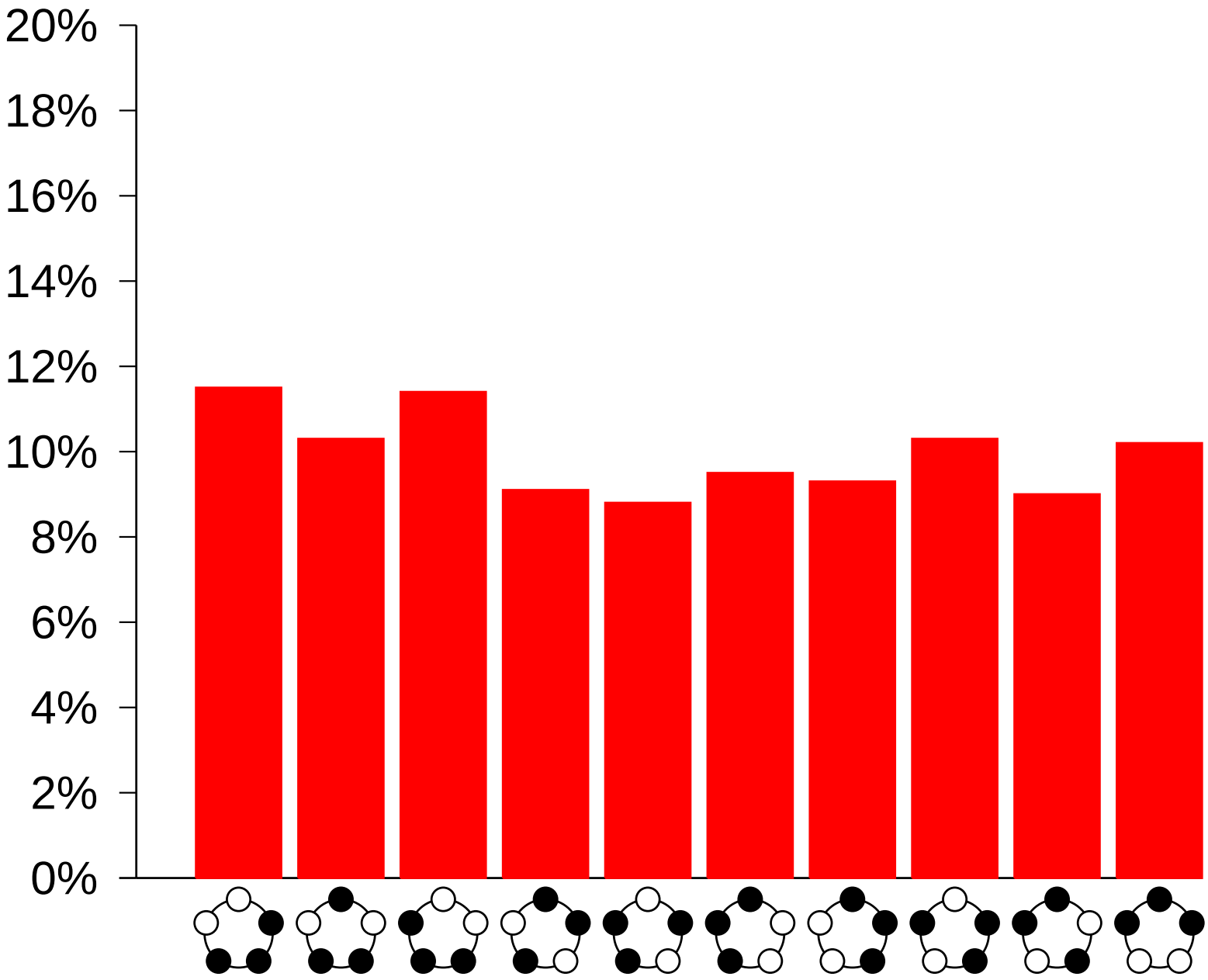
Stationary distribution



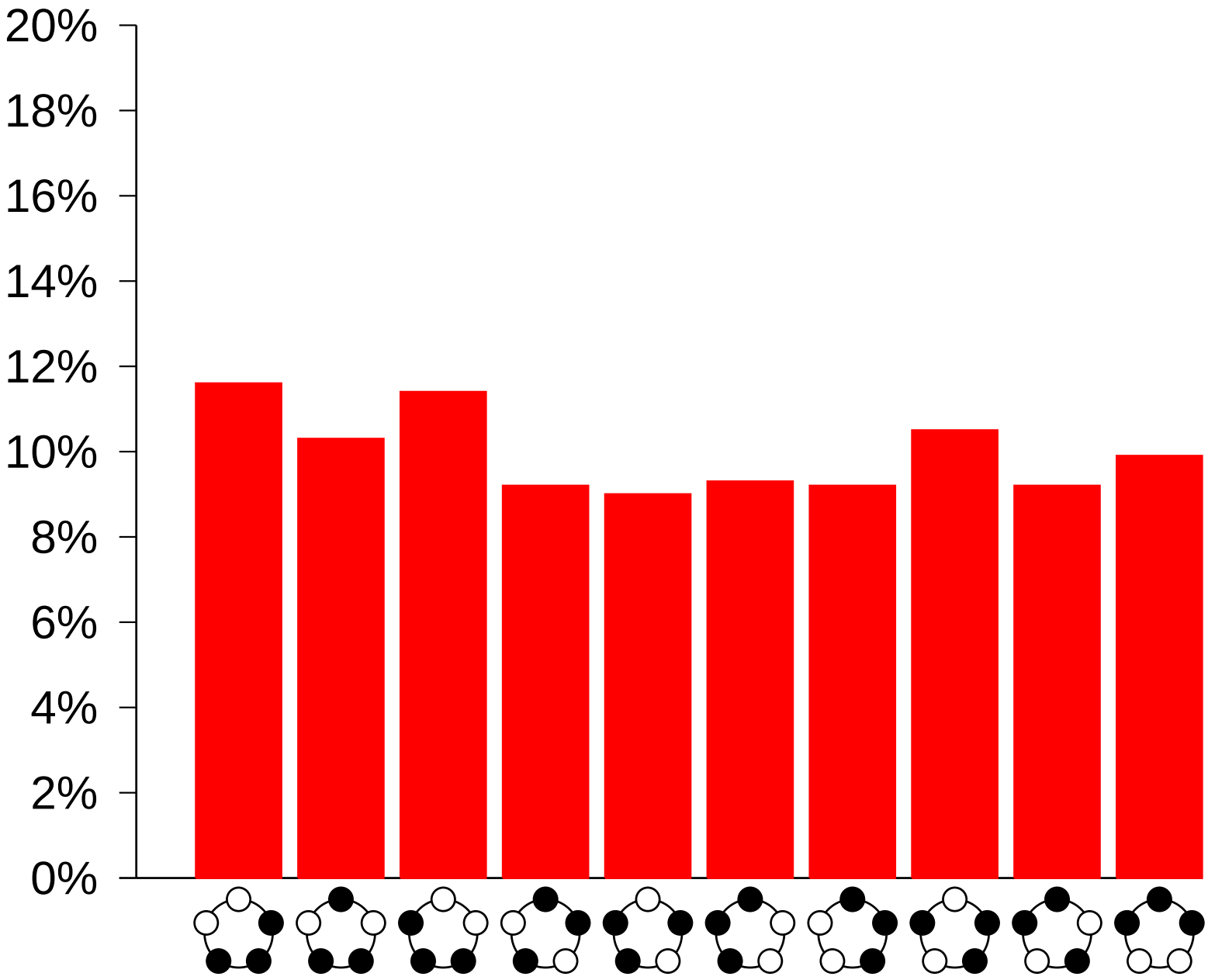
Stationary distribution



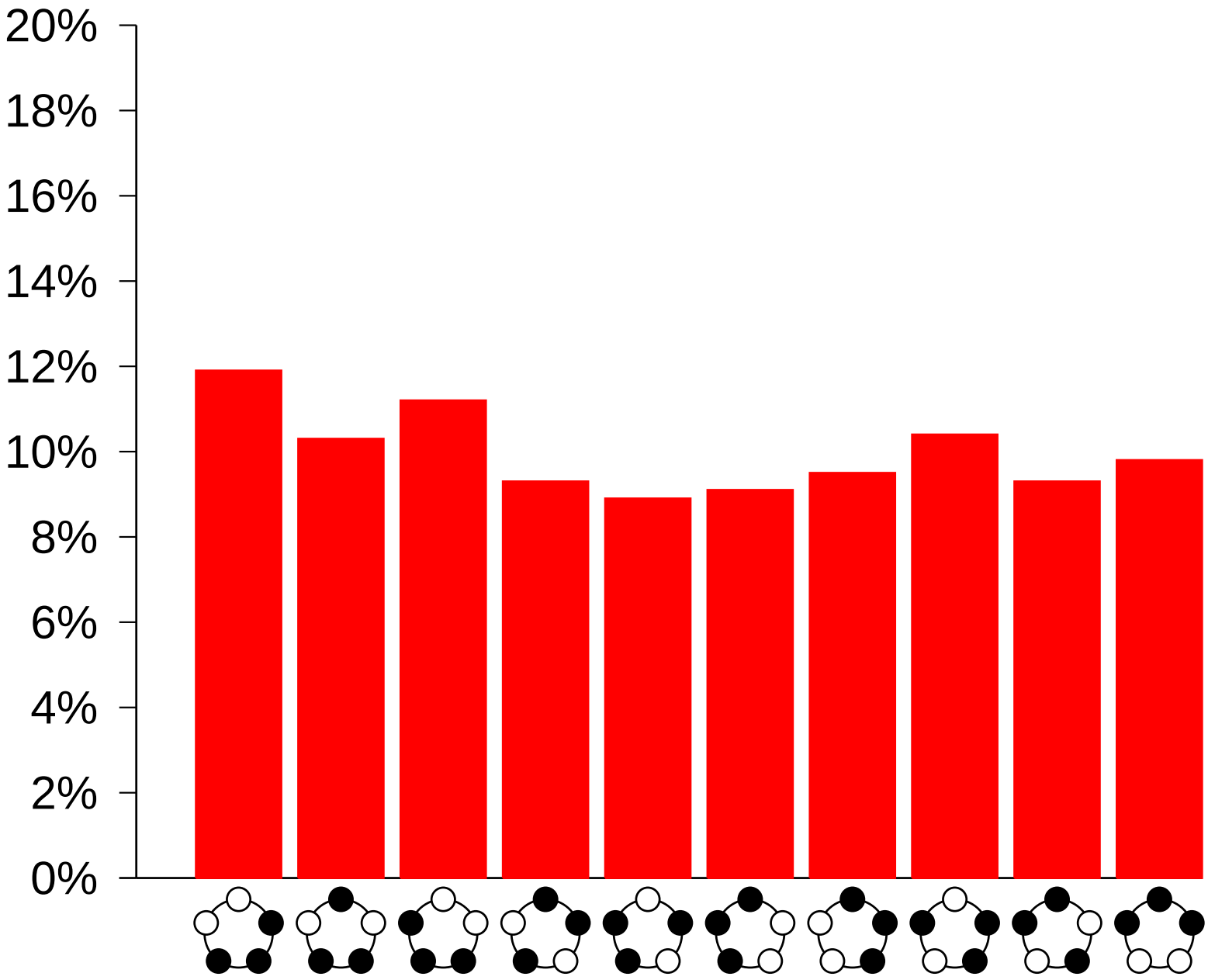
Stationary distribution



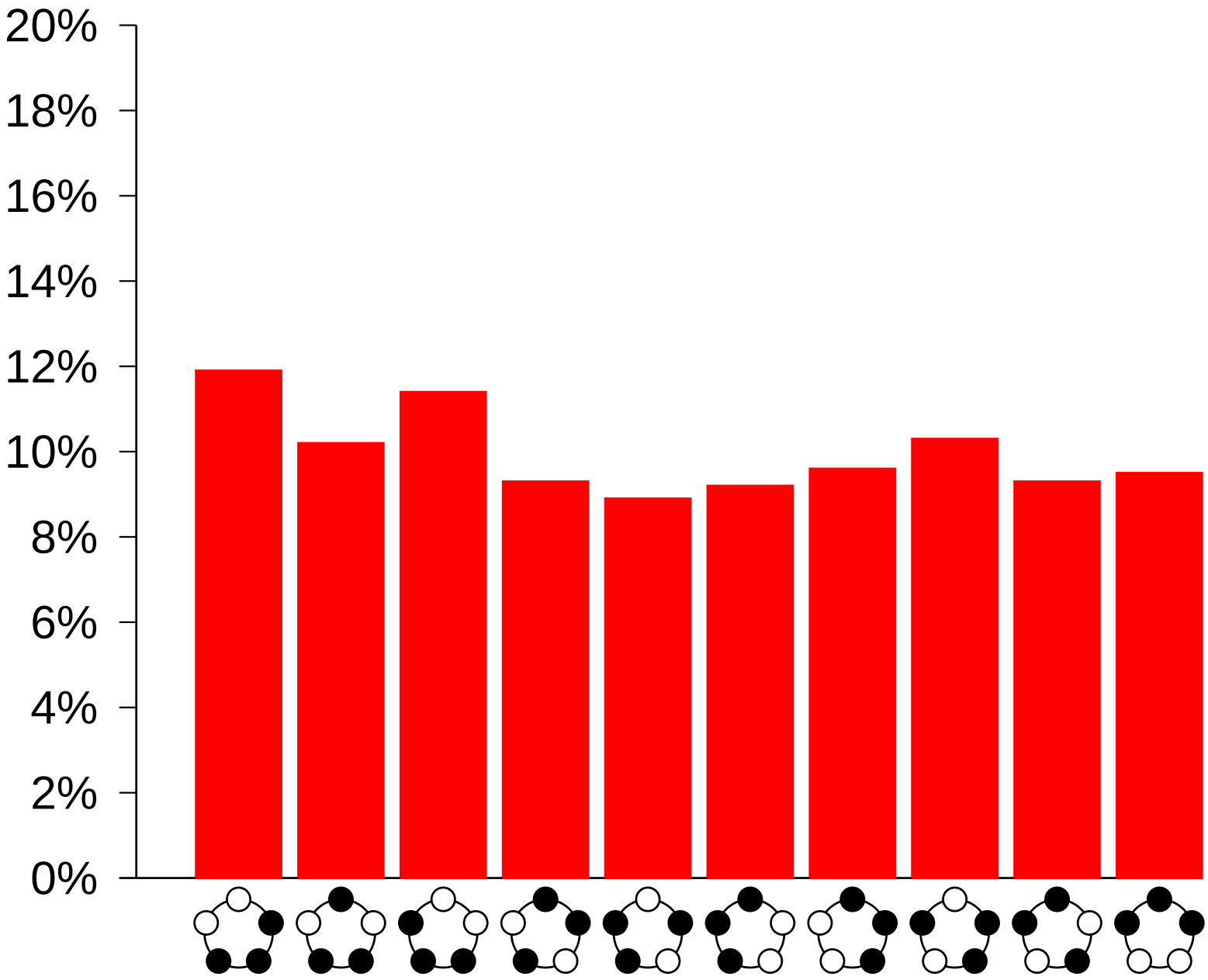
Stationary distribution



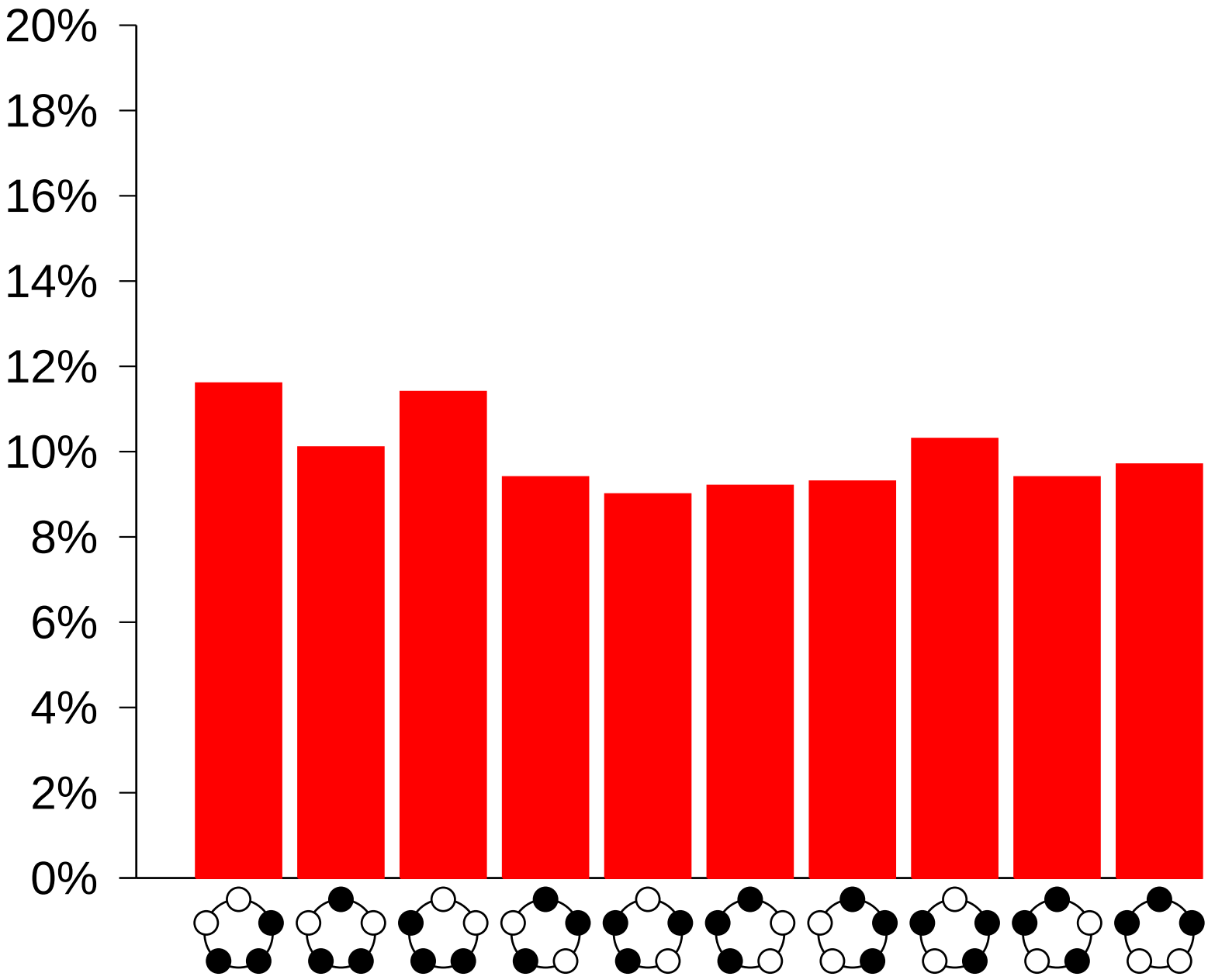
Stationary distribution



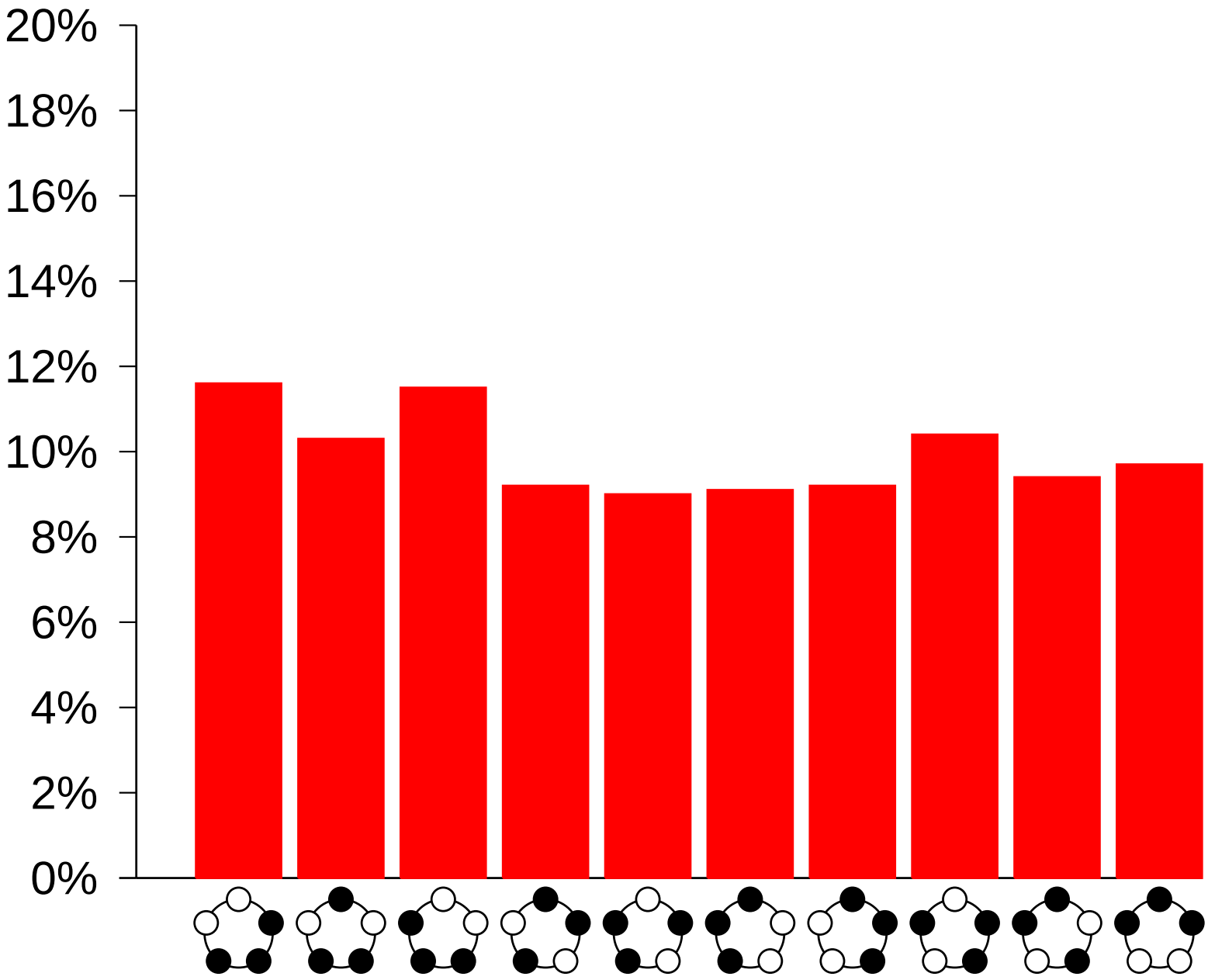
Stationary distribution



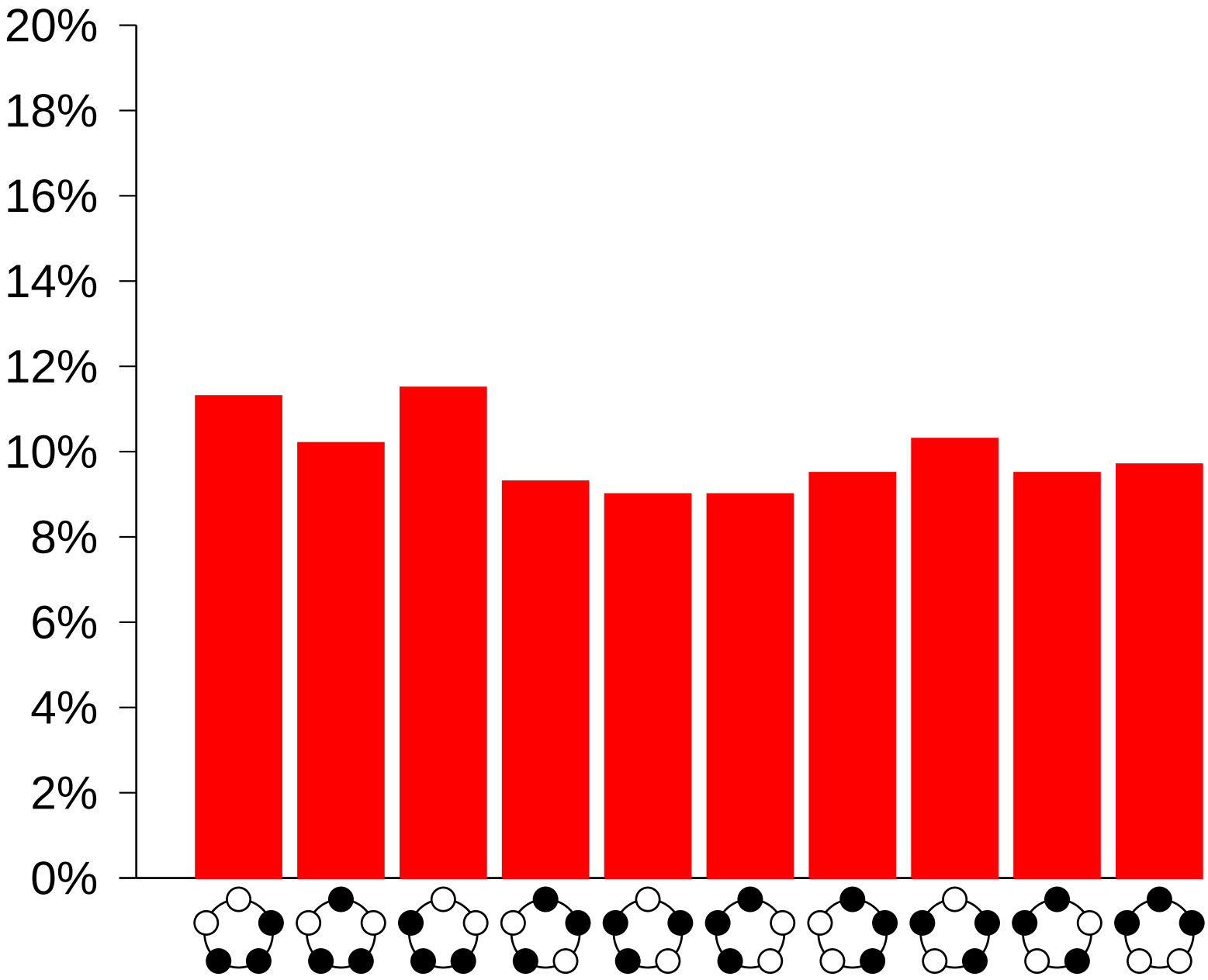
Stationary distribution



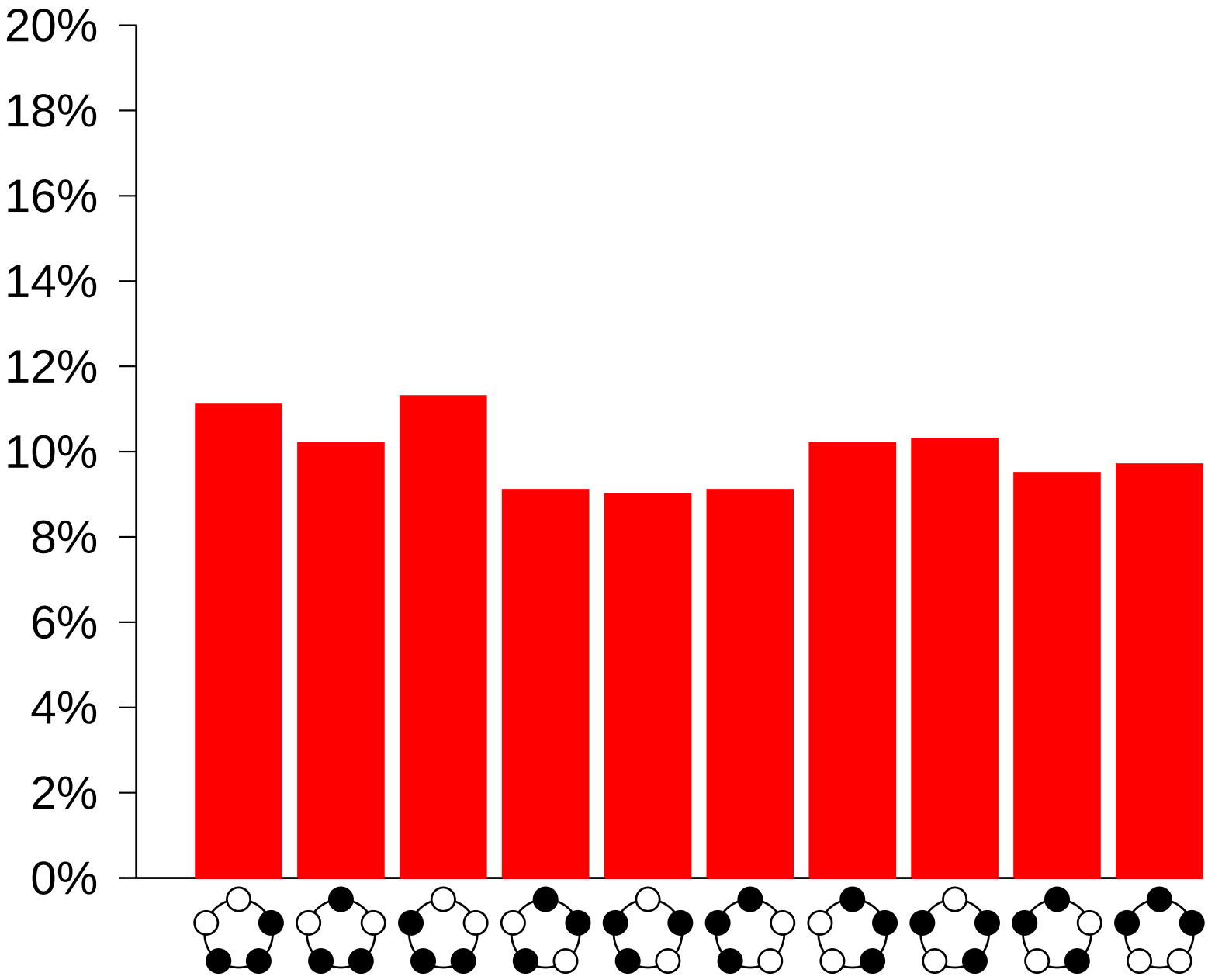
Stationary distribution



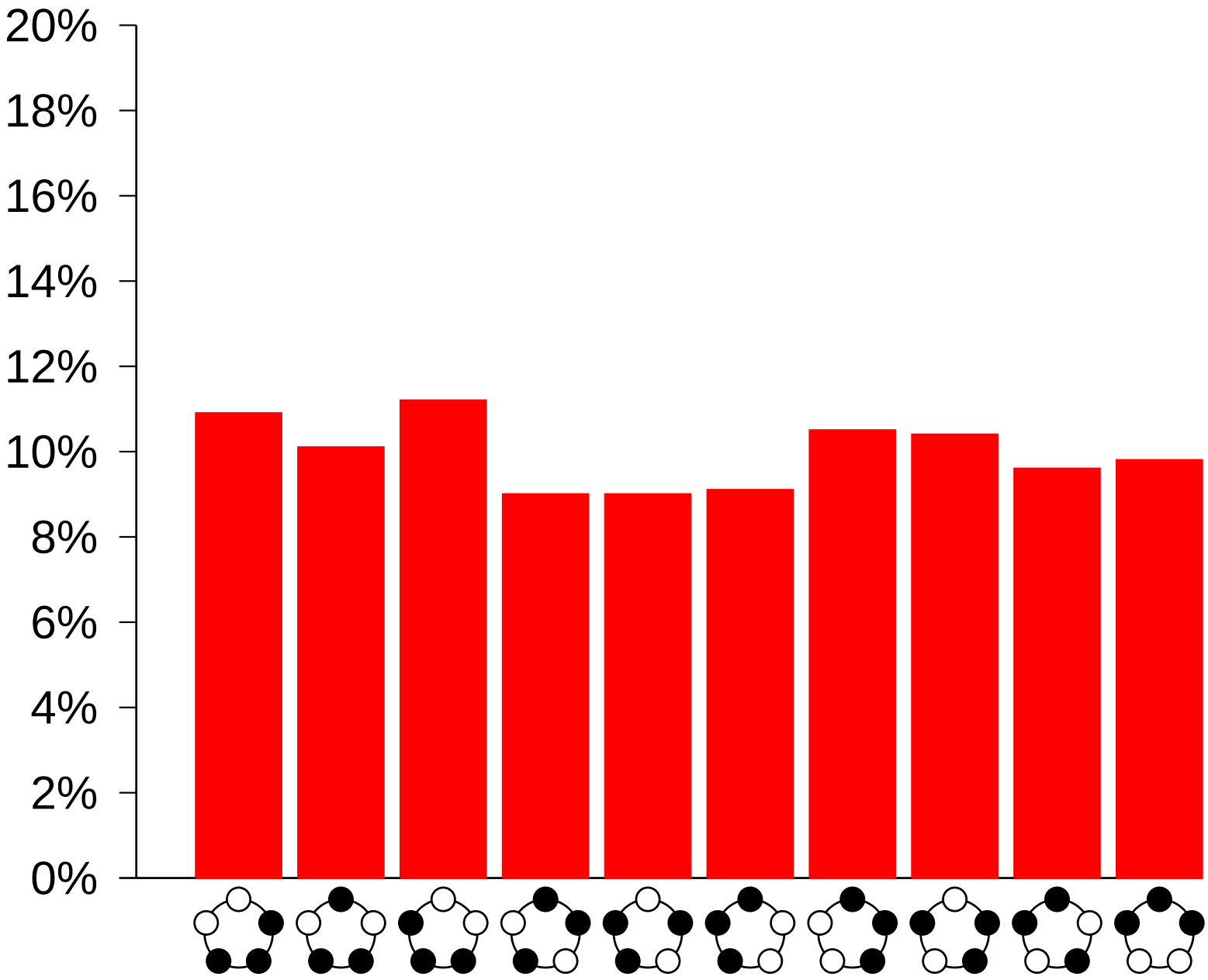
Stationary distribution



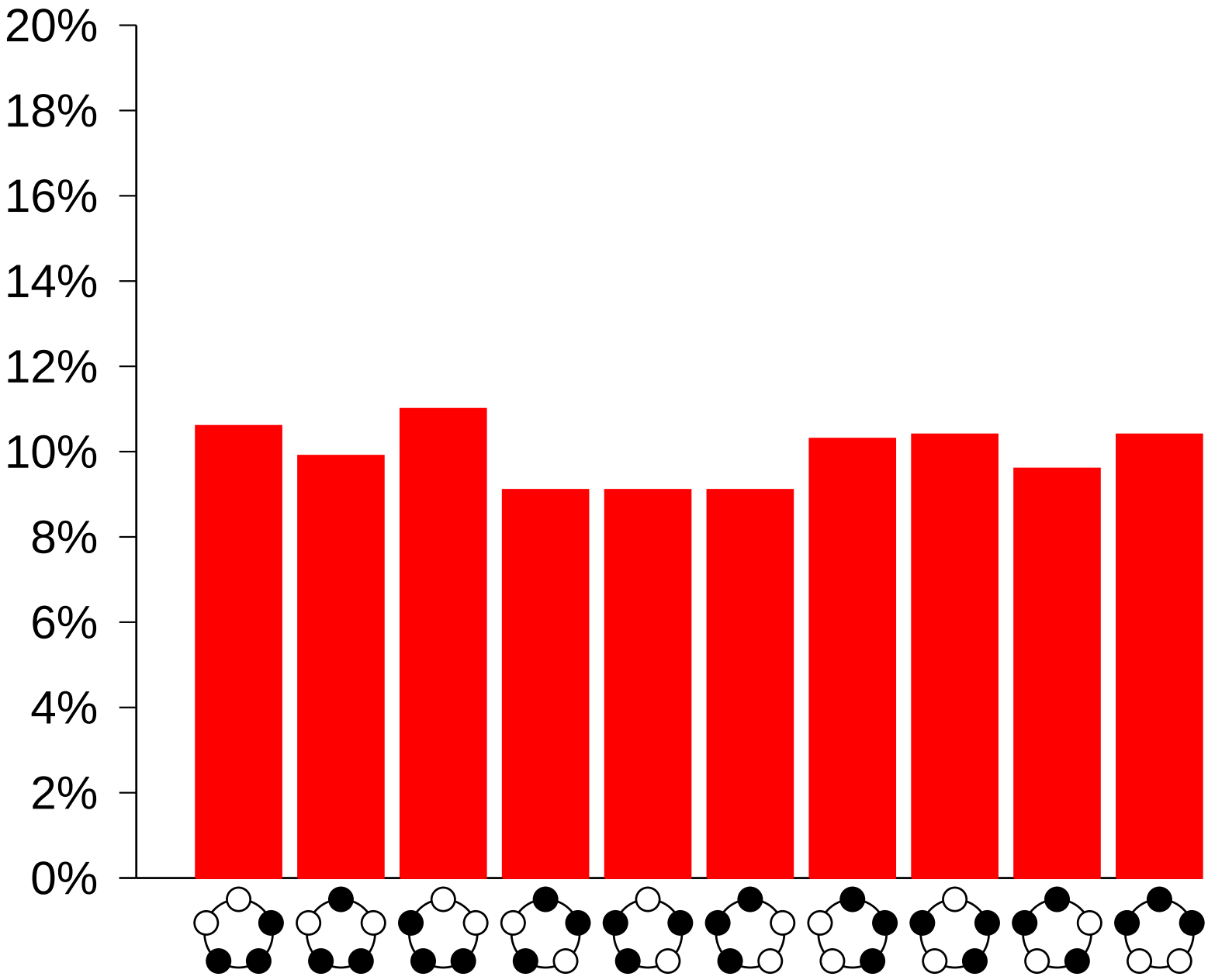
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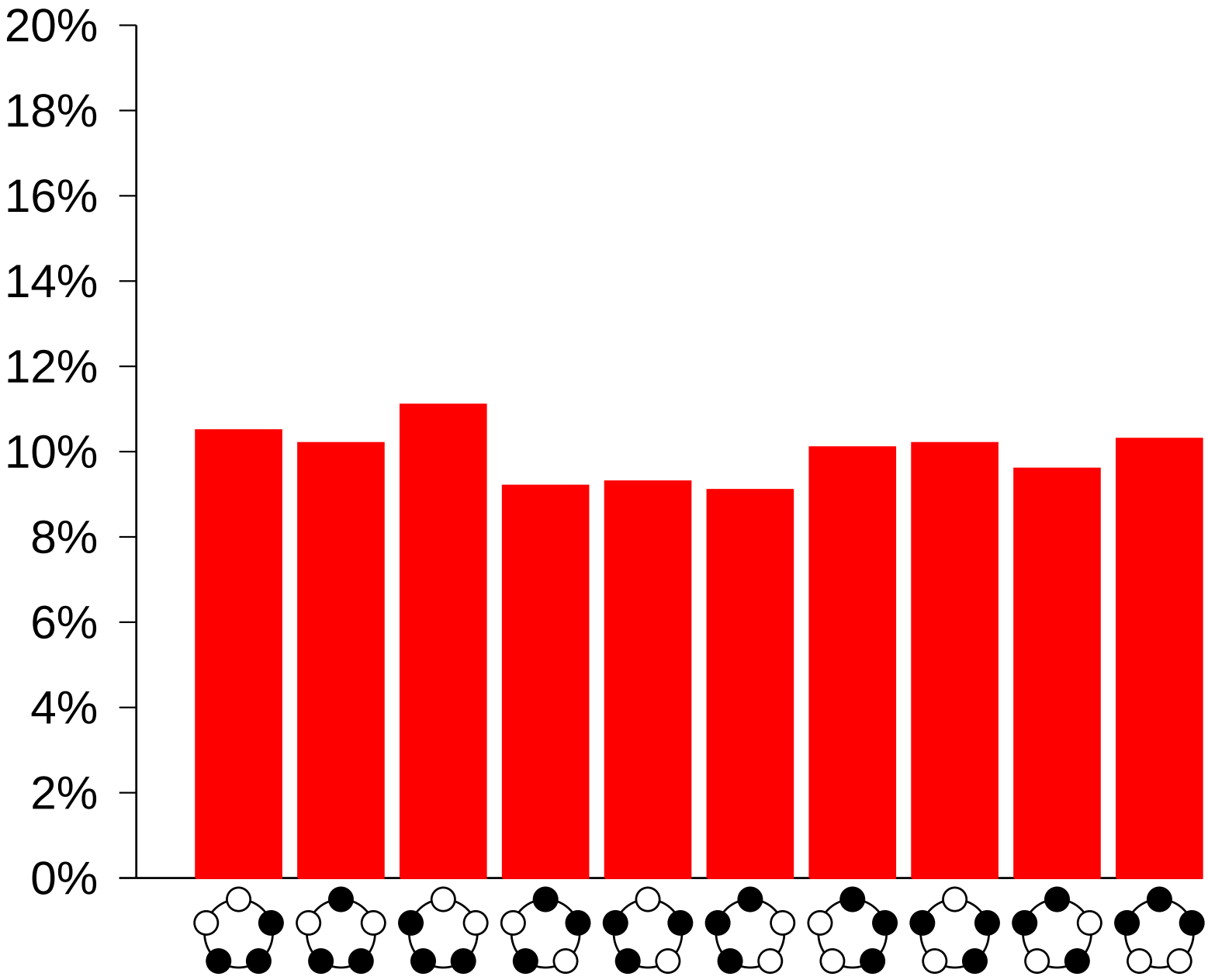
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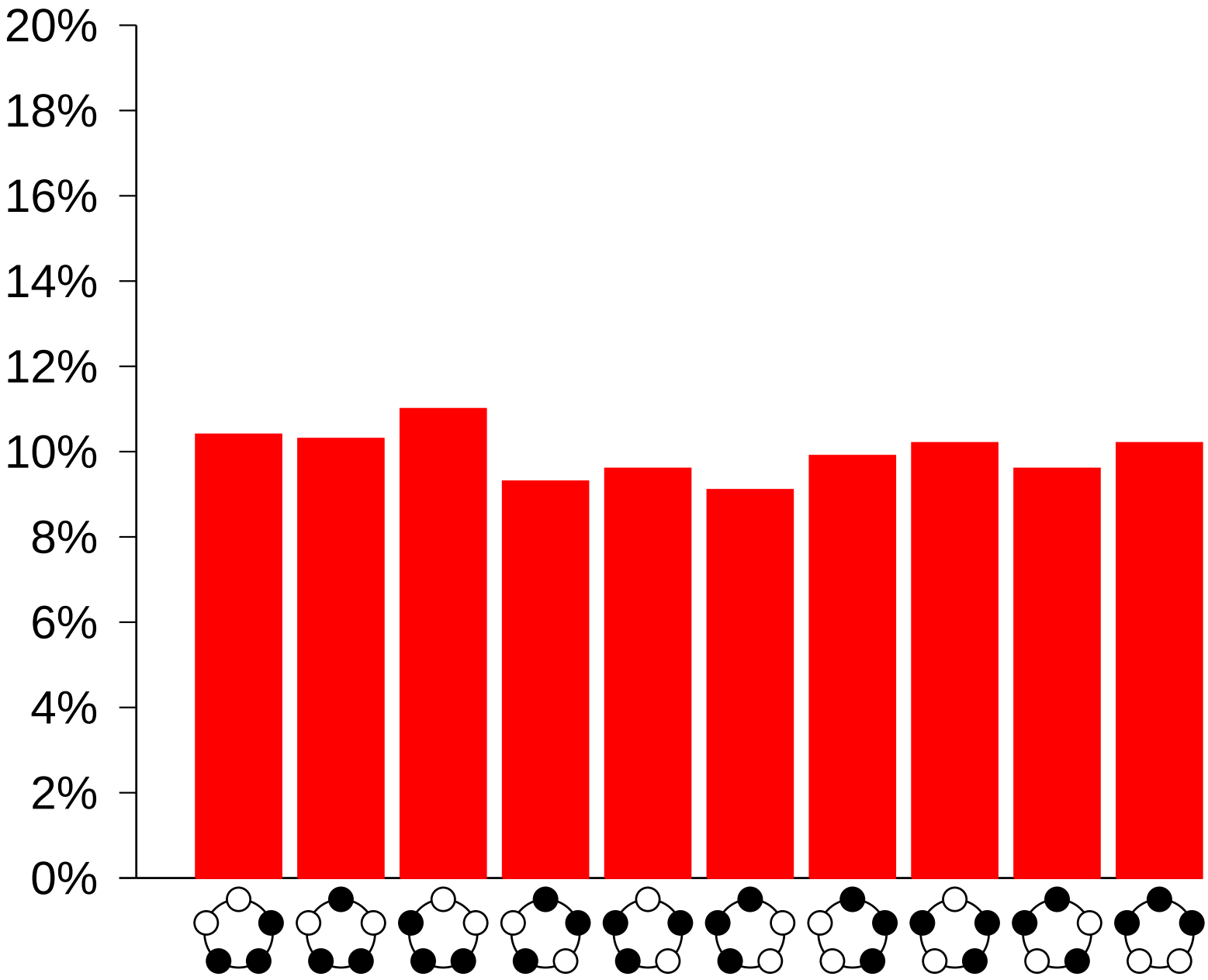
Stationary distribution



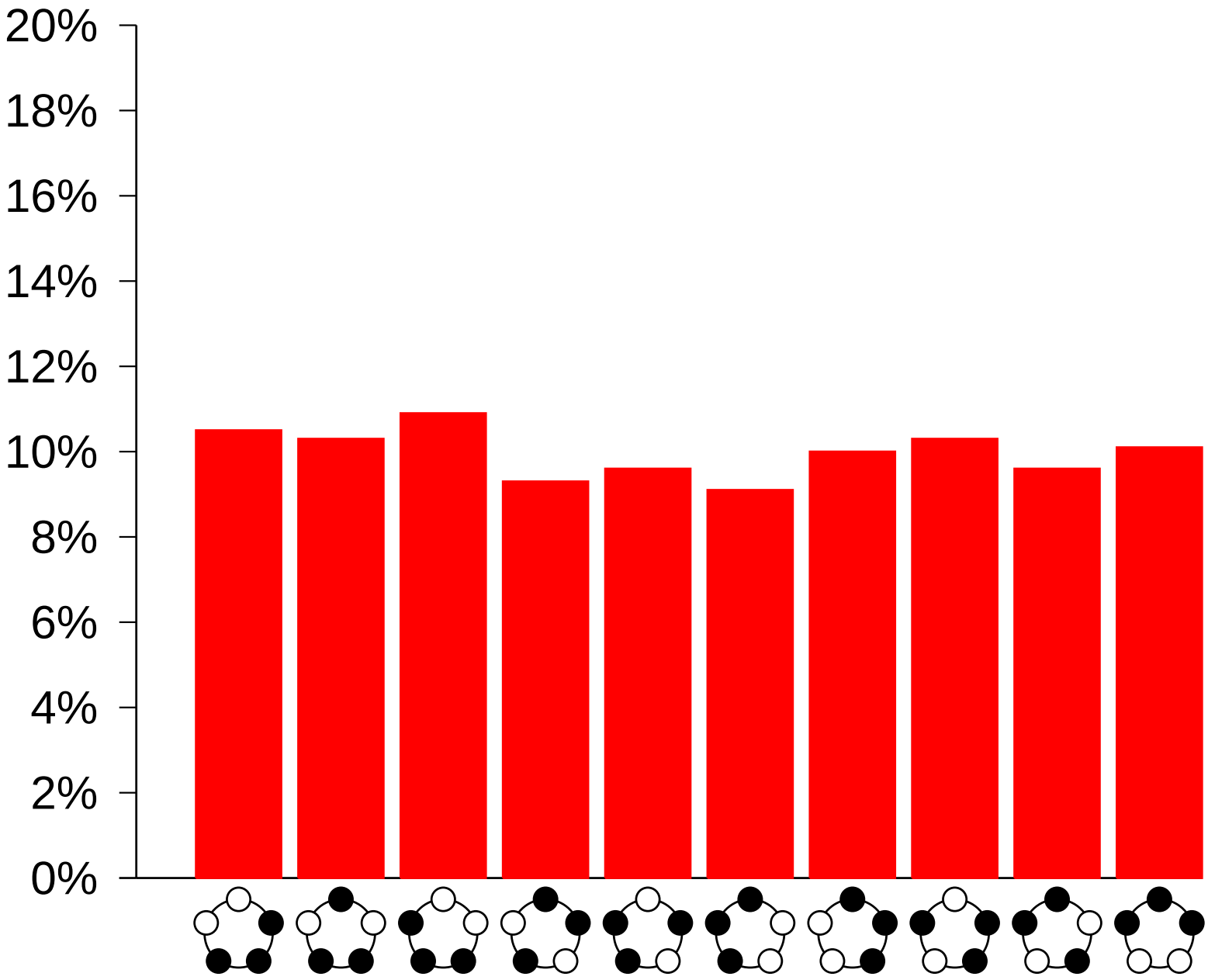
Stationary distribution



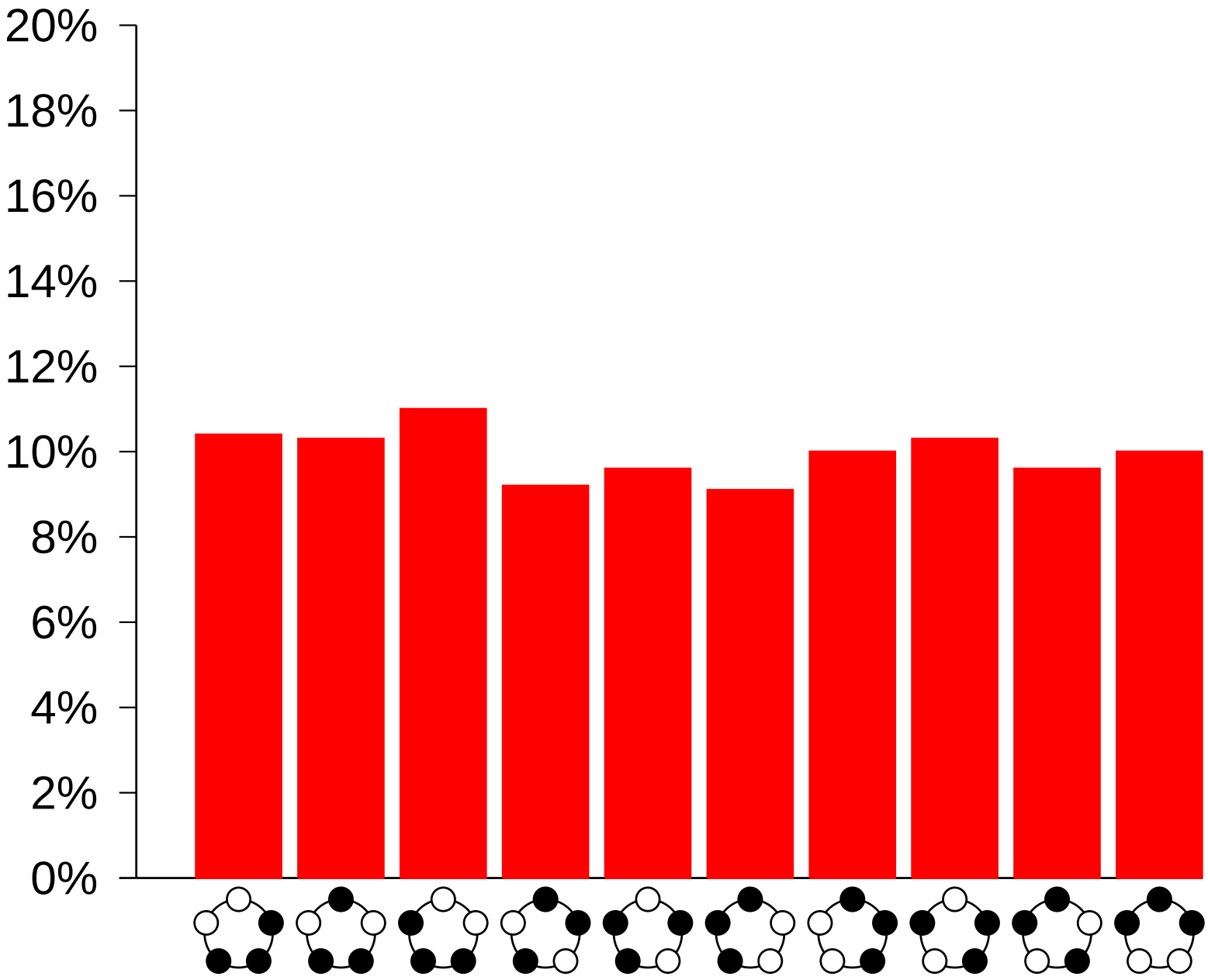
Stationary distribution



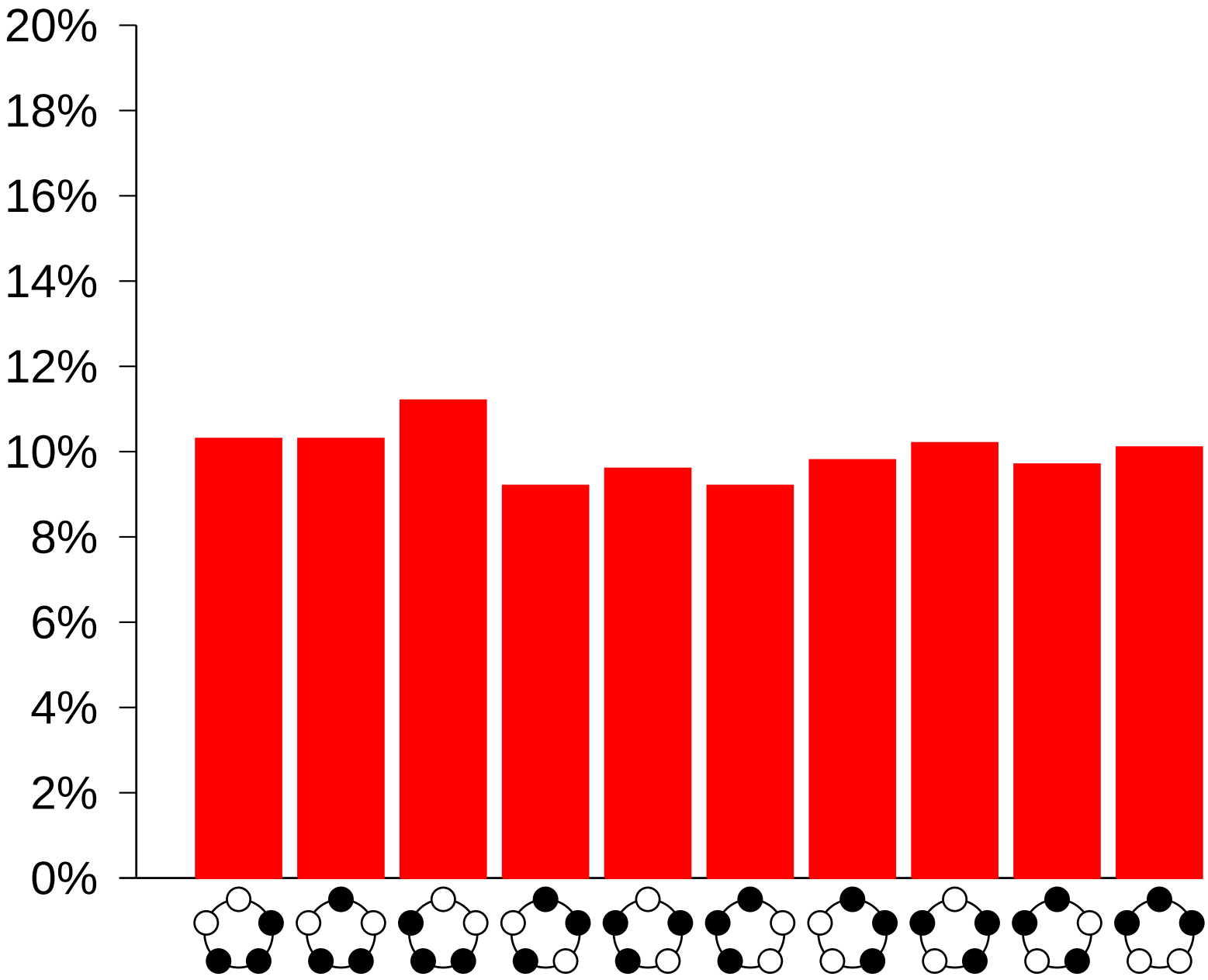
Stationary distribution



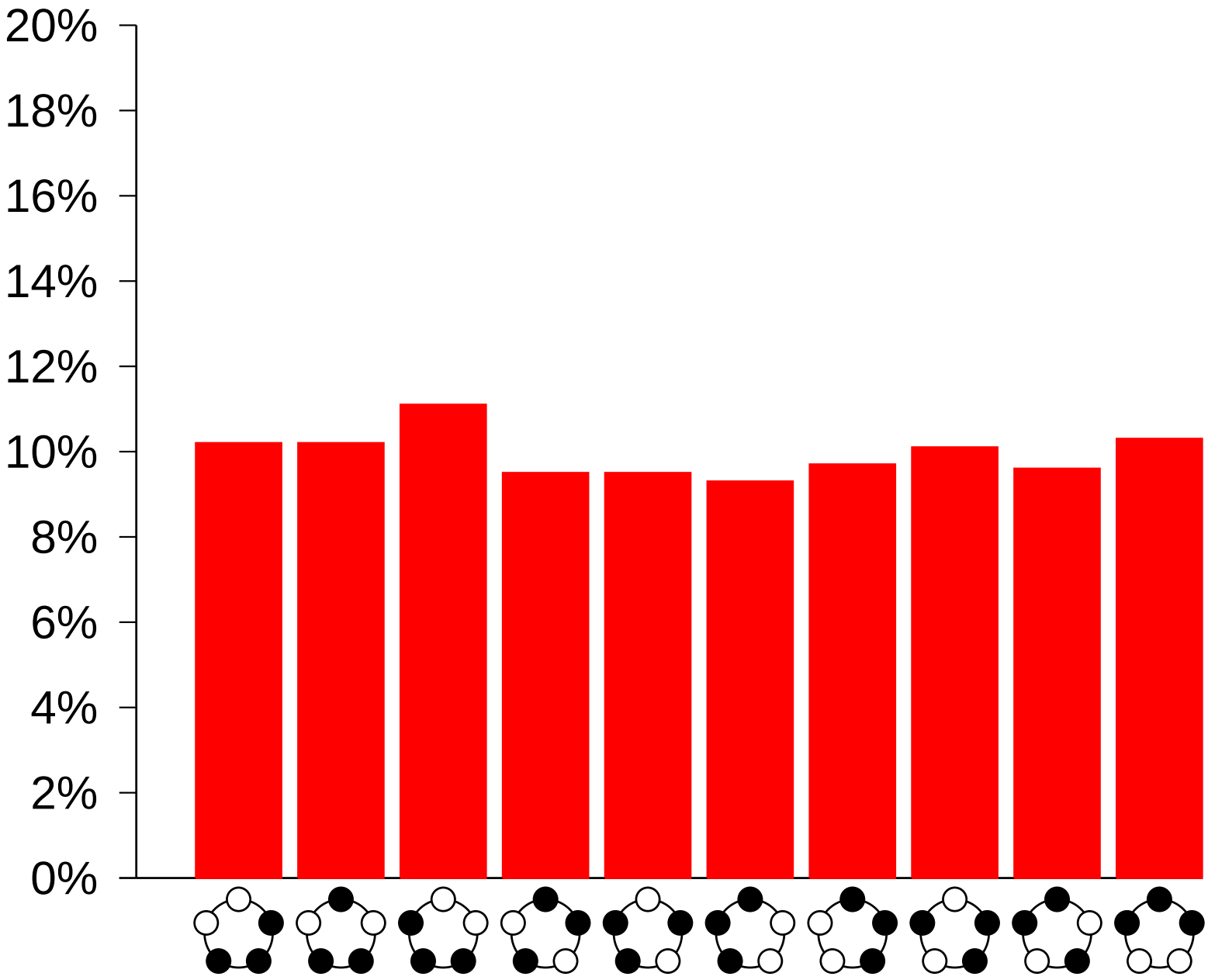
Stationary distribution



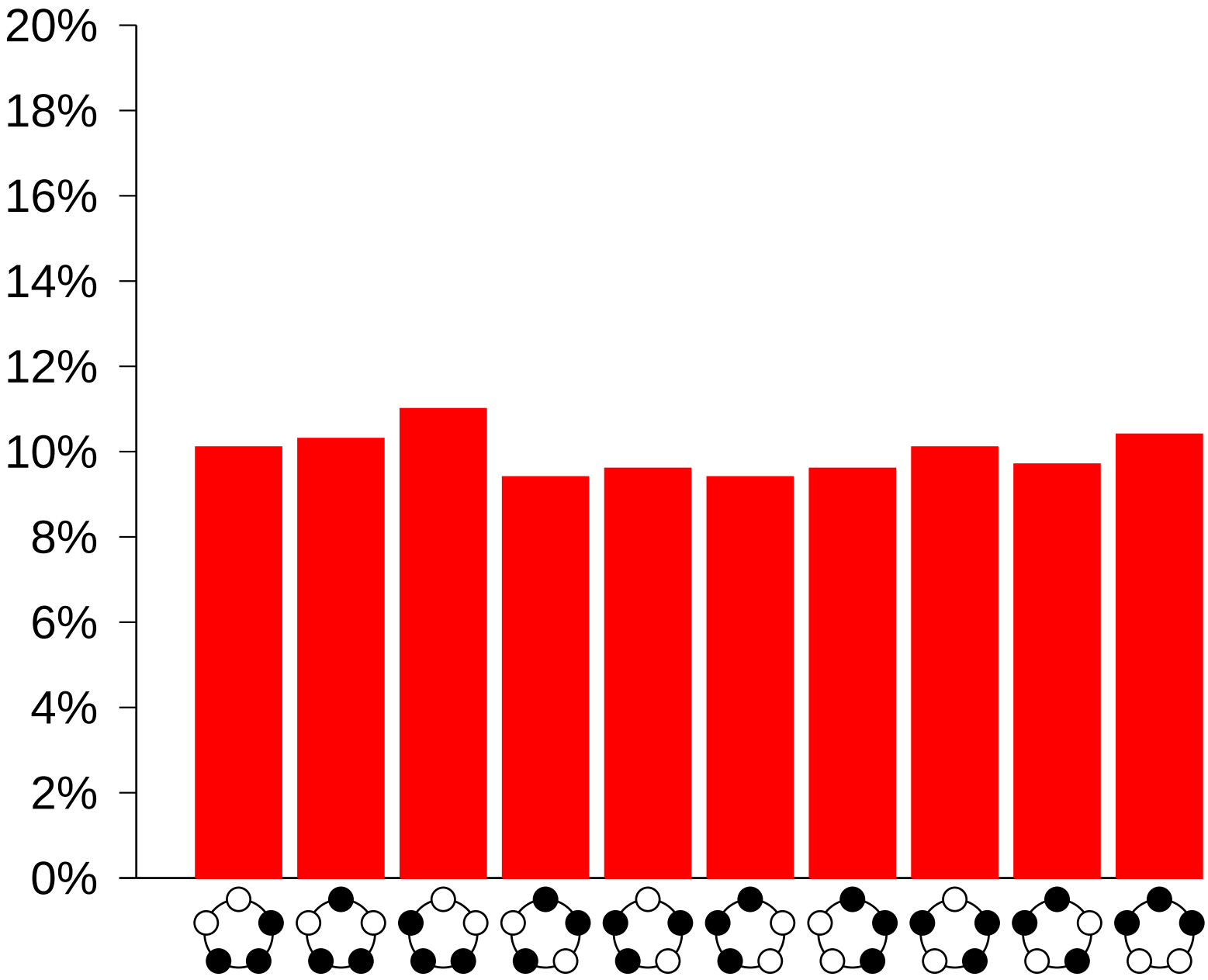
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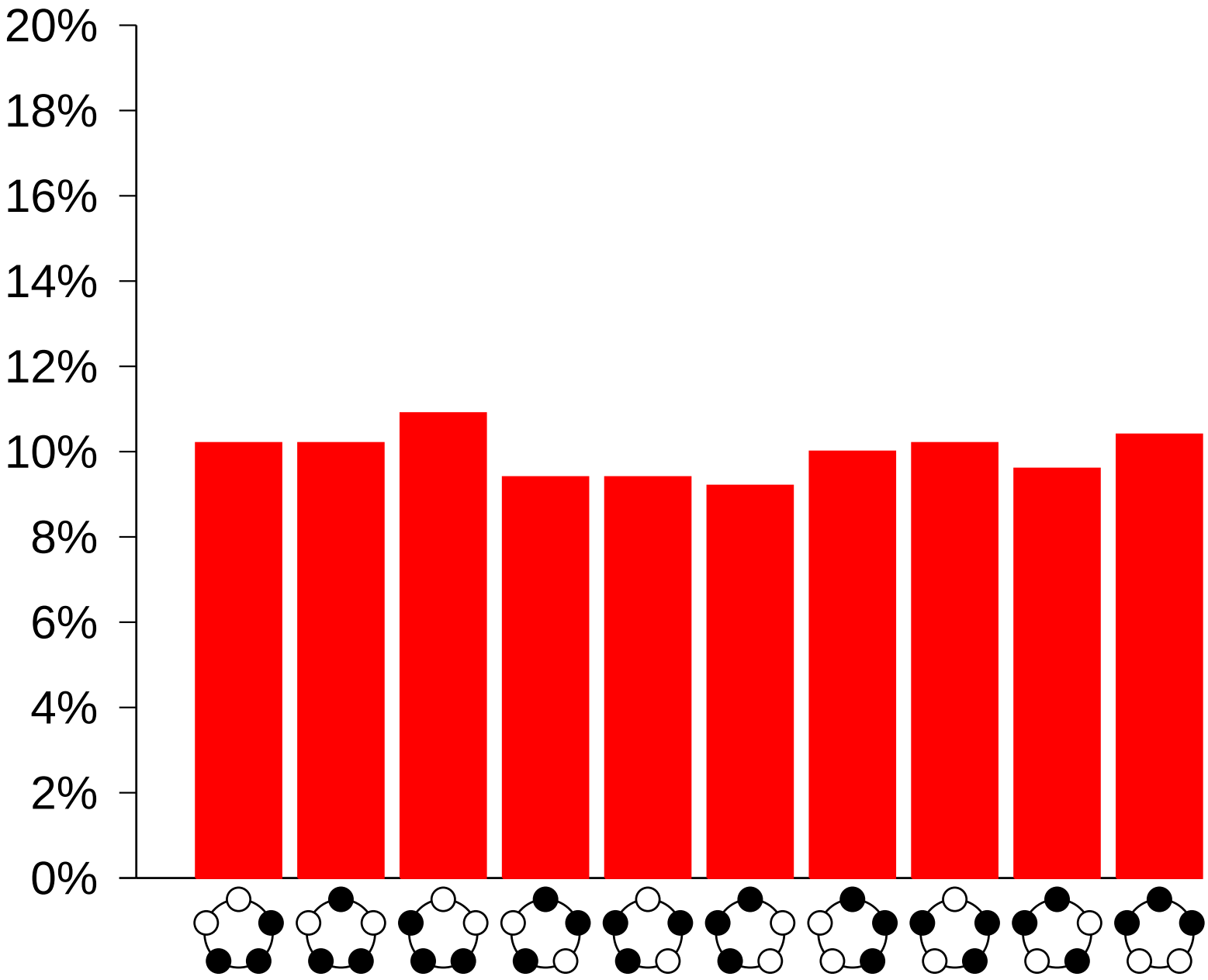
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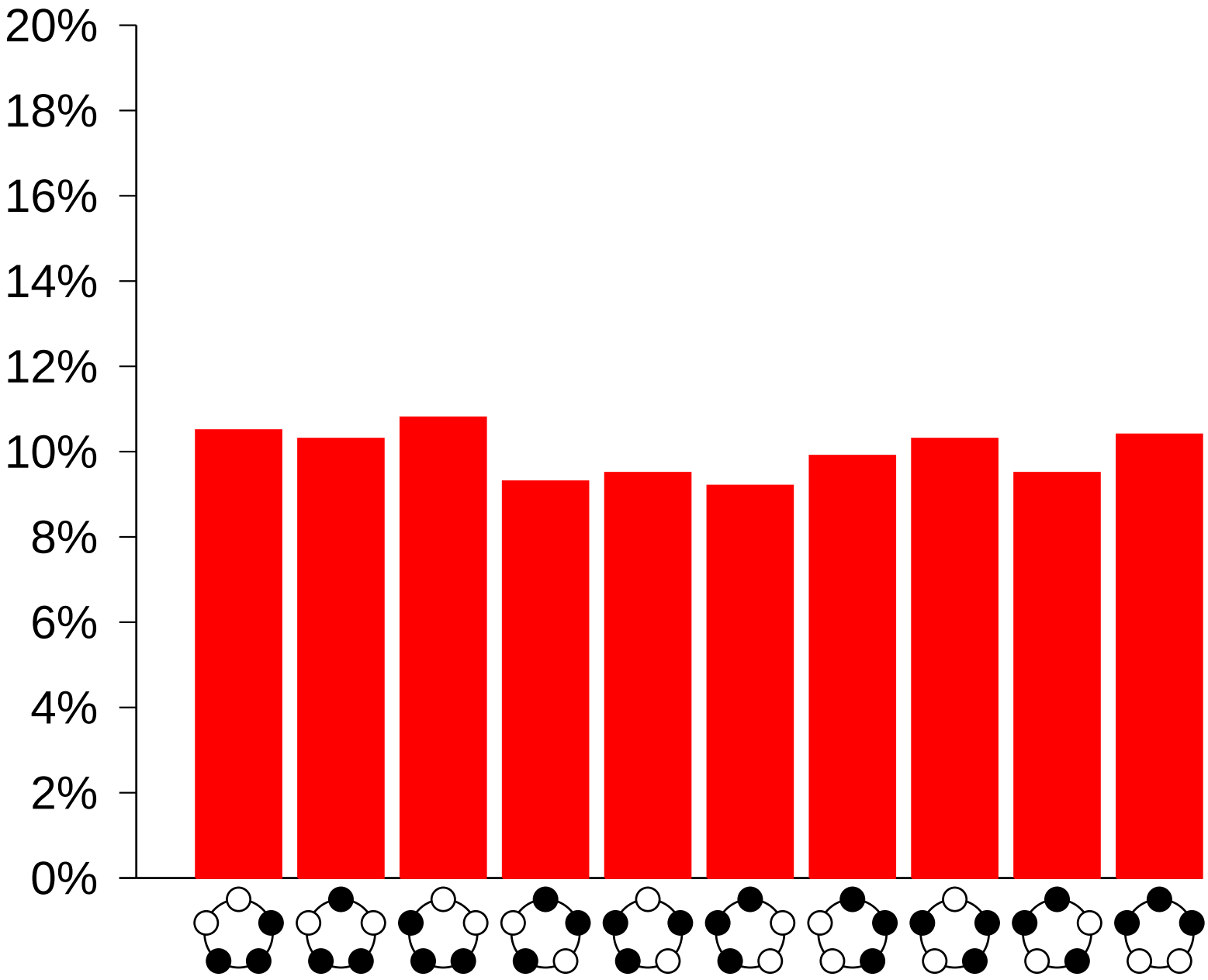
Stationary distribution



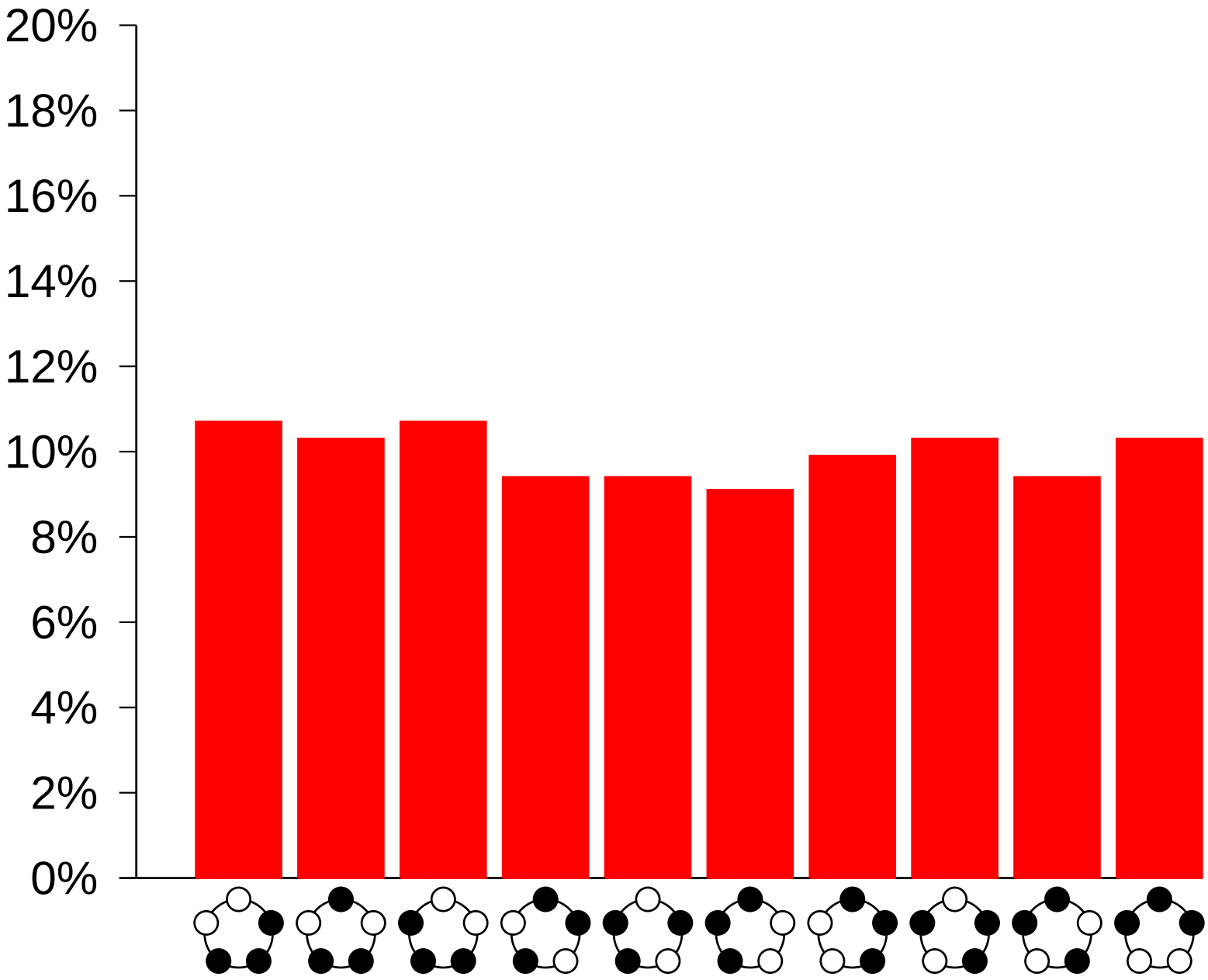
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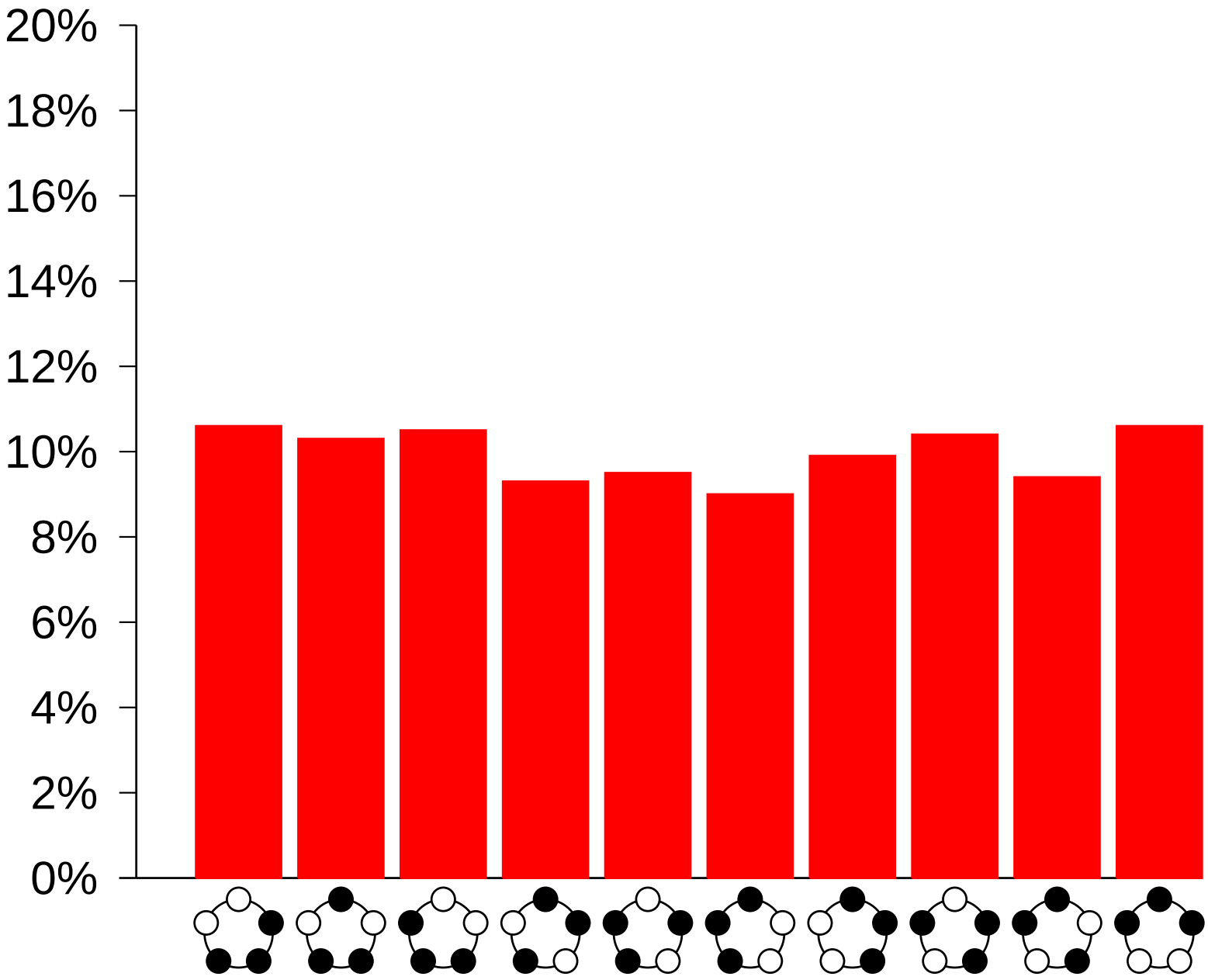
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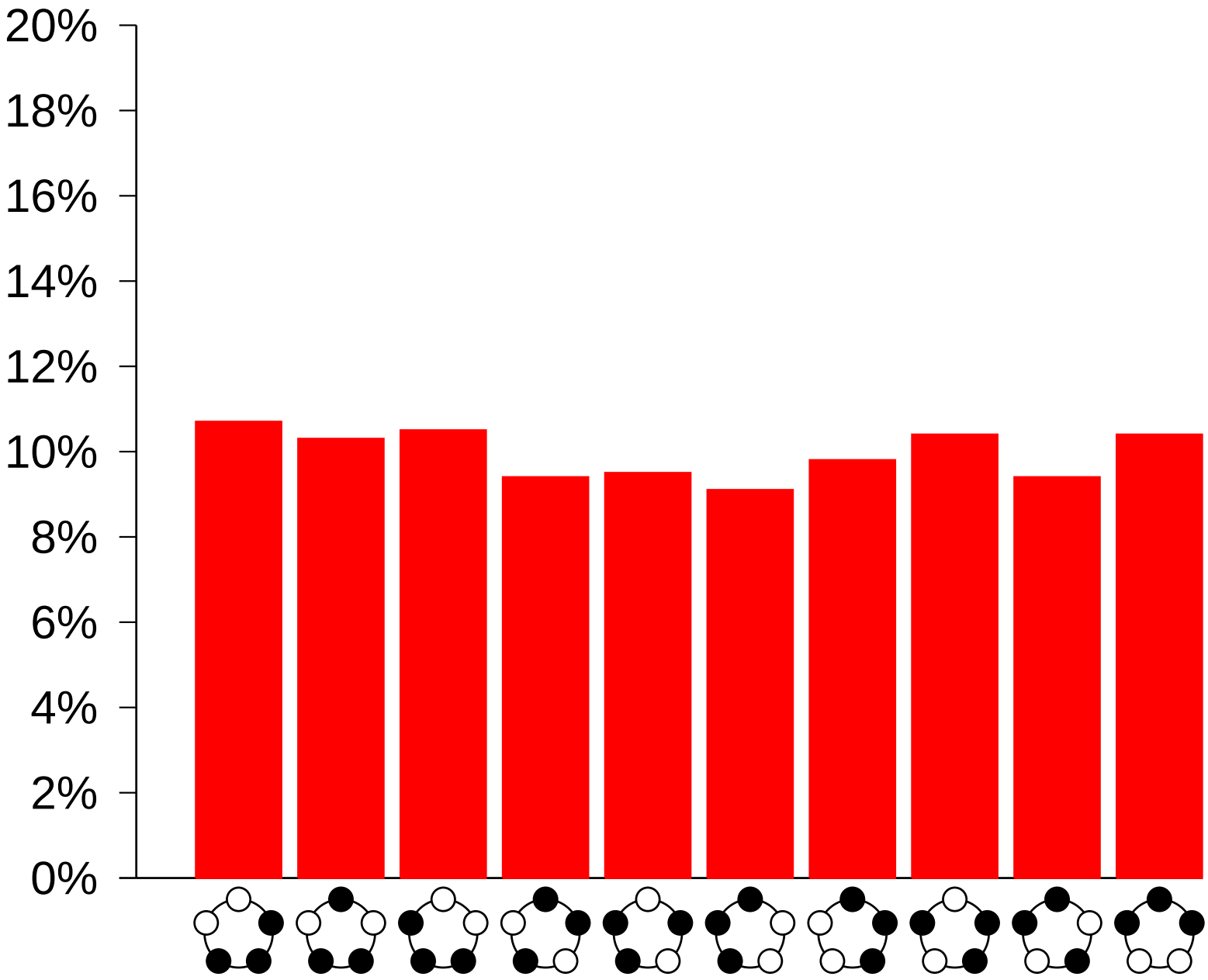
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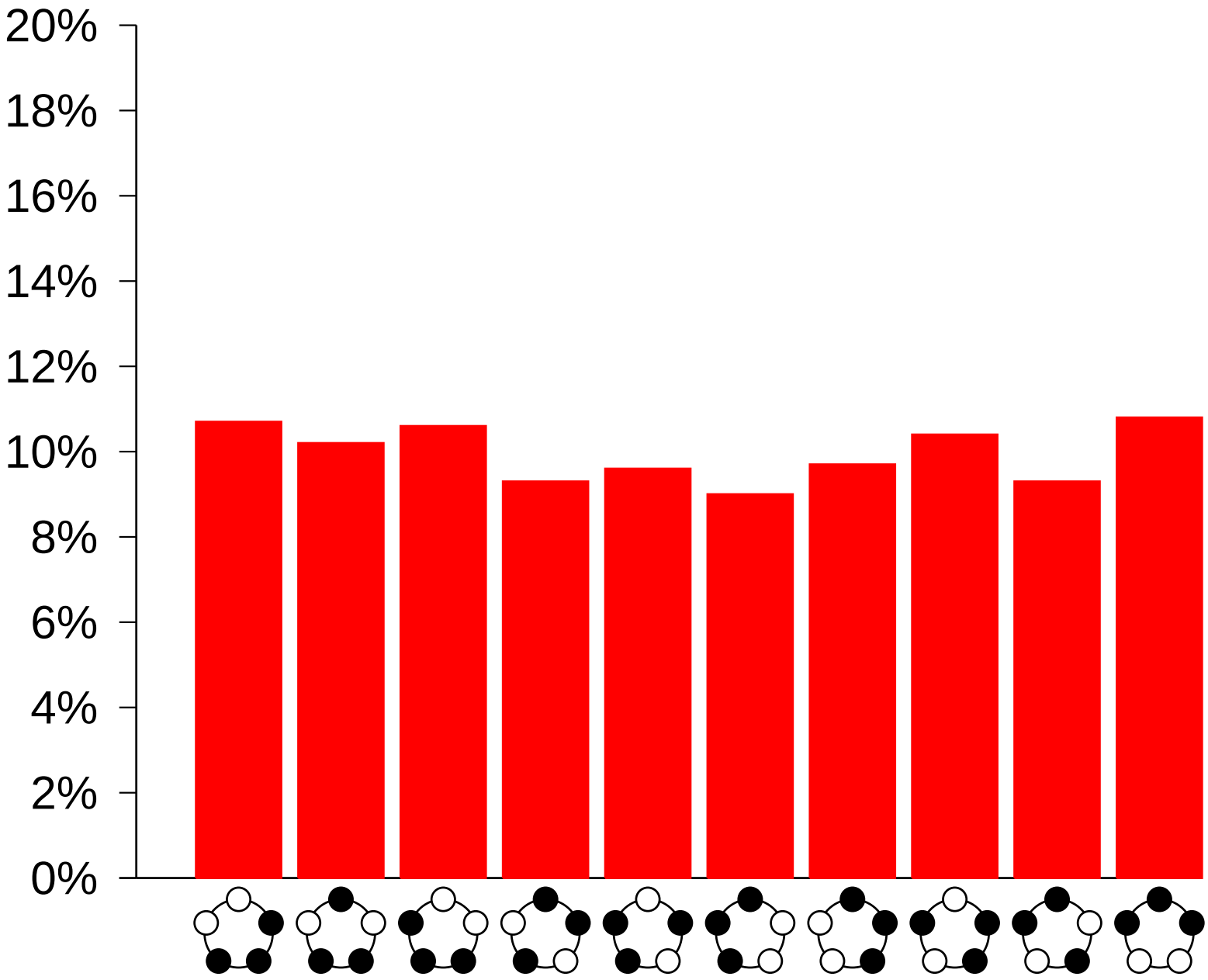
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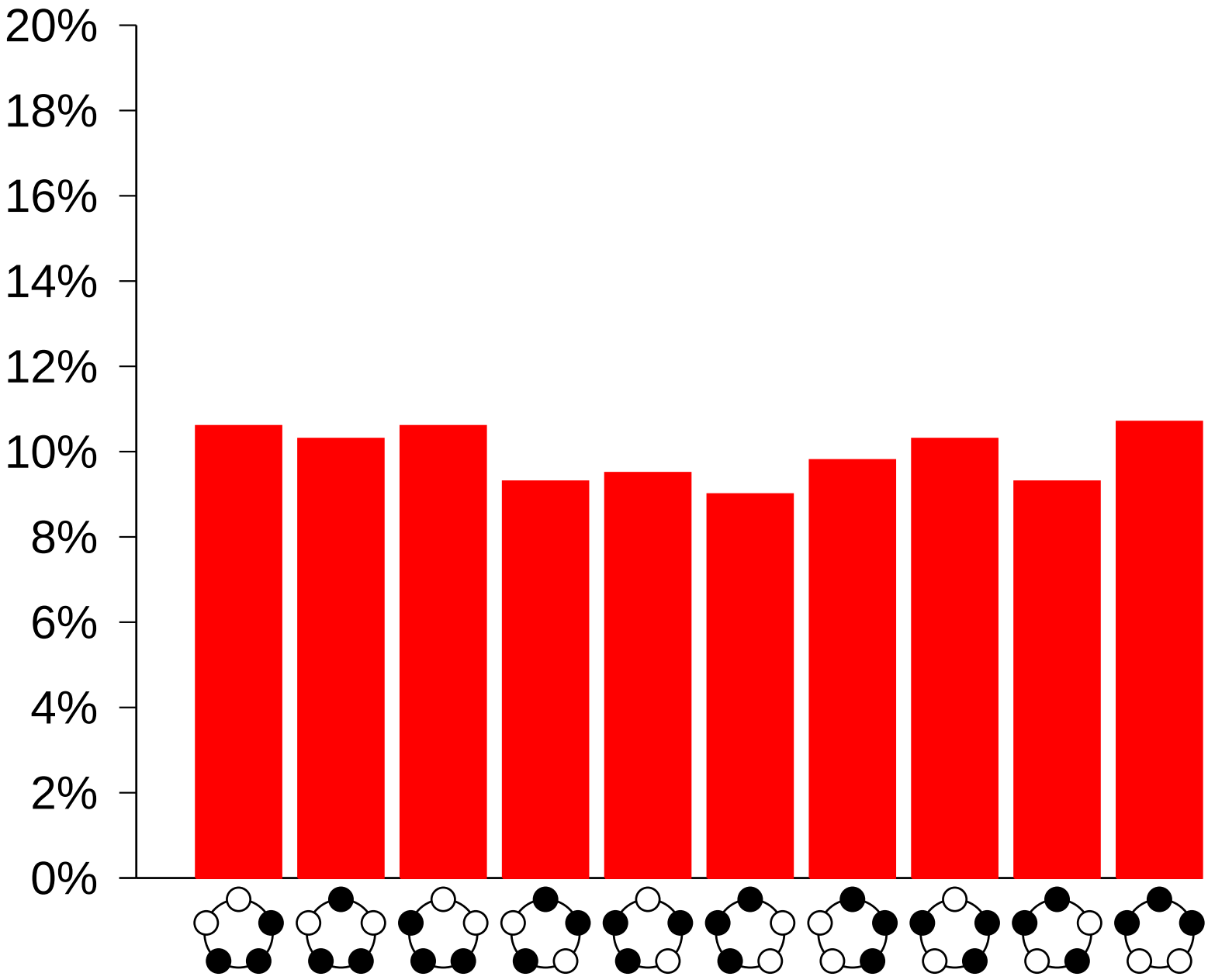
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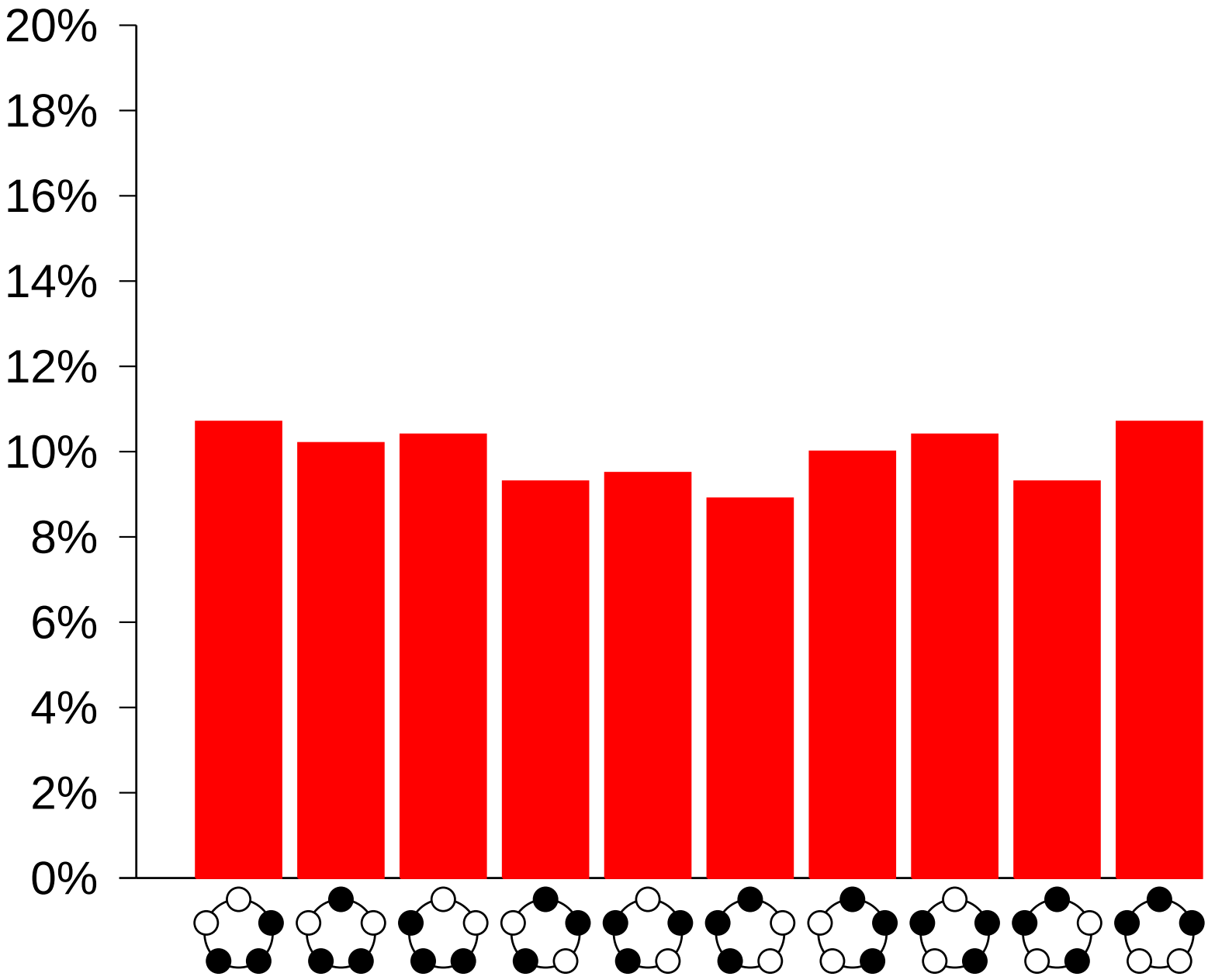
Stationary distribution



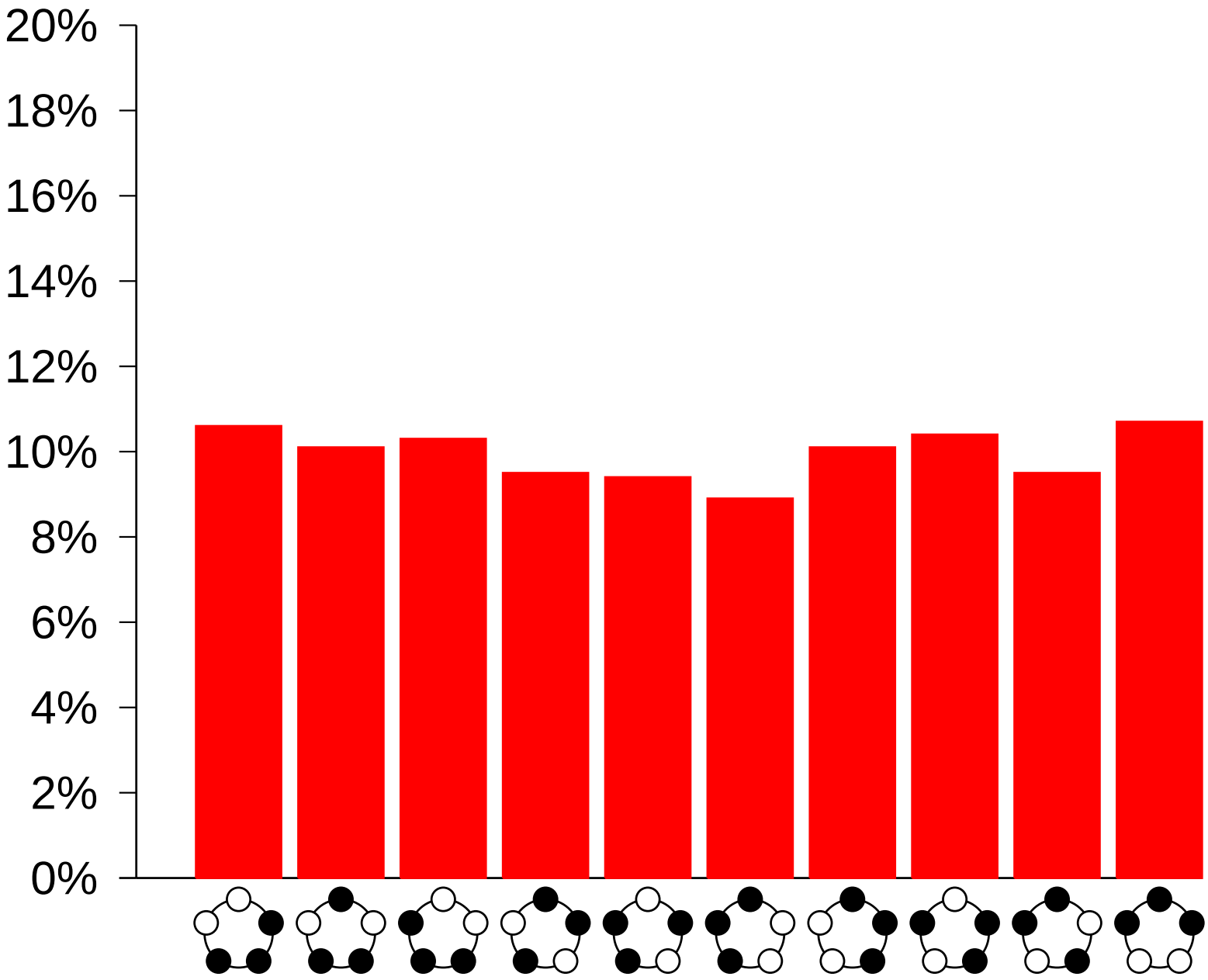
Stationary distribution



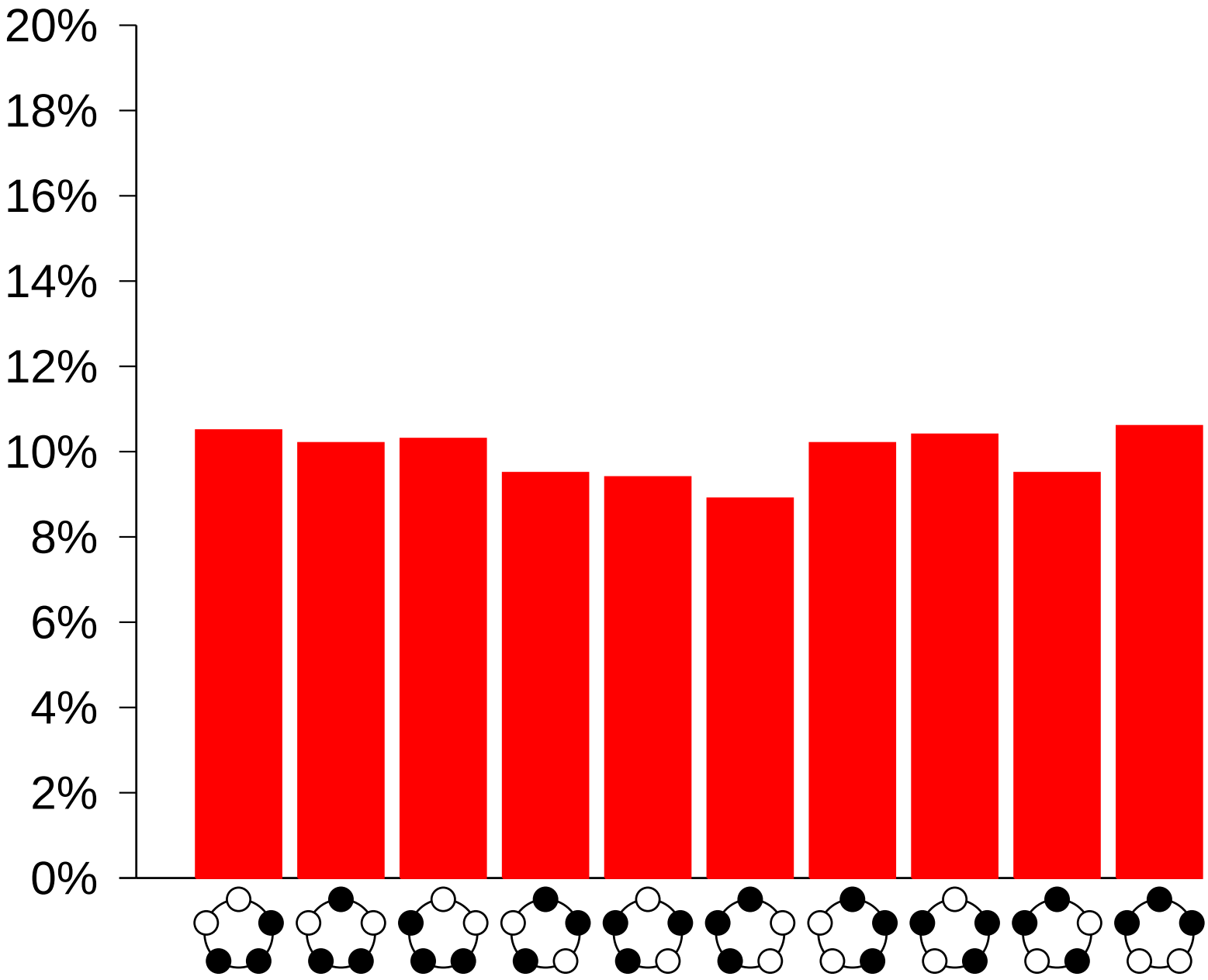
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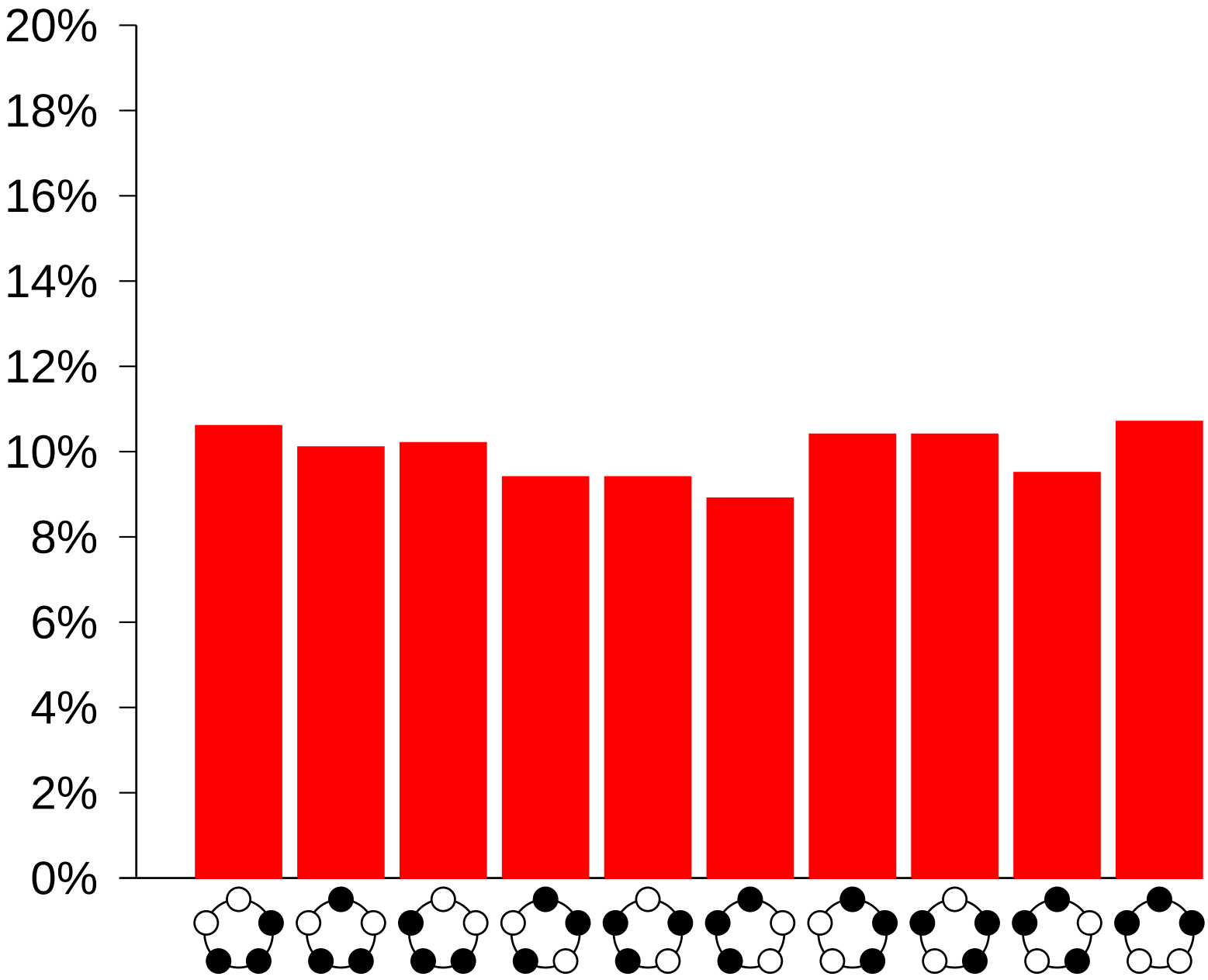
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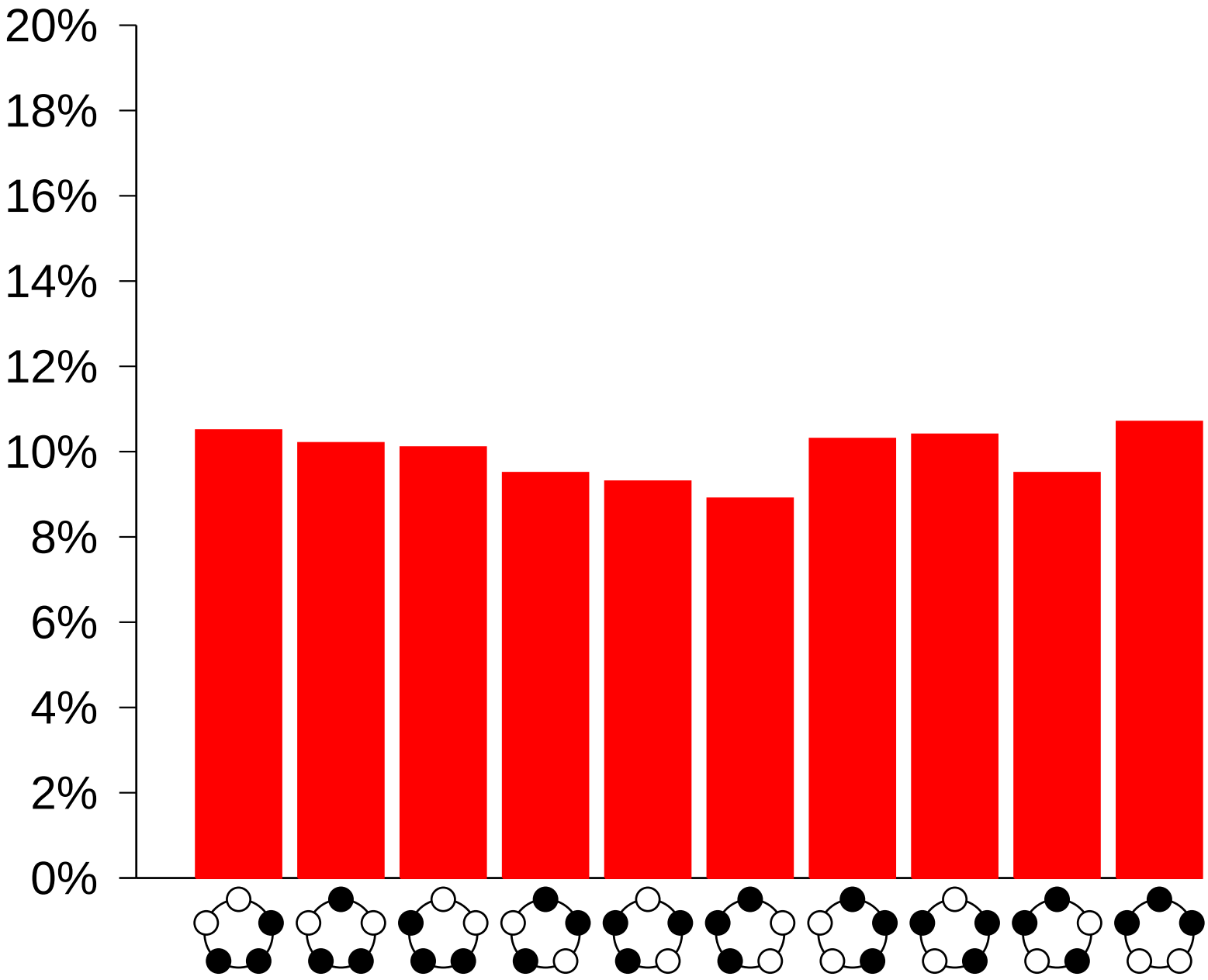
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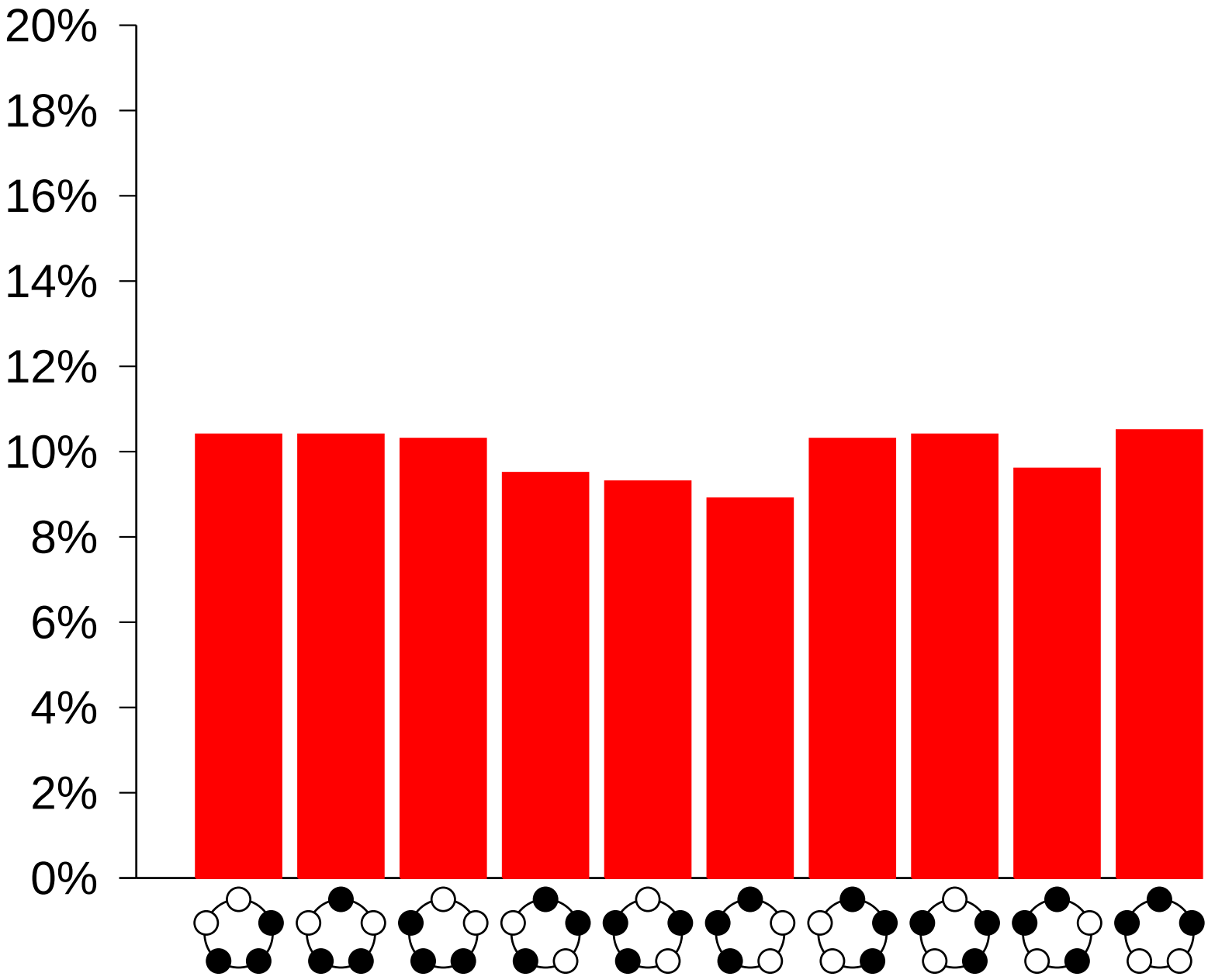
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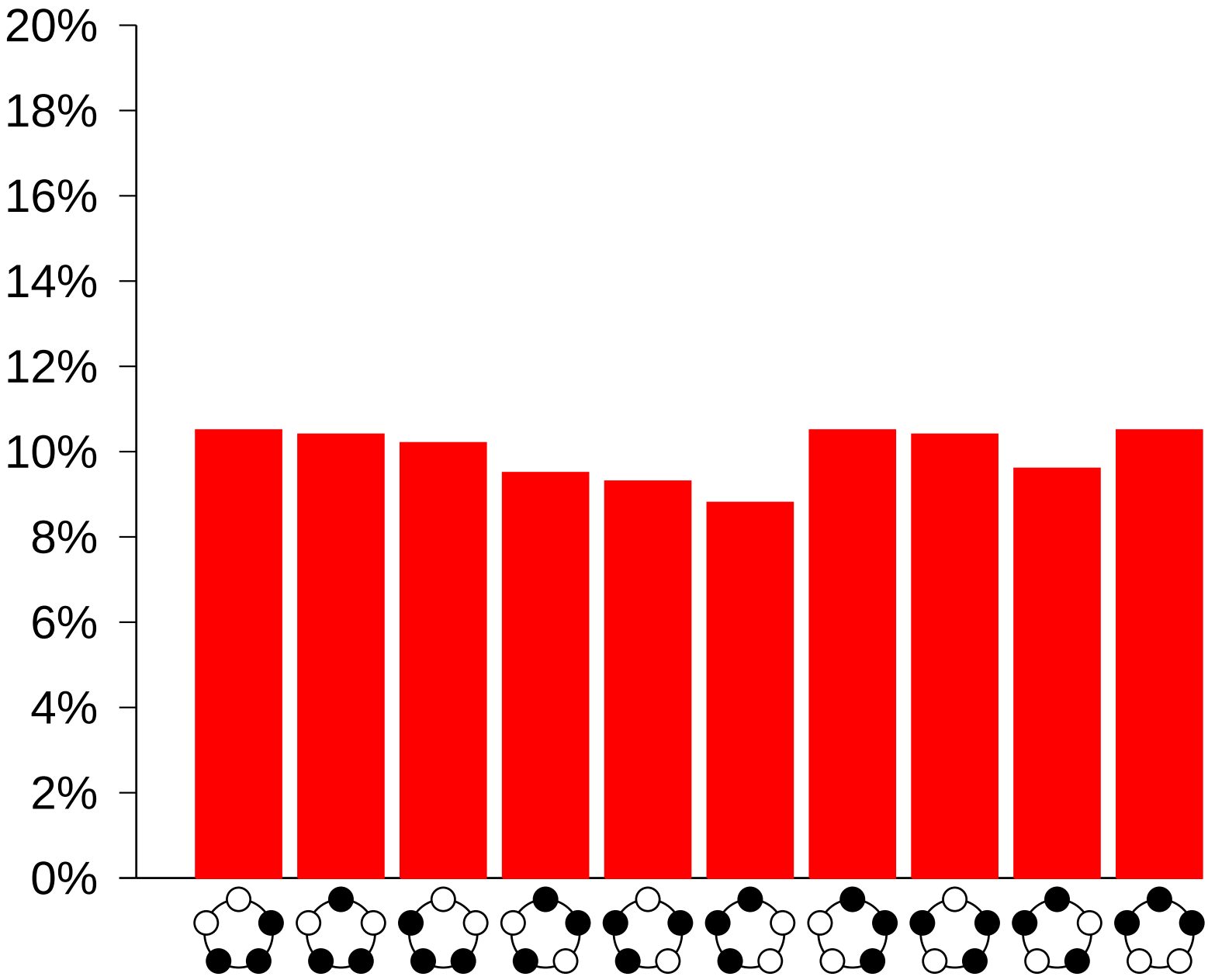
Stationary distribution



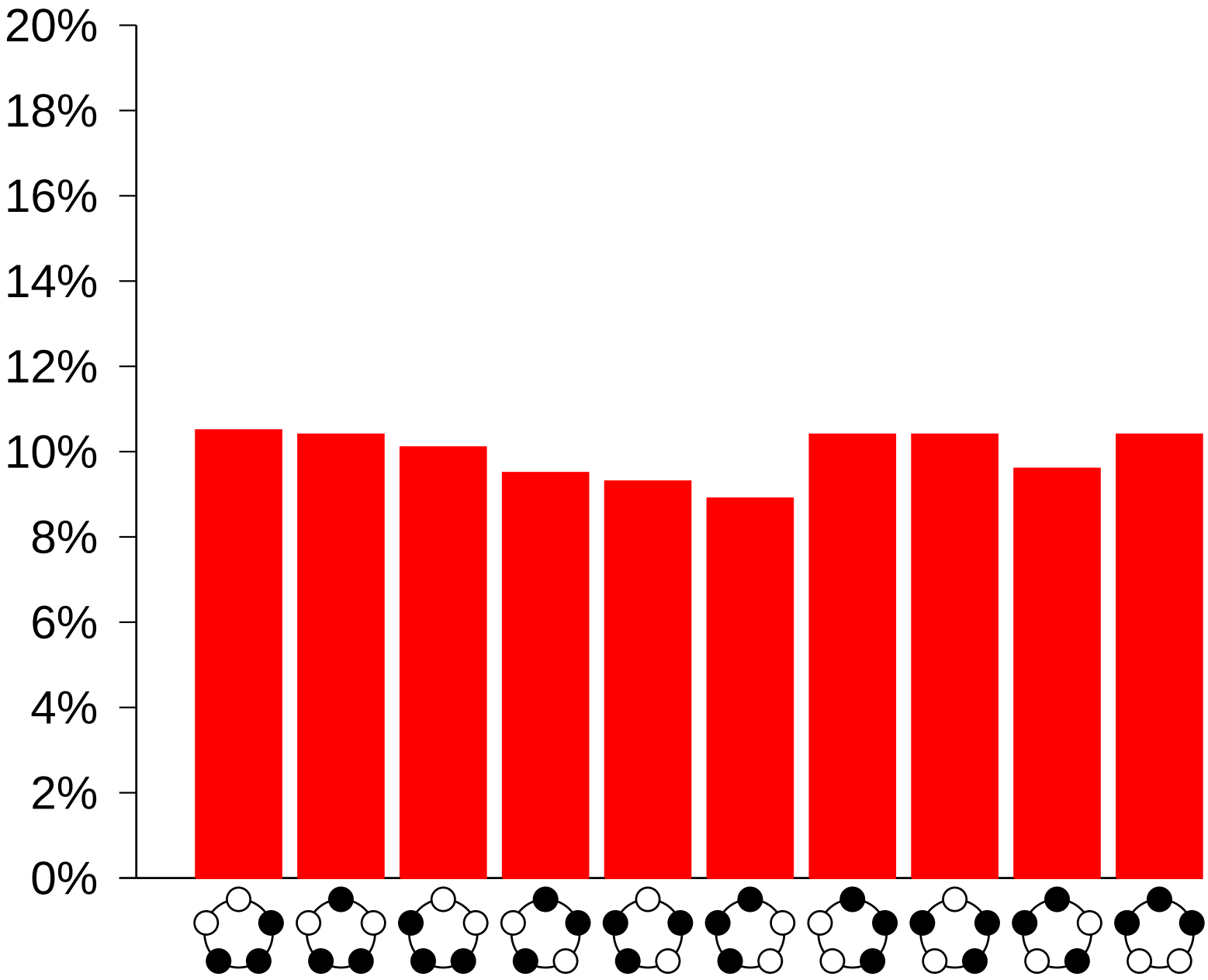
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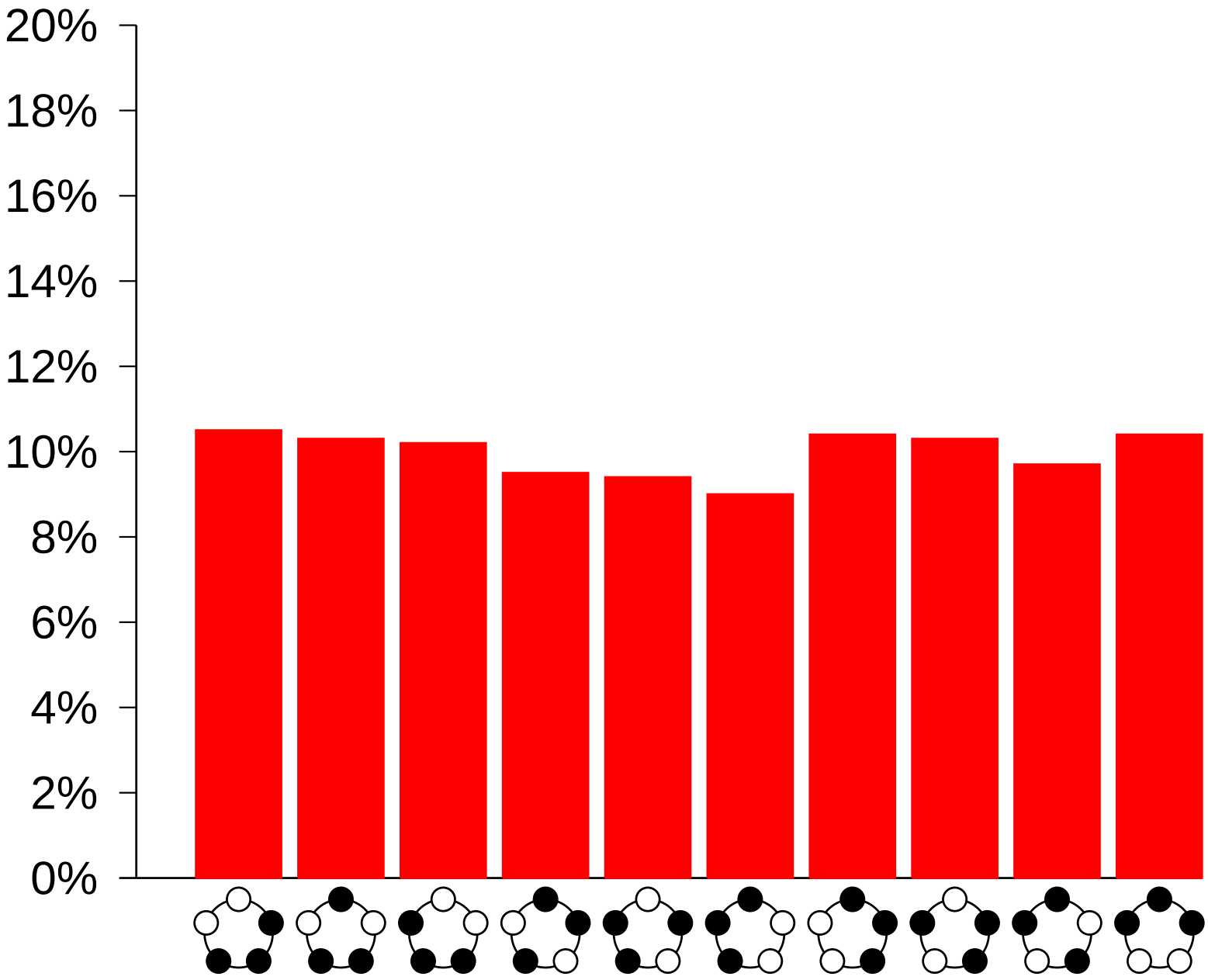
Stationary distribution



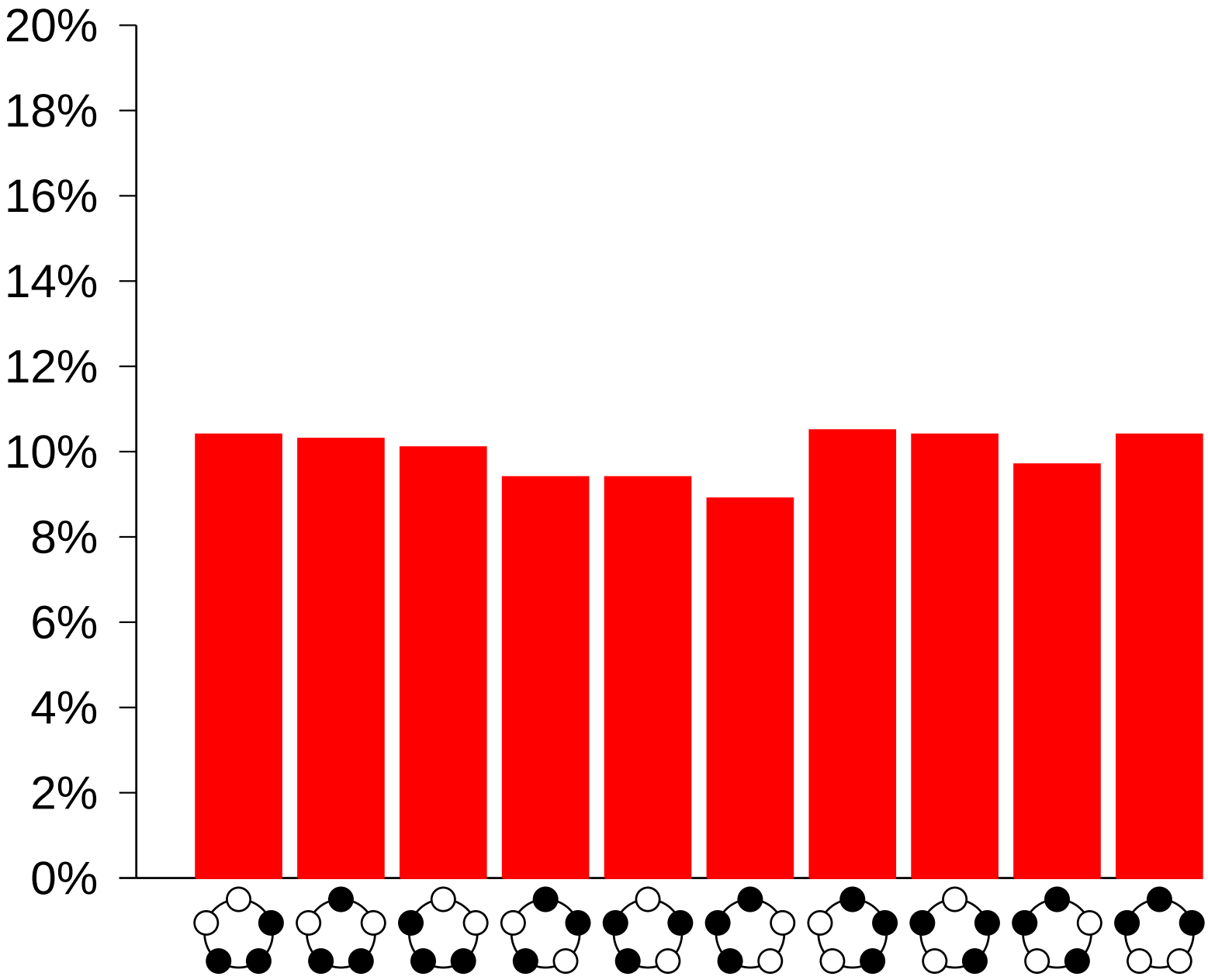
Stationary distribution



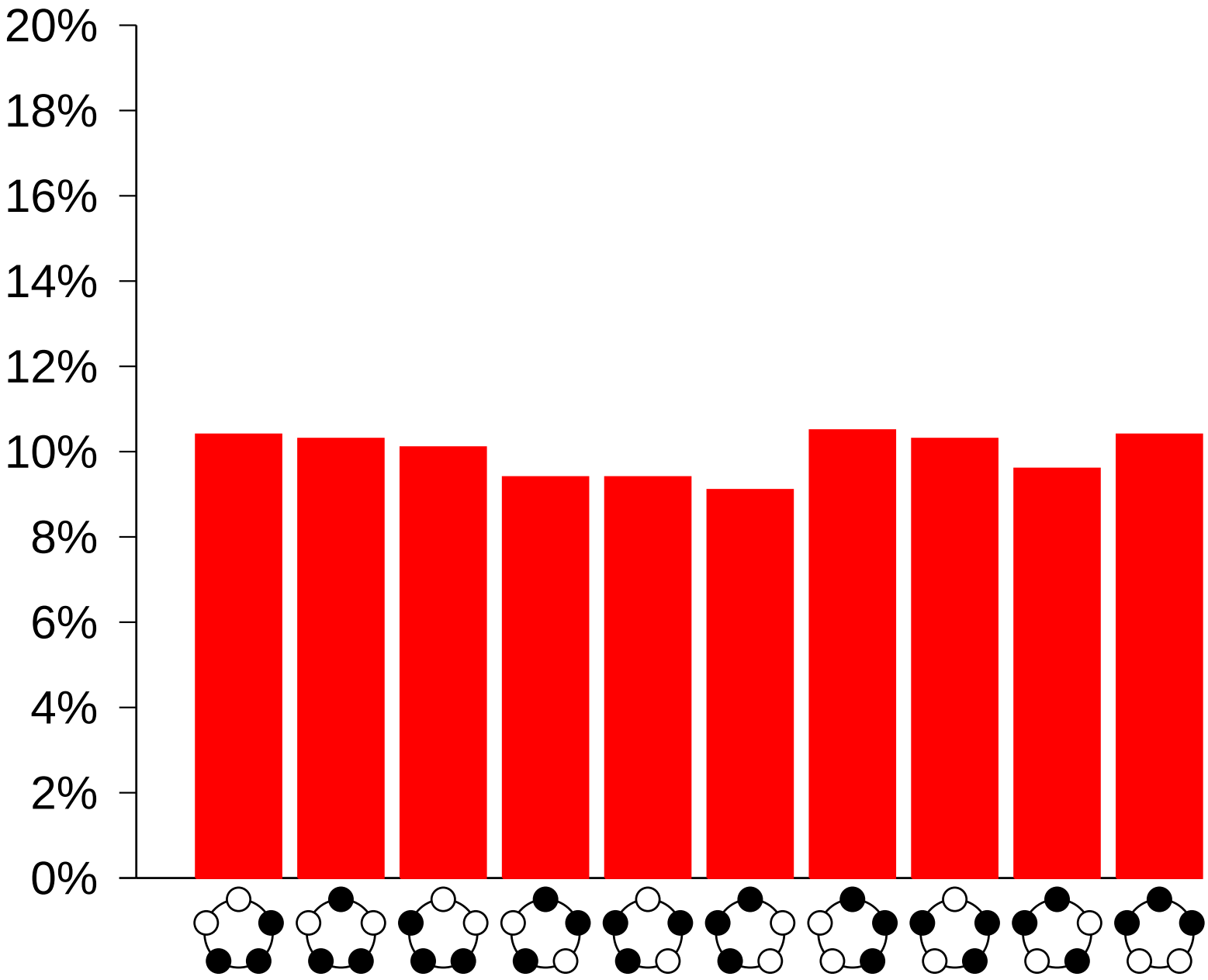
Stationary distribution



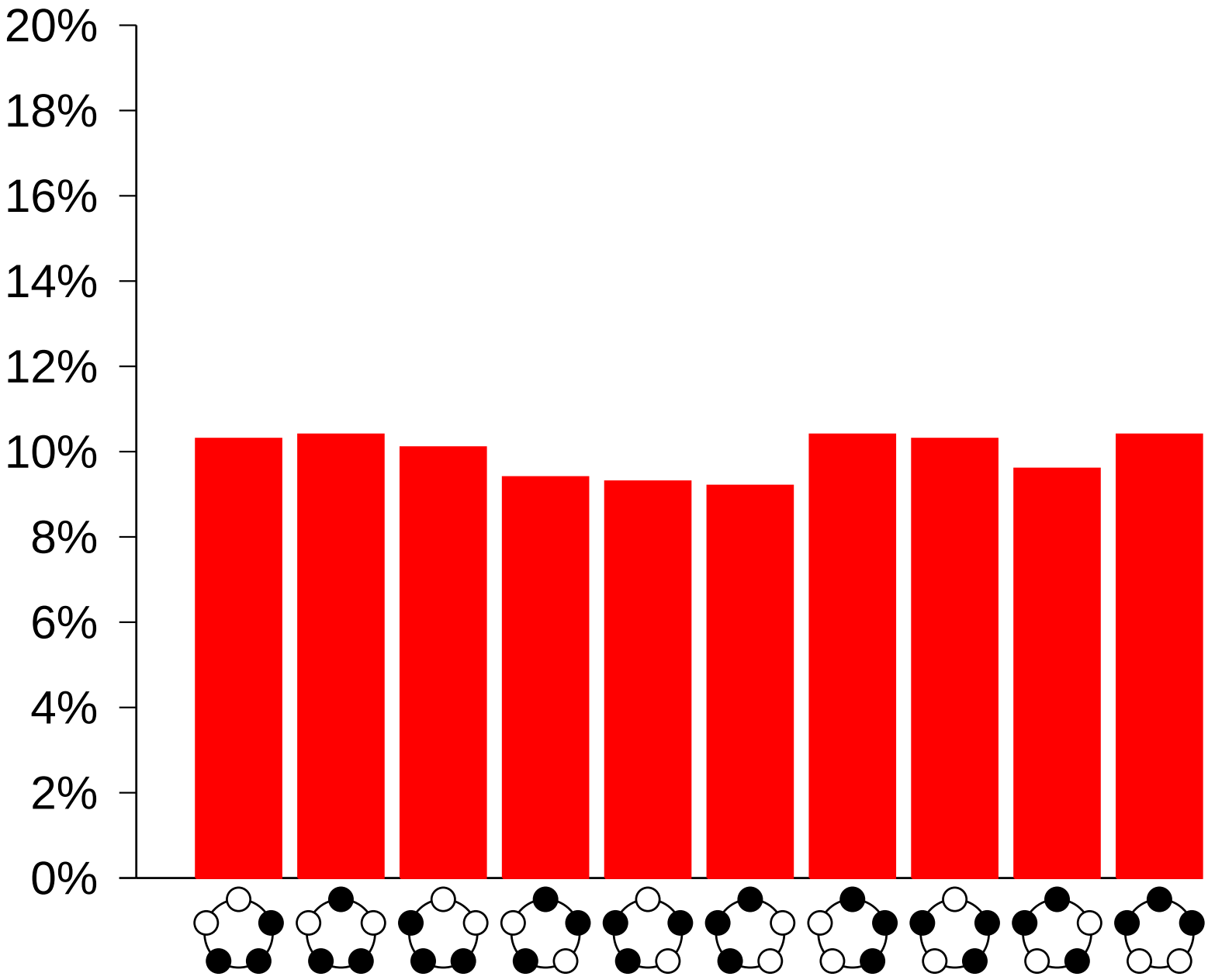
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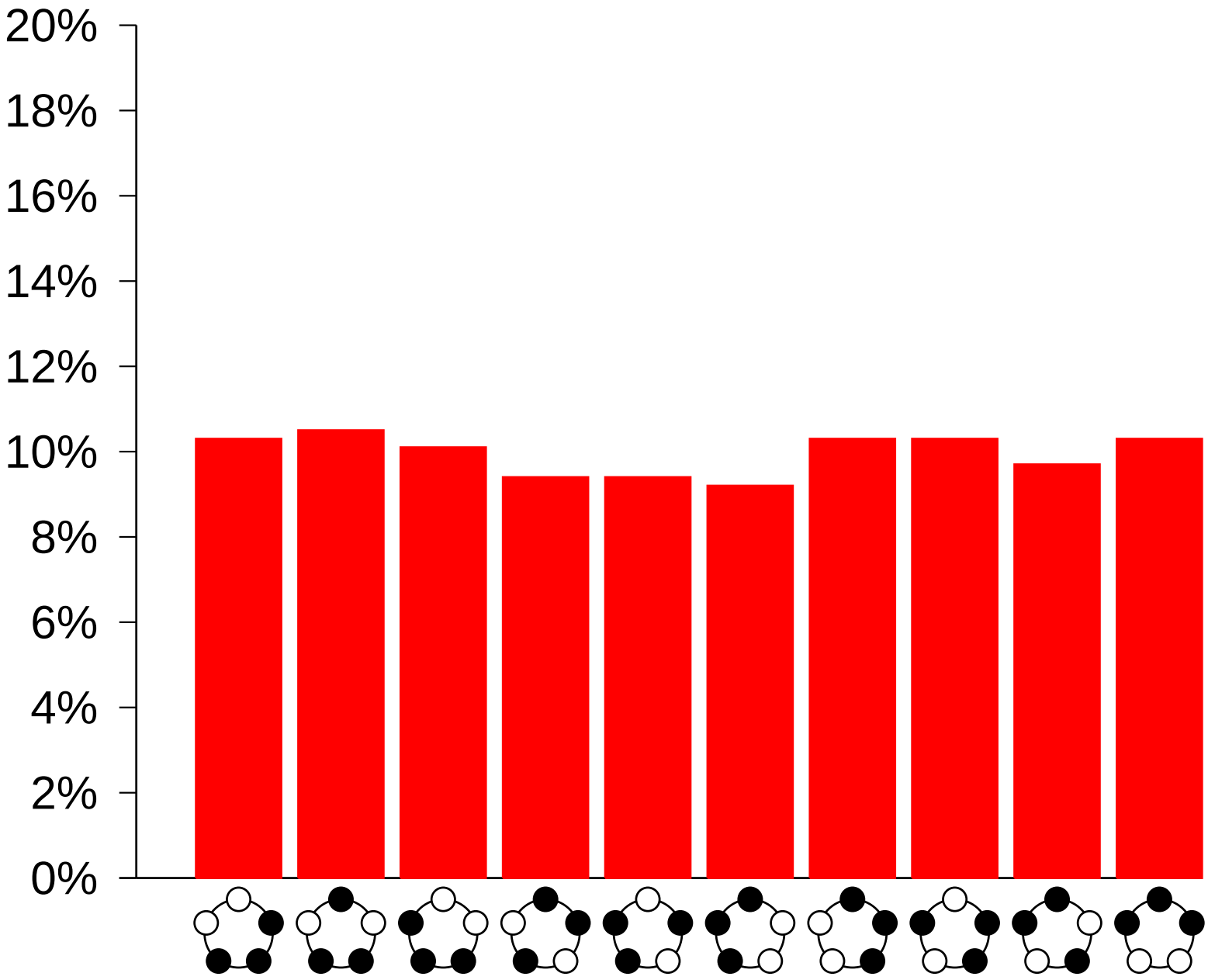
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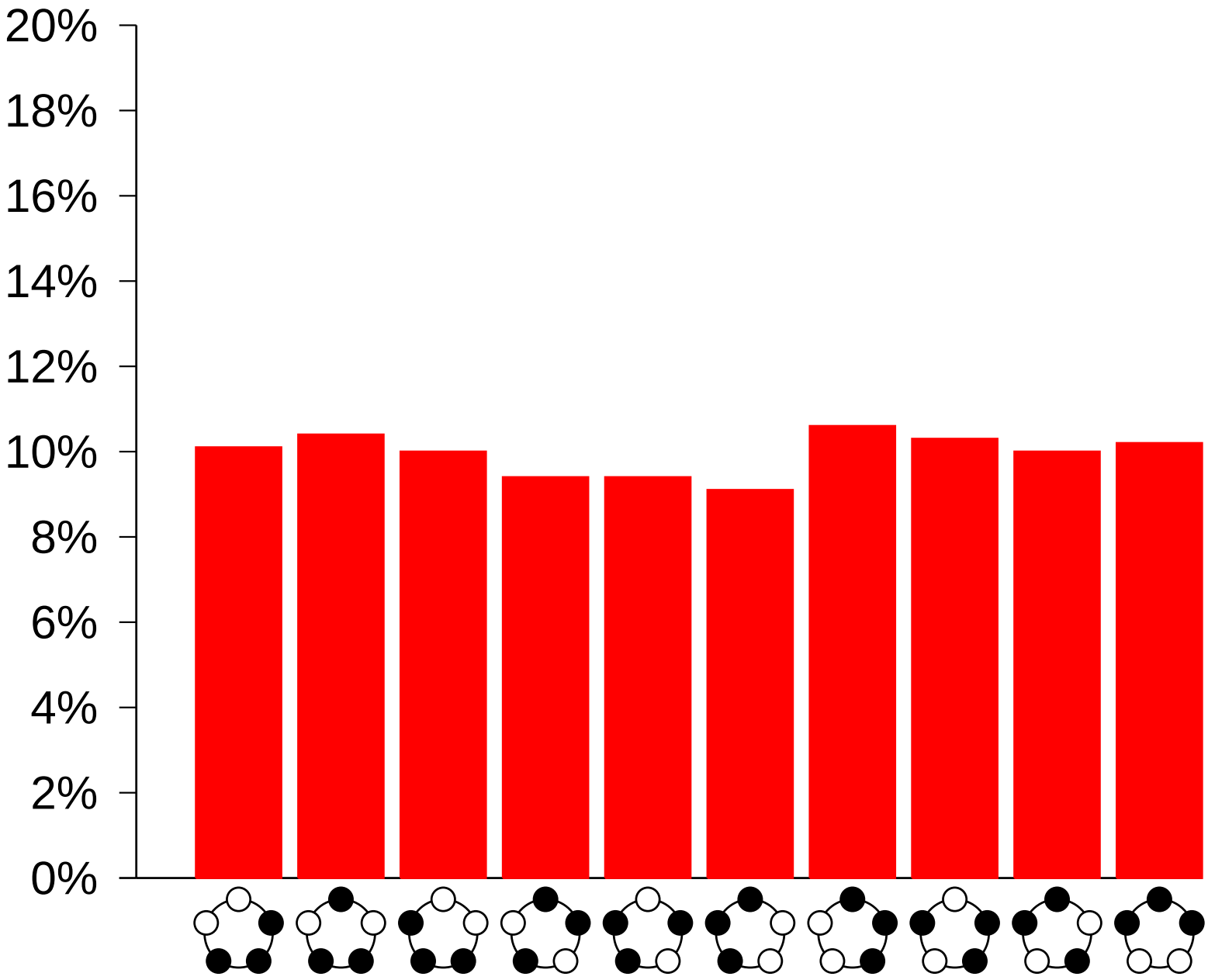
Stationary distribution



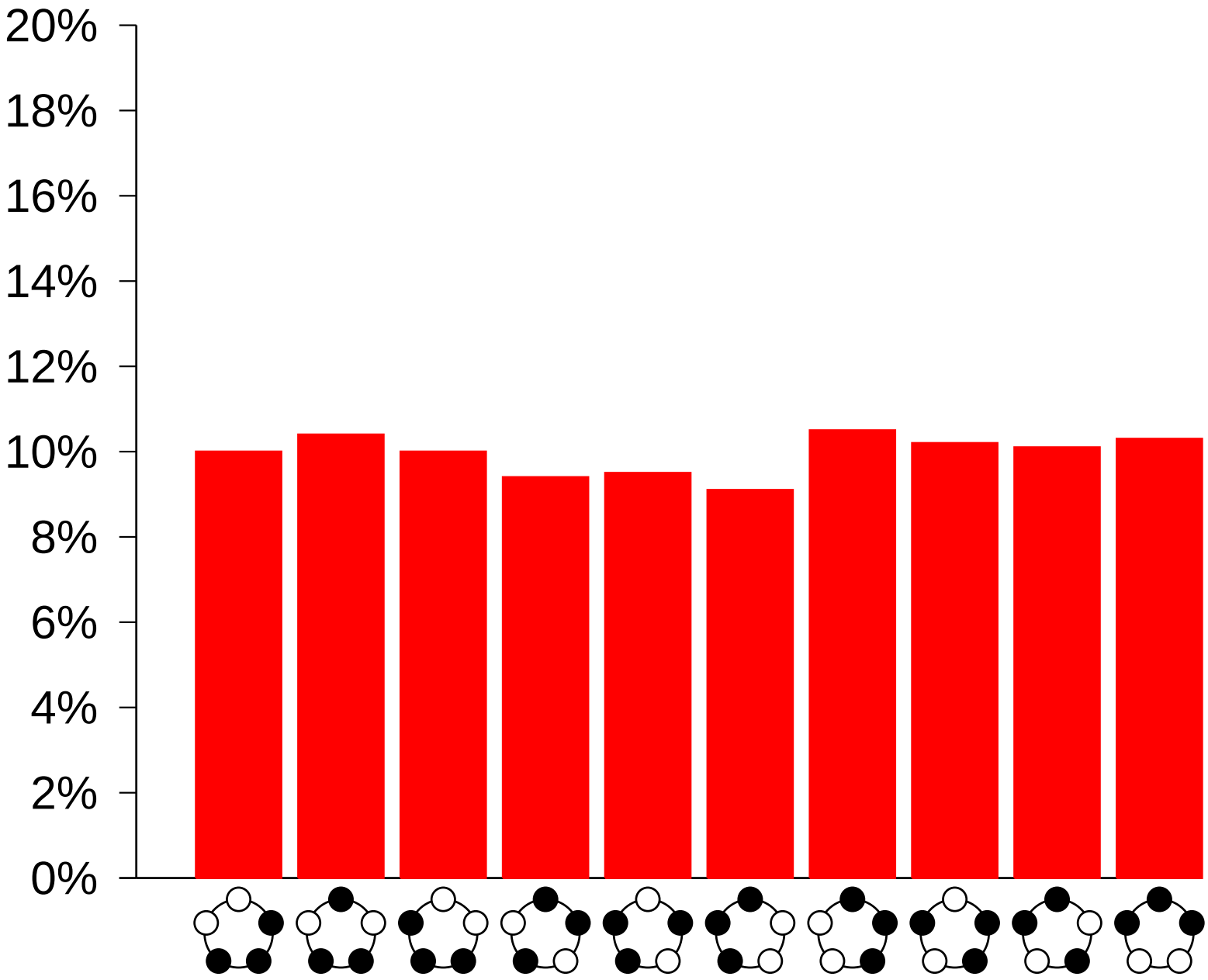
Stationary distribution



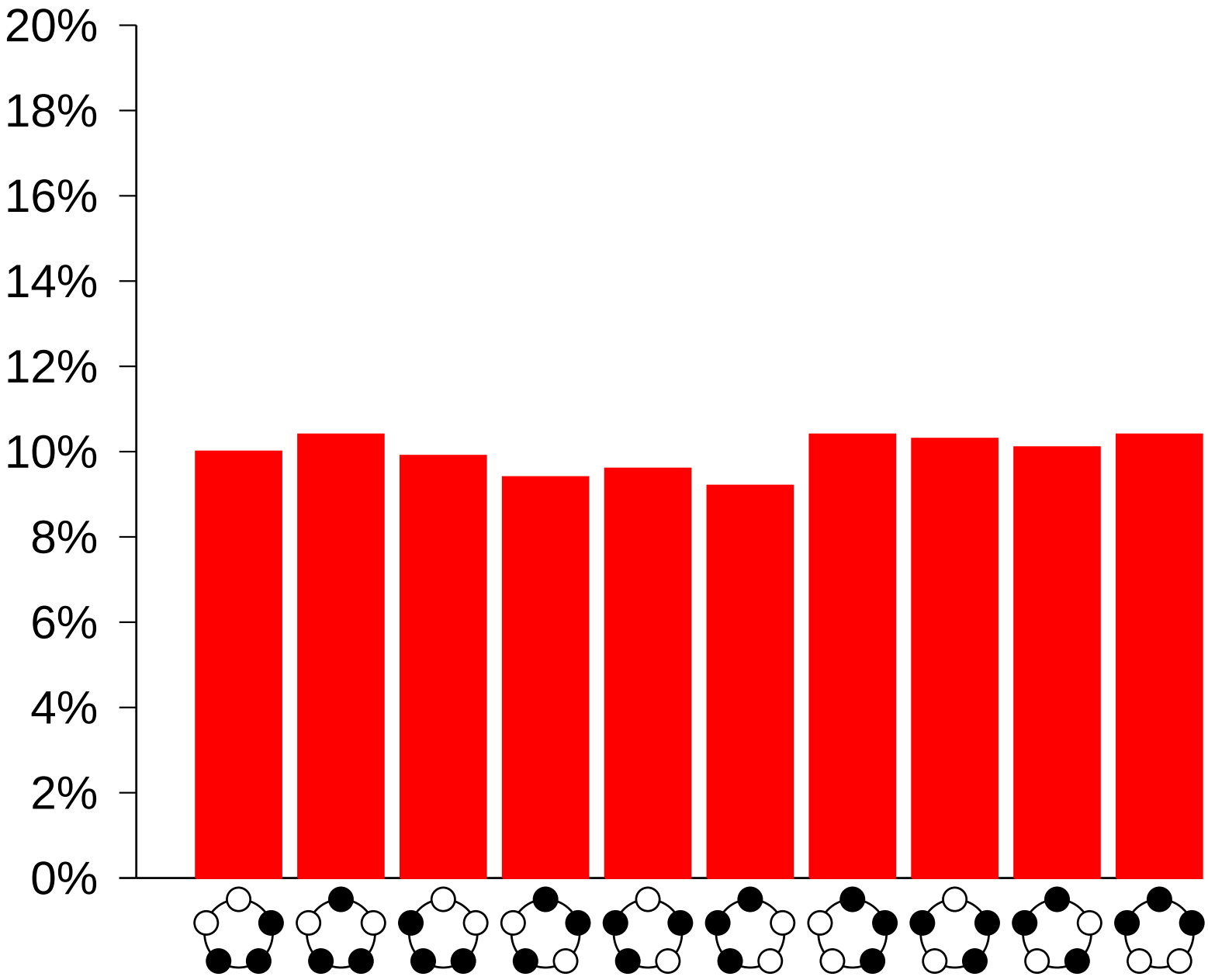
Stationary distribution



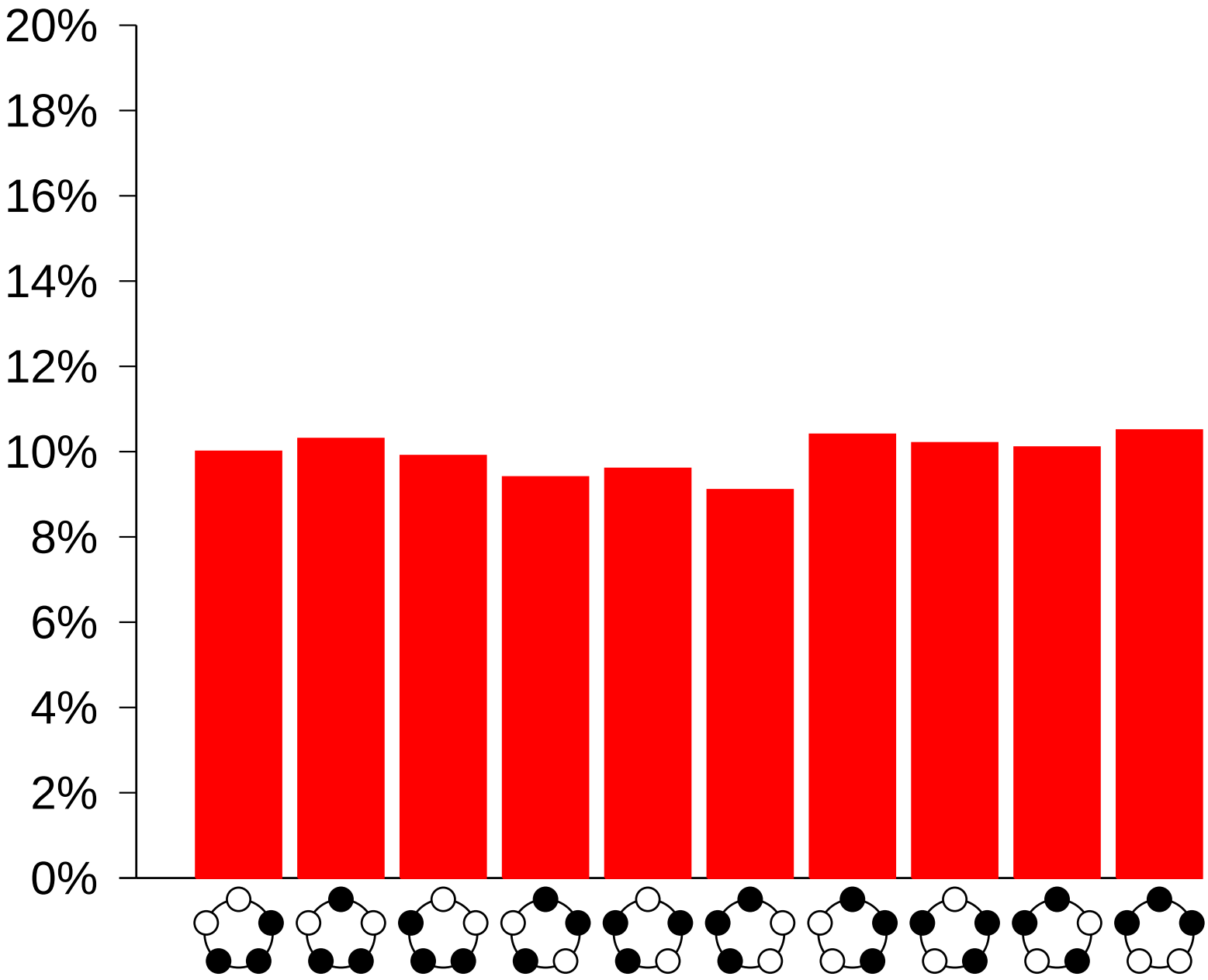
Stationary distribution



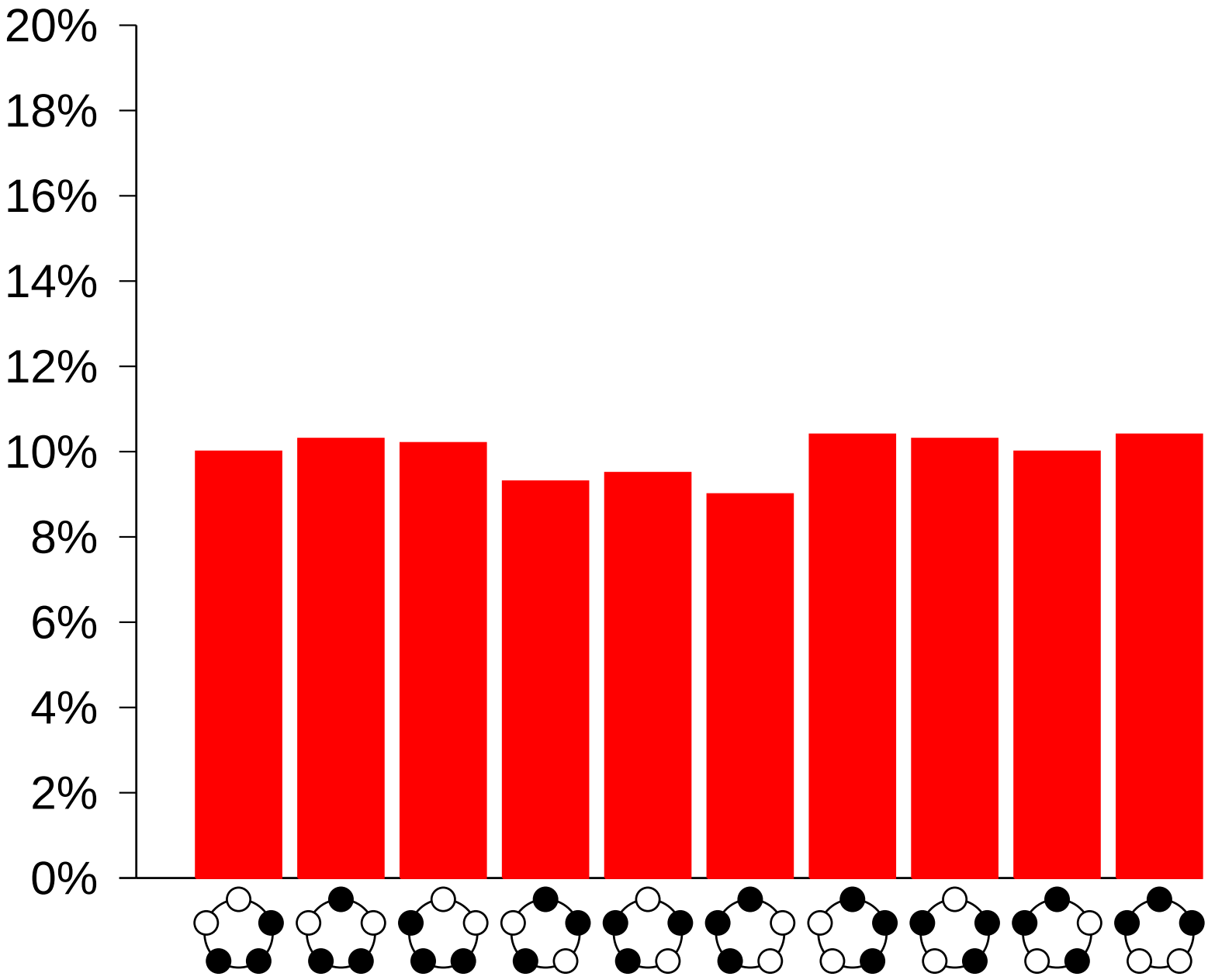
Stationary distribution



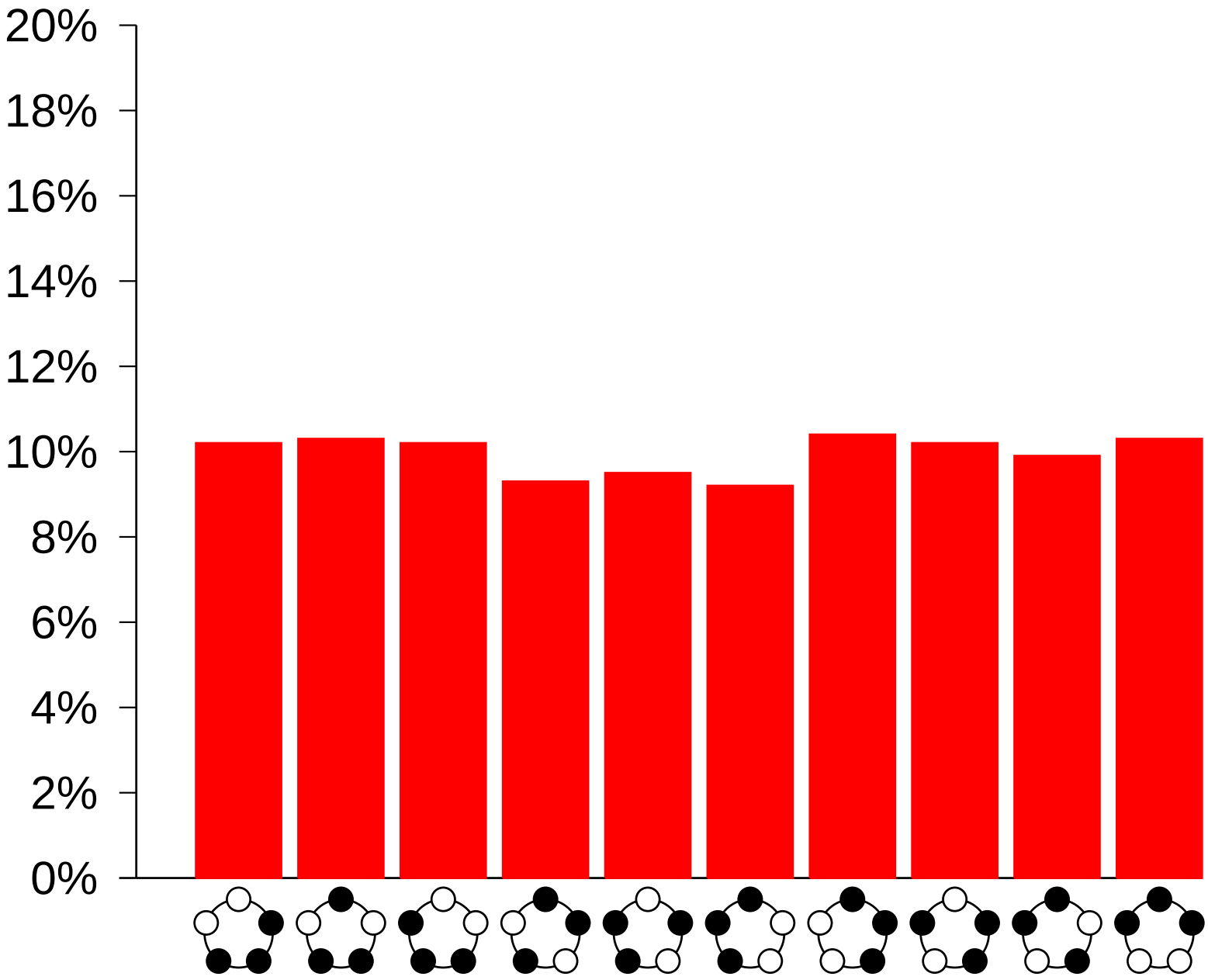
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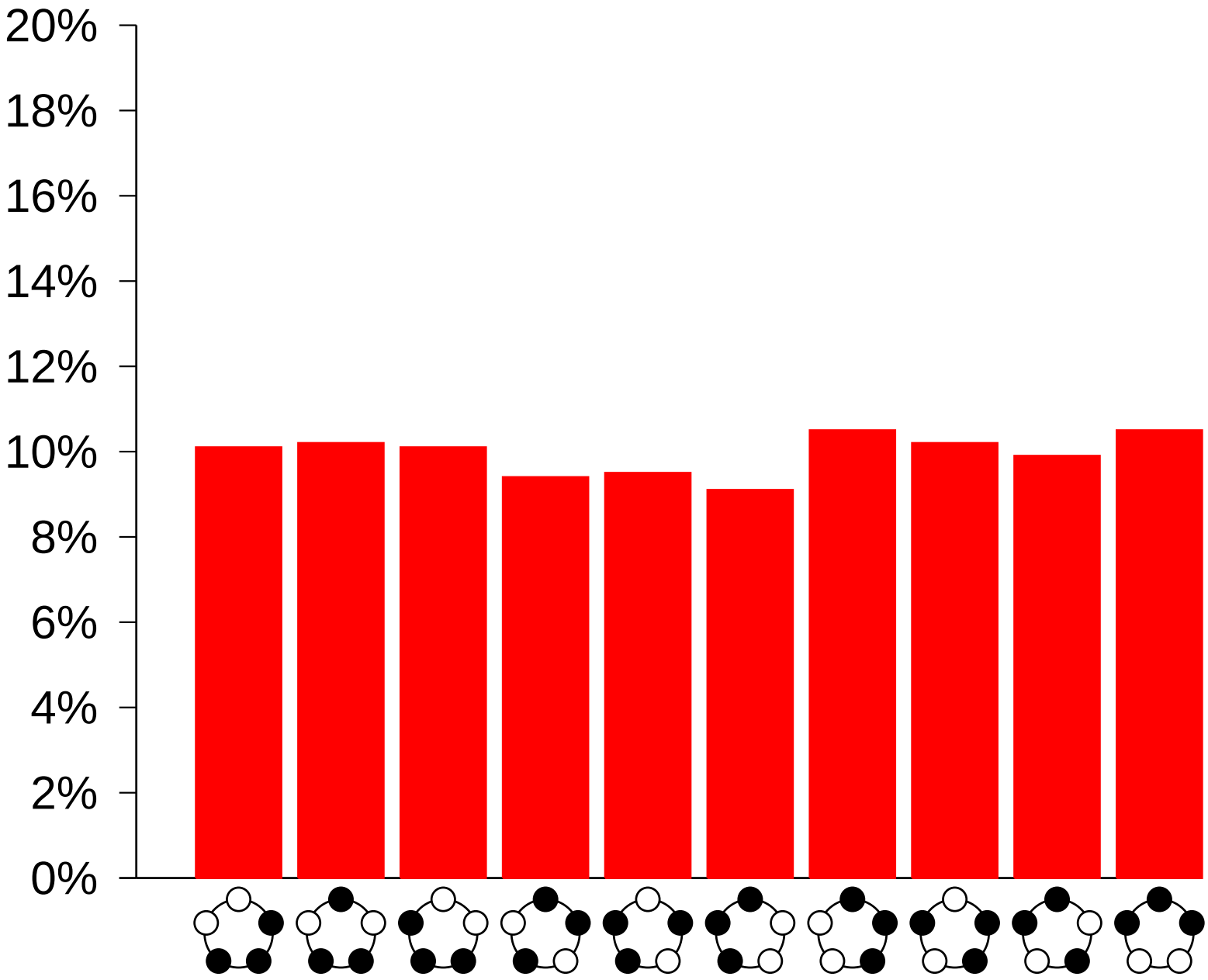
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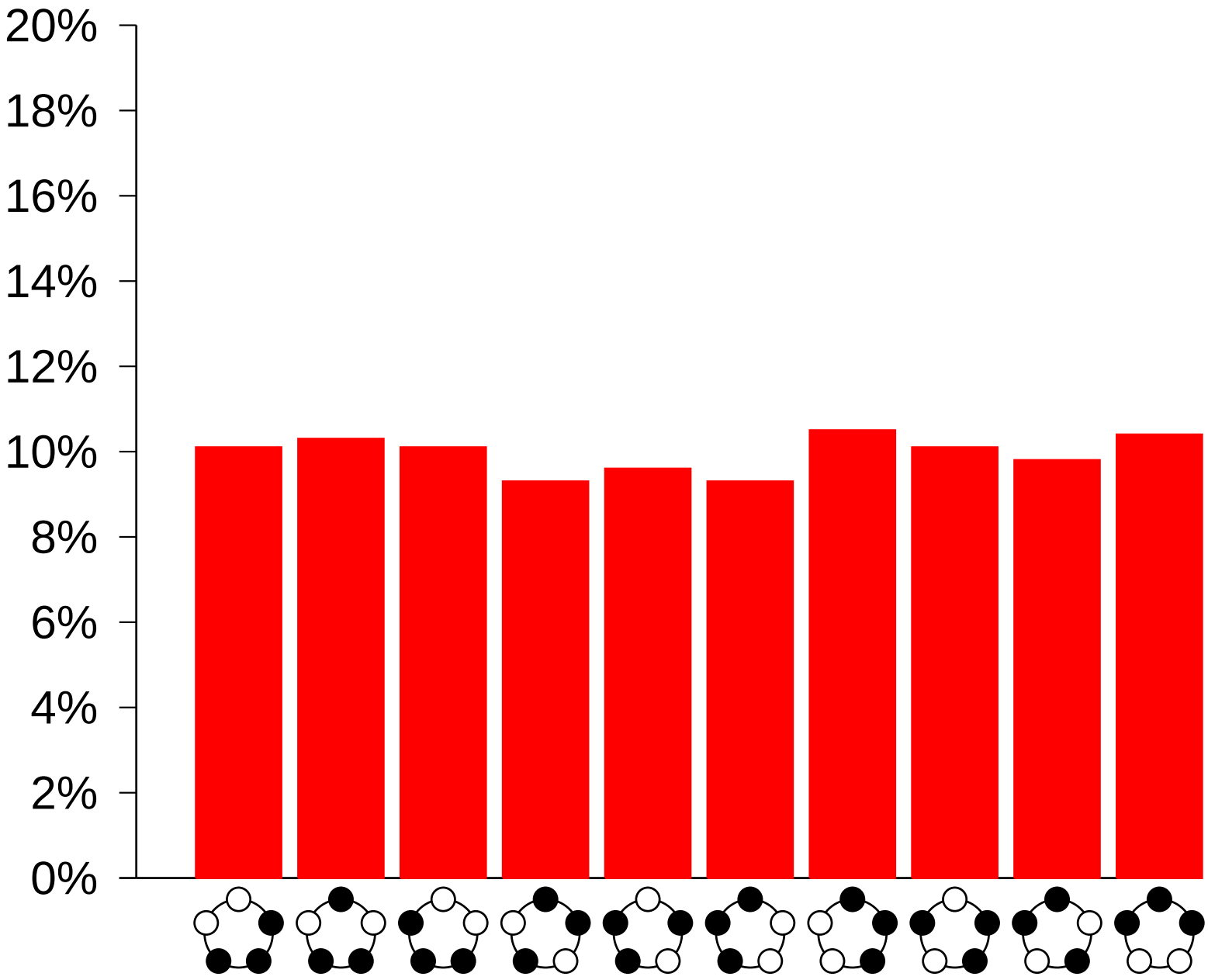
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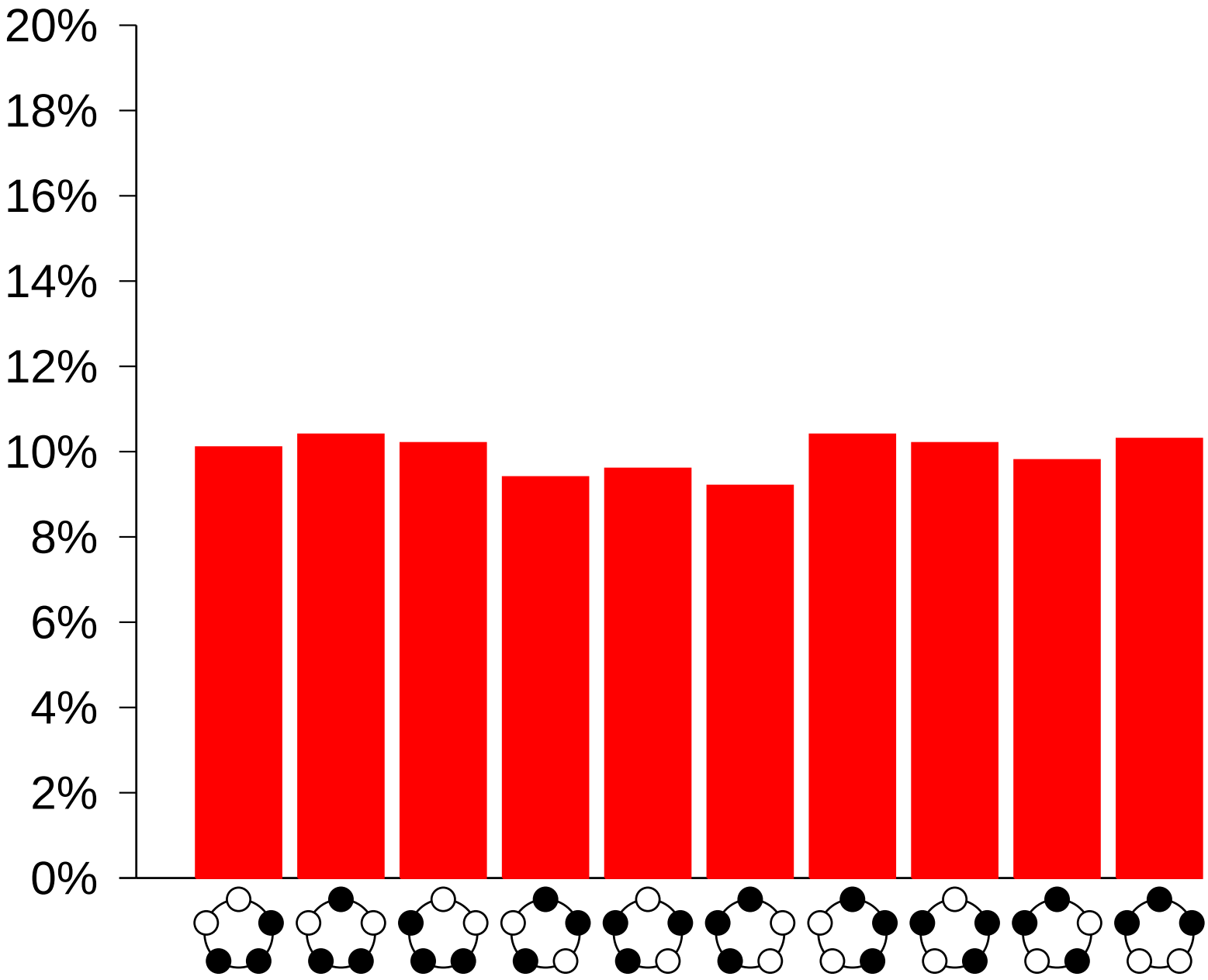
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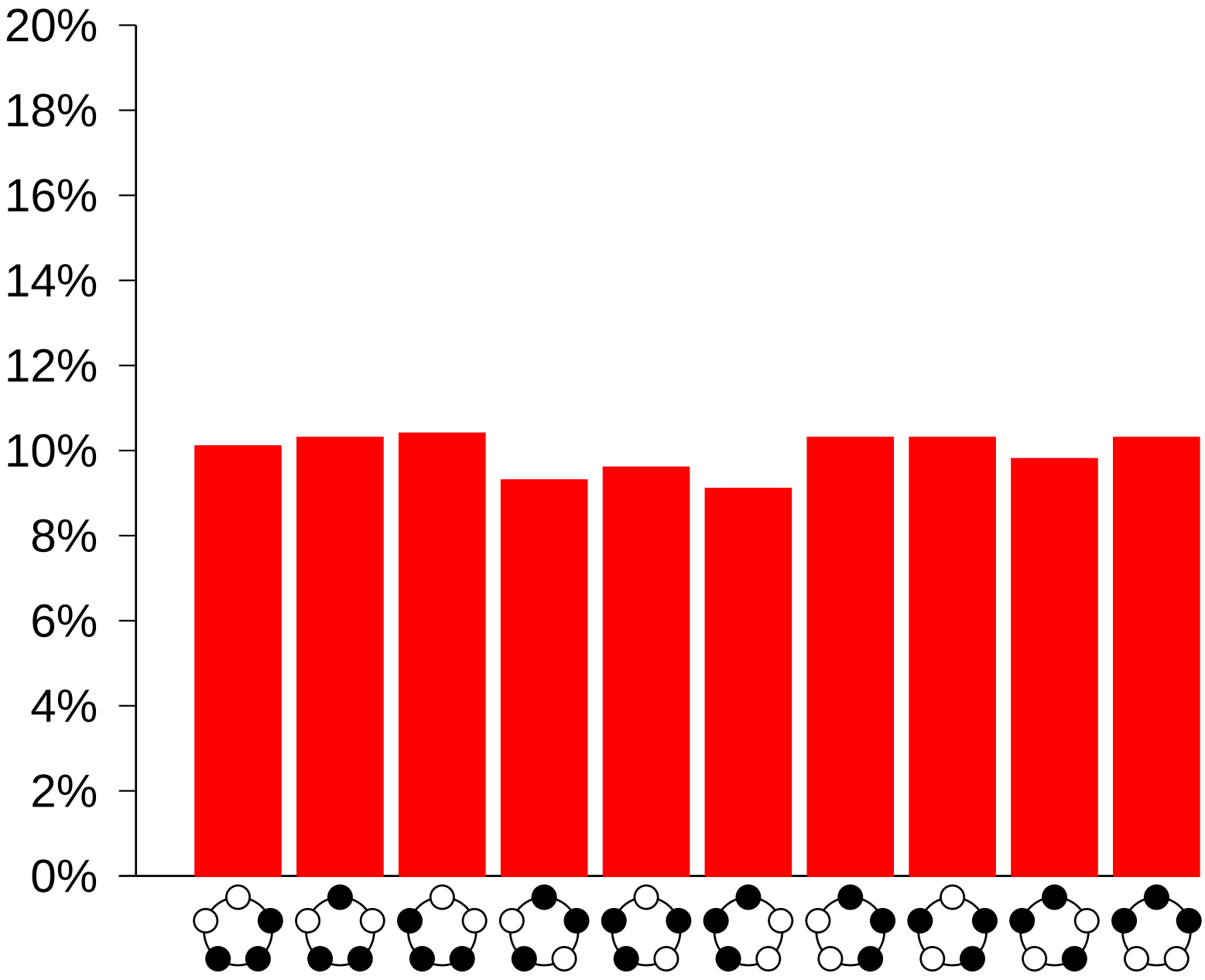
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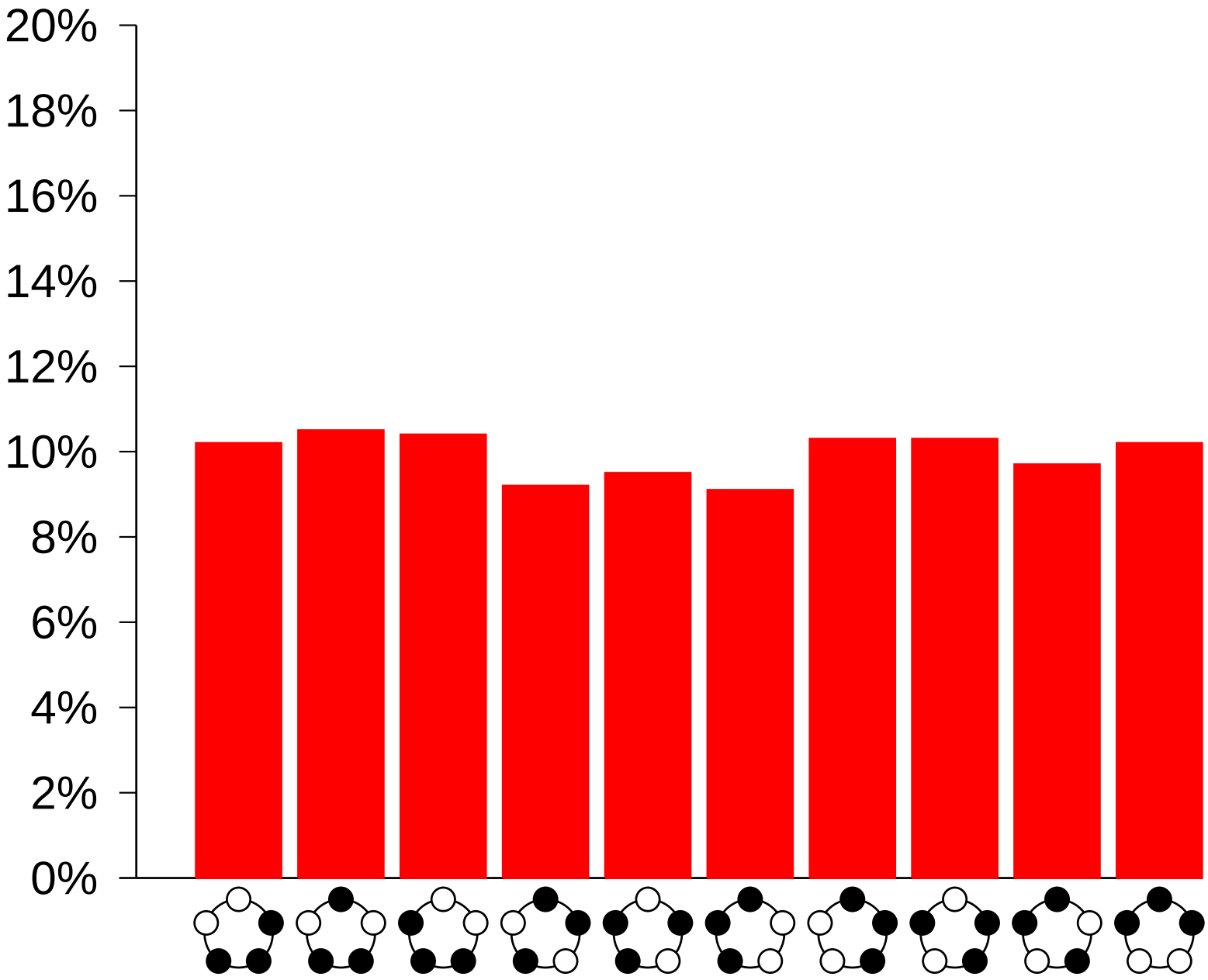
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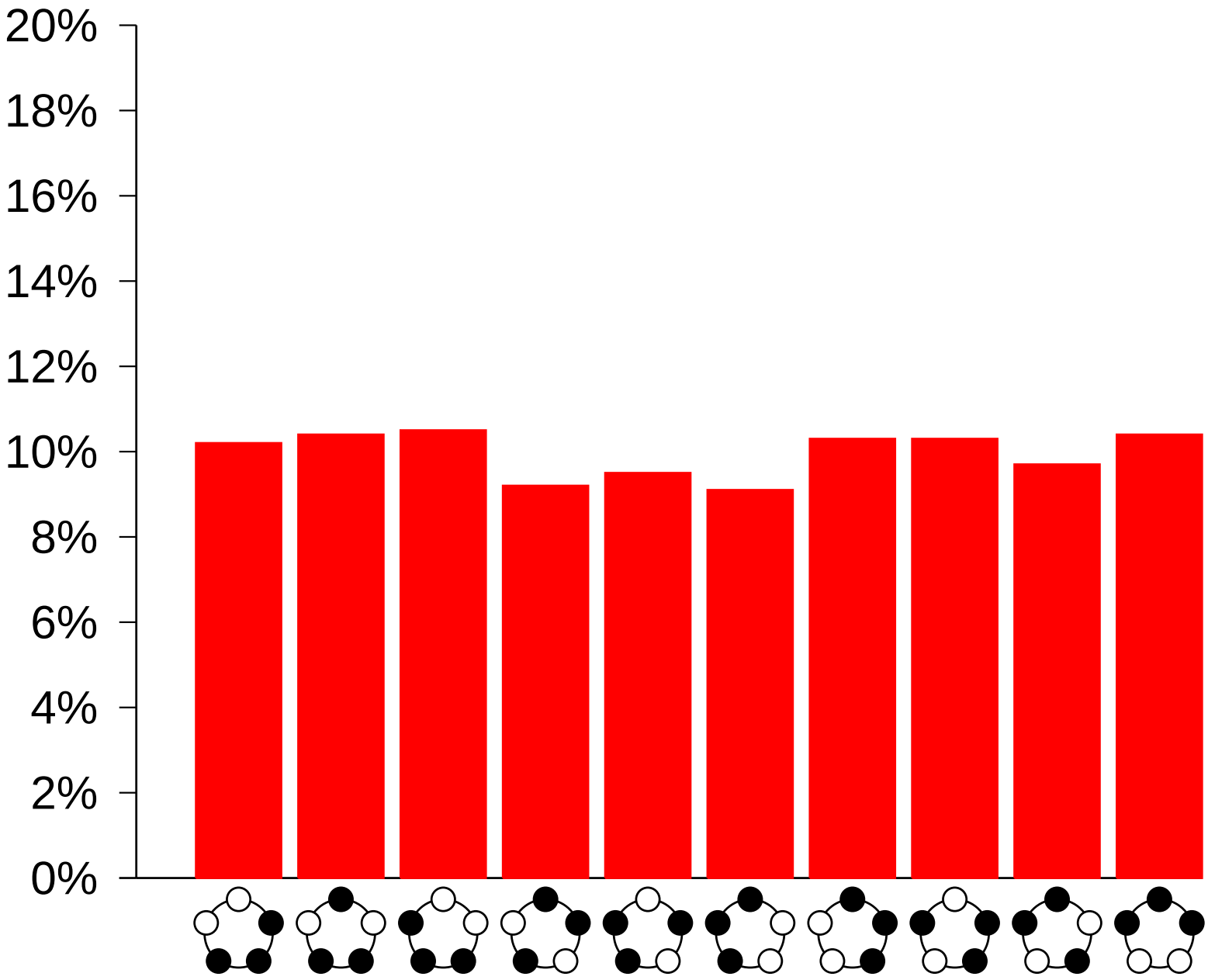
Stationary distribution



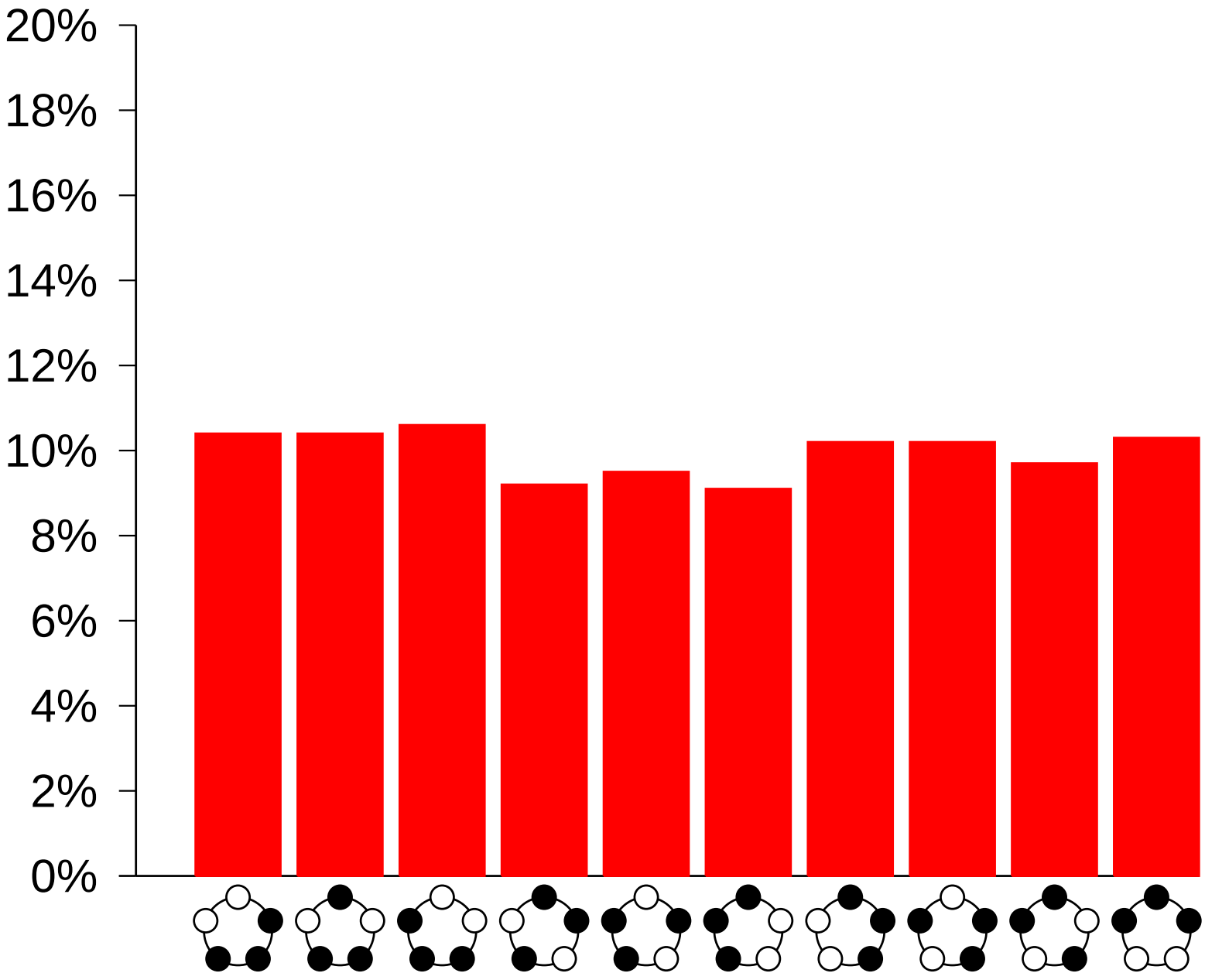
Stationary distribution



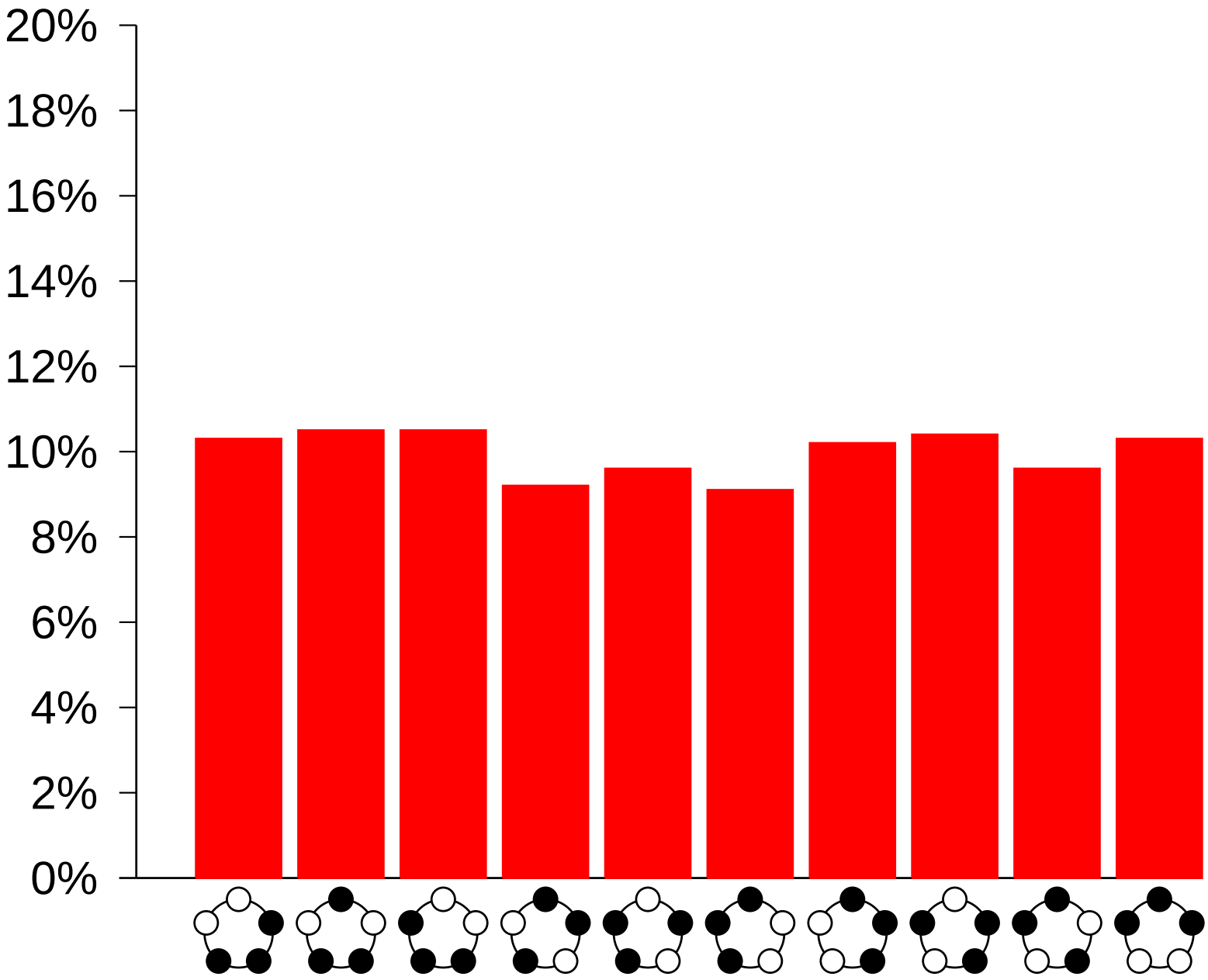
Stationary distribution



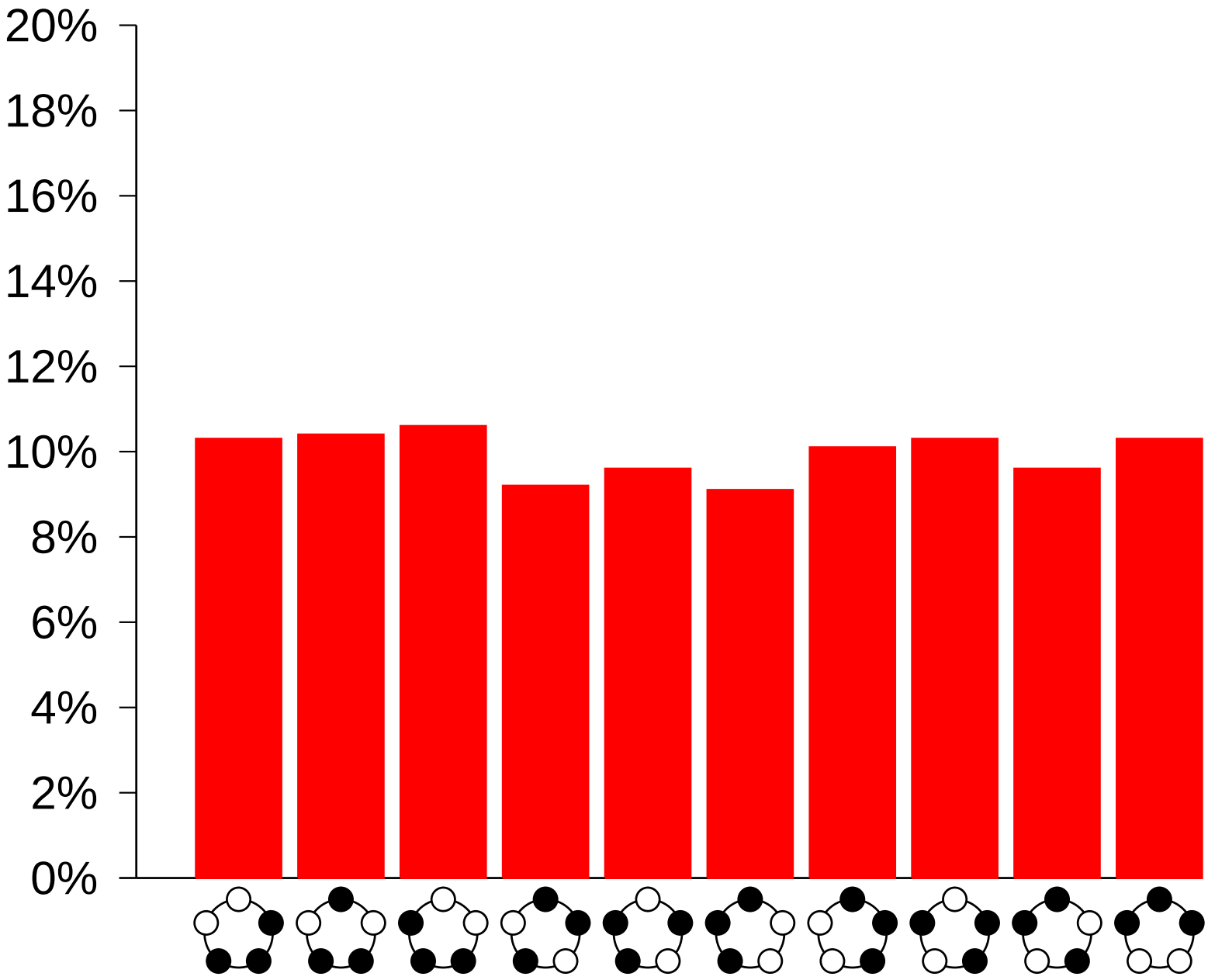
Stationary distribution



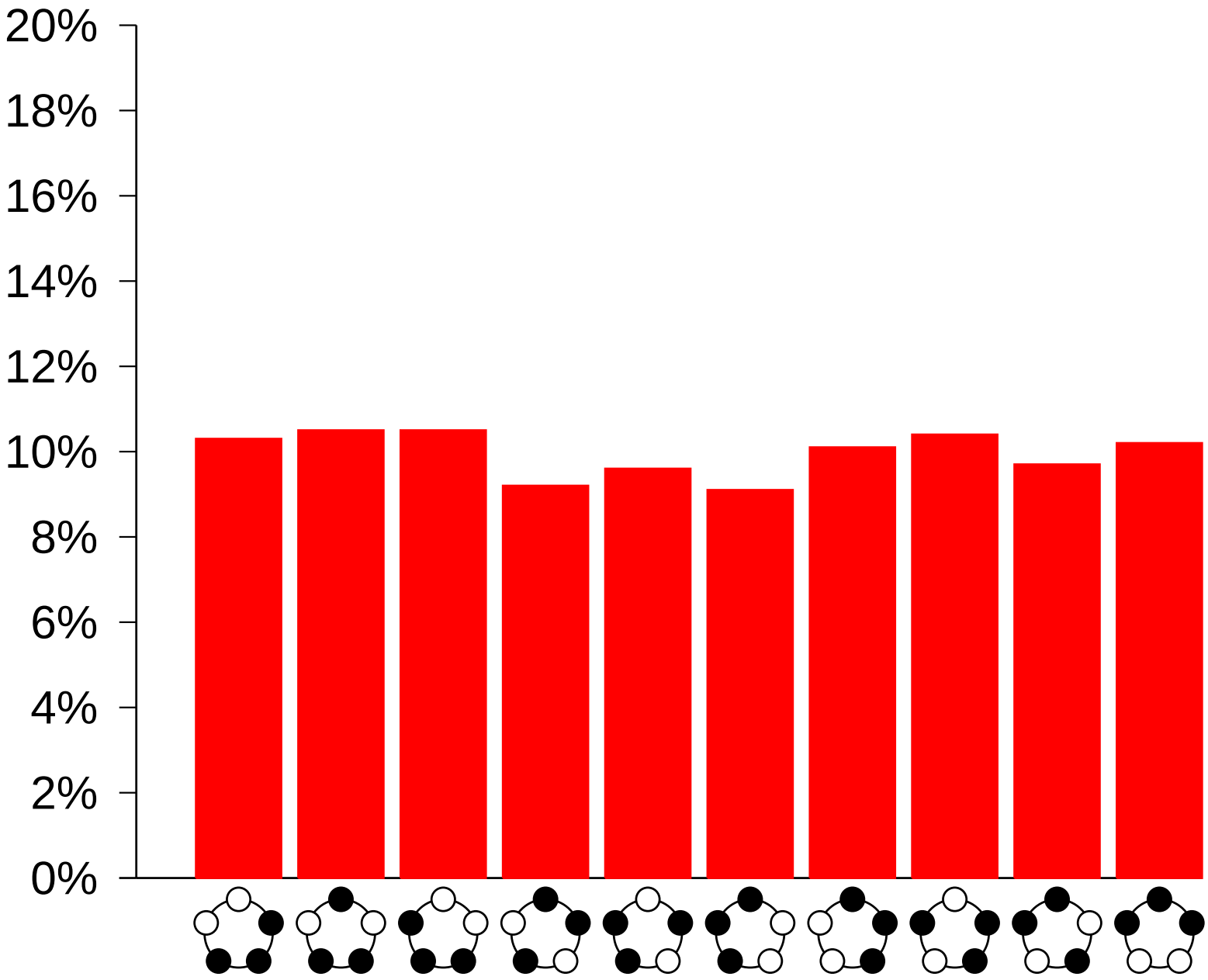
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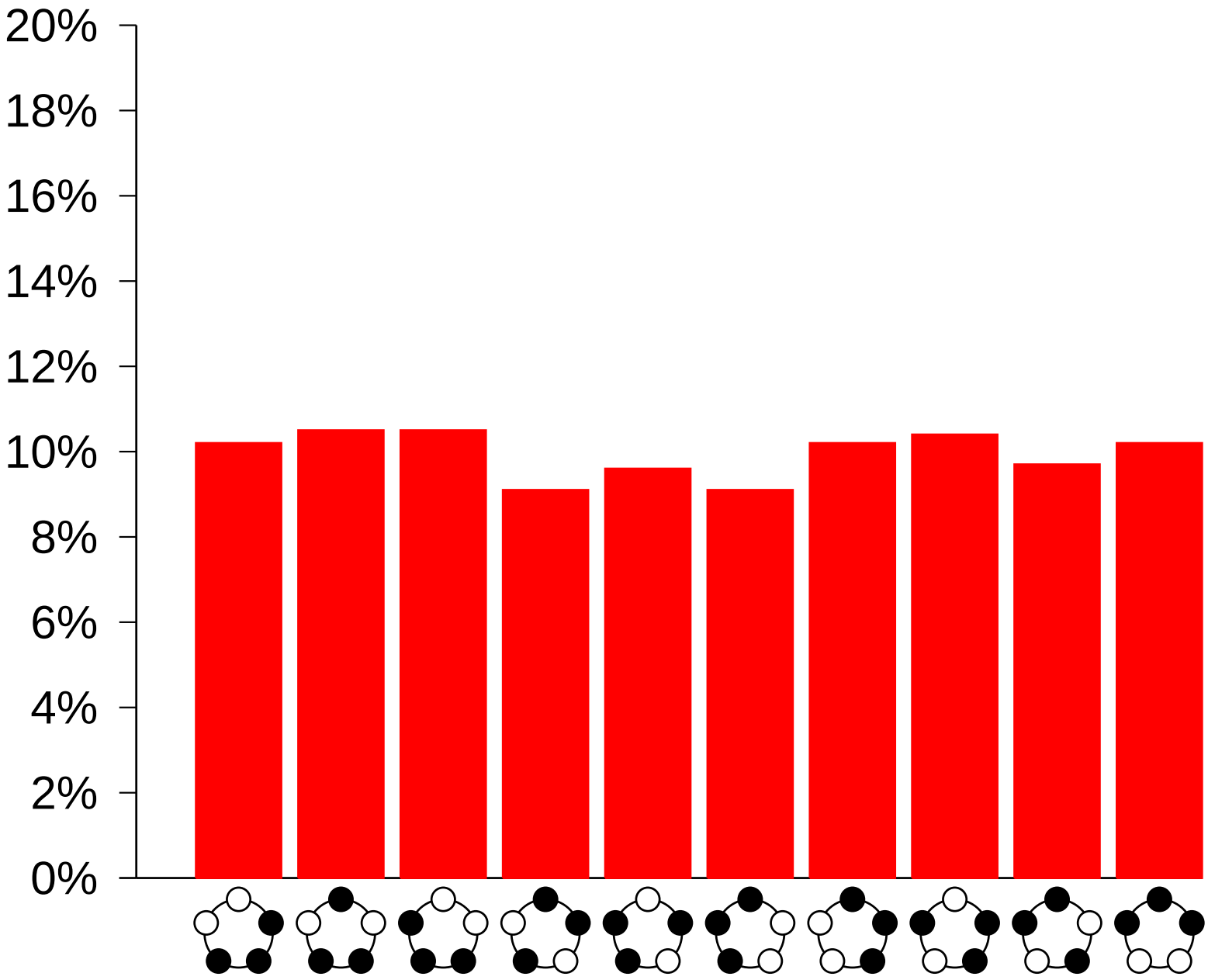
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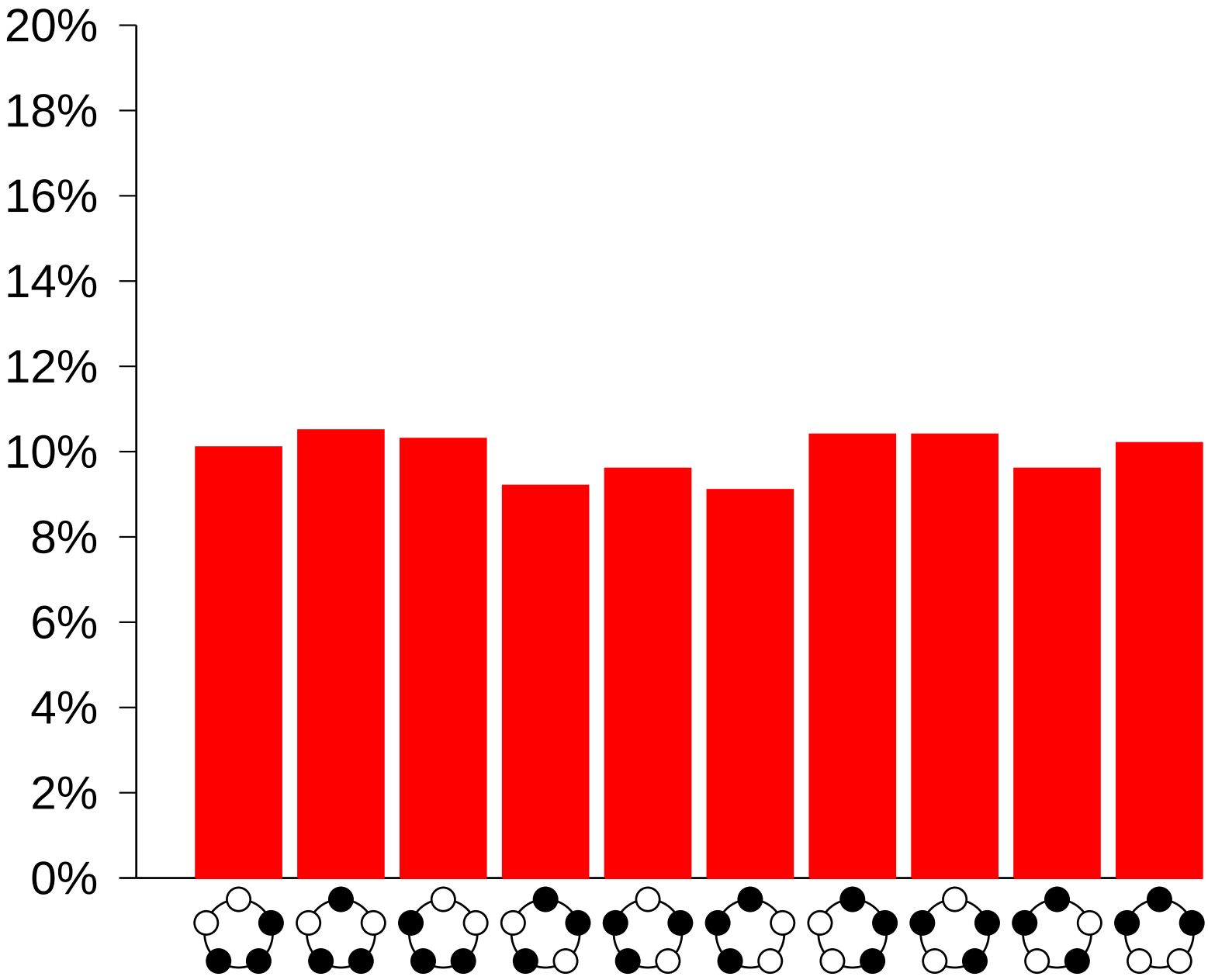
Stationary distribution



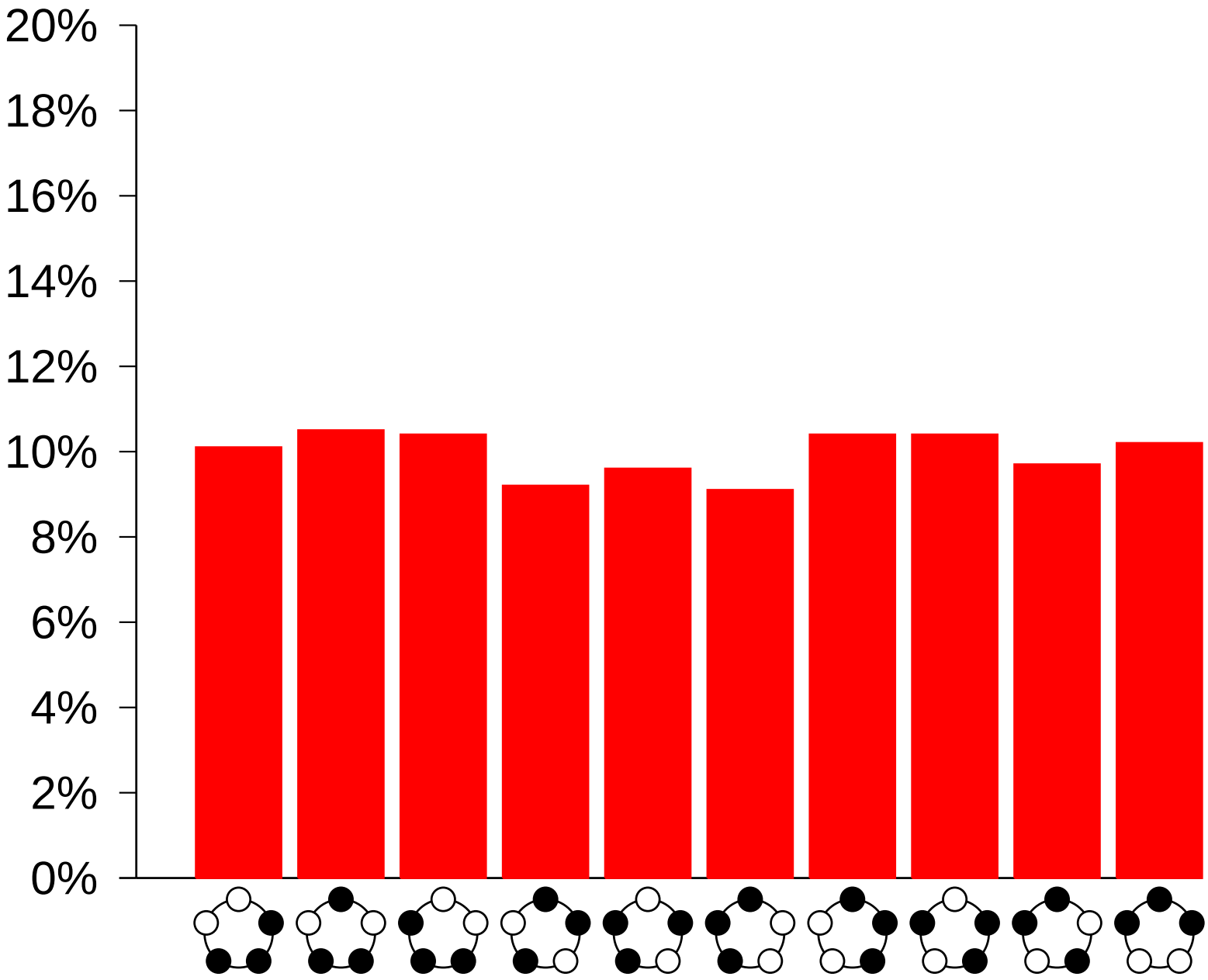
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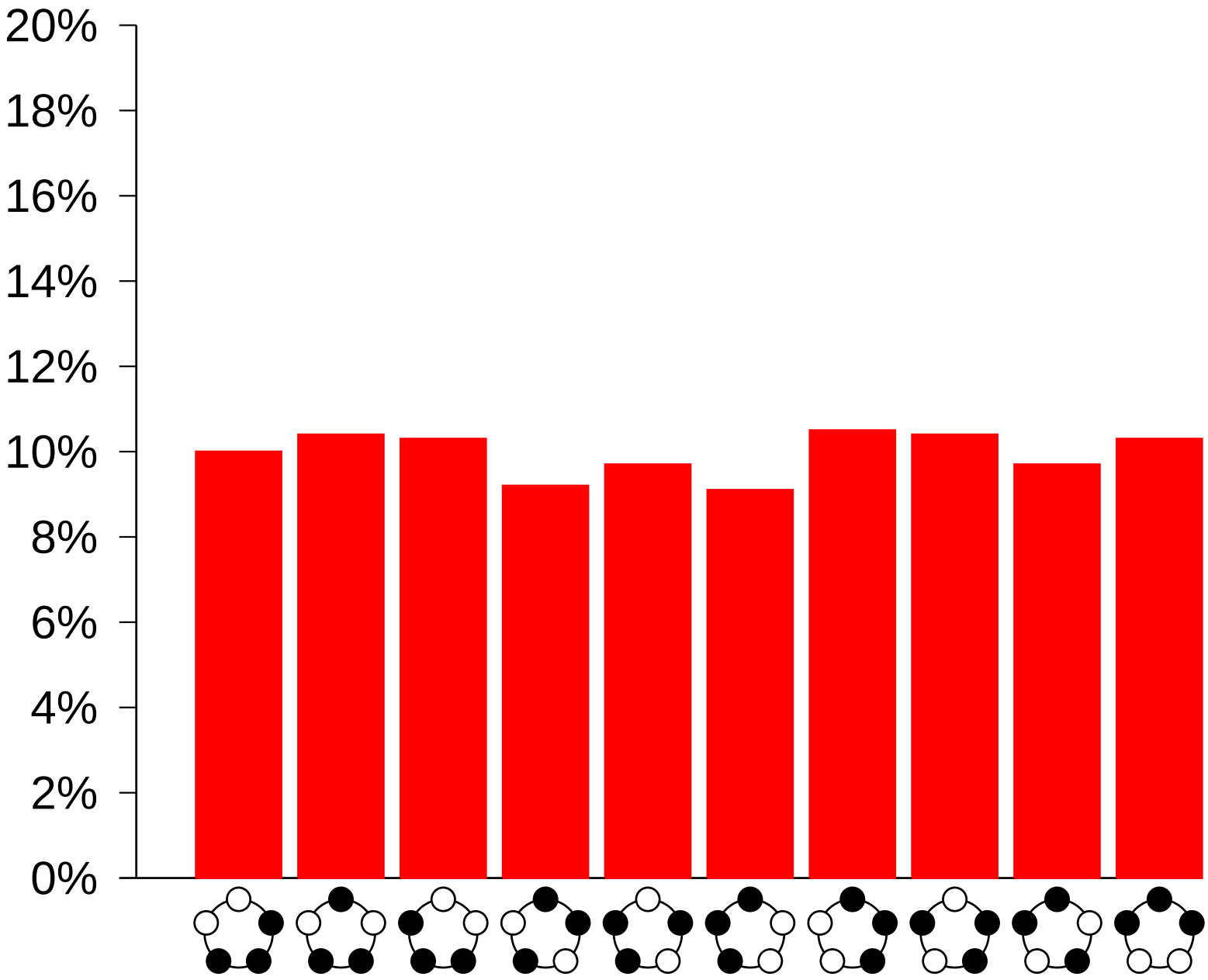
Stationary distribution



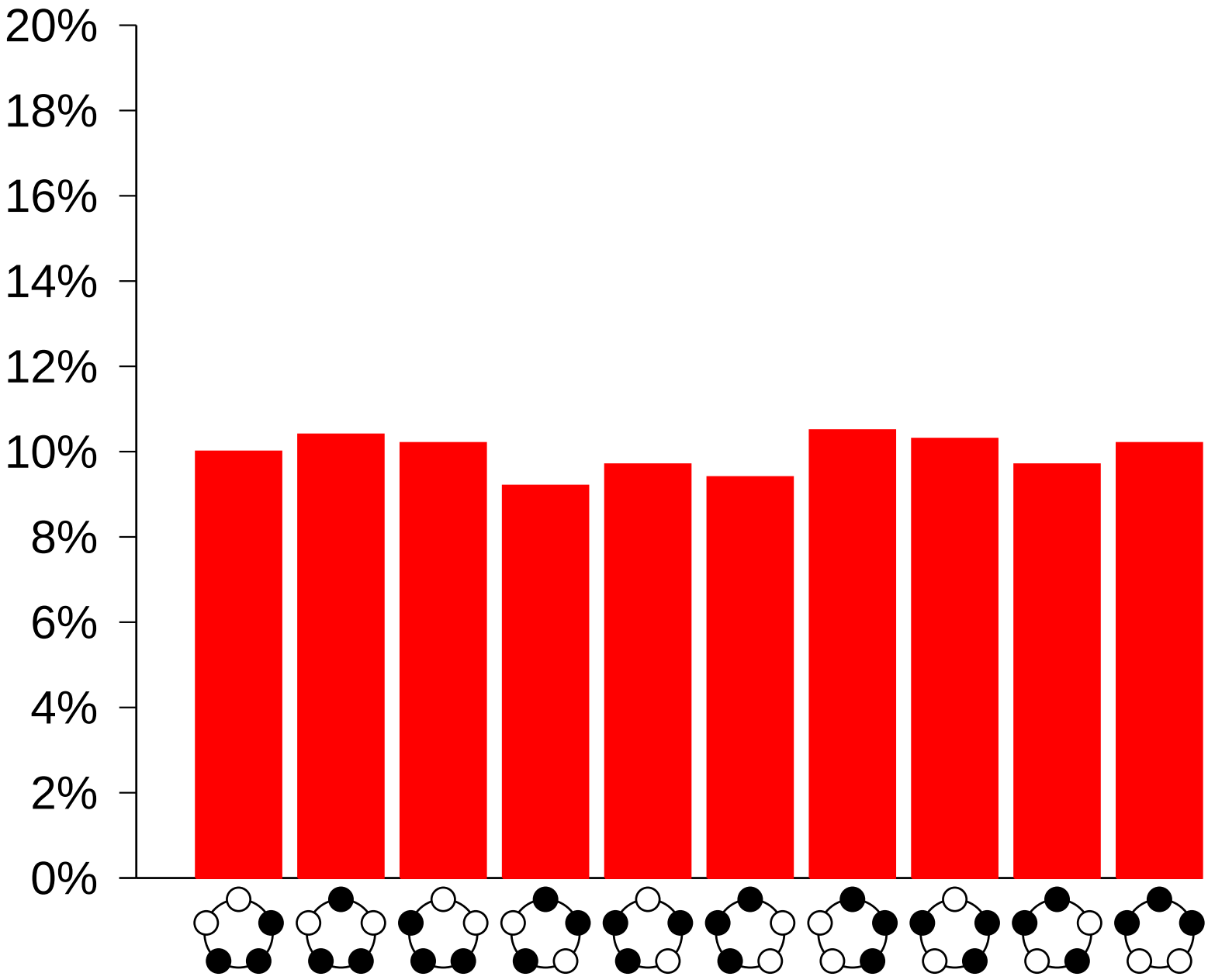
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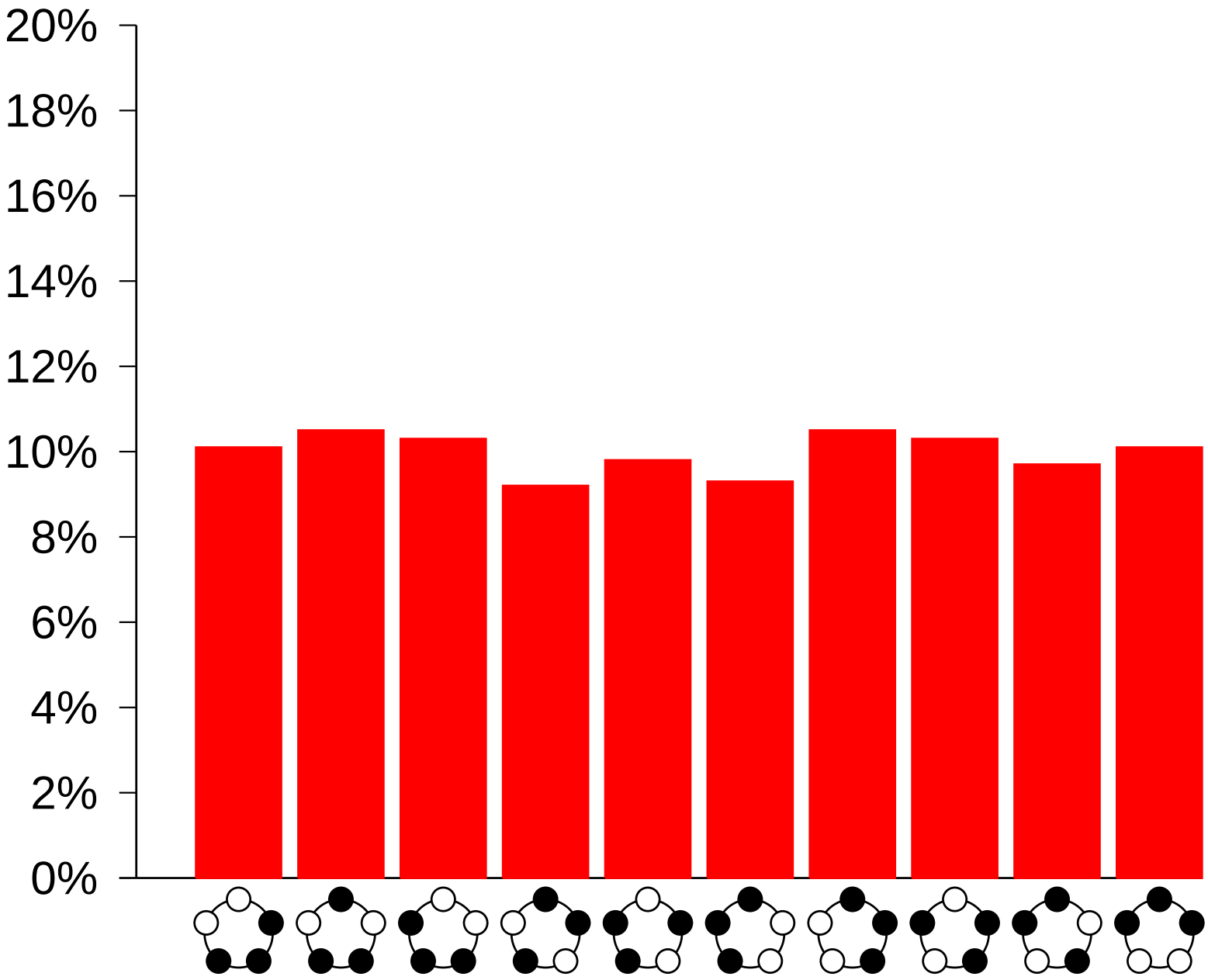
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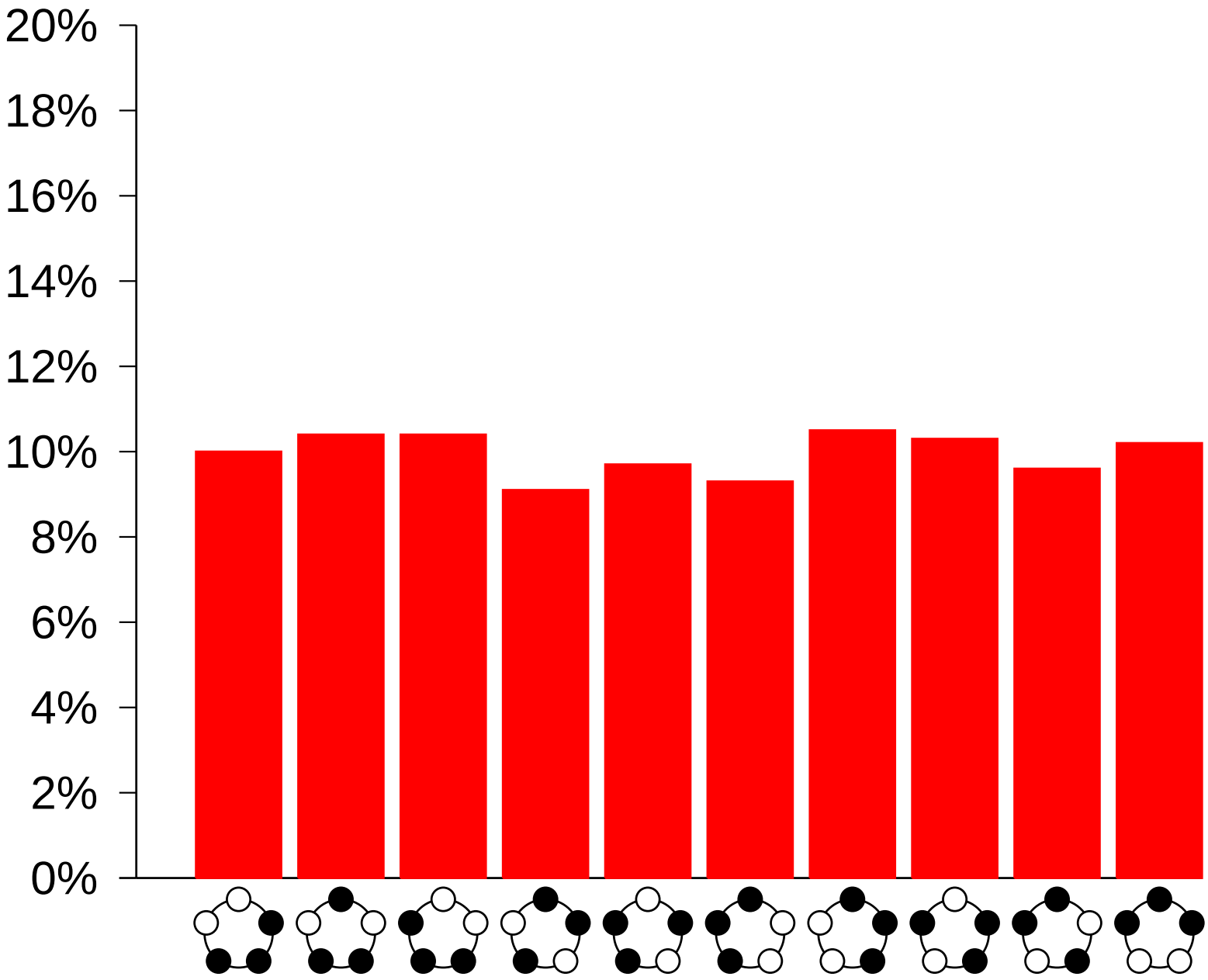
Stationary distribution



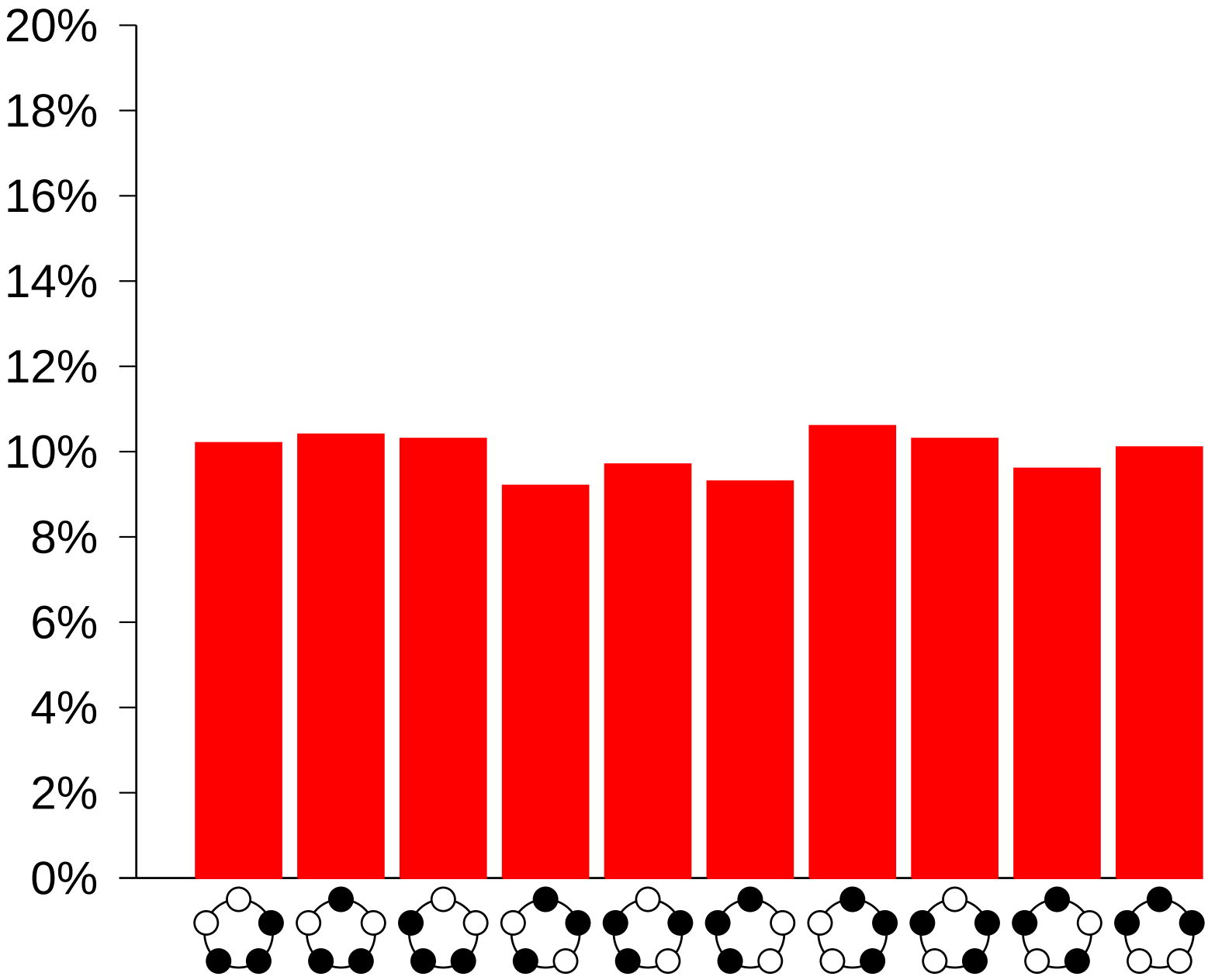
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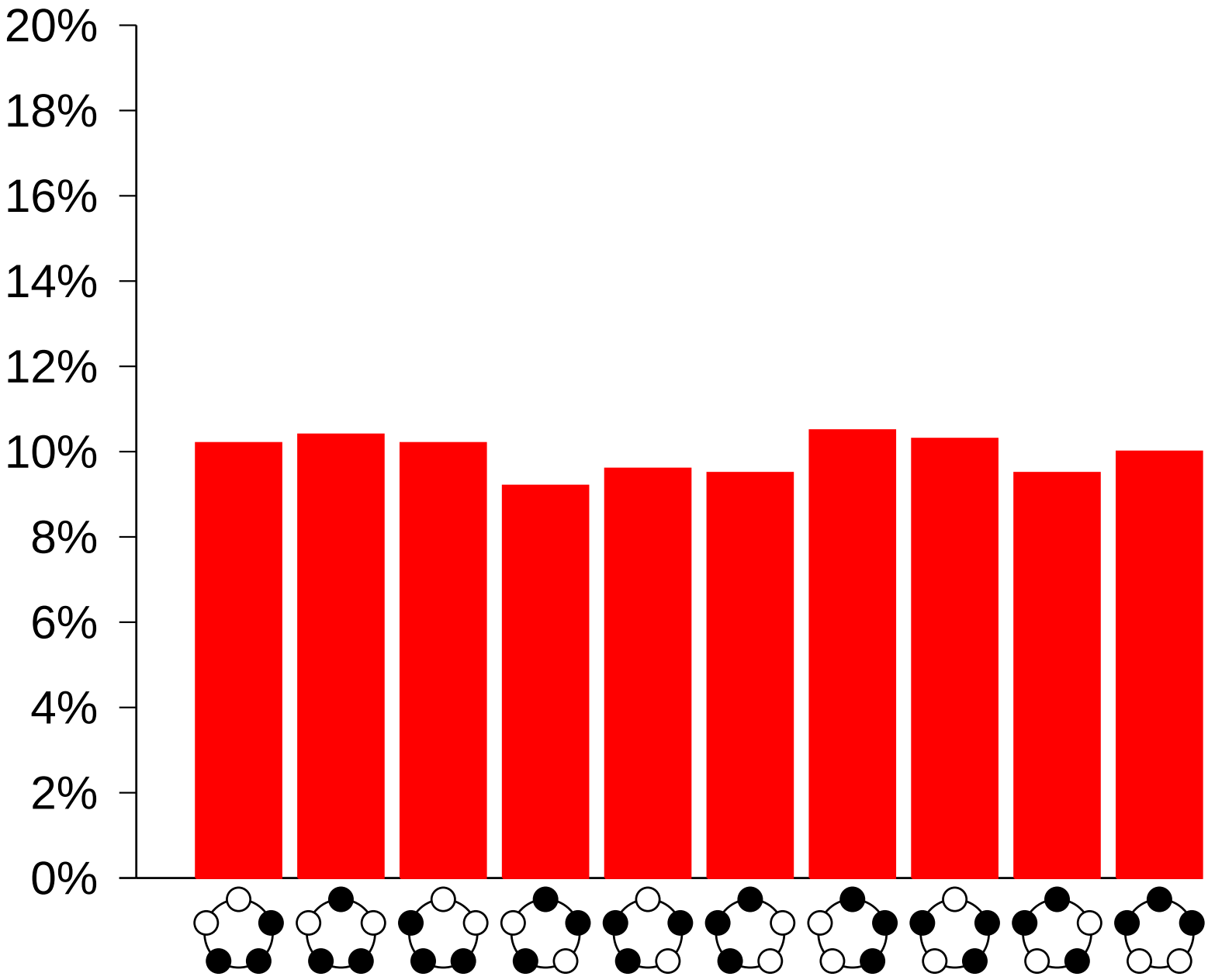
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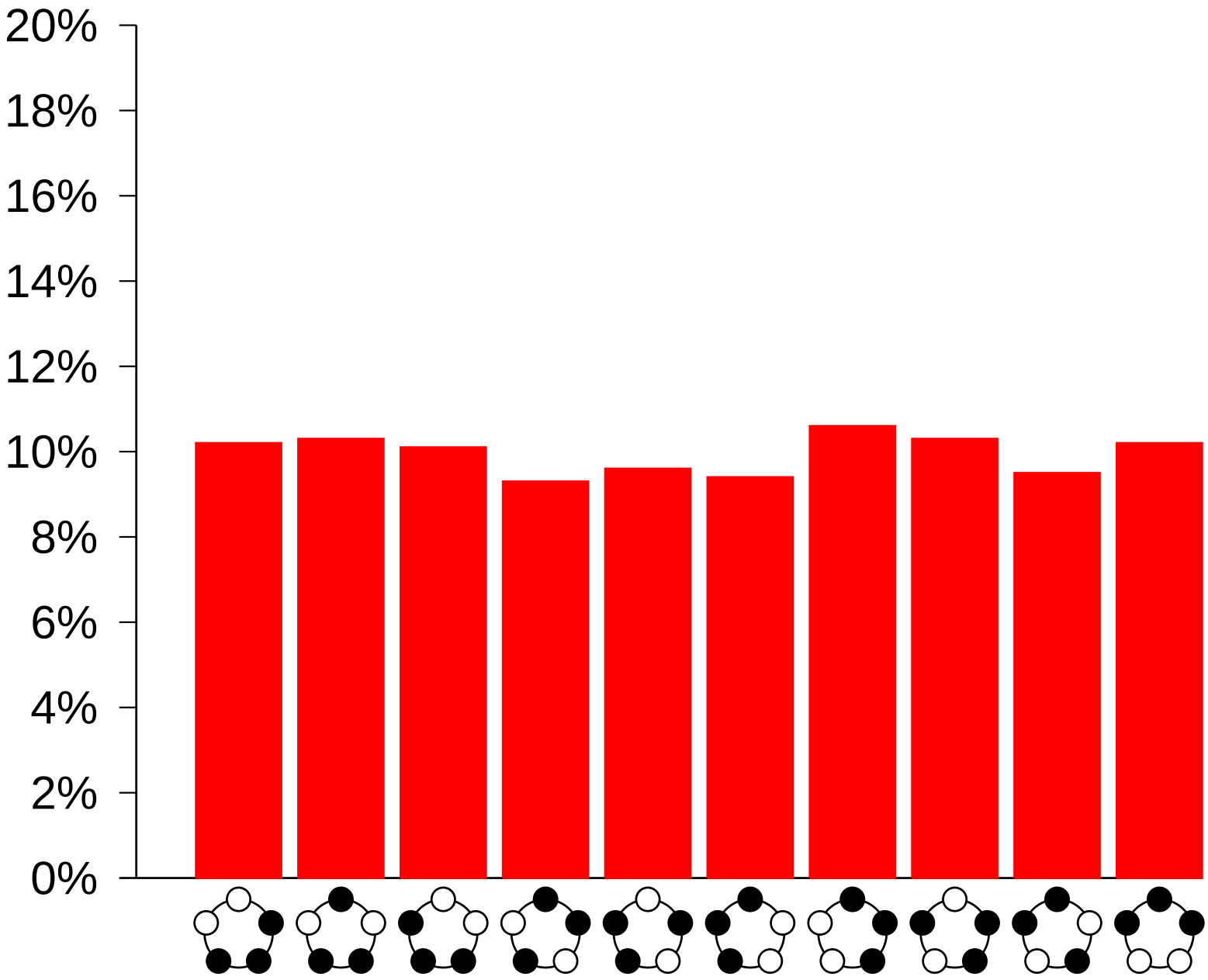
Stationary distribution



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The infinite model

Take now N (the number of slots) and m (the number of balls) to infinity such that $m/N \simeq \varrho$.

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In the limit we obtain a model on $\mathbb{Z}$. In its stationary distribution we have a ball with probability ϱ, and don't have one with probability $1 - \varrho$ independently for each slot.

On large scales

Let us now look at the infinite model on the large scale, and let it evolve for a long time. If we change the initial density ϱ on the large (X) scale, then the process will not be stationary anymore. Instead, its density will change on the large time scale (T).

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Theorem

The density $\varrho(T, X)$ as a function of the large scale variables satisfies the differential equation

$$\frac{\partial}{\partial T}\varrho + \frac{\partial}{\partial X}[\varrho(1-\varrho)] = 0$$

(*Burgers equation*).

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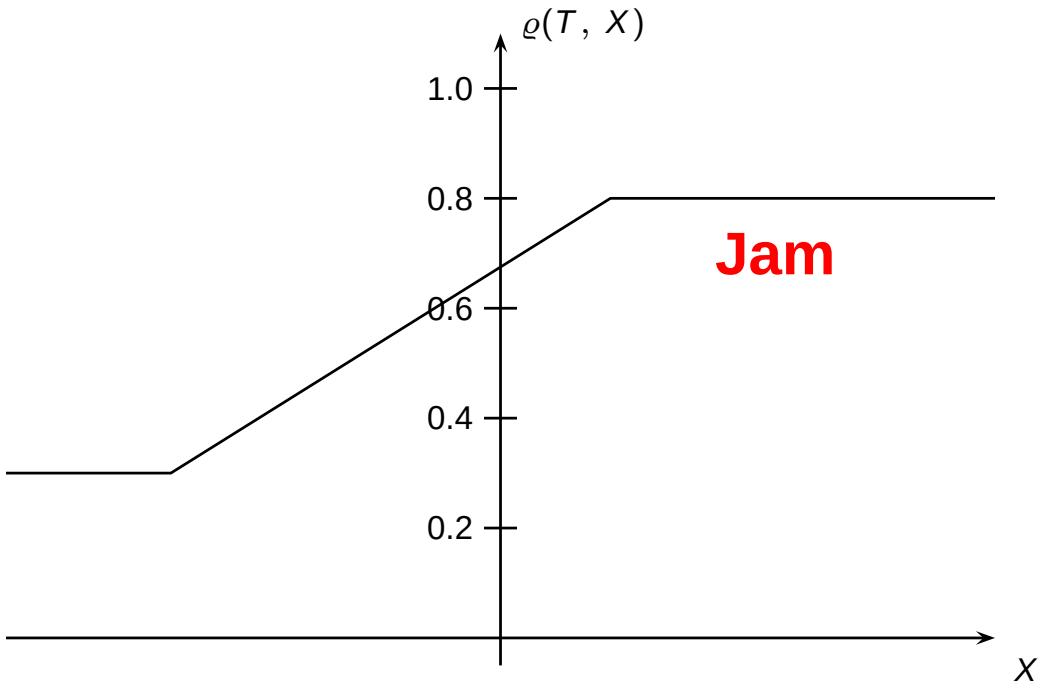
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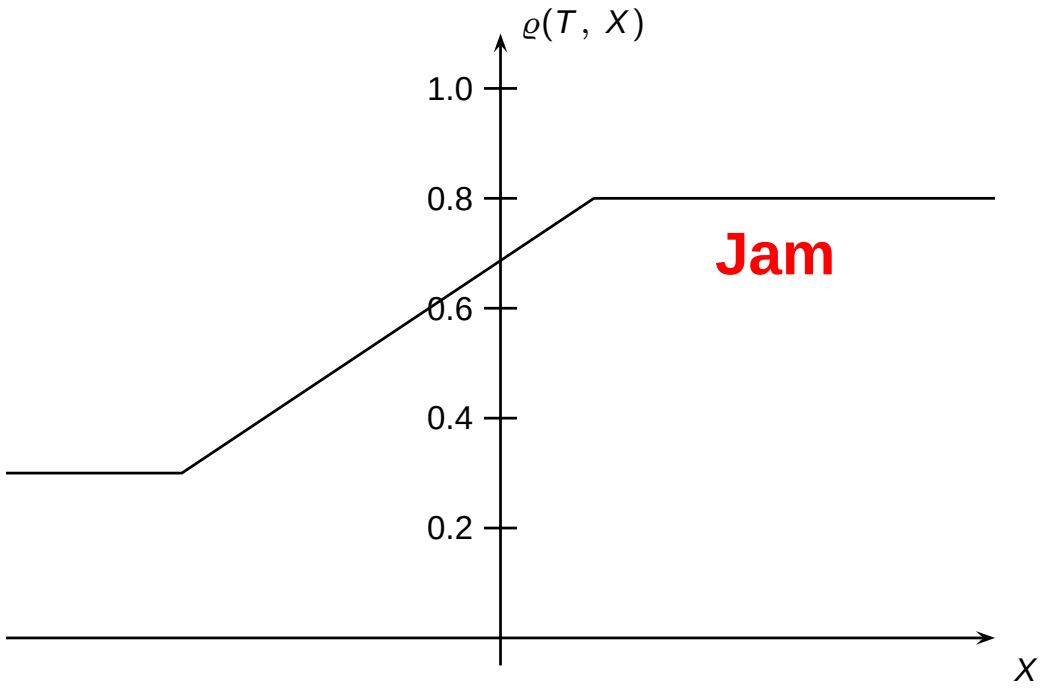
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The following are solutions of this equation:

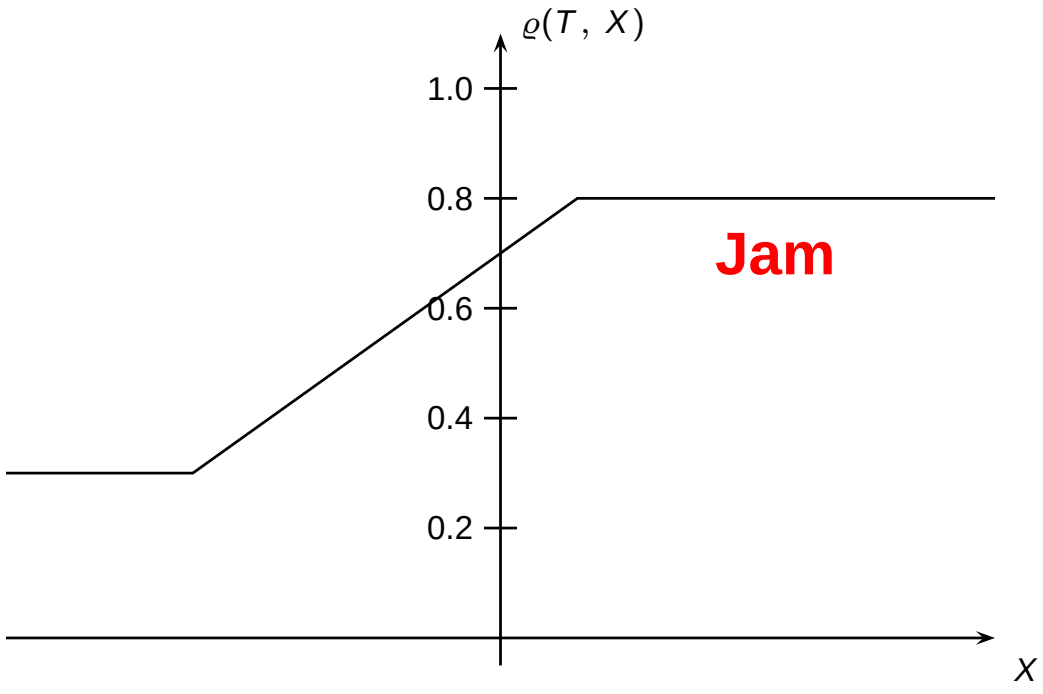
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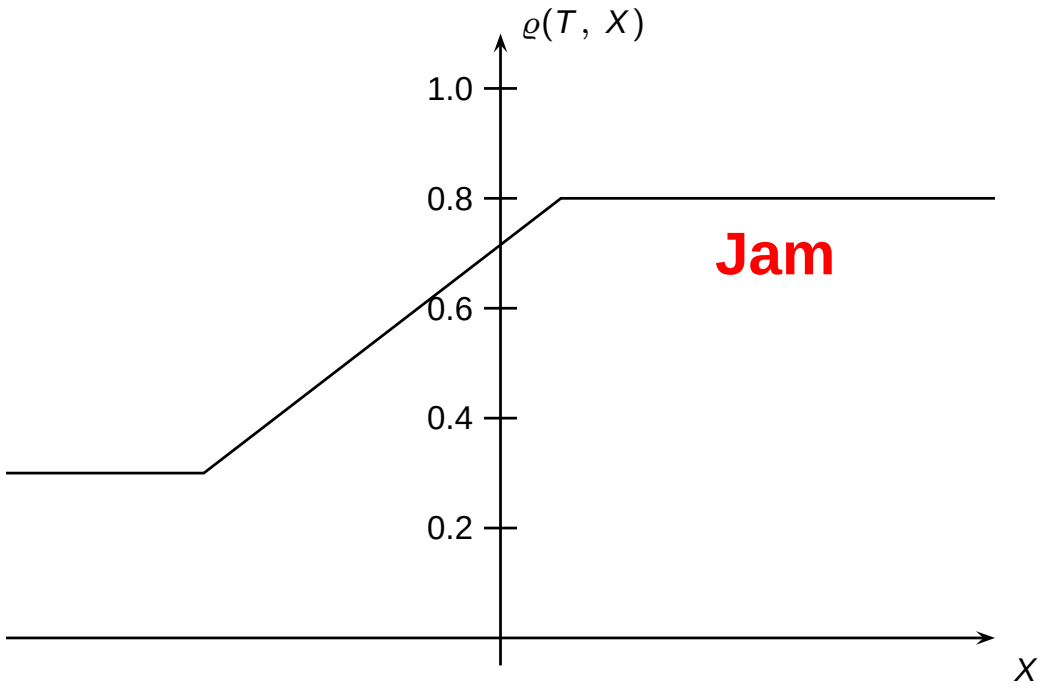
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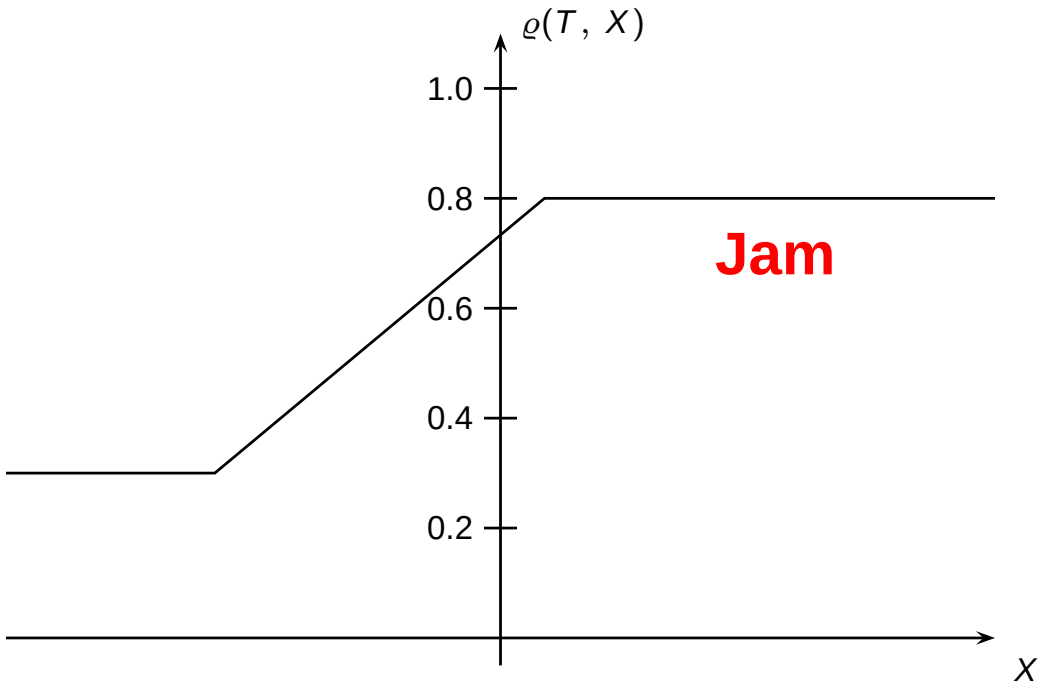
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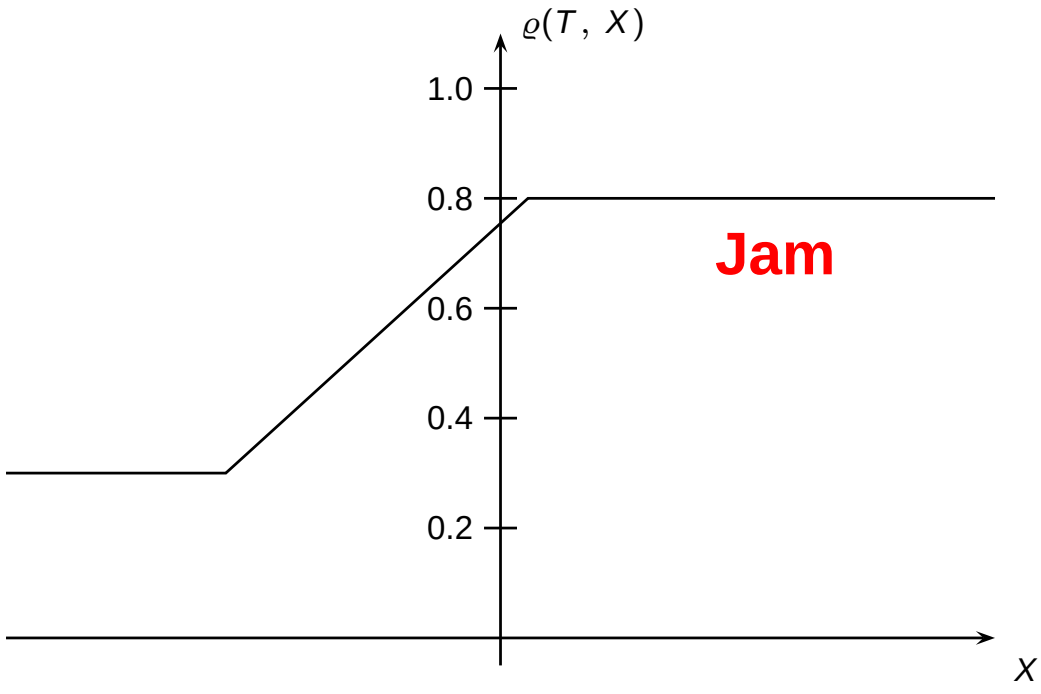
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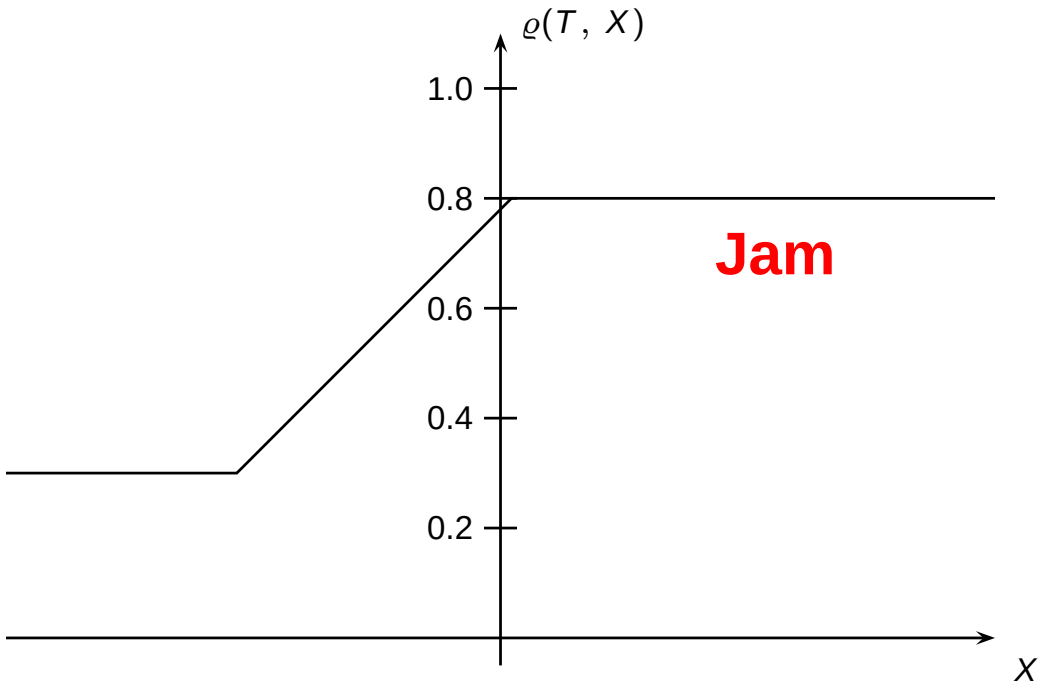
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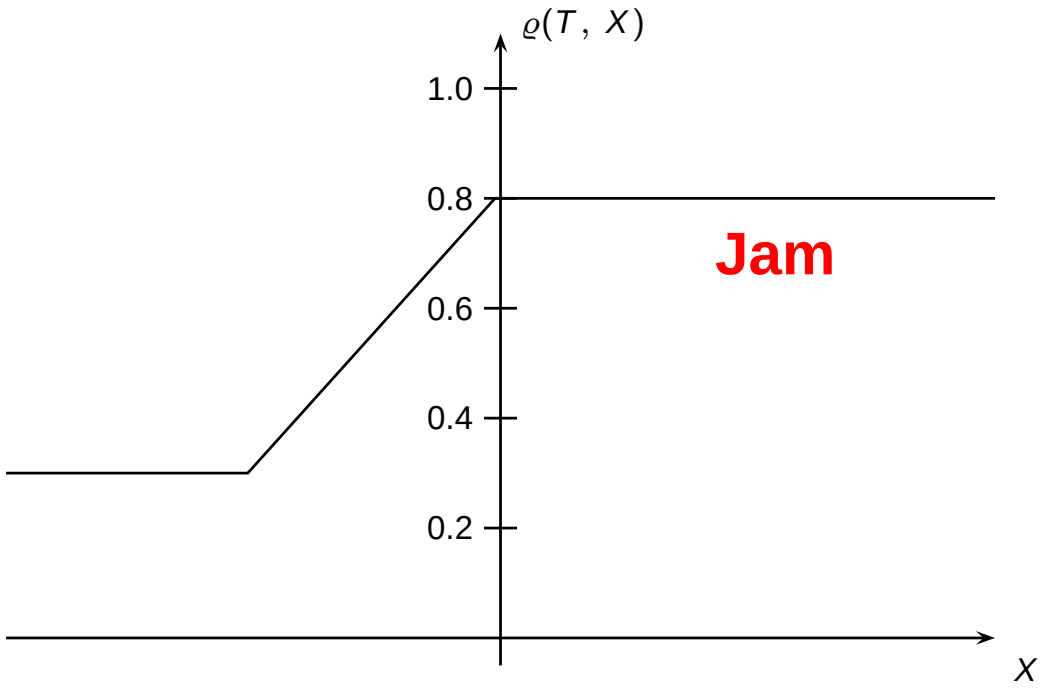
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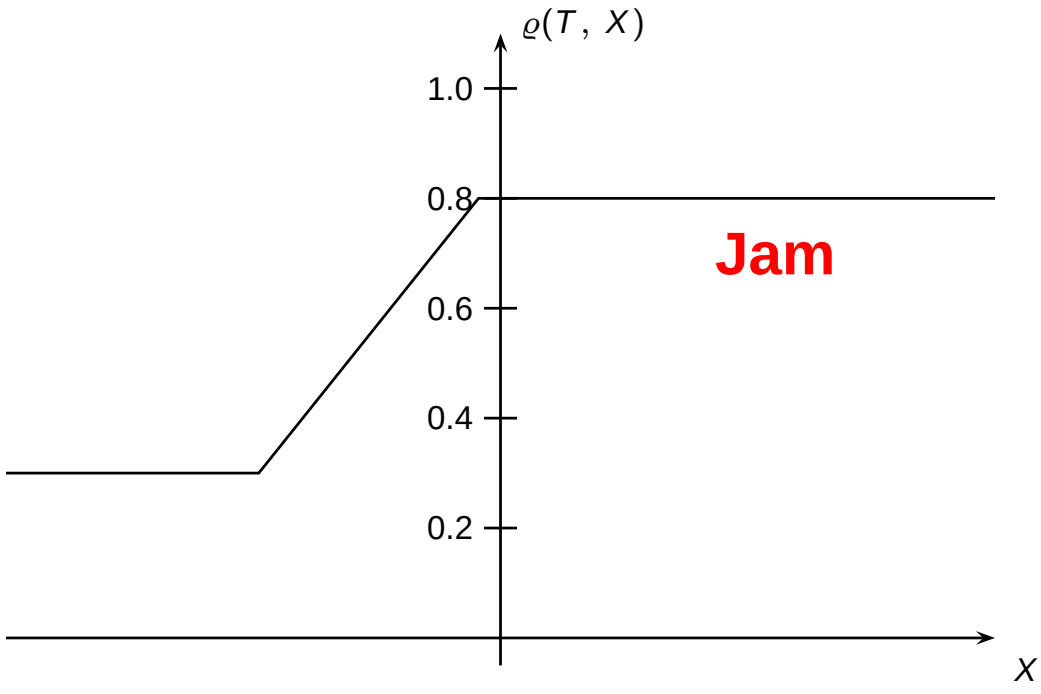
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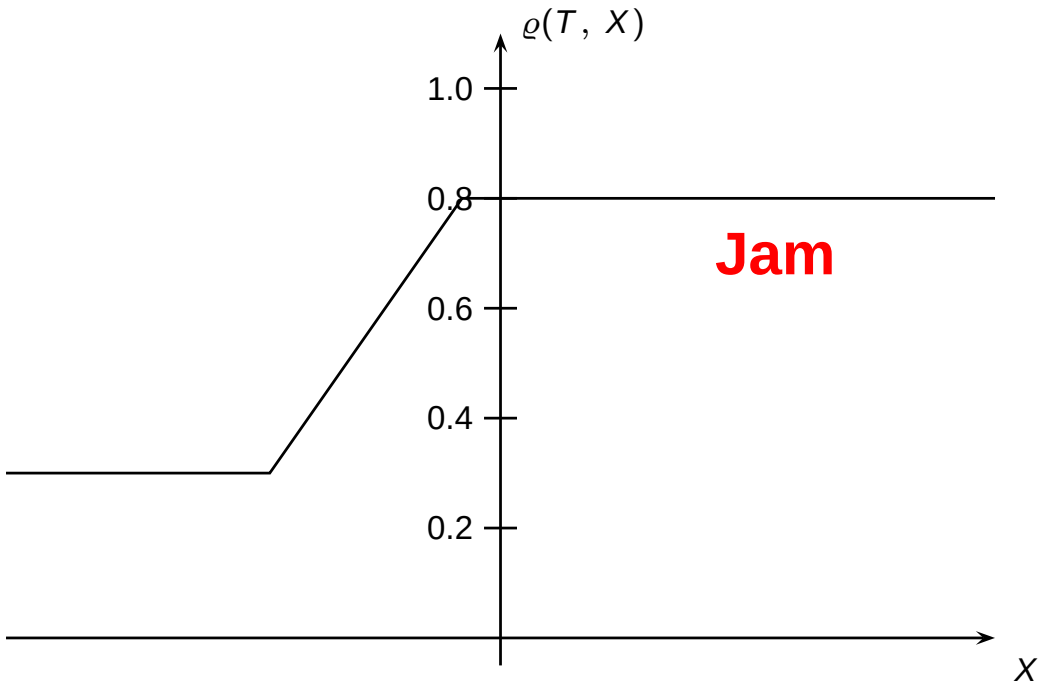
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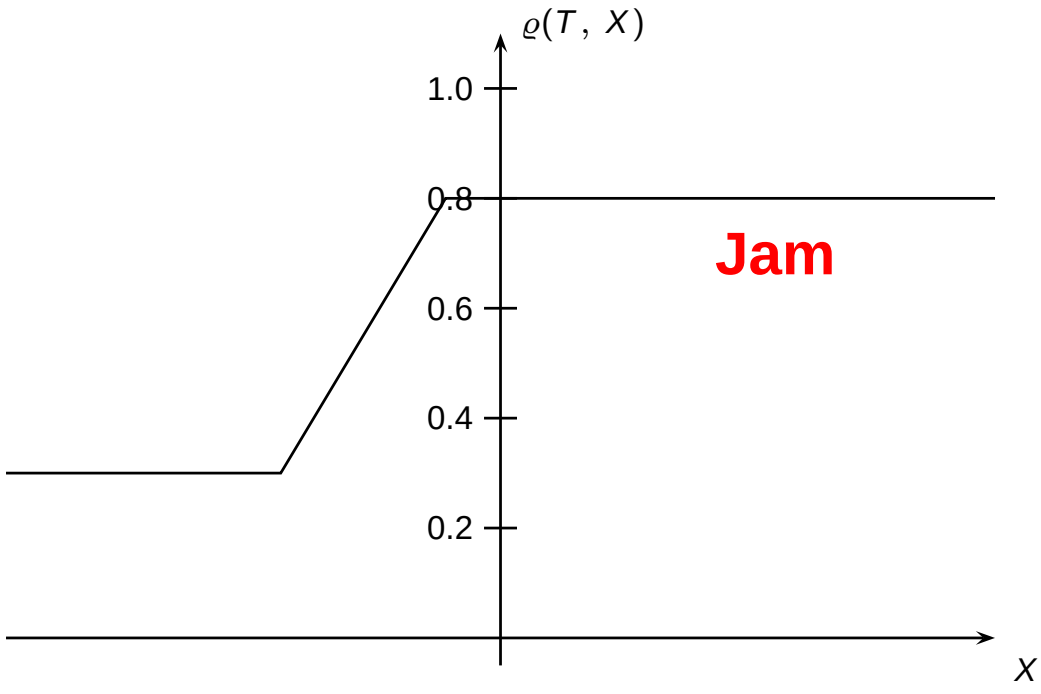
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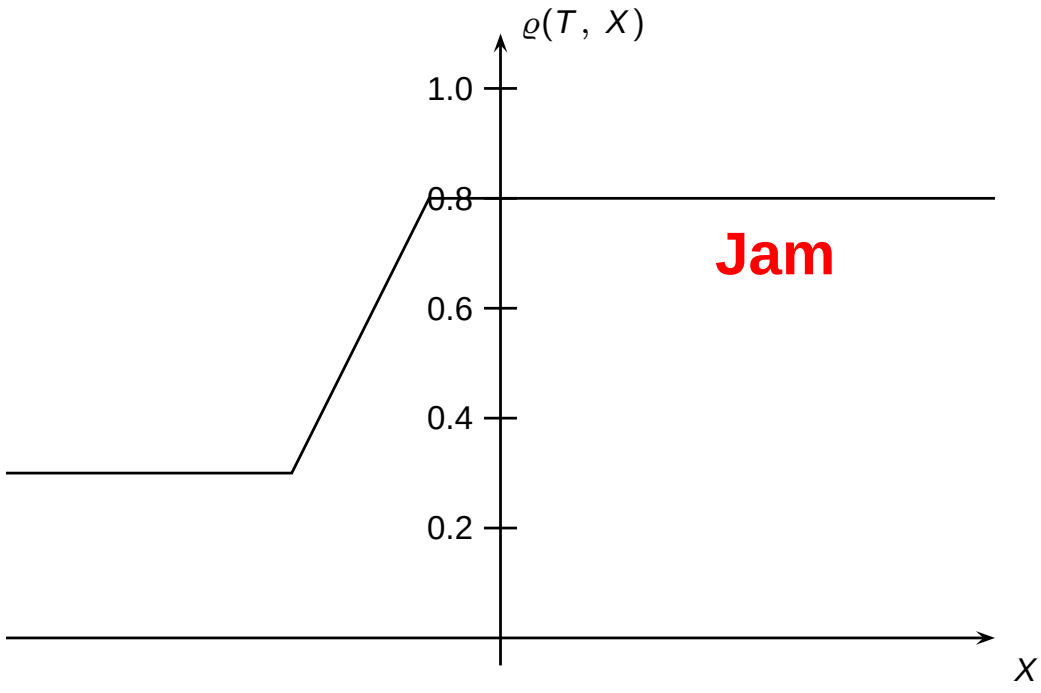
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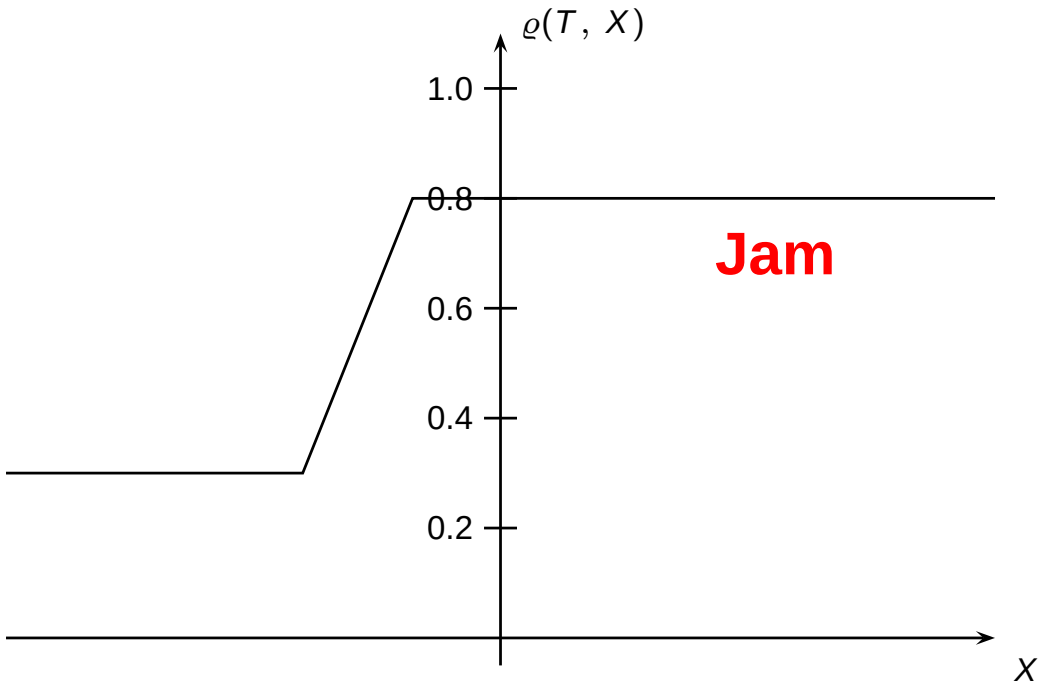
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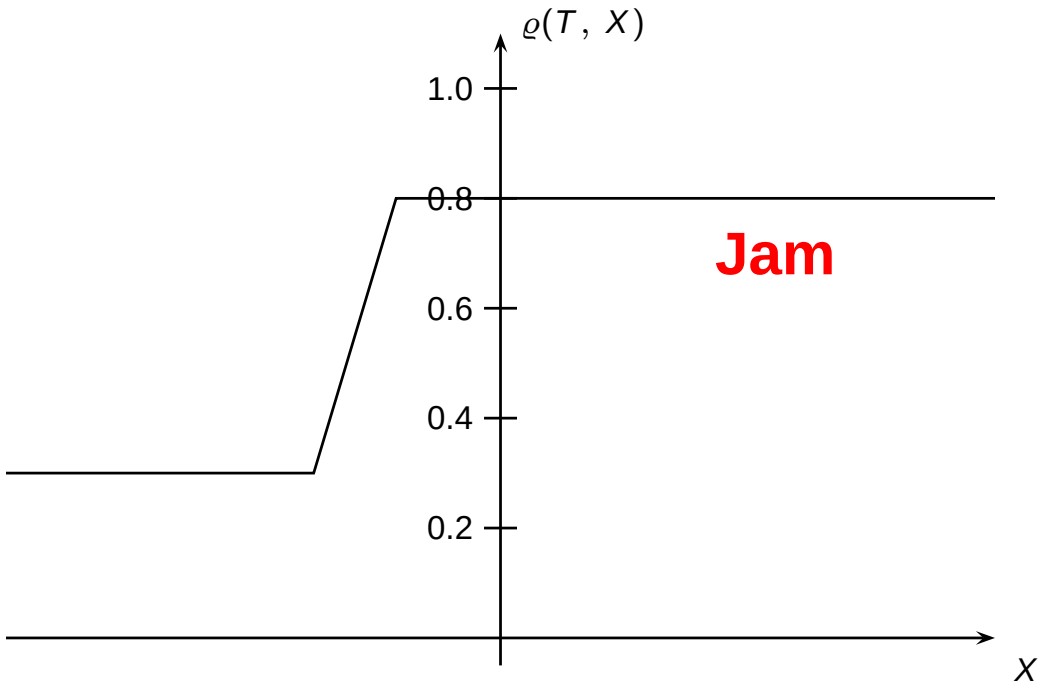
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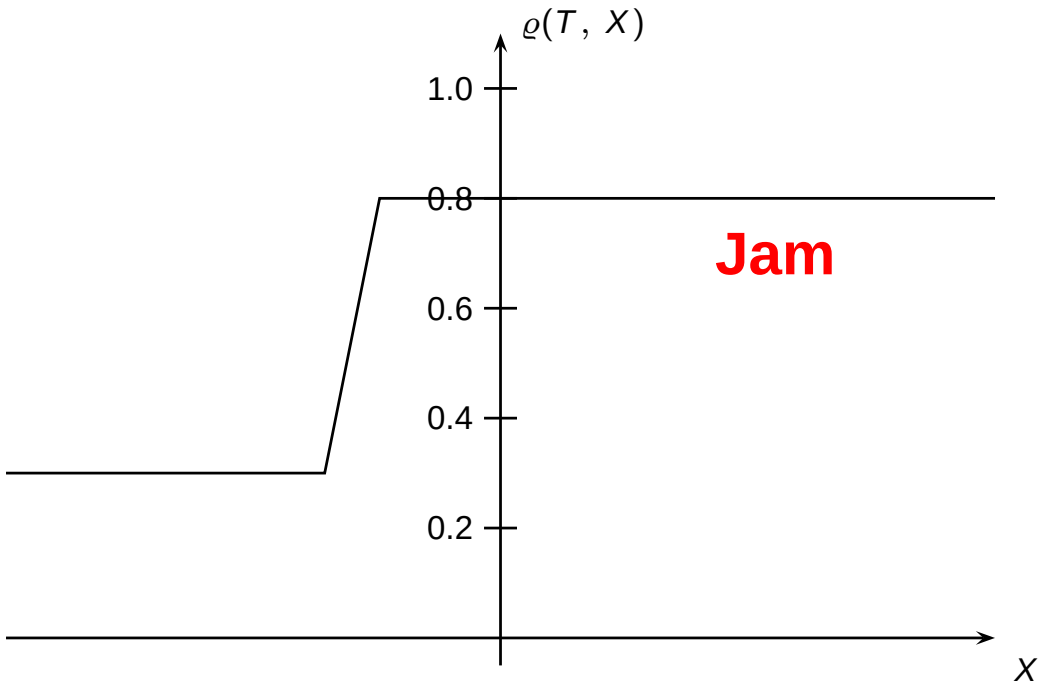
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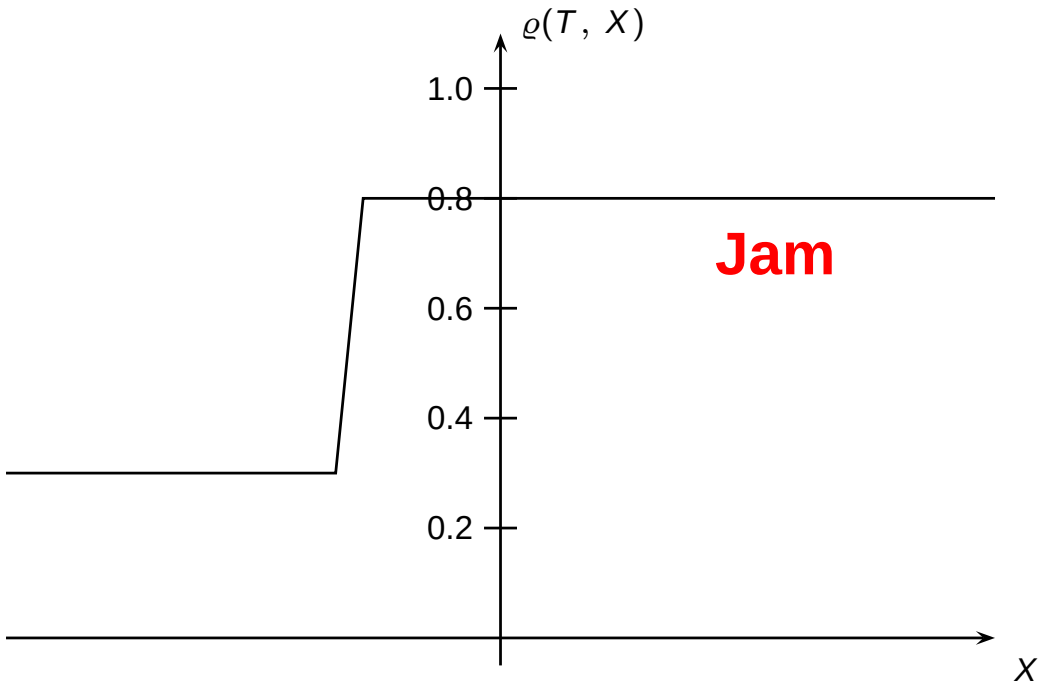
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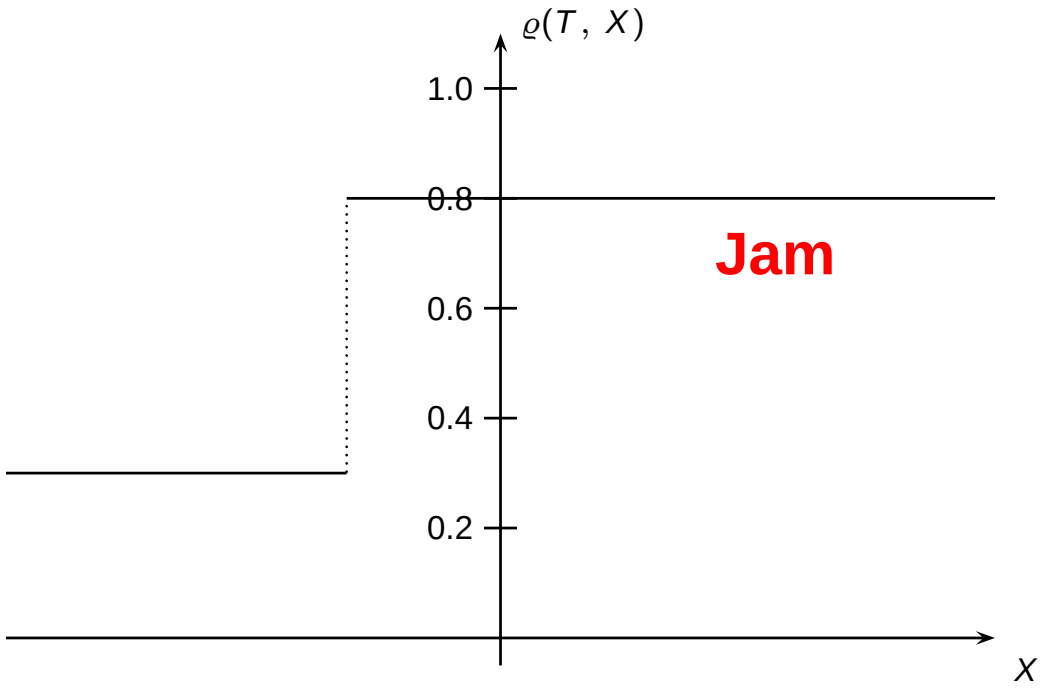
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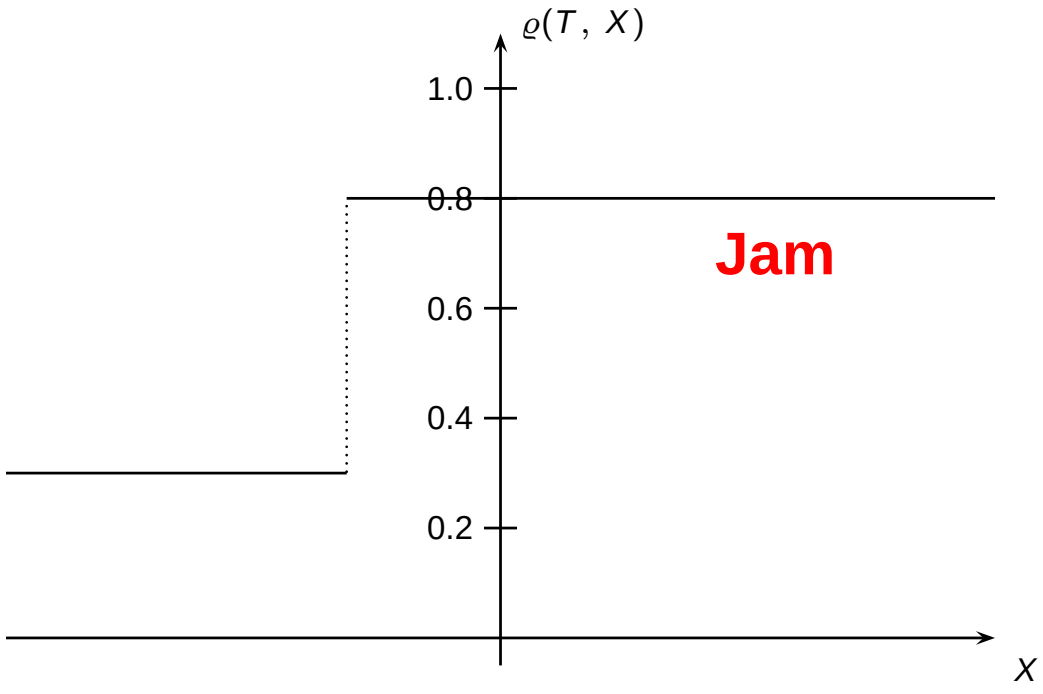
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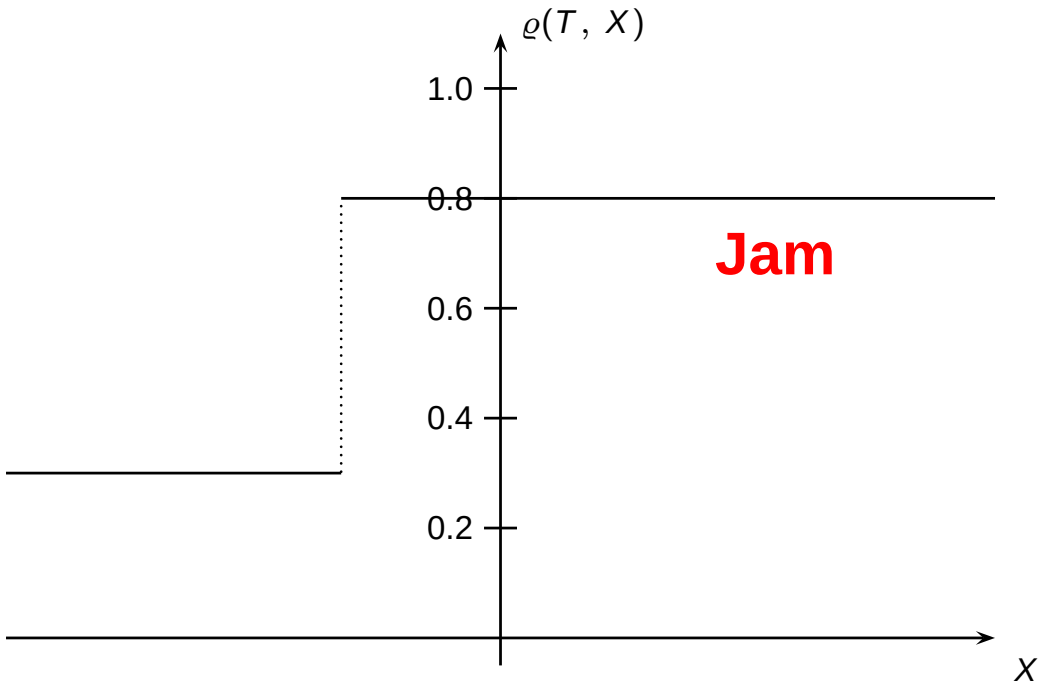
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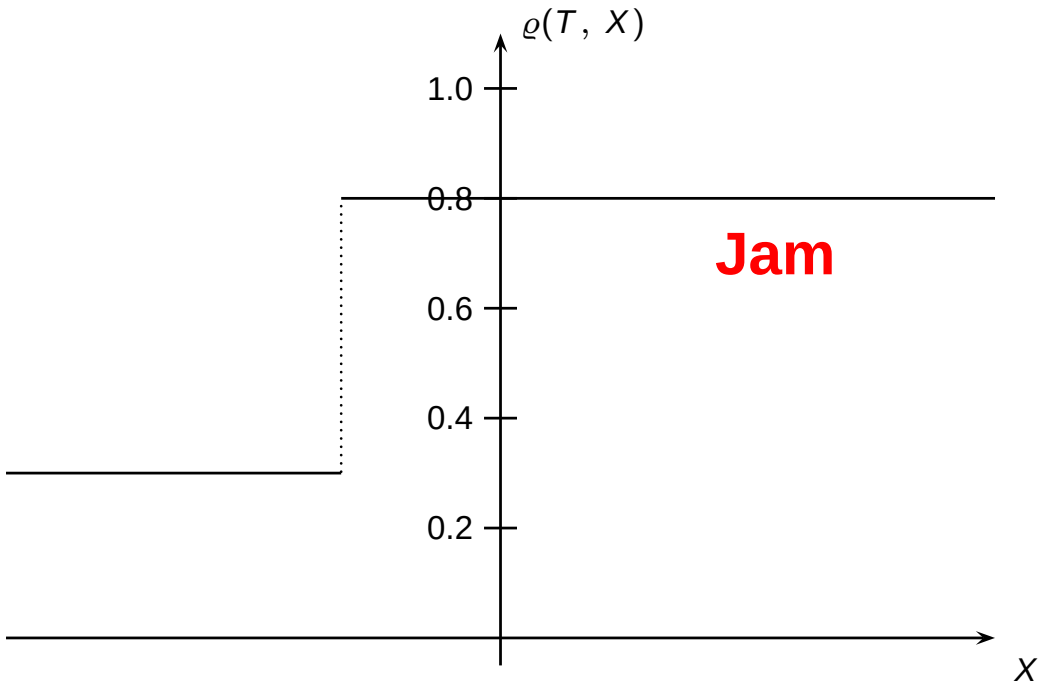
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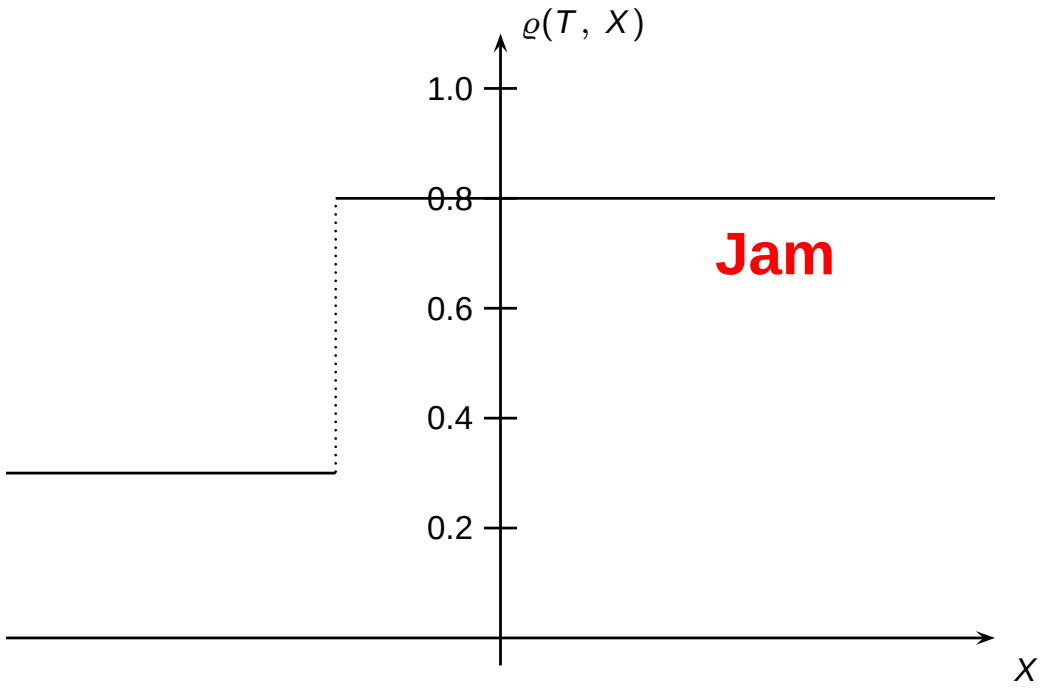
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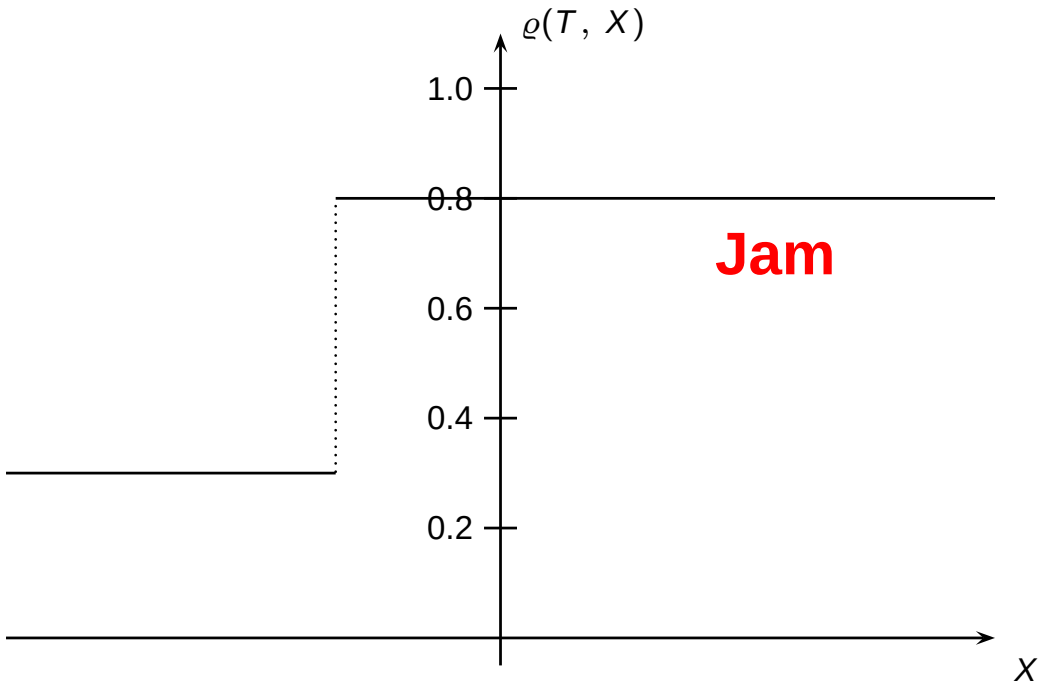
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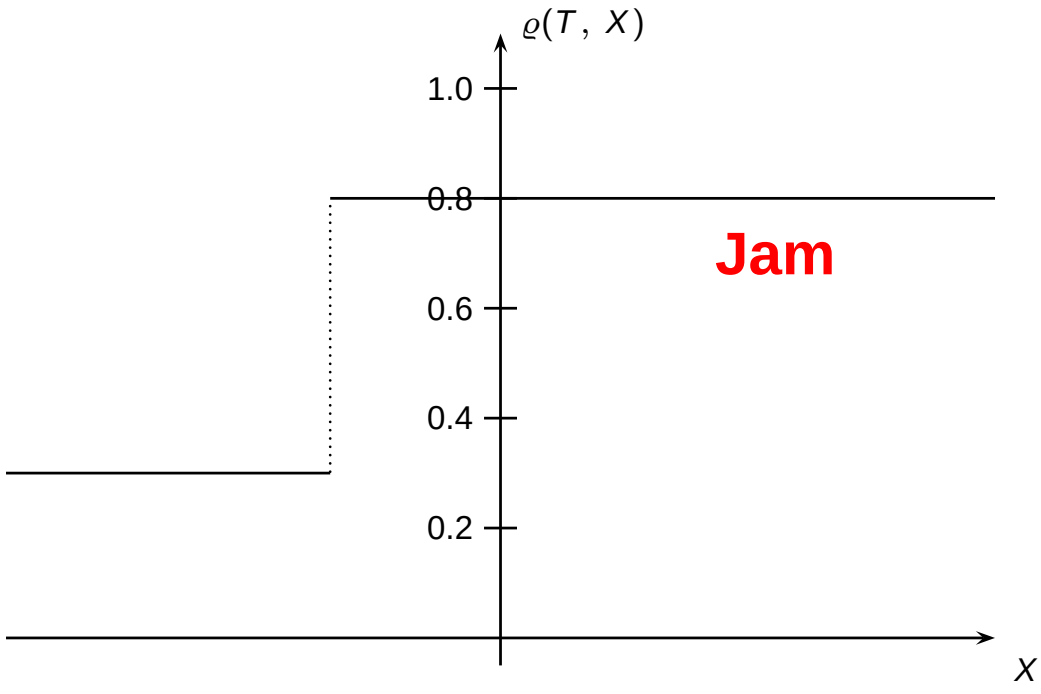
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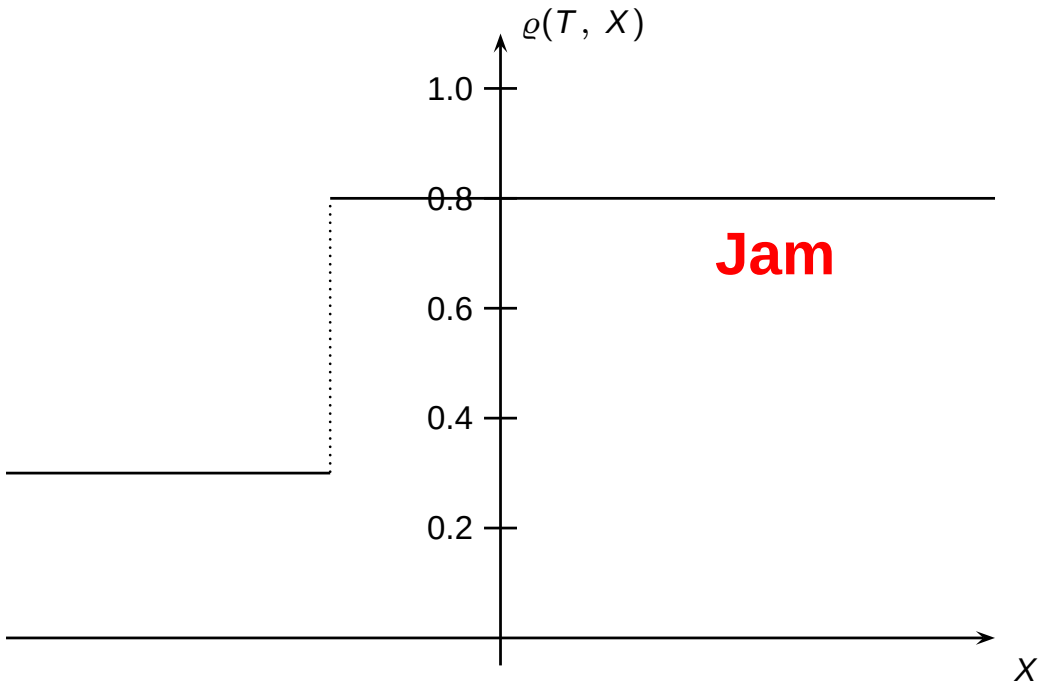
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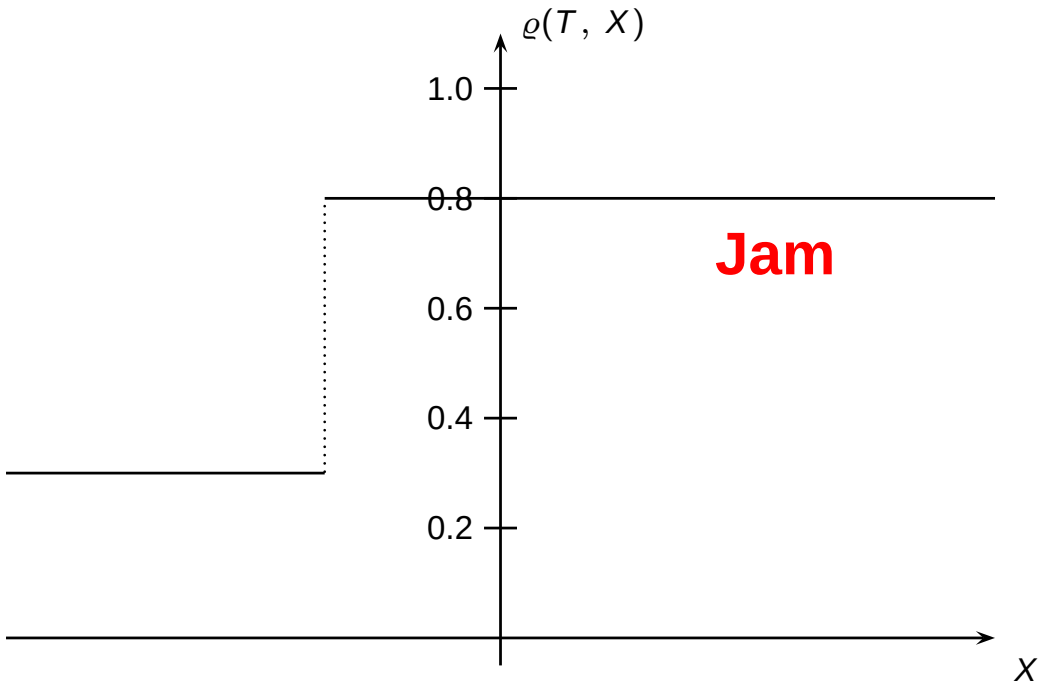
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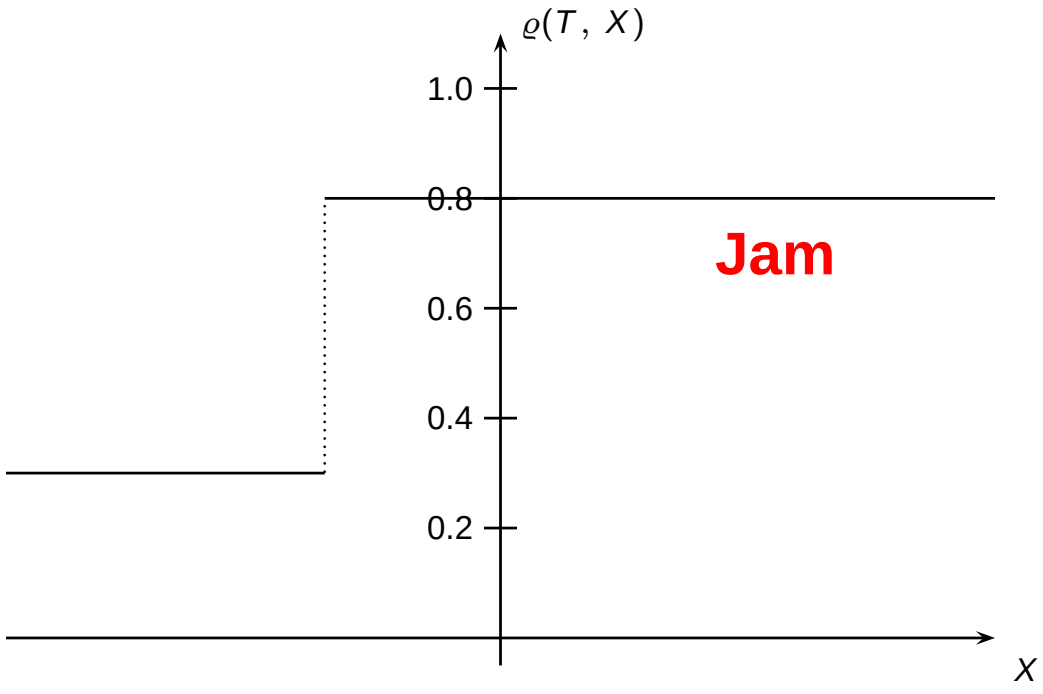
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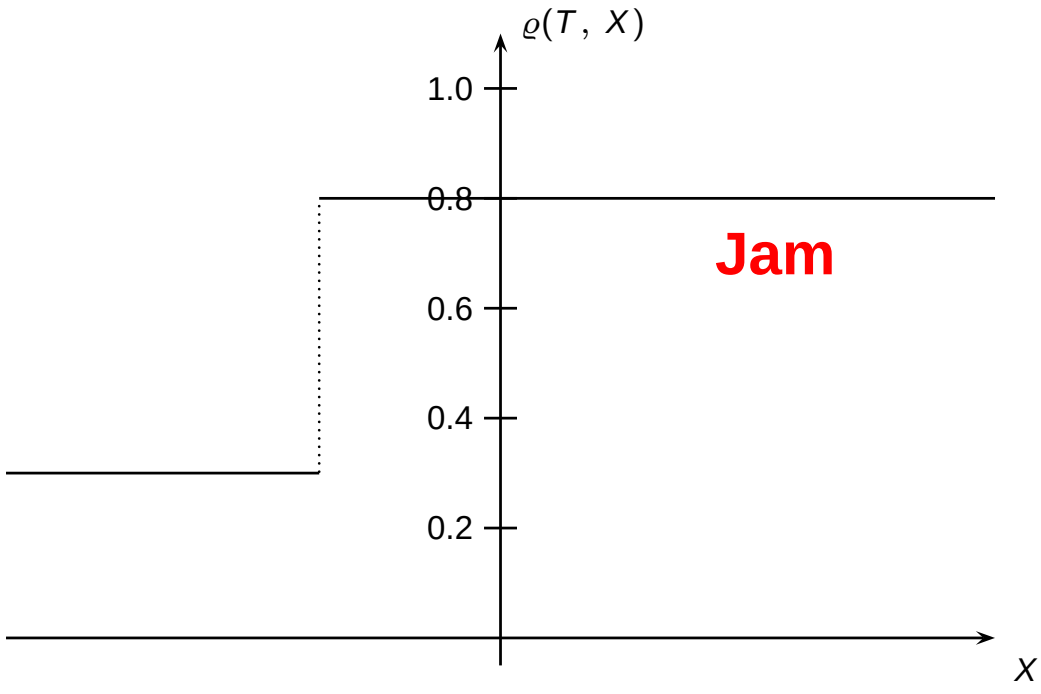
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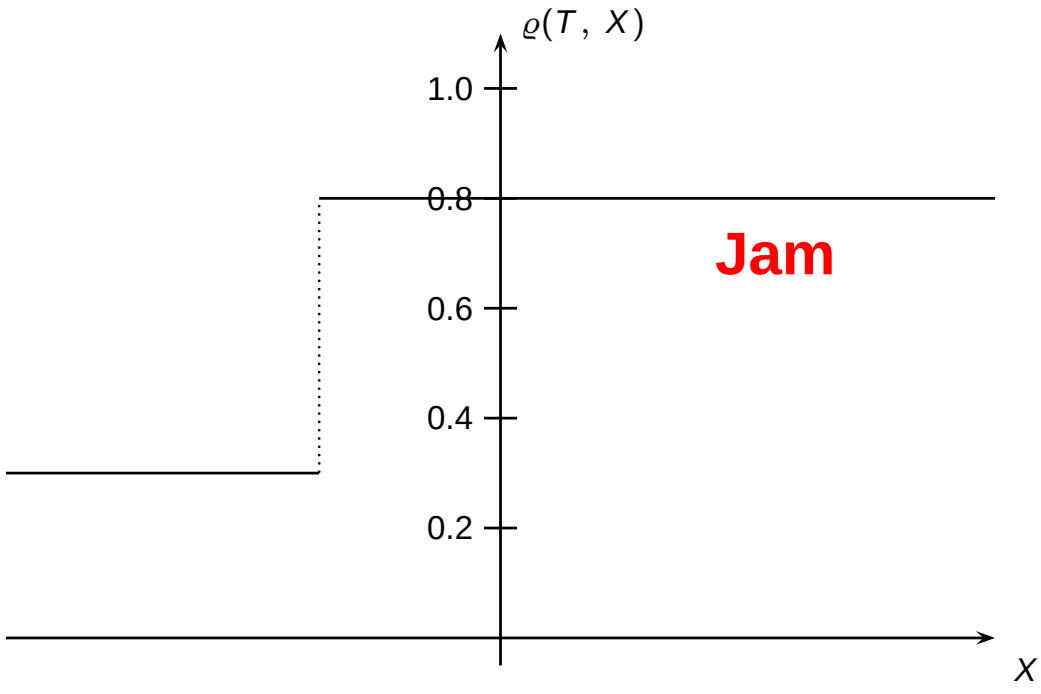
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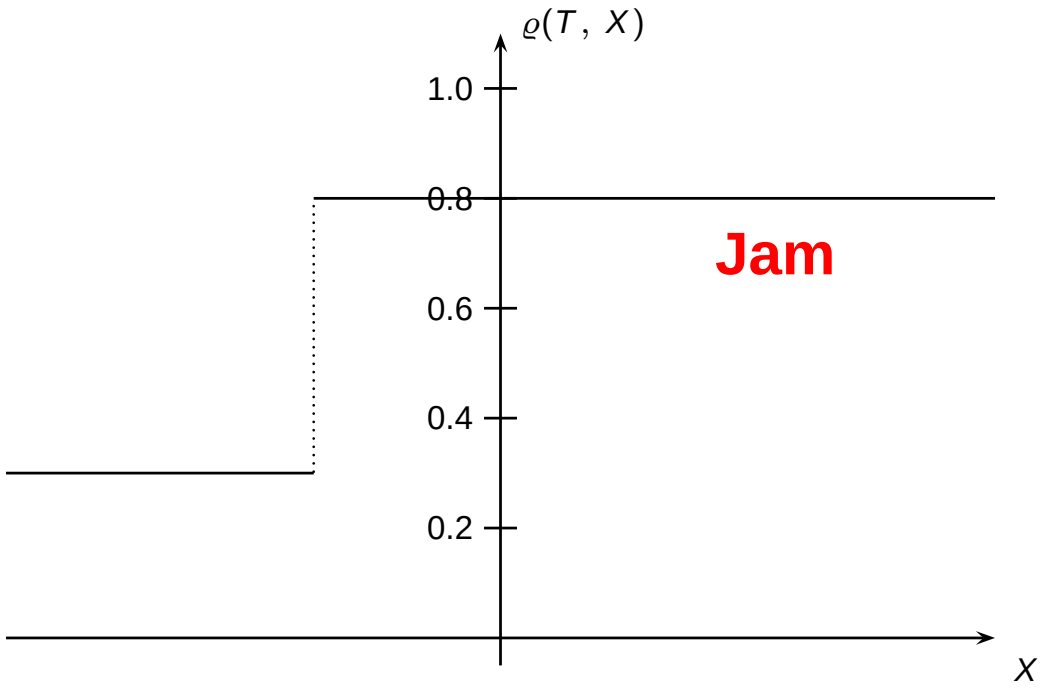
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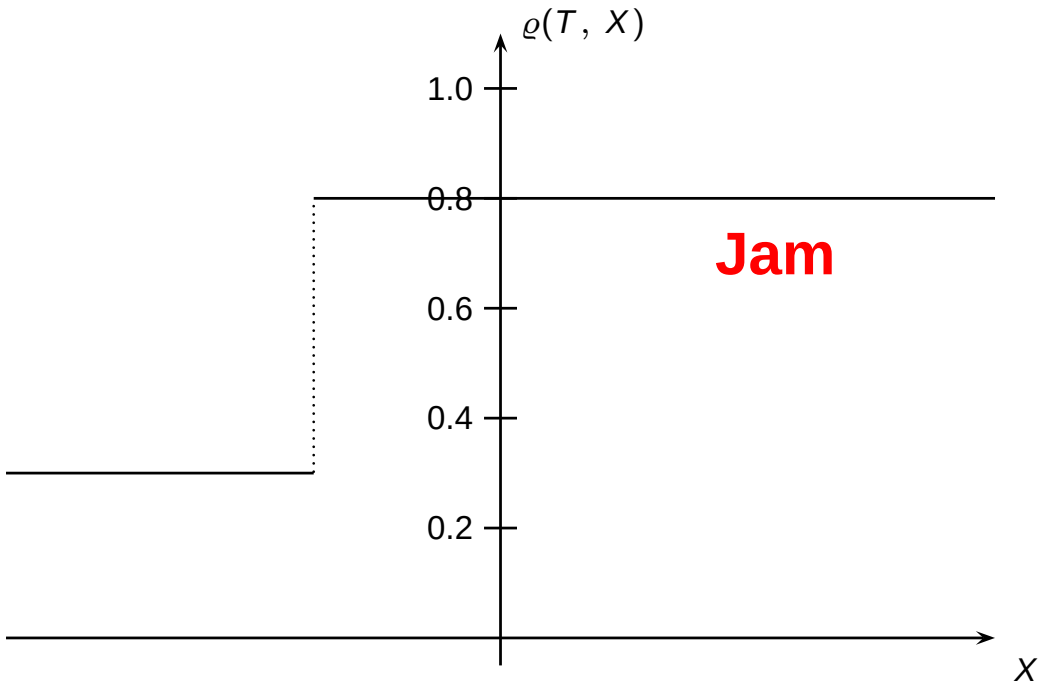
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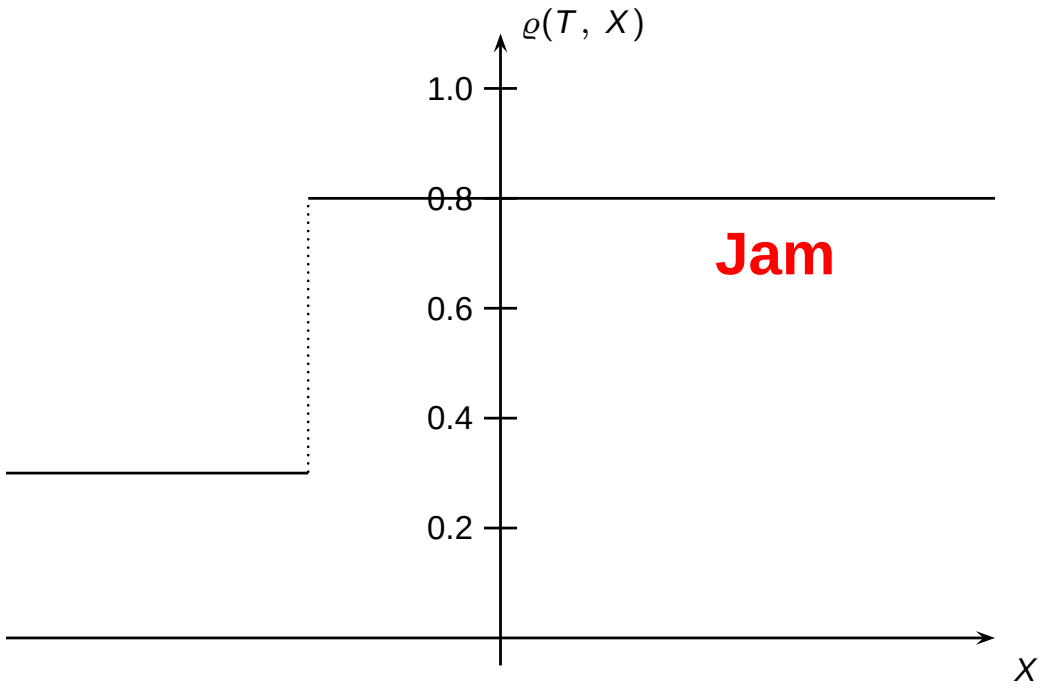
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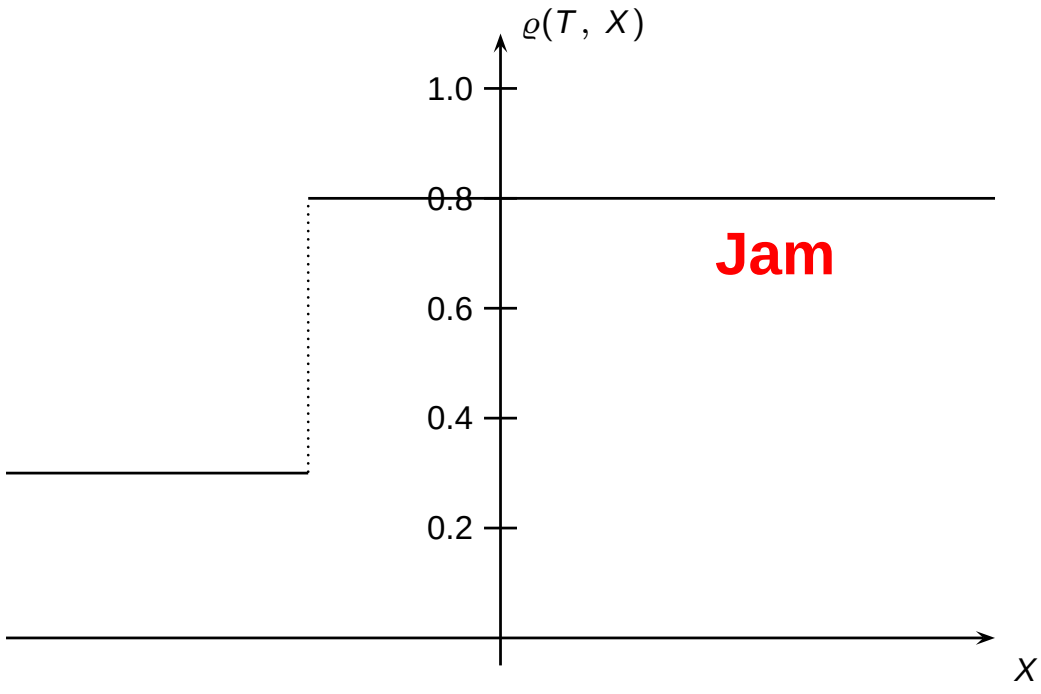
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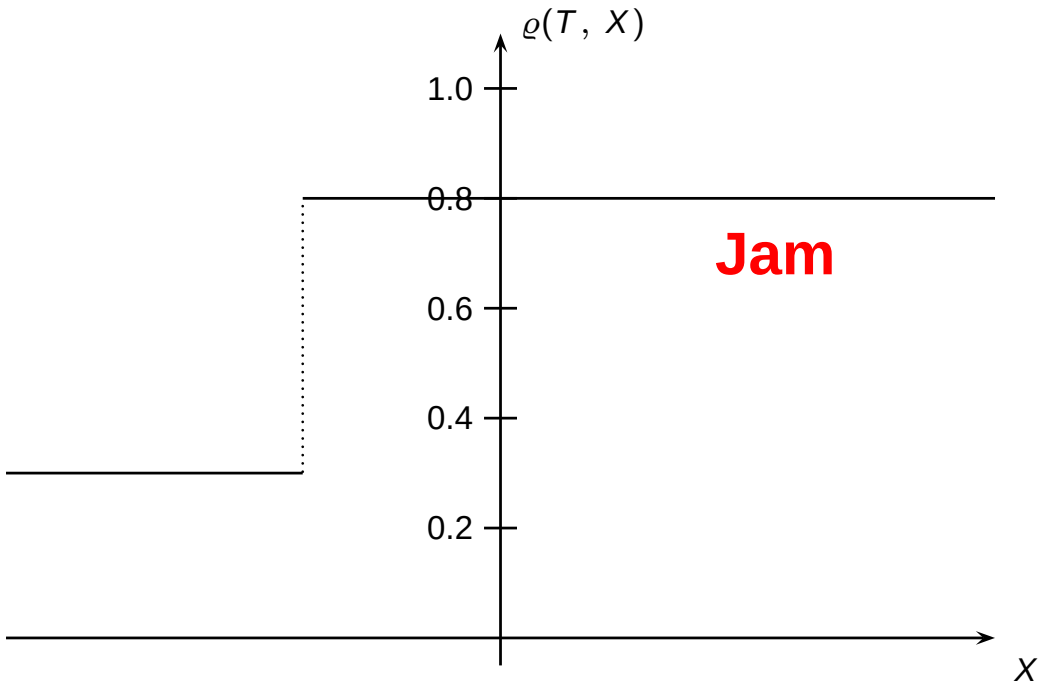
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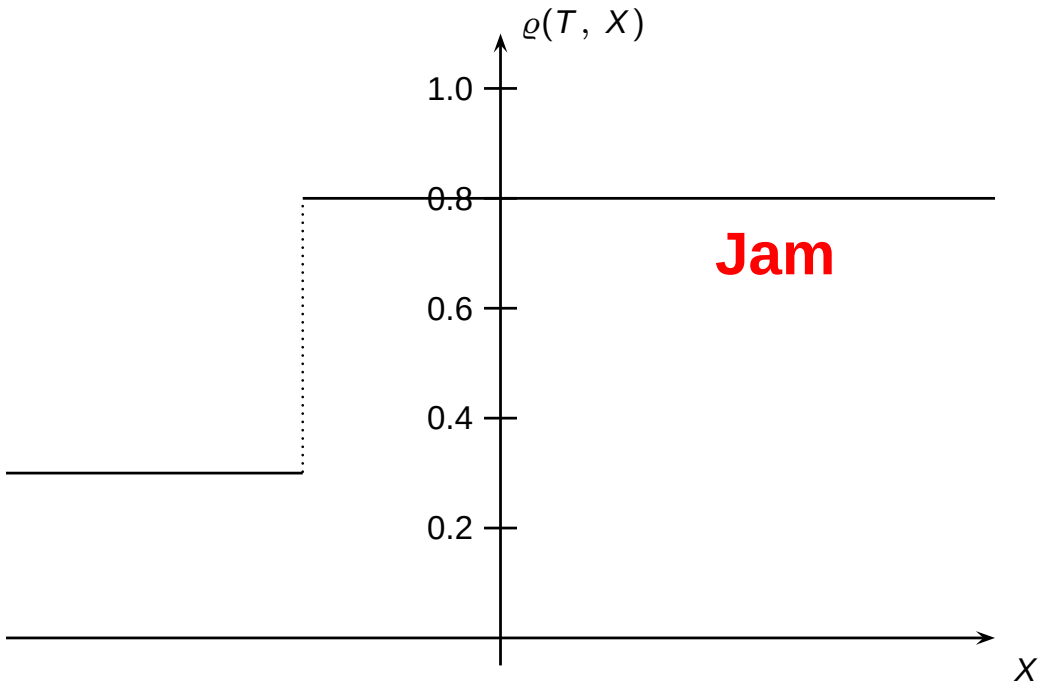
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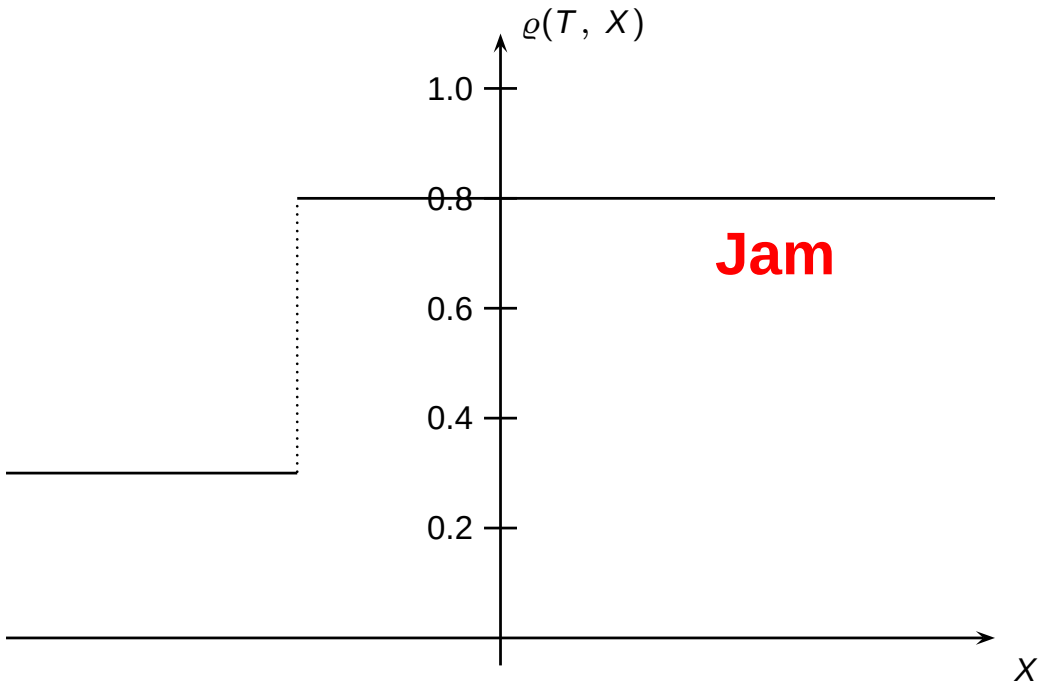
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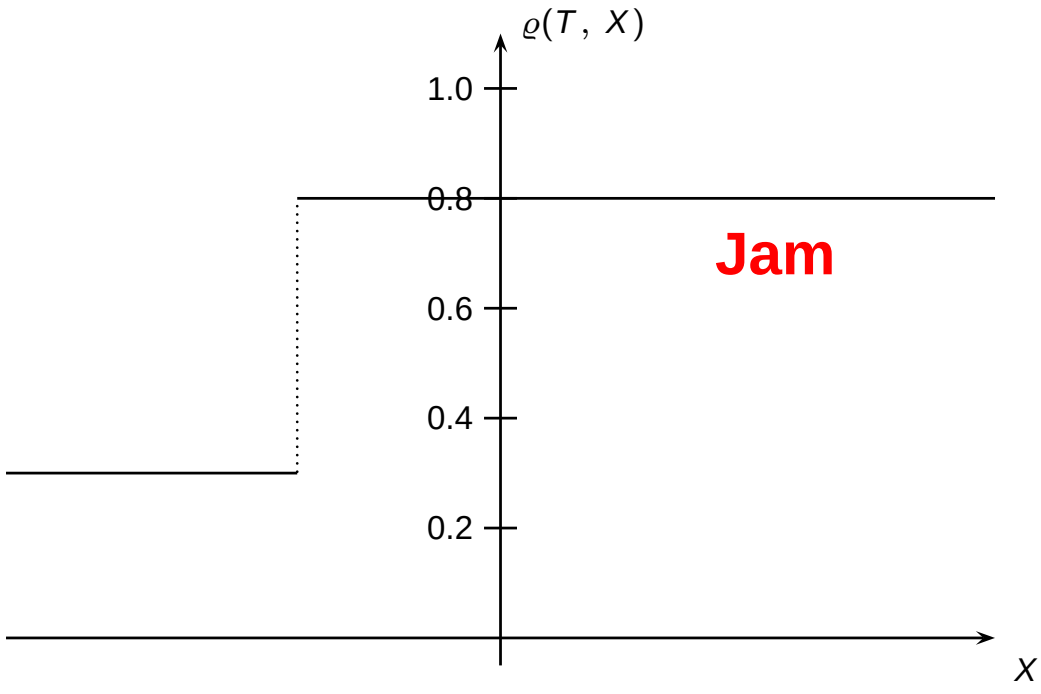
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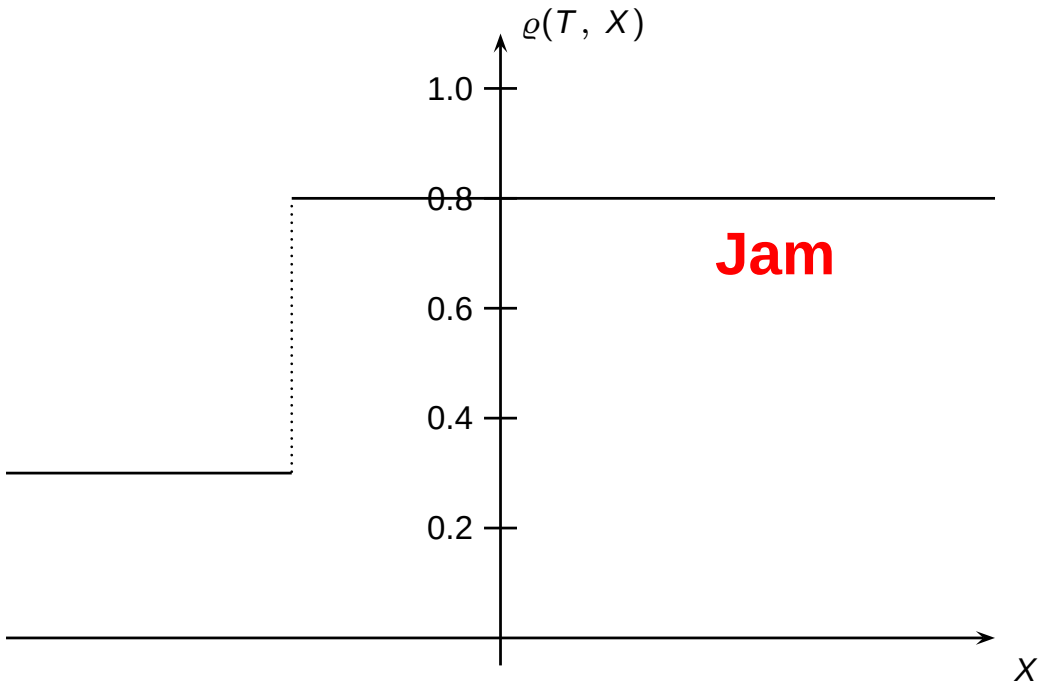
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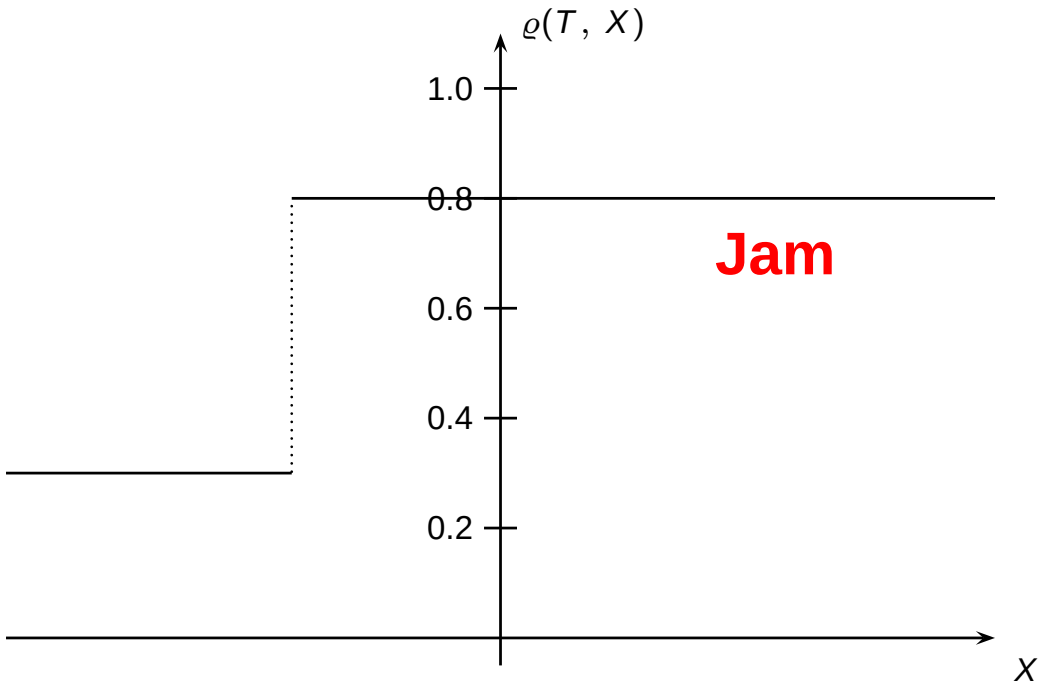
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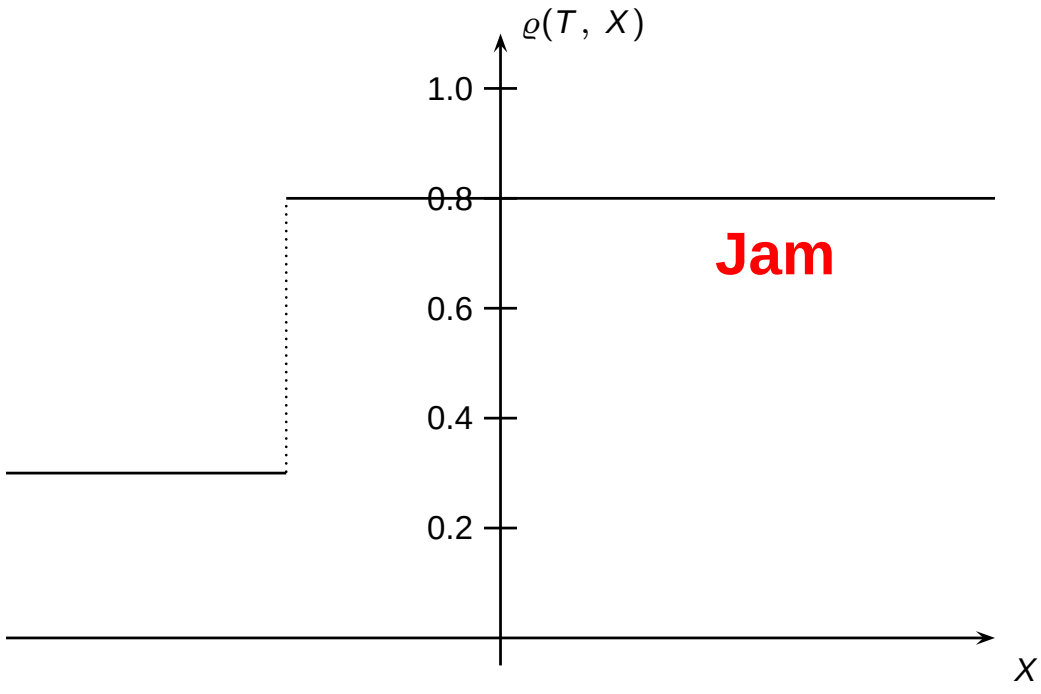
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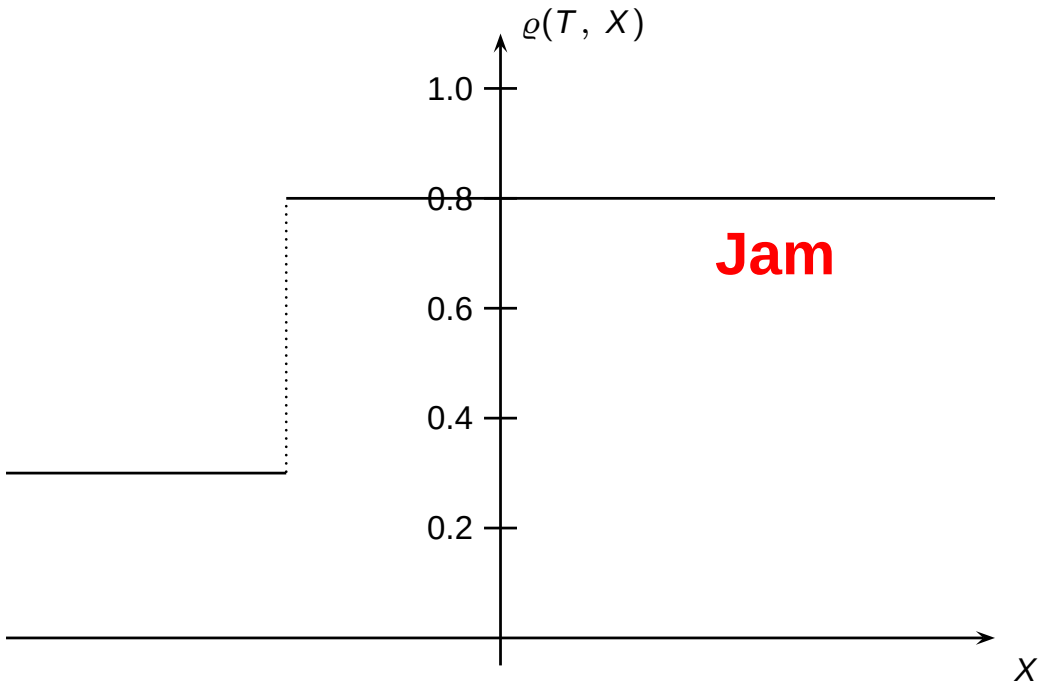
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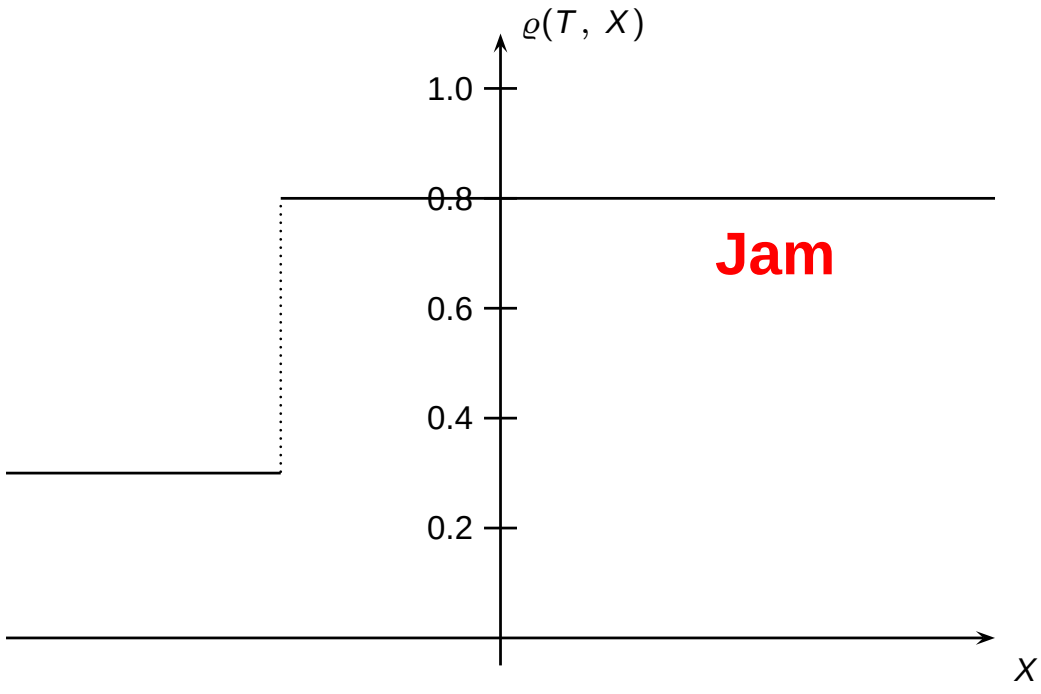
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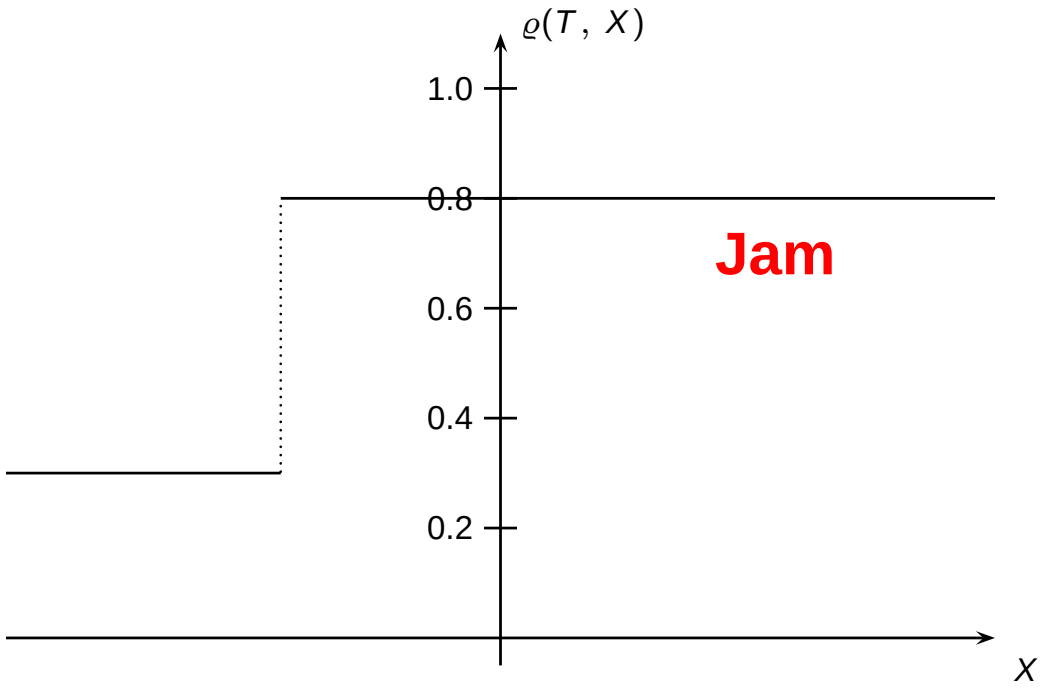
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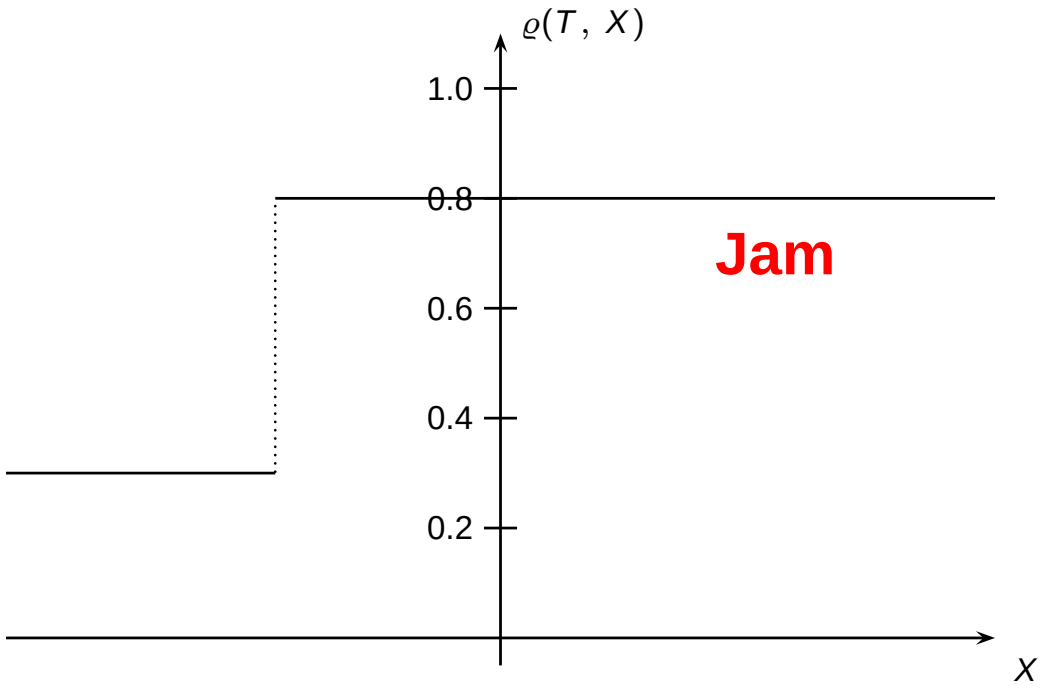
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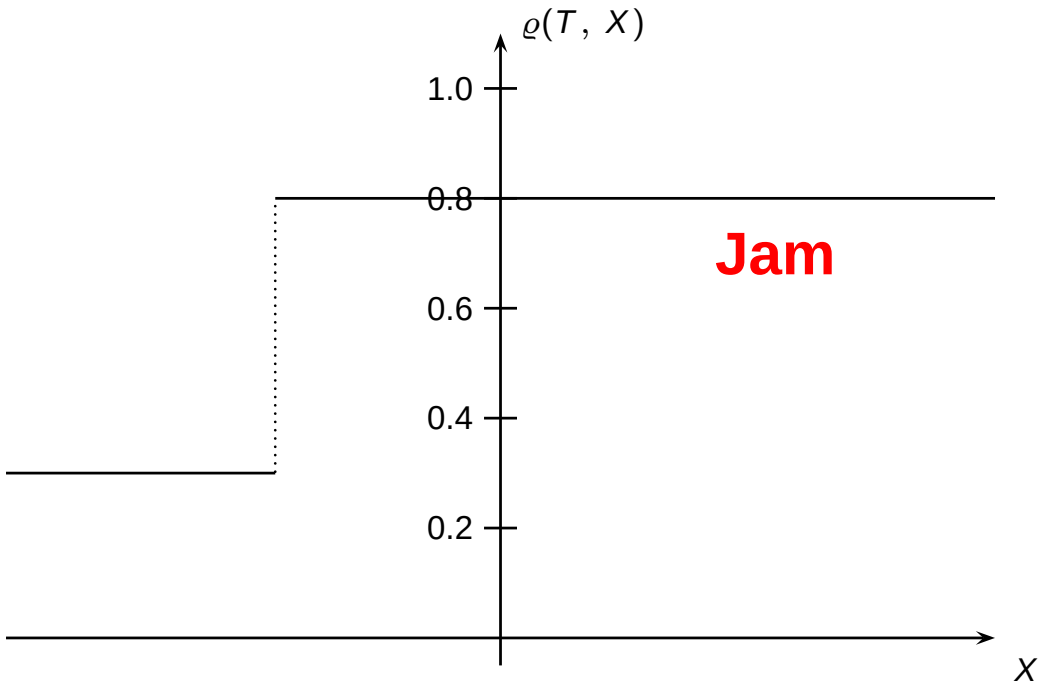
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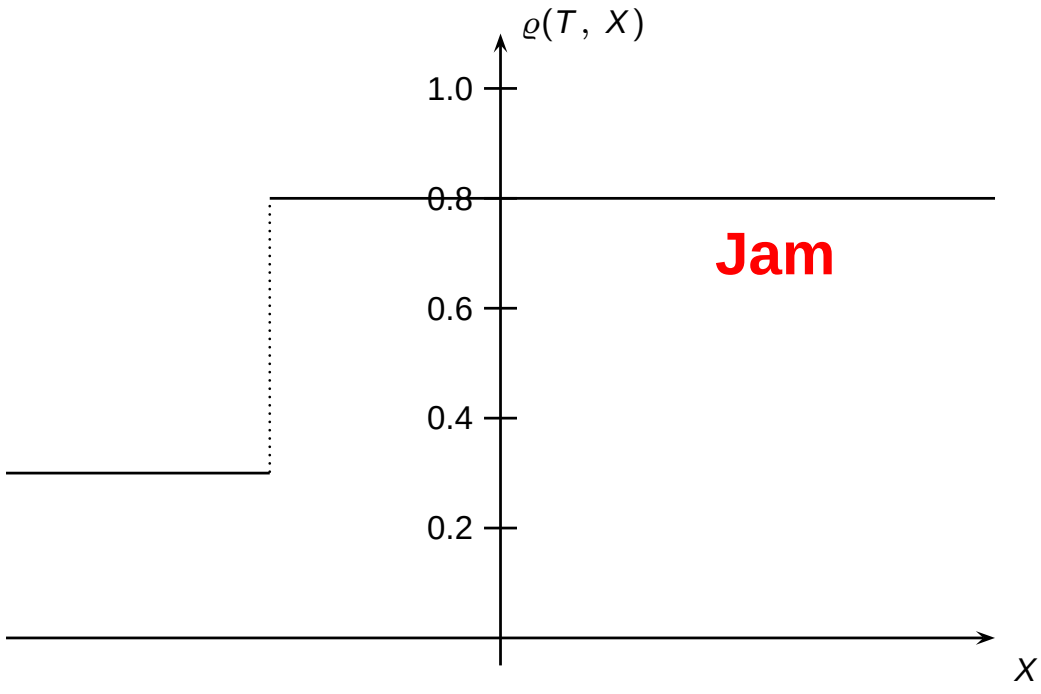
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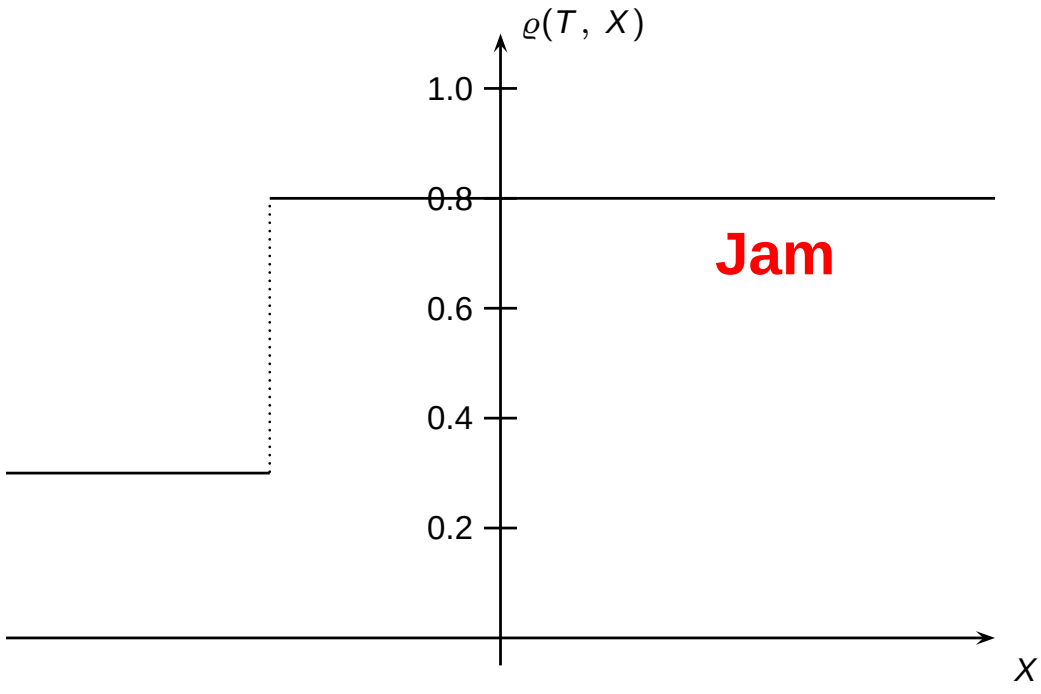
On large scales



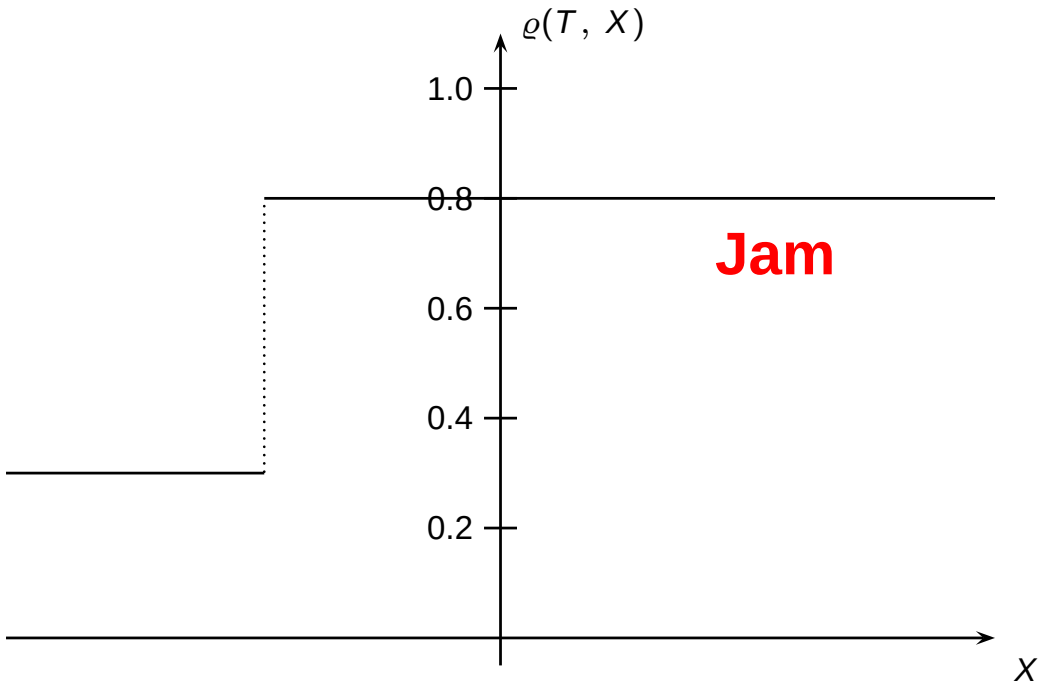
On large scales



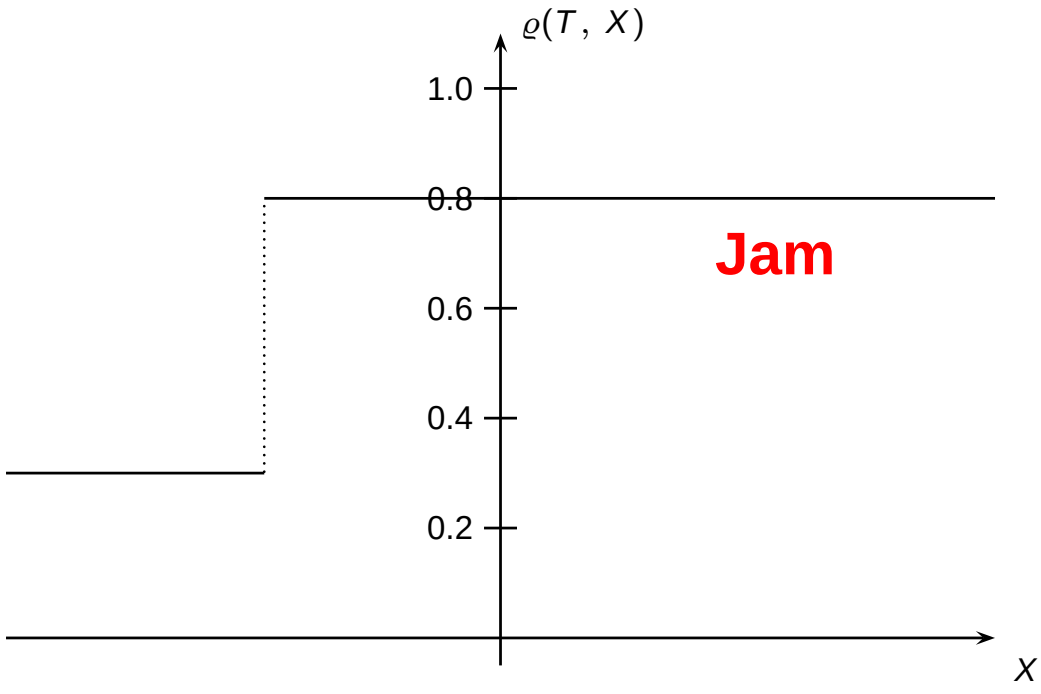
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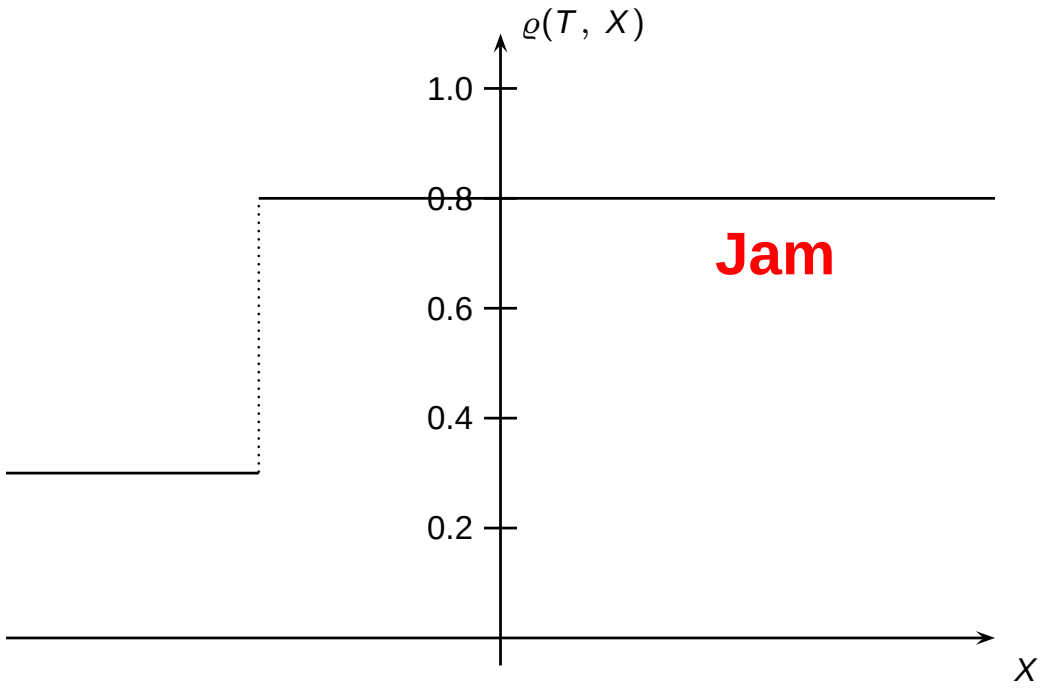
On large scales



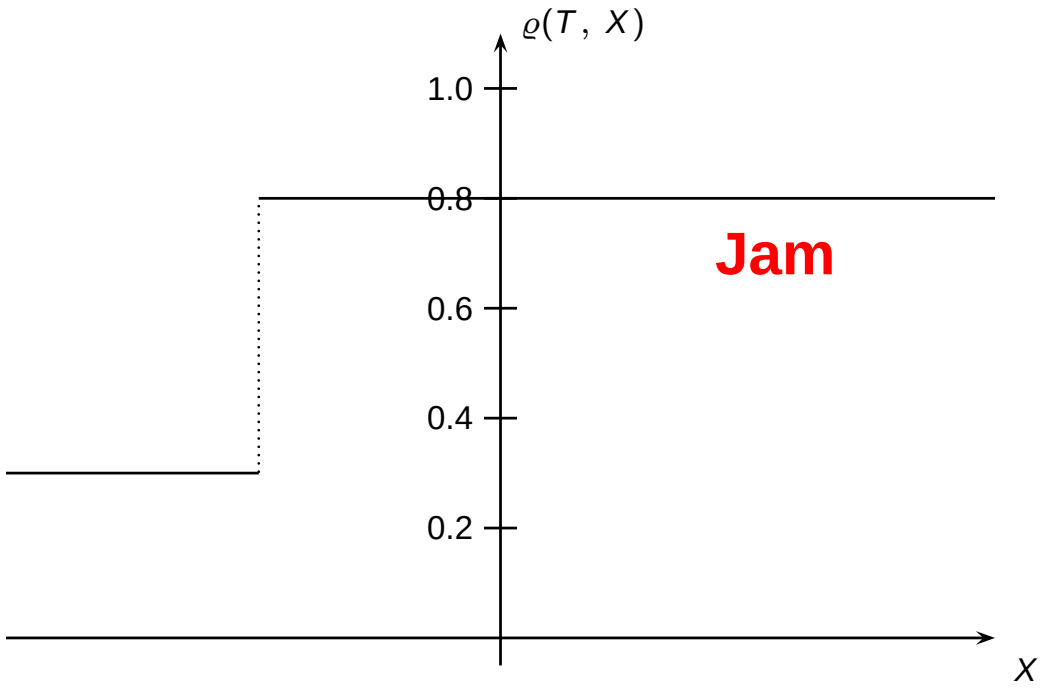
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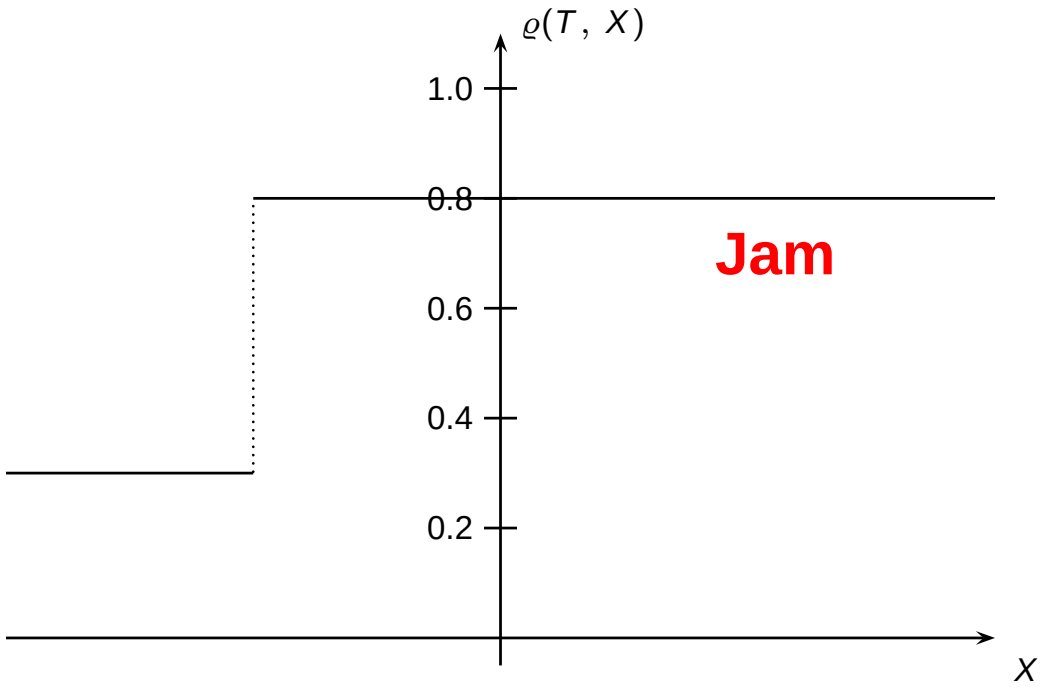
On large scales



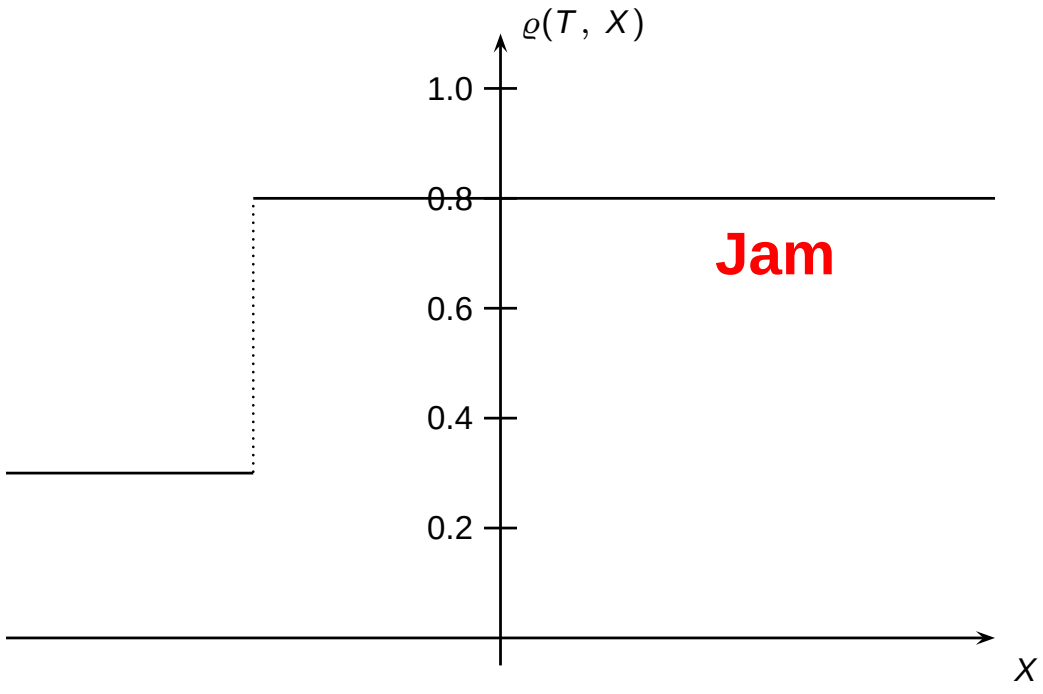
On large scales



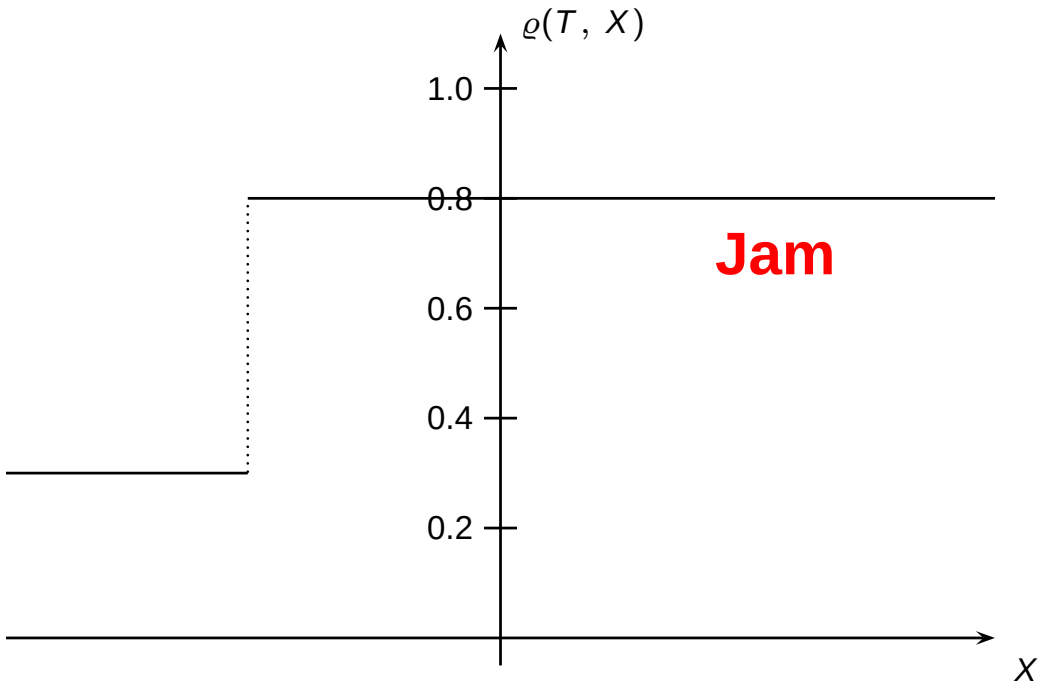
On large scales



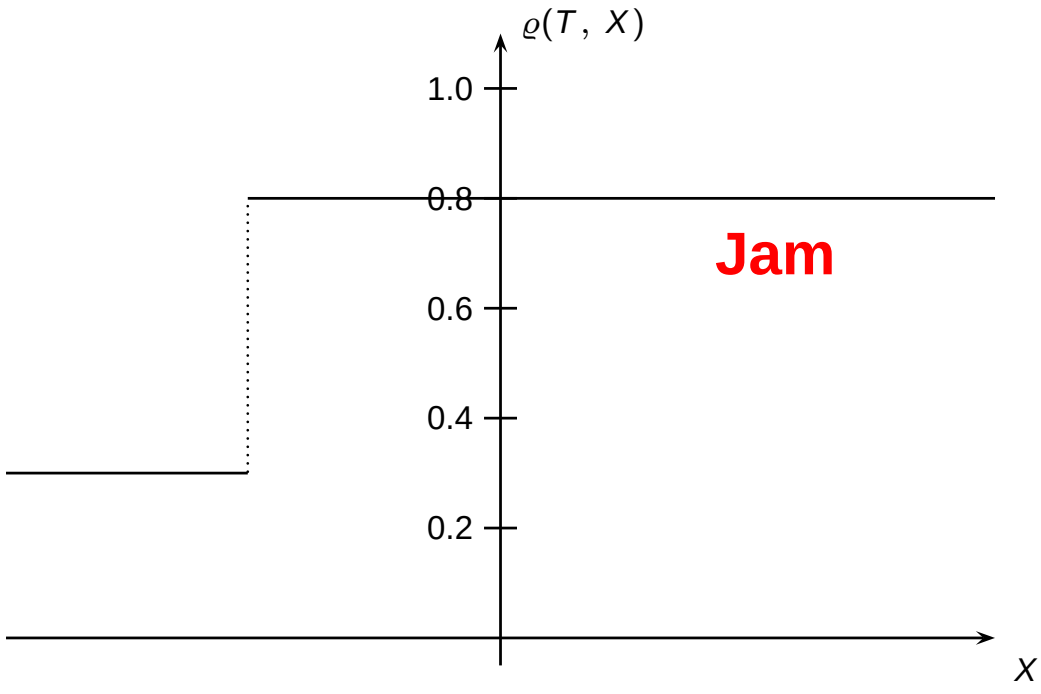
On large scales



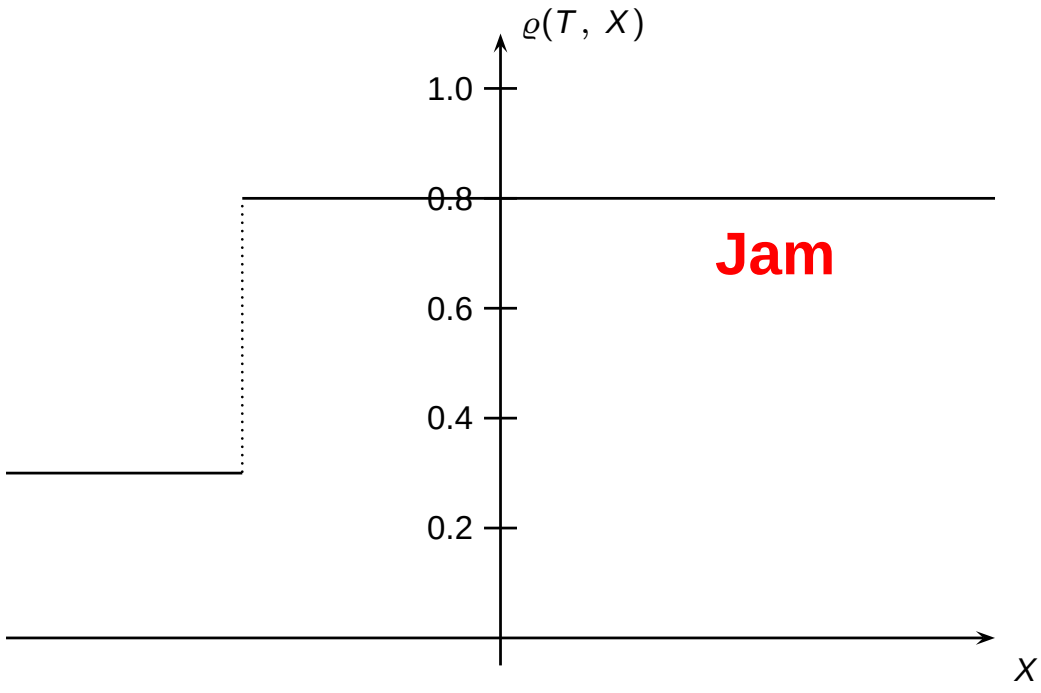
On large scales



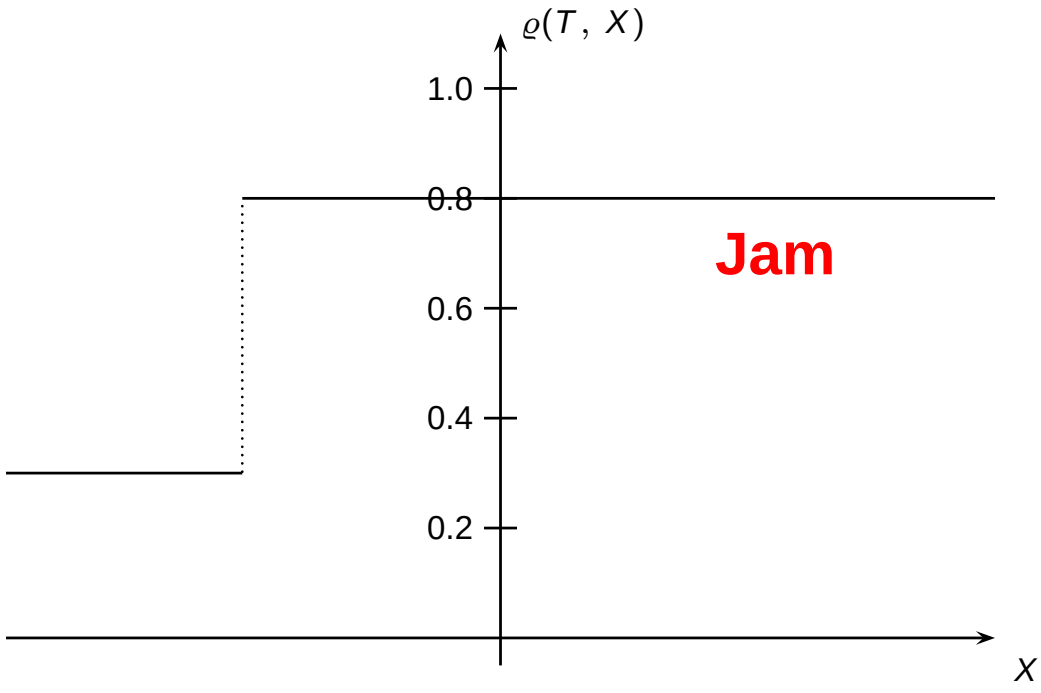
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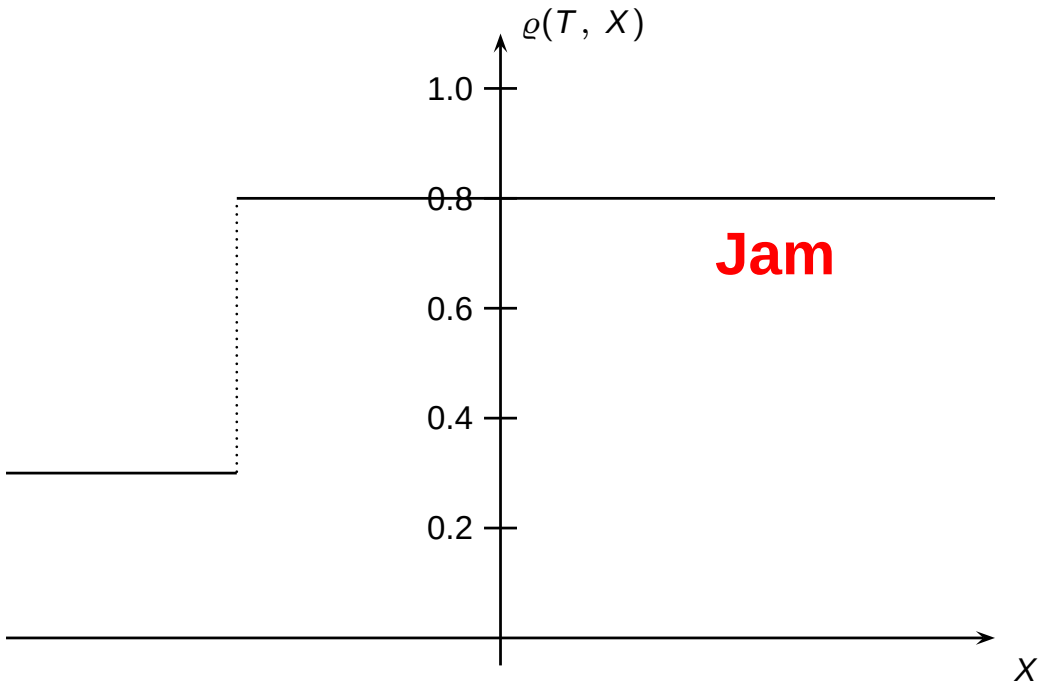
On large scales



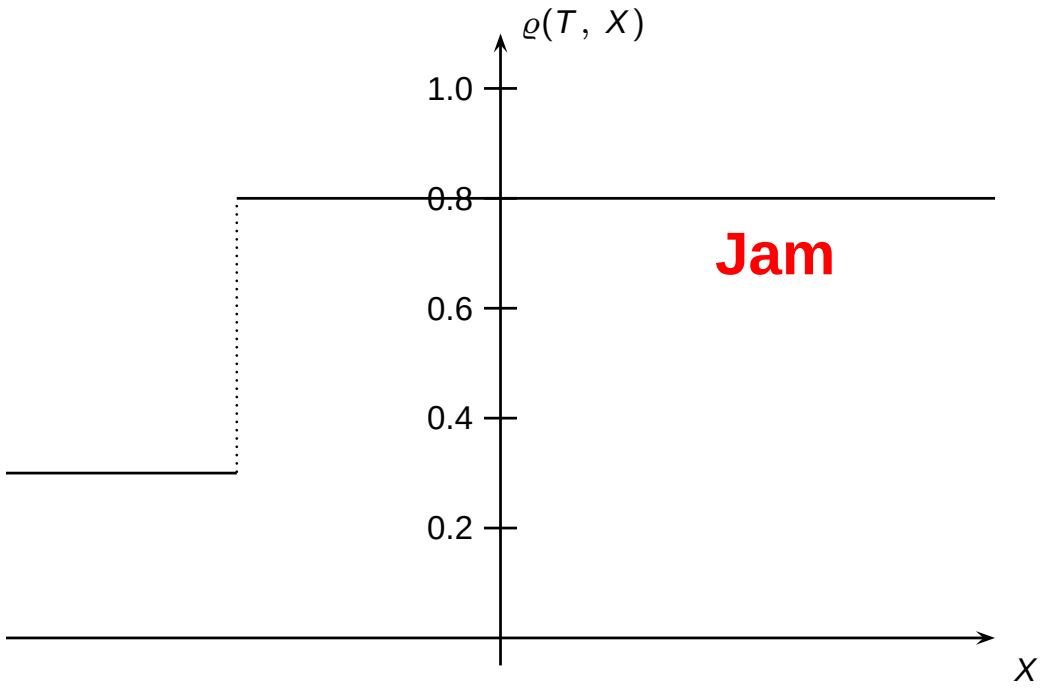
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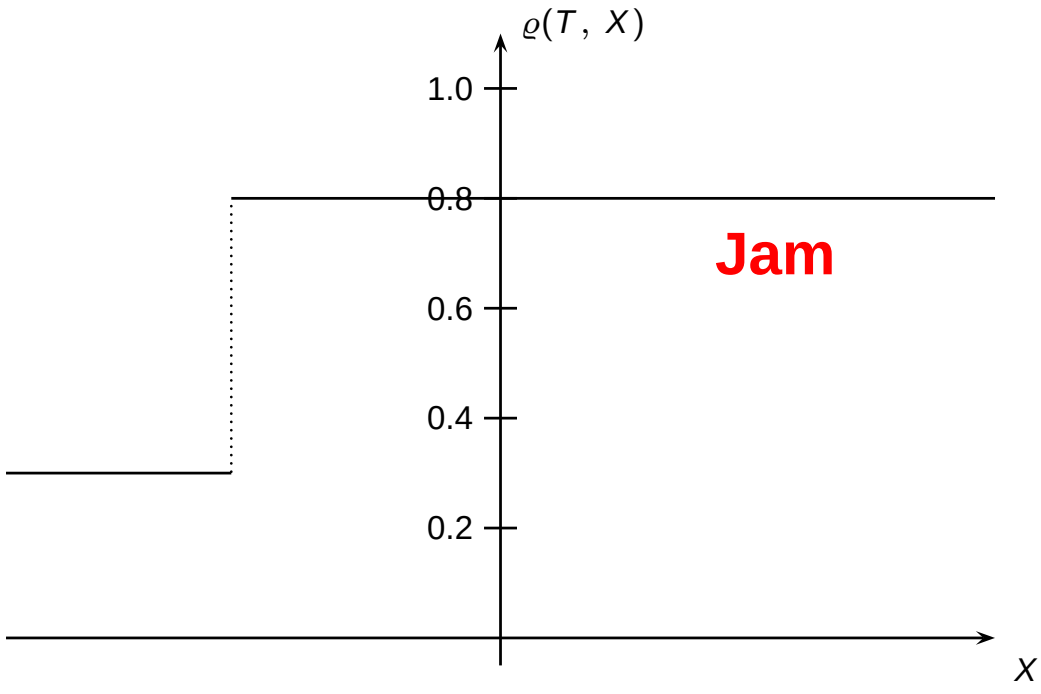
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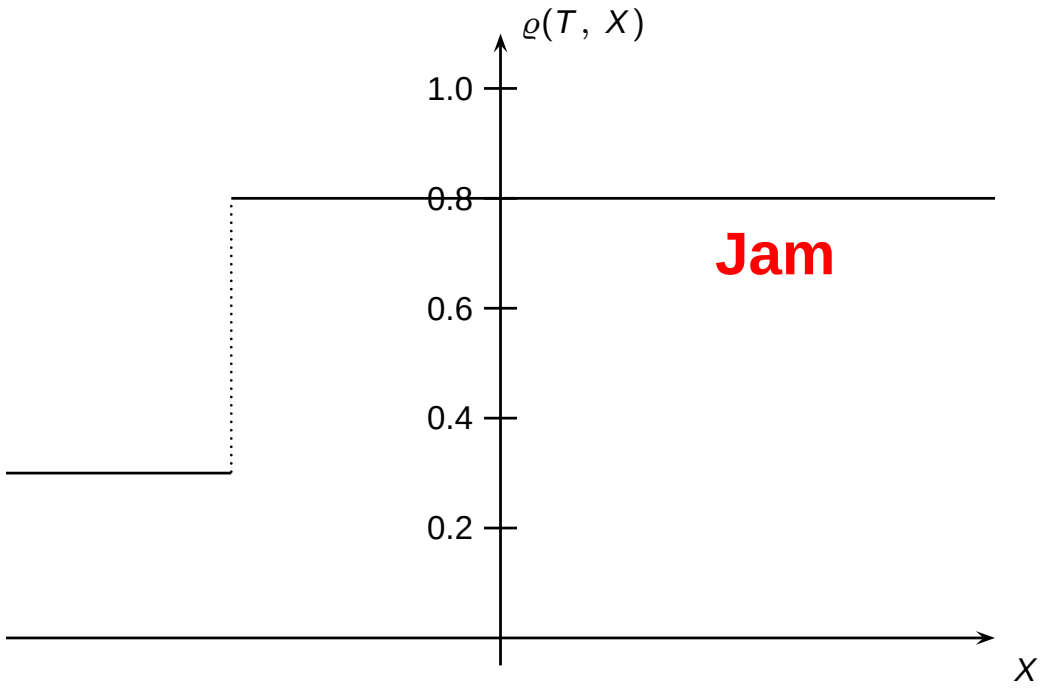
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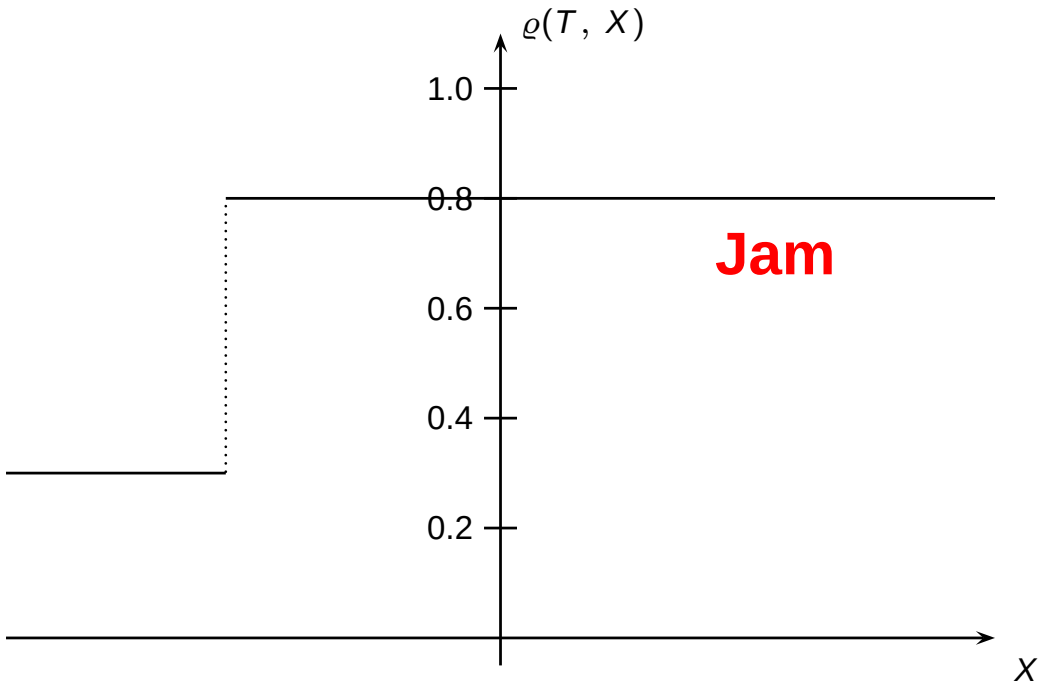
On large scales



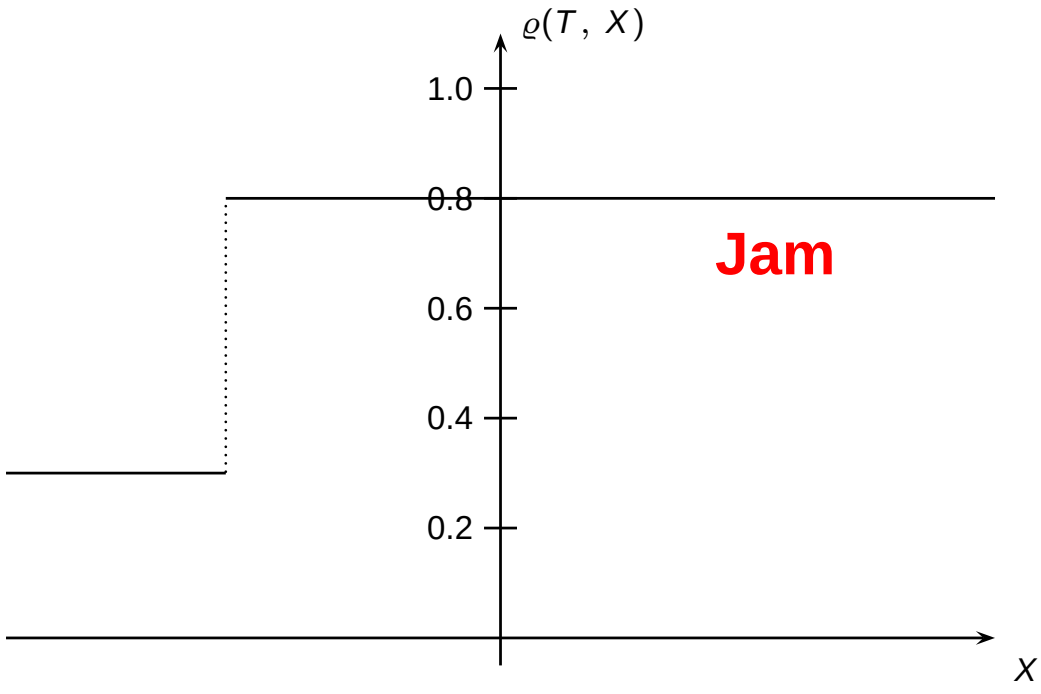
On large scales



On large scales

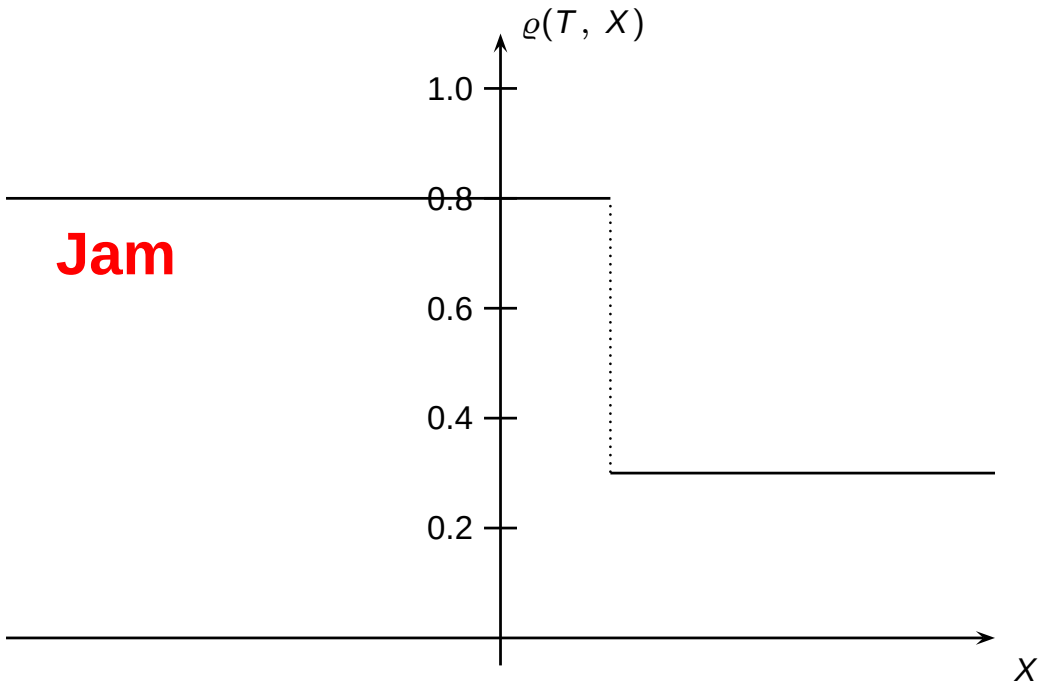


On large scales

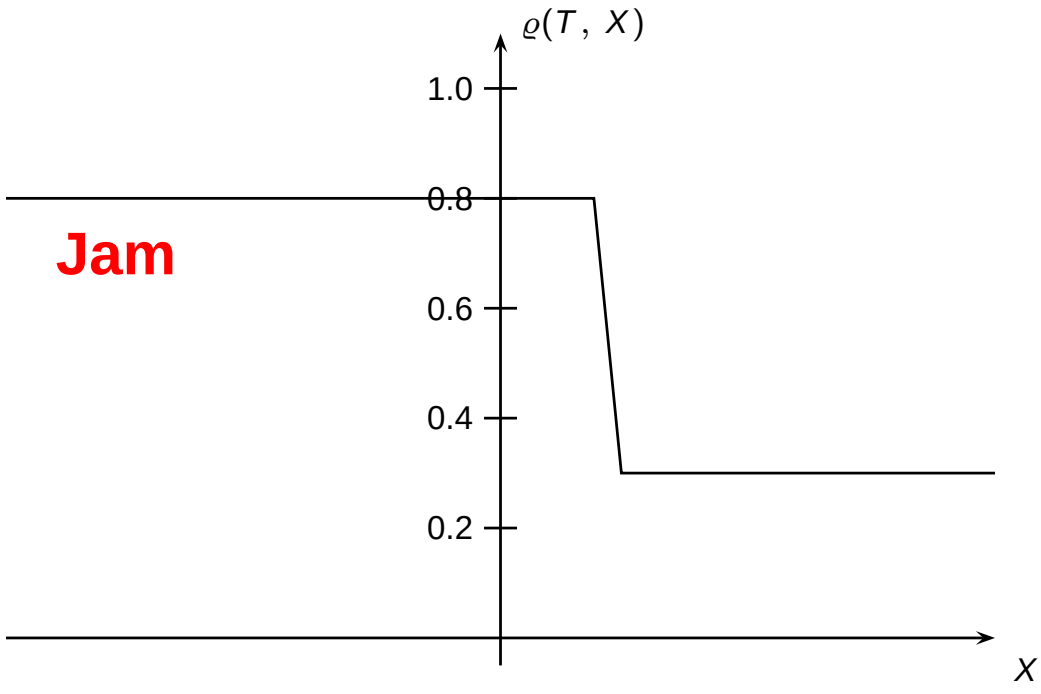


The start of the jam: sharpens.

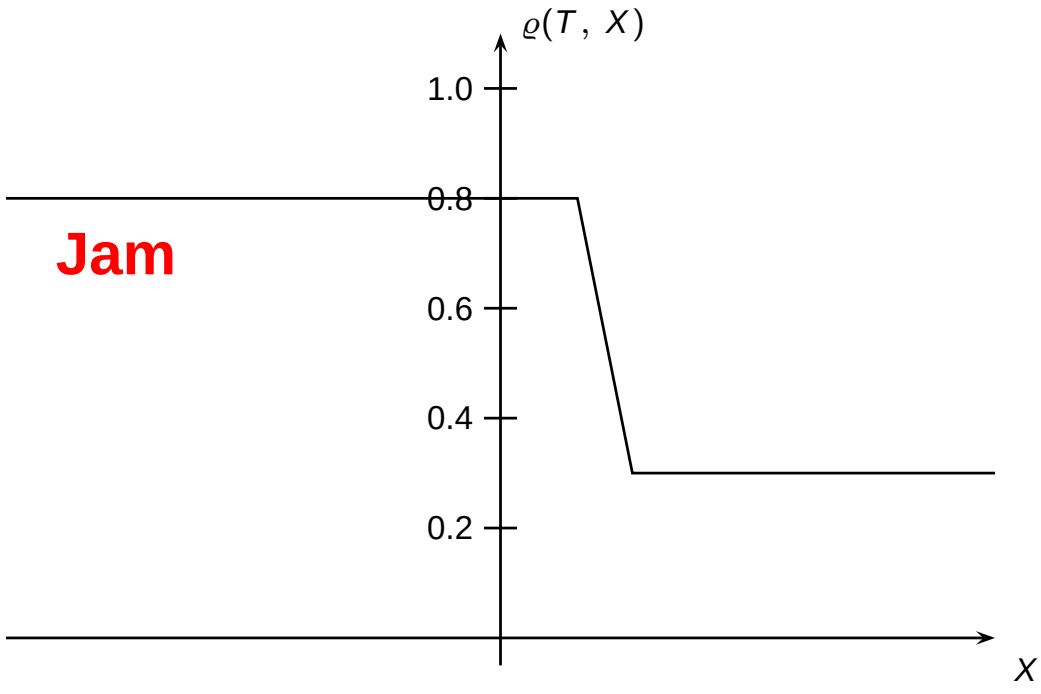
On large scales



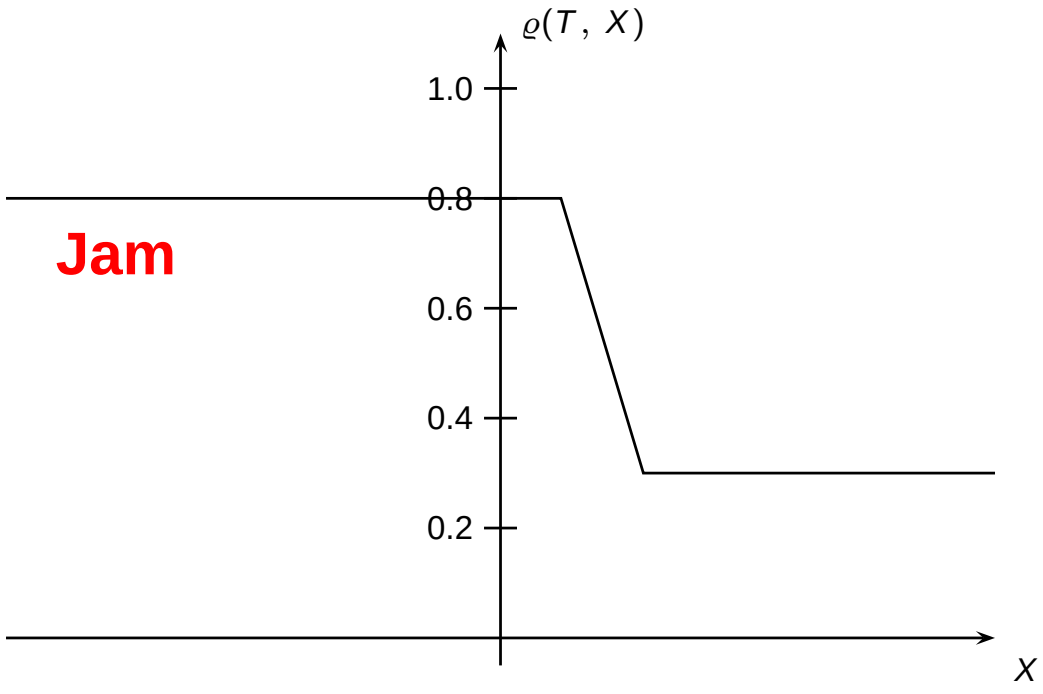
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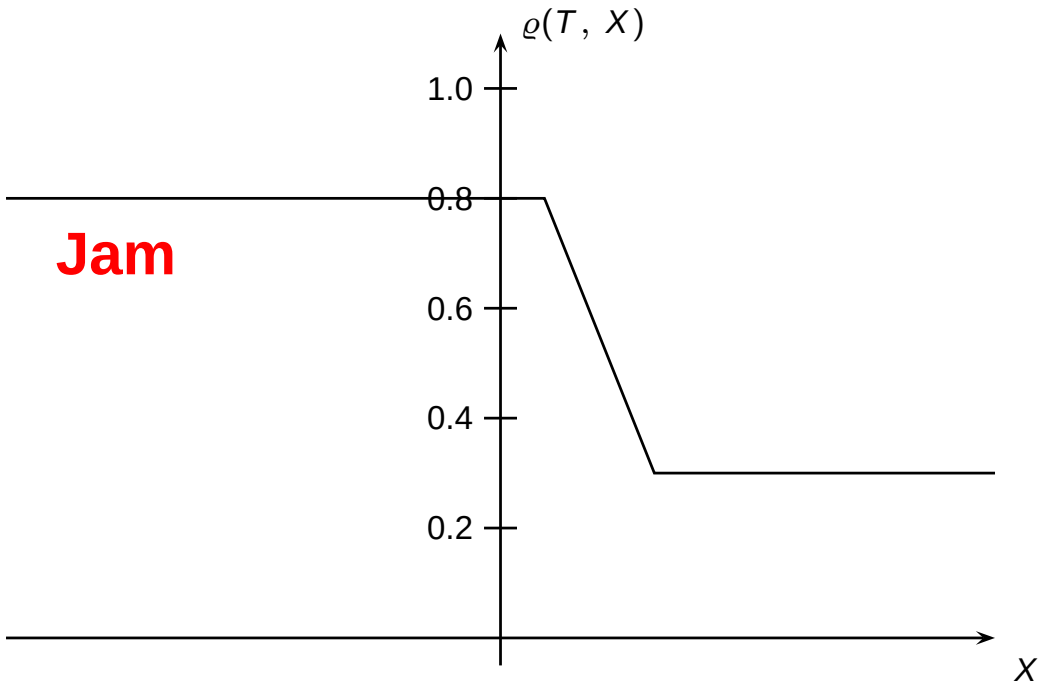
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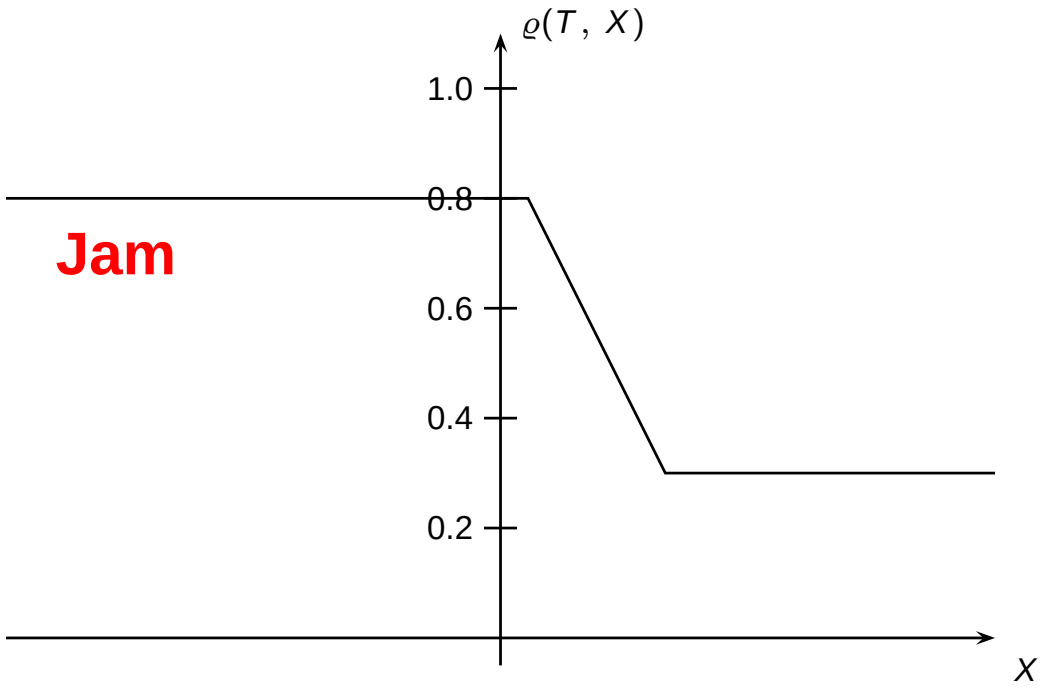
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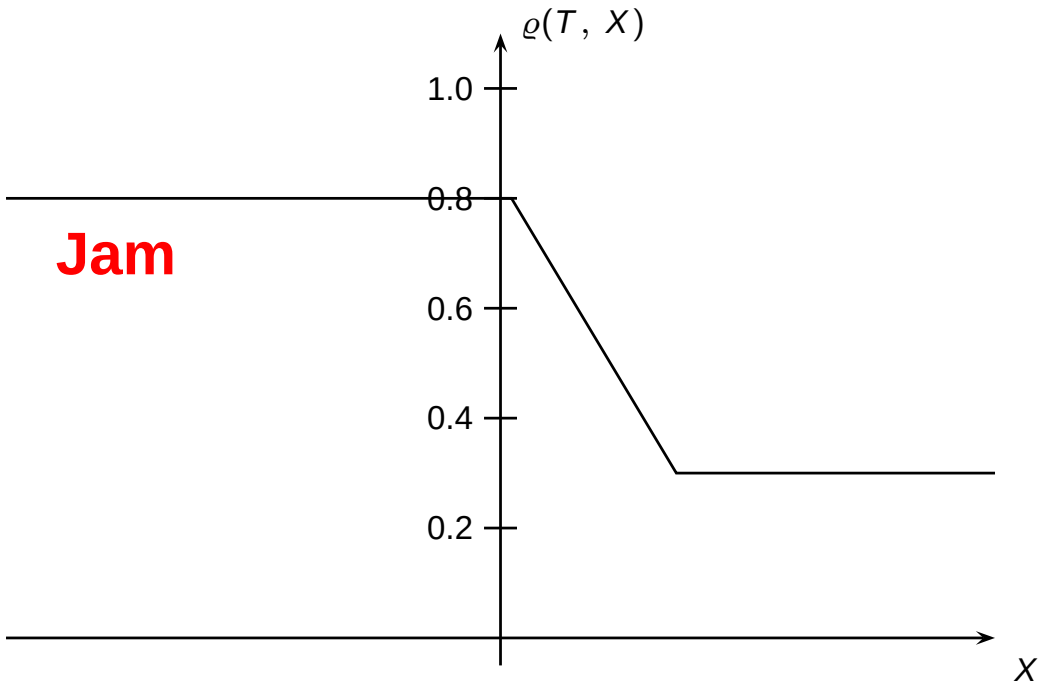
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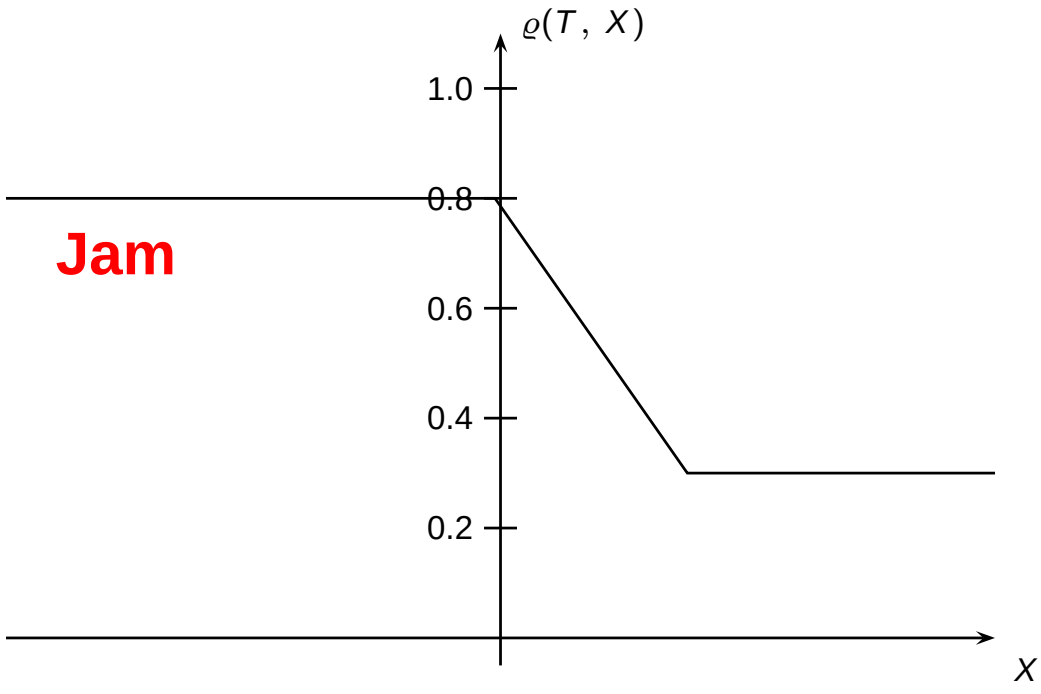
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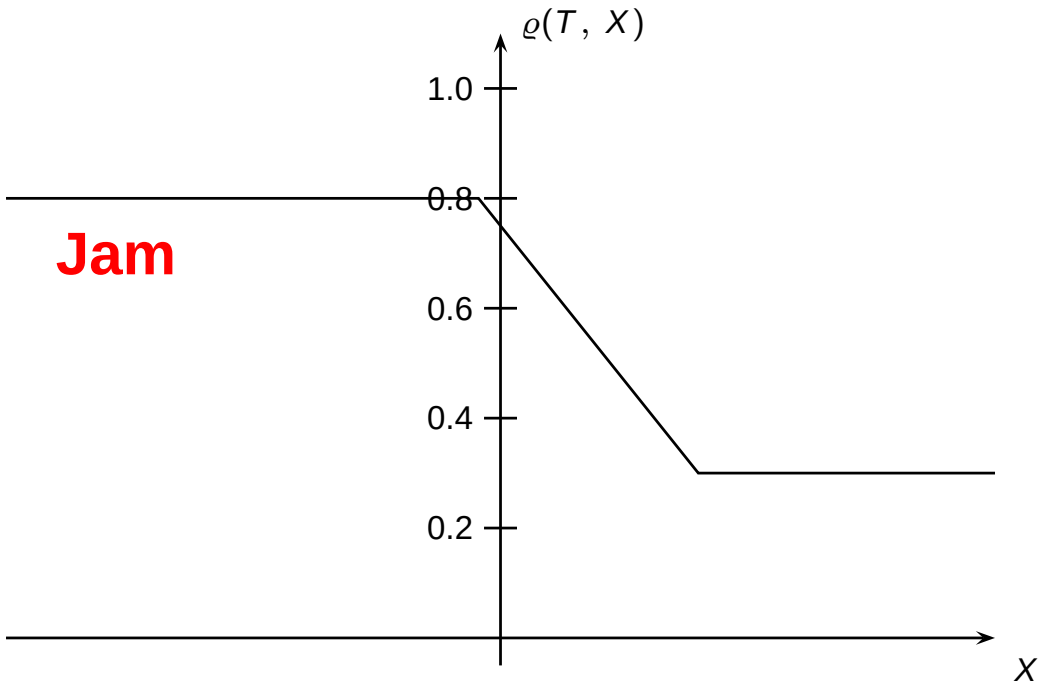
On large scales



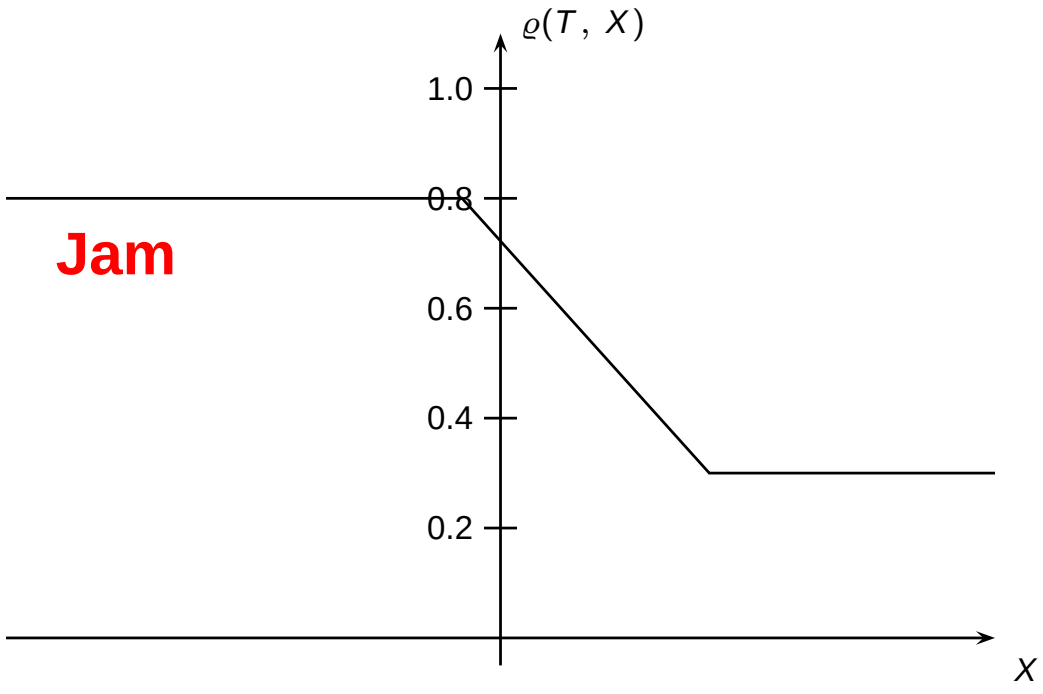
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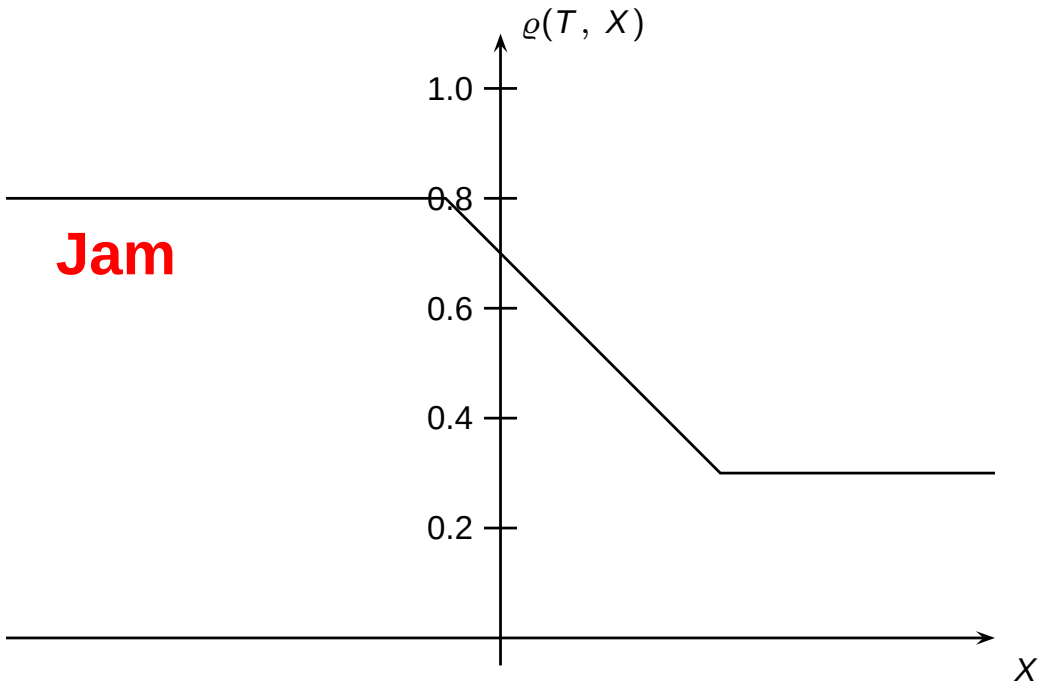
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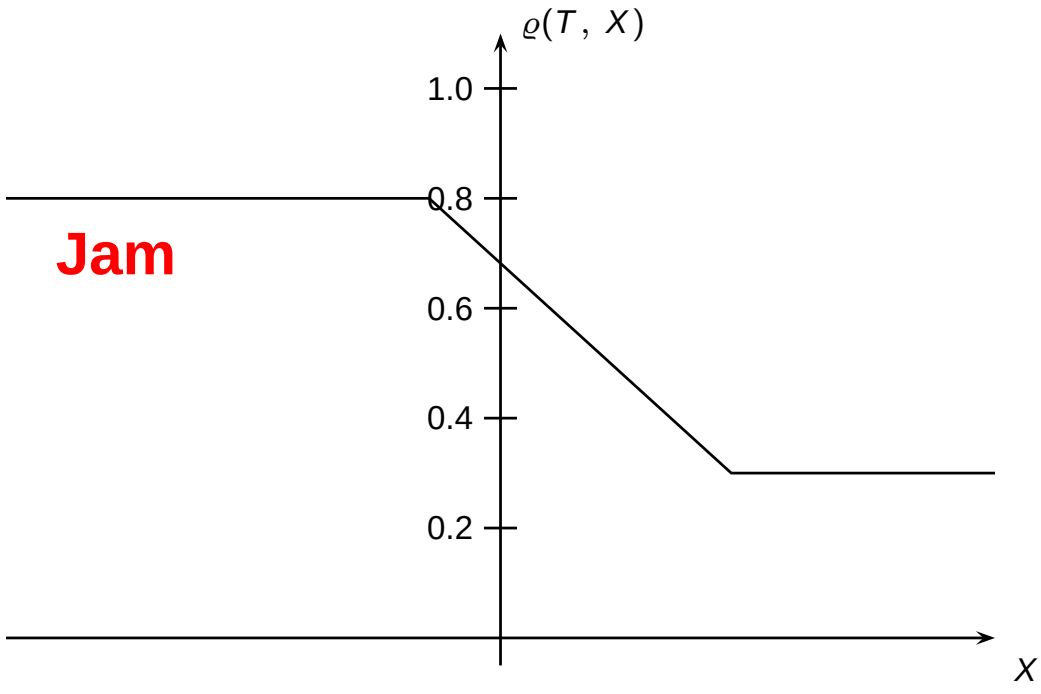
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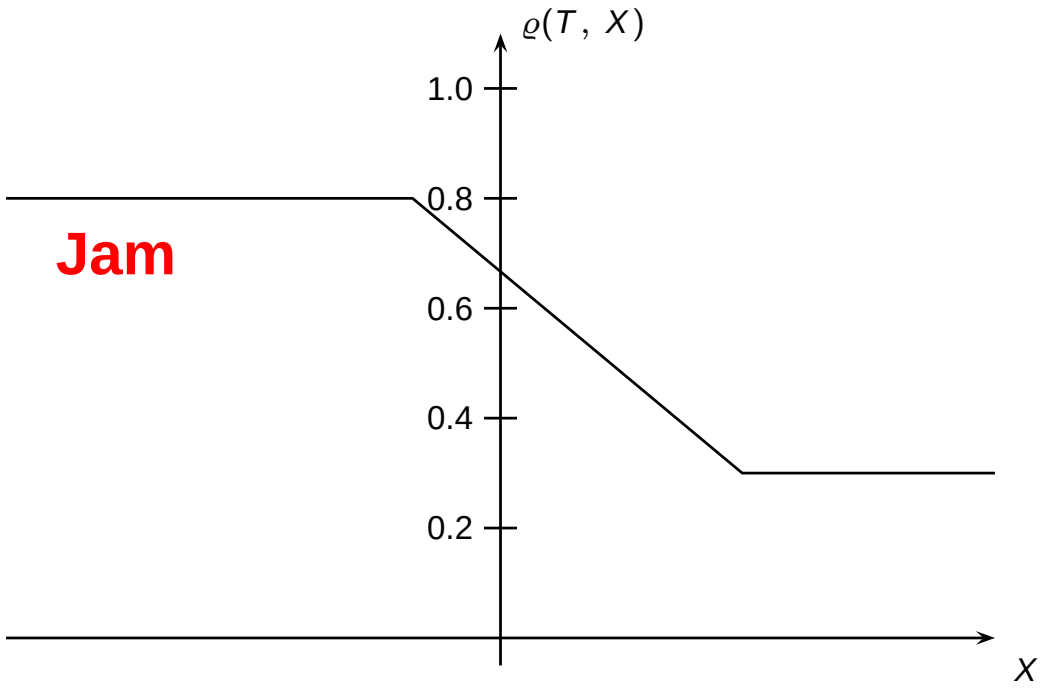
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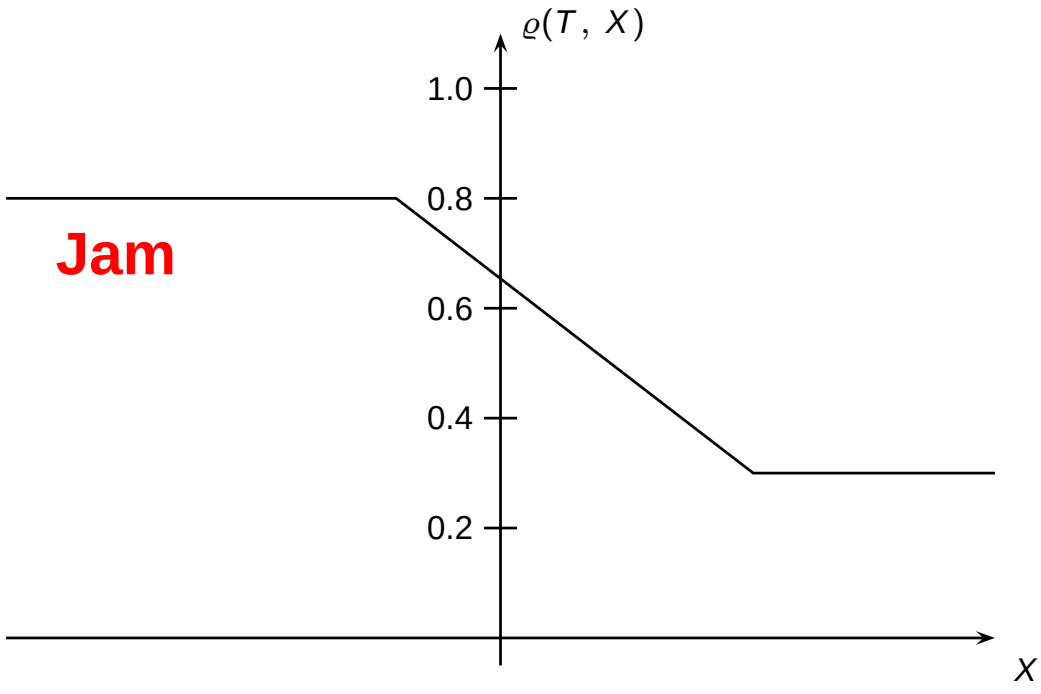
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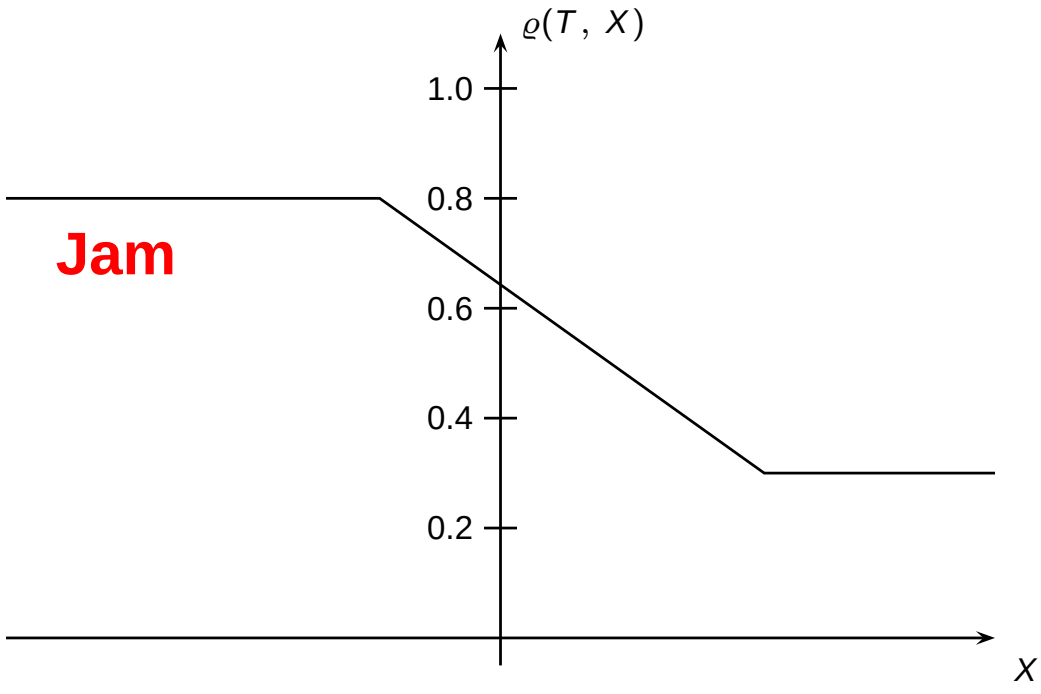
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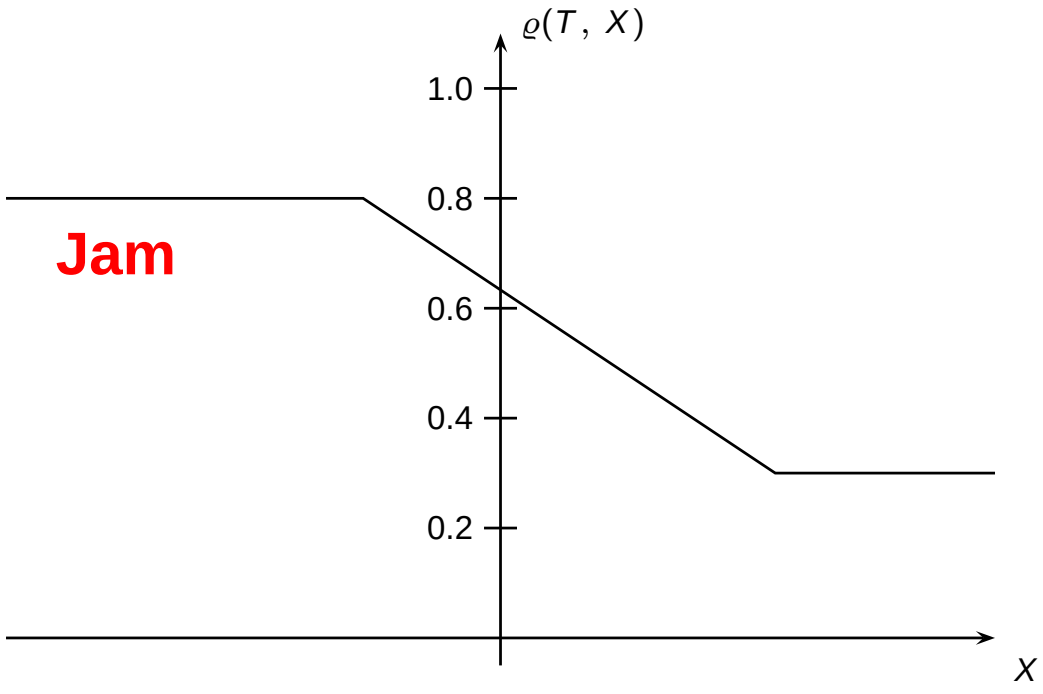
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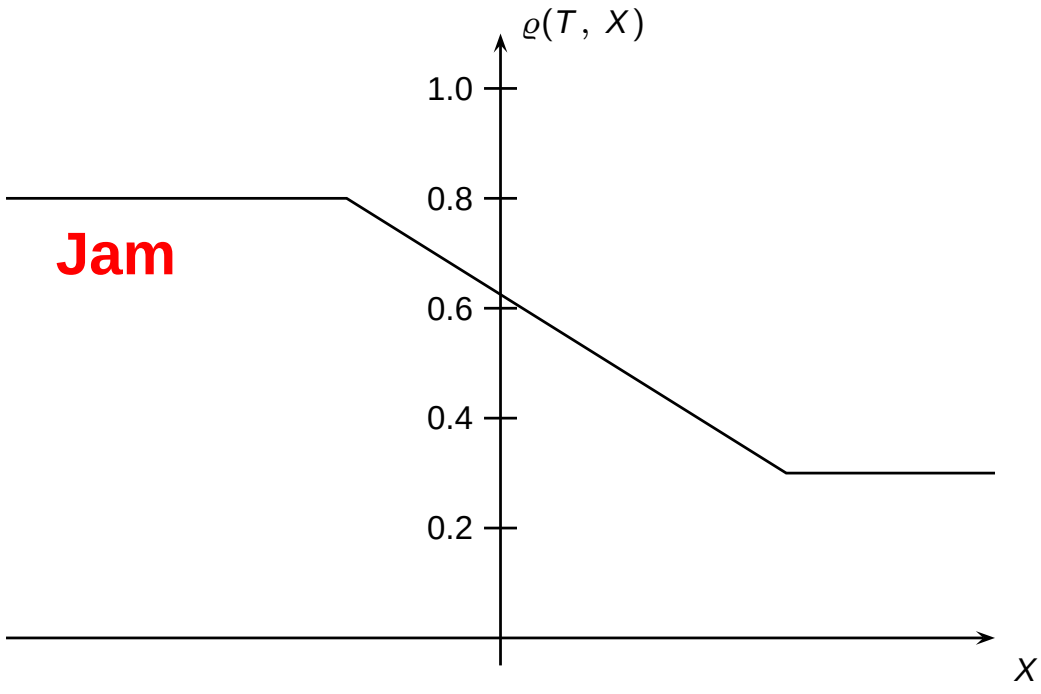
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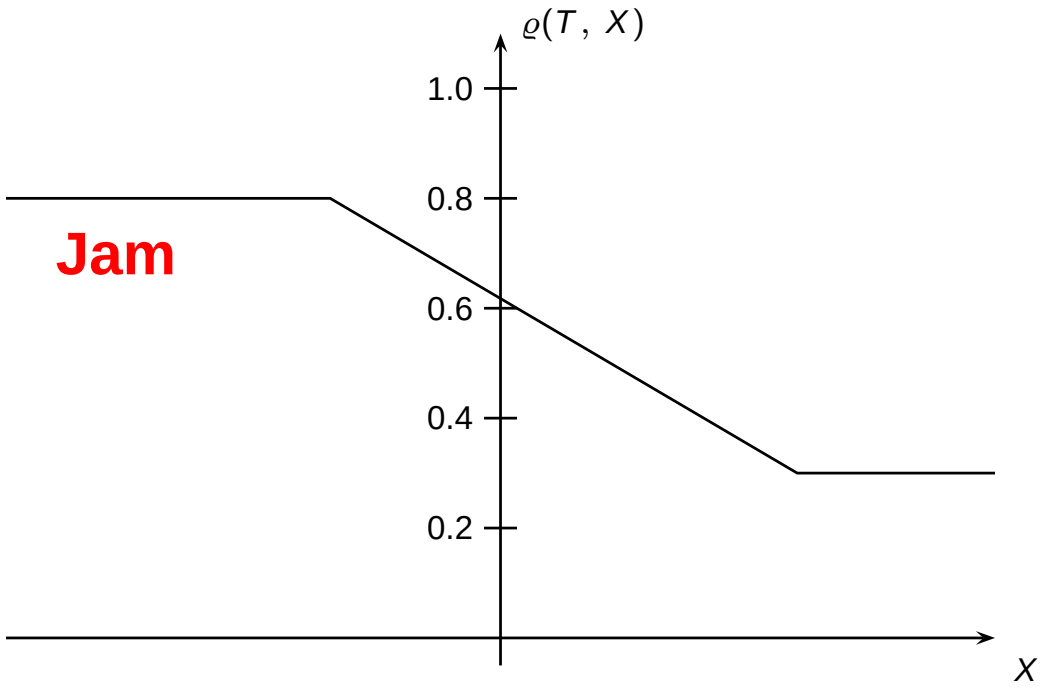
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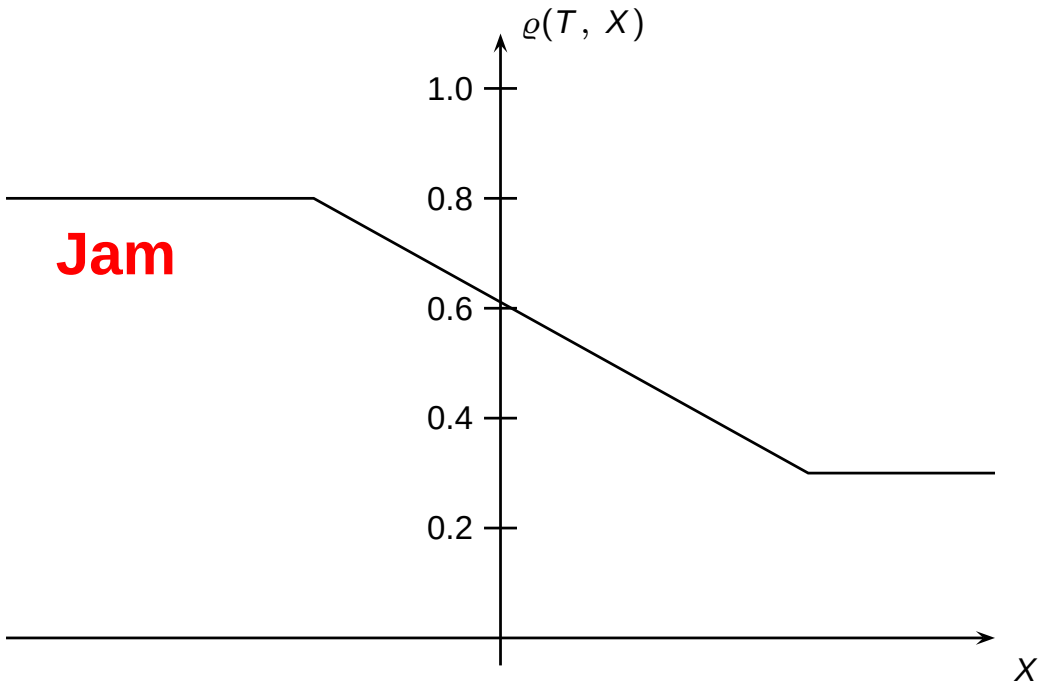
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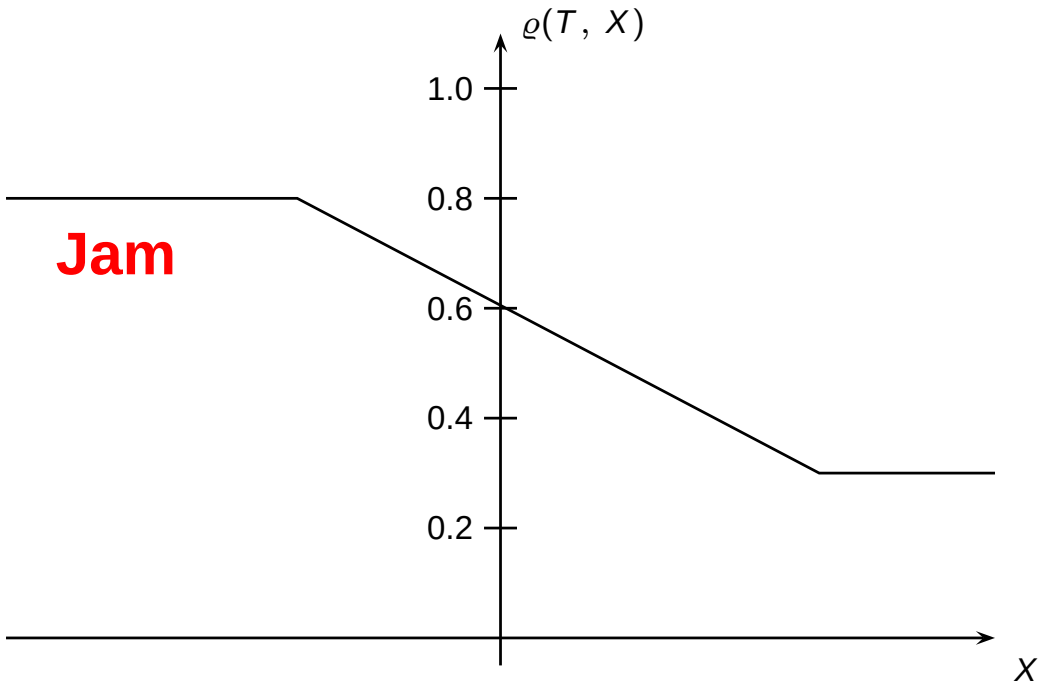
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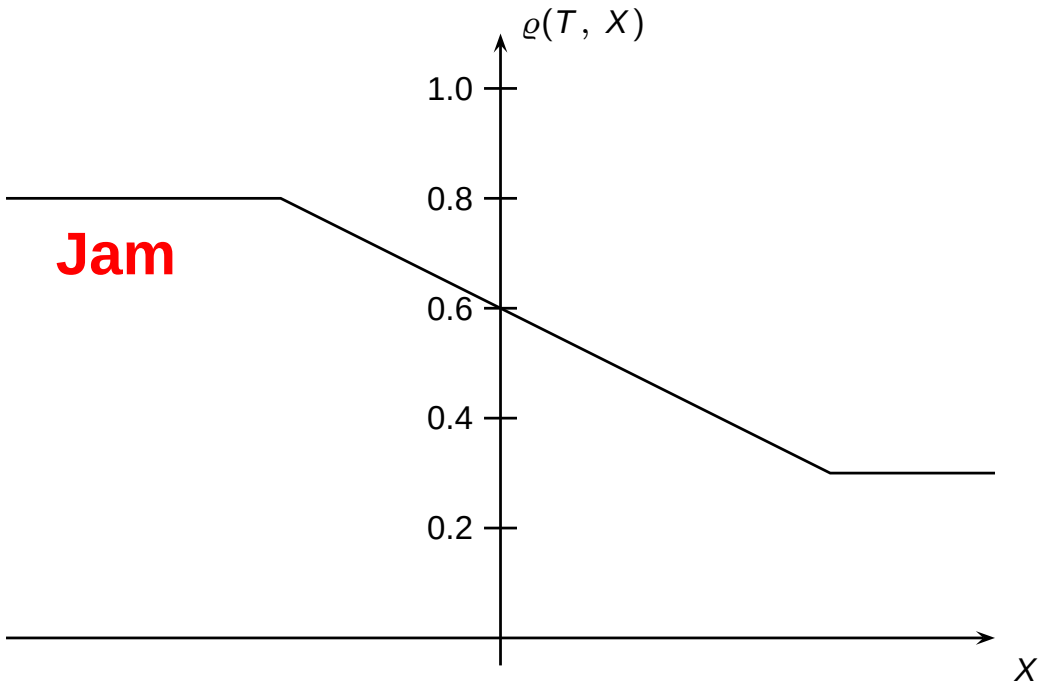
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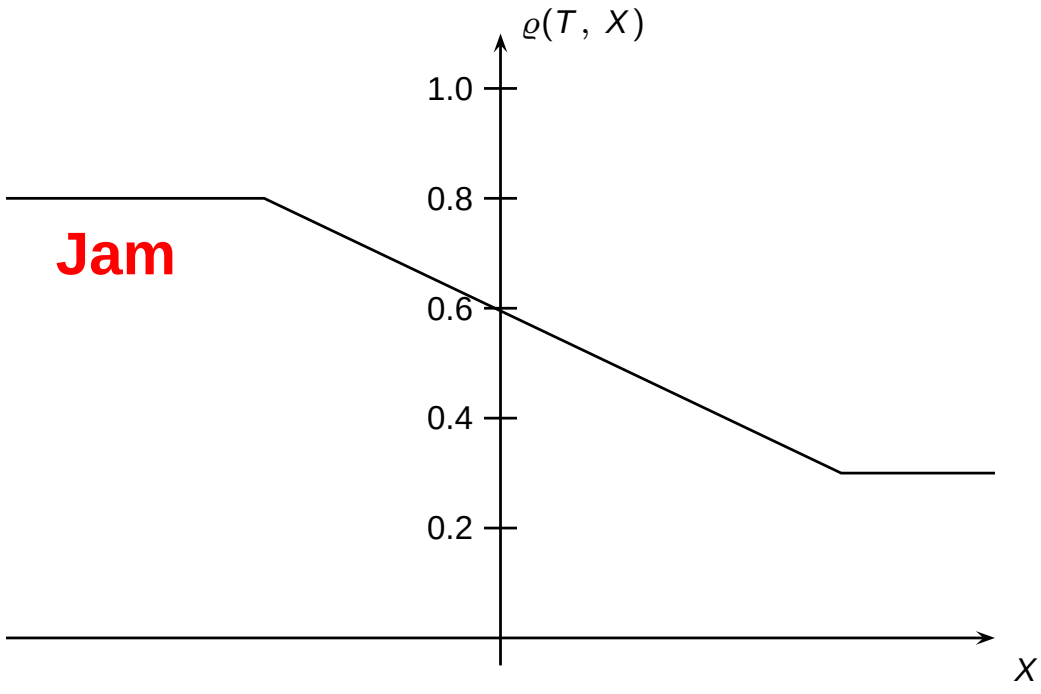
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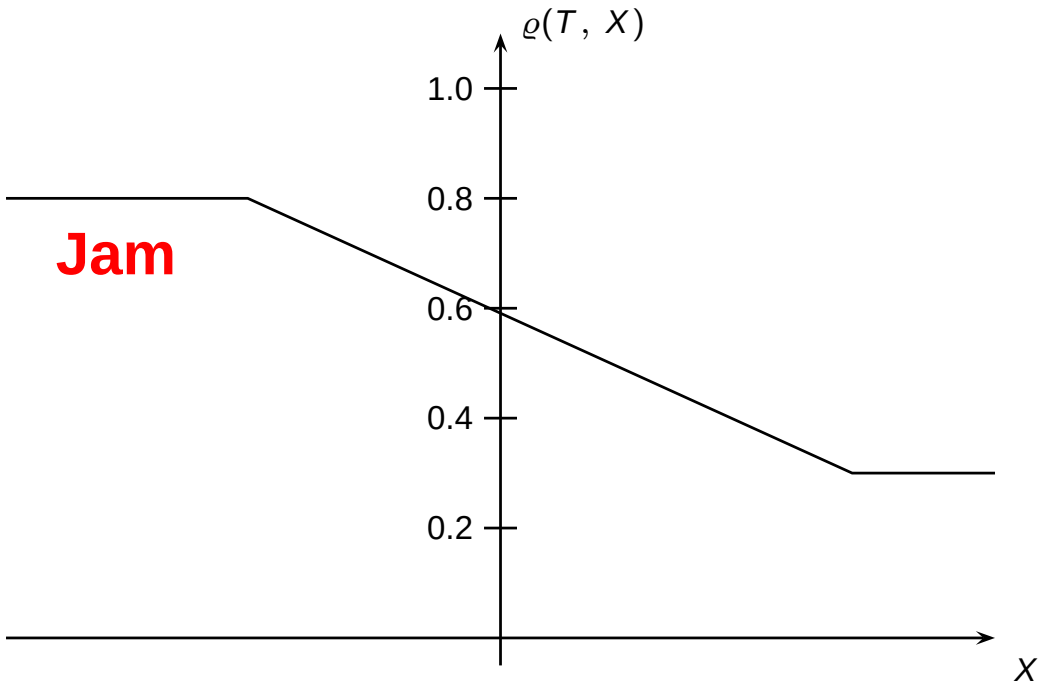
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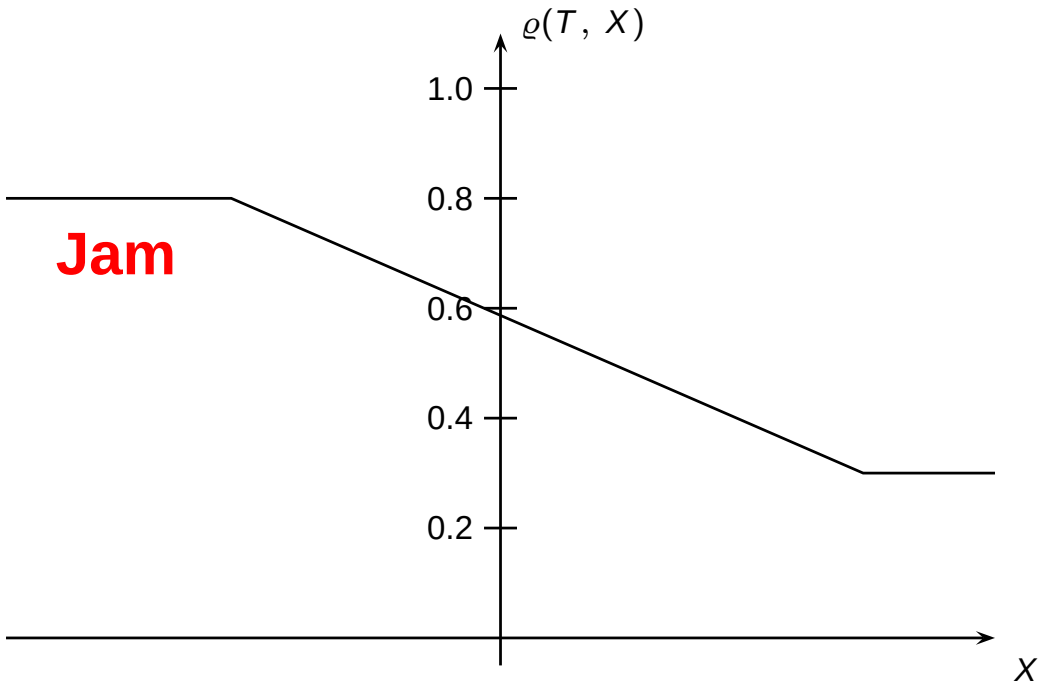
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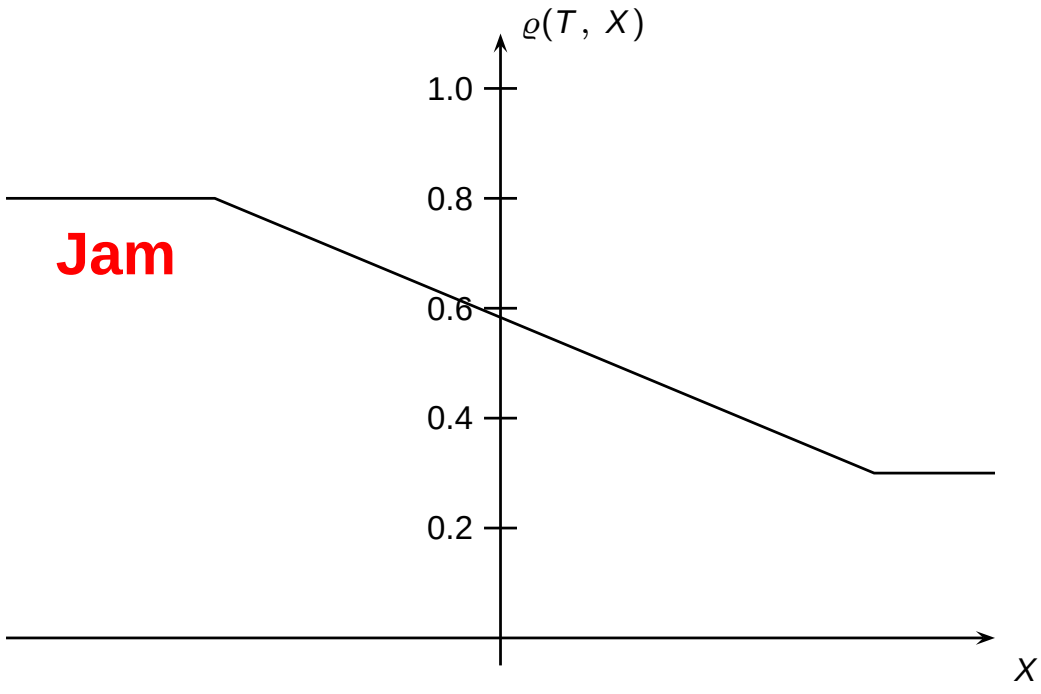
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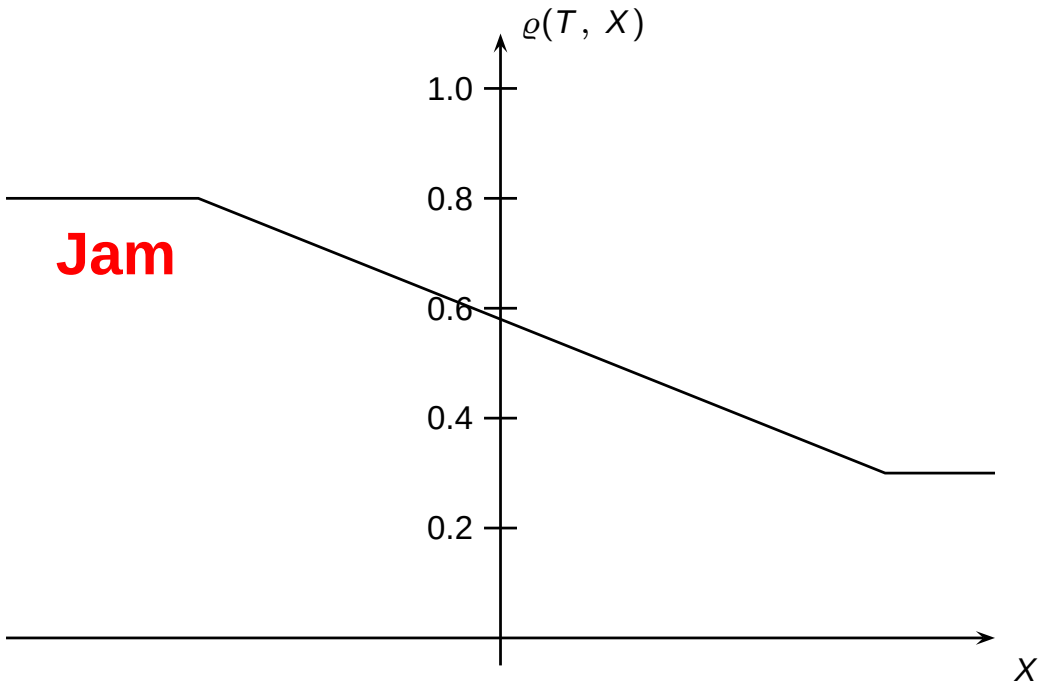
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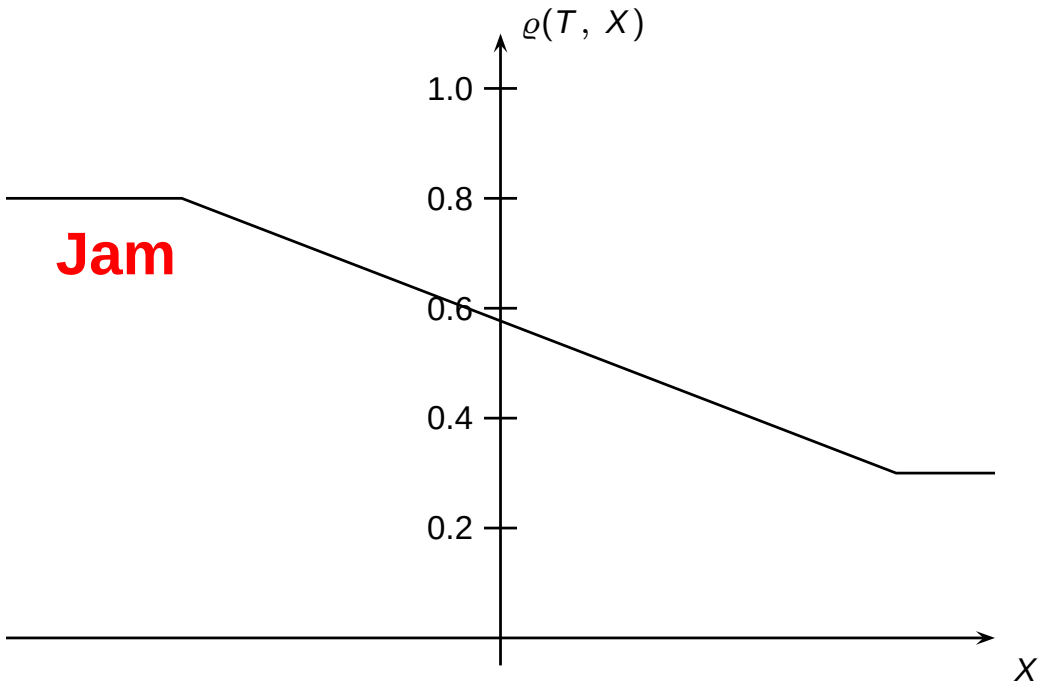
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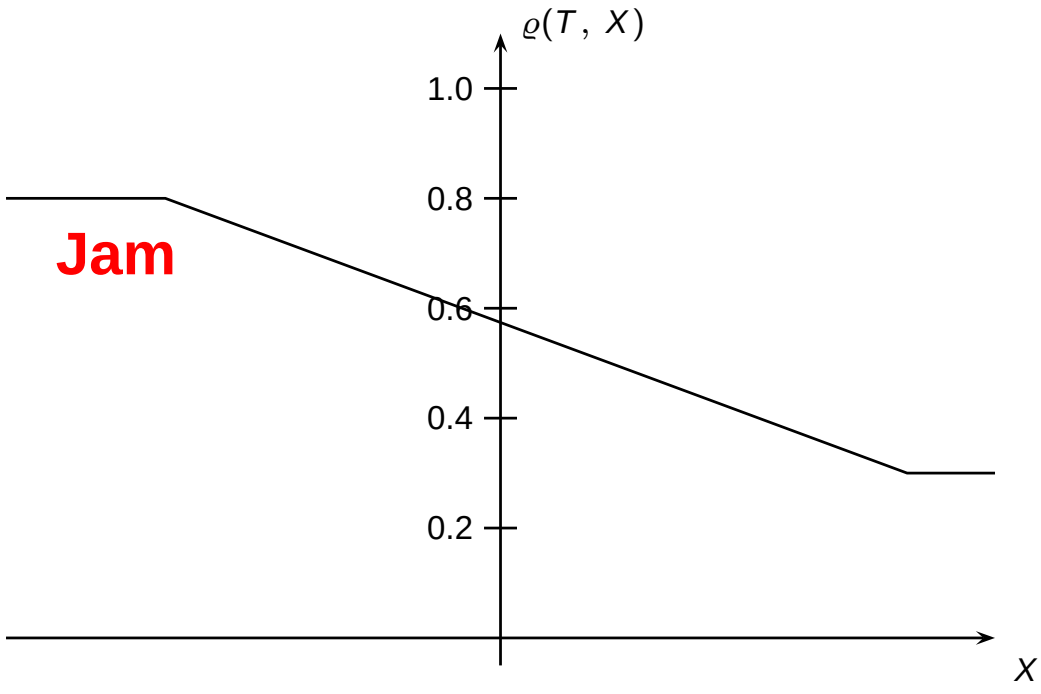
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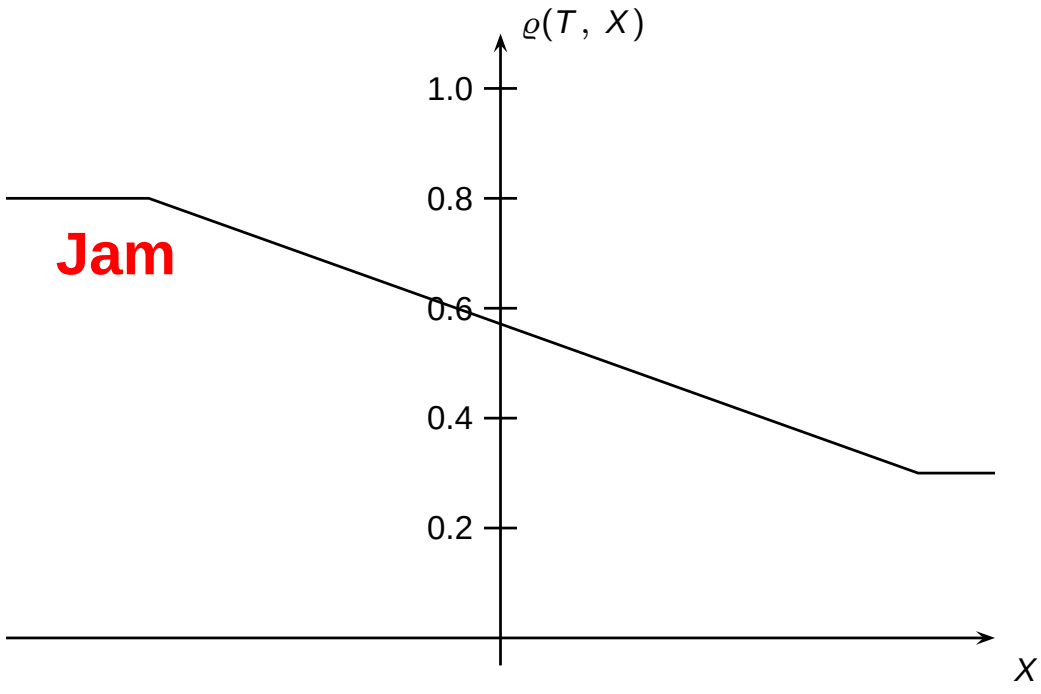
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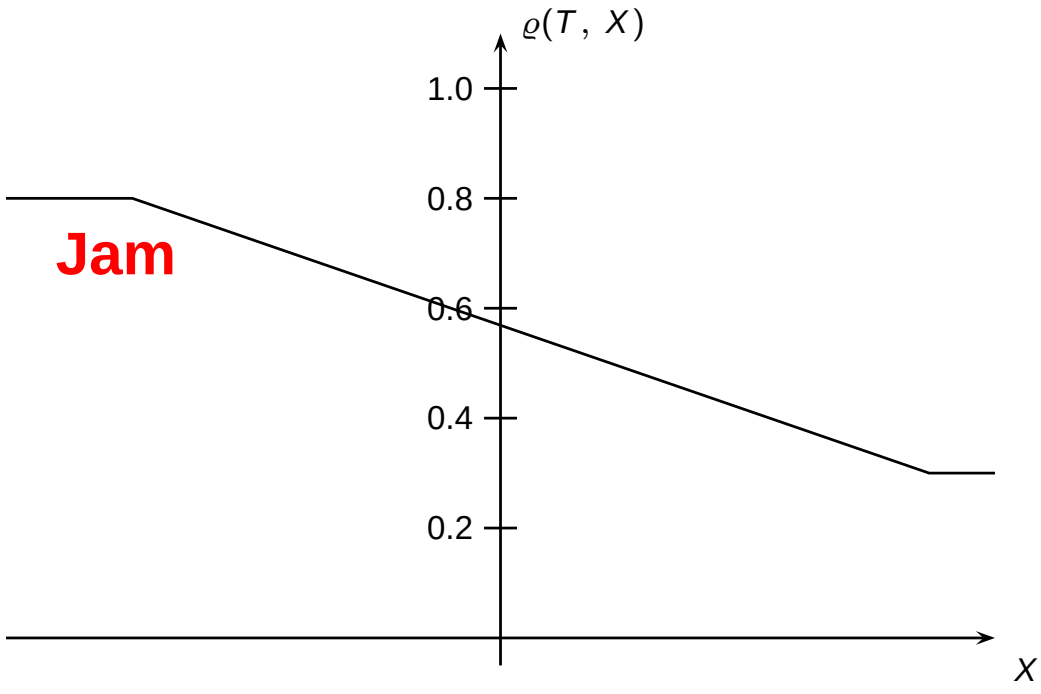
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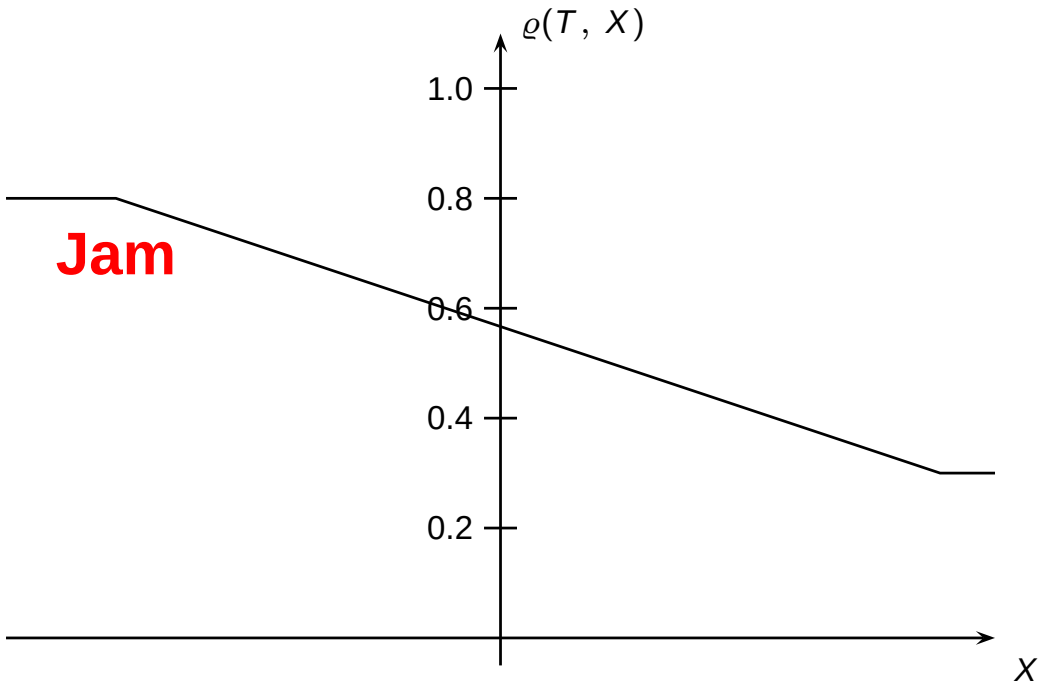
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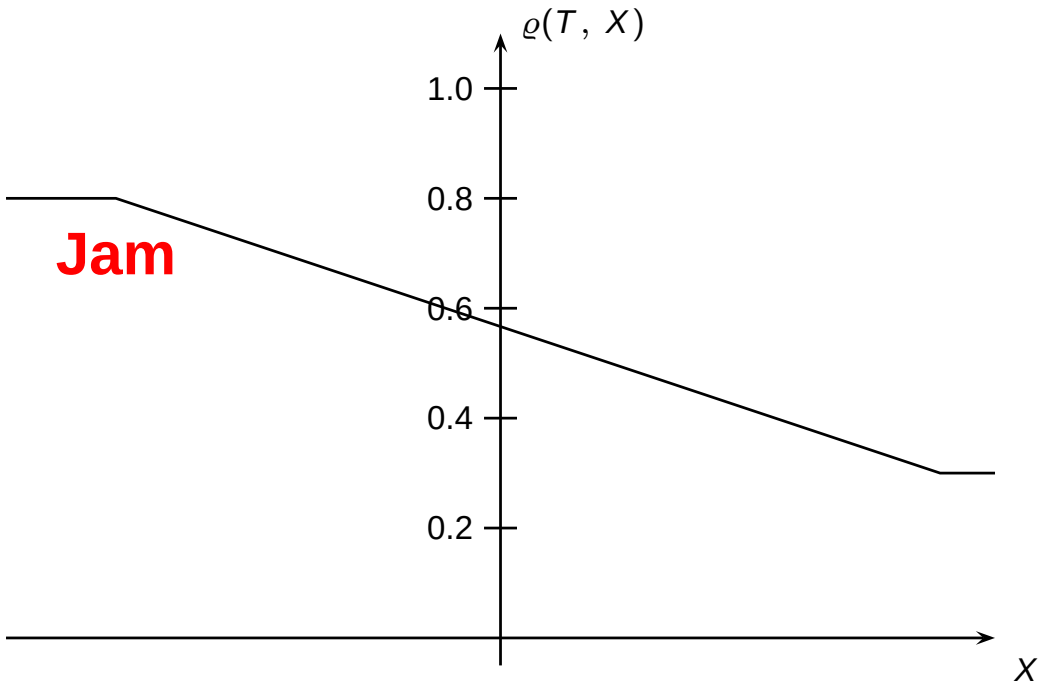
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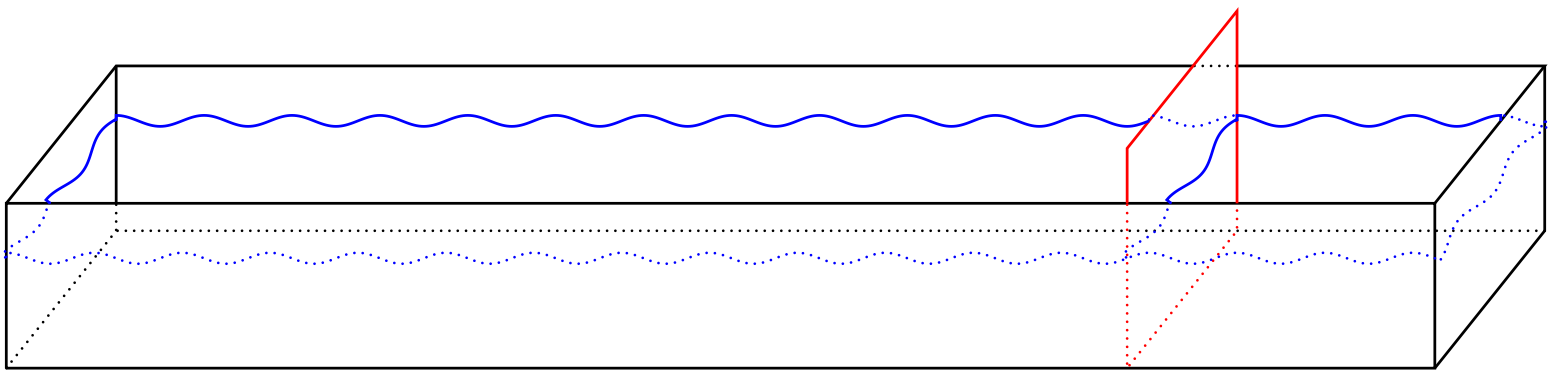
On large scales



End of the jam: smoothens.

Remarks.

In general, non-linear differential equations are fun. (And difficult.)
E.g., solitary waves were discovered by John Scott Russell in 1834: he chased one along a channel for miles!



`http://youtu.be/MADng1fqECY`

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Thank you.